

INDIAN MINERALS YEARBOOK

2024

GENERAL REVIEWS
METALS & ALLOYS
MINERAL REVIEWS



INDIAN BUREAU OF MINES



**Government of India
Ministry of Mines
Indian Bureau of Mines**

INDIAN MINERALS YEARBOOK 2024



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INDIAN BUREAU OF MINES

Pankaj Kulshrestha

Controller General

V. Jayakrishna Babu

Chief Controller of Mines (MES)

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Deputy Mineral Economists(Int.)

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Abdul Samad Khan

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Devesh Mali

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PREFACE

Indian Minerals Yearbook–2024 (IMYB–2024) is the 63rd edition in its series and constitutes three major sections – General Reviews, Metals & Alloys, and Mineral Reviews. The ‘General Reviews’ comprise topics that encompass country-wide & macro-level information on minerals. Attempts have been made to comprehensively depict the Indian Mineral Industry and its contribution to the National Economy, along with concise portrayal of Policy & Legislation, Research & Development and Exploration & Development that have relevance to the year 2023-24. Besides, an extensive chapter on State Reviews elaborately delves into the Mineral-wealth Status, Production and Industry of each of the States/UTs of the country. Topics on Prices, Production, Foreign Trade, Mineral-based Industries and Status of Mineral Concessions in India, form part of the Reviews contained in the General Reviews.

The second part of the book comprises Reviews on Metals & Alloys. This section primarily focuses on important metalliferous minerals, metals and alloys produced in the country and comprehensively cover details of the country’s ferrous, non-ferrous, strategic and precious metals and their related metallurgical industries. Each of the 19 Reviews provides valuable insights on resource/reserve positions, usage, consumption, R & D, trade, world scenario and future outlook of the metalliferous minerals/metals/alloys.

The last and final section of the Yearbook comprises ‘Mineral Reviews’ and contains 26 Reviews of important minerals produced in the country which are arranged in an alphabetical sequence. Each Review provides valuable insights on resource/reserve positions; production, stock & prices, mining, marketing & transport; usage & specifications; trade policy; world review; foreign trade; and future outlook of the minerals. The data coverage of this Edition, i.e., IMYB–2024 pertains to the year 2023-24.

It has been our continuous endeavour to improve upon the coverage and the content of the Yearbook and to present the entire spectrum of Minerals and Metals Sector to the maximum extent possible. Every update and nuances specific to Mineral Sector in terms of policy changes/guidelines issued/laws enacted and amendments made to them are carefully monitored and incorporated in the relevant chapters. The National Mineral Policy, 2019 which was introduced with an aim to ensure transparency, better regulation and enforcement, balanced socio-economic growth as well as sustainable mining practices have been assimilated into the content under relevant topics. Besides, legislative changes for Mines and Minerals Sector effected to make the sector growth-oriented and vibrant have been assigned special focus. Recently the Government of India classified Barytes, Felspar, Mica and Quartz as Major Minerals vide S.O. 924 (E) dated 20-02-2025

Indian Minerals Yearbook is a flagship title of Indian Bureau of Mines and its publication & release is the outcome of collective & coordinated efforts of the Bureau’s Mineral Economics Division and Mining & Mineral Statistics Division. The preparatory process of the yearbook encompasses assimilation of materials & inputs resourced from the reports of various Divisions of IBM. Various survey reports/annual reports, technical journals, periodicals of various organisations, including the affirmative responses received from the Mineral Industry on statutory and non-statutory basis have also been referred and inputs from related websites were drawn and incorporated wherever necessary to impart a sense of holism to the information covered under this Title.

The Bureau is indebted to Central and State Government Departments, Public Sector Undertakings, Public and Private Companies and Research Organisations, Mineral-based Industries & Associations concerned with mines and minerals for their support & cooperation in lending and sharing information. I firmly believe that the present edition of Indian Minerals Yearbook, i.e., IMYB–2024 will serve the interest of all its referring/reading clientele, who in the past have reposed such unshakable faith in the authenticity of the data/information published in this series.

Nagpur

Date: 19th January 2026

Pankaj Kulshrestha

Controller General
Indian Bureau of Mines

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EXPLANATORY NOTES AND SOURCES

The statistics presented in this publication are in metric units and the prices quoted are in the Indian currency unless otherwise stated.

The stage of measurement of quantity is normally the mine output which refers to the form in which the minerals are extracted. It also includes the usual processing operations done at the mine site to render the ore marketable. Exceptions to the above definition are gold and silver for which the metal output is considered, and for copper, lead & zinc, the concentrates.

The value of the mineral is reckoned in terms of the Ex-Mine Price which represents the sale value of the mineral at the mine site. The value of production of minerals is calculated by multiplying in each case the quantity of production and pit's mouth value per unit as furnished by the mine owners in the returns under MCDR 1988 in all cases except captive mines where the value is calculated on the basis of the cost of production. In case of fuel minerals, the production value figures in respect of coal & lignite are supplied by the Office of the Coal Controller, Kolkata, on annual basis. Regarding petroleum and natural gas (utilised), value published by the National Accounts Division, Central Statistical Office, is used. Value of sulphur produced as by-product from fertilizer plants and oil refineries is not included in the value of mineral production. The value of fuel minerals, non-ferrous metals is furnished by the respective units. The export valuation is on the basis of free on board (f.o.b.) inclusive of export duty, wherever such duty is levied. The basis of valuation of imports is the cost, insurance and freight (c.i.f.) value.

Break-ups may not add to total in some tables due to rounding-off.

Sources

The statistical data presented in this publication have been taken from a large number of sources as listed below:

Minerals other than fuels, atomic minerals and 'minor minerals'	The basic data relating to major minerals except coal, petroleum and natural gas are collected by IBM under Rule 45 of the MCDR, 1988 framed under the Mines and Minerals (Development and Regulation) Act, 1957. These Rules cover all the States and Union Territories of the Indian Union and apply to all minerals except i) petroleum and natural gas; ii) coal, lignite and sand for stowing; iii) minor minerals; and iv) any mineral declared as prescribed substance by Atomic Energy Act, 1962. Data on sulphur are collected from fertilizer plants and oil refineries.
Ilmenite, rutile, monazite, rare earths and zircon	Indian Rare Earths Ltd; Kerala Minerals and Metals Ltd; Department of Atomic Energy, Mumbai, and Private Sector processors.

Trade statistics

DGCI & S, Kolkata

Prices

a) Minerals	i) Principal producers and exporters ii) Coal Controller, Kolkata iii) Industrial Minerals (UK) iv) Basic Statistics on Indian Petroleum & Natural Gas, Ministry of Petroleum & Natural Gas, Government of India.
b) Metals	v) DGCI & S, Kolkata (Import Value) i) Producers and exporters ii) Reserve Bank of India Bulletin iii) World Metal Statistics (WBMS) iv) London Metal Exchange (Website) v) Minerals & Metals Review (Monthly/Yearly)

World information & statistics

- i) Mineral Commodity Summaries (USGS)
- ii) World Mineral Production (BGS)
- iii) Minerals Yearbook (USGS)
- iv) World Metal Statistics (WBMS)
- v) Mineral Industry Surveys (USGS)
- vi) Canadian Minerals Yearbook

Minerals Consumption

Data obtained on statutory and non-statutory basis from industrial units Consuming minerals/ores. Data have also been obtained in some cases from Central Government Ministries. The consumption indicated relates to the number of reporting units in the organised sector only. Estimated consumption data is based on statistical norms in vogue.

Reserves/ Resources

Reserves/resources of minerals in India have been taken from National Mineral Inventory prepared by IBM as per UNFC system. The source of information for the world resources of minerals is given against each mineral.

Port facilities

Annual Report of the Ministry of Shipping, Indian Ports Association, Major and Minor Port Authorities and exporters of minerals.

Research and Development

IBM's Ore Processing Laboratory, National Laboratories under the Council of Scientific & Industrial Research, and Ore Dressing Division of BARC and R&D laboratories in the Public/Private Sector.

Besides, Annual Reports of various Ministries of Government of India, Annual Reports, Brochures and Websites of Public Sector undertakings and private companies, Bulletins concerned with minerals and mineral-based industries, etc. were also referred.

Information/ data Liability Disclaimer

The reviews as presented in the Indian Minerals Yearbook are the product of the concerted efforts of in-house authors. In preparation of manuscripts, the authors resourced data/ information from various sources, such as, published information on the internet, various publications, Annual Reports, etc. Major chunks of information have actually been collated internally from the different Divisions of Indian Bureau of Mines which regularly and routinely are in the cycle of gathering data/information through correspondences.

All these sourced information/data that get included in the General/Metals & Alloys/Mineral Reviews are subjected to analyses, interpretations and sometimes extrapolations in the case of paucity of data. There have been instances when the data have been used 'as it is' hence it is recommended that the readers apply discretion in discerning the data for their further utilisation for general or scientific purposes.

Indian Bureau of Mines while processing of the sourced data/information undertakes its best efforts to ensure accuracy and to verify that the data published have been selected on the basis of thorough scientific judgement. However, IBM would make no warranties to that effect, and shall not be liable for any consequent damage that may result from errors or omissions in the data base contained therein.

ABBREVIATIONS

The abbreviations and symbols /units used in the Publication are as follows:

AMD	Atomic Minerals Directorate for Exploration and Research	IREL	Indian Rare Earths Ltd
APMDC	Andhra Pradesh Mineral Development Corp. Ltd	ISRO	Indian Space Research Organisation
BALCO	Bharat Aluminium Company Ltd	JPC	Joint Plant Committee
BARC	Bhabha Atomic Research Centre	JV	Joint Venture
BGML	Bharat Gold Mines Limited	KCC	Khetri Copper Complex
BGS	British Geological Survey, UK	KMML	Kerala Minerals & Metals Ltd
BIS	Bureau of Indian Standards	LAPL	Large Area Prospecting Licence
BISAG	Bhaskaracharya Institute of Space Applications & Geo-Informatics	LME	London Metal Exchange
BOT	Build, Operate, Transfer	MALCO	Madras Aluminium Company Ltd
BSMDC	Bihar State Mineral Development Corp. Ltd	M(A)R	The Mineral (Auction) Rules
CAPEXIL	Chemical and Allied Export Promotion Council	MCDR	Mineral Conservation and Development Rules
CBM	Coal Bed Methane	M(EMC)R	The Minerals (Evidence of Mineral Contents) Rules
CCI	Cement Corporation of India Ltd	MCR	Mineral Concession Rules
c.i.f.	Cost, Insurance and Freight	MECL	Mineral Exploration Corporation Ltd
CMDC	Chhattisgarh Mineral Development Corporation	ML	Mining Lease
CMPDI	Central Mine Planning & Design Institute	MMDR Act	Mines & Minerals (Development & Regulation) Act
CSO	Central Statistical Office	MMTC	Minerals and Metals Trading Corp. Ltd
DAE	Department of Atomic Energy	MoEFCC	Ministry of Environment, Forest and Climate Change
DES	Directorate of Economics & Statistics	MoU	Memorandum of Understanding
DGCI&S	Director General of Commercial Intelligence and Statistics	MSS	Mining Surveillance System
DGH	Directorate General of Hydrocarbons	MTS	Mining Tenement System
DGM	Directorate of Geology and Mining	MSTC	Metal Scrap Trade Corp. Ltd
DGPS	Differential Global Positioning System	NA	Not Available
DMG	Directorate of Mining and Geology	NAS	Not Available Separately
DMF	District Mineral Foundation	NALCO	National Aluminium Co. Ltd
EEZ	Exclusive Economic Zone	NCMT	National Centre of Mineral Targeting
EU	European Union	ND	Not Determined
FDI	Foreign Direct Investment	NELP	New Exploration Licensing Policy
FIMI	Federation of Indian Mineral Industries	NES	Not Elsewhere Stated
f.o.b.	free on board	NFL	National Fertilizers Ltd
f.o.b.t.	free on board trimmed	NLC	Neyveli Lignite Corporation Ltd
f.o.r.	free on rail	NMDC	National Mineral Development Corp. Ltd
GMDC	Gujarat Mineral Development Corp. Ltd	NMET	National Mineral Exploration Trust
GSI	Geological Survey of India	NMEP	National Mineral Exploration Policy
GVA	Gross Value Added	NMI	National Mineral Inventory
HCL	Hindustan Copper Ltd	NML	National Metallurgical Laboratory
HGML	Hutti Gold Mines Co. Ltd	NRSC	National Remote Sensing Centre
Hindalco	Hindalco Industries Ltd	NTPC	National Thermal Power Corp. Ltd
HZL	Hindustan Zinc Ltd	NQ	Not Quoted
IBM	Indian Bureau of Mines	N/v	Near Village/s
IMMT	Institute of Minerals & Materials Technology (Formerly RRL, Bhubaneswar)	OIL	Oil India Ltd
		OMC	Orissa Mining Corporation Ltd
		ONGC	Oil and Natural Gas Corporation Ltd

PMKKKY	Pradhan Mantri Khanij Kshetra Kalyan Yojana	TSL	Tata Steel Ltd (formerly Tata Iron and Steel Co. Ltd)
PL	Prospecting Licence	TW	Territorial Waters
PPP	Public Private Partnership	UAE	United Arab Emirates
RP	Reconnaissance Permit	UK	United Kingdom
RRL	Regional Research Laboratory	UNFC	United Nations Framework Classification
RSMML	Rajasthan State Mines and Minerals Ltd	USA	United States of America
SAIL	Steel Authority of India Ltd	USGS	United States Geological Survey
SBICAP	SBI Capital Markets Limited	UT	Union Territory
SCCL	Singareni Collieries Company Ltd	VE	Visual Estimate
SDF	Sustainable Development Framework	VISL	Visvesvaraya Iron & Steel Ltd
SEZ	Special Economic Zone	w.e.f.	with effect from
SMC	Sikkim Mining Corporation Ltd	(e)	Estimated
STD	Standard (Code of UNFC)	(P)	Provisional
TAMIN	Tamil Nadu Minerals Ltd	(R)	Revised
TAMRA	Transparency, Auction Monitoring and Resource Augmentation	(U)	Under reference
TERI	The Energy and Resources Institute	--	Nil
tpd	tonnes per day	++	Negligible
tpy	tonnes per year		

Units

cm	centimetre	t	tonne
m	metre	'000	tonnes thousand tonnes
mm	millimetre	lkm	line kilometre
cu m	cubic metre	ct	carat
'000 cu m	thousand cubic metres	g	gram
m cu m	million cubic metres	kg	kilogram
sq m	square metre	₹	Indian rupees
km	kilometre	₹ '000	thousand rupees
ha	hectare	kWh	kilowatt-hour
sq km	square kilometre	s	second

Conversion Table

Troy oz	31.1035 g	cwt	112 lb
kg	2.2046 lb	foot	0.3048 m
tonne	Metric ton of 2,204.6 lb	Crore	Ten million
ton	Long ton of 2,240 lb	Lakh	Hundred thousand

Classification of Reserves/Resources of Various Minerals as per United Nations Framework Classification (UNFC) System

The classification of reserves/resources of various minerals based on UNFC system were first prepared by IBM as on 1.4.2000 and later, as on 1.4.2005. Reserves/resources are furnished mineralwise in State Reviews and gradewise and statewise in Mineral Reviews. Quinquennially updated resources for 43 minerals as on 1.4.2010, for 70 minerals as on 1.4.2015 and for 46 minerals as on 1.4.2020 have been included in this edition of Indian Minerals Yearbook in State Reviews and Mineral Reviews. The amendment to Mineral Conservation & Development Rules, 1988 vide Gazette Notification No.185 dated 17.4.2003 makes it statutory for all non-coal major mineral mine-owners to report their reserves data as per UNFC and for Mining Lease applications to submit mining plans accordingly. Detailed guidelines, definitions, etc. concerning UNFC were issued by IBM on 3rd June 2003 and published in the edition of Mineral Conservation & Development Rules, 1988.

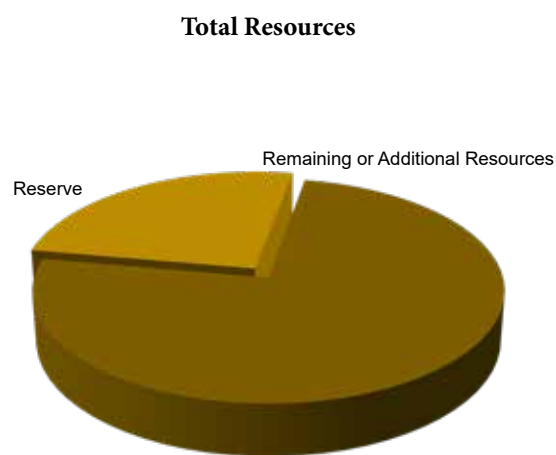
The UNFC consists of a three-dimensional system with the following three axes: Geological Assessment, Feasibility Assessment and Economic Viability. The process of geological assessment is generally conducted in stages of increasing details. The typical successive stages of geological investigation, i.e., reconnaissance, prospecting, general exploration and detailed exploration, generate resource data with a clearly defined degree of geological assurance. These four stages are, therefore, used as geological assessment categories in the classification. Feasibility assessment studies form an essential part of the process of assessing mining project. The typical successive stages of feasibility assessment, i.e, Geological study as initial stage followed by prefeasibility study and feasibility study/mining report are well-defined. The degree of economic viability (economic or sub-economic) is assessed in the course of prefeasibility and feasibility studies. A prefeasibility study provides a preliminary assessment with a lower level of accuracy as compared to that of a feasibility study which assess the economic viability in detail.

It is a three-digit-code-based system, the economic viability axis representing the first digit, the feasibility axis, the second digit and the geologic axis, the third digit. The three categories of economic viability have codes 1, 2 and

3 in decreasing order. Similarly, the three categories of feasibility study have also codes 1, 2 and 3 while the four stages of geological assessment are represented by 4 codes, i.e., 1 (detailed exploration), 2 (general exploration), 3 (prospecting) and 4 (reconnaissance). Thus, the highest category of resources under UNFC system will have the code (111) and lowest category, the code (334). The various terms used in this classification and their definitions in brief are as follows

Total Mineral Resources

Reserve plus Additional or Remaining Resource comprise the Total Resource, or Total Resource minus Reserve gives the Remaining Resource



Diagrammatic Representation of Reserve and Resource

A. Mineral Reserve

Economically mineable part of measured and/or indicated mineral resource.

(i) Proved Mineral Reserves (111)

Economically mineable part of Measured Mineral Resource.

(ii) Probable Mineral Reserves (121 & 122)

Economically mineable part of indicated or in some cases, a measured mineral resource.

B. Mineral Resource

A Mineral Resource (Remaining or Additional Resource) is the balance of the Total Mineral Resources that have not been identified as Mineral Reserve.

(i) Measured Mineral Resource (331)

That part of mineral resource for which tonnage, density, shape, physical characteristics, grade and mineral content can be estimated with a high level of confidence, i.e., based on detailed exploration.

(ii) Indicated Mineral Resource (332)

Tonnage, density, shape, physical characteristics grade and mineral content can be estimated with reasonable level of confidence based on exploration, sampling and testing information, location of borehole, pits etc.

(iii) Inferred Mineral Resource (333)

Tonnage, grade and mineral content can be estimated with low level of confidence inferred from geological evidence.

(iv) Reconnaissance Mineral Resource (334)

Estimates based on regional geological studies and mapping, airborne and indirect methods, preliminary field inspections as well as geological inference and extrapolation.

(v) Prefeasibility Mineral Resource (221 and 222)

That part of an indicated and in some circumstances measured mineral resource that has been shown by prefeasibility study as not economically mineable or can become economically viable subject to changes in technological, economic, environmental and/or other relevant conditions.

(vi) Feasibility Mineral Resource (211)

That part of measured mineral resource, which after feasibility study has been found to be economically not mineable.

Definition of Uneconomic Occurrence

Materials of estimated quantity, that are too low in grade or for other reasons are not considered potentially economic. Thus, Uneconomic Occurrence is not part of a mineral resource. If quantity and quality are considered worthy of reporting, it should be recognised that an Uneconomic Occurrence cannot be exploited without major technological and/or economic changes, which are not currently available.

Mineral Occurrence

A mineral occurrence is an indication of mineralization that is worthy of further investigation. The term mineral occurrence does not imply any measure of volume/tonnage or grade/quality and is thus not part of a mineral resource.



General Reviews

1. Indian Mineral Industry and National Economy

7.5 %

increase in Index of Mineral Production (excluding atomic and minor minerals) during the year 2023-2024 over the year 2022-23

1,359

mines (excluding Fuel Minerals, Atomic and Minor Minerals) were reported in the Country during 2023-24.

Mining & Quarrying Sector accounted for approximately 1.97% of the total GVA in 2023-24.

India displayed reasonably steady economic growth in 2025. As per the projections made of the advance estimates of national accounts, India's real GDP is expected to grow by 6.4 per cent in FY25. Growth estimates were largely on account of impressive performances seen in Agriculture and Services sectors. With rural demand improving on the back of favourable agricultural conditions, the forecasts of positive momentum in growth prospects were apparently made visible in the first quarter performance of FY 2025. The Manufacturing Sector faced constraints as a result of weak global demand and domestic fluctuations. Private consumption remained positive, reflecting steadiness in domestic demand. The overall macroeconomic stability witnessed was mainly an outcome of strict fiscal discipline and strong external balance supported by a services trade surplus and healthy remittance growth. The cumulative effect of these factors contributed to the circumstances for sustained growth amid inherent uncertainties.

With strong fundamentals in place, India's economic prospects for FY26 are poised to head toward balanced path amid headwinds in the form of elevated geopolitical and trade uncertainties and possible commodity price shocks. Domestically, the higher revenue prospects through private capital goods sector will ensure sustained investment pick-up, improvements in consumer confidence, and corporate wage pick-up and these will be key to promoting growth. Besides, rural demand backed by a rebound in agricultural production, an anticipated easing of inflationary pressures and a stable macro-economic environment provide an upside to near-term growth. It is crucial for India to implement structural reforms at the grassroot level in order to be globally competitive and to initiate deregulation to

reinforce its medium-term growth potential.

The National Statistical Office, Ministry of Statistics & Programme Implementation (MoSPI), in its first advance estimates has predicted the real gross domestic product (GDP) growth for FY25 to be 6.4 per cent. From the purview of aggregate demand in the economy, Private Final Consumption Expenditure (PFCE) at constant prices is poised to grow by 7.3 per cent, driven mainly by a rebound in rural demand. PFCE as a share of GDP (at current prices) is estimated to increase from 60.3 per cent in FY24 to 61.8 per cent in FY25. This share is the highest since FY03. Gross fixed capital formation (GFCF) (at constant prices) is estimated to grow by 6.4 per cent.

The supply side perspective, on the other hand, reveal that in India the real gross value added (GVA) is also expected to grow by 6.4 per cent primarily due to improved performances of the Agriculture Sector which expected to rebound to a growth of 3.8 per cent in FY25. Similarly, the Industrial Sector is estimated to grow by 6.2 per cent in FY25—strongly driven by growth in construction activities and electricity, gas, water supply and other utility services. Growth in the Services Sector is expected to remain robust at 7.2 per cent, propelled by positive activity in financial, real estate, professional services, public administration, defence and other services.

The widespread disruptions caused by COVID-19 pandemic to economies worldwide have had disastrous impact in India as well. Economic Survey 2023-24 in its comparative analysis of the post-pandemic trends until Q4 FY24 with the pre-pandemic scenario revealed that the economy grew in a resilient manner and absorbed the shocks to avert any permanent loss of output. The details of the study though presaged caution, the recovery trajectory was indeed encouraging. Aggregate GVA surpassed its pre-pandemic trend in Q1 FY25, and it now hovers above the trend in the H1 FY25. The Agriculture Sector recovered briskly, consistently performing and regaining grounds. The Industrial Sector also managed to find its footing. The Services Sector after experiencing massive downturn has now turned the corner and is showing rapid recovery and expansion prospects.

The standpoint of the Industrial sub-sectors is that of a string of high performances with its core segments, i.e., Construction performing at exceptional levels soaring approximately 15 per cent above its pre-pandemic levels chiefly as a consequence of robust infrastructure development and housing demand. The Utilities Sector, that includes electricity, gas, water supply and other services, regained momentum and levelled its pre-pandemic trend by the end of FY23 and has consistently performed at reasonably high order. Manufacturing, though showed steady recovery, remained marginally below its pre-pandemic trajectory. Mining, however, lagged below its pre-pandemic trend.

From a demand perspective, Private Final Consumption Expenditure (PFCE) during H1 FY25 reflected growth of 6.7 per cent YoY with indicators, such as, upward trend of 2-wheeler and 3-wheeler sales and tractor sales signalling clearly that rural demand contributed to private consumption growth. The National Bank for Agriculture and Rural Development (NABARD), in its Rural Economic Conditions and Sentiments Survey too reflected similar trends, i.e., increase in consumption expenditure by 78.5 per cent of rural households. The rural demand is expected to continue with the much anticipated favourable agriculture conditions.

On the flipside, urban demand indicators presented mixed trends. According to data from the Federation of Automobile Dealers Associations (FADA), there was decline in growth of passenger vehicle sales by about 4.2 per cent YoY in April–November 2024 as compared to 9.2 per cent in the corresponding period of the previous year. Fast-Moving Consumer Goods (FMCG) sales in urban areas, reflected a moderate growth in H1 FY25.

The Industrial Sector (which includes four sub-sectors, i.e., mining and quarrying; electricity, gas, water supply & utilities; manufacturing; and construction) that suffered significant contraction in FY21 on account of the pandemic showed signs of growth fluctuations in subsequent years. The industrial growth in FY25 was expected to rise and consistent with the predictions grew by 6.2 per cent in (Q1) FY25, driven by robust growth in electricity and construction segments.

However, on account of sharp slowdown in manufacturing exports mostly as a consequence of intensified trade competitions and unfavourable industrial policies of trading nations, industrial growth slumped by 3.6% in the second quarter of FY25. The cascading effect of the slowdown was noticeably reflected in the downturn seen in the mining, construction and to an extent in manufacturing.

The Mining Sector contributes about 8 per cent to the total industrial output. The mining activity concentrated mostly among five States, that is, Assam, Chhattisgarh, Gujarat, Maharashtra and Odisha, account for about 60 per cent of the all-State mining Gross State Value Added (GSVA).

The Mining & Quarrying Sector accounted for approximately 1.97% of the total GVA in 2023-24. The Gross Value Added at Basic Price in 2023-24 (at current prices) reached an estimated ₹ 5,25,881 crore registering a 6.3% growth from that of the previous year's estimate.

However, as per GVA of Basic Price (at 2011-12 prices) as on 31.05.2024 for the Mining & Quarrying Sector registered an increase of 7.1% at ₹ 3,37,623 crore as compared to ₹ 3,15,256 crore registered in the previous year.

In terms of degree of self-sufficiency, India, in the Principal Minerals & Metals category, is nearly 100% in respect of Iron Ore, Aluminium & Zinc and to a large extent in limestone.

India's Merchandise Trade

India's trade sector has demonstrated remarkable stability and growth, achieving milestones despite global economic headwinds. Following a dip in FY20 amid the global downturn and the pandemic, overall exports rebounded strongly in FY22, reaching a record high in FY23. This

momentum continued into FY24, with overall exports surpassing the previous year's record, even as imports moderated slightly.

Export

India's total exports (merchandise+services) have shown positive momentum in the first nine months of FY25, reaching USD 602.6 billion, witnessing a YoY growth of 6 per cent. This increase demonstrates the resilience of exports, which have been building on a steady upward trend in recent years despite global economic challenges. Total imports during April–December 2024 reached USD 682.2 billion, registering a YoY growth of 6.9 per cent. This positive import growth indicates a steady demand for goods in the Indian market, supporting domestic consumption and production needs. A more significant increase in overall imports compared to exports led to a rise in the overall trade deficit from USD 69.7 billion during April–December 2023 to USD 79.5 billion in the corresponding period of FY25.

Import

Merchandise imports grew by 5.2 per cent during April–December 2024. This increase was largely due to a rise in non-oil, non-gold imports, which reached USD 352.1 billion in the first nine months of FY25 compared to USD 340.5 billion during the same period last year, indicating a rebound in domestic consumption despite the inflationary impact. Gold imports increased due to higher international prices, driven by frontloading ahead of festival spending and demand for safe-haven assets. Among the major non-oil, non-gold imports, non-ferrous metals, machine tools, machinery, electrical & non-electrical goods, and transport equipment witnessed remarkable growth in the first nine months of FY25, reflecting growing demand for capital goods. Electronic goods also maintained consistent

momentum, signalling a rise in discretionary consumer spending.

Foreign Direct Investment (FDI)

India remains a strong destination for FDI, ranking high in greenfield project announcements and international project finance deals. However, the country has to pay heed to numbers. As per the data published by RBI, net FDI to India during the first eight months of FY25 stood at USD 0.48 billion compared to USD 8.5 billion in the corresponding period of FY24. Similarly, for FY23, the figure stood at USD 19.8 billion. For FY24 as a whole, the net FDI was USD 10.1 billion. The last two financial years have indeed seen much larger repatriation from India. The amounts were USD 29.3 billion and USD 44.5 billion, respectively, in FY23 and FY24. In the current year, up to November, the repatriation amount is USD 39.6 billion. At this rate, the full-year figure might exceed last year's figure.

As per DPIIT report statement on sector-wise FDI equity inflows from April 2000 to March 2025 in Mining Sector was ₹ 21,451.11 crore.

MINING INDUSTRY

The index of mineral production (excluding atomic and minor minerals) (with base year 2011-12=100) for 2023-24 at 128.9 displayed an increase of 7.5 % as compared to the previous year.

The value of Metallic Minerals in 2023-24 at ₹ 1,28,632 crore increased by about 16.1 % over that of the previous year.

The value of production of non-metallic minerals at ₹ 13,171 crore during 2023-24 increased by 5.5 % as compared to that of the previous year (Table-1).

**Table – 1: Indian Mineral Industry: Value of Mineral Production*
2021-22 to 2023-24**

Sector	(In ₹ crore)		
	2021-22 (R)	2022-23 (R)	2023-24 (P)
Total: All Minerals	137421	123317	141803
Metallic minerals	126428	110829	128632
Non-metallic minerals	10993	12488	13171

* Excluding the minerals declared as prescribed substances under the Atomic Energy Act, 1962; fuel minerals and minor minerals.

Reporting Mines

Reporting mine is defined as “A mine reporting production or reporting ‘nil’ production during a year but engaged in developmental work, such as, overburden removal, underground driving, winzing, sinking work, exploration by pitting, trenching or drilling as evident from the MCDR returns”.

There were 1,359 reporting mines (excluding Fuel Minerals, Atomic and Minor Minerals) in India located in different States and UTs during 2023-24. Among them, 543 belong to Metallic Minerals and 816 to Non-metallic Minerals. There were 171 mines in Public Sector and the rest of 1,188 mines were in Private Sector (Table-2).

**Table – 2: Number of Reporting Mines#
2022-23 and 2023-24**

Sector	2022-23	2023-24 (P)
All Minerals	1408	1359
I (i) Public sector	161	171
(ii) Private sector	1247	1188
II (i) Metallic minerals	577	543
(ii) Non-metallic minerals	831	816

Note: #: Excluding atomic, fuel and minor minerals.

Role of Public Sector

The Public Sector has played important role in the overall mineral production in 2023-24. The entire production of copper ore & conc., among Metallic Minerals and diamond, rock-salt & sulphur in respect of Non-metallic Minerals was reported from the Public Sector. By and large, the entire production of gold, tin conc. and phosphorite came from Public Sector during 2023-24.

Gross Value Added from Mining & Quarrying Sector

The Ministry of Statistics & Programme Implementation

has released the provisional estimates of national income, revising the base year from 2004-05 to 2011-12 in the year 2015. The industry-wise estimates are now presented as Gross Value Added (GVA) at basic prices. Certain changes have been made in this series including for Mining & Quarrying Industry. During 2023-24, Mining and Quarrying Industry accounted for about 1.97% of the GVA at current prices. The GVA at current and constant prices for the period from 2021-22 to 2023-24 is furnished in Tables 3 and 4.

**Table – 3: Gross Value Added at Basic Price, 2021-22 to 2023-24
(At Current Prices) (31.05.2024)**

(Value in ₹ Crore)

Industry	2021-22	2022-23 (FRE)	2023-24 (PE)	"% Change in 2023-24 over the previous year"
GVA (All)	21635584	24659041	26762147	8.5
Mining & Quarrying	439339	494602	525881	6.3

Source: NAD, MoSPI FRE: First Revised Estimates PE: Provisional Estimates

**Table – 4: Gross Value Added at Basic Price, 2021-22 to 2023-24
(At 2011-12 prices) (31.05.2024)**

(Value in ₹ Crore)

Industry	2021-22	2022-23 (FRE)	2023-24 (PE)	"% Change in 2023-24 over the previous year"
GVA (All)	13876840	14804901	15873751	7.2
Mining & Quarrying	309276	315256	337623	7.1

Source: NAD, MoSPI FRE: First Revised Estimates PE: Provisional Estimates

Employment

The average daily employment of labour engaged in Mining Sector (excluding Fuel Minerals, Atomic and Minor Minerals) was 1,28,970 in 2023-24. Metallic Minerals accounted for 80.77% (1,04,165) and Non-metallic Minerals 19.23% (24,805) of the total labour force during the year.

Rank of India in World Mineral Production

As per World Mineral Production, 2019-23, British Geological Survey, India's ranking in world production in terms of quantity in the year 2023 was 2nd in aluminium (primary) & steel (crude/liquid); followed by 3rd in chromite, iron ore & zinc (slab); 5th in manganese ore; 6th in bauxite; 10th in copper (refined) & lead (refined); 15th in magnesite and 18th in apatite & rock phosphate (Table- 5).

Table – 5: Contribution and Rank of India in World Production of Principal Minerals & Metals, 2023

Sector	Unit of Commodity	Production quantity		Contribution (Percentage)	India's rank in World order ^{\$}
		World	India*		
Metallic Minerals					
Bauxite	'000 tonnes	416300	23968	5.75	6 th
Chromite	'000 tonnes	37500	3148	8.39	3 rd
Iron ore	million tonnes	2507	277	11.05	3 rd
Manganese ore	'000 tonnes	53000	3381	6.38	5 th
Industrial Minerals**					
Magnesite	'000 tonnes	24200	132	0.55	15 th
Apatite & Rock phosphate	'000 tonnes	237000	1558	0.66	18 th
Metals					
Aluminium (primary)	'000 tonnes	69900	4159 [#]	5.95	2 nd
Copper (refined)	'000 tonnes	27600 ^a	509 [#]	1.84	10 th
Steel (crude/liquid)	million tonnes	1877	141	7.51	2 nd
Lead (refined)	'000 tonnes	15000 ^b	216 [#]	1.44	10 th
Zinc (slab)	'000 tonnes	14200	817 [#]	5.75	3 rd

Source: World Mineral Production, 2019-2023, British Geological Survey for world production and MCDR returns & individual plants for production with respect to India.

Note: Data in respect of World Mineral Production is on calendar year basis, however the data on India's production is based on financial year.

* Figures relate to financial year 2023-24 (P) and MCDR returns for production data.

** As per Government of India Notification S.O. 423(E) dated 10th February, 2015, following minerals have been declared as minor minerals: i) barytes ii) dolomite iii) felspar iv) fireclay v) quartz/silica sand and vi) talc/steatite/soapstone & pyrophyllite, hence not included in the table due to non-availability of production data with respect to India.

\$: India's rank is based on world's production mentioned in World Mineral Production 2019-23; British Geological Survey and considering India's production as per MCDR returns.

#: As per World Mineral Production, 2019-23, the production of Aluminium (primary), Copper (refined), lead (refined) and Zinc (Slab) in India during 2023 are 3306 thousand tonnes, 509 thousand tonnes, 1038 thousand tonnes and 748 thousand tonnes, respectively.

a: Figures relate to both primary and secondary refined copper, whether electrolytic or fire refined. Metal recovered from secondary materials by remelting alone is excluded.

from secondary materials by remelting alone is excluded.

b: Figures related to both primary & secondary refined lead and include the lead content of antimonial lead.

POLICY

National Mineral Policy

National Mineral Policy, 2019, was approved by the Union Cabinet, on 28th February 2019. The aim of National Mineral Policy, 2019, is to have a more effective, meaningful and implementable policy that ensures greater transparency, better regulation and enforcement, balanced social and economic growth as well as sustainable mining practices.

The National Mineral Policy, 2019, includes provisions which aim to boost the Mining Sector, such as,

- introduction of Right of First Refusal for RP/PL holders,
- encouraging the Private Sector to take up exploration,
- auctioning of virgin areas for composite RP-cum- PL-cum-ML on revenue share basis,
- encouragement of merger and acquisition of mining entities,
- transfer of mining leases and creation of dedicated mineral corridors to boost Private Sector mining areas,
- proposal to grant status of industry to mining activity to boost financing of mining for Private Sector and for

acquisitions of mineral assets in other countries by Private Sector,

- proposal to auction mineral blocks with prebedded clearances to give fillip to auction process,
- proposal to make efforts to harmonise taxes, levies & royalty with global benchmarks to help Private Sector.

The NMP-2019 will ensure more effective regulation and promote sustainable development of the mining sector, while addressing the issues of project-affected persons especially those residing in tribal areas.

Star Rating of Mines

Ministry of Mines, in its endeavour to ensure exhaustive and universal implementation of the Sustainable Development Framework (SDF) in mining, has developed a system of Star Rating of Mines.

The Ministry of Mines instituted the Sustainable Development Framework (SDF) for taking up mining activity, encompassing inclusive growth, without adversely affecting the social, economic and environmental well-being of the present and future generations. It has been instituted as a two-tier system consisting of self-evaluation templates to be filled in by the mine operator followed by

validation through Indian Bureau of Mines.

The evaluation templates for Star Rating was notified vide Notification dated 23.05.2016 for major minerals.

Based on the performance of the mining lease, 1 to 5 star rating would be awarded. The prospect of getting higher Star Rating is expected to drive miners to quickly adopt sustainable mining practices. In the Mineral Conservation & Development Rules, 2017, Star Rating for mines has been included as statutory provision for achieving a minimum of 3 stars.

A web enabled online system for evaluation of measures has been developed and launched on 18th August, 2016 as a vital step for ensuring compliance of environmental protection and social responsibility by the Mining Sector. A template for star rating of minor minerals is also being prepared.

During the year 2022-23, so far a total of 1,254 SDF templates for the assessment have been filed by the lessees. Validation of the submitted template for final evaluation is under progress and so far field verification of 803 leases has been completed.

LEGISLATIVE FRAMEWORK

The details of Legislative Framework are provided in the Review on "Mineral Policy and Legislation" under "General Review".

Auction of Mineral Blocks

As per information available on the website of Ministry of Mines as on 31.03.2024, a total of 354 mineral blocks have been auctioned successfully across 11 States.

Measures for Controlling Illegal Mining

Illegal mining means any reconnaissance or prospecting or mining operation undertaken by any person or a company in any area without holding a reconnaissance permit or a prospecting licence or, as the case may be, a mining lease as required under Sub-section (1) of Section 4 of the MMDR Act. Section 23C of Mines and Minerals (Development and Regulation) Act 1957, empowers the State governments to frame rules to prevent illegal mining and the State Government may by notification in the official gazette, make such rules for preventing illegal mining, transportation & storage of minerals and for the purposes connected therewith in the State.

There is a three-pronged strategy for prevention of illegal mining viz. constitution of task force by the State Government at State and District Levels, framing of rules under Section 23C of the MMDR Act, 1957 and furnishing of quarterly returns on illegal mining for review of the Central Government. The details of States who have constituted task force at State level, framed Rules under Section 23C of the MMDR Act, 1957 and have furnished quarterly returns on illegal mining to IBM are as follows:

Twenty-two State Governments have constituted the

task forces. The function of the task force is to review the action taken by member departments for checking the illegal mining activities in their respective jurisdictions.

Twenty-one State Governments have framed the rules under Section 23C of MM (D&R) Act, 1957 to curb illegal mining. The State Governments submit quarterly returns on prevention of illegal mining to IBM. These returns contain details, such as, number of cases detected and action taken there on etc. IBM on receipt of the returns from the various State Governments, consolidates the information and sends it to the Ministry at the end of each quarter.

The Mineral Conservation and Development Rules, 2017 (MCDR) provide measures to ensure systematic & scientific mining. Rule 45 of the MCDR provides for the mining companies to submit periodic reports on the extraction and disposal of the mined material. Rule 45 of MCDR also facilitates end-to-end national-scale accounting of all minerals produced in the country from the pit head to its end-use, reducing the scope for illegal mining, royalty evasion, etc. The amended Rule 45 now makes it mandatory for all miners, traders, stockists, exporters and end-users of minerals to register and report on the production, trade and utilisation of minerals to the State Government (s) and Indian Bureau of Mines.

The MMDR Act, 1957 was amended through the MMDR Amendment Act, 2015. Through MMDR Amendment Act, 2015, Central Government has inserted the following provisions under the MMDR Act with the objective to curb illegal mining in the country:

- a) Penalty for illegal mining has been made more stringent by amendment of the MMDR Act in 2015. Penalties for contravention of Section 4(1) and 4(1A) of the Act have been increased from ₹ 25 thousand per hectares to ₹ 5 Lakh per hectare and the term of imprisonment has been increased from 2 years to 5 years. [Section 21(1) & Section 21(2)]
- b) Section 30B of the MMDR Act, 1957 provides for constitution of Special Courts by State Governments for speedy trial of illegal mining/transportation/storage cases and 30C of the Act provides that such Special Courts shall be deemed to be a Court of Session.

Space Technology for Checking Illegal Mining

Indian Bureau of Mines (IBM) has entered into an MoU with National Remote Sensing Centre (NRSC), for a pilot project "Sudoor Drishti" to demonstrate the feasibility of using High Resolution Satellite Imagery and Digital Elevation Model (DEM) in monitoring mining activities / changes over selected group of mines.

Application of Drone Technology in Mining

Furthering the efforts to utilize new technology, the Ministry of Mines has explored the applicability of the Unmanned Aerial Vehicles (UAV) Technology or commonly referred

to as 'UAVs' for the Mining Sector. The UAV-based remote sensing is an emerging technology, increasingly used in agriculture, environmental, geology, mining, town planning and forestry applications and other applications. UAVs, typically operate at lower altitudes than manned aircraft and are also able to provide unique data with regard to spatial resolution angle of view. Compared to manned fixed-wing aircraft, typically used in aerial remote sensing, UAVs can provide lower ground sample distances (GSD) or higher spatial resolutions on the ground. After a successful pilot project during 2020-21 to ascertain the efficacy of using Unmanned Aerial Vehicles (UAV) to monitor the mining activities in the country, necessary amendments were made in the Mineral Conservation and Development Rules 2017 to include submission of digital images to IBM. The MCDR, 2017 vide Rule 34A now mandates the mineral concession holders to submit digital aerial images to IBM on an annual basis. The digital images to be submitted to IBM will be based on drone (UAV) survey of the mines having production capacities of more than 1 million tonne or lease area of 50 hectares or more based on the Standard Operating Procedure laid down by IBM. For other mines the lessee will be required to submit satellite images based on the SOP laid down by IBM.

The regular survey of the mines using drones will bring in more transparency and create more awareness to work following the proposals in the mining plans. The miners can self-regulate themselves to ensure that all their activities are restricted within the lease area and meet the requirements of the mining plan. In effect, all future submissions of mining plans or modifications will be based on drone survey carried out in the mines resulting in better and scientific mining plans reflecting the actual ground realities. The mine surveys can be taken up at a faster pace using drone surveys thereby increasing accuracy and also reducing the time of survey.

To facilitate the IBM's Regional Offices in processing the drone and satellite image data, necessary hardware & software infrastructure is being provided progressively in a phased manner.

During the year 2023-24, 1,513 digital images have been received on DDMS portal.

Mining Surveillance System (MSS)

MSS Project using satellite remote sensing technology together with information technology has been developed and rolled out for major & minor minerals to curb cases of illegal mining. In the 4th phase of 2022-23, a total of 138 preliminary triggers across the country were generated for Major Minerals and uploaded on the portal for further transmission to the State Governments. During 2022-23, field verification reports in respect of 40 triggers for Major Minerals have been received, out of which unauthorised mining in 7 cases have been confirmed by the State Governments.

District Mineral Foundation / Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)

District Mineral Foundation (DMF) established by contributions from the mining companies, came into force specially for addressing the long-time grievances of the neglected communities affected by mining activities. Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY) scheme formulated for the welfare and development of the mining affected areas and people under DMF was also launched. About ₹ 90,730.07 crore have been collected till 31.03.2024. Under the PMKKKY 3,29,775 projects have been sanctioned. For these projects, funds to the tune of ₹ 49,844.92 crore have been utilised and about ₹ 82,030.33 crore have been allocated.

Mining Tenement System (MTS)

The Mining Plan, Star Rating and OAS modules are under testing. The SRS V3.1 of Phase II Modules, i.e., Grant and Execution of Concession, Inspection Module, GIS Module, IBM existing databases, ML WMIMP, NMI, MCP and Final Mine Closure Plan modules are under examination.

Indian Bureau of Mines (IBM)

IBM plays the role of National Repository of mineral data, maintaining a data bank of mines and minerals by developing advanced IT-based Mineral Information System. IBM also carries out mining research project on need-based aspects of mining; and conducts mineral beneficiation studies, including mineralogical testing and chemical analysis; and preparation of mineral maps. Indian Bureau of Mines (IBM), as a facilitator to the Mineral Industry, performs multifarious functions, such as, providing technical consultancy services for conducting feasibility studies, environment impact assessments, environment management plans, etc. acting as a storehouse of data.

A Remote Sensing Centre was set up at IBM in 2018. Multi-mineral leasehold maps are updated on ARC-GIS platform. All the maps viz lease boundaries, Geological layer and toposheet layer have been integrated for the States of Goa & Maharashtra. During 2021-22, the vectorisation of 61 toposheets and plotting of 273 mining leases and attachment of Assam, Meghalaya, Manipur, Kerala, Haryana, Himachal Pradesh, West Bengal States and Jammu & Kashmir (UT) were completed. Up to March 2022, georeferencing, projection and vectorisation of all the 561 identified toposheets having Major Mineral leaseholds were completed.

Mineral beneficiation studies were carried out by IBM to encourage value addition, conservation and development of mineral resources. During 2023-24, 48 Ore dressing investigations, 24,150 chemical analyses and 3,004 mineralogical examinations were completed.

The Project on Mining Surveillance System (MSS) was undertaken by Indian Bureau of Mines, Ministry of Mines, and BISAG (Bhaskaracharya Institute for Space Applications

and Geo-informatics) of Ministry of Electronics and Information Technology (MEITY) to develop a system for detection of incidences of illegal mining by use of space technology and Surveillance of area up to 500 m outside the lease boundary to check instances of illegal mining. The deterrence effect enabled by advanced technology applications, such as, Mining Surveillance System (MSS) have been extremely useful in curbing instances of illegal mining.

IBM undertakes preparation of National Inventory of mineral resources on a quinquennial basis. Under this programme, implementation of UNFC system was adopted in 2002 replacing the earlier resource classification based on Indian system. The last National Mineral Inventory (NMI) was updated as on 01.04.2020 for 46 minerals. The preparatory work towards updation of National Mineral Inventory (NMI) as on 01.04.2020 for 46 major minerals was completed.

RESEARCH & DEVELOPMENT

The Science and Technology (S&T) programmes of the Ministry of Mines, Government of India, cover the disciplines of Geology, Exploration, Mining, Beneficiation & Mineral Processing, Rock Mechanics, Ground Control & Non-ferrous Metallurgy and Environmental issues related to Mining & Metallurgy.

During the 23rd PERC meeting held on 11th December 2023, a total of 82 project proposals were received under S&T Program Scheme of Ministry of Mines. After screening, 45 proposals covering three areas, namely (i) Metallurgy — Alloys, Rare Earths, Specialty materials and product ; Metal Extraction (Metallurgical processes); Circular Economy and Recycling of Non ferrous metals (ii) Mining — rock mechanics, design, equipment, energy, environment, safety; Geoscience and Exploration; and Decarbonisation: Green Technology in Mineral-based Industry (iii) Mineral Processing — Beneficiation, Ore Dressing, Mineral Processing & Recovery from waste; Extraction of strategic, critical minerals and REE were considered by the PERC. The project proposals were shortlisted for further presentation by the respective Principal Investigators (PIs). Based on the detailed review and evaluation, the PERC recommended 10 Project Proposals with or without changes to SSAG.

The Research & Development (R&D) work in the field of Ores & Minerals is being carried out by IBM, JNARDDC, CSIR & allied laboratories, other research organisations relating to mineral/metal and various mining & mineral-based industries. The research & development details are covered in the Review on "Research & Development" under "General Review".

FOREIGN TRADE

As per the World Trade Statistics Review 2024, India's ranking amongst the leading exporters in the world merchandise trade improved from 30 in 2004 to 18 in

2024 with a global export share of 1.80%. Similarly, India's ranking amongst the leading importers in world merchandise trade was 8 in 2024 as compared to 23 in 2004 with a share of 2.9 per cent.

Exports

During the year 2023-24, the value of exports (including re-exports) of ores and minerals was ₹ 2,13,740 crore. The export value which had decreased from ₹ 2,57,640 crore in 2021-22 to ₹ 2,42,009 crore in 2022-23, further decreased to ₹ 2,13,740 crore in 2023-24. The value of mineral exports showed a decrease of 11.68% in 2023-24 as compared to that in the previous year. Diamond continued to be the largest constituent item with a share of 65.27% in the total value of mineral exports in 2023-24. Next in order of share was Iron ore with contribution of 15.17% followed by Granite 5.26 %, Alumina 3.01% and Salt (Other than Common Salt & Rock Salt) 1.14%. The individual share of remaining minerals in the total value of exports of ores and minerals from India during the year under review was around or less than one per cent. The value of exports of ores & minerals (including re-exports) showed a mixed trend for most of the minerals in 2023-24 as compared to that of the previous year. A significant increase was also noticed in some cases. The exports value of minerals which have shown significant growth are Sand (excl. Metal Bearing) (1,457%), Zirconium ores & conc. (217%) and Quartz & Quartzite (155%). On the other hand, the exports value recorded significant decline in the cases of Niobium or Tantalum ores and conc. (100%), Silver ores and conc. (100%) and Tin ores and conc. (100%) with respect to that in the previous year.

The value of exports (including re-exports) of metals & alloys stood at ₹ 2,80,311 crore in the year 2023-24. The export value which had decreased from ₹ 3,48,475 crore in 2021-22 to ₹ 2,90,375 crore in 2022-23 further decreased to ₹ 2,80,311 crore in 2023-24. The value of export of metals & alloys showed a decrease of 3% in 2023-24 as compared to the previous year. In terms of value of exports, Iron & Steel has the largest share (55.3%), followed by Aluminum and Alloys Incl. Scrap (22.6%), Ferroalloys (8.43%) and Copper & Alloys (Incl. Brass & Bronze) (6.84%). The contributions of Lead & alloys including scrap and Zinc & alloys including scrap were 2.31% and 2.15% respectively. The individual share of other remaining metals and alloys was less than one per cent.

Imports

During the year 2023-24, the value of imports (including re-imports) of ores and minerals was ₹ 18,63,067 crore. The import value which had increased from ₹ 15,51,614 crore in 2021-22 to ₹ 21,63,881 crore in 2022-23, decreased to ₹ 18,63,067 crore in 2023-24. The value of mineral imports showed a decrease of 13.90% in 2023-24 as compared to that in the previous year. Petroleum (crude) continued to be the largest constituent item with a share of 61.92% in the total value of mineral imports in 2023-24. Next in order

of share was Coal (except Lignite) with the contribution of 16.65% followed by Diamond 9.60% and Natural Gas 5.96%. The combined share of value of these four minerals was 94.12% in 2023-24 as against 93.87% in the previous year. The value of imports of some ores & minerals in 2023-24 has recorded significant increase in the cases of Tripoli Earth (49,250%), Garnet (abrasive) (7,159%), Corundum (natural) (4,333.33%), Kieselguhr (364.04%), Iron ore (175.71%), as compared to that in the previous year.

The value of imports (including re-imports) of metals & alloys was ₹ 7,91,646 crore in the year 2023-24. The import value which had increased from ₹ 6,27,313 crore in 2021-22 to ₹ 6,69,639 crore in 2022-23, further increased to ₹ 7,91,646 crore in 2023-24. The value of metal & alloys imports showed an increase of 18.22% in 2023-24 as compared to the previous year. In terms of value of imports, Gold (Non-monetary & monetary): total has the largest share of 47.65%, Iron & Steel 23.55%, Copper & Alloys (Incl. Brass & Bronze) 9.66%, Aluminum & Alloys Incl. Scrap 7.18% and Silver 5.71%. The individual share of remaining metals was less than two per cent of the total value of metals & alloys. The value of imports of the metals & alloys that showed significant growth included Gold clad Metals / Base metal NES (29,426.32%), Tellurium (116.53%), Molybdenum & Scrap (66.82%), Pig & Cast Iron (incl. Speigeliessen) (63.57%) and Platinum clad base/Precious Metal (56.31%) in 2023-24 as compared to the previous year. On the other hand, during 2023-24, the value of imports of metals & alloys that showed a declining trend as compared to the previous year was mainly in the case of Precious metal/metal clad with precious metal (85.48%), Platinum alloys & related metals (80.20%) and Tantalum & scrap (33.39%).

VALUE-ADDED EXPORT TRADE

India's foreign trade includes exports of minerals, both in the raw form and semi-processed & processed forms like mineral-based primary manufactured products.

Ores and minerals contributed significantly to India's exports trade in 2023-24 with a share of about 5.90% (i.e., ₹ 2,13,741 crore) in the total value of Exports of all merchandise. The contribution of minerals in exports in raw/unprocessed forms was about ₹ 44,302 crore and in semi-processed/processed forms was about ₹ 1,69,438 crore. The manufactured mineral-based commodities (final stage of transformation) contributed about ₹ 28,76,54 crore to the total value of exports of all merchandise. The value-added semi-processed/processed minerals figuring in India's foreign trade included cut & polished diamond/emerald, pulverised barytes, steatite, felspar (cut), garnet, calcined magnesite, magnesite (fused), magnesite (dead-burnt), magnesium oxide, slate (worked), processed mica & manufactured mica products, coke, cut & polished dimension stones, alumina, etc. The manufactured mineral-based commodities included metals & alloys and products thereof, cement, firebricks & other refractory materials, clay-bonded graphite crucibles & silicon carbide crucibles, manganese dioxide, asbestos-cement products, inorganic chemicals like lime & fluorine chemicals, refined borax & borates, elemental phosphorus & phosphoric acid, titanium dioxide, petroleum products, phosphatic & potash fertilizers, etc. Table-6 provides data on contribution of various value-added minerals and mineral-based products to India's exports during 2021-22 to 2023-24.

Table – 6: Contribution of Value-added (Processed) Minerals & Mineral-based Products in India's Export* Trade, 2021-22 to 2023-24

Sl. No.	Commodity group	Value of exports (₹ million)			Contribution (percentage)		
		2021-22	2022-23 (R)	2023-24 (P)	2021-22	2022-23	2023-24 (P)
1	All Merchandise	31470214	36215498	36189522	100	100	100
2	Ores & Minerals	2576402	2420095	2137405	8.18	6.68	5.90
	2.1 Raw/Unprocessed form	357484	335036	443021	1.13	0.92	1.22
	2.2. Semi-processed/processed forms intermediate stages of processing)"	2218918	2085059	1694384	7.05	5.75	4.68
3	Manufactured Mineral-based Commodities (final stage of transformation)	3547108	2981950	2876547	11.27	8.23	7.94
	3.1 Metals/Alloys	3484754	2903758	2803112	11.07	8.01	7.74
	3.2 Others	62354	78192	73435	0.19	0.21	0.20

Figures rounded off.

* Including re-exports.

INFRASTRUCTURE

Infrastructure

In order to achieve the GDP of \$5 trillion by 2024-25, India needs to spend about \$1.4 trillion over these years on infrastructure. During FYs 2008–2017, India invested about

US\$1.1 trillion on infrastructure. However, the challenge is to step up infrastructure investment substantially.

Keeping this objective in view, National Infrastructure Pipeline (NIP) was launched with projected infrastructure investment of around ₹ 111 lakh crore (US\$ 1.5 trillion) during FY 2020–2025 to provide world-class infrastructure

across the country, and improve the quality of life for all citizens. It also envisages to improve project preparation and attract investment, both domestic and foreign in infrastructure. NIP was launched with 6,835 projects, which later was expanded to over 9,000 projects covering 34 infrastructure sub-sectors. During the fiscals 2020 to 2025, sectors, such as, energy (24 per cent), roads (19 per cent), urban (16 per cent) and railways (13 per cent) accounted for around 70 per cent of the projected capital expenditure in infrastructure in India. NIP took under its fold all the stakeholders for a coordinated approach to infrastructure creation in India to boost short-term as well as the potential GDP growth.

NITI Aayog has developed the 'National Monetisation Pipeline (NMP Volumes 1&2)' in consultation with infrastructure line ministries. Asset monetisation, entails a limited period licence/ lease of an asset, owned by the Government or a public authority, to a Private Sector entity for an upfront or periodic consideration. The Private Sector entity is expected to operate and maintain the asset based on the terms of the contract/concession, generating returns through higher operating efficiencies and enhanced user experience. Funds, so received by the public authority, are reinvested in new infrastructure, or deployed for other public purposes. Such contracts include provision for transfer of asset back to the authority at the end of the period.

A robust asset pipeline has been prepared to provide a comprehensive view to investors and developers of the investment avenues in infrastructure. The pipeline includes selection of de-risked and brownfield assets with stable revenue generation profile (or long rights) which will make for an attractive investment option. Total indicative value of NMP for core assets of the Central Government has been estimated at ₹ 6.0 lakh crore over 4-year period (5.4 per cent of total infrastructure investment envisaged under NIP).

Coal

The Country has witnessed the highest ever coal production in the year 2023-24. Coal production (provisional) at 997.25 million tonnes in 2023-24 increased by 11.65% from that of 893.19 million tonnes in 2022-23.

Electricity

Electricity is essential for powering economic activity. The Power Sector has witnessed substantial transformation from both the demand and supply-side.

The installed capacity was 442 GW as on 31.04.2024. During the year 2023-24, the total generation of energy (including imports and renewable sources of energy) was 1,739.09 BU. During the year 2023-24, peak shortage was 1.4% and the energy shortage was 0.3% as compared to 4.0% and 0.5%, respectively in the previous year.

Transport

Railways

Indian Railways (IR) with over 68,000 route km is the third largest network in the world under single management. During the year 2023-24, Indian Railways carried 1,589.95 million tonnes of freight traffic and 6,905 million passengers making it the world's largest passenger carrier and 4th largest freight carrier.

Civil Aviation

India is one of the fastest growing market for civil aviation in the world. It is expected to become the third largest overall (including domestic and international traffic) by the year FY25.

India's domestic traffic has more than doubled from around 61 million in FY-14 to around 153.7 million in 2023-24.

Ports and Shipping

The Major Ports in the country have an installed capacity of 2,690 MTPA and handled traffic of 819.23 MT during 2023-24. While increasing the capacity of major ports, Ministry of Shipping has been striving to improve the operational efficiencies through mechanisation, digitisation and process simplification. As a result, key efficiency parameters have improved considerably. The Average Turnaround Time in 2023-24 improved to 48.06 hrs as against 52.43 hrs in 2022-23. The Average Output Per Ship Berthday has increased from 17,605 tonnes in 2022-23 to 18,905 tonnes in 2023-24.

Roads

Road infrastructure in the form of a network of national highways, state highways, district roads, rural roads and urban roads acts as a major mode of transportation and connectivity for the country's diverse population of consumers and businesses. Roads supplement the other modes of transport through last-mile connectivity to the far-flung regions of the country.

There has been an increase in the construction of National Highways (NHs)/roads over time, with 12,349 km of roads constructed in FY-24 as compared to 6,061 km in FY-16. Pace of NH construction increased 2.8 times to 33.8 km/day from 12.1 km/day in 2014-15. Total budgetary support for investment in the sector has been increasing rapidly in the last four years and stood at around ₹ 2.17 lakh crore during FY23-24.

PERFORMANCE OF SELECTED MINERAL-BASED INDUSTRIES

Steel

Globally, India is the second largest producer of crude steel in the world after China. During 2023-24, crude steel production stood at 144.299 million tonnes. During 2023-24, the total export of finished steel reached the highest volume of 7.49 million tonnes.

Cement

As per DIPP Annual Report, production of cement during 2023-24 was 426.29 million tonnes as against 391.40 million tonnes in 2022-23 and registered an increase of about 8.9% per cent. The induction of advanced technology has helped the industry immensely to improve its efficiency by conserving energy, fuel and addressing the environmental concerns. Cement Industry has been undergoing a transition with modernisation and upgradation of technology particularly with a view to conserve energy. India exports cement including white cement and other cement clinker. The exports of cement (total) increased to 0.81 million tonnes in 2023-24 from 0.67 million tonnes in 2022-23.

Petroleum Oil and Refineries

Crude oil production & condensate in 2023-24 at 29.36 million tonnes registered a nominal increase of 0.60% as compared to that in 2022-23. The production of natural gas (utilised) was at 36.44 billion cubic metres in 2023-24, 5.77% higher than 34.45 billion cubic metres achieved in 2022-23. The total refining capacity in the country was about 256.62 MMTA in 2023-24. Production of petroleum products (including LPG production from natural gas) was 276.05 million tonnes in 2023-24 as compared to 266.54 million tonnes in 2022-23.

SELF-RELIANCE IN MINERALS & MINERAL-BASED PRODUCTS

India continued to be wholly or largely self-sufficient in minerals which constitute primary mineral raw materials that are supplied to industries, such as, iron & steel, cement etc. India is self-sufficient or near to self-sufficient in bauxite, chromite, iron ore, kyanite, limestone, aluminium (primary), lead (primary) and zinc. In spite of significant production, some ores/minerals are also imported to meet the demand either for blending with locally available mineral raw materials and/or to meet special requirement for manufacturing special qualities of mineral based products. India is deficient in magnesite, manganese ore, rock phosphate and copper, which were imported to meet the domestic demand. To meet the increasing demand of uncut diamonds, emerald and other precious & semi-precious stones by the domestic Cutting & Polishing Industry, India is dependent on imports of raw uncut stones for their value-added re-exports. The Degree of Self-sufficiency in respect of various principal minerals and metals in 2023-24 is furnished in Table -7.

Table – 7: Degree of Self-sufficiency in Principal Minerals & Metals, 2023-24 (P)

Sl. No.	Commodity	"Apparent Consumption" ('000 tonnes)	"Supply/Domestic supply" ('000 tonnes)	"Order of self-sufficiency (%)
Minerals*				
1	Bauxite	28296.58	23967.86	84.70
2	Chromite	3318.24	3147.95	94.86
3	Iron ore	235916	277332	100.00
4	Kyanite	4.81	3.32	69.02
5	Limestone	484156	452009 ^a	93.36
6	Magnesite	607.68	132	21.72
7	Manganese ore	8973.5	3380.89	37.67
8	Rock phosphate (including apatite)	10368.2	1557.78	15.02
Metals				
9	Aluminium (primary)	4134.47	4158.84	100.00
10	Copper (cathode) ^b	1144.81	509.43	44.50
11	Lead (primary) ^c	276.81	215.98	78.02
12	Zinc ^d	781.74	817.06	100.00

Source: MCDR Returns for production and DGCI&S, Kolkata for export & import.

#: Apparent Consumption: (production+ import-export)

*: Minerals declared as minor mineral vide Government of India Notification S.O. 423(E) dated 10th February, 2015, are not included due to non-availability of production for the year 2023-24.

a: Excludes production of limestone as a minor mineral, calcite & chalk and includes limeshell, limekankar & marl.

b: Based on production of copper cathode and imports & exports of copper & alloys.

c: Based on production of lead (primary), and imports & exports of lead & alloys.

d: Based on production of zinc (ingots) and imports & exports of zinc & alloys.

Note: Even in cases where almost entire domestic demand is satisfied by domestic supplies, some quantities of certain special quality/types of minerals and metals/ferroalloys are imported to meet the requirement in certain specific end-uses.



2. Mineral Policy & Legislation

The Central Government vide Gazette Notification, S.O. 3054 (E), proposes to undertake prospecting and mining operations in respect of Gold in the State of Karnataka.

Vide Gazette Notification, S.O. 2077 (E), The Central Government reserves an area of 48.493 in favour of NMDC Limited in Chhattisgarh

POLICY

1. National Mineral Policy, 2019

National Mineral Policy, 2019 was approved by the Union Cabinet on 28th February, 2019.

Objective

The aim of National Mineral Policy, 2019 is to have a more effective, meaningful and implementable policy that brings in further transparency, better regulation and enforcement, balanced social and economic growth as well as sustainable mining practices.

Details

The National Mineral Policy, 2019 includes provisions which will give boost to the Mining Sector such as:

- Introduction of Right of First Refusal for RP/PL holders;
- Encouraging the Private Sector to take up exploration;
- Auctioning in virgin areas for composite RP-cum- PL-cum-ML on revenue share basis;
- Encouragement of merger and acquisition of mining entities;
- Transfer of mining leases and creation of dedicated mineral corridors to boost Private Sector mining areas;
- Proposes to grant status of industry to mining activity to boost financing of mining for Private Sector and for acquisitions of mineral assets in other countries by Private Sector;
- Proposes to auction mineral blocks with pre embedded clearances to give fillip to auction process; and
- Proposes to make efforts to harmonise taxes, levies & royalty with global benchmarks to help Private Sector.

LEGISLATION

Notifications

Important Notifications notified/issued during the period under review are furnished below:

Ministry of Mines

1. Ministry of Mines, Notification, G.S.R. 264(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and

Minerals (Development and Regulation) Act, 1957 (67 of 1957), the Central Government hereby rescinds the notification of the Government of India in the Ministry of Mines number G.S.R. 40(E), dated the 18th January, 2018, published in Gazette of India, Extraordinary, Part II, Section 3, Sub-section (i), dated the 18th January, 2018 except in respect of things done or omitted to be done before such rescission, with effect from the date of publication of this notification.

Source: The Gazette of India, Extraordinary, No. 243 Part II, Section 3-Sub section (i), dated 30.04.2024.

2. Ministry of Mines, Notification, S.O. 1872(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957), the Central Government hereby notifies the M/s NTPC Mining Limited, a Central Public Sector Enterprise under the control of the Ministry of Power, for the purposes of the said proviso subject to the condition that the said Public Sector Enterprise shall make available the data generated by it, in respect of the prospecting operations undertaken by it, to the concerned State Government.

2. This notification shall come into force on the date of its publication in the Official Gazette.

Source: The Gazette of India, Extraordinary, No. 1780 Part II, Section 3-Sub section (ii), dated 30.04.2024.

3. Ministry of Mines, Notification, S.O 1900(E) — In exercise of the powers conferred by Sub-sections (1) and (2) of section 16 read with Section 17 and Sub-section (3) of Section 25 of the Bureau of Indian Standards Act, 2016 (11 of 2016), the Central Government is of the opinion that it is necessary so to do in the public interest and after consulting the Bureau of Indian Standards, hereby makes the following Order to amend the Nickel (Quality Control) Order, 2023, namely: –

1. Short title and commencement – (1) This Order may be called the Nickel (Quality Control) Amendment Order, 2024.

(2) It shall come into force on the date of its publication in the Official Gazette.

2. In the Nickel (Quality Control) Order, 2023, in paragraph 1, for sub-paragraph (2), the following sub- paragraph shall be substituted, namely: –

“(2) This Order shall come into force with effect from 01.12.2024.”

Source: The Gazette of India, Extraordinary, No. 1808 Part II, Section 3-Sub section (ii), dated 03.05.2024.

4. Ministry of Mines, Notification, S.O 1901(E)— In exercise of the powers conferred by Sub-sections (1) and (2) of Section 16 read with Section 17 and Sub-section (3) of Section 25 of the Bureau of Indian Standards Act, 2016 (11 of 2016), the Central Government is of the opinion that it is

necessary so to do in the public interest and after consulting the Bureau of Indian Standards, hereby makes the following Order to amend the Copper (Quality Control) Order, 2023, namely: –

1. Short title and commencement – (1) This Order may be called the Copper (Quality Control) Amendment Order, 2024.

(2) It shall come into force on the date of its publication in the Official Gazette.

2. In the Copper (Quality Control) Order, 2023, in paragraph 1, for sub-paragraph (2), the following sub- paragraph shall be substituted, namely: –

“(2) This Order shall come into force with effect from 01.12.2024.”

Source: The Gazette of India, Extraordinary, No. 1809 Part II, Section 3-Sub section (ii), dated 03.05.2024.

5. Ministry of Mines, Notification, S.O 1902(E)— In exercise of the powers conferred by Sub-sections (1) and (2) of Section 16 read with Section 17 and Sub-section (3) of Section 25 of the Bureau of Indian Standards Act, 2016 (11 of 2016), the Central Government is of the opinion that it is necessary so to do in the public interest and after consulting the Bureau of Indian Standards, hereby makes the following Order to amend the Aluminium and Aluminium Alloys (Quality Control) Order, 2023, namely: –

1. Short title and commencement – (1) This Order may be called the Aluminium and Aluminium Alloys (Quality Control) Amendment Order, 2024.

(2) It shall come into force on the date of its publication in the Official Gazette.

2. In the Aluminium and Aluminium Alloys (Quality Control) Order, 2023, in paragraph 1, for sub-paragraph (2), the following sub-paragraph shall be substituted, namely: –

“(2) This Order shall come into force with effect from 01.12.2024.”

Source: The Gazette of India, Extraordinary, No. 1810 Part II, Section 3-Sub section (ii), dated 03.05.2024.

6. Ministry of Mines, Notification, S.O. 2077(E)— In exercise of the powers conferred by Sub-section (1A) of Section 17A of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957), the Central Government, after consultation with the State Government of Chhattisgarh, hereby reserves an area of 48.493 hectares located near the western lease boundary of Deposit No. 11 ML for undertaking mining operations, including, dumping of waste material and allied infrastructure facilities in relation to existing Deposit No. 14 NMZ mining lease for mineral iron ore in favour of NMDC Limited, a Central Public Sector Enterprise under the administrative control of the Ministry of Steel, lying within the boundary (demarcated by latitude and longitude) of such reserved area as specified below:–

District	Forest Division	Forest Compartment Number	Area (in hectares)	Coordinates (Toposheet No. E44J2 and J6)		
				Point	Longitude	Latitude
South Bastar	Bailadilla Reserve	1855	47.931	A64	81°13'54.6335" E	18°37'34.6006" N
Dantewada	Forest	1860	0.562	A	81°13'47.6912" E	18°37'32.8557" N
		Total	48.493	B	81°13'42.8663" E	18°37'37.4589" N
				C	81°13'35.8232" E	18°38'00.1730" N
				D	81°13'35.8232" E	18°38'10.6256" N
				E	81°13'52.9019" E	18°38'11.2296" N

Source: The Gazette of India, Extraordinary, No. 1978 Part II, Section 3-Sub section (ii), dated 22.05.2024.

7. Ministry of Mines, Notification, S.O. 2379(E)—In exercise of the powers conferred under the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India (QCI-NABET), the Central Government hereby notifies the following entities as specified in the guidelines for notification of accredited private exploration agencies issued by the Government of India in the Ministry of Mines vide order number M.VI-16/15/2021-Mines VI, dated the 12th August, 2021, namely:—

Sr. No.	Exploration agency.	Category of exploration agency
1	M/s Critical Mineral Trackers.	A
2	M/s Mining Tech Consultancy Services Limited.	A

2. The agencies shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall come into force on the date of its publication in the Official Gazette and shall remain valid for a period of three years from the date of publication or till expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 2259 Part II, Section 3-Sub section (ii), dated 20.06.2024.

8. Ministry of Mines, Notification, G.S.R. 345(E)—In exercise of the powers conferred by Sub-section (1A) of Section 17A of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957), the Central Government, after consultation with the Government of Rajasthan, hereby makes the following amendment in the notification of the Government of India, Ministry of Mines, published in the Gazette of India, Extraordinary, Part II, Section 3, Sub-section (i), vide number G.S.R. 1220(E), dated the 19th December, 2018, namely:—

In the said notification, in the table, serial number 1 and the entries relating thereto shall be omitted.

2. This notification shall come into force on the date of its

publication in the Official Gazette.

Source: The Gazette of India, Extraordinary, No. 322 Part II, Section 3-Sub section (i), dated 24.06.2024.

9. Ministry of Mines, Notification, S.O. 3648(E)—In exercise of the powers conferred by Clause (a) of Section 4 of the Offshore Areas Mineral (Development and Regulation) Act, 2002 (17 of 2003) and in supersession of the notification of the Government of India, Ministry of Mines, published in the Gazette of India, Extraordinary, Part-II, Section 3, Sub-Section (ii), vide S.O. 2628 (E), dated the 04th July, 2024, except in respect of things done or omitted to be done before such supersession, the Central Government hereby notifies Shri Vivek Kumar Bajpai, Joint Secretary, Ministry of Mines as the administering authority for the purpose of the said Act.

Source: The Gazette of India, Extraordinary, No. 3331 Part II, Section 3-Sub section (ii), dated 28.08.2024.

10. Ministry of Mines, Notification, S.O. 2628(E)—In exercise of the powers conferred by Clause (a) of Section 4 of the Offshore Areas Mineral (Development and Regulation) Act, 2002 (17 of 2003) and in supersession of the notification of the Government of India, Ministry of Mines, published in the Gazette of India, Extraordinary, Part-II, Section 3, Sub-Section (ii), vide S.O. 1523(E), dated the 06th April, 2018, except in respect of things done or omitted to be done before such supersession, the Central Government hereby notifies Dr Veena Kumari Dermal, Joint Secretary, Ministry of Mines as the administering authority for the purposes of the said Act.

Source: The Gazette of India, Extraordinary, No. 2493 Part II, Section 3-Sub section (ii), dated 04.07.2024.

11. Ministry of Mines, Notification, S.O. 2936(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies M/s Bhushilp Mines and Minerals Pvt Ltd under 'Category A Exploration Agencies' as specified in the guidelines for notification of accredited private exploration

agencies issued by the Government of India in the Ministry of Mines vide Order Number M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2. The agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall come into force on the date of its publication in the Official Gazette and shall remain valid for a period of three years from the date of publication or till expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 2799 Part II, Section 3-Sub section (ii), dated 24.07.2024.

12. Ministry of Mines, (Indian Bureau of Mines), F. No. T-43010/2/2024-TS-IBM-HQ—In exercise of the powers conferred by Rule 66 of the Mineral Conservation and Development Rules, 2017, the Notification No. T-43010/CGBM/91, dated the 19th February, 1998 published in the Gazette of India, Part III, Section 1, dated the 7th March 1998, specifying the limits of territorial jurisdiction of the Controller of Mines and the Regional Controller of Mines of Indian Bureau of Mines for the purposes of the said rules is hereby amended as follows, namely—

(i) In the table specifying the territorial jurisdiction of the Controller of Mines and the Regional Controller of Mines of Indian Bureau of Mines, in the entry related to Zonal Office of 'Controller of Mines (South Zone)' in the first column, for the entry related to Regional Office of 'Hyderabad Region' in the second column and corresponding entries in the third and fourth columns, the following entries shall be substituted, namely—

Regional Office	State / UT	District
(4) Hyderabad Region:	Telangana	All districts
	Maharashtra	Latur, Nanded and Osmanabad
(5) Vijaywada Region:	Andhra Pradesh	All districts

2. The above shall come into force with effect from the date of publication of this notification in the Official Gazette.

ORDER

Ordered that these territorial jurisdictions of the Controller of Mines and the Regional Controllers of Mines of Indian Bureau of Mines, be published in the Gazette of India, Part III Section 1 for general information of all.

Source: The Gazette of India, Extraordinary, No. 08 Part III, Section 1, dated 25.07.2024.

13. Ministry of Mines, Notification, S.O. 3847(E)—Whereas in exercise of the powers conferred by Clause (a) of Sub-section (1) of Section 26 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and in supersession of the notifications of the Government of

India, Ministry of Mines, published in the Gazette of India, vide number G.S.R. 860(E), dated the 16th December, 2021 and vide number S.O. 3890(E), dated the 1st September, 2023, except in respect of things of done or omitted to be done before such supersession, the Central Government hereby directs that on and from the date of publication of this notification in the Official Gazette, the powers of the Central Government under the provisos to sub-sections (3) and (4) of Section 10B, the provisos to Sub-sections (4) and (5) of Section 11 and section 11D of the said Act and under the Mineral (Auction) Rules, 2015, shall also be exercisable by Smt. Farida M. Naik, Joint Secretary to the Government of India in the Ministry of Mines.

2. For the purposes of exercising the above said powers, the said officer shall be known as the "Designated Officer".

3. The said officer shall be bound by the written directions as may be given by the Central Government from time to time.

Source: The Gazette of India, Extraordinary, No. 3517 Part II, Section 3-Sub section (ii), dated 09.09.2024.

14. Ministry of Mines, Notification, S.O. 3929(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies M/s GMMCO Technology Services Limited under 'Category A Exploration Agencies' as per the 'Guidelines for notification of accredited private exploration agencies' issued by the Government of India in the Ministry of Mines vide Order No. M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2. The said agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall remain in force for a period of three years from the date of its publication or till the expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 3594 Part II, Section 3-Sub section (ii), dated 13.09.2024.

15. Ministry of Mines, Notification, S.O. 4561(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies M/s Envirogreen Consultants (India) Private Limited under "Category 'A' Exploration Agencies" as per the 'Guidelines for notification of accredited private

exploration agencies' issued by the Government of India in the Ministry of Mines vide Order No. M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2.The said agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3.This notification shall remain in force for a period of three years from the date of its publication or till the expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 4197 Part II, Section 3-Sub section (ii), dated 17.10.2024.

16. Ministry of Mines, Notification, S.O. 4760(E)—In exercise of the powers conferred by Sub-section (2) of Section 16 of the Offshore Areas Mineral (Development and Regulation) Act, 2002 (17 of 2003), the Central Government hereby makes the following amendments in the First Schedule to the said Act, namely—

2. In the First Schedule to the Offshore Areas Mineral (Development and Regulation) Act, 2002,—

(i) in item 1, for the words “sale price”, the words “average sale price” shall be substituted;

(ii) after item 1 and the entries relating thereto, the following item and entries shall be inserted, namely—

“1A.	Construction Sand	Forty rupees per tonne.”;
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(iii) for item 2 and the entries relating thereto, the following item and entries shall be substituted, namely—

“2.	Dolomite	Fifty rupee per tonne.”;
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(iv) in item 3, for the words “sale price”, the words “average sale price” shall be substituted;

(v) for item 5 and the entries relating thereto, the following item and entries shall be substituted, namely—

“5.	Limestone and Lime mud	Fifty rupee per tonne.”;
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(vi) in item 6, for the words “sale price”, the words “average sale price” shall be substituted;

(vii) after item 7 and the entries relating thereto, the following item and entries shall be inserted, namely—

“7A.	Polymetallic Nodules and Crusts	Three per cent. of the average sale price on ad valorem basis.”;
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(viii) in item 8, for the words “sale price”, the words “average sale price” shall be substituted;

(ix) after item 9 and the entries relating thereto, the following item and entries shall be inserted, namely—

“9A.	Overburden or waste	Ten rupees per tonne.”;
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(x) in item 10, for the words “sale price”, the words “average sale price” shall be substituted.

Source: The Gazette of India, Extraordinary, No. 4383 Part II, Section 3-Sub section (ii), dated 29.10.2024.

17. Ministry of Mines, Notification, S.O. 5069(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies the Synergy Geotech Private Limited under “Category ‘A’ Exploration Agencies” as per the guidelines for notification of accredited private exploration agencies issued by the Government of India in the Ministry of Mines vide Order No. M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2. The said agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall remain in force for a period of three years from the date of its publication or till the expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 4679 Part II, Section 3-Sub section (ii), dated 26.11.2024.

18. Ministry of Mines, Notification, S.O. 924 (E)—In exercise of the powers conferred by Clause (e) of Section 3 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957), the Central Government hereby makes the following amendment in the notification of the Government of India, in the Ministry of Mines, published in the Gazette of India, Extraordinary, Part II, Section 3, Sub-section (ii), vide number S.O. 423(E), dated the 10th February, 2015, namely—

In the said notification, minerals specified in items (iii), (xiv), (xxii) and (xxv) shall be omitted.

2. This notification shall come into force on the date of its publication in the Official Gazette.

Source: The Gazette of India, Extraordinary, No. 916 Part II, Section 3-Sub section (ii), dated 20.02.2025.

19. Ministry of Mines, Notification, S.O. 1328(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies the M/s Geo Marine Solutions Private Limited under “Category ‘A’ Exploration Agencies” as per the guidelines for notification of accredited private exploration agencies issued by the Government of India in the Ministry of Mines vide Order No. M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2. The said agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall remain in force for a period of three years from the date of its publication or till the expiry or termination of the accreditation granted, whichever is earlier.

4. This notification supersedes the earlier Notification S.O. 4596(E) dated 29th September 2022 in respect of M/S Geo Marine Solutions Pvt. Ltd.

Source: The Gazette of India, Extraordinary, No. 1310 Part II, Section 3-Sub section (ii), dated 19.03.2025.

20. Ministry of Mines, Notification, S.O. 1327(E)—In exercise of the powers conferred by the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India, the Central Government hereby notifies the M/s Mining Associates Private Limited under “Category „A’ “Exploration Agencies” as per the guidelines for notification of accredited private exploration agencies issued by the Government of India in the Ministry of Mines vide Order No. M.VI-16/15/2021-Mines VI, dated the 12th August, 2021.

2. The said agency shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall remain in force for a period of three years from the date of its publication or till the expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 1309 Part II, Section 3-Sub section (ii), dated 19.03.2025.

21. Ministry of Mines, Notification, S.O. 967(E)—In exercise of the powers conferred under the second proviso to Sub-section (1) of Section 4 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and consequent upon accreditation provided by the National Accreditation Board for Education and Training of the Quality Council of India (QCI–NABET), the Central Government hereby notifies the following entities as

specified in the guidelines for notification of accredited private exploration agencies issued by the Government of India in the Ministry of Mines vide Order Number M.VI-16/15/2021-Mines VI, dated the 12th August, 2021, namely–

Sr. No.	Exploration agency.	Category of exploration agency
1	M/s Enkay Enviro Services Pvt. Ltd.	A
2	M/s Ocean Drilling and Exploration Pvt. Ltd.	A
3	M/s Eartnenviro Lab Private Limited.	A

2. The agencies shall carry out prospecting operations in compliance with the conditions specified in the said guidelines.

3. This notification shall come into force on the date of its publication in the Official Gazette and shall remain valid for a period of three years from the date of publication or till expiry or termination of the accreditation granted, whichever is earlier.

Source: The Gazette of India, Extraordinary, No. 959 Part II, Section 3-Sub section (ii), dated 24.02.2025.

22. Ministry of Mines, Notification, S.O. 3054(E)— In exercise of the powers conferred by Sub-sections (2) and (4) of Section 17 of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and after consultation with the State Government of Karnataka, the Central Government hereby declares that–

(i) the Central Government proposes to undertake prospecting and mining operations in respect of mineral Gold and any other minerals produced during the course of such prospecting or mining operations in the State of Karnataka as per the boundaries specified in the Schedule hereto annexed; and

(ii) no mineral concession shall be granted in respect of any land specified in the said Schedule.

2. This notification shall come into force on the date of its publication in the Official Gazette.

S No	Owner of the land	Village	Survey number	Extent (in Acres)	Coordinates				Toposheet No	Block ID
					LAT DD	LAT DMS	LONG DD	LONG DMS		
1.	Bharat Gold Mines Limited	Bangardagani Village, Kolar Gold Fields Taluk, Kolar District	2	374	12.92102	12° 55' 15.672"	78.26807	78° 16'5.052"	57L/2	D 1 and 2
					12.92052	12° 55'13.872"	78.27362	78° 16'25.032"		
					12.90567	12° 54'20.412"	78.27383	78° 16'25.788"		
					12.9059	12° 54' 21.24"	78.26655	78° 15'59.58"		
					12.91187	12° 54'42.732"	78.26515	78° 15'54.54"		
					12.91926	12° 55' 9.336"	78.26442	78° 15'51.912"		
2.	Bharat Gold Mines Limited	Bangardagani Village, Kolar Gold Fields Taluk, Kolar District	2	95.4	12.98597	12° 59' 9.492"	78.26017	78° 15'36.612"	57L/2	D13 A and B
					12.98456	12° 59' 4.416"	78.26533	78° 15'55.188"		
					12.98241	12° 58' 56.676"	78.26543	78° 15'55.548"		
					12.97904	12° 58' 44.544"	78.26388	78° 15'49.968"		
					12.97924	12° 58' 45.264"	78.25946	78° 15'34.056"		
					12.98597	12° 59' 9.492"	78.26017	78° 15'36.612"		

Table (Cont.)

S No	Owner of the land	Village	Survey number	Extent (in Acres)	Coordinates				Toposheet No	Block ID
					LAT DD	LAT DMS	LONG DD	LONG DMS		
3	Bharat Gold Mines Limited	Bangardagani Village, Kolar Gold Fields Taluk, Kolar District	2	534	12.96124	12° 57'40.464"	78.266	78° 15'57.6"	57L/2	D 3, 4, 5, 6, 7, 8,9,10,11,12
					12.96208	12° 57'43.488"	78.26768	78° 16'3.648"		
					12.95978	12° 57'35.208"	78.2699	78° 16'11.64"		
					12.95827	12° 57'29.772"	78.26919	78° 16'9.084"		
					12.9571	12° 57' 25.56"	78.26825	78° 16'5.7"		
					12.95578	12° 57'20.808"	78.26545	78° 15'55.62"		
					12.95537	12° 57'19.332"	78.26552	78° 15'55.872"		
					12.95241	12° 57' 8.676"	78.2663	78° 15'58.68"		
					12.95015	12° 57' 0.54"	78.26635	78° 15'58.86"		
					12.94855	12° 56' 54.78"	78.26485	78° 15'53.46"		
					12.94825	12° 56' 53.7"	78.26383	78° 15'49.788"		
					12.94674	12° 56'48.264"	78.26221	78° 15'43.956"		
					12.94598	12° 56'45.528"	78.26232	78° 15'44.352"		
					12.94585	12° 56' 45.06"	78.26189	78° 15'42.804"		
					12.94575	12° 56' 44.7"	78.26384	78° 15'49.824"		
					12.94206	12°56'31.416"	78.26443	78°15'51.948"		
					12.94054	12°56'25.944"	78.26482	78°15'53.352"		
					12.93919	12° 56'21.084"	78.26595	78°15'57.42"		
					12.93857	12°56'18.852"	78.26659	78°15'59.724"		
					12.93742	12°56'14.712"	78.26729	78°16'2.244"		
					12.93668	12°56'12.048"	78.26779	78°16'4.044"		
					12.93655	12° 56'11.58"	78.266	78° 15'57.6"		
					12.93625	12° 56'10.5"	78.26334	78°15'48.024"		
					12.93671	12°56'12.156"	78.26321	78°15'47.556"		
					12.93704	12°56'13.344"	78.26414	78° 15'50.904"		
					12.93733	12°56'14.388"	78.26421	78°15'51.156"		
					12.93764	12° 56'15.504"	78.26437	78°15'51.732"		
					12.93776	12°56'15.936"	78.26492	78°15'53.712"		
					12.93895	12° 56'20.22"	78.26467	78°15'52.812"		
					12.93958	12°56'22.488"	78.26446	78°15'52.056"		
					12.94024	12° 56'24.864"	78.2642	78°15'51.12"		
					12.94081	12°56'26.916"	78.26306	78°15'47.016"		
					12.9405	12° 56'25.8"	78.26086	78°15'39.096"		
					12.94103	12°56'27.708"	78.26034	78°15'37.224"		
					12.94222	12°56'31.992"	78.2571	78°15'25.56"		
					12.9412	12° 56'28.32"	78.25526	78°15'18.936"		
					12.94078	12°56'26.808"	78.25333	78°15'11.988"		
					12.93875	12° 56'19.5"	78.25246	78°15'8.856"		
					12.93676	12°56'12.336"	78.25243	78°15'8.748"		
					12.93215	12° 55'55.74"	78.25407	78°15'14.652"		
					12.9315	12° 55'53.4"	78.24632	78°14'46.752"		
					12.94016	12°56'24.576"	78.24597	78°14'45.492"		
					12.94024	12°56'24.864"	78.25053	78°15'1.908"		
					12.9412	12° 56'28.32"	78.25169	78°15'6.084"		
					12.94306	12° 56'35.016"	78.25216	78°15'7.776"		
					12.94471	12°56'40.956"	78.25211	78°15'7.596"		
					12.94471	12°56'40.956"	78.25507	78°15'18.252"		
					12.94434	12°56'39.624"	78.2569	78° 15'24.84"		
					12.94215	12°56' 31.74"	78.25896	78°15'32.256"		
					12.94297	12°56'34.692"	78.26049	78°15'37.764"		
					12.94531	12°56'43.116"	78.26017	78°15'36.612"		

S No	Owner of the land	Village	Survey number	Extent (in Acres)	Coordinates				Toposheet No	Block ID
					LAT DD	LAT DMS	LONG DD	LONG DMS		
					12.94558	12°56'44.088"	78.26077	78°15'38.772"		
					12.94663	12°56'47.868"	78.26054	78°15'37.944"		
					12.94656	12°56'47.616"	78.25866	78°15'31.176"		
					12.94727	12°56'50.172"	78.25882	78°5'31.752"		
					12.94958	12°56'58.488"	78.2592	78°15'33.12"		
					12.95026	12°57'0.936"	78.2609	78°15'39.24"		
					12.95174	12°57' 6.264"	78.26177	78°15'42.372"		
					12.95201	12°57' 7.236"	78.26329	78°15'47.844"		
					12.95332	12°57'11.952"	78.26495	78°15'53.82"		
					12.95477	12°57'17.172"	78.26498	78°15'53.928"		
					12.95554	12°57'19.944"	78.26493	78°15'53.748"		
					12.95568	12°57'20.448"	78.26234	78°15'44.424"		
					12.95613	12°57'22.068"	78.26094	78°15'39.384"		
					12.95964	12°57'34.704"	78.26154	78°15'41.544"		
					12.96015	12°57' 36.54"	78.26394	78°15'50.184"		
					12.96124	12°57'40.464"	78.266	78° 15'57.6"		

Source: The Gazette of India, Extraordinary, No. 2913 Part II, Section 3-Sub section (ii), dated 31.07.2024.

23. Ministry of Mines, Notification, S.O. 4819(E)—In exercise of the powers conferred by Section 10 of the Offshore Areas Mineral (Development and Regulation) Act, 2002 (17 of 2003), the administering authority hereby declares that parts of the offshore area along with the coordinates of the boundary points as specified in the Table below, shall be available for grant of composite licence.

TABLE

1. Kollam Construction Sand Block-1, Off-Kerala, West Coast, Arabian Sea of Area - 79 square kilometre for construction sand mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	76°05'40.192"E	09°00'06.406"N
B	76°07'41.151"E	09°00'06.012"N
C	76°07'41.508"E	09°01'06.241"N
D	76°08'41.988"E	09°01'06.059"N
E	76°08'42.372"E	09°02'06.288"N
F	76°11'43.812"E	09°02'05.711"N
G	76°11'43.812"E	09°01'05.485"N
H	76°12'44.291"E	09°01'05.303"N
I	76°12'43.907"E	08°59'04.849"N
J	76°13'44.387"E	08°59'04.632"N
K	76°13'44.387" E	08°58'04.403"N
L	76°05'39.835"E	08°58'05.952"N

2. Kollam Construction Sand Block-2, Off-Kerala, West Coast, Arabian Sea of Area - 78 square kilometre for construction sand mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	76°06'40.314"E	08°58'05.773"N
B	76°14'44.866"E	08°58'04.225"N
C	76°14'44.509"E	08°57'03.996"N
D	76°15'45.374"E	08°57'03.779"N
E	76°15'44.989"E	08°56'03.554"N
F	76°13'44.03"E	08°56'03.948"N
G	76°13'43.672"E	08°55'03.719"N
H	76°09'41.396"E	08°55'04.512"N
I	76°09'41.396"E	08°54'04.283"N
J	76°08'40.56"E	08°54'04.465"N
K	76°08'40.917"E	08°55'04.691"N
L	76°07'40.437"E	08°55'04.873"N
M	76°07'40.794"E	08°57'05.328"N
N	76°06'39.957"E	08°57'05.544"N

3. Kollam Construction Sand Block-3, Off-Kerala, West Coast, Arabian Sea of Area – 85 square kilometre for construction sand mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	76°09'41.396"E	08°55'04.512"N
B	76°13'43.672"E	08°55'03.719"N
C	76°13'44.03"E	08°56'03.948"N
D	76°15'44.989"E	08°56'03.554"N
E	76°15'44.632"E	08°55'03.325"N
F	76°16'45.469"E	08°55'03.108"N
G	76°16'45.112"E	08°54'02.879"N
H	76°17'45.591"E	08°54'02.701"N
I	76°17'45.591"E	08°53'02.471"N

Table (Concl'd.)

Block End Point	Longitude	Latitude
J	76°15'44.275"E	08°53'02.866"N
K	76°15'44.275"E	08°52'02.641"N
L	76°13'42.958"E	08°52'03.035"N
M	76°13'42.958"E	08°51'02.844"N
N	76°10'41.162"E	08°51'03.421"N
O	76°10'41.519"E	08°53'03.876"N
P	76°09'41.039"E	08°53'04.058"N

4. Porbandar Limemud Block-1, Off-Gujarat, West Coast, Arabian Sea of Area – 66.64 square kilometre for Limemud mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	68°51'29.88"E	21°19'25.49"N
B	68°53'37.97"E	21°19'25.58"N
C	68°53'38.01"E	21°18'29.63"N
D	68°55'46.1"E	21°18'29.7"N
E	68°55'46.12"E	21°17'33.76"N
F	68°57'54.19"E	21°17'33.8"N
G	68°57'54.22"E	21°15'41.9"N
H	68°56'50.2"E	21°15'41.88"N
I	68°56'50.24"E	21°13'49.99"N
J	68°55'46.23"E	21°13'49.97"N
K	68°55'46.2"E	21°14'45.91"N
L	68°54'42.19"E	21°14'45.88"N
M	68°54'42.15"E	21°15'41.83"N
N	68°52'34.11"E	21°15'41.75"N
O	68°52'34.06"E	21°16'37.7"N
P	68°51'30.04"E	21°16'37.65"N
Q	68°51'29.98"E	21°17'33.59"N
R	68°50'25.95"E	21°17'33.54"N
S	68°50'25.89"E	21°18'29.48"N
T	68°51'29.93"E	21°18'29.54"N

5. Porbandar Limemud Block-2, Off-Gujarat, West Coast, Arabian Sea of Area –69.81 square kilometre for Limemud mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	68°49'21.92"E	21°17'33.48"N
B	68°51'29.99"E	21°17'33.60"N
C	68°51'30.04"E	21°16'37.65"N
D	68°52'34.07"E	21°16'37.70"N
E	68°52'34.12"E	21°15'41.76"N
F	68°54'42.16"E	21°15'41.84"N
G	68°54'42.19"E	21°14'45.89"N
H	68°55'46.21"E	21°14'45.92"N
I	68°55'46.29"E	21°11'58.08"N
J	68°53'38.30"E	21°11'58.01"N
K	68°53'38.26"E	21°12'53.96"N
L	68°52'34.26"E	21°12'53.91"N

Table (Concl'd.)

Block End Point	Longitude	Latitude
M	68°52'34.21"E	21°13'49.86"N
N	68°50'26.19"E	21°13'49.75"N
O	68°50'26.13"E	21°14'45.70"N
P	68°48'18.11"E	21°14'45.57"N
Q	68°48'17.96"E	21°16'37.46"N
R	68°49'21.99"E	21°16'37.53"N

6. Porbandar Limemud Block-3, Off-Gujarat, West Coast, Arabian Sea of Area – 66.64 square kilometre for Limemud mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	68°47'14.09"E	21°14'45.49"N
B	68°50'26.13"E	21°14'45.69"N
C	68°50'26.20"E	21°13'49.76"N
D	68°52'34.21"E	21°13'49.86"N
E	68°52'34.26"E	21°12'53.71"N
F	68°53'38.26"E	21°12'53.96"N
G	68°53'38.53"E	21°11'58.11"N
H	68°54'42.29"E	21°11'58.26"N
I	68°54'42.15"E	21°11'01.10"N
J	68°48'18.40"E	21°11'01.10"N
K	68°48'18.32"E	21°11'57.72"N
L	68°47'14.40"E	21°11'57.70"N
M	68°47'14.20"E	21°12'53.50"N
N	68°46'10.48"E	21°12'53.50"N
O	68°46'10.45"E	21°13'49.45"N
P	68°47'14.12"E	21°13'49.51"N

7. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-1, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°02'54.035"E	07°36'23.673"N
B	94°07'53.757"E	07°36'22.920"N
C	94°02'52.975"E	07°31'22.896"N
D	94°02'53.311"E	07°31'23.640"N

8. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-2, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°03'26.494"E	07°31'21.848"N
B	94°08'26.157"E	07°31'21.097"N
C	94°08'25.377"E	07°26'21.072"N
D	94°03'25.771"E	07°26'21.815"N

9. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-3, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°03'28.279"E	07°26'19.533"N
B	94°08'27.884"E	07°26'18.790"N
C	94°08'27.113"E	07°21'18.765"N
D	94°03'27.564"E	07°21'19.499"N

10. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-4, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°03'49.728"E	07°21'18.337"N
B	94°08'49.276"E	07°21'17.598"N
C	94°08'48.510"E	07°16'17.572"N
D	94°03'49.018"E	07°16'18.302"N

11. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-5, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°02'51.203"E	07°16'18.071"N
B	94°07'50.698"E	07°16'17.351"N

Table (Concl.)

Block End Point	Longitude	Latitude
C	94°07'49.951"E	07°11'17.322"N
D	94°02'50.512"E	07°11'18.033"N

12. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-6, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°10'24.322"E	07°13'16.833"N
B	94°15'23.770"E	07°13'16.035"N
C	94°15'22.947"E	07°08'16.020"N
D	94°10'23.553"E	07°08'16.808"N

13. West Sewell Ridge, Polymetallic Nodules and Crusts Mineral Block-7, Off-Great Nicobar Island, Andaman Sea of Area – 84.63 square kilometre for Polymetallic nodules and crusts mineral:

Block End Point	Longitude	Latitude
(1)	(2)	(3)
A	94°10'53.919"E	07°08'14.165"N
B	94°15'53.311"E	07°08'13.372"N
C	94°15'52.492"E	07°03'13.356"N
D	94°10'53.154"E	07°03'14.140"N

Source: The Gazette of India, Extraordinary, No. 4438 Part II, Section 3-Sub section (ii), dated 06.11.2024.



3. Status of Mineral Concession in India

2,995

Mining Leases were in force in the country in 23 States and 2 Union Territories covering an area of 2,93,811 hectares during the year 2023-24.

15.97

per cent of mining lease area are in Odisha followed by Madhya Pradesh (14.86%), Rajasthan (12.34%) and Karnataka (10.53%).

644

mining leases were registered in Madhya Pradesh followed by Tamil Nadu (455).

INTRODUCTION

Under the Mines & Minerals (Development and Regulation) Act, 1957 and the Minerals (Other than Atomic and Hydrocarbons Energy Minerals) Concession Rules, 2016, the State Governments grant mineral concessions. A “mineral concession” means either a reconnaissance permit, prospecting licence, mining lease, composite licence or a combination of any of these and the expression “concession” shall be construed accordingly.

A “reconnaissance permit” (for the holder of a reconnaissance permit granted prior to January 12, 2015) means a permit granted for the purpose of undertaking reconnaissance operations. “Reconnaissance Operations” means any operations undertaken for preliminary prospecting of a mineral through regional, aerial, geophysical or geochemical surveys and geological mapping and include pitting, trenching, drilling and sub-surface excavation.

A “prospecting licence” (for the holder of a reconnaissance permit granted prior to January 12, 2015) means a licence granted for the purpose of undertaking prospecting operations. “Prospecting Operations” means any operations undertaken for the purpose of exploring, locating or proving mineral deposit.

A “composite licence” means the prospecting licence-cum-mining lease which is a two stage concession granted for the purpose of undertaking prospecting operations followed by mining operations in a seamless manner.

Also, the State Governments are required to submit a copy of every mineral concession granted or renewed under the Act and rules made thereunder, within two months of such grant or renewal, to the Controller

General, Indian Bureau of Mines and the Director General, Directorate General of Mines Safety under Rule 59 (1) of Minerals (Other than Atomic and Hydrocarbons Energy Minerals) Concession Rules, 2016. Additionally, the State Governments also have to submit a consolidated annual return of all mineral concessions granted or renewed under the Act and rules made thereunder, to the Controller General, Indian Bureau of Mines, in such form as may be specified for the purpose, and a copy shall also be supplied to the Director General, Directorate General of Mines Safety under Rule 59 (2) of Minerals (Other than Atomic and Hydrocarbons Energy Minerals) Concession Rules, 2016, not later than the 30th day of June following the year to which the return relates.

RECONNAISSANCE PERMITS, PROSPECTING LICENCES AND COMPOSITE LICENCES

No information from the State Governments/Union Territories was reported regarding grant of reconnaissance permit/ prospecting licence/ composite licence for minerals (other than Atomic Minerals, Coal, Lignite, Petroleum, Natural Gas and Minor Minerals) during the period.

MINING LEASES

Section 3(c) of the Mines & Minerals (Development & Regulation) Act 1957, defines “Mining Lease” (ML) is a lease granted for the purpose of undertaking mining operations, and includes a sub-lease granted for such purpose. The Act defines “mining operations” as any operations undertaken for the purpose of winning any mineral.

The status of mining leases of 37 Metallic and Non-metallic minerals (excluding Atomic Minerals, Coal, Lignite, Petroleum, Natural Gas and Minor Minerals) as on 31.3.2024 (P) indicates that 2995 mining leases were in force in the country across 21 States and 2 Union Territories, covering an area of 2,93,811.5373 hectares.

The state-wise summary of existing mining leases as on 31st March 2022, 2023 and 2024 (P) is detailed in Table-1. The mineral-wise summary of existing mining leases as on 31.3.2024 (P) is furnished in Table-2. Sectorwise distribution of mining leases as on 31.3.2024 (P) is furnished in Table-3. Areawise distribution of Leases (frequency in Hect.) as on 31.3.2024 (P) is furnished in Table-4.

The state-wise break up of mining leases as on 31.3.2024 as reflected in Table-1 indicates that Madhya Pradesh was leading with 644 mining leases followed by Tamil Nadu (455), Gujarat & Andhra Pradesh (377 each), Karnataka (262), Chhattisgarh (158), Odisha (153), Rajasthan (137), Maharashtra (111), Jharkhand (101) and Telengana (78).

Of the total mining lease area covered by different States/Union Territories, Odisha accounts for 15.97%, Madhya Pradesh (14.86%), Rajasthan (12.34%), Karnataka (10.53 %), Gujarat (9.10%), Andhra Pradesh (8.73%), Chhattisgarh (8.45%), Jharkhand (6.16%), Maharashtra (3.74%) and Telangana (3.55%). These ten States accounted for about 93.43% of the total mining lease area granted and the remaining 6.57% was accounted for by the rest of the 11 states and 2 UTs.

During 2023-24, mining leases in force were in both Private and Public Sectors which included Central and State Government Undertakings. Out of the total 2995 mining leases in force in the country, 2744 (91.62%) mining leases with an area of 2,10,661.4783 hectares (71.69%) are in the Private Sector and the remaining 251 (8.38%) with an area of 83,150.059 hectares (28.3%) are in the Public Sector.

In the Metallic Minerals, Ferrous group of minerals includes iron ore, manganese ore, chromite, while the Non-ferrous group of minerals comprises bauxite, copper ore, lead & zinc ores, molybdenum, nickel, tin and the noble metals which include gold, silver and platinum group of metals. Mining Leases and Composite Licences granted through auction during 2023-24 is given in Table-5.

Table – 1: Existing Mining Leases*as on 31st March 2022, 2023 and 2024 (P)
(By States)

State/UT	As on 31.03.2022		As on 31.03.2023		As on 31.03.2024 (P)	
	No.	Area (ha)	No.	Area (ha)	No.	Area (ha)
India	3095	278008.94	3007	282356.54	2995	293811.5373
Andhra Pradesh	379	25450.49	376	25411.77	377	25650.8434
Assam	7	889.5	7	889.5	6	858.5
Bihar	1	53.38	1	53.38	5	53.378
Chhattisgarh	175	26210.55	158	25491.13	158	24825.281
Goa	9	442.49	9	439.79	10	918.3144
Gujarat	440	26836.82	432	27300.96	377	26732.8153
Himachal Pradesh	40	2448.58	40	2448.61	39	2430.489
Haryana	4	46.85	4	46.85	4	46.85
Jammu & Kashmir**	37	2020.43***	38	2146.76	37	1330.231
Jharkhand	100	18068.24	100	18068.28	101	18099.9169
Karnataka	298	33047.05	293	32830.72	262	30945.005
Kerala	5	421.65	5	421.65	5	421.654
Ladakh(UT)***					1	159
Meghalaya	21	789.34	23	1061.39	23	824.6102
Madhya Pradesh	613	37838.75	598	41196.77	644	43658.856
Maharashtra	130	12872.74	113	11149.17	111	10990.034
Odisha	117	33872.45	118	36066.91	153	46921.1818

Table- 1 (Concl'd.)

State/UT	As on 31.03.2022		As on 31.03.2023		As on 31.03.2024 (P)	
	No.	Area (ha)	No.	Area (ha)	No.	Area (ha)
Rajasthan	168	34528.73	145	35007.52	137	36253.10025
Tamil Nadu	459	9158.7	456	9097.96	455	9096.255
Telangana	80	10227.38	79	10442.61	78	10429.733
Uttar Pradesh	4	2579.56	4	2579.55	4	2960.21
Uttarakhand	7	191.79	7	191.79	7	191.809
West Bengal	1	13.47	1	13.47	1	13.47

* Excluding Atomic Minerals, Coal, Lignite, Petroleum and Natural Gas & Minor Minerals

Source: Data received from various State Governments, respective State Government (DGMS/DMGS etc).

** formed a new Union Territory to be known as the Union Territory of Jammu & Kashmir vide Gazette Notification No. 53, New Delhi, the 9th August, 2019/Shravana 18, 1941 (Saka)

*** Including mining lease in Ladakh (U/T).

Note:- The data received from respective regional offices of IBM have also been taken in account wherever necessary.

(P): Provisional

Table – 2: Existing Mining Leases* as on 31.03.2024 (P)
(By Minerals)

Sl. No.	Mineral	No. of Leases	Lease Area (ha)
1	Amethyst	2	5.83
2	Apatite	1	13.47
3	Bauxite	310	23524.363
4	Borax	1	159
5	Chromite	22	5620.859
6	Copper Ore	9	3916.851
7	Diamond	2	275.963
8	Emerald	2	125.64
9	Epidote	1	4.0515
10	Fluorite	7	416.158
11	Garnet	20	118.7838
12	Gold	10	6934.26
13	Graphite	27	1199.0369
14	Iolite	2	16.417
15	Iron Ore	337	56920.617
16	Kyanite	15	288.982
17	Lead and Zinc Ore	11	6890.4875
18	Limeshell	18	873.958
19	Limestone	1812	168264.605
20	Magnesite	33	2301.721
21	Manganese Ore	215	11482.96
22	Moulding Sand	6	39.232
23	Perlite	1	144.8831
24	Rock Phosphate	7	1534.237
25	Rock Salt	1	8.122
26	Semi Precious Stone	14	164.214
27	Selenite	4	624.3466
28	Siliceous Earth	17	168.4086
29	Sillamanite	1	4.64
30	Stibnite/Antimony	1	40.468
31	Tin	17	338.183
32	Vermiculite	50	890.9683
33	Wollastonite	12	254.1042

Table- 2 (Concl'd.)

Sl. No.	Mineral	No. of Leases	Lease Area (ha)
34	Gemstone Cats Eye	3	82.357
35	Beryl	2	9.8
36	Fluorspar	1	63.2
37	Aluminous Laterite	1	90.36
Total		2995	293811.54

* Excluding Atomic Minerals, Coal, Lignite, Petroleum and Natural Gas & Minor Minerals

Source: Data received from various State Governments

Note:- The data received from respective regional offices of IBM have also been taken in account wherever necessary.

(P): Provisional

Table – 3: Existing Mining Leases* as on 31.3.2024 (P)

(By Sectors)

Sector	No. of Leases	Percent (%)	Area (ha)	Percent (%)
Total	2995	100	293811.5373	100
Public	251	8.38	83150.059	28.30
Private	2744	91.62	210661.4783	71.70

*Excluding Atomic Minerals, Coal, Lignite, Petroleum and Natural Gas & Minor Minerals

Source: Data received from various State Governments

Table – 4: Areawise Status of Lease (Frequency In Hect.)(Other than Atomic, Hydrocarbons Energy & Minor Minerals) as on 31.3.2024 (P)

(All India)

Frequency In (ha)	No. of Leases	Lease Area (ha)
0 to 2	337	398.30
>2 to 5	777	2931.42
>5 to 10	392	2817.74
>10 to 20	303	4400.40
>20 to 50	400	12891.16
>50 to 100	223	15944.85
>100 to 200	189	27245.18
>200 to 500	204	65804.03
Above 500	170	161378.45
Total	2995	293811.54

Source:- Respective State Government (DGMS/DMGs etc.)

Note:- The data received from respective regional offices of IBM have also been taken in account wherever necessary.

(P): Provisional

Table – 5: Mining leases and Composite Licences Granted through Auction during the Year 2023-24

S No	State	Name of the Block	Mineral	Date of auction	ML / CL	Area (in Ha)
1	Andhra Pradesh	Adakula	Semi-Precious stone	19.03.2024	CL	36.997
2	Andhra Pradesh	Addankivariapalem	Iron Ore	28.07.2023	CL	N.A.
3	Andhra Pradesh	Lakshmakapalle (North)	Iron Ore	28.07.2023	CL	N.A.
4	Andhra Pradesh	Lakshmakapalle (South)	Iron Ore	28.07.2023	CL	N.A.
5	Andhra Pradesh	Mincheri RF	Iron Ore	11.03.2024	CL	1327
6	Chhattisgarh	North of Arjunda	Glauconite (Potash)	04.07.2023	CL	600
7	Chhattisgarh	Saloni Block	Limestone	21.09.2023	CL	600
8	Chhattisgarh	Devri Limestone Block	Limestone	19.09.2023	CL	630
9	Chhattisgarh	Kareli Chandi Block	Limestone	20.09.2023	CL	500
10	Chhattisgarh	Giroud Uprani Block	Glauconite	27.09.2023	CL	N.A.
11	Chhattisgarh	Tumrisur Garda II	Gold	20.12.2023	CL	240

Table 5 (Cont.)

S No	State	Name of the Block	Mineral	Date of auction	ML / CL	Area (in Ha)
12	Gujarat	Mevasa Block-1	Bauxite (Aluminous Laterite) and Laterite (minor mineral)	18.07.2023	ML	5.99
13	Gujarat	Mevasa Block	Bauxite, and Marl	18.07.2023	ML	5.53
14	Gujarat	Kukaras block (Private)	Limestone and Marl	18.07.2023	ML	29.17
15	Gujarat	Nandana Block	Bauxite Limestone and Marl	19.07.2023	ML	29.17
16	Gujarat	Kodidra Block	Limestone and Marl	19.07.2023	ML	29.17
17	Gujarat	Virpur Lusari Block	Bauxite (Aluminous Laterite), Limestone and Marl	21.07.2023	ML	29.17
18	Jharkhand	Chiropat Bauxite Block/ Gumla&Lohardaga	Bauxite	20.04.2023	ML	63
19	Jharkhand	Baraiburu - Tatiba Iron and Manganese Ore Block	Iron and Manganese Ore	18.08.2023	CL	258.99
20	Jharkhand	Meralgara-Barabaljori Iron Ore Block	Iron Ore Block	10.08.2023	ML	115.22
21	Karnataka	Block No. 04, HRG	Iron Ore	24.07.2023	ML	40.04
22	Karnataka	Thimmanahalli Gold block	Gold	21.08.2023	CL	N.A.
23	Karnataka	Jaisinghpura North Block	Iron Ore	21.08.2023	ML	298.59
24	Karnataka	Basavanagudda	Gold	21.08.2023	CL	2501.9
25	Karnataka	Nidodi Bauxite Block	Bauxite	21.09.2023	CL	52.82
26	Karnataka	Kuddarka Bauxite Block	Bauxite	22.09.2023	CL	52
27	Madhya Pradesh	Katni Limestone Block*(10A2b Case)	Limestone	05.09.2023	ML	120.949
28	Madhya Pradesh	Pahari Limestone Block (10A2d Case)	Limestone	11.09.2023	ML	100.144
29	Madhya Pradesh	Pipartola Manganese Ore Block*(10A2b Case)	Manganese Ore	12.09.2023	ML	4.653
30	Madhya Pradesh	Chorgadi-Puraina Limestone Block	Limestone	06.09.2023	ML	378.404
31	Madhya Pradesh	Bhilapar Manganese Ore & Dolomite Block(10A2b Case)	Manganese Ore & Dolomite	05.09.2023	CL	9.037
32	Madhya Pradesh	Dhamani Nana Manganese Ore Block*(10A2b Case)	Manganese Ore	08.09.2023	CL	4.7
33	Madhya Pradesh	Guvali Manganese Ore Block*(10A2b Case)	Manganese Ore	05.09.2023	CL	4.61
34	Madhya Pradesh	Katangjhari Manganese Ore Block*(10A2b Case)	Manganese Ore	08.09.2023	CL	5.101
35	Madhya Pradesh	Lingaponar Manganese Ore Block(10A2b Case)	Manganese Ore	11.09.2023	CL	9.712
36	Madhya Pradesh	Naganwat-Chhoti Manganese Ore Block*(10A2b Case)	Manganese Ore	08.09.2023	CL	11.86
37	Madhya Pradesh	Uberao Manganese Ore Block*	Manganese Ore	06.09.2023	CL	4.68
38	Madhya Pradesh	Bamanbardi Limestone Block	Limestone	05.09.2023	CL	913.3954
39	Madhya Pradesh	Doter Guvali Manganese Ore Block	Manganese Ore	09.09.2023	CL	736.4
40	Madhya Pradesh	East of Dabri Limestone Block	Limestone	08.09.2023	CL	815
41	Madhya Pradesh	Garhi-Upcha Limestone Block	Limestone	09.09.2023	CL	1001.95
42	Madhya Pradesh	Nawapara Rampura Manganese Ore Block	Manganese Ore	11.09.2023	CL	725.7

Table 5 (Cont.)

S No	State	Name of the Block	Mineral	Date of auction	ML / CL	Area (in Ha)
43	Madhya Pradesh	Piploda Phosphorite Block	Phosphorite	05.09.2023	CL	464
44	Madhya Pradesh	Shitalpani Copper Block*	Copper	09.09.2023	CL	220.971
45	Madhya Pradesh	Siluwa and Jhansi Iron Ore Block	Iron Ore	13.09.2023	ML	4.59
46	Madhya Pradesh	Modri Phosphorite Block	Phosphorite	13.09.2023	CL	699.55
47	Madhya Pradesh	Pindrai Iron Ore Block	Iron Ore	14.09.2023	ML	6
48	Madhya Pradesh	Makra Block	Graphite Vanadium	14.09.2023	CL	37.5
49	Maharashtra	DevalmariKatepalli Limestone Block	Limestone	09.05.2023	ML	537.54
50	Maharashtra	Surjagad – 1 Iron Ore Block	Iron Ore	09.05.2023	CL	1526
51	Maharashtra	Surjagad – 2 Iron Ore Block	Iron Ore	09.05.2023	CL	886
52	Maharashtra	Surjagad – 3 Iron Ore Block	Iron Ore	09.05.2023	CL	640
53	Maharashtra	Surjagad – 4 Iron Ore Block	Iron Ore	12.05.2023	CL	397
54	Maharashtra	South Padve	Bauxite	09.05.2023	CL	500
55	Maharashtra	Surjagad – 6 Iron Ore Block	Iron Ore	17.05.2023	CL	658
56	Maharashtra	Kondhala	Limestone	19.05.2023	ML	105
57	Maharashtra	Minzhari Copper Block	Copper	21.11.2023	CL	1743
58	Maharashtra	Savali Manganese Block	Manganese	22.11.2023	CL	1200
59	Rajasthan	Nayorana-Dandela	Iron Ore	11.07.2023	CL	16.78
60	Rajasthan	Hariyav Jashpura	Limestone	10.07.2023	ML	94.62
61	Rajasthan	Khakhiliya Khera Block Rajasmand, CL	Basemetal & Associated Minerals	24.08.2023	CL	642.45
62	Rajasthan	Pipliyani Block, Udaipur, CL	Basemetal	25.08.2023	CL	518.65
63	Rajasthan	Manpura Block, Banera, CL	Basemetal & Associated Minerals	28.08.2023	CL	1150
64	Rajasthan	Ladi Ka Bas Block Tehsil Neem ka Thana & District Sikar	Iron Ore	28.08.2023	CL	38.75
65	Rajasthan	Kalakota Block Tehsil Neem ka Thana & District Sikar	Iron Ore	29.08.2023	CL	34.44
66	Rajasthan	Ladanan/v Ladana-Ladani, Tehsil Mavli, District - Udaipur(Raj)	Basemetal	31.08.2023	CL	300
67	Rajasthan	Bhabhriya ka Kheda Block n/v Lakha Ka Khera District Chittaurgarh	Basemetal	04.09.2023	CL	970
68	Rajasthan	Toda Block n/v Toda, Tehsil Neem ka Thana, District Sikar	Iron Ore	12.09.2023	CL	18.43
69	Rajasthan	PSB02 n/V Harima Tehsil, District Nagaur	Limestone	12.09.2023	ML	4.9
70	Rajasthan	PSB01 n/V Tehsil & District Nagaur	Limestone	11.09.2023	ML	4.9
71	Rajasthan	PSB06 n/V Tehsil & District Nagaur	Limestone	18.09.2023	ML	4.8
72	Rajasthan	PSB07 n/V Tehsil & District Nagaur	Limestone	19.09.2023	ML	4.8
73	Rajasthan	HPB 19 n/V Harima Khetolaw, Pithasia, Sarasani	Limestone	03.10.2023	ML	476.8
74	Rajasthan	HPB 20 n/V Harima Khetolaw, Pithasia, Sarasani	Limestone	04.10.2023	ML	547.55
75	Rajasthan	PSB03 n/v Harima Tehsil & District Nagpur	Limestone	21.02.2024	ML	4.8

Table- 5 (Concl'd.)

S No	State	Name of the Block	Mineral	Date of auction	ML / CL	Area (in Ha)
76	Rajasthan	PSB04 n/v Harima Tehsil & District Nagpur	Limestone	22.02.2024	ML	4.8
77	Rajasthan	PSB05 n/v Harima Tehsil & District Nagpur	Limestone	23.02.2024	ML	4.8
78	Rajasthan	PSB08 n/v Harima Tehsil & District Nagpur	Limestone	26.02.2024	ML	4.8
79	Rajasthan	PSB09 n/v Harima Tehsil & District Nagpur	Limestone	27.02.2024	ML	4.8
80	Rajasthan	PSB010 n/v Harima Tehsil & District Nagpur	Limestone	28.02.2024	ML	4.8
81	Rajasthan	PSB011 n/v Harima Tehsil & District Nagpur	Limestone	29.02.2024	ML	4.8
82	Rajasthan	PSB012 n/v Harima Tehsil & District Nagpur	Limestone	01.03.2024	ML	4.8
83	Rajasthan	PSB013 n/v Harima Tehsil & District Nagpur	Limestone	04.03.2024	ML	4.8
84	Rajasthan	PSB 014 n/v Sarsani tehsil & district Nagaur	Limestone	05.03.2024	ML	4.8
85	Rajasthan	PSB 015 n/v Sarsani tehsil & district Nagaur	Limestone	06.03.2024	ML	4.8
86	Rajasthan	PSB016 n/v Harima Tehsil & District Nagaur	Limestone	07.03.2024	ML	4.8
87	Rajasthan	PSB017 n/v Harima Tehsil & District Nagaur	Limestone	11.03.2024	ML	4.8
88	Rajasthan	PSB 018 n/v Khetolow Tehsil & District Nagaur	Limestone	12.03.2024	ML	4.8
89	Rajasthan	HPB21 n/v Khetolow Tehsil & District Nagaur	Limestone	13.03.2024	ML	307
90	Uttar Pradesh	Girar	Iron Ore & Gold	23.11.2023	CL	271.18
91	Uttar Pradesh	Bharhari	Iron Ore	24.11.2023	CL	134.77
92	Uttar Pradesh	Sona Pahari	Gold	23.11.2023	CL	79
93	Goa	Block V- Advalpale-Thivim Mineral Block	Iron Ore	21.04.2023	ML	36.22
94	Goa	Block VI- Cudnem-Cormoalem Mineral Block	Iron Ore	25.04.2023	ML	38.51
95	Goa	Block VII-Cudnem Mineral Block	Iron Ore	26.04.2023	ML	75.3
96	Goa	Block VIII-Thivim-Pirna Mineral Block	Iron Ore	27.04.2023	ML	72.05
97	Goa	Block IX-Surla-Sonshi Mineral Block	Iron Ore	28.04.2023	ML	254.51

Source:- Ministry of Mines website.



4. Exploration & Development

18,591.5 sq. km Total area of Large Scale Mapping (LSM) was carried out by GSI, in 2023-24.	216.212 sq. km Total area of Detailed Mapping (DM) was carried out by GSI, during 2023-24	1,15,470.92 m Drilling was carried out by Geological Survey of India (GSI) in 2023-24.
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NATIONAL MINERAL POLICY

The Hon'ble Supreme Court in its judgment dated 02.08.2017 in the Writ Petition (Civil) No.114 of 2014, inter alia, directed the Union of India to revisit the National Mineral Policy (NMP), 2008 and formulate a fresh, more effective and meaningful policy.

In compliance with the directions of the Hon'ble Supreme Court, Ministry of Mines (MoM) vide its Order No. 15/1/2017-MV dated 14.08.2017 had constituted a Committee. The Committee included representatives from Central Ministries, State Governments, Industry Associations, Professional Bodies and it also consulted NGOs and other Stakeholders. Adopting a problem-solving approach, the committee held four meetings wherein stakeholders' concerns were discussed in detail.

The Committee submitted its report to the Ministry on 31.12.2017. Based on the report submitted by the committee, Ministry of Mines prepared a draft National Mineral Policy (NMP), 2018 and uploaded it on the official website of the Ministry on 10.01.2018 for seeking comments/suggestions from the stakeholders.

Based on the Committee's Report and the inputs received from stakeholders during subsequent consultations, the Ministry of Mines prepared the National Mineral Policy, 2019. The Union Cabinet in its meeting held on 28.02.2019 approved the "National Mineral Policy, 2019". The salient features of the "National Mineral Policy, 2019" are as follows:

1. It proposes to increase the production of major minerals by 200% in 7 years. It also proposes to reduce trade deficit in Mineral Sector by 50% in 7 years.
2. It aims to attract private investment through incentives like financial package, right of first refusal at the time of auction etc. and any other appropriate incentives as per international practice.
3. Introduces the concept of Exclusive Mining Zones having in-principle statutory clearances for grant of mining lease. It also proposes to identify critically fragile ecosystem and declare such areas as 'no-go areas'/inviolable areas.
4. It emphasises implementation of all relevant Acts/ Rules related to rehabilitation, resettlement and welfare

of tribal communities at the time of grant of mineral concessions.

5. Encourages States to auction mineral blocks with pre-embedded statutory clearances.
6. To institutionalise the mechanism for ensuring sustainable growth of Mining Sector, an interministerial body is proposed.
7. Efforts shall also be made to grant mining the status of an Industry.
8. In case of small deposits of precious metals and basemetals, the establishment of common smelting and refining facilities shall be encouraged.
9. It seeks to align downstream regulations for the exploration, development and acquisition of overseas mineral assets to ensure an adequate supply of minerals that are not available in the country.
10. It focuses on a long-term export-import policy for the Mineral Sector to provide stability for investing in large-scale commercial mining activities.
11. Efforts shall be made to benchmark and harmonise royalty and all other levies and taxes with mining jurisdictions across the world.
12. It also introduces the concept of Inter-Generational Equity which is also recognized by the Hon'ble Supreme Court in various judgments.

MINERAL INVESTIGATIONS & EXPLORATION

Geological Survey of India (GSI)

Geological Survey of India (GSI) is an attached office of the Ministry of Mines (MoM) and is responsible for mineral resource assessment of the country in addition to the updation of national geoscience information. Natural

resource assessment is achieved through the mineral investigation and exploration by acquisition of surface and subsurface data along with the available baseline geoscience data. The exploration programmes of GSI for the field Season 2023-2024, were prioritized based on the recommendations of CGPB meeting, CGPB committee meetings and SGPB meetings of different states and time to time guidelines received from MoM. Mission II: Natural Resources Assessment is categorized into (1) Mission II A: Mineral Resource Assessment (Non-energy) (2) Mission II B: Natural Energy Resources (Coal, lignite, shale gas, geothermal etc.). GSI carries out 'reconnaissance survey' [G4], 'preliminary exploration' [G3] and 'general exploration' [G2], following the guidelines of United Nations Framework Classification (UNFC) and Mineral Evidence and Mineral Content Rules [(MEMC)-2015] and subsequent amendments.

During FS 2023-24, total 360 programmes (including 11 exploration programmes of M&CSD) were taken up under Mission-II.

The achievement made under Large Scale Mapping (LSM), Detailed Mapping (DM) and drilling pertaining to FS 2023-24 carried out by GSI is tabulated below.

Activity	Achievement
LSM (sq.km)	18591.5
DM (sq. km)	216.212
Drilling (meter)	115470.92

The details of exploration work carried out by GSI and various Central Government Undertakings, State DMGs & State Government Undertakings are furnished in Tables-1 & 2 respectively.

Table – 1: Exploration Carried out by GSI (F.S.: 2023-2024)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
IRON ORE							
West Singhbhum District, Jharkhand	Bhangaon Block	-	-	6	508.05	Sampling & Drilling	Preliminary exploration (G3) in Bhangaon Block, West Singhbhum, Jharkhand established iron ore mineralization within IOG rocks of the Bonai–Kendujhar belt, mainly hosted in BHJ and hilltop lateritic/laminated ores. Drilling (508.05 m in 6 boreholes) and trenching confirmed ore zones with Fe values ranging from 21.07–63.85% (avg. 59.54%).
West Singhbhum District, Jharkhand	Khasjamda block	1:4000	3	8	450	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Khasjamda Block, West Singhbhum, Jharkhand covered 3.0 sq km through drilling (450 m), pitting/trenching, and mapping. Iron ore zones (0.32 MT @ Fe 36.50%) and manganese ore zones (2.05 MT @ Mn 12.56%) were intersected, placing the resources in the UNFC 333 (inferred) category.
Sundargarh District, Odisha	Anandapur Block	1:4000	7	17	1450.75	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Anandapur Block, Sundargarh, Odisha involved mapping (7 sq km), drilling (1450.75 m in 17 boreholes), and trenching/pitting. Out of 17 boreholes, 11 were mineralized—7 in the north intersecting manganese—indicating subsurface potential for manganese, iron ore, and bauxite.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Kendujhar District, Odisha	Gadadharpur Block	1:4000	2	6	685.25	Mapping, Sampling & Drilling	Preliminary exploration (G-3) in Gadadharpur Block, Kendujhar, Odisha covered 2.00 sq km and established iron ore mineralization over 1.63 sq km. Drilling of 685.25 m in six boreholes intersected ore zones up to 137.15 m thick with an average grade of 45.09% Fe, indicating good potential for hematitic iron ore.
Sundargarh District, Odisha	Barpada South Block	1:2000	0.60	6	551.90	Mapping, Sampling & Drilling	Preliminary exploration (G-3) in Barpada South Block, Sundargarh, Odisha, included 0.60 sq km mapping and 551.90 m drilling. OSBS-7 intersected manganese (15% Mn), and earlier drilling (2007–2016) indicated iron ore zones averaging 45 m thickness with 59% Fe, showing good potential for iron and manganese.
Phek District, Nagaland	Purr-Sutsu	1:4000	-	3	66	Mapping, Sampling & Drilling	Preliminary exploration (G-3) around Purr-Sutsu, Phek District, Nagaland, included 1:4000 scale mapping, 15 L.km magnetic survey, and 66 m drilling (3 boreholes). Two types of magnetite were identified: bouldery pods/lenses in serpentinized peridotite (Fe ₂ O ₃ 53.8–65.6%, Cr 2.2–4%, Ni 0.68–0.75%) and lenses along the ophiolite–Phokphur contact (Fe ₂ O ₃ 30–58.8%, Cr 1.35%, Ni 0.4%, Co 0.03%). Drilling confirmed limited depth persistence (1–2 m) of the magnetite pods, indicating local enrichment of magnetic minerals.
Sonbhadra District, Uttar Pradesh	Block 'D', North of Khoriya, Bharhari	1:4000	2	6	487.15	Mapping, Sampling & Drilling	Preliminary exploration (G-3) in Mahakoshal Group, Block 'D', Sonbhadra, UP, mapped 2 sq km and drilled 487 m in six boreholes. BHC/J and ferruginous phyllite host iron ore with 3.45–43.90 m thickness (37.59% Fe) at 30% cut-off and 2.75–20.65 m (39.49% Fe) at 35% cut-off. Block 'C' resources are estimated at 15.98 MT (30% Fe) and 12.21 MT (35% Fe).
Namakkal District, Tamil Nadu	Nachukkaradu – Kavakkarapatti areas	-	-	-	-	Sampling	Reconnaissance survey in Nachukkaradu–Kavakkarapatti, Namakkal District, TN, identified banded magnetite quartzite (BMQ) within dunite, charnockite, granulite, and gneiss. Four major bands—Vadavattur, Valasiramani lower, Papappatti upper, and Panangadu—show weighted Fe ₂ O ₃ averages of 46.45%, 42.82%, 36.9%, and 32.44%, respectively, while five minor bands were also delineated, indicating the area's potential for iron ore.
MANGANESE							
Balaghat District, Madhya Pradesh	Bhagoli, Hirapur, Payali and Servi Area	-	-	-	-	Sampling	Reconnaissance survey in Bhagoli, Hirapur, Payali, and Servi, Balaghat, MP, identified manganese in garnet-bearing phyllite (up to 28.77% Mn) and limonitic iron in laterite, occurring as syn-sedimentary and hydrothermal deposits, indicating potential for both Mn and Fe.
Balaghat District, Madhya Pradesh	Kanki, Balaghat, Garra, Dongariya and Kaspur area	-	-	-	-	Sampling	Reconnaissance survey in Kanki, Balaghat, Garra, Dongariya, and Kaspur, Balaghat District, MP, identified manganese mineralization in small Gondite bands (MnO 3–5%) at phyllite–granite contacts, and minor Cu (565 ppm) in pyrite-chalcopryrite-bearing phyllite. Lateritic cappings over granite and phyllite are limonitic/goethitic, with Fe ₂ O ₃ 17–36.5% and MnO 1–2.8%, indicating localized potential for Mn and Fe.
Seoni and Balaghat Districts, Madhya Pradesh	Piparwani, Agri and Koylari areas	-	-	-	-	Mapping & Sampling	Reconnaissance survey in Piparwani, Agri, and Koylari areas, Seoni and Balaghat Districts, MP, identified three potential manganese zones—Dulhapur, Pathrapet, and Koylari—occurring as narrow lenses within quartz-mica schist and biotite gneiss. Mn mineralization is mostly near-surface (1–1.5 m), pinch-and-swell along strike and dip, and largely soil-covered. Analytical results show MnO up to 35.76% (Dulhapur) and 21.84% (Pathrapet), with elevated Ba values, indicating promising Mn potential.
Simdega District, Jharkhand	Kambakera area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) in and around Kambakera area, Simdega, Jharkhand, covered 100 sq km with 1:12500 mapping, 50 cum of pitting/trenching, and 50 bedrock samples. Mn mineralization was recorded at Gidhiburu, Donagpani, Ginikera, and Bhukumunda as lenses, pockets, thin bands, and laminations within quartz-tourmaline veins, with Mn ranging 0.008–0.12% (mean 0.047%). Iron occurs in ferruginous quartzite with Fe 10.02–43.06% (mean 17.30%), Cr 44–153 ppm (mean 92.87 ppm), and Zn 10–368 ppm.
Sundargarh District, Odisha	Jhirpani-Panchra block	1:2000	2	7	390	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Jhirpani-Panchra block, Sundargarh, Odisha, covered 2.00 sq km mapping (1:2000 scale) and 390 m drilling in 7 boreholes, with 80 pitting/trenching, 50 channel, and 55 core samples. Mn mineralization occurs as two discontinuous NE-SW bands—Southern (1.3 km × 2–4 m) and Northern (600 m × 2–3 m)—with 10.10–32.43% Mn. Borehole intersections show cumulative thickness 1–2 m (avg. 1.5 m), and indicated resources are 0.038–0.039 Mt at ~14.3% Mn.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Bolangir District, Odisha	Dumerjharan-Banipali block	-	-	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Dumerjharan-Banipali block, Bolangir, Odisha, identified manganese and graphite zones within the Precambrian khondalite meta-sedimentary sequence. A major E-W trending Mn band (~2.5 km × 15–30 m) south of Telkuchhapar shows 19.30–37.41% Mn (avg. 30.57%), while another N-S/NW-SE band (~1–1.2 km × 20–70 m) has Mn 9.10–35% with sub-zones averaging 17.55% and 25.40% Mn. Associated graphite occurs as a 16 m thick zone with 1.02–6.30% fixed carbon (avg. 2.48% FC). Mn ore is strata-bound, conformable, and structurally controlled, classified as a stratiform deposit.
Vizianagaram District, Andhra Pradesh	Gadabavalasa Block	-	-	15	1164.65	Sampling & Drilling	G-2 stage exploration in Gadabavalasa Block, Vizianagaram, Andhra Pradesh, involved 1164.65 m drilling in 15 boreholes (7 first-level, 8 second-level). Considering intersected mineralized boreholes from G-3 and G-2 stages, the total manganese resource (secondary supergene and primary silicate-carbonate ore) is estimated at 0.475–0.525 Mt at 10% Mn cut-off, with an average grade of ~11.9% Mn.
Vizianagaram and Srikakulam Districts, Andhra Pradesh	Palavalasa Block	-	-	7	650.50		G-2 stage exploration in Palavalasa Block, Vizianagaram and Srikakulam, Andhra Pradesh, involved 650.50 m drilling in 1st- and 2nd-level boreholes. Manganese resources are estimated at 0.37–0.42 Mt with an average grade of 11.50–11.56% Mn, and the block is classified as 332 under UNFC based on geological, feasibility, and economic assessments.
Vizianagaram District, Andhra Pradesh	Manapuram-Marrivalasa area	1:12500	105	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Manapuram-Marrivalasa area, Vizianagaram, Andhra Pradesh, covered 105 sq km mapping (1:12500) over Khondalite and Migmatite rocks. Mn mineralization occurs as bands, veins, and disseminations along S1, with occurrences near Chinna Kottavalasa (100 m × 1.5 m) and Patarega (70 m × 1 m). BRS samples show Fe ₂ O ₃ 5.26–66.9% (avg. 22.45%) and MnO 0.02–12.21% (avg. 1.93%), while PTS samples show Mn up to 16.6% (avg. 1.24%), indicating limited strike and low-grade manganese.
Vizianagaram, Anakapalli and Visakhapatnam Districts, Andhra Pradesh	Kantakapalle-Sunkarapalem area	1:12500	143	-	-	Mapping & Sampling	Reconnaissance survey in Kantakapalle-Sunkarapalem area, Vizianagaram, Anakapalli, and Visakhapatnam, Andhra Pradesh, covered 143 sq km (1:12500 scale). Mn occurs as vugs and cavity fillings in host rocks, with MnO 0.05–6.58% (mean 1.5%), and graphite occurs as crystalline flaky graphite in garnet-sillimanite gneiss and pegmatite veins (FC 0.02–5.46%). Notable occurrences include MnO >6% near Relli village and Co 2289 ppm in a BRS sample. PTS samples show MnO 0.09–0.17%, FC 0.043–4.8%, and V 38–350 ppm.
Vizianagaram District, Andhra Pradesh	Madduvalasa-Ungarada Area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey in Madduvalasa-Ungarada, Vizianagaram, Andhra Pradesh, identified a Mn zone at Ungaradametta within Khondalite rocks. Mn occurs as pyrolusite and psilomelane lenses with MnO 0.01–22.04% (avg. 4.07%), Ni 2–503 ppm, Co <1–837 ppm, and FC 0.07–3.79%. Trenching (15 nos., 600 m) confirmed MnO 0.02–8.47% (avg. 2.08%) in 65 samples.
Banswara District, Rajasthan	Panchal-Amba-Gamaniya-Moti areas	-	-	-	-	Mapping & Sampling	G4 survey in Panchal-Amba-Gamaniya-Moti, Banswara, Rajasthan, identified manganiferous quartzite of Bhawanpura Formation (~900 m strike, 20 m avg. thickness) with NW-SE to WNW-ESE trend and NE dip. MnO ranges 4.37–16.25% (avg. 8.78%), blast furnace grade. Key prospective zones are near Patiya Kodar and Leelwani villages.
CHROMITE							
West Singhbhum District, Jharkhand	Mahaburu	-	-	-	-	Sampling	Reconnaissance survey (G4) in Mahaburu, West Singhbhum, Jharkhand, mapped altered ultramafic rocks/serpentinite along stream sections (~60 m wide) with laterite cover. NE-SW granite ridge and tremolite-rich bodies occur nearby. BRS and PTS analyses show high Cr (max 25,902 ppm) and Ni (max 2,993 ppm) in peridotite, serpentinite, and tremolite-bearing rocks. Laterites near Akashmani Plantation also yield encouraging Cr (9,059 ppm) and Ni (2,166–2,993 ppm).
Sukinda Ultramafic Complex, Jaipur District, Odisha	Tungeisuni Block	1:12500	-	-	-	Mapping & Sampling	G4 survey in Tungeisuni Block, Jaipur, Odisha, identified three chromite zones: Kansa and Tungeisuni ultramafics (50×100 m each) and Kendupura laterite (20×20 m). Cr ranges 1.6–10.82% (Kansa) and 1.977–2.967 ppm (Tungeisuni), Ni 599–4,411 ppm; laterite Cr 2.6–3%, Ni 316–348 ppm. Trenches show avg. Cr ₂ O ₃ 6% (Kansa) and 2.48% (Kendupura). EPMA: Cr ₂ O ₃ 57–62%, FeO 14–15%, Al ₂ O ₃ 10%.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Kargil District of Ladakh (UT)	Thasgam area	1:12500	50	-	-	Mapping & Sampling	G4 reconnaissance in Thasgam, Kargil (50 sq km), mapped ultramafic-mafic rocks of the Shilakong Ophiolite and Dras Volcanic Complex. Chromite occurs mainly in serpentinite (100×20 m) near Khiber village, with Cr 9.8–15% and Ni 1,511–1,829 ppm; peridotite shows minor mineralization. BRS samples show Ni 1,912–2,214 ppm. PGE values are low (Pd ≤25 ppb, Pt ≤35 ppb). Mineralization is lithologically controlled, confined to ultramafics, likely from mantle-derived magmatic processes and fractional crystallization.
Kodagu District, Karnataka	Kundahalli-Takeri block	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Kundahalli-Takeri, Kodagu (100 sq km), mapped lateritic profiles over serpentinitized ultramafics (dunite/peridotite) hosting Ni-Cr-Co-Sc mineralization. Trench, bedrock, and regolith samples show Cr 47–38,095 ppm, Ni 47–30,479 ppm, Co 47–797 ppm, and Sc 7–163 ppm. SEM studies confirm Ni-bearing phases: pentlandite, millerite, siegenite, and willemseite.
GOLD							
Sarangarh-Bilaigarh District, Chhattisgarh	Bilaigarh and surrounding area	1:12500	106	-	-	Mapping & Sampling	G4 reconnaissance in Sonakhan Greenstone Belt, Sarangarh-Bilaigarh, Chhattisgarh (106 sq km, 1:12500) mapped meta-basics (Sonakhan Group) and Chhattisgarh Supergroup sediments. Meta-basalt (massive and pillowed) and gabbro show quartz-carbonate-epidote veins with pyrite and base metal sulphides. Channel samples from gabbro near Tergidipa recorded Au 0.06–0.31 ppm; shale-siltstone intercalations near Arjuni showed K ₂ O 4.16–5.03 %.
Baloda Bazar and Mahasamund District, Chhattisgarh	Chandranagar and Kavtapali area	-	-	-	-	Mapping & Sampling	G4 reconnaissance in Chandranagar-Kavtapali, Chhattisgarh, mapped Sonakhan Greenstone Belt. Hydrothermal gold occurs in smoky quartz veins within sheared meta-rhyolite, meta-basalt, and talc-chlorite schist, associated with pyrite, chalcopyrite, galena, and arsenopyrite. Au up to 2060 ppb; Cu 1–2.9%, Pb 0.17–0.83%, Ni 0.19%, Cr 0.17%.
Mahasamund District, Chhattisgarh	Karankhol-Rachpalpur block	-	-	4	697	Mapping, Sampling & Drilling	G3 exploration in Karankhol-Rachpalpur block, Mahasamund, Chhattisgarh (1.5 sq km) identified gold-base metal mineralization within a 1.3 km shear zone in granophyre-metabasalt contact. Drilling (697 m in 4 BHs) intersected multiple sulfide-rich zones with Au values averaging 1.04 ppm (CMK-01) and 0.24 ppm (CMK-02).
Mahasamund District, Chhattisgarh	Tulsidipa-Remratola block	-	-	5	729.10	Sampling & Drilling	G3 exploration in Tulsidipa – Remratola block, Mahasamund, Chhattisgarh (Sonakhan Greenstone Belt) targeted sulfide mineralization hosted in brecciated chert. Drilling (729.10 m in 5 BHs) intersected multiple sulfide-rich zones (pyrite, arsenopyrite, galena) within silicified-brecciated chert and quartz-carbonate veins. Channel samples returned Au values of 0.05–0.09 ppm.
Bilaspur and Janjgir Champa Districts, Chhattisgarh	Bitkuli-Khondhra area	-	-	-	-	Sampling	A G4 reconnaissance survey was carried out in the Bitkuli-Khondhra area, Bilaspur and Janjgir Champa Districts, Chhattisgarh for gold, copper and tungsten. The study mapped quartzo-feldspathic veins in meta-basalt with two mineralized bands. Band-I (300 m × 3 m) yielded Au (0.17–1.06 ppm), W (11.29–403.50 ppm) and Cu (135–405 ppm), while Band-II (200 m × 2 m) recorded Au (0.16–0.31 ppm), W (138.67–1008.18 ppm) and Cu (430–460 ppm). The mineralization is associated with a volcano-igneous plumbing system and indicates mesothermal porphyry/IOCG affinity.
Jamui District, Bihar	Karmatiya, Boutha and Mahugaon, Sono area	LSM & DM	50 & 1	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold was undertaken in and around Karmatiya, Boutha and Mahugaon, Sono area, Jamui District, Bihar. The study included 50 sq km of LSM, 1 sq km of detailed mapping, geochemical sampling, pitting and trenching. Surface indications of mineralization were noted in quartz veins, schistose amphibolite and ferruginous quartzite with pyrite and chalcopyrite. Analytical results show anomalous Au values up to 200 ppb in bedrock, pit/trench and soil samples, particularly in ferruginised quartzite, smoky quartz veins and silicified amphibolite.
Ranchi & Saraikela-Kharsawan Districts, Jharkhand	Birdih-Harat areas	1: 12500	-	-	-	Mapping & Sampling	A G4 reconnaissance survey in Birdih-Harat areas, Ranchi and Saraikela-Kharsawan Districts, Jharkhand, delineated potential gold zones within the North Singhbhum Mobile Belt. Mapping and sampling revealed surface evidences like silicification, sulphides, ferruginization and visible gold grains. Gold values up to 200 ppb in stream sediments, 100–500 ppm in pyrite (EPMA), and 360 ppb in bedrock samples led to identification of three potential mineralized blocks for further exploration.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
West Singhbhum and Saraikela Kharsawan Districts, Jharkhand	Itihasa-Rolamarcha area	1: 12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey in Itihasa-Rolamarcha sector, West Singhbhum and Saraikela Kharsawan Districts, Jharkhand, covered 100 sq km to assess gold and associated mineralisation. A 10 km × 1.5 km ENE–WSW trending promising zone from Moradih to Itihasa was delineated, hosted in chloritised mica schist, cherty quartzite and quartz veins. The zone shows ferruginisation, silicification, sericitization, chloritisation and disseminated sulphides including pyrite, chalcopyrite and bornite.
Mayurbhanj District, Odisha	Suriagora block	1:1000	2.5	11	1238.90	Mapping, Sampling & Drilling	Preliminary exploration (G3) for gold in Suriagora block, Mayurbhanj, Odisha covered 2.5 sq km with mapping, pitting, geophysical surveys, and 11 boreholes (1238.9 m). A NE–SW mineralized zone (~500 m strike) in meta-chert and quartz–sericite schist with pyrite, chalcopyrite, and arsenopyrite was identified, with one borehole intersecting 0.86 ppm Au over 3.05 m. EPMA confirmed auriferous pyrite.
Sundergarh District, Odisha	Sargipali-Tangarpali-Khamarimunda area	-	-	-	-	Sampling	Reconnaissance survey (G4) for gold and associated minerals in Sargipali-Tangarpali-Khamarimunda, Sundergarh, Odisha used LANDSAT/ASTER imagery, mapping, sampling, pits, and trenches. A 10–15 m wide brecciated, ferruginized zone along the NNW–SSE Tangarpalli-Katra fault hosts pyrite, chalcopyrite, galena, and sphalerite, with gold up to 0.07 ppm, silver up to 2.3 ppm, and elevated Cu, Pb, and Zn. Pan concentrates from Budabahal showed gold between 0.48–6.6 ppm.
Purulia, Bankura and Jhargram Districts, West Bengal	Sirisgora, Laupal and Kalapati area	-	-	-	-	Sampling	Reconnaissance survey (G4) for gold in Sirisgora, Laupal, and Kalapati area, West Bengal covered Chaibasa Formation rocks with quartz, mica-bearing quartzite, calc-silicate, and amphibolites intruded by Kuilapal and Kalapati granites. Panning of streams confirmed gold flakes and grains. EPMA and petrography revealed base metal and telluride mineralization in veins. Gold values ranged from 0.33–0.71 ppm in calc-silicate and 0.61 ppm in pegmatite near Sirisgora.
Purulia District, West Bengal	Chatarma-Malti-Burudih area	1: 12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) for gold and base metals in Chatarma-Malti-Burudih area, Purulia, West Bengal covered 100 sq km, mapping granite gneiss/biotite gneiss, mica schist, acid volcanics, cherty quartzite, pegmatite, and quartz veins. Sampling (BRS, soil, trenches, streams) indicated elevated Pb (up to 0.22%), Cu (up to 912 ppm), Au (~0.1 ppm), and Ag (34–74 ppm) in ferruginous quartz veins. The area also shows potential for rare and critical elements like Sc, Rb, Nb, Ti, and Ni
Purulia District, West Bengal	Pargela and Dandudih area	1: 12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) for gold and REE in Pargela–Dandudih area, Purulia, West Bengal covered 100 sq km with 1:12500 mapping and systematic sampling. The area comprises Singhbhum Group, Chhotanagpur Gneissic Complex, and Dalma meta-volcanics. Altered and mineralized zones include quartz-magnetite-apatite rock (Nb 3283 ppm, Au in soil), silicified brecciated cherty quartzite (bastnaesite), and Kutni carbonatite (monazite and REE minerals). Gold grains were also recovered from panned stream sediments in Totko nala and tributaries.
Kendujhar and Jajpur Districts, Odisha	Daitari-Alutama area	-	-	-	-	Sampling	Reconnaissance survey (G4) for gold and associated minerals in Daitari–Alutama area, Kendujhar and Jajpur, Odisha covered parts of Toposheet 73G/16. The area comprises Tomka Daitari Iron Ore Group supracrustals, including BIF, banded meta-chert, and various schists, intruded by Singhbhum Granite and dolerite dykes, with polyphase deformation and shearing. Mineralization occurs in meta-basalt, meta-chert, and quartz veins, showing pyrite, silicification, oxidation, and ferruginization. Maximum gold values of 4.64 ppm were recovered from higher-order streams, with 0.11 ppm in BRS from schistose metabasalt.
North Lakhimpur and Sonitpur Districts, Assam	Dikrang River basin	-	-	-	-	Sampling	G4 reconnaissance survey for placer gold in the Dikrang River basin, Assam, identified gold in pebbly horizons of Quaternary and Kimin Formation deposits. Panning recovered 4957 grains (0.122 g) from 67 pits, with Newer Alluvium showing sporadic concentrations and bulk sampling averaging 0.00867 g/tonne.
Gangtok District, Sikkim	Lingdum and Tumin area	1:12500 & 1:2000	-	-	-	Mapping & Sampling	G4 reconnaissance survey for gold and associated minerals in Lingdum–Tumin, Gangtok, Sikkim, involved large-scale and detailed mapping. The area exposes quartzite, quartz–chlorite phyllite/schist, quartz–sericite schist, and granite gneiss, with profuse quartz and amphibolite veins. Sulphides including chalcopyrite, pyrrhotite, pyrite, and arsenopyrite were observed, along with malachite staining, and alteration features like kaolinitization, epidotization, sericitization, ferrugination, and limonitization.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Chittoor District, Andhra Pradesh	Rajagollapalli Block	-	-	11	1335	Sampling & Drilling	G3 reconnaissance survey for gold and associated minerals in Rajagollapalli Block, Chittoor District, Andhra Pradesh, was carried out. Resources of this block for Au is estimated by the cross-section method, are 258,781.2 tons with an average grade of 1.25 g/t and are classified under UNFC 333
Anantpur and Sri Sathyasai Districts, Andhra Pradesh	Togarakunta-Bhadrapuram area	1:12500	110	-	-	Mapping & Sampling	G4 reconnaissance survey for gold and associated minerals in Togarakunta-Bhadrapuram area, Anantapur and Sri Sathya Sai Districts, Andhra Pradesh, covered 110 sq km with bedrock, trench, stream sediment, petrochemical, and petrographic sampling. Gold values in the area are generally low, with isolated anomalous occurrences up to 620 ppb, along with minor concentrations of Ag, As, Mo, Sb, W, Bi, and Cu. Despite these sporadic gold occurrences, no continuous mineralized zone could be delineated in the area.
Kurnool District, Andhra Pradesh	Aspari-Hosuru area	-	-	-	-	Sampling	G4 reconnaissance survey in Aspari-Hosuru, Kurnool, Andhra Pradesh, confirmed gold (50–130 ppb) and identified high Cu (2001 ppm) and Cr (514 ppm) in schist and granodiorite along a 17 km × 3.5 km NNW-SSE shear zone, indicating strong potential for gold and sulphide mineralization and the need for detailed exploration.
Haveri District, Karnataka	Savanur and Krishnapura areas	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance survey in Savanur–Krishnapura, Haveri, Karnataka, mapped 100 sq km and identified syngenetic and epigenetic sulphide mineralization. Four potential gold-bearing blocks—Hurlikoppa, Sirbadgi, Krishnapura West, and Krishnapura East—were demarcated, with Au values in BRS and trench samples ranging from 25 ppb to 5.8 ppm, indicating significant gold potential.
Eastern Dharwar Craton, Karnataka	Amrapur area, Hungund-Kushtagi Schist Belt	-	-	5	800	Sampling & Drilling	G3 preliminary exploration in Amrapur area, Hungund-Kushtagi Schist Belt, Karnataka, identified structurally controlled gold mineralization in sheared felsic volcanics, metabasalt, and BIFs. A total of 800 m of Au-bearing zones were delineated across seven boreholes, with grades ranging from 0.07 g/t to 4.57 g/t over varying widths of 1–2.4 m.
Raichur District, Karnataka	Goldinni and adjoining area	1:12500 & 1:2000	50 & 2	-	-	Mapping & Sampling	Reconnaissance survey for gold in Goldinni and adjoining areas, Raichur District, Karnataka, covered 50 sq km (large-scale) and 2 sq km (detailed mapping). Structurally controlled gold mineralization was observed along the sheared contact between the Hutti-Maski Schist Belt and biotite granite, with sulphides (pyrite, chalcocopyrite, arsenopyrite, pyrrhotite) showing Au values in bedrock and stream sediments ranging from <25 to 130 ppb. Copper values varied from 10 to 585 ppm in bedrock.
Koppal District, Karnataka	Tondsihal area	1:12500	125	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Tondsihal, Koppal District, Karnataka, targeted gold, copper, nickel, and associated mineralization in the Hungund-Kushtagi Schist Belt. Large-scale mapping (125 sq km) identified shear zones, lineaments, and hydrothermal quartz-carbonate, tourmaline, and epidote veins with secondary copper minerals (malachite, azurite, cuprite). Gold assays were mostly below detection, except six samples (30–158 ppb) and one quartz vein (1.263 ppm). Base metals showed Cu up to 17.68% in chlorite schist and Ni up to 911 ppm in talc-tremolite schist.
Ballari District, Karnataka and Ananthapur District of Andhra Pradesh	Chellakuriki area	-	-	-	-	Sampling	Reconnaissance survey (G4) in Chellakuriki, Ballari (Karnataka) and Ananthapur (Andhra Pradesh), targeted gold and associated metals in the Hungund-Kushtagi Schist Belt. A 6 km NW–SE trending shear zone, averaging 75 m width, was delineated in three segments. Bedrock samples from Zone-I yielded up to 38 ppb Au and 1625 ppm Cu, while Zone-II samples showed up to 1.8 ppm Au and 850 ppm Cu. IP and resistivity surveys (11.5 L km) identified anomalies south of Zone-II.
Chitradurga and Davanagere Districts, Karnataka	Bharamsagara-Halekallu-Holuvudara areas	1:12500	147	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold and associated minerals was carried out over 147 sq km in Chitradurga–Davanagere Districts, Karnataka, delineating four alteration zones. Gold values up to 170 ppb (with Cu 1245 ppm) were recorded in Zone-1, while Zone-3 showed 94 ppb Au with high Mn (20% MnO, Co 185 ppm). Zone-4 yielded high Cr (5378 ppm) and Ni (1450 ppm) but no significant PGE. However, no consistent gold-bearing zones were delineated.
Raichur District, Karnataka	Buddini and adjoining area	1:12500	-	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold and associated mineralization in Buddini block, Raichur District, Karnataka, part of the Hutti-Maski schist belt, confirmed structurally controlled gold in quartz-ankerite veins along N-S shear zones. Scout drilling and trenching returned values up to 3.4m × 0.81 g/t Au and 3m × 0.5 g/t Au, indicating southern continuity of Buddini gold lodes, though drilling near Pamankallur was not encouraging.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Wayanad District, Kerala	Vythiri area, Wayanad Gold Field	1:12500	110	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold was carried out over 110 sq km in the Vythiri area, Wayanad Gold Field, Kerala, to delineate potential mineralized zones. The study revealed quartz and quartz-carbonate veins with hydrothermal alterations and sulphide mineralization (pyrite, chalcopyrite, bornite, malachite). Old gold workings were mapped, with a dump sample from Thariyode showing 21.05 ppm Au, while other samples recorded up to 256 ppb Au. Significant Cu values (up to 2.83%) were also obtained, and SEM-EDX confirmed Au-Cu-Mo, Au-Ag-Te, and Au-Cu assemblages, indicating an epigenetic hydrothermal origin.
Ballari District, Karnataka	Kurugodu	1:12500	112	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold and associated metals was carried out south of Kurugodu, Bellary District, Karnataka, covering 112 sq km with LSM, geophysical surveys (IP & resistivity), trenching, and multi-sample studies. The area comprises meta-basalts, schists, banded ferruginous chert, granite gneiss, dolerite, pegmatite, and quartz veins. Three alteration zones were delineated, with the highest gold value of 104 ppb obtained from a trench sample (KT-3) near Dammuru village.
Tiruvannamalai District, Tamil Nadu	Vengikkal-Vediappanmalai areas	1:12500	-	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold and associated minerals in Vengikkal-Vediappanmalai, Tiruvannamalai District, Tamil Nadu, identified sulphide-bearing, shear-controlled zones in banded magnetite quartzite (BMQ) and other litho-units. Out of 100 BRS, 41 samples showed gold values up to 382 ppb. Two anomalous gold zones were delineated: Vengayavelur (90–100 m × 3–5 m) and Sorakulathur (300 m × 35–50 m), with associated Cu, Ni, Co, Zn, and high Fe content.
Salem and Namakkal Districts, Tamil Nadu	Singiliyankombai area	-	-	-	-	Sampling	A G4 reconnaissance survey for gold and associated minerals in Singiliyankombai, Salem and Namakkal Districts, Tamil Nadu, mapped BMQ, biotite gneiss, ultramafics, Charnockite, migmatite, and intrusive rocks. Sulphide mineralization occurs as disseminated and vein/fracture-filled types. Four major BMQ bands, three dunite bands, and a pyroxenite band were delineated. Gold values ranged 26–292 ppb, with BMQ Fe content 19.66–42.15% and dunite MgO up to 11.10%.
Vellore District, Tamil Nadu	Rangasamuthiram-Kavalur area (Javadi hills)	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold, molybdenum, and iron ore quality was carried out over 100 sq km in Rangasamuthiram-Kavalur area, Javadi Hills, Vellore District, Tamil Nadu. The area comprises BMQ, charnockite, pyroxene granulite, hornblende gneiss, and garnetiferous charnockite, with quartz, pegmatite, and dolerite intrusives. BMQ occurs as NE-SW trending, 3–10 m thick bands with Fe content of 32–40%. Sulphide mineralization and graphite streaks were noted, but no significant gold or molybdenum values were recorded.
Nilgiri District, Tamil Nadu	Devala-Pandalur area	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold in Devala-Pandalur, Nilgiri District, Tamil Nadu, mapped BMQ, biotite gneiss, charnockite, pyroxene granulite, pyroxenite, and quartz veins. Gold values ranged 0.09–19.3 ppm in quartz veins and pyroxene granulite, while BMQ was barren. Trench samples yielded 0.05–1.5 ppm Au, and stream sediments contained gold specks. Extraction potential is limited due to terrain and forest cover.
Sikar District, Rajasthan	Saidala block	1:2000	1.5	3	370	Mapping, Sampling & Drilling	In 2022–23, a G3 preliminary exploration for gold, base metals, and associated minerals was conducted in Saidala Block, Sikar District, Rajasthan. Gold and copper mineralization occurs in quartzofeldspathic veins within albite, with sulphides as pyrite, chalcopyrite, and minor chalcocite. Channel sampling showed Cu 0.26–1.90% and Au 0.05–0.43 ppm, while boreholes encountered fine, sporadic mineralization. Only 370 m of the planned 800 m drilling was completed.
Sirohi District, Rajasthan	Watera-Bhimana-Sangwara areas	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey for gold and base metals in Watera-Bhimana-Sangwara, Sirohi District, Rajasthan, mapped Basantgarh Formation and Phulad Ophiolites with quartz veins, calc-silicates, marbles, schists, and ultramafics. Malachite, ferruginous zones, skarns, and oxidized pyrites indicate mineralization. Channel samples showed encouraging Te (0.03–0.21 ppm) and Sc (4–35 ppm) values, supporting potential gold zones.
Banswara District, Rajasthan	Jagpura North Block	-	-	16	3659.35	Sampling & Drilling	A G-2 stage exploration for gold and associated base metals in Jagpura North Block, Banswara District, Rajasthan, over 1.556 sq km to 180 m depth, mapped staurolite-mica schist, dolomite marble, quartz-mica ± magnetite schist, with quartz veins and pegmatites. Sixteen boreholes and 1318 core samples revealed sulphides (pyrrhotite, pyrite, chalcopyrite, arsenopyrite) as disseminations, veins, and massive bodies. Gold ranged 0.025–3.175 ppm, with copper up to 5.3% and nickel present.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sirohi District, Rajasthan	Pipela-Swaroopganj areas	-	-	-	-	Sampling	A G4 reconnaissance survey for gold and base metals in Pipela-Swaroopganj, Sirohi District, Rajasthan, mapped metasediments, metavolcanics, calc-silicates, schists, amphibolites, and intrusives. Three deformation phases and metamorphism affected the area. Gossans, malachite, chalcopyrite, and a 1 km × 200 m gossanized zone near Mandwaea village indicate potential gold and base metal mineralization.
DIAMOND							
In parts of Ashok Nagar, Chhatrapur, Damoh and Sagar Districts, Madhya Pradesh and Lalitpur District of Uttar Pradesh		-	-	-	-	Sampling	A reconnaissance survey in Malthone, Niwar, and Pali blocks of Madhya Pradesh and Uttar Pradesh targeted primary diamond source rocks in the Panna Diamond Belt. Geological mapping, remote sensing, and aeromagnetic data identified potential kimberlite sites. Fieldwork covered 1,400 sq km, locating 16 ultramafic dykes and collecting 200 stream sediment samples for indicator mineral studies, with anomalies and geophysical blocks delineated for magnetic surveys
Nabarangpur District, Odisha	Umerkote area	-	-	-	-	Sampling	A G4 reconnaissance survey around Umerkote, Nabarangpur (50–60 km south of Mainpur Kimberlite Field) targeted diamond source rocks in granite gneiss, porphyritic granite, and Bengapal enclaves intruded by mafic-ultramafic dykes. Mapping and 150 stream sediment samples yielded magnetite and non-magnetic heavies, with garnets identified as almandine (G0–G5), while other indicators were non-kimberlitic.
Nuapara & Bargarh Districts, Odisha	Beltukri-Kurlapali-Jagdarpur sector	-	-	-	-	Sampling	A G4 reconnaissance survey was conducted in the Beltukri-Kurlapali-Jagdarpur sector of Nuapara and Bargarh, Odisha, within the Bastar Craton-Chhattisgarh basin, part of the East-Indian Diamond Province. Remote sensing, geological traverses, and stream sediment sampling over 380 sq km targeted kimberlite/lamproite bodies. Heavy minerals including garnet, ilmenite, and diopside were recovered and analyzed, with garnets mostly falling below the G5 field, indicating non-kimberlitic affinity.
Banda & Chitrakoot Districts of Uttar Pradesh and Satna District of Madhya Pradesh	Nakaila, Kolhua Mafi, Dadwa Manpur areas	1:12500	105	2	486.0	Mapping, Sampling & Drilling	A G4 survey over 105 sq km in Nakaila, Kolhua Mafi, and Dadwa Manpur targeted diamondiferous bodies in undeformed Kaimur Sandstone. Two boreholes (486 m) intersected mafic rocks with garnet, eskolaite, and calc-alkaline lamprophyre, while XRD, Raman, and EPMA analyses confirmed celadonite, rutile, zircon, sphene, apatite, and plagioclase.
Krishna River Basin, Guntur and Palnadu Districts, Andhra Pradesh	Bellamkonda and Sattenapalli Area	1:50,000	750	-	-	Mapping & Sampling	AG4 reconnaissance survey over 750 sq km in Bellamkonda and Sattenapalli, Andhra Pradesh, targeted primary and secondary diamond sources in the Krishna River Basin. Geological traverses and 120 stream sediment samples yielded heavy minerals such as garnet, ilmenite, pyroxene, and zircon, but EPMA confirmed no kimberlitic or lamproitic affinity. The garnets, mainly almandine (G4 category), and other minerals were of crustal origin.
In parts of Tumakuru and Chikballapur Districts, Karnataka and parts of Anantapur District, Andhra Pradesh	Wajrakarur Kimberlite field in Puravara- Byalya-Holvanahalli areas	-	-	-	-	Sampling	A G4 survey over 760 sq km in the southern extension of the Wajrakarur Kimberlite Field (Karnataka-Andhra Pradesh) involved traverses and 175 stream sediment samples. While no definitive KIMs were found, Mg-ilmenite, eclogitic garnet, and chrome spinel indicated kimberlitic affinity. Xenotime-rich zones with high thorium were also identified, suggesting HREE potential, especially in the Hosuru block.
In parts of Tumakuru District, Karnataka and parts of Ananthapur District Andhra Pradesh	Southern extension of the Wajrakarur Kimberlite field in Madhugiri Kesturu areas	-	-	-	-	Sampling	A G4 reconnaissance survey over 748 sq km in Madhugiri-Kesturu (Tumakuru, Karnataka and Anantapur, Andhra Pradesh) targeted the southern extension of the Wajrakarur Kimberlite Field. Geological traverses and 175 stream sediment samples yielded heavy minerals including xenotime, zircon, monazite, garnet, spinel, and suspected KIMs (under EPMA study). Radiometric surveys and mapping identified sheared/alteration zones, with xenotime-rich areas proposed for detailed REE exploration in the next field season.
In Part of Krishna River Basin in Nalgonda and Nagarkurnool Districts, Telangana	Devarkonda-Siddapur area	1:50,000	740	-	-	Sampling	A G4 reconnaissance survey over 740 sq km in the Devarkonda-Siddapur area (Nalgonda and Nagarkurnool, Telangana) investigated potential diamond source rocks within Eastern Dharwar Craton formations. Geological traverses, stream sediment sampling, and SEM-EPMA analysis were carried out, identifying non-kimberlitic garnets (G0 category) and amphiboles from amphibolite enclaves. Suspected mafic dykes resembling lamproites/lamprophyres were noted, but no KIMs or diamonds were recovered. The study concluded the area lacks primary diamondiferous rocks, with only minor quartz, pegmatite, and dolerite dykes observed.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
PLATINUM GROUP OF ELEMENTS							
Arvalli District, Gujarat	Bhamrechina muwada and Gulab Khan Kampa area	-	-	10	1220	Sampling & Drilling	A G3 preliminary exploration was carried out in the Dadhaliya ultramafics of Aravalli District, Gujarat, covering 2 sq km based on earlier G4 results. The area comprises quartzite and phyllite of the Samlaji Formation intruded by ultramafics (birbrite, listvenite, serpentinite) within a folded synformal structure. Ten boreholes totaling 1220 m were drilled, with core samples showing sulphides (pyrrhotite, pentlandite, chalcopyrite, pyrite). Surface samples yielded 50–264 ppb PGE (avg. 176.66 ppb), and borehole samples 30–236 ppb (avg. 124.45 ppb). Further PGE and base metal assay results from core samples are awaited.
BASEMETALS							
Balrampur District, CG	Dhulangi area	-	-	6	870	Sampling & Drilling	A G3 exploration for Copper, Silver, and Lead-Zinc (Base Metals) was carried out in Dhulangi area, Balrampur District, Chhattisgarh, delineating three mineralized zones (MZ-I, II, III) within calc-silicate, amphibolite, mica schist, and quartzite along shear zones. Surface and drill core studies (870 m in 6 boreholes) revealed sulphide mineralization as disseminations, vein fillings, and stringers of chalcopyrite, bornite, galena, sphalerite, pyrite, and pyrrhotite. Resource estimation indicates 0.46 MT copper @ 0.48% Cu, 0.60 MT silver @ 9–10 ppm, and 0.19 MT Pb+Zn ore @ 4.9% grade.
Bhandara District, Maharashtra	Silli-Makardhokra-Vaghabori area	-	-	1	150	Sampling & Drilling	A G4 reconnaissance survey for copper and associated minerals was conducted in the Silli-Makardhokra-Vaghabori area, Bhandara District, Maharashtra, within Sakoli Group schists. Copper occurs along NE–SW shear zones, with malachite, quartz veins, and sulphide enrichment. Bedrock samples show up to 2.2% Cu, 1.95 ppm Au, 45 ppm Ag, and 600 ppm Bi.
Gadchiroli District, Maharashtra	Chamorshi Block	-	-	6	718.50	Sampling & Drilling	A G3 preliminary exploration for copper and associated mineralization was conducted in Chamorshi Block, Gadchiroli District, Maharashtra, within the western margin of the Bastar Craton. NNW–SSE trending quartz reefs intruding Bengpal Group granitic gneisses host copper in the form of chalcopyrite, pyrite, and malachite/azurite stains. Channel sampling (30 samples) showed 26–1700 ppm Cu, and six boreholes (718.5 m total) confirmed quartz–chlorite veins with sulphides. Resource estimates indicate 0.39–0.41 million tonnes of copper at an average grade of 0.33% Cu.
Chandrapur District, Maharashtra	Palasgaon Wadhona area	-	-	-	-	Sampling	A G4 reconnaissance survey for base metals and associated mineralization was carried out in Palasgaon–Wadhona area, Chandrapur District, Maharashtra, within the western Bastar Craton. The area exposes Archean supracrustals of the Sukma Group and basement granite gneisses intruded by granite, mafic dykes, and quartz veins. Five mineralized quartz veins occur along major NNW–SSE and conjugate ENE–WSW and NE–SE shears, showing ferruginization, silicification, chlorite alteration, and malachite staining. Analytical results show copper up to 0.60% (Dongargaon, Awalgaoon, Kharkhara, Kargata), lead up to 0.84% and zinc 0.29% (Denwari), with gold 170–244 ppb. Ore microscopy confirmed chalcopyrite and galena as primary sulphides, identifying two blocks with potential base metal mineralization.
Chandrapur District, Maharashtra	Motegaon -Aregaon area	-	-	6	-	Sampling & Drilling	A G3 preliminary investigation for copper and associated mineralization was conducted during FS 2022–2024 in Motegaon–Aregaon area, Chandrapur District, Maharashtra, covering two blocks. In Aregaon, a sub-vertical N–S to NNE–SSW trending silicified quartz reef (strike 600 m, width up to 8 m) was drilled with six boreholes, yielding estimated copper resources of ~0.18 million tonnes @ 0.39% Cu (0.2% cut-off) and ~0.29 million tonnes @ 0.37% Cu (0.1% cut-off). In Motegaon, surface mineralization occurs along N40°W–S40°E to NW–SE quartz veins, but subsurface mineralization was found to be not encouraging.
Shivpuri District, Madhya Pradesh	Burera Tal and Nayagaon areas	-	-	-	-	Sampling	Reconnaissance survey (G4) in Burera Tal and Nayagaon areas, Shivpuri District, Madhya Pradesh, identified copper and lead mineralization associated with quartz veins and mafic dykes within Bundelkhand Granitoids. Field evidence shows malachite, azurite, pyrite, chalcopyrite, bornite, and covellite. Quartz veins (1–25 cm wide) extend up to 250 m, with Cu and Pb values up to 8880 ppm and 4200 ppm respectively, indicating hydrothermal activity and sulphide mineralization.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Balaghat District, Madhya Pradesh	Baigatola Chakarwahi area	-	-	-	-	Sampling	Reconnaissance survey in Baigatola–Chakarwahi area, Balaghat District, Madhya Pradesh, identified Cu mineralization (up to 2000 ppm) in quartz and quartz-carbonate veins within Malanjkhand Granitoids, along with molybdenite. BIF of the Chilpi Group shows potential for low- to medium-grade iron ore, while gold grains and bauxite with high vanadium and gallium were also recorded.
Kabirdham District, Chhattisgarh	Pathratola-Pateltola block	1:2000	2.75	3	365	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Pathratola–Pateltola Block, Kabirdham District, Chhattisgarh, identified copper mineralization within Malanjkhand Granitoid Complex, mainly hosted in pink granites and associated quartz–carbonate veins along NNE–SSW to NNW–SSE shear zones. Chalcopyrite and bornite occur as disseminations and vein fillings with malachite stains. Drilling (365 m in 3 boreholes) intersected mineralized zones, with core samples showing up to 23,000 ppm Cu and 100 ppb Au, confirming significant subsurface sulphide mineralization potential.
Shahdol District, Madhya Pradesh	Dhandhokui and Karahi area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey in Dhandhokui–Karahi area, Shahdol District, Madhya Pradesh, covering 100 sq km, revealed base metal and gold mineralization within rocks of the Agori Formation of the Mahakoshal Group. Mineralization occurs as sulphide veinlets and disseminations of pyrite, chalcopyrite, bornite, cuprite, and galena within phyllite, carbonaceous phyllite, dolomite, and metabasites. Analytical results reported up to 70 ppb Au, 2030 ppm Zn, 800 ppm Pb, and 705 ppm Cu, indicating hydrothermal sulphide and gold mineralization controlled by multiple deformation phases.
Hesatu-Belbathan Polymetallic Belt, Jamui and Banka Districts, Bihar	Baghmari-Ambakola-Bhorsar-Phaga-Chandaipati areas	-	92	-	511.05 m	Mapping, Sampling & Drilling	Reconnaissance survey (G4) in Baghmari–Ambakola–Bhorsar–Phaga–Chandaipati areas, Jamui and Banka Districts, Bihar, revealed polymetallic sulphide mineralization hosted in talc–tremolite–actinolite schist and calc-silicate rocks. Notable results include up to 0.97% Cu and 5.10% Pb–Zn from drill cores, confirming significant Cu–Pb–Zn mineralization zones.
Hesatu-Belbathan Polymetallic Belt, Banka District, Bihar	Dhawa-Biharbari area	1:1000	2.5	8	933.55	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Dhawa–Biharbari area, Banka District, Bihar, revealed Cu–Pb–Zn mineralization hosted in tremolite–actinolite schist and amphibolite. Surface samples show up to 6.78% Cu, 10.17% Pb, and 1.27% Zn. Drilling (933.55 m in 8 boreholes) intersected zones with up to 0.95% Cu and 2.20% Pb–Zn, confirming polymetallic sulphide mineralization controlled by fracture and vein systems.
East Singhbhum District, Jharkhand	Singapura-Angarpara block	1:12500	-	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Singapura–Angarpara block, East Singhbhum District, Jharkhand, identified Cu–Ni–Au and associated mineralization within schists and quartzites of the Dhanjori and Chaibasa formations in the North Singhbhum Mobile Belt. Surface indications include malachite, azurite, bornite, chalcopyrite, pyrite, and magnetite. Analytical results show up to 5010 ppm Cu, 11412 ppm Cr, and 2611 ppm Ni, confirming hydrothermal and magmatic sulphide mineralization potential.
Giridih District, Jharkhand	Kusmarja-Gobindpur area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Kusmarja–Gobindpur area, Giridih District, Jharkhand, identified Pb–Zn–Cu mineralization hosted in talc–tremolite schist, mica schist, and amphibolite of the Chhotanagpur Gneissic Complex. Sulphides occur as minor disseminations with malachite–azurite staining near Pipradih. Analytical results show up to 13,753 ppm Pb, 10,627 ppm Zn, and 6,319 ppm Cu, though mineralization is discontinuous and lacks significant strike continuity.
Giridih District, Jharkhand	Bhikhodih-Phaphundi-Aurangaro area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Bhikhodih–Phaphundi–Aurangaro area, Giridih District, Jharkhand, identified Pb–Zn–Cu mineralization hosted mainly in calc-silicate rocks, amphibolite, and quartzite. Mineralization occurs as disseminated, massive, and vein-type ores, with galena, sphalerite, malachite, and azurite observed. Key zones include Kurkuri Pahar and Aurangaro village, with mineralization controlled by foliation and fracture planes in a highly deformed and metamorphosed Chhotanagpur Gneissic Complex.
Deogarh District, Odisha	Western extension of Adash Block	-	-	7	1523.75	Sampling & Drilling	Preliminary exploration (G3) in the western extension of Adash Block, Deogarh District, Odisha, identified copper and associated sulphide mineralization in pyroxene granulite and silicified pyroxene granulites of the Eastern Ghats Supergroup. Drilling (1,523.75 m in 7 boreholes) intersected chalcopyrite, pyrite, and pyrrhotite as disseminations, stringers, and veinlets, often associated with graphite. A significant fixed carbon zone of 8.75 m × 5.69% FC was intersected, with mineralized thickness up to 28.2 m along boreholes, confirming subsurface potential of copper and associated metals.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Mayurbhanj District, Odisha	Ambapunja block	1: 2000	2	9	2206.10	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Ambapunja block, Mayurbhanj District, Odisha, identified multiple Cu lodes within hornblende–chlorite–biotite and quartz–biotite–chlorite schists. Drilling (2,206.1 m in 9 boreholes) intersected chalcopyrite, pyrite, pyrrhotite, magnetite, and minor pentlandite as stringers and disseminations. Copper lodes extend 250–1,150 m with widths of 2–8.1 m and average grades of 0.20–0.32% Cu; a massive quartz–sulphide vein recorded 5% Cu with 0.35 ppm Au, confirming subsurface potential of copper and associated precious metals.
Mayurbhanj District, Odisha	Badasarasposi block	1: 2000	1.7	7	1530.50	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Badasarasposi block, Mayurbhanj District, Odisha, included geological mapping, pitting/trenching, geophysical surveys, and 1,530.5 m drilling in 7 boreholes. Sulphide mineralization (chalcopyrite with pyrite and pyrrhotite) was intersected in schist and granite, with copper lodes of 3–3.7 m thickness and 0.22–0.58% Cu.
Mayurbhanj District, Odisha	Bhuasuni-Kripaduma area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Bhuasuni–Kripaduma area, Mayurbhanj District, Odisha, covered 100 sq km with large-scale mapping. The area comprises Singhbhum Group metavolcanics, Bhuasuni granite variants, leucocratic granite, pegmatite and quartz veins, and ultramafic-schist units. Hydrothermal alterations with pyrite, chalcopyrite, and bornite were observed, and silicified veins (5–10 m wide, 10–30 m long) were systematically sampled for base and precious metals.
Deogarh District, Odisha	Iswarpali-Gundiapali-Brahmanipali area	1:12500	-	3	-	Mapping, Sampling & Drilling	Reconnaissance survey (G4) in Iswarpali–Gundiapali–Brahmanipali area, Deogarh District, Odisha, covered large-scale mapping, geophysical studies, sediment and bedrock sampling, trenching, and scout drilling. The area, at the contact of Eastern Ghat Mobile Belt and Iron Ore Supergroup, comprises quartzite, mafic granulite, amphibolite, migmatite gneiss, hornblende/charnockite gneisses, and dolerites, structurally complex with multiple folding and shearing phases. Surface mineralization was limited, with minor pyrrhotite, pyrite, iron oxides, and malachite stains. Scout drilling intersected thin sulphide zones in all three boreholes.
East Khasi Hills District, Meghalaya	Mawlong block	1:12500	50	-	-	Mapping & Sampling	G4 reconnaissance in Mawlong block, East Khasi Hills, Meghalaya (50 sq km) identified base metal mineralization in Shillong Group rocks. Sulfides include pyrite, arsenopyrite, chalcopyrite, sphalerite, galena, and molybdenite. Key anomalies: Umtyrniut (Cu 10–41 ppm, Pb 11–74 ppm, Zn 18–63 ppm), Lyngkien (Cu 5775 ppm), Rangshken (Pb 0.72%, Zn 0.39%, As 2955 ppm).
Doda District of Jammu & Kashmir	Bhaderwah area	1:12500	70	-	-	Mapping & Sampling	G4 reconnaissance in Bhaderwah, Doda (70 sq km) for basemetals mapped Chinta granite, Bhalla granite, calc-silicate, and schists. Mineralization of Sn, W, Rb, Sr, Li, and Th is mostly low to moderate and appears disseminated. One XRD sample confirmed scheelite; no continuous ore bodies observed.
Tehri Garhwal District, Uttarakhand	Paluni, Koti and Sumeri areas	-	-	-	-	Sampling	G4 reconnaissance for basemetal and associated mineralization in Paluni, Koti, and Sumeri, Tehri Garhwal (Lesser Himalayas) mapped ILH and OLH rocks. Sporadic malachite staining, sulpho-salts, and minor pyrite, chalcopyrite, bornite, and covellite were observed. Geochemical values were mostly low, with isolated highs (Zn 880 ppm, Cu 5637 ppm, Cu 890 ppm), but no continuous mineralized zones were identified.
Pauri Garhwal District, Uttarakhand	Odgaon, Pali Mald and Khurgad area	-	58	-	-	Sampling	G4 reconnaissance for basemetal and associated mineralization in Odgaon, Pali, Mald, and Khurgad, Pauri Garhwal, mapped 58 sq km of Paleoproterozoic to Holocene rocks. Sulphide mineralization, including bornite, chalcopyrite, and pyrite, occurs along quartz veins in phyllite around Sanglakoti-Odgaon. BRS results show Cu 10–12, 176 ppm, Pb 10–4,536 ppm, Zn 25–1,152 ppm, Hg up to 6,858 ppb, and Au <50–410 ppb.
Chamoli District, Uttarakhand	Niti area	1:12500	50 (L Km)	-	-	Mapping & Sampling	G4 reconnaissance for base metal, gold, and graphite mineralization around Niti, Chamoli, involved 50 L km traverse mapping (1:12500) across Higher Himalayan Crystalline and Tethyan Himalayan rocks. Graphite schist bands (20 cm–1 m thick, 5–25 m long) occur within Mana gneiss near Niti. Sulphide mineralization (pyrite, chalcopyrite, bornite) is noted in quartzite and quartz-carbonate veins. BRS/CS results show V 66–2,052 ppm, Ba 326–16,189 ppm in graphite schist; Cu, Pb, Zn values are low. EPMA detected subhedral gold in graphite schist (0.062–23.495 wt%).

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Mahendragarh District, Haryana	Said Alipur, Donkhera, Dholera and Begopur block	1:2000	2	1	150	Mapping, Sampling & Drilling	G3 preliminary exploration for copper and associated minerals in Said Alipur, Donkhera, Dholera, and Begopur blocks, Mahendragarh, Haryana, involved 1:2000 scale mapping over 2 sq km. Rocks of the Delhi Supergroup (calcareous quartz-biotite schist, amphibole marble/ limestone, biotite-muscovite schist, quartzite) were studied. Maximum copper values were 101 ppm (soil) and 64 ppm (channel/impure marble). Geophysical surveys (20 L km) guided a single 150 m borehole (HMDBH01), where chalcopryrite, pyrite, and ferruginisation were observed. Chemical analyses for Cu, Pb, Zn, Ni, Co, Cd, Mn, Ag, Au, and trace elements showed no significant mineralized zones.
Tons valley, Sirmaur District, H.P.	Chamri-Anyar-Ambota-Khamiara area	1:25000	50	-	-	Mapping & Sampling	G4 reconnaissance survey for lead-zinc in Chamri-Anyar-Ambota-Khamiara, Sirmaur, HP, covered 50 sq km with mapping, 90 channel, 22 BRS, 20 PCS, and 15 petro samples. Mineralization occurs in Minas Formation (host) along a 10 km NW-SE belt at Ambota, Anyar, Amtyargad, and Chamri, with galena, sphalerite, malachite, and pyrite. Grades range: Cu 586 ppm-1 %, Pb 313-792 ppm, Zn 0.1-3.1 %, low-grade at Ambota, higher at other blocks.
Annamayya & Sri Satyasai Districts, Andhra Pradesh	Chaudasamudram-Gopidine area	-	-	-	-	Sampling	G4 reconnaissance survey for base metal, gold, and REE in Chaudasamudram-Gopidine, Andhra Pradesh, identified two east-west ferruginous quartz veins (Vein 1: 1500 m × 5 m; Vein 2: 300 m × 5-8 m) with minor sulphides (pyrite, pyrrhotite, chalcopryrite). Five BRS samples collected show Cu >2000 ppm and Fe ₂ O ₃ up to 29.71 %. Systematic channel and soil sampling was carried out to delineate mineralized zones.
Tirupati District, Andhra Pradesh	Kondakuru Block	1:2000	1.4	7	964	Mapping, Sampling & Drilling	G3 preliminary exploration in Kondakuru Block, Tirupati, Andhra Pradesh, involved 964 m drilling in 7 boreholes, detailed 1:2000 mapping (1.4 sq km), and 18.42 L km of geophysical surveys. Hosted in silicified gabbro and basalt of the Kandra Ophiolite Complex, mineralization includes chalcopryrite, pyrite, pyrrhotite, covellite, chalcocite, with minor pentlandite and bravoite. Resource estimates: 0.639-0.706 Mt at 0.56-0.59 % Cu (0.1 % cut-off, 333 UNFC).
Yadgir District, Karnataka	Wajhal, Thanda, and Hebbal Buzurg	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance survey in Wajhal, Thanda, and Hebbal Buzurg, Yadgir District, Karnataka, targeted hydrothermal breccia. The brecciated mafic dyke yielded a maximum of 1.43 % Cu (10 out of 32 samples >2000 ppm Cu). Channel sampling in the western brecciated mafic dyke returned 0.812 % Cu over 5 m. The ENE-WSW trending, steeply north-dipping hydrothermal breccia cuts felsic and mafic gabbro dykes. High-grade Cu is hosted in the brecciated mafic dyke, while low-grade occurs in felsic breccia near dyke contacts. Alteration zones: proximal—quartz + chlorite + carbonate + pyrite + chalcopryrite; distal—epidote + carbonate + pyrite ± chalcopryrite.
Davanagere District, Karnataka	Harappanahalli West Block and adjoining areas	1:12500 & 1:2000	97 & 3.26	3	370	Mapping, Sampling & Drilling	G4 reconnaissance survey around Harappanahalli West Block, Davanagere District, Karnataka, covered 97 sq km by LSM and 3.26 sq km by DM, with 370 m of inclined scout drilling (3 boreholes). Rocks include Sargur Group supracrustals, biotite gneiss, migmatite, Giant Quartz Veins (GQVs), brecciated zones, and felsic/ mafic intrusives. Copper mineralization occurs mainly in GQVs as disseminations and veins of chalcopryrite, pyrite, covellite, malachite, and bornite, with alteration zones of ferruginization, chloritization, silicification, carbonatization, and brecciation. BRS samples show Cu 10 ppm-5.5 %, Pb 10-85 ppm, Zn 10-420 ppm. One channel sample returned 1.857 % Cu over 9 m; a 3 m inclined copper-bearing zone (0.21 % Cu at 0.2 % cutoff) was intersected in BH KDHWO4.
Yadgir District, Karnataka	Balashettihal and adjoining areas	-	5	-	-	Mapping & Sampling	G4 reconnaissance survey around Balashettihal, Yadgir District, Karnataka, identified a 100 m wide, 5 km long ENEWSW trending brecciated silicified zone associated with a dolerite dyke, showing malachite staining. Borehole KYB1 (218.5 m) intersected localized copper mineralization with core values from <10 to 2425 ppm, mainly between 99.2-106.2 m. Bedrock samples also show limited Cu near maficbreccia contacts. Geophysical mapping over 5 sq km recorded high resistivity (1000-10000 Qm) and low chargeability (3-8 mV/V) zones, but no significant sulphide-bearing silicified zone. Overall, Cu mineralization is localized with no major deposit indicated.
Dharmapuri District, Tamil Nadu	Pagalalahalli - Gettipatti shear zone of Dharmapuri Suture Rift Zone	-	-	-	-	Sampling	G4 reconnaissance survey along PagalahalliGettipatti shear zone, Dharmapuri, Tamil Nadu, targeted copper and associated metals. Mineralization occurs as pyrite, pyrrhotite, chalcopryrite, covellite, bornite, pentlandite, and sphalerite along fractures and altered zones in pyroxene granulite, banded magnetite quartzite, and occasionally quartzofeldspathic gneiss.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Bhadradi-Kothagudem District, Telangana	Reddigudem, Kinnersani, Sarakal and Venkatapuram areas	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance survey in Reddigudem, Kinnersani, Sarakal, and Venkatapuram, Bhadradi-Kothagudem, Telangana, targeted copper and associated mineralization. Mapping (100 sq km, 1:12500) identified mineralized zones in old workings: Sarakal shows a 200 m × 40–50 m Cu zone (dumps up to 2.46 % Cu), while Venkatapuram has a 400 m × 70 m zone (dumps ~0.3 % Cu). Quartzite, amphibolite, BIF, and schist are the host lithologies, with mineralization largely near old workings.
Rangareddy District, Telangana.	Chinna Madgul and Girikottapalli areas	-	-	-	-	Sampling	G4 reconnaissance survey in Chinna Madgul and Girikottapalli, Rangareddy, Telangana, targeted basemetal, rare metal (RM), REE, and associated mineralization. The area comprises monzogranite, syenogranite, granite gneiss, pegmatites, and mafic dykes with shear zones trending NW-SE and NNW-SSE. Mineralization is generally insignificant, with REE mostly confined to allanite-bearing monzogranite. ΣREE values range 21–1288 ppm (highest in BRS-51), and trench/regolith samples are low. Two REE zones were delineated: Zone 1 (0.6 km × 0.2 km, avg ΣREE 695 ppm) and Zone 2 (0.5 km × 0.2 km, avg ΣREE 352 ppm). Pegmatites showed negligible REE potential.
Udaipur District, Rajasthan	Semari Block	-	-	-	-	Sampling	G4 reconnaissance survey in Semari Block, Udaipur, Rajasthan, targeted basemetal, REE, gold, and associated mineralization. The area lies in the Aravalli Fold Belt and Mangalwar Complex, exposing gneisses, migmatites, dolomitic marble, quartzite, phyllite, and schists. Sulphide mineralization (pyrite, chalcopyrite) occurs at contacts between dolomitic marble and migmatite, while graphite is associated with carbonaceous phyllite (strike 1.5–1.7 km, thickness 2–3 m). A total of 23 bedrock, 26 pitting/trenching, 12 PCS, and 86 channel samples were collected, with 27 samples from graphite zones analyzed for fixed carbon.
Bhilwara District, Rajasthan	Barana-Malesari area	-	-	-	-	Sampling	G4 reconnaissance survey in Barana-Malesari area, Bhilwara, Rajasthan, targeted basemetal and associated mineralization. The area exposes Archean basement rocks of Sandmata and Mangalwar Groups, including granites (magnetite-bearing, orthoclase-rich, sheared, gneissic). Bedrock and channel samples were collected to assess REE mineralization. Three mineralized zones were identified: MZ-I near Ambesar (strike 400 m, width 15–20 m) with malachite/azurite staining and pyrite/chalcopyrite in quartzite and calc-silicate rocks; MZ-II near Barana (strike 6 km, width 30–40 m) showing malachite staining in amphibolite with chalcopyrite and bornite.
Sikar District, Rajasthan	Manda area	-	-	-	475	Sampling & Drilling	G4 reconnaissance survey in Manda area, Sikar, Rajasthan, targeted basemetal and REE mineralization within Paleoproterozoic Delhi Supergroup (Rajgarh Formation, Alwar Group). The area is largely covered by sand dunes and alluvium, with isolated NE-SW trending hills of feldspathic and ferruginous quartzite. Bedrock chemical analyses were generally low, though Zn in water reached 927 ppb. A borehole (475 m) was drilled, 100 core samples submitted, and pyrite chunks were observed in quartz veins within amphibolites.
Sikar & Nagaur Districts, Rajasthan	Savant Gadh area	1:12500	100	3	484	Mapping, Sampling & Drilling	G4 reconnaissance in Savant Garh, Sikar & Nagaur, Rajasthan, targeted basemetal, REE, and rare metals over 100 sq km. The area exposes Rajgarh quartzites, mica schists, albitite, and intrusives with NE–SW foliation and polyphase deformation. LSM, 22 channel samples, 26 BRS, 484 m scout drilling, and 100 m core sampling were conducted. Basemetal values in BRS/channel samples are low (Cu, Pb <100 ppm; Zn 163 ppm), though water samples show elevated Cu (max 477.64 ppb). Boreholes intersected quartz-mica schist, tremolitic dolomite, albitized granite/gneiss, and albitite with minor pyrite/chalcopyrite, ferruginization, and limonitization. Hydrothermal mineralization is indicated; REE sampled from quartz, pegmatite, and albitite zones.
Banswara District, Rajasthan	Ghatiyana block	-	-	15	2399	Sampling & Drilling	G-2 exploration in Ghatiyana block, Banswara, Rajasthan, targeted copper and associated gold mineralization within Jaggura Formation (Aravalli Supergroup) comprising phyllite, metabasics, quartz-chlorite-albite-epidote rock, impure marble, and quartz veins. During FS 2023-24 and spillover 2024-25, 15 boreholes (RJBGN-12 to RJBGN-26) totaling 2399 m were drilled at 100 m spacing targeting 60–180 m depth. Copper lode-1d was the main target; mineralization in 2nd and 3rd level boreholes was weaker than earlier 1st level boreholes (FS 2021-22). Five boreholes (RJBGN-13, 16, 17, 18, 19) intersected copper lodes (>0.2% Cu, 2 m stoping width). Gold values were notable in RJBGN-12, 13, 15, 17 (max 0.80 ppm, RJBGN-13), and molybdenum was recorded in RJBGN-17, 18, 19 (max 1105 ppm, RJBGN-17).

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Banswara District, Rajasthan	Ghatiyana North block	1:2000	1.035	8	923	Mapping, Sampling & Drilling	G-3 exploration in Ghatiyana North block, Banswara (1.035 sq km) mapped phyllite, quartz-chlorite-albite-epidote rock, dolomitic marble, and quartz/pegmatite veins. Mineralization occurs at phyllite-marble contacts in quartz-chlorite-albite-epidote rock (700 m × 4–10 m), with malachite, ferruginous veins, hydrothermal alteration, and sulphides (chalcopyrite, bornite, covellite, minor chalcocite). Drilling (923 m, 8 boreholes) confirmed sulphides; main ore: chalcopyrite > bornite > chalcocite > covellite > magnetite > digenite.
Ajmer District, Rajasthan	Kajipura Block	1:12500	50	-	-	-	A G4 reconnaissance survey for base metals (Pb, Zn, Cu), rare earth elements (REE), and associated minerals was conducted over 50 sq km in Kajipura Block, Ajmer, covering rocks of the Delhi Supergroup (Beawar and Ajmer Formations), Sendra-Ambaji Granite, pegmatites, amphibolite dykes, and Quaternary sediments. The area is structurally complex, exhibiting multiple deformation phases, folding, faulting, and regional metamorphism (greenschist to amphibolite facies). Analytical results show Pb up to 1,370 ppm and Zn up to 10,522 ppm in impure/dolomitic marble near Somalpura, while granite samples yield TREE values up to 1,263 ppm. Hydrothermal veins in quartzite and calc-gneiss show elevated Fe ₂ O ₃ , Cr, Pb, Zn, and Ba, with graphite occurring in calc-schist/calc-gneiss. Mineralization is mainly associated with dolomitic/impure marble, calc-schist/calc-gneiss, and ferruginized quartzite along hydrothermal zones, and sulphide-bearing zones have been delineated in the southeastern part of Somalpura.
Bharatpur District, Rajasthan	Nithar-Kankhera areas	-	100	-	-	Mapping & Sampling	A reconnaissance survey for base metals (Cu, Pb, Zn) was conducted in Nithar-Kankhera, Bharatpur District, Rajasthan, to assess their potential. The area exposes Bhilwara and Delhi Supergroup rocks, with the Nithar Formation hosting quartzite, conglomerate, and basic metavolcanics. Large-scale mapping over 100 sq km included 50 bedrock, 100 channel, 10 petrochemical, and 15 PS/OM samples. Three mineralized zones in Nithar, Kankhera, and Kair show chalcopyrite and malachite. Of six analytical results received, Cu values of 1%, 0.34%, and 1.10% were recorded; other results are awaited.
Udaipur District, Rajasthan	Chanavda - Paduna area	-	-	-	-	Sampling	Reconnaissance survey for Pb, Zn, Cu was conducted in Chanavda-Paduna, Udaipur, Rajasthan (G4). The area exposes Aravalli Supergroup rocks with granite/quartz veins, trending N-S and folded. Mineralization occurs as malachite staining, chalcopyrite, pyrite, and galena in dolomite and dolomitic conglomerate. Channel sampling shows Pb+Zn up to 3.29% and Cu up to 0.30%. The block is considered favorable for Pb, Zn, and Cu exploration.
Sikar District, Rajasthan	Surera area	1:12500	100	3	506.60	Mapping, Sampling & Drilling	A reconnaissance survey (G4) covering 100 sq. km in the Surera area, Sikar District, Rajasthan, was conducted to identify potential base metal and REE mineralization zones. The area comprises Bhilwara Supergroup (mica schist, quartzite) and Delhi Supergroup rocks, with post-Delhi intrusives like mafic dykes and pegmatites. Geochemical sampling and scout drilling (506.60 m in 3 boreholes) revealed lean sulphide zones, with highest Cu (424 ppm) and notable REE values in Kankra area, but initial analytical results were not encouraging. Final evaluation awaits complete base metal and REE assay results.
Sikar District, Rajasthan	Northern part of Dariba Block	1:2000	0.50	6	850	Mapping, Sampling & Drilling	Preliminary exploration (G3) in the northern part of Dariba Block, Sikar District, Rajasthan, covered 0.5 sq. km during 2023–24 to assess base metal potential. The area mainly comprises dolomitic marble with visible malachite staining, specularite, and chalcocite veins. Channel sampling delineated a 900 m long and 2–18 m wide mineralized zone (MZ-I) with copper grades between 0.2% and 0.53% Cu. Subsequent drilling (850 m across six boreholes) confirmed copper sulfide mineralization in the form of disseminations and vein fillings containing chalcocite, bornite, chalcopyrite, and covellite, with intersected zones showing up to 0.39% Cu.
Ajmer and Pali Districts, Rajasthan	Chang-Samel block	-	-	-	-	Sampling	A reconnaissance survey (G4) in the Chang-Samel block, Ajmer and Pali Districts, Rajasthan, was conducted to identify base metal and associated mineralization. The area comprises Delhi Supergroup rocks intruded by granite, pegmatite, and quartz veins. Surface signs like malachite staining and sulphide minerals indicate copper mineralization along a 1260 m long, 3–5 m wide NE-SW carbonate horizon near Ravaniya. Chemical analyses show up to 1.10% Cu and high REE values (4705 ppm) in pegmatite near Chhoti Roopnagar.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Alwar District, Rajasthan	Kohrana-Raiwana area	1:12500	100	-	-	Mapping & Sampling	A reconnaissance survey (G4) in the Kohrana-Raiwana area, Alwar District, Rajasthan, covered 100 sq km to locate potential zones of base metal, REE, and RM mineralization. The area comprises rocks of the Delhi Supergroup — mainly phyllite, quartzite, and carbonaceous phyllite of the Arauli and Bharkol formations — showing complex folding and deformation. Surface indications like malachite staining and rare pyrite-chalcocopyrite occurrences were observed. High vanadium values (600–1000 ppm) were recorded in carbonaceous phyllite west of Khapariya and Tasing villages, indicating lithologically controlled vanadium mineralization.
Jaipur and Sikar Districts of Rajasthan	Baneti-Bhopiya area	1:12500	100	-	-	Mapping & Sampling	A reconnaissance survey (G4) in the Baneti-Bhopiya area, Jaipur and Sikar Districts, Rajasthan, covering 100 sq km, aimed to identify potential base metal, REE, and RM mineralization zones. Surface indications such as malachite staining, ferruginisation, and sulphide minerals (pyrite, chalcocopyrite, chalcocite) occur in banded dolomitic marble of the Kushalgarh Formation. Ancient mine workings, 650–700 m long, were discovered southwest of Mothoka with strong copper staining. Channel sampling around these workings revealed copper values ranging from 0.13%–2.10%, leading to delineation of two mineralized zones—MZ-I (1.8 km long) and MZ-II (1 km long).
Alwar & Dausa Districts Rajasthan	Jaisinghpura Block	1:2000	1.7	7	940	Mapping, Sampling & Drilling	Preliminary exploration (G3) in the Jaisinghpura Block, Alwar and Dausa Districts, Rajasthan, covered 1.7 sq km to delineate base metal and associated precious metal zones. The area comprises Dogeta Formation rocks of the Raialo Group within the Delhi Supergroup. Integrated geological, geochemical, and geophysical studies led to the identification of two mineralized zones—MZ-I (700 m long, 10–40 m wide) and MZ-II (300 m long, 10–30 m wide). Drilling (940 m in seven boreholes) confirmed copper mineralization in six holes, recording values up to 0.30% Cu, along with traces of Ag (5–30 ppm) and Au (0.05–0.17 ppm), indicating significant polymetallic potential.
Udaipur and Chittorgarh Districts, Rajasthan	Fatehnagar block	-	-	-	-	Sampling	A reconnaissance survey (G4) in the Fatehnagar block, Udaipur and Chittorgarh Districts, Rajasthan, was conducted to assess copper and associated base metal mineralization. The area comprises rocks of the Bhilwara Supergroup, including schists, quartzite, and dolomitic marble intruded by Untala Granite. A mineralized zone about 10 m wide and 300 m long (N30°E trend) was identified near Jagat Shing Ji Ka Kheda, marked by malachite staining and sulphide dissemination. Trench and bedrock samples show copper values ranging from 0.12% to 0.41%, indicating a promising copper-bearing zone within the silicified granite.
Udaipur District, Rajasthan	Jetpura-Jamun-Savna areas	1:12500	100	-	-	Mapping & Sampling	A reconnaissance survey (G4) in the Jetpura-Jamun-Savna areas, Udaipur District, Rajasthan, covering 100 sq km, was conducted to assess base metal and associated mineralization. The area comprises rocks of the Mangalwar Complex of the Bhilwara Supergroup, including granite gneisses, migmatites, ultramafic bodies, amphibolites, and ferruginous quartzites intruded by grey and pink granites. Structurally, it has undergone three deformation phases. Surface indications such as malachite staining, ferruginization, and sulphide specks were observed, along with old workings near Jamun showing high chromium and nickel values, suggesting potential ultramafic-hosted mineralization.
Udaipur District, Rajasthan	Nauva Block	1: 2000	2	5	750	Mapping, Sampling & Drilling	G3 Preliminary exploration in the Nauva Block, Udaipur District, Rajasthan, covering 2 sq km, aimed to assess copper and associated mineralization along the tectonic contact between the Mangalwar Complex (Bhilwara Supergroup) and the Aravalli Supergroup. Surface indications like malachite staining, ferruginization, and sulphides (chalcocopyrite, bornite, pyrite) were observed. Drilling of 750 m in five boreholes tested subsurface continuity, with copper values up to 2.62% Cu recorded from bedrock and channel samples. Significant Cu zones of 5 m × 0.28% Cu, 2 m × 0.30% Cu, and 3 m × 0.58% Cu were delineated, supported by high chargeability anomalies from ground geophysical data.
Udaipur District, Rajasthan	Ramaj Block	-	-	3	370	Sampling & Drilling	G3 preliminary exploration in the Ramaj Block, Udaipur District, Rajasthan, aimed to assess copper and associated mineralization within the Mangalwar Complex. The area comprises granite gneiss, micaceous quartzite, schist, and amphibolite intruded by Gingla Granite. Surface mineralization features include malachite staining, silicification, and sulphides (pyrite, chalcocopyrite), mainly within meta-sedimentary rocks and shear zones. Drilling of 370 m in three boreholes revealed weak sulfide mineralization (0.05–0.1% Cu) with analytical results up to 330 ppm Cu, indicating low potential despite minor surface enrichment (800 ppm Cu in one channel).

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sikar District, Rajasthan	Palsana area	-	-	-	-	Sampling	G4 reconnaissance survey in the Palsana area, Sikar District, Rajasthan, covered Rajgarh Formation lithologies of the Delhi Supergroup, focusing on feldspathic quartzite, mica schist, and intrusive pink granite and albitite veins. Structural mapping revealed complex folding and multiple foliation phases. Borehole sampling identified pyrite and minor sulphides within schists, but surface base metal mineralization is absent, and preliminary analyses show base metals below 100 ppm and REE values under 500 ppm; further assay results are pending.
Sikar District, Rajasthan	Ganeshpura area	-	-	3	404.6	Sampling & Drilling	G4 reconnaissance survey in the Ganeshpura area, Sikar District, Rajasthan, was carried out over the Rajgarh Formation of the Alwar Group (Delhi Supergroup) comprising arkose, conglomerate, mica schist, and quartzite intruded by acidic and basic dykes. The area shows NE-SW trending, folded, and faulted rocks with prominent joints. Sulphide mineralization, mainly pyrite and chalcopyrite, is spatially associated with the Albitite line. Three scout boreholes totaling 404.6 m of drilling were completed, with two intersecting visible sulphide zones. Core, petrochemical, and petrological samples have been submitted for geochemical analysis to assess the area's base metal and REE potential.
Bhilwara District, Rajasthan	Kesarpura Block	-	-	7	1223	Sampling & Drilling	G3 preliminary exploration for reassessment of base metal mineralization in the Kesarpura Block, Bhilwara District, Rajasthan, involved 1,223 m of drilling in seven boreholes (RJKB-9 to RJKB-15) to test copper continuity along IP anomalies. Significant copper mineralization was intersected in four boreholes within silicified feldspathic mica schist of the Potla Formation and quartz-amphibole mica schist of the Rewara Formation. Sulphides include chalcopyrite, pyrite, pyrrhotite, and bornite occurring as disseminations and stringers. Cobalt occurrences up to 1200 ppm were also noted, possibly within the chalcopyrite lattice. Final resource estimation awaits complete analytical results.
Ajmer District, Rajasthan	West of Bhojyawas area	1:2000	-	6	790	Mapping, Sampling & Drilling	G3 preliminary exploration west of Bhojyawas, Ajmer District, Rajasthan, covered mapping and 790 m of drilling to follow up on earlier G4 recommendations. The area belongs to the Ajmer Formation of the Ajabgarh Group within the Delhi Supergroup and is mostly soil-covered, with limited exposures of chlorite schist, quartzite, and dolomite. Sulphide mineralization, dominated by pyrite with minor chalcopyrite and pyrrhotite, occurs in intercalated schist-quartzite sequences as veins, stringers, and disseminations. Despite surface indications like malachite staining, drilling did not reveal significant copper mineralization.
Udaipur District, Rajasthan	Kotri block	1:2000	1.5	5	660	Mapping, Sampling & Drilling	G3 preliminary exploration in the Kotri block, Udaipur District, Rajasthan, covered 1.5 sq km of mapping and 660 m drilling in five boreholes targeting copper within basic metavolcanics and quartzite of the Delwara Formation. Surface signs included silicification, ferruginisation, malachite staining, and sulphides in quartz veins. Core logging confirmed pyrite, pyrrhotite, and chalcopyrite occurrences, but analytical results showed low base metal concentrations (Cu up to 0.14%) and no significant Pb-Zn values. Mineralization occurs mainly as disseminations, stringers, and fracture fillings, likely formed by hydrothermal fluids.
Sirohi District, Rajasthan	Sagna-Deldar-Bharja area	1:12500	50	-	450	Mapping, Sampling & Drilling	G4 reconnaissance survey in the Sagna-Deldar-Bharja area, Sirohi District, Rajasthan, covered 50 sq km through large-scale mapping, pitting, trenching, and 450 m of scout drilling with comprehensive sampling. The area comprises impure marble and amphibolite bodies intersected by basic intrusives. Though analytical results for gold and base metals were unpromising, notable scandium (~90 ppm) and higher strontium concentrations (>5000 ppm) were recorded in altered impure marble near basic contacts. Pyrite was observed along fractures in drill cores, and hydrothermal alteration features with ferruginisation and calcification were noted along the Kui-Chitrasani shear zone.
Jhunjhunu District, Rajasthan	Fatehpura Mandri area	-	-	4	500	Sampling & Drilling	G4 reconnaissance survey in the Fatehpura-Mandri area, Jhunjhunu District, Rajasthan, targeted copper and associated mineralization within the Alwar and Ajabgarh Groups of the Delhi Supergroup. The area comprises quartzite, amphibolite, schist, and granite with minor albititic intrusions along structural weak zones. Four scout boreholes totaling 500 m were drilled, each intersecting base metal mineralization. Visual estimation and logging indicate mineralized zones containing chalcopyrite, pyrite, pyrrhotite, and occasional bornite between depths of 88 m and 139 m, suggesting subsurface copper-bearing potential.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Ajmer District, Rajasthan	Rajgarh-Chainpura-Banewara area	-	-	-	-	Sampling	G4 reconnaissance survey in the Rajgarh-Chainpura-Banewara area, Ajmer District, Rajasthan, was carried out to assess base metal and associated mineralization within the Mangalwar Complex (BGC) and Delhi Supergroup. The area features quartzites, schists, dolomitic marble, calc-silicates, and ultramafic Phulad ophiolite bodies intruded by pegmatites and quartz veins. Mineralization mainly occurs along a NE-SW ductile shear zone between the BGC and Delhi Supergroup, marked by gossans, malachite staining, ferruginization, and old workings near Bithur. Significant values include up to 1.6% Cu, 0.38% Ni, 0.29% Cr, 0.5% Ba, and 942 ppm Li, indicating multimetallic potential within ultramafic and pegmatitic zones.
Around Karera Bhilwara and Rajsamand Districts, Rajasthan	Around Karera Bhilwara and Rajsamand	-	-	-	-	Artificial Intelligence and Machine Learning analysis	Verification study (ML-based targeting). A machine learning-based geoscientific data integration using Random Forest and CNN identified a prospective base metal zone near Karera, Bhilwara District, Rajasthan. Follow-up gravity, magnetic, and AMT surveys delineated a high magnetic anomaly (~500 m × 400 m) at depths around 250–271 m, with an associated conductive zone and an inferred fault. The area is largely soil-covered, with scarce outcrops, and a vertical test borehole is being drilled to investigate the anomaly's causative source and validate the ML-derived prospectivity map. Analytical results are awaited following geophysical logging and sampling.
REE & RM							
Balrampur District, Chhattisgarh	Chainpur-Sundarpur area	-	-	-	-	Sampling & Trenching	A reconnaissance survey in the Chainpur-Sundarpur area of the Chhotanagpur Gneissic Complex (CGC), Chhattisgarh, identified complexly deformed lithounits including gneisses, schists, granites, and pegmatites. Rare earth element (REE) values were elevated in some granitic and schistose rocks but not sufficient for extensive exploration. Four potential zones for graphite mineralization were mapped, with trench and bedrock samples showing promising fixed carbon content up to 27.25%, warranting further exploration and geophysical surveys.
Raigarh District, Chhattisgarh	Sardab - Ganjaipali area	-	-	-	-	Sampling	A reconnaissance survey in the Sardab-Ganjaipali area, Raigarh, Chhattisgarh, of the Gondwana Supergroup identified ferruginous and feldspathic sandstones containing REE-bearing minerals like zircon, monazite, rutile, and hematite. Geochemical studies show ΣREE values in various media, with stream sediments exhibiting the highest concentrations (up to 25,671 ppm). Despite localized enrichments, overall REE concentrations are not sufficiently promising for further exploration.
Nagpur District, Maharashtra	Kawalapur block,	1: 4,000	04	53	494.5	Mapping, sampling & Drilling	Preliminary exploration (Stage-G3) of Rare Earth Elements (REE) and Rare Metals (RM) in the Kawalapur block, Nagpur, Maharashtra, over 4 sq km involved detailed mapping, auger drilling (53 boreholes, 494.5 m), and collection of bedrock, soil, stream sediment, pit/trench, and auger samples. Petrographic and geochemical studies indicate significant in-situ REE mineralization, with tREE in heavy fractions ranging from 197.51 ppm to 48,508.86 ppm, dominated by monazite, along with xenotime and zircon. Anomalous REE zones, particularly in eastern and southeastern parts, trend NW-SE and cover ~25% of the area. Pegmatites and dolomitic marble are the main lithologies, with stream sediments showing ΣREE up to 11,985 ppm and pit samples up to 35,523 ppm, suggesting strong exploration potential in pegmatite-hosted zones.
Chandrapur District, Maharashtra	Chionda-Kanhargaon area	-	-	-	-	Mapping & sampling	A reconnaissance survey (Stage-G4) in the Chionda-Kanhargaon area, Chandrapur, Maharashtra, targeted secondary REE mineralization in Gondwana sandstones, calc arenites, conglomerates, ferruginous sandstones, and mudstones. Analytical results show ΣREE in soil, stream sediment, and pit samples ranging from 90 ppm to 4.2%, with heavy mineral concentrates consistently exceeding 1% ΣREE in several samples. Monazite was identified as the primary REE-bearing mineral, indicating localized but significant REE enrichment.
Chandrapur District, Maharashtra	Dabha-Chakbapur area	1:12500	100	-	-	Mapping & sampling	A reconnaissance survey (G4 stage) in the Dabha-Chakbapur area, Chandrapur, Maharashtra, covering 100 sq km, mapped Gondwana sandstones and alluvial deposits. Geochemical analysis shows TREE values in heavy mineral-concentrated stream sediments ranging from 0.07% to 8.45% (average 1.2%), with LREE dominant. REE mineralization occurs mainly as secondary enrichment in alluvial placers. Heavy minerals identified include monazite (dominant), zircon, and xenotime, contributing to elevated TREE and zirconium in stream sediment concentrates. Soil, pit, and bedrock samples also indicate moderate REE values, supporting localized enrichment.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sindhudurg District, Maharashtra	Mangaon-Kumbharwada area	-	-	-	-	Sampling	A reconnaissance survey (G4 stage) in the Mangaon-Kumbharwada area, Sindhudurg, Maharashtra, mapped granitoids of the Peninsular Gneissic Complex, including plagioclase-rich granodiorite and alkali feldspar granite. Analytical results from regolith, bedrock, and groove samples show tREE values up to 1,913 ppm, with alkali feldspar granite displaying the highest REE enrichment, particularly in weathered B and C horizons. A potential 13 sq km zone around Humraswadi and Sawantwadi was identified for further exploration.
Shahdol District, Madhya Pradesh	Gohrsa-Barkachh-Tihki-Sota and Kalleh area,	-	-	-	-	Sampling	A reconnaissance survey (G4 stage) in the Gohrsa-Barkachh-Tihki-Sota-Kalleh area, Shahdol, Madhya Pradesh, covering the South Rewa Gondwana basin, shows ΣREE values in bedrock (71–1,448 ppm), regolith/soil (166–857 ppm), stream sediments (247–7,406 ppm), and heavy mineral concentrates (4,912–18,974 ppm). High REE concentrations, particularly in Tiki Formation sandstones, along with thorium enrichment up to 4,000 ppm, indicate that areas south of Tihki, Sota, and Barkachh are promising for further REE exploration.
Jamui District, Bihar	Betiah and Barmasiya areas	1:12500	100	-	-	Mapping & sampling	A reconnaissance survey (G4 stage) in the Betiah and Barmasiya areas, Jamui, Bihar, covering 100 sq km, mapped granite gneisses, enclave suites of the Chhotanagpur Gneissic Complex, Bihar Mica Belt metasediments, and younger intrusives including pegmatites. Heavy mineral analysis identified coltan, monazite, zircon, allanite, tourmaline, apatite, and sphene. Geochemical results show tREE values in bedrock (5–1,305 ppm), pit/trench (76–4,404 ppm), and stream sediments (230–6,373 ppm), with high concentrations associated with intrusive granite, granite gneiss, migmatite, and zoned pegmatites, indicating promising REE and rare metal mineralization zones.
Jamui District, Bihar and Giridih District, Jharkhand	Dumakola and Paharpur areas	1:12500	100	-	-	Mapping & sampling	A reconnaissance survey (G4 stage) in Dumakola and Paharpur areas, Jamui (Bihar) and Giridih (Jharkhand), covering 100 sq km, mapped Chhotanagpur Gneissic Complex and Bihar Mica Belt lithologies, including granulite, amphibolite, quartz-mica schist, migmatite gneiss, biotite gneiss, and intrusives like porphyritic granite and pegmatites. Heavy mineral studies identified REE-bearing minerals such as allanite, monazite, and zircon. Geochemical analysis of 187 samples shows tREE values ranging from 16.99 ppm to 693.18 ppm in bedrock, with highest averages in amphibole-bearing granite (≈506 ppm) and migmatite gneiss (≈426 ppm), indicating moderate REE potential for further exploration.
Giridih District, Jharkhand	Duha area	1:12500	100	-	-	Mapping & sampling	A reconnaissance survey (G4 stage) in the Duha area, Giridih, Jharkhand, covering 100 sq km, mapped the northern Bihar Mica Belt with psammatic-pelitic sequences over CGC basement. Lithologies include quartzites, pelitic schists, amphibolite, and pegmatites, with pegmatites hosting major minerals (quartz, feldspar, micas) and accessory minerals such as tourmaline, beryl, cassiterite, and garnet. Lithium-bearing minerals (elbaite and possibly lepidolite) contribute anomalous values in bedrock samples, with Li up to 1.61%, Cs 0.76%, Rb 0.66%, Sn 258 ppm, Ta 277 ppm, and Nb 194 ppm, indicating pegmatites as fertile zones for lithium and associated rare metals. Based on these results, a G3 stage project is planned for 2024-25.
Koderma and Giridih Districts, Jharkhand	Jhalakdiha area	1:12500	100	-	-	Mapping & sampling	A reconnaissance survey (G4 stage) REE and associated Rare Metals (Li, Cs) in the Jhalakdiha area, Koderma and Giridih, Jharkhand, mapped 100 sq km of Chhotanagpur Gneissic Complex, Bihar Mica Belt, metamorphics, and intrusives (monzogranite, alkali granite, pegmatite, quartz veins). Structural features include NW-SE foliation, folding, and sinistral shearing. Out of 153 samples, 17 show tREE >1,000 ppm, with Li up to 94 ppm and Cs up to 65 ppm. Highest tREE (5,403 ppm) occurs in stream sediment over streaky gneiss, indicating localized REE and rare metal enrichment.
Gumla District, Jharkhand	Dargaon, Mundatoli and Nawatoli area	-	-	-	-	Sampling	A G4 reconnaissance survey for REE and rare metals in Dargaon, Mundatoli, and Nawatoli, Gumla, Jharkhand, mapped Chhotanagpur Gneissic Complex lithologies with pegmatite and aplitic granite intrusions. Heavy minerals include monazite, allanite, ilmenite, zircon, garnet, and magnetite. tREE values reach 4,211 ppm in stream sediments, 964 ppm in bedrock, and 707 ppm in soil, highlighting localized REE enrichment.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Simdega District, Jharkhand	Bodisera area	-	-	-	-	Sampling	A G4 reconnaissance survey for REE, rare metals, and fluoride in Bodisera, Simdega, Jharkhand, mapped granite variants intruded by pegmatites, syenite, and quartz veins. Pegmatites were classified based on trend and composition, including muscovite-rich, two-mica with garnet/tourmaline/beryl, and magnetite-bearing types. Fluoride (F) concentrations in bedrock, stream sediments, and pit samples range from 178 ppm to 4,720 ppm, with several samples >1,000 ppm. Heavy minerals identified include monazite, xenotime, zircon, and allanite, highlighting REE, rare metal, and fluoride-enriched zones for further investigation.
Ganjam District, Odisha	Kumbharajholi-Borada area	1:12500	-	-	-	Mapping & sampling	A G4 reconnaissance survey for REE and rare metals was conducted in Kumbharajholi-Borada, Ganjam, Odisha, covering Khondalite, Charnockite, and porphyroblastic granitoid gneiss of the Eastern Ghats Mobile Belt. Key lithologies include quartz-garnet-sillimanite gneiss, pyroxene granulite, and pegmatite veins with quartz, feldspar, biotite, garnet, and tourmaline. Sampling of 157 sites, including NGCM unit cells, bedrock, soil, pits, and streams, yielded TREE up to 7,808 ppm, with pink alkali feldspar granite and abyssal pegmatites identified as primary REE-hosting rocks.
Angul District, Odisha	Debinagar area	1:12500	100	-	-	Mapping & sampling	A G4 reconnaissance survey for REE mineralization in Debinagar, Angul District, Odisha covered 100 sq km with mapping, pitting-trenching, and collection of 161 samples. Geological formations belong to the Gondwana Supergroup, mainly Kamthi, Barren Measures, and Barakar. Analytical results show high REE values in stream sediments (up to 15,054 ppm) with LREE enrichment, along with anomalous Zr values (up to 13,820 ppm) in stream sediments and soils.
Nayagarh District, Odisha	Khuntapada-Arkhapal Block	1:4000	1	7	-	Mapping, sampling & Drilling	Preliminary (G3) exploration for REE & RM in Khuntapada-Arkhapal Block, Nayagarh District, Odisha involved detailed mapping and drilling of seven boreholes. The area hosts granite gneiss, charnockite, leptynite, syenite-granite pegmatites, and a syenite-pyroxenite suite with titanite and apatite veins. Analytical results from 227 core samples show consistently high ΣREE values, reaching up to 33,211 ppm in apatite. REE enrichment is mainly in pyroxene-bearing syenite and pyroxenite associated with titanite/apatite. EPMA confirmed allanite, bastnaesite, monazite, titanite, apatite, and thorite as REE phases. The main mineralized band (~220 m × 40 m) consists of apatite-titanite-bearing pyroxenite, flanked by pyroxene-bearing syenite.
Purulia District, West Bengal	Ukma-Noahatu-Bhandhhat area	1:12500	100	-	-	Mapping & sampling	Reconnaissance (G4) survey for REE covered 100 sq km with mapping, pitting, trenching, and sampling. The area lies in the northern Chhotanagpur Gneissic Complex, dominated by granite gneiss (biotite and alkali feldspar variants), with schistose rocks, amphibolite, calc-silicate, pegmatite, quartz veins, and laterite. Coarse-grained pegmatites, mainly of quartz, feldspar, and muscovite with minor garnet, tourmaline, and biotite, are widespread and trend mostly NE-SW. Structural features include NE-ESE foliations and shearing. Samples were collected for chemical, petrographic, and heavy mineral studies to assess REE potential.
Purulia District, West Bengal	Golamara-Baruadih block	-	-	-	-	Sampling & Drilling	Preliminary (G3) exploration for REE & RM in Golamara-Baruadih block identified porphyroclastic granite gneiss with quartz-feldspathic and pegmatite veins hosting vein-type mineralization. Petrographic, SEM, and EPMA studies confirmed enrichment in LREE (La, Ce, Nd) with allanite, monazite, apatite, bastnaesite, zircon, thorite, rutile, and uraninite. A 4.5 m thick zone (TREE avg. 0.14%) was intersected in WBGB-3; WBGB-9 intersected three zones up to 14 m thick (TREE avg. 0.11%), with maximum TREE 1.5% in core and 0.32% in channel samples. Mineralization is associated with magmatic-hydrothermal processes, indicating promising REE-RM potential.
Purulia District, West Bengal	Kathaldiha-Bhabanipur-Bhangra area	-	-	-	-	Sampling	Reconnaissance (G4) survey for REE & RM in Kathaldiha-Bhabanipur-Bhangra mapped CGC lithounits with pegmatite intrusions. Geochemical studies reveal REE-bearing phases (xenotime, monazite, allanite, bastnaesite, sphene) and TREE values up to >2500 ppm. A potential but sporadic REE zone is delineated around Mahara-Pathardi-Dubcharka-Chriumacha.
Purulia District, West Bengal	Kanadungri-Chauniya-Pakhi Pahar area	-	-	-	-	Sampling	Reconnaissance (G4) survey for REE & RM in Kanadungri-Chauniya-Pakhi Pahar mapped CGC gneiss, granite, migmatite, and mica schist intruded by pegmatite veins. The area shows multiple deformation phases and greenschist-amphibolite facies metamorphism. Surface indications include metallic stains, sulphides, and ferruginization. Petrography and EPMA confirmed REE-bearing phases like xenotime, allanite, and bastnaesite.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Kamle District, Arunachal Pradesh	Dha Kills and Kamporijo area	1:12500	50	-	-	Mapping & sampling	Reconnaissance (G4) survey for REE in Dha Hills–Kamporijo mapped 50 sq km of Bomdila Gneiss, including granite gneiss, biotite granite gneiss, and quartzofeldspathic veins. 74 bedrock, 40 pit, 22 petrochemical, and additional EPMA/XRD/ORE samples were collected. tREE ranged 106–786 ppm in bedrock and 135–599 ppm in soil, with LREE dominating. The highest tREE (786 ppm) occurred in weakly foliated granite gneiss, but no significant surface mineralization was observed.
West Karbi Anglong District, Assam	Krogaon block, Jashora Alkaline Complex	1:2000	1	14	892.71	Mapping, sampling & Drilling	G3 exploration in Krogaon Block, Jashora Alkaline Complex mapped 1 sq km and drilled 14 boreholes. The complex hosts pyroxenite, gabbro, carbonatite, syenite, nepheline syenite, and intrusives with perovskite-magnetite in topsoil, regolith, and fenitized pyroxenite. Analyses of 316 samples show ΣREE up to 10,643 ppm, with high Nb, Sr, and Ti. EPMA/SEM identified perovskite, aegirine, strontianite, monazite, bastnaesite, and pyrochlore, indicating significant REE and associated mineral potential.
West Karbi Anglong District, Assam	Chelear-Rongpham area	1:12500	50	-	-	Mapping & Sampling	G4 reconnaissance in Chelear-Rongpham mapped 50 sq km and collected bedrock, regolith, petrochemical, petrographic, and pit samples. Dominant lithology includes porphyritic granite of the Assam Meghalaya Gneissic Complex and quartzite of the Shillong Group. Partial analytical results show REE concentrations of 324–1046 ppm in bedrock, 52–1120 ppm in petrochemical samples, 401–1428 ppm in pitting/trenching samples, and 389–1291 ppm in regolith. Petrography identified REE-bearing phases: sphene, apatite, zircon, and monazite in granitoids.
Tinsukia District, Assam	Lido-Tikok-Tipangpani area	-	-	-	-	Sampling	G4 reconnaissance in Lido-Tikok-Tipangpani targeted REE in coal and clay beds of the Makum Coalfield. Analysis of 14 bedrock samples shows total REE values ranging from 35.2 to 282.7 ppm.
East Khasi Hills District, Meghalaya	Mawphir area of South Khasi Batholith	1:12500	50	-	-	Mapping & Sampling	G4 reconnaissance in Mawphir mapped 50 sq km to assess REE, Sn, and Li mineralization. Pegmatite and aplite veins in the South Khasi Batholith host REE-bearing minerals including allanite, monazite, apatite, fluorite, sphene, and zircon. Sulfide minerals such as molybdenite, chalcocopyrite, pyrite, bornite, and azurite are also observed.
Eastern West Khasi Hills District, Meghalaya	Kyllang, Nongthymmai and Mawjaskaw	1:12500	-	-	-	Mapping & Sampling	G4 reconnaissance in Kyllang–Nongthymmai–Mawjaskaw shows ΣREE in regolith samples ranging 1223–2161 ppm, with REE hosted in monazite, allanite, zircon, and apatite; Li occurs in lepidolite veins.
South Khasi Hills and Eastern West Khasi Hills Districts, Meghalaya	Parts of South Khasi Batholith	-	-	-	-	Sampling	Reconnaissance survey in the South Khasi Batholith (G4) indicates total REE (ΣREE) values in soil ranging from 220 to 2,067 ppm (mean 1,319 ppm), in bedrock samples from 199 to 1,548 ppm (mean 848 ppm), and in pit samples from 485 to 2,580 ppm. Associated elements include fluorine (F) from 115 to 6,200 ppm and gold (Au) from 50 to 280 ppb. The REE mineralization is predominantly hosted in allanite, monazite, zircon, and sphene within alkali feldspar porphyritic granite, along with pegmatite and aplite veins.
Ri-Bhoi District, Meghalaya	Purduwa area	1:12500	50	-	-	Mapping & Sampling	A G4 reconnaissance survey in the Purduwa area recorded REE values up to 1600 ppm, with xenotime, monazite, bastnaesite, and REE-bearing apatite, sphene, and zircon as the principal minerals.
Lalitpur District, UP	Ramnagar-Gangora area	1:12500	100	-	-	Mapping & Sampling	A G4 exploration in the Ramnagar–Gangora area (100 sq. km, Lalitpur District, UP) revealed encouraging REE mineralization within regolith and Bundelkhand granitoids. Regolith samples returned up to 1734 ppm tREE, with 29 showing >50 ppm HREE, while 11 out of 25 bedrock samples exceeded 500 ppm tREE. Pit samples yielded up to 1339 ppm tREE, and two boreholes confirmed subsurface REE zones with 500–1029 ppm tREE in regolith and granite. Results indicate a positive correlation between regolith and parent rock, with localized enrichment in both LREE and HREE.
Sonbhadra District, Uttar Pradesh	Intrusive granite with in Mahakoshal belt	1:12500	50	-	-	Mapping & Sampling	A G4 exploration for Rare Metals in the Mahakoshal belt (50 sq. km, Sonbhadra District, UP) mapped Semri Group rocks, Jhirkadandi granites, and Parsoi Formation schists with contact aureole rocks. Geophysical surveys recorded magnetic anomalies up to 1200 nT and gravity variations of 6 mgal. Chemical analyses showed REE values ranging from 26.13 to 829.54 ppm (avg. 167.5 ppm), with andalusite noted in the Parsoi Formation. The Jhirkadandi granites were classified as A-type, ferroan alkali-calcic to alkalic, indicating possible mafic–acidic magma mixing in a continental crustal setting.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Prakasam District, Andhra Pradesh	Podili Konda-Kanigiri Area	-	-	-	-	Sampling	A G4 reconnaissance survey in the Podili Konda–Kanigiri area (Prakasam District, Andhra Pradesh) identified REE- and RM-enriched syenogranites within the Nelore Schist Belt and associated intrusives. Analytical results from Podili Konda show Nb+Ta up to 1950 ppm and total REE up to 3564 ppm, while Kanigiri Konda records Nb+Ta up to 1136 ppm and REE up to 1865 ppm. Both granites are high-K, peraluminous, and anorogenic, with chondrite-normalized patterns showing strong LREE enrichment and negative Eu anomaly. Key REE-bearing minerals include monazite, xenotime, bastnaesite, zircon, and fergusonite, leading to the delineation of two potential blocks: Podili Konda (6.5 sq km) and Kanigiri Konda (4.5 sq km) for further exploration.
Prakasam District, Andhra Pradesh	Chalivendra-Marlapalem area	1:12500	-	-	-	Mapping & Sampling	A G4 survey in the Chalivendra–Marlapalem area (Prakasam, AP) revealed strong REE–RM mineralization in younger biotite and hornblende–biotite granites, with TREE values up to 25,176 ppm in regolith and 2,832 ppm in bedrock. Key minerals identified include bastnaesite, monazite, euxenite, zircon, and ilmenite, leading to delineation of potential zones for further exploration
Anantapur District, Andhra Pradesh.	Kambaduru-Aiparsipalli area	1:12500	100	-	-	Mapping & Sampling	A G4 survey over 100 sq. km delineated REE–RM mineralization mainly as allanite, occurring in alkali feldspar granite, Closepet granite, migmatite, and late-stage pegmatites, with zircons also identified. The most significant mineralization is south of Gollapalli village, where three zones (MZ-I to MZ-III) yielded TREE up to 2.62%. Of 89 bedrock samples, 13 exceeded 0.2% TREE, while soil and regolith sampling confirmed secondary enrichment. Pitting and trenching established mineralization continuity, highlighting Gollapalli as a potential REE–RM zone. Sulfide mineralization was also noted in schist belt rocks and granitoids
Annamayya District, Andhra Pradesh and Kolar District, Karnataka	Surapuvripalle-Kottapalli area	-	-	-	-	Sampling	A G4 reconnaissance survey in the Surapuvripalle–Kottapalli area (Annamayya, AP and Kolar, Karnataka) identified REE mineralization within PGC-II granodiorite, monzogranite, and syenogranite, with allanite, zircon, and apatite as key REE-bearing phases. Of the samples collected, one stream sediment sample yielded a very high TREE value of 12,579 ppm, while four soil samples also exceeded 1000 ppm. Sulfide mineralization associated with pegmatite veins was noted, comprising chalcopyrite, pyrite, bornite, and molybdenite.
Eluru District, Andhra Pradesh	Mulagalampalli area	-	-	-	-	Mapping & Sampling	A G4 reconnaissance survey in the Mulagalampalli area (Eluru, AP) identified REE mineralization in Khondalite Group gneisses and Gondwana sandstones, with monazite and xenotime as principal REE-bearing phases. Bedrock, stream sediment, soil, PCS, and pitting samples showed TREE up to 3,665 ppm, with LREE dominant over HREE. The highest REE concentrations occur in the central and northern parts of the area, leading to the delineation of a 10 sq. km anomalous REE zone around Ankampalem village.
Chittoor and Annamayya Districts, Andhra Pradesh	Sorakayalapeta-Tummanaguntapalle Area	1:12500	144	-	-	Mapping & Sampling	A G4 reconnaissance survey in the Sorakayalapeta–Tummanaguntapalle area (Chittoor and Annamayya, AP) identified REE–RM mineralization in PGC-II granodiorite, monzogranite, and syenogranite, with allanite-bearing pegmatite veins, particularly in the northern Timmapuram area. Of 204 samples analyzed, TREE values reached up to 7,632 ppm in bedrock and 3,668 ppm in regolith, with several samples exceeding 1,000 ppm. Based on geological, petrological, and chemical data, a 32 sq. km zone near Mathamapalle–Timmapuram has been delineated as REE anomalous.
Chamrajnagar District, Karnataka	Gundlupet Block	-	-	44	6979.45	Sampling & Drilling	A G2 general exploration in the Gundlupet block (Chamarajanagar, Karnataka) targeted REE and RM mineralization in sovite carbonatites and syenites located at the junction of Kollegal, Moyar, and Sargur shear zones. Carbonatites occur as linear and discontinuous bodies (strike length 2.6 km) with monazite, allanite, apatite, bastnaesite, and britholite as REE phases, showing TREE 3,148–12,478 ppm (avg. 7,193 ppm) dominated by Ce>La>Nd>Pr. Borehole drilling (6,979 m) confirmed depth continuity of ore zones. At G3 stage, Gundlupet REO resources are estimated at 7.38 million tonnes (0.25% REO cut-off, avg. 0.50% TREO, 0.42% TREE) and 3.64 million tonnes (0.5% REO cut-off, avg. 0.67% TREO, 0.57% TREE).
Trivandrum District, Kerala	Kallar-Bonacaud area	1:12500	110	-	-	Mapping & Sampling	A G4 survey in Kallar–Bonacaud, Kerala, identified REE–RM enrichment in N55°–N70°W pegmatites within Khondalite, Charnockite, and PGC-II gneisses, with bedrock REE up to 12,771 ppm. Key minerals include columbite, monazite, and graphite (up to 67.85% C). Two zones, Zone I (3.25 sq. km) and Zone II (2.47 sq. km) were delineated for further exploration.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Kollam District, Kerala	Punnala-Valiyakavu, Punalur Mica Belt	-	110	-	-	Mapping & Sampling	A G4 survey around Punnala-Valiyakavu, Kollam, Kerala, mapped 110 sq. km over diverse lithologies, identifying REE enrichment in granite and pegmatites (Σ REE up to 4,060 ppm) and high-grade graphite in garnet-biotite gneiss and phlogopite schist (fixed C up to 95.3%). Pegmatites also show elevated Th (593 ppm), while laterites and ultramafic rocks contain Cr, Ni, and V anomalies. Mineralization is structurally controlled along WNW-ESE deep-seated fractures, with mica and graphite extending to depth.
Theni District, Tamil Nadu	Kambam Area	1:2000	2.6	18	~1080	Mapping, Sampling & Drilling	A G3 survey in Kambam, Theni (2.6 sq. km), identified ENE-WSW trending carbonatite bodies with primary and secondary REE mineralization. Boreholes confirmed resources of 10.05 MT Σ REE in bedrock and 3.17 MT Σ REE in topsoil/quaternary sediments, with high grades up to 26,478 ppm P2O5.
Kanyakumari District, Tamil Nadu	Near Putteti	1:12500	100	-	-	Mapping & Sampling	A G4 survey in Putteti, Kanyakumari (100 sq. km), identified REE mineralization in Pan-African syenites within Archean Khondalite, Charnockite, and PGC-II gneisses. Three mineralized zones were delineated: Zone I (Σ REE up to 11,655 ppm), Zone II (up to 6,107 ppm), and Zone III (up to 4,541 ppm), based on bedrock, trench, and soil analyses.
Jayashankar-Bhupalpally, Telangana	Rudraram Kalan area	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey in Rudraram Kalan, Jayashankar-Bhupalpally, Telangana (100 sq. km), identified REE mineralization in Gondwana sediments (Upper Kamthi, Yerrapalli, Bheemaram, Maleri formations) as monazite in bedrock, soil, and stream sediments. Stream sediments and soil heavy fractions show high Σ REE (up to 89,817 ppm), with LREE dominant over HREE, alongside elevated Ti, Zr, Th, and Hf. Two REE potential zones were delineated for further exploration.
Bhadradi-Kothagudem District, Telangana	Annapureddipalli area	-	140	-	-	Mapping & Sampling	A G4 reconnaissance survey in Annapureddipalli, Bhadradi-Kothagudem, Telangana (140 sq. km), identified REE mineralization as monazite in Upper Gondwana sandstones (Kota & Maleri Formations), mainly enriched in stream sediments and heavy minerals (Σ REE up to 40,429 ppm, avg. 16,820 ppm) with LREE dominant. Soil regolith shows moderate REE (avg. 624 ppm), and actinides (U, Th), Zr, Hf are also present. Three potential REE zones were delineated for further exploration.
Kawant Taluka, Chhota Udepur District, Gujarat	Eastern Extension of Central Block of Ambadungar Carbonatite Complex	-	-	20	11983	Sampling & Drilling	A G2 exploration in the eastern extension of the Central Block, Ambadungar Carbonatite Complex, Gujarat, involved 20 vertical boreholes (total 11,983 m) to delineate REE, Nb, and RM beneath Deccan Trap basalts. Carbonatite (soelite, brecciated, ferro-carbonatite) occurs at 55–90 m depth, with 10,500 m of REE/Nb-favorable zones intersected. Pyrochlore, sulphides, magnetite, vanadium, apatite, zircon, and barite were observed, and resource estimation is pending.
Parts of Mehsana and Banaskantha Districts of Gujarat	Sudasna-Jawanpura areas	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Sudasna-Jawanpura, Gujarat (100 sq.km) mapped Sendra-Ambaji granitoids and Aravalli metasediments, collecting bedrock samples from granite, pegmatite, microgranite, and syenite. TREE values ranged 597–1509 ppm, with Rb 402–1277 ppm, while pegmatites were tourmaline- and biotite-rich, indicating REE and rare metal potential.
Chhota Udepur District, Gujarat	Koliyathar-Gunata	1:4000	11	72	233.55	Mapping, Sampling & Drilling	G3 preliminary exploration in Koliyathar-Gunata, Chhota Udepur (11 sq.km) involved detailed mapping and 233.55 m of auger drilling over Godhra granite and regolith (1–7 m thick). Analyses show LREE enrichment (La, Ce, Nd) with TREE 532–1876 ppm in regolith, 894–1364 ppm in BRS, and REE-bearing minerals like monazite and zircon identified, indicating notable REE/RM potential.
Chhota Udepur District, Gujarat	Aniyadri-Gondariya area	-	-	-	-	Sampling	G4 reconnaissance in Aniyadri-Gondariya, Chhota Udepur, targeted granite-gneiss, granite, and late-stage intrusives (pegmatite, aplite) for REE/RM. Lamprophyre dykes (Camptonite type) and regolith (2–4 m thick) are potential zones for secondary ion-adsorption REE, while heavy minerals from channel bars reveal xenotime, monazite, zircon, and apatite. Sulphide mineralization was also noted; complete interpretation awaits full chemical analyses.
Chhota Udepur District, Gujarat	Panvad-Kawant area in and around Panvad and Sajwa areas	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Panvad-Kawant, Chhota Udepur (100 sq.km) mapped 218 alkaline dykes (lamprophyre, trachyte, tinguaitite, carbonatite). Analyses show Nb 102–453 ppm, Ba 270–5831 ppm, Sr 254–3781 ppm; REE 126–846 ppm. XRD identified andradite, polybasite, wulfenite, perovskite, confirming REE/RM potential.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Barmer District, Rajasthan	Nakoda, Malani Igneous Suite	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Nakoda, Barmer (100 sq.km) mapped rhyolites, pyroclastic breccias, tuffs, lapilli, and basalt dykes. REE analysis shows LREE > HREE with moderate negative Eu anomaly. Maximum concentrations: TREE 1,999 ppm, TREE+Y 2,288 ppm; Zr 3,765 ppm; Nb 200 ppm; Sn 42 ppm; Hf 95 ppm; Sc 53 ppm. Hydrothermal breccias and ferruginised veins also show significant REE, Zr, Nb, Mo, indicating REE and rare metal potential.
Around Jaswantpura, Jalore & Sirohi Districts, Rajasthan	Around Jaswantpura, Jalore & Sirohi	-	-	-	-	Sampling	G4 reconnaissance in Jaswantpura, Jalore & Sirohi (area not specified) mapped Erinpura, Siwana, and Jalore granites along with younger dolerite and rhyolite dykes of Malani Igneous Suite. Preliminary analysis of 13 BRS and 6 PCS samples shows TREE values ranging from 55 ppm to 803 ppm, indicating low to moderate REE mineralization. Most chemical results are pending, and further exploration is required to substantiate REE potential.
Sirohi District, Rajasthan	Punawa-Ranela-Kooma areas	-	-	-	-	Sampling	G4 reconnaissance in Punawa-Ranela-Kooma, Sirohi mapped Reodar Formation metasediments, Erinpura granites, and mafic intrusives including dolerite and quartz porphyry. Ferro-carbonatite dyke/vein near Punawa shows skarn-related garnet, but no surface REE signatures. TREE values in channel and BRS samples range from 401 ppm to 1,984 ppm. Disseminated sulphides are present, and further chemical results are awaited to confirm REE and rare metal potential.
Dungarpur District, Rajasthan	Barwasa-Lodowal area	-	-	-	-	Sampling	G4 reconnaissance in Barwasa-Lodowal, Dungarpur mapped Bhilwara and Aravalli Supergroup rocks, along with Palaeo-Proterozoic intrusives. Lithologies include garnet-staurolite schist, quartz-mica schist, quartzite, wollastonite-bearing carbonate, dolomite, polymictic conglomerate, and granites. NW-SE trending skarn zones developed near Barwasa and Barwasa Mafi villages, 1.8–3 km long and 50–350 m wide. Wollastonite-bearing carbonate samples show 73.44–82.99% CaSiO ₃ .
Sirohi District, Rajasthan	Wan-Mochhal-Bhev area	1:12500	50	-	-	Mapping & Sampling	G4 reconnaissance in Wan-Mochhal-Bhev, Sirohi (50 sq. km) mapped mica schist/phyllite intruded by Erinpura granite and Malani rhyolite. Granite is coarse to very coarse grained, monzogranite, enriched in ΣLREE over ΣHREE (LREE/HREE = 7.27) with high Rb/Sr (14.90) and low K/Rb (0.01), Ba/Rb (0.80). Surface mineralisation occurs as greisen zones and tourmaline clots. Channel samples show significant Sn, Rb, Pb, Cu, and sporadic W enrichment, indicating REE and rare metal potential.
Udaipur District, Rajasthan	Padrara-Sayra area	1:12500	129	4	450	Mapping, Sampling & Drilling	G4 reconnaissance in Padrara-Sayra, Udaipur (129 sq.km) mapped Delhi Supergroup rocks, Sendra-Ambaji granite, and Phulad Ophiolite Suite. Surface mineralisation includes skarn-hosted sulphides, magnetite, graphite, calcite, and tourmaline-bearing pegmatites. Channel samples show TREE up to 2,457.78 ppm, Ba 19,529 ppm, Th 340 ppm, and Sc 84 ppm. REE phases and further analytical results from core and trench samples are awaited.
Ajmer Districts, Rajasthan	Piloda-Nagola areas	-	-	-	-	Sampling	G4 reconnaissance in Piloda-Nagola, Ajmer mapped Sandmata Complex gneisses, granite intrusives, Rapakivi and leucocratic porphyritic granites, pegmatites, albitites, and mafic intrusives over low-lying and hilly areas. REE sampling from granite, albitite, and pegmatites shows ΣREE up to 3,727 ppm with LREE-dominance (La 201–1,010 ppm, Ce 379–1,850 ppm, Nd 162–475 ppm). Key mineral phases include allanite, tourmaline, and beryl, indicating REE and rare metal potential.
Alwar District, Rajasthan	The Northern Aravalli Orogeny, Dadikar and Khairthal Area	-	-	-	-	Sampling	G4 reconnaissance in Dadikar-Khairthal, Alwar mapped A-type granites, pegmatites, and quartz ± tourmaline, magnetite, and fluorite veins. Structural analysis records three deformation phases (F1–F3) with dominant F2 folds controlling outcrop patterns. REE and rare metal (W, Sn, Nb, Be, Ta, Hf) mineralization is indicated in pegmatitic and quartz-feldspathic veins, with surface ferruginisation and quartz-tourmaline ± magnetite assemblages highlighting mineralized zones.
Udaipur District, Rajasthan	Seriya Block	-	-	-	-	Sampling	G4 reconnaissance in Seriya Block, Udaipur (Aravalli Fold Belt and Mangalwar Complex) mapped dolomitic marble, magnetite-bearing bands, pegmatites, quartz veins, and granitic gneisses. Base metal and gold mineralization is indicated by malachite-stained quartz/calcite veins and ferruginized dolomitic marble. REE potential is targeted in tourmaline-bearing pegmatites and granites. Channel sample CH6 from dolomitic marble near Saleya shows Cu 0.27 %.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Barmer District, Rajasthan	East of Gugrot	1:2000	1.8	19	2344.30	Mapping, Sampling & Drilling	G3 preliminary exploration east of Gugrot, Barmer (1.8 sq. km) mapped Siwana Ring Complex with Phase-I rhyolites and basalts, Phase-II granites, and Phase-III microgranite dykes. Σ REE+Y in bedrock ranges 178–4,063 ppm; channel samples 976–3,160 ppm. Core samples from 19 vertical boreholes show Nb 52–517 ppm, Σ REE+Y 955–3,798 ppm, and Zr 195–5,616 ppm, indicating significant REE and rare metal potential.
SCANDIUM							
Jaisalmer District, Rajasthan	Manpiya-Mulsagar-Lodorva area		120	-	-	Mapping & Sampling	G4 reconnaissance in Manpiya-Mulsagar-Lodorva, Jaisalmer (120 sq.km) mapped Jaisalmer Formation carbonates and Baisakhi Formation sandstones with near-horizontal bedding. Sedimentary structures include laminations, cross-bedding, ripples, and organic traces. Channel sampling (165 nos.) indicates anomalous scandium in marl and fossiliferous limestone, with the highest Sc values recorded in Jaisalmer Formation.
NIObIUM							
Purulia District, West Bengal	Around Shusina area	1:12500	-	-	-	Mapping & Sampling	G4 reconnaissance in Shusina, Purulia (Chhotanagpur Gneissic Complex & North Singhbhum Mobile Belt) mapped nepheline syenite, fenitized rocks, mica schist, tuffaceous phyllite, amphibolite, and gneiss-migmatites. Ferruginous brecciated zones (1 km $\times$ 40–50 m) along the South Purulia Shear Zone indicate gossan formation. Analytical results show Nb 155 ppm–0.32%, Cu 75 ppm–0.69%, Zn 11 ppm–0.47%, and TREE 2,228–4,272 ppm, highlighting REE and Nb potential.
Jhunjhunu and Sikar Districts, Rajasthan	Sior-Dabla-Lakha-ka-Nangal area	-	-	2	200	Sampling & Drilling	G4 reconnaissance in Sior-Dabla-Lakha-ka-Nangal, Jhunjhunu & Sikar (Alwar & Ajabgarh groups) mapped quartzite, calc-silicate, marble, post-Delhi granites, and albitite-iron skarn zones. Key magnetite-albitite zones near Ramkumarpura, Nayi Dhani, and Sior show Fe_2O_3 up to 84.9%, Co 256 ppm, Cr 608 ppm, V 301 ppm, and Sn 275 ppm. Scout drilling (2 boreholes of 100 m each) intersected 16–23 m thick magnetite-albitite bands, confirming subsurface continuity and significant Nb and associated rare metal potential.
LITHIUM							
Korba District, Chhattisgarh	Raha and Ramakachhar area	1:12500	126	-	-	Mapping & Sampling	G4 reconnaissance in Raha-Ramakachhar, Korba (126 sq. km, Chhotanagpur Gneissic Complex) mapped gneisses, granitoids, metasediments, pegmatites, aplites, and quartz veins. The area shows polyphase deformation (F1–F3 folds) with NE-SW shearing and NW-SE brecciation. Geochemical analysis of 125 BRS and 50 pit-trench samples indicates Li up to 2,348 ppm, Cs up to 948 ppm, and Nb up to 446.5 ppm, highlighting significant lithium and rare metal potential.
South Garo Hills and East Garo Hills District, Meghalaya	Darang area	1:12500	67	-	-	Mapping & Sampling	G4 reconnaissance in Darang, South and East Garo Hills (67 sq.km, Assam Meghalaya Gneissic Complex and Rongjeng Granitoids) mapped gneiss, granite, pegmatite, amphibolite, calc-silicate, and Shella Formation sediments. Alteration features (ferruginization, kaolinization, epidotization, silicification, chloritization) and epigenetic mineralization (sulphide, magnetite, wolframite, allanite) were observed near microcracks and diastrophic structures. Preliminary analyses show Li up to 80.9 ppm, Nb 73 ppm, Zr 916–1107 ppm, tREE 1947 ppm, Fe_2O_3 32.8–42%, and Hg 233 ppm, indicating significant lithium and associated mineral potential.
Jammu, Ramban, Reasi, Rajouri and Udhampur Districts, UT: J&K	Subathu Formation	-	-	-	-	Sampling	G4 reconnaissance in Subathu Formation, parts of Jammu, Ramban, Reasi, Rajouri, and Udhampur (UT: J&K) mapped cherty dolomite, brecciated quartzite, aluminous laterite/ bauxite, carbonaceous and khaki shales, and Murree Group sandstones and siltstones. A total of 107 BRS, 12 channel, and 32 PCS samples were collected. Analyses indicate lithium and vanadium are not anomalous, and the limited bauxite outcrops (2 locations, 200–250 m strike, 10–50 m width) suggest no further detailed exploration potential.
Raichur District, Karnataka	Amareshwar block, Lingsugur taluk	-	-	-	-	Sampling	Preliminary exploration in Amareshwar block, Lingsugur, Raichur (Karnataka) mapped Sargur-equivalent Gurgunta Schist Belt intruded by acidic and basic bodies, with pegmatites classified as spodumene-bearing and spodumene-poor. Spodumene-bearing pegmatites occur as tabular bodies and veins (10 cm–2 m width; 15–350 m strike) with green tabular-to-needle crystals. Chemical analyses show Li 10–6,750 ppm, Cs 12–375 ppm, Nb 9–98 ppm, Ta 1.11–402.17 ppm, TREE 37.7–224.3 ppm, and Rb 29.13–4,387 ppm. Talus samples also show anomalous Li, Nb, Ta. Spodumene and columbite-tantalite are primary Li and Nb-Ta carriers.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
TUNGSTEN							
Balrampur District, Chhattisgarh	Adhaura area	1:12500	107	-	-	Mapping & Sampling	Reconnaissance survey in Adhaura, Balrampur (Chhattisgarh, G4) covered 107 sq km with 65 BRS, 24 channel, 21 pit-trench, 11 PCS, and 27 petro-mineralogical studies. Chemical analyses show W 0.5–1,080.65 ppm, Sn 0.3–51.62 ppm, Ta <0.2–287.21 ppm, Nb <0.5–275.97 ppm, and Be <1–1,181.61 ppm. Highest W occurs in calc-silicate skarn; Sn, Ta, Nb, and Be are highest in beryl pegmatites. Scheelite identified as W ore; columbite-tantalite and W-bearing Ta-Nb phases dominate rare metals. Tungsten and rare metal zones mapped (~750×500 m and ~1,250×500 m, respectively).
Bankura District, West Bengal	Tepra- Mashak Pahar- Baghalkhulia area	-	-	-	-	Sampling	A G4-stage reconnaissance survey in the Tepra–Mashak Pahar–Baghalkhulia area, Bankura, West Bengal, targeted tungsten and REE-related mineralization. The study area includes North Singhbhum Mobile Belt metavolcano-sedimentary rocks and Chhotanagpur Gneissic Complex gneisses and granites, with quartz–tourmaline veins, shear zones, and ferruginized quartz–carbonate veins. Bedrock, soil, and stream sediment sampling identified rutile, xenotime, monazite, and zircon, with high Th (1194 ppm), Zr (550–1496 ppm), W (76–113 ppm), and Nb (98 ppm), highlighting potential G4 mineralization zones.
Sonbhadra, Uttar Pradesh	Garapathar-Kurkuti-Semada areas	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey in Garapathar–Kurkuti–Semada, Sonbhadra, U.P., mapped 100 sq km of Mahakoshal Group rocks. Bedrock, trench, and stream sediment samples showed low tungsten (<0.5–37.4 ppm) with minor sulphides, while arsenic, lead, and gold occurred in trace to moderate amounts, indicating limited tungsten potential.
Nagaur District, Rajasthan	Miyan Ki Dhani East Block, Degana	-	-	13	2,470	Sampling & Drilling	A G3-stage exploration in Miyan Ki Dhani East Block, Degana, Rajasthan, investigated tungsten, lithium, and associated mineralization to assess strike continuity, depth extension, and resource potential. Thirteen inclined boreholes totaling 2,470 m were drilled, generating 550 core samples. Analyses show high tungsten (up to 6628.86 ppm), with notable tin, molybdenum, rubidium, lithium, and presence of Nb+Ta, highlighting promising zones for further resource assessment.
Nagaur District, Rajasthan	NE of Jalsu Khurd block, Degana	-	-	2	-	Sampling & Drilling	A G3-stage exploration in the NE Jalsu Khurd block, Degana, Rajasthan, involved two inclined boreholes, which did not reveal significant tungsten or rare metal mineralization. Consequently, no tungsten resource was estimated. However, rubidium resources were assessed: 0.083 Mt @ 0.062% Rb (200 ppm cut-off) and 0.059 Mt @ 0.073% Rb (400 ppm cut-off), indicating limited but notable rubidium potential.
MOLYBDENUM							
Chitrakoot & Banda Districts of Uttar Pradesh and Satna District of Madhya Pradesh	-	1: 12500	100	-	-	Mapping & Sampling	A G4 survey over 100 sq km in Chitrakoot–Banda (U.P.) and Satna (M.P.) mapped Bundelkhand Granitoid Complex and Vindhyan sediments. While gold anomalies up to 1090 ppb and traces of Cu, Cr, Zn, and Fe were recorded, geophysical studies showed no significant molybdenum mineralization.
Dharmapuri District, Tamil Nadu	Marudipatti North B-block, Harur-Uthangarai molybdenum belt	-	-	5	-	Sampling & Drilling	A G2 exploration in Marudipatti North-B block, Dharmapuri, delineated shear- and lithology-controlled molybdenum mineralization within epidotized hornblende–quartzofeldspathic gneisses and quartz veins. Eighteen boreholes outlined 16 Mo lodes (100 ppm cut-off) with assays up to 2300 ppm. The block hosts 0.344 Mt @ 346 ppm Mo (100 ppm cut-off), including 0.092 Mt @ 759 ppm (300 ppm) and 0.056 Mt @ 987 ppm (500 ppm), confirming significant resource potential.
Tirupattur District, Tamil Nadu	Nachiyarkuppam Block, North-Eastern Extension of Harur-Uthangarai Belt (Hub), part of ts no. 571/10 & 14	1:12500	-	3	382	Mapping Sampling, Pitting, Trenching, & Scout Drilling.	A G4 reconnaissance survey in Nachiyarkuppam block, Tirupattur, Tamil Nadu, mapped charnockite, epidote–hornblende gneiss, quartz–feldspathic gneiss, and quartz–baryte veins, the main mineralized target. Scout drilling (382 m in 3 boreholes) and trenching revealed molybdenite, galena, pyrite, and chalcopyrite as disseminations within mineralized quartz veins (2 cm–1 m thick). Geochemical results show Mo up to 1156 ppm, Pb up to 42,465 ppm, and Ba up to 38.5 wt%, leading to the demarcation of six blocks and delineation of a Mo lode at 100 ppm cut-off with 1.5 m stoping width.
NICKEL							
Katni District, Madhya Pradesh	Laterite of Jhalwara, Majhgawan, Padariya and Bijauri area	-	-	-	-	Sampling	A G4 survey in Jhalwara–Majhgawan–Padariya–Bijauri, Katni (M.P.), found ferruginous laterites (60% area) over Vindhyan and Mahakoshal rocks. Sampling (n=253) showed significant values of V (up to 2847 ppm), Cr (5125 ppm), Co (400 ppm), Ni (465 ppm), and Ga (51 ppm), indicating promising multi-metal potential.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sundargarh District, Odisha	Bonai Granitoid Complex around Roxy area	-	-	-	-	Sampling	A G4 survey in the Roxy area, Sundargarh District (Odisha), assessed Ni-PGE potential within mafic-ultramafic enclaves of the Bonai Granitoid Complex. Mapping and image analysis identified pyroxenite, amphibolite, and ultramafic exposures with sharp granite contacts, alteration to serpentine/asbestos, and talc. Of 180 samples collected, results from 49 show nine with >2000 ppm Ni, indicating appreciable nickel mineralization and encouraging scope for further exploration.
Phek District, Nagaland	Naga Hills Ophiolite near Mollen	-	-	-	-	Sampling	A G3 preliminary exploration near Mollen, Phek (Nagaland), identified ~1 sq km of nickeliferous laterite over serpentinized peridotites and dunite pods in the Naga Hills Ophiolite. The regolith, up to 5 m thick, shows two weathering horizons: limonitic topsoil and saprolitic base. Limited analyses indicate high Ni (2.07%) with Co up to 802 ppm in the saprolite zone of the laterite regolith.
Leh District, Ladakh (UT)	Tso Morari area	-	50	-	-	Mapping & Sampling	A G4 survey in Tso Morari, Leh (UT), mapped 50 sq km of metabasic rocks and Karzok Ophiolite chromitite. Analyses show high Cr (35.85–41.95%), Ni (1098–1570 ppm), V (up to 1288 ppm), and significant PGE values. Estimated chromite ore is ~2400 m ³ (9156 t) at 39.37% Cr.
Lalitpur District of Uttar Pradesh and Sagar District of Madhya Pradesh	Sonrai basin under the Vindhyan sediments	1:12500	50	-	-	Mapping & Sampling	A G4 reconnaissance survey was conducted over 50 sq km in the western extension of the Sonrai Basin across Lalitpur (U.P.) and Sagar (M.P.), targeting concealed magmatic Ni-Cu mineralization beneath Vindhyan sediments. The area comprises Kaimur and Rewa Group rocks (Mesoproterozoic-Neoproterozoic) with gentle NW-SE dipping, overlain locally by Deccan basalts. Petrochemical analyses of basaltic flows show Ni ranging 51–97 ppm and Cu 190–419 ppm, while groundwater samples exhibit trace Ni (<1.71 ppb) and Cu (<1 ppb). Cr concentrations in basaltic flows are low, ranging <4–29.18 ppb, indicating limited but measurable Ni-Cu potential in the mapped extension.
Mysuru District, Karnataka	South of Sindhuvali and Dura area	-	-	-	-	Sampling	A reconnaissance survey in the Sindhuvali-Talur Schist Belt, Mysuru (Karnataka), assessed Ni and Cr mineralization across serpentinite, metapyroxenite, amphibolite, and talc-tremolite schist. Serpentinite showed Ni 161–3229 ppm and Cr up to 10,400 ppm, while metapyroxenite and amphibolite had Ni up to 2694 ppm and 2292 ppm, and Cr up to 6547 ppm and 3254 ppm, respectively. Five zones were delineated, with Zones I, II, IV, and V identified as highly prospective, showing Ni 369–2336 ppm and Cr 1252–10,400 ppm, highlighting significant exploration potential.
Kodagu & Mysuru Districts Karnataka	Siddalingapura-Ambalare areas	-	-	-	-	Sampling	A G4 reconnaissance survey in Siddalingapura-Ambalare, Kodagu and Mysuru, Karnataka, identified two Ni-PGE prospective zones. Zone I (Gaddehosahalli-Argal) shows serpentinized and dolomitized ultramafics with Ni 890–4476 ppm and Co up to 221 ppm. Zone II (Jainkalbetta RF) features limonitic nickeliferous laterites with Ni 2250–9950 ppm (avg. 0.4% over 10 m) and Co up to 195 ppm. Minor Cu, Mo, and V anomalies were also recorded, indicating multi-metal potential.
North Goa District, Goa and Sindhudurg District, Maharashtra	Alorna area	-	-	-	-	Sampling	A reconnaissance survey in Alorna, North Goa (Goa) and Sindhudurg (Maharashtra), targeted Ni and associated mineralization in ultramafic rocks and laterites of the Goa Group. Bedrock samples show Ni up to 0.31% and PGE totals 10–322 ppb (Pt 141 ppb, Pd 77 ppb), while laterite analyses reveal Ni 508–2295 ppm, Co 28–282 ppm, and Cr 1004 ppm–1.78%. Three aluminous laterite blocks (~23 sq km) were demarcated, with G3-stage vertical drilling recommended to assess depth continuity.
Udaipur District, Rajasthan	Bagdunda block	-	-	-	-	Sampling	A G4 reconnaissance survey in Bagdunda, Udaipur (Rajasthan), over Aravalli and Delhi Supergroup rocks, identified Ni-Cr mineralization associated with serpentinite intrusions. Structural mapping recorded three deformation phases with folds, foliations, and faults. Bedrock analyses show significant Ni (0.24–0.45%) and Cr (0.16–0.45%), with disseminated sulphides, magnetite, and ferruginization indicating promising mineralization zones.
HEAVY MINERALS							
Sonbhadra District, Uttar Pradesh	areas of Dudhi Gneissic Complex (DGC)	1:12500	100	-	-	Mapping & Sampling	A G4 survey in Dudhi Gneissic Complex, Sonbhadra (U.P.), identified abundant zircon, ilmenite, tourmaline, garnet, mica, and REE minerals in stream sediments and soils. Total zircon resources across 88.56 sq km are estimated at 4079 t (4.83% of heavy minerals), with placer accumulation in regolith and sand bars.

Table 1 (Cont.)

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
VANADIUM							
Barwani District, Madhya Pradesh	Bomia-Kathora-Kospur Area	-	-	-	-	Sampling	A G4 reconnaissance survey in Bomia-Kathora-Kospur, Barwani (M.P.), targeted vanadium and REE in carbonate dykes within Deccan Trap basalts. Dykes (ash grey, yellow limonitic, and pure calcite) show high silica, alumina, and iron, indicating magmatic origin with alteration. Vanadium anomalies of 500–600 ppm were recorded, structurally controlled by the ENE–WSW Narmada Lineament.
Lower subansiri District, Arunachal Pradesh	Kugitago-Miya areas	1: 2000	1.5424	5	370.6	Mapping, Sampling & Drilling	A G3 preliminary exploration in Lower Subansiri District, Arunachal Pradesh, focused on vanadium, graphite, and associated minerals. Targeted lithologies exhibit features indicative of magmatic origin with alteration, and vanadium anomalies of were recorded.
West Siang District, Arunachal Pradesh	Daba Gamlin block	1: 2000	2.0086	-	-	Mapping & Sampling	A G3 preliminary exploration for Vanidium, Graphite and Limestone is dpne. 600 m drilling for F.S. 2024-25 is proposed.
Sirmaur District, Himachal Pradesh	Khil block	1:4000	2	3	349	Mapping, Sampling & Drilling	A G3 preliminary exploration in Khil block, Sirmaur (H.P.), aimed to assess vanadium resources and subsurface continuity. Detailed 1:4000 mapping (2 sq km) targeted the carbonaceous shale–siltstone and black chert–shale–siltstone of the Lower Tal (Shaliyan Formation) within the Nigalidhar syncline. Eight channels (82 samples) and three boreholes (349 m) were completed, with additional petrology and EPMA studies for vanadium mineralogy. Analytical results are pending for resource estimation.
Lalitpur District, Uttar Pradesh	Dhandraul sandstone of Kaimur Group	-	-	-	-	Sampling	A G4 reconnaissance survey in Lalitpur (U.P.) targeted vanadium, titanium, and associated rare metals in lateritic profiles over Dhandraul Sandstone of the Kaimur Group. Lithologies include Archean supracrustals, Bundelkhand Gneissic Complex, Vindhyan Supergroup, laterite/ bauxite, and Banda alluvium. Chemical analyses indicate high Fe ₂ O ₃ (avg. ~31.5 wt%) and moderate TiO ₂ (avg. ~1.5 wt%), with vanadium ranging 225–2111 ppm (avg. 707 ppm) in bedrock and 384–1039 ppm (avg. 725 ppm) in trenches. Barium shows local enrichment, while REE and base metals are generally low.
TIN							
Bhiwani District, Haryana	Tosham – Khanak – Jamalpur – Dulhera - Ratera areas	-	-	-	-	Sampling	A G4 reconnaissance survey was conducted in Tosham–Khanak–Jamalpur–Dulhera–Ratera areas, Bhiwani District (Haryana) to explore tin, tungsten, copper, and associated mineralization. The area is largely covered by Upper Pleistocene–Holocene aeolian and alluvial sediments, with nearest exposures at Tosham Hill and Khanak comprising quartzite, micaceous quartzite, and andalusite-bearing quartz-mica schist of the Delhi Supergroup, intruded by granitic batholiths. Geophysical surveys (IP, SP, Magnetic) identified two major anomaly zones near Kirawar and Paposa, with causative sources at depths of 250–460 m and 300–500 m, respectively. Planned drilling to test these anomalies (1000 m in two boreholes) could not proceed due to landowner opposition, leading to project termination.
Kasaragod District, Kerala	Canom block	1:1000	-	-	-	Sampling	G3 exploration in Canom Block, Kerala, targeted TVM-hosted V–Cr and potential PGE mineralization. Two main TVM bands (1–10 m wide, 800 m strike; 5 m wide, 200 m strike) were mapped. Drilling (658 m, 8 boreholes) intersected magnetite, ulvospinel, ilmenite, and sulfides, yielding TiO ₂ 2.5–4.07%, V 0.13–0.26%, Cu 0.203–0.31%, Ni 0.02–0.129%, and PGE up to 414 ppb, confirming 350–450 m subsurface continuity.
COBALT							
NTR & Palnadu Districts Andhra Pradesh	Jayantipuram-Vedadri area	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Jayantipuram–Vedadri, Andhra Pradesh, identified cobalt-rich ironstones over Narji Limestones. Seven pockets (50–450 m × 5–200 m × 5 cm–3 m) show Co 84–836 ppm, Ni 100–778 ppm, Zn 100–2703 ppm, and Fe ₂ O ₃ 1.25–65.3%.
Shimoga District, Karnataka	Tarlagatta area	-	0.52	8	-	Mapping, Sampling & Drilling	G2 exploration in Tarlagatta, Shimoga, Karnataka, targeted cobalt and manganese over 0.52 sq.km. Drilling was not conducted due to lack of forest clearance. Trench and BRS analyses indicate MnO 13–17% and Co 123–733 ppm, confirming low-grade enrichment consistent with previous G3-stage resource estimates.
LIMESTONE							
Sagar and Damoh District, Madhya Pradesh	Garhakota and Pathria area	-	-	-	-	Sampling	Reconnaissance survey in Garhakota–Pathria (Sagar & Damoh, MP) targeted Nagod and Lameta limestones. Nagod Formation (375–400 m MSL) shows CaO 35–48%, SiO ₂ 9.5–35%, MgO 0.7–10%; Lameta limestone (420–440 m) has CaO 38–46%, SiO ₂ 14–24%, MgO 1.7–2.2%. Overlying Deccan basalts occur above 440 m. Lithologies exhibit characteristic sedimentary structures, jointing, and weathering, indicating potential limestone zones.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sagar and Damoh Districts, Madhya Pradesh	Sasa Sajnipur Bamhorikhurd and Khahamau area	-	-	-	-	Sampling	Reconnaissance survey in Sasa-Sajnipur-Bamhorikhurd-Khahamau (Sagar & Damoh, MP) assessed Nagod and Lameta limestones. Nagod limestone is stromatolitic to massive grey with low silica; Lameta limestone shows higher silica. 155 samples indicate potential for white, Portland, and blendable cement with balanced MgO.
Damoh District, Madhya Pradesh	Botrai Block (parts of toposheet no. 55M/01)	1:4000	3.68	5	134.20	Mapping & Sampling	Preliminary exploration in Botrai block, Patharia tehsil (Damoh, MP) over 3.68 sq km involved 5 boreholes, detailed mapping, pitting, and 89 core samples. Limestone is predominantly siliceous, fine-grained, stromatolitic, grey to pinkish-brown, interbedded with shales and sandstone, showing low dip (2°–6°) and NNE–SSW strike. CaO ranges 1.59–49.73%, MgO 0.16–16.45%, and SiO ₂ 5.92–92.34%, indicating high-silica limestone at shallow depths.
Damoh, District, Madhya Pradesh	Kanaktala block	-	-	5	-	Sampling & Drilling	Preliminary exploration in Kanaktala block, Damoh (MP) intersected Nagod Limestone—hard, bluish-grey to yellowish-grey, massive to bedded, locally stromatolitic, with fractures and stylolites. three boreholes (800 m spacing) yielded 15.48 Mt limestone of different grades (5.57 Mt white cement, 5.57 Mt blendable, 2.38 Mt Portland, and 1.94 Mt blendable cement grade). Units trend NE–SW, dipping 20°–50° SE.
Panna District, Madhya Pradesh	Itauri-Jharkua Block	LSM of 1:12500 & DM of 1:4000	50 & 3.5	5	146.7	Mapping, Sampling & Drilling	G-3 exploration in Itauri-Jharkua, Panna (MP) mapped Nagod Formation limestone (50 sq km). Boreholes intersected 1–10 m thick bands of stromatolitic, grey, and shaly limestone with CaO 26.8–50.3%. Resources estimated: 9.61 Mt Cement Blendable/Beneficiable (CaO 39.16%), 11.03 Mt Blendable (CaO 35.08%).
Satna and Rewa Districts, Madhya Pradesh	Khajuri-Deora Block	LSM of 1:12500 & DM of 1:4000	50 & 3.5	5	220	Mapping, Sampling & Drilling	G-3 exploration in Khajuri-Deora, Satna & Rewa (MP) mapped 50 sq km of Vindhyan rocks. Five boreholes (29–54.5 m) yielded 156 core samples. Total limestone resources: 72.65 MT (CaO 43.47%, MgO 2.48%). White Cement: 12.96 MT (CaO 50.57%), Portland: 24.84 MT (CaO 50.71%), Beneficiable/Blendable: 21.41 MT (CaO 50.71%), Blendable: 7.86 MT (CaO 36.71%), Unclassified: 5.18 MT.
Satna District, Madhya Pradesh	Bamhauri - Goraia Block	-	-	-	-	Mapping, Sampling & Drilling	G-3 preliminary exploration in Bamhauri-Goraia, Satna (MP) mapped Nagod Limestone, Ganurgarh & Sirbu Shales. Five boreholes (800×800 m spacing) and pit sampling delineated three limestone units: dark grey hard, dark grey stromatolitic, and light grey interbedded with shale. BRS and PTS analyses show average CaO 41.72–41.76%, MgO 2.77–3.10%, SiO ₂ 13.95–15.13%. Limestone thickness and quality improve SSE, with dark grey varieties reaching 51.18% CaO. Total inferred resource: 45.67 MT — Portland: 5.42 MT, Blendable/Beneficiable: 12.75 MT, Blendable: 13.39 MT, Unclassified: 14.11 MT, following IBM 2020 and UNFC G3 guidelines (2 m stoping width)
Rewa District, Madhya Pradesh	Chhpta block	1:4000	3.5	5	142.95	Mapping, Sampling & Drilling	G-3 preliminary exploration in Chhpta block, Rewa (MP) covered 3.5 sq km with five vertical boreholes (16.7–45 m depth, 800×800 m spacing). Boreholes intersected limestone, shale-limestone, sludge, and soil, with 80–95% core recovery. A total of 113 core and 20 bedrock samples were analyzed. Blendable/beneficiable cement-grade limestone resource estimated at 36.99 MT, averaging CaO 39.98% and MgO 3.56%, classified as UNFC 333.
Satna District, Madhya Pradesh	Kuria-Sivpur Block	1:12500 & 1:4000	50 & 3.5	5	227.75	Mapping, Sampling & Drilling	Kuria-Sivpur block, Satna (MP) G-3 exploration: total limestone resource 106.73 MT (White cement 56.22 MT, Portland 35.65 MT, Blendable/Beneficiable 9.55 MT, Blendable 5.36 MT), CaO 34.16–46.16%, MgO 1.85–2.35%, UNFC 333.
Upper Siang District, Arunachal Pradesh	Puging-Paling-Janbo-Likor areas	1:12500	50	-	-	Mapping & Sampling	Reconnaissance survey for limestone in Puging-Paling-Janbo-Likor areas, Upper Siang District, Arunachal Pradesh (G4): A 50 sq km area was mapped to evaluate limestone/dolomite zones. Five dolomite zones were identified with CaO ranging from 23.34% to 28.12% and MgO from 14.62% to 20.25%. Most of the dolomite is below flux grade, with only samples from the Likor area meeting flux-grade potential.
Kurung Kumey District, Arunachal Pradesh	Hugu and Sakehugu area	1:12500	50	-	-	Mapping & Sampling	A G4 reconnaissance survey of the Chilleipam Formation, Bomdila Group, was conducted over 50 sq km in Hugu, Pipa, Sakehugu, and surrounding areas, Kurung Kumey District, Arunachal Pradesh. The study identified three discontinuous limestone bands, with Hugu and Pipa showing high-grade limestone (CaO up to ~54%), while Chiya exhibited lower-grade limestone with higher MgO and SiO ₂ . Sixty bedrock and 18 trench samples confirmed these variations, and two natural caves with botryoidal structures were documented.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Dima Hasao District, Assam	Krungming Block	-	-	5	407.75	Sampling & Drilling	Preliminary G3 exploration in Krungming Block, Assam, intersected Shella Formation limestone up to 93 m thick. Chemical analysis (99 samples) shows CaO 34–47%, SiO ₂ 5.7–15%, MgO 1.8–4.7%. Draft resource: 175.78 Mt, including 23.82 Mt Portland and 22.53 Mt Blendable/Beneficiable grade.
South Garo Hills District, Meghalaya	West of Siju Area	-	-	-	-	Sampling	Reconnaissance G4 survey west of Siju, South Garo Hills, Meghalaya, identified limestone within the Siju Limestone Member of the Shella Formation. Beds are 40–80 m thick, richly fossiliferous, and indicate a warm marine environment. Chemical analysis classifies the limestone as suitable for Cement (Portland, Blendable/Beneficiable) and SMS (OH) grade, showing good economic potential.
South Garo Hills District, Meghalaya	Chokpot Area	-	-	-	-	Mapping & Sampling	Reconnaissance G4 survey around Chokpot, South Garo Hills, Meghalaya, identified fossiliferous Siju Limestone of the Shella Formation. The beds, 80–100 m thick and medium to dark grey, are massive, medium-grained, and contain Nummulites, Discocyclina, Assilina, and Alveolina fossils, indicating a Mid–Late Eocene shallow marine origin. The limestone extends over 20 km strike and ~1 km width, with CaO 40–52 wt %, suitable for Portland and blendable cement, showing significant resource potential.
East Jaintia Hills District, Meghalaya	Khunti Block, Litang Valley	1:4000	3.78	5	607.6	Mapping, Sampling & Drilling	G3 exploration in Khunti Block, East Jaintia Hills, Meghalaya, covered 3.78 sq km. Five boreholes (607.6 m) intersected Prang Limestone, 91–121 m thick (avg. 104 m) with ~97% core recovery. CaO 32.5–53.9 %, SiO ₂ 1–14 %, MgO 0.77–4.5 %, suitable for SMS (OH), Portland, and Blendable/Beneficiable cement.
Anantnag District, Jammu and Kashmir	Naubag-Halan Area	1:4000	2.5	5	472	Mapping, Sampling & Drilling	G3 exploration in Naubag-Halan, Anantnag, J&K, over 2.5 sq km with five boreholes (472 m) identified three promising limestone members of the Upper Triassic Wuyan Formation. Member A (dark grey oolitic limestone) has 2 km strike, up to 460 m thick, CaO 47–55 %. Member C (yellowish limestone) and Member D (massive grey limestone) both show 1.9 km strike with CaO 46–53 %. Borehole results confirm high CaO and significant Sc content (avg. 208 ppm in D, 188 ppm in C), indicating good potential for cement and specialty uses.
Anantnag District, Jammu and Kashmir	Sop Block	-	-	4	500	Mapping, Sampling & Drilling	G3 exploration in Sop Block, Anantnag, J&K, targeted limestone units of the Upper Triassic Wuyan Formation within the regional Sop syncline. Detailed mapping revealed four members (A–D), with A, C, and D as limestone units (thickness 240 m, 60 m, 260 m; strike 3.2 km, 1.1 km, 0.76 km). Fifty-five channel samples showed CaO >40% in 42 samples and Sc anomalies (59–667 ppm, avg. 201 ppm). Based on these results, four inclined boreholes (500 m spacing, 120–130 m depth) were drilled, generating 325 core samples for further analysis.
Chitrakoot District, Uttar Pradesh	Kolgadhiya-Rehutia-Takhatapur-Biyur areas	-	-	-	-	Sampling	G4 reconnaissance in Kolgadhiya-Rehutia-Takhatapur-Biyur, Chitrakoot, UP, identified flux-grade dolomitic limestone in 2.8 km long belts. Analyses of 85 samples show CaO 0.14–28.65% (avg. 17.82%) and MgO 0.82–20.86% (avg. 12.14%). Dolomites are micritic with sparry cement clasts, detrital inputs, and dolomitization increasing MgO content.
Kalaburagi District, Karnataka	North Kadaboore Block	1:4000	3.47	7	-	-	Preliminary exploration in North Kadaboore Block, Kalaburagi, Karnataka, involved detailed 1:4000 mapping and 7 boreholes over 3.47 sq. km to assess limestone of the Shahabad Formation. The area hosts stylolitic Wadi Limestone (Blast Furnace grade), grey Sedam Member limestone (blendable/Portland cement grade), and intercalated shale. Boreholes intersected Blast Furnace grade up to 50.12% CaO, Portland cement grade 45–52% CaO, and blendable limestone. Eastern parts show shallower, thinner Blast Furnace grade, while western parts have thicker deposits at greater depths, indicating promising resource potential with low overburden.
Kalaburagi District, Karnataka	Nalwar Block	1:4000	2.5	4	175.10	Mapping, Sampling & Drilling	Preliminary G3 exploration in Nalwar Block, Kalaburagi, Karnataka, covered 2.5 sq. km with detailed 1:4000 mapping and four vertical boreholes at 800 m spacing. Drilling (175.1 m) and 159 core samples delineated Wadi Member limestone of the Shahabad Formation, showing NE–SW strike with <2° dip. Estimated resources total 227.71 Mt, including 89.44 Mt Blast Furnace grade, 72.55 Mt Portland cement grade, 34.03 Mt blendable cement grade, and 31.69 Mt above threshold grade.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Virudhunagar District, Tamil Nadu	Meenachipuram block, Kariyapatti Area	1:4000	6.5	6	770	Mapping, Sampling & Drilling	Preliminary G3-stage exploration in Meenachipuram Block, Kariyapatti area, Virudhunagar District, Tamil Nadu covered 6.5 sq. km through detailed mapping and 770 m of drilling in six inclined boreholes. The area hosts crystalline limestone, calc-granulite, charnockite, biotite gneiss, and younger felsic intrusives, with 25 limestone bands delineated across three blocks. Individual bands measure 56–490 m in strike length and 3–36 m in width, with cumulative limestone thickness ranging from 30–100 m. Analytical results of samples indicate CaO (40.87–53.59%), MgO (0.29–15.67%), and SiO ₂ (2.24–20.53%), with resource estimation pending core analysis.
Dindigul and Karur Districts, Tamil Nadu	Salikkarai area	1:4000	3	7	652.05	Mapping, Sampling & Drilling	G3-stage exploration in Salikkarai, Dindigul–Karur Districts (3 sq. km) delineated six crystalline limestone bands through mapping, trenching, and 652 m of drilling. Major bands III (2.6 km) and IV (1.48 km) were targeted. Resource estimation indicates 19.68 Mt of Portland cement grade limestone (47.3% CaO, 11.66% SiO ₂ , 1.02% MgO), 1.63 Mt of blendable grade, and 1.80 Mt of threshold grade.
Adilabad District, Telangana	Jamni block	1:4000	5.07	5	330	Mapping, Sampling & Drilling	G3 exploration in Jamni block, Adilabad (5.07 sq. km) established 786.98 Mt of inferred limestone resources (UNFC 333). This includes 199.08 Mt Portland Cement Grade, 505.58 Mt Blendable Grade, and 82.32 Mt Threshold Grade. With low overburden (avg. 1:1.7), the block is economically viable for cement and iron & steel industries.
Jaisalmer District, Rajasthan	North of Bhojaniyon ki Dhani block, Nibh Dungar area	1: 2000	2	5	250	Mapping, Sampling & Drilling	G3 exploration in north of Bhojaniyon ki Dhani block, Jaisalmer, covered 2 sq. km with 250 m drilling (5 boreholes, 198 samples). The Jaisalmer Formation hosts biosparitic, intrasparitic, micritic, and biomicritic limestone. Initial analyses confirm cement-grade limestone, with resource estimation awaited from complete data.
Jaisalmer District, Rajasthan	South east of Dedha block, Nibh Dungar area	1: 2000	1	3	150	Mapping, Sampling & Drilling	G3 exploration in south-east of Dedha block, Jaisalmer, covered 1 sq. km with 150 m drilling in three boreholes. The Jaisalmer Formation hosts intrasparitic, biosparitic, micritic, and biomicritic limestone with interbeds of clay and sandstone. Initial analysis confirms cement-grade limestone, with resource estimation pending further analytical results.
Jaisalmer District, Rajasthan	North-east of Jogiyon ki Dhani Block, Nibh Dungar area	1: 2000	1	3	150	Mapping, Sampling & Drilling	G3 exploration in north-east of Jogiyon ki Dhani block, Jaisalmer, covered 1 sq. km with 150 m drilling in three boreholes, yielding 116 core samples along with petrographic, petrochemical, and channel samples. The area, part of the Jaisalmer Formation, hosts cyclic deposits of siliceous, oolitic, and fossiliferous limestone of Jurassic age. Boreholes intersected yellow oolitic fossiliferous limestone, siliceous limestone, fine-grained compact limestone, and grey fossiliferous limestone, confirming its potential for cement-grade limestone resource estimation.
Jaisalmer District, Rajasthan	West of Jogiyon ki Dhani Block, Nibh Dungar area	1: 2000	2	5	204	Mapping, Sampling & Drilling	G3 exploration in west Jogiyon ki Dhani, Jaisalmer (2 sq. km), drilled 5 boreholes (204 m) and collected 154 core samples. The area near Jaisalmer–Baisakhi formation contact hosts oolitic, fine-grained, compact, and fossiliferous limestone, confirming cement-grade potential.
Jaisalmer District, Rajasthan	Joyan north block	1: 2000	2	5	250	Mapping, Sampling & Drilling	G3 exploration in Joyan North block, Jaisalmer (2 sq. km), involved 250 m drilling in five boreholes (119 core samples) and detailed mapping. The Jaisalmer Formation hosts hard compact, grey, and localised bioclastic limestone suitable for low-silica SMS/cement grade. Additional clay, shale, sand, and alluvium were also observed. Resource estimation is pending chemical analysis, with the work classified as UNFC G3 (333).
Jaisalmer District, Rajasthan	Southern part of Jaisalmer south block	1: 2000	2	5	270	Mapping, Sampling & Drilling	G3 exploration in southern Jaisalmer South block (2 sq. km) involved 270 m drilling in five boreholes and collection of 118 core, 20 channel, and 5 petrochemical samples. The area, part of the Kuldhra Member of Jaisalmer Formation, hosts calcarenite, calcilutite, grey limestone, and calcareous sandstone. Cement-grade limestone was intersected in all boreholes, with cumulative thicknesses ranging 9.40–19.90 m. Resource estimation is pending chemical analysis.
Jaisalmer District, Rajasthan	Northern part of Jaisalmer south block	1: 2000	2	5	250	Mapping, Sampling & Drilling	G3 exploration in northern Jaisalmer South block (2 sq. km) involved 250 m drilling in five boreholes and 150 core samples for chemical analysis. The Kuldhra Member of Jaisalmer Formation hosts calcareous sandstone, calcarenite, and calcilutite. Cement-grade limestone was intersected in four boreholes, with cumulative thicknesses of 5.10–15.70 m, while one borehole intersected only a minor 3.50 m band. Initial analytical results are available for 18 core samples.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Jaisalmer District, Rajasthan	Vijay Nagar block	1: 2,000	2	5	250	Mapping, Sampling & Drilling	G3 exploration in Vijay Nagar block, Jaisalmer (2 sq. km), involved 250 m drilling in five boreholes with 104 core, 5 petrochemical, 10 petrological, and 8 channel samples collected for chemical and REE analysis. The area, part of the extensive Jaisalmer sedimentary basin, hosts horizontal to sub-horizontal Mesozoic-Tertiary rocks. Limestone bands were identified in boreholes based on colour, hardness, and effervescence, with chemical results pending.
GLAUCONITE							
Baloda Bazar District, Chhattisgarh	Asnid and Bakla area	-	-	-	-	Sampling	G4 reconnaissance survey in Asnid-Bakla, Baloda Bazar, Chhattisgarh, mapped parts of the Lohardih, Chaporadih, and Kansapathar formations of the Chandarpur Group for glauconite mineralization. K ₂ O values in BRS, PCS, and pit/trench samples ranged from 0.09% to 7.31%, with higher values in the Chaporadih Formation. Three potential zones—Jhalpani, North Jhalpani, and Chitiya Pathar—were identified for further G3-stage drilling to evaluate economic viability.
Mahasamund District, Chhattisgarh	Chhaporadih and Sirpur area	-	-	-	-	Mapping & Sampling	Reconnaissance in Chhaporadih-Sirpur, Mahasamund mapped glauconitic shale/siltstone (4.5 km × 500 m) and Lohradih sandstone/siltstone (2 km × 600 m) with K ₂ O 5–6.79%, highlighting the Chhaporadih unit for further exploration.
Balod District, Chhattisgarh	Kotera, Reghai and Kohangatola	-	-	-	-	Sampling	G4 reconnaissance in Kotera-Reghai-Kohangatola, Balod, mapped glauconitic sandstone of the Lohardih Formation overlying Dongargarh granitoids. The sandstone is medium-grained, Fe-rich, and arenitic, with E-W strike and 4°–10° dip. K ₂ O values from 85 BRS, 15 PCS, 50 PTS, and 15 channel samples range 0.3%–3.24%.
Baloda Bazar District, Chhattisgarh	Pairaguda and Jhalpani	-	-	-	-	Sampling	G4 reconnaissance in Pairaguda-Jhalpani, Baloda Bazar, mapped glauconitic sandstone and shale/siltstone in Lohardih, Chaporadih, and Kansapathar formations of the Chandarpur Group. Glauconite occurs mainly in Kansapathar quartz arenite and Chaporadih shale/siltstone. K ₂ O values range 0.09%–6.74%, with higher values in Chaporadih units.
Sheopur District, Madhya Pradesh	Agra, Dubera and Magarda area	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance in Agra-Dubera-Magarda, Sheopur, mapped 100 sq km of Govindgarh Sandstone (Rewa Group) for glauconite. The sandstone is fine-to-medium grained, well-sorted, with NNW-SSW to NNE-SSW strike and 3°–5° dips. Glauconite occurs as grains, pellets, and rims around quartz, indicating authigenic and reworked nature. K ₂ O values range 0.01%–6.53%, with highest values in nala and dug well sections, though no continuous high-K ₂ O band was found; trace Zn up to 929 ppm was also recorded.
Sonbhadra District, Uttar Pradesh	Chitikpurwa area	-	-	4	271.40	Sampling & Drilling	G3 exploration in Chitikpurwa, Sonbhadra (UP) drilled 4 boreholes (271.4 m) with 189 core samples. Glauconitic sandstone and Kheinjua shale show K ₂ O 3.29–12.18% (avg. 4.82–7.93%). Thickness: 0.9–5.25 m; depth: 0.5–57.55 m. Inferred resource: 59.32 MT K ₂ O at 6.33% (4% cut-off).
POTASH							
Hanumangarh District, Rajasthan	Jakharanwali block, Bharusari sub-basin, Nagaur-Ganganagar Evaporite Basin	1:12500	100	2	1394.50	Mapping, Sampling & Drilling	G4 reconnaissance in Jakharanwali block, Hanumangarh, Rajasthan (100 sq km, 1:12500) drilled 2 scout boreholes (RJNGJW01: 716.5 m; RJNGJW02: 678 m; total 1,394.5 m) in the Bharusari sub-basin of Nagaur-Ganganagar Evaporite Basin. Boreholes intersected all seven halite cycles (H1–H7). Sylvite and polyhalite occur within H2: RJNGJW01 (573.24–575.4 m, 600.32–603.51 m, 624.4–624.7 m; thickness 0.3–3.16 m), RJNGJW02 (540.92–543.7 m, 549.36–553.66 m; thickness 2.78–4.3 m)
Hanumangarh District, Rajasthan	Chistiyan block, Satipura sub-basin, Nagaur-Ganganagar Evaporite Basin	1: 4000	6	-	-	Mapping & Sampling	Preliminary G3 exploration in Chistiyan block, Satipura sub-basin, Hanumangarh (6 sq km) mapped the Marwar Supergroup. Drilling (8 boreholes, 800 m grid) is planned for FS 202425. HEG contains cyclic evaporites—halite, polyhalite, anhydrite, claystone, dolomite, magnesite, and sylvite, with mineralization mainly in H1 and H2 cycles.
Hanumangarh District, Rajasthan	Jorkian North block, Satipura sub-basin, Nagaur-Ganganagar Evaporite Basin	1: 4000	6	-	-	Mapping & Sampling	Preliminary G3 exploration in Jorkian North block, Satipura sub-basin, Hanumangarh (6 sq km) mapped the Marwar Supergroup. Drilling (8 boreholes, 800 m grid) is planned for FS 202425. HEG comprises cyclic evaporites—halite, polyhalite, anhydrite, claystone, dolomite, magnesite, and sylvite, with all seven halite cycles present.
Hanumangarh District, Rajasthan	Dabli Ratan area, Satipura sub-basin, Nagaur-Ganganagar Evaporite Basin	-	50	4	3100	Mapping, Sampling & Drilling	Reconnaissance G4 survey in Dabli Ratan, Hanumangarh (50 sq km) drilled 4 boreholes (3,100 m) with 200 cores. Potash (sylvite/polyhalite) occurs in lower H1–H2 halite cycles of HEG as discontinuous pockets.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Hanumangarh District, Rajasthan	Chistiyan South Block, Satipura Sub-Basin, Nagaur - Ganganagar Evaporite Basin	-	-	6	4,634	Sampling & Drilling	G2 exploration in Chistiyan South Block, Satipura sub-basin, Hanumangarh District, involved 4,634 m of drilling in six boreholes to assess potash (sylvite, polyhalite) and halite resources. The indicated resource (UNFC-332) without stopping width is 22.98 Mt at 2% cut-off (6.18% K), 9.40 Mt at 5% cut-off (11.25% K), and 6.61 Mt at 8% cut-off (13.56% K). With a 1 m stopping width, it is 20.45 Mt at 2% cut-off, 7.34 Mt at 5%, and 4.31 Mt at 8%, hosted in sylvite-polyhalite interbedded halite sequences.
GRAPHITE							
Balrampur District, Chhattisgarh	Jogni Sonhara area	1:12500	100	-	-	Mapping & Sampling	A reconnaissance survey over 100 sq km in Jogni-Sonhara, Balrampur (Chhattisgarh) identified E-W trending graphite bands within granite gneisses and calc-silicate rocks. Graphite occurs as massive crystalline lenses (30–150 m) often intercalated with calc-silicate and locally associated with pyroxene. The area shows high-grade metamorphism from amphibolite to granulite facies, with pegmatite veins and younger granite intrusions also present.
Jhabua District, Madhya Pradesh	Warlipara-Sajeli-Umradara-Jamuri area	-	-	-	-	Mapping & Sampling	Reconnaissance in Warlipara–Jamuri, Jhabua District (G4) identified two graphite–vanadium bands in carbonaceous phyllite/graphite schist: Band-I (1 km × 6–20 m) with 2.41% FC, 304 ppm V; Band-II (700 m × 15–50 m) with 2.57% FC, 206 ppm V. Mineralization is lithologically controlled.
Palamu District, Jharkhand	Karma block	-	-	-	-	Sampling	General exploration in Karma block, Palamu District (G2) identified graphite in lensoidal bodies within granite gneiss, quartzite, and calc-silicate, structurally controlled and metamorphosed to upper amphibolite–granulite facies. Estimated resources: 8.35 MT graphite (7.01% FC) and 6.19 MT vanadium (avg. 853 ppm).
Palamu District, Jharkhand	Gurha-Manasoti Block	-	-	4	-	Mapping, Sampling & Drilling	Preliminary exploration in Gurha-Manasoti block, Palamu District (G3) identified flake- and vein-type graphite hosted in graphite gneiss within the Proterozoic Chhotanagpur Granite Gneissic Complex. Five graphite bands were mapped (lengths 335–525 m; widths 2–100 m), though four bands lie within the Betla National Park eco-sensitive zone, limiting drilling. Drilling intersected mineralized zones up to 15.3 m. Estimated resources: graphite 0.056 MT at 4.46% FC and vanadium 0.07 MT at 892 ppm (0.15% V ₂ O ₅). Analytical samples show FC 0.61–37.4% (avg. 10.93%) and V 73–3135 ppm (avg. 649 ppm).
Palamu District, Jharkhand	Nawadih-Hanumanr-Gaura Block	-	-	11	-	Sampling & Drilling	Preliminary exploration in the Nawadih-Hanumanr-Gaura block, Palamu District (G3) identified graphite hosted mainly in granite gneiss, migmatite gneiss, and quartzofeldspathic rocks of the Chhotanagpur Gneissic Complex. Graphite occurs as lenses and bands with strike lengths of 150–800 m and widths of 45–75 m, often associated with aplite and pegmatite veins. At Hanumanr and Gaura, bands form antiformal structures with steep to moderate dips. Eleven boreholes were drilled, and ore zones correlated for graphite and vanadium. Resources were estimated using cross-section and LV panel methods (2% FC cutoff, 500 ppm V, 3 m stoping width), yielding a total graphite resource of 24.54 MT at 7.4% FC.
Nayagarh District, Odisha	Tumandi East Block	1:2000	2.5	13	1495.15	Mapping, Sampling & Drilling	Preliminary exploration (G3) in Tumandi East Block, Nayagarh, Odisha targeted graphite in migmatized khondalite of the Central Migmatite Zone. Mapping (2.5 sq km) and 1,495 m drilling identified two sub-parallel lensoidal bands. Band-4 shows continuous graphite (FC > 2%) over 1.3 km, ~10 m thick, averaging 2.7% FC. Band-5 was intersected in two boreholes; strike length remains undefined.
Nayagarh District, Odisha	Tumandi West Block	-	1.5	7	500	Mapping, Sampling & Drilling	Preliminary exploration in Tumandi West Block, Nayagarh, Odisha (G3) targeted graphite in migmatized khondalite of the Eastern Ghats Mobile Belt. Detailed mapping (1.5 sq km), 500 m drilling, P&T work, and geophysical surveys (25 L km) delineated two interpreted bands (VI & VII). Band-VI showed no significant graphite, while Band-VII intersected graphite in two boreholes. Raman and SEM analyses confirm the graphite's organic nature.
Kalahandi District, Odisha	Dangarigan area	-	-	-	-	-	Reconnaissance survey (G-4) around Dangarigan, Kalahandi, Odisha, indicates graphite mineralization in the Eastern Ghats Mobile Belt and adjacent anorthosite complex. Major lithounits include Khondalite, Charnockite, and Migmatite groups. Graphite occurs in pegmatite-associated khondalite and quartz-feldspathic-sillimanite schist, with key occurrences at Sana Karli (5 m × 5 m × 10 m; amorphous and flaky graphite) and Sindhkoti (>15 sq m; mainly amorphous graphite).

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Alluri Sitharamaraju District, Andhra Pradesh	Utla Block	1:2000	2	9	1035.65	Mapping, Sampling & Drilling	Preliminary exploration (G-3) in Utla Block, Andhra Pradesh, covered 2 sq km with nine boreholes totaling 1,035.65 m, SP/resistivity surveys, and geophysical logging. Graphite occurs as veins and lenses in garnetiferous Khondalite gneiss, 5–9 m wide, ~100 m strike, dipping 78° SSE, with 2–10% fixed carbon. SP anomalies range -620 mV to -298 mV over 200 m width and extend 20 m depth.
East Godavari and Alluri Sitaramaraju Districts, Andhra Pradesh	Narsapuram-Nellipudi areas	-	-	-	-	Sampling	Preliminary exploration (G-3) at Pata Ramavaram Hill, Andhra Pradesh, identified graphite in khondalite and calc-silicate granulite, extending 50 m strike and 20–25 m width. Rocks show polyphase deformation with steep dips. Graphite occurs as flakes, fracture fillings, and disseminations at multiple locations. Analysis of 92 samples shows 0.7%–4.28% fixed carbon (avg. 2.35%). Limited extent prevented drilling.
Kottayam and Idukki Districts, Kerala	Purapuzha-Neeloor area to southeastern extension of Thodupuzha-Muvattupuzha Graphite belt	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G-4) in Purapuzha-Neeloor, Kerala, mapped 100 sq km of the Southern Granulite Terrain, including khondalite, quartz-feldspathic gneiss, charnockite, pyroxene granulite, biotite gneiss, granite, dolerite, and pegmatite. Graphite occurs as disseminated flakes (1 mm–2 cm) and veins, mainly in khondalite, granulite contacts, and gneisses. Fixed carbon (FC) values range 0.01–3.89% (mean 0.47%) in bedrock and up to 3.83% in trench samples. REE, molybdenum, lithium, and vanadium are mainly associated with granites, pegmatites, and quartzofeldspathic veins; TREE values reach 5534 ppm. Raman studies confirm both flaky/massive and vein-type graphite.
Sivagangai District, Tamil Nadu	Ulaganathapuram block in the Sivaganga Graphite belt	1: 2000	1.70	-	-	Mapping & Sampling	Preliminary exploration (G3) in Ulaganathapuram, Sivagangai, Tamil Nadu, mapped 1.7 sq km, excavated three trenches, and collected 76 samples. Graphite occurs as flake and amorphous types in sheared gneisses, forming lensoidal/pocket-type zones over 500 m strike. SP and resistivity surveys showed E-W anomalies (-50 to -200 mV). Most samples exceeded 10% fixed carbon, up to 22.64% FC.
Banswara District, Rajasthan	Bhongra-Bargun area	1:12500	75	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Bhongra-Bargun, Banswara, Rajasthan, covered 75 sq km at 1:12500 scale. Mapping, pitting, trenching (50 cu m), and 200 samples (BRS, PTS, channel, and PCS) were collected to evaluate graphite and associated RM mineralisation. The graphite-bearing schist belt trends NNWSSE for ~17 km within the Debari Group of the Aravalli Supergroup. Graphite occurs mainly in schist as flaky, lenticular bands along foliation planes, with fixed carbon (FC) ranging <1–21% in trench samples. Vanadium averages ~308 ppm, peaking at 681 ppm near Sompur village.
Ajmer District, Rajasthan	Bhaonta-Amba and Dumra area	1:12500	100	-	-	Mapping & Sampling	Reconnaissance survey (G4) in Bhaonta-Amba and Dumra, Ajmer, Rajasthan, covered 100 sq km at 1:12500 scale. The area is largely covered by sand, with mapped lithounits including Delhi Supergroup metasediments, Sendra granite, and younger intrusives (amphibolite, ultramafic, pegmatite). Two graphite-bearing schist bands near Amba village, 900 m and 400 m long and 1.5–5 m thick, were delineated. Sampling included 100 bedrock, 50 channel, 10 petrochemical, 20 petrological, 5 XRD, and 3 Raman analyses. Trenches (20 cu m) exposed graphite in concealed zones, and chemical analysis of four samples showed up to 6% fixed carbon.
BAUXITE/ALUMINOUS LATERITE							
Jashpur District, Chhattisgarh	North of Rouni block	1:4000	2.72	78	1096.10	Mapping, Sampling & Drilling	G2 exploration for bauxite in North of Rouni block, Jashpur (2.72 sq km) involved 78 boreholes (1,096 m) and detailed mapping. Bauxite occurs as massive and pisolitic lenses within laterite overlying gneisses and Deccan basalts. Al_2O_3 in bauxite ranges 30.36–47.5% (avg. 37.89%), SiO_2 3.53–6.97%; in aluminous laterite, Al_2O_3 24.2–42.23% (avg. 30.88%), SiO_2 1.46–31.48%. Gallium: 1–142 ppm (avg. 71 ppm), vanadium: 182–3,061 ppm.
Tumsar, Bhandara District, Maharashtra	Hardoli Gondikhari area	1:12500	62	-	-	Mapping & Sampling	G4 reconnaissance survey for bauxite and aluminous laterite in Hardoli-Gondikhari, Tumsar, Bhandara (62 sq km) involved large-scale mapping, collection of 53 bedrock, 16 soil/regolith, and 33 trench samples. Laterite analyses show Al_2O_3 : 13.02–27.07%, Fe_2O_3 : 24.1–45.37%.
Sindhudurg District, Maharashtra	Phanasgoan area	-	-	-	-	Mapping & Sampling	G4 reconnaissance survey in Phanasgoan, Sindhudurg, Maharashtra mapped 66 sq km of Deccan Trap basalts with bauxite and aluminous laterite. Bauxite occurs as gibbsite-rich nodules; Al_2O_3 : 32–54.97%, Fe_2O_3 : 3.51–44.23%, SiO_2 : 1.19–6.85%; aluminous laterite: Al_2O_3 22.62–42.43%, Fe_2O_3 20.06–50.02%, SiO_2 3.75–33.71%. Vanadium: 79–1137 ppm, Gallium: 45–84 ppm, REE: 94.92–257.92 ppm. Potential bauxite: 6 sq km; aluminous laterite: 60 sq km.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sindhudurg District, Maharashtra	Achra Area	1:12500	100	-	-	Mapping & Sampling	G4 reconnaissance survey in Achra, Sindhudurg, Maharashtra mapped 100 sq km of TTG gneiss and Kaladgi quartzite/shale capped by laterite. Bauxite occurs as pockety/lensoidal bodies; aluminous laterite is extensive. 100 bedrock and 50 pit samples analyzed: Al_2O_3 22.54–54.45%, SiO_2 2.84–34.8%, Gallium 39–86 ppm. Potential area: 5 sq km bauxite, 21 sq km aluminous laterite (Munage, Vayangani, Tonadavali, Maldi).
Dindori District, Madhya Pradesh	Around Dadargaon and Chauradadar Areas	-	-	-	-	Mapping & Sampling	Reconnaissance survey in Dadargaon and Chauradadar, Dindori, MP mapped 100 sq km of Deccan Basalt overlain by laterite and bauxite. Bauxite occurs as pockets, lenses, massive and tabular bodies on flat-topped mounds at 980–1040 m elevation, with thickness 0.5–3 m. Chemical analysis of 100 BRS shows Al_2O_3 25.95–62.57%, SiO_2 0.23–21.59%, Fe_2O_3 2.46–44.46%, TiO_2 1.82–14.59%, V 96.56–2410 ppm, Ga 9.69–166.98 ppm. Bauxite is enriched in Al_2O_3 , Ti, Zr, Nb; SiO_2 and Fe_2O_3 are lower. Ore types: pisolitic, vermicular, and massive; laterite mostly aluminous with interspersed bauxitic patches. Mineralized zones are lenticular, making precise delineation challenging.
Belgavi District, Karnataka	Bailur-Tirth Kunde area, Khanapur Taluk	-	-	-	-	Sampling	Reconnaissance survey, Bailur-Tirth Kunde, Belgavi (G4): Deccan Basalt capped by pockety bauxite/laterite; argillite and granite-gneiss at contacts. Aluminous laterite: pink/grey, vesicular, pisolitic; ferruginous laterite: brick-red, hard, vermicular. Bauxite: bouldery, sporadic. XRD: gibbsite dominant. Bedrock: Al_2O_3 17–68%, SiO_2 2–57.6%, Fe_2O_3 3.73–33.3%, Sc 11–152 ppm, V 34–2459 ppm, Ga 8–116 ppm, Ni 20–577 ppm, Cr 110–3043 ppm.
Kannur District, Kerala	Kolattur area	1:12500	120	-	-	Mapping & Sampling	Reconnaissance Survey, Kolattur, Kannur (G4): Large-scale mapping (120 sq km) and sampling delineated bauxite and aluminous laterite potential. Area: Coorg Block, SGT, exposing Wayanad Group, Charnockite Group, Peninsular Gneiss, intrusives, and Quaternary laterites. Laterite: dominant, hard duricrust, 5–14.65 m thick, ferric transitioning to aluminous (off-white to reddish-brown), containing goethite, hematite, magnetite, iron tubes, and bauxite pockets. Pisolitic, nodular, and vermicular textures common. Bedrock samples (139) Al_2O_3 : 10.10–47.27%; most aluminous laterite, one bauxite sample. Pit samples: Al_2O_3 15.39–49.15%; regolith: 7.21–45.57%. Chromium in laterite: 0.10–1.39%, with chromiferous zones identified. Area promising for G3-stage exploration of aluminous laterite and chromium.
Kasaragod District, Kerala	Yetadka block	1:4000	5	14	276	Mapping, Sampling & Drilling	Yetadka Block, Kasaragod (G3): Preliminary exploration over 5 sq km included 20 cu m pitting, 276 m drilling (14 boreholes), and collection of BRS, PTS, core, and petrochemical samples. Area exposes duricrust, aluminous laterite, bauxite patches, garnetiferous hornblende gneiss, quartzo-feldspathic gneiss, pyroxenite/granulite, and dolerite. Pit sample Al_2O_3 : 21–32%, SiO_2 : 9–23%, Fe_2O_3 : 28–45%, Cr: 0.1–0.7%, V: 0.07–0.6%; bedrock Al_2O_3 : 21–41%, SiO_2 : 4.6–43.5%, Fe_2O_3 : 6–35%, Cr: 0.04–0.09%, V: 0.02–0.1%. Drilling intersected mineralized zones for core sampling
Sangareddy & Vikarabad Districts, Telangana	Digwal area	1:12500	150	-	-	Mapping & Sampling	Digwal, Telangana (G4): Reconnaissance over 150 sq km mapped laterite, aluminous laterite, and saprolite. 125 BRS and 25 trench samples show Al_2O_3 13.9–40.4% (mean 28.6%), Fe_2O_3 5.7–53.3%, TiO_2 1.5–8.8%, with Ga, V, Li enrichment. Most samples classify as aluminous laterite; trenches average 22.3% Al_2O_3 . Geochemical plots confirm predominant lateritic/saprolitic character with minor bauxite/boehmite. Area shows strong potential for aluminous laterite and bauxite.
Vikarabad District, Telangana	Velchal area	1:12500	150	-	-	Mapping & Sampling	Velchal, Telangana (G4): Reconnaissance over 150 sq km mapped Cenozoic laterites and Deccan Trap basalts with intertrappeans. Laterite occurs as cappings and blankets, 1–30 m thick; aluminous zones 1–16 m, forming lenses/pockets within duricrust and saprolite. Analytical results from 160 samples show Al_2O_3 15.0–41.5% (avg. 29.7%), Fe_2O_3 6.2–44.2%, TiO_2 1.3–10.2%, Ga 20–64 ppm, V 306–1460 ppm, Sc 38–180 ppm; reactive silica 4.6–7.9%. Profile near Ekmamidi: Al_2O_3 32.8–37.9%. Area shows good potential for aluminous laterite.
Vikarabad & Kalaburagi Districts, Telangana & Karnataka	Tattepalli areas	1:12500	100	-	-	Mapping & Sampling	Tattepalli Block (100 sq km, G4) shows low- to high-alumina laterite beneath iron-rich duricrust, with saprolite at depth. Mineralization occurs as vug- and stringlet-type pockets. Analyses of 100 BRS: Al_2O_3 2.3–37.2% (43 >30%), SiO_2 8.4–53.9%, Fe_2O_3 3.1–45.2%, TiO_2 0.8–4.6%, Ga 5–67 ppm, V 59–1535 ppm. Geochemical and petrological studies indicate bauxitic clay, Laterite.

Table 1 (Cont.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Sangareddy District, Telangana	Parvathapur block	1:5000	3.35	5	130.8	Mapping, Sampling & Drilling	Preliminary exploration in Parvathapur block, Sangareddy District (3.35 sq km, G3 stage) involved detailed mapping, 130.8 m drilling in 5 boreholes, and collection of 10 BRS and 131 core samples. The area, part of Deccan Trap overlying Archaean Gneiss-Granite, comprises laterite, aluminous laterite, bauxite, saprolite, and basalt, with bauxite/aluminous laterite occurring between 626–661 MSL. Chemical analyses show Al_2O_3 ranging from 12.98–41.78%, SiO_2 19.98–46.37%, Fe_2O_3 4.19–40.98%, and vanadium up to 2876 ppm. Inferred resources are estimated at 34.44 Mt of high-grade bauxite (33.95% Al_2O_3), 50 Mt aluminous bauxite (26.03% Al_2O_3), and 91 Mt for vanadium (1171 ppm).
Sangareddy District, Telangana	Govindpur block	1:5000	3.70	5	88.10	Mapping, Sampling & Drilling	Preliminary G3 exploration in Govindpur block (3.7 sq km) identified laterite, aluminous laterite, and bauxite over Deccan Trap basalts. Drilling, mapping, and analyses show Al_2O_3 18–35%, TiO_2 1–8.5%, V up to 2023 ppm. Inferred resources: 26 Mt bauxite (31.6% Al_2O_3), 47.3 Mt aluminous bauxite (27.9% Al_2O_3), 34.8 Mt TiO_2 , and 22.4 Mt vanadium.
BARYTE							
Anantapur District, Andhra Pradesh	Venkatampalle-Mutssukota area	-	110	-	-	Mapping & Sampling	A G4-stage reconnaissance survey over 110 sq km in Venkatampalle-Mutssukota, Anantapur (AP) identified Baryte and lithium mineralization. Lithologies include stromatolitic dolomite, shale, tuff beds (5–35 cm), polymictic conglomerates, and serpentinized basic sills. Three Baryte zones were delineated: Zone I (SW of Mutssukota, 500 m strike, up to 65% Ba), Zone II (SE of Venkatampalle, ~1 km discontinuous, up to 60% Ba), and Zone III (NE of Maddalapalle, quartz-baryte band, 1.3 km strike, 11–47% Ba). Lithium occurs in tuffaceous beds interlayered with dolomite and shale.
YSR Kadapa District, Andhra Pradesh	Rajupalem-Tummaluru area	-	110	-	-	Sampling	A G4-stage survey in Rajupalem-Tummaluru, Kadapa, targeted barite and lithium in Vempalle Formation rocks. BRS analyses show Ba 12.6–66.77% with minor high Cu. A 70 m scout borehole revealed variable core recovery; crystalline dolomite was barite-free.
Annamaya District, Andhra Pradesh	Idugupalle area	1:12500	100			Mapping & Sampling	A G4-stage survey in Idugupalle, Annamaya (100 sq km, 1:12500), identified dolomite, shale, tuff, and quartzite. Structural trends are NNW–SSE with NE dips. Black shale near Reddypalle (1–12 m) shows REE and Li potential; tuffaceous shale (1–8 m) hosts quartz–calcite veins. Baryte is minor, with values up to 5168 ppm; a 12 m trench near Idugupalle quarry shows 0.95–7.5% Ba.
Hungund-Kushtagi Schist Belt, Eastern Dharwar Craton, Karnataka	Gadi Sunkapur Block	1: 2000	3.5	6	551.25	Mapping, Sampling & Drilling	A G3-stage exploration in Gadi Sunkapur Block, Hungund-Kushtagi Schist Belt, Karnataka (3.5 sq km, 1:2000) focused on barite, gold, silver, and associated mineralization. Detailed mapping identified metavolcanics, intrusive granite, and hydrothermal quartz–barite–chlorite±calcite±epidote veins. Ground geophysical surveys (IP and resistivity, 30 L km) highlighted lensoid IP anomalies coinciding with ENE–WSW to WNW–ESE trending barite veins. Six boreholes (551.25 m) in the western domain intersected barite veins 0.5–1.5 m thick with 16–65.57% $BaSO_4$. No significant Au or Ag mineralization was detected.
PHOSPHATE/PHOSPHORITE							
Balod District, Chhattisgarh	Bhedi block	-	2.56	30	1260.25	Mapping, Sampling & Drilling	A general exploration in Bhedi Block, Balod District, Chhattisgarh (2.56 sq km) targeted phosphorite and associated mineralization. Thirty vertical boreholes totaling 1260.25 m revealed lithologies including soil, laterite, clay, phosphorite-bearing clay, cherty limestone, siltstone, and sandstone. Analytical results of 526 clay samples showed P_2O_5 ranging from 0.28–27.06%. Indicated phosphorite resources are 12.52 MT ($\geq 5\%$ P_2O_5 , non-beneficiable), 11.88 MT (5–10% P_2O_5 , beneficiable), and 0.30 MT ($\geq 16\%$ P_2O_5 , soil reclamation). Cement-grade limestone resources are 32.73 MT (CaO 44–52%) and 0.04 MT blendable limestone (CaO 34–44%).
Durg and Balod Districts, Chhattisgarh	Khaira Nawagaon area	1:12500	50	5	287	Mapping, Sampling & Drilling	A G4 reconnaissance in Khaira-Nawagaon, Chhattisgarh (50 sq km, 287 m drilling) identified phosphorite, limestone, glauconite, and vanadium-rich aluminous laterite. Laterite, phosphatic clay, cherty limestone, and glauconitic sandstone were mapped. Estimated resources: 38.27 MT phosphorite (5.57% P_2O_5), 2,808.1 MT limestone, 314.19 MT glauconitic sandstone (5.44% K_2O), and 540.8 MT vanadium-bearing laterite (700 ppm V_2O_5).

Table- 1 (Concl'd.)

District/ State	Location	Geological mapping		Drilling		Details of work done	Result obtained/Remark
		Scale	Area (sq. km)	Bore holes (Nos.)	Meterage (m)		
Durg and Balod Districts, Chhattisgarh	Labor-Dewinawagaon area	1:12500	53	5	-	Mapping, Sampling & Drilling	A G4 reconnaissance in Labor-Dewinawagaon, Chhattisgarh (53 sq km, 5 boreholes) mapped clay and limestone of the Charmuria Formation. Estimated inferred resources include 3,562.46 MT cement-grade limestone (499.36 MT Portland, 3,063.10 MT blendable) and 345.78 MT glauconite (>4% K ₂ O, 5.47% avg).
Balod District, Chhattisgarh	Sambalpur Block	1:5000	3	32	1586.30	Mapping, Sampling & Drilling	A G2 general exploration in Sambalpur Block, Balod (3 sq km, 32 boreholes, 1,586 m drilling) mapped clay, cherty limestone, sandstone, and soil of the Charmuria Formation. Phosphatic clay shows P ₂ O ₅ 0.14–26.77%, and cherty limestone CaO 0.47–52.05%. Beneficiation of phosphorite yielded 14.9% P ₂ O ₅ (28.5% recovery). Estimated resources include 9.44 MT phosphorite (5% cut-off), 52.36 MT Portland-grade limestone (CaO 44–52%), and 0.16 MT blendable/beneficiable limestone (CaO 34–44%).
Chhatrapur District, Madhya Pradesh	Chopra-Panchhi-Padro Area	1:12500	100	-	-	Mapping & Sampling	A G4 reconnaissance survey over 100 sq km in Chopra-Panchhi-Padro area, Chhatrapur, mapped Bijawar and Vindhyan Supergroup rocks. Phosphorite mineralization occurs mainly in thin ferruginous sandstone and cherty sandstone lenses, with P ₂ O ₅ estimated up to 5–18%. Chemical analysis of 100 BRS and 50 PTS samples mostly shows <5% P ₂ O ₅ , though Shapiro test indicates higher potential due to ferruginous and quartz content. Iron-rich zones (Fe ₂ O ₃ up to 86.5%) and laterite patches were also noted. Additional findings include vanadium in laterites (up to 1,458 ppm V), dolomitic CaO/MgO, and minor sulphide mineralization in gabbro. Glauconite was observed in Chorhat Formation sandstones.
Lower Siang District, Arunachal Pradesh	Garu-Gensi area	-	-	-	-	Sampling	A G4 reconnaissance survey in Garu-Gensi, Lower Siang, Arunachal Pradesh, identified 15 nodule-bearing zones within carbonaceous shale of the Bichom Formation (Lower Gondwana). Nodules are hard, compact, siltstone-cherty with calcareous cores, 3–45 cm in size, often aligned along bedding planes, some showing concentric layering and pyrite. Analysis of 67 nodules shows P ₂ O ₅ content of 0.14–8.6% (average 1.71%), while surrounding shale contains <0.5% P ₂ O ₅ . Petrography indicates the nodules are primarily cryptocrystalline phosphate minerals.
Dehradun and Tehri Garhwal Districts, Uttarakhand	Manjgoan block	1:12500	50	-	-	Mapping & Sampling	A G4 survey in Manjgoan, Uttarakhand, mapped 50 sq km for phosphorite and rare metals. A 0.5–1.5 m thick phosphatic limestone band in the Kauriyala Formation shows P ₂ O ₅ up to 43.92%. Rare metals in shale/slate are low, e.g., Sc 13.9 ppm, Li 24 ppm, V 122 ppm, and ΣLREE 158 ppm.
Banswara District, Rajasthan	Rupgarh area	-	-	-	-	Sampling	A reconnaissance survey in Rupgarh, Banswara (Rajasthan) identified NW–SE to N–S trending Aravalli Supergroup rocks, including phosphatic dolomite/stromatolitic dolomite, phyllite, quartz-mica schist, and quartzite. Two major dolomite bands show phosphorite indications in field Shapiro tests, though laboratory confirmation of P ₂ O ₅ content is pending.
PYROPHYLLITE							
Bhandara District, Maharashtra	Khapa - Nawargaon area	-	-	-	-	Sampling	The G4-stage reconnaissance in Khapa-Nawargaon, Bhandara District, identified pyrophyllite and kyanite mineralization within schistose units of the Sakoli Group. Mineralization is structurally controlled along shear zones and quartz reefs, with pyrophyllite occurring in foliated and granular forms, and kyanite as dark blue blades. Hydrothermal activity influenced the mineral assemblage, including quartz, muscovite, paragonite, and kaolinite. Chemical analyses show Al ₂ O ₃ 31.8–48.2% and SiO ₂ 39.8–54.2%. The mineralized zone extends 540 m northeast of the existing Nawargaon mines.

Table – 2: Exploration carried out by Central Government Undertakings, State DMGs and State Government Undertakings in 2023-24

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)		
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Balaghat Mine For Lease 182.3004 Ha. At post Bharweli, Tehsil & District: Balaghat (M.P.)	NIL	NIL	26	11140	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 1898492 tonnes 122 – 11678706 211 – 10105595 tonnes 332 – 1179766 tonnes Total – 24862559 tonnes
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Balaghat Mine For Lease 76.409 Ha. At post Bharweli, Tehsil & District: Balaghat (M.P.)	NIL	NIL	30	14010	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 321788 tonnes 222 – 265475 tonnes Total – 587263 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Beldongri Mine For Lease 26.66 Ha. Parseoni tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	Reserves (as on 01.04.2024) as per UNFC classification: 111 – 109523 tonnes 221 – 245668 tonnes Total – 355191 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Beldongri Mine For Lease 12.99 Ha. Parseoni tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	Reserves (as on 01.04.2024) as per UNFC classification: 111 – 0 122 – 31767 tonnes Total – 31767 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	New Satak Mine For Lease 16.84 Ha. Parseoni tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	Reserves (as on 01.04.2024) as per UNFC classification: 111 – 30787 tonnes 221 – 3885 tonnes Total – 34672 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Old Satak Mine For Lease 8.68 Ha. Parseoni tehsil, Nagpur District						Reserves (as on 01.04.2023) as per UNFC classification: 111 – 249951 tonnes 221 – 652234 tonnes Total – 902185 tonnes
MOIL/ Manganese ore/ Maharashtra/ Bhandara	Chikla Mine	NIL	NIL	03	850	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification are 4814904 tonnes. Type of drill – Core Drill; NX and BX size
MOIL/ Manganese ore/ Maharashtra/ Bhandara	Dongri Buzurg Mine For Lease 174.80 Ha. Tehsil: Tumsar, District: Bhandara, State: Maharashtra	NIL	NIL	03	924	NIL	In-situ ore reserves & resources (as on 31.03.2024) as per UNFC classification: 111 – 2986230 tonnes 121- 32508 tonnes 122 – 389365 tonnes 211 – 5142700 tonnes 222 – 2942757 tonnes 331 – 598080 tonnes 332 – 477236 tonnes Total – 14583826 tonnes (14.58 million tonnes)
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Gumgaon Mine For Lease 85.90 Ha. Gumgaon village, Khapa Post, Saoner Tehsil, Nagpur District	NIL	NIL	20	6094	NIL	In-situ ore reserves as per UNFC classification: 111 – 2775045 tonnes 221 – 5670141 tonnes 332 – 218750 tonnes Total – 8663936 tonnes Exploratory mining: Total = 960.08 m (ore drive, haulage road, x-cut, winze) Shaft sinking = 425.91 m
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Munsar mine For Lease 108.63 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	06	1474	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 2020132 tonnes 221 – 3045133 tonnes Total – 5065265 tonnes Type of drill is NQ.
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Munsar mine For Lease 25.15 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 11235 tonnes 221 – 0 Total - 11235 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Munsar mine For Lease 0.97 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 2466 tonnes 221 – 0 Total - 2466 tonnes

Table 2 (Cont.)

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)		
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Munsar mine For Lease 5.74 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	04	802	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 53595 tonnes 221 – 164518 tonnes Total - 218113 tonnes Type of drill is NQ.
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Parsoda mine For Lease 53.75 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 1030314 tonnes 221 – 286561 tonnes Total - 1316875 tonnes
MOIL/ Manganese ore/ Maharashtra/ Nagpur	Kandri mine For Lease 83.06 Ha. Ramtek Tehsil, Nagpur District	NIL	NIL	01	550	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 3005233 tonnes 221 – 8807750 tonnes Total - 11812983 tonnes Type of drill is HQ.
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Sitapatore-Sukli mine For Lease 43.353 Ha. Tirodi Tehsil, Balaghat District, Madhya Pradesh	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 331205.84 tonnes 211 – 95893.47 tonnes Total – 427099.31 tonnes
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Sitapatore-Sukli mine For Lease 79.55 Ha. Tirodi Tehsil, Balaghat District, Madhya Pradesh	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 4676 tonnes 121 – 34570.39 tonnes 333 – 57000 tonnes Total –96246.39 tonnes
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Tirodi mine Tirodi Tehsil, Balaghat District, Madhya Pradesh	NIL	NIL	NIL	NIL	NIL	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 506083 tonnes 122 – 266115 tonnes Total – 772198 tonnes
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Ukwa mine For Lease 199.0673 Ha. Baihar Tehsil, Balaghat District, Madhya Pradesh	NIL	NIL	02	308	04	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 982826 tonnes 122 – 1174187 tonnes 221 - 1398105 tonnes Total - 2555118 tonnes Type of drill is Diamond drill.
MOIL/ Manganese ore/ Madhya Pradesh/ Balaghat	Ukwa mine For Lease 69.581 Ha. Baihar Tehsil, Balaghat District, Madhya Pradesh	NIL	NIL	40	2255	223	In-situ ore reserves (as on 01.04.2024) as per UNFC classification: 111 – 440308 tonnes 221 – 167125 tonnes 331 – 3093563 tonnes 332 – 2804200 tonnes Total - 6505196 tonnes Type of drill is Diamond drill.
NALCO/ Bauxite/ Odisha/ Koraput	Panchpatmali Bauxite Mine Central & North Block, Koraput Dist, Odisha	NIL	NIL	NIL	NIL	NIL	The Mineable part of measured resource of bauxite in Panchpatmali Central and North Block is kept under proved category as on 31.03.2024 while remaining resource is kept under feasibility resource category as per below:
		Reserve/ resources	Type	U N F C Code	Quantity of bauxite (Mill.MT)		
		Reserves	Proved	111	75.72		
			Probable	121	---		
		Total (a)	---	---	75.72		
		Remaining resources	Feasibility	211	12.17		
			Pre-feasibility	222	---		
			Measured	331	---		
			Indicated	332	---		
			Inferred	333	---		
			Reconnaissance	334	---		
		Total (b)	---	---	12.17		
		Total (a+b)	---	---	87.89		

The above reserve data is subject to change as per 25mX25m grid pre-production drill data.
The entire deposit has been covered in (100m X 100m) grid and the reserve of entire bauxite deposit has been estimated based on 100mX100m grid drilling data.

Table 2 (Cont.)

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated																																																																		
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)																																																																				
NALCO/ Bauxite/ Odisha/ Koraput	Panchpatmali Bauxite Mine South Block Koraput Dist, Odisha	NIL	NIL	NIL	NIL	NIL	The Mineable part of measured resource of bauxite in Panchpatmali South Block is kept under proved category as on 31.03.2024 while remaining resource is kept under feasibility resource category as per the following table: <table><tr><td>Reserve/ resources Reserve/ resources</td><td>Type</td><td>UNFC Code</td><td>Quantity of bauxite (Mill.t)</td><td>Grade (Al₂O₃%/ SiO₂%)</td><td>Cut-off grade (Al₂O₃% / SiO₂%)</td></tr><tr><td rowspan="2">Reserves</td><td>Proved</td><td>111</td><td>76.70</td><td>42.37/3.88</td><td>+20 / -7</td></tr><tr><td>Probable</td><td>121</td><td>---</td><td>---</td><td>---</td></tr><tr><td>Total (a)</td><td>---</td><td>---</td><td>76.70</td><td>---</td><td>---</td></tr><tr><td rowspan="6">Remaining resources</td><td>Feasibility</td><td>211</td><td>9.38</td><td>---</td><td>---</td></tr><tr><td>P r e - feasibility</td><td>222</td><td>---</td><td>---</td><td>---</td></tr><tr><td>Measured</td><td>331</td><td>---</td><td>---</td><td>---</td></tr><tr><td>Indicated</td><td>332</td><td>---</td><td>---</td><td>---</td></tr><tr><td>Inferred</td><td>333</td><td>---</td><td>---</td><td>---</td></tr><tr><td>R e c o n n - aissance</td><td>334</td><td>---</td><td>---</td><td>---</td></tr><tr><td>Total (b)</td><td>---</td><td>---</td><td>9.38</td><td>---</td><td>---</td></tr><tr><td>Total (a+b)</td><td>---</td><td>---</td><td>86.08</td><td>42.52/3.59</td><td>+20 / -7</td></tr></table> The above reserve data is subject to change as per 25mX25m grid preproduction drill data. The entire deposit has been covered in (100m X 100m) grid and the reserve of entire bauxite deposit has been estimated based on 100mX100m grid drilling data.	Reserve/ resources Reserve/ resources	Type	UNFC Code	Quantity of bauxite (Mill.t)	Grade (Al ₂ O ₃ %/ SiO ₂ %)	Cut-off grade (Al ₂ O ₃ % / SiO ₂ %)	Reserves	Proved	111	76.70	42.37/3.88	+20 / -7	Probable	121	---	---	---	Total (a)	---	---	76.70	---	---	Remaining resources	Feasibility	211	9.38	---	---	P r e - feasibility	222	---	---	---	Measured	331	---	---	---	Indicated	332	---	---	---	Inferred	333	---	---	---	R e c o n n - aissance	334	---	---	---	Total (b)	---	---	9.38	---	---	Total (a+b)	---	---	86.08	42.52/3.59	+20 / -7
Reserve/ resources Reserve/ resources	Type	UNFC Code	Quantity of bauxite (Mill.t)	Grade (Al ₂ O ₃ %/ SiO ₂ %)	Cut-off grade (Al ₂ O ₃ % / SiO ₂ %)																																																																				
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Total (a+b)	---	---	86.08	42.52/3.59	+20 / -7																																																																				
NMDC/ Iron ore (Hematite)/ Karnataka/ Bellary	Donimalai Iron Ore Mine – 2E, 3E, SB & BMMX Deposits. Lease area: 597.54 Ha.	NIL	NIL	13	1032	894	Reserves/ Resources estimated during the year 2023-24 as per UNFC Classification as on 01.04.2024: 111 - 86.90 million tonnes 121 & 122 - 31.17 million tonnes 211 - 4.10 million tonnes 333- 14.20 million tonnes																																																																		
NMDC/ Iron ore (Hematite)/ Karnataka/ Bellary	Kumaraswamy Iron Ore Mine - B-Block and C-Block- Deposits Lease area 639.80 Ha.	NIL	NIL	08	937	347	Reserves/ Resources estimated during the year 2023-24 as per UNFC Classification as on 01.04.2024: 111-159.67 million tonnes 121 & 122 - 28.51 million tonnes 211- 20.67 million tonnes 333- 20.04 million tonnes																																																																		
NMDC/ Iron ore (Hematite)/ Chhattisgarh/ South Bastar	Village-Bacheli ,Post Office -Bacheli,Taluka- Badee Bacheli, District - South Bastar Dantewada, Toposheet No. - 65F/2,65F/6 Deposit 10 ML Lease area: 309.34 Ha	1:2000	3 . 0 9 3 4 sq km	15	1333	407	Balance as on 01.04.2024: Reserves: 111- 182871764.00 tonnes Total - 182871764.00 tonnes Resources: 211- 44320000 tonnes 221- 620000 tonnes 331- 37040000 tonnes 332- 15050000 tonnes 333- 2440000 tonnes Total -99470000 tonnes Exploratory drilling is under progress. Reserves estimation is under process.																																																																		
NMDC/ Iron ore (Hematite)/ Chhattisgarh/ South Bastar	Village-Kirandul, Post Office- Kirandul, Taluka - Bade Bacheli, District - South Bastar Dantewada, Forest Compartment No -1857, 1858, 1859, 1860, 1861, 1863 , 1864 Coordinates: Latitude: 18°36'44.0492" - 18°38'31.8650" Longitude: 81°13'54.6335" - 81°15'17.3110" Toposheet No. - E44J2 Dep-14 ML Lease area: 322.368 Ha	1:2000	3.22368 sq km	20	1883	546	Balance as on 01.04.2024: Reserves: 111- 356599481 tonnes, Fe - 64.77% 122-37364000 tonnes, Fe- 63.56% 121-0.00, Fe - 0.00% Total - 393963481 tonnes, Fe 64.66% Resources: 211-58663000 tonnes Fe - 64.25% 221-0, Fe-0.00% 222-18074000 tonnes, Fe - 62.17% 331-5400000 tonnes, Fe - 59.62% 332-1688000 tonnes, Fe - 62.41% 333-31378000 tonnes, Fe - 55.18% 334-5411000 tonnes, Fe - 54.78% Total -120614000 tonnes, Fe - 60.92% Exploratory drilling is under progress. Reserves estimation is under process.																																																																		

Table 2 (Cont.)

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)		
NMDC/ Iron ore (Hematite)/ Chhattisgarh/ South Bastar	Village- Kirandul, Post Office- Kirandul, Taluka- Bade Bacheli, District- South Bastar Dantewada, Forest Compartment No.-1857, 1858, 1859, 1860, 1861 Toposheet No.- E44J2 & E44J6 Dep-14 NMZ Lease area: 506.742 Ha	1:2000	5.06742 sq km	12	1080	107	Balance as on 01.04.2024: Reserves: 111- 199001219 tonnes, Fe - 64.94% 122-7777000 tonnes, Fe 64.33% 121-0,Fe- 0.00% Total -206778219 tonnes, Fe 64.92% Resources: 211 - 34415000 tonnes, Fe 62.66% 221 - 0, Fe - 0.00% 222 - 3700000 tonnes, Fe 64.19% 331 - 12743000 tonnes, Fe 58.30% 332 - 1140000 tonnes, Fe 59.41% 333 - 14461000 tonnes, Fe 58.96% 334 - 12150000 tonnes, Fe 60.31% Total- 78609000 tonnes, Fe 60.93% Exploratory drilling is under progress. Reserves estimation is under process.
NMDC/ Iron ore (Hematite)/ Chhattisgarh/ South Bastar	Village-Kirandul, Post Office-Kirandul, Taluka- Bade Bacheli, District- South Bastar Dantewada Forest Compartment No.- 1843, 1844, 1845, 1846, 1847, 1848, 1849, 1850, 1852, 1853, 1854, 1855, 1860. Dep-11 ML Lease area: 874.924 Ha	1:2000	8.74924 sq km	16	1859	851	Balance as on 01.04.2024: Reserves: 111 – 279215793 tonnes, Fe 65.29% 122 – 39680000 tonnes, Fe 63.61% 121- 0, Fe-0.00% Total 318895793 tonnes, Fe 65.08% Resources: 211-41800000 tonnes, Fe 62.70% 221- 0, Fe- 0.00% 222-9350000 tonnes, Fe 61.66% 331-18060000 tonnes, Fe 59.54% 332-6870000 tonnes, Fe 63.86% 333-128060000 tonnes, Fe 62.22% 334-16950000 tonnes, Fe 64.88% Total - 221090000 tonnes, Fe 62.32% Exploratory drilling is under progress. Reserves estimation is under process.
RINL/ Limestone (BF grade)/ Andhra Pradesh/ Krishna	Budawada Village, Jaggayyapeta Mandal, Krishna District, Andhra Pradesh Latitude : 16°52'45.14" N Longitude : 80°03'26.92" E	G2 scale	12.95 sq km	NIL	NIL	NIL	Proved Mineral Reserves (111) - 60.53 MT Probable Mineral Reserves (121& 122) - 268.40 MT Prefeasibility Mineral Resources (221 & 222) – 272.74 MT Inferred Mineral Resources (333) – 451.21 MT
RINL/ Manganese/ Andhra Pradesh/ Vizianagaram	Garbham Manganese Mine, Merakamudidam Mandal, Vizianagaram District, Andhra Pradesh	NIL	NIL	NIL	NIL	NIL	28 nos. of boreholes with 1574 m of core drilling carried out in the FY 2018-19 and the potential areas of Central and West Garbham leases are drilled and proved to be mineralized in G1 axis. The total resources of 3 million tonnes are estimated, out of which total reserves stand at 2.011 million tonnes and balance resources are estimated at 0.99 million tonnes.
DGM (J&K)/ Limestone/ Jammu & Kashmir/ Poonch	N-W extension of Dharamora and Faislabad Limestone Blocks, Surankote tehsil, Poonch District	1:12500	25 sq km	NIL	NIL	NIL	Reserves estimated during the year 2023-24 was 62,50,000 MTs [Reconnaisance reserves (334)]
DMG (Karnataka)/ Aluminous Laterite/ Karnataka/ Belgaum	Navage, Belgaum	1:2000	0.6 sq km	NIL	NIL	NIL	Reserves estimated during the year 2023-24 was 6,37,935 tonnes [Reconnaisance reserves (334)]. 39 nos. of samples were chemically analysed.
DMG (Karnataka)/ Limestone/ Karnataka/ Kalburgi	Itga 1, Kalburgi	1:4000	6.03 sq km	05	225	NIL	Reserves estimated during the year 2023-24: (333): 217.11 MMT (334): 320.60 MMT 262 nos. of samples were chemically analysed.

Table 2 (Cont.)

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)		
DMG (Karnataka)/ Limestone/ Karnataka/ Kalburgi	Itga 2. Kalburgi	1:4000	4.57	06	244	NIL	Reserves estimated during the year 2023-24: (333): 202.7 MMT (334): 184.72 MMT 253 nos. of samples were chemically analysed.
DMG (Karna- taka)/ China clay/ Karnataka/ Chickballapura	Bairaganahalli, Chickballapura	1:4000	0.72	NIL	NIL	NIL	Reserves estimated during the year 2023-24: (333): 14,01,882 MT 21 nos. of samples were chemically analysed.
GMDC/ Baux- ite/ Gujarat/ Kutch	Wandh-2 Baux- ite mine, PO: Gadhsisa, Kutch District	NIL	NIL	NIL	NIL	NIL	Pitting is done for confirmation of qualitative baux- ite in total mines pit area for mine plan renewal. 21 nos. of samples were chemically analysed.
GMDC/ Fluorspar/ Gujarat/ Chhota Udaipur	Ambadungar Mine Village: Ambadungar, District: Chhota Udaipur	NIL	NIL	08	2435	NIL	Not assessed.
GMDC/ Bauxite/ Gujarat/ Dwarka	Mevasa-I Bauxite Mine Village: Mevasa, District: Devbhumi Dwarka	NIL	NIL	NIL	NIL	NIL	Pitting (07 in number; 1.83 m x 0.91 m x 4.27 m) is done for the confirmation of qualitative Bauxite within Mine lease area. 07 nos. of samples were chemically analysed.
GMDC/ Lead- Zinc-Copper/ Gujarat/ Banaskantha	Ambaji, Banaskantha District	NIL	NIL	12	4931	NIL	Reserves estimated during the year 2023-24: (333): 1176763 tonnes 95 nos. of samples were chemically analysed.
GMDC/ Lignite/ Gujarat/ Bhavnagar	Surka (N) Lignite Mine Tagadi village, Bhavnagar District	NIL	NIL	23	2670	NIL	Drilling was carried out to establish lignite boundary and confirmation of lignite seams, hard strate etc.
MECL/Coal/ Chhattisgarh/ Raigarh	Batati Kolga North East Block, Chira South East block, Batati Kolga East & West Block, Nawagaon Block	NIL	NIL	NIL	NIL	NIL	A total of 9220.47 million tonnes of coal resources were established during 2023-24 in Batati Kolga North East Block, Chira South East block, Batati Kolga East & West Block, Nawagaon Block, Raigarh, Charbata East, West of Chhal, Chhattisgarh.
MECL/ Lignite/ Tamil Nadu/ Cuddalore- Ramanathapuram	Veeranam Block-IV, Cuddalore & RS Mangalam block, Ramanathapuram	NIL	NIL	NIL	NIL	NIL	814.82 million tonnes of lignite resources were established in Veeranam Block-IV, Cuddalore & RS Mangalam block, Ramanathapuram, Tamil Nadu .
MECL/ Iron ore/ Odisha & Karnataka/ Sundargarh, Keonjhar & Tumakuru	Malangtoli G sub block, Malangtoli E Sub block & Malangtoli D sub block, District Sundargarh & Keonjhar, Odisha and Habbigegudda Iron ore mine, Chitradurga, Tumakuru District, Karnataka.	NIL	NIL	NIL	NIL	NIL	187.86 million tonnes of iron ore resources were established in Malangtoli G sub block, Malangtoli E Sub block & Malangtoli D sub block, District Sundargarh & Keonjhar, Odisha and Habbigegudda Iron ore mine, Chitradurga, Tumakuru District, Karnataka.
MECL/ Copper/ Jharkhand, Chhattisgarh & Rajasthan/ East Singhbhum, Jahpur & Jhunjhunu	Rakha Mining Lease (Phase-I), Indian Copper Complex Ghatsila, Jaspur, Chhattisgarh & Dhanota Block, Jhunjhunu, Rajasthan.	NIL	NIL	NIL	NIL	NIL	85.20 million tonnes of copper resources were established in Rakha Mining Lease (Phase-I), Indian Copper Complex Ghatsila, Jaspur, Chhattisgarh & Dhanota Block, Jhunjhunu, Rajasthan.
MECL/ Potash/ Rajasthan/ Hanumangarh	Bharusari Block, Bharusari North block, Hanumangarh, Rajasthan	NIL	NIL	NIL	NIL	NIL	84.34 million tonnes of potash resources were established in Bharusari Block, Bharusari North block, Hanumangarh, Rajasthan.

Table 2 (Concl.)

Agency/ Mineral/ State/ District	Location	Geological mapping		Drilling		Sampling (nos)	Remarks Reserves/resources estimated
		Scale	Area (sq. km)	Boreholes (Nos.)	Meterage (m)		
MECL/ Limestone/ Ariyalur, Gulbarga & Satna/ Tamil Nadu, Karnataka & M.P.	Periyakurichi, Ariyalur, Tamil Nadu, Malkhed-1 & 2, Karnataka, Kiranagi-1 & 2, Kalaburgi, Naddisonur-1 & 2, Gulbarga, Karnataka and Piprahat-Kubri block, Satna, Madhya Pradesh	NIL	NIL	NIL	NIL	NIL	958.60 million tonnes of limestone resources were established in Periyakurichi, Ariyalur, Tamil Nadu, Malkhed-1 & 2, Karnataka, Kiranagi-1 & 2, Kalaburgi, Naddisonur-1 & 2, Gulbarga, Karnataka and Piprahat-Kubri block, Satna, Madhya Pradesh.
MECL/ Graphite/ Odisha/ Bolangir	Salepali block and Larambha- Kanaital & Pudapadar Bharatbahal block, Dist. Bolangir, Odisha	NIL	NIL	NIL	NIL	NIL	1.97 million tonnes of graphite resources were established in Salepali block and Larambha-Kanaital & Pudapadar Bharatbahal block, Dist. Bolangir, Odisha.
MECL/ Phosphorite/ Uttar Pradesh/ Lalitpur	Pahadi Kalan- Gora Kalan block, Dist. Lalitpur, Uttar Pradesh	NIL	NIL	NIL	NIL	NIL	19.08 million tonnes of phosphorite resources were established in Pahadi Kalan-Gora Kalan block, Dist. Lalitpur, Uttar Pradesh.
MECL/ Glauconitic sandstone/ Madhya Pradesh/ Satna	Pindra SW Extn. Dist. Satna, Madhya Pradesh	NIL	NIL	NIL	NIL	NIL	1918.03 million tonnes of Glauconitic sand stone resources were established in Pindra SW Extn. Dist. Satna, Madhya Pradesh.



5. Research & Development

48

Ore Dressing investigations were carried out at various Laboratories of Mineral Processing Division of IBM in 2023-24.

24,150

Chemical analysis and 3,004 mineralogical examinations were also completed at MP Division of IBM during the period.

Research and Development (R&D) is vital for advancing exploration technologies, improving mining practices and promoting sustainable utilization of mineral resources. In view of evolving sectoral challenges and strategic mineral requirement, the Government of India has prioritized R&D in the mining sector through targeted policy and programmer interventions.

The Ministry of Mines, Government of India provides funds to academic institutions, universities, national institutes and R&D institutions recognised by the Department of Scientific and Industrial Research, Government of India, for implementing R&D projects under the Science and Technology Programme Scheme of the Ministry of Mines with the vision to promote research in applied geosciences, mineral exploration, mining and allied areas, mineral processing, optimum utilisation and conservation of the mineral resources of the country, for the benefit of the nation and its people.

The broad thrust areas for supporting research in mining are furnished below:

1. Prospecting / exploration for strategic, rare and rare earth minerals.
2. Development of new technologies for mineral exploration and mining– both on land and in deep sea – to locate and exploit new mineral resources.
3. Research on mining methods. This includes rock mechanics, mine designing, mining equipment, energy conservation, environmental protection and mine safety.
4. Improvement of efficiency in processes, operations, recovery of by-products and reduction in specification and consumption norms.
5. Research in metallurgy and mineral beneficiation techniques to utilise lower grade and finer size ores.
6. Extraction of value-added products from mine waste, plant tailings etc.
7. Development of new alloys and metal related products, etc.
8. Evolution of low capital and energy saving processing systems.
9. Production of high purity materials.
10. Promotion of Cooperative research among organisations associated with the Mineral Sector.

SCIENCE AND TECHNOLOGY PROGRAMME OF MINISTRY OF MINES

Recognising the paramount importance of safety, economy, speed and efficiency in extraction of mineral resources and in their eventuality into viable economic alloys and metals, the National Mineral Policy has accorded higher priority to Research and Development (R&D) programmes. The Ministry of Mines, Government of India, has been funding research and development projects (R&D projects) of many research institutions in Mining and Metallurgy Sector since 1978.

Science and Technology Programme of the Ministry of Mines has the following three components—

1. Research and Development (R&D) component
2. Information, Education and Communication (IEC) component
3. Promotion of Research and Innovation in Startups and MSMEs in Mining, Mineral Processing, Metallurgy and Recycling Sector (S&T-PRISM) component.

A brief description of these components is mentioned below:

1. Research and Development (R&D) Component

Under this component, funds are released to academic institutions, universities, national institutes and R&D institutions recognised by the Department of Scientific and Industrial Research, Government of India for undertaking research and development projects.

2. Information, Education and Communication (IEC) Component

Under this component, funds are released to industry associations, recognised Universities, recognised academic and research bodies having at least three years' experience in the Mining and Mineral Sector by organising or being associated with promotional events

Promotion of Research and Innovation in Startups and MSMEs in Mining, Mineral Processing, Metallurgy and Recycling Sector (S&T-PRISM) component: Under this component, funds are released to Startups to ensure timely availability of the seed support to eligible and deserving startups.

R&D REPORT OF INDIAN BUREAU OF MINES

Research & Development activities carried out by the mineral processing Division of Indian Bureau of Mines (IBM) during the year 2023-24 are as below:

Mineral Beneficiation

Mineral beneficiation studies, including mineralogical testing and chemical analysis are closely related to both conservation and development of mineral resources. During the year 2023-24, 48 ore dressing investigations,

24,150 chemical analysis, 3,004 mineralogical examinations were completed.

Some of the salient achievements of ore dressing investigations are as follows:

1. Iron Ore

Bench-scale Beneficiation Studies were carried out on an iron ore (G-2 Stage Exploration) sample received from the Mankarnacha Iron Ore Project located in Keonjhar and Sundargarh District, Odisha (routed through Directorate of Mines and Geology, Odisha).

The as-received sample assayed 51.96% Fe(T), 5.59% SiO₂, 9.89% Al₂O₃, 10.10% LOI, 0.32% FeO, 0.59% K₂O, 0.25% Na₂O, 0.09% CaO and 0.10% P₂O₅.

Two process routes were developed, viz.

- a) Jigging on coarse fraction, followed by magnetic separation and
- b) Gravity separation employing tabling at different size fractions.

The best result were achieved by gravity separation employing tabling and concentrate assaying 63.54% Fe(T) 1.17% SiO₂, 3.53% Al₂O₃ and 4.45% LOI with Fe (T) recovery of 46.0% (wt.% yield : 37.7 was obtained).

2. Bauxite Ore

Aluminous laterite ore: Bench-scale beneficiation studies on aluminous laterite ore, (G-2 Exploration Sample) recieved from Chichili Block, Jashpur, SU: Chhattisgarh for GSI, Raipur.

The as received sample that assayed 53.91% Al₂O₃, 10.45% Fe₂O₃, 1.32% SiO₂ (T), 0.96% SiO₂(R), 10.49% TiO₂, 1.08% CaO, 1.11% MgO, 23.98% L.O.I. was subjected to the following routes.

Route- I: The as received sample was firstly crushed passed through -10 mesh and screened at - 100 mesh size, followed by magnetic separation of both the -10+100 and -100 mesh fractions, the result yield assayed 56.99 % Al₂O₃, 0.89% SiO₂ (T), 8.25% TiO₂, 7.82% Fe₂O₃, 24.74% L.O.I with recovery of 73.6 % and weight per cent yield of 71.2.

Route- II: The as received sample was firstly crushed and passed through -10 mesh and later subjected to stage grinding and screening at 100 mesh size. The ground product obtained was subjected to gravity separation. The composite concentrate (Table Middling +Table Lights + Table slimes) obtained assayed 55.21% Al₂O₃, 9.5% Fe₂O₃, 10.14 % TiO₂, 1.22% SiO₂, and 23.76% LOI with Al₂O₃ recovery of 98.2% and weight per cent yield of 97.4.

The Table Heavies-I assayed 0.147 % V with wt% yield of 0.5% whereas T-Heavies- II assayed 0.143% V with wt% yield of 2.1%.

3. Copper & Gold

Copper & Ore: Bench-scale beneficiation studies carried out on a low-grade copper ore sample of G-2 Stage Exploration, from Kesharpur East Copper Block, Mayurbhanj District, Odisha for Geological Survey of India, Bhubaneswar, SU: Odisha, Eastern Region.

The as received sample with chalcopyrite as the main copper-bearing mineral and that which assayed 0.61% Cu, 293 ppm Ni, 131 ppm Co, 0.153 ppm Au, 6.18 ppm Ag, 49 ppm Cd, 57 ppm Pb, 115 ppm Zn, 54.87% SiO₂, 9.86% Al₂O₃, 62.13% Acid Insoluble, 11.85%, Fe(T), 1.86% Na₂O, 4.86% K₂O, 3.33% CaO, 2.44% MgO, 0.98% S(T). After being subjected to flotation test was followed by the three floats being combined and subjected to two stage cleaning resulted in a copper concentrate that assayed 29.89% Cu, 0.72% Ni, 0.36% Co, 2.67% SiO₂, 3.46% Acid insoluble, 5.89% Al₂O₃ and 28.28%, Fe(T) with the recovery of copper, nickel and cobalt as 85.3% 43.4 % and 38.6%, respectively (wt% yield 1.8).

The beneficiation study revealed that the low-grade copper ore sample is amenable to beneficiation and the grade of the 4 concentrate obtained is suitable for metallurgical use.

b) Copper Ore: Bench scale beneficiation studies were carried out on a copper Ore of G-2 Stage Exploration Sample received from Kesharpur East Copper Block, Mayurbhanj District, for Geological Survey of India, Bhubaneswar (State Unit: Odisha, Eastern Region).

The as received sample that assayed 0.61% Cu, 0.153 ppm Au, 293 ppm Ni, 131 ppm Co, 6.18 ppm Ag, 49 ppm Cd, 57 ppm Pb, 115 ppm Zn, 350 ppm As, 54.87% SiO₂, 9.86% Al₂O₃, 62.13% Acid Insoluble, 11.85%, Fe(T), 1.82 P₂O₅, 1.86% Na₂O, 4.86% K₂O, 3.33% CaO, 2.44% MgO, 0.98% S(T) after being subjected to beneficiation study employing flotation test at - 200 mesh size produced a copper concentrate assaying 29.89% Cu, 0.72% Ni, 0.36% Co, 2.67% SiO₂, 3.46% Acid insoluble, 5.89% Al₂O₃ and 28.28%, Fe(T) with the recovery of 85.3% Cu, 43.4 %Ni and 38.6%Co respectively (wt% yield 1.8).

The laboratory scale beneficiation study revealed that the copper ore is amenable to beneficiation. Suitable copper concentrate could be produced with high recovery (Au, Ni and Co were associated valuables present in the sample).

c) Copper Ore: Bench-scale beneficiation studies were conducted on a lean-grade copper ore sample of G-2 Stage Exploration from Lakhnauti and Ainti Block, Gwalior District, Madhya Pradesh for Geological Survey of India, Bhopal, Madhya Pradesh.

The as received sample that assayed 0.094%

Cu, 88 ppm Zn, 50 ppm Pb, 34.5 ppm Ni, 21 ppm Co, 77.75% SiO₂, 7.89% Al₂O₃, 1.22%, Fe(T), 0.84% CaO, 1.4% MgO, 92.18% Acid Insoluble and 1.30% LOI after being subjected to beneficiation study employing three stages of rougher flotation followed by two-stage cleaning yielded a copper concentrate that assayed 16.43% Cu, 24.34% Acid insoluble with 77.6% recovery of copper (wt% yield 0.5).

d) Copper Ore: A copper ore sample received from Madhya Pradesh that assayed 0.91% Cu, 0.3 ppm Au, 10.02 ppm Ag, 78.5 ppm Ni, 31.5 ppm Co, 123 ppm Zn, 102 ppm Pb, 21 ppm Cd, 10.5 ppm Li, 583 ppm As, 1.60% S(T), 67.46% SiO₂, 4.14% Fe(T), 7.58% Al₂O₃, 2.23% K₂O, 1.71% CaO, 1.59% MgO, 1.41% Na₂O₃ and 1.60% LOI was subjected to a process route that involved crushing and grinding followed by 3 stages of rougher flotation and 2 stages of cleaner flotation. The copper concentrate yielded after the second cleaner float assayed 21.11% Cu, 2.35 ppm Au, 34.15 ppm Ag, 368 ppm Co, 134 ppm Ni, 12 ppm Cd with Cu recovery of 91.2% and wt% yield of 3.9.

4. Base Metals Ore

a) Base Metals Ore: Bench-scale beneficiation studies were conducted on a base metal (lead and zinc) mineralisation (G-2 stage) sample received from Jaukulapalle, YSR Kadapa District, Andhra Pradesh (for Geological Survey of India, Andhra Pradesh).

The as received sample that assayed 1.61% Zn, 0.29% Pb, 10.15% SiO₂, 2.17% Al₂O₃, 2.52% Fe₂O₃, 25.25% CaO, 11.78% MgO, 0.25% BaO, 1.90% S, Co, Ni & Cu 60 ppm each and 33.38% LOI was subjected to beneficial from stools.

Beneficiation study revealed that:

- (i) Pb 1st cleaner float yielded a lead concentrate assaying 20.27% Pb, 5.86% Zn with 87.5% Pb recovery (wt% yield: 1.3)
- (ii) Zn 2nd cleaner float yielded a zinc concentrate assaying 40.86% Zn, 0.14% Pb with 84.5% Zinc recovery (wt% yield: 3.4).
- (iii) Composite zinc concentrate yielded after Zn 2nd cleaner float and Zn scavenger float assayed 39.93% Zn, 0.28% Pb with 87.5% zinc recovery (wt% yield: 3.6).

The beneficiation study revealed that the base metal sample is amenable to beneficiation by employing flotation process.

b) Base Metal: Bench-scale beneficiation studies were carried out on a sample from Adarshile–Dariba Block, Sikar District, Rajasthan, received from the Geological Survey of India, Western Region, Jaipur, Rajasthan.

The as received sample that assayed 0.50% Cu, 4.71% Fe(T), 38.58% SiO₂, 9.45% Al₂O₃, 18.92% CaO, 6.25% MgO, 0.11% S(T), 0.05% TiO₂, 0.012% Zn, 0.14% MnO, 1.50% Na₂O, 2.16 % K₂O, 0.007% Ni, 0.008% Co and traces of Pb & Mo after subjecting to Beneficiation study, employing froth flotation process yielded a copper concentrate assaying 38.69% Cu, with 81.15% copper recovery and a weight yield of 1.05%.

Beneficiation study was carried out to recover calcite as a by-product from copper tailings. The flotation on rougher tails yielded a calcite concentrate assaying 47.46% CaO with 56.31% recovery (weight yield 22.45%).

The bench-scale beneficiation study of revealed that the copper ore sample is amenable to beneficiation. The study also revealed that calcite in the sample is upgradable and is recoverable as a by-product.

5. Manganese Ore:

a) Manganese Ore: Bench-scale beneficiation studies were conducted on a lean-grade manganese ore sample of G-2 Stage Exploration received from Kendudihi–Parulipada Block, Kendujhar District, Odisha (For Geological Survey Of India, SU: Odisha).

The as received sample assayed 10.87% Mn(T), 25.75% Fe (T), 29.69% SiO₂, 6.93% Al₂O₃, 0.01% CaO, 1.00% TiO₂, 0.09% P₂O₅, 0.04% K₂O and S(T) in traces.

Two different process routes were developed but the best result was obtained by crushing the as received sample to - 10 mm size followed by wet screening, jigging, tabling and wet high intensity magnetic separation of composite slimes.

The combined composite concentrate assayed 21.72% Mn, 30.26% Fe(T), 10.81% SiO₂ and 4.04% Al₂O₃ with Mn recovery of 44.1 % and wt% yield 19.9.

b) Manganese Ore: Bench-scale beneficiation studies were conducted on a low-grade manganese ore (Iron & Manganese) sample of G-2 Stage Exploration received from Tatiba Block, West Singhbhum District, Jharkhand for Geological Survey of India, State Unit: Jharkhand, Ranchi.

The as received sample assayed 14.71% Mn(T), 22.14% MnO₂, 22.86% Fe(T), 25.6% SiO₂, 2.83% Al₂O₃, 0.41% TiO₂, 0.38% P₂O₅, 1.22% CaO, 1.06% MgO & 0.21% K₂O.

Bench-scale beneficiation studies were carried out employing different techniques by varying various parameters to upgrade manganese values present

in the sample. The process route evolved consisted of magnetic and gravity separation processes of wet sieve products of the as received fines without grinding. The composite concentrate obtained assayed 27.14% Mn(T), 39.15% MnO₂, 9.14% SiO₂ and 26.90% Fe(T) with distribution of 27.0% Mn with weight yield of 15.5%.

6. Platinum Group Minerals

Bench-scale beneficiation studies were carried out on Platinum-Group-Minerals-bearing drill core sample of G-2 stage exploration received from Vellamari, Palakkad Attapdi valley, Kerala, through Geological Survey of India, State unit : Kerala and Lakshadweep.

The as received original samples assayed 1,423 ppb Pd, 360 ppb Pt, 17 ppb Ru, 20 ppb Rh, 14 ppb Ir. Apart from PGMs, the sample also assayed 0.05 ppm Au, 5.1 ppm Ag, 0.06% Cu, 0.08% Ni, 0.35% As, 0.02% Zn, 0.01% Pb, 42.56% SiO₂, 14.44% Al₂O₃, 8.0% Fe₂O₃, 8.88% MgO, 8.53% CaO, 0.65% S(T).

Two process routes were developed for pre-concentration of PGE—

Route- I: Flotation at -300# yielded a PGM-bearing float concentrate assaying 42 ppb Ru, 70 ppb Rh, 5,947 ppb Pd, 51 ppb Ir and 1,483 ppb Pt with recovery of 58.1% Ru, 69.5% Rh, 76.1%Pd, 70.5% Ir and 72.4% Pt. (wt.% yield :21.8).

Route - II: Gravity separation at differ size followed by flotation a produced composite concentrate (combining Conc. I+II+III) assaying 51 ppb Ru, 59 ppb Rh, 7,141 ppb Pd, 39 ppb Ir and 1,226 ppb Pt with overall recovery of 59.3% Ru, 63.9% Rh, 81.2%Pd, 57.4% Ir and 64.6% Pt. (wt.% yield :20.8).

7. Tungsten Ore and Associated Minerals

Bench-scale beneficiation studies were conducted on tungsten ore and associated lithium, rubidium, bismuth and molybdenum-bearing minerals in a G-2 Stage exploration sample received from rewat Hill Block, Degana, Nagaur Dist., Rajasthan (For Geological Survey of India, Jaipur).

The as received sample assayed 0.33% WO₃, 894 ppm Li₂O, 91 ppm Mo, 923 ppm Rb, 52 ppm Cs, 110 ppm Bi, 44 ppm Nb, 67.23% SiO₂, 13.47% Al₂O₃, 3.94 Fe(T), 2.96% K₂O and 1.6% LOI.

Detailed beneficiation studies were carried out employing two process routes. The best result was obtained for recovery of tungsten, lithium and rubidium was by gravity separation followed by combination of magnetic and gravity separation.

The beneficiated concentrate assayed 54.29% WO₃ and 1,121 ppm Bi with recovery of 54.7% WO₃ and 2.7% Bi along with weight yield of 0.35%.

Lithium values recovered from gravity product (middling, tabling & slimes) was subjected to flotation. The composite float assayed 2,732 ppm Li_2O with recovery of 65.9% along with weight % yield of 24.6.

8. Limestone

Bench-scale beneficiation studies were conducted on limestone sample received from Tamil Nadu. The sample assayed 37.73% CaO , 21.46% SiO_2 , 2.78% MgO , 4.02% Fe_2O_3 , 3.90% Al_2O_3 and 26.78% LOI .

The objective of the study was to reduce the SiO_2 content in the sample to below 12.0% and increase the CaO content to more than 41.0% to make the ore suitable for cement grade applications, either through simple dry methods of beneficiation or wise by flotation process.

The original sample was ground in Pebble, Ceramic and Autogenous Mill (AG Mill) and then the ground product was dry screened. The best grade and recovery was achieved in case of autogenous mill (where in the lumpy ore sample was used as grinding media). The minus- 200 mesh obtained from autogenous mill assayed 43.98% CaO and 13.28% SiO_2 with recovery of 64.1 % and wt yield 54.9%. The minus- 500 mesh product obtained by screening the minus- 200# product of AG Mill assayed 46.29% CaO and 11.01% SiO_2 with recovery of 37.2 % and wt yield 30.3%.

The study demonstrated that the desired grade of limestone concentrate could be obtained through differential grinding and screening.

9. Graphite

a) Low-grade Graphite Ore: Bench-scale beneficiation studies were conducted on a low grade graphite ore of G2 Stage Exploration Sample received from Oranga–Revatipur Area, Balrampur District, SU: Chhattisgarh for GSI, Raipur.

The proximate analysis of the as received sample assayed 4.90% Fixed Carbon, 3.05% Volatile matter, 0.32% moisture and 91.73% ash. The Ash assayed 4.36% Fe(T) , 63.16% SiO_2 , 9.69% Al_2O_3 , 4.49% CaO , 6.75% MgO , 0.81% TiO_2 , 0.29% Na_2O , 1.15 % K_2O , 8.62% LOI .

The beneficiation techniques adopted to obtain best results involved stage crushing the as received sample to all passing 10 mesh size. The all passing 10 mesh size sample was ground to all passing 150 mesh size in ball mill followed by rougher flotation. The rougher float obtained was subjected to 8 stages of cleaner flotation followed by re-grinding the cleaner float after every second stage of cleaning. The 8th cleaner float produced a graphite concentrate assaying 68% FC with 66.43% recovery (wt% yield 4.79).

Bench-scale beneficiation study of low-grade graphite ore revealed that the sample is amenable to beneficiation and the Fixed Carbon value is upgradable to higher grade with satisfactory recovery.

b) Graphite Ore: Bench-scale beneficiation studies were conducted on a lean-grade graphite ore sample received from Khalspadar Graphite Exploration Project, Rayagada (for Director of Mines and Geology, Odisha)

The proximate analysis of the as received sample assayed 1.6% FC , 3.89% VM , 0.55% moisture and 94.24% ash. The ash analysis of as received sample established 68.34% SiO_2 , 11.39% Al_2O_3 , 6.91% Fe_2O_3 , 2.02% CaO , 1.60% MgO , 0.017% V_2O_5 and 0.19% S(T) , 0.21% Na_2O and 2.0% K_2O .

Beneficiation route employing minus 10 mesh size sample stage ground to all passing 30 mesh followed by flotation yielded a graphite concentrate assaying 52.8 % FC , 6.56% VM , 39.82% Ash , 0.86% moisture with FC recovery of 97.3 % (wt.% yield: 3.5).

c) Graphite Ore: Bench-scale beneficiation studies were conducted out on a graphite ore Sample of G2 Stage Exploration received from Adhmaniya Block, Palamu District, Jharkhand (for Geological Survey of India, Ranchi, Jharkhand).

The proximate analysis of the as received sample assayed 10.78% Fixed Carbon, 0.20% V_2O_5 , 2.62% volatile matter, 0.39% moisture and 86.21% ash. The ash assayed 0.028% Cu , 8.59% Fe(T) , 56.37% SiO_2 , 8.72% Al_2O_3 , 2.40% CaO , 1.72% MgO , 0.54% TiO_2 , 0.073% Zn , 0.032% MnO , 0.23% Na_2O , 2.67 % K_2O , 4.25% S(T) , 0.024% Ni , 14.01% LOI & 59.72% AI

The beneficiation process involved crushing, grinding, screening and various flotation techniques. The results of the process were as follows:

- (i) The 5th cleaner float produced a graphite concentrate that assayed 77.50% FC with 83.89% recovery and 0.23% V_2O_5 with 13.11% recovery and wt% yield of 11.67%.
- (ii) The magnetic separation of the graphite concentrate (5th cleaner float) produced a non-magnetic graphite concentrate assaying 86.17% FC with 77.69% recovery (wt% yield 9.72%).
- (iii) Vanadium value was recovered from tailings of graphite by conducting flotation on combined rejects (RT+1st cleaner tails) which produced float assaying 0.66% V_2O_5 with 55.63% recovery at 17.26 wt% yield.

The composite value of vanadium recovery was 0.49% V_2O_5 with a 68.74 % recovery.

d) **Graphite:** Bench-scale beneficiation studies on two-low grade graphite ore samples of foreign origin were carried out.

- (i) The as received sample assayed 10.4% F.C, 87.53% Ash, 79.41% SiO₂, 9.53% Al₂O₃, 0.86% S(T), 2.46% K₂O, and 0.42% TiO₂.

Beneficiation study employing froth flotation process with regrinding of rougher float, 1st and 3rd cleaner float separately yielded an 8th cleaner float assaying 83.02% F.C with recovery of 62.7% and 7.5% weight yield.

- (ii) The as received sample assayed 9.20% F.C, 89.30% Ash, 64.23% SiO₂, 13.39% Al₂O₃, 0.36% S(T), 3.90% K₂O, and 0.91% TiO₂.

OTHER MAJOR R&D STUDIES

CSIR-IMMT (Institute of Minerals and Materials Technology)

Research & Development activities carried out by CSIR-IMMT are as follows:

Mineral Processing Objectives

To maximize the recovery of mineral values from lean and low-grade resources as well as waste generated from mineral-based industries by developing eco-friendly processes through basic and applied research for accelerating industrial growth in the areas of bulk, strategic and precious minerals.

Core Area Expertise of CSIR-IMMT:

- Process development from bench-scale to pilot-scale study for beneficiation of low- and lean-grade ores and minerals through physical beneficiation and reduction roasting/reduction processes to enrich the mineral values
- Agglomeration of fine concentrate generated through the beneficiation process
- DRI and smelting of oxide ores (iron ore, chromite ore, manganese ore and ilmenite ore)
- Simulation and optimisation of the process of the existing mineral-based industries
- Equipment development for mineral-based industries
- Preparation of TEFR for mineral beneficiation process
- HRD in the area of Mineral Processing

Design and Development of Flotation Column System for the Recovery of Monazite from IREL Mineral Separation Plant

Objectives

- To design and develop a flotation column system for the recovery of monazite from the IREL mineral separation plant.

Highlights

1. Successfully installed and commissioned a pilot-scale (100 mm diameter) flotation column test rig at IREL, Chatrapur, in February 2024 to recover monazite values from the IREL mineral separation plant.
2. Preliminary investigations on the pilot-scale flotation column could enrich the monazite content to more than 90% from a feed containing about 49% monazite.

Pelletisation and DRI Studies of Low-grade Chromite Overburden

Objectives

The aim of the project was to create DR-grade pellets from a low-grade chromite overburden sample and study the reduction behaviour of the developed pellets.

Highlights

The chromite overburden used in the study was rich in iron (52%) and contained a significant amount of nickel (0.8%). High-quality pellets from low-grade ferruginous chromite overburden were produced through various optimisation techniques. The optimal conditions for making the best pellets were a basicity value of 1.0, an induration temperature of 1,300 °C, and an induration time of 10 minutes. These pellets exhibited excellent metallurgical properties and can be used as a burden for subsequent DRI production.

Beneficiation Study of Iron Ore From Hahaladdi Mine, North Bastar Kanker

Objectives

Process flow sheet development to upgrade Fe (T) of low-grade iron ores

Highlights

A developed flow sheet integrates findings that indicate a systematic approach to recover lumpy +3 mm particles with Fe(T) content of 63% or above.

Characterisation Study of Low-grade Iron Ore Fines

Objectives

To study and characterise two iron ore samples supplied by M/s Penguin Trading & Agencies Ltd, Keonjhar, Odisha

Highlights

Detailed chemical analysis of the feed samples revealed that PTA-1 & PTA-2 contained 50.25% and 52.27% (wt%) iron content respectively. Silica & alumina content in PTA-1 & PTA-2 were 8.75%, 10.50% & 7.92%, 8.50% respectively.

From the particle size analysis, it was observed that d50 and d80 of both the samples were around 0.5 mm & 1.4 mm. Size fractions below 0.045 mm contained less iron and higher silica & alumina.

Study the Physical and Metallurgical Characteristics of Iron Ore Pellets and Analysis of its Suitability for Iron Making

Objectives

The objective is to study the physical and metallurgical characteristics of iron ore pellets and analysis of its suitability for iron making.

Highlights

The following R&D works carried out for 12 samples with respect to the pellet quality parameters— Physical Characteristics Sampling and Size analysis of the pellet samples and Porosity Tumbler & Abrasion indices (ISO-3271). Cold crushing strength (ISO-4700). Chemical analysis: Total chemical analysis (Major Fe, SiO₂, Al₂O₃, LOI, Minor and Trace elements) of iron ore pellets Metallurgical testing • Reducibility by the rate of reduction index (dR/dt) /Fe=0.9 (ISO-4695) • Reducibility by the final degree of reduction index R180 (ISO-7215) • Swelling index (ISO-4698) • Low-temperature reduction disintegration indices RDI-1(ISO-4696-1) • Low-temperature reduction disintegration indices RDI-2(ISO-4696-2)

Petrographic Study of Some Indian Coal Sample

Objectives

The aim is to conduct detailed petrographic studies for maceral identification and determination of vitrinite reflectance of various coal samples. Petrographic analysis plays the vital role to maintain process parameter to improve the quality of coal. Petrographic studies of approximately 200 samples in total are to be carried out. This data will enable deciding about the metallurgical properties and rank of the coal for their industrial applications.

Highlights

The petrographic data decides the rank as well as grade of the coal for commercial and industrial uses.

Selective Removal of CO₂ from the Gas produced from Coal/Biomass Using Suitable Media for Gas Enrichment

Objectives

- Preparation of media for selective removal of CO₂ from input gas stream
- Development of indigenous media cartridge for effective use in laboratory scale

Highlights

The scope of the study is CO₂ removal using suitable media and preparation of media cartridge in laboratory scale.

The objectives are formed based on the present need

of the Indian Steel Industry and status of the technology available.

1. Preparation of media for selective removal of CO₂ from input gas stream:

- Identification, characterisation and preparation of the industrial wastes as CO₂ capturing media have been completed
- Kinetic study of CO₂ absorption by the solid media has been completed and optimisation study is in progress
- Kinetic study of CO₂ absorption by the liquid media is in progress

2. Development of indigenous media cartridge for effective use in laboratory scale:

- Composition of the solid media cartridge has been developed using indigenously designed inhouse experimental set-up
- Optimisation of the solid media cartridge is under progress

Characterisation Study on Magnetite Samples of Gorumahisani Iron Ore Mine

Objectives:

The specific objective was to carry out (i) Petrographical analysis of the rock/ore samples for identification of all ore minerals, gangue minerals and their incidence; (ii) Magnetic attractability with respect to pure iron at different Gauss values; and (iii) Complete chemical analysis of major elements viz. FeO, Fe₂O₃, Fe₃O₄, SiO₂, Al₂O₃, P, S, Mn and other gangue minerals including trace elements.

Highlights

The study provided insights into the BMQ samples using Petrographic Study, Magnetic Separation and Chemical Compositional study.

Beneficiation Study and Development of Process Flow Sheet for Upgradation of LD Slag for Utilisation in Cement Making

Objectives

To upgrade the quality of LD slag so that it can be used in cement-making

Highlights

The characterisation studies revealed that LD slag majorly contains iron, silica and lime. From microscopic analysis, it was found that LD slag had less glass content (around 15 vol.%) in comparison to BF slag (around 93 vol.%). Processed LD-Slag obtained had higher glass content (147% increase), basicity factor (2 to 2.5% increase) and yield (>85%). This indicates the increase in reactivity of this processed LD slag.

Characterisation and Analysis of Graphite Samples

Objectives

The specific objective was to carry out:

- 1) Petrographical analysis of the graphite samples
- 2) Physico-chemical and chemical compositional study of graphite samples
- 3) XRD study of graphite samples

Highlights

Determination and evaluation of graphite properties using proximate analysis, chemical compositional analysis, petrographic analysis and XRD analysis.

Chemical Compositional Study of Graphite-bearing Samples

Objectives

Carrying out detailed physico-chemical and chemical compositional study of 3,200 graphite-bearing samples.

Highlights

Determination and evaluation of graphite properties, such as, Ash %, Fixed Carbon %, Volatile Matter/LOI %, Moisture %, SiO_2 % and Al_2O_3 %, etc.

Pilot-scale Induration Studies of Low-grade Chromite Overburden

Objectives

This study aimed to investigate the induration process of low-grade chromite overburden samples on a pilot-scale.

Highlights

The purpose of this study was to investigate the induration process of low-grade chromite overburden samples on a pilot scale. The samples were provided by TATA Steel Ltd, Jamshedpur. At first, lab-scale experiments were conducted to determine the optimal conditions for pelletisation. The optimal conditions were then used in a pilot-scale study of 100 kg. Initially, green pellets were prepared with the optimal balling conditions. These pellets were then tested for induration using a potgrate test at the optimal temperature and time. The pellets possessed the recommended properties required for the subsequent direct reduction process. The information obtained from this study is crucial for ensuring the quality and efficiency of the entire pelletisation process.

Pelletisation Studies of Nickel-rich Chromite Overburden Sample

Objectives

The objective of the project was to produce pellets suitable for DRI (Direct Reduced Iron) production from a low-grade chromite overburden sample with a significant amount of nickel of approximately 0.8%.

Highlights

Various pelletisation factors, such as, basicity, induration temperature and time, were optimised to attain the objective of the project. The pellets that were produced under the optimal conditions of basicity of 0.25, a temperature of 1,250 °C and a time of 10 minutes met the recommended metallurgical standards and were found suitable for the subsequent direct reduction process.

Study on Effectiveness of Sinter Aid on Iron Ore Sintering Process

Objectives

The objective of the study is to improve mixing and nodulising in a sinter making process using sinter aid thereby reducing return fines and improving the sinter characteristics.

Highlights

The objective of M/s Abhitech Energycon Ltd the effectiveness of their sinter aid product for sinter making with the iron ore fines (-10 mm).

Mineralogical Study of Iron ore from Koira Block, Sundargarh, Odisha

Objectives

The primary objective of receiving these iron ore samples at CSIR-IMMT is to conduct extensive laboratory investigations with respect to their mineralogical parameters. These investigations involve physical, chemical, and mineralogical analyses, which provide crucial insights into the characteristics of the ore samples. Some of the common analyses conducted on iron ore samples include moisture content, particle size distribution, chemical and mineralogical composition analyses. By conducting these comprehensive analyses, CSIR-IMMT aim to provide a detailed characterisation of the iron ore samples from Nadidihi and Nadikasira mines. The results of these investigations will assist in assessing the ore's quality, understanding its potential uses, and determining the most appropriate processing and beneficiation techniques for extracting valuable metals from the ore.

Highlights

The laboratory investigations conducted on the iron ore samples from Nadikasira and Rengalbeda mining areas of Koira block provided valuable insights into their mineralogical and chemical characteristics. The ore samples exhibited a high iron grade, with haematite being the predominant mineral. The particle size distribution analysis and mineralogical variation assessment shed light on the physical properties and mineral composition of the ores. The findings from these investigations will contribute to a better understanding of the ore's quality, potential uses and appropriate processing techniques for extracting valuable metals.

Feasibility Studies on Recovering Valuable and Rare Earth Elements From Coal Gasification Ash, CGP

Objectives

Conduct feasibility studies on recovering valuable and rare-earth elements from the coal gasification ash sample of India's only operational commercial-scale coal gasification unit, located at JSPL Angul, Odisha.

Highlights

Significant improvements in the REEs concentration in the product sample were achieved based on gravity, flotation, magnetic separation and other unit operations, as well as acid-leaching experiments.

Studies on the Utilization of Rock Phosphate Flotation Tailings

Objectives

Recovery of the phosphate concentrate from the tailing Rock Phosphate samples of RSMML, Rajasthan, and its holistic utilisation towards the construction material development.

Highlights

The required phosphate concentrations in the tailing and concentrate samples have been successfully achieved during the lab-scale flotation experiments.

CSIR–CECRI (Central Electrochemical Research Institute)

Research & Development activation carried out by CSIR–CECRI are as follows:

1. High-purity Graphite from Low-grade Graphite Ore

CSIR–CECRI has developed a method for the separation of graphite and metals from the secondary source generated by Hindustan Zinc Limited during the electro-refining of lead process. The graphite content was enriched from 33% of fixed carbon (FC) to 95% and the dissolved metals were recovered from the sludge by precipitation obtained from graphite purification. To attain the above, the graphite ore was froth floated with Methyl isobutyl ketone followed by treatment with aqua regia. The aqua regia treated ore was then treated with hydrogen fluoride (HF) to remove the silicates. The silicate-free graphite ore was chlorinated to eliminate the impurities to their respective chlorides. The components (aqua regia, HF, Cl_2 gas) used for graphite separation were cheap, abundant and could be recycled. The metal ions present in the graphite ore were separated as a by-product by precipitation method.

2. Laboratory-scale Hydrometallurgical Process to Recover Lithium from Spent Lithium-Ion Batteries (LIB).

CSIR–CECRI has developed a laboratory-scale hydrometallurgical process to recover lithium

from Spent Lithium-ion Batteries (LIB). This is a hydrometallurgical process based on the precipitation methods to produce Li_2CO_3 . From the recovered lithium carbonate, coin cells were made and the electrochemical performances were evaluated. A specific capacity of 135 mAh/g was achieved, confirming its utility in new LIBs. The recovery of Li from the secondary sources, like spent LIB, can reduce the dependency on the primary reserves as well as the import dependencies.

3. Recovery of Metal Values from Ferrochrome Slag

A process was developed for the recovery of Mg from ferrochrome slags obtained from the ferrochrome industries. In this process, Mg from the slag was recovered in the form of MgCl_2 with a purity of ~ 98%. The slag treated in this method was also free from Cr^{6+} ions, which are detrimental to the environment. From the recovered MgCl_2 , Mg metal could be produced through molten salt electrolysis.

4. Extraction of RE Values from Spent NdFeB Magnets

CSIR–CECRI is developing electrometallurgical routes for the recycling of spent NdFeB magnets and recovery of RE values. Considering the process complexity and economics, Molten Salt Electrowinning process parameters were established with simulated composition that mimics recycled spent magnets and obtained RE products are presented. Electrowinning using actual RE mixtures recovered from spent magnets is under progress during 2024-25.

5. Development of Alternate Flux from Red Mud for Dephosphorisation

Red mud is an industrial waste generated during alumina production in the Bayer process. Laboratory-scale study was conducted to evaluate the dephosphorisation capability of red mud. In this study, along with dephosphorisation, the thermal stability of sodium-containing compounds in red mud, which is critical for its utilisation in the iron and steel industry, was also studied and confirmed. The dephosphorisation experiments conducted at 1,320 °C demonstrated effective phosphorus removal from pig iron, with the value of 31.85% for the hot metal containing 0.42%. This performance surpasses the traditional sodium-based flux used for this purpose. This study proved that red mud can be a viable alternative for the traditional sodium-based fluxes used in Iron & Steel industries.

6. Recovery of MnO_2 from Anode Sludge

Electrolytic grade MnO_2 was obtained from the anode sludge provided by Hindustan Zinc Limited. The anode sludge primarily consisted of a mixture of ZnSO_4 , MnSO_4 and MgSO_4 . XRF of the anode deposit showed 100% MnO_2 with no impurities, XRF of the cathode after deposition showed metallic zinc. The metallic Zn

deposited at the cathode seemed to produce H_2 when dipped in 0.5 M H_2SO_4 . The current efficiency of the electrolytic process was found to be at its maximum value of 57% at a current density of 5.95 mA/cm² using TSIA as the anode and SS as cathode with bath temperature at 80 °C. The electrolytic grade MnO_2 was confirmed using XRD and the cathode deposit at SS was found to be 100% Zinc.

7. Metallic Magnesium from Spent Grignard Reagent Solution

Agro based chemical companies utilise Grignard reagent for some chemical processes, which produces $MgCl_2$ solution as a by-product. $MgCl_2$ was purified from Grignard reagent solution at a separation efficiency of 99% and 99.6% purity. The purified $MgCl_2$ was converted to Mg metal through molten salt electrowinning at a current efficiency of 60–70% and product recovery of 70%.

8. Extraction of Zn and Zinc Oxide from Galvaniser Ash

The zinc present in the galvaniser ash was selectively leached using a suitable lixiviant. From the leach liquor, the zinc was extracted through electrowinning process. The conversion and current efficiency was estimated to be more than 90%. The developed technology was transferred to Deep constructions, Gujarat (MSME).

9. R&D in Building Materials: Minerals and Mineral-based Products in Construction Activities, Substitution, etc.

Synthetic coarse aggregates were successfully manufactured using industrial by-products, such as, fly ash and copper slag. The process involved pelletising these materials with cementitious binders under controlled conditions to produce high-strength, durable aggregates suitable for use in structural concrete. The manufacturing process was optimised by controlling critical parameters, such as, particle size, pelletiser drum inclination, rotation speed, moisture content and curing conditions, all of which contribute to enhanced mechanical performance and long-term durability of the aggregates. The developed synthetic aggregates met the essential quality criteria for strength, density and water absorption, while also offering a sustainable solution by reducing environmental hazards related to the disposal of industrial by-products. The process is scalable and aligns with circular economy principles by transforming potential waste into a valuable resource for the Construction Industry.

10. Synthesis of Silicon Carbide for Defense Applications

A carbothermal reduction process was developed to produce silicon carbide using environmentally friendly precursors like Resorcinol-formalin and silica derived from rice husk ash. The SiC produced through this process showed a crystallite size of 15.4 nm with a band gap of 2.36 eV. The developed process can be a scalable and sustainable solution for the production of armor and refractory-grade silicon carbide.

11. Production of Magnesium from Sea Bitterns

The sea bitterns, which are a by-product obtained in the salt industries, have magnesium concentrations of approximately 13.5%. A process to extract Mg in the form of $MgCl_2$ from the sea bittern was developed. From the $MgCl_2$, magnesium metal was produced using molten salt electrolysis. The developed process was demonstrated on a scale of 1 kg per day. The commissioning of a pilot-scale demonstration plant with the capacity to produce 10 kg/day of magnesium was in progress.

12. Indigenous Molten Salt Electrowinning Processes for Rare Earths at CSIR–CECRI

Rare Earths find applications in permanent magnets, electrical, electronics, metallurgy additives, hydrogen storage, batteries and most significantly, aerospace and defence sectors. At present, there is no commercial production of RE metals/alloys in the country due to non-availability of indigenous process for RE metal extraction. Rare Earths are extremely reactive forming stable compounds which makes them difficult to be produced in high-purity metallic form. Molten Salt Electrowinning process is a commercially viable strategy for the production of RE metals. CSIR–CECRI is developing Molten Salt Electrowinning protocols for RE metals (La, Ce, Nd), their alloys (NdPr, NdFe, NdPrFe and hexaborides, REB6 (La, Sm, Ce). Process optimisation is underway to further enhance the purity of niche RE products. CSIR–CECRI is now augmenting its capabilities in Molten Salt Electrowinning of Rare Earths by modernisation and scaling-up suitably for commercial translation to interested stakeholders.

Jawaharlal Nehru Aluminium Research Development and Design Centre (JNARDDC)

Patents Filed and Granted

The following patent applications were granted under the Patents Act, 1970 for various indigenous R&D process developed by JNARDDC under various research projects.

S.N.	Details of the Granted Patent
1.	A process for preparation of aluminium hydroxide with low soda content. Granted vide no. 455707 on 29.09.2023 Dr Suchita Rai , M. T. Nimje, A Agnihotri
2.	Development of process for revitalisation of high alumina castables from residual Aluminium dross. No. 471016 on 21.11.2023 Dr Upendra Singh , M T Nimje, Shoeb Ansari, Anupam Agnihotri, (JNARDDC) & A S P Mishra (VEDANTA)
3.	A process for removal of iron oxide, silica and enrichment of alumina in ferruginous, siliceous and aluminous laterite by physical separation process No. 483133 on 15.12.2023 Dr Pravin Bhukte , S P Puttewar, Anupam Agnihotri
4.	A process for conversion of saprolite into refractory aggregate. No. 507474 on 06.02.2024. Dr P G Bhukte , S P Puttewar, M T Nimje, M J Chaddha, M Najar, U Singh and A Agnihotri

Project Undertaken and completed by JNARDDC during the Year 2022-24 are as below:

Instrument for Real-time Measurement of Anode Current Distribution of Aluminium Electrolysis Cell: DST, New Delhi

Objective

To develop a Wi-Fi – enabled real-time anode current distribution measurement system for aluminium electrolysis cell.

Outcome

- Developed a system which automates the measurement of anode current distribution, eliminating the need for manual measurements using probes. This automation leads to significant labour savings, improved precision and the ability to perform real-time monitoring.
- As it allows continuous, real-time monitoring of current flow in individual anode stems, operators can observe current variations in real-time & make adjustments as needed, enhancing process control and stability in aluminium production.
- Data is wirelessly transmitted from slave units to the master unit and subsequently to the control room via Wi-Fi. It minimises wiring complexity and enables remote monitoring, reducing manual efforts and enhancing operational efficiency.
- The use of TEG for energy harvesting allows continuous recharging of the Ni-MH batteries in the slave units, eliminating the need for manual charging. This results in continuous system operation without downtime for battery replacement or charging there by reducing maintenance efforts and improving system uptime. TEG modules harvest energy from the temperature difference

between the aluminium pot and the surrounding environment. This technique makes the system more energy-efficient and cost-effective. The system provides high-precision data and allows for accurate monitoring of voltage drop and temperature across anodes.

- No electromagnetic interference issues faced during continuous operation in the aluminium smelter plant. This ensures the system's reliability and consistency in transmitting accurate data even in challenging industrial environments. On the whole considerable improvement in the overall efficiency of the smelting process was recorded.
- System was tested and validated in NALCO's aluminium smelter, demonstrating its accuracy & reliability in measuring anode current distribution. The validation of the system ensures that it meets industry standards for accuracy & reliability for its use in smelter.

Geo-technological Evaluation of Bauxite and Laterite Deposits of Chhattisgarh State by using Geospatial Technology under Smart Mining 4.0 (With Chhattisgarh Council of Science & Technology, Government of Chhattisgarh, S&T (Mines))

Objective

- Creation of district-wise digital database of laterite and bauxite deposits for Chhattisgarh State using geo-referenced cadastral maps and high-resolution satellite imageries
- Geotechnological evaluation of bauxite and laterite deposits of Chhattisgarh State under Smart Mining 4.0 programme.

Outcome

- Geo-technological database in respect of (geology, chemical, mineralogy, REE, petrology, morphology, Bond work index, bulk density, moisture content, available alumina, reactive silica, digestion efficiency) were generated. Geoinformatics database, such as, cadastral boundaries, geomorphology, pre-field investigation perception, identification of prospective deposits/ area, demarcation of rock/ore bodies, etc. that which could be of use to entrepreneurs participating in mining auction processes have been generated. The database provides a deep insight about the various deposits the information data that could be of immense use to the entrepreneurs of metallurgical and non-metallurgical applications and State Mining Department for auction-related purposes.
- Generation of exhaustive and comprehensive digital database of laterite and bauxite deposits in geospatial domain. Multiple unique outputs like GeoPDF, KML, and GIS projects have been generated.

- Entrepreneurs could identify deposits suitable for metallurgical and non-metallurgical applications.
- The developed geotechnological database on bauxite deposits are expected to be vital for Bauxite miners, Alumina Industry and State Mining Department to facilitate the auction process.

Solid-state Recycling of Aluminium Chips (waste) for Production of Billets for Pilot-scale Extrusion: S&T (Mines)

Objective

Utilisation of aluminium swarf/ chips of AA2024 for the production of aluminium billets for extrusion, its validation at pilot scale using extrusion press, testing and characterisation to understand the feasibility of the process.

Outcome

Developed a novel pilot-scale solid-state process, which is environmental friendly and low melt loss process, for the recycling of AA2024 aluminium swarf/chips to replace conventional re-melting. The following are the highlights of the findings achieved in this project:

- Among the cleaning solvents, like DM water, ethanol and acetone that were tried to remove dirt, coolant and oil residues adhered to the AA2024 swarf, acetone was found to be the best cleaning solvent for the same time duration.
- The pilot-scale compacted billets produced using 1400 Ton extrusion press attained an average density of more than 85% and these compacts were successfully extruded with an extrusion ratio of 62 into 19 mm round bars using the same extrusion press.
- The peak load observed in solid state recycled billet extrusion was 26% lesser than that of commercial billet extrusion.
- The strength of solid-state recycled billet extrudate was 8% higher than that of commercial billet extrudate.
- The cost-benefit and equivalent CO₂ emission analysis disclosed approximately 35% and 92% savings when the billet is produced via solid-state recycling as compared to commercial AA2024 billets.
- The C-Channel profile extrudate produced from semi-solid cast homogenised billet (with swarf to base metal scrap ratio of 1:1) was found to have mechanical properties similar to that of commercial billet extrudate.
- The peak load of extrusion did not change with the production of billet by semi-solid casting.
- The knowhow will be shared with the relevant manufacturing companies for exploring the utility of this process.

Techno-Economic Survey of Copper / Lead / Zinc Scrap Recycling in India; S&T (Mines)

Objective

- To establish a techno-economic scenario of copper, lead and zinc scrap recycling industries in India
- To support and promote sustainable recycling practices and circular economy for these metals.
- To provide insights for policy formulation by the Ministry of Mines, Government of India, aiding in the development of competitive and environmentally sound recycling practices.

Common Outcomes

The combined insights from the surveys on copper, lead and zinc recycling highlighted several key outcomes:

1. Industry Overview and Growth

Copper: The Indian Copper Industry, valued at \$8 billion, supports 50,000 jobs and is expected to see a significant increase in consumption by 2030, driven by Sectors like Electric Vehicles and Infrastructure.

Lead: The recycling activity is widespread, with numerous recyclers across the nation handling lead-acid batteries as the primary scrap source.

Zinc: Despite the low domestic recycling rate, the industry showed potential for growth, especially with increased focus on recovering zinc from galvanising residues.

2. Dependency on Imports

All three metal recycling industries rely significantly on imported scrap, adhering to ISRI specifications. For copper and lead, domestic recycling is complemented by imports, though capped by regulations. Zinc recycling also depends on imported materials due to limited availability of domestic scrap.

3. Technology Adoption and Challenges

Large and medium-sized units have adopted advanced sorting and beneficiation technologies, such as, eddy current separators, magnetic separators and briquetting machines to improve scrap quality. However, smaller units lag in technology penetration, leading to inefficiencies and environmental concerns.

Direct melting is common in the Copper Recycling sector, but the lack of proper smelting and refining processes results in quality issues. For zinc, the recovery of zinc from Steel Industry dust remains a challenge, with a need for wider adoption of processes like the Waelz process.

4. Environmental Concerns

The surveys underline the environmental challenges posed by improper handling of non-metal components in scrap, particularly, in lead and zinc recycling. Eco-friendly practices are limited to a few large players, while

smaller units often lack the necessary infrastructure for safe disposal.

5. Government Initiatives and Policy Support:

The Ministry of Mines is actively working on initiatives like the National Non-Ferrous Metal Scrap Recycling Policy and Circular Economy frameworks. JNARDDC has contributed significantly by providing various standards and recommendations, including:

- **Process Standards for Recycling/Remelting:** Guidelines for scientifically improved recycling practices for copper, lead and zinc, focusing on energy-efficient and environmentally friendly methods.
- **Minimum Infrastructure Requirements:** Specifications for the necessary infrastructure that recycling facilities must have, especially for ensuring safety efficiency, and environmental compliance.
- **Minimum Recycled Content Requirements:** Defined thresholds for the percentage of recycled content in products made from copper, lead and zinc to promote the use of secondary materials.
- **Rating of Recycling Industries:** A mechanism for rating recycling units based on their adherence to best practices, technological adoption and environmental performance.
- **Green Rating of Products:** A system for rating products made from recycled materials, helping consumers make environmentally conscious choices.

JNARDDC is also developing a dynamic web portal to monitor the Recycling Industry, facilitating real-time data collection and enabling a ranking mechanism to transform the Sector into a more organised and structured system.

Future Opportunities and Policy Contributions

The outcomes of the survey have already contributed to the preparation of key policy documents for the Ministry of Mines, including:

- **Policy Measure:** Guidelines for Processing of End-of-Life (EoL) Copper / Lead / Zinc Scrap.
- **Zero Waste Management Policy:** For sustainable waste utilisation in the zinc secondary, downstream, galvanising and chemical industries.
- **Extended Producer Responsibility (EPR) Provisions:** For the effective utilisation of non-ferrous scraps and wastes.

- **Standard Operating Procedures (SOPs) for Air Pollution Norms:** Specific to zinc recovery from Electric Arc Furnace (EAF) or Induction Furnace (IF) dust in the Steel Industry.

CONCLUSION

The combined findings from the techno-economic surveys of copper, lead and zinc recycling underscore the need for modernisation, regulatory support and infrastructure development to unlock the full potential of these industries. By embracing advanced technologies, adopting best practices, and implementing the recommended standards and policies, India can not only meet its growing demand for these critical metals but also contribute to global sustainability goals through efficient resource management. The findings will play a pivotal role in the formulation of policies pertaining to the Non-ferrous Recycling Industry, aligning with the Ministry's vision for sustainable and competitive growth.

Demonstration-cum-heat Treatment, Leaching-recycling and Liming Study of JNARDDC-NALCO process (by utilising 50–60 kg batch of 1st cut SPL), NALCO Bhubaneswar

Objectives

JNARDDC had developed a simple and cost-effective process on bench scale (1 kg capacity) for detoxification of 1st cut SPL. Based on these successful results, it was decided to undertake the batch scale study for 50–60 kg as a scale up initiative.

Outcome

The results/observations of batch-scale studies were found interesting. More than 98% of leachable cyanide gets destructured during heat treatment. The concentration of caustic product $[\text{Na}_2\text{O}(\text{t})]$ reached up to 61.4 gpl by repeated recycling. The carbon product obtained was free from leachable cyanide, fluoride and sodium and can be used in (i) a captive power plant as fuel (by mixing a certain proportion with present raw material/coal), (ii) In anode manufacturing (carbon is mostly in the form of graphite and may increase electrical conductivity), (iii) Ramming paste preparation or (iv) Cement Industry (as fuel and mineraliser). The calcium fluoride product obtained was 65% pure with the remaining can be used as a flux in the Steel Industry.



6. Mineral-based Industries

139.153 million tonnes
finished steel were produced during the
year 2023-24.

4,159 thousand tonnes
aluminium were produced in India
during 2023-24.

3.78 million tonnes
copper were produced during the year
2023-24.

Minerals are vital raw materials for many basic industries and are major components for growth and industrial development. The management of mineral resources, hence, has to be closely integrated with the overall strategy for development and exploitation of minerals, with an aim towards long-term national goals. In tune with the Economic Liberalisation Policy adopted in July 1991, the National Mineral Policy announced in March 1993 has opened the Mineral Sector for private entrepreneurs, both domestic and foreign. The changing global scenario necessitated revision in the National Mineral Policy which led to notifying of, National Mineral Policy 2019, with an objective to have more effective, meaningful and implementable policy for more transparency, better regulation and enforcement, balanced social and economic growth as well as sustainable mining practices.

The National Mineral Exploration Policy (NMEP) approved by Government of India in June, 2016, aims at accelerating the exploration activity in the country through enhanced participation of the Private Sector and these

policy initiatives are expected to bring about a turnaround to the entire Mineral Sector across the country. Capacity and production of important mineral-based products are detailed in Table-1.

Table – 1: Capacity and Production of Important Mineral-based Products, 2022-23 to 2023-24

Mineral-based product	Unit	Annual Installed Capacity	Production	
			2022-23	2023-24
Ferrous Metals				
Sponge iron#	million tonnes	60.52	43.62	51.56
Crude/liquid steel		179.51	127.19	144.29
Ferro-Alloys#				
Ferrochrome/Chargechrome	'000 tonnes	1690*	1126	1332
Ferromanganese	tonnes	3160*	NA	NA
Silicomanganese	'000 tonnes	-	356	354
Ferrosilicon	'000 tonnes	250	NA	NA
Magnesium Ferro-Silicon	'000 tonnes	-	15	17
Ferromolybdenum	tonnes	-	581	766
Ferrotitanium	tonnes	-	346	387
Ferrovandium	tonnes	-	839	856
Ferroaluminium	tonnes	-	965	752
Non-ferrous Metals				
Aluminium	million tonnes	4.126	4.06	4.15
Copper (Cathode)	'000 tonnes	785	554	509
Lead (Primary) *	'000 tonnes	210	210	215
Zinc Ingote*	tonnes	951	820	817
Silver*###	tonnes	966	713	745
Cement^	million tonnes	600	391.4	426.29
Fertilizers\$				
Complex Fertilizer	lakh tonnes	116.33	92.95	95.48
SSP	lakh tonnes	120.85	56.44	44.44
DAP	lakh tonnes	53.88	43.47	42.93

Source:

*: MSMP, March, 2023 published by IBM

#: Data on Ferro Alloys taken from MSMP, March, 2023 published by IBM

\$: Annual Report 2021-22 and 2022-23, Department of Fertilizer & Fertilizer statistics (upto Dec., 2022)- from Annual Report of DPIIT, 2023 Installed capacity is as on 01.11.2023

##: Includes production reported by Hindalco Industries.

FERROUS METALS

India is poised for brownfield expansion of existing steel plants, backward integration of re-rollers, and forward integration of DRI or pig iron producers and unfolding of a few Greenfield projects. The National Steel Policy, 2017 projected a target of 300 million tonnes of domestic steel production by 2030. The total production of finished steel during 2023-24 stood at 139.153 million tonnes of which the contribution of SAIL, RINL, TSL, AM/NS, and JSWL & JSPL accounted for a major integrated share.

In view of the long-term demand projection for steel, the Government adopted a two-pronged strategy for increasing steel production in the country through modernisation and expansion of existing Public Sector steel plants in the country and encouraging creation of new steel capacities in Private Sector.

Pig Iron

Pig iron is one of the basic raw materials required by the Foundry & Casting Industry for manufacturing various types of castings for the engineering section. The main sources of pig iron have traditionally been the integrated steel plants of SAIL, besides, plants of Tata Steel Ltd and Rashtriya Ispat Nigam Ltd. Domestic production of pig iron lags behind and is not in tandem with the demand. Efforts were, therefore, made to increase pig iron manufacturing facilities in the Secondary Sector. In 2023-24, about 7.36 million tonnes of pig iron was produced in the country as compared to 5.86 million tonnes in the previous year. Plant-wise capacity details as available in respect of major pig iron units are furnished in Table-2.

Table –2: Location and Capacity of Principal Pig Iron Units

(In thousand tonnes)

Sl.No.	Plants/unit	Location	Capacity
1	Adhunik Metaliks Ltd, Odisha	Odisha	70
2	Ankit Metal and Power Ltd	West Bengal	12
3	Aparant Iron and Steel Pvt. Ltd	Goa	125
4	Arcelor Mittal Nippon Steel India Ltd, Surat	Gujarat	3490
5	Arjas Steel Pvt Ltd (Gerdau Steel)	Andhra Pradesh	300
6	Atibir Industries Co. Ltd (Unit ii)	Jharkhand	600
7	B R G Iron and Steel Co. Pvt. Ltd	Odisha	120
8	Balmukund Sponge and Iron Pvt. Ltd	Jharkhand	40
9	Bhushan Power and Steel Ltd, Odisha	Odisha	2500
10	Electro Steels Ltd, Jharkhand	Jharkhand	1450
11	Electrosteel Castings Limited, Khardah	West Bengal	250
12	Electrotherm (India) Ltd	Gujarat	277
13	Ispat Damodar Ltd	West Bengal	15
14	JSW Steel Ltd, Salem (Siscol)	Tamil Nadu	1000
15	JSW Steel Ltd, Vijaynagar	Karnataka	12000
16	Jai Balaji Industries Ltd West Bengal Unit-3	West Bengal	429
17	Jai Balaji Industries Ltd West Bengal Unit-4	West Bengal	81
18	Jai Balaji Industries Ltd - I	West Bengal	30
19	Jayaswals Neco Inds Ltd	Chhattisgarh	650
20	Jindal Steel and Power Ltd, Chhattisgarh	Chhattisgarh	2125
21	Jindal Steel and Power Ltd, Odisha	Odisha	3200
22	Jsw Ispat Special Products Ltd, Raigarh	Chhattisgarh	613
23	JSW Steel Ltd, Dolvi	Maharashtra	3500
24	K I C Metaliks Ltd	West Bengal	165
25	Kalyani Steels Ltd	Karnataka	480
26	Kirloskar Ferrous Inds Ltd	Karnataka	385
27	Kohinoor Steels Ltd	Jharkhand	48
28	Makers Casting India Pvt Ltd	Jharkhand	2
29	Mideast Integrated Steels Ltd	Odisha	460
30	Narsingh Ispat Ltd	Jharkhand	83
31	Neelachal Ispat Nigam Ltd	Odisha	1099
32	Neo Metaliks Ltd	West Bengal	188
33	Niranjan HiTech Ltd	Jharkhand	15
34	Rashmi Metaliks Limited	West Bengal	170
35	SAIL–Bhilai Steel Plant	Chhattisgarh	3925
36	SAIL–Bokaro Steel Plant	Jharkhand	4360
37	SAIL– Durgapur Steel Plant	West Bengal	1802
38	SAIL– IISCO Steel Plant	West Bengal	2500
39	SAIL–Rourkela Steel Plant	Odisha	4400
40	SAIL–Visveswaraya Iron And Steel Ltd, Bhadravati	Karnataka	118
41	Sathavahana Ispat Ltd	Andhra Pradesh	240
42	Satyam Ferro Tech Ltd	Jharkhand	30
43	Shyam SEL and Power Ltd, Jamuria	West Bengal	60
44	SLR Metaliks Ltd	Karnataka	240
45	Sree Metaliks Ltd	Odisha	36
46	Sri Kalahasthi Pipes Limited (Lanco)	Andhra Pradesh	300
47	Suraj Products Ltd	Odisha	24
48	Swati Concast and Power Pvt Ltd	Jharkhand	43

Table- 2 (Concl'd.)

(In thousand tonnes)

Sl.No.	Plants/unit	Location	Capacity
49	Tata Metaliks Ltd, West Bengal	West Bengal	600
50	Tata Steel BSL Ltd, Odisha	Odisha	3919
51	Tata Steel Long Products Limited	Jharkhand	650
52	Tata Steel Ltd, Jamshedpur Works	Jharkhand	9600
53	Tata Steel Ltd, Kalinganagar Works	Odisha	3000
54	Uttam Galva Metalics Ltd	Maharashtra	600
55	Vedanta Limited	Goa	625
56	VISA Steel Ltd	Odisha	225
57	VIZAG Steel Plant	Andhra Pradesh	6300

Sponge Iron

Commercial production of sponge iron in India commenced in 1980. Sponge Iron India Ltd was first to set-up a plant in 1980 at Palwancha, district Khammam in Telangana.

In the last few years, combined use of hot metal and sponge iron in electric arc furnace have been in practice

for production of liquid steel. Consequently, production of sponge iron increased, substantially. The installed capacity of sponge iron in 2023-24 was 60.52 million tonnes and the production was 34.70 million tonnes as compared to 30.51 million tonnes in the previous year. Plant-wise details as available in respect of major sponge iron units are furnished in Table-3.

Table – 3: Capacities of Principal Sponge Iron (DRI) Plants

(In lakh tonnes)

Unit	Location	Capacity
Gas-based		
Arcelormittal NIPPON Steel India Ltd	Hazira, Gujarat	55
JSW Steel Ltd (Salav) (formerly Welspun Maxsteel Ltd)	Salav, Raigad, Maharashtra	9
JSW Steel (formerly Ispat Industries Ltd)	Geetapuram, Dolvi, Raigad, Maharashtra	16
Coal-based		
Aarti steel Ltd	Ludhiana, Punjab	3.3
Godawari Power & Ispat Ltd	IGC Siltara, Raipur, Chhattisgarh	5.94
Electro Therm (India Ltd)	Bhachau, Kacch, Gujarat	1.8
Jaiswal NECO Industries Ltd	Siltara, Raipur, Chhattisgarh	2.7
Monnet Ispat & Energy Ltd	Naharpalli, Raigarh, Chhattisgarh	8
Lloyd Metals & Engg.	Ghugus, Chandrapur, Maharashtra	2.7
Rungta Mines Ltd	Karakola and Kamando, Sundergarh, Odisha	3.3
NMDC Ltd	Khamman, Odisha	0.6
Tata Steel Long product Ltd	Joda, Kendujhar, Odisha	3.9
Visa Steels Ltd	KIC, Jajpur Road, Odisha	3
Sunflag Iron & Steel Co. Ltd	Bhandara, Maharashtra	1.5

IGC: Industrial Growth Centre.

Source: Sponge Iron Manufacturers' Association (SIMA) and individual plants.

Finished Steel/Saleable Steel

Some significant facts on Indian Steel Industry are as follows:

1. The National Steel Policy (NSP) was announced in 2017. The New Steel Policy, 2017 aspires to achieve 300 million tonnes of steel making capacity by 2030.

2. The total exports of finished steel increased to 7.48 million tonnes in 2023-24 from 6.71 million tonnes in 2022-23 and the imports increased to 8.32 million tonnes in 2023-24 from 6.02 million tonnes in 2022-23.

Details about capacity and production of crude liquid steel & hot metal by main producers are furnished in Table-4.

Table – 4: Capacity and Production of Hot Metal and Crude/Liquid Steel, 2022-23 and 2023-24

(By Principal Producers)

(In '000 tonnes)

Unit	Annual installed capacity Crude/Liquid steel	Production			
		Hot metal		Crude/Liquid steel	
		2022-23	2023-24	2022-23	2023-24 (P)
Public Sector					
SAIL	20632	19409	20496	18292	19241
Rashtriya Ispat Nigam Ltd	7300	4407	4700	4137	4411
NMDC	3000	-	-	-	540
Private Sector					
JSW Group	28080	22476	23549	23623	25608
TSL Group	21500	19835	21434	19805	20783
Am/NS	9600	3375	3573	6688	7683
Jindal Steel & Power Ltd	9600	6165	6116	7509	7645
Others	82803	5496	7177	47143	58388

Source: JPC Annual Statistics, 2023-24.

Steel Companies in Public Sector

Steel Authority of India Ltd (SAIL)

SAIL is a Public Sector Company that operates five integrated steel plants at Bhilai in Chhattisgarh, Bokaro in Jharkhand, Durgapur & Burnpur in West Bengal and Rourkela in Odisha. SAIL has three special and alloy steel plants viz Alloy Steel Plant at Durgapur (West Bengal), Salem Steel Plant at Salem (Tamil Nadu) Visvesvaraya Iron & Steel Plant at Bhadravati (Karnataka). Crude steel production from SAIL plants during the year 2023-24 was 16.25 million tonnes as against 15.28 million tonnes during the year 2022-23.

National Mineral Development Corporation (NMDC)

NMDC has demerged Steel Plant at Nagarnar, Bastar District in Chhattisgarh. NSL plant has been commissioned and started production during the year 2023-24.

NMDC has taken up an ambitious project of laying slurry pipeline from Bailadila to Jagdalpur and further up to Visakhapatnam. The capacity of the line would be 15 MTPA and this will be associated with facilities like beneficiation plant at Kirandul & Bacheli and the pellet plant at Nagarnar and Vizag.

NMDC is strategic management plan, 'Vision 2025' envisions is to take the Company's total iron ore production capacity from 43 MTPA to 67 MTPA, with a judicious mix of Greenfield and Brownfield expansion of earmarked assets.

Major Steel Companies in Private Sector

Tata Steel Ltd (formerly, TISCO)

The Company has been rechristened as Tata Steel Ltd (TSL). The Company has an integrated steel plant located

at Jamshedpur, Jharkhand, with annual crude steel making capacity of 21.50 million tonnes and variety of finishing mills. TSL has produced 20.18 million tonnes of crude steel in 2023-24 as compared to 19.80 million tonnes in 2022-23.

The production of total finished steel in 2023-24 was 21.15 million tonnes as against 19.45 million tonnes in 2022-23. TSL has set target of 35-40 mtpa crude steel capacity by 2030.

JSW Steel Ltd

JSW Steel Ltd's combined installed capacity of crude steel from its plants at Karnataka, Tamil Nadu and Maharashtra was 28.08 million tpy with value-added products spread across six locations- Toranagallu (Vijayanagar Works), Salem (Salem Works), Vasind, Tarapur (downstream units), Dolvi and Kalmeshwar (Maharashtra). Vijayanagar Works by its existing operations produces flat and long steel products; Salem Works has its focus only in long products while the downstream units produce CR/galvanised, colour-coated, value-added flat products.

JSW Steel has initiated a strategic capex plan of ₹ 64,434 crore spread across three financial years towards augmenting the installed capacity to 42 mtpa. The significant investments are allocated for expanding the dolvi phase-III project, to enhance the plant's capacity by an additional 5 MTPA.

The 5 MTPA integrated steel plant being set-up by JSW metallic Ltd is progression on schedule. Salem Works is India's largest special Steel plant with 1MTPA capacity and produces about 850 special grades of steel.

Jindal Steel & Power Ltd (JSPL)

JSPL has manufacturing facilities for steel products in three locations; Raigarh in Chhattisgarh, Angul in Odisha and

Patratu in Jharkhand. Its combined working capacity was 9.60 MTPA. It has also set up a rail and universal beam plant. The sponge iron plant at Raigarh, Chhattisgarh, has capacity of 1.37 million tpy. Facilities at Raigarh also include a Sinter plant of 2.84 MTPA and Steel melt shop of 3.25 MTPA.

JSPL is in advance stage of completing Angul Greenfield project with installation of 12.3 MTPA integrated steel plant in Odisha.

BF/BOF & EAF/IF Industry

Performance of the BF/BOF & EAF/IF Industry is summarised below:

Basic Oxygen Furnace (BOF)

Presently, there are around 20 Basic Oxygen Furnace units with an installed capacity of 71.19 million tonnes. The production of steel through this route was around 61.60 million tonnes (43%).

Electric Arc Furnace (including corex & MBF/EOF)

Steel produced in the Electric Arc Furnace (including corex & MBF/EOF) is mostly by recycling of steel scrap using Electric Arc Furnace (EAF). Presently, there are 37 EAF based steel plants that are operational in the country with an aggregate working capacity of around 39.52 million tonnes per annum. The production of steel through EAF route and IF route was around 36.11 million tonnes (22%) and 51.08 million tonnes (35%), respectively.

The recent developments in EAF technology, viz, to increase oxygen consumption, to reduce power consumption and to reduce tap time have led to increase in metal production. The development of thin slab casting

has made EAF route more productive. This route enables slab strips rolling at lesser cost, facilitating production of cheaper strips/sheets than those that can be achieved through BF/BOF route.

Induction Furnace (IF)

Presently, in India, EAF based industries are yet to switch over to induction furnace route. An induction furnace is an electrical furnace in which heat is generated through electro-magnetic induction in an electrically conductive medium. Induction furnaces use steel melting scraps, sponge iron and pig iron/cast iron. On an average, the proportion of these items is 40% sponge iron + 10% cast iron or pig iron and the remaining is steel melting scraps. There are around 1,126 ground induction furnaces with an aggregate working capacity of about 39.53 million tonnes.

FERRO ALLOYS

The Indian Ferroalloy Industry was established during the second Five-year plan as an ancillary Industry to cater to the growing needs of the domestic Steel Industry. As a deoxidant and alloying agent, ferroalloys are in demand for crude steel and alloy steel production.

Bulk ferroalloys of high-carbon category are produced by large-scale industries. The Noble ferroalloys are of low-carbon category and include ferro-vanadium, ferrotungsten, ferroniobium, ferro-molybdenum and ferrotitanium. There are also a number of units under the Small-scale Sector for the manufacture of ferroalloys, particularly ferrosilicon, ferrochrome and ferromanganese.

India is the net exporter of ferroalloys. About 25% to 30% production is usually exported. India is an established regular exporter of high-carbon ferro manganese, silicomanganese and highcarbon ferrochrome (Table-5).

Table – 5: Capacity of Ferroalloys Industry

(In tonnes per annum)	
Ferroalloys	Installed capacity
Total	5150000
Bulk Ferroalloys	5100000
Manganese alloys	3160000
Chrome alloys	1690000
Ferro silicon	250000
Noble Ferroalloys	50000^(e)

Source: Indian Ferro-alloys Producers' Association (IFAPA), Mumbai.

Bulk Ferroalloys

Ferromanganese and Silicomanganese

The country's total installed capacity for ferromanganese is around 42 lakh tonnes. The total consumption of bulk ferroalloys accounts for over 98% of all ferroalloys. Also about 80% of total ferroalloys production is used in steel making, while the balance is used for the manufacture of casting, superalloys, aerospace and other special applications.

The total production of silicomanganese was 3,54,938 tonnes in the year 2023-24, as compared 3,56,095 tonnes in the previous year.

Ferrochrome and Charge chrome

The total combined capacity of ferrochrome and charge chrome is around 16 lakh tpy. Stainless and Alloy-steel Industry are the chief consumers of ferro-chrome.

The total production of ferrochrome in 2023-24 was about 13,22,000 tonnes.

Noble Ferroalloys

Noble Ferroalloys are one of the vital inputs required for producing special types of steel & alloy. The total capacity of noble ferroalloys was 50,000 tpy and they majorly include ferromolybdenum, ferro vanadium, ferrotungsten, ferrotitanium, ferro silico-magnesium, ferroaluminium,

ferroboron, etc. Mishra Dhatu Nigam (A Govt. of India Undertaking), with a capacity of 2,729 tpy produced different types of super alloy, chiefly, cobalt, molybdenum, titanium and tungsten-based super alloys and products. The production of various noble ferroalloys is furnished in Table-6.

Table – 6: Production of Noble Ferroalloys, 2023-24

		(In tonnes)
Ferroalloys	Quantity	
Ferromolybdenum	766	
Ferrotitanium	387	
Ferrovandium	856	
Ferroaluminium	752	

Electrolytic Manganese Dioxide (EMD)

EMD is consumed along with natural manganese dioxide for the manufacture of dry battery cells. There are two units, one owned by MOIL in Bhandara district of Maharashtra, having a capacity of 1,000 tpy and the other by the then Union Carbide Ltd (now Eveready Ltd) at Thane, Maharashtra, having a capacity of 2,500 tpy. The capacity of the plant was increased from 1,000 tpy to 1,500 tpy during 2018-19. The production of EMD by MOIL was 1,413 tonnes in 2023-24 as against 1,100 tonnes in 2022-23.

NON-FERROUS METALS

Aluminium

There were four companies with a total installed capacity of 4.1 million tonnes in operation. NALCO, the only Public Sector Company in aluminium & alumina segment, has an installed capacity of 0.46 million tpy at Angul. BALCO has an installed capacity of 0.57 million tpy at Korba. The three

companies with six plants in the Private Sector have a total capacity of 3.6 million tpy in operation. One unit at Korba of BALCO at Korba and a plant of MALCO have suspended operations.

The production of aluminium in 2023-24 was 4.16 million tonnes as compared to 4.07 million tonnes in the previous year. The installed capacity and production of aluminium in 2023-24 and 2022-23 are enumerated in Table-7.

Alumina

The production of alumina was 7.41 million tonnes in 2023-24 as compared to 7.26 million tonnes in the previous year. The details of alumina producers in the country, their capacities and production are provided in Table-8.

Hindalco's Renukoot Integrated Smelter uses alumina produced in their plant for producing aluminium.

Table – 7: Capacity and Production of Aluminium, 2022-23 and 2023-24

				(In million tonnes)
Producer	Annual capacity	Production		
		2022-23	2023-24 (P)	
Total	4.12	4.06	4.145	
Public Sector				
National Aluminium Co. Ltd (Angul) (NALCO)	0.46	0.46	0.46	
Private Sector				
Bharat Aluminium Co. Ltd (Korba) (BALCO)	0.57	0.56	0.59	
Hindalco Industries Ltd	1.35	1.32	1.345	
Vedanta Aluminium Ltd (Jharsuguda)	1.74	1.72	1.75	

Figures rounded off.

Source: Information received from individual plants/Annual reports.

Note: MALCO Plant is closed since December 2008, hence it is removed from the table.

Table – 8: Capacity and Production of Alumina, 2022-23 and 2023-24

(In million tonnes)

Producer	Annual capacity	Production	
		2022-23	2023-24 (P)
Total	7.47	7.26	7.475
Public Sector			
National Aluminium Co. Ltd (Damanjodi) (NALCO)	2.275	2.02	2.275
Private Sector			
Bharat Aluminium Co. Ltd (BALCO)	0.200 [#]	0.2	0.2
Hindalco Industries Ltd	3	3.45	3
Vedanta Aluminium Ltd (Lanjigarh)	2	1.79	2

Figures rounded off.

Source: Information received from individual plants/Annual Reports.

[#]: Plants remained non-operational during the year.

Note: MALCO Plant is closed since December 2008, hence it is removed from the table.

National Aluminium Co. Ltd

NALCO has undertaken projects for capacity addition both in refinery and bauxite mines as an integrated project. The project include 5th stream addition (Capacity 1 MTPA) to the existing Alumina Refinery and operational of pottangi Bauxite mine with total estimated capacity outlay of ₹ 8,254.40 crore. The Company has port facilities at Visakhapatnam to export alumina at the rate of 1.4 million tpy.

Vedanta Aluminium

Vedanta Aluminium is a Private Sector Company with an integrated alumina/aluminium complex at Korba in Bilaspur district in Chhattisgarh and in Jharsuguda, Odisha. The Company has two captive bauxite mines. The Company's two alumina refineries are located at Korba, Chhattisgarh and Lanjigarh, Odisha, with installed capacities of 2.0 lakh tpy and 35.00 lakh tpy, respectively. The total capacity of the Korba and Jharsuguda smelter is 0.57 million tpy and 1.7 million tpy, with total smelter capacity of 23.20 lakh tpy, respectively. The Company also has the capacities to produce ingots, wire-rods billets, bushbars and rolled products. The state-of-the art alumina refinery at Lanjigarh, feeds the aluminium smelters at Jharsuguda and BALCO and forms a crucial link in the value chain.

Hindalco Industries Ltd

Hindalco Industries Ltd has a total aluminium production capacity of around 1.35 million tonnes. Aluminium smelting operations are located at Renukoot (Uttar Pradesh), Aditya Aluminium (Odisha), Mahan Aluminium (Madhya Pradesh) and Hirakud (Odisha). Aditya and Mahan Aluminium smelters are operating on state-of-the-art AP36 technology. In addition to aluminium, Renukoot (Uttar Pradesh), Intergrated Aluminium Complex also produces semi-fabricated products viz. conductor redraw rods, sheet, extrusion, etc. Hindalco's plants are equipped

with sophisticated rolling mills and finishing equipment. The plants are located at Hirakud (Odisha), Belur (West Bengal), Mouda (Maharashtra), Renukoot (Uttar Pradesh) and Taloja (Maharashtra). Hindalco's finished products include, alumina, primary aluminium in the form of ingots, billets & wire rods, value-added products, such as, rolled products, extrusion and foils. Hindalco is the largest manufacturer of entire range of Flat Rolled Products. The Hirakud Flat Rolled Products (FRP), produce rolled products, extrusions products and wire rods.

Hindalco has signed MoU with Odisha Mining Corporation Ltd (OMC) for long-term linkage of raw material for upcoming 2 MTPA alumina refinery project at Kansariguda, Rayagada district, Odisha.

Recycling

Aluminium is recyclable without any loss of properties and consumes only 5% of the total energy requirement compared with primary metal production. Under the Organised Sector, Hindalco is the only company operating 25,000 tpy capacity recycling plant at Taloja in Maharashtra.

Copper

The production of copper ore at 3.78 million tonnes in 2023-24 increased by 13.70% as compared to that in the previous year. The metal content in the ore produced in 2023-24 works out to 31,292 tonnes as against 27,728 tonnes in 2022-23

Hindustan Copper Ltd produces copper metal from the ore produced at their captive mines. Sterlite Industries (India) Ltd and Hindalco Industries Ltd produce copper metal from imported copper concentrates. Details regarding capacity and production of copper are furnished in Table-9.

Production of refined copper (cathodes) in 2023-24 and 2022-23 was 509 thousand tonnes and 554 thousand tonnes, respectively.

Table – 9: Capacity and Production of Copper (Cathodes)

(In million tonnes)

Producer	State	Annual capacity	Production	
			2022-23	2023-24 (P)
Total		0.78	0.55	0.5
Hindustan Copper Ltd**	Jharkhand	0.06	-	-
Vedanta Ltd	Dadra & Nagar Haveli	0.22	0.14	0.14
Hindalco Industries Ltd	Gujarat	0.5	0.41	0.36

Figures rounded off.

** : Metal capacity. However, the cathode capacity of HCL is 51,500 tonnes.

Hindustan Copper Ltd

Hindustan Copper Ltd is a Mini Ratna Government of India Enterprise under the administrative control of the Ministry of Mines. Copper is produced at two smelters of HCL at Indian Copper Complex (ICC), Ghatsila, East Singhbhum district in Jharkhand and Khetri Copper Complex (KCC), Khetrinagar, district Jhunjhunu, Rajasthan. The aggregate capacity of the two smelters for copper cathode production is 51,500 tpy. Refinery at ICC also has a Wire Bar Casting Plant with a capacity of 8,400 tpy and a Brass Rolling Mill that manufactures brass sheets by using copper produced at ICC. The aggregate installed capacity of wire bars is 39,400 tpy and wire rod capacity is 60,000 tpy at HCL. It also has a precious metal recovery plant for the recovery of gold, silver, selenium, tellurium and nickel sulphate and copper sulphate at Ghatsila. Though HCL, has an installed capacity of 390 tonnes in respect of nickel sulphate, no production of nickel sulphate was reported since 2004-05.

HCL has a target to enhance ore production capacity to 12.20 MTPA by FY 2028-29. The capacity of Khetri Copper Complex (KCC) smelter is 31,000 tpy. However, HCL has shut down the Khetri smelting refining plant due to economic reasons. KCC has a concentrator plant at Khetri in Jhunjhunu district, Rajasthan, having a capacity of 2.02 million tpy. KCC & ICC Ghatsila, Jharkhand with 1.55 million tpy capacity each and Malanjkhand, Madhya Pradesh with two million tpy capacity also operate sulphuric acid plant.

Chhattisgarh Copper Ltd (CCL) established in the year 2018 is a joint venture company between Hindustan Copper Ltd and Chhattisgarh Mineral Development Corporation Ltd. The Company was established for exploration, mining and beneficiation of copper and its associated minerals in the State of Chhattisgarh.

Continuous Cast Copper Wire Rods Project, (TCP) Talaja, and Maharashtra: This project has a capacity of 60,000 tpy continuous cast copper wire rods (CCWR). The plant is based on the Southwire SCR-2000 technology of the USA, which uses natural gas as fuel and imported copper cathodes.

Vedanta Ltd (Sterlite Copper)

Vedanta Ltd is India's largest mining company with

interests and operations in iron ore, aluminium, copper, zinc, lead and power. The smelter and refinery of Sterlite Industries (India) Ltd are located at Thoothukudi in coastal belt of Tamil Nadu and Silvassa, Dadra & Nagar Haveli and has a total installed capacity of 6.16 lakh tpy. The Unit is based on 'ISASMELT' technology from MIM, Australia, using imported concentrates. Sterlite closed its operations at Thoothukudi plant w.e.f May, 2018 on the State Government order. The 2,16,000 tpy copper cathode refinery of Sterlite is located in Chinchpada at Silvassa in the Union Territory of Dadra & Nagar Haveli predominantly caters to the domestic market. The copper anodes at Sterlite were refined into cathodes at Silvassa for domestic markets. The technology for refineries and Continuous Cast Copper Rod Plant is of MIM, Australia and Continuous Properzi, Italy, respectively. The imported copper concentrates for smelters are obtained from captive mines in Australia through long-term contracts with producers in Chile and Indonesia, as also through spot purchases.

Hindalco Industries Ltd (Birla Copper)

The Company's copper smelters located at Dahej, Lakhigam, district Bharuch, Gujarat, has an installed capacity of 5,00,000 tpy. The operation consists of producing copper through smelting, refining copper from imported copper concentrates and converting refined copper cathode into continuous cast rod. It is now one of the world's largest smelter at a single location. It is based on Outokumpu Technology. The Company also produces continuous cast copper rods (CCR) with an annual capacity of 0.54 million tonnes. In the process of extraction of copper metal, by-products are recovered and include sulphuric acid (1.67 million tpy), phosphoric acid (1,80,000 tpy), di-ammonium phosphate (DAP) & complex fertilizers (4,00,000 tpy), gold (15 tpy) and silver (150 tpy). The entire requirement of copper concentrates is met through imports.

Gujarat Copper Project (formerly Jhagadia Copper Ltd)

Gujarat Copper Project is located at Jhagadia in Bharuch district, Gujarat. HCL acquired the assets of Jhagadia Copper Ltd and renamed it as GCP. It is a scrap-based electrolytic smelter that produces cathodes with a capacity of 50,000 tpy and additional 20,000 tpy of copper anodes.

The plant was in technical collaboration with Outokumpu Technology (formerly Boliden contech AB), Sweden. The precious metals like gold, silver, platinum, palladium, etc. are also recovered as part of anode slime during the refinery process. The refinery is based on ISA-Technology from Mount ISA Mines Ltd, Australia.

Lead

The total installed capacity of primary lead smelting was 2,05,000 tpy. The smelting capacity of HZL for lead is distributed between two smelters at Chanderiya (85,000 tpy) and Dariba (1,20,000 tpy). Primary lead was produced entirely by HZL at lead-zinc smelter at Chanderiya, district Chittorgarh, and Rajpura-Dariba Plant, district Udaipur, Rajasthan.

Zinc

HZL is the major producer of Zinc. The smelting capacity of HZL for zinc is distributed between three smelters at Debari 1.1 MTPA, Chanderiya and Dariba. Edyar Zinc Ltd's plant at Binanipuram, Kerala, has a capacity of 38,000 tpy.

The primary product of Debari and Vizag smelters is high-grade zinc, while cadmium is recovered as by-product. Chanderiya smelter complex with a total capacity of 5,58,000 tpy of zinc is the world's largest single location zinc smelting complex. Besides lead and zinc, HZL also produces silver as by-products at its Pant Nagar plant in Uttarakhand whose capacity is 800 tonnes per year. The data on total capacity and production of primary lead and zinc ingots in 2022-23 and 2023-24 are furnished in Table-10.

Table – 10: Capacity and Production of Primary Lead and Zinc Ingots, 2022-23 and 2023-24

(In tonnes)

Producer	Lead capacity (tpy)	Production		Zinc capacity (tpy)	Production	
		2022-23	2023-24 (P)		2022-23	2023-24 (P)
Total	210000	210690	215983	951000	820899	817058
Hindustan Zinc Ltd	210000	210690	215983	913000	820899	817058
Edayar Zinc Ltd (formerly, Binani Zinc Ltd)	-	-	-	38000	-	-

ABRASIVES

Natural abrasives, which include calcite, emery, diamond, zircon, corundum, novaculite, pumice, etc., are generally sold as dressed stones. Synthetic abrasives include borazon, ceramic, dryice, glass powder, silica carbide, etc. Commercial abrasives are manufactured in many shapes as bonded or coated abrasives including belt discs, wheels, sheets, blocks, rods and loose grains. A large number of units exist in the Unorganised Sector. However, important producers of coated abrasives were: Grindwell Norton Ltd, Mora, Uran, district Raigad, Maharashtra; Flexoplast Abrasives (India) Ltd, Aurangabad, Maharashtra; Associated Abrasives Ltd, Nashik, Maharashtra; Carborundum Universal Ltd, Chennai, Tamil Nadu, Cutfast Abrasives Tools Pvt. Ltd, Chennai, Tamil Nadu and John Oakey and Mohan Ltd, Ghaziabad, Uttar Pradesh. Important producers of bonded abrasives (grinding wheels) are Associated Abrasives Ltd, Nashik, Maharashtra; Carborundum Universal Ltd, Chennai, Tamil Nadu; Cutfast Abrasives Tools Pvt. Ltd, Chennai, Tamil Nadu; and K.L. Thirani & Company Ltd, Kolkata, West Bengal.

Silicon Carbide (SiC)

Silicon Carbide (SiC) is a synthetic material most commonly produced by the so called Arcean process in electrical resistance furnaces. SiC does not occur naturally except in some types of pre-solar meteorites, along with diamonds. SiC can be produced either in black colour or green colour depending on the raw material. SiC products

have applications in metallurgical refractories, abrasives, slurry wire sawing, and for technical ceramics.

Major producers of silicon carbide are: Grindwell Norton Ltd, at Renigunta, Andhra Pradesh and at Bengaluru, Karnataka; Indian Metals & Carbide Ltd, Therubali, Odisha; Carborundum Universal Ltd, Tiruvottiyur, district Chennai, Tamil Nadu; and Speedfam (India) Pvt. Ltd, Navi Mumbai, Maharashtra.

CEMENT

The Cement Industry which is one of the key infrastructure industries recorded exponential growth pattern in successive years since the introduction of partial decontrol in 1982, total decontrol in 1989 and post delicensing of the Industry and Policy Reforms initiated in 1991.

As per DIPP, the total installed capacity of Indian Cement Industry is about 600 million tonnes. The Cement Industry comprises around 217 large cement plants manufacturing all types of cement in the country. Cement consumption in India is around 260 kg per capita against global average of 540 kg per capita, which shows significant growth potential for the Industry. The total production of cement (all kinds) in 2023-24 was about 426.29 million tonnes as compared to 391.40 million tonnes in the previous year. The Cement Industry produces a variety of cement, such as, Ordinary Portland Cement (OPC) Portland Pozzolana Cement (PPC), Portland Blast Furnace Slag Cement (PBFC), Oil Well Cement, White Cement, etc. to suit a host of applications.

ASBESTOS-CEMENT PRODUCTS

By virtue of the high tensile strength and bonding properties with cement, it is used in the manufacture of fibre cement products.

Industries that deal with asbestos-cement products include Everest Building Products Ltd which has units located at Kymore in Madhya Pradesh and at Podanur in Tamil Nadu. Similarly, Hyderabad Industries Ltd has three plants at Sanatnagar, Ranga Reddy district in Andhra Pradesh; Jasidih in Jharkhand; and Ballabhgarh in Haryana. Ramco Industries Ltd has three plants at Arakkonam, district Vellore, Tamil Nadu; Karur in district Dharwad, Karnataka; and Maksi in district Shajapur, Madhya Pradesh. Southern Asbestos Cement Ltd has two plants at Karur in district Dharwad, Karnataka; and Arakkonam, district Vellore in Tamil Nadu. Shree Pipes Ltd Hamirgarh, district Bhilwara, Rajasthan; Malabar Building Products Ltd, Malakunnathukavu, district Thrissur, Kerala; Konark Cement and Asbestos Industries Ltd at Bhubaneswar in Odisha; Shri Digvijay Cement Co. Ltd, Digvijay nagar, Ahmedabad in Gujarat; Uttar Pradesh Asbestos Ltd, Mohanlalganj, district Lucknow, Uttar Pradesh; Assam Asbestos Ltd, Bonda, Narangi, district Guwahati, Assam; Utkal Asbestos Ltd, Dhenkanal in Odisha; and Visaka Asbestos, Pattencheru (Medak) in Andhra Pradesh are some of the other industries that produce asbestos cement products.

Besides, Swastik Industries, Pune, in Maharashtra; Kalani Asbestos, a Division of Kalani Industries Pvt. Ltd, Pitampur, district Dhar in Madhya Pradesh; Tamil Nadu Asbestos (Pipes), a unit of Tamil Nadu Cement Corp. Ltd, Mayanur, district Tiruchirapalli in Tamil Nadu; and Ganga Asbestos Cement Ltd, Raebareli in Uttar Pradesh produced only asbestos pressure pipes. The present status of all these asbestos cement units is not available with Indian Bureau of Mines.

REFRACTORY INDUSTRY

Refractory Units fall under Medium and Small-scale Sectors. Steel Industry is the biggest group of customers of this Industry, which consumes about 70% of total refractory

production, followed by 12% in Cement, 5-6% in non-ferrous, 3% in Glass and balance in other industries. There are more than 100 refractories in India, out of which around 14 are major, 33 are medium-sized and the rest are relatively smaller in respect of production. The estimated annual installed capacity of all types of refractory was 2.15 Million tonnes. Bharat Refractories Ltd (BRL), a Government of India Undertaking, has four units that are engaged in the manufacture and supply of various kinds of refractories not only to the integrated steel plants but also to smaller steel plants. The important refractory producers are Calderys India Refractories Ltd, Associated Ceramic Ltd, Dalmia Bharat Ltd, IFGL Refractories Ltd, Orient Refractories Ltd, TRL Krosaki Refractories Ltd, Vesuvius India Ltd, Maithan Ceramic Ltd, National Refractories, etc.

With the modernisation and renovation of steel plants, the requirements for various types of refractories have undergone revolutionary changes. The stress is now on for more sophisticated products like precast monolithics. The domestic Refractory Industry, taking cue of this change, has acquired the technical know-how for production of sophisticated refractories, such as, magnesia carbon bricks, new generation sliding-gate plate refractories, for ladles, gunning materials and castables. Manufacture of carbon bonded silicon carbide crucible and clay graphite foundry products is continuously done with constant upgradation for production of improved products. The use of these special refractories has brought down the consumption of refractories per tonne of steel production. However, the customers are benefited by way of improved performance, lower shut down time and savings on energy. The specific consumption of refractories at present in integrated steel plants varies from 8 to 10 kg/tonne of crude steel as compared to 6-8 kg/tonne of crude steel in advanced countries.

The price and supply of imported raw materials are subjected to international demand and supply situation and most of the refractory makers are completely dependent on imported raw materials, especially for making high-end products. Refractory production in India during 2019-20 & 2020-21 is furnished in Table-11.

Table – 11: Refractory Production in India, 2019-20 and 2020-21

(In tonnes)

Item	Production	
	2019-20	2020-21
Total	1208926	1264064
Fireclay Bricks & Shapes	197803	197148
High Alumina Bricks & Shapes	230319	225911
Silica Bricks & Shapes	79259	65437
Basic Bricks & Shapes	180299	173386
Monolithics/ Castables/ Pre-cast Blocks	392837	470256
Special Products	60802	61964
Others	67607	69962

Source: IRMA Journal

CERAMIC & GLASS INDUSTRY

Ceramic Industry

Ceramic Industry in India is about 100 years old. The main product segments are the Wall tile, Floor tile, Vitrified tile and Industrial tile segments. The market shares (in value terms) are 20%, 23% 50% and 7% for Wall, Floor, Vitrified and Industrial tiles respectively. Ceramic products are made from clay and felspar and are manufactured in Large and Small-scale Sectors with wide variations in type, range, quality and standard. Ceramic items have properties, such as, glassy smooth finish, high thermal shock resistance, poor thermal electrical conductivity, high abrasion resistance, acid resistance and weather resistance. During the last two decades, there has been a phenomenal growth in the field of ceramics to meet specific demands of the industry, such as, high alumina ceramics, cutting tools and other structural ceramics. The state-of-the-art technology of international standards are adopted for production of high quality, ceramic goods in the country. The major industries include Kajaria Ceramics, Somani Ceramics, Asian Granite India, Orient Ceramics & Industries, Nitco, Regency Ceramics, Euro Ceramics, Bell Ceramics, etc. The per capita consumption of ceramic tiles in the country was about 0.50 sq. m which is comparatively lower as compared to 2.6 sq. m in China and 5-6 sq. m in Europe. Ceramics Technological Institute (CTI), Bengaluru, a National Level Institute for R&D in BHEL, offers the much-needed technical support for product development by enabling the Indian Ceramic Industry to adopt a modernised technology for development of new and advanced ceramics. Areas of research are nano-technology, separation technology, microwave processing, etc.

Ceramic Tiles

Following the development and growth of the Building Industry, ceramic glazed tiles producing industries too flourished considerably during the last decade. As per ICCTAS, there were 14 units in the National Sector, which accounted for 40% of production of ceramic tiles. Besides, there are about 200 units in Regional Sector, which accounted for 60% of production of ceramic tiles. The domestic Ceramic Tile Industry has been growing at about 15% per annum. Indian tiles are competitive in the international market and are chiefly exported to East and West Asian countries. In India, both traditional methods of manufacturing (tunnel) as well as the latest single fast firing methods are in vogue in manufacturing of ceramic tiles.

Sanitarywares

The India's sanitarywares installed capacity is already in excess of 40 million pieces/year, India is the world's second largest sanitaryware producer after China. Around 75% of the country's entire production comes from Gujarat. The basic raw materials for sanitaryware are felspar, ball clay, kaolin and quartz. There were 7 units with installed capacity of 143 thousand tpy in the organised sector, while around 210 plants with a capacity of 53,000 tpy exist in

the small-scale sector. Some units have either been closed or merged with the other existing ones. This Industry has been reporting a growth rate of about 10% per annum. The major manufacturers of sanitaryware include Hindustan Sanitaryware Industries Ltd, Parryware Roca Bathroom Products, CeraSanitaryware, Neycer India, Kohler India, Toto, RAK Ceramics India, DuravitSanitarywarePvt. Ltd, Golf Ceramics, etc.

Potterywares

Potterywares include crockery and tableware and its manufacturers are a part of an age old handicraft industry in the country. Produced both in the large-scale and the small-scale sectors, there were 16 units in the Organised Sector with a total installed capacity of about 43,000 tpy,

while in the small-scale sector, there were over 1,400 plants with a capacity of 3 lakh tpy. Out of these, over 600 units are located in Uttar Pradesh.

Glass Industry

The Glass Industry includes manufacturing unit that makes glass products, such as, glass containers and hollow-ware, tablewares, flat glass (including float, sheet, figured, wired and safety, mirror glass), speciality glass (such as, electronics, optics, lighting, ophthalmic lenses) vacuum flasks, refills, laboratory glasswares, fibre glass, kitchen glassware, glass bangles, etc. Principal raw materials used in the manufacture of glass are silica sand, soda ash, calcite, dolomite, etc.

Glass Industry comes under the category of deli-censed industry and manufacturing units are spread all over India. The large-scale producers are located mostly at Mumbai, Kolkata, Bengaluru, Hyderabad and in Gujarat and are equipped mostly with modern melting furnace technology. The medium and Small-scale Industries, on the other hand, include cottage industries that still use outdated technology for production of glass products. The share of organised sector in the Glass industry is dominant at about 55% whereas, the unorganised sector accounts for about 45%. There is considerable scope and demand for glass fibre products, particularly due to growth in Petrochemical Sector, solar products, packaging industry and allied products. Glass Industry in India remained in the form of Cottage Industry till the beginning of 20th century. First glass plant in India was set-up in August 1908 by freedom fighter & Bharat Ratna Lokmanya Bal Gangadhar Tilak at Talegaon in the state of Maharashtra. Glass Industry in India has made a steady progress since then, particularly after independence. Firozabad, known as glass city of India, continues to be a place of master craftsmen and entrepreneurs, where traditional processes are still used for production of a wide variety of glass items. About, 70% of the total glass production in the unorganised sector in the country is contributed by Firozabad Glass Industry.

The production in terms of quantity/value of Sheet Glass, Fibre Glass, Glass Bangles and Glassware during

2018-19 was 91,748.07 thousand square metres, 1,16,743.29 tonnes, 39.76 `crore, 3973.05 `crore respectively.

The per capita consumption of glass in India is about 1.1 kg, which is on the lower side when compared to 15 kg in China. Indian Glass market is estimated to increase at a CAGR of 15% in future. Most of the glass demand in India comes from container glass which accounts for 50% of country's glass consumption by value. The market share of Indian Glass Industry consists of architectural (45%), automotive (15%), value-added glass (15%), mirrors (10%) & furniture segment (15%).

Glass Containers and Hollow-wares

About 43 units in the organised Sector are engaged in the manufacture of glass containers and hollow-wares, with an in-installed capacity of around 9,305 tonnes per day. Glass containers are ideal packaging medium, but are increasingly being replaced by other packaging materials like plastic, PET, aluminium and tetrapack. The per capita consumption of container glass in India is 1.8 kg as compared to 27.5 kg in USA and 9 kg in China. The major producers include Hindustan National Glass & Industries, Piramal Glass, Haldy Glass Gujarat, La Opala RG, Mohan Meakin, Gujarat Glass, Associated Glass Industries (AGI), etc.

Laboratory Glasswares

There were six units in this Sector which manufacture neutral glass tubing, laboratory glasswares and chemical process equipment. The installed capacity of neutral glass tubing was 46,600 tpy. The data on production are not available. The demand for neutral glass tubing has not picked up due to sizeable switch over from glass items to plastic items.

Flat Glass

Silica sand, dolomite, limestone are some of the mineral ingredients used in the manufacture of flat glass. The term flat glass includes float glass, sheet glass or plate glass, figured and wired glass. These are further processed into mirror, toughened glass, laminated glass, double glazing, etched glass, glass doors, etc. The total size of Flat Glass Industry in India is about 2.05 lakh tonnes per month, Hindustan National Glass and Industries Ltd, (HNG) has a new plant being set-up at Naidupeta in Andhra Pradesh and is undertaking expansion of capacity at its Nashik (Maharashtra) plant. The major consumers of flat glass are Architectural (80%) and Automotive (15%). The per capita consumption of float glass in India is 1.90 kg as against 12 kg in China, 9 kg in Thailand, 13 kg in Malaysia and 4 kg in Indonesia. There has been growing acceptability of the Indian flat glass products in the global market.

Vacuum Flasks and Refills

There were eight units in the organised sector that manufacture vacuum flasks and refills, with an installed capacity of 36 million numbers per annum.

Fibre Glass (Glass-reinforced plastic)

Silica sand, limestone, kaolin, fluorspar, dolomite, etc. are some of the important minerals used in manufacturing fibre glass. Fibre glass is highly capital and technology-intensive Industry. Fibre glass is lighter than aluminium but stronger than steel. Moreover, being an inorganic material, it does not pose any health hazard. There are five units with production capacity of 55,000 tpy, while the production hovered around 39 thousand tonnes. Presently, India exports about 80% of its glass fibre production.

GRANITE INDUSTRY

Major production of granite in raw as well as processed form is generally from Andhra Pradesh, Rajasthan, Karnataka, Tamil Nadu and Gujarat. Granite is used in monuments, building slabs, tiles, surface plates, etc. Over 160 varieties of granite with exotic colours/shades have been identified as products that could be exported after processing.

Granite is a minor mineral as defined under Section 3(e) of MMDR Act, 1957, and as per Section 15 of MMDR Act, 1957, all powers to make rules and grant of Mineral Concessions for minor minerals have been entrusted with concerned State Government. Granite Conservation and Development Rules, 1999 were notified separately on 1.6.1999 for ensuring systematic/scientific exploitation and conservation of granite resources of the country. The deposits are dispersed widely in all parts of the country. Major production of granite in raw as well as in processed form is generally from Andhra Pradesh, Rajasthan, Karnataka, Tamil Nadu and Gujarat.

Granite is a Non-scheduled Industry and the processing of granite is a phenomenon that was started in 1930s. The mining and processing techniques of granite adopted in the country have improved over the years. Looking at its export potential, the Government of India has been encouraging setting up of 100% EOU in this Sector to promote export of value-added granite products.

CHEMICALS

Caustic Soda (Sodium hydroxide)

Caustic soda is a basic inorganic chemical prepared by electrolysis of salt brine and is consumed in textile, organic chemicals, alumina, paper & pulp, soaps & detergents, inorganic chemicals and for water treatment.

A significant quantity of caustic soda is used in the manufacture of other inorganic chemicals and dyestuffs, in metallurgical operations and in petroleum refining. The major Indian producers are Gujarat Alkalies & Chemicals, Grasim Industries, Nirma, Shriram Alkali & Chemicals, Reliance Industries, Aditya Birla Chemicals (India), etc. Aditya Birla Management Corp. Pvt Ltd, Gujarat Alkalies & Chemicals Ltd, DCM Shriram Ltd, Nirma Ltd and Reliance Industries Ltd. together accounted for about 57.6% of the total Caustic Soda capacity in India. Some of the major upcoming projects are GACL-NALCO Alkalies & Chemicals

Dahej Caustic Soda Plant, Kutch Chemical Industries Gandhidham Caustic Soda Plant and Gujarat Alkalies & Chemicals Bhavnagar Caustic Soda Plant, with a capacity of 0.270 mtpa, 0.216 mtpa and 0.198 mtpa, respectively. India's Caustic Soda Plant capacity is forecasted to grow at a compound annual growth rate (CAGR) of 6.1% from 2017 to 2022, according to a survey.

Soda Ash

Soda ash is an important chemical used widely as a raw material in the manufacture of glass and glassware, sodium silicate, textile, paper & pulp, in metallurgical industries, desalination plants and in the preparation of a host of chemicals. Soda ash is an essential ingredient in the manufacture of detergent, soap, sodium salts and dyes. The major soda ash producers are Tata Chemicals, Gujarat Heavy Chemicals Ltd, Nirma, Saurashtra Chemicals, DCW, etc. The manufacture of soda ash in India started in 1932 at Dhrangadhra in Gujarat with installed capacity of 50 tpd.

RSPL Jamnagar Greenfield Soda Ash Plant, Gujarat, is a project that involves construction of a Greenfield soda ash plant with a production capacity of 5,00,000 tonnes of light soda ash per year at Village Kuranga in Dwarka, district Jamnagar. The installed capacity of soda ash was 34.64 lakh tonnes.

Calcium Carbide

Calcium carbide is used in the manufacture of flammable acetylene gas for Rubber, Synthetic and Plastic Industry. It is used as a raw material for manufacturing various rubber goods. It is self-reinforcing filler. It is also used for cutting & welding of metals besides its use in manufacturing various chemical substances. The major Indian producers are Birla Carbide, TECIL, ICML and Panyam. The installed capacity of calcium carbide was 1.12 lakh mtpa.

Navin Fluorine Industries, Bhestan, Gujarat, is an important producer of synthetic cryolite. Other producers are Tanfac Industries Ltd, Cuddalore, Tamil Nadu; (Aditya Birla Group) and Adarsh Chemicals and Fertilizers Ltd, Udhna, Gujarat. GMDC, Gujarat has 500 tpd fluorite beneficiation plant at Village Kadipani that produces 96% CaF_2 acid-grade & 90% CaF_2 metallurgical grade concentrate. The acid-grade finds use in aluminium fluoride, synthetic rutile and fluorine chemicals.

Aluminium Fluoride

The important units that produce aluminium fluoride include Navin Fluorine Industries, Maya Rasayan Ltd, Mumbai, Tanfac Industries Ltd, SPIC and Aegis Chemical Industries Ltd. The installed capacity of aluminium fluoride was about 25,600 tpy.

Titanium Dioxide

Titanium Dioxide (TiO_2) is a chemically inert white pigment used in a wide range of consumer products from paints, paper and toothpaste to plastics and cement. The key raw materials used in the production of TiO_2 are ilmenite and rutile found mainly in coastal regions of Tamil Nadu, Kerala,

Andhra Pradesh and Odisha. Anatase and rutile are the two major types of TiO_2 which are manufactured by the sulphate and chloride process, respectively. The 4 key players in the Indian Titanium Dioxide Industry are Kerala Minerals and Metals Limited (KMML), Travancore Titanium Products Limited (TTPL), Kilburn Chemicals (VVTi Pigments Ltd) and Kolmak Chemicals Ltd. According to the recently published Tech Sci Research report "India Titanium Di-Oxide Market Study, 2011-2025", the Titanium di-oxide Market in India is projected to exhibit a CAGR of 3.98% during 2016-2025, owing to broad growing applications of titanium di-oxide in paints, rubbers, plastics, textiles, cosmetics, pare & printings, etc. As per Ministry of Chemicals & Fertilizers report the production of titanium di-oxide was 57.06 thousand tonnes in 2018-19 against the installed capacity of 82.50 thousand tonnes as compared to 57.82 thousand tonnes in the previous year.

Sulphuric Acid

There were 140 (130 sulphur based and 10 smelter gas-based) units with an annual capacity of more than 12 million tonnes that manufacture sulphuric acid in the Organised Sector based on sulphur as a raw material. In addition, sulphuric acid is also recovered at HCL, Hindalco & Sterlite and at HZL & BZL during lead-zinc smelting.

Phosphoric Acid

Important units that produce phosphoric acid of various grades, such as, pharma-grade, food-grade, technical-grade, analytical reagent grade, etc. include Gujarat State Fertilizer Company, Vadodara, Gujarat; Fertilizers and Chemicals Travancore Ltd (FACT), Udyogmandal, Kochi, Kerala; Sterlite Industries India Ltd (Vedanta); HCL, Khetri, Rajasthan; HZL, Udaipur, Rajasthan; Southern Petrochemical Industries Corp. Ltd, Thoothukudi, Tamil Nadu; EID Parry (India) Ltd, Ennore, Tamil Nadu; Star Chemical Ltd, Mumbai, Haldia, West Bengal; Ballarpur Industries Ltd, Karwar, Karnataka; Hindalco Industries Ltd, Dahej, Gujarat; and Paradeep Phosphates Ltd, Paradeep, Odisha. The important uses of phosphoric acid are in the manufacture of phosphatic fertilizers, agricultural feed, waxes, polishes, soaps & detergents, and in waste water treatment, tea-leaf processing, sugar refining, as well as anodising & stabilising agent. According to Tech Sci Research report, "India Phosphoric Acid Market by Type, by application, competition forecast and opportunities, 2011-2021", Phosphoric acid (H_3PO_4) market in India is projected to grow at a CAGR of 7.2% during 2016-2021, on account of increasing demand for phosphoric acid in various applications including fertilizer, sugar, oil, detergent, food & beverage, etc.

Ferrophosphorus (FeP)

Ferro-phosphorus is obtained as a by-product during steel manufacturing, during the production of yellow phosphorus or is smelted by phosphate rock & ferro-rock in blast furnace. It is used as an ingredient in high strength low-alloy steel, foundry products, as de-oxidiser in Metallurgy Industry

& as a brake liner with 23% minimum phosphorus and 1% maximum carbon. Ferro-phosphorus is also used as a drying agent and as an additive in metallic paints.

Red Phosphorus

Star Chemicals (Bombay) Pvt. Ltd and United Phosphorus Ltd, Gujarat are the leading manufacturers and suppliers of red phosphorus in the country. It is mainly consumed in the Match Industry for making strike plate of match box. Besides, in Agriculture Industry, it is used as fumigant and in the making of pesticides. Red phosphorus finds application in the manufacture of phosphoric acid, semi-conductors and also as flame retardant for polymers. It is also used in pharmaceuticals for synthesis of drugs. The installed capacity of red phosphorus was 1.68 thousand tpy. The production of red phosphorus during the year 2018-19 was 1.03 thousand tpy as compared to 0.88 thousand tonnes in the previous year.

Borax

Borax is used as a component of glass, ingredient in enamel glazes, pottery & ceramics. The main manufacturers of borax are Borax Morarji Ltd with an installed capacity of 24,000 tpy at Dahej, GIDC in the State of Gujarat. The plant uses imported crude sodium borate concentrates (rasorite) and crude calcium borate (colemanite) as these are not produced indigenously. Indo-Borax & Chemical Ltd also operates borax and boric acid plants at Pithampur, Madhya Pradesh. As a thumb rule, for one tonne production of boric acid about 2 tonnes of boro-gypsum is produced. However, boro-gypsum does not have ready market for its disposal.

Besides the above listed chemicals, activated bleaching earth, fluorochemicals, alumina ferric and sodium silicofluoride were the other mineral-based products.

CHEMICAL FERTILIZERS

In India, the Agricultural Sector plays a vital role in the economic development of the country as securing food for 1.2 billion plus population is a mammoth task. To maximise agricultural output, it is imperative that better agricultural methods, and greater, but judicious use of fertilizers be put to effect. The application of fertilizers has been known well over a hundred years, but the use of chemical fertilizer started in the beginning of this century. The first phosphate fertilizer plant in India went on stream in 1906. Since then, the Phosphatic Fertilizer Industry has grown considerably, but, the growth has not been able to keep pace with the ever increasing demand.

Presently, there are around 37 units manufacturing urea, 10 units for DAP, 20 units for Complex fertilizers and 110 unit for SSP in India.

As per Fertilizer Association of India, the total installed capacity of P_2O_5 almost stood at 7.31 million tonnes. The major raw materials for Single Super Phosphate (SSP) are rock phosphate and sulphur. Besides containing 14%-16% phosphorous, Single Super Phosphate (SSP) also contains 11%-12% sulphur and 16%-21% calcium. This provides an advantage in the form of improving agricultural productivity since large areas in the country are deficient in sulphur and calcium.

Different types of straight and complex fertilizers are manufactured from rock phosphate, such as, SSP, DAP, nitrophosphate, urea ammonium phosphate etc. In the year 2017-18, the consumption of straight fertilizer and complex fertilizer contributed about 8% and 92% of total P_2O_5 in the country. However, the trend is towards the production of complex fertilizers having the total P_2O_5 in water soluble form along with other nutrients. In the absence of commercially exploitable resources of potash in the country, the entire demand of potassic fertilizers is met through imports. The capacity and production of different types of fertilizers are provided in Table-12.

The principal list of fertilizer plants is furnished in Table-13.

Types of fertilizers produced in India are detailed below:

- A) Straight Nitrogenous Fertilizers:
 - 1) Ammonium Sulphate (AS)
 - 2) Calcium Ammonium Nitrate (CAN)
 - 3) Ammonium Chloride
 - 4) Urea
- B) Straight Phosphatic Fertilizers:
 - 1) Single Super Phosphate (SSP)
 - 2) Triple Super Phosphate (TSP)
- C) NP/NPK Complex Fertilizers:
 - 1) Urea Ammonium Phosphate
 - 2) Ammonium Phosphate Sulphate
 - 3) Diammonium Phosphate (DAP)
 - 4) Mono Ammonium Phosphate (MAP)
 - 5) Nitro Phosphate
 - 6) Nitro Phosphate with Potash
 - 7) NP/NPK

Table – 12: Installed Capacity and Production of Various Types of Fertilizers

(In lakh tonnes)

Products	No. of Units	Installed Capacity (as on 01.04.2024)	Production	
			2022-23	2023-24 (E)
Urea	36	283.77	284.94	314.09
DAP	10	53.88	43.47	42.93
Complex Fertilizers	21	116.33	92.95	95.48
SSP	110	120.85	56.44	44.44

Source: Annual Report 2023-24, Department of Fertilizer.

Table – 13: Principal Fertilizer Plants

Sl.No.	Plant	Location
Public Sector		
1	National Fertilizer Ltd	Nangal- I & II and Bhatinda (Punjab), Panipat (Haryana), Vijaipur, Vijaipur Expansion (Madhya Pradesh)
2	Brahmaputra Valley Fertilizer Corp. Ltd	Namrup- II and III (Assam)
3	Fertilizers & Chemicals Travancore Ltd	Udyogmandal and Cochin-II (Kerala)
4	Rashtriya Chemicals & Fertilizers Ltd	Trombay and Trombay IV, V and Thal (Maharashtra)
5	Madras Fertilizers Ltd	Chennai (Tamil Nadu)
6	Steel Authority of India Ltd	Rourkela (Odisha)
7	Hindustan Fertilizer Corp. Ltd	Khetrinagar (Rajasthan)
Private Sector Large Units		
8	Gujarat State Fertilizers Co. Ltd	Vadodara and Sikka I & II (Gujarat)
9	Shriram Fertilizers & Chemicals	Kota (Rajasthan)
10	DIL (Duncan Industries Ltd)	Kanpur (Uttar Pradesh)
11	Zuari Agro Chemicals Ltd	Zuari Nagar (Goa)
12	Coromandal Fertilizers Ltd	Visakhapatnam and Kakinada (Andhra Pradesh), Ennore (Tamil Nadu)
13	Mangalore Chemicals & Fertilizers Ltd	Mangaluru (Karnataka)
14	Gujarat Narmada Valley Fertilizers Company Ltd	Bharuch (Gujarat)
15	Southern Petrochemicals Industrial Corp.	Thoothukudi (Tamil Nadu)
16	Tata Chemicals Ltd	Haldia (West Bengal), Babrala (Uttar Pradesh)
17	Punjab National Fertilizers and Chemicals Ltd	Nangal (Punjab)
18	Deepak Fertilizers & Petrochemicals Corporation	Taloja (Maharashtra)
19	Tuticorin Alkali	Thoothukudi (Tamil Nadu)
20	Indo-Gulf Fertilizers & Chemicals Corp. Ltd	Jagdipshpur (Uttar Pradesh)
21	Nagarjuna Fertilizers & Chemicals Ltd	Kakinada I & II (Andhra Pradesh)
22	Godavari Fertilizers & Chemicals Ltd	Kakinada (Andhra Pradesh)
23	Hin. Ind. Ltd	Dahej (Gujarat)
24	Chambal Fertilizers & Chemicals Ltd	Gadepan I & II kota (Rajasthan)
25	KSF Ltd	Shahjahanpur (Uttar Pradesh)
26	Paradeep Phosphates Ltd	Paradeep (Odisha)
Co-operative Sector		
27	Indian Farmers' Fertilizers Co-operative Ltd	Kalol and Kandla (Gujarat), Aonla I & II, Phulpur I & II (Uttar Pradesh), Paradeep (Odisha)
28	Krishak Bharti Co-operative Ltd	Hazira (Gujarat)

PAPER & PAPER BOARD INDUSTRY

The Indian Paper Industry accounts for about 5.1% of the world's total production of paper. There are around 850 units which are manufacturing pulp, paper, paper board and newsprint with an installed capacity of nearly 30 million tonnes out of which 5.50 million tonnes are lying idle. As on date around 550 mills are in operation with a total operating capacity of around 30 million tonnes. The total production for the year 2022-23, stood at 20.997. The Indian Paper Industry is in a fragmented structure, consisting of small, medium and large paper mills having capacity ranging from 5 to 2,000 tonnes per day. The Sector uses wood, agro residues and waste paper as input substrates for production. Presently, in the total production, the share of wood, agro and waste paper based mills stand at 18-20%, 5-8%, and 74-76%, respectively. Domestic paper and paperboard segment produces all the main varieties of paper that are in demand in the market viz. writing and printing (35%), packaging grade paper (55%), newsprint (6%) and others/specialty paper (<4 %). The Newsprint Sector in the country is governed by the Newsprint Control Order (NCO),

2004. At present, there are 125 mills registered under the Schedule to the NCO. However, due to prevalent market conditions, only 125 mills are under production. The per capita consumption of paper in India is 15-16 kg, which is far behind the global average of 57 kg. As a thumb rule, in Paper Industry, cost of energy is nearly 25% of the cost of production. Hence, energy management is an important aspect in this Sector. Minerals like china clay, limestone, talc, salt, sulphur, etc. besides coal as fuel are used for purposes such as filler, coating & surface sizing, etc., in this industry and also play vital role in quality control.

PAINT & ALLIED PRODUCTS INDUSTRY

The Paint & Allied Products Industry comprises paints, enamels, varnishes, pigments, synthetic resins, printing inks, etc. Approximately, 65% of the production is contributed by the organised sector. The Indian Paint Industry is expected to grow at a rate of 12-13% annually. India is self-sufficient in the production of paints. Barytes, bentonite, calcite, china clay, mica powder, rutile, talc/steatite/soapstone, ochre, silica & dolomite powder are some of the important minerals consumed in the Paint Industry.



7. Production

7.5%

increase in mineral production (excluding Atomic Minerals and Minor Minerals) in India during 2023-24.

₹ 1,28,632 crore

value of production of Metallic Minerals were reported in 2023-24.

₹ 13,172 crore

value of production of Non-Metallic Minerals were reported in 2023-24.

MINERALS

The index of mineral production for Mining and Quarrying Sector (excluding Atomic Minerals and Minor Minerals) in India increased by 7.5 % (as per index of mineral production base year 2011-12) during 2023-24 as compared to the previous year due to increase in the production of raw Coal, natural gas and petroleum (crude) among Fuel Minerals; bauxite, copper conc., gold, iron ore, lead conc., zinc conc., manganese ore, etc. among Metallic Minerals and fluorite (graded), graphite (r.o.m.), kyanite, limeshell, limestone, magnesite, marl & selenite, etc. among Non-metallic Minerals.

Metallic Minerals

The value of Metallic Minerals in 2023-24 at ₹ 1,28,632 crore increased by about 16.1 % over that of the previous year.

Non-Metallic Minerals

The value of production of Non-metallic Minerals at ₹ 13,172 crore during 2023-24 increased by 5.5 % as compared to that of the previous year.

Reporting Mines

Reporting mine is defined as “A mine reporting production or reporting ‘nil’ production during a year but engaged in developmental work, such as, overburden removal, underground driving, winzing, sinking work, exploration by pitting, trenching or drilling as evident from the MCDR returns”.

There were 1,359 reporting mines (excluding Fuel Minerals, atomic and Minor Minerals) in India located in different States and UTs during 2023-24. Among them, 543 belong to Metallic Minerals and 816 to Non-metallic Minerals. There were 171 mines in Public Sector and the rest of 1,188 mines were in Private Sector.

Employment

The average daily employment of labour engaged in Mining Sector (excluding Fuel Minerals, Atomic and Minor Minerals) was 1,28,970 in 2023-24. Metallic Minerals accounted for 80.77% (1,04,165) and Non-metallic Minerals 19.23% (24,805) of the total labour force during the year.

Role of Public Sector

The Public Sector has played important role in the overall mineral production in 2023-24. The entire production of

copper ore & conc., among Metallic Mineral and diamond, rock-salt and sulphur in respect of Non-metallic Minerals was reported from the Public Sector. By and large, the entire production of gold, tin conc. and phosphorite came from Public Sector during 2023-24.

Index of Mineral Production

The index of mineral production (excluding atomic and minor minerals) (with base year 2011-12=100) for 2023-24 at 128.9 displayed a increase of 7.5 % as compared to the previous year.

**Table – 1: Mineral Production
(Excluding Atomic, Minor & Fuel Minerals) in India, 2021-22 to 2023-24**

(By Mineral Groups & Minerals)

(Value in ₹ '000)

Minerals	Unit	2021-22		2022-23		2023-24	
		Quantity	Value	Quantity	Value	Quantity	Value
All Minerals			1374219051		1233176808		1418034586
Metallic Minerals			1264284201		1108287792		1286319428
Bauxite	t	22494049	25284169	23844337	27371722	23967866	26903449
Chromite	t	3785625	47969167	3557289	46633415	3147954	48863434
Copper Ore	t	3569632	-	3326337	-	3782256	-
Copper Conc.	t	115313	11024312	112745	10208275	125230	13141478
Gold Ore	t	474994	-	633514	-	760252	-
Gold	kg	1407	6761018	1437	7655220	1603	9670820
Iron Ore	'000t	254099	1008264608	258312	834001723	277332	998247994
Lead & Zinc Ore	t	16338564	-	16744082	-	16521284	-
Lead Conc.	t	368040	22025747	376665	24488505	380800	25473522
Manganese Ore	t	2692408	22067826	2826964	22842633	3380889	24787707
Silver	kg	647156	42126921	713768	43814376	745980	53559386
Tin Conc.	kg	26301	32619	45432	52285	22345	17341
Zinc Conc.	t	1594085	78727814	1670208	91219638	1710144	85654297
Non-metallic Minerals			109934850		124889016		131715158
Diamond	ct	266	18051	388	61473	349	23858
Fluorite (graded)	t	1237	9430	1155	6224	6042	5716
Garnet (abrasive)#	t	8182	24660	9859	36217	23838	95837
Graphite (r.o.m.)	t	62888	122147	94789	180152	168341	224542
Iolite	kg	27	191	-	-	-	-
Kyanite	t	9320	17267	2765	5319	3324	10515
Limeshell	t	100	231	250	656	891	2633
Limestone	'000t	392034	102022622	405664	111239856	452009	118909140
Magnesite	t	113497	554772	107525	502587	132005	611500
Marl	t	1853481	326498	1461668	316762	1606380	301431
Moulding Sand	t	17583	5256	17260	5373	24494	7386
Phosphorite	t	1394959	6648070	1978450	12354618	1557783	11309762
Rock-Salt	t	286	712	1002	4849	-	-
Selenite	t	716	902	327	442	14136	19566
Siliceous Earth	t	33898	22837	32070	17360	58047	33735
Sillimanite	t	3432	8283	1437	3757	771	2648
Sulphur*	t	880858	-	925663	-	976615	-
Vermiculite	t	3060	3765	2303	1610	1129	781
Wollastonite	t	108335	149156	110793	151761	110596	156108

(P) : Provisional

*: Obtained as by-product from fertiliser plants and oil refineries. The coverage relates to PSUs only

#: Other than BSM (Beach Sand Minerals)

**Table – 2: Mineral Production (Excluding Atomic, Minor & Fuel Minerals) in India
2022-23 and 2023-24**

(By Sector)

(Value in ₹ '000)

Minerals	Unit	All India		Public		Private		Share of Public Sector in Total Production		Overall increase or decrease in production in 2023-24 over 2022-23
		2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	
Metallic Minerals										
Bauxite	t	23844337	23967866	10993539	10966582	12850798	13001284	46.11	45.76	0.52
Chromite	t	3557289	3147954	1043173	1311497	2514116	1836457	29.32	41.66	-11.51
Copper Ore	t	3326337	3782256	3326337	3782256	-	-	100	100	13.71
Copper Conc.	t	112745	125230	112745	125230	-	-	100	100	11.07
Gold Ore	t	633514	760252	593677	716486	39837	43766	93.71	94.24	20.01
Gold	kg	1437	1603	1415	1575	22	28	98.47	98.25	11.55
Iron Ore	'000t	258312	277332	107144	113794	151168	163538	41.48	41.03	7.36
Lead & Zinc Ore	t	16744082	16521284	-	-	16744082	16521284	-	-	-1.33
Lead Conc.	t	376665	380800	-	-	376665	380800	-	-	1.1
Manganese Ore	t	2826964	3380889	1309285	1756219	1517679	1624670	46.31	51.95	19.59
Silver	kg	713768	745980	148	150	713620	745830	0.02	0.02	4.51
Tin Conc.	kg	45432	22345	44899	21946	533	399	98.83	98.21	-50.82
Zinc Conc.	t	1670208	1710144	-	-	1670208	1710144	-	-	2.39
Non-metallic Minerals		-	-	-	-	-	-	-	-	-
Diamond	ct	388	349	388	349	-	-	100	100	-10.05
Fluorite (graded)	t	1155	6042	1010	782	145	5260	87.45	12.94	423.12
Garnet (abrasive)#	t	9859	23838	-	-	9859	23838	-	-	141.79
Graphite (r.o.m.)	t	94789	168341	54216	79263	40573	89078	57.2	47.08	77.6
Iolite	kg	-	-	-	-	-	-	-	-	-
Kyanite	t	2765	3324	2405	1924	360	1400	86.98	57.88	20.22
Limeshell	t	250	891	-	-	250	891	-	-	256.4
Limestone	'000t	405664	452009	20142	20963	385522	431046	4.97	4.64	11.42
Magnesite	t	107525	132005	60462	69288	47063	62717	56.23	52.49	22.77
Marl	t	1461668	1606380	-	-	1461668	1606380	-	-	9.9
Moulding Sand	t	17260	24494	-	-	17260	24494	-	-	41.91
Phosphorite	t	1978450	1557783	1878580	1459919	99870	97864	94.95	93.72	-21.26
Rock-Salt	t	1002	-	1002	-	-	-	100	-	-100
Selenite	t	327	14136	327	8506	-	5630	100	60.17	4222.94
Siliceous Earth	t	32070	58047	-	-	32070	58047	-	-	81
Sillimanite	t	1437	771	-	-	1437	771	-	-	-46.35
Sulphur*	t	925663	976615	925663	976615	-	-	100	100	5.5
Vermiculite	t	2303	1129	-	2	2303	1127	-	0.18	-50.98
Wollastonite	t	110793	110596	-	-	110793	110596	-	-	-0.18

(P) : Provisional

* Obtained as by-product from fertiliser plants and oil refineries. The coverage relates to PSUs only

Other than BSM (Beach Sand Minerals)

**Table – 3: Index of Mineral Production, 2021-22 to 2023-24
(Excluding Atomic & Minor Minerals)**

(Base year 2011-12=100)

Year	Index of mineral production (1000)	Coal & lignite (306.854)	Crude petroleum & natural gas (444.318)	Metallic minerals (230.004)	Non-metallic minerals (18.824)
2021-22	113.3	143.5	74.4	146.8	130.1
2022-23	119.9	163.6	74.1	148.6	137.7
2023-24	128.9	181.8	76.2	158.4	149

Note: Figures in parentheses indicate the weight attached to respective groups.

Gross Value Added from Mining & Quarrying Sector

The Ministry of Statistics & Programme Implementation has released the provisional estimates of national income, revising the base year from 2004-05 to 2011-12 in the year 2015. The industry-wise estimates are now presented

as Gross Value Added (GVA) at basic prices. Certain changes have been made in this series including for Mining & Quarrying Industry. During 2023-24, Mining and Quarrying Industry accounted for about 1.97% of the GVA at current prices. The GVA at current and constant prices for the period from 2021-22 to 2023-24 is furnished in Tables-4 and 5.

Table – 4: Gross Value Added at Basic Price, 2021-22 to 2023-24
(At Current Prices) (31.05.2024)

(Value in ₹ Crore)				
Industry	2021-22	2022-23 (FRE)	2023-24 (PE)	% Change in 2023-24 over the previous years
GVA (All)	21635584	24659041	26762147	8.5
Mining & Quarrying	439339	494602	525881	6.3

Source: NAD, MoSPI

FRE: First Revised Estimates

PE: Provisional Estimates

Table – 5: Gross Value Added at Basic Price, 2021-22 to 2023-24
(At 2011-12 Prices) (31.05.2024)

(Value in ₹ Crore)				
Industry	2021-22	2022-23 (FRE)	2023-24 (PE)	% Change in 2023-24 over the previous years
GVA (All)	13876840	14804901	15873751	7.2
Mining & Quarrying	309276	315256	337623	7.1

Source: NAD, MoSPI

FRE: First Revised Estimates

PE: Provisional Estimates

Non-Ferrous Metals

The production of aluminium at 4,158 thousand tonnes in 2023-24 registered a increase of 2.27% as compared to that in the previous year.

Smelting and refining of copper is carried out by Hindustan Copper Ltd in their existing plants located at Ghatshila and Raigad. Copper metal is also produced from imported copper concentrates at the plants of Vendanta Ltd. (formerly Sterlite Industries (India) Ltd) and Hindalco Industries Ltd. No production of copper blister reported during 2023-24 as well as in the previous year. The production of copper cathodes at 5,09,429 tonnes in 2023-24 decreased by 8.08% as compared to that in the previous year. The production of copper continuous cast wire rods at 4,69,414 tonnes in 2023-24 increased by 10.89% as compared to that in the previous year.

The production of lead (primary) at 2,15,983 tonnes in 2023-24 increased by 2.5% as compared to that in the previous year. No production of lead (secondary) was

reported during the last ten years. The production of zinc ingots in 2023-24 was 8,17,058 tonnes, showing decrease of 0.46%.

Precious Metals

Gold primary is produced from gold ore by HGML in the State of Karnataka. Gold is also recovered as by-product from copper slime of Hindalco Industries Ltd in Gujarat. The total production of gold bullion during the year at 16,639 kg increased by 20% as compared to previous year.

Production of silver in India is reported as a by-product from lead and zinc concentrates and copper slime and as a co-product of gold refining.

The production of silver at 8,19,725 kg registered an increase of 3.18% as compared to that in the previous year.

Other Metals

Production of cadmium is reported as a by-product of zinc smelting and no production of cadmium was reported during 2023-24 and in the previous year (Table-6).

Table – 6: Production and Value of Non-Ferrous Metals,
2021-22 to 2023-24

(Value in ₹ '000)

Metal	Unit of Quantity	2021-22		2022-23		2023-24(p)	
		Quantity	Value	Quantity	Value	Quantity	Value
Aluminium	tonnes	4016621	714320396	4066459	809764986	4158844	755025581
Alumina	tonnes	5207600	130475653	4928449	151965578	4970572	144832952
Cadmium	tonnes	-	-	-	-	-	-
Copper (Blister)	tonnes	-	-	-	-	-	-
Copper (Cathode)	tonnes	483994	363507671	554241	390034135	509429	362507497
Copper (Continuous cast wire rod)	tonnes	351464	261303186	423283	304546760	469414	341516354
Gold	kilograms	10087	48426018	13854	73341570	16639	100994820
Lead (Primary)	tonnes	191185	34944605	210690	39567243	215983	40696202
Silver	kilograms	713653	46424521	794431	48789075	819725	58767385
Tin	kilograms	4868	7307	17160	35435	9249	19885
Zinc Ingot	tonnes	775808	202092090	820899	241626520	817058	180301114

(P) : Provisional

Source: Data for Metals and Alloys are obtained from individual producers/units.



8. Prices

In the domestic markets of Mumbai and Delhi, the prices of various grades of Brass, Copper and Tin rose, while those of Aluminium, Lead, Nickel and Zinc dipped in 2023-2024 as compared to previous year.

Prices of most Steel grades items remained elevated in last two years in comparison to prices in 2021-22. The prices of steel have been volatile with an upward bias in last few years.

MINERALS

Domestic Markets

Fuel Minerals

The pit head prices of different varieties of coking coal, semi-coking coal and the weighted average of crude oil prices are furnished in Table –1.

METALS

Domestic Markets

In the domestic markets of Mumbai and Delhi, mostly upward trend was observed in the prices of various grades of Brass, Copper, and Tin. In contrast, prices of different grades of Aluminium, Lead, Nickel and Zinc slightly dipped in the year 2023-24 as compared to 2022-23. Prices of metals (non-ferrous) during 2022-23 & 2023-24 in domestic markets are given in Table –2.

The prices of most of the steel grade items have remained elevated in the last two years compared to prices in year 2021-22, although, there was a slight moderation

seen in the year 2023-2024. The prices of steel have been volatile with an upward bias in last few years. Prices of steel in the domestic market during the period 2021–22 to 2023–24 are given in Table –3.

Foreign Markets

In the International market, the prices of metals for Aluminium HG, Copper Grade A, Tin HG, and Zinc SHG showed an upward trend during the years 2022 to 2024, while Lead and Nickel exhibited a mixed trend in the same period. Prices of metals in the foreign market during 2022 to 2024 are given in Table – 4.

**Table – 1: Pit head (Run-of-Mine) Price (Rupees per Tonne)
of Coking Coal applicable for Consumers other than Power Utilities (including IPPs),
Fertiliser and Defence [Price-Rs/Tonne] & Weighted Average Crude Oil Prices.
(2021-22 to 2023-24)**

	(In ₹ per tonne)		
Grade	2021-22	2022-23	2023-24
South Eastern Coalfields Ltd			
Semi-Coking grade I	2260	2260	2260
Semi-Coking grade II	1890	1890	1890
Eastern Coalfields Ltd (Ranigunj)			
Semi-Coking grade I	2800	2800	2800
Semi-Coking grade II	2330	2330	2330
Bharat Coking Coal Ltd *			
Washery grade I	5028	5028	5028
Washery grade II	3840	3840	3840
Washery grade III	3060	3060	3060
Washery grade IV	2892	2892	2892
Western Coalfields Ltd			
Washery grade II	2220	2220	2220
Washery grade III	1830	1830	1830
Washery grade IV	1680	1680	1680
Central Coalfields Ltd			
Washery grade I	3450	3450	3450
Washery grade II	3210	3210	3210
Washery grade III	2750	2750	2750
Washery grade IV	2300	2300	2300
Crude Oil (weighted average price)			
Basic Price Offshore	43745	56837	51425
Basic Price Onshore	41088	52719	47600
Basic Price Offshore & onshore	42874	55436	50116

Sources : 1. Coal directory of India 2023-24, Ministry of Coal, for prices of coal

2. Indian Petroleum & Natural Gas Statistics, 2023-24, Ministry of Petroleum & Natural Gas, for basic prices of crude oil.

*: Prices of BCCL (unspecified) has been substituted with BCCL due to non availability of prices

**Table – 2: Prices of Metals (Non-ferrous), 2022-23 & 2023-24
(Domestic markets)**

	(In ₹ per quintal)		
Grade	Market	2022-23	2023-24
Aluminum			
Utensil Scrap	Mumbai	16485	16700
Ingot	Mumbai	21905	21067
Wire Scrap	Mumbai	20419	19767
Sheets (Scrap)	Delhi	21972	15883
Utensil Scrap	Delhi	16653	16733
Wire Scrap	Delhi	20576	19911
Brass			
Honey	Mumbai	47819	49861
Utensil Scrap	Mumbai	48490	49333
Sheet Cutting	Mumbai	49180	48922
Gun Metal Scrap	Delhi	53695	53817
Copper			
Wire Scrap	Mumbai	71000	72889
Utensil Scrap	Mumbai	67561	68544
Wire Rod	Mumbai	75147	74378

Table – 2 (Concl'd.)

(In ₹ per quintal)

Grade	Market	2022-23	2023-24
Wire Bar	Mumbai	73933	75117
Cathode	Mumbai	76133	74594
Wire Scrap	Delhi	71133	73011
Wire	Delhi	74709	74167
Mixed Scrap	Delhi	71809	72006
Lead			
Ingot	Mumbai	19414	18633
Soft	Delhi	19576	18750
Hard (4%)	Delhi	19680	18888
Nickel			
Ingot	Mumbai	205538	149506
Ingot (4x4)	Delhi	205714	149633
Tin			
Ingot	Mumbai	215057	247367
Ingot	Delhi	215247	247528
Zinc			
Ingot	Mumbai	26257	22172
Ingot	Delhi	26395	22278
Dross	Delhi	26552	22383

Source: Minerals and Metals Review.

Table – 3: Prices of Steel, 2021-22 to 2023-24

(Domestic markets)

(In ₹ per tonne)

Grade	Market	2021-22	2022-23	2023-24 [#]
TMT Bars (ISI, 8mm)	Delhi	46180	45044	61650
MS Squares (8mm)	Delhi	49383	63694	61666
MS Angles (25 x 3mm)	Delhi	51011	63289	61751
Channels (75 x 40 mm)	Delhi	51335	64184	61609
Joists (150 x 75 mm)	Delhi	52416	64492	60489
Melting Scrap	Delhi	40515	42856	37785
Induction Ingots	Delhi	47258	50911	46581
TMT Bars (local 8 mm)	Mumbai	50682	63725	61495
MS Rounds (8 mm)	Mumbai	48358	61391	59432
MS Angles(40 x 6 mm)	Mumbai	50893	63665	61429
Joists (150 x 75 mm)	Mumbai	49986	61070	58209
Melting Scrap (Foundry Grade)	Mumbai	45980	52645	47712
Melting Scrap (Steel Grade)	Mumbai	43625	51779	47944
Melting Scrap (CRCA)	Mumbai	46201	50103	44467
Induction ingots	Mumbai	47950	51953	46748
Arc Ingots	Mumbai	47827	51700	46381
Concast Billet ingots	Mumbai	46627	51906	46848
TMT Bars (ISI, 8 mm)	Kolkata	51752	59127	57508
MS Squares (8 mm)	Kolkata	51161	58572	55178
MS Angles (25 x 3 mm)	Kolkata	51530	63410	61310
Channels (75 x 40 mm)	Kolkata	51318	63313	61742
Joists (150 x 75 mm)	Kolkata	50396	61082	57495
Induction Ingots	Kolkata	47740	51462	46606
Arc Ingots	Kolkata	48231	52311	46573
Concast Billet Ingots	Kolkata	47525	51730	46881

Table – 3 (Concl'd.)

(In ₹ per tonne)

Grade	Market	2021-22	2022-23	2023-24 [#]
Induction ingots (round)	Gobind (Punjab)	46456	50325	45550
Blooms (SAIL, 150 mm)	Gobind (Punjab)	47052	51192	46610
Old Ship Breaking Scrap	Gobind (Punjab)	39898	49351	49099
Melting Scrap (rolling)	Gobind (Punjab)	41169	45768	40246
MS Rounds (10 mm)	Gobind (Punjab)	58165	57108	49385
MS Squares (8 mm)	Gobind (Punjab)	59723	59819	54509
MS Angles (25 x 3 mm)	Gobind (Punjab)	56954	58903	51055
MS Sponge Iron	Gobind (Punjab)	37985	40042	33224
MS Flat (3 x 20 mm)	Gobind (Punjab)	60044	59549	53026
Pig Iron Foundry Grade – A*	Mumbai	49710	57416	50513
Pig Iron Foundry Grade – B**	Punjab	48285	54195	46833
Pig Iron Steel Grade	Punjab	46050	50813	41556

Source: Minerals & Metals Review.

#: Calculated by averaging monthly prices from April 2023 to March 2024

A*: Low Sulphur/Phosphorus i.e. 0.09% Max which is used in Critical automotive engine components and specialize casting.

B**: High Sulphur / Phosphorus i.e. above 0.09% which is used in Non-Critical castings.

Table – 4 : Prices of Metals, 2022 to 2024

(Foreign Markets)

Metal/Grade	Unit	2022*	2023*	2024*
Aluminum HG	\$/tonne	1958.29	2249.48	2561.35
Antimony (min 99.65 %)		NA	NA	NA
Bismuth		NA	NA	NA
Cadmium (min 99.95%)		NA	NA	NA
Cadmium (min 99.99%)		NA	NA	NA
Chromium (min 99%)		NA	NA	NA
Cobalt (99.80%)		NA	NA	NA
Copper Grade A	\$/tonne	6555.57	8477.77	8999.05
Germanium Dioxide		NA	NA	NA
Gold (London)		NA	NA	NA
Indium		NA	NA	NA
Iridium (min 99.9 %)		NA	NA	NA
Lead	\$/tonne	2081.16	2138.18	1991.17
Magnesium (min 99.9 %)		NA	NA	NA
Manganese (min 99.7%)		NA	NA	NA
Mercury		NA	NA	NA
Molybdenum Molybdic Oxide		NA	NA	NA
Nickel	\$/tonne	13359.89	21474.38	15608.66
Palladium (min 99.9%)		NA	NA	NA
Platinum (min 99.9%)		NA	NA	NA
Rhodium (min 99.9%)		NA	NA	NA
Ruthenium (min 99.9%)		NA	NA	NA
Selenium		NA	NA	NA
Silver (London)		NA	NA	NA
Silicon 99.9%		NA	NA	NA
Tungsten		NA	NA	NA
Tin HG	\$/tonne	21409.03	25959	29333.9
Vanadium Pentoxide, min 98%V		NA	NA	NA
Zinc SHG	\$/tonne	2636.64	2646.57	3020.5

Source: Minerals and Metals Review.

*: Annual average based on calendar year

NA: Not Available; Tr OZ: Troy Ounce



9. Foreign Trade

₹ 2,13,740 crore
value of exports (including re-exports)
of ores and minerals in 2023-24.

₹ 2,80,311 crore
value of exports (including re-exports)
of metals & alloys during 2023-24.

₹ 7,91,646 crore
value of imports (including re-imports)
of metals & alloys in the year 2023-24.

EXPORTS

Ores & Minerals

During the year 2023-24, the value of exports (including re-exports) of ores and minerals was ₹ 2,13,740 crore. The export value which had decreased from ₹ 2,57,640 crore in 2021-22 to ₹ 2,42,009 crore in 2022-23, further decreased to ₹ 2,13,740 crore in 2023-24. The value of mineral exports showed a decrease of 11.68% in 2023-24 as compared to that in the previous year. Diamond continued to be the largest constituent item with a share of 65.27% in the total value of mineral exports in 2023-24. Next in order of share was Iron ore with contribution of 15.17% followed by Granite 5.26 %, Alumina 3.01% and Salt (Other than Common Salt & Rock Salt) 1.14%. The individual share of remaining minerals in the total value of exports of ores and minerals from India during the year under review was around or less than one percent. The value of exports of ores & minerals (including re-exports) showed a mixed trend for most of the minerals in 2023-24 as compared to that of the previous year. A significant increase was also noticed in some cases. The exports value of minerals which have shown significant growth are Sand (excl. Metal Bearing) (1457%), Zirconium ores and conc. (217%), Quartz and Quartzite (155%). On the other hand, the exports value recorded significant decline in the cases of Niobium or Tantalum ores and Conc. (100%), Silver Ores and Conc. (100%) and Tin Ores and Conc. (100%) with respect to that in the previous year. Ores and minerals were exported to various countries in 2023-24. About (86%) of the total value of exports of ores and minerals (including re-exports) was confined to only ten countries. During 2023-24, USA occupied the top position in terms of value accounting for 24% of the total value of mineral exports, China was at the second place and contributed 18.11% followed by Hong Kong (17.69%), UAE (8.86%), Belgium (7.84%), Israel (2.96%), Thailand (2.4%), UK (1.7%), Oman (1.26%), Japan (1.19%) and Switzerland (1.10%). The individual share of remaining countries was less than one per cent. Countrywise analysis of value of mineral exports revealed that exports of minerals in 2023-24 witnessed significant positive growth as compared to the previous year for Armenia (251%), Kenya (121.1%), China (92%), Egypt (64.4%), Russia (52.2%), Indonesia (49.41%), Qatar (36.4%), Algeria (20.29%) and Oman (19.29%). However, value of export of ores & minerals declined significantly in respect of Singapore (84%), South Africa (65%), Turkey (53.91%), Morocco (49.38%), Botswana (47.19%), Thailand (38%) and Israel (36.47%) (Tables -1 to 3).

**Table – 1: Exports of Ores & Minerals (including Re-exports)
2021-22 to 2023-24**

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	2576402469	**	2420095296	**	2137405283
Abrasive (natural)	tonne	14106	236450	4689	42219	6791	89537
Alabaster	tonne	++	11	52	1732	23	916
Alumina	tonne	1487035	47334417	2014428	64237704	2088162	64431546
&alusite	tonne	18	1233	19	1515	9	894
Antimony Ores & Conc.	tonne	--	--	--	--	4	2415
Arsenic Sulphide (natural)	tonne	51	505	++	++	--	--
Asbestos: Total	tonne	1906	68026	3286	217449	1206	81470
Asbestos (others Nes)	tonne	1	65	1	7	--	--
Asbestos Fibre raw, beaten or washed or graded	tonne	1905	67961	3220	217275	1206	81465
Asbestos Flakes or powder	tonne	--	--	65	164	--	--
Asbestos Other In Rock Form	tonne	--	--	++	3	++	5
Ball Clay	tonne	266680	664037	993811	3779211	974719	3754998
Barytes	tonne	1874837	11075666	2445691	16613885	2310272	17945276
Bauxite	tonne	378081	1005256	177535	479019	164177	243541
Bentonite	tonne	1585962	5850483	1375350	6607488	1660359	7791069
Borax: Total	tonne	4725	656472	4151	818305	3190	627776
Borax:other Borates	tonne	3147	548877	3163	724781	2345	539599
Natural Borate	tonne	44	3489	94	6417	76	5099
Sodium Borate	tonne	1534	104106	894	87107	769	83078
Building & Monumental Stones Nes	tonne	13295779	24912268	12276938	20638218	10919664	22191218
Calcite	tonne	24789	163372	27892	208639	35328	283078
Chalk	tonne	1129	7557	1057	7330	857	6374
Chromite: Total	tonne	2625	89710	33881	904154	2117	99434
Chrome Ore Others	tonne	2614	76859	4303	123468	2117	99434
Chrome Ore Conc.	tonne	--	--	29578	780686	--	--
Chrome Ore Lumps	tonne	11	12851	--	--	--	--
Clay (others)	tonne	49635	361745	34138	314150	28156	298809
Coal (excl. lignite)	'000 tonnes	1314	11233701	1165	15001652	1543	16434833
Coal:lignite	'000 tonnes	1	203336	1	90369	1	137261
Coke	tonne	1299461	41017403	365897	15566657	137166	3827703
Copper Ores & Conc.	tonne	34827	3964549	26336	2435371	23188	2430311
Corundum (natural)	tonne	89	241	59	144	++	15
Diamond: Total		**	1893641728	**	1837280521	**	1395132973
Diamond (industrial)	CTM	12125962	1162179	12527927	901704	10217771	596444
Diamond (mostly cut)	CTM	40571616	1892188684	32096573	1836070859	26579149	1394245154
Diamond Powder		**	290865	**	307958	**	291375
Diatomite	tonne	1964	58544	1901	62622	2658	69569
Dolomite	tonne	113380	391843	77124	307324	66496	292056
Earth Clay	tonne	3775	53782	9432	96235	17841	173214
Emerald (cut & uncut): Total		**	10808861	**	17096852	**	19355783
Emerald (cut)	CTM	5947237	10596194	7406273	17002198	7255717	19254658
Emerald (uncut)	tonne	++	212667	2	94654	++	101125

Table- 1 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Felspar (cut & uncut): Total		**	254062	**	275473	**	321240
Felspar (cut)	CTM	16943380	231801	24143815	246802	15025099	295573
Felspar (uncut)	tonne	33	22261	66	28671	84	25667
Felspar (natural)	tonne	763219	4194510	716776	4292153	872814	5129625
Fire Clay	tonne	4473	45578	8654	48207	7686	38043
Flint	tonne	570	5837	976	8675	688	6274
Fluorspar	tonne	844	43463	1024	65899	1358	81974
Garnet(cut & uncut): Total		**	378455	**	484011	**	555370
Garnet (cut)	CTM	12263033	359152	60229611	452710	58423044	518357
Garnet (uncut)	tonne	29	19303	3	31301	4	37013
Garnet (abrasive)	tonne	81270	1433741	103398	1903765	146915	2771501
Granite: Total	tonne	7572368	126460352	6632215	125512876	6026862	112450691
Granite (cut blocks/slabs)	tonne	895870	11084741	699533	9359466	785703	10290842
Granite (others)	tonne	2145266	74148740	2164020	78489779	1878555	68713184
Granite (polished blocks/tiles)	tonne	241062	5419911	363358	7482645	264661	5474988
Granite (crude or roughly trimmed)	tonne	4290170	35806960	3405304	30180986	3097943	27971677
Graphite (natural)	tonne	764	46963	2239	120029	708	63308
Gypsum & Plaster	tonne	220634	765738	191717	574560	97329	356889
Iron Ore: Total	'000 tonnes	26494	241480427	21168	144299670	46376	324332446
Iron Ore Conc: Non Agglomerated	'000 tonnes	100	669724	102	913160	230	1510622
Iron Ore Fines	'000 tonnes	14570	83677314	14457	71125032	31380	183311744
Iron Ore Lumps	'000 tonnes	433	1876849	183	696000	2524	16063986
Iron Ore Pellates	'000 tonnes	11391	155225838	6426	71532986	12242	123425961
Iron Ore Pyrites	'000 tonnes	++	30702	++	32492	++	20133
Kaolin	tonne	339591	2398327	490071	3440275	595762	4116922
Kieselguhr	tonne	28	516	41	3329	26	584
Kyanite	tonne	1655	15376	297	8631	199	8627
Lead Ores & Conc.	tonne	12	1595	137	7521	58	2867
Limestone	tonne	12160342	4551537	2515714	3124865	1662127	1899997
Magnesite: Total	tonne	5384	173809	5263	239139	3842	240820
Dead-burnt Magnesias	tonne	208	6829	862	37034	412	15717
Magnesias (fused)	tonne	2	585	1	781	3	2102
Magnesite (calcined)	tonne	117	5014	401	14462	375	14634
Magnesite (not calcined)	tonne	436	11483	460	10164	1096	22982
Magnesite (other)	tonne	2735	39107	2279	36936	663	20667
Magnesium Oxide	tonne	1886	110791	1260	139762	1293	164718
Manganese Ore: Total	tonne	113606	588189	1557	21566	271	2204
Manganese Ore (35% or more but Below 40% Mn)	tonne	--	--	1367	20378	--	--
Manganese Ore (44% or more but below 44% Mn)	tonne	18379	321424	--	--	46	396
Manganese Ore (46% or more Mn)	tonne	21	1059	++	24	--	--

Table- 1 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Manganese Ore (ferruginous, 10 % or more but below 30% Mn)	tonne	24570	59450	--	--	--	--
Manganese Ore (others)	tonne	70636	206256	190	1164	225	1808
Marble: Total	tonne	324267	11352007	265229	11941092	289787	14068344
Marble (dresssed)	tonne	158060	2361734	111125	2170658	141422	2510711
Marble (others)	tonne	166207	8990273	154104	9770434	148365	11557633
Mica: Total	tonne	151706	6594832	115898	5557843	118778	6309988
Mica (unmanufactured)	tonne	150805	5113041	114945	4333392	117677	5034037
Mica (blocks)	tonne	1469	390254	1433	428594	3793	487805
Mica (condenser Films)	tonne	--	--	1	486	102	5636
Mica (powder)	tonne	110005	3535632	80825	2844915	76315	2976919
Mica (splittings)	tonne	23581	672270	19931	597198	24098	994973
Mica (waste & scrap)	tonne	15750	514885	12755	462199	13369	568704
Mica (worked): Total	tonne	901	1481791	953	1224451	1101	1275951
Mica (bricks)	tonne	++	512	1	436	++	162
Mica (cond. Films, Plates, Cuts Nes)	tonne	44	14811	39	15279	37	15281
Mica (sheets & strips)	tonne	11	94814	13	78794	9	73717
Mica (washers & discs)	tonne	4	23513	9	40987	28	47359
Mica Worked (others)	tonne	823	1338217	877	1083457	1021	1134554
Micanite & Other Built-up Mica	tonne	19	9924	14	5498	6	4878
Molybdenum Ores & Conc.	tonne	++	120	60	223955	76	185324
Natural Gas	tonne	3	883	23	93848	836	112182
Nickel Ores & Conc.	tonne	20	5183	--	--	19	2971
Niobium Or Tantalum Ores & Conc.	tonne	++	18	105	1007	--	--
Ochre: Total	tonne	31512	1590819	32701	2259503	60662	2447562
Ochre: Earth Colours	tonne	30586	1550778	31840	2207363	59984	2328427
Ochre: yellow Ochre	tonne	153	7416	119	4459	33	2603
Ochre: Persian Red	tonne	--	--	--	--	++	998
Ochre: Red Oxide	tonne	773	32625	742	47681	645	115534
Other Minerals Nes	tonne	27918	1808669	41838	3552794	26117	2533169
Other Preciuos Metal Ores & Concentrates	kg	26253	1220	70	49	29	13
Precious & Semi-precious Stones (cut & uncut): total		**	50616826	**	20516238	**	22933748
Precious & Semi precious Stones (uncut)	tonne	7420	31012547	7233	1690206	8872	1834793
Precious & Semi Precious Stones Cut	CTM	1291581389	19604279	1184790443	18826032	2285579087	21098955
Quartz & Quartzite: Total	tonne	965159	7559270	1069301	9221459	1485741	23512490
Quartz (natural)	tonne	863212	6770721	938962	8321576	1378289	21983387
Quartzite (natural)	tonne	101947	788549	130339	899883	107452	1529103
Rock Phosphate	tonne	540	11316	532	19359	534	3171
Rock salt	tonne	139164	277009	225595	621445	310132	811417
Salt (other than common salt & rock salt)	tonne	8724616	13116708	13406235	23673306	13473826	24463925
Sand (excl. metal bearing)	tonne	619	17107	2761	4882	57961	76035
Sandstone	tonne	691771	11282294	406815	5822370	415344	5956257
Silica Sand	tonne	825	4928	767	5944	187	2255
Sillimanite	tonne	3120	64355	2161	73954	3293	115853
Silver Ores & Conc.	kg	++	45	26000	269	--	--
Slate: Total	tonne	68569	2733090	50909	2505184	48643	2403474
Slate (others)	tonne	49656	1206391	33317	889041	33022	860608
Slate (worked)	tonne	18913	1526699	17592	1616143	15621	1542866

Table- 1 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Steatite: Total	tonne	324593	5201974	338297	5626721	281381	4643882
Steatite Blocks	tonne	292	5970	136	2927	808	8298
Steatite Lumps	tonne	136104	2316878	136957	2246370	97404	1420572
Steatite Powder & Others	tonne	188197	2879126	201204	3377424	183169	3215012
Sulphur (exc. sublimed precipitated & colloidal)	tonne	1290620	21010532	1554999	33267617	1185636	10346076
Tin Ores & Conc.	tonne	--	--	++	17	--	--
Titanium Ores & Conc.: Total	tonne	215910	6155343	143607	7288953	224756	7915287
Titanium Ores & Conc. (ileminite)	tonne	215857	6148952	143607	7288953	220344	7437190
Titanium Ores & Conc. (others)	tonne	1	152	--	--	112	20786
Titanium Ores & Conc. (rutile)	tonne	52	6239	--	--	4300	457311
Tripoli Earth	tonne	--	--	10	556	--	--
Tungsten Ores & Conc.	tonne	13	7139	--	--	--	--
Vermiculite	tonne	1263	21780	989	12888	843	17551
Witherite	tonne	++	128	11	762	--	--
Wollastonite	tonne	11705	282266	12847	365656	12285	348673
Zinc Ores & Conc.	tonne	1762	46757	43877	182437	19902	64126
Zirconium Ores & Conc.	tonne	++	180	43	15779	198	50059

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

Table – 2: Value of Exports of Ores & Minerals (including Re-exports) 2021-22 to 2023-24
(By Principal Ores & Minerals)

(Value in ₹ '000)

Ores & Minerals	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in '000	% share in total value	Value in '000	% share in total value	Value in '000	% share in total value	
All	2576402469	100.00	2420095296	100.00	2137405283	100.00	-11.68
Diamond	1893641728	73.50	1837280521	75.92	1395132973	65.27	-24.07
Iron Ore	241480427	9.37	144299670	5.96	324332446	15.17	124.76
Granite	126460352	4.91	125512876	5.19	112450691	5.26	-10.41
Alumina	47334417	1.84	64237704	2.65	64431546	3.01	0.30
Salt (other than common salt & rock salt)	13116708	0.51	23673306	0.98	24463925	1.14	3.34
Quartz & Quartzite	7559270	0.29	9221459	0.38	23512490	1.10	154.98
Precious & Semi-precious Stones (cut & uncut):total	50616826	1.96	20516238	0.85	22933748	1.07	11.78
Building & Monumental Stones Nes	24912268	0.97	20638218	0.85	22191218	1.04	7.52
Emerald (cut & uncut)	10808861	0.42	17096852	0.71	19355783	0.91	13.21
Barytes	11075666	0.43	16613885	0.69	17945276	0.84	8.01
Coal (Excl. lignite)	11233701	0.44	15001652	0.62	16434833	0.77	9.55
Marble	11352007	0.44	11941092	0.49	14068344	0.66	17.81
Sulphur (exc. sublimed precipitated & colloidal)	21010532	0.82	33267617	1.37	10346076	0.48	-68.90
Titanium Ores & Conc.	6155343	0.24	7288953	0.30	7915287	0.37	8.59
Bentonite	5850483	0.23	6607488	0.27	7791069	0.36	17.91
Mica	6594832	0.26	5557843	0.23	6309988	0.30	13.53
Sandstone	11282294	0.44	5822370	0.24	5956257	0.28	2.30

Table- 2 (Cont.)

(Value in ₹ '000)

Ores & Minerals	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in `000	% share in total value	Value in `000	% share in total value	Value in `000	% share in total value	
Felspar (natural)	4194510	0.16	4292153	0.18	5129625	0.24	19.51
Steatite	5201974	0.20	5626721	0.23	4643882	0.22	-17.47
Kaolin	2398327	0.09	3440275	0.14	4116922	0.19	19.67
Coke	41017403	1.59	15566657	0.64	3827703	0.18	-75.41
Ball Clay	664037	0.03	3779211	0.16	3754998	0.18	-0.64
Garnet (abrasive)	1433741	0.06	1903765	0.08	2771501	0.13	45.58
Other Minerals Nes	1808669	0.07	3552794	0.15	2533169	0.12	-28.70
Ochre	1590819	0.06	2259503	0.09	2447562	0.11	8.32
Copper Ores & Conc.	3964549	0.15	2435371	0.10	2430311	0.11	-0.21
Slate	2733090	0.11	2505184	0.10	2403474	0.11	-4.06
Limestone	4551537	0.18	3124865	0.13	1899997	0.09	-39.20
Rock salt	277009	0.01	621445	0.03	811417	0.04	30.57
Borax	656472	0.03	818305	0.03	627776	0.03	-23.28
Garnet(cut & uncut)	378455	0.01	484011	0.02	555370	0.03	14.74
Gypsum & Plaster	765738	0.03	574560	0.02	356889	0.02	-37.88
Wollastonite	282266	0.01	365656	0.02	348673	0.02	-4.64
Felspar (cut & uncut)	254062	0.01	275473	0.01	321240	0.02	16.61
Clay (others)	361745	0.01	314150	0.01	298809	0.01	-4.88
Dolomite	391843	0.02	307324	0.01	292056	0.01	-4.97
Calcite	163372	0.01	208639	0.01	283078	0.01	35.68
Bauxite	1005256	0.04	479019	0.02	243541	0.01	-49.16
Magnesite	173809	0.01	239139	0.01	240820	0.01	0.70
Molybdenum Ores & Conc.	120	0.00	223955	0.01	185324	0.01	-17.25
Earth Clay	53782	0.00	96235	0.00	173214	0.01	79.99
Coal: lignite	203336	0.01	90369	0.00	137261	0.01	51.89
Sillimanite	64355	0.00	73954	0.00	115853	0.01	56.66
Natural Gas	883	0.00	93848	0.00	112182	0.01	19.54
Chromite	89710	0.00	904154	0.04	99434	0.00	-89.00
Abrasive (natural)	236450	0.01	42219	0.00	89537	0.00	112.08
Fluorspar	43463	0.00	65899	0.00	81974	0.00	24.39
Asbestos	68026	0.00	217449	0.01	81470	0.00	-62.53
Sand (excl. metal Bearing)	17107	0.00	4882	0.00	76035	0.00	1457.46
Diatomite	58544	0.00	62622	0.00	69569	0.00	11.09
Zinc Ores & Conc.	46757	0.00	182437	0.01	64126	0.00	-64.85
Graphite(natural)	46963	0.00	120029	0.00	63308	0.00	-47.26
Zirconium Ores & Conc.	180	0.00	15779	0.00	50059	0.00	217.25
Fire Clay	45578	0.00	48207	0.00	38043	0.00	-21.08
Vermiculite	21780	0.00	12888	0.00	17551	0.00	36.18
Kyanite	15376	0.00	8631	0.00	8627	0.00	-0.05
Chalk	7557	0.00	7330	0.00	6374	0.00	-13.04
Flint	5837	0.00	8675	0.00	6274	0.00	-27.68
Rock Phosphate	11316	0.00	19359	0.00	3171	0.00	-83.62
Nickel Ores & Conc.	5183	0.00	--	0.00	2971	0.00	--
Lead Ores & Conc.	1595	0.00	7521	0.00	2867	0.00	-61.88
Antimony Ores & Conc.	--	0.00	--	0.00	2415	0.00	--
Silica Sand	4928	0.00	5944	0.00	2255	0.00	-62.06
Manganese Ore	588189	0.02	21566	0.00	2204	0.00	-89.78
Alabaster	11	0.00	1732	0.00	916	0.00	-47.11
& alusite	1233	0.00	1515	0.00	894	0.00	-40.99
Kieselguhr	516	0.00	3329	0.00	584	0.00	-82.46

Table- 2 (Cont.)

(Value in ₹ '000)

Ores & Minerals	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹'000	% share in total value	Value in ₹'000	% share in total value	Value in ₹'000	% share in total value	
Corundum (natural)	241	0.00	144	0.00	15	0.00	-89.58
Other Precious Metal Ores & Concentrates	1220	0.00	49	0.00	13	0.00	-73.47
Arsenic Sulphide (natural)	505	0.00	0	0.00	--	0.00	--
Niobium or Tantalum Ores & Conc.	18	0.00	1007	0.00	--	0.00	-100.00
Silver Ores & Conc.	45	0.00	269	0.00	--	0.00	-100.00
Tin Ores & Conc.	--	0.00	17	0.00	--	0.00	-100.00
Tripoli Earth	--	0.00	556	0.00	--	0.00	-100.00
Tungsten Ores & Conc.	7139	0.00	--	0.00	--	0.00	--
Witherite	128	0.00	762	0.00	--	0.00	-100.00

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

Table – 3: Value of Exports of Ores & Minerals (including Re-exports), 2021-22 to 2023-24
(By Principal countries)

(Value in ₹ '000)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	2576402469	100.00	2420095296	100.00	2137405283	100.00	-11.68
USA	775883929	30.12	690380198	28.53	513367467	24.02	-25.64
China	269369069	10.46	201745905	8.34	387086810	18.11	91.87
Hong Kong	522968453	20.30	456438219	18.86	378186361	17.69	-17.14
UAE	179093280	6.95	212331516	8.77	189305275	8.86	-10.84
Belgium	199682425	7.75	208832352	8.63	167526083	7.84	-19.78
Israel	107492826	4.17	99134302	4.10	63339041	2.96	-36.11
Thailand	85468616	3.32	83742722	3.46	51966352	2.43	-37.95
UK	36289789	1.41	34082583	1.41	37015150	1.73	8.60
Oman	17562052	0.68	22593116	0.93	26950642	1.26	19.29
Japan	37664560	1.46	32823221	1.36	25328495	1.19	-22.83
Switzerland	9863912	0.38	22830038	0.94	23599696	1.10	3.37
Indonesia	19267492	0.75	13356223	0.55	19955077	0.93	49.41
Italy	24454889	0.95	19939144	0.82	16748943	0.78	-16.00
Bangladesh	20891041	0.81	19916857	0.82	16633205	0.78	-16.49
Nepal	12796502	0.50	16528323	0.68	15817369	0.74	-4.30
Russia	2472709	0.10	10193149	0.42	15521221	0.73	52.27
France	15286111	0.59	14709476	0.61	15105282	0.71	2.69
Vietnam	25428859	0.99	23754114	0.98	15091843	0.71	-36.47
Saudi Arabia	8181523	0.32	11035402	0.46	13143300	0.61	19.10
Korea, Rep. of	18680016	0.73	12349683	0.51	12685631	0.59	2.72
Australia	17023348	0.66	15296470	0.63	11585637	0.54	-24.26
Germany	13672669	0.53	13520397	0.56	9285659	0.43	-31.32

Table- 3 (Cont.)

(Value in ₹ '000)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
Malaysia	13857026	0.54	8382134	0.35	8685971	0.41	3.62
Singapore	8126311	0.32	51883670	2.14	8395065	0.39	-83.82
Canada	9831802	0.38	8318921	0.34	7369610	0.34	-11.41
Turkey	6636755	0.26	14696414	0.61	6774297	0.32	-53.91
Qatar	5298310	0.21	4531817	0.19	6184639	0.29	36.47
Netherlands	7156518	0.28	5755127	0.24	5976842	0.28	3.85
Taiwan	5126196	0.20	6161599	0.25	5011835	0.23	-18.66
Botswana	17209700	0.67	9253885	0.38	4886836	0.23	-47.19
Poland	7895620	0.31	4923576	0.20	4601671	0.22	-6.54
Sri Lanka	2088796	0.08	4544535	0.19	4285799	0.20	-5.69
Egypt	5727873	0.22	2390118	0.10	3930920	0.18	64.47
Bhutan	4083341	0.16	4788163	0.20	3694073	0.17	-22.85
Spain	2533713	0.10	3124565	0.13	3131508	0.15	0.22
South Africa	15291101	0.59	8549309	0.35	2993365	0.14	-64.99
Armenia	699152	0.03	848841	0.04	2981883	0.14	251.29
Algeria	1499304	0.06	1999710	0.08	2405362	0.11	20.29
Kuwait	1525212	0.06	1968616	0.08	1636079	0.08	-16.89
Bahrain	2374535	0.09	1308399	0.05	1560808	0.07	19.29
Morocco	1345495	0.05	3054004	0.13	1546055	0.07	-49.38
Maldives	1089380	0.04	1528374	0.06	1540929	0.07	0.82
Lebanon	1147912	0.04	1441300	0.06	1399537	0.07	-2.90
Kenya	615238	0.02	629065	0.03	1390871	0.07	121.10
Libya	1354887	0.05	1860198	0.08	1383370	0.06	-25.63
Other Countries	34394222	1.33	32619546	1.35	20393419	0.95	-37.48

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

Metals & Alloys

The value of exports (including re-exports) of metals & alloys stood at ₹ 2,80,311 Crore in the year 2023-24. The export value which had decreased from ₹ 3,48,475 crore in 2021-22 to ₹ 2,90,375 crore in 2022-23 further decreased to ₹ 2,80,311 crore in 2023-24. The value of export of metals & alloys showed a decrease of 3% in 2023-24 as compared to the previous year. In terms of value of exports, Iron & Steel has the largest share (55.3%), followed by Aluminum and Alloys Incl. Scrap (22.6%), Ferroalloys (8.43%) and Copper & Alloys (Incl. Brass & Bronze) (6.84%). The contributions of Lead & alloys including scrap and Zinc & alloys including scrap were 2.31% and 2.15% respectively. The individual share of other remaining metals and alloys was less than one percent. As compared to previous year, the value of exports for different important metals had shown a mixed trend in 2023-24. The export value of Gold clad Metals/base metals NES registered a huge spike of 1076% Substantial increase was also observed in Cadmium (incl. waste & Scrap) 494%,

Platinum-clad base/Precious metal 367%, Platinum alloys and Related Metal 264% and Tellurium 62%. However, the export values showed significant negative growth during 2023- 24 as compared to that of the previous year in the cases of Copper (cement copper Precipitated) (99.78%) and Mercury (91.5%). India exported metals & alloys to various countries in 2023-24. Bulk of the metals and alloys having share of more than 2% to the total value were exported to 16 countries. These countries together accounted for 67% of the total value of exports during 2023-24. USA led the group with a share of 13.58% followed by Italy (7.2%), Saudi Arabia (5.31%), Rep. of Korea (5.2%), UAE (4.8%), Malaysia (4.1%), China (3.6%) and Nepal (3.03%). The individual share was less than 3% in respect of the remaining countries. The countries that recorded significant increase in value of exports of metals & alloys during the year 2023-24 were Morocco (333.18%), Saudi Arabia (72.46%) and Russia (71.26%) (Tables - 4 to 6).

Table – 4: Exports of Metals and Alloys (including Re-exports), 2021-22 to 2023-24

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	3484754703	**	2903758013	**	2803112523
Aluminium & Alloys Incl. Scrap	tonne	3454121	791688112	2870455	709309716	2761484	633466261
Antimony Alloys & Scrap	tonne	1918	1507988	1590	1554406	1406	1318411
Bismuth & Scrap	tonne	11	45115	17	9605	++	1319
Boron	tonne	++	6093	++	6402	1	1497
Cadmium (incl. waste & scrap)	tonne	169	23278	33	6115	222	36328
Chromium & Alloys(including scrap)	tonne	284	228803	157	164945	215	210654
Cobalt & Alloys (incl waste & scrap): Total	tonne	1340	895452	2705	2216211	2616	1771899
Cobalt & Alloys	tonne	1309	812402	2219	1821280	2526	1481920
Cobalt & Scrap	tonne	31	83050	486	394931	90	289979
Copper (cement Copper Precipitated)	tonne	501	25047	293	19034	++	41
Copper & Alloys (incl. brass & bronze): Total	tonne	246973	171375285	185017	134690313	257490	191725863
Brass & Bronze	tonne	57394	41628109	39656	24434423	26512	18610723
Brass & Bronze (scrap)	tonne	2020	1002659	2261	1134445	4016	1870949
Copper (scrap)	tonne	18244	8312550	12995	7965012	14859	7780926
Copper Alloys & Scrap	tonne	++	2	--	--	--	--
Copper & Alloys	tonne	169315	120431965	130105	101156433	212103	163463265
Ferroalloys: Total	tonne	2537463	271246477	2564465	269482234	2590862	236362325
Ferro Silicochrome	tonne	20	8932	1059	194107	48	30191
Ferro Silicomanganese	tonne	1112950	113600728	1232727	117179220	1111364	82548623
Ferroalloys (others)	tonne	3690	659380	1823	400058	2135	287833
Ferrobobron	tonne	33	11024	15	6010	13	4864
Ferrochrome	tonne	753389	76899138	745829	83050361	763354	88276327
Ferrocobalt	tonne	++	250	++	36	++	125
Ferrocolumbium	tonne	1	1787	--	--	1	4273
Ferromanganese	tonne	638850	74044066	532543	56487483	651429	51701166
Ferromolybdenum	tonne	324	529440	974	2588532	760	2030374
Ferronickel	tonne	1	2822	1	1976	24240	5247296
Ferroniobium	tonne	25	69528	11	32186	5	17150
Ferrophosphorous	tonne	156	16804	108	14912	81	13110
Ferroselenium	tonne	5	3653	++	720	++	736
Ferrosilicomagnesium	tonne	6407	1098652	11422	2278094	11183	1752704
Ferrosilicon	tonne	19567	3246485	36069	6084574	24140	3641995
Ferrotitanium	tonne	1860	689076	1765	909712	2044	719361
Ferrotungsten	tonne	1	4368	2	10724	1	4332
Ferrovanadium	tonne	168	353217	85	233314	26	69330
Ferrozirconium	tonne	16	7127	32	10215	38	12535
Gold (non-monetary & monetary): total	kg	126	463986	2623	11920184	1521	8162909
Gold-monetary	kg	--	--	--	--	++	13
Gold-nonmonetary	kg	126	463986	2623	11920184	1521	8162896
Gold, non-onetary, Other Unwrought Forms	kg	114	423280	2623	11917576	1005	5483281

Table- 4 (Cont.)

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Gold,non-monetary, Powder	kg	++	565	--	--	++	1
Gold, non-monetary other Semimanufactured forms	kg	12	40141	++	2608	516	2679614
Gold-clad Metals/base Metals Nes	tonne	++	770	30	1925	3	22631
Iron & Steel: Total		**	2044501042	**	1550875589	**	1550719423
Iron & Steel (other finished steel, Nes)	tonne	4505929	2195779	3085262	2024879	569079239	192437945
Iron & Steel (finished steel Inc. Cr sheet)	tonne	6798430	607736761	5226893	524940706	4278693	487304517
Iron & Steel (other finished steel, Nes)	tonne	2789854	462760769	2308191	524145305	1737673	358105275
Iron & Steel (scrap)	tonne	11492	941199	14610	1547783	9306	1112446
Iron & Steel (semi-finished steel Incl. Stl Ingo	tonne	13492292	874972590	4545495	383068371	5665550	406065232
Iron & Steel (sponge ilron)	tonne	789189	25256226	1096806	38009852	1323458	39551146
Iron & Steel (steel wire)	tonne	250944	48353501	182786	43621658	176995	37812544
Iron & Steel Material	tonne	134773	11850110	162394	15327828	173718	16676622
Iron & Steel (stainless Steel)	tonne	80501	10322040	95970	18120498	63120	11564385
Iron & Steel: Alloy Steel (granules)	tonne	678	45269	487	39652	575	42857
Iron & Steel: Alloy Steel (powder)	tonne	129	66798	78	29057	155	46454
Lead & Alloys Incl. Scrap	tonne	229864	40542449	202248	38752553	342724	64700569
Lead & Alloys	tonne	229727	40523557	202238	38750995	342723	64700457
Lead Waste & Scrap	tonne	137	18892	10	1558	1	112
Magnesium & Scrap	tonne	7269	1314210	5069	1400550	3480	680737
Manganese & Alloys (incl waste & scrap): Total	tonne	616	631089	432	539489	428	562771
Manganese & Alloys Nes	tonne	280	298469	12	3841	1	186
Manganese & Alloys:(wrought & Unwrought)	tonne	288	307908	380	514599	427	561897
Manganese Waste & Scrap	tonne	48	24712	40	21049	++	688
Mercury	tonne	18	99467	17	63865	++	5414
Molybdenum & Scrap	tonne	365	342257	305	521412	50	759707
Nickel & Alloys Incl. Scrap	tonne	5893	9410607	6885	16098361	5917	13667911
Nickel & Alloys	tonne	4199	8107346	4996	13842957	3918	11378655
Nickel Waste & Scrap	tonne	1694	1303261	1889	2255404	1999	2289256
Other Base Metals Nes	tonne	3	16751	44	9153	28	5788
Pig & Cast Iron Incl. Spiegeliesen	tonne	1250907	49495194	672227	36969622	429042	17001783
Platinum Alloys & Related Metals: Total	kg	1194	4795081	1328	769761	1044	2804912
Other Metals of Platinum Group	kg	325	3877873	137	312645	383	1468078
Platinum (powder, unwrought & others)	kg	869	917208	1191	457116	661	1336834
Platinum - Clad Base / Precious Metal	kg	70	6	3	9	++	42
Precious Metals / Metals Clad With Precious Metals	tonne	3045	12713613	3187	11668802	2075	8983744
Selenium	tonne	39	81781	184	295421	165	299169

Table- 4 (Cont.)

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Silicon	tonne	365	119140	322	92247	58	15222
Silver	tonne	89	3633631	130	5737293	132	5252653
Silver Clad Base Metals	kg	2979	14664	2861	13326	2777	14655
Tantalum & Scrap: Total	tonne	6	80531	9	171014	6	53533
Tantalum Alloys Other	tonne	--	--	9	170552	6	52449
Tantalum Alloys Unwrought	tonne	3	40332	++	460	++	928
Tantalum & Scrap : Total	tonne	3	40199	++	2	++	156
Tellurium	tonne	++	17	2	197	++	320
Tin & Alloys Incl. Scrap: Total	tonne	1191	1494894	797	975946	1025	1374631
Tin (scrap)	tonne	1	223	++	153	8	6445
Tin & Alloys	tonne	721	1432470	361	914572	762	1327129
Tin & Alloys : Worked	tonne	469	62201	436	61221	255	41057
Titanium & Alloys (incl. waste & scrap)	tonne	131	386027	252	576880	392	625949
Tungsten & Alloys Incl. Scrap	kg	1209171	2600160	1117211	2746384	623881	2298988
Zinc & Alloys Incl. Scrap: Total	tonne	283204	74950497	354217	106078261	263413	60177420
Zinc (scrap)	tonne	4	501	1	245	40	6056
Zinc & Alloys	tonne	283200	74949996	354216	106078016	263373	60171364
Zirconium & Scrap	tonne	3	25189	9	20788	++	27044

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil ' ; ++: Negligible

Table – 5: Exports of Metals and Alloys (including Re-exports), 2021-22 to 2023-24
(By Principal Metals & Alloys)

(Value in ₹ '000)

Metals & Alloys	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	3484754703	100.00	2903758013	100.00	2803112523	100.00	-3.47
Iron & Steel	2044501042	58.67	1550875589	53.41	1550719423	55.32	-0.01
Aluminium & Alloys Incl. Scrap	791688112	22.72	709309716	24.43	633466261	22.60	-10.69
Ferroalloys	271246477	7.78	269482234	9.28	236362325	8.43	-12.29
Copper & Alloys (incl. brass & bronze)	171375285	4.92	134690313	4.64	191725863	6.84	42.35
Lead & Alloys Incl. Scrap	40542449	1.16	38752553	1.33	64700569	2.31	66.96
Zinc & Alloys Incl. Scrap	74950497	2.15	106078261	3.65	60177420	2.15	-43.27
Pig & Cast Iron Incl. Spiegeliesen	49495194	1.42	36969622	1.27	17001783	0.61	-54.01
Nickel & Alloys Incl. Scrap	9410607	0.27	16098361	0.55	13667911	0.49	-15.10
Precious Metals / Metals Clad with Precious Metals	12713613	0.36	11668802	0.40	8983744	0.32	-23.01
Gold (non-monetary & monetary): total	463986	0.01	11920184	0.41	8162909	0.29	-31.52
Silver	3633631	0.10	5737293	0.20	5252653	0.19	-8.45
Platinum Alloys & Related Metals	4795081	0.14	769761	0.03	2804912	0.10	264.39

Table- 5 (Cont.)

(Value in ₹ '000)

Metals & Alloys	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
Tungsten & Alloys Incl. Scrap	2600160	0.07	2746384	0.09	2298988	0.08	-16.29
Cobalt & Alloys (incl waste & scrap)	895452	0.03	2216211	0.08	1771899	0.06	-20.05
Tin & Alloys Incl. Scrap	1494894	0.04	975946	0.03	1374631	0.05	40.85
Antimony Alloys & Scrap	1507988	0.04	1554406	0.05	1318411	0.05	-15.18
Molybdenum & Scrap	342257	0.01	521412	0.02	759707	0.03	45.70
Magnesium & Scrap	1314210	0.04	1400550	0.05	680737	0.02	-51.40
Titanium & Alloys (incl. waste & scrap)	386027	0.01	576880	0.02	625949	0.02	8.51
Manganese & Alloys (incl waste & scrap)	631089	0.02	539489	0.02	562771	0.02	4.32
Selenium	81781	0.00	295421	0.01	299169	0.01	1.27
Chromium & Alloys (including scrap)	228803	0.01	164945	0.01	210654	0.01	27.71
Tantalum & Scrap	80531	0.00	171014	0.01	53533	0.00	-68.70
Cadmium (incl. waste & scrap)	23278	0.00	6115	0.00	36328	0.00	494.08
Zirconium & Scrap	25189	0.00	20788	0.00	27044	0.00	30.09
Gold-clad Metals/Base Metals Nes	770	0.00	1925	0.00	22631	0.00	1075.64
Silicon	119140	0.00	92247	0.00	15222	0.00	-83.50
Silver-clad Base Metals	14664	0.00	13326	0.00	14655	0.00	9.97
Other Base Metals Nes	16751	0.00	9153	0.00	5788	0.00	-36.76
Mercury	99467	0.00	63865	0.00	5414	0.00	-91.52
Boron	6093	0.00	6402	0.00	1497	0.00	-76.62
Bismuth & Scrap	45115	0.00	9605	0.00	1319	0.00	-86.27
Tellurium	17	0.00	197	0.00	320	0.00	62.44
Platinum - clad Base / Precious metal	6	0.00	9	0.00	42	0.00	366.67
Copper (cement copper precipitated)	25047	0.00	19034	0.00	41	0.00	-99.78

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

Table – 6: Exports of Metals & Alloys (including Re-exports), 2021-22 to 2023-24
(By Principal Countries)

(Value in ₹ '000)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	3484754703	100.00	2903758013	100.00	2803112523	100.00	-3.47
USA	376145691	10.79	440166533	15.16	380598360	13.58	-13.53
Italy	218489523	6.27	206246193	7.10	204312872	7.29	-0.94
Saudi Arabia	49876656	1.43	86299295	2.97	148831973	5.31	72.46
Korea, Rep. of	199629384	5.73	133347246	4.59	147314472	5.26	10.47
UAE	186793558	5.36	157111617	5.41	134711038	4.81	-14.26
Malaysia	50818501	1.46	95942700	3.30	116401615	4.15	21.32
China	300073618	8.61	97819623	3.37	102894797	3.67	5.19

Table- 5 (Cont.)

(Value in ₹ '000)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
Nepal	124141885	3.56	94835039	3.27	84942421	3.03	-10.43
Belgium	131292286	3.77	93045562	3.20	77466321	2.76	-16.74
Netherlands	83895854	2.41	126694059	4.36	74665164	2.66	-41.07
UK	77348299	2.22	63997380	2.20	74238015	2.65	16.00
Spain	72746464	2.09	55638443	1.92	70127877	2.50	26.04
Germany	72976459	2.09	74469474	2.56	69961641	2.50	-6.05
Bangladesh	67824957	1.95	63079237	2.17	62157033	2.22	-1.46
Mexico	57812067	1.66	85145102	2.93	61660203	2.20	-27.58
Japan	61058933	1.75	56404947	1.94	59286207	2.12	5.11
Vietnam	141449641	4.06	79066863	2.72	54594243	1.95	-30.95
Turkey	160408510	4.60	91606086	3.15	53355309	1.90	-41.76
Thailand	65160606	1.87	49229842	1.70	49701455	1.77	0.96
Singapore	51491556	1.48	41596303	1.43	48391427	1.73	16.34
Taiwan	74805393	2.15	41990241	1.45	42676571	1.52	1.63
Canada	40867323	1.17	49032279	1.69	39247109	1.40	-19.96
Brazil	39217886	1.13	26706652	0.92	38836068	1.39	45.42
Indonesia	63285763	1.82	58549924	2.02	37583017	1.34	-35.81
Poland	42333801	1.21	23210815	0.80	34070973	1.22	46.79
Australia	31110696	0.89	29953972	1.03	31841988	1.14	6.30
Russia	22158700	0.64	17399693	0.60	29798031	1.06	71.26
Greece	43771434	1.26	18170926	0.63	29138877	1.04	60.36
Qatar	26939943	0.77	18436943	0.63	22790364	0.81	23.61
Oman	28430087	0.82	31377603	1.08	21275107	0.76	-32.20
France	19794456	0.57	21330974	0.73	20957263	0.75	-1.75
Egypt	43780187	1.26	23237780	0.80	19781107	0.71	-14.88
South Africa	31271976	0.90	14011920	0.48	18985428	0.68	35.49
Tanzania	11205563	0.32	12337025	0.42	18855057	0.67	52.83
Portugal	14227837	0.41	10097507	0.35	16749821	0.60	65.88
Morocco	3859639	0.11	3747148	0.13	16231866	0.58	333.18
Sri Lanka	39052634	1.12	11881888	0.41	14877532	0.53	25.21
Other Countries	359206937	10.31	300543179	10.35	273803901	9.77	-8.90

Source: DGCIS, Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible

Value of exports of selected mineral-based products in the previous year. Exports of selected mineral-based products during 2021-22 to 2023-24 are decrease of 2% as compared to that of ₹ 10,450 crore furnished in Table-7.

**Table – 7: Exports of Selected Mineral Based Products for the Period
2021-22 to 2023-24**

(By Principal Metals & Alloys)

(Value in ₹ '000)

Mineral Based Product	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	82453123	**	104496135	**	101919814
Aluminium Fluoride	tonne	984	24720	391	50222	2115	50925
Asbestos Cement Products	tonne	117844	1944963	136028	2530491	119975	2345999
Asbestos Fibre Products	tonne	49044	6902347	42269	6866139	47128	7689004

Table- 7 (Cont.)

(Value in ₹ '000)

Mineral Based Product	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Bleaching Powder	tonne	30919	931596	28840	1014781	33028	1038960
Boric Acid	tonne	1384	176162	3456	414727	1383	194800
Briquets of Coke, Semi-coke of Coal	tonne	272	2410	398	1202	++	++
Calcium Carbide	tonne	787	97371	735	84151	450	40597
Caustic Soda	tonne	343042	13196705	458237	24294052	465258	16958585
Cement: Total	tonne	1901076	6187851	670528	3281107	817546	3879760
Cement (others)	tonne	152375	722743	196882	1057979	332571	1558059
Cement (portland grey)	tonne	995557	3314929	455037	2038722	463435	2133791
Cement (portland white)	tonne	18285	177863	14590	160360	9351	119104
Cement Clinker	tonne	734859	1972316	4019	24046	12189	68806
Cryolite & Chiolite (artificial)	tonne	126	12127	29	4274	343	12043
Graphite (artificial)	tonne	28218	4090984	29522	4136910	33484	4901183
Graphite Bricks & Shapes	tonne	59	4099	231	7097	115	2186
Graphite Crucibles	tonne	397	49894	--	--	--	--
Hydrofluoric Acid	tonne	1325	132129	824	113899	1291	167582
Manganese Oxide: Total	tonne	21148	986863	20784	1270422	33911	1911843
Manganese Dioxide	tonne	5297	258822	3395	250032	9255	604262
Manganese Oxides (other than Mn Dioxide)	tonne	15851	728041	17389	1020390	24656	1307581
Non-ferrous Ash & Residues	tonne	278914	2224354	333837	3841934	261133	2677004
Other Refractory Manufactures	tonne	551138	8958764	293054	9033963	403009	9591041
Phosphoric Acid	tonne	9187	1228169	3762	672208	1508	259429
Phosphorus (elemental)	tonne	627	362922	671	552440	528	322904
Phosphoric Fertilizers	tonne	16390	238335	52239	1031749	118493	1887065
Potash Fertilisers	tonne	5877	209314	5347	237161	29114	1224475
Potassium Nitrate	tonne	917	187910	936	251715	1639	340277
Refractory Bricks	tonne	830258	13283270	844982	17751765	624143	19340460
Silicon Carbide Crucibles	tonne	30137	888930	4926	861075	3339	966116
Slag (dross etc. from iron & steel exc. granules)	tonne	3681957	6467916	5026219	9321018	2690435	8672120
Soda Ash	tonne	250630	5373921	204198	6945863	422206	10311467
Sodium Nitrate	tonne	427	25774	343	36512	112	8091
Sodium Nitrite	tonne	17044	1001187	18964	1868789	19659	1401145
Sulphur (Sublimed Precipitated & Colloidal): Total	tonne	17607	1941916	14515	2059103	14599	1922318
Sulphur (colloidal)	tonne	++	9	++	312	++	307
Sulphur Precipitated	tonne	++	16	++	5961	1	576
Sulphur Sublimed	tonne	17607	1941891	14515	2052830	14598	1921435
Titanium Oxide & Dioxide Total	tonne	48003	5320220	35393	5961366	31871	3802435
Titanium Dioxide	tonne	6049	1538174	4930	1276988	4828	1061591
Titanium Oxides (other than titanium dioxides)	tonne	41954	3782046	30463	4684378	27043	2740844

Source: DGCIS, Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

IMPORTS

Ores & Minerals

During the year 2023-24, the value of imports (including re-imports) of ores and minerals was ₹ 18,63,067 crore. The import value which had increased from ₹ 15,51,614 crore in 2021-22 to ₹ 21,63,881 crore in 2022-23, decreased to ₹ 18,63,067 crore in 2023-24. The value of mineral imports showed a decrease of 13.90% in 2023-24 as compared to that in the previous year. Petroleum (crude) continued to be the largest constituent item with a share of 61.92% in the total value of mineral imports in 2023-24. Next in order of share was Coal (Except Lignite) with the contribution of 16.65% followed by Diamond 9.60% and Natural Gas 5.96%. The combined share of value of these four minerals was 94.12% in 2023-24 as against 93.87% in the previous year. The value of imports of some ores & minerals in 2023-24 has recorded significant increase in the cases of Tripoli Earth (49250%), Garnet (Abrasive) (7159%), Corundum (natural) (4333.33%), Kieselguhr (364.04%), iron ore

(175.71%), as compared to that in the previous year. During 2023-24, ores and minerals were imported from various countries. Bulk of ores & minerals having share more than 1% to total value were imported from 17 countries. These 17 countries, accounted for about 88.44% of the total value of ores & minerals imported in 2023- 24. Russia occupied the top position in 2023-24 in terms of imports value and accounted for 22.92% of the total mineral imports value. Next in the order was Iraq with 12.85% share followed by Saudi Arabia (9.67%), UAE (9.44%), USA (6.36%), Australia (5.33%), Indonesia (5.13%), Qatar (3.29%), Kuwait (1.92%), and South Africa (1.95%), the individual share of remaining countries was less than two percent. Country wise analysis of value of imports of ores and minerals in 2023-24 revealed that imports value from Rep. of Korea reported increase of about 227.72%, Malaysia 101.62%, Russia 42.02%. However, imports value showed a decline in respect of Other Countries (31.32%) (Tables - 8 to 10).

**Table – 8: Imports of Ores & Minerals (Including Re-imports),
2021-22 to 2023-24**

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	15516147570	**	21638813090	**	18630671856
Abrasive (natural)	tonne	3718	78311	1994	65016	1601	51218
Alabaster	tonne	1572	59681	2388	113198	1729	94617
Alumina	tonne	2549567	82447635	2407867	89152924	2411907	83475392
Andalusite	tonne	10419	344730	14168	599795	14454	847210
Antimony Ores & Conc.	tonne	4555	1499919	6460	2708869	4069	1623894
Arsenic Sulphide (natural)	tonne	18	2119	12	448	5	272
Asbestos: Total	tonne	437511	16631390	404154	22940349	465449	23260379
Asbestos (others Nes)	tonne	534	18171	382	19756	--	--
Asbestos Fibre raw, beaten or washed or graded	tonne	436122	16579872	403293	22894231	464688	23213394
Asbestos Flakes or powder	tonne	630	25594	479	26362	582	37040
Asbestos other In Rock Form	tonne	225	7753	--	--	179	9945
Ball Clay	tonne	93855	1134427	34761	715449	33137	654590
Barytes	tonne	16577	523859	14241	476940	16642	503402
Bauxite	tonne	3009079	18963241	3596098	23423723	4492899	28197412
Bentonite	tonne	142010	2131123	119280	1861091	135448	2058659
Borax	tonne	223368	7973967	198719	10161598	202250	10661154
Borax: Total other Borates	tonne	10821	874142	11489	1248036	9407	1174537
Natural Borate	tonne	101337	2910832	70532	2894632	91918	3514354
Sodium Borate	tonne	111210	4188993	116698	6018930	100925	5972263
Building & Monumental Stones Nes	tonne	44482	422105	545687	926601	1990580	1711400
Calcite	tonne	41688	290721	55362	381332	94848	592070
Chalk	tonne	64	2197	41	2191	11	975
Chromite: Total	tonne	245710	4232459	111291	2969916	172411	4563915
Chrome Others	tonne	130179	2296377	84361	2179213	115752	2904035
Chrome Conc	tonne	4987	126819	7911	215038	7776	240354

Table- 8 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Chrome Lump	tonne	110544	1809263	19019	575665	48883	1419526
Clay (others)	tonne	14245	279441	7152	355004	13231	263789
Coal (excl. lignite)	"000 tonne	208636	2288189160	237678	3836139804	264545	3102453301
Coal:lignite	"000 tonne	1	9495	2	19282	1	16920
Cobalt Ores & Conc.	tonne	1	6917	++	1766	1	820
Coke	tonne	2501153	81047701	3639296	136398611	3972082	115539531
Copper Ores & Conc.	tonne	1018934	223814328	1178921	273744293	1016303	259512343
Corundum (natural)	tonne	++	10	++	9	2	399
Diamond: Total		**	2056382187	**	2077695939	**	1787312382
Diamond (industrial)	CTM	16457278	10785779	20011091	16783936	17981091	15377560
Diamond (mostly cut)	CTM	155672092	2042729864	120861334	2057858821	137631888	1771687398
Diamond Powder		**	2866544	**	3053182	**	247424
Diatomite	tonne	1787	94373	3722	243082	2396	166678
Dolomite	tonne	5510404	9682992	5672340	9944292	5754760	10302045
Earth Clay	tonne	2	247	--	--	--	--
Emerald (cut & uncut): Total		**	16560915	**	32333990	**	34739081
Emerald (cut)	CTM	3177205	2553209	2651674	9054590	94568931	11991030
Emerald (uncut)	tonne	58	14007706	64	23279400	63	22748051
Felspar (cut & uncut): Total		**	22005	**	43069	**	53818
Felspar (cut)	CTM	774936	8386	1102583	31621	1330702	47283
Felspar (uncut)	tonne	37	13619	64	11448	13	6535
Felspar (natural)	tonne	2795	54485	2107	52870	2002	50966
Fire Clay	tonne	898	58418	1137	61115	237	16112
Flint	tonne	8611	83205	7874	82200	9616	72884
Fluorspar	tonne	286224	7792038	310173	11509548	256110	10416654
Garnet (cut & uncut)		**	183591	**	303072	**	488685
Garnet (cut)	CTM	1299992	42311	1102403	90487	1990073	142256
Garnet (uncut)	tonne	23	141280	52	212585	13	346429
Garnet (abrasive)	tonne	140	1789	30	929	3086	67440
Gold Ores & Conc.	KG	799178	2376524	2464467	7336354	3619767	8675743
Granite	tonne	35032	1279464	38427	1660364	36150	1480546
Granite (cut blocks/slabs)	tonne	5980	176076	7939	259235	5773	192189
Granite (others)	tonne	1375	188932	2528	367367	2379	283940
Granite (polished blocks/tiles)	tonne	733	22736	1238	42361	816	30295
Granite (crude or roughly trimmed)	tonne	26944	891720	26722	991401	27182	974122
Graphite (natural)	tonne	54047	2651642	45994	2939533	54784	3203146
Gypsum & Plaster	tonne	5632758	11823817	6128694	13798597	7397750	15730513
Iron Ore: Total	"000 tonne	6683	35389345	1790	15947473	4960	43968382
Iron Ore Conc Non Agglomerated	"000 tonne	1446	9469224	1316	13603285	845	9672717
Iron Ore Fines	"000 tonne	4075	15630812	337	943068	3934	32400087
Iron Ore Lumps	"000 tonne	927	8903226	131	1262348	176	1782376
Iron Ore Pellates	"000 tonne	170	874034	--	--	--	--
Iron Ore Pyrites	"000 tonne	65	512049	6	138772	5	113202

Table- 8 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Kaolin	tonne	223127	5048415	220124	7423589	176944	5748095
Kieselguhr	tonne	++	64	6	520	26	2413
Kyanite	tonne	1668	53418	1354	63110	1691	81361
Lead Ores & Conc.	tonne	5325	255224	5558	294381	4877	363290
Limestone	tonne	27582767	49014650	29736036	62593636	33809129	66157693
Magnesite: Total	tonne	510898	13106490	502837	15431256	479521	14872734
Dead-burnt Magnesite	tonne	170744	5340576	179510	6135407	203528	5522817
Magnesite (fused)	tonne	42306	2614441	47972	3095460	67441	3240102
Magnesite (calcined)	tonne	44231	940743	48967	1109909	76606	2095096
Magnesite (not calcined)	tonne	136894	317688	27279	140613	7215	73505
Magnesite (other)	tonne	49392	1825575	135823	2506278	56980	1905344
Magnesium Oxide	tonne	67331	2067467	63286	2443589	67751	2035870
Manganese Ore	tonne	6500149	96424799	4641809	82621009	5592880	77503284
Manganese Ore (30% or more but below 35% Mn)	tonne	929453	9970098	189188	2136109	730981	7393098
Manganese Ore (35% or more but below 40% Mn)	tonne	1393012	18449749	1654178	23587098	1741076	20184175
Manganese Ore (44% or more but below 44% Mn)	tonne	1127494	18744814	796039	16758263	1103971	16255347
Manganese Ore (44% or more but below 46% Mn)	tonne	1943657	32617971	1334096	28828240	1380005	23460875
Manganese Ore (46% or more Mn)	tonne	185816	3797474	47028	1042963	27418	520328
Manganese Ore (ferruginous, 10 % or more but below 30% Mn)	tonne	307008	2929378	180387	1667126	66037	552002
Manganese Ore (others)	tonne	613709	9915315	440893	8601210	543392	9137459
Marble: Total	tonne	1073654	21110673	1453668	29835923	1372012	28750974
Marble (dressed)	tonne	1038849	18724997	1410696	26642956	1320578	25007541
Marble (others)	tonne	34805	2385676	42972	3192967	51434	3743433
Mica: Total	tonne	3338	1319896	4068	1594113	5348	2006895
Mica (unmanufactured): Total	tonne	1569	157461	2030	184553	2983	248301
Mica (blocks)	tonne	79	12727	301	16600	217	46293
Mica (powder)	tonne	508	61955	607	70281	728	86334
Mica (splittings)	tonne	981	82548	1122	97608	2013	114337
Mica (waste & scrap)	tonne	1	231	++	64	25	1337
Mica (worked): Total	tonne	1769	1162435	2038	1409560	2365	1758594
Mica (bricks)	tonne	++	43	--	--	--	--
Mica (cond. Films, Plates, cuts Nes)	tonne	4	6768	++	1439	10	9324
Mica (sheets & strips)	tonne	238	49539	306	71974	182	44572
Mica (washers & discs)	tonne	++	542	++	949	++	858
Mica Worked (others)	tonne	1527	1105543	1732	1335198	2173	1703840
Molybdenum Ores & Conc.	tonne	9114	15470962	10436	23316669	13481	34312389
Natural Gas	tonne	23417029	1005206968	19852386	1372095889	23996378	1109940174
Nickel Ores & Conc.	tonne	106	16165	20	369	++	11
Niobium Or Tantalum Ores & Conc.	tonne	2	488	1	242	--	--
Ochre: Total	tonne	25314	2502368	25243	2976109	27385	3132508
Ochre: Earth Colours	tonne	24975	2397990	24878	2830181	27208	3035397
Ochre: yellow Ochre	tonne	9	6219	6	4221	20	18014
Persian Red	tonne	3	473	++	75	++	116

Table- 8 (Cont.)

(Value in ₹ '000)

Ores & Minerals	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Red Oxide	tonne	327	97686	359	141632	157	78981
Other Minerals Nes	tonne	678353	4311144	1154973	4703439	1701205	5211599
Other Precious Metal Ores & Concentrates	kg	++	120	--	--	203	672
Petroleum (crude)	000 tonne	220034	9139168005	236665	13027071596	231462	11535300059
Precious & Semi-precious Stones (cut & uncut): Total		**	128825205	**	229220056	**	25535318
Precious & Semi precious Stones (uncut)	tonne	4323	10225806	5791	10919741	3531	12936491
Precious & Semi precious Stones Cut	CTM	383631408	118599399	736129040	218300315	3772305854	12598827
Quartz & Quartzite: Total	tonne	7792	211384	5854	306024	47859	467916
Quartz (natural)	tonne	1049	28661	1515	39142	1697	93233
Quartzite (natural)	tonne	6743	182723	4339	266882	46162	374683
Rock Phosphate	tonne	9659818	104667349	9091591	151370002	8810954	126829600
Rock salt	tonne	42396	258506	45106	363389	37760	339688
Salt (other than common Salt & rock salt)	tonne	27153	253890	53880	545064	94868	926054
Sand (excl. metal bearing)	tonne	5121	348319	5717	319717	1776	57520
Sandstone	tonne	130	2691	28	1709	222	2860
Silica Sand	tonne	57095	527973	162087	747250	145715	845219
Sillimanite	tonne	801	13972	58	3424	13	3128
Silver Ores & Conc.	kg	--	--	77361	10001	1	++
Slate: Total	tonne	109	5741	83	6744	40	4026
Slate (others)	tonne	24	1385	--	--	++	21
Slate (worked)	tonne	85	4356	83	6744	40	4005
Steatite: Total	tonne	9978	407255	10395	545770	10525	563345
Steatite Blocks	tonne	2	246	++	5	3	731
Steatite Lumps	tonne	641	2117	489	1961	147	721
Steatite Powder & Others	tonne	9335	404892	9906	543804	10375	561893
Sulphur (exc. sublimed precipitated & Colloidal)	tonne	1895211	35362092	1337789	26766634	1708664	17365951
Tin Ores & Conc.	tonne	++	299	++	366	++	242
Titanium Ores & Conc.: Total	tonne	111653	5292058	68000	4969119	89920	6409133
Titanium Ores & Conc. (ileminite)	tonne	77003	2326341	32126	1398364	44530	1533498
Titanium Ores & Conc. (others)	tonne	11690	485783	16406	796611	19865	1154201
Titanium Ores & Conc. (rutile)	tonne	22960	2479934	19468	2774144	25525	3721434
Tripoli Earth	tonne	19	1174	++	2	23	987
Tungsten Ores & Conc.	tonne	151	14800	423	142414	74	25679
Vanadium Ores & Conc.	tonne	5869	436744	3661	209238	325	77069
Vermiculite	tonne	1096	25520	2292	73873	627	19588
Wollastonite	tonne	30625	675667	41436	877192	62410	1037583
Zinc Ores & Conc.	tonne	720	24772	1041	39415	455	10666
Zirconium Ores & Conc.	tonne	94839	11260337	82831	15163201	82257	13937396

Source: DGCIS Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible

Table – 9: Value of Imports of Ores & Minerals (Including Re-imports), 2021-22 to 2023-24

(By Principal Minerals)

(Value in ₹ '000)

Ores & Minerals	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	15516147570	100.00	21638813090	100.00	18630671856	100.00	-13.90
Petroleum (crude)	9139168005	58.90	13027071596	60.20	11535300059	61.92	-11.45
Coal (excl. lignite)	2288189160	14.75	3836139804	17.73	3102453301	16.65	-19.13
Diamond	2056382187	13.25	2077695939	9.60	1787312382	9.59	-13.98
Natural Gas	1005206968	6.48	1372095889	6.34	1109940174	5.96	-19.11
Copper Ores & Conc.	223814328	1.44	273744293	1.27	259512343	1.39	-5.20
Rock Phosphate	104667349	0.67	151370002	0.70	126829600	0.68	-16.21
Coke	81047701	0.52	136398611	0.63	115539531	0.62	-15.29
Alumina	82447635	0.53	89152924	0.41	83475392	0.45	-6.37
Manganese Ore	96424799	0.62	82621009	0.38	77503284	0.42	-6.19
Limestone	49014650	0.32	62593636	0.29	66157693	0.36	5.69
Iron Ore	35389345	0.23	15947473	0.07	43968382	0.24	175.71
Emerald (cut & uncut)	16560915	0.11	32333990	0.15	34739081	0.19	7.44
Molybdenum Ores & Conc.	15470962	0.10	23316669	0.11	34312389	0.18	47.16
Marble	21110673	0.14	29835923	0.14	28750974	0.15	-3.64
Bauxite	18963241	0.12	23423723	0.11	28197412	0.15	20.38
Precious & Semi-precious Stones (cut & uncut): total	128825205	0.83	229220056	1.06	25535318	0.14	-88.86
Asbestos	16631390	0.11	22940349	0.11	23260379	0.12	1.40
Sulphur (exc. sublimed precipitated & colloidal)	35362092	0.23	26766634	0.12	17365951	0.09	-35.12
Gypsum & Plaster	11823817	0.08	13798597	0.06	15730513	0.08	14.00
Magnesite	13106490	0.08	15431256	0.07	14872734	0.08	-3.62
Zirconium Ores & Conc.	11260337	0.07	15163201	0.07	13937396	0.07	-8.08
Borax	7973967	0.05	10161598	0.05	10661154	0.06	4.92
Fluorspar	7792038	0.05	11509548	0.05	10416654	0.06	-9.50
Dolomite	9682992	0.06	9944292	0.05	10302045	0.06	3.60
Gold Ores & Conc.	2376524	0.02	7336354	0.03	8675743	0.05	18.26
Titanium Ores & Conc.	5292058	0.03	4969119	0.02	6409133	0.03	28.98
Kaolin	5048415	0.03	7423589	0.03	5748095	0.03	-22.57
Other Minerals Nes	4311144	0.03	4703439	0.02	5211599	0.03	10.80
Chromite	4232459	0.03	2969916	0.01	4563915	0.02	53.67
Graphite (natural)	2651642	0.02	2939533	0.01	3203146	0.02	8.97
Ochre	2502368	0.02	2976109	0.01	3132508	0.02	5.26
Bentonite	2131123	0.01	1861091	0.01	2058659	0.01	10.62
Mica	1319896	0.01	1594113	0.01	2006895	0.01	25.89
Building & Monumental Stones Nes	422105	0.00	926601	0.00	1711400	0.01	84.70
Antimony Ores & Conc.	1499919	0.01	2708869	0.01	1623894	0.01	-40.05
Granite	1279464	0.01	1660364	0.01	1480546	0.01	-10.83
Wollastonite	675667	0.00	877192	0.00	1037583	0.01	18.28
Salt (other than common salt & rock salt)	253890	0.00	545064	0.00	926054	0.00	69.90

Table- 9 (Cont.)

(Value in ₹ '000)

Ores & Minerals	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
Andalusite	344730	0.00	599795	0.00	847210	0.00	41.25
Silica Sand	527973	0.00	747250	0.00	845219	0.00	13.11
Ball Clay	1134427	0.01	715449	0.00	654590	0.00	-8.51
Calcite	290721	0.00	381332	0.00	592070	0.00	55.26
Steatite	407255	0.00	545770	0.00	563345	0.00	3.22
Barytes	523859	0.00	476940	0.00	503402	0.00	5.55
Garnet (cut & uncut)	183591	0.00	303072	0.00	488685	0.00	61.24
Quartz & Quartzite	211384	0.00	306024	0.00	467916	0.00	52.90
Lead Ores & Conc.	255224	0.00	294381	0.00	363290	0.00	23.41
Rock salt	258506	0.00	363389	0.00	339688	0.00	-6.52
Clay (others)	279441	0.00	355004	0.00	263789	0.00	-25.69
Diatomite	94373	0.00	243082	0.00	166678	0.00	-31.43
Alabaster	59681	0.00	113198	0.00	94617	0.00	-16.41
Kyanite	53418	0.00	63110	0.00	81361	0.00	28.92
Vanadium Ores & Conc.	436744	0.00	209238	0.00	77069	0.00	-63.17
Flint	83205	0.00	82200	0.00	72884	0.00	-11.33
Garnet (abrasive)	1789	0.00	929	0.00	67440	0.00	7159.42
Sand (excl. metal bearing)	348319	0.00	319717	0.00	57520	0.00	-82.01
Felspar (cut & uncut)	22005	0.00	43069	0.00	53818	0.00	24.96
Abrasive (natural)	78311	0.00	65016	0.00	51218	0.00	-21.22
Felspar (natural)	54485	0.00	52870	0.00	50966	0.00	-3.60
Tungsten Ores & Conc.	14800	0.00	142414	0.00	25679	0.00	-81.97
Vermiculite	25520	0.00	73873	0.00	19588	0.00	-73.48
Coal: lignite	9495	0.00	19282	0.00	16920	0.00	-12.25
Fire Clay	58418	0.00	61115	0.00	16112	0.00	-73.64
Zinc Ores & Conc.	24772	0.00	39415	0.00	10666	0.00	-72.94
Slate	5741	0.00	6744	0.00	4026	0.00	-40.30
Sillimanite	13972	0.00	3424	0.00	3128	0.00	-8.64
Sandstone	2691	0.00	1709	0.00	2860	0.00	67.35
Kieselguhr	64	0.00	520	0.00	2413	0.00	364.04
Tripoli Earth	1174	0.00	2	0.00	987	0.00	49250.00
Chalk	2197	0.00	2191	0.00	975	0.00	-55.50
Cobalt Ores & Conc.	6917	0.00	1766	0.00	820	0.00	-53.57
Other Precious Metal Ores & Concentrates	120	0.00	--	0.00	672	0.00	--
Corundum (natural)	10	0.00	9	0.00	399	0.00	4333.33
Arsenic Sulphide (natural)	2119	0.00	448	0.00	272	0.00	-39.29
Tin Ores & Conc.	299	0.00	366	0.00	242	0.00	-33.88
Nickel Ores & Conc.	16165	0.00	369	0.00	11	0.00	-97.02
Silver Ores & Conc.	--	0.00	10001	0.00	0	0.00	-100.00
Earth Clay	247	0.00	--	0.00	--	0.00	--
Niobium or Tantalum Ores & Conc.	488	0.00	242	0.00	--	0.00	-100.00

Source: DGCIS Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

Table – 10: Value of Imports of Ores & Minerals (Including Re-imports), 2021-22 to 2023-24
(By Principal Countries)

(Value in ₹ '000)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	15516147570	100.00	21638813090	100.00	18630671856	100.00	-13.90
Russia	376212689	2.42	3007260310	13.90	4270945171	22.92	42.02
Iraq	2266196177	14.61	2688547525	12.42	2394024431	12.85	-10.95
Saudi Arabia	1721278333	11.09	2337429207	10.80	1802426914	9.67	-22.89
UAE	1843988123	11.88	2517375480	11.63	1759061701	9.44	-30.12
USA	1586195660	10.22	1701099747	7.86	1184280243	6.36	-30.38
Australia	1076064891	6.94	1290849160	5.97	993302406	5.33	-23.05
Indonesia	574654486	3.70	1281466702	5.92	955899126	5.13	-25.41
Qatar	527094184	3.40	828373534	3.83	613437976	3.29	-25.95
South Africa	405484896	2.61	390562923	1.80	362414066	1.95	-7.21
Kuwait	596318462	3.84	646652875	2.99	358043391	1.92	-44.63
Nigeria	747501535	4.82	510151563	2.36	330844328	1.78	-35.15
Angola	203521139	1.31	289007940	1.34	289389351	1.55	0.13
Hong Kong	250333161	1.61	368644456	1.70	284522430	1.53	-22.82
Malaysia	92681696	0.60	118938570	0.55	239798011	1.29	101.62
Belgium	439151544	2.83	382809078	1.77	236611593	1.27	-38.19
Singapore	134462362	0.87	273326860	1.26	205019598	1.10	-24.99
Korea, Rep. of	2026115	0.01	60077960	0.28	196885090	1.06	227.72
Colombia	175546001	1.13	150490633	0.70	184744325	0.99	22.76
Turkey	52311108	0.34	181198262	0.84	173318520	0.93	-4.35
Other Countries	2445125008	15.76	2614550305	12.08	1795703185	9.64	-31.32

Source: DGCIS, Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible

Metals & Alloys

The value of imports (including re-imports) of metals & alloys was ₹ 7,91,646 crore in the year 2023-24. The import value which had increased from ₹ 6,27,313 crore in 2021-22 to ₹ 6,69,639 crore in 2022-23, further increased to ₹ 7,91,646 crore in 2023-24. The value of metal & alloys imports showed an increase of 18.22% in 2023-24 as compared to the previous year. In terms of value of imports, Gold (Non-monetary & Monetary): total has the largest share of 47.65%, Iron & Steel 23.55%, Copper & Alloys (Incl. Brass & Bronze) 9.66%, Aluminum and Alloys Incl. Scrap 7.18%, and Silver 5.71%. The individual share of remaining metals was less than two percent of the total value of metals & alloys. The value of imports of the metals & alloys that showed significant growth included Gold clad Metals / Base metal NES (29426.32%), Tellurium (116.53%), Molybdenum and Scrap (66.82%), Pig and Cast Iron (incl. Speigeliessen) (63.57%) and Platinum clad base/ Precious Metal (56.31%) in 2023-24 as compared to the previous year. On the other hand, during 2023-24, the value of imports of metals and alloys that showed a declining

trend as compared to the previous year was mainly in the case of Precious metal/metal clad with precious metal (85.48%), Platinum alloys and related metals (80.20%) and Tantalum and scrap (33.39%). India imported metals & alloys from various countries in 2023-24. Bulk of the metals and alloys having share of more than 1% of the total value were imported from 22 countries which accounted for 85.76% of the total value of metals & alloys imported in 2023-24. Switzerland occupied the top position with a share of 18.96% of the total value of metal imports in 2023-24 followed by UAE in second place with a share of 11.92%. The countries next in order were China in third place with a share of 7.65%, South Africa (5.33%) followed by Japan at the 5th place with share of 4.77%, Rep. of Korea at 6th place with 4.33% and USA at 7th place with 4.32%. The remaining countries are those with a share less than four percent. During the year under review, imports in terms of value from Uganda increased manifold by 92.38%, Australia (85.10%) and UAE (78%). On the other hand, the value of imports of metals & alloys showed a declining trend in the case of many countries, the most prominent among

them were Bolivia (56.14%), Guinea (41.14%) and Brazil (37.48%). Imports of selected mineral-based products during 2021-22 to 2023-24 are furnished in Table-14. Value of imports of selected mineral-based products at ₹ 42,565

crore in the year 2023-24 registered a decrease of 21% as compared to that of ₹ 54,113 crore in the previous year (Tables 11 to 14).

Table – 11: Imports of Metals and Alloys (Including Re-imports), 2021-22 to 2023-24

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	6273137692	**	6696396003	**	7916460461
Aluminium & Alloys Incl. Scrap	tonne	2334438	452887747	2542127	562908946	2737109	568263506
Antimony Alloys & Scrap	tonne	1173	998247	1579	1548561	1660	1635339
Bismuth & Scrap	tonne	286	189060	372	267681	338	264542
Boron	tonne	++	2818	++	6094	++	8793
Cadmium (incl. waste & scrap)	tonne	6787	1218010	12258	3321224	10736	3589719
Chromium & Alloys (including scrap)	tonne	1349	899805	1344	1382782	1976	1670445
Cobalt & Alloys (incl waste & scrap)	tonne	1130	5852552	1186	7612854	1822	8409176
Cobalt & Alloys	tonne	1128	5850055	1184	7611201	1822	8409176
Cobalt & Scrap	tonne	2	2497	2	1653	--	--
Copper (cement copper precipitated)	tonne	281	8758	1039	7114	1188	9618
Copper & Alloys (incl. brass & bronze)	tonne	823597	526294673	950010	612858387	1179030	764661761
Brass & Bronze	tonne	20223	15783278	21873	17403629	21522	16481816
Brass & Bronze (scrap)	tonne	133181	65116768	134801	66231070	112806	49181958
Copper (scrap)	tonne	116755	53221910	135906	66026351	197224	89443372
Copper & Alloys	tonne	553438	392172717	657430	463197337	847478	609554615
Ferroalloys	tonne	608617	123345004	398067	98050366	420822	87560715
Ferroalloys: Charge-chrome	tonne	8832	769273	1494	151335	15891	1350873
Ferro silicochrome	tonne	100	12778	262	31057	4520	562613
Ferrosilico-manganese	tonne	26676	2741398	32779	2785678	30677	2112598
Ferroalloys (others)	tonne	8140	2258637	6922	1921216	14575	3317785
Ferroboron	tonne	1046	215732	2008	609479	1028	290907
Ferrochrome	tonne	44631	8360932	32307	7573631	27476	6104676
Ferrocobalt	tonne	--	--	++	1467	1	4277
Ferromanganese	tonne	40881	7059207	29021	4753503	13192	845436
Ferromolybdenum	tonne	2558	6140376	4498	12093687	2994	8766969
Ferronickel	tonne	249315	55890205	65680	20467621	102191	23575596
Ferroniobium	tonne	3982	8642844	5245	12336415	5079	12895411
Ferrophosphorous	tonne	3746	148545	3507	136457	2937	90739
Ferrosilico-magnesium	tonne	4299	631424	6048	1051977	13510	2007710
Ferrosilicon	tonne	212256	28265119	205830	30855315	183435	22344260
Ferrotitanium	tonne	731	300529	564	252271	1528	530315
Ferrotungsten	tonne	2	4892	33	71825	60	124378
Ferrovanadium	tonne	900	1806932	1211	2795214	1206	2519991
Ferrozirconium	tonne	522	96181	658	162218	522	116181
Gold (non-monetary & Monetary) total	kg	879010	3440928249	678299	2804792414	795246	3772489677
Gold-monetary	kg	--	--	1	2321	--	--

Table- 11 (Cont.)

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Gold-non-monetary	kg	879010	3440928249	678298	2804790093	795246	3772489677
Gold, non-monetary, other unwrought forms	kg	868489	3395591272	672402	2778174273	792828	3760345436
Gold, non-monetary, Powder	kg	++	5	1	70	--	--
Gold,non-monetary:other Semi manufactured forms	kg	10521	45336972	5895	26615750	2418	12144241
Gold-clad Metals/Base Metals Nes	tonne	++	52	++	19	++	5610
Iron & Steel		**	1163362737	**	1706382409	**	1864604839
Iron & Steel (finished Steel Inc. Cr sheet)	tonne	3186329	400431501	3980055	546607218	4510768	558374413
Iron & Steel (other finished steel, Nes)	tonne	501617	2119643	355515	2355133	361517157	40086650
Iron & Steel (other finished steel, Nes)	tonne	904569	237700907	923610	259739982	2648023	249002161
Iron & Steel (scrap)	tonne	4976246	323514680	10375501	542058265	11165464	522554515
Iron & Steel (semi finished Steel Incl. Steel Ingot	tonne	1504120	116884064	2932868	265021615	5119221	392271023
Iron & Steel (sponge Iron)	tonne	37451	1088387	304299	9750521	581563	19752020
Iron & Steel (steel wire)	tonne	149312	21164209	149052	22150739	170246	22056913
Iron & Steel Material	tonne	537016	48043695	441732	43172388	598887	47191909
Iron & Steel (stainless Steel)	tonne	47633	10040638	57317	12897261	64315	10296537
Iron & Steel: Alloy Steel (granules)	tonne	18629	1463023	18666	1586757	25155	1871320
Iron & Steel: Alloy Steel (powder)	tonne	3450	911990	3473	1042530	3289	1147378
Lead & Alloys Incl. Scrap	tonne	303573	52969860	350294	62709661	498355	90626318
Lead & Alloys	tonne	225448	40056256	269596	49768026	403554	74782799
Lead Waste & Scrap	tonne	78125	12913604	80698	12941635	94801	15843519
Magnesium & Scrap	tonne	28845	10442217	32696	12691108	35204	9607721
Manganese & Alloys (incl waste & scrap)	tonne	40463	12752665	46715	10775920	52038	8914889
Manganese & Alloys Nes	tonne	621	350188	374	323699	97	15745
Manganese & Alloys (wrought & unwrought)	tonne	39842	12402477	46341	10452221	51941	8899144
Mercury	tonne	93	417082	100	375260	85	316647
Molybdenum & Scrap	tonne	554	2453374	413	2415813	602	4029973
Nickel & Alloys Incl. Scrap	tonne	51520	79431113	48896	104929668	53574	111013637
Nickel & Alloys	tonne	48438	77420105	45782	102238388	49224	107346581
Nickel Waste & Scrap	tonne	3082	2011008	3114	2691280	4350	3667056
Other Base Metals Nes	tonne	2345	1271516	646	1037729	854	1056937
Pig & Cast Iron Incl. Spiegeliesen	tonne	43732	4890464	152200	9641932	371345	15771043
Platinum Alloys & Related Metals	kg	9603	37563471	31498	122892033	10410	24329799
Other Metals of Platinum Group	kg	3583	22921897	6458	21504594	5092	15377842
Platinum (powder, unwrought & others)	kg	6020	14641574	25040	101387439	5318	8951957
Platinum Clad Base / Precious Metal	kg	178	72575	7665	17093	9315	26718
Preciuos Metals / Metals Clad with Precious Metals	tonne	320	2723812	711	10188846	128	1479307

Table- 11 (Cont.)

(Value in ₹ '000)

Metals & Alloys	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Selenium	tonne	508	712159	387	689860	506	887589
Silicon	tonne	77220	17371458	73832	17980477	85326	16189864
Silver	tonne	4422	244542763	8156	422370924	7337	451647495
Silver-clad Base Metals	kg	4862	22436	1342	26471	2060	24771
Tantalum & Scrap	tonne	5	99544	6	201090	1	133946
Tantalum Alloys other	tonne	--	--	4	165736	1	127185
Tantalum Alloys unwrought	tonne	5	99544	2	35354	++	6761
Tellurium	tonne	3	18935	2	20915	7	45287
Tin & Alloys Incl. scrap	tonne	10809	28696669	13792	32096144	12672	28051073
Tin (scrap)	tonne	--	--	20	19136	--	--
Tin & Alloys	tonne	10333	28331809	13507	31791629	12473	27799595
Tin & Alloys : Worked	tonne	476	364860	265	285379	199	251478
Titanium & Alloys (incl. waste & scrap)	tonne	9369	7998954	9970	10442098	11022	11138422
Tungsten & Alloys Incl. Scrap	kg	364880	1805805	422270	2478727	601162	3362809
Zinc & Alloys Incl. Scrap	tonne	227424	50722547	279972	73134370	293657	64537932
Zinc (scrap)	tonne	79048	14135488	91406	18814024	65612	11726817
Zinc & Alloys	tonne	148376	36587059	188566	54320346	228045	52811115
Zirconium & Scrap	tonne	45	170561	40	141013	32	94544

Source: DGCIS, Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible

Table – 12: Value of Imports of Metals & Alloys (Including Re-imports), 2021-22 to 2023-24
(By Principal Metals & Alloys)

(Value in ₹ '000)

Metals & Alloys	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	6273137692	100.00	6696396003	100.00	7916460461	100.00	18.22
Gold (non-monetary & Monetary): Total	3440928249	54.85	2804792414	41.89	3772489677	47.65	34.50
Iron & Steel	1163362737	18.55	1706382409	25.48	1864604839	23.55	9.27
Copper & Alloys (incl. brass & bronze)	526294673	8.39	612858387	9.15	764661761	9.66	24.77
Aluminium & Alloys Incl. Scrap	452887747	7.22	562908946	8.41	568263506	7.18	0.95
Silver	244542763	3.90	422370924	6.31	451647495	5.71	6.93
Nickel & Alloys Incl. Scrap	79431113	1.27	104929668	1.57	111013637	1.40	5.80
Lead & Alloys Incl. Scrap	52969860	0.84	62709661	0.94	90626318	1.14	44.52
Ferroalloys	123345004	1.97	98050366	1.46	87560715	1.11	-10.70
Zinc & Alloys Incl. Scrap	50722547	0.81	73134370	1.09	64537932	0.82	-11.75
Tin & Alloys Incl. Scrap	28696669	0.46	32096144	0.48	28051073	0.35	-12.60
Platinum Alloys & Related Metals	37563471	0.60	122892033	1.84	24329799	0.31	-80.20
Silicon	17371458	0.28	17980477	0.27	16189864	0.20	-9.96

Table- 12 (Cont.)

(Value in ₹ '000)

Metals & Alloys	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
Pig & Cast Iron Incl. Spiegeliesen	4890464	0.08	9641932	0.14	15771043	0.20	63.57
Titanium & Alloys (incl. waste & scrap)	7998954	0.13	10442098	0.16	11138422	0.14	6.67
Magnesium & Scrap	10442217	0.17	12691108	0.19	9607721	0.12	-24.30
Manganese & Alloys (incl waste & scrap)	12752665	0.20	10775920	0.16	8914889	0.11	-17.27
Cobalt & Alloys (incl Waste & Scrap)	5852552	0.09	7612854	0.11	8409176	0.11	10.46
Molybdenum & Scrap	2453374	0.04	2415813	0.04	4029973	0.05	66.82
Cadmium (incl. waste & scrap)	1218010	0.02	3321224	0.05	3589719	0.05	8.08
Tungsten & Alloys Incl. scrap	1805805	0.03	2478727	0.04	3362809	0.04	35.67
Chromium & Alloys (Including scrap)	899805	0.01	1382782	0.02	1670445	0.02	20.80
Antimony Alloys & Scrap	998247	0.02	1548561	0.02	1635339	0.02	5.60
Precious Metals / Metals Clad with Precious Metals	2723812	0.04	10188846	0.15	1479307	0.02	-85.48
Other Base Metals Nes	1271516	0.02	1037729	0.02	1056937	0.01	1.85
Selenium	712159	0.01	689860	0.01	887589	0.01	28.66
Mercury	417082	0.01	375260	0.01	316647	0.00	-15.62
Bismuth & Scrap	189060	0.00	267681	0.00	264542	0.00	-1.17
Tantalum & Scrap	99544	0.00	201090	0.00	133946	0.00	-33.39
Zirconium & Scrap	170561	0.00	141013	0.00	94544	0.00	-32.95
Tellurium	18935	0.00	20915	0.00	45287	0.00	116.53
Platinum Clad Base / Precious Metal	72575	0.00	17093	0.00	26718	0.00	56.31
Silver clad Base Metals	22436	0.00	26471	0.00	24771	0.00	-6.42
Copper (cement copper precipitated)	8758	0.00	7114	0.00	9618	0.00	35.20
Boron	2818	0.00	6094	0.00	8793	0.00	44.29
Gold-clad Metals/Base Metals Nes	52	0.00	19	0.00	5610	0.00	29426.32

Source: DGCIS Kolkata

P: Provisional

*: Not additive

--: Nil

++: Negligible

**Table – 13: Value of Imports of Metals and Alloys (Including Re-imports),
2021-22 to 2023-24**

(By Principal Countries)

Country	2021-22		2022-23		2023-24 (P)		% change in 2023-24 over 2022-23
	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	Value in ₹ '000	% share in total value	
All	6273137692	100.00	6696396003	100.00	7916460461	100.00	18.22
Switzerland	1564912576	24.95	1038839681	15.51	1500620546	18.96	44.45
UAE	591086989	9.42	530121863	7.92	943639489	11.92	78.00
China	379915515	6.06	560218892	8.37	605528349	7.65	8.09
South Africa	319969889	5.10	365349540	5.46	421700987	5.33	15.42
Japan	206956505	3.30	290528257	4.34	378010319	4.77	30.11
Korea, Rep. of	288358274	4.60	326659599	4.88	342586698	4.33	4.88
USA	223442010	3.56	410396557	6.13	342347797	4.32	-16.58
UK	194015551	3.09	367219618	5.48	311397275	3.93	-15.20
Peru	173847784	2.77	144239075	2.15	237512739	3.00	64.67
Tanzania	125279262	2.00	174978416	2.61	203209323	2.57	16.13
Australia	89277620	1.42	109001591	1.63	201757631	2.55	85.10
Indonesia	120637568	1.92	163917554	2.45	174782520	2.21	6.63
Hong Kong	137896241	2.20	143447966	2.14	161657032	2.04	12.69
Singapore	144126525	2.30	142010072	2.12	160862230	2.03	13.28
Vietnam	61293256	0.98	101564980	1.52	122484691	1.55	20.60
Malaysia	82134889	1.31	101807800	1.52	119011930	1.50	16.90
Saudi Arabia	57541485	0.92	80969103	1.21	103189065	1.30	27.44
Thailand	76222636	1.22	94288658	1.41	100689273	1.27	6.79
Germany	99632672	1.59	102659413	1.53	99809982	1.26	-2.78
Bolivia	153988602	2.45	203266092	3.04	89149559	1.13	-56.14
Ghana	80349443	1.28	115546541	1.73	88600059	1.12	-23.32
Guinea	248657333	3.96	137392312	2.05	80868093	1.02	-41.14
Uganda	5755028	0.09	36364762	0.54	69958888	0.88	92.38
Colombia	43209967	0.69	56344834	0.84	66169150	0.84	17.44
Netherlands	51426979	0.82	60836547	0.91	58321004	0.74	-4.13
Belgium	38780340	0.62	46845856	0.70	47808584	0.60	2.06
Russia	53612494	0.85	50524637	0.75	47362866	0.60	-6.26
Brazil	62749799	1.00	70866899	1.06	44307958	0.56	-37.48
Italy	36638398	0.58	42060560	0.63	43494579	0.55	3.41
Other Countries	561422062	8.95	628128328	9.38	749621845	9.47	19.34

Source: DGCIS, Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible

**Table – 14: Imports of Selected Mineral-based Products for the Period
2021-22 to 2023-24**

(Value in ₹ '000)

Mineral-based Product	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
All		**	364020772	**	541134099	**	425649072
Aluminium Fluoride	tonne	74348	5287781	55141	5621747	54864	5120665
Asbestos Cement Products	tonne	20957	618789	24361	646965	23687	549428
Asbestos Fibre Products	tonne	3353	3130952	2838	3237762	2952	3671629
Bleaching Powder	tonne	31	6228	17	5096	3	2318

Table- 14(Cont.)

(Value in ₹ '000)

Mineral-based Product	Unit	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
Boric Acid	tonne	7412	378403	5478	409638	2562	230157
Briquets of Coke, Semi-coke of Coal	tonne	108	3681	2	204	20000	238796
Calcium Carbide	tonne	22008	1576090	22133	1715011	25223	1334734
Caustic Soda	tonne	203275	7734743	135822	7422730	221351	7745256
Cement	tonne	2028180	9220479	1798578	9063366	2359188	12169042
Cement (others)	tonne	283181	2183376	223780	1819921	360030	2408961
Cement (portland & grey)	tonne	345549	1635014	196614	1094540	343909	1885874
Cement (portland & white)	tonne	187661	1473250	212205	2027762	273401	2615157
Cement Clinker	tonne	1211789	3928839	1165979	4121143	1381848	5259050
Cryolite & Chiolite (artificial)	tonne	4167	152877	1747	121744	1944	122318
Graphite (artificial)	tonne	75657	8125841	82721	12008937	101889	12796588
Graphite Bricks & Shapes	tonne	200053	23733	1017	13039	391	109315
Graphite Crucibles	tonne	296	38483	--	--	--	--
Hydrofluoric Acid	tonne	1122	133130	3729	440788	6663	734419
Manganese Oxide	tonne	29224	2103412	19809	2056547	24228	1912785
Manganese Dioxide	tonne	11268	1267664	7522	1207803	11041	1311349
Manganese Oxides(other than Mn Dioxide)	tonne	17956	835748	12287	848744	13187	601436
Non-ferrous Ash & Residues	tonne	68776	10876447	102886	11588866	130436	11390170
Other Refractory Manufactures	tonne	529721	5908835	1191604	6501448	1728808	7880804
Phosphoric Acid	tonne	6442448	186191658	2697479	289035035	2264546	181523337
Phosphorus (elemental)	tonne	52577	17336452	42176	19536933	47860	16059022
Phosphoric Fertilizers	tonne	223	37752	847	57769	2215	94383
Potash Fertilizers	tonne	3020152	77176503	2563414	120342931	3430279	104684672
Potassium Nitrate	tonne	153	16481	524	59500	1473	139736
Refractory Bricks	tonne	14632269	12745613	11943527	24402335	15645385	25307497
Silicon Carbide Crucibles	tonne	918	135424	1237	82534	64023	177254
Slag (dross etc. from Iron & steel exc. granulated)	tonne	95006	545846	91839	528734	160773	997068
Soda Ash	tonne	563139	9700159	577480	19535859	1041140	25197524
Sodium Nitrate	tonne	1207	39664	2193	162404	122	7178
Sodium Nitrite	tonne	13368	744245	10395	736934	5801	320456
Sulphur Sublimed Precipitated & Colloidal	tonne	959	335922	1036	290426	881	270187
Sulphur (colloidal)	tonne	27	10666	27	10921	19	9661
Sulphur Precipitated	tonne	39	6196	252	40331	19	3356
Sulphur Sublimed	tonne	893	319060	757	239174	843	257170
Titanium Oxide & Dioxide Total	tonne	15233	3695149	18654	5508817	17786	4862334
Titanium Dioxide	tonne	15135	3623000	18503	5392495	17584	4720880
Titanium Oxides (other than Titanium Dioxides)	tonne	98	72149	151	116322	202	141454

Source: DGCIS Kolkata

P: Provisional

**: Not additive

--: Nil

++: Negligible



10. State Reviews

7.5%

Increase recorded in the index of mineral production for Mining & Quarrying Sector (excluding Atomic and Minor Minerals) in India during 2023-24.

₹ 1,41,803 crore

value of MCDR minerals produced in the States.

1,359

mines (excluding Atomic, Fuel and Minor Minerals) reported in 2023-24.

Summary

The index mineral production for Mining & Quarrying Sector (excluding Atomic and Minor Minerals) in India showed an increase of 7.5% (as per index of mineral production base year 2011-12) during 2023-24 as compared to the previous year.

Mineral production (excluding Atomic, Fuel and Minor Minerals) was reported from 20 States and Union Territories during the year 2023-24. The value of MCDR minerals produced in the States was reported at ₹ 1,41,803 crore. The major part of the value of MCDR mineral production was confined to eight States, namely, Odisha, Rajasthan, Chhattisgarh, Karnataka, Jharkhand, Maharashtra, Madhya Pradesh and Andhra Pradesh (Table-1).

The total number of reporting mines (excluding Atomic, Fuel and Minor Minerals) at all India level were 1,359 in 2023-24. Out of these, 304 were from Madhya Pradesh, 164 from Gujarat, 126 from Karnataka, 120 from Odisha, 117 from Andhra Pradesh, 108 from Chhattisgarh, 96 from Maharashtra, 94 from Rajasthan and remaining mines were reported from other states (Table-2).

Mineral wise reserves/resources in the country as on 01.04.2020 are furnished in Table-3.

Table – 1: Value* of Mineral Production, 2021-22 to 2023-24

(By States & Union Territory)

(Value ₹ '000)

State / Union Territory	2021-22	2022-23	2023-24 (P)
India	1374219051	1233176808	1418034586
Andhra Pradesh	12220092	13532234	15007669
Assam	540157	609080	634849
Bihar	348399	327984	396419
Chhattisgarh	235140157	185214973	201642376
Goa	-	-	17421
Gujarat	7038388	7148745	7637705
Himachal Pradesh	2984595	2756602	3118011
Jammu & Kashmir	286892	297134	310646
Jharkhand	57687063	51602914	57962656
Karnataka	178162467	133754048	161035005
Kerala	370969	317936	304893
Madhya Pradesh	33579947	33729906	38637608
Maharashtra	19372221	33596406	55237866
Meghalaya	2974757	4127730	4765751
Odisha	625422945	542501227	639421108
Rajasthan	183986669	208949157	216709400
Tamil Nadu	7208720	8204069	8819511
Telangana	5993983	5744002	5565237
Uttar Pradesh	851937	689063	696538
Uttarakhand	48693	73598	113917

* Excluding the value of Atomic, Fuel and Minor Minerals.
(P) provisional

Table – 2: Number of Reporting Mines*, 2021-22 to 2023-24

(By States)

State / Union Territory	2021-22	2022-23	2023-24 (P)
India	1323	1317	1359
Andhra Pradesh	114	110	117
Assam	2	3	4
Bihar	1	1	1
Chhattisgarh	109	111	108
Goa	35	10	2
Gujarat	139	161	164
Himachal Pradesh	23	23	27
Jammu & Kashmir	10	16	19
Jharkhand	47	41	38
Karnataka	132	125	126
Kerala	1	1	1
Madhya Pradesh	263	275	304
Maharashtra	74	83	96
Meghalaya	16	17	17
Odisha	129	119	120
Rajasthan	93	92	94
Tamil Nadu	94	88	83
Telangana	38	37	33

Table- 2 (Concld.)

State / Union Territory	2021-22	2022-23	2023-24 (P)
Uttar Pradesh	2	3	3
Uttarakhand	1	1	2

* Excluding Atomic minerals, Fuel minerals and Minor minerals.

(P) provisional

Table – 3: Reserves/Resources of Minerals as on 01.04.2020: India

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Andalusite	'000 tonnes	-	126050	126050
Antimony				
Ore	tonne	7503	11180	18683
Metal	tonne	75	179.92	254.92
Apatite	tonne	29395	21080904	21110299
Asbestos	tonne	-	22908067	22908067
Bauxite	'000 tonnes	646493	4311754	4958248
Borax	tonne	-	74204	74204
Chromite	'000 tonnes	78535	253150	331685
Cobalt (Ore)	million tonnes	-	44.91	44.91
Copper				
Ore	'000 tonnes	163891	1496979	1660870
Metal	'000 tonnes	2161.57	10035.52	12197.09
Diamond	carat	847559	30876432	31723991
Diatomite	'000 tonnes	-	2885	2885
Emerald	kilogram	-	55869	55869
Fluorite	tonne	404241	20588239	20992480
Garnet	tonne	8590472	47416654	56007126
Gold				
Ore				
(Primary)	tonnes	23728100	494506270	518234370
Metal				
(Primary)	tonnes	92.76	514.5	607.26
Ore				
(Placer)	tonnes	-	26121000	26121000
Metal				
(Placer)	tonnes	-	5.86	5.86
Graphite	tonne	8563411	203060176	211623587
Iron Ore				
(Haematite)	'000 tonnes	6209034	17848870	24057905
Iron Ore				
(Magnetite)	'000 tonnes	202823	11024791	11227614
Kyanite	tonne	846865	104835455	105682321
Lead & Zinc				
Ore	'000 tonnes	103275	663222	766497
Lead Metal	'000 tonnes	1900.19	10969.8	12869.99
Zinc Metal	'000 tonnes	7438.05	25732.32	33170.37
Lead+Zinc				
Metal	'000 tonnes	-	143.13	143.13
Limestone	'000 tonnes	19028470	208560789	227589259
Magnesite	'000 tonnes	66070	393047	459118

Table- 3 (Concl'd.)

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Manganese Ore	'000 tonnes	75041	428583	503624
Marl				
Molybdenum	tonne	68145000	31053477	99198477
Ore				
Contained MoS ₂	tonne	-	27203398	27203398
Nickel Ore	million tonnes	-	16890.56	16890.56
Perlite	'000 tonnes	-	189	189
Platinum Group		-	2406	2406
of Metals	tonnes of metal			
(PGM)	contained	-	20.92	20.92
Potash	million tonnes	-	23091	23091
Pyrite	'000 tonnes	-	1674401	1674401
Rare-earth				
Elements (REE)	tonne	-	459727.49	459727.49
Rock Phosphate	tonne	30876093	280377392	311253485
Rock Salt	'000 tonnes	3860	8920	12780
Ruby	kilogram	-	5349	5349
Sapphire	kilogram	-	450	450
Sillimanite	tonne	8262300	64005091	72267391
Silver				
Ore	tonne	170446020	398197732	568643752
Metal	tonne	7707.07	22560.84	30267.91
Sulphur				
(Native)	'000 tonnes	-	210	210
Tin				
Ore	tonne	2101	83720794	83722895
Metal	tonne	973.99	102782.91	103756.9
Titanium	tonne	15998625	411108526	427107150
Tungsten				
Ore	tonne	-	89432464	89432464
Contained				
WO ₃	tonne	-	144650.07	144650.07
Vanadium				
Ore	tonne	-	24633855	24633855
Contained				
V ₂ O ₅	tonne	-	64594.01	64594.01
Vermiculite	tonne	1590996	765227	2356223
Wollastonite	tonne	2680978	22427488	25108466
Zircon	tonne	669466	1674435	2343901



Andhra Pradesh

230 thousand tonnes
of apatite resources have been reported
in Andhra Pradesh

125 thousand tonnes
of iron ore was produced in Andhra
Pradesh in 2023-24.

117
mines in the case of MCDR minerals
reported in 2023-24.

MINERAL RESOURCES

Andhra Pradesh is the leading producer of barytes, ball clay, dolomite, garnet (abrasive), laterite, limestone, quartz, quartzite, silica sand and vermiculite. It accounts for 92% barytes, 40% calcite, 41% mica, 31% each kyanite & garnet, 19% titanium minerals, 16% bauxite, 15% dolomite, 13% sillimanite and 12% each vermiculite & limestone resources of the country. Andhra Pradesh is endowed with the internationally known black, pink, blue and multicoloured varieties of granites. Krishna–Godavari basin areas in this State have emerged as new promising areas for hydrocarbons, especially natural gas.

Important minerals occurring in Andhra Pradesh are: apatite in Visakhapatnam district; asbestos in Cuddapah district; ball clay in West Godavari district; barytes in Anantapur, Cuddapah, Krishna, Kurnool, Nellore & Prakasam districts; calcite in Anantapur, Cuddapah, Kurnool & Visakhapatnam districts; china clay in Anantapur, Chittoor, Cuddapah, East Godavari, West Godavari, Guntur, Kurnool, Nellore & Visakhapatnam districts; coal in Godavari Valley Coalfield; corundum in Anantapur districts; dolomite in Anantapur & Kurnool districts; felspar in Anantapur, Cuddapah, West Godavari, Nellore & Vizianagaram districts; fireclay in Chittoor, Cuddapah, East Godavari, West Godavari, Kurnool & Srikakulam districts; garnet in East Godavari, Nellore & Srikakulam districts; granite in Anantapur, Chittoor, Cuddapah, Guntur, Krishna, Nellore, Prakasam, Srikakulam & Vizianagaram districts; iron ore (haematite)

in Anantapur, Cuddapah, Guntur, Krishna, Kurnool & Nellore districts; iron ore (magnetite) in Prakasam district; lead-zinc in Cuddapah, Guntur & Prakasam districts; limestone in Anantapur, Cuddapah, East Godavari, West Godavari, Guntur, Krishna, Kurnool, Nellore, Srikakulam, Visakhapatnam & Vizianagaram districts; manganese ore in Srikakulam & Vizianagaram districts; mica in Nellore & Visakhapatnam district; ochre in Anantapur & Cuddapah, West Godavari, Guntur, Kurnool & Visakhapatnam districts; pyrophyllite in Anantapur, Chittoor & Cuddapah districts; quartz/silica sand in Anantapur, Chittoor, Cuddapah, West Godavari, Guntur, Krishna, Kurnool, Nellore, Prakasam, Srikakulam, Visakhapatnam & Vizianagaram districts; quartzite in Kurnool, Srikakulam, Visakhapatnam & Vizianagaram districts; talc/soapstone/steatite in Anantapur, Chittoor, Cuddapah & Kurnool districts & vermiculite in Nellore & Visakhapatnam districts. Petroleum & natural gas deposits of importance

are located in the onshore and offshore areas of Krishna-Godavari basin of the State.

Other minerals that occur in the State are bauxite in East Godavari & Visakhapatnam districts; chromite in Krishna district; copper in Guntur, Kurnool & Prakasam districts; diamond in Anantapur, Krishna & Kurnool districts; gold in Anantapur, Chittoor & Kurnool districts; graphite in

East Godavari, West Godavari, Srikakulam, Visakhapatnam & Vizianagaram districts; gypsum in Guntur, Nellore & Prakasam districts; kyanite in Nellore & Prakasam districts; magnesite in Cuddapah district; pyrite in Kurnool district; sillimanite in West Godavari & Srikakulam district; silver in Guntur district; titanium minerals in East Godavari, Krishna, Nellore, Srikakulam & Visakhapatnam districts; and tungsten in East Godavari district (Tables-1 & 2).

Table –1: Reserves/Resources of Minerals as on 01.04.2020: Andhra Pradesh

Mineral	Unit	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	29395	200163	229558
Asbestos	tonne	-	797995	797995
Bauxite	000' tonnes	-	615267	615267
Chromite	000' tonnes	-	#	#
Copper				
Ore	000' tonnes	-	7582	7582
Metal	000' tonnes	-	113.7	113.7
Diamond	Crt	-	1822955	1822955
Garnet	tonne	-	17267129	17267129
Gold				
Ore (Primary)	tonne	3258100	12472898	15730998
Metal (Primary)	tonne	5.3	41.87	47.17
Graphite	tonne	-	1138275	1138275
Iron Ore (Haematite)	000' tonnes	44744	350872	395616
Iron Ore (Magnetite)	000' tonnes	-	1472383	1472383
Kyanite	tonne	-	32004228	32004228
Lead-Zinc Ore				
Ore	000' tonnes	-	22689	22689
Lead metal	000' tonnes	-	836.88	836.88
Zinc metal	000' tonnes	-	63.16	63.16
Limestone	000' tonnes	3256690	26582132	29838822
Magnesite	000' tonnes	-	80	80
Manganese Ore	000' tonnes	8088	21756	29844
Pyrite	000' tonnes	-	880	880
Sillimanite	tonne	1670025	9395739	11065764
Silver				
Ore	tonne	-	16950000	16950000
Metal	tonne	-	128.13	128.13
Titanium	tonne	-	76733874	76733874
Tungsten				
Ore	tonne	-	14802300	14802300
Contained WO ₃	tonne	-	20262.57	20262.57
Vermiculite	tonne	74193	117908	192101

Figures rounded off

Negligible

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Andhra Pradesh

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total/Godavari Valley	1 0 2 5	2 3 6 9	77 8	4 1 7 2

Source: Coal Directory of India, 2023-24.

EXPLORATION & DEVELOPMENT

The exploration & development details, if any, are covered in the Review on “Exploration & Development” under “General Reviews”.

PRODUCTION

Many important minerals are produced in Andhra Pradesh. The principal minerals produced in the state were Iron Ore, Manganese Ore, Limestone, etc. The number of reporting mines in the state was 117 in the year 2023-24 in case of MCDR minerals (Table-3).

Table – 3: Mineral Production in Andhra Pradesh, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24(p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		114	-	12220092	110	-	13532234	117	-	15007669
Gold Ore	t	-	856	-	-	31028	-	-	33687	-
Gold	kg	1*	-	-	1	4	582	1	22	25957
Iron Ore	'000t	15	319	237855	8	185	126441	9	125	107592
Manganese Ore	t	20	204002	1115138	18	213790	1387367	21	297537	1748784
Limestone	'000t	74	50278	10865618	76	53526	12016234	78	60877	13124556
Vermiculite	t	4	2370	1481	7	2303	1610	8	1127	780

(p) provisional

* Only labour reported.

MINERAL-BASED INDUSTRY

The present status of each mineral-based industry is not readily available. However, the principal mineral-based

industries in the Organised Sector in the State are provided in Table-4.

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Asbestos/Cement sheets	
Hyderabad Industries Ltd, Ibrahimpatnam, Distt Krishna	3000
Ramco Industries, Ibrahimpatnam, Distt Krishna	NA
Cement	
ACC Ltd, (formerly, Encore cement), Vishakhapatnam (G)	300
Andhra Cements Ltd, (Visaka Cement Works), Durga Nagar, Distt Visakhapatnam (G)	540
Andhra Cements Ltd, Durga Cement Works, Dachepalli, Distt Guntur	2310
Anjani Portland Cement Ltd, Chintalapalem, Mellacheruvu	1925
Bharthi Cement Corp. Pvt. Ltd, Nallingayapalli, Distt Cuddapha	5000
Bhavya Cement, Thangeda, Distt Guntur	1400
BMM Cement Ltd, Gudipadu, Yediki	950
Dalmia Cement (Bharat) Ltd, Cuddapha	4060
	2600 (Clinker)
Deccan Cement Ltd, Ravipahad, Nareducherla	1800
Deccan Cement Ltd, Ravipahad, Palakeedu Nalgonda	1800
Greygold Cement Ltd, Hyderabad	91
Himadri Cement Ltd, Vedadri, Jaggyyapet	247.5
India Cements Ltd, Chilamkur, Distt Cuddapha	1460
India Cements Ltd, Malkapur, Tandur	2900
India Cements Ltd, Vishnupuram Work, Wadapally, Mariyalaguda	3500
India Cements Ltd, Yeraguntla, Distt Cuddapha	1000
	540 (Clinker)
My home Industries Pvt. Ltd, Mellacheruvu, Nalgonda	3200
UltraTech Cements Ltd, Jaypee Balaji Cement, Budawada, Distt Krishna	5000
JSW Cement Ltd, Nandyal, Distt Kurnool	4800

Table 4 (Cont.)

('000 tpy)

Industry/plant	Capacity
JSW Cement Ltd, Bilakalagudur, Gadivemula	4800
KCP Ltd, Macherla, Distt Guntur	825
KCP Ltd, Muktyala, Distt Krishna	1860
KCP Ltd, Muktyala, Jaggayyapeta Unit II	3520
Kakatiya Cement Sugar and Industries Ltd, Dondapadu, Melacheruvu	297
My Home Cement Industries Ltd, Mulakapalli, Distt Visakhapatnam (G)	2000
NCL Industries Ltd, Kondapalli, Distt Krishna (G)	990
Orient Cement Ltd, Devapur, Kasipet	5000
Panyam Cements & Mineral Industries Ltd, Cement Nagar, Distt Kurnool	1000
Parashakti Cement, Jettipalem, Distt Guntur	1260
Penna Cement Industries Ltd, Talaricheruvu	2200
Penna Cement Industries Ltd, Boyareddypalli	2000
Penna Cement Industries Ltd, Ganeshpahad	1200
Prism Cement Ltd, Kotapadu, Kolimigundla	4800
Rain Commodities Ltd, (Rain Cements), Boincheruvupalli, Peapully, Distt Kurnool	2770
Rain Cements, Ltd, Ramapuram, Mellacheruvu	1500
Ramco Cement Ltd, (formerly, Madras Cements), Jayantipuram, KS Rajanagar, Distt Krishna	3650
Ramco Cement Ltd, Vizag Grinding Unit, Distt Visakhapatnam	950
Shree Jayajothi (Subs of Myhome Cement Ind) Yanakandala, Distt Kurnool	3200
Shri Chakra Cements Ltd, Alamada, Distt Vizianagaram (G)	260
Shri Chakra Cements Ltd, Narsimhapuri, Distt Guntur	310
Sagar Cements Bayyavaram, Distt Visakhapatnam (G)	200
Sagar Cements, Mattampally	2350
Sagar Cement Ltd, BMM Cement Anantapur	1000
UltraTech Cements Ltd, (APCW), Tadipatri, Distt Anantapur	9000
Zuari Cement, Krishnanagar, Yerranguntala, Distt Cuddapha	3800
Zuari Cement, Ltd, Sitapuram Dondapadu Mellacheruvu	1200
Chemical	
Andhra Sugars Ltd, Saggonda, Distt West Godavari	400 TPD (Caustic soda) 99 (H ₂ SO ₄)
Shree Rayalseema Alkalies & Allied Chem Ltd, Gondiparla, Distt Kurnool	156.95 (Total) 69.5 (Caustic soda) 49.8 (Cl) 24.7 (HCl) 23.1 (KOH)
Shree Rayalseema High Strength Hypo Ltd, Gondiparla, Distt Kurnool	14.85 (Bleaching powder) 49.5 (H ₂ SO ₄) 15 (Oleum)
Ceramic	
Sentini Ceramica Pvt. Ltd, Kanukollu, Distt Krishna (JV with H R Johnson (I) Ltd)	58 mill. sq.m
Spartek Ceramics India Ltd, Narsingapuram, Distt Chittoor	NA
Kajaria Ceramics Ltd, Vijayawada	2.9 mill. sq m
Electrode	
Indus Elctrode Gundlapalli, Maddiapdumandal	0.90
Magnarc Elcetodes Pvt. Ltd, Pendurthy	1.8
Fertilizer	
Agri Green Fertilizers & Chemicals Pvt. Ltd, Cuddapah	30 (SSP)
Bhaskar Fertiliser (P) Ltd, Anantapur	45 (SSP)
Coromandel International Ltd, Visakhapatnam.	1300 (NP/NPKs)
Coromandel International Ltd, Kakinada, Distt East Godavari	1925 (DAP)
GDS Chemicals & Fert. Pvt. Ltd, Anakapalli, Visakhapatnam	36 (SSP)

Table- 4 (Concl'd.)

('000 tpy)

Industry/plant	Capacity
K. P. R. Fertilizers Ltd, Biccavolu, E. Godavari	11300 (SSP)
Krishna Industrial Corpn. Ltd, Nidadavole, Distt West Godavari	45 (SSP) 33.5 (H ₂ SO ₄)
Nagarjuna Fertilizers & Chemicals Ltd, Kakinada, Distt East Godavari (Unit I & II)	1500 (Urea)
NG Fertilizers & Chemicals Pvt. Ltd, Kodurupadu, Distt Krishna	200 (SSP)
Prathyusha Chems and Fertilisers Ltd, Parwada, Visakhapatnam	100 (SSP)
Subhodaya Chemicals Ltd, Gauripatnam, Distt West Godavari	42.9 (SSP)
Pesticides	
Jayalakshmi Fertilizers, Tanuku, Distt West Godavari	2000
Glass	
Triveni Glass Ltd, Kondagudem, Distt West Godavari	10 mill. sq. m
Iron & Steel	
Visakhapatnam Steel Plant, Visakhapatnam	8856 (sinter) 3400 (pig iron) 6300 (Crude/liquid steel)
Sponge Iron	
Amoda Iron and Steel Pvt. Ltd, Jayanthipuram, Jaggayyapet Mandal, Aggayyapet	60
Apple Industries Ltd, Dhiral, Anantapur	150
Maa Mahamaya Industries Ltd, Vizianagaram	112
Pushpit Steel Pvt. Ltd, Merlapaka, Yerpandu, Chittoor	86.4
SLV Steels and alloys Pvt. Ltd, Anantapur	60
Sri Sai Sindhu Industries Ltd, Tadpatri	52.5
Steel exchange India Ltd, Srirampuram, Visakhapatnam	250
Sree Rayalseema Green Steloy Ltd, Gooty, Distt Anantapur	36
Pig Iron	
Rishrtriya Ispat Nigam Ltd, Vishakhapatanam, Andhra Pradesh	6300
Sathavahana Ispat Ltd, Haresamudram, Distt Anantapur	210
Pellets	
Essar Steel Ltd, Visakhapatnam	8000
Ferroalloys	
Berry Alloys Ltd, Kothavalasa, Distt Vizianagaram	"40 (Fe-Mn) 32 (Si-Mn)"
Deccan Ferro alloys (P) Ltd, Pendurthi, Visakhapatnam	"30 (Si-Mn) 10(Fe-Mn)"
FACOR Alloys Ltd, Shreeramnagar, Distt Vizianagaram	90.3
Jindal Stainless (Hisar) Ltd, Kothavalasa, Distt Vizianagaram	40
Maithan Alloy Ltd, Atchutapuram	120
Hira Elector Smelters Ltd, Bobbili, Distt Vizianagaram	NA
Nava Bharat Ventures Ltd, Paloncha	125
Rhodium Ferro-alloy Pvt. Ltd, Gollapuram	8
Siri Smelters & Energy Pvt. Ltd, Bobbili	8.5
Petroleum Refinery	
HPCL, Vizag	8300
ONGC, Tatipaka, Distt East Godavari	66

Note: Data, not readily available for fertilizer and cement Industries on respective website. Therefore, data is sourced from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.



Arunachal Pradesh

76 million tonnes
of Graphite Resources have been
established in Arunachal Pradesh

53 m cu. m
Natural gas (ut.) and 47 thousand tonnes
petroleum (crude) were produced in
Arunachal Pradesh in 2022-23

₹ 56 crore
value of Minor Minerals production
were reported in 2022-23

MINERAL RESOURCES

The most important mineral resource of the State is petroleum & natural gas and its chief occurrence is reported in Ningru and Dam Duma areas. These hydrocarbon deposits are located in the Assam Arakan Fold Belt (AAFB) and Upper Assam basin in the State. The State also reports resources of coal in Namchick Namphuk and Miaobum Coalfields; copper in East Kameng district, dolomite in West Kameng district; fuller's earth in Tirap district; graphite in Lohit, East Siang and Upper Subansiri districts; limestone in Dibang Valley, Lohit, East Siang and Upper Subansiri districts and quartzite in West Kameng district (Tables-1 & 2).

EXPLORATION & DEVELOPMENT

The exploration & development details, if any, are covered in the Review on "Exploration & Development" under "General Reviews".

PRODUCTION

Natural gas (ut.) and petroleum (crude) were the important mineral items produced in Arunachal Pradesh. The value of Minor Minerals' production was estimated at 56 crore for the year 2022-23 (Table - 3).

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Arunachal Pradesh

Mineral	Unit	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Copper				
Ore	'000 tonnes	10	10	10
Metal	'000 tonnes	0 .02	0 .02	0 .02
Graphite	tonne	73118257	76318257	76318257
Limestone	'000 tonnes	1	482796	482796

Figures rounded off.

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Arunachal Pradesh

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	31	40	19	90
Namchik- Namphuk	31	40	13	84
Miao Bum	–	–	6	6

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Arunachal Pradesh, 2020-21 to 2022-23

(Excluding Atomic Minerals)

(Value in ₹ '000)

Mineral	Unit	2020-21			2021-22			2022-23 (P)		
		No. of mines	Quantity	Value ^{\$}	No. of mines	Quantity	Value ^{\$}	No. of mines	Quantity	Value
All Minerals		-	-	397944	-	-	416302	-	-	558770
Natural Gas (ut.)	m c m	-	56	-	-	58	-	-	53	-
Petroleum (crude)	'000t	-	54	-	-	48	-	-	47	-
Minor Minerals		-	-	397944	-	-	416302	-	-	558770

\$: Excludes the value of Fuel minerals.



Assam

Coal, petroleum & natural gas, limestone, fuller's earth, sillimanite and minor minerals are the chief mineral resources of the State

₹ 63.48 crore
Estimated value of limestone production was reported during the year 2023-24.

4
mines reported in Assam in case of MCDR minerals in 2023-24.

MINERAL RESOURCES

Coal, petroleum & natural gas, limestone, fuller's earth, sillimanite and minor minerals are the chief mineral resources of the State. Coal occurs in Mikirs Hills, Dilli-Jeypore, Makum and Singrimari coalfields. Coal extracted from the State is friable and contains high sulphur. Petroleum & natural gas occurs in Digboi oilfields, Lakhimpur district and at Moran Rudrasagar oilfields in Sivasagar district located in Assam Arakan Fold Belt (AAFB), Upper Assam and Assam basins. Limestone occurs in Karbi Anglong, North Cachar Hills and Nowgaon districts. Besides, china clay occurs in Karbi Anglong and North Lakhimpur districts; fireclay in Dibrugarh, Karbi Anglong, North Cachar Hills & North Lakhimpur districts; fuller's earth in Nalbari district; granite in Goalpara, Kamrup & Karbi Anglong districts, iron ore (haematite) in Kokrajhar district; iron ore (magnetite) in Dhubri, Goalpara & Kokrajhar districts; quartz/silica sand in Nowgaon district; and sillimanite in Karbi Anglong & Nowgaon districts. The reserves / resources of coal and minerals are furnished in (Tables-1 & 2).

EXPLORATION & DEVELOPMENT

The Exploration & Development details for the State of Assam during the year 2023-24 are covered under the Review on Exploration & Development.

PRODUCTION

Limestone and Sulphur (recovered as by-product from oil refinery) were the principal minerals produced in Assam in 2023-24. The value of limestone production was reported at

₹ 63.48 crore for the year 2023-24. There were 4 reporting mines of limestone in 2023-24 in Assam in case of MCDR minerals (Table-3).

MINERAL-BASED INDUSTRY

The present status of each mineral-based industry is not readily available. However, as per the available information, the principal mineral-based industries in the organised sector in the State are furnished in (Table- 4).

Table – 1: Reserves/Resources of Coal as on 01.04.2024: Assam

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
	464.78	57.21	3.02	525.01
Singrimari	0	14.49	0	14.49
Makum	432.09	20.7	0	452.79
Dilli-Jeypore	32	22.02	0	54.02
Mikir Hills	0.69	0	3.02	3.71

Source: Coal Directory of India, 2023-24.

Table – 2: Reserves/Resources of Minerals as on 01.04.2020: Assam

Mineral	Unit	Reserves	Remaining resources	Total Resources
Iron Ore	'000 tonnes	-	30890	30890
(Haematite)				
Iron Ore	'000 tonnes	-	15380	15380
(Magnetite)				
Limestone	'000 tonnes	188130	1683540	1871670
Sillimanite	tonne	-	4604700	4604700

Figures rounded off

Table – 3: Mineral Production in Assam, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		2	-	540157	3	-	609080	4	-	634849
Limestone	'000t	2	1681	540157	3	1812	609080	4	1996	634849
Sulphur #	t	-	6545	-	-	7705	-	-	9148	-

(p) provisional

Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

(Excluding Atomic, Fuel and Minor Minerals)

('000 tpy)

Industry/plant	Capacity
Asbestos Products	
Assam Roofing Ltd, Bonda, Distt. Kamrup	58
Cement	
Barak Valley Cements Ltd, Jhoom basti,	330
Badarpurghat, Distt. Karimganj.	
Calcom Cement (Dalmia Subsidiary),	1720
Distt. Nagaon	
CCI Ltd, Bokajan, Distt. Karbi Anglong.	200
Cement Manufacturing Co. Ltd, Chamata	2000
Pathar, P. O. Sonapur, Distt. Kamrup (G)	
Purbanchal Cement, Vill. Sarutari,	360
Distt Kamrup	
Topcem Cement Gauripur Kamrup	660
Fertilizer	
Assam State Fertilizer & Chemicals Ltd	33 (SSP)

Table- 4 (Concl'd.)

Industry/plant	Capacity
Chandrapur, Distt. Kamrup	16.5
H_2SO_4	
Brahmaputra Valley Fertilisers Corpn. Ltd, (Urea)	510
Namrup (Namrup II & III), Distt Dibrugarh Progressive Fertichem Pvt. Ltd, Topatoli, Kamrup	45 (SSP)
Iron & Steel	
Shri Ganapati Ispat Pvt Ltd, Tinsukia.	NA
Petroleum Refinery	
Indian Oil Corporation, Bongaigao.	2350
Indian Oil Corporation, Moonmati, Guwahati	1000
Indian Oil Corporation, Digboi	650
NRL, Numaligarh, Golaghat	3000

Note: Data, as not readily available for fertilizer and cement industries on respective websites, was taken from Indian Fertilizer Scenario, FAI Statistics and Survey of Cement Industry & Directory, respectively.

As per Rule-45 of MCDR, 2017, a total of 36 companies are registered with IBM as given in <https://ibm.gov.in/writereaddata/files/05102020114426ASSAMALL.pdf>



Bihar

94%

Pyrite resources of India is located in Bihar.

₹ 39.64 crore

value of Minerals Production were reported in Bihar in 2023-24.

1

Mine reported limestone production in Bihar in 2023-24.

MINERAL RESOURCES

Bihar principally holds the country's major pyrite resources and is acknowledged to possess 94% of pyrite resources. The important mineral occurrences in Bihar are coal in Rajmahal coalfield; limestone in Kaimur (Bhabhua), Monghyr & Rohtas districts; mica in Nawada district; quartz /silica sand in Bhagalpur, Jamui, Monghyr & Nalanda districts; quartzite in Lakhisarai, Monghyr & Nalanda districts; and talc/ soapstone/steatite in Monghyr district. Besides, occurrences of bauxite in Monghyr & Rohtas districts; china clay in Bhagalpur & Monghyr districts; felspar in Gaya, Jamui & Monghyr districts; fireclay in Bhagalpur & Purnea districts; gold in Jamui district; granite in Bhagalpur, Gaya, Jahanabad & Jamui districts; iron ore (haematite) in Bhagalpur district; iron ore (magnetite) in Gaya & Jamui districts; lead-zinc in Banka & Rohtas districts; and pyrites in Rohtas district have been reported (Tables - 1 & 2).

PRODUCTION

In 2023-24, the mineral production in the State of Bihar was mainly reported for limestone. The production of limestone increased by 7.53% to 9,99,000 tonnes in 2023-24 from 9,29,000 tonnes in the preceding year. Furthermore, the value of all minerals in the State increased by 20.86% to ₹ 39.64 crore in 2023-24 from ₹ 32.79 crore in the preceding year. There was only one mine in Bihar for MCDR minerals

that reported production of limestone. Sulphur was also produced as a by-product from oil refinery in the state. (Table- 3).

MINERAL-BASED INDUSTRY

The present status of each mineral-based industry is not readily available. However, the principal mineral-based industries in the Organised Sector in the State with their total installed capacities are furnished in Table- 4.

Table – 1: Reserves/Resources of Mineral as on 01.04.2020: Bihar

Mineral	Unit	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	'000 tonnes	-	4114	4114
Gold				
Ore	tonne	-	222884860	222884860
Metal	tonne	-	37.6	37.6
Iron Ore (Haematite)	'000 tonnes	-	55	55
Iron Ore (Magnetite)	'000 tonnes	-	49439	49439
Lead-Zinc Ore				
Ore	'000 tonnes	-	11435	11435
Lead metal	'000 tonnes	-	24	24
Zinc metal	'000 tonnes	-	38.75	38.75
Limestone	'000 tonnes	11807	994188	1005995
Potash	Million tonnes	-	230	230
Pyrite	'000 tonnes	-	1574561	1574561
Rare-Earth Elements	tonne	-	1459	1459

Figures rounded off

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Bihar

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total/ Rajmahal	2346.36	3014.65	36.66	5397.67

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in Bihar, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		1	-	348399	1	-	327984	1	-	396419
Limestone	'000t	1	987	348399	1	929	327984	1	999	396419
Sulphur #	t	-	8160	-	-	14652	-	-	16240	-

(P): provisional

#: Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

Industry/plant	"Capacity ('000 tpy)"
Cement	
Eco cement Durgawati Bhabhua	1000
Kalyanpur Cements Ltd, Banjari, Distt Rohtas	1000
Kanodia Cement Bhabhua Bangar Cement	1200
Shree Cement Ltd, Jasoia Aurangabad Grinding Unit, Aurangabad	3600
Shree Cement Ltd, New Bihar Cement plant, Aurangabad	2000
UltraTech Cement plant, Patliputra	1900
Petroleum Refinery	
Indian Oil Corporation, Barauni	6000



Chhattisgarh

36%

of country's tin ore resources is located in Chhattisgarh.

46

million tonnes of iron ore and 49 million tonnes of limestone were produced in the State in 2023-24

108

Reporting mines in case of MCDR minerals were reported in 2023-24.

MINERAL RESOURCES

Chhattisgarh is the sole producer of tin concentrates and moulding sand. It is one of the leading producers of coal, dolomite, bauxite and iron ore. The State accounts for about 36% tin ore, 20% iron ore (haematite), 18% coal, 11% dolomite and 4% each of diamond & marble resources of the country. Important mineral occurrences in the State are bauxite in Bastar, Bilaspur, Dantewada, Jashpur, Kanker, Kawardha (Kabirdham), Korba, Raigarh & Sarguja districts; china clay in Durg & Rajnandgaon districts; coal in Korba, Raigarh & Sarguja districts; dolomite in Bastar, Bilaspur, Durg, Janjgir–Champa, Raigarh & Raipur districts; and iron ore (haematite) in Bastar district, Bailadila deposit in Dantewada district, Chhote Dongar deposit in Kanker district, Rowghat, Chargaon, Metabodeli & Hahaladdi deposits in Rajnandgaon district and Boria Tibbu deposits in Dalli–Rajhara area, Durg district. Bailadila–Rowghat hill ranges in the State are considered to be one of the biggest iron ore fields in India. Limestone occurs in Bastar, Bilaspur, Durg, Janjgir–Champa, Kawardha (Kabirdham), Raigarh, Raipur & Rajnandgaon districts; quartzite in Durg, Raipur, Rajnandgaon & Raigarh districts; and talc/soapstone/ steatite in Durg & Kanker districts.

Other minerals found in the State are corundum in Dantewada district; diamond and other gemstones in Raipur, Mahasamund & Dhamtari districts; fire clay in Bilaspur, Raigarh & Rajnandgaon districts; fluorite in Rajnandgaon district; garnet & marble in Bastar district; emerald & gold in Raipur district; granite in Bastar, Kanker & Raipur districts; quartz/silica sand in Durg, Jashpur, Raigarh, Raipur & Rajnandgaon districts; and tin in Bastar & Dantewada districts (Table-1). The reserves/ resources of coal are furnished in Table-2.

EXPLORATION & DEVELOPMENT

The Exploration & Development details for the State of

Chhattisgarh during the year 2023-24 are covered under the Review on Exploration & Development.

PRODUCTION

Bauxite, Iron Ore, Tin Conc., Graphite (r.o.m.), Limestone and Moulding Sand are the major minerals produced in Chhattisgarh. There were 108 reporting mines in 2023-24 for MCDR minerals. The production value of all minerals (except atomic, fuel & minor minerals) was estimated at ₹ 20,164 crore for the year 2023-24 (Table-3).

MINERAL-BASED INDUSTRY

The principal mineral-based industries in the Organised Sector in the State are furnished in Table- 4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Chhattisgarh

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	000' tonnes	23695	968860	992555
Diamond	carat	-	1304000	1304000
Fluorite	tonne	-	545455	545455
Garnet	tonne	-	28800	28800
Gold				
Ore (Primary)	tonne	-	4841033	4841033
Metal (Primary)	tonne	-	5.51	5.51
Graphite	tonne	5282	1330	6612
Iron Ore (Haematite)	000' tonnes	1593732	2998379	4592111
Iron Ore (Magnetite)	000' tonnes	75876	30045	105921
Limestone	000' tonnes	1486351		13211218
Tin				
Ore	tonne	2101	29795176	29797277
Metal	tonne	973.99	15909.58	16883.57

Figures rounded off.

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Chhattisgarh

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	40078	41093	1495	82666
Sohagpur	94	434	-	529
Sonhat	950	1988	2	2940
Jhilimili	228	39	-	267
Chirimiri	320	11	31	362
Bisrampur	2014	678	5	2698
East Bisrampur	-	629	-	629
Lakhanpur	456	3	-	459
Panchbahini	-	11	-	11
Hasdeo-Arand	2032	3273	223	5529
Sendurgarh	153	126	-	279
Korba	8769	4212	49	13030
Mand-Raigarh	22933	26623	1099	50656
Tatapani-Ramkola	2128	3064	85	5278

Figures rounded off.

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Chhattisgarh, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24(p)		
		No. of Mines	Quantity	Value	No. of mines	Quantity	Value	No. of Mines	Quantity	Value
All Minerals		109	-	235140157	111	-	185214973	108	-	201642376
Bauxite	t	13	968248	1079736	11	1056576	1215499	15	1033176	1230964
Iron Ore	'000t	21	41333	222498723	20	42487	170877445	21	46043	186363948
Tin Conc.	kg	6	26301	32619	4	45432	52285	4	22345	17341
Graphite (r.o.m.)	t	1*	-	-	1*	-	-	1	1007	1316
Limestone	'000t	64	41889	11523823	72	44777	13064371	63	49279	14021421
Moulding Sand	t	4	17583	5256	3	17260	5373	4	24494	7386

(p) provisional

* Only labour reported.

Table – 4: Principal Mineral-based Industries

(’000 tpy)

Industry/plant	Capacity
Aluminium	
Bharat Aluminium Co. Ltd (Unit I & II), Korba.	200 # (Alumina)
(# Plants remained non-operational during the year)	570 (Aluminium)
Cement	
ACC Ltd, Jamul, Distt Durg.	2400
Ambuja Cements Ltd, Bhatapara, Distt Raipur	3500
Bhilai Jaypee Cement Ltd, Bhilai, Distt Durg (G)	2200
Century Cement, Baikunth, Distt Raipur	2100
Century Textile & Industries Ltd, Tandwa, Tilda	2400
Emami Ltd, Baloda Bazar, Distt Raipur	2500
	3200 (clinker)
Emami Ltd, Risda Baloda Bazar, Distt Raipur	2500
Emami Cement Ltd, Risda Baloda Bazar	3000
J. K. Laxmi, Durg	2.7
J. K. Laxmi Cement Ltd, Malpurikhurd, khasadhe, Dhamdha	2400
Lafarge India Pvt. Ltd, Arasmeta, Distt Janjgir-Champa.	1.8
Lafarge India Pvt. Ltd, Sonadih, Distt Raipur	550
NUVOCO Vistas Co. Ltd	1000
Sonadih Cement Plant, Rasedi, Baloda Bazar	
Shree Cement, Baloda Bazar, Distt Raipur	3000
Shree Cement Ltd, Khapradih Simga, Balrampur	3000
UltraTech Cement Ltd, Hirni, Distt Raipur	1900
UltraTech Cement Ltd, Rawan, Distt Raipur	2500
Chemical	
Indu Ragukul Food & Chemical Pvt. Ltd	1.5 (Sodium dicromate)
	2.7 (Sodium chromate)
Electrode Rajghatta, Kharsia	1.35 (Sodium sulphate)
Bhanpuri, Raipur Calcutta Electrode Pvt. Ltd	7.4
Fertilizer	
Dharamsi Morarji Chemical Co. Ltd, Kumhari, Distt Durg.	183 (SSP & H ₂ SO ₄)
	66 (SSP)
Khaitan Chemicals & Fertilizers Ltd, Distt Rajnandgaon	49.5 (H ₂ SO ₄)
Iron & Steel	
Bhilai Steel Plant, Bhilai	6334 (Sinters)
	4700 (Pig iron)
	3925 (crude/liquid steel)
	30 (Refractory bricks)
Jindal Steel & Power Ltd, Raigarh	2500 (Sinters)
	1320 (Sponge iron)
	8600 (Crude/liquid)
Jayaswal NECO Industries Ltd, Siltara, Distt Raipur	650 (Pig iron)
	255 (Sponge iron)
	1200 (Pellets)
	1200 (Steel)

Table 4 (Cont.)

Industry/plant	Capacity (^{'000 tpy})
Monnet Ispat & Energy Ltd, Naharpalli, Raigarh	962.3 (Sinters)
	612.5 (Pig iron)
	750 (MS billet)
	450 (TMT Bar)
Sarda Energy & Minerals Ltd, (formerly Raipur Alloys & Steel Ltd), IGC,	600 (Pellets)
	210 (Sponge iron)
Shri Bajrang Power & Ispat Ltd, Borjhara, Distt Raipur	130 (Steel)
	1200 (Pellets)
Sponge Iron	
A.P.I. Ispat & Power Tech. Pvt. Ltd, Siltara Billets, Raipur	210
Alliance Integrated Metallica Ltd, Bemta, Distt Raipur.	500
Anjani Steel Ltd, Ujalpur, Distt Raigarh	108
Arti Sponge & Power Ltd, Siltara, Distt Raipur	60
Ambika Ispat (I) Pvt Ltd, Tarainal, Distt Raigarh	30
Baldev Alloys Pvt. Ltd, Siltara, Raipur	30
Bhagavati Power & Steel Pvt Ltd, Siltara, Distt Raipur	60
B.S. Sponge Pvt Ltd, Taraimal, Raigarh	90
Crest Steel & Power Pvt. Ltd, IGC Borai, Distt Durg	231
Devi Iron & Power Pvt Ltd, Tandira, Distt Raipur	90
Drolia Electro Steel Pvt Ltd, Siltara, Raipur	66
Euro Pratik Ispat Pvt Ltd, Charoda, Distt Raipur	30
Gravity Treksim Pvt Ltd, Siltara, Distt Raipur	30
Godavari Power & Ispat Ltd, Siltara, Distt Raipur	495
	2100 (Pellets)
Gopal Sponge & Power Pvt Ltd, Siltara, Distt Raipur	30
Gitanjali Ispat & Power Pvt Ltd, Sirgititi, Distt Bilaspur	10
GR Sponge & Power Ltd, Siltara, Distt Raipur	72
Shree Hare Krishna Sponge Iron Ltd, Siltara, Distt Raipur	12000
Jai Shree Balaji Steel Pvt Ltd (HEG Ltd), Borai, Distt Durg	120 (Sponge iron)
Hi-Tech Power & Steel Ltd, Parsada, Distt Raipur	60
Khetan Sponge & Infrastructure Pvt. Ltd, Sarora, Distt Raipur	30
Maa Kali Alloys (Ind.) Pvt Ltd, Pali, Distt Raigarh	60
MSP Steel & Power Ltd, Raigarh	192
	900 (pellets)
Monnet Ispat & Energy Ltd, Hasaud, Raipur	300
	250 (Semi-finished Steel)
	150 (Finished Steel)
Monnet Ispat & Energy Ltd, Naharpalli, Raigarh	500
NR Sponge Pvt. Ltd, Raipur	90
Nalwa Steel & Power Ltd, Taraimal, Raigarh	198
Nakoda Ispat Ltd, Siltara, Raipur	171
Niros Ispat Pvt. Ltd, Hathkhoj, Bhilai	97.5
Nova Iron & Steel Ltd, Dagori, Bilaspur	150
Nutan Ispat & Power Ltd, Jaroda, Raipur	60
PD Industries Pvt. Ltd, Siltara, Raipur	60
Prakash Industries Ltd, Hathenewra, Janjgir-Champa	1000
Raigarh Ispat & Power Ltd, Delari, Distt Raigarh	60
Rameswaram Steel & Power Ltd, Gharghoda, Distt Raigarh	72
Real Ispat and Power Ltd, Borjhara, Raipur.	60

Table 4 (Cont.)

Industry/plant	Capacity (^{'000 tpy})
	460 (Finished Steel)
Sarda Energy & Minerals Ltd, Mandhar, Raipur	150
Hanumant Alloys (India) Pvt. Ltd, Hardikala, Distt Bilaspur	16.5
Shivalaya Ispat & Power Pvt Ltd, Guma, Distt Raipur	90
Shivshakti Steel Pvt. Ltd, Chakradharpur, Distt Raigarh	97.5
Shree Shyam Ispat (India) Pvt. Ltd, Taraimal, Raigarh	120
Singhal Enterprises Pvt Ltd, Taraimal, Distt Raigarh	253.5
Singhal Energy Pvt. Ltd, Taraimal, Raigarh	60
Sree Nakoda Ispat Ltd, Siltara, Distt Raipur	66
Sunil Sponge Iron Ltd, Chiraipani, Distt Raigarh	60
Sunil Sponge Pvt. Ltd, Munrethi, Dharsiwa	60
Topworth Steel Pvt Ltd, Rasmada, Distt Durg	165
Trimula Sponge Iron Pvt Ltd, Siltara, Raipur	30
Vandana Global Ltd, Siltara, Distt Raipur	216
Vasvani Industries Ltd, Siltara, Distt Raipur	90
Vidhyan Minerals India Pvt. Ltd, Bilaspur	30
Ferroalloys	
Alok Ferro Alloys Ltd, Urla, Raipur	18
Deepak Ferro Alloys Ltd, Urla, Distt Raipur	5
Indsil Energy & Electro Chemical Ltd, Urla, Distt Raipur	19.2
Hira Ferro alloys Ltd, Urla, Distt Raipur	61.5
Jindal Steel & Power Ltd, Kharsia, Distt Raigarh	36
Sarda Energy & Minerals Ltd, (merged Chhattisgarh Electricity Co. Ltd)	600 (Pellets) 360 (Sponge Iron) 410 (Finished Steel)
Siltara, Distt Raipur	150 (Silico & Ferro- manganese)
Nav-chrome Ltd, Urla, Distt Raipur	50
Orion Ferroalloy Pvt. Ltd Punjipathra, Gharghora	8
V.A. Power & Steel Pvt. Ltd Punjipathra, Gharghoda	8.1(Fe-Si) [#]
Manganese Oxide	
Vandana Allied Minerals and alloy Bodegaon, Durg	3.6
Refractory	
SAIL Refractory Unit (formerly Bharat Refractories Ltd), Bhilai, Distt Durg	60
Vishva Vishal Engineering Ltd, Bhilai, Distt Durg	8.2

(G): Grinding Unit

Note: Data, not readily available for fertilizer and cement industries on respective websites, is taken from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.

[#] plant remained unoperational during the year



Delhi

5,289 thousand tonnes

Total Resources of China clay have been established in Delhi

Fireclay and silica material known as Badarpur sand/quartzite occurrences have also been reported from Delhi.

MINERAL RESOURCES

Kaolin deposits are found in areas west of Qutub Minar at Mehrauli, Masoodpur, Kusumpur and Mahipalpur. In addition, occurrences of fireclay and silica material known as Badarpur sand/quartzite have also been reported from Delhi (Table - 1).

MINERAL-BASED INDUSTRY

The principal Mineral-based Industries in the Organised Sector with their total installed capacities are furnished in Table-2.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Delhi

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
China clay [#]	'000 tonnes	-	5289	5289
Fireclay [#]	'000 tonnes	-	64	64

Figures rounded off.

#: Declared as Minor Mineral vide Gazette Notification dated 10.02.2015.

Table – 2: Principal Mineral-based Industries

Industry	No. of units	Total installed capacity (tpy)
Abrasive	1	120
Activated earth	1	3600
Alum	1	35500
Asbestos products	1	25000
Bleaching powder	1	10000
Caustic soda	1	15080
Ceramic & stoneware pipes	2	18500
S. R. Industries Electrode, Narela	1	1800
Refractory	2	39000



Goa

1,198 million tonnes
Total Resources of Iron Ore (Haematite) have been established in State.

₹ 1.74 crore
value of major mineral (Iron ore) production was reported in 2023-24.

2
Mines in case of MCDR minerals reported production in 2023-24.

MINERAL RESOURCES

Goa is well-known for its iron and manganese ores. Bauxite and laterite are the other minerals produced in the State. Iron and manganese ore belts extend from south-east to north-west of the state. Manganese ores are associated with iron ores and occur as pockets of various sizes in the form of concretionary pebbles in shales. Important iron ore and manganese ore deposits are located at Bicholim, Sanguem and Satari talukas. Bauxite occurs in the North and South Goa districts; kaolin reportedly occurs in South Goa district, while quartz/silica sand deposits occur in both North and South Goa districts (Table-1).

PRODUCTION

Iron ore was the only major mineral produced in Goa in the year 2023-24. Its value of production was estimated at ₹ 1.74 crore. There were 02 reporting mines in 2023-24 in case of MCDR minerals (Table-2).

MINERAL-BASED INDUSTRY

The present status of each Mineral-based Industry is not readily available. However, the principal Mineral-based Industries in the Organised Sector in the state are provided in Table-3.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Goa

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	'000 tonnes	9613	58244	67857
Iron ore (Haematite)	'000 tonnes	117235	1080322	1197557
Iron ore (Magnetite)	'000 tonnes	4990	261345	266336
Manganese Ore	'000 tonnes	65	34436	34501

Figures rounded off.

Table – 2: Mineral Production in Goa, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹'000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		35	-	-	10	-	-	2	12	17421
Iron Ore	'000t	33**	-	-	8*	-	-	1	12	17421
Manganese Ore *	t	2	-	-	2	-	-	1	-	-

(P) Provisional

** Only labour reported, production activity stopped by S.C. Order.

* Only labour reported.

Table – 3: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Pellets	
Mandovi Pellets Ltd, Mandovi, Shiroda.	NA
Chowgule & Co. Ltd	NA
Pig Iron	
Sesa Goa Ltd, Bicholim.	625
Aparant Iron & Steel Pvt. Ltd, Sanguem	160
Vedanta Ltd, Amona, Bicholim	832 (Sinter)
Sponge Iron	
Ambey Metallic Ltd, Pissurlem, Sattari.	36
Goa Sponge & Power Ltd, Santona.	90
Shraddha Ispat Pvt. Ltd, Santona, Sanguem.	72
Ferroalloys	
Karthik Alloys Ltd, Cuncalim.	3.2

Note: Data, for fertilizer industries, is taken from Indian Fertilizer Scenario, FAI Statistics.



Gujarat

Gujarat is the sole holder of the country's chalk, marl and perlite resources.

₹ 763.77 crore
value of mineral production in Gujarat.

164
mines reported in 2023-24 in case of MCDR minerals.

MINERAL RESOURCES

Gujarat is the sole producer of chalk and is the principal producer of clay (others), fluorite (graded), kaolin, silica sand, lignite, petroleum & natural gas and marl in the country. The State is the sole holder of the country's chalk, marl and perlite resources and possesses 66% fluorite, 28% diatomite, 25% bentonite, 18% granite, 12% wollastonite, 10% limestone and 9% bauxite resources.

The important mineral occurrences in the State are: Bauxite in Amreli, Bhavnagar, Jamnagar, Junagadh, Kheda, Kachchh, Porbandar, Sabarkantha & Valsad districts; Ball clay in Banaskantha, Bharuch, Kachchh & Patan districts; Bentonite in Amreli, Bhavnagar, Jamnagar, Kachchh & Sabarkantha districts; China clay in Amreli, Banaskantha, Bhavnagar, Jamnagar, Junagadh, Kachchh, Mahesana & Sabarkantha districts; Chalk in Porbandar district; Diatomite in Bhavnagar district; Dolomite in Bhavnagar & Vadodara districts; Fireclay in Bharuch, Kachchh, Mahesana, Rajkot, Sabarkantha, Surat & Surendranagar districts; Fluorite in Vadodara & Bharuch districts; Gypsum in Bhavnagar, Jamnagar, Junagadh, Kachchh and Surendranagar districts; Lignite in Bharuch, Bhavnagar, Kachchh & Surat districts; Limestone in Amreli, Banaskantha, Bharuch, Bhavnagar, Jamnagar, Junagadh, Kheda, Kachchh, Panchmahals, Porbandar, Rajkot, Sabarkantha, Surat, Vadodara & Valsad districts; Marl in Amreli, Junagadh & Porbandar districts; Ochre in Banaskantha, Bhavnagar & Kachchh districts; Perlite in Rajkot district; Petroleum and Natural Gas in oil fields of Ankaleshwar, Kalol, Navgam, Balol & Cambay in Cambay onshore and offshore basins; Quartz/Silica sand in Bharuch, Bhavnagar, Dahod, Kheda, Kachchh, Panchmahals, Rajkot, Sabarkantha, Surat, Surendranagar, Vadodara & Valsad districts; and Talc/ Soapstone/Steatite in Sabarkantha district.

Other minerals that occur in the State are: Apatite and Rock Phosphate in Panchmahals district; Calcite in Amreli & Bharuch districts; Copper ore in Banaskantha district; Granite in Banaskantha, Mahesana & Sabarkantha districts; Graphite in Panchmahals district; Lead-Zinc and Marble in Banaskantha & Vadodara districts; Manganese ore in Panchmahals & Vadodara districts; Vermiculite in Vadodara district; and Wollastonite in Banaskantha district. The Lignite resources are located in Bharuch, Bhavnagar, Kachchh and Surat districts (Tables- 1 & 2).

Table – 1 : Reserves/Resources of Minerals as on 01.04.2020 : Gujarat

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	-	351000	351000
Bauxite	'000 tonnes	101230	295797	397027
Copper				
Ore	'000 tonnes	-	12613	12613
Metal	'000 tonnes	-	200.74	200.74
Diatomite	'000 tonnes	-	811	811
Fluorite	tonne	-	14355280	14355280
Graphite	tonne	-	3355805	3355805
Lead-zinc Ore				
Ore	'000 tonnes	-	5682	5682
Lead metal	'000 tonnes	-	208.45	208.45
Zinc metal	'000 tonnes	-	261.4	261.4
Lead-zinc metal	'000 tonnes	-	0.9	0.9
Limestone	'000 tonnes	903115	21929169	22832284
Manganese Ore	'000 tonnes	695	2180	2875
Marl	tonne	68145000	31053477	99198477
Perlite	'000 tonnes	-	2406	2406
Rare-earth Elements	tonne	-	424000	424000
Rock Phosphate	tonne	-	314820	314820
Vermiculite	tonne	-	1960	1960
Wollastonite	tonne	-	1990000	1990000

Figures rounded off

Table – 2: Reserves/Resources of Lignite as on 01.04.2024: Gujarat

(In million tonnes)

District	Proved	Indicated	Inferred	Total
Total	1278.65	283.7	1159.7	2722.05
Kachchh	335.61	56.4	33.09	425.1
Bhavnagar	0	0	299.17	299.17
Bharuch	724.76	118.59	491.23	1334.58
Surat	218.28	108.71	336.21	663.2

Source: Coal Directory of India, 2023-24.

Production

In 2023-24, the mineral production in the State of Gujarat was mainly reported for bauxite, limestone etc. The production of bauxite marginally decreased by 1.43% to 16,43,130 tonnes in 2023-24 from 16,67,043 tonnes in the preceding year. On the other hand, the production of limestone marginally increased by 2.42% to 23.40 million tonnes in 2023-24 from 22.85 tonnes in the preceding year. The value of all minerals in the State increased by 7% to ₹ 763.77 crore in 2023-24 from ₹ 714.87 crore in the preceding year. There were 164 reporting mines in 2023-24 in case of MCDR minerals (Table- 3).

Further, as per the information available on the website of the Ministry of Mines, during 2023-24, a total of six mining leases, i.e., two for 'limestone & marl' and one each for 'bauxite (aluminous laterite) and laterite (Minor Mineral)'; 'Bauxite (Aluminous Laterite), Limestone and Marl'; 'Bauxite, Limestone and Marl'; and 'Bauxite and Marl', were granted in the State.

Mineral-based Industry

The present status of each mineral-based industry is not readily available. However, the important Mineral-based Industries in the Organised Sector in the State are furnished in Table- 4.

Table –3: Mineral Production in Gujarat, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24(p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		139	-	7038388	161	-	7148745	164	-	7637705
Bauxite	t	59	2018462	1692854	75	1667043	1470137	75	1643130	1677753
Fluorite (graded)	t	-	-	-	2	100	330	-	-	-
Limestone	'000t	80	23250	5212323	84	22852	5493989	89	23405	5767770
Marl %	t	-	900560	133211	-	875308	184289	-	1081896	192182
Sulphur #	t	-	117588	-	-	146401	-	-	135697	-

(p): provisional

%: Associate with Limestone

#: Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

(Excluding Atomic, Fuel and Minor Minerals)

Industry/ Plant	Capacity ('000 tpy)
Abrasives	
Bombay Mineral Limited, Jam Khambhalia	86.4 (Abrasive Grain)
Carborandum Universal Ltd, Okha, Distt Jamnagar.	NA
Carborandum Universal Ltd, Bhatia, Distt Jamnagar	NA
Flexo-Plast Abrasives, Ahmedabad	NA
Orient Abrasive Ltd, Porbandar	75 (Abrasive Grain)
	150 (Calcined Bauxite)
	30 (Castable Refractory)
Asbestos Products	
Ramco Industries Ltd, Singura, Distt Kachchh.	72
Sanghi Industries Ltd, Sanghipuram, Distt Kachchh.	36
U.P. Asbestos Ltd, Valsad.	36
Cement	
Ambuja Cements Ltd, Ambuja Nagar, Distt Junagadh.	5700
Ambuja Cement Ltd, Magdalla, Distt Surat (G).	1560
Mehta Group Gujarat Sidhee Cement, Sidheegram, Sutrapada Distt Junagadh.	1200
Mehta Group Saurashtra Cement Ltd, Porbandar, Distt Junagadh.	1500
Saurashtra Cement Ltd, Ranavav Porbandar,	3063
Hi Bond Cement, Gondal.	1200
J.K. Laxmi, Kalol, Distt Ganghinagar (G).	1000
J.K. Laxmi, Surat	1350
Sanghi Industries Ltd, Sanghipuram, Distt Kachchh.	4000
Shree Digvijay Cement Co. Ltd, Digvijaygram, Sikka Distt Jamnagar.	1200 (43 Gr.)
	1200 (53. Gr.)
	1200 (PPC)
	1200 (Oil well cement)
	1200 (Sulphate Resisting P.C.)
Tata Chemicals Ltd, Mithapur, Distt Jamnagar.	500
UltraTech Cement Co. Ltd, Pipavav, Distt Amreli.	6400
UltraTech Cement Ltd, (Narmada Cement), Jafrabad, Distt Amreli.	1450
UltraTech Cement Ltd, (Gujarat Cement), Kovaya, Babarkot, Rajula Jafrabad,	6400
UltraTech Cement (formerly a unit of JCCL), Sewagram, Abdasa, Distt Kachchh.	2400

Table 4 (Cont.)

Industry/ Plant	Capacity ('000 tpy)
UltraTech Cement (formerly a unit of JCCL), Wanakbori, Distt Kheda (G).	2400
UltraTech Cement Ltd, Magdalla (G).	750
Sparta Cements & Infra Ltd. Bhuj	1000
Vadraj Cement, Mora, Surat	6000
Ceramic	
Unifrax India, Lakhtar	7.5 (Ceramic fiber product)
Orient Glazes Ltd, OGPL, Kheda Unit Radhu	35.53
Chemical	
Baroda Rayon Corpn. Ltd, Surat.	2.2 (sodium sulphate)
Century Chemicals, Nava Nanga, Distt Jamnagar.	108 (refined salt)
Gujarat Alkalies & Chemicals Ltd, Baroda.	14.9 (caustic soda)
Gujarat Alkalies & Chemicals Ltd, Dahej, Distt. Bharuch.	33.408 (phosphoric acid)
GHCL Limited, Sutrapada.	1100 (Soda Ash)
	71 (Sodium bicarbonate)
Indian Rayon Industries Ltd,	21 (yarn)
Veraval, Distt Junagadh.	35.7 (H ₂ SO ₄)
	10 (carbon disulphide)
	9.3 (sodium sulphate)
	91.3 (caustic soda)
Kamadhenu Nutrients Pvt.Ltd. Panoli, Ankleshwar	10.8 (Dicalcium phosphate)
Kohler India Corp. Pvt. Ltd, Jhagadia, Talodara	15.02 (2Pc B)
	8.29 (lav)
	2.25 (Pedestal)
	4.73 (tank)
Navin Fluorine Industries Ltd, Surat.	22 (HF)
Kalatalav, Bhavnagar	648 (Soda Dense)
	144 (Refined Sodium Bicarbonate)
	Vaccun Salt (864)
Saurashtra Chemicals Ltd, Porbandar, Distt Porbandar	365 (soda ash)
	20.4 (caustic soda)
	26.4 (refined bicarbonate)
Shree Sulphurics Pvt. Ltd, Ankleshwar, Distt Bharuch.	58 (H ₂ SO ₄)
	12 (chloro-sulphuric acid)
Tata Chemicals Ltd, Mithapur, Distt Jamnagar. Copper Smelter	875 (soda ash)
Hindalco Industries Ltd,	500 (copper smelting)
Birla Copper, Dahej, Distt Bharuch.	1670 (H ₂ SO ₄)
	15 tonnes (Au)
	150 tonnes (Ag)
HCL, Gujarat Copper Project, Jhagadia, Distt. Bharuch.	50 (electrolytic copper)
	20 (copper anodes)
Electrode	
Power Electrode Varaval Shapar, Kotda Sangani	0.6
Fertilizer	
Aarti Fertilizers, Vapi, Valsad	132 (SSP)
Coromandel International Ltd (Formerly	100 (SSP)
Liberty Phosphate Ltd), Nandesari, Vadodara	367 (urea)
GSFC, Vadodara	108 (DAP)
	200 (complex)
	196 (AS)

Table 4 (Cont.)

Industry/ Plant	Capacity ('000 tpy)
GSFC, Sikka (Sikka - I & II), Jamnagar	326 (DAP)
GNFC, Bharuch	636.9 (urea)
	142.5 (complex)
Hindalco Industries Ltd, Dahej, Distt Bharuch	400 (DAP/complex)
IFFCO Ltd, Kandla, Distt. Kachchh	2420
IFFCO Ltd, Kalol, Distt. Gandhinagar	602 (urea)
Khaitan Chemicals & Fertilizers Ltd, Dahej, Bharuch	200 (SSP)
KRIBHCO Ltd, Hazira, Distt. Surat	2195 (urea)
Nirma Ltd, Moraiya, Ahmedabad	100 (SSP)
Sona Phosphates Ltd, Sarigam, Valsad	15 (SSP)
T J Agro Fertilizers Pvt. Ltd, Navsari	22 (SSP)
Foundry	
Steelcast Ltd, Ruvapuri Road, Bhavnagar	30
Intolcast Pvt. Ltd, 16, 17 & 19 Ankur Industrial Complex, Rajkot Gundal Road Shaper, Rajkot	2.4 (steel casting)
Intricast Pvt. Ltd, 25/28 Galaxy	1.08 (steel casting)
Industrial Estate, Rajkot Gundal Road	
Shaper, Rajkot	
Invac Cast Pvt. Ltd, 444, 453 & 455 Nana Fofadia Road Bamangam, Vadodara	2.4 (steel casting)
Gujarat Intuxt Ltd.184/P, Rajkot Gundal Road Shaper, Rajkot	1.8 (steel casting)
Iron & Steel	
Essar Steel Ltd, Hazira, Distt Surat	6700 (sponge iron)
	10000 (crude/liquid steel)
Jindal Saw Ltd, Samaghogha, Mundra	900 (Sinter)
	580 (Pig Iron)
Ferroalloys	
Baroda Ferro Alloys Ltd, Panchmahals.	3.5
Essel Mining & Industries Ltd, Vapi, Distt Valsad.	9
Electro Ferro Alloys Ltd, Ahmedabad.	0.3
Sponge Iron	
Electrotherm India Pvt. Ltd, Samakhalli, Distt Kachchh	75
Global Hi-Tech Industries Ltd, Bhuj, Distt Kachchh	105
Welspun Steel Ltd, Versamedi, Anjar	144
Glass	
Alembic Glass Industries Ltd, Baroda.	35
Bhagwati Glass Containers Ltd, Kalol.	8.7
Bharat Glass Tube Ltd, Bharuch.	7.2
Gobind Glass & Industries Ltd, Kadi.	NA
Gopal Glass Works Ltd, Budasan, Distt Mahesana.	40.6
Gujarat Borosil Ltd, Govali, Distt. Bharuch.	62.5
Piramal Glass Ltd, Jambusar.	355 (tpd)
Piramal Glass Ltd, Kosamba.	340 (tpd)
Haldyn Glass (Gujarat) Ltd, Padra, Vadodara.	320 TPD
Prestige Glass Industries Pvt Ltd, Vagra.	11.5
Petroleum Refinery	
IOCL, Koyali.	13700
RPL, Jamnagar	33000
RPL, Jamnagar (SEZ).	27000
Essar Oil Ltd, Vadinar.	20000
Refractory	
Calders India Refractorie Ltd, Bhayati Jambudiya, Wankaner	42

Table- 4 (Concl'd.)

Industry/ Plant	Capacity ('000 tpy)
Lilanand Magnesite Pvt. Ltd, Dharmpur, Ranavav	10.8
Synthetic Gas	
Reliance Industries Ltd, JG-DTA Gasification Area, Kunalus Lalpur	13122.48
Calcined Bauxite	
Birla VXL Ltd, Porbandar	36
Bombay Minerals Ltd, Jamkhambhaliya	96
Gujarat Credo Mineral Industries Ltd,	500 (dry beneficiated)
Naredi, Abdasa	10 (processed bauxite)
Saurashtra Calcine Bauxite & Allied Industries Ltd, Bhatia	39
Shri Natraj Ceramics & Chemical Industries Ltd, Khambhaliya	24

G: Grinding Unit

Data, not readily available for fertilizer and cement industries on respective websites, is taken from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.



Haryana

Sulphur was the principal mineral produced as a byproduct from fertilizer plant in Haryana

0.16

million tonne of Sulphur production was reported in 2023-24

Mineral Resources

The State is the principal holder of country's resources of slate (85%), tin ore (64%), quartz-silica sand (42%) and quartzite (53%). The principal minerals that are found to occur in Haryana are china clay in Faridabad, Gurgaon & Rewari districts; limestone in Ambala, Bhiwani, Mahendragarh & Panchkula districts; quartz/silica sand in Bhiwani, Faridabad, Gurgaon & Mahendragarh districts; quartzite in Faridabad & Gurgaon districts; and slate in Mahendragarh & Gurgaon districts. Other minerals, such as, barytes, calcite, felspar & marble occur in Mahendragarh district; copper in Bhiwani & Mahendragarh districts; dolomite in Ambala & Mahendragarh districts; granite in Bhiwani district; and tin & tungsten mineralisations in Tosham area of Bhiwani district have also been reported (Table-1).

Production

Sulphur was the principal mineral produced as a by-product from fertilizer plant in Haryana. The production of sulphur at 0.16 million tonnes in 2023-24 increased by 7% as compared to 0.15 million tonnes in 2022-23 (Table-2).

Mineral-based Industry

The important large and medium-scale Mineral-based Industries in the Organised Sector in the State are furnished in (Table-3).

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Haryana

Mineral	Unit	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Copper Ore	'000 tonnes	-	53816	53816
Metal	'000 tonnes	-	179.01	179.01
Limestone	'00 tonnes	-	74677	74677
Tin Ore	tonne	-	53910000	53910000
Metal	tonne	-	86220.6	86220.6
Tungsten Ore	tonne	-	2230000	2230000
Contained WO ₃	tonne	-	3568	3568

Figures rounded off.

Table – 2 : Mineral Production in Haryana, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

Mineral	Unit	2021-22			2022-23			2023-24(p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		-	-	-	-	-	-	-	-	-
Sulphur #	t	-	178740	-	-	157228	-	-	163930	-

(p) provisional

Recovered as by-product from fertilizer plant.

Table – 3: Principal Mineral-based Industries

Industry/plant	"Capacity ('000 tpy)"
Asbestos Products	
Hyderabad Industries Ltd, Ballabgarh.	2160 MT
Cement	
Shree Cement (formerly, unit of Jaypee Cement), Panipat (G).	1500
J K Cement Ltd, Jharli, Distt Jhajjar (G).	1500
J K Laxmi, Bijitpur, Distt Jhajjar (G).	1300
UltraTech Cement, Panipat (G).	2000
UltraTech Cement, Jhajjar (G).	2250
Ceramic/Sanitaryware	
Hindustan Sanitaryware & Industries Ltd, Bahadurgarh.	14
SPL Ltd, Bahadurgarh.	35000 (sq. m/day)
Somany Ceramics Ltd, Kassur, Distt Jhajjar.	10.4 (Mill sq m)
Chemical	
Oriental Carbon & Chemicals Ltd, Dharuhera, Distt Rewari.	28500
Bhalla Chemical Works Pvt Ltd, Ballabgarh.	10 (Zirconium derivatives) 5 (Zirconium silicate opacifiers)
Saint Gobain Gyproc India Ltd (formerly India Gypsum Ltd), Jind.	NA
S. B. Zircon Pvt. Ltd. Sikri, Ballabgarh	3 (Zirconium opacifiers)
Varun Electrode Pvt Ltd, Panipat.	3.6
Electrode	
Devay Udyog, Charkhi Dadri	0.4
Fertilizer	
NFL, Gohana Road, Panipat.	511.5 (Urea) 8.70 (S)
Kisan Phosphates Pvt. Ltd, Gawar, Hisar.	132 (SSP)
Nitin Chemicals & Fertilizers Ltd, Rukri, Ambala.	20 (SSP)
Iron & Steel	
Jindal Stainless Ltd, Hisar.	780 (stainless steel)
Ferroalloys	
Haryana Ferro Alloys Ltd,	2.5
Glass	
Haryana Sheet Glass Ltd, Sevli, Distt Sonipat.	89.5
Hindustan National Glass & Industries Ltd, Ballabgarh.	690 TPD
Petroleum Refinery	
IOCL, Panipat.	15000
Refractory	
Bhaskar Refractories & SW Pipes (P) Ltd, Amar Nagar.	12

Note: Data, not readily available for Fertilizer and Cement Industries on respective websites, is taken from Indian Fertilizer Scenario, FAI Statistics and Survey of Cement Industry & Directory, respectively. G: Grinding Unit



Himachal Pradesh

Limestone and Rock Salt were the principal minerals reporting production in the State

₹ 311 crore
Value of major mineral production were reported during the year 2023-24.

27
Mines in Himachal Pradesh reported in case of MCDR minerals in 2023-24.

The State is the sole holder of country's antimony and rock salt ore resources. Limestone and shale are the important minerals produced in the State. Barytes occurs in Sirmaur district; limestone in Bilaspur, Chamba, Kangra, Kullu, Mandi, Shimla, Sirmaur & Solan districts; and rock salt in Mandi district. Other minerals that occur in the State are antimony in Lahaul & Spiti districts; gypsum in Chamba, Sirmaur and Solan districts; magnesite in Chamba district; pyrite in Shimla district; and quartz, quartzite & silica sand in Una district (Table - 1).

PRODUCTION

Limestone and Rock Salt were the principal minerals reported production in the state during the year 2023-24. The value of minor mineral production was estimated at ₹ 292 crore for the year 2023-24. There were 27 reporting mines during 2023-24 of which 26 were of limestone and 1 of rock salt. The only rock salt mine in the state reported

nil production and had a closing stock of 295 tonnes during the year 2023-24 (Tables - 2 to 4).

MINERAL-BASED INDUSTRY

The principal mineral-based industries in the Organised Sector in the State are furnished in Table-5 based on Survey of Cement Industry & Directory.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020

(Himachal Pradesh)

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Antimony				
Ore	tonne	-	10588	10588
Metal	tonne	-	174	174
Limestone	'000 tonnes	1022012	5597134	6619146
Magnesite	'000 tonnes	-	298	298
Pyrite	'000 tonnes	-	2560	2560
Rock salt	'000 tonnes	3860	8920	12780

Figures rounded off.

Table – 2: Mineral Production in Himachal Pradesh, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		23	-	2984595	23	-	2756602	27	-	3118011
Limestone	'000t	22	13810	2983883	22	11879	2751753	26	13706	3118011
Rock Salt	t	1	286	712	1	1002	4849	1	-	-

(P) provisional

Table – 3: Mine-head closing stocks of Rock salt

(2022-23 and 2023-24)

(In tonnes)

State	2022-23	2023-24 (P)
India	561	295
Himachal Pradesh	561	295

Table – 4: Production Data of Minor Minerals for the Financial Year 2022-23 and 2023-24

Sl. No.	Minor Minerals	No. of mines Pvt./Public	Total		No. of mines Pvt./Public	Total		
			Quantity In tonnes	Value in Rupees		Quantity In tonnes	Value in Rupees	
1	Bajri		3843229	538057100		8100771	1134107940	
2	Sand		2915478	408166920		4163676	582914640	
3	Boulders		4840692	677701220	498 Pvt.	6740615	943686100	Mine of sand, stone & Bajri have also been granted to cater the demand of Stone Crusher units for manufacturing Aggregates and M Sand
4	Building Stone	443 Pvt. 343/ Public-100	53669	11907110	433/ Public-65	77838	112865110	
5	Slate	Slate-11	1957	2490000	Slate-10	1106	3318000	
6	Silica Sand	stone-12 Sand-43 Shale-1 Silica	18961	3981810	stone/ Boulder-67	4000	840000	
7	Aggregate	Sand- 03 Brick Earth-8	40684.48	5695760	Sand-33	45465	6365100	
8	ough Stone		101252	10125200	Building Stone/ Cut	691425	69142500	
9	Shale (Minor)		59001	2950050	Stone-11	0	0	
10	Clay/ Soil		95377	4768850		0	0	
11	Shale (Major)		836867	58580690		1005231	70366170	
TOTAL			1724424710			2923605560		

Source : Data as received from Department of Industries, Himachal Pradesh.

Table – 5: Principal Mineral-based Industries

Industry/plant	Capacity ('000 tpy)
ACC Ltd, Gagal (Gaggal I & II), Distt Bilaspur	4640
	3000 (Clinker)
Ambuja Cement, Suli, P.O. Darlaghat, Distt Solan	1600 Cement & 2600 (Clinker)
Ambuja Cement, Nalagarh, Distt Solan	1500
Asian Concretes and Cements Pvt Ltd, Bir Palsi, Distt Solan	1300
CCI Ltd, Rajban, Distt Sirmaur	240
Ultra Tech Cement Ltd, Bagga, Distt Solan (Jaypee Himachal Cement Plant)	2540 Cement & 2970 (Clinker)
UltraTech Cement Ltd, (Blending & Grinding), Bagheri Solan	2000

Note: Data, not readily available for cement industries on respective websites, is taken from Survey of Cement Industry & Directory



Union Territory of Jammu & Kashmir

UT of Jammu & Kashmir is the sole holder of country's borax, sapphire and sulphur (native) resources

Limestone was the principal mineral reporting production in the State.

19

mines reported in case of MCDR of minerals in 2023-24.

MINERAL RESOURCES

Union Territory of Jammu & Kashmir is the sole holder of country's borax, sapphire and sulphur (native) resources and possesses 30% graphite and 14% of gypsum. Coal occurs in Kupwara district; gypsum in Baramulla & Doda districts; limestone in Anantnag, Baramulla, Kathua, Leh, Poonch, Pulwama, Rajauri, Srinagar & Udhampur districts; and magnesite in Leh & Udhampur districts. Other minerals that occur in the U.T. are bauxite & china clay in Udhampur district; bentonite in Jammu district; borax & sulphur in Leh district; diaspore in Rajouri & Udhampur districts; graphite in Baramulla district; lignite & marble in Kupwara district; quartz & silica sand in Anantnag, Doda & Udhampur districts; quartzite in Anantnag district; and sapphire in Doda district (Tables - 1 & 2).

PRODUCTION

Limestone was the only principal mineral reported production in the Union Territory of Jammu & Kashmir. There were 19 reporting mines in 2023-24 in case of MCDR of minerals in the U.T. (Table-3).

MINERAL-BASED INDUSTRY

Jammu & Kashmir Cements Ltd, a State Government Undertaking, operates a cement plant of 4.00 lakh tpy capacity at Khrew in Pulwama district and 1.00 lakh tpy capacity at Samba Jammu. The Company also owns a small

cement plant of 20,000 tpy capacity located at Wuyan in Srinagar district, besides two other tiny cement plants that have a total capacity of 5,20,000 tpy. Khyber Indus. (P) Ltd operates a cement plant of 3,30,000 tpy in the U.T. The U.T. also has a 1,800 tpy capacity unit that manufactures ceramic and refractory products in Kathua District. A 3,000 tpy capacity calcium carbide plant is situated at Pulwama District. J.K. Minerals Ltd has a plant of 30,000 tpy of DBM and 75,000 tpy plant of sized magnesite at Chipprian deposit near Village Panthal in Udhampur district in the Union Territory (Table-4).

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: U.T. of Jammu & Kashmir

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	'000 tonnes	0	2725	2725
Borax	tonne	0	74204	74204
Graphite	tonne	0	62740555	62740555
Limestone	'000 tonnes	185490	2242071	2427561
Magnesite	000 tonnes	0	4145	4145
Sapphire	kg	0	450	450
Sulphur (Native)	'000 tonnes	0	210	210

Figures rounded off

Table – 2: Reserves/Resources of Lignite as on 01.04.2024

(Excluding Atomic Fuel and Minor Minerals)

(In million tonnes)

District	Proved	Indicated	Inferred	Total
Total	–	20.25	7.3	27.55
Kupwara	–	20.25	7.3	27.55

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in U.T. of Jammu & Kashmir, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		10	-	286892	16	-	297134	19	-	310646
Limestone	'000 tonnes	10	1156	286892	16	1237	297134	19	1185	310646

(p) provisional

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Jammu & Kashmir Cement Ltd, Khrew, Pulwama	400
Jammu & Kashmir Cement Ltd, Samba Jammu.	100
Jammu & Kashmir Cement Ltd, Wuyan Srinagar.	200
Khyber Indus (P) Ltd	330
Ceramic & Refractory Product, Kathua.	1.8
Calcium Carbide Plant, Pulwama	3
J. K. Mineral Ltd, Chipprian,	30 (DBM)
Panthal, Udhampur	75 (Magnesite)
Nayyar Electrode Pvt. Ltd, Barri Brahmana	4.45



Jharkhand

22.89 million tonnes
of iron ore were produced in the State
in 2023-24

₹ 5,796 crore
Value of production of major minerals
(other than Atomic, fuel and Minor
mineral) in the year 2023-24.

38
Reporting mines in case of MCDR
minerals in 2023-24.

MINERAL RESOURCES

Jharkhand is one of the major mineral producing States. It is the sole producer of flint stone in the country and is one of the leading producers of coal, gold, graphite, bauxite, iron ore & limestone. Uranium ore is mined and processed by Uranium Corporation of India Ltd (UCIL) for supply as fuel to the country's nuclear power reactors through six underground mines, one opencast mine, and two processing plants. Jharkhand has the sole resources of emerald mineral. It accounts for about 34% rock phosphate, 20% iron ore (haematite), 34% apatite, 9.3% andalusite, 20% cobalt ore, 20% copper ore, 9% graphite and 4% silver ore resources of the country.

Important minerals that occur in the State are bauxite in Dumka, Gumla, Latehar, Lohardaga & Palamu districts; china clay in Dumka, Hazaribagh, Lohardaga, East & West Singhbhum, Sahebganj & Ranchi districts; coal in Bokaro, Deoghar, Dhanbad, Giridih, Godda, Hazaribagh, Palamau, Pakur & Ranchi districts; copper in Hazaribagh & East Singhbhum districts; dolomite in Garhwa & Palamu districts; felspar in Deoghar, Dhanbad, Dumka, Giridih, Hazaribagh, Jamtara, Koderma, Latehar, Palamu & Ranchi districts; fireclay in Dhanbad, Dumka, Giridih, Godda, Hazaribagh, Latehar, Palamu, Ranchi & West Singhbhum districts; gold in East Singhbhum district; graphite in Palamu district; iron ore (haematite) in West Singhbhum district; iron ore (magnetite) in Gumla, Hazaribagh, Latehar, Palamu & East Singhbhum districts; kyanite in Saraikela-Kharsawan & West Singhbhum districts; limestone in Bokaro, Dhanbad, Garhwa, Giridih, Hazaribagh, Palamu, Ranchi, East & West Singhbhum districts; manganese ore in East & West Singhbhum districts; mica in Giridih and

Koderma districts; ochre in West Singhbhum district; dunite/pyroxenite in East Singhbhum district; quartz/silica sand in Deoghar, Dhanbad, Dumka, Giridih, Godda, Hazaribagh, Jamtara, Koderma, Latehar, Palamu, Ranchi, Sahebganj, Saraikela-Kharsawan & West Singhbhum districts; and quartzite in East & West Singhbhum districts.

Other minerals that occur in the State are andalusite and rock phosphate in Palamu district; apatite, chromite, cobalt, nickel, gold & silver in East Singhbhum district; asbestos in East & West Singhbhum districts; barytes in Palamu & East Singhbhum districts; bentonite in Pakur & Sahebganj districts; garnet in Hazaribagh district; granite in Deogarh, Dhanbad, Dumka, Giridih, Godda, Gumla, Hazaribagh, Koderma, Lohardaga, Palamu, Ranchi & East Singhbhum districts; sillimanite in Hazaribagh district; talc/steatite/soapstone in Giridih, Koderma, Palamu, East & West Singhbhum districts; pyrophyllite in Saraikela-Kharsawan district; titanium minerals in Ranchi and East Singhbhum

districts; and vermiculite in Giridih & Hazaribagh districts (Table-1). The reserve/resources of coal and the various coalfields located in Jharkhand are furnished in Table – 2.

MINERAL-BASED INDUSTRY

The principal large and medium-scale mineral-based industries in the Organised Sector in the State are furnished in Table - 4.

PRODUCTION

Coal is the principal mineral item for which production is

reported in the State. The other important minerals produced are bauxite, copper ore & concentrate, iron ore, graphite etc. The value of Major Minerals' production (other than Atomic, Fuel and Minor Minerals) was estimated at ₹ 5,796 crore for the year 2023-24. There were 38 reporting mines in 2023-24 in case of MCDR minerals. Some of the major mineral production data of Jharkhand during 2023-24 are as follows- 2 million tonnes of bauxite, 4.9 thousand tonnes of copper ore, 22.8 million tonnes of iron ore, 20 thousand tonnes of graphite and 28 kg of gold metal (Table – 3).

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Jharkhand

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Andalusite	'000 tonnes	-	11800	11800
Apatite	tonne	-	7270000	7270000
Asbestos	tonne	-	154893	154893
Bauxite	'000 tonnes	39972	249272	289244
Chromite	'000 tonnes	-	736	736
Cobalt	Million tonnes	-	9	9
Copper				
Ore	'000 tonnes	9150	242313	251463
Metal	'000 tonnes	107.45	2672.21	2779.66
Emerald	Kilogram	-	55869	55869
Garnet	tonne	-	110071	110071
Gold				
Ore (Primary)	tonne	-	10076527	10076527
Metal (Primary)	tonne	-	15.43	15.43
Graphite	tonne	2604079	17402288	20006367
Iron Ore (Haematite)	'000 tonnes	534677	4175469	4710146
Iron Ore (Magnetite)	'000 tonnes	-	10667	10667
Kyanite	tonne	331193	7943367	8274560
Limestone	'000 tonnes	10687	610078	620765
Manganese Ore	'000 tonnes	1059	13691	14749
Nickel	Million tonnes	-	9	9
Potash	Million tonnes	-	152	152
Rare-earth Elements	tonne	-	4	4
Rock Phosphate	tonne	-	107370000	107370000
Sillimanite	tonne	-	83000	83000
Silver				
Ore	tonne	-	23840000	23840000
Metal	tonne	-	5.22	5.22
Titanium	tonne	-	26603767	26603767
Vermiculite	tonne	-	30048	30048

Figures rounded off.

Table – 2: Reserves/Resources of Coal as on 01.04.2024

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	59876.88	27135.39	4799.3	91811.57
Raniganj	1594.21	444.64	0	2038.85
Jharia	17735.43	2059	35.25	19829.68
East Bokara	3977.19	3552.66	761.88	8291.73
West Bokara	3922.75	1278.59	17.05	5218.39
Ramgarh	936.65	911.77	58.05	1906.47
North Karanpura	13372.85	5955.93	1864.96	21193.74
South Karanpura	6044.83	1266.6	1082.78	8394.21
Aurangabad	1391.49	1743.56	172.55	3307.6

Table- 2 (Concl'd.)

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Hutar	190.79	26.55	32.48	249.82
Dalongunj	83.86	60.1	0	143.96
Deogarh	326.24	73.6	0	399.84
Rajmahal	10300.59	9762.39	774.3	20837.28

Source : Coal Directory of India, 2023-24

Table – 3: Mineral Production in Jharkhand, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		47	-	57687063	41	-	51602914	38	-	57962656
Bauxite	tonne	19	1808750	2352953	21	2214879	3108731	21	2043734	2946196
Copper Ore	tonne	-	25834	-	-	53375	-	-	4936	-
Copper Conc.	tonne	2	-	-	2	-	-	2	-	-
Gold Ore	tonne	-	3682	-	-	8809	-	-	10079	-
Gold	kg	1	12	56268	1	22	119607	1	28	166140
Iron Ore	'000 tonnes	17	24728	55168902	12	23091	48359178	9	22896	54829400
Graphite (r.o.m.)	tonne	4	54	59	2	12162	10615	2	20756	19533
Kyanite	tonne	1	2785	5152	1	2071	3831	1	210	389
Limestone	'000 tonnes	3	72	103729	2	3	952	2	3	998

(p) provisional

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Alumina Hindalco Industries Ltd, Muri	450 KTPA
Asbestos Products	
Hyderabad Industries Ltd, Jasidihi, Distt Deogarh	72
Cement	
ACC Ltd, Chaibasa, Distt Singhbhum	1200
ACC Ltd, Sindri, Distt Dhanbad (G)	2500
Bokaro Cement Plant (formerly JV of Jaypee Cement & SAIL), Bokaro (G)	2100
Lafarge, Jojobera, Distt Singhbhum	4600
Burnpur Cement Patratu Ramgarh	300
Ceramic	
Maithan Ceramics Pvt Ltd, Dhanbad	80
Chemicals	
Bihar Caustic & Chemicals Ltd, Garhwa Road, Distt Palamu	92.75 (caustic soda lye)
Copper Smelter	
HCL, ICC, Ghatsila, Distt Singhbhum (East)	19 (refined copper) 20.5 (copper smelting) 18.5 (copper cathode) 84 (fabricated wire bar) 54 (H ₂ SO ₄), 390 t (NiSO ₄) 480 kg (CuSO ₄) 14.6 kg (selenium) 9868 kg (Ag), 698 kg (Au)
Foundry	
Grind chem, Adityapur	15 (Foundry fluxes)
Jharkhand Grid chem Pvt. Ltd, Adityapur, Gamharua	25 (Foundry fluxes)

Note: Data, not readily available for cement industries on respective websites, is taken from Survey of Cement Industry & Directory



Karnataka

1,553 kg
of gold was produced in the state in 2023-24. Karnataka is principal producer of gold in country.

₹ 16,103.5 crore
value of mineral production were reported during the year 2023- 24.

126
mines in case of MCDR of minerals were reported in the State during 2023-24

MINERAL RESOURCES

Karnataka has the distinction of being the principal Gold producing State in the country. The State is the sole producer of felsite and one of the leading producers of Iron ore, Chromite and Kyanite. Karnataka hosts the country's 79% Vanadium ore, 69% Iron ore (magnetite), 65% Corundum, 41% Tungsten ore, 36% Asbestos, 25% Limestone, 20% Gold ore (primary), 25% Manganese ore, 17%, Kyanite and 7% PGMs resources.

The important mineral-occurrence found in the State are bauxite in Belagavi, Chikkamagaluru, Uttara & Dakshina Kannada and Udupi districts; chromite in Chikkamagaluru, Hassan & Mysuru districts; feldspar in Bengaluru, Belagavi, Chitradurga & Hassan districts; gold in Chitradurga, Dharwad, Gadag, Kalaburagi, Hassan, Haveri, Kolar, Raichur & Tumakuru districts; iron ore (haematite) in Bagalkot, Ballari, Vijayapura, Chikkamagaluru, Chitradurga, Dharwad, Gadag, Uttara Kannada, Shivamogga & Tumakuru districts; Iron ore (magnetite) in Chikkamagaluru, Hassan, Uttara & Dakshina Kannada and Shivamogga districts; kyanite in Chikkamagaluru, Chitradurga, Coorg, Mandya, Mysuru, Shivamogga & Dakshina Kannada districts; Limestone in Bagalkot, Belagavi, Ballari, Vijayapura, Chikkamagaluru, Chitradurga, Davangere, Gadag, Kalaburagi, Hassan, Mysuru, Uttara & Dakshina Kannada, Shivamogga, Tumakuru & Udupi districts; magnesite in Coorg, Mandya & Mysuru districts; manganese ore in Belagavi, Ballari,

Chikkamagaluru, Chitradurga, Davangere, Uttara Kannada, Shivamogga & Tumakuru districts; quartz/silica sand in Bagalkot, Bengaluru, Belagavi, Ballari, Chikkamagaluru, Chitradurga, Davangere, Dharwad, Gadag, Kalaburagi, Hassan, Haveri, Kolar, Koppal, Mandya, Mysuru, Uttara & Dakshina Kannada, Raichur, Shivamogga, Tumakuru & Udupi districts. Other minerals that occur in the state are asbestos in Chikkamagaluru, Hassan, Mandya, Mysuru and Shivamogga districts; barytes & pyrite in Chitradurga district; copper in Chikkamagaluru, Chitradurga, Kalaburagi, Hassan, Uttara Kannada, Raichur & Shivamogga districts; graphite in Kolar & Mysuru districts; molybdenum in Kolar & Raichur districts; nickel in Uttara Kannada district; Platinum Group of Metals in Davangere district; sillimanite in Hassan, Mysuru & Dakshina Kannada districts; silver in Chitradurga & Raichur districts; titanium minerals in Hassan, Uttara Kannada & Shivamogga districts; tungsten in Gadag, Kolar & Raichur districts; vanadium in Hassan, Uttara Kannada & Shivamogga districts; and vermiculite in Hassan, Mandya & Mysuru districts (Table-1).

Table No – 1: Reserves/Resources of Minerals as on 01.04.2020: Karnataka

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Asbestos	tonne	0	8282457	8282457
Bauxite	'000 tonnes	5207	41242	46449
Chromite	'000 tonnes	499	1317	1817
Copper				
Ore	'000 tonnes	0	41499	41499
Metal	'000 tonnes	0	245.86	245.86
Gold				
Ore (Primary)	tonne	20470000	82538506	103008506
Metal (Primary)	tonne	87.46	163.71	251.17
Graphite	tonne	0	992632	992632
Iron Ore (Haematite)	'000 tonnes	1043212	1792781	2835992
Iron Ore (Magnetite)	'000 tonnes	318	7801853	7802171
Kyanite	tonne	181600	12991460	13173060
Limestone	'000 tonnes	2271221	53899236	56170457
Magnesite	'000 tonnes	1027	4516	5543
Manganese Ore	'000 tonnes	15464	108508	123972
Molybdenum				
Ore	tonne	0	1320900	1320900
Contained MoS ₂	tonne	0	1718.7	1718.7
Nickel Ore	million tonnes	0	0.23	0.23
Pt. Group Of Metals	tonne	0	1.5	1.5
Pyrite	'000 tonnes	0	3000	3000
Rare Earth Elements	tonne	0	3734	3734
Sillimanite	tonne	0	982725	982725
Silver				
Ore	tonne	22120000	3813612	25933612
Metal	tonne	5.43	4.29	9.72
Titanium	tonne	0	13862094	13862094
Tungsten				
Ore	tonne	0	36677818	36677818
Contained WO ₃	tonne	0	6235	6235
Vanadium				
Ore	tonne	0	19384430	19384430
Contained V ₂ O ₅	tonne	0	49497.55	49497.55
Vermiculite	tonne	0	162240	162240

Figures rounded off

PRODUCTION

Gold ore and bullion (metal), Iron Ore, Manganese ore, Limestone and Magnesite were the important minerals produced in Karnataka State in 2023-24. The value of MCDR minerals' production was estimated at 16,103.5

crore for the year 2023-24. There were 126 reporting mines in 2023-24 in case of MCDR minerals (Table-2).

MINERAL-BASED INDUSTRY

The important mineral-based industries in Organised Sector in the State are furnished in Table-3.

Table – 2: Mineral Production in Karnataka, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		132	-	178162467	125	-	133754048	126	-	161035005
Chromite	t	2	-	-	-	-	-	-	-	-
Gold Ore	t	-	470456	-	-	593677	-	-	716486	-
Gold	kg	4	1395	6704750	3	1411	7535031	4	1553	9478723
Iron Ore	'000t	56	40333	160324539	51	39822	115421232	51	40996	140433482
Manganese Ore	t	8	380004	3167390	10	344731	2592147	11	355057	2934535
Silver #	kg	-	143	11503	-	148	9299	-	150	10841
Graphite (r.o.m.)*	t	2	-	-	1	-	-	-	-	-
Kyanite	t	1	5075	-	1	360	644	1	-	-
Limestone	'000t	54	39395	7893558	53	38942	8129116	53	40832	8129362
Limeshell	t	1	100	231	2	250	656	2	891	2633
Magnesite	t	3	7057	51412	3	8195	65923	3	6816	45429
Vermiculite*	t	1	-	-	1	-	-	1	-	-

(p) provisional

Recovered at Raichur & Tumkur during refining of gold.

* Only labour reported.

Table – 3: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Abrasives	
Grindwell Norton Ltd, Bengaluru	NA
Alumina	
Hindalco Industries Ltd, Belagavi	350 (alumina) 40 (paste) 0.090(Vanadium)
Cement	
ACC Ltd, Wadi (Wadi & Wadi New), Distt Kalaburagi	5450
ACC Ltd, Kudithini, Ballari (G)	1100
ACC Ltd, Thondebhavi, Distt Chikaballapur (G)	1660
Bagalkot Cement Industries Ltd, Distt Bagalkot	600
Chettinad Cement, Kallur, Distt Kalaburagi	2500
Dalmia Cement, Yadwad, Distt Belagavi	4000
	2600 (Clinker)
Heidelberg Cement India Ltd, (Formerly Mysore Cements Ltd) Ammasandra, Distt Tumakuru	510
J K Cement Ltd, Muddapur, Distt Bagalkot	3000
JSW Cement, Vijaynagar, Distt Ballari	3200
Kesoram Industries, Vasavadatta Cement, Sedam, Distt Kalaburagi	8565 (OPC) 8565 (PPC)
Kalaburagi Cement Pvt Ltd (formerly Viratsagar) Gulbargha, Distt Kalaburagi	2750
Kalaburagi Cement Pvt Ltd Karchikhed, Chincholi	3500
	2750 (Clinker)
Orient Cement LtdItagi, Chittapur	3000
Ramco Cement Ltd, Mathodu, Distt Chitradurga	290
Shree Cement LtdBenekanahalli, Kodla Sedam, Kalaburagi	3000
Ultratech Cement, Raj Shree Cement, Malkhed, Distt Kalaburagi	6100

Table 3 (Cont.)

Industry/plant	Capacity (^{'000 tpy})
Ultratech Cement, Ginigera, Distt Koppal (G)	1300
Orient Cement Chittapur, Kalaburagi	3000
Ceramic	
Ceramic Products Ltd, Khanapur, Distt Belagavi	NA
H&R Johnson (India) Ltd, Hubballi	47.72
Murudeshwar Ceramics Ltd, Dharwad	8.4 mill.sqm
The Mysore Spongware Pipes Potteries Ltd, Solandavanahalli, Bengaluru	NA
Chemical	
Solaris Chem Tech Industries Ltd, Bhinga, Distt Uttara Kannada	59.4 (caustic soda), 52.3 (Cl), 133.7 (HCl)
	24.0 (H ₃ PO ₄)
Magnesium & allied Product Hurugalavadi , Mandya	3 (Magnesium Carbonate)
	1.875 (Magnesium Oxide)
Shivam Minerals , Honaga Belgaum	4.6 (Magnesium Carbonate)
	4.6 (Magnesium Oxide)
Fertilizer	
K P R Fertilizers Ltd Halvarthi, Koppal	60 (SSP)
Mangalore Chemical & Fertilizers Ltd, Panambur, Mangaluru	379.5 (Urea)
	260 (DAP)
	40 (Complex)
Tungabhadra Fertilizers & Chemicals Ltd, Munirabad, Koppal	45 (SSP)
Iron & Steel	
JSW Steel Ltd, Tornagallu Sandur Distt Ballari	12100 (pig iron)
	9200 (pellets)
	12000 (crude/liquid steel)
	12950 (sinter)
	4618(Coke)
Visvesvaraya Iron & Steel Ltd,	205 (pig iron)
Bhadravati, Distt Shivamogga	118(crude/liquid steel)
	4.8 (refractory bricks)
Sunvik Steels Pvt Ltd,	60 (sponge iron)
Jodidevarahally, Distt Tumakuru	60 (TMT bar)
	36 (Pellets)
BMM Ispat, Danapur, Distt Ballari	2400 (pellets)
KIOCL, Mangaluru	3500 (pellets)
	6700 (conc.)
Minera Steel & Power Pvt Ltd, Sandur	600
SLR Metalliks Ltd Narayan Devera Kera Hagari Bommanahalli	343.2 (Sinter)
Xindia Steel, Koppal	800 (pellets)
Uni-Metal Ispat Ltd, Ballari	75 Pig Iron
Kalyani Ferrous Ind Ltd, Koppal	500 (Sinter)
	289.6

Table- 3 (Concl'd.)

('000 tpy)

Industry/plant	Capacity
Kirloskar Ferrous Industries Ltd, Bevinahalli,	500 (Sinter)
Distt Koppal	720
Mukund limited, Ginigera, Koppal	500 (Sinter)
	410.3 (sponge Iron)
Agrawal Sponge & Energy (P) Ltd, Kuduthini, Distt Ballari	90
Balakundi Premium Steels Pvt Ltd, Halakundi, Distt Ballari	34
Bellary Ispat (P) Ltd, Halakundi Distt Ballari	52.5
Ballary Steel & Alloys Ltd, Ballari	60
Benaka Sponge Iron Pvt Ltd, Belagal, Distt Ballari	84
BMM Ispat Ltd, Danapur	600
	2400 (pellet)
BRU Industries, Anekal Taluk	1.2 (cast Iron)
Dhruvdesi Metasteel Pvt Ltd, Hirebaganal, Distt Koppal	72
Divya Jyoti Steel Ltd, Taranagar, Distt Ballari	30
Gayatri Metals Pvt Ltd, Belagal, Distt Ballari	5000
Hindustan Calcined Metal Pvt Ltd, Janekunnte Ballari	60
Jairaj Ispat Limited Belagal village	60
Haryana Steel and Power, Shanthigrama, Distt Hassan	35
Hare Krishna Metalics Pvt Ltd, Hire Baganal, Distt Koppal	144
Hospet Ispat Pvt Ltd, Allanagar Bagnal Road, Distt Koppal	60
Hothur Ispat Pvt Ltd, Veniveerapur, Distt Ballari	300 TPD
Minera Steel & Power Pvt Ltd, Yerabanahally, Distt Ballari	120
MSMetals & Steels PVT Ltd Hirebagnal Koppal	105
	109.5 (TMT Bars)
Noble Distillaries & Powers Ltd, Sirivar, Distt Ballari	200 TPD
PGM Ferro Steel Pvt Ltd, Hariganadani, Distt Ballari	60
Popuri Steels Ltd, Halakundi, Distt Ballari	30
Padmawati Ferrous Metal, Chikantpur Sandur, Ballari	150
Rayon Steel Pvt Ltd, Veniverapur, Distt Ballari	60
Rengineni Steel Pvt Ltd, Halakundi, Distt Ballari	25.5
Shree Venkateshwara Sponge & Power Ltd, Halakundi, Distt Ballari	60
Yashshvi Steel & Alloys Ltd, Halakundi, Distt Ballari	30
Ferroalloys	
Ani Smelters Yaradakatta, Hariyur	1.5
Dandeli Steel & Ferro Alloys Ltd, Dandeli	6
Padmawati Ferrous Metal, Chikantpur Ballari	30
	5 (Ferro-manganese)
	5 (Silico-manganse)
	2 (Ferro-silicon)
Sandur Manganese & Iron Ore Ltd, Mariyammanahalli Hospet	36 (SiMn)
Refractories	
T SRanganath & Company, Keshavapurahuliyar, Chikkanayakanahalli	1.0 (Clay tiles & Block)
SR Chemicals & Ferro Alloys Ltd, Honaga, Distt Belagavi	0.3
Thermit Alloys Pvt Ltd, Shivamogga	1.2
Petroleum Refinery	
MRPL, Mangaluru	15000

G: Grinding Unit

Note: Data for fertilizer and cement industries is taken from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.



Kerala

10%

of Country's sillimanite resource lie in Kerala.

381 thousand tonnes

limestone production were in the State in the year of 2023-24.

1

reported production mines in case of MCDR minerals in the State during 2023-24

MINERAL RESOURCES

Kerala is endowed with a diverse mineral resource base, dominated by china clay (kaolin) and coastal placer deposits rich in ilmenite, rutile, zircon, sillimanite, garnet, leucosene and monazite. The State contributes about 23% of India's china clay and 10% of sillimanite resources, and is the principal producer of limeshell and sillimanite. As per the Atomic Minerals Directorate for Exploration and Research, Kerala holds 144.02 million tonnes of ilmenite, 7.83 million tonnes of rutile and 7.96 million tonnes of zircon, with major concentrations in the Chavara belt. Other significant resources include bauxite, limestone, silica sand, graphite and iron ore (magnetite), along with minor occurrences of gold, garnet, fire clay, kyanite, lignite, magnesite, steatite and dimension stone granite. While Kerala hosts a wide spectrum of mineral occurrences, its mineral economy is strongly anchored by the well-established sectors of heavy mineral sands and china clay, with other deposits offering scope for future development (Tables – 1 & 2).

PRODUCTION

During 2023-24, limestone was the only important mineral produced in the state of Kerala. Its production stood at 381 thousand tonnes in 2023-24, registering an increase of 9.8% over the preceding year. During 2023-24, there was only one reporting mine of limestone reported under the MCDR minerals category (Table-3).

MINERAL-BASED INDUSTRY

Comprehensive data on the current status of each mineral-based industry in the state is not readily available. However, the key mineral-based industries operating in the organised sector are presented in Table-4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Kerala

Mineral	Unit	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	'000 tonnes	0	19449	19449
Garnet	tonne	0	198861	198861
Gold				
Ore (Primary)	tonne	0	558460	558460
Metal (Primary)	tonne	0	0.2	0.2
Ore (Placer)	tonne	0	26121000	26121000
Metal (Placer)	tonne	0	5.86	5.86
Graphite	tonne	15443	1419532	1434975
Iron Ore (Magnetite)	'000 tonnes	0	83435	83435
Kyanite	tonne	0	184733	184733
Limestone	'000 tonnes	10540	184059	194599
Magnesite	'000 tonnes	0	40	40
Pt. Group Of Metals	tonne	0	0.18	0.18
Sillimanite	tonne	553000	6366167	6919167
Titanium	tonne	2370712	117607659	119978371
Zircon	tonne	156509	1240355	1396864

Figures rounded off

Table – 2: Reserves/Resources of Lignite as on 01.04.2024: Kerala

(In million tonnes)

District	Proved	Indicated	Inferred	Total
Total/Kannur	–	–	9.65	9.65

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Kerala, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		1	-	370969	1	-	317936	1	-	304893
Limestone	'000t	1	379	370969	1	347	317936	1	381	304893
Sulphur#	t	-	182352	-	-	182398	-	-	216097	-

(P): provisional

#: Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

Industry/plant	Capacity ('000 tpy)
Abrasives	
Carborandum Universal Ltd, Ernakulam	NA
Carborandum Universal Ltd, Thrissur	NA
Carborandum Universal Ltd, Pattanamthitta	NA
Asbestos Products	
Hyderabad Industries Ltd (formerly, Malabar Building Products Ltd) Mulagunnathukavu, Diat. Thrissur	84
Cement	
J K Tex Coats Nadama, Kanayannur	0.030 (Cerastone)
	0.025 (Rock tiles)
	0.35 (Others)
Malabar Cements, Walayar, Distt. Palakkad	660

Table- 4 (Cont.)

Industry/plant	Capacity ('000 tpy)
Malabar Cement, Cherthala, Distt. Alappuzha (G)	200
The Travancore Cements Ltd, Nattakom, Distt. Kottayam	81
Ceramic & Clays	
Kerala Ceramics Ltd, Kundara, Distt. Kollam	18000
Tata Ceramics, Kozhikode	NA
FACR-RCF Building Product Ltd (FRBL), Kochi.	NA
Kundara Ceramics, Kollam	NA
English Indian Clays Ltd. (EICL), Thiruvananthapuram	NA
Kerala Clays and Ceramic Products Ltd., Palayangadi, Kannur	NA
Chemical	
Tecil Chemicals and Hydro Power Ltd, Chingavanam, Distt. Kottayam	30 (calcium carbide) 2 (acetylene black) 7.5 (ferrosilicon)
Cochin Minerals and Rutile Ltd, Kadungalloor, Alwaye	50 (Synthetic Rutile) 82.5 (Ferrous chloride) 30 (Ferric chloride) 8 (Recovered TiO ₂) 6 (Recovered Upgraded Ilmenite)
Electrode	
Super Electrode, Patlla	0.6
Synthetic Rutile	
CMRL, Edayar, Distt. Ernakulam	50
KMML, Chavara, Distt. Kollam	50
TiO₂ Pigment	
TTPL, Kochuveli, Distt. Thiruvananthapuram	1.8
KMML, Chavara, Distt. Kollam	40
Fertilizer	
FACT Ltd, Udyogmandal, Distt. Ernakulam	148.5 (Complex) 225 (AS)
FACT Ltd, Ambalamedu (Cochin II), Distt. Ernakulam	485 (NP/NPKs)
Ferroalloys	
INDSIL Electrosmelts Ltd, Pallatheri, Distt. Palakkad.	14
The Silcal Metallurgic Ltd, Wayalur.	3.6
Foundry	
HMT Machine Tools Ltd, Bengaluru.	1500
Glass	
Excel Glass Ltd, Pathirapally, Distt. Alappuzha.	72
Lead-Zinc	
BZL Zinc Ltd, Binanipuram (Edayar Zinc Ltd)	38 (Zn ingot) 0.08 (Cd ingot) 50 (H ₂ SO ₄)
Petroleum Refinery	
BPCL, Kochi.	12400

Note: Data for Fertilizer Industries is taken from Indian Fertilizer Scenario, FAI Statistics.



Madhya Pradesh

8,47,559 carats
Reserves (as on 01.04.2020) of diamond
have been estimated in the State

25.46 lakh tonnes
Copper Ore production have reported
in the State in 2023-24

304
mines reported production of MCDR
Minerals in Madhya Pradesh during
2023-24

MINERAL RESOURCES

Madhya Pradesh is the only diamond producing State in the country and is the leading producer of copper conc., diaspore, pyrophyllite, manganese ore, limestone and clay (others). The State hosts the country's 90% diamond, 74% diaspore, 55% laterite, 48% pyrophyllite, 41% molybdenum, 27% dolomite, 19% copper ore, 18% fireclay, 12% manganese and 8% rock phosphate ore resources.

Important mineral occurrences in the State are: bauxite in Balaghat, Guna, Jabalpur, Katni, Mandla, Rewa, Satna, Shahdol, Shivpuri, Sidhi & Vidisha districts; calcite in Barwani, Jhabua, Khandwa & Khargone districts; china clay in Betul, Chhatarpur, Chhindwara, Gwalior, Hoshangabad, Jabalpur, Khargone, Narsinghpur, Raisen, Satna, Shahdol & Sidhi districts; copper in Balaghat, Betul & Jabalpur districts; coal in Betul, Shahdol & Sidhi districts; diamond in Panna district; diaspora & pyrophyllite in Chhatarpur, Shivpuri & Tikamgarh districts; dolomite in Balaghat, Chhindwara, Damoh, Dewas, Harda, Hoshangabad, Jabalpur, Jhabua, Katni, Mandla, Narsinghpur, Sagar & Seoni districts; fireclay in Betul, Chhindwara, Jabalpur, Katni, Narsinghpur, Panna, Sagar, Shahdol & Sidhi districts; iron ore (haematite) in Betul, Gwalior, Jabalpur & Katni districts; limestone in Balaghat, Chhindwara, Damoh, Dhar, Hoshangabad, Jabalpur, Jhabua, Khargone, Katni, Mandsaur, Morena, Narsinghpur, Neemach, Rewa, Sagar, Satna, Sehore, Shahdol & Sidhi districts; manganese ore in Balaghat and Jhabua districts; ochre in Dhar, Gwalior, Jabalpur, Katni, Mandla,

Rewa, Satna, Shahdol & Umaria districts; pyrophyllite in Chhatarpur, Sagar, Shivpuri & Tikamgarh districts; quartz/silica sand in Balaghat, Dewas, Dhar, Jabalpur, Khandwa, Khargone, Morena, Rewa & Shahdol districts; talc/steatite/soapstone in Dhar, Jabalpur, Jhabua, Katni, Narsinghpur & Sagar districts and vermiculite in Jhabua district.

Other minerals that occur in the State are: barytes in Dewas, Dhar, Shivpuri, Sidhi & Tikamgarh districts; calcareous shales (used in slate pencil) in Mandsaur district; feldspar in Jabalpur & Shahdol districts; fuller's earth in Mandla district; gold in Jabalpur & Sidhi districts; granite in Betul, Chhatarpur, Chhindwara, Datia, Jhabua, Panna, Seoni & Shivpuri districts; graphite in Betul & Sidhi districts; gypsum in Shahdol district; lead-zinc in Betul district; molybdenum in Balaghat district; potash in Panna district; quartzite in Sehore district; rock phosphate in Chhatarpur, Jhabua & Sagar districts; and sillimanite in Sidhi district (Table-1). The reserves/resources of coal along with various coalfields in Madhya Pradesh are furnished in Table-2.

PRODUCTION

Madhya Pradesh is the only state that produces diamond. Apart from this, Bauxite, Copper Ore & Concentrate, Iron Ore, Manganese Ore, Phosphorite and Limestone are the principal minerals produced in Madhya Pradesh. There were 304 reporting mines in 2023-24 in case of MCDR minerals Table-3.

MINERAL-BASED INDUSTRY

The present status of each Mineral-based Industry is not readily available. However, the important large and medium-scale mineral-based industries in the Organised Sector in the State are furnished in Table-4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Madhya Pradesh

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Antimony				
Ore	tonne	7503	592	8095
Metal	tonne	75	5.92	80.92
Bauxite	'000 tonnes	18564	167695	186259
Copper				
Ore	'000 tonnes	120353	266312	386665
Metal	'000 tonnes	1571.04	2095.82	3666.86
Diamond	carat	847559	27749477	28597036
Gold				
Ore (Primary)	tonne	-	7692934	7692934
Metal (Primary)	tonne	-	8.25	8.25
Graphite	tonne	-	12640000	12640000
Iron Ore (Haematite)	'000 tonnes	54129	302870	356999
Lead-Zinc				
Ore	'000 tonnes	-	19067	19067
Lead Metal	'000 tonnes	-	36.29	36.29
Zinc Metal	'000 tonnes	-	470.53	470.53
Limestone	'000 tonnes	1692431	7960747	9653178
Manganese Ore	'000 tonnes	19558	40499	60057
Molybdenum				
Ore	tonne	-	8000000	8000000
Contained MoS ₂	tonne	-	5020	5020
Potash	Million tonnes	-	1244	1244
Rock Phosphate	tonne	9031093	49425938	58457031
Sillimanite	tonne	-	101600	101600
Silver				
Ore	tonne	-	3216000	3216000
Metal	tonne	-	159.86	159.86
Vermiculite	tonne	-	329	329

Figures rounded off

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Madhya Pradesh

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	15425.17	12378.97	5010.99	32818.13
Johilla	185.08	262.59	32.83	480.5
Umaria	177.7	3.59	-	181.29
Pench-Kanhan	2112.46	922.85	1903.13	4938.44
Pathakhara	290.8	88.13	68	446.93
Gurgunda	-	84.92	53.39	138.31
Mohpani	7.83	-	-	7.83
Sohagpur	3630.13	4992.59	289.78	8912.5
Singrauli	9021.17	6024.3	2663.86	17709.33

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in Madhya Pradesh, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		263	-	33579947	275	-	33729906	304	-	38637608
Bauxite	tonne	18	608848	486524	23	601911	552530	24	597278	640503
Copper Ore	tonne	-	2442459	-	-	2165042	-	-	2545820	-
Copper Conc.	tonne	1	65914	5560337	1	63860	5193489	1	69873	7151077
Iron Ore	'000 tonne	23	7375	4727826	24	4365	2475698	25	6047	3812315
Manganese Ore	tonne	46	845351	6842478	50	855874	7437348	54	1033424	7703313
Phosphorite	tonne	6	113730	111393	6	444985	483255	6	236964	331722
Diamond	crt	3	266	18051	3	388	61473	3	349	23858
Limestone	'000 tonne	166	49807	15833338	168	50597	17526113	191	62556	18974820

(P): Provisional

Table – 4: Principal Mineral-based Industries

Industry/plant	Capacity ('000 tpy)
Aluminium/Alumina	
Hindalco Industries Ltd, Mahan	360 (Aluminium)
Aluminium, Bargwan, Distt Singrauli	
Asbestos Products	
Everest Building Products Ltd, Kymore	NA
Kalani Industries Pvt. Ltd, Pitampur, Dhar	NA
Ramco Industries Ltd, Maksi, Distt Shajapur	NA
Calcined Lime	
Rekha Harlalka, Jukehi, Maihar	11
Padampani Tripathi, Mamalime Industries	9.6
Rajarwara, Katni	
Cement	
ACC Ltd, Kymore, Distt Katni	2720
Bhilai Jaypee Cement Ltd, Babupur, Satna	1300
Birla Corpn. Ltd, (Satna Cement Works & Birla Vikas Cement), Satna	2200
Birla Corporation Ltd, (Erstwhile Reliance Cement Pvt. Ltd, Maihar, Distt Satna	3000
Century Textiles & Ind. Ltd, Maihar Cement, Maihar (Unit I&II), Distt Satna	4200
Heidelberg Cement (I) Ltd, Narsingarh, Distt Damoh	2000
Jaiprakash Power Ventures, Singrauli (G)	2000
Jaypee Rewa Cement Plant, Distt Rewa	2500
Jaypee Bela Cement Plant, Distt Rewa	2600
KJS Cement, Rajnagar, Distt Satna	2200
Prism Cement Ltd, (Unit I & II), Satna	6600
Satguru Cement Pvt. Ltd, Ghursal, Gandhawani	95
UltraTech Cement Ltd, Sidhee	2300
UltraTech Cement, Dhar Cement Plant,	3500
Tonki, Temarni sounul, Golpura Manawar	
UltraTech Cement, Vikram Cement Plant,	4500 (OPC)
Khor, Distt Neemuch	4500 (PPC)
UltraTech Cement Ltd, Majhigawan, Rampur Naikin Ceramic	3000
Roca Bathroom Products Ltd, Dewas	NA
Govind Tiles Pvt. Ltd, Garra, Distt Balaghat	NA
Calcined lime	
Som Lime Work, Jukehi, Katni	21.6
Jai Mata Lime Industries Pathra, Katni	15.2
Dharampal Industries Pathra, Katni	6

Table- 4 (Concl'd.)

Industry/plant	Capacity ('000 tpy)
Sampuran Singh Saluja Patra, Katni	6.07
Fertilizer	
Agro Phos. (India) Ltd, Dewas	45 (SSP)
Arihant Ferts. & Chems. India Ltd, Kanawati, Neemuch	66 (SSP)
Basant Agro Tech (India) Ltd, Jawad, Neemuch	45 (SSP)
Coromandel International Ltd, (Formerly, Liberty Urvarak Ltd,) Nirmani Khargone	100 (SSP)
Indra Industries Ltd, (Formerly, Swastik Ferts & Chems Ltd), Indore, Dhar	66 (SSP)
KMN Chemicals & Fertilizers Ltd, Diwanganj, Raisen	60 (SSP)
Khaitan Chemical & Fertilizers Ltd,	400 (SSP)
Nimrani, Distt Khargone	115.5 (H ₂ SO ₄)
NFL, Vijaipur (Unit I & II), Distt Guna	2066.1 (Urea)
Krishna Phoschem Ltd, Meghnagar, Jhabua	120 (SSP)
Madhya Bharat Agro Products Ltd, Rajoa, Sagar	60 (SSP)
Madhya Bharat Phosphate Pvt. Ltd, (Unit I),	132 (SSP)
Diwanganj, Sanchi, Raisen	
Madhya Bharat Phosphate Pvt. Ltd, (Unit II),	165 (SSP)
Meghnagar, Jhabua	
Mexican Agro Chemical Ltd, (Formerly, Asha Phosphates Ltd.), Jaggakhedi, Mandsaur	60 (SSP)
Mukteswar Fertilizers Ltd, Narayankhedi, Ujjain.	60 (SSP)
Rama Phosphates Ltd, Indore	250 (SSP)
	102 (H ₂ SO ₄)
Suman Phosphates and Chemicals Ltd, Indore	330 (SSP)
Varun Fertilizers Pvt. Ltd, Dewas	100 (SSP)
Ferroalloys	
Crescent Alloys Pvt. Ltd, Seoni	4.5
Jalan Ispat Castings Ltd, Meghnagar, Distt Jhabua	12
MOIL Ferro Manganese Plant, Bharveli, Distt Balaghat	10
Petroleum Refinery	
Bharat Oman Refineries Ltd, Bina, Distt Sagar	6000
Refractory	
ACC Refractories, Katni	65
Calderys India Refractories Limited	78
Katni Refractory Works, Katni	30 (Binder)
Murwara	9 (Grout)
Mahakoshal Refractories Pvt. Ltd, Katni	61.09
Mahakoshal Refractories Pvt. Ltd, Gudri, Bohariband	31
Premier Refractories India Pvt. Ltd, Katni.	50

G: Grinding Unit



Maharashtra

3,808 million tonnes
of Limestone Resources along with vast resource base of bauxite, kyanite and manganese ore have been established in the State.

₹ 5,523.79 crore
value of production of MCDR minerals (excluding Atomic, Fuel and Minor Minerals) were reported in the year 2023-24.

96
Mines reported production in 2023-24 in case of MCDR minerals.

MINERAL RESOURCES

Maharashtra is the sole producer of fluorite (graded) and the principal producer of bauxite, kyanite, manganese ore, quartzite and sand (others). The principal mineral-bearing belts in Maharashtra are Vidarbha area in the east and Konkan area in the west. Important mineral occurrences are: bauxite in Kolhapur, Raigad, Ratnagiri, Satara, Sindhudurg & Thane districts; china clay in Amravati, Bhandara, Chandrapur, Nagpur, Sindhudurg & Thane districts; chromite in Bhandara, Chandrapur, Nagpur & Sindhudurg districts; coal in Nagpur, Chandrapur & Yavatmal districts; dolomite in Chandrapur, Nagpur & Yavatmal districts; fireclay in Amravati, Chandrapur, Nagpur & Ratnagiri districts; fluorite & Shale in Chandrapur district; iron ore (haematite) in Chandrapur, Gadchiroli & Sindhudurg districts; iron ore (magnetite) in Gondia district; kyanite in Bhandara & Nagpur districts; laterite in Kolhapur district; limestone in Ahmednagar, Chandrapur, Dhule, Gadchiroli, Nagpur, Nanded, Pune, Sangli & Yavatmal districts; manganese ore in Bhandara, Nagpur & Ratnagiri districts; corundum & pyrophyllite in Bhandara district; quartz & silica sand in Bhandara, Chandrapur, Gadchiroli, Gondia, Kolhapur, Nagpur, Ratnagiri & Sindhudurg districts; quartzite in Gondia & Nagpur districts; and sillimanite in Chandrapur district.

Other minerals that occur in the State are: barytes in Chandrapur & Gadchiroli districts; copper in Bhandara, Chandrapur, Gadchiroli & Nagpur districts; feldspar in Sindhudurg district; gold in Bhandara & Nagpur districts; granite in Bhandara, Chandrapur, Dhule, Gadchiroli, Nagpur, Nanded, Nashik, Sindhudurg & Thane districts; graphite & mica in Sindhudurg district; lead-zinc & tungsten in Nagpur district; marble in Bhandara & Nagpur districts; ochre in Chandrapur & Nagpur districts; silver & vanadium in Bhandara district; steatite in Bhandara, Ratnagiri & Sindhudurg districts; and titanium minerals in Gondia & Ratnagiri districts (Table-1). As per the AMD of the Department of Atomic Energy India, Maharashtra state accounted for 5.50 million tonnes of ilmenite resources and 0.01 million tonnes of rutile resources. The coal reserves and resources along with the various coalfields located in the State are shown in Table-2.

Table –1: Reserves/Resources of Minerals as on 01.04.2020: Maharashtra

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Bauxite	"000 tonnes	38472	193958	232430
Chromite	"000 tonnes	5	533	538
Copper				
Ore	"000 tonnes	-	17755	17755
Metal	"000 tonnes	-	158.08	158.08
Fluorite	tonne	386142	1'00000	486142
Gold				
Ore (Primary)	tonnes	-	1627'000	1627'000
Metal (Primary)	tonne	-	3.64	3.64
Graphite	tonne	-	116'0000	116'0000
Iron ore (Haematite)	'000 tonnes	15241	286304	301544
Iron ore (Magnetite)	'000 tonnes	578	1210	1788
Kyanite	tonne	332389	3063615	3396004
Lead-zinc				
Ore	"000 tonnes	-	9272	9272
Zinc metal	"000 tonnes	-	589.67	589.67
Limestone	'000 tonnes	701349	3107044	3808392
Manganese ore	'000 tonnes	17733	41303	59036
Rare-Earth Elements	tonne	-	2090	2090
Sillimanite	tonne	181748	30580	212328
Silver				
Ore	tonne	-	235'000	235'000
Metal	tonne	-	0.23	0.23
Titanium	tonne	303551	3980786	4284337
Tungsten				
Ore	tonne	-	10122250	10122250
Contained WO ₃	tonne	-	18590.72	18590.72
Vanadium				
Ore	tonne	-	384630	384630
Contained V ₂ O ₅	tonne	-	1538.52	1538.52

Figures rounded off

Table – 2: Reserves/Resources of Coal as on 01.04.2024 : Maharashtra

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	8163.11	3371.82	1816.7	13351.63
Wardha Valley	5107.02	1837.72	1411.09	8355.83
Kamptee	2046.24	937.91	107.21	3091.36
Umrer Makardhokra	308.41	—	160.7	469.11
Nand Bander	691.44	596.19	117.7	1405.33
Bokhara	10	—	20	30

Source: Coal Directory of India, 2023-24.

PRODUCTION

Coal, Bauxite, Iron Ore, Manganese Ore, Fluorite, Kyanite, Sillimanite and Limestone are the principal minerals produced in Maharashtra State. The value of all minerals production (Excluding Atomic, Fuel and Minor Minerals)

was estimated at ₹ 5,523.79 crore for the year 2023-24. There were 96 reporting mines in 2023-24 in case of MCDR minerals. The details of mineral production in the state are shown in Table-3.

Table – 3 : Mineral Production in Maharashtra, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		74	-	19372221	83	-	33596406	96	-	55237866
Bauxite	t	12	640345	387799	14	900608	590438	19	1058581	619010
Iron Ore	'000t	11	1984	6653399	10	6109	20708375	10	10727	40921814
Manganese Ore	t	28	732018	8425784	29	751104	8288184	31	1032992	9117126
Fluorite (graded)	t	1	1237	9430	1	910	5850	2	782	4658
Kyanite	t	3	1460	3031	3	334	844	3	3114	10126
Sillimanite %	t	-	3432	8283	-	1437	3757	-	771	2648
Limestone	'000t	19	15758	3884495	26	16656	3998958	31	19320	4562484
Sulphur #	t	-	53165	-	-	46283	-	-	50577	-

(P) Provisional

% Associate with Kyanite

Recovered as by-product from oil refinery.

Further as per the information available on the Ministry of Mines website, during 2023-24, a total of two mining leases for 'Limestone' was granted and a total of eight composite licence, i.e. five for 'Iron Ore' and one each for 'Bauxite', 'Copper' and 'Manganese Ore' were granted in the state.

MINERAL-BASED INDUSTRY

The present status of each Mineral-based Industry is not readily available. However, the important mineral-based industries in the Organised Sector in the State are given in Table-4.

Table – 4: Principal Mineral-based Industries

(Excluding Atomic, Fuel and Minor Minerals)

Industry/ Plant	Capacity ('000 tpy)
Abrasives	
Grindwell Norton Ltd, Mora, Uraon, Raigad	NA
Aluminium Products	
Hindalco, Recycling plant, Taloja	60
Hindalco, Mouda, Distt. Nagpur	30 (rolling mill) 14 (conductor rod)
Asbestos Products	
Everest Building Products Ltd, Mulund	NA
Hyderabad Industries Ltd, Musarane	60
Newkem Products Corp, Mumbai	9.9
Swastik Industries, Pune	NA
Cement	
ACC Ltd, Ghugus, Distt. Chandrapur	3800
Ambuja Cement Ltd, (Maratha Cement Works), Upparwahi, Chandrapur	4750
India Cement, Vajinath, Parli, Distt Beed (G)	1100
JSW Cement, Dolvi, Distt. Raigad	1000 (slag cement)
Manikgarh Cement, (I) Korpana, Distt. Chandrapur	2000
Manikgarh Cement, (II) Korpana, Distt. Chandrapur	4000
Murli Industries Ltd, Naranda, Distt. Chandrapur.	3000
Orient Cement, Jalgaon (G)	2000
Birla Corpn. Ltd, Butibori, Distt. Nagpur (G)	500
UltraTech Cement, Hotgi, Distt. Solapur (G)	4000
UltraTech Cement Ltd, Awarpur, Distt. Chandrapur	6000
UltraTech Cement Ltd, Ratnagiri Works (G), Distt. Ratnagiri	4500 (Clinker) 480

Table-4 (Cont.)

Industry/ Plant	Capacity ('000 tpy)
UltraTech Cement Ltd, Nagpur	2000
Zuari Cement, Solapur	1200
Ceramics	
H & R Johnson (India) Ltd, Pen	154.8
Joglekar Refractory & Ceramics Pvt. Ltd, Rabale, Distt. Thane.	364.8
Jyoti Ceramic Industries Pvt. Ltd, Satpur	0.16 (Ref. coating) 1.0 (Ceramic Product)
NITCO Tiles Ltd, Raigad	66 lakh (sq. m)
Chemicals	
Borax Morarji Ltd, Ambarnath	25 (borax) 8 (boric acid)
Century Rayon, Shahad, Distt. Thane	25 (rayon yarn) 20 (caustic soda)
Foseco India Ltd, Sanswadi	15 (foundry chemicals)
Gargi Huttenes Albertus Pvt.Ltd, Kukshet, Navi Mumbai	12 (Foundry Chemical)
National Peroxide Ltd, Kalyan, Distt Thane.	1.4 (sodium per borate)
Star Earth Minerals Pvt. Ltd, Tanjola, Panvel	0.6 (zirconium basic carbonet)
Sudarshan Chemical Ind. Ltd, Roha, Distt Raigad	5.2 (pigments)
Tecil Chemical & Hydro Power Ltd, Mumbai.	30 (calcium carbide)
Zirconium Chemicals Pvt. Ltd, Taloja, Distt. Raigad	0.3 (zirconium salt)
Copper Wire Rods	
HCL, Copper Project, Taloja	60
Electrode	
GEE Ltd, Thane.	4.02 (Mill. m)
Weldfast Electrode Pvt. Ltd, Nagpur	15.9
Weldstrong Electrode Pvt. Ltd, Butibori, Hihgna Nagpur	0.9 0.15 (Welding flux)
Electrolytic Manganese Dioxide	
MOIL, Dongri Buzurg, Distt. Bhandara	1
Fertilizers	
Balaji Fertilisers Pvt. Ltd, Nanded	20 (SSP)
Basant Agro Tech (India) Ltd, Barshi Takli, Akola	120 (SSP)
Basant Agro Tech (India) Ltd, Jalgaon.	132 (SSP)
BEC Fertilizer (Unit of Bhilai Engg. Corpn. Ltd), Gunjakheda, Wardha	66 (SSP)
Bharat Agri Fert & Realty Ltd, Kharivali, Thane	132 (SSP)
Coromandel International Ltd, (Formerly, Liberty Phosphate Ltd), Pali, Raigad	66 (SSP)
Deepak Fertilizers & Petrochemical Corporation Ltd, Taloja	230 (ANP)
Rama Krishi Rasayan (A division of Rama Phosphates Ltd), Loni Kalbhor, Pune	132 (SSP)
Shiva Global Agro Industries Ltd, (Formerly, Shiva Fertilizers Ltd), Nanded	120 (SSP)
Shri Bhavani Mishra Fertilizers Pvt. Ltd, Vazirabad, Nanded	30 (SSP)

Table-4 (Cont.)

Industry/ Plant	Capacity ('000 tpy)
Shree Pushkar Chems & Fertiliser Ltd, Lote Porshuram, Khed, Ratnagiri	100 (SSP)
Zuari Fertilizers and Chemicals Ltd, Mahad, Distt. Raigad	216 (SSP)
RCF, Trombay	330 (Urea)
	690 (Complex)
RCF, Thal, Distt. Raigad	2000 (Urea)
Pesticides	
Hindustan Insecticides Ltd, Rasaini, Distt. Raigad	13.2
Paint	
Jespco, Irechwara, Miraj	8 (Zircon Paint)
Glass	
Ace Glass Containers Ltd, Pimpri, Distt. Nashik	NA
Empire Industries Ltd, (Vitrum Glass), Vikroli, Mumbai	37.5
Hindustan National Glass & Industries Ltd, Nashik	320 TPD
Iron & Steel	
JSW Ispat Steel Ltd, Dolvi, Raigad	5400 (Sinter)
	1600 (Sponge iron)
	5040 (Crude/Liquid steel)
	3500 (pig iron)
Lloyds Steel Ltd, Wardha	600 (HRC)
	350 (CRC)
	250 (GPC)
Indian Seamless Steel & Alloys Ltd, Jejuri, Distt Pune	450 (seamless tubes)
Sunflag Iron & Steel Co. Ltd, Warrthy, Mohadi	350 (alloy & carbon steel)
	262 (sponge iron)
	250 (Pig iron)
	250 (sinter)
	505 (Finished steel)
Uttam Galva Metallics Ltd, Bhugaon, Wardha	886.95 (Sinter)
	525 (pig iron)
Lime	
Hetendra Lime Products, Rajur, Wani	5.5
Swastic Lime Factory, Rajur, Wani	5.5
Swastic Mineral & Lime Industries, Rajur, Wani	5.5
Pellet	
Amba River Coke Ltd, Dolvi, Pen	4000
Pig Iron	
Ispat Metallics India Ltd, Dolvi, Raigad.	2000
Lint Export Pvt. Ltd, Chincholi, Mohol	0.25
Tata Metaliks Ltd, (Usha Ispat Ltd, Redi), Distt Sindhudurg.	300
Sona Alloys Pvt. Ltd, Satara.	314
Usha Ispat Ltd, Redi.	300
Uttam Galva Metallics Ltd, Bhugaon, Wardha	225
Gopani Iron Ore Ltd, Chandrapur.	389.95 (Sinter)
	144
	75 (Semi-Finished Steel)
Lloyds Metals & Engineers, Ghugus, Chandrapur.	300
JSW Steel Salav Ltd,	900

Table- 4 (Concl'd.)

Industry/ Plant	Capacity ('000 tpy)
Welspun Max Steel Ltd, (formerly Vikram Ispat), Distt. Raigad	
Ferroalloys	
Chandrapur Ferro Alloys Plant (SAIL), (formerly Maharashtra Elektros melt Ltd), Chandrapur	100
Minex Metallurgical Co. Ltd, Nimji, Kalmeshwar	0.250 (Fe-Ti)
Natural Sugar & Allied Industries Ltd, Sai Nagar, Ranjani, Distt. Osmanabad	16.5 (Si-Mn) 16.5 (H. C.Si-Mn)
SRC Chemical Pvt. Ltd, Borieandi, Daund, Pune	6
Welspun Maxsteel Ltd, Salav, Raigad.	90
Refractory	
ACE Refractories, Nagpur.	60
NECO Ceramics	NA
Ceraflux India Pvt. Ltd, Gokul Shirgaon, Kolhapur	2.7 (Ref. Die releasing Agent) 2.7 (Ref. Coating)
Calderys India Refractories Limited	58 (castable)
Nagpur Refractory Works, Ruikhairi, Butibori, Nagpur	
Joglekar Refractories Pvt. Ltd, Rabale, Navi Mumbai	4.8 (Ramming Mass) 0.54 (Chrome Ore +60) 0.15 (Chrome Ore -60)
Petroleum Refinery	
BPCL, Mumbai.	12000
HPCL, Mumbai.	9500

(G) : Grinding units.



Manipur

Chromite and Limestone are important mineral occurrences in the State.	6.66 million tonnes resources of chromite have been established in the State.	46 million tonnes resources of limestone have been established in the State.
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MINERAL RESOURCES

Important mineral occurrences in the State are chromite and limestone. The total resources of chromite and limestone in the State are 6.66 million tonnes and 46 million tonnes respectively (Table-1).

PRODUCTION

No major mineral production was reported from Manipur in the year 2023-24.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Manipur

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Chromite	'000 tonnes	-	6657	6657
Limestone	'000 tonnes	-	46053	46053

Figures rounded off



Meghalaya

23,835 million tonnes
of total Resources of limestone have
been established in the State as on
01.04.2020

10,070 thousand tonnes
of Limestone were produced in the
year 2023-24

17
Limestone mines reported production
in the State during 2023-24

MINERAL RESOURCES

Coal and limestone are the major minerals commercially mined in Meghalaya. Coal occurs in the Mikir, Khasi, Jaintia, and Garo Hills, while limestone is found in West Garo, East and West Khasi, and Jaintia Hills. Other mineral occurrences include apatite (Jaintia Hills), china clay (East and West Garo, Jaintia, East Khasi Hills), base and precious metals such as copper, lead-zinc, silver, and titanium (East Khasi Hills), feldspar and rock phosphate (East Garo, Jaintia Hills), fireclay (East Khasi, West Garo Hills), granite (West Khasi Hills), iron ore – magnetite (East Garo Hills), quartz and silica sand (East and West Garo, East Khasi Hills), and sillimanite (West Khasi Hills). While these are not under active mining, they hold future development potential. The reserves/resources as per NMI as on 01.04.2020 is furnished in Table-1 whereas coalfield-wise reserves and resources are detailed in Table-2.

PRODUCTION

Limestone was the principal MCDR mineral produced in Meghalaya during 2023-24, with output reaching at 10,070 thousand tonnes registering an increase of 24% over the previous year's production of 8,111 thousand tonnes. A total of 17 limestone mines were reported in the state during 2023-24 (Table-3).

MINERAL-BASED INDUSTRY

The current status of individual mineral-based industries in the state is not readily available. However, key mineral-based industries operating in the organised sector are listed in Table-4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020 : Meghalaya

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	-	1300000	1300000
Bauxite	'000 tonnes	-	4300	4300
Copper				
Ore	'000 tonnes	-	880	880
Metal	'000 tonnes	-	9	9
Iron Ore (Haematite)	'000 tonnes	-	225	225
Iron Ore (Magnetite)	'000 tonnes	-	3380	3380
Lead-Zinc Ore				
Ore	'000 tonnes	-	880	880
Lead metal	'000 tonnes	-	16.5	16.5
Zinc metal	'000 tonnes	-	14	14
Limestone	'000 tonnes	251043	23583945	23834988
Rock Phosphate	tonne	-	1311035	1311035
Sillimanite	tonne	82512	55807	138319
Silver				
Ore	tonne	-	880000	880000
Metal	tonne	-	19.8	19.8
Titanium	tonne	-	3345000	3345000

Figures rounded off

Source: NMI

Table – 2: Reserves/Resources of Coal as on 01.04.2024 : Meghalaya

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	96	17	471	583
West-Darangiri	65	0	60	125
East Darangiri	0	0	34	34
Balphakram-Pendengurr	0	0	107	107
Siju	0	0	125	125
Langrin	17	17	106	140
Mawlong Shelia	2	0	4	6
Khasi Hills	0	0	10	10
Bapung	11	0	23	34
Jayanti Hill	0	0	2	2

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Meghalaya, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		16	-	2974757	17	-	4127730	17	-	4765751
Limestone	'000 tonnes	16	6399	2974757	17	8111	4127730	17	10070	4765751

(P): Provisional

Table – 4 : Principal Mineral-based Industries

Industry/plant	Capacity ('000 tpy)
Cement	
Amrit Cement Industries Ltd, Khleriat, Distt Jaintia Hills	3000
Cement Manufacture Co. Ltd, Lumshnong, Distt Jaintia Hills	792
DCBL Meghalaya Cements Ltd, Thangskai, Narpuh Distt Jaintia Hills	1500
Green Valley Industries, Nongsning, Jowai, Distt Jaintia Hills.	1000
JUD Cement Ltd, Narpuh, Distt Jaintia Hills	500
Mawmluh Cherra Cements Ltd, Cherrapunjee, Distt East Khasi Hills	185
Meghalaya Cements Ltd, Thangskai, Distt Jaintia Hills	860
Megha Technical & Engineering (P) (MTEPL), Lumshnong, Distt Jaintia Hills	700
Hills Cement, Jaintia Hills	1000
RNB Cement, East Khasi	400
Ferroalloys	
Jaintia Ferro Alloys Pvt. Ltd, Byrnihat.	6
Maithan Alloys Ltd, Ri-Bhoi	15 MVA
Maithan Alloys Ltd, Raja Bagan	28
Nalari Ferro alloys Pvt Ltd, Norbhog	11
Khasi alloys Pvt. Ltd, EPIP Meghalaya	4.1
Iron & Steel	
Jai Kamakhya Alloy Pvt. Ltd	815 tpd

Source: Data sourced from respective websites of cement industries as well as Survey of Cement Industry & Directory.



Mizoram & Nagaland

Important mineral occurrences in Mizoram are lignite, sandstone and pyrites

Important mineral occurrences in Nagaland are coal, iron ore (magnetite), cobalt, dunite, nickeliferous chromite and limestone.

MIZORAM

Mineral Resources

Occurrences of lignite, sandstone and pyrites are reported from the State. Major deposits of economic importance have not been reported so far in the State.

Production

No major mineral production was reported from Mizoram during 2023-24.

NAGALAND

Mineral Resources

Important mineral occurrences in the State are: coal in Borjan, Jhanzi-Disai, Tiesang, Tiru Valley, Changki and DGM Coalfields; iron ore (magnetite), cobalt, dunite and nickeliferous chromite in Tuensang district and limestone in Phek and Tuensang districts (Table-1). The various coalfields and their reserves/resources are furnished in Table-2

Production

No major mineral production was reported from Nagaland during 2023-24.

Table – 1 : Reserves/Resources of Minerals as on 01-04-2020 : Nagaland

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Chromite	'000 tonnes	-	3200	3200
Cobalt Ore	Million tonnes	-	5	5
Copper Ore	'000 tonnes	-	2000	2000
Copper Metal	'000 tonnes	-	15	15
Iron Ore (Magnetite)	'000 tonnes	-	5280	5280
Limestone	'000 tonnes	-	1752200	1752200
Nickel Ore	Million tonnes	-	5	5

Figures rounded off.

Table – 2 : Reserves/Resources of Coal as on 1.4.2024 : Nagaland

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Nagaland (Total)	8.76	21.83	447.72	478.31
Borjan	5.50	–	4.50	10.00
Jhanzi-Disai	2.00	21.83	109.26	133.09
Tiensang	1.26	–	2.00	3.26
Tiru Valley	–	–	6.60	6.60
Changki	–	–	31.89	31.89
DGM	–	–	293.47	293.47

Source: Coal Directory of India, 2023-24.



Odisha

27,987 lakh tonnes
of Iron Ore Reserves (Haematite) have been established in the State. Odisha has reported vast Resources of Bauxite, Chromite, Iron Ore, Manganese Ore, Graphite and Limestone.

120
mines in case of MCDR minerals reported production in 2023-24.

₹ 63,942 crore
value of mineral production were reported in the State during the year of 2023-24.

MINERAL RESOURCES

Odisha is the leading producer of chromite, garnet (abrasive), bauxite, manganese ore, iron ore, sillimanite, quartzite and dolomite. The State hosts the country's sole resources of ruby. It accounts for the country's 96% chromite, 93% nickel ore, 68% PGM metal, 69% cobalt ore, 41% bauxite, 34% manganese, 39% iron ore (haematite), 25% sillimanite, 24% fireclay, 23% pyrophyllite, 20% vanadium ore, 17% mica, and 10% dolomite resources. As per AMD of the Department of Atomic Energy, Odisha, accounted for 150.62 million tonnes of rutile resources.

Important minerals that occur in the State are: bauxite in Balangir, Kalahandi, Kandhamal, Kendujhar, Koraput, Malkangiri, Rayagada & Sundargarh districts; china clay in Bargarh, Boudh, Balangir, Kendujhar, Koraput, Mayurbhanj, Sambalpur & Sundargarh districts; and chromite in Balasore, Cuttack, Dhenkanal, Jajpur & Kendujhar districts. Chromite deposits of Sukinda and Nuasahi ultramafic belt constitute 95% of the country's chromite resources. Besides, coal occurs in Ib river valley and Talcher coalfield, Dhenkanal district; dolomite in Bargarh, Kendujhar, Koraput, Sambalpur & Sundargarh districts; dunite/pyroxenite in Kendujhar and Sundargarh districts; fireclay in Angul, Cuttack, Dhenkanal, Jharsuguda, Khurda, Puri, Sambalpur & Sundargarh districts; garnet in Ganjam, Kalahandi & Sambalpur districts; graphite in Bargarh, Boudh,

Balangir, Kalahandi, Koraput, Nuapada & Rayagada districts; iron ore (haematite) in Dhenkanal, Jajpur, Kendujhar, Koraput, Mayurbhanj, Sambalpur & Sundargarh districts; iron ore (magnetite) in Mayurbhanj district; limestone in Bargarh, Koraput, Malkangiri, Nuapada, Sambalpur & Sundargarh districts; manganese ore in Balangir, Kendujhar, Koraput, Rayagada, Sambalpur & Sundargarh districts; Pyrophyllite in Kendujhar district; quartz/silica sand in Boudh, Balangir, Kalahandi, Sambalpur & Sundargarh districts; quartzite in Balangir, Dhenkanal, Jajpur, Jharsuguda, Kendujhar, Mayurbhanj, Sambalpur & Sundargarh districts; sillimanite in Ganjam & Sambalpur districts; talc/steatite/soapstone in Mayurbhanj, Sundargarh & Sambalpur districts; titanium minerals in Dhenkanal, Ganjam, Jajpur & Mayurbhanj districts; and zircon in Ganjam district.

Other minerals that occur in the State are asbestos in Kendujhar district; cobalt in Cuttack & Jajpur districts; copper in Mayurbhanj & Sambalpur districts; granite in Angul, Boudh, Balangir, Cuttack, Deogarh, Dhenkanal, Ganjam, Kendujhar, Khurda, Koraput, Mayurbhanj, Nuapada, Rayagada & Sambalpur districts; lead in Sargipalli area, Sundargarh district; mica in Sonapur district and nickel in Cuttack, Kendujhar & Mayurbhanj districts. Occurrences of ruby and emerald are reported from Balangir and Kalahandi districts, respectively. Platinum Group of Metals occur in Kendujhar district; silver in Sundargarh district; tin in Koraput

& Malkangiri districts; and vanadiferous magnetite occurs in Balasore & Mayurbhanj districts (Table-1). The various coalfields along with their reserves/resources are given in Table-2.

PRODUCTION

During 2023-24, the important minerals produced in the state were Bauxite, Chromite, Iron Ore, Manganese Ore, Graphite and Limestone. The number of reporting mines in 2023-24 was 120 in case of MCDR minerals (Table-3).

MINERAL-BASED INDUSTRY

The principal mineral-based industries in the Organised Sector in the State of Odisha are furnished in Table-4.

Table –1: Reserves/Resources of Minerals as on 01.04.2020: Odisha

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Asbestos	tonne	-	56700	56700
Bauxite	'000' tonnes	409740	1647284	2057024
Chromite	'000' tonnes	78031	240237	318269
Cobalt	Million tonnes	-	31	31
Copper				
Ore	'000 tonnes	-	11991	11991
Metal	'000 tonnes	-	97.03	97.03
Garnet	tonne	8330046	1177318	9507364
Graphite	tonne	2838414	17142707	19981121
Iron Ore (Haematite)	'000 tonnes	2798749	6610582	9409331
Iron Ore (Magnetite)	'000 tonnes	-	242	242
Lead-Zinc Ore	'000 tonnes	-	1750	1750
Lead Metal	'000 tonnes	-	76.96	76.96
Limestone	'000 tonnes	468580	1727424	2196004
Manganese Ore	'000 tonnes	11470	160058	171528
Nickel Ore	Million tonnes	-	175	175
Pt. Group of Metals	tonne	-	14.2	14.2
Sillimanite	tonne	5640985	12062208	17703193
Silver				
Ore	tonne	-	1749500	1749500
Metal	tonne	-	64.91	64.91
Tin				
Ore	tonne	-	15618	15618
Metal	tonne	-	652.73	652.73
Titanium	tonne	12654141	53019062	65673203
Vanadium				
Ore	tonne	-	4864795	4864795
Metal	tonne	-	13557.94	13557.94
Zircon	tonne	476672	390247	866919

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Odisha

(In million tonnes)				
Coalfield	Proved	Indicated	Inferred	Total
Total	53799.43	39053.01	6351.39	99203.83
Ib-River	19259.25	21612.98	2345.66	43217.89
Talcher	34540.18	17440.03	4005.73	55985.94

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in Odisha, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24(P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		129	-	625422945	119	-	542501227	120	-	639421108
Bauxite	tonne	6	16449396	19284303	6	17403320	20434387	6	17591967	19789023
Chromite	tonne	18	3785625	47969167	18	3557289	46633415	18	3147954	48863434
Iron Ore	'000 tonnes	69	136792	553095800	64	140932	469762857	66	148959	564457411
Manganese Ore	tonne	21	512591	2429734	15	644218	3033883	16	650856	3222884
Graphite (r.o.m.)	tonne	6	26620	69216	6	28411	103302	6	67315	149102
Iolite	kg	3	27	191	3*	-	-	1*	-	-
Limestone	'000 tonnes	6	7059	2574534	7	6551	2533383	7	8019	2939254
Sulphur #	tonne	-	207831	-	-	228518	-	-	257338	-

(P) Provisional

*: Only labour reported.

#: Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

(’000 tpy)

Industry/plant	Capacity
Aluminium/Alumina	
Hindalco Industries Ltd, Hirakud	215 (aluminium)
Hindalco Industries Ltd, Aditya Aluminium, Lapanga, Distt. Sambalpur	360 (aluminium)
NALCO, Damanjodi, Dist. Koraput	2275 (alumina)
NALCO, Angul	460 (aluminium)
Utkal Alumina International Limited, Rayagada	1500 (alumina)
Vedanta Aluminium Ltd, Lanjigarh, Dist. Kalahandi	2000 (alumina)
Vedanta Aluminium Ltd, Jharsuguda, Dist. Sambalpur	1750 (aluminium)
Asbestos Products	
UAL Industries Ltd, Korian, Dist. Dhenkanal	NA
Konark Cement & Asbestors Industries Ltd, Bhubaneswar	NA
Cement	
ACC Ltd, Bargarh Cement Ltd, Bargarh	2140
Ultra-Tech Cement Ltd, Jharsuguda (G)	2600
OCL India Ltd, Rajgangpur, Distt. Sundargarh	4000
	1064 (Refractory)
	2900 (Clinker)
OCL India Ltd, Kapilas (G), Cuttack	1350
Toshali Cements Pvt Ltd, Ampavalli, Distt. Koraput	200
Ceramics	
Prabhu Ceramics & Minerals Pvt Ltd, Majhipali, Rengali, Sambalpur	24 (Acidic Ramming Mass)
	9.6 (EBT Filling Mass)
Chemical	
Arrow Minerals & Metals Pvt. Ltd, Vejidihi, Banspal	1.8 (Manganese Oxide)
	2.25 (Manganese dioxide powder)
Chrome Concentrate	
K L Resources Pvt. Ltd	74.7



Punjab

Quartz / silica sand and quartzite are the important minerals found in the State

81,912 thousand tonnes
of quartzite resources were established in the State

Mineral Resources

Minerals reported to occur in the State are quartz and silica sand in Hoshiarpur district; and quartzite in Hoshiarpur & Ropar (Rupnagar) districts (Table - 1).

Production

During 2023-24, no production of major mineral was reported in the State of Punjab.

Mineral-based Industry

The important Mineral-based Industries in the Organised Sector in the State are furnished in Table-2.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Punjab

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Quartz/silica sand [#]	'000 tonnes	-	3927	3927
Quartzite [#]	'000 tonnes	-	81912	81912

[#] Figures rounded off

[#] Declared as Minor mineral vide Gazette Notification dated 10.02.2015

Table – 2: Principal Mineral-based Industries

(^{'000 tpy})

Industry/plant	Capacity
Alloy Steel	
Antarctic Industries Ltd, Ludhiana	120
Cement	
Ambuja Cement, Ropar (Rupnagar) ^(G)	2500
Asian FCPL, Patiala	1500
Ambuja Cement, Bathinda ^(G)	1200
UltraTech Cement, Bathinda ^(G)	1750
Chemical	
Siel Chemical Complex, Charatrampur	74.3 (NaOH)
	65.8 (Cl)
	18 (bleaching powder)
	66.0 (HCl)
VTPL (Vardhaman Industries Ltd), Distt Patiala	60
Fertilizer	
NFL, Nangal, Distt. Ropar (Rupnagar)	478.5 (Urea)
	22 (methanol)
NFL, Sibian Road, Distt. Bathinda	511.5 (Urea)
	8.7 ^(S)
Petroleum Refinery	
HPCL Mittal energy Ltd, Bathinda	9000
Sponge Iron	
Vallabh Steels Ltd, Sahnewal, Ludhiana	120

(G): Grinding Units

Note:

Data, not readily available for Fertilizer and Cement Industries on their respective website, hence it is taken from Indian Fertilizer Scenario, FAI Statistics and Survey of Cement Industry & Directory, respectively.



Rajasthan

22

major minerals were produced in the State during 2023-24.

94

mines reported production in case of MCDR minerals in 2023-24.

₹ 21,670

crore

value of mineral production was estimated in State for the year of 2023-24.

MINERAL RESOURCES

Rajasthan is the richest state in terms of availability and variety of minerals in the country and produces 22 major minerals during 2023-24. Rajasthan is the sole producer of lead & zinc ores, selenite and wollastonite. Rajasthan was the sole producer of garnet (gem) till 2004-05. Almost entire production of silver in the country comes from Rajasthan. The State is a major producer of copper ore/conc., limestone, ochre, phosphorite/rock phosphate and talc/soapstone/steatite. The State is also an important producer of marble of various shades. Makrana area is the world famous center for marble mining.

Rajasthan is a major hub for mineral production in India, produces major minerals i.e. lead, zinc, copper, silver, lignite, limestone, iron ore. The State also holds promising reserves of precious, strategic and critical minerals like gold, potash, rare earth elements (REE), tungsten, and lithium.

Important minerals that are found to occur in the State are: asbestos (amphibole) in Ajmer, Bhilwara, Dungarpur, Pali, Rajsamand & Udaipur districts; ball clay in Bikaner, Nagaur & Pali districts; barytes in Alwar, Bharatpur, Bhilwara, Bundi, Chittorgarh, Jalore, Pali, Rajsamand, Sikar & Udaipur districts; calcite in Ajmer, Alwar, Bhilwara, Jaipur, Jhunjhunu, Pali, Sikar, Sirohi & Udaipur districts; china clay in Ajmer, Barmer, Bharatpur, Bhilwara, Bikaner, Bundi, Chittorgarh, Dausa, Jaipur, Jaisalmer, Jhunjhunu, Kota, Nagaur, Pali, Sawai Madhopur & Udaipur districts; and copper in Khetri belt in Jhunjhunu district & Dariba in Alwar district. Deposits of copper are also reported at

Ajmer, Bharatpur, Bhilwara, Bundi, Chittorgarh, Dausa, Dungarpur, Jaipur, Jhunjhunu, Pali, Rajsamand, Sikar, Sirohi and Udaipur districts. Occurrence of other minerals, namely, Dolomite in Ajmer, Alwar, Bhilwara, Chittorgarh, Dausa, Jaipur, Jaisalmer, Jhunjhunu, Jodhpur, Sikar & Udaipur districts; felspar in Ajmer, Alwar, Bhilwara, Jaipur, Pali, Rajsamand, Sikar, Tonk & Udaipur districts; fireclay in Alwar, Barmer, Bharatpur, Bhilwara, Bikaner, Dausa, Jaisalmer, Jhunjhunu & Sawai Madhopur districts; fluorspar in Ajmer, Dungarpur, Jalore, Jhunjhunu, Sikar, Sirohi & Udaipur districts; garnet in Ajmer, Bhilwara, Jhunjhunu, Sikar & Tonk districts; gypsum in Barmer, Bikaner, Churu, Sri Ganganagar, Hanumangarh, Jaisalmer, Jalore, Nagaur & Pali districts; iron ore (haematite) in Alwar, Dausa, Jaipur, Jhunjhunu, Sikar & Udaipur districts; iron ore (magnetite) in Bhilwara, Jhunjhunu & Sikar districts; and lead-zinc in Zawar in Udaipur district, Bamnia Kalan, Rajpura-Dariba

in Rajsamand & Rampura/Agucha in Bhilwara district. Lead-zinc occurrences have also been reported from Ajmer, Chittorgarh, Pali and Sirohi districts. Lignite deposits are found to occur in Barmer, Bikaner, Jaisalmer, Jalore, Nagaur and Pali districts. Flux grade limestone occurs in Jodhpur and Nagaur districts and Chemical-grade limestone in Jodhpur, Nagaur and Alwar districts. Cement grade deposits of limestone are widespread in Ajmer, Alwar, Banswara, Bhilwara, Bikaner, Bundi, Chittorgarh, Churu, Dungarpur, Jaipur, Jaisalmer, Jodhpur, Jhunjhunu, Kota, Nagaur, Pali, Sawai Madhopur, Sikar, Sirohi and Udaipur districts. Magnesite in Ajmer, Dungarpur, Pali & Udaipur districts; marble in Ajmer, Alwar, Banswara, Bhilwara, Bundi, Chittorgarh, Dungarpur, Jaipur, Nagaur, Sikar, Sirohi & Udaipur districts; mica in Ajmer & Bhilwara districts; ochre in Baran, Bharatpur, Bhilwara, Bikaner, Chittorgarh, Jaipur, Sawai Madhopur & Udaipur districts; pyrite in Sikar district; pyrophyllite in Alwar, Bhilwara, Jhunjhunu, Rajsamand & Udaipur districts; quartz/silica sand in Ajmer, Alwar, Bharatpur, Bhilwara, Bikaner, Bundi, Chittorgarh, Dausa, Jaipur, Jaisalmer, Jhunjhunu, Jodhpur, Kota, Pali, Rajsamand, Sawai Madhopur, Sikar, Sirohi, Tonk & Udaipur districts; quartzite in Ajmer, Alwar, Jhunjhunu & Sawai Madhopur districts; rock phosphate in Alwar, Banswara, Jaipur, Jaisalmer & Udaipur districts; talc/steatite/soapstone in Ajmer, Alwar, Banswara, Bharatpur, Bhilwara, Chittorgarh, Dausa, Dungarpur, Jaipur, Jhunjhunu, Karauli, Pali, Rajsamand, Sawai Madhopur, Sirohi, Tonk & Udaipur districts; vermiculite in Ajmer & Barmer districts; and wollastonite in Ajmer, Dungarpur, Pali, Sirohi & Udaipur districts.

Other important minerals that occur in the State are: apatite in Udaipur & Sikar districts; bauxite in Kota

district; bentonite in Barmer, Jaisalmer & Jhalawar districts; corundum in Tonk district; diatomite in Barmer & Jaisalmer districts; emerald in Ajmer & Rajsamand districts; fuller's earth in Barmer, Bikaner & Jodhpur districts; gold in Banswara, Bhilwara, Dausa, Sirohi & Udaipur districts; granite in Ajmer, Alwar, Banswara, Barmer, Bhilwara, Chittorgarh, Jaipur, Jaisalmer, Jalore, Jhunjhunu, Jodhpur, Pali, Rajsamand, Sawai Madhopur, Sikar, Sirohi, Tonk & Udaipur districts; graphite in Ajmer, Alwar & Banswara districts; kyanite & sillimanite in Udaipur district; manganese ore in Banswara, Jaipur & Pali districts; potash in Jaisalmer & Nagaur districts; silver in Ajmer, Bhilwara, Jhunjhunu, Rajsamand, Sikar & Udaipur districts; and tungsten in Nagaur & Sirohi districts (Table -1). District-wise reserves/resources of lignite in the State are provided in Table-2.

Deposits of petroleum are located in the Bikaner-Nagaur and Barmer-Sanchore basin and those of natural gas in Jodhpur and Jaisalmer basins in the State.

Production

Many important minerals are produced in the State of Rajasthan. The principal minerals produced in the State were Copper Ore & Conc., Iron Ore, Lead and Zinc Ore & Conc., Manganese Ore, Silver, Phosphorite, Wollastonite, etc. The number of reporting mines of MCDR minerals in the State was 94 in the year 2023-24 (Table-3).

Mineral-based Industry

The present status of each mineral-based industry is not readily available. However, the important mineral-based industries in the organised sector in the State are furnished in Table - 4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Rajasthan

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	-	1067521	1067521
Asbestos	tonne	-	13615710	13615710
Bauxite	'000 tonnes	-	528	528
Copper				
Ore	'000 tonnes	34388	833461	867849
Metal	'000 tonnes	483.08	4152.52	4635.6
Diatomite [#]	'000 tonnes	-	2074	2074
Fluorite	tonne	18099	5587504	5605603
Garnet	tonne	207888	842923	1050811
Gold				
Ore (Primary)	tonne	-	125913720	125913720
Metal (Primary)	tonne	-	234.56	234.56
Graphite	tonne	-	1913554	1913554
Iron ore				
(Haematite)	'000 tonnes	7314	28166	35480
(Magnetite)	'000 tonnes	121060	673866	794926
Kyanite	tonne	-	23703	23703
Lead-Zinc				
Ore	'000 tonnes	103275	581381	684656

Table 1 (Concl'd.)

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Lead metal	'000 tonnes	1900.19	9431.73	11331.92
Zinc metal	'000 tonnes	7438.05	23827.97	31266.02
Lead-Zinc metal	'000 tonnes	-	142.23	142.23
Limestone	'000 tonnes	4804154	24157095	28961249
Magnesite	'000 tonnes	-	54091	54091
Manganese Ore	'000 tonnes	568	1790	2359
Potash	million tonnes	-	20572	20572
Pyrite	'000 tonnes	-	90876	90876
Rock Phosphate	tonne	21845000	72003769	93848769
Sillimanite	tonne	-	819	819
Silver				
Ore	tonne	148326020	342383997	490710017
Metal	tonne	7701.64	21940.57	29642.21
Tungsten				
Ore	tonne	-	23928294	23928294
Contained WO ₃	tonne	-	93707.94	93707.94
Vermiculite	tonne	-	104125	104125
Wollastonite	tonne	2680978	20433955	23114933

Figures rounded off

Note: As of 1st April 2024, the total 2P reserves at the national level are estimated at 434 Million Metric Tonnes (MMT) of oil and 643 Billion Cubic Metres (BCM) of natural gas, while the total 2C contingent resources stand at 237 MMT of oil and 451 BCM of gas.
(India's Hydrocarbon Outlook, 2024-25)

Table – 2: Reserves/Resources of Lignite as on 01.04.2024: Rajasthan

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total	1203.85	3108.55	2273.84	6586.24
Bikaner	560.3	230.33	309.19	1099.82
Barmer	495.23	2509.46	1555	4559.69
Jaisalmer & Bikaner	-	-	11.47	11.47
Jaisalmer	-	-	70.44	70.44
Jaisalmer & Barmer	-	-	13.80	13.80
Jalore	-	-	76.08	76.08
Nagaur	148.32	368.26	219.17	735.75
Nagaur & Pali	-	0.5	18.69	19.19

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in Rajasthan, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		93	-	183986669	92	-	208949157	94	-	216709400
Copper Ore	tonne	-	1101339	-	-	1107920	-	-	1231500	-
Copper Conc.	tonne	2	49399	5463975	1	48885	5014786	2	55357	5990401
Iron Ore	'000 tonnes	10	1235	5557564	10	1321	6270497	10	1527	7304611
Lead & Zinc Ore	tonne	-	16338564	-	-	16744082	-	-	16521284	-
Lead Conc.	tonne	10	368040	22025747	10	376665	24488505	10	380800	25473522
Zinc Conc. %	tonne	-	1594085	78727814	-	1670208	91219638	-	1710144	85654297
Manganese Ore	tonne	1	8008	25626	1	6437	22528	1	4968	20661
Silver #	kg	-	647013	42115418	-	713620	43805077	-	745830	53548545
Phosphorite	tonne	1	1281229	6536677	2	1533465	11871363	2	1320819	10978040
Fluorite (graded)	tonne	-	-	-	1	145	44	1	5260	1058

Table- 3 (Concl'd.)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
Garnet (abrasive)	tonne	5	8182	24660	5	9859	36217	8	23838	95837
Limestone	'000t	41	87483	23336293	41	93964	26050938	44	103644	27433019
Magnesite	tonne	1*	-	-	1	1	1	-	-	-
Selenite	tonne	4	716	902	3	327	442	3	14136	19566
Siliceous Earth	tonne	13	33898	22837	12	32070	17360	10	58047	33735
Wollastonite	tonne	5	108335	149156	5	110793	151761	3	110596	156108

(P) provisional

% Number of mines covered under lead concentrates.

Recovered at Chanderiya Lead-Zinc Smelter of HZL (as by product) from lead concentrates produced in Rajasthan.

* Only labour reported

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Cement	
ACC Ltd, Lakheri, Dist. Bundi	1500
Ambuja Cements Ltd, Rabriyawas, Dist. Pali	3600
Binani Cement, Binanipuram, Dist. Sirohi	4850
Binani Cement, Neem Ka Thana, Sikar (G)	1400
Birla Corporation Ltd, (Birla Cement Works &Chandaria Cement Works), Chittorgarh	4000
India Cements Ltd, Jhalo ka garha Garhi	1800
J.K. Cement, Nimbahera, Dist. Chittorgarh	3250
J.K. Cement, Mangrol, Dist. Chittorgarh	2500
J.K. Cement, Gotan, Dist. Nagaur	500
J.K. White Cement Works, Gotan Merta, Dist. Nagaur	610 (White cement) 500 (White putty)
J.K. Laxmi Cement, Banas, Dist. Sirohi	8700
NUVOCO Vistas (Lafarge) India Ltd, Nimbahera, Dist. Chittorgarh	2600
Mangalam Cement (Mangalam Cement & Neer Shree Cement), Morak, Dist. Kota	3250
Nirma Limited, Nimbol, Jaitaran	2280
Shree Cement Ltd, Beawar, Dist. Ajmer	3000
Shree Cement Ltd, Andherideori, , Masuda, Ajmer	3600
Shree Cement Ltd, Ras, Dist. Pali	3000
Shree Cement Ltd, Ras, Jaitaran, Dist. Pali	4000
Shree Cement Ltd, Kushkhera, Dist. Alwar (G)	3500
Shree Cement Ltd, Suratgarh, Dist. Sri Ganganagar (G)	1800
Shree Cement Ltd, Suratgarh, Rohi, Udaipur-Udasar, Dist. Sri Ganganagar (G)	3600
Shree Cement Ltd, Jobner, Dist. Jaipur (G)	1500
Shriram Cement Works, Kota	400
Trinetra Cement (Subsidiary of India Cement), Nokhala, Dist. Banswara	1800
Udaipur Cement Works (Subsidiary of JKCL), Udyog Ltd, Udaipur	1240
UltraTech Cement (Birla White Cement Division), Kharia Khangar, Bhopalgarh	680 (White cement) 400 (Putty)
UltraTech Cement Nathdwara	4850 (Cement)
Binnani Cement Ltd, Aml, Pindwara	-
UltraTech Cement (Aditya I & II), Shambhupura, Dist. Chittorgarh	8000
UltraTech Cement, Kotputali, Dist. Jaipur	4000
Wonder Cement, Nimbahera, Dist. Chittorgarh	8000
Chemical	

Table 4 (Cont.)

Industry/plant	Capacity
DCM Shriram Industries Ltd. Dist. Kota	9 (Rayon/yarn)
Modi Alkalies & Chemicals Ltd. Dist. Alwar	7.7 (Sodium sulphate) 84.2 (caustic soda) 50.3 (Cl), 39.6 (HCl)
Ceramics/Chemicals	
Bikaner Ceramics Pvt, Ltd Bikaner	9 (Insulators)
Kajaria Ceramics Ltd, Gailpur	6.5 mill. sq m
Kajaria Ceramics Ltd, Malootana	24.5 mill. sq m
Bhalla Chemical Works Pvt. Ltd	10 (Zirconium oxychloride & special zirconia)
Roca Bathroom Product Pvt Ltd, Alwar	12.9
Roca Bathroom Product Pvt Ltd, Alwar	2 mill. pc.
Fertilizer	
Adheeshaa Phosphate, Umarada, Udaipur	132 (SSP)
Arawali Phosphate Ltd, Umra, Udaipur	40 (SSP)
Arihant Phosphate & Fertilizers Ltd, Nimbaheda, Chittorgarh	66 (SSP)
Bohra Industries Ltd, Umra, Udaipur	200 (SSP)
Chambal Fertilizers & Chemicals Ltd, Gadepan, Kota	180 (SSP)
Coromandel International Ltd, (Formerly) Liberty Phosphate Ltd), Jagpura, Kota	132 (SSP)
Devyani Phosphate Pvt. Ltd, Udaipur	60 (SSP)
Dharamsi Morarji Chemical Co. Ltd, Khemli, Udaipur	66 (SSP)
Gayatri Spinners Ltd, Hamirgarh, Bhilwara	30 (SSP)
Indian Phosphate Ltd, Umrada, Udaipur	130 (SSP)
Jagdamba Phosphate, Kota	132 (SSP)
Jubilant Agri and Consumer Products Ltd, Singhpur, Kapasan, Chittorgarh	264 (SSP)
Khaitan Chemical & Fertilizers Ltd, Dhinwa, Dist. Chittorgarh	198 (SSP)
Mangalam Phosphates Ltd, Hamirgarh, Bhilwara	72 (SSP)
Ostwal Phoschem (India) Ltd, Hamirgarh, Bhilwara	132 (SSP)
Patel Phoschem (P) Ltd, Umarda, Udaipur	100 (SSP)
Prem Sakhi Fertx. Ltd, Lakadwas, Udaipur	66 (SSP)
Rama Phosphates Ltd, Umra, Udaipur	181 (SSP)
Sadhana Phosphates & Chems Ltd, Gudli, Udaipur	120 (SSP)
Shriram Fertilizers & Chemicals Ltd, Shriramnagar, Dist. Kota	379.5 (Urea)
	113.8 (Caustic soda)
	13.2 (Bleaching powder)
	61.2 (HCl)
	61.2 (Cl)
Shri Ganapati Fertilizers Ltd, Kapasan, Chittorgarh	99 (SSP)
Shurvi Colour Chem Ltd, Madri, Udaipur	12 (SSP)
Plaster of Paris	
Abhishek Plaster Industries, Baramsar, Dist. Hanumangarh	6.1
Agrawal Industries, Nohar, Dist. Hanumangarh	6.3
Balaji Plaster Industries, Taranagar, Dist. Churu	6
Balaji Industries, Taranagar, Dist. Churu	6.5
Ganesh Plaster Industries, Taranagar, Dist. Churu	6
Gil Brothers, Taranagar, Dist. Churu	7.1
Hind Plaster Industries, Taranagar, Dist. Churu	6
Jaishri Plaster Industries, Taranagar, Dist. Churu	6.3
Jagdamba Plaster Industries, Rawatsav, Dist. Hanumangarh	7
Coromandel International Ltd, (Formerly Liberty Phosphate Ltd), Jagpura, Kota	132 (SSP)
Devyani Phosphate Pvt. Ltd, Udaipur	60 (SSP)

Table- 4 (Concl'd.)

Industry/plant	Capacity
Dharamsi Morarji Chemical Co. Ltd, Khemli, Udaipur	66 (SSP)
Jai Bhavani Plaster Industries, Baramsar, Dist. Hanumangarh	6
Jai Sriram Plaster Industries, Taranagar, Dist. Churu	7.1
M.G. Plaster Pvt Ltd, Taranagar, Dist. Churu	6.2
Mahabir Plaster Industries, Taranagar, Dist. Churu	6
Multani Industries, Nohar, Dist. Hanumangarh	8.4
R.D. Plaster Industries, Nohar, Dist. Hanumangarh.	8.4
R.N. Industries, Bikaner, Dist. Bikaner	18
Shalimar Plaster & Chemical Industries, Sardarshahar, Dist. Churu	14
Shri Lakshmi Gypsum, Chak, Dist. Hanumangarh	6
Shriram Plaster, Taranagar, Dist. Churu	6.3
SS Plaster Industries, Taranagar, Dist. Churu	6
Shiv Bhakti Industries, Nohar, Dist.. Hanumangarh	8.4
Tiger Plaster, Sardarshahar, Dist. Churu	11
The Sardarshahar Plaster & Minerals, Sardarshahar, Dist. Churu	19.4
Updesh Industries Ltd, Chak, Dist. Hanumangarh	9
Pellet	
Jindal Saw Limited, Pur, Bilwara	1500
Power generation	
JSW Energy Barmer Ltd, Bhadresh.	1080 MW
Copper Smelters	
HCL, KCC, Jhunjhunu.	31 (Cu cathode)
Rajpura Dariba Lead & Zinc Mine Dariba, Rajsamand	76.827 (Zinc conc.)
	17.506 (Lead conc.)
Lead & Zinc Smelters	
HZL Zinc Smelter, Debari, Dist. Udaipur.	88 (Zn)
HZL Lead-zinc Smelter, Chanderiya, Dist. Chittorgarh.	85 (Pb)
	525 (Zn)
	0.833 (Cd)*
	168 tonnes (Ag)
HZL, Dariba Smelting Complex, Dariba Dist. Rajsamand.	100 (Pb)
	210 (Zn)

*: Total for all smelters of HZL

(G): Grinding Units

Note: Data, not readily available for fertilizer and cement industries on respective websites, is taken from Indian Fertilizer Scenario, FAI Statistics and Survey of Cement Industry & Directory respectively.



Sikkim

Copper-Lead-Zinc and Silver are the important mineral resources of the State.

Other minerals reported in the State are: dolomite, quartzite, talc, steatite, soapstone, Limestone and marble

MINERAL RESOURCES

The important mineral resources of the State are copper-lead-zinc and silver, reported in Bhotang, Rangpo and Dikchu in East Sikkim districts. Occurrences of other minerals reported in the State include dolomite, quartzite and talc/steatite/soapstone in West Sikkim district; limestone in North Sikkim district and marble in East Sikkim & North Sikkim districts (Table-1). The reserves/resources of coal and the coalfield located in Sikkim are reflected in Table-2.

PRODUCTION

MCDR minerals production was not reported in the State during the year 2023–24.

MINERAL-BASED INDUSTRY

SMC, a joint venture of Government of Sikkim and Government of India was established for the purpose of development of Bhotang polymetallic ore deposit at

Rangpo. Sikkim's Mines & Geology Department had set up a pilot dimension rock cutting unit and pilot lime making unit to ascertain the feasibility of setting up of commercial lime plant and dimension rock cutting plant in the State. A ferroalloys plant, namely, Akshay Ispat & Ferro Alloys Ltd with an installed capacity of 6,000 tpy is located at Mamring, Namchi district, Sikkim.

Table – 1: Reserves/Resources of Minerals as on 01-04-2020: Sikkim

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Copper				
Ore	'000 tonnes	0	958	958
Metal	'000 tonnes	0	21.47	21.47
Lead-Zinc				
Ore	'000 tonnes	0	950	950
Lead metal	'000 tonnes	0	8.58	8.58
Zinc metal	'000 tonnes	0	20.07	20.07
Limestone	'000 tonnes	0	2380	2380
Silver				
Ore	tonnes	0	949623	949623
Metal	tonnes	0	56.69	56.69

Figures rounded off

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Sikkim

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total/Rangit Valley	-		42.98	101.23

Source: Coal Directory of India, 2023-24



Tamil Nadu

Tamil Nadu is the leading holder of country's resources of vermiculite, molybdenum, dunite, rutile, garnet, and ilmenite.

₹ 881.95 crore
value of major minerals reported in
Tamil Nadu in 2023-24

83
Reporting mines in case of MCDR
minerals were reported in 2023-24.

MINERAL RESOURCES

Tamil Nadu is the leading holder of country's resources of vermiculite, molybdenum, dunite, rutile, garnet and ilmenite. The State accounts for the country's 79% vermiculite, 65% dunite, 46% garnet, 66% molybdenum, 24% sillimanite and 16% fire clay resources. As per AMD of the Department of Atomic Energy, Tamil Nadu accounted for 167.70 million tonnes of ilmenite resources and 7.85 million tonnes of rutile resources.

Important minerals that are found to occur in the State are: bauxite in Dindigul, Namakkal, Nilgiris & Salem districts; dunite/pyroxenite in Salem district; feldspar in Coimbatore, Dindigul, Erode, Kanchipuram, Karur, Namakkal, Salem & Tiruchirapalli districts; fireclay in Cuddalore, Kanchipuram, Perambalur, Pudukottai, Sivaganga, Tiruvallur, Tiruchirapalli, Vellore & Villupuram districts; garnet in Ramanathapuram, Tiruchirapalli, Tiruvarur, Kanyakumari, Thanjavur & Tirunelveli districts; granite in Dharmapuri, Erode, Kanchipuram, Madurai, Salem, Tiruvannamalai, Tiruchirapalli, Tirunelveli, Vellore & Villupuram districts; graphite in Madurai, Ramanathapuram, Sivaganga & Vellore districts; and gypsum in Coimbatore, Perambalur, Ramanathapuram, Tiruchirapalli, Tirunelveli, Thoothukudi & Virudhunagar districts. Similarly, occurrences of minerals, such as, lignite deposits are located in Cuddalore, Ariyalur, Thanjavur, Tiruvarur, Nagapattinam, Ramnad, Sivaganga & Ramanathapuram districts; limestone in Coimbatore, Cuddalore, Dindigul, Kanchipuram,

Karur, Madurai, Nagapattinam, Namakkal, Perambalur, Ramanathapuram, Salem, Tiruvallur, Tiruchirapalli, Tirunelveli, Vellore, Villupuram & Virudhunagar districts; magnesite in Coimbatore, Dharmapuri, Karur, Namakkal, Nilgiri, Salem, Tiruchirapalli, Tirunelveli & Vellore districts; quartz/silica sand in Chennai, Coimbatore, Cuddalore, Dharmapuri, Dindigul, Erode, Kanchipuram, Karur, Madurai, Namakkal, Periyar, Perambalur, Salem, Tiruvallur, Tiruvarur, Nagapattinam, Tiruchirapalli, Villupuram, Virudhunagar & Vellore districts; talc/steatite/soapstone in Coimbatore, Salem, Tiruchirapalli & Vellore districts; titanium minerals in Kanyakumari, Nagapattinam, Ramanathapuram, Tiruvallur, Tirunelveli & Thoothukudi districts; vermiculite in Dharmapuri, Tiruchirapalli & Vellore districts; and zircon in Kanyakumari district have been established.

Other minerals that occur in the State are: apatite in Dharmapuri & Vellore districts; barytes in Erode, Madurai,

Perambalur, Tirunelveli & Vellore districts; bentonite in Chengai-Anna district; calcite in Salem district; chinaclay in Cuddalore, Dharmapuri, Kanchipuram, Nilgiris, Sivaganga, Tiruvallur, Tiruvannamalai, Tiruchirapalli & Villupuram districts; chromite in Coimbatore & Salem districts; copper, lead-zinc & silver in Villupuram district; corundum & gold in Dharmapuri district; dolomite in Salem & Tirunelveli districts; emerald in Coimbatore district; iron ore (magnetite) in Dharmapuri, Erode, Nilgiris, Salem, Tiruvannamalai, Tiruchirapalli & Villupuram districts; kyanite in Kanyakumari & Tirunelveli districts; molybdenum in Dharmapuri, Dindigul & Vellore districts; pyrite in Vellore district; sillimanite in Kanyakumari, Karur & Tirunelveli districts; tungsten in Madurai & Dindigul districts; and wollastonite in Dharmapuri & Tirunelveli

districts (Table-1). Area/field wise reserves/resources of lignite are provided in Table-2.

In addition to the above, petroleum and natural gas deposits are found to be located in Cauvery basin area.

PRODUCTION

The important minerals produced in the State during 2023–24 were graphite (r.o.m.), limestone, magnesite, etc. The number of reporting mines for MCDR minerals was 83 in 2023–24, compared to 88 in 2022–23 and 94 in 2021–22 (Table-3).

MINERAL-BASED INDUSTRY

The important mineral-based industries in the Organised Sector in the State are furnished in Table -4.

Table – 1: Reserves/Resources of Minerals as on 1.4.2020: Tamil Nadu

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	0	240000	240000
Bauxite	'000 tonnes	0	24112	24112
Chromite	'000 tonnes	0	282	282
Copper				
Ore	'000 tonnes	0	790	790
Metal	'000 tonnes	0	3.81	3.81
Garnet	tonne	52538	25831356	25883894
Gold				
Ore (Primary)	tonne	0	67000	67000
Metal (Primary)	tonne	0	1	1
Graphite	tonne	3100193	6605086	9705279
Iron Ore (Magnetite)	'000 tonnes	0	528901	528901
Kyanite	tonne	1683	247829	249512
Lead-Zinc Ore				
Ore	'000 tonnes	0	790	790
Lead metal	'000 tonnes	0	7.74	7.74
Zinc metal	'000 tonnes	0	36.52	36.52
Limestone	'000 tonnes	547024	1577025	2124049
Magnesite	'000 tonnes	55084	100402	155486
Molybdenum				
Ore	tonne	0	17882498	17882498
Contained MOS ₂	tonne	0	10151.86	10151.86
Pt. Group of Metals	tonne	0	1.69	1.69
Pyrite	'000 tonnes	0	24	24
Sillimanite	tonne	134030	17218747	17352777
Silver				
Ore	tonne	0	790000	790000
Metal	tonne	0	42.55	42.55
Titanium	tonne	670221	113677284	114347505
Tungsten				
Ore	tonne	0	250000	250000
Contained WO ₃	tonne	0	50	50
Vermiculite	tonne	1516803	343051	1859854
Wollastonite	tonne	0	3533	3533
Zircon	tonne	36285	43833	80118

figure rounded off.

Table – 2: Reserves/Resources of Lignite as on 1.4.2024: Tamil Nadu

(Qty. in MT)

Coalfield	Proved	Indicated	Inferred	Total
Area/Field	Proved	Indicated	Inferred	Total
Total	5476	21412.16	10635.49	37523.65
Neyveli lignite field	5405.39	1182.36	1908.63	8496.38
Mannargudi lignite field	0	19913.44	4780.46	24693.90
Ramanthapuram lignite field	70.61	316.36	3946.40	4333.37

Source: Coal Directory of India, 2023-24

Table – 3: Mineral Production in Tamil-Nadu, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24(p)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		94	-	7208720	88	-	8204069	83	-	8819511
Garnet (abrasive)	t	2*	-	-	-	-	-	-	-	-
Graphite (r.o.m.)	t	1	36214	52872	1	54216	66235	1	79263	54591
Limestone	'000 t	85	21323		81	24434	7642296	76	26777	8203516
Magnesite	t	5	81013	454667	5	62123	363065	5	78085	452154
Marl %	t	-	952921	193287	-	586360	132473	-	524484	109249
Vermiculite	t	1	690	2284	1	-	-	1	2	1

(p) provisional

* Only labour reported.

% Associate with Limestone

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity ('000 tpy)
Abrasives	
Carborandum Universal Ltd, Chennai	NA
Cutfast Abrasives Tools Pvt. Ltd, Chennai	NA
Asbestos Products	
Hyderabad Industries Ltd, Kannigaiper	100
Ramco Industries Ltd, Arakkonam, Distt. Vellore	NA
Southern Asbestos Cement Ltd, Arrakonam, Distt. Vellore	NA
Tamil Nadu Asbestos, Alangulam, Distt. Virudhunagar	28.5
Cement	
ACC Ltd, Madukkarai, Distt.. Coimbatore	1000
Chettinad Cement Corpn. Ltd, Puliya, Distt. Karur	1700
Chettinad Cement Corpn. Ltd, Karikalli, Distt. Dindigul	3500
Chettinad Cement Corpn. Ltd, Ariyalur	5500
Dalmia Cements, Dalmiapuram, Distt. Tiruchirapalli	3400
Dalmia Cements, Ariyalur	3000
Dhandhapani Cement Pvt. Ltd, Thathamangalam, Manachanallur	225
India Cements Ltd, Sankarnagar, Distt. Tirunelveli	2050
India Cements Ltd, Sankaridurg, Distt. Salem (G)	860
India Cements Ltd, Dalavoi, Distt. Ariyalur	2160
India Cements Ltd, Vallur, Distt. Chennai (G)	1100
India Cements Ltd, Panaiveedu, Thiruchengodu	1400
My Home Industries Ltd. Tuticorin	1500
UltraTech Cement Ltd, Reddipalayam, Distt. Ariyalur	1600
UltraTech Cement Works (ARCW), Arakkonam (G)	1100

Table 4 (Cont.)

Industry/plant	Capacity ('000 tpy)
Ramco Cement (formerly Madras Cement), R.S. Raja Nagar, Distt. Virudhunagar	2000
Ramco Cement (formerly Madras Cement), Alathiyur Works, Distt. Ariyalur	3050
Ramco Cement (formerly Madras Cement), Ariyalur Plant, Govindpuram, Distt. Ariyalur	3500
Ramco Cement (formerly Madras Cement), Chengalpet Grinding Unit, Uthiramerur, Distt. Kanchipuram (G)	750
Ramco Cement (formerly Madras Cement), Valapady, Distt. Salem (G)	2000
Tamil Nadu Cements, Alangulam, Distt. Virudhunagar	400
Tamil Nadu Cements, Ariyalur, Distt. Ariyalur	1500
Tamil Nadu News -print & Paper Ltd., Kagithapuram, Manmangalam	328.5
Vijay Cements Trichy	75
Zuari Cements Ltd, Chennai Grinding Unit, Attipattu, Tiruvallur (G)	1000
Ceramics	
Carborandum Universal Ltd, Hosur	NA
Murugappa Morgan Thermal Ceramics Ltd, Ranipet, Distt. Vellore	5.44
Neycer India Ltd, Vadalur, Distt. Cuddalore	9
Roca Bathroom Product Pvt Ltd, Ranipet, Distt. Erode	12.6
Roca Bathroom Product Pvt Ltd, Perundurai, Distt. Vellore	24
Spartek Ltd, Chennai	NA
Copper Smelter	
Sterlite Industries (I) Ltd, Thoothukudi	400 (Cu smelting)
	205 (Cu cathode)
	90 (wire rods)
	1200 (H ₂ SO ₄)
Chemicals	
Tanfac Industries Ltd, Cuddalore	16.5 (anhydrous HF)
	16.5 (AlF ₃)
	67.5 (H ₂ SO ₄)
	14 (Hydrofluoric acid)
	3.4 (speciality fluorides)
Tuticorin Alkali Chemicals & Fertilizers Ltd, Thoothukudi	180 (Soda Ash)
	162 (AlCl ₃) (Fertilizer Grade)
	12 (AlCl ₃) (Technical pure Grade)
	7.2 (Ammonium Bicarbonate)
	14.4 (Sodium Bicarbonate)
Vaiyapuri Shanthi Ferric alum, Sellipalayam, Namakal	3.4(Ferric alum)
Electrode	
AVR Electrodes, No1, SIDCO,Indl.Estate, Rajapalyam, Virudhnagar	250
Fertilizer	
Coimbatore Pioneer Fertilizer Ltd, Muthugoundanpudur, Distt. Coimbatore.	66 (SSP)
	30 (H ₂ SO ₄)
	3 (oleum)
Coramandal International Ltd, (Formerly EID Parry), Ranipet, Distt. N. Arcot	132 (SSP)
	33 (H ₂ SO ₄)
Coramandal International Ltd, Ennore, Distt. Thiruvallur.	330 (Complex)
Kothari Industrial Corp. Ltd, Ennore.	66 (SSP)
Madras Fertilizer Ltd, Manali, Distt. Thiruvallur.	486.8 (Urea)
	840 (NP/NPKs)
Greenstar Fertilizers Ltd, Guindy.	115 (SSP)
Southern Petrochemical Industries Corpn. Ltd, Thoothukudi.	620 (Urea)

Table- 4 (Concl'd.)

('000 tpy)

Industry/plant	Capacity ('000 tpy)
Ferroalloy	
Electralloy Special Steel Casting Pvt. Ltd.	1.0 (alloy , Stainless steel casting)
Synthetic Rutile	
DCW Ltd, Sahupuram, Distt. Thoothukudi.	48
TiO2 Pigment	
VVTi Pigments (P) Ltd, (formerly, Kilburn Chemicals), Distt. Thoothukudi	14.5 (Titanium Di-oxide) 30 (Ferrous Sulphate Heptahydrate)
Iron & Steel	
Salem Steel Plant (SAIL), Salem.	180 (Crude/Liquid steel)
JSW Steel Plant (acquired Southern Iron & Steel Co. Ltd), Salem.	1180 (sinter) 1000 (pig iron) 1000 (specialised alloy steel) 18000 (Crude/Liquid steel)
Sponge Iron	
Akshara Industries Ltd, Eguvarpalayam, Distt. Thiruvallur.	100
Kaushik Steel Industries Ltd, Pappen Kuppam, Distt. Thiruvallur.	60
Agni Steels Pvt Ltd, Olappalayam Road, Ingur, Distt. Erode.	36
Refractory	
ABREF Pvt. Ltd, Gummidipoondi, Distt. Thiruvallur.	1.3
Sharda Ceramics Pvt. Ltd, Ambattur, Chennai.	9.9
Shri Natraj Ceramic & Chemical Industries Ltd, Dalmiapuram, Distt. Tiruchirapalli.	42
VRW Refractories, Vanagaram.	21.6
Zirconium Complex, Pazhakayal, Thoothukudi.	0.5 (Zr-Oxide) 0.25 (Zr sponge)
DBM & Calcined Magnesite	
SAIL Refractory Co. Ltd (formerly Burn Standard Co. Ltd), Salem.	18 (calcined magnesite) 61 (DBM) 15 (refractory bricks) 24 (Dunite fracton)
Dalmia Magnesite Corpn., Chettichavadi, Distt. Salem.	72 (DBM)
Ramkrishna Magnesite Mines, Salem.	3 (calcined)
Tamil Nadu Magnesite Ltd, Kurumbapatty, Distt. Salem.	19.5(calcined magnesite) 30(DBM) 26.5 (DBM)
Sri Pon Kumar Magnesite Ltd, Salem.	
Silicon Carbide	
Carborandum Universal Ltd, Tiruvottiyur.	NA
Petroleum Refinery	
CPCL, Manali, Dist. Thiruvallur.	10500
CPCL, Narimanam.	1000

Note: Data, not available for Fertilizer and Cement Industries on respective website, is taken from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.
(G): Grinding unit.



Telangana

Telangana is the leading holder of country's resources of barytes, dolomite, feldspar, laterite, limestone and quartz

Manganese ore and limestone were the important minerals produced in the State during 2023–24

33

mines in case of MCDR minerals reported production in 2023-24

Telangana was formed on 2nd of June 2014 with ten districts, namely, Hyderabad, Adilabad, Khammam, Karimnagar, Mahabubnagar, Medak, Nalgonda, Nizamabad, Rangareddy and Warangal. Telangana is surrounded by Maharashtra and Chhattisgarh in the North, Karnataka in the West and Andhra Pradesh in the South and East directions.

MINERAL RESOURCES

Telangana is the leading holder of country's resources of barytes, dolomite, feldspar, laterite, limestone, Quartz and others. It accounts for 46% kyanite, 29% corundum, 10% fuller's earth and 7.75% limestone resources of the country. Telangana is endowed with internationally known black, pink, blue and multicolored varieties of granites.

Important minerals occurring in Telangana are: barytes in Khammam, district; china clay in Adilabad, Mahabubnagar, Nalgonda, Rangareddy and Warangal districts; coal in Adilabad, Karimnagar, Khammam and Warangal districts; corundum in Khammam district; dolomite in Khammam and Warangal districts; feldspar in Hyderabad, Khammam, Mahabubnagar, Medak and Rangareddy districts; fireclay in Adilabad and Nalgonda districts; garnet in Khammam district; granite in Karimnagar, Khammam, Mahabubnagar, Medak, Nalgonda, Rangareddy and Warangal districts; iron ore (haematite) in Khammam district; iron ore (magnetite) in Adilabad and Warangal districts; limestone

in Adilabad, Hyderabad, Karimnagar, Mahabubnagar, Nalgonda, Rangareddy, districts; manganese ore in Adilabad district; mica in Khammam districts; quartz/silica sand in Hyderabad, Khammam, Mahabubnagar, Medak, Nalgonda, Rangareddy and Warangal districts; and talc/soapstone/steatite in Khammam district. Other minerals that occur in the State are chromite, copper, graphite and kyanite in Khammam district; fuller's earth in Medak and Rangareddy districts; and marble in Khammam district (Table-1), while coal deposits are primarily located in the Godavari Valley (Table-2).

PRODUCTION

The important minerals produced in the State during 2023–24 were Manganese ore and limestone. The number of reporting mines for MCDR minerals was 33 in 2023–24, compared to 37 in 2022–23 and 38 in 2021–22 (Table-3).

MINERAL-BASED INDUSTRY

The important mineral based industries in the Organised Sector in the State are given in Table-4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Telangana

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Chromite	'000 tonnes	0	186	186
Copper				
Ore	'000 tonnes	0	666	666
Metal	'000 tonnes	0	9.12	9.12
Garnet	tonne	0	1960196	1960196
Graphite	tonne	0	219455	219455
Iron ore				
(Haematite)	'000 tonnes	0	105627	105627
Iron ore				
(Magnetite)	'000 tonnes	0	87366	87366
Kyanite	tonne	0	48350000	48350000
Limestone	'000 tonnes	1214127	16438327	17652454
Manganese ore	'000 tonnes	342	4162	4503

Figures rounded off.

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Telangana

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total/Godavari Valley	11256.78	8496.57	3452.17	23205.52

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Telangana, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		38	-	5993983	37	-	5744002	33	-	5565237
Manganese Ore	tonne	8	10434	61676	7	10810	81176	3	6055	40404
Limestone	'000 tonnes	30	28499	5932307	30	26464	5662826	30	26082	5524833

(P) provisional

Table – 4: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Aluminium Foil	
Hindalco, Kollur, Medak	4
Asbestos Products	
Bhagyanagar Wood Plast Ltd, Nandikandi, Distt. Medak	60
Hyderabad Industries Ltd, Sanathnagar, Distt. Rangareddy	160
Hyderabad Industries Ltd, Thimmapur	230
J.J. Spun Pipe Industries, Arsapalli, Distt. Nizamabad	4.5
Visaka Industries Ltd, Medak	900
Bleaching Clay	
Ashapura Clay Tech. Ltd, Dharur, Distt. Rangareddy	20 (Fuller's earth granules) 15 (Bentonite granules)
Cement	
Anjani Portland Cements Ltd (Subs. of Chettinad Cement), Anjanipuram, Distt. Nalgonda	1160
CCI Ltd, Tandur, Distt. Rangareddy	1000
Bheema Cement, Nalgonda	900
Greygold Cement, Nalgonda	90

Table- 4 (Concl'd.)

Industry/plant	Capacity
Deccan Cements Ltd, Bhavanipuram, Distt. Nalgonda	2300
India Cement Ltd, Malkapur, Distt. Rangareddy	2400
India Cement Ltd's (formerly Raasi Cements), Vishnupuram, Distt. Nalgonda	3500
Keerthi Industries Ltd, Mellacheruvu, Distt., Nalgonda	594
Kesoram Cement, Basantnagar, Distt. Karimnagar	6000
Mancherial Cement Co. (P) Ltd, Mancherial, Distt. Adilabad	330
My Home Cement Industries Ltd, Mellacheruvu, Distt. Nalgonda	3300
NCL Industries Ltd, Simhapuri, Distt. Suryapet	2000
Orient Cement, Devapur, Distt. Mancherial	3000
Penna Cement Industries Ltd, Tandur, Distt. Rangareddy	2000
Penna Cement Industries Ltd, Ganeshpahad, Distt. Nalgonda	1200
Rain Commodities Ltd (Rain Cements), Ramapuram, Distt. Nalgonda	4000
Sagar Cements Ltd, Mattampally, Distt. Nalgonda	3000
Sri Lalita Cement, Mattampally, Distt. Nalgonda	1000
Zuari Cements Ltd (Sri Vishnu Cements Works), Dondapadu, Sitapuram, Distt. Nalgonda	1200
Ceramic/Sanitaryware	
Hindustan Sanitaryware & Industries Ltd, Bibinagar, Distt. Nalgonda	1.8
Montana International Ltd, Faralwadi, Distt. Medak	3.6
Restile Ceramics Ltd, Malkapur, Distt. Medak	1.4
Fertilizer	
Chemtech Fertilizers Ltd, Kazipalli, Medak	33 (SSP)
Sponge Iron	
Ashirwad Steels & Ind. Ltd, Veliminedu, Distt. Nalgonda	60
Anand Metallics & Power Pvt. Ltd, Kodi Cherla, Distt. Mahabubnagar	24
Binjusaria Sponge & Power Pvt. Ltd, Farooq Nagar, Distt. Mahabubnagar	30
Lakshmi Gayatri Iron & Steel, Kethepally, Distt. Nalgonda	60
NMDC (Sponge Iron Division), Paloncha, Khammam.	60
Reactive Metals of India Ltd, Appajipally, Distt. Mahabubnagar.	36.5
Sunder Steels Ltd, S.D. Road, Secunderabad.	36
Ferroalloys	
Nav Bharat Ferro Ventures Ltd, Paloncha, Distt. Khammam.	125
Shree Raghvendra Ferro alloys Pvt Ltd, Nalgonda.	15
VBC Ferro Alloys Ltd, Rudraram, Distt. Medak.	48 (silico-manganese)
	32.4 (ferro-manganese)
Refractory	
MPR Refractories Ltd, Medak.	9.5
Raasi Refractories, Narketapally, Distt Nalgonda.	35

Note: Data, not readily available for fertilizer and cement Industries on respective websites, is therefore taken from FAI Statistics and Survey of Cement Industry & Directory, respectively



Tripura

Natural gas is the most important mineral resource in Tripura

3,000 tonnes per year
lime-pozzolana mixture factory has been in operation in north Tripura district.

150 tonnes per year
glazed pottery unit is also in operation in West district.

MINERAL RESOURCES

Natural gas is the most important mineral resource in Tripura located in the Assam–Arakan Fold Belt (AAFB) basin. Other minerals of significance found to occur in the State are glass sands, limestone, fireclay, plastic clay, shale and quartz-silica sand used particularly for building/construction purposes (Table-1).

PRODUCTION

Production of MCDR minerals was not reported in the State of Tripura during the year 2023-24.

MINERAL-BASED INDUSTRY

A 3,000 tpy lime–pozzolana mixture factory has been in operation at Kumarghat in North Tripura district. A 150 tpy glazed pottery unit of Tripura Khadi and Village Industrial Board is also in operation at Anandnagar in Tripura (West)

district. M/s DP Group has set up a cold Steel Rolling Plant at Bodhjunnagar Industrial Complex. The State Government is actively wooing Private Sector investment for establishment of gas-based industries. Besides, Private Sector's involvement in setting up of Ceramic tiles units and other mineral-based industries are also being actively pursued. Private participation in setting up plastic clay and glass sand industries is also under consideration by the State Government.

Table – 1: Reserves/Resources of Minerals as on 01.04.2015: Tripura

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Fireclay [#]	'000 tonnes	-	370	370
Quartz-silica sand [#]	'000 tonnes	-	490	490

Figures rounded off.

#: Declared as Minor Mineral vide Gazette Notification dated 10.02.2015.



Uttar Pradesh

86 million tonnes
Iron Ore (Haematite) Resources along with vast resources of andalusite bauxite, limestone and rock phosphate have been established in the State.

2,878 thousand tonnes
limestone production were reported in the State in the year 2023-24.

3
Reporting mines in case of MCDR minerals in the State during 2023-24.

MINERAL RESOURCES

Uttar Pradesh is the principal holder of country's andalusite and diaspore resources & possesses 91% andalusite, 37% diaspore, 16% sillimanite & 10% pyrophyllite, of the country. Important minerals occurring in Uttar Pradesh are-coal in Singrauli coalfields, Sonbhadra district; and diaspore & pyrophyllite in Hamirpur, Jhansi, Lalitpur & Mahoba districts. Naini area of Prayagraj district has good resources of high quality silica sand, an important source of glass sand, that contains 98% SiO_2 and a very low proportion of Fe_2O_3 . It is found at Shankargarh and Lohargarh in Prayagraj district and also Bargarh in Banda district. Silica sand is also found in Aligarh and Chitrakoot districts.

Other minerals that occur in Uttar Pradesh are andalusite & calcite in Mirzapur district; bauxite in Banda, Varanasi & Lalitpur districts; china clay and dolomite in Banda & Sonbhadra districts; felspar in Jhansi district; fireclay, limestone, potash & sillimanite in Sonbhadra district; ochre in Banda district; granite in Banda, Hamirpur, Lalitpur & Mahoba districts; iron ore (haematite) and rock phosphate in Lalitpur district (Table-1). The reserves/resources of coal along with details of coalfield are provided in Table-2.

PRODUCTION

Limestone was the principal mineral produced in Uttar Pradesh in 2023-24. There were 3 reporting mines of MCDR minerals in Uttar Pradesh during 2023-24 (Table-3).

MINERAL-BASED INDUSTRY

The important mineral-based industries in the Organised Sector in Uttar Pradesh are furnished in Table-4.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: Uttar Pradesh

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Andalusite	'000 tonnes	-	114250	114250
Bauxite	'000 tonnes	-	18908	18908
Gold				
Ore (Primary)	tonne	-	13000000	13000000
Metal (Primary)	tonne	-	2.08	2.08
Iron ore (Haematite)	'000 tonnes	-	86330	86330
Limestone	'000 tonnes	3720	439723	443443
Potash	Million tonnes	-	893	893
Platinum Group of Metals	tonne	-	3.35	3.35
Rare-Earth Elements	tonne	-	2948	2948
Rock Phosphate	tonne	-	25773444	25773444
Sillimanite	tonne	-	11450000	11450000

Figures rounded off

Table – 2: Reserves/Resources of Coal as on 01.04.2024: Uttar Pradesh

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total/Singrauli	884.04	177.76	–	1061.8

Source: Coal Directory of India, 2023-24.

Table – 3: Mineral Production in Uttar Pradesh, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		2	-	851937	3	-	689063	3	-	696538
Limestone	'000 tonnes	2	2809	851937	3	2583	689063	3	2878	696538
Sulphur #	Tonne	-	60307	-	-	55411	-	-	53193	-

(P) Provisional

Recovered as by-product from oil refinery.

Table – 4: Principal Mineral-based Industries

Industry/Plant	Capacity ('000 tpy)
Abrasives	
John Oakey and Mohan Ltd, Ghaziabad	NA
Micro Lapping Abrasive, Near Munni, Near Munni Devi Temple, Seohra, Dhampur	0.08 (Abrasive powder)
Aluminium	
Hindalco Industries Ltd, Renukoot	700 (alumina) 410 (aluminium)
Asbestos Product	
Asbestos Cement Ltd, Raebareli	NA
Uttar Pradesh Asbestos Product Ltd, Mohanlalganj, Lucknow	NA
Cement	
ACC Ltd, Tikaria, Distt Sultanpur ^(G)	3910
Ambuja Cement, Dadri, Gautam Budh Nagar ^(G)	1800
Raebareli Cement Works, Birla Corporation ^(G)	1240
Heidelberg Cement (Diamond Cement), Madora, Jhansi ^(G)	3250
Jaypee Cement, Sadvakhurd (Blending Unit)	600
Kanodia Cement, Bulandsahar	330
Shree Cement, Sikandarabad ^(G) , Bulandsahar	2000

Table- 4 (Concl'd.)

Industry/Plant	Capacity ('000 tpy)
Jaypee Cement, Chunar Cement Factory, Chunar, Distt Mirzapur ^(G)	2500
Jaypee Cement, Tanda, Ayodhya, Distt Ayodhya ^(G)	1000
Jaypee Cement, Churk, Mirzapur ^(G)	1500
Birla Corp. Ltd. (Erstwhile Reliance Cement), Kundanganj, Distt Raebareli	2210
UltraTech Cement, Dadri ^(G)	1300
UltraTech Cement, Aligarh ^(G)	1300
UltraTech Cement Ltd. Ayodhya, Ambedkar Nagar	1000
UltraTech Cement Ltd. Bara , Prayagraj (formerly Prayagraj)	4000
UltraTech Cement Ltd. Dalla (Dallakolta), Robertsonganj, Sonebhadra	1800
Mangalam Cement, Aligarh ^(G)	750
Electrode	
Ankit Sangal, Sujroo, Muzaffarnagar	0.85 (electrode no 10)
	0.15 (electrode no 8)
Fertilizer	
Asian Fertilizers Ltd, Deokahia, Gorakhpur	66 (SSP)
Coromandel International Ltd (Formerly Liberty Phosphate Ltd), Raebareli	132 (SSP)
Jubilant Agri and Consumer Products Ltd (Formerly Vam Organic ChemiscsIs Ltd), Bhartiagram, Gajraula, Uttar Pradesh	165 (SSP)
IFFCO, Phulpur (Unit I & II), Distt Prayagraj	1697.8 (Urea)
IFFCO, Aonla (Unit I & II)	1999.8 (Urea)
Indo Gulf Fertilizer Ltd. (a unit of Aditya Birla Nuva Ltd), Jagdishpur	1200 (Urea)
Kanpur Fertilizer & Cement, (formerly Duncan India Ltd), Kanpur	722 (Urea)
Khaitan Chemicals & Fertilizers Ltd., Goramachhia, Jhansi	132 (SSP)
Khaitan Chemicals & Fertilizers Ltd., Malwan, Fatehpur	115 (SSP)
KRIBHCO Shyam Fertilizer, Piprola Shahjahanpur	864.6 (Urea)
Madan Madhav Fertilizers & Chems Pvt. Ltd., Fetezgarh, Uttar Pradesh	24 (SSP)
Natraj Organics Ltd, Muzaffarnagar	60 (SSP)
Yara Fertilisers India Private Limited (formerly known as Tata Chemicals Limited's Fertilizer Division, Babrala), Distt Badaun	1155 (Urea)
V. K. Phosphates Ltd, Bartara, Shahjahanpur	33 (SSP)
Natraj Organics Ltd, Muzaffarnagar	60 (SSP)
Ferroalloys	
Hindustan Ferro Alloys, Hamirpur	3.2
The India Thermit Corpn. Ltd, Kanpur	0.3
Jagdishpur SAIL Unit (JSU) (Formerly Known as Iron & Steel, Malvika Steel Ltd) Jagdishpur	511 (pig iron)
	600 (saleable steel)
Sponge Iron	
RLJ Concast Pvt.Ltd, Baragaon, Chunar	60
S. A.Iron & Alloy Pvt. Ltd. Jeewanthpur, Mughalsarai	90
Petroleum Refinery, IOCL, Mathura	8000

(G): Grinding Unit

Note: Data, not readily available for fertilizer and cement Industries on respective websites, is taken from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory, respectively.



Uttarakhand

Magnesite was the principal mineral produced in State during 2023-24.	₹ 11.39 crore value of mineral production were reported in Uttarakhand in the year 2023-24.	2 mines reported production of MCDR minerals in the State during 2023-24.
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MINERAL RESOURCES

Important minerals found in Uttarakhand include high-grade limestone in Almora, Bageshwar, Dehradun, Nainital, Pauri-Garhwal, Pithoragarh & Tehri-Garhwal districts; magnesite & steatite in Almora, Bageshwar, Chamoli & Pithoragarh districts; and tungsten in Almora district.

Other minerals that occur in Uttarakhand are asbestos in Chamoli district; barytes & marble in Dehradun district; copper in Almora, Dehradun & Pithoragarh districts; dolomite in Dehradun, Nainital & Tehri-Garhwal districts; graphite in Almora district; gypsum in Dehradun, Pauri-Garhwal & Tehri-Garhwal districts; lead-zinc & silver in Dehradun & Pithoragarh districts; and rock phosphate in Dehradun & Tehri-Garhwal districts (Table-1).

PRODUCTION

Magnesite was the only important mineral produced in Uttarakhand during 2023-24. There were 2 reporting mines in Uttarakhand, both of which were magnesite mines. (Table-2).

MINERAL-BASED INDUSTRY

The important medium and large-scale mineral-based industries in the organised sector in Uttarakhand are furnished in Table-3.

Table – 1: Reserves/Resources of Minerals as on 1.4.2020: Uttarakhand

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Asbestos	tonne	-	311	311
Copper				
Ore	000 tonnes	-	4220	4220
Metal	000 tonnes	-	60.04	60.04
Graphite	tonne		10700	10700
Lead-Zinc				
Ore	'000 tonnes	-	5620	5620
Lead metal	'000 tonnes	-	182.6	182.6
Zinc metal	'000 tonnes	-	266.83	266.83
Limestone	'000 tonnes	-	1575771	1575771
Magnesite	'000 tonnes	9959	229476	239434
Rock Phosphate	tonne	-	24178386	24178386
Silver				
Ore	tonne	-	3390000	3390000
Metal	tonne	-	138.59	138.59
Tungsten				
Ore	tonne	-	658000	658000
Contained WO ₃	tonne	-	705	705

Figures rounded off

Table – 2: Mineral Production in Uttarakhand, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

(Value in ₹ '000)

Mineral	Unit	2021-22			2022-23			2023-24 (P)		
		No. of mines	Quantity	Value	No. of mines	Quantity	Value	No. of mines	Quantity	Value
All Minerals		1	-	48693	1	-	73598	2	-	113917
Magnesite	tonne	1	25427	48693	1	37206	73598	2	47104	113917

(P) provisional

Table – 3: Principal Mineral-based Industries

('000 tpy)

Industry/plant	Capacity
Abrasives	
Tirupati Microns, Bhagwantpur, Kashipur, Udham Singh Nagar	0.15 (Abrasives Powder)
Cement	
Ambuja Cement, Roorkee, Distt Haridwar ^(G)	1200
Shree Cement, Roorkee, Distt Haridwar ^(G)	1800
Shree Cement, Laskar Grinding unit Akbarpur-Oud, Laskar	1800
DBM	
Almora Magnesite Ltd, Village-Matela Distt., Bageshwar	30 (DBM, calcined & semi calcined magnesite)
Minerals & Refractories. Haldwani Pithoragarh	3 (DBM)
Ramesh Chandra Binjola, Kumaon Refractories, Narsingh Talla, Haldwani	9 (DBM, calcined magnesite)
Glass	
Hindustan National Glass & Industries Ltd, Rishikesh	155

G: Grinding Unit



West Bengal

44.71 million tonnes
of Limestone Resources have been
established in the State.

3.93 million tonnes
Reserves/Resources of Lignite have been
estimated in the State.

74,395 tonnes
of Sulphur production were reported
in the State during 2023-24.

MINERAL RESOURCES

West Bengal is the principal holder of country's apatite resources. It possesses 50% apatite resources of the country. Important minerals that occur in West Bengal are: apatite in Purulia district; coal in Bardhaman, Bankura, Birbhum, Darjeeling, Jalpaiguri & Purulia districts; china clay in 24-Parganas, Bankura, Birbhum, Bardhaman, Hooghly, Midnapur & Purulia districts; and fireclay in Bankura, Birbhum, Bardhaman & Purulia districts.

Other minerals that occur in West Bengal are barytes, copper, gold, kyanite, pyrite & titanium minerals in Purulia district; dolomite in Jalpaiguri district; felspar in Bankura & Purulia districts; granite in Bankura, Birbhum & Purulia districts; lead-zinc in Darjeeling district; limestone in Bankura & Purulia districts; manganese ore & sillimanite in Midnapur district; quartz/silica sand in Bankura, Hooghly & Purulia districts; and tungsten & vermiculite in Bankura district (Table-1). Reserves/resources of coal and lignite along with details of coalfields/ districts are provided in Tables- 2 & 3.

PRODUCTION

Sulphur was the main mineral for which production was reported in the State of West Bengal in the year 2023-2024. Mineral production in West Bengal from 2021-22 to 2023-24 is furnished in Table- 4.

MINERAL-BASED INDUSTRY

The important mineral-based industries located in the State with their total installed capacities are furnished in Table- 5.

Table – 1: Reserves/Resources of Minerals as on 01.04.2020: West Bengal

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Apatite	tonne	-	10652220	10652220
Copper				
Ore	'000 tonnes	-	113	113
Metal	'000 tonnes	-	2.09	2.09
Gold				
Ore (Primary)	tonne		12833333	12833333
Metal (Primary)	tonne		0.65	0.65
Kyanite	tonne	-	26520	26520
Lead-Zinc		-		
Ore	'000 tonnes	-	3706	3706
Lead metal	'000 tonnes	-	140.07	140.07
Zinc metal	'000 tonnes	-	143.42	143.42
Limestone	'000 tonnes	-	44706	44706
Manganese ore	'000 tonnes	-	200	200
Pyrite	'000 tonnes	-	2500	2500
Sillimanite	tonne	-	1653000	1653000
Titanium	tonne	-	2279000	2279000
Tungsten				
Ore	tonne	-	763802	763802
Contained WO ₃	tonne		1530.84	1530.84
Vermiculite	tonne	-	5566	5566

Figures rounded off.

Table – 2: Reserves/Resources of Coal as on 01.04.2024: West Bengal

(In million tonnes)

Coalfield	Proved	Indicated	Inferred	Total
Total (All Coalfields)	18752.19	11432.59	3773.29	33958.07
Raniganj	17093.34	6468.82	2856.95	26419.11
Barjora	200.79	—	—	200.79
Birbhum	1458.06	4963.77	901.34	7323.17
Darjeeling	—	—	15	15

Source: Coal Directory of India, 2023-24

Table – 3: Reserves/Resources of Lignite as on 01.04.2024: West Bengal

(In million tonnes)

District	Proved	Indicated	Inferred	Total
Total (All districts)	—	1.13	2.8	3.93
Bardhaman	—	0.29	1.82	2.11
Birbhum	—	0.84	0.98	1.82

Source: Coal Directory of India, 2023-24

Table – 4: Mineral Production in West Bengal, 2021-22 to 2023-24

(Excluding Atomic, Fuel and Minor Minerals)

Mineral	Unit	2021-22	2022-23	2023-24 (p)
All Minerals		—	—	—
Sulphur [#]	tonne	66170	87067	74395

(p) provisional

[#] Recovered as by-product from oil refinery.

Table – 5 : Principal Mineral-based Industries

(’000 tpy)

Industry/plant	Capacity
Asbestos Products	
Everest Building Products Ltd, Kolkata	NA
Ramco Industries Ltd, Haratara, Distt Paschim Medinipur, Kharagpur	NA
UAL Industries Ltd, Tungadhowa, new Jhargram district	250
Abrasives	
Carborandum Universal Ltd, Gopalpur	NA
K.L. Thirani & Co. Ltd., Kolkata	NA
Satya Narayan Roller Floor Mill Pvt. Ltd., Chandur Tarkeshwar	MOG-303 0.35 MOG-302 0.40 MOG-M2 0.40 MOG-220 0.20 Fine-0.15
Cement	
ACC Ltd (Damodar Cement), Purulia (G)	750
Ambuja Cement Ltd, Sankrail, Distt Howrah (G)	2400
Ambuja Cement Ltd, Farakka, Distt Murshidabad (G)	1250
Birla Corporation Ltd (Durgapur Cement Works & Durga Hitech Cement), Durgapur (G)	2300
Burnpur Cement, Asansol, Distt Burdwan (G)	300
UltraTech Cement Lmt., Sonar Bangala Grinding Unit, Distt Murshidabad	3300
Nu Vista Limited (Formerly Emami Cement Limited), Panagarh, Burdwan	2000
Nu Vista Limited (Formerly Emami Cement Limited), Kotagram, Ausgram-II	2000
JSW Cement, Salboni, Paschim Medinipur	2400
Lafarge Cement Pvt. Ltd, Mejia (G)	1650
Maa Chandi Cement Bamunara Burdwan	330
NUVOCO Vistas Corp. Ltd, Amdang, Bakura	1650
OCL India Ltd, Bengal Work Mednapore	1350
Ramco Cement (formerly Madras Cement), Kolaghat, Distt Purba Medinipur (G)	1950
Shristi Cement Ltd., Mangalpur, Burdwan	363
Swasata Cement Ltd., Purulia	1500
Ultra-Tech Cement Works, Dankuni, Hooghly, West Bengal	2400
Ultra-Tech Cement Ltd, WBCW(G), Burdwan	1800
Ceramics	
WBCDC Ltd (West Bengal Comprehensive Area Development Corporation), Kolkata	0.18
Chemical	
Hindustan Heavy Chemicals Ltd, Khardah, Distt 24-Parganas	14.8 (caustic soda) 6 (Cl), 9.8 (HCl) 4.5 (ferric alum) 18.7 (H ₂ SO ₄)
Alchrome Chemical Industries, Kalyani Industrial Estate	1.2 (sodium bicarbonate) 0.6 (sodium sulphate)
Electrodes	
GEE Ltd, Kandua, Sankrail, Howrah	18
Graphite India Ltd, Kolkata	NA
Radix Arc Pvt. Ltd, Dangadighila	4
Shield Arc Equipment Pvt. Ltd, Rasputja	2.5
Fertilizer	
Tata Chemicals Ltd, Haldia	675 (DAP)
Tata Chemicals Ltd (Phosphatic Division), Haldia, Medinipur	160 (SSP)
Teesta Agro Industries Ltd, Rajganj, Jalpaiguri	165 (SSP)
Jay Shree Chemicals & Fertilizers, Khardah, 24 Parganas	132 (SSP)

Table 5 (Cont.)

Industry/plant	Capacity (^{'000 tpy})
	62.70 (H ₂ SO ₄)
The Phosphate Company Ltd, Rishra, Hooghly	112.8 (SSP)
Sai Fertilizers Pvt. Ltd, Dewanmara, West Midnapore	132 (SSP)
Flux	
Priyanka Tradelink Pvt. Ltd	0.6
Glass	
Hindustan National Glass & Industries Ltd, Rishra	805 TPD
Iron & Steel	
Durgapur Steel Plant, SAIL, Durgapur	3880 (sinter)
	2450 (pig iron)
	2200 (crude/liquid steel)
	19.2 NH ₄ SO ₄
IISCO Steel Plant, SAIL, Burnpur, Distt Bardhaman	2695 (pig iron)
	2500 (crude/liquid steel)
	3800 (sinter)
Alloy Steel Plant, SAIL, Durgapur	234 (crude/liquid steel)
Tata Steel Mining Ltd (Formerly Rohit Ferrotech Ltd), Bishnupur	100 (stainless steel)
	71.4 (ferroalloys)
Rashmi Metaliks Ltd, Gokulpur, West Midnapore	1200 (pellets)
	1440 (sinter)
Pig Iron	
Electrosteel Castings Ltd, Khardah	250
	365 (sinter)
Jai Balaji Industries Ltd, Banskopa, Distt Bardhaman	428.7
	908 (sinter)
Kajaria Iron Castings Ltd, Durgapur	110
	360 (sinter)
KIC Metaliks Ltd, Raturia, Angadpur, Durgapur	336 (sinter)
	165 (pig iron)
Orissa Metallic Pvt. Ltd, Mathurakismat, Gokulpur, Shyamraipur Pachmi medinipur Unit -I & Dhekia & Chaksonadhar, Pachmi Medinipur Unit-II	1370 (pellets)
	300 (pig iron)
	329.72 (sponge Iron)
Rashmi Metaliks Ltd, Shyamraipur, Gokulpur, Medinipur	600 (sinter)
	900 (pellets)
	180 (pig iron)
Tata Metaliks Ltd, Kharagpur	345
	528 (sinter)
Tata Metaliks Ltd, Gokulpur, Maheshpur	600
Neo Metaliks Ltd, Gopalpur, Durgapur	300 (Sinter)
	187.9
Sponge Iron	
Adhunik Corporation Ltd, Angadpur, Durgapur	72
Ankit Metal & Power Limited Jorehira Chhatna	210
	600 (pellet)
Aryavrata Steel Pvt. Ltd, Lohamelya, Distt West Medinipur.	36
C. P. Sponge Iron Pvt. Ltd, Raturia Angadpur, Industrial Area Durgapur	60
C. P. Ispat Pvt. Ltd, G.T. Road Bhirigee	60
Divya Jyoti Sponge Pvt. Ltd, Nandanpur	60
Electrosteel Castings Limited, Haldia	60
Howrah Gasses Ltd, Raniganj, Distt Bardhaman	60

Table- 5 (Concl'd.)

('000 tpy)

Industry/plant	Capacity
Jai Balaji Sponge Ltd, Raniganj, Mangalpur, Distt Bardhaman. I	105
Jai Balaji Sponge Ltd, Banskopa, Distt Bardhaman. IV	120
Kunj Bihari Steel Pvt. Ltd, Jamuria Nandi	30
M. B. Sponge & Power Limited, Hijalgola Jamuria	60
Maithan Steel & Power Ltd, Bora, Bonra	60
Rashmi Ispat (Pvt.) Ltd, Jhargram, Distt W. Medinipur	150
Rashmi Cement Ltd, Jetusole Jhargram, Distt W. Medinipur	492.7
	33 (Ferroalloys)
Ravindra Enterprise Pvt. Ltd, Digha, Purulia	30
Rishabh Sponge Ltd, Barjora, Bankura	90
SRS Sponge Pvt. Ltd, Dantia, Balampur	15
Satyam Iron & Steel co. Pvt. Ltd, Asansol	60
Sen Ferro-alloys Pvt. Ltd, Dejudi	15
Shyam Sel Ltd, Dewabdighi, Burdwan	100
Shyam Steel Industries Ltd, Anandpur, Durgapur	79
Sunil Sponge Iron Ltd, Kolkata	115
Ferroalloys	
Kartik Alloys Ltd, Durgapur	10.7
Bhaskar Shrachi Alloys Ltd, Durgapur	40
Corporate Ispat Alloys Ltd (Abhijit Group), Durgapur	74
Dimension Steel and Alloys Pvt. Ltd, Bakura	38
Gagan Ferro Tech. Ltd, Jamuria	138.6
Jai Balaji Industries Ltd, Durgapur (JBIL Group)	106
Jai Balaji Industries Ltd, Unit IV, Banskopa	76.5
Jai Balaji Industries Ltd, Mangalpur	30.1
Maithan Alloys Ltd, Kulti Bardhaman	94.6
Modern India Con-Cast Ltd, Bishnupur, Distt Bankura	75
Modern India Cone-Cost Ltd, Haldia	100
Nilkanth Ferro Ltd, Radhamadhavpur	39.9 (silico-manganese)
Rohit Ferro-Tech Ltd, Haldia	100.6
Shyam Ferro Alloys Ltd, Burdwan	100 (50 MVA)
Shyam Ferro Alloys Ltd, Durgapur	100 (50 MVA)
Shri Vasavi Industries Ltd, Bishnupur, Distt Bankura	45
Srinivasa Ferro Alloys Ltd, Durgapur, Distt Bardhaman	84.2
Shri Goyatri Minerals Pvt. Ltd, Bishnupur, Distt Bankura	24
Refractory	
Alcoa-ACC Industrial Chemicals Ltd, Kalatalahat	10
Barakar Refractories (P) Ltd, Barakar, Distt Bardhaman	3.6
Kero Rajendra Monolithics Ltd, Banjora	NA
National Refractories Prop. Snowtex Udyog Ltd, Salanpur	43.2
Saswat International Ltd, Kulti, Distt Bardhaman	NA
Vesuvius India Ltd, Kolkata	96.5
Coke Oven Batteries	
IISCO Burnpur Works, Burnpur, Distt Bardhaman	1084
Petroleum Refinery	
IOCL, Haldia	7500
TiO₂ Pigment	
Kolmak Chemicals Ltd, Kalyani, Distt Nadia	4.8

(G): Grinding units.

Note: Data, not readily available for fertilizer and cement Industries on respective websites. Therefore, the data is sourced from Indian Fertilizer Scenario, FAI Statistics, and Survey of Cement Industry & Directory.



Union Territories

Geological Survey of India has carried out 50 sq. km mapping on 1:12,500 scale in Thasgam, Kargil, delineating chromite-bearing ultramafics	Aluminium, Abestos Products, Copper and Alloy is the principal mineral-based industry in Puducherry
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ANDAMAN & NICOBAR ISLANDS

The MCDR mineral production was not reported from Andaman & Nicobar Islands during the year 2023-24.

MINERAL-BASED INDUSTRY

The principal mineral-based industries in the Organised Sector in Andaman & Nicobar Islands are furnished in Table - 1.

Table – 1: Principal Mineral-based Industries at Andaman & Nicobar Islands	
Industry/plant	Capacity ('000 tpy)
Cement	
India Cement Ltd Andaman & Nicobar Islands	1650
Sponge Iron	
Gallantt Metal Limited Andaman & Nicobar Islands	225

CHANDIGARH

Production of MCDR minerals was not reported from Chandigarh during 2023-24.

DADRA & NAGAR HAVELI

Production of MCDR minerals was not reported from Dadra & Nagar Haveli during 2023-24. The principal mineral-based industries in the Organised Sector located in Dadra & Nagar Haveli are furnished in Table - 2.

Table – 2: Principal Mineral-based Industries in Dadra & Nagar Haveli

Industry/plant	Capacity ('000 tpy)
Aluminium	
Hindalco Industries Ltd, Silvassa	30 (Al foil)
Asbestos Products	
Ramco Industries Ltd, Golanda, Silvassa	NA
Copper	
Sterlite Industries (India) Ltd,	180 (copper cathode)
Chinchpada, Silvassa	150 (CC copper rod)
Alloy	
Hindustan Alloys Mfg. Co Ltd, (HAMCO), Silvassa	3 (tin ingot)
	38 (Al alloy ingot)
	2.8 (Sn-Pb solder wire)

DAMAN & DIU

Production of MCDR minerals was not reported from Daman & Diu during 2023-24. Reserves/ Resources of mineral as on 1.04.2020 are furnished in Table-3.

Table – 3: Reserve/Resource of Mineral as on 01.04.2020 : Daman & Diu

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Limestone	'000 tonnes	-	128670	128670

Figures rounded off

LAKSHADWEEP

Production of MCDR minerals was not reported from Lakshadweep during 2023-24.

PUDUCHERRY

Production of MCDR minerals was not reported from Puducherry during 2023-24. Reserves/ Resources of mineral as on 01.04.2020 are furnished in Table-4. Reserves/ Resources of lignite as on 01.04.2024 are furnished in Table-5. The principal mineral-based industries in the Organised Sector located in Puducherry are furnished in Table-6.

Table – 4: Reserves/Resources of Minerals as on 01.04.2020 : Puducherry

Mineral	Unit	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Limestone	'000 tonnes	-	15732	15732

Figures rounded off

Table – 5: Reserves/Resources of Lignite as on 01.04.2024: Puducherry

(In million tonnes)				
District	Proved	Indicated	Inferred	Total
Total/ Bahur & West of Bahur of Neyveli Lignite Field	-	405.61	11	416.61

Source: Coal Directory of India, 2023-24.

Table – 6: Principal Mineral-based Industries in Puducherry

Industry/plant	Capacity ('000 tpy)
Ceramics	
Regency Ceramics Ltd, Yanam.	150
H&R Johnson (India) Ltd, Karaikal.	24.2
Ferroalloys	
The Silcal Metallurgic Ltd.	12.4
VSK Ferro Alloys Ltd, Tutthipet.	3
Snam Alloys Pvt. Ltd, Kariamanikam.	12
Iron & Steel	
Sumangala Steel (P) Ltd.	NA
Glass	
Hindustan National Glass & Industries Ltd, Puducherry.	350 TPD



Offshore Regions

671.4 million tonnes balance recoverable reserves of crude oil were estimated as on 01.04.2024.	1,094.2 billion cu. m balance recoverable reserves of natural gas were estimated as on 01.04.2024.	9 Placer mineral exploration programmes were executed to identify and estimate economic placer minerals within the territorial waters of East and West Coast of India.
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The Government of India notified the Offshore Areas Minerals (Development & Regulation) Act, 2002 (OAMDR Act), No. 17 of 2003 in the Gazette of India, Extraordinary, Part-II, Section-1, dated 31.1.2003. The purpose of the Act is to provide for development and regulation of mineral resources in the territorial waters, continental shelf, exclusive economic zone and other maritime zones of India and to provide for matters connected therewith or incidental thereto. The Act is applicable to all minerals in offshore areas including minerals prescribed under Atomic Energy Act, 1962, but excludes oils and related hydrocarbons as there is separate legislation in force. The Act came into effect from 15.1.2010 vide S.O. 338 (E), dated 11.2.2010 as notified by the Central Government.

The Act makes it mandatory to undertake reconnaissance, exploration or production operation in the offshore areas in accordance with the prescribed terms and conditions for Reconnaissance Permit (RP), Exploration Licence (EL) or Production Lease (PL) granted under the Act and the rules made thereunder. The Act further states that availability of the areas for grant of RP, EL or PL shall be notified within six months from the commencement of the Act, and subsequently at such times as considered necessary. The Act empowers the Central Government to make rules for the purpose of the Act including terms and conditions under the RP, EL, PL, etc. The Rules, namely, the Offshore Areas Mineral Concession Rules, 2006 have been framed and notified on 3.11.2006 by G.S.R.691(E) published in the Gazette of India, Extraordinary, Part II, Section 3 (i), No. 539, dated 4.11.2006. The Rules came into effect on the date on which the Offshore Areas Mineral (Development and Regulation) Act, 2002 came into force, i.e, 15.1.2010.

As per S.O.1341(E) dated 7.6.2010, the Controller General, Indian Bureau of Mines had notified the mineral-bearing offshore blocks available for grant of Exploration Licence. As per the attached Schedule to the said Notification, there were 26 offshore areas available in offshore waters of Bay of Bengal and 37 offshore areas in the offshore waters of Arabian Sea for grant of Exploration Licence.

The orders for grant of exploration licences were issued by the Administering Authority on 05.04.2011 for the 62

exploration blocks (the bounding latitude and longitude of Block Nos. 3 & 32 falling in the Arabian Sea were same and therefore these were considered as a single block and granted as Block No. 3). Before execution of deed granting such licence, the grant of exploration licences in 62 blocks was challenged through the writ petition in the judicature of various High Courts. Due to interim orders passed by various Hon'ble High Courts on the writ petition and non-disposal of the said petition, the offshore exploration licences granted have not been executed. Besides, it was brought to the notice of the Administering Authority that some of the exploration blocks notified for grant of offshore exploration licences vide Notification dated 07.06.2010 overlapped with areas other than offshore area, to which the OAMDR Act did not apply.

The Central Government vide S.O.19 (E) dated 06.01.2011, published in the Official Gazette, declared the extent of the Coastal Regulation Zone (CRZ) and also imposed certain restrictions on the setting up and expansion of industries, operations or processes and the like in the CRZ. The said statutory order also did state that CRZ shall apply to the water and the bed area between the Low Tide Line to the territorial water limit (12 Nm) in case of seas and has prohibited in the area so identified as CRZ, inter alia, the mining of sand, rocks and other substrata materials except those rare minerals not available outside the CRZ area. In the context of the said notification, all the 62 offshore blocks lie within the area identified as CRZ which attracts the prohibition of mining (operation undertaken for the purpose of winning any mineral).

The OAMDR Act provides that the holder of an exploration licence for offshore area shall have the exclusive right to a production lease for winning of a mineral. In view of the effect of the CRZ Notification dated 06.01.2011, the purpose of executing the 62 offshore exploration licences could not be realised as the applicants could not undertake operations for winning of minerals in spite of grant of Production Lease after successful completion of exploration operations.

Therefore, taking into consideration all the above

stated facts, the Controller General, IBM and administering authority Offshore Areas Minerals (Development & Regulation) vide S.O.19 (E) dated 6th January, 2011, published in the Official Gazette, annulled the Notification issued vide S.O.1341(E) dated 7th June 2010 with effect that all subsequent actions undertaken for grant of the 62 exploration licences hereby would stand rescinded. As per S.O. 1523(E) dated 06.04.2018, the Additional Director General, National Mission Head-II, Geological Survey of India, has been notified as the "Administering Authority" for the purpose of the said Act by Clause (a) of Section (4) of the Offshore Area Mineral Development and Regulation Act, 2002, 17 of 2003 and in supersession of the notification published in Gazette of India, Extraordinary Part II, Section 3, Sub-section (ii) vide S.O. 339(E) dated 11th February 2010.

The Central Government vide G.S.R 315(E) dated 6th June, 2024 notified offshore Areas (Existence of Mineral Resources) Rules, 2024. The Central Government vide G.S.R 490 (B) dated 9th August, 2024 notified Offshore Areas Mineral Trust Rule 2024. The Central Government vide G.S.R502 (E) dated 14th August, 2024 notified Offshore Areas Mineral (Auction) Rules, 2024. The Central Government vide G.S.R. 79 (E) dated 31st December, 2024 notified Offshore Areas Mineral Conservation and Development Rules, 2024.

The Government of India further signed 360 contracts under NELP (New Exploration and Licensing Policy) regime with National Oil Companies and Private (both Indian and foreign)/ Joint Venture companies. At present, 229 contracts are operational out of the total 602 contracts [360 (NELP), 139 (OALP), 103 (DSF Round)] signed so far under various bidding rounds.

The awarded 254 blocks under NELP regime are at locations in inland (114), offshore shallow water (59) and deepwater (81) areas. As a result of exploratory activities, several unexplored and poorly explored areas, in particular, offshore and deepwater areas, have been appraised through geophysical surveys and exploratory drilling. Details of exploration block awarded/relinquished/operational are provided in Table -1.

Table – 1: Details of Exploration Block Awarded (as on 01.04.2024)

Round	No. of blocks awarded	No. of blocks relinquished	No. of blocks active	Present Area (sqkm)
NELP-I	24	21	3	231527
NELP-II	23	22	1	267883
NELP-III	23	19	4	204596
NELP-IV	20	17	3	192810
NELP-V	20	16	4	115180
NELP-VI	52	46	6	306426
NELP-VII	41	36	5	112947
NELP-VIII	32	31	1	52573
NELP-IX	19	14	5	26441
Total NELP	254	222	34	1510383
OALP-I	55	6	49	59283
OALP-II	14	1	12	29233
OALP-III	18	3	13	29765
OALP-IV	7	-	7	18510
OALP-V	11	-	11	19789
OALP-VI	21	0	21	35346
OALP-VII	8	0	7	15766
OALP-VIII	10	0	6	34364.53
Total OALP	144	11	126	242056
DSF-I	30	11	15	777
DSF-II	24	5	16	3004
DSF-III	30	1	24	12727
Total DSF	84	17	55	16508

Source: IPNG Statistics 2023-24, Ministry of Petroleum and Natural Gas.

In order to explore and produce new sources of natural gas from coal-bearing areas, the Government had formulated a CBM Policy in 1997, wherein CBM being Natural Gas is explored and exploited under the provisions of OIL Fields (Regulation & Development) Act, 1948 (ORD Act, 1948) and Petroleum & Natural Gas Rules, 1959 (P & NG Rules 1959) administered by Ministry of Petroleum & Natural Gas (MOP & NG). CBM policy was aimed to provide attractive fiscal and contractual framework for exploration and production of CBM which is an environment-friendly clean gas fuel similar to conventional natural gas. In order to harness CBM (Coal-bed Methane) potential in the country, CBM blocks were offered through international competitive bidding for exploration and production for the first time in the year 2001. Under the CBM policy till date,

four rounds of CBM bidding have been implemented by MoP&NG, resulting in award of 33 CBM blocks [including 2 blocks on Nomination and 1 block through Foreign Investment Promotion Board (FIPB) route]. Till date, most CBM exploration and production activities in India are pursued by domestic Indian companies. These CBM blocks are in the States of Andhra Pradesh, Assam, Chhattisgarh, Gujarat, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, Rajasthan, Tamil Nadu and West Bengal.

RESERVES/RESOURCES

As on 01.04.2024, balance recoverable reserves of crude oil were estimated at 671.4 million tonnes. The balance recoverable reserves of natural gas as on 01.04.2024 were placed at 1,094.2 billion cu. m. (Table-2).

Table – 2: Balance Recoverable Reserves of Crude Oil & Natural Gas in India including Offshore Areas (as on 01.04.2024)

(Crude oil in million tonnes; Natural gas in billion cu. m)

Area	Crude oil	Naturalgas
India	671.4	1094.2
1. Onshore	417.4	488.6
2. Offshore	254	605.6
a) Western Offshore	213.3	342.3
b) Eastern Offshore	40.7	263.3

Source: Indian Petroleum and Natural Gas Statistics, 2023-24, Ministry of Petroleum and Natural Gas, Government of India.

Note: * Proved and indicated balance recoverable reserves. In case of Natural Gas, reserves includes coal-bed Methane.

EXPLORATION ACTIVITIES

Conventional Hydrocarbon

ONGC, GSI and other Public & Private Sector companies continued their efforts in respect of exploration for hydrocarbon in offshore region, both shallow and deep water, during 2023-24.

Marine and Costal Survey

The Annual Programme of Marine & Coastal Survey Division (MCSD) for the FS 2023-24 included multidisciplinary offshore survey and mineral exploration within the Exclusive Economic Zone (EEZ) of India including Territorial Waters (TW) and International Waters. Nine placer mineral exploration programmes both preliminary and close grid were executed to identify and estimate economic placer minerals within the territorial waters and beyond in the east coast and west coast of India. Preliminary mineral exploration for lime mud/lime sand were carried out in the continental shelf off Gujarat and Maharashtra coast to search for the extent of lime mud/lime sand distribution, its nature and quality in the selected area of the continental shelf off Gujarat and Maharashtra coast. Close grid mineral exploration for phosphorite, in the east coast margin of India, off Point Calimere, Tamil Nadu was carried out to delineate the High Potential Area (HPA) for phosphorite and to evaluate the resource potential of phosphorite in the phosphatic zone. Mapping programme was taken up in the Ninety East Ridge near Equator, Indian Ocean, for base line data generation with multiple objectives like 1) study the morphology, structure and tectonic history by using geological as well as geophysical mapping techniques, 2) to investigate the possible occurrences of cobalt-rich ferromanganese encrustations with trace metals & Rare-earth elements, 3) to study palaeo-oceanographic condition and mode of origin of ridge using ferromanganese encrustations and volcanic samples. Two programmes were taken up to fill up the shallow seismic data gap areas between Dwaraka Sector and Jakau–Okha Sector in TW off West Coast, Arabian Sea, under baseline geoscientific data generation.

Two multi-thematic mapping programmes were taken up within the contiguous zone off Tamil Nadu to bring out the bathymetry, surface & sub-surface geology of the area through geological and geophysical studies. Systematic seabed mapping was carried out in an area of 1,507 sqkm within TW and CZ and 21,020 sqkm area was mapped beyond EEZ of India under the baseline geoscience data collection programme. Preliminary mineral investigation over an area of 15,573 sqkm and close grid mineral investigation to the tune of 610 sqkm were also carried out in TW and beyond for the augmentation of offshore mineral potentials in Indian EEZ. Along with these, a total of 4,243 lkm of shallow seismic surveys were also carried out as part of baseline geoscientific data generation and to study the sub-surface disposition of sediment sequences in TW

and contiguous zone and to identify the possible locales of offshore minerals through advanced processing techniques. The marine geoscientific programme taken up during the period comprised 3 cruises onboard RV Samudra Ratnakar in deep waters. In shallow water domain, seven cruises were undertaken by RV Samudra Kaustubh and seven cruises by RV Samudra Shaudhikama. Besides these, one coastal item using mechanised boat, one RP item and another continuous RP item of FS-2022 was also taken up during the period.

1. R. V. SAMUDRA RATNAKAR

Preliminary mineral exploration of lime mud/lime sand in the continental shelf off Gujarat and Maharashtra coast (SR-085) (FSP ID: MIAMCS-MME/NC/SR/MCSDWC- 1/2023/48555)

The survey area is located at the northernmost extension of the Fifty Fathom Flat (FFF) platform, situated on the outer continental shelf off the western coast of India. The survey aimed to reveal the three-dimensional arrangement of lime mud layers in a 14,000-square-kilometer area on the continental shelf off Gujarat. This was accomplished through a systematic approach involving sub-bottom profiling (SBP) conducted at 5-kilometer intervals. Additionally, gravity measurements and single-beam bathymetric surveys were performed. To further validate the projected extent and depth continuity of the lime mud, a total of 30 core samples were gathered. These samples comprised 5 piston cores, 15 VC cores and 10 GC cores. The high-density gravity contours from the gravity survey observed in the western part of the survey area were the continental shelf break. The FAA satellite map revealed that the depth of flood basalts beneath the Eocene to Holocene sediment layers within the surveyed region ranged from 2,000 meters in the Northeastern Sector to 4,000 meters, sloping towards the Southwest. Sub-bottom profiling studies have revealed the seabed structure characterised by a series of intermittent reflectors, running parallel to the seabed and ranging in thickness from 2 to 20 meters. These reflectors have been verified through ground truth validation, where sediment cores have shown them to consist of pure lime mud or lime sand layers with varying compaction levels, overlaid by terrigenous clay. In the middle shelf region, there is an approximately 2-meter-thick layer of oolitic lime sand exposed near the seabed, with no overlying sediment. Most of the samples in this sector have a low abundance of terrigenous materials, may be due to morphology of the area. Beneath the lime mud layer, there exists a high-density acoustic layer approximately 2 meters thick consistently identified across profiles. This layer is confirmed to be an indurated formation found at depths ranging from 5 to 20 meters below the seabed. It tapers as it approaches the shelf break. Below the indurated layer, there are sporadic, faint and occasionally inclined sets of reflectors, with an estimated thickness of approximately

5 meters. These reflectors are detected at depths ranging from 20 to 25 meters below the seabed.

Close grid mineral exploration for phosphorite in the east coast margin of India, off Point Calimere, Tamil Nadu (Block-II) (Cruise SR-086) (FSP ID: M1AMCS-MME/C/ SR/MCSD-EC-2/2022/46689)

The close grid mineral exploration for phosphorites in the East Coast Margin of India (ECMI), off Point Calimere, Tamil Nadu (Block-II) was taken up in an area of 514 sqkm by sub-bottom profiling, gravity and magnetic surveys, single beam survey, CTD profiling and seabed sampling. The water depth of the investigation area varied from 130 m to 400 m. The sediment thickness in the survey area varied from 0 to a maximum of 15 m and the maximum thickness observed was in the middle portion and towards the northeastern part of the survey area. The free-air gravity anomaly showed the total variation of 48 mGal, ranging from 4 mGal to 52 mGal. The trend of the contour exhibited NW-SE and NE-SW orientations. The Magnetic anomaly variation of about 565 nT, ranging from -665 nT to -100 nT was observed in the study area. The general trend of the contours was NW-SE direction in the central part of the surveyed area, while in the southern and northeastern parts, the contours were aligned in a predominantly NE-SW direction. A broad NE-SW trending magnetic high anomaly with a long wavelength was observed in the southern part of the surveyed area, with a sharp gradient on either side, indicating the presence of a fault contact. In the central part of the study area, a zone of positive magnetic anomaly trending in a NW-SE direction was bounded by negative magnetic anomalies on either side, suggesting the presence of horst & graben structures in the magnetic basement. Another magnetic high was identified in the northern part, trending in a nearly NE-SW direction, with a sharp gradient towards the north, suggesting a faulted contact. Phosphatic materials occur in the form of concretions, micro concretions, bio-clasts, steinkerns and as phosphatised fish bones and studded to *Xenophora* sp. gastropod shell. The size of phosphatic nodules varied from small size of 0.5 cm x 0.5 cm to as big as 6 cm x 4 cm. In addition to concretions, phosphatised rock pieces, phosphatised coral pieces and granules also were recovered from the samples. Phosphatic material was mainly concentrated in the middle of the survey area within the water depths from 200 m to 348 m trending north to south. The phosphatic material which was recovered in the western part (below 200 m water depth) and eastern part (beyond 350 m water depth) were mainly xenophoric studds and fish bones. The sub-surface distribution of the phosphatic concretions observed as thin layers starting from 2 cm bsf followed by 20–23 cm bsf, 40 cm bsf, 50 cm bsf, 80–90 cm bsf, 1.20 m–1.29 m, 1.56 m, 1.78 m, 2.25 to 2.58 m, 3.17 m, 3.28 m, 3.56 m, 5.64 m and further downwards up to 10m bsf in the piston core. In the survey area, the phosphatic materials were mainly concentrated between the drape of two rocky exposures and both sides of rocky exposures.

R. V. Samudra Kaustubh

Preliminary search for placer minerals in the territorial waters off Pentakota–Pudimadaka sector (off Revu Polavaram-Ontimamidi), Andhra Pradesh, Bay of Bengal, ST-302 (south block) (FSP ID: IAMCS-PMIWEZ/NC/SR/MCSD-EC-2/2022/47025)

A total of 56 vibrocore sediment samples were collected in 2 km x 3 km grid covering an area of 278 sqkm within the water depths between 11.50 and 41.1 m. The length of the vibro cores varied from 0.31 m (VC-88) to 4.33 m (VC- 93) with an average of 1.89 m. The granulometric analysis of surface sediment showed that sediments could be placed under sand, silty sand, sandy silt, clayey silt and silt regimes. Sand content in the sediments varied from 8.01 wt% to 99.06 wt% with an average of 76.544 wt%. Among sand classification, the majority of the area was occupied by medium sand (28.57%) and coarse sand (17.52%) followed by fine sand (14%), very fine sand (9%) and very coarse sand (7%). Granule content when added to sand content, the granulometric metric varied from 0 to 18.39 wt% with an average of 3.85 wt%. Granules mainly comprised shells and shell fragments of gastropod, bivalves, calcrites, quartz grains and clay gals. Silt was seen dominant in the southern part of the survey area. Heavy mineral studies were carried out at 56 sample locations and generated a total of 123 sub-samples at 1 m interval. THM distribution from 0 to 1 m ranged from 0.08 wt% to 6.71 wt% with an average value of 1.88 wt%, majority of the part were with THM less than 3 wt% except at few locations towards western and south-western part of the surveyed area. The THM wt% ranged from 0.47 to 5.89 with an average 1.81 for 1 to 2 m; from 2 to 3 m level, the THM wt% ranged from 0.24 to 2.91 with an average 1.44; from 3 to 4 m level, the THM wt% ranged from 0.14 to 3.35 with an average of 1.71. Grain counting of some of the samples revealed that the economic minerals like ilmenite, sillimanite and garnet are as major minerals and monazite, rutile and zircon as a minor mineral.

Geophysical (magnetic) surveys within the contiguous zone off Pulicat, Andhra Pradesh and Tamil Nadu coasts, Bay of Bengal (Block-II) (ST-303) (FSP ID: M1AMCSAPS/NC/SR/MCSD-EC-2/2022/47013)

Geophysical surveys, including magnetic and bathymetric surveys, were conducted off Pulicat during Cruise ST-303 onboard R.V. Samudra Kaustubh. The surveys were carried out along twenty-three coast-perpendicular transects (L-1 to L-23) in an E-W direction and two coast-parallel transects (CL-1 and CL-2) in an NNW-SSE direction, covering an approximate area of 3,780 sqkm with a cumulative line length of 840 km. The water depth in the surveyed area ranged from 59 to 2,752 m. The seafloor morphology revealed a wide continental shelf, extending from 30 to 45 km in the south and narrowing to 25 to 30 km in the north due to the influence of the River Swarnamukhi. The shelf break, sub-parallel to the coastline,

occured at a depth of around 400 m in the southern part and approximately, 500 m in the northern part. A magnetic anomaly variation of approximately 624 nT was observed ranging from a minimum of -513 nT to a maximum of 112 nT. In the southern part of the area off Pulicat, the magnetic anomalies exhibit positive and negative patterns, bounded by a high on the northern side. This suggests that the crust in the Palar basin is highly fragmented and separated by several faults/fractures. Further south, bipolar magnetic anomalies suggest potential isolated intrusions within the basement or shallow causative bodies, possibly due to shallow basement undulations. Upward continuation maps indicate that the central part of the study area has a more pronounced crustal root compared to other regions. The residual anomaly map preserves the NNW-SSE and N-S trends of anomalies, and the compiled magnetic anomaly map revealed the extension of several magnetic anomaly highs/lows from onshore to offshore. An inferred fault has been identified off Chennai, cutting across E-W trending shear zones onshore, which may be related to the action of the River Korttalalyar extending into the Cauvery offshore.

Preliminary Mineral Exploration (PME) of placer minerals in the territorial waters off Pentakota, Andhra Pradesh (ST-304) (FSP ID: MIAMCS-MME/NC/SR/MCSD-EC-2/2023/46676)

The cruise was taken up by covering an area of 258 sqkm within the water depths of 7.33 m to 36.6 m with an average of 19.08 m. It is observed that bathymetric contours were conspicuously bifurcating in south-eastern and south-western directions by 13–18 m and 18–23 m respectively due to the development/protruding of sand pit. One prominent paleostrandline was observed at around 21 m contour. One channel like feature was also observed at NW part of the study area and that may be due to the influence of River Pampa causing presence of coarse-medium and fine sand. At shallower depths around 10 m towards SW part, heap of disturbed sediment was observed causing irregular contour pattern. In the remaining part, bathymetric contours were seen comparatively smoothened and dipping towards South-easterly directions. High resolution single channel shallow seismic survey revealed that the seabed is smooth and gently dipping towards the sea. A total of 58 vibrocore sediment samples were collected in 3 km X 2 km grid pattern within the water depths of 9 m to 35 m. The length of the cores varied from 0.62 m to 4.20 m, with an average core length of 1.76 m. In the surveyed area, surface of the seabed was observed to be mainly carpeted with variants of sand (v. fine sand, clayey sand and medium sand in order of abundance) followed by clay and silty clay. The bottom sediments mainly consisted of variegated compact clay often mixed with sand and very coarse sand with pebbles and granules at places (mainly Northern part). THM wt% in surface sediments varied from 0 to 9.08 with an average of 1.68 wt%. Processing of seismic, bathymetric, HM sand and coarse fraction data and drafting report for L-1 was under progress.

Preliminary Mineral Exploration for placer minerals in the Territorial waters between Cuddalore and Portonova, Nagapattinam District, Tamil Nadu (ST-305) (FSP ID: IAMCS-PMIWEZ/NC/SR/SU-TNP/2023/47063)

An Area of 250 sqkm between water depth of 7.7 and 39 m was covered during the survey. A total of 72 vibrocore sediment samples were collected in 3 km X 2 km grid pattern within the water depths of 7.46 m to 38.3 m. The length of the cores varied from 0.22 m to 4.34 m, with an average core length of 1.34 m. Surface of the seabed was seen mainly carpeted with sand up to 49% (fine sand, coarse sand and medium sand in order of abundance) followed by clay 27% and silt 23%. A total of 128 sub-samples generated from 72 VC for Heavy Mineral studies, 127 sub-samples generated for Size Analysis, 75 samples for Geochemical studies were pulverised and submitted to Chemical Division, OPWC-1, MCSD, GSI, Mangaluru, for Major oxides and Trace elements studies. Heavy mineral processing was completed for all the samples and total Heavy Mineral Concentration THM wt% was evaluated and revealed as Minimum 0.001, Maximum 9.31 and Average 1.87. The total Economic Heavy Mineral Concentration TEHM wt% was evaluated which revealed Minimum 0.001, Maximum 8.70 and Average 1.54 while Total Magnetic Heavy Mineral concentration TMag-HM wt% that was evaluated revealed Minimum 0.00 wt%, Maximum 2.76 and Average 0.3. Heavy mineral concentration increased with depth [specifically beyond 20 m water depth]. In the middle-eastern and north-eastern part, higher concentration of heavy mineral could be observed in the seabed sediments.

R.V. Samudra shaudhikama

Multi-thematic mapping of contiguous zone beyond Territorial Waters, off Dharmapuram, Tamil Nadu (SD-318) (FSP ID: IAMCS- SCWTEEZ/NC/SR/MCSDWC-2/2022/47203)

During the cruise SD-318, 757 sq km area was covered in the Territorial Waters of Dharmapuram, Tamil Nadu to bring out the bathymetry, surface and sub-surface geology of the area through systematic Geological and Geophysical studies and generation of baseline data. The water depth varied from 47.7 m to 69.5 m. A sand deposit was marked in the seismic profiles along L-6 line which is bounded by a small mound like structure in the deeper side. Beyond that, two inclined interfaces were observed that were dipping towards the sea. While towards land, indistinguishable interface supports, probable disposition of hard rock basement overlain by very thin layer of sediments were observed. Moreover, no significant traces of the sediments in the sea floor were noticed except small “sand deposit” in the southernmost end of the line. The extension of this “sand deposit” can be observed in the adjacent lines (L-5 and L-7). A total of 80 grab samples were collected (G-1 to G-80) in 5 km x 2 km interval along 7 survey lines. Based on visual estimation, sediments were found to be fine to coarse sand and silty sand at places with shells and

shell fragments. Coral pieces and opaque grains were also occasionally observed. The vibro cores were taken at 5 km x 5 km interval. The maximum core recovery was 1.25 m and minimum core recovery was 16 m. The top sediments of the cores varied from fine to medium sand and silty sand. The bottom part of VC-03, VC-06 and VC-09 was indurated with hard calcareous material and dark grey coloured carbonaceous sticky clay for VC-12 (Bottom). The core recovery is very poor in many vibrocores because of hard, indurated, calcareous material at the bottom and high swells (up to 2.5 m) due to rough sea state. Core logging and sub-sampling were completed for all the collected vibrocores and 18 samples for granulometric studies, 17 samples for geochemical analysis, 10 samples for heavy mineral studies and 22 samples for micro paleontological studies were generated.

Multi-thematic mapping of contiguous zone beyond Territorial Waters off Ulasekharappattinam, Tamil Nadu, Cruise SD-319. (FSP ID: IAMCS-SCWTEEZ/NC/SR/MCSD-WC-2/2022/47201)

The cruise SD-319 was taken up, with an objective to bring out the bathymetry, surface and sub-surface geology of the area through geology and geophysical studies and to generate the base line data. Single beam bathymetry was carried out for 178 lkm along with seismic and magnetic survey (188 lkm). The shallowest depth was 13 m and the deepest depth was 123.5 m. Magnetic and single channel shallow seismic surveys were carried out along 9 coast perpendicular and 01 coast parallel line with an interval of 5 km. The map showed that magnetic anomaly values varied from -308 nT to 244 nT with a total variation of 552 nT. The magnetic highs indicated the NE-SW oriented basement high and magnetic lows indicated the basement depression in the SW part. Two bipolar magnetic anomaly pairs were observed in the northeastern part which indicated presence of shallow intrusive bodies. Two reflectors R1 & R2 were observed along with seafloor. The material between seafloor and R1 was probably of unconsolidated sediments with an average thickness of around 2 m and the material between R1 & R2 could be sediments with an average thickness of 4 m. During geological survey, 80 grab samples at an interval of 2 x 5 km and 12 vibrocore samples at an interval of 5 km x 5 km were collected. The main sediment types observed in the survey area were fine and medium sand. The top of the core sample was characterised by fine sand whereas bottom sediments consisted of medium sediments with shells and shell fragments. Elevated Zn, Zr and Cr values were also observed. Traced pellets of green clay and Fe Oxy-Hydroxides coatings on sediments were observed during coarse fraction studies. The THM percentage in the sub-samples was less than 1.5% in all the samples. The heavies included mainly sillimanite, ilmenite, mica, zircon and garnet. The survey area contained 9–94% biogenic components which constituted benthic foraminifera, planktic foraminifera, bivalves, coiled & non-coiled pteropods, microgastropods, bryozoans, ostracods, coral

and shell fragments, etc. Benthic foraminifera included mainly *Amphistegina* Sp., *Quinqueloculina* Sp. *Operculina* Sp., etc. Major planktic foraminifera observed included *Globigerina* Sp. and *Globorotalia* Sp.

Preliminary mineral exploration for placer minerals in the contiguous zone off Kodungallur, Central Kerala (Cruise SD-320) (FSP ID: M1AMCS-MME/NC/SR/MCSDWC- 1/2023/47264)

Bathymetric and shallow seismic data were collected along 10 coast-perpendicular transects (approximately, 14 km each) and 2 coast-parallel transects (approximately, 18 km each). Water depths within the survey area ranged from 51 to 280 meters. Generally, deeper depths were observed in the north-western part of the area, where the shelf break was located at approximately 120 meters. In contrast, the southern part exhibited a wider shelf with less variation in depth, as evidenced by the sparse contour pattern on the bathymetry map. The slope angle and gradient of the seafloor along the east-west profiles varied from 0.07° to 7.6° and 1:785 to 1:7, respectively. In the northern part, a very gentle slope angle (0.11°) prevailed until reaching the shelf break, after which it steepened to 7.6°. In the middle and southern parts of the survey area, a consistently gentle slope (0.07° to 0.57°) was observed throughout. Seismic data analysis revealed a top layer consisting of sediment deposits throughout the survey area. The seismic signature above reflector R1 suggested the presence of uniformly sized sediments with low clay content. The reflectors exhibited a high degree of undulation in the central portion of the survey lines (coast-perpendicular) and resembled those of a paleo-tranline in the eastern part. A total of 80 vibrocore samples were collected at a 2 km x 2 km grid pattern. The maximum core length recovered was 3.05 meters (VC-38), while the minimum was 0.25 meters (VC-41), with an average core length of 1.50 meters. Surface sediments primarily consisted of sand and silty sand, while sub-surface sediments varied, including sand, silty sand, sand-silt-clay, hard compact clay, or hard rock. A total of 149 sub-samples were generated for heavy mineral studies, 321 sub-samples for granulometric studies, 164 sub-samples for coarse fraction studies and 229 sub-samples for chemical analysis. The granulometric studies, chemical analysis & THM separation of the samples were completed, and individual heavy mineral counting & coarse fraction studies are in progress. The THM percentage of the analysed samples varied from 0.3% to 6.56%. The highest concentration of heavy minerals was observed near the shelf break in the north-western part of the survey area.

Preliminary mineral exploration for placer minerals in the contiguous zone off Puvar, Southern Kerala. (Cruise SD-321) (FSP ID: M1AMCS-MME/NC/SR/MCSDWC- 1/2023/46963)

The survey area covered by 276 lkm bathymetric survey was along 19 coast perpendicular lines at 1 km interval and two coast parallel lines. The study area falls within

the inner shelf region off Puvar, Southern Kerala. The water depth in the study area varied between 38 m and 58 m. The bathymetric map revealed that along the North-western portion of the survey area, the seabed was gently sloping marked by closely spaced coast parallel contours in between water depth of 38 m and 47 m. The south-western portion of the survey area was represented by more widened bathymetric contours ranging between 47 m and 58 m of water depth. A total of 77 vibrocore samples were collected from the survey area at 2 km X 2 km grid interval in 134 attempts. The maximum core length recovered was 2.57 m and minimum core recovered was 0.14 m. The average core length was 0.8 m and the total core length was 63 m. The survey area was seen carpeted with very fine to medium sand with minor heavy mineral assemblages. The bottom part of the cores was marked by rock fragments, concretions and gravelly sediments which was outlined within a water depth of 38 m to 48 m. The samples collected from the deeper portions of the survey area was intersected by oxidised clayey sediments in the bottom.

Preliminary mineral exploration for placer minerals in the contiguous zone off Chavara, Kerala (SD-322) (FSP ID: IAMCS-PMIWEEZ/NC/SR/MCSD-WC-1/2023/47238)

The Cruise SD-322 was carried out with an objective for preparation of heavy mineral distribution map and demarcation of potential area of heavy mineral resource. The water depth varied from 43 m to 57 m with the isobaths trending NNW-SSE parallel to the coastline. The slope is gentle and have an approximate gradient of 1:1500 with slope angle $\theta = 1.7^\circ$ dipping in SW-NE direction. The isobaths were relatively broader between 46 m and 50 m, defining a surface with gentle dip of 5 km width probably indicating a palaeobeach. Seismic data analysis revealed three reflectors — R1, R2 and R3 indications of river channels and associated erosional features in the survey area. The missing reflectors in a particular region were indications of buried channels. R1 seem to be the boundary of transgressive–regressive phases. Thus, there were two sedimentary units, Unit-I and Unit-II and were separated by a surface of erosion/prominent marker reflector. Unit-1 was formed from Holocene sediment and Unit-2 was formed from Pleistocene sediment and the surface separating them was the erosional surface (R1) designated as LGM. A total of 77 vibrocore samples were collected at a 2 km x 2 km grid pattern with a core recovery of 3.98 m as maximum and 0.26 m as minimum. Surface sediments primarily consisted of sand and silty sand, while sub-surface sediments seemed varied and included sand, silty sand, sand-silt-clay and hard compact clay, or hard rock. A total of 122 sub-samples were generated for heavy mineral studies, 227 sub-samples for granulometric studies, 39 core samples for coarse fraction studies and 199 sub-samples for chemical analysis. The granulometric studies, chemical analysis & THM separation were in various stages of completion, and individual heavy mineral counting & coarse fraction studies were in progress.

Parametric (shallow seismic) survey in the gap area off Dwaraka sector, Arabian Sea (SD-323) (FSP ID: M1AMCS-APS/NC/SR/MCSD-WC-1/2022/47079)

An area of 3,650 sqkm was covered with nineteen coast perpendicular lines and two coast parallel lines with a total of 1,063 km in the continental shelf off Dwaraka, Gujarat. The present survey area was within the Saurashtra basin, which is one of the five Pre-cratonic basins bounded by Saurashtra Arch in the north and Narmada-Son Lineament in the south. The Miliolitic limestone of Early to Late Pleistocene to Holocene was commonly found to occur in the Saurashtra coast. Chaya formation of Late Pleistocene to Holocene was observed to rest unconformably over the Miliolite Formation or the Dwarka Formation which was extend towards the sea and was seen overlain by recent clay/mud (reworked Miliolite limestone), separated by an Erosional Surface (ES) representing the LGM (Last Glacial Maxima) surface. The seafloor appeared very smooth and gently dipping towards SW direction with occasional presence of sharp pinnacle indicating the presence of Holocene reef structure. The seismic sections in the northwestern part showed two highly irregular unconformity surfaces represented by highly irregular ES. The lower unconformity surface was not present in seismograms that indicated higher depth of this layer in the southeastern part of the survey area. Many shore facing reflectors were observed between these two unconformity surfaces indicating the marine transgression/regression. These unconformities are understood to be related to lowest glacial phases (peak of Dry Age) with upper (shallower) one as LGM (20K years) and lower one with around 150K years. The upper sedimentary layer above the shallower ES of Holocene Age was clayey in nature with presence of limemud. The layer between these two ES seemed to have indurated limemud of late Pleistocene Age. This layer was marked by many shore facing inclined effectors. These inclined reflectors also showed some evidences of neotectonic activities like folding and slumping structures.

Parametric (shallow seismic) survey in the gap area off Jakhau–Okha sector, Arabian Sea (SD-324) (FSP ID: M1AMCS-APS/NC/SR/MCSD-WC-1/2022/47115)

Shallow seismic along with bathymetry surveys were carried out in the gap area off Jakhau–Okha sector, West Coast, Arabian Sea, covering an area of 4,200 Sqkm. A total of 1,006 line km of shallow seismic and bathymetry surveys were carried out during this cruise. Shallow seismic data along with bathymetry were collected along 25 NE-SW transects with line interval of 5 km and 3 cross lines along NW-SE direction. The depth of the seabed varied from 5.6 m to 67 m in the study area. The seabed was found smooth and gently dipping towards the western end of each coast perpendicular transect from L–1 to L–18 and no drastic variation was observed over these transects. The seafloor was more undulated and uneven over the Transects L–19 to L–25 as it cuts across the different morphological features which were heavily affected by corals. There were three

general trends of the bathymetric contours — (a) NW-SE – presenting the shelf parallel to coast line— the contours are almost aligned to present day coast line (L1 to L-18); (b) NE-SW – presenting elongated depression/ channel in the south western part of the area and it further extends toward the gulf. The seafloor was observed to be more undulated and uneven over the Transect L-19 to L-25 as it cuts across the different morphological features that seemed to be heavily affected by corals. Coral pinnacles were observed at several places. Deep channels, abrupt truncations, ridges, etc. could be attributed to the tectonic disturbances and the postglacial strandlines formed during the standstill positions of recent Holocene transgression. The Transect L-18 seemed as a pivotal line that separates tectonically disturbed Domain “B” from relatively stable Domain “A”. It seems to form a transition boundary that limited the continuation of some geomorphological features between the above mentioned domains. In the NW part of Transect L-20, a vertical, chaotic disordered vertical disturbances were observed on seismic data, where the reflectors were discontinuous and reflection amplitudes seemed weaker (blanking).

BOAT SURVEY

Preliminary search for placer minerals in the nearshore area off Manamelkudi, Pudukottai District, Tamil Nadu (ITEM-150) (FSP ID: IAMCS-PMIWEEZ//SR/SUTNP/2023/47068)

An area of 125 sqkm was investigated and systematic sampling was carried out all along and across the spit axis in 1 km X 1 km grid pattern with grab sampler engaged to study the spatial distribution and relative abundance of major placer minerals. The bathymetry survey had been carried out along 6 coast parallel lines [L-1 to L-6] and 6 Cross Lines. A total of 167 LKM had been completed. The water depth varied from 1.46 m to 11.23 m. The bathymetry survey revealed a smooth topography and the depth appeared

to increase gently towards the western part of the area. The eastern part was shallow in nature with water depths ranging from 1.46 m to 2 m. The depths slowly increased towards deeper side and maximum depth was noted at sample location G-29 (11.23 m). A narrow sand ridge was observed along L-3 (G-1 to G-9), wherein the water depth was very shallow and survey had been carried out with the help of fiber boat. The sudden increase in water depth, after the sandy ridge indicated a probable neotectonic activity. A total of 139 grab sediment samples were collected during the survey which mainly comprised medium-coarse sand, fine-medium sand, clayey silt and little clay-rich zones. The terrigenous content comprised mainly of quartz grains, rock fragments, heavy minerals and carbonate materials. Heavy minerals observed were mostly ilmenite (opaques), garnet, rutile, sillimanite, zircon, yroxeenes/amphiboles, mica, etc. Nature of sand varied from very fine to fine at the eastern margin to medium sand towards the deeper eastern part of the area. Presence of significant sand domain in the survey area demarcated a good amount of construction grade sand resource in addition to the heavy mineral concentration. The fine to medium sandy domain showed more presence of heavies than the coarser fragments. To understand the physico-chemical properties of sea water and to reassess the prevalence and intensity of the neotectonic activity around, 6 offshore and 6 onshore samples were collected at 12 stations. Water samples from the borewells drilled around the survey area in the hinterland for ground water experienced artesian flowing condition. A total of 06 borewell water samples were collected in the vicinity of Manamelkudi. Bathymetric profile study showed the area as flat with a gentle slope towards east. Prominent ridge like feature was observed in the eastern part of the study area. The THM percentage varied from 0.01% to 4.59 % (G-2/07) with an average of 0.77%. Preliminary study indicated that the heavy minerals were mainly found concentrated in the medium sand.



Metal & Alloys

11. Aluminium and Alumina

4,159 thousand tonnes
production of aluminium was reported
in 2023-24.

7,410 thousand tonnes
production of alumina was reported
in 2023-24.

41.45 lakh, tonnes
installed capacity of aluminium plants
in the country in 2023-24.

Aluminium Industry in India is strategically well-placed and is one of the largest producers in the world, with discernible growth plans and prospects for the future. India's rich bauxite mineral base gives it a competitive edge over the industry in other countries. The Indian Aluminium Industry scaled significant heights since the establishment of the first manufacturing company, namely, Indian Aluminium Company (INDAL) in 1938. In 2004, all business activities of INDAL were merged with HINDALCO Industries Limited (HINDALCO). Four major primary producers, National Aluminium Co. Ltd, HINDALCO Industries Ltd, Bharat Aluminium Co. Ltd and Vedanta Aluminium Ltd (VAL) are at the forefront of aluminium production. The primary producers have a strong presence in the sheet business and are expanding their roles in the foil segment. The primary producers are also in the extrusion segment in which a large number of secondary manufacturers participate with fragmented capacities. The overall total annual installed capacity of aluminium plants in the country decreased to 41.45 lakh tonnes in 2023-24 from 41.65 lakh tonnes in previous year. The production of aluminium comes from the plants viz, NALCO, HINDALCO & Vedanta. Producer-wise capacity of aluminium is furnished in (Table-1). The installed capacity of alumina plants in the country was 74.75 lakh tpy. Producerwise capacity of alumina is furnished in (Table-2).

Table – 1: Installed Capacity of Aluminium
(By Producers)

(In '000 tonnes)

Producer	Plant	Annual capacity
Total		4145
Public Sector		
National Aluminium Co. Ltd	Angul (Odisha)	460
Private Sector		
Bharat Aluminium Co. Ltd	Korba (CG) - I Korba (CG)- II	590

Table- 1 (Concl'd.)

(In '000 tonnes)

Producer	Plant	Annual capacity
Hindalco Industries Ltd	Aditya (Odisha)- 360 Hirakud (Odisha)- 216 Mahan (M.P) - 359 Renukoot(U.P) - 410	1345
Vedanta Aluminium Ltd	Jharsuguda-I (Odisha) - 500 Jharsuguda-II (Odisha) - 1250	1750

Source: Information received from the companies/Annual Reports.

Table – 2: Installed Capacity of Alumina
(By Producers)

(In '000 tonnes)

Producer	Plant	Annual capacity
Total		7475
Public Sector		
National Aluminium Co. Ltd	Damanjodi (Odisha)	2275
Private Sector		
Bharat Aluminium Co. Ltd	Korba (Chhattisgarh)	200
Hindalco Industries Ltd	Renukoot (Uttar Pradesh)	700
	Belagavi (Karnataka)	350
	Muri (Jharkhand)	450
	Utkal Alumina (Odisha)	1500
Vedanta Aluminium Co. Ltd	Lanjigarh (Odisha)	2000

Source: Information received from the companies/Annual Reports.

PRODUCTION

Aluminium

The production of aluminium at 4,159 thousand tonnes in 2023-24 registered an increase of 2.27% as compared to the previous year. Seven plants reported production of aluminium during the year. (Tables - 3 & 4).

Alumina

The production of alumina at 7,410 thousand tonnes in 2023-24 increased by 0.85% as compared to that in the previous year. NALCO continued to be the leading producer of Alumina accounting for 41% of the total production during the the year under review. (Tables-5 & 6).

Table – 3: Production of Aluminium, 2021-22 to 2023-24

(Quantity in tonnes; Value in ₹ '000)

Year	Production	
	Quantity	Value
2021-22	4016621	714320396
2022-23	4066459	809764986
2023-24 ^(P)	4158844	755025581

(p): Provisional

Table – 4: Production of Aluminum, 2022-23 and 2023-24

(By Plants)

(In tonnes)

Producer	Plant	Production	
		2022-23	2023-24 (P)
National Aluminium Co. Ltd	Angul	459564	463427
Hindalco Industries Ltd	Aditya	368058	370855
	Hirakud	175693	177749
	Mahan	368781	372162
	Renukoot	409047	410661
Bharat Aluminium Co. Ltd	Korba	568549	582987
Vedanta Ltd	Jharsuguda	1716767	1781003

(p) : Provisional

Table – 5: Production of Alumina 2021- 22 to 2023-24

(Quantity in tonnes; Value in ₹ 000)

Year	Production	
	Quantity	Value
2021-22	7229508	159576853
2022-23	7264449	191159578
2023-24 ^(P)	7410072	176891452

(p): Provisional

Table – 6: Production of Alumina, 2022-23 and 2023-24

(By Plants)

(In tonnes)

ZProducer	Plant	Production	
		2022-23	2023-24 (P)
National Aluminium Co. Ltd	Damanjodi	2024000	2045900
Hindalco Industries Ltd	Belgaum	262400	272900
	Muri	272523	233210
	Renukoot	576782	561459
	Utkal Alumina	2336000	2439500
Vedanta Ltd.	Lanjigarh	1792744	1857103

(p): Provisional

INDUSTRY

Aluminium is the fastest growing non-ferrous metal in India and the same is evident by its growing and widespread use. Most of the Aluminium Smelters are located near their respective alumina refineries. Ten aluminium smelters are operated by four companies viz. NALCO, BALCO, HINDALCO & Vedanta. NALCO is the only company in the Public Sector with an installed capacity of 4,60,000 tpy. BALCO, earlier a Public Sector company, is now in the Private Sector. The aluminium plants of NALCO and BALCO have their alumina-aluminium complexes at Damanjodi-Angul (Odisha) and Korba (Chhattisgarh), respectively.

NALCO is one of the largest integrated Bauxite-Alumina- Aluminium- Power Complexes in the country. The Company has 4,60,000 tpy Aluminium Smelter at Angul and 22,75,000 tpy Alumina Refinery located at Damanjodi in Koraput, Odisha.

HINDALCO Aluminium smelting operations are located at Renukoot (Uttar Pradesh), Aditya Aluminium (Odisha), Mahan Aluminium (Madhya Pradesh) and Hirakud (Odisha). Newly installed smelters at Aditya Aluminium and Mahan Aluminium have state-of-the-art AP36 technology. The HINDALCO's total primary aluminium (metal) capacity has enhanced to around 13,45,000 tpy from earlier capacity of 5,62, 000 tpy. This increase was primarily on account of production from Mahan and Aditya smelter commissioned during the year 2014-15. In addition to aluminium, Renukoot Integrated Aluminium Complex also produces semi-fabricated products viz. conductor redraw rods, sheet, extrusion, etc. The Alupuram (Kerala) smelter is closed but extrusion unit currently operates at a capacity of 8,000 tpy.

HINDALCO's plants are equipped with sophisticated rolling mills and finishing equipment. The plants are located at Hirakud (Odisha), Belur (West Bengal), Mouda (Maharashtra), Renukoot (Uttar Pradesh) & Taloja (Maharashtra). HINDALCO's finished products include, alumina, primary aluminium in the form of ingots, billets & wire rods, value-added products, such as, rolled products, extrusion and foils. HINDALCO is the largest manufacturer of entire range of Flat Rolled Products. Flat Rolled Products facilities at Hirakud (Odisha) and Mouda (Maharashtra) are being modernised to produce world class Can Body stock and Ultra Thin Gauge Foils, respectively. The Hirakud plant produces Flat Rolled Products (FRP), rolled products, extrusions products and wire rods. HINDALCO has a conductor redraw capacity of 56,400 tpy at Renukoot plant and sheet rolling capacity of 2,05,000 tpy spread over Renukoot (80,000 tpy), Belur (45,000 tpy), Taloja (50,000 tpy) and Mouda (30,000 tpy) plants. The Company also has two plants for aluminium extrusion with a combined capacity of 31,000 tpy, comprising units at Renukoot (23,000 tpy) and Alupuram (Kerala) (8,000 tpy).

HINDALCO's foil unit located at Silvasa (Dadra & Nagar Haveli) has an installed capacity of 30,000 tpy and produces foils with thickness varying from 9 microns to 200 microns. Kollur plant in Medak district (Andhra Pradesh) has a capacity of 4,000 tpy and produces an array of high quality foils, from cigarette and blister foil to lidding foil in thicknesses from 50 to 7 microns.

The overall smelter capacity of BALCO increased to 5,90,000 tpy with the commissioning of Korba-II smelter, which has capabilities to produce ingots, wire-rods billets, bushbars and rolled products. The Korba-I plant showed

consistent production during the year. The Company has two power plants for commercial power generation, of which one produces power for captive consumption, refining, smelting, fabrication, etc. and the other 1,200 MW power plant is under construction. The ramp up of BALCO-II smelter was completed in 2018. With the ramping up of the Jharsuguda-II smelter with capacity of 1.25 million tpy, the total smelter capacity of Vedanta in Odisha, has increased to 1.75 million from 50,000 tpy. MALCO is under care and maintenance by Vedanta since May, 2017. It generates 100 MW power from 4 units of 25 MW each through power plants located at Mettur (Tamil Nadu), and is one of the largest Private Sector power suppliers in Tamil Nadu. Jindal Aluminium Ltd (JAL) established its factory for manufacture of aluminium extrusion in 1968 at Bengaluru. JAL has 11 aluminium extrusion presses with an installed capacity of 1,20,000 tpy. The Company is the leader in aluminium extrusions, meeting about 30% of the country's demand.

JAL has taken a step forward by diversifying into Aluminium rolled products. Jindal commissioned the state-of-the-art Aluminium sheet and foil manufacturing facility with an installed capacity of 40,000 tpy, at Dabaspur, Bengaluru, 35 km from the existing extrusion plant.

DEVELOPMENT & EXPANSION

National Aluminium Company Limited (NALCO) : NALCO is in the process of setting up the 5th Stream expansion of its existing Alumina Refinery which would reportedly add 10.0 lakh tonnes to its existing installed capacity of 22.75 lakh tonnes per year (total capacity 32.75 lakh tonnes). This expansion would be based on improved Medium Pressure Digestion Technology of M/s Rio Tinto Alcan International Limited (RTAIL).

Sourcing of bauxite for 5th Stream expansion of alumina refinery has been envisaged from Pottangi Bauxite Mines. The availability of bauxite from Pottangi Mines is expected beyond the scheduled commissioning of 5th Stream expansion. Hence, sourcing of bauxite from South Block of Panchpatmali Mines has been planned through setting up of a crushing & conveying system.

The Pottangi Bauxite mine (75 million tonnes) has been reserved by the Government of India in favour of NALCO for meeting the bauxite requirement of 1 million tonnes alumina refinery under expansion. The mining plan has already been approved. The pre-project activities are underway. The mine is likely to be operational in the 1st quarter of financial year 2026-27.

Mining Lease of Utkal-D coal block has been granted by the Govt of Odisha over an area of 301.28 Ha and the lease was executed on 25.03.2021. thereafter Utkal coal block will be operational after obtaining of statutory clearance.

NALCO, as a JV partner with Mishra Dhatu Nigam Ltd (MIDHANI), formed Utkarsha Aluminium Dhatu

Nigam Ltd (UADNL) in August 2019 for establishment of a 60,000 TPA High End Aluminium Alloy Plant in JV mode for Defence, Aerospace & Automobile Sectors. The plant is expected to be commissioned by financial year 2024-25. The Company has formed JV with Neelachal Ispat Nigam Ltd (NINL) to set up a Coal Tar Distillation plant of 20,000 TPA based on the Coal Tar generated in NINL's Coke Oven Plant.

NALCO and Odisha Industrial Infrastructure Development Corporation (IDCO) have formed a JV Company for establishment of Angul Aluminium Park Pvt. Ltd (AAPPL) for promotion of downstream industries in the State of Odisha.

HINDALCO : In the field of aluminium, HINDALCO operates in both primary aluminium and downstream aluminium segments and is one of the world's largest integrated aluminium producers. HINDALCO's subsidiary Novalis with 61% recycling capability, is the largest aluminium recycler and producer of rolled aluminium products in the world. HINDALCO launched India's first indigenous lightweight and eco-friendly aluminium bulker in India. This bulker saves up to 13,000 litres of fuel, generates 20 tonnes less GHGs and is BS-VI compliant.

HINDALCO is planning to expand its Aluminium downstream business with a focus on Value added products over the next 3-7 years. The Products would cater to customised requirement for varied and complex applications of Aluminium. HINDALCO plans to invest around ₹ 8,000-10,000 crore in expanding flat rolling capacity at Hirakud, new extrusion plant at Silvassa and in a greenfield site at Mundra with a recycling facility. The Hirakud plant capacity for flat rolled products is estimated to be 340 KTPA. The planned capacity of the extrusion plant at Silvassa is 34 KTPA, which would have three extrusion presses to service premium customers in building and construction, automobile and transport, electrical, consumer and industrial good sectors. In addition, the new extrusion and recycling unit at Mundra is awaiting land acquisition and would have a capacity of 93 KTPA.

VEDANTA: Vedanta Aluminium Ltd (VAL) is the largest aluminium producer in India with a capacity of 1.75 million tonnes per annum. The aluminium smelting unit at Jharsuguda has established itself as the world's largest single-location smelter with an already installed capacity of 1.75 mtpa, backed by two smelters – 0.5 million tonnes per annum and 1.25 million tonnes per annum (SEZ) and two power plants with a combined capacity of 3,615 MW, which is over 1.3 million tonnes per annum. This increase has been achieved by scaling up the production capability of the smelter and the refinery capacity has been significantly enhanced in the last few years. With the ramp up of both smelters, production of aluminium has increased during the last couple of years. Similarly, production of Alumina has also increased due to debottlenecking of the refinery

operations. The bauxite requirements for alumina refinery are met from captive mines, domestic sources and imports.

BALCO, a subsidiary of Vedanta, operates through its plant at Korba in Chhattisgarh with a smelter capacity of 0.57 million tonnes per annum and power generation capacity of 2,010 MW. The state-of-the-art alumina refinery at Lanjigarh feeds the aluminium smelters at Jharsuguda and BALCO and forms a crucial link in the value chain. It is one of the world's largest one-site integrated alumina refining complexes with a current capacity of 2 million tonnes per annum that can be ramped up to 5 million tonnes per annum.

USES & CONSUMPTION

Aluminum is the second most used metal in the world after steel. Aluminium is more environment-friendly than steel, plastic and other materials. The metal that made flying possible is sustainable, light and energy efficient. Aluminium has widespread uses throughout the economy and is equally important to both the industrial and consumer sectors. Aluminium is used in the Aerospace Industry as well as other industries requiring light weight metal. On the industrial side, aluminium is heavily used in electrical power transmission, machinery & equipment and construction. Aluminium usage in automobiles is rising and is expected to increase globally. India's auto sector consumes about 4% aluminium. Over the past five years considerable progress has been made in aluminium intensive vehicle production.

Housing, in particular, makes heavy use of the lightweight material as a substitute for steel and wood in doors, windows and sidings. Aluminium is also used in a variety of retail products including cans, packaging, air conditioners, furniture and automobiles. In addition, India has pioneered the replacement of copper by aluminium in power transmission & distribution which has enhanced the demand for aluminium. There are nearly 600 cable and conductor manufacturing units in the country, having a total capacity of about 4,00,000 tpy. The major end-use of aluminium is as rolled sheets, extrusions and foils. India Foils, Pennar Aluminium and Century Extrusions are the major players in the extrusion & foil market. Key sectors to drive aluminium consumption in India are Auto, Power, Electronics, Railways, Aerospace & Defence Construction, Solar Energy and Aluminium packaging.

China was the largest producer as well as consumer during the year, contributing about 57% of the world production (36.15 million tonnes) and 55% of the world consumption (35.71 million tonnes) of aluminium. The world excluding China is expected to see aluminium consumption growth of around 1% in the year 2019 from around 2% in the year 2018 due to likely moderation in demand from North America and Europe.

In India, the power, packaging, transport, construction,

machinery and equipment sectors are key demand drivers of aluminium. The domestic imports of aluminium products, including scrap, are growing significantly, which is a major concern for the domestic aluminium producers. In advanced economies, aluminium is increasingly replacing wood and steel in Building Sector. Aluminium cans and containers are used extensively, world over. Aluminium is also the ideal packaging material for pharmaceuticals and processed foods.

In India, aluminium was consumed mainly in the Electrical sector (48%), followed by Automobile & Transport sector (15%), Construction (13%), Consumer Durables (7%), Machinery & Equipment (7%), Packaging (4%) and others (6%). In the Electrical sector, aluminium is used in overhead conductor, and power cable for generation, transmission and distribution of electricity. Aluminium is also used in switchboards, coil windings, capacitors, etc.

As per Technology Vision Document 2035, the per capita consumption of aluminium in India is among the lowest in the world with only 2.2 kg as compared to the world average of roughly 8 kg and with that of the developed nations which is 22-25 kg.

Alumina is produced from bauxite. About one tonne of alumina is produced from 3 to 3.5 tonnes of bauxite and about one tonne of aluminium is produced from about two tonnes of alumina.

RECYCLING

Aluminium is 100% recyclable with no loss of quality on properties. Compared to primary aluminium production, recycled aluminium consumes 95% less energy and emits 95% fewer greenhouse gases. Products of aluminium, such as, UBC (Used Beverages Can), aluminium foils, plates and automotive components can be easily recycled, thereby, saving energy and reducing greenhouse emissions. Aluminium recycling process is less capital intensive than primary metal production as the process requires only 5% of energy, i.e., 13–15 thousand units of power for producing one tonne of aluminium through primary route. Recycling of aluminium saves about 6 kg of bauxite/kg and 14 kWh of electrical energy /kg of primary aluminium. Besides, it keeps the emission levels of greenhouse gases as low as 5% from the actual emission experienced during primary production. Further, recycling facilitates reduced stress on the use of bauxite thereby preserving about six lakh tonnes of bauxite resources every year.

India's metal recycling rate is about 25%. Altogether the rise in aluminium production from old scrap has grown from one million tonnes in 1980 to 20 million tonnes in 2019. All the activities related to aluminium scrap recovery

are limited to the Unorganised sectors, catering mostly to the utensil and casting industries. The proportion of recycled aluminium has been increasing over the years. It is expected that in the years to come, it will reach a figure of about 35–40% of total aluminium consumption. Currently, there is only one recycling unit of HINDALCO in the Organised Sector at Talaja with 25,000 tonnes annual capacity. Although the plant at Talaja was facing challenges due to limited availability of scrap, the production from the unit has improved and the plant is now operating at 80% of the rated capacity as against earlier capacity of 60%. HINDALCO is planning to set up greenfield recycling unit at Mundra with a capacity of 93 KTPA.

Most recycling units in India operate on outdated, or primitive technology which leads to high levels of

pollution and energy consumption. This is an area that needs to be addressed by the Indian Aluminium Industry. Due recognition of recycling could encourage users of aluminium particularly in transport, housing, packaging and durable sectors to broaden the organised markets for the scrap generated.

WORLD REVIEW

World production of alumina was 138 million tonnes in 2023. China continued to be the leading producer with a share of about 60% which is followed by Australia (14%), Brazil (8%), India (5%) and Russia & UAE (2% each). World production of aluminum was at 69.90 million tonnes in 2023. China continued to be the leading producer with a share of about 59.51% which is followed by Russia (6%), Canada & India (5% each). (Tables-7 & 8).

Table – 7: World Production of Alumina
(By Principal Countries)

Country	(Tonnes of Al_2O_3 content)		
	2021	2022	2023
World: Total (rounded off)	140400000	140200000	137900000
China	77480000	81862000	82441000
Australia	20624088	19576174	18802465
Brazil	10992300	10871500	10443400
India ^(a)	7325000	7441000	6763000
Russia	3054000	3080000	3022000
UAE	2300000	2430000	2480000
Saudi Arabia	1879000	1745000	1825000
Canada	1471000	1478000	1501000
Vietnam	1456068	1428503	1491300
Ireland	1878000	1629000	1380333
Other countries	13818544	10287823	9130835

Source: BGS, World Mineral Production, 2019-2023.

(a) Years ended 31st March following that stated.

Note (s):

(1) Wherever possible figures in this table show the alumina equivalent (Al_2O_3) of total hydrate produced, whether or not calcined.

(2) The figures in this table represent alumina produced from bauxite. Alumina-rich materials can also be derived from salt slags formed during the production of secondary aluminium.

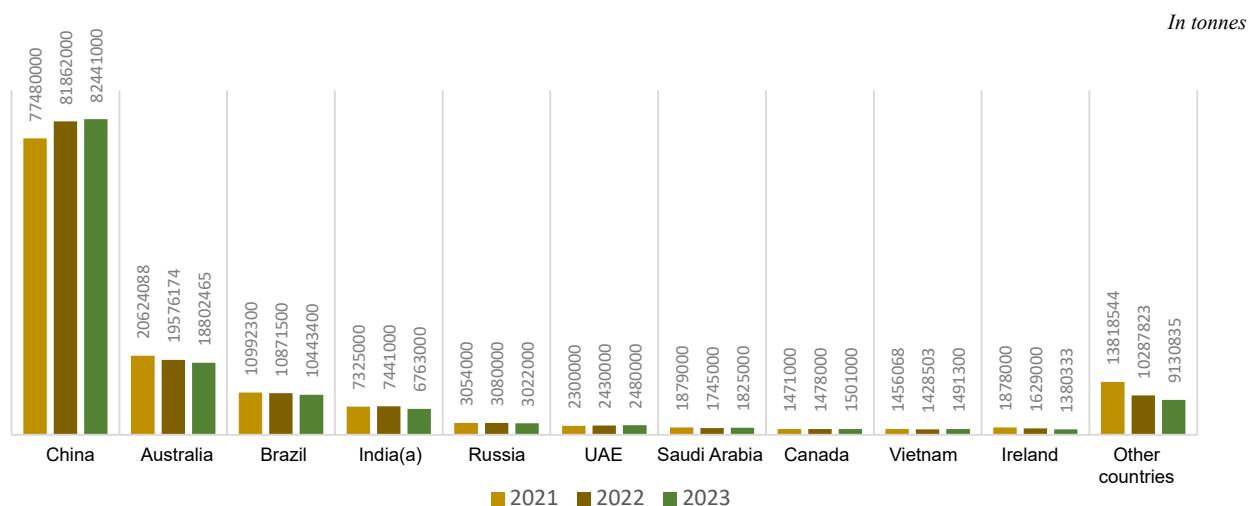


Fig 1: Countrywise Production of Alumina

Table – 8: World Production of Aluminium (Primary)

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World: Total (rounded off)	67800000	68300000	69900000
China	38502600	40214000	41594000
Russia	3937000	4014000	4026000
India ^{*(c)}	4016621	2322270	3305910
Canada	3157762	3026236	3301844
UAE	2540000	2720000	2750000
Bahrain	1561222	1600111	1620665
Australia	1558529	1509646	1555474
Norway	1419000	1424600	1321200
Brazil	771700	1629900	1021700
USA	907846	872300	735000
Other countries	9427720	8966937	8668207

Source: BGS, World Mineral Production, 2019-2023

(c) Years ended 31st March following that stated.

*estimated

Note (s) :

(1) The figures in this table are for primary aluminium only. Aluminium can also be produced from secondary sources and may be recovered from salt slags.

In tonnes

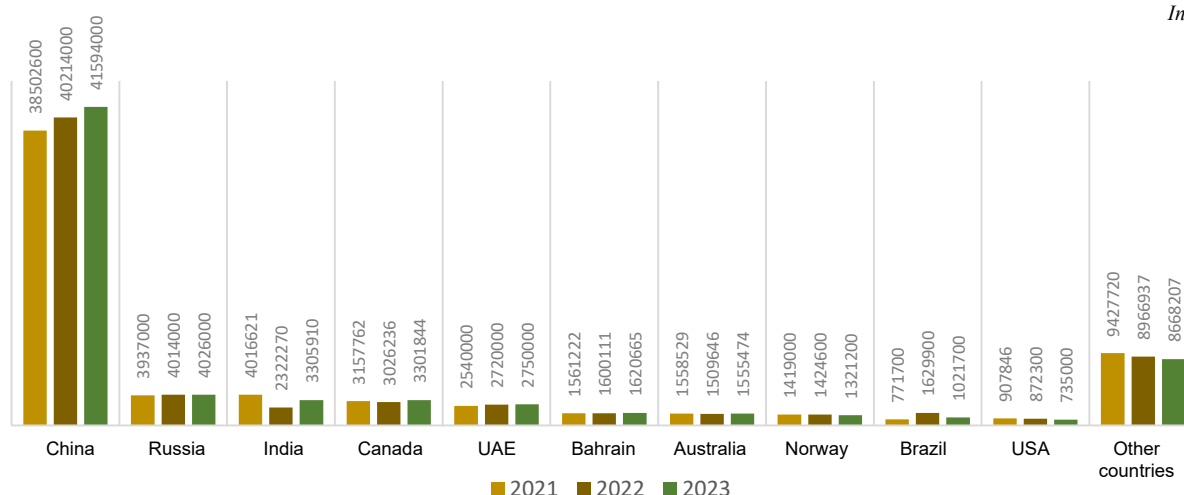


Fig. 2: Countrywise Production of Aluminium

FOREIGN TRADE

Exports

Export of alumina increased by 4% to 2,088 thousand tonnes in 2023-24 from 2,014 thousand tonnes in the previous year. Exports were mainly to Oman (31%), Russia (22%), UK (18%) and UAE (10%).

Export of aluminium and alloy Incl.Scrap decreased by 4% to 2,761 thousand tonnes in 2023-24 from 2,870 thousand tonnes in the previous year. Exports were mainly to Malaysia (17%), Republic of Korea (14%), USA (8%), & Japan(6%) (Tables-9 to 11).

Table – 9: Exports of Alumina

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2014428	64237704	2088162	64431546
Oman	613138	18789515	651777	20231872
Russia	289499	8261955	467168	13626855
UK	246221	7348356	368490	10698107
UAE	687641	22093403	217404	6281318
Indonesia	1984	101649	96030	2944651

Table- 9 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
China	32978	1001625	91630	2703380
Qatar	73	11077	61344	1822677
Egypt	5	4022	61896	1777790
Italy	13313	843262	12574	734905
Taiwan	18145	1037353	12248	693666
Other Countries	111431	4745487	47601	2916325

Figures rounded off

Table – 10: Exports of Aluminium and Alloys Incl. Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2870455	709309716	2761484	633466261
Malaysia	291396	63526272	462028	90744330
USA	244144	88115029	233566	78307884
Korea, Rep. of	345808	74170229	382070	75558056
Japan	114462	23347527	176944	34265723
China	89434	17825342	145262	28206639
Mexico	231688	52697568	118528	24979669
Bangladesh	55726	20439941	74816	23561219
Netherlands	306686	68245571	104028	22471455
Greece	56992	12288023	99410	20188067
Vietnam	103833	22824215	97029	19549797
Other Countries	1030286	265829999	867803	215633422

Figures rounded off

Table – 11: Exports of Aluminium

(By Items)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All items	2870455	709309716	2761484	633466261
Aluminium & Alloys : Unwrought	2314182	504636153	2099613	414746023
Aluminium Alloys Unwrought	226365	56610660	309333	64090011
Aluminium Ingots	2067465	443904894	1787752	350167982
Aluminium Unwrought Nes	20352	4120599	2528	488030
Aluminium & Alloys : Worked	542660	201592647	647031	215656662
Aluminium And Alloys: worked (bars, rods, plates etc)	293915	90407711	374740	98684457
Aluminium And Alloys, worked, nes	248745	111184936	272291	116972205
Aluminium Powders and Flakes	4004	1362281	4188	1352349
Aluminium Waste and Scrap	9609	1718635	10652	1711227

Figures rounded off

Imports

Import of alumina marginally increased by 0.2% to 2,411 thousand tonnes in 2023-24 from 2,407 thousand tonnes in the previous year. Imports were mainly from Indonesia (47%), Australia (27%), Vietnam (16%), Saudi Arabia & China (4% each).

Imports of aluminium & alloys including scrap also increased by 8% to 2,737 thousand tonnes in 2023-24 from 2,542 thousand tonnes in the previous year. The imports were mainly from USA (17%), China (15%), UAE (8%), UK (6%), and Saudi Arabia (5%) (Tables- 12 to 14).

Table – 12: Imports of Alumina

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2407867	89152924	2411907	83475392
Indonesia	933578	32214885	1134509	35623162
Australia	662736	22115636	659153	20759209
Vietnam	674076	23338494	396825	12774322
China	66267	5737210	91952	6538741
Saudi Arabia	31500	1003743	93728	3024113
USA	2708	729111	2589	859770
Germany	5372	1109670	2776	844189
Netherlands	9744	918560	6408	700335
Japan	681	428159	1703	657472
Canada	4056	422020	3158	437379
Other Countries	17149	1135436	19106	1256700

Figures rounded off

Table – 13: Imports of Aluminium Alloys Incl. Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2542127	562908946	2737109	568263506
China	336026	112046665	408611	122020465
USA	488236	84316262	469984	79847596
UAE	195549	40787914	222778	43871920
Malaysia	77480	19324115	130711	28692143
UK	203966	33052532	161421	27033187
Saudi Arabia	161537	32560837	142853	26721621
Thailand	80191	28026278	72414	20622135
Netherlands	73869	12779643	106513	17737604
Australia	61294	10912121	84553	15396063
Singapore	79100	14940479	74653	13535110
Other Countries	784879	174162100	862618	172785662

Figures rounded off

Table – 14: Imports of Aluminium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All items	2542127	562908946	2737109	568263506
Aluminium & Alloys : Unwrought	274056	63633827	362980	76283696
Aluminium Alloys Unwrought	62285	15331313	63988	13965339
Aluminium Ingots	210095	47912663	282396	59081050
Aluminium Unwrought Nes	1676	389851	16596	3237307
Aluminium & Alloys : Worked	531170	201270989	604502	201592908
Aluminium and Alloys :worked (bars, rods, plates etc)	446845	154637499	461688	142443830
Aluminium and Alloys, worked, nes	84325	46633490	142814	59149078
Aluminium Powders and Flakes	2385	831908	2581	847966
Aluminium Waste and Scrap	1734516	297172222	1767046	289538936

Figures rounded off

FUTURE OUTLOOK

Aluminum is one of the most abundant metals found in the Earth's crust accounting for nearly 8% of its weight. The wide availability and numerous properties of aluminum make it a widely used metal across the world. Aluminum is used in various applications such as packaging, household products, electronics, and transportation.

It possesses several properties such as high conductivity, ease of recycling, and corrosion resistance; however, aluminum carries drawbacks such as moderate tensile strength and moderate machineability. This limits its direct usage in various end-user industries.

Aluminum is an ideal substitute for materials such as steel and iron, primarily due to its lower self-weight and high strength. Prices of aluminum fluctuate primarily due to its application in several end-user industries.

Over the past few years, the metal has been used in combination with diverse alloying elements, including silicon, copper, magnesium, and zinc.

This combination helps in advancing the properties of aluminum, including high tensile strength, high fatigue strength, and high-temperature sustainability.

Alumina and Aluminium Market is expected to witness significant growth in the coming years, primarily driven by the growing demand for Refractory, Metallurgy and Other. Based on the type, the market can be segmented into Metallurgical Grade, Refractory Grade, Grinding Grade, Cement Grade, Other. As per Research and Markets report, the Aluminum Furniture Market is projected to reach USD 101.03 billion by 2030 from USD 63.13 billion in 2022, at a CAGR of 6.05% during the forecast period.



12. Antimony

18,683 tonnes

Total reserves/resources of antimony have been established as on 1st April, 2020.

4 tonnes

antimony ores & concentrates were exported in 2023-24.

4,069 tonnes

antimony ores & concentrates were imported in 2023-24.

Antimony is a strategic metal. The predominant ore of antimony is stibnite composed of antimony trisulphide, Sb_2S_3 (Sb-71.4%). The other important ores of antimony are Jamesonite ($\text{Pb}_4\text{FeSb}_6\text{S}_{14}$) and Senarmonite/Valentinite (Sb_2O_3). Antimony in its elemental form is a silvery white, brittle, fusible, crystalline solid that exhibits poor electrical and heat conductivity properties. Commercial forms of antimony are generally traded in the form of ingots, broken pieces, and granules or cast cake. Other forms are powder, shots and single crystals. Occurrence of antimony in the earth crust ranges from 0.2 to 0.5 parts per million. Antimony is geochemically categorised as a chalcophile, occurring with sulphur and associated with heavy metals, such as, lead, copper and silver.

RESERVES AND RESOURCES

As per the NMI database based on UNFC system, as on 01.04.2020, the total reserves/resources of antimony ore has been estimated at 18,683 tonnes & the total reserves/resources of antimony metal has been placed at 255 tonnes. A total of 18,683 tonnes of antimony ore is distributed

among Himachal Pradesh (57%) and Madhya Pradesh (43%) (Table-1).

The Stibnite and its decomposition products, Cervantite and Kermesite occur as veins, stringers and specks. Occurrences of antimony ores are also reported from the States of Andhra Pradesh, Jharkhand, Karnataka, Uttarakhand and Union Territory of Jammu & Kashmir.

Table –1: Reserves/Resources of Antimony as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
Ore	7503	11180	18683
Metal	75	179.92	254.92
By States			
Himachal Pradesh			
Ore	0	10588	10588
Metal	0	174	174
Madhya Pradesh			
Ore	7503	592	8095
Metal	75	5.92	80.92

Figures rounded off.

USES

Antimony and its alloys find numerous applications in a wide range of high technology industries like electronic, space, defence, photographic materials, electroplating, besides cosmetic, paint, plastics and textile industries. Traditionally, it is used in type metal in Printing Industry and other alloys. It is now used extensively worldwide to harden and increase the mechanical strength of lead, particularly in Battery Industry. Antimony trioxide is the most important of the antimony compounds and is primarily used in flame-retardant applications, including such markets, as children's clothing, toys as well as in manufacturing aircraft and automobile seat covers. Antimony sulphide is one of the ingredients used in safety matches. It is used in solar panels to improve stability of the solar performance of the glass upon exposure to UV radiations or sunlight and also as a decolorizing and refining agent in Glass Industry. Antimony compounds also find use in pharmaceutical applications. It is also used in semiconductors for making infrared detectors, diodes & acoustic devices and in plastic production as heat stabiliser in PVC.

INDUSTRY

The metal is obtained commonly as a by-product in lead-zinc-silver smelting. As part of its R&D programme, HZL successfully implemented antimony dust treatment flow sheet at Ancillary Industry. Antimony dust at Pantnagar Metal Plant (PMP) was leached in controlled conditions to recover antimony as Potassium Antimony Tartarate (PAT) reagent which is used in Zinc Hydro plants purification section and enriched lead silver residue. Presently, there is no production of antimony in India. The entire requirement of antimony in the country is met through imports of its ore and concentrates. HZL is currently operating metal plant having 1,400 TPA antimony concentrate (by-product). The antimony slag is recovered as antimony trioxide with more

than 95% purity, which is commercially accepted with high demand as flame retardant.

SUBSTITUTES

Combination of tin, calcium, copper, selenium, cadmium, strontium and sulphur is among the substitutes used as hardeners for lead used in batteries. Low maintenance batteries have started using calcium as additive to substitute antimony. Antimony can be replaced by organic compounds or hydrated aluminium oxide in flame retardants and by tellurium and selenium in rubber manufacturing. Compounds of titanium, zinc, chromium, tin and zirconium are substituted for antimony chemicals in paints, pigments and enamels.

TECHNICAL POSSIBILITIES

Antimony products can be used as stabilizers in specialized plastics. Development of electric vehicles could lead to the use of high antimonial lead batteries because of their deep cycling characteristics. Antimony semiconductors have possible use in aircraft night vision systems and in space based astronomy. Antimony has also been found to be used in the manufacture of DVDs.

RECYCLING

Traditionally, the bulk of secondary antimony has been recovered at secondary lead smelters as antimonial lead, most of which was generated and then consumed by the Lead-acid Battery Industry.

WORLD REVIEW

The world reserves are estimated at approximately 2 million tonnes of Antimony content. Reserves are located mainly in China (32%), Russia (18%), Bolivia (16%), Kyrgyzstan (13%), Australia (7%), Myanmar (7%), Turkey (5%), Canada (4%), United States & Tajikistan (3% each) and Pakistan & Mexico (1% each) (Table-2).

Table – 2: World Reserves of Antimony

(By Principal Countries)

(In tonnes)

Country	Reserves
World: Total (rounded off)	2000000
China	640000
Russia	350000
Bolivia	310000
Kyrgyzstan	260000
Australia*	140000
Myanmar	140000
Turkey	99000
Canada	78000
USA	60000
Tajikistan	50000
Pakistan	26000
Mexico	18000

Source: USGS, Mineral Commodity Summaries, 2024.

*: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 20,000 tonnes

The world production of Antimony in 2023 was 108 thousand tonnes in terms of Antimony content as against 104 thousand tonnes in 2022. China is the leading producer of Antimony with 38% share in total production in 2023,

followed by Russia (28%), Tajikistan (19%), Iran, Bolivia & Turkey (4% each), Australia (2%) and Myanmar & Mexico (less than 1%) (Table-3).

Table – 3: World Production of Antimony Oxides

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World total	100000	104000	108000
China*	42622	39000	41000
Russia*	9000	11000	30000
Tajikistan	28531	30238	20000
Iran(a)*	2000	2000	3900
Bolivia	3093	3453	3884
Turkey	4210	10692	3816
Australia	3380	2292	1860
Myanmar*	6400	3500	1100
Mexico	413	632	700
Laos*	242	250	500

Source : BGS, World Mineral Production, 2019-23.

*Estimated

(a): Year ending 31st March following that stated.

(In tonnes of metal content)

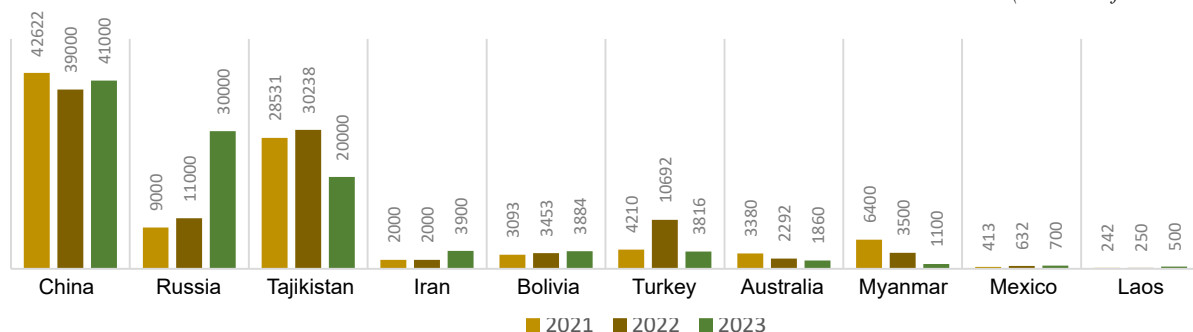


Fig. 1: Countrywise Production of Antimony Oxides

FOREIGN TRADE

Exports

Exports of antimony ores & concentrates were 4 tonnes for the years 2023-24. Exports of antimony alloys and scrap decreased slightly by 12 % to 1,406 tonnes in 2023-24 as against 1,590 tonnes in the previous year. Exports were mainly to USA (88%), Netherlands (4%), Italy & Brazil (2% each) and South Africa, Sri Lanka & Nigeria (1% each). Exports of Antimony and Articles NES were 2 tonnes

for the years 2023-24. Exports of Antimony Waste and Scrap were nil for the years 2023-24. Exports of antimony (unwrought), powders decreased by 12% to 1,404 tonnes in 2023-24 as against 1,588 tonnes in 2022-23. Exports of antimonial lead were increased by 18% to 26,882 tonnes in 2023-24 as compared to 22,852 tonnes in 2022-23. Exports were mainly to Korea (39%), UAE (18%), Bangladesh (14%), Vietnam (10%), Lebanon (9%) and Japan (5%) (Tables - 4 to 9).

Table – 4: Export of Antimony Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	4	2415
Turkey	--	--	2	1461
South Africa	--	--	1	919
Bangladesh	--	--	1	35
USA	--	--	++	++

Figures rounded off.

Table – 5: Export of Antimony Alloys and Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1590	1554406	1406	1318411
USA	1363	1317890	1238	1147415
Netherlands	25	22350	50	44852
Italy	50	50926	25	26280
Brazil	37	35922	22	20611
South Africa	52	50502	21	20131
Sri Lanka	17	21867	14	18119
Nigeria	1	1237	14	15029
Nepal	4	5008	6	6800
Bangladesh	10	10838	5	5358
Egypt	--	--	5	5258
Other Countries	31	37866	6	8558

Figures rounded off.

Table – 6: Export of Antimony and Articles Nes

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2	4742	2	2668
Philippines	--	--	1	1509
Nepal	--	--	1	1154
Nigeria	--	--	++	3
Kazakhstan	--	--	++	2
Germany	1	2660	--	--
Thailand	1	2054	--	--
USA	++	26	--	--
Azerbaijan	++	1	--	--
Korea, Rep. of	++	1	--	--
UAE	++	++	--	--

Figures rounded off.

Table – 7: Export of Antimony Waste and Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	++	69
UAE	--	--	++	69

Figures rounded off.

Table – 8: Export of Antimony (unwrought), Powders

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1588	1549664	1404	1315674
USA	1363	1317864	1238	1147415
Netherlands	25	22350	50	44852
Italy	50	50926	25	26280
Brazil	37	35922	22	20611
South Africa	52	50502	21	20131
Sri Lanka	17	21867	14	18119
Nigeria	1	1237	14	15026
Nepal	4	5008	5	5646
Bangladesh	10	10838	5	5358
Egypt	--	--	5	5258
Other Countries	29	33150	5	6978

Figures rounded off.

Table – 9: Export of Antimonial Lead

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	22852	4426688	26882	5236315
Korea, Rep. of	10230	2102614	10503	2087596
UAE	5421	988087	4831	923065
Bangladesh	1943	354466	3897	738054
Vietnam	2346	432230	2808	548719
Lebanon	--	--	2388	440953
Japan	1279	232131	1258	241073
USA	163	36543	294	65002
China	--	--	143	39923
Italy	86	16987	148	26184
Malaysia	50	9421	128	22957
Other Countries	1334	254209	484	102789

Figures rounded off.

Imports

Imports of antimony ores and concentrates were decreased by 37% to 4,069 tonnes in 2023-24 as compared to 6,460 tonnes in the previous year. Imports were mainly from China (66%), Chile (15%), Oman (12%), Italy & Turkey (2% each) and Canada (1%). Imports of antimony alloys and scrap were increased by 5% to 1,660 tonnes in 2023-24 from 1,579 tonnes in the previous year. Imports of antimony alloys and scrap were mainly from Thailand (38%), Oman (33%), Vietnam (21%), China (4%), UK (2%) and Hong Kong & Japan (1% each). Imports of antimony and articles NES were nil in the year 2023-24. Import of antimony waste

and scrap were nil in the year 2023-24. Imports of antimony (unwrought), powders increased by 5% to 1,660 tonnes in 2023-24 as compared to 1,579 tonnes in the preceding year. Imports were mainly from Thailand (38%), Oman (33%), Vietnam (21%), China (4%), UK (2%) and Hong Kong & Japan (1% each). Imports of antimonial lead were increased substantially by 76% to 1,00,416 tonnes in 2023-24 as compared to 56,996 tonnes in 2022-23. Imports were mainly from Malaysia (49%), Singapore (20%), Republic of Korea (12%), UAE (5%), Japan & Luxembourg (3% each), Saudi Arabia & Thailand (2% each) and Vietnam & USA (1% each) (Tables - 10 to 15).

Table – 10: Import of Antimony Ores and Concs.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	6460	2708869	4069	1623894
China	1970	762883	2692	996010
Oman	119	45476	490	250273
Chile	661	305049	627	244692
Italy	87	69656	98	77481
Turkey	--	--	90	31161
Canada	2876	1227218	52	20617
Peru	--	--	20	3658
Kazakhstan	--	--	++	2
Russia	497	188121	--	--
Lithuania	250	110466	--	--

Figures rounded off.

In tonnes

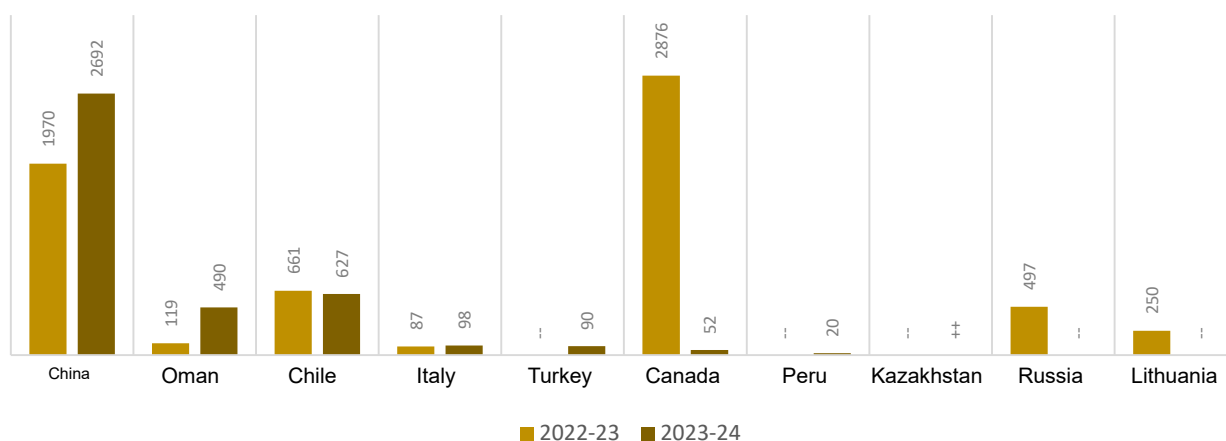


Fig. 2: Countrywise Exports of Antimony Ores and Concs

Table – 11: Import of Antimony Alloys and Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1579	1548561	1660	1635339
Thailand	214	210622	623	640585
Oman	792	757119	549	514941
Vietnam	261	251297	356	351557
China	297	311659	74	73448
UK	++	1368	25	24129
Hong Kong	15	16091	22	21067
Japan	--	--	11	9251
USA	++	242	++	361
Germany	++	128	--	--
Netherlands	++	32	--	--
Other Countries	++	3	++	++

Figures rounded off.

Table – 12: Import of Antimony and Articles Nes

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	741	++	49
UK	++	452	++	43
USA	++	87	++	6
Germany	++	128	--	--
China	++	74	--	--

Figures rounded off.

Table – 13: Import of Antimony Waste and Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	++	2
UK	--	--	++	2

Figures rounded off.

Table – 14: Import of Antimony (unwrought), Powders

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1579	1547820	1660	1635288
Thailand	214	210622	623	640585
Oman	792	757119	549	514941
Vietnam	261	251297	356	351557
China	297	311585	74	73448
UK	++	916	25	24084
Hong Kong	15	16091	22	21067
Japan	--	--	11	9251
USA	++	155	++	355
Netherlands	++	32	--	--
Belgium	++	3	--	--

Figures rounded off.

Table – 15: Import of Antimonial Lead

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	56996	10271277	100416	18339052
Malaysia	27433	4848198	49633	9065306
Singapore	4553	822440	20086	3777561
Korea, Rep. of	7245	1370061	12151	2407218
UAE	2904	558741	5189	791519
Japan	--	--	2596	469371
Luxembourg	3706	664681	3004	383739
Saudi Arabia	1341	250884	1719	315656
Thailand	622	118405	1502	279367
Vietnam	104	17402	1328	246983
USA	562	104564	850	158915
Other Countries	8526	1515901	2358	443417

Figures rounded off.

FUTURE OUTLOOK

Antimony has been in a list of 30 critical minerals for India, released by Ministry of Mines, Government of India on 24.07.2023. The future growth in demand for antimony will be much dependent on the level of requirement from the Flame-retardant Sector which accounts for about 48% primary antimony consumption worldwide and for about 90% global antimony trioxide consumption.

In the Flame-retardant Sector, antimony trioxide is used as a synergist normally with bromine and chlorine. Currently, antimony-based catalysts account for around 90% usage worldwide in polyethylene terephthalate (PET) production.

A new chip, based on germanium-antimony-telluride was developed abroad for 'Phase-change' Random Access Memory chips (PRAMS) which can process data faster than flash memory chips and unlike silicon are non-flammable. The chips have been commercialised and are expected to

find applications in mobile phones and digital cameras. In contrast, antimony metal consumption in metallurgical and battery markets could show a declining trend. The recent research and development programmes initiated by lead-acid battery manufacturers have led to significant changes in lead-acid battery design that have yielded substantial performance improvement which is bound to make lead-acid batteries a better and viable option as compared to its counterparts. This would eventually result in reduced use of antimony in lead-acid batteries diminishing the prospect of use of antimony in Battery Markets. The world supplies of antimony are expected to rise to an extent sufficient enough to meet the prospective demand. However, as per USGS, global consumption of antimony is expected to increase owing to projected increase in the use of antimony in flame retardants, lead-acid batteries and plastics, primarily in Asia. Antimony is a versatile element with a range of industrial application, although its use has become more regulated due to concern about its environmental and health impacts.



13. Cadmium

913 tonnes per year

Installed Capacity for recovery of cadmium in the country in 2023-24.

23,600 tonnes

world production of cadmium have been reported in 2023.

Cadmium, first discovered in Germany in 1817, is a soft, malleable, ductile, bluish-white metal that is produced almost exclusively as a byproduct of zinc smelting. USGS data confirms that over 95% of primary cadmium output is recovered during zinc refining, with additional contributions from lead and copper ores. Valued for its corrosion resistance, ductility, and low melting point, it is used in nickel–cadmium batteries, coatings, pigments, stabilisers and cadmium telluride (CdTe) solar panels. Although it poses environmental and health risks, cadmium remains vital for clean energy technologies like thin-film solar cells and rechargeable batteries. Currently, a large percentage of global cadmium metal production takes place in Asia. Global supply is limited and depends on zinc production, so recycling and responsible management are becoming increasingly important. In June 2023, the Ministry of Mines, Government of India recognised cadmium as a critical mineral, highlighting its strategic importance and supply-chain risks.

MINING LEASES AND PRODUCTION

Currently, there are no working mining leases for cadmium in the country for production purpose. Production of cadmium is generally reported as a by-product of zinc smelting; however, no production has been reported since 2018-19. The foreign market prices of cadmium are furnished in the General Review on "Prices".

INDUSTRY

The total annual installed capacity for recovering cadmium

was 913 tonnes per year (tpy), of which HZL accounted for 833 tpy. Binani Zinc Ltd (Edayar Zinc Ltd) reported the remaining 80 tpy (Table-1). HZL produces high-quality cadmium at its zinc smelters, which is cast in the form of pencils weighing between 250 g and 500 g. The purity levels are 99.95% Cd (max.) at Debari, 99.97% Cd (max.) at Vizag, and 99.99% Cd (max.) at the Chanderiya plant. HZL also has plans to undertake R&D for the production of high-purity cadmium, which is typically used in nuclear shielding applications

Table – 1: Installed Capacity for Recovery of Cadmium

Unit	Location	Installed capacity (tpy)
Total		913
1. HZL, Debari Zinc Smelter	Debari, Distt. Udaipur, Rajasthan	250
2. HZL, Vizag Zinc Smelter*	Visakhapatnam, Andhra Pradesh	115
3. HZL, Chanderiya Lead-Zinc Smelter	Chanderiya, Distt. Chittorgarh, Rajasthan	468
4. Binani Zinc Ltd** (Edayar Zinc Ltd)	Binanipuram, Distt. Ernakulam, Kerala	80

* Operation suspended since 2002

** Operation suspended since April, 2014

USES & CONSUMPTION

Despite being toxic and strictly regulated due to its environmental and health hazards, cadmium remains essential in specific industrial and technological applications. Its properties, such as, corrosion resistance, low melting point and high electrical conductivity make it difficult to substitute, particularly in high-performance uses. The most prominent application of cadmium is in nickel-cadmium (NiCd) rechargeable batteries, widely used in industrial backup power systems, aviation, railways, emergency lighting and cordless tools. NiCd batteries provide durability, long cycle life and reliable performance in extreme conditions, maintaining demand despite the rise of lithium-ion alternatives.

Cadmium is also extensively employed in corrosion-resistant coatings, especially in aerospace and defence, where it protects steel and aluminium components from marine and atmospheric corrosion. In the chemical and pigment industries, cadmium compounds like cadmium sulphide and cadmium selenide produce stable bright yellow, orange and red pigments for ceramics, plastics, glass and paints. It is also used in low-melting-point alloys, electronics, quantum dots, infrared detectors, and thin-film cadmium telluride (CdTe) solar panels, contributing to clean energy applications. Radioactive cadmium-109 finds use in scientific measurements, X-ray calibration and nuclear reactors due to its high neutron absorption.

Global cadmium use is tightly regulated under frameworks, such as, the EU RoHS directive and the Minamata Convention, confining it to essential or high-performance applications. With growing emphasis on recycling, safe handling and responsible sourcing, cadmium continues to be a strategically important metal, balancing industrial and technological relevance with environmental and health considerations.

RECYCLING

National Waste and Recycling Associations (NWRAs) have been established around the world to promote the collection and recycling of all types of batteries, both from the general public and industrial consumers. Nickel-cadmium batteries are recyclable in excess of 75% by weight once they have been collected. The recovery of cadmium from cadmium products through recycling not only ensures that cadmium be kept out of the waste stream and out of the

environment, but also that there is efficient consumption of this valuable natural resource as well. In alignment with the National Critical Mineral Mission (NCMM), notified vide notification No. F. No. 28/15/2024-CMM-Part (1), the government has introduced research initiatives, Centres of Excellence, and ₹ 1,500 crore Incentive Scheme to promote and strengthen recycling of cadmium and other critical minerals.

Exide Industries Ltd, India's largest manufacturer of lead-acid storage batteries and power storage solutions provider, has invested in building a battery recycling plant at Haldia (West Bengal) and has plans to expand its existing facility to manufacture Nickel-Cadmium batteries. The plant in Haldia will have a monthly capacity of 15,000 tonnes, making it the country's largest lead recycling facility. The project would give the company's recycling capacity a tremendous boost. It already operates two lead recycling plants near Pune and Bengaluru, which have a combined monthly capacity of 11,500 tonnes of recycled lead. The expansion project in Haldia is aimed at producing high-end Nickel-Cadmium batteries in technical collaboration with Furukawa of Japan. These batteries find applications in bullet trains, metro rail and other critical installations. This plant will also produce lead-acid batteries.

As per JMK Research estimates, the lithium-ion battery market in India is expected to increase from 2.9 GWh in 2018 to about 132 GWh by 2030 at a CAGR of 35.5%. Raasi Solar has announced plans to set up a 300 MW plant focussing on lithium battery recycling along with battery assembling and cell manufacturing facility.

SUBSTITUTES

Cadmium substitution is largely driven by environmental, health and regulatory concerns due to its toxicity. Though cadmium offers unique properties in batteries, pigments, coatings and solar cells, many alternatives are now available, each with trade-offs in cost, performance or compatibility. In batteries, NiCd cells are widely replaced by lithium-ion and NiMH batteries, offering higher efficiency and lower environmental impact. However, NiCd use persists in aviation and military applications due to specific performance needs. For plating, cadmium is being replaced by zinc-nickel, tin-zinc and aluminium-based coatings, especially where its unique surface properties are not critical. In pigments, cadmium compounds are

used for bright, stable colours, butcerium sulphide and bismuthvanadate are gaining ground, especially in plastics and consumer products. In PVC stabilisers, cadmium compounds have been phased out in favour of calcium-zinc, barium-zinc and organotin systems, particularly in Europe. In solar photovoltaics, CdTe thin-film remains prominent. Alternatives like CIGS and perovskites are under development.Cadmium use is expected to decline further as safer and more sustainable substitutes become more viable across industries.

WORLD REVIEW

Cadmium is generally recovered from zinc ores and concentrates. Sphalerite, economically the most significant zinc ore mineral, commonly contains minor amounts

of cadmium, which shares certain similar chemical properties with zinc and often substitutes for zinc in the sphalerite crystal lattice. The cadmium mineral greenockite is frequently associated with weathered sphalerite and wurzite. Quantitative estimates of reserves are not available. Cadmium content of typical zinc ore averages about 0.03%.

The world production of cadmium was estimated at about 23,600 tonnes in 2023, rising approximately 1% from 23,400 tonnes in the previous year. China accounted for 38% of the total production, followed by the Republic of Korea with 19%, Japan and Canada, each with 8%, Russia with 6%, Mexico and Kazakhstan each with 4%, Netherlands with 3%, and Uzbekistan, Germany and Norway each with 2%. The remaining 4% was contributed by other countries (Table-2).

Table- 2: World Production of Cadmium
(By Principal Countries)

Country	(In tonnes)		
	2021	2022	2023
World Total (Rounded off)	25700	23400	23600
China	10349	8700*	9000*
Korea, Rep. of	4500*	4500*	4500*
Japan	1900	1900*	1900*
Canada(a)	1800*	1800*	1800*
Russia	1400*	1500*	1300*
Mexico	1051	1174	1053
Kazakhstan	1500*	1000*	1000*
Netherlands	854*	574*	750*
Uzbekistan	470	500	500*
Germany	400*	400*	400*
Norway	390*	400*	400*
Other Countries	1086	952	997

Source: BGS, World Mineral Production, 2019-2023.

*estimated

(a)Including cadmium sponge and/or secondary metal

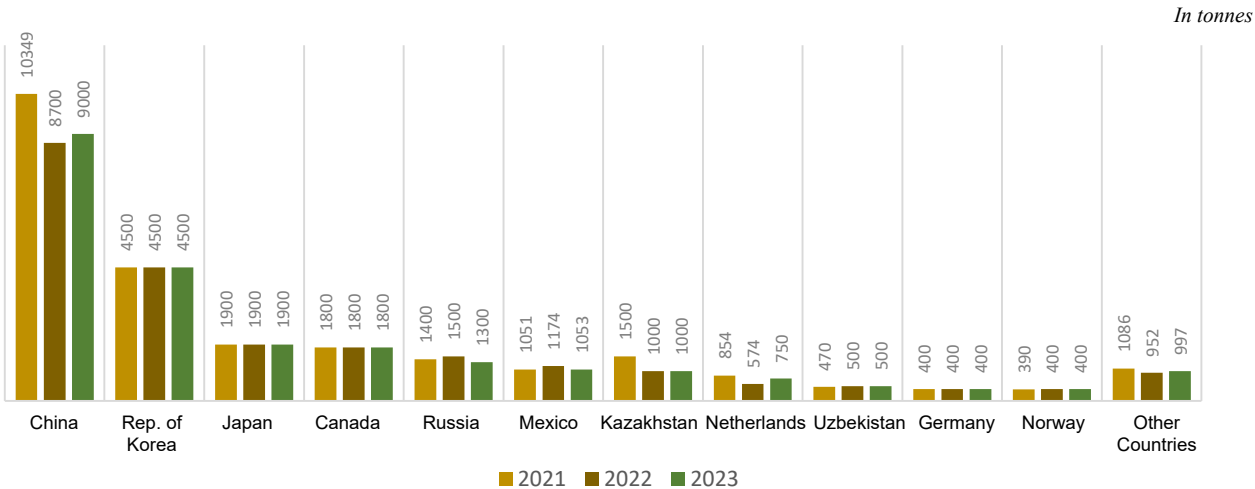


Fig. 1: Countrywise Production of Cadmium

FOREIGN TRADE

Exports

Exports of cadmium: unwrought and powders and cadmium & alloys were also negligible during 2023-24. However, exports of cadmium (including waste and scrap) during 2023-24 amounted to 222 tonnes, representing a

drastic increase of 573% from 33 tonnes in the previous year (Table-3). Similarly, exports of cadmium and scrap were also 222 tonnes during 2023-24, compared to 33 tonnes in the previous year (Table-4). Exports of cadmium (including waste and scrap) and cadmium & scrap were mainly to Bangladesh (94%), followed by the UK and Saudi Arabia (2% each), and the UAE (1%).

Table – 3: Export of Cadmium (Incl. Waste and Scrap)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	33	6115	222	36328
Bangladesh	31	4574	209	30618
UK	--	--	5	2360
Saudi Arabia	--	--	5	1878
UAE	2	845	3	1429
Lebanon	--	--	++	38
South Africa	--	--	++	3
Tunisia	--	--	++	2
Russia	--	--	++	++
Sri Lanka	--	--	++	++
Israel	++	602	--	--
Other Countries	++	94	++	++

Figures rounded off

Table – 4: Export of Cadmium and Scrap

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	33	6115	222	36328
Bangladesh	31	4574	209	30618
UK	--	--	5	2360
Saudi Arabia	--	--	5	1878
UAE	2	845	3	1429
Lebanon	--	--	++	38
South Africa	--	--	++	3
Tunisia	--	--	++	2
Russia	--	--	++	++
Sri Lanka	--	--	++	++
Israel	++	602	--	--
Other Countries	++	94	++	++

Figures rounded off

Imports

Imports of cadmium: unwrought and powders and cadmium & alloys were also negligible during 2023-24, just as in the previous year. However, imports of cadmium (including waste and scrap) during 2023-24 amounted to 10,736 tonnes, representing a decrease of 12% from 12,258 tonnes in the previous year (Table-5). Similarly, imports

of cadmium and scrap were also 10,736 tonnes during 2023-24, compared to 12,258 tonnes in the previous year (Table-6). Imports of cadmium (including waste and scrap) and cadmium & scrap were mainly from Republic of Korea (25%), followed by China (16%), Japan (15%), Latvia (8%), Belgium (6%), UAE (5%), Peru (4%), France, Russia, and Mexico (3% each) and other countries (12%).

Table – 5: Import of Cadmium (Incl. Waste and Scrap)

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	12258	3321224	10736	3589719
Korea, Rep. of	2572	703303	2693	907900
China	3400	955054	1672	645035
Japan	1934	481470	1641	572000
Latvia	330	112574	859	291737
UAE	838	216288	588	176043
Belgium	470	99079	635	157986
Peru	308	80284	431	128071
France	295	86956	318	111534
Russia	464	121237	303	106208
Mexico	436	125202	305	102104
Other Countries	1211	339777	1291	391101

Figures rounded off

Table – 6: Import of Cadmium and Scrap

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	12258	3321224	10736	3589719
Korea, Rep. of	2572	703303	2693	907900
China	3400	955054	1672	645035
Japan	1934	481470	1641	572000
Latvia	330	112574	859	291737
UAE	838	216288	588	176043
Belgium	470	99079	635	157986
Peru	308	80284	431	128071
France	295	86956	318	111534
Russia	464	121237	303	106208
Mexico	436	125202	305	102104
Other Countries	1211	339777	1291	391101

Figures rounded off

FUTURE OUTLOOK

The global cadmium market shows a fluctuating yet resilient trend, reflecting the interplay between environmental constraints and sustained industrial demand. Market surveys estimate its value at around USD 39.54 million in 2024, projected to reach USD 44.81 million by 2033, growing at a CAGR of about 1.4%, while global production is expected to rise from 24,756.9 tonnes in 2024 to 35,851.3 tonnes by 2033 (CAGR ~3.99%). Although primary cadmium supply is declining, production from recycling sources is showing a modest rise.

Despite concerns over cadmium's toxicity and environmental hazards, its production will continue as long as lead and zinc mining persists, since cadmium is obtained as a by-product. The nickel-cadmium (NiCd) battery sector remains a major driver, with rising demand from industrial and renewable-energy applications. While lithium-ion batteries have replaced NiCd batteries in most consumer

electronics, NiCd systems retain importance in heavy-duty, hybrid, and off-grid energy storage applications owing to their high reliability, stability, and durability, especially under harsh weather or offshore conditions.

Cadmium also maintains niche demand in pigments, stabilizers, and specialized materials used in plastics, glass, ceramics and enamels. These applications benefit from cadmium's ability to produce bright, durable colors and long service life, with the metal being present in a chemically stable, insoluble form embedded within the product matrix. India has led in volume, the overall regional consumption although the growth is expected to be modest moving forward. Some growth is anticipated due to recycling and possibly battery-related use. Overall, the cadmium market is expected to experience slow but stable growth, supported by recycling initiatives, industrial battery applications, and specialty material uses, even as substitution trends and stricter environmental norms continue to shape its long-term trajectory.



14. Cobalt

44.91 million tonnes

Total reserves/resources of cobalt have been established as on 1st April, 2020.

2,616 tonnes

of cobalt and alloys (including waste & scrap) were exported in 2023-24.

1,822 tonnes

of cobalt and alloys (including waste & scrap) were imported in 2023-24.

Cobalt is an important ferromagnetic strategic alloying metal having irreplaceable industrial applications. It is a chemical element with the symbol Co and atomic no. 27. Cobalt is associated mostly with copper, nickel and arsenic ores. Cobalt is a silvery gray metal that has diverse uses based on certain key properties, including ferromagnetism, hardness and wear-resistance when alloyed with other metals, low thermal and electrical conductivity, high melting point, multiple valences, and production of intense blue colors when combined with silica. Cobalt is used mostly in cathodes in rechargeable batteries and in superalloys for turbine engines in jet aircraft. Cobalt is extracted as a by-product of copper, nickel, zinc or precious metals. Lateritic/limonitic nickel ore usually is found to contain 0.08 to 0.15% Co along with 1.5 to 4% Ni in many parts of the world. In 2023, the Democratic Republic of Congo (DRC) was the world's leading producer of cobalt.

RESERVES/RESOURCES

Cobalt resources are predominantly associated with copper-cobalt ore, nickel-cobalt ore, arsenic-cobalt ore and pyrite deposits. The mineralogy of cobalt deposits is diverse which includes both primary (hypogene) and secondary (supergene) phases. Principal terrestrial (land-based) deposit types, which represent most of world's cobalt mine production, include primary magmatic Ni-Cu (-Co-PGE) sulfides, primary and secondary stratiform sediment-hosted Cu-Co sulfides and oxides, and secondary Ni-Co laterites. While independent cobalt minerals are rare, seabed manganese nodules emerge as a promising long-term resource for cobalt.

In India, cobalt deposits have been reported in the Singhbhum district of Jharkhand; the Kendujhar and Jajpur districts of Odisha; the Jhunjhunu district of Rajasthan;

the Tuensang district of Nagaland; and the Jhabua and Hoshangabad districts of Madhya Pradesh. In the Sukinda area of the Jajpur district, Odisha, cobalt is found associated with nickeliferous limonite/laterite, and copper slags produced by Hindustan Copper Limited also present another potential source. Additionally, seabed polymetallic nodules, which contain an average of 0.3% cobalt along with other minerals, represent another source of cobalt.

According to NMI data based on the UNFC system, reserves/resources of cobalt in terms of ore as on 1.4.2020 have been estimated at 44.91 million tonnes under Remaining Resources category of which about 69%, i.e., 30.91 million tonnes are estimated in Odisha. The remaining 31% resources are in Jharkhand (9 million tonnes) and Nagaland (5 million tonnes). The reserves/resources of cobalt as per UNFC system are furnished in Table-1.

Table –1: Reserves/Resources of Cobalt Ore as on 01.04.2020

(By Grades/States)

(In Million tonnes)

State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India	0	44.91	44.91
By States			
Odisha	0	30.91	30.91
Jharkhand	0	9	9
Nagaland	0	5	5

USES & CONSUMPTION

Batteries

- ◆ Cobalt is used as precursors (cobalt compounds) for cathodes in rechargeable batteries. Largest demand for cobalt has been from the Rechargeable Battery Industry.
- ◆ It was initially used in NiCd and NiMH cells, however, post the invention of the Lithium-ion battery, there was phenomenal growth in cobalt consumption in the Battery Sector (CRU).

Industry

- ◆ Cobalt is alloyed with aluminium and nickel to manufacture powerful magnets that find application in automobiles & aircraft industries. Cobalt powder finds an important application as a binder in the production of cemented tungsten carbides for heavy-duty and high-speed cutting tools. It is also used on bonded tools for Diamond Industry.
- ◆ Cobalt application improves the coating/adhesive property of enamel in steel appliances and is used in manufacturing steel-belted tyres.

Super alloys/Alloys

- ◆ Cobalt based super alloys normally contain 45% or more cobalt, while nickel and iron-based superalloys contain 8 to 20% cobalt.
- ◆ Superalloys made of cobalt have improved strength and wear & corrosion-resistance characteristics at elevated temperatures. Another use of cobalt-based super alloys is in turbines for pipeline compressors and jet aircraft engine.
- ◆ Hard-facing or cutting tools with cobalt alloys provide greater resistance to wear, heat, impact and corrosion.

Chemical Industry

- ◆ Cobalt oxide is used in chemical applications, such as, catalyst, dyes & pigments, paint driers/ adhesives and glass & ceramics.
- ◆ Cobalt catalyst, mostly cobalt acetate is used in the manufacture of Terephthalic acid (TPA) and Dimethyl terephthalate (DMT).
- ◆ Cobalt-molybdenum-alumina compound is used as catalyst in hydrogenation and for petroleum de-sulphurisation.

Magnets

- ◆ Cobalt can retain ferromagnetic property upto a temperature of 1,100 °C, highest for any metal.
- ◆ It is used in the manufacturing of Alnico magnets, magnetic recording media, soft magnetic material, alloys for spacecraft, etc.
- ◆ Cobalt is alloyed with aluminium and nickel to manufacture powerful magnets. Permanent magnets are used in wind turbines and electric motors for automobiles & aircraft.

MINING LEASE AND PRODUCTION

India does not currently mine or produce cobalt from primary ore sources. The demand for cobalt is usually met through imports. Some cobalt is recovered and reprocessed from scrap, including cemented carbide and spent catalysts, by domestic processors. Refining capacity of cobalt in India is estimated at about 2,060 tonnes per year. Nicomet Industries Ltd., located in Cuncolim, Goa, and Rubamin Ltd., situated in Vadodara, Gujarat, were India's leading producers of cobalt cathodes and compounds. Installed capacity for cobalt metal and different cobalt salts at Nicomet is 1,000 tpy. Nicomet Industries Ltd., located in Mumbai, Maharashtra, produces nickel cathodes, sodium sulfate, and cobalt cathodes that meet LME-approved criteria under the NICO brand. Vedanta Group is also exploring ways to produce cobalt for batteries as the Group has become the latest entrant among companies that seeks to capitalise on

the anticipated electric vehicle boom. According to reports, Sandvik Asia Ltd. recovers cobalt metal powder from trash cemented carbide at its pilot plant in Pune, Maharashtra. In addition, spent cobalt catalyst from plants producing DMT, TPA and oxo alcohols are also understood to be reprocessed by several small cobalt chemical processors. However, information on reprocessing of cobalt from scrap is not available. It is expected that recycled cobalt would continue to be used for domestic supply.

SUBSTITUTES

Cobalt is used in specialized applications and is challenging to substitute. Potential alternatives include barium or strontium ferrites, neodymium-iron-boron, or nickel-iron alloys in magnets; nickel, cermets, or ceramics in cutting and wear-resistant materials; nickel-based alloys or ceramics in jet engines; nickel in petroleum catalysts; rhodium in hydroformylation catalysts; and cerium, lead, manganese, iron, or vanadium in paints. Currently, about one-third of cobalt is replaced by cobalt-manganese-nickel in lithium-ion batteries. In some applications, substitution for cobalt would result in a loss in product performance. Other potential substitutes include cerium, iron, lead, manganese, or vanadium in paints; cobalt-iron-copper or iron-copper in diamond tools; copper-iron-manganese for curing unsaturated polyester resins; iron, iron-cobalt-nickel, iron-phosphorus, manganese, nickel-cobalt-aluminum, or nickel-cobalt-manganese in lithium-ion batteries.

RECYCLING

Recycling technologies for recovery of cobalt especially from waste Li-ion batteries have been an evolving process. The need for technologies which can recover valuable metals and the commercialisation of that technology by the industry is highly desirable. The technology related to “Recovery of cobalt from Li-ion batteries of mobile phones” developed by National Metallurgical Laboratory (NML), Jamshedpur, reportedly claims up to 95% recovery of pure cobalt from Li-ion batteries. Considering the need and significance of the problem related to energy materials

like Ni and Co, CSIR-IMMT has developed suitable process flow sheets for the processing of secondary resources, such as, alloy scrap and spent catalyst to produce Ni/Co based precursor material that can be used for battery applications particularly in preparing electrodes of Li-ion batteries. In addition to this CSIR-IMMT has taken up another project from MIDHANI to produce high purity cobalt metal from impure cobalt hydroxide. The Mobility Mission held consultations with industry to develop battery recycling as a sustainable method for ensuring up to 95% recovery of critical minerals, such as, lithium, nickel, cobalt etc. from spent batteries, thereby ensuring regular supply of raw materials for battery maintenance.

TRADE POLICY

As per the ITC (HS), 2022 Scheduled Import Policy, imports of cobalt ores & concentrates under ITC (HS) code no. 26050000 and the cobalt alloys and its products under Heading No. 8105 are allowed freely, except cobalt waste & scrap (ITC-HS Code No. 8105 3000) which is restricted.

WORLD REVIEW

The world cobalt reserves are estimated at 11 million tonnes of cobalt metal content. Cobalt reserves are mainly in the Democratic Republic of Congo which contributes (55%) to the total reserves followed by Australia (15%). Besides, major reserves are also located in Indonesia & Cuba (5% each) and Philippines, Russia & Canada (~2% each). The world reserves of cobalt are provided in Table-2.

As per British Geological Survey’s World Mineral Production (2019-2023), the world of metal content in 2022 mine production increased by 22% from 165,000 tonnes of metal content in 2022 to 2,02,000 tonnes of metal content in 2023. Principal producer Democratic Republic of Congo contributed about (69%) followed by Indonesia (11%), Russia (4.45%), Australia and Canada (3% each) and Cuba, Madagascar, Philippines & Papua New Guinea contributed ~2% each (Table-3).

Table – 2: World Reserves of Cobalt
(By Principal Countries)

<i>(In tonnes of metal content)</i>	
Country	Reserves
World: Total (rounded off)	1,10,00,000
Congo (Kinshasa)	60,00,000
Australia ^a	17,00,000
Indonesia	5,00,000
Cuba	5,00,000
Philippines	2,60,000
Russia	2,50,000
Canada	2,30,000
Madagascar	1,00,000
Turkey	91,000

Table- 2 (Concl'd.)

		(In tonnes of metal content)
Country		Reserves
USA		69,000
Papua New Guniea		49,000
Other Countries		7,80,000

Source: USGS Mineral Commodity Summaries, 2024

a: for Australia, Joint Ore Reserves Committee-compliant reserves were 6,10,000 tonnes.

Table – 3: World Mine Production of Cobalt
(By Principal Countries)

		(In tonnes of metal content)		
Country		2021	2022	2023
World total		133000	165000	202000
Congo, Democratic Republic		93144	115371	139838
Indonesia		2100	9500	21600
Russia		8000	9000	9000*
Australia		5232	5793	5222
Cuba		4400	4200	3200*
Madagascar		2111	3428	3500*
Philippines		3200	3400	3300*
Canada		3953	3534	5099
Papua New Guinea		2953	2987	3072
New Caledonia		1500	1750	2200*
Other Countries		6585	6396	6523

Source: BGS, World Mineral Production, 2019-23

* estimated

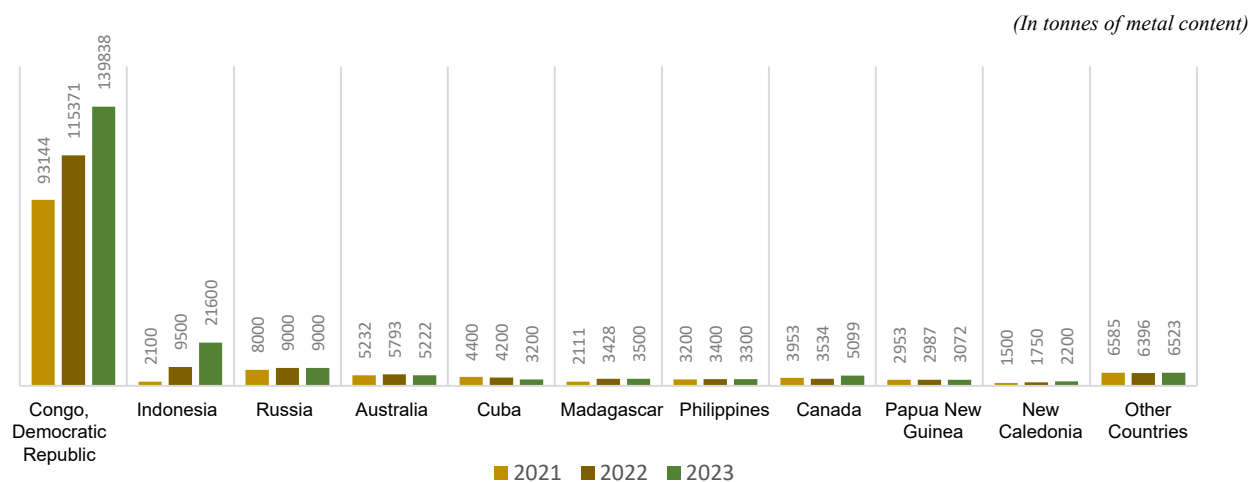


Fig. 1: Countrywise Production of Cobalt

FOREIGN TRADE

Exports

Cobalt ores & concentrates exports were nil during the last two years. Exports of cobalt and alloys including waste and scrap decreased by 3% to 2,616 tonnes in 2023-24 from 2,705 tonnes in the previous year. Exports were mainly to Republic of Korea (50%), Malaysia (15%) and Philippines and Morocco (~10% each).

Out of the total exports in 2023-24, exports of cobalt and alloys were at 2,526 tonnes and those of cobalt & scrap were at 90 tonnes. Similarly, during 2023-24 exports of cobalt powder were at 2,435 tonnes, cobalt (other articles) were at 74 tonnes and of cobalt unwrought at 17 tonnes (Tables-4 to 9).

Table – 4 Export of Cobalt and Alloys(incl Waste and Scrap)

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2705	2216211	2616	1771899
Korea, Rep. of	1608	1055839	1305	583044
USA	39	126911	94	344191
Malaysia	237	251104	382	191227
Morocco	--	--	259	184634
Thailand	--	--	180	139000
Philippines	409	307576	264	72498
Netherlands	--	--	20	53044
UK	45	80827	15	52667
Singapore	135	99988	64	30480
Germany	1	7923	17	24092
Other Countries	231	286043	16	97022

Figures rounded off

Table – 5: Export of Cobalt and Alloys

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2219	1821280	2526	1481920
Korea, Rep. of	1464	985911	1305	583044
Malaysia	27	134039	382	191227
Morocco	--	--	259	184634
Thailand	--	--	180	139000
Philippines	409	307576	264	72498
USA	18	51999	24	70141
Netherlands	--	--	20	53044
UK	31	48907	11	45600
Singapore	115	90516	64	30480
Saudi Arabia	++	3033	1	19692
Other Countries	155	199299	16	92560

Figures rounded off

Table – 6: Export of Cobalt Waste and Scrap

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	486	394931	90	289979
USA	21	74912	70	274050
Germany	--	--	16	8862
UK	14	31920	4	7067
Malaysia	210	117065	--	--
Korea, Rep. of	144	69928	--	--
Belgium	17	57792	--	--
Japan	60	33828	--	--
Singapore	20	9472	--	--
Nepal	++	14	--	--

Figures rounded off

Table – 7: Export of Cobalt Powders

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2156	1536090	2435	1203828
Korea, Rep. of	1464	985111	1305	583044
Malaysia	--	--	382	191227
Morocco	--	--	259	184634
Thailand	--	--	180	139000
Philippines	409	307576	241	61547
Singapore	115	90516	62	25900
UAE	--	--	5	13612
UK	21	14654	1	3994
USA	--	--	++	412
Turkey	41	30333	++	243
Other Countries	106	107900	++	215

Figures rounded off

Table – 8: Export of Cobalt Other Articles

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	36	149659	74	231835
Netherlands	--	--	20	53044
USA	18	51962	13	42647
UK	10	34253	10	41606
Saudi Arabia	++	2504	1	19692
Germany	1	7923	1	15230
Israel	++	5514	++	14531
Finland	++	5452	1	11442
Philippines	--	--	23	10951
Japan	3	20284	2	5333
Singapore	--	--	2	4580
Other Countries	4	21767	1	12779

Figures rounded off

Table – 9: Export of Cobalt Unwrought

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	27	135531	17	46257
USA	++	37	11	27082
Japan	--	--	4	12484
Australia	--	--	2	6191
Kuwait	--	--	++	419
Colombia	--	--	++	54
Romania	--	--	++	27
Malaysia	27	134039	--	--
Korea, Rep. of	++	793	--	--
Saudi Arabia	++	529	--	--
Taiwan	++	83	--	--
Other Countries	++	50	++	++

Figures rounded off

Imports

Imports of cobalt & alloys including waste and scrap increased about 54% to 1,822 tonnes in 2023-24 from 1,186 tonnes in the previous year. Imports in 2023-24 were mainly from Belgium (39%) and other contributed as follows China (21%), USA (17%) and Japan & UK (less than 6%).

Out of the total imports in 2023-24, imports of cobalt & alloys were at 1,822 tonnes and those of cobalt & scrap were negligible. Besides, imports of cobalt in the form of cobalt powder, other articles and unwrought cobalt were at 428 tonnes, 1,076 tonnes and 318 tonnes, respectively (Tables-10 to 16).

Table – 10: Import of Cobalt and Alloys (incl Waste and Scrap)

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1186	7612854	1822	8409176
China	262	1407052	387	2013868
USA	143	1048116	311	1797968
UK	115	1653266	100	1758226
Belgium	261	890514	712	998314
France	13	382897	26	468291
Japan	102	447491	110	463787
Germany	9	309036	12	411948
Norway	73	378121	40	151224
Morocco	40	175661	30	81946
Turkey	28	152295	18	67369
Other Countries	140	768405	76	196235

Figures rounded off

Table – 11: Import of Cobalt Powders

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	340	1846554	428	1445768
Belgium	109	586733	194	572306
Japan	52	281252	52	259230
China	49	230616	81	256060
France	11	69897	20	85020
USA	14	55266	20	75236
Turkey	28	151881	18	67003
Finland	13	78341	9	33037
South Africa	11	54819	8	29013
Germany	2	17246	3	27095
UK	8	49676	4	18731
Other Countries	43	270827	19	23037

Figures rounded off

Table – 12: Import of Cobalt Other Articles

Country	By Countries			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	672	5044465	1076	6068212
UK	104	1586285	96	1739300
USA	124	967815	291	1722274
China	135	867652	209	1517388
Germany	7	291727	9	384303
France	2	312022	1	368737

Table- 2 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Morocco	40	175661	30	81946
Japan	19	43945	17	76081
Belgium	152	303776	385	71376
Singapore	31	156311	8	34722
Hong Kong	++	434	2	30962
Other Countries	58	338837	28	41123

Figures rounded off

Table – 13: Import of Cobalt Unwrought

By Countries				
Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	172	720182	318	895196
Belgium	++	5	133	354632
China	78	308784	97	240420
Norway	20	75635	40	151224
Japan	31	122294	41	128476
France	++	978	5	14534
Netherlands	7	32994	2	4707
Germany	++	63	++	550
USA	5	25035	++	458
UK	3	17305	++	195
UAE	27	131139	--	--
Other Countries	1	5950	++	++

Figures rounded off

Table – 14: Import of Cobalt and Alloys

By Countries				
Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1184	7611201	1822	8409176
China	262	1407052	387	2013868
USA	143	1048116	311	1797968
UK	115	1653266	100	1758226
Belgium	261	890514	712	998314
France	13	382897	26	468291
Japan	102	447491	110	463787
Germany	9	309036	12	411948
Norway	73	378121	40	151224
Morocco	40	175661	30	81946
Turkey	28	152295	18	67369
Other Countries	138	766752	76	196235

Figures rounded off

Table – 15: Import of Cobalt and Scrap

Country	By Countries		2023-24 (P)	
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2	1653	++	++
Saudi Arabia	2	1653	--	--

Figures rounded off

Table – 16: Import of Cobalt Ores and Conc.

Country	By Countries		2023-24 (P)	
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	1766	1	820
UK	++	1766	++	780
China	--	--	1	40

Figures rounded off

FUTURE OUTLOOK

Cobalt bearing minerals are identified as critical and strategic mineral by Ministry of Mines, Government of India as per Part-D, Schedule 1 of Mines and Minerals (Development and Regulation) Act, 1957. Cobalt with their various forms are essential for the uses in green technologies like solar panels, wind turbines, batteries, and electric vehicles which are integral to the transition toward clean energy and a low-carbon economy. Also, it is used in electric vehicle, batteries, glassware, ceramics, fuel manufacturing and lubricant. India does not have any primary cobalt resources. However, two possible secondary resources are nickel-bearing laterite deposits in Odisha and copper slag produced by HCL, which have been under R&D studies for commercial applications over the years.

The cobalt refiners in India have catered to the market for chemical applications or where the cobalt metal or salt is dissolved and converted to cobalt oxide for cutting tools application. In India, cobalt is expected to have significant uses in metallurgy because of the increased need for special alloys/super alloys, cutting tools, and as an alloy in permanent magnets. The demand for cobalt powder will keep increasing due to its widespread use in manufacturing bonded tools for the Diamond Industry.

When it comes to cobalt, the Indian Industry is relatively small, but it is steadily expanding in different

areas, particularly in aerospace. The Aerospace Industry relies primarily on importing cobalt. Other sectors are experiencing steady growth. Chemical industries primarily utilize cobalt sulfate.

The Battery Industry's rising demand for cobalt, particularly for personal electronics, has caused global yearly growth rates of cobalt consumption to surpass the growth rates for the global gross domestic product. Between 2018 and 2025, it was predicted that the rate would rise to 9.5%, primarily due to cobalt's utilisation in rechargeable lithium-ion batteries for electric cars. For this, critical minerals, especially those with no known domestic resources, mineral-wise strategies are required to ensure robust access for India's manufacturing needs and climate change mitigation ambitions.

The discovery of Afanasy Nikitin Seamount (ANS) in the Indian Ocean Region has opened the doors of energy-hungry countries that fall under its jurisdiction of marine lines. Afanasy Niketan Seamount, which has huge deposits of Cobalt, falls under its Exploration license zone, which was earlier granted by the International Seabed Authority. Earlier, Sri Lanka had urged the ISA to halt India's exploration bid while citing its own UN-CLCS claims over the ANS. India, which is mostly dependent on Cobalt imports, needs to prioritise the threefold strategy of refining, recycling, and legal diplomacy with regional cooperation.



15. Copper

3.78 million tonnes
production of copper ore have been
reported in 2023-24.

23,188 tonnes
copper ores & concentrates were
exported in 2023-24.

10,16,303 tonnes
copper ores & concentrates were
imported in 2023-24.

Copper is a soft, malleable, and ductile metal with very high thermal and electrical conductivity. It is one of the few metals that occurs in nature in directly usable metallic form (native metals) and is an important non-ferrous base metal having wide industrial applications, ranging from defence, space programme, railways, power cables, mint, telecommunication cables, etc. India is not self-sufficient in the production of copper ore. In addition to domestic production of ore and concentrates, India imports copper concentrates for its smelters. The domestic demand for copper and its alloys is met through domestic production, recycling of scrap and by imports.

Hindustan Copper Limited (HCL), a Public Sector Undertaking, is the only integrated Company in the country that is involved in mining & beneficiation of ore and is engaged in smelting, refining and casting of refined copper. Hindalco Industries Ltd and Vedanta Limited are the major copper producers in the Private Sector that mainly rely on imported copper concentrates. These companies own copper mines in other countries.

In June 2023, the Ministry of Mines, Government of India recognized copper as one of the critical minerals, highlighting its strategic importance and supply-chain risks.

RESERVES/ RESOURCES

As per NMI database based on UNFC system as on 01.04.2020, all India total reserves/resources of copper ore have been placed at about 1.66 billion tonnes. Of the total resources of copper ore, only 163.89 million tonnes (9.87%) are classified as Reserves while 1.50 billion tonnes (90.13%) fall under the 'Remaining Resources' category.

Grade-wise, ore with (+) 0.50% to below 1% Cu accounts for 48.04% followed by ore with 1.00% to below 1.85% Cu for 35.33%, ore with (-) 0.50% Cu for 16.13% and ore with 1.85% & above Cu for 0.50%.

All India total reserves/resources of copper metal have been placed at about 12.20 million tonnes. Of the total resources of copper metal, only 2.16 million tonnes (17.72%) are classified as Reserves while 10.04 million tonnes (82.28%) fall under the 'Remaining Resources' category.

By states, largest reserves/resources of total copper ore are in Rajasthan (52.25%) followed by Madhya Pradesh (23.28%) and Jharkhand (15.14%). The remaining 9.33% are distributed across Andhra Pradesh, Arunachal Pradesh, Gujarat, Haryana, Karnataka, Maharashtra, Meghalaya, Nagaland, Odisha, Sikkim, Tamil Nadu, Telangana,

Uttarakhand and West Bengal.

Similarly, by states, largest reserves/resources of total copper metal are in Rajasthan (38.01%) followed by Madhya Pradesh (30.06%) and Jharkhand (22.79%).

The remaining 9.14% are distributed across Andhra Pradesh, Arunachal Pradesh, Gujarat, Haryana, Karnataka, Maharashtra, Meghalaya, Nagaland, Odisha, Sikkim, Tamil Nadu, Telangana, Uttarakhand and West Bengal (Table-1).

Table – 1: Reserves/Resources of Copper as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
Ore	163891	1496979	1660870
Metal	2161.57	10035.52	12197.09
By Grades			
Ore with 1.85% & Above Cu	-	8283	8283
Ore With 1.00 % to below 1.85% Cu	163891	422903	586795
Ore with (+) 0.50% to below 1.85% Cu	-	797906	797906
Ore with (-) 0.50% Cu	-	267886	267886
Metal	2161.57	10035.52	12197.09
By States			
Andhra Pradesh			
Ore	-	7582	7582
Metal	-	113.7	113.7
Arunachal Pradesh			
Ore	-	10	10
Metal	-	0.02	0.02
Gujarat			
Ore	-	12613	12613
Metal	-	200.74	200.74
Haryana			
Ore	-	53816	53816
Metal	-	179.01	179.01
Jharkhand			
Ore	9150	242313	251463
Metal	107.45	2672.21	2779.66
Karnataka			
Ore	-	41499	41499
Metal	-	245.86	245.86
Madhya Pradesh			
Ore	120353	266312	386665
Metal	1571.04	2095.82	3666.86
Maharashtra			
Ore	-	17755	17755
Metal	-	158.08	158.08
Meghalaya			
Ore	-	880	880
Metal	-	9	9
Nagaland			
Ore	-	2000	2000
Metal	-	15	15
Odisha			
Ore	-	11991	11991

Table- 1 (Concl'd.)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Metal	-	97.03	97.03
Rajasthan			
Ore	34388	833461	867849
Metal	483.08	4152.52	4635.6
Sikkim			
Ore	-	958	958
Metal	-	21.47	21.47
Tamil Nadu			
Ore	-	790	790
Metal	-	3.81	3.81
Telangana			
Ore	-	666	666
Metal	-	9.12	9.12
Uttarakhand			
Ore	-	4220	4220
Metal	-	60.04	60.04
West Bengal			
Ore	-	113	113
Metal	-	2.09	2.09

Figures rounded off.

MINING LEASES AND PRODUCTION

As on 31.03.2024 (Provisional), a total of nine mining leases had been granted for copper ore, covering a mining lease area of 3,916.85 hectares. As per the auction data

available on Ministry of Mines website, five copper mineral blocks have been successfully auctioned since 2015 (as on 27.05.2025), as given in Table-2.

Table – 2: Auctioned Blocks of Copper (as on 27.05.2025)

S. No.	State	Name of the Block	Mineral	Date of auction	ML/ CL	Area (in Ha)	Reserves (in MT)	Final Bid (in %)	Preferred Bidder
1	Madhya Pradesh	Shitalpani Copper Block	Copper	09.09.2023	CL	220.971	0.36	22.1	The Commodity Hub
2	Maharashtra	Thanewasana	Copper	08.05.2019	CL	768.62	8.02	3.6	Vedanta Limited
3	Maharashtra	Dubarpeth	Copper	09.05.2019	CL	816.29	1.343	2.1	Vedanta Limited
4	Maharashtra	Ghanpur Mudholi (West)	Copper & Gold	31.03.2022	CL	917.42	Cu- 0.22 Au- 0.11	2.05	M/s Vedanta Ltd
5	Maharashtra	Minzhari Copper Block	Copper	21.11.2023	CL	1743	1.13	7.25	Hindalco Industries Limited

Source: Ministry of Mines website.

Copper Ore and Concentrates

The production of copper ore at 3.78 million tonnes in 2023-24 increased by 13.7% as compared with the previous year.

The metal content in the ore produced in 2023-24 works out to 31,292 tonnes as against 27,728 tonnes in previous year. During the year under review, 3.59 million tonnes of ore was treated for obtaining copper concentrates

as against 3.26 million tonnes in previous year.

Production of copper concentrates at 1,25,230 tonnes in 2023-24 increased by about 11% as compared with the previous year. Madhya Pradesh was the leading producer of copper concentrates, accounting for about 56% of the total production during 2023-24, followed by Rajasthan (44%). The number of reporting mines was 5 in 2023-24 and 4 in previous year (Tables- 3 to 7).

Table – 3: Producer of Copper Concentrates, 2023-24

Name & Address of Producer	Location of the mine	
	State	District
Hindustan Copper Ltd, Tamra Bhavan, 1, Ashutosh Choudhury Avenue, Kolkata - 700019	Madhya Pradesh	Balaghat
	Rajasthan	Jhunjhunu

Table – 4: Production of Copper Ore, 2022-23 and 2023-24

(By States)

(In tonnes)

State	2022-23			2023-24 (P)		
	Ore Produced	Cu%	Metal Content	Ore Produced	Cu%	Metal Content
India	3326337	0.83	27728	3782256	0.83	31292
Jharkhand	53375	0.88	471	4936	1.03	51
Madhya Pradesh	2165042	0.91	19702	2545820	0.81	20499
Rajasthan	1107920	0.68	7555	1231500	0.87	10742

(P): Provisional

Table – 5: Copper Ore Treated, 2022-23 and 2023-24

(By States)

(In tonnes)

State	2022-23			2023-24 (P)		
	Ore Produced	Cu%	Metal Content	Ore Produced	Cu%	Metal Content
India	3257190	0.83	27035	3599725	0.84	30281
Jharkhand	-	-	-	-	-	-
Madhya Pradesh	2144220	0.83	17797	2375235	0.82	19587
Rajasthan	1112970	0.83	9238	1224490	0.87	10694

(P): Provisional.

Table – 6: Production of Copper Concentrates, 2021-22 to 2023-24

(By States)

(Quantity in tonnes; value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	115313	11024312	112745	10208275	125230	13141478
Jharkhand	-	-	-	-	-	-
Madhya Pradesh	65914	5560337	63860	5193489	69873	7151077
Rajasthan	49399	5463975	48885	5014786	55357	5990401

(P): Provisional .

Table – 7: Production of Copper Concentrates, 2022-23 and 2023-24

(By Sector/States/Districts)

(Quantity in tonnes; Value in ₹ '000)

State/District	No. of Mines	2022-23		No. of Mines	2023-24 (P)	
		Quantity	Value		Quantity	Value
India	4	112745	10208275	5	125230	13141478
Public Sector	4	112745	10208275	5	125230	13141478
Jharkhand	2	-	-	2	-	-
Singhbhum (East)	2	-	-	2	-	-
Madhya Pradesh	1	63860	5193489	1	69873	7151077
Balaghat	1	63860	5193489	1	69873	7151077
Rajasthan	1	48885	5014786	2	55357	5990401
Jhunjhunu	1	48885	5014786	2	55357	5990401

(P): Provisional

Grade Analysis

During the year, average copper content in the ore produced was 0.83% Cu, same as in the previous year. All India average metal content of ore treated during the 2023-24 works out to 0.84% Cu as compared with 0.83% Cu in previous year. The copper content in the ore treated varies from state to state. The average metal content in the concentrate produced works out to 23.24% Cu in 2023-24 as against 23.23% Cu in the previous year.

The average daily employment of labour in copper mines in 2023-24 was 3,302 as against 1,187 in the preceding year.

Copper Metal

Hindustan Copper Ltd. produces copper metal from the ore produced at their captive mines. Vedanta Limited [formerly known as Sterlite Industries (India) Ltd.] and Hindalco Industries Ltd. produce copper metal from imported copper concentrates (Table-8).

No production of copper blister was reported during 2023-24. Copper continuous cast wire rods amounted to 4,69,414 tonnes, representing an increase of 10.89% in 2023-24 from 4,23,283 tonnes in 2022-23. The production of copper cathodes during 2023-24 at 5,09,429 tonnes registering a decrease of 8.09% from 5,54,241 tonnes in the previous year (Tables- 9 to 12). Prices of copper are furnished in the General Review on 'Prices'.

Table – 8: Producers of Copper Metal, 2023-24

Name & Address of Producer	Location of the mine	
	State	District
Hindustan Copper Ltd, Tamra Bhavan, 1, Ashutosh Chowdhury Avenue, Post Box No.10224, Kolkata-700 019.	Jharkhand	Singhbhum (East)
Hindalco Industries Ltd, Century Bhawan, Dr Annie Besant Road, Mumbai – 400 030, Maharashtra.	Maharashtra	Raigad
Vedanta Limited, Sesa Ghor, 20 EDC Complex, Patto, Panaji – 403 001, Goa.	Gujarat	Bharuch
	Tamil Nadu	Thoothukudi
	Dadra & Nagar Haveli	Chinchpada (Silvassa)

Table – 9: Production of Copper Metal, 2021-22 to 2023-24

(In tonnes)

Year	Copper Blister	Copper Cathodes	Copper Electrolytic Wirebars	Copper CCWR
2021-22	-	483994	-	351464
2022-23	-	554241	-	423283
2023-24 (P)	-	509429	-	469414

(P): Provisional

Table – 10: Production of Copper (Blister), 2022-23 and 2023-24

(Bv State/Plant)

(Quantity in kg; Value in ₹ '000)

State	Plant	2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value
India		-	N.A.	-	N.A.
Jharkhand	Surda ICC	-	N.A.	-	N.A.

(P): Provisional

Table – 11: Production of Copper (CCWR), 2022-23 and 2023-24

(By State/Plant)

(Quantity in tonnes ; Value in ₹ '000)

State	Plant	2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value
India		423283	304546760	469414	341516354
Gujarat	Hindalco	267995	196596400	282855	208475700
Maharashtra	HCL Taloja	-	-	-	-
Tamil Nadu	Vedanta Ltd	-	-	-	-
Dadra & Nagar Haveli	Vedanta Ltd	155288	107950360	186559	133040654

(P): Provisional

Table – 12: Production of Copper (Cathodes), 2022-23 and 2023-24

(By State/Plant)

(Quantity in tonnes ; Value in ₹ '000)

State	Plant	2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value
India		554241	390034135	509429	362507497
Gujarat	Hindalco	407056	288340600	368103	263690000
Jharkhand	Surda ICC	-	-	-	-
Tamil Nadu	Vedanta Ltd	-	-	-	-
Dadra & Nagar Haveli	Vedanta Ltd	147185	101693535	141326	98817497

(P): Provisional

MINING & MILLING

Hindustan Copper Limited (HCL), established in 1967, is a Central Public Sector Undertaking under the administrative control of Ministry of Mines, Government of India. The Company is domiciled in India and has its registered office in Kolkata. HCL is the only vertically integrated copper producer in the country which produces refined copper from its own mined ore. The principal activities of the company are exploration, exploitation, mining of copper and copper ore including beneficiation of minerals, smelting and refining. The Company is primarily engaged in the business of mining and processing of copper ore into Metal-In-Concentrate (MIC), which has been grouped as a single segment in accordance with the 'Ind AS 108 - Operating Segments'. The Company is listed with BSE Ltd. and National Stock Exchange of India Ltd.

HCL's mines and plants are spread across five operating units, namely, the Indian Copper Complex (ICC) at Ghatsila in Jharkhand, the Khetri Copper Complex (KCC) at Khetrinagar in Rajasthan, Malanjkhanda Copper Project (MCP) at Malanjkhanda in Madhya Pradesh, Talaja Copper Project (TCP) at Talaja in Maharashtra and Gujarat Copper Project (GCP) at Jhagadia in Gujarat. HCL operates four underground mines and one opencast mine, with a combined ore production capacity of about 3.78 million tonnes per year.

The Company has copper mines & concentrator plants at Malanjkhanda Copper Project in Madhya Pradesh, Khetri Copper Complex (KCC) in Rajasthan and Indian Copper Complex (ICC), Ghatsila in Jharkhand. The Company has facilities of Smelter & Refinery plant at ICC and Gujarat Copper Project (GCP), Gujarat for production of copper cathode and thereafter conversion of cathode to Copper wire rod at Talaja Copper Project, Talaja (TCP), Maharashtra.

As per the UNFC classification, HCL holds around two-fifth of the copper ore reserves and resources in India. As on 1.4.2023, HCL has total reserves and resource (R&R) of 6.71 million tonnes in terms of copper metal (i.e. 698.44 million tonnes of ore with average grade of 0.96% based on UNFC system). HCL has plans to increase its mining capacity from current level of around 4.0 million tonnes per

annum to 12.20 million tonnes per annum by FY 2028-29 in phase-I (under implementation) through expansion of existing mines, re-opening of closed mines and opening of new mines.

The status of different continuing mine expansion projects is as under:

1. Malanjkhanda Mine (Madhya Pradesh)

The ongoing capacity expansion project i.e. development of underground mine below the existing open cast mine at Malanjkhanda Copper Project (MCP) will augment the ore production capacity from 2.5 MTPA to 5.0 MTPA. The ore production from open cast mine has now exhausted and underground mine is operating below the existing open cast mine leaving some barrier pillar. The project has achieved seamless transition from open cast mine to underground mine in FY 2023-24 with ramping-up of ore production from Malanjkhanda Underground mine to 22.48 Lakh tonnes which is about 60.8% increase with respect to previous FY 2022-23. Further, for continuity of ore production, two more Mine development contracts for both North and South Sections of the MCP Mine have been awarded.

To complete the residual work of underground mine construction for capacity expansion, a consultancy contract was entrusted on M/s MECON, a reputed CPSE Consultancy Organization, for estimation of value of residual work, preparation of NIT document and evaluation of tenders which have been divided in multiple parts (Mine Excavation at North Side, Mine Excavation at South Side, Shaft Furnishing, Men & Material Hoisting system, Crushing & Pumping system, Power system, Main Mechanical Ventilator) to complete the work in an efficient manner. Out of the above multiple parts, the contracts for completion of mine development work at North & South side have been awarded and started in FY 2022-23 and development work is under progress.

The contract awarded for construction of 3.00 MTPA Paste Fill Plant for back-filling the voids of underground mine at MCP to the EPC contractor (M/s Shapoorji Pallonji and Company Private Ltd) is under progress and the construction work was expected to be completed by June/ July, 2024. The contract for other enablers of paste fill plant

i.e. transport of paste from surface to underground mined out stope – (underground distribution system); surface bore hole drilling; operation and maintenance work etc. has also been awarded and the work is under progress.

The Company has also installed underground mine communication system using leaky feeder cable with traffic management system at Malanjkhand Copper Underground mine. This will enhance safety and productivity of mine.

2. Khetri & Kolihan Mine (Rajasthan)

The proposed expansion of mines at western sector would increase ore production capacity from

existing 1 MTPA to 3 MTPA. Mine-wise status is as under:

a) Kolihan Mine: Shaft sinking and creation of ore handling facilities below 0 mRL (meter Reduced Level) has been undertaken to augment the production capacity to 1.5 MTPA for which EC is already in place. The study report of Geophysical Exploration work, taken up in FY 2021-22, has shown possible extension of ore body up to -300 mRL. Based on the outcome of above work, the validation drilling has been taken up which confirmed the continuity of ore body extension up to (-) 300mRL, i.e. about 750 m vertical depth. Scheme of detailed exploration work is under preparation. Further, new contracts for Production & Definition Drilling has also been awarded to M/s EF Mining.

b) Khetri Mine: Execution of the earlier awarded contract to augment ore production capacity at the mine from 0.5 MTPA to 1.5 MTPA through deepening of existing shafts and other related activities, could not be completed due to extremely bad ground/fault zone encountered and as a result the contract had to be terminated. To sustain the ore production from mine, another contract has been awarded for conversion of track mining to trackless mining at 0 mRL and below in September, 2022 and the work is under progress. For Banwas deposit of Khetri Mine, Company appointed the contractual agency which has produced 4,15,808 tonnes of ore in FY 2023-24. Further, a depth exploration of 58,000 drilling meterage has been taken up to prove the continuity ore body extension upto (-) 300 mRL and estimation of reserve at deeper level.

3. Surda Mine (Jharkhand)

The plan envisages sinking of shaft, deepening of various winzes to increase production capacity from 0.4 MTPA to 0.9 MTPA in Surda mine. The validity of Surda Mining Lease has been extended till 31.3.2040 by the Government of Jharkhand. EC was granted by the Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi for 0.9 MTPA ore production over 323.16 ha on 30.5.2022. Subsequently, for execution of Mining Lease deed, the Government of Jharkhand intimated to submit EC over total lease area i.e. 388.68 ha. Accordingly, the EC amendment application was made to MoEF&CC, Delhi. On 28.11.2022, MoEF&CC recommended amendment in

EC area from 323.16 ha. to 388.68 ha. subject to grant of Stage-I Forest Clearance (FC) over balance 65.52 ha. forest area within Surda mining lease. The MoEF&CC, New Delhi has now granted Stage-I FC over balance 65.52 ha. forest area on 15.06.2024 and the same has been submitted to EC division of MoEF&CC for early issuance of amended EC.

4. Re-opening of Closed Mines at Indian Copper Complex (ICC) Ghatsila (Jharkhand)

The Company has initiated action to re-open the closed mines, development of new underground mine at Singhbhum Copper Belt of ICC namely, Kendadih and Rakha mines. Mine-wise status is given below:

a) Kendadih Mine: Kendadih mine was reopened in December, 2017 with commissioning of winders after completion of mine-dewatering. The Production and mine development contract was awarded on 06.07.2021. The development activities and ore production were started, but the performance of the contract was not satisfactory and as a result the said contract was terminated on 25.05.2023.

b) Rakha Mine: Rakha Mining Lease expired on 28.08.2021 and extension of mining lease period is under progress with the Government of Jharkhand. Further, the application for grant of Stage-I FC over remaining forest area within mining lease has been accepted by Project Screening Committee (PSC)-I. The site inspection by various authorities of State Forest department has been completed and the meeting of PSC-II was held on 30.05.2024 which has recommended the proposal to PCCF (Nodal) on 31.05.2024. PCCF (Nodal) recommended the proposal to Principal Secretary,

Forest, Government of Jharkhand on 21.06.2024. Further, recommendation of Principal Secretary, Forest, Government of Jharkhand to MoEF&CC, New Delhi is awaited.

For engagement of MDO (Mine Developer cum Operator) for re-opening and expansion of Rakha Copper Mine, development of a new underground mine at Chapri Block to produce 3.00 MTPA of ore and erection & commissioning of a matching capacity new Concentrator Plant at ICC, the Company has web-hosted tender in MSTC platform again in the month of May, 2024.

SMELTING

Hindustan Copper Limited (HCL) was the sole producer of refined copper till 1995 and the focus was on vertical integration so that the entire quantity of ore produced in its mines was converted into copper cathode and ultimately, wire rod. After liberalisation of the economy, the copper segment of industry has transformed significantly.

Currently, three major players dominate the Indian copper industry. Hindustan Copper Limited (HCL) in Public Sector, M/s Hindalco Industries Ltd. and Vedanta Limited (formerly M/s Sesa Sterlite Ltd) in private sector having current total installed refined copper capacity in the country at 0.785 million tonnes. It is reported that M/s

Adani Group is installing custom copper smelter refinery complex named as Kutch Copper Limited of capacity 1 million tonnes in a phased manner. Details regarding capacity of copper smelter are reflected in Table-13.

HCL has two primary smelting & refining plants at KCC and ICC. However, due to economic considerations the Company suspended KCC's smelting and refinery operation from December, 2008. HCL has one secondary copper smelter in Bharuch district, Gujarat. HCL also has one continuous casting plant of copper wire rod, namely,

Taloja Copper Project (TCP) with 60,000 tonnes per annum capacity at Taloja, Maharashtra.

Hindalco at Dahej in Gujarat and Sterlite Industries in Thoothukudi (plant at Tuticorin in Tamil Nadu – which is closed since May 2018) have set up port-based smelting and refining plants which depend on imported copper concentrates either from their own mines abroad or other overseas sources. Besides, there are a few small companies which produce Electrowon copper but their capacities are very low and production is inconsistent.

Table – 13: Capacity of Copper Smelters 2021-22 to 2023-24

(Quantity in '000 tonnes)

S.No.	Company/Location	Annual Refining Capacity
	TOTAL	784.5
1	Hindustan Copper Ltd.	68.5
	i) Indian Copper Complex, Distt. East Singhbhum, Ghatsila, Jharkhand	18.5*
	ii) Gujarat Copper Project, Distt. Bharuch, Gujarat	50
2	Hindustan Industries Ltd, Pahej, Distt Bharuch, Gujarat	500
3	Vedanta Ltd. (Formerly Sesa Sterlite Industries Ltd.), Thoothukudi, Tamil Nadu,	216

Production under Temporary Suspension.

RECYCLING OF COPPER

Copper scrap is traded in the form of new scrap generated from copper smelters, copper workings as well as old scrap recovered from electrical motors, electronic equipment, cables, wires, utensils, etc.

Copper is one of the most recycled metals of all the metals. The recycling of copper scrap is gaining importance worldwide simply because of the fact that recovery of copper metal from scrap requires much less energy than its recovery made from primary source. Besides, it enables conservation of natural resources.

In Indian condition, however, collection of scrap is in the Unorganized Sector and there is paucity of factual data in this regard. Still, as per the licences granted by Central Pollution Control Board as on 13.05.2010, there were 35 units operating in different states with a combined capacity of 2.42 lakh tonnes per annum for handling different types of scrap.

In addition, there are 132 units with combined capacity of 5.17 lakh tonnes per annum which recover copper along with other metals. As per the estimates made in the Market Survey on Copper published by IBM, production of 1.07 lakh tonnes per annum of secondary copper was reported and all of which have been from the Organized Sector in the country.

USES & CONSUMPTION

Electrical/Electronic Industry is by far the largest consumer of copper. Copper is used in the form of cables and winding wires because it is the best non-precious metal conductor of electricity. It offers low resistance and ensures safety

and efficiency in electrical distribution system—from high-voltage transmission cables to micro-circuits applications. Copper also has relatively high creep strength as compared to other commonly used materials.

In Electronic Industry, semi-conductor manufacturers have launched a revolutionary 'copper chip'. By using copper for circuitry in silicon chips, microprocessors are able to operate at higher speeds using less energy. Copper heat sinks help remove heat from transistors and enable computer speeds using less energy, and processors operate at peak efficiency.

Copper is used in Construction Industry as plumbing, taps, valves and fittings components.

The corrosion-resistant properties of copper and copper alloys (such as brass, bronze and copper-nickel) make them especially suitable for use in marine and other demanding environments. Vessels, tanks and piping exposed to seawater, propellers, oil platforms and coastal power stations, all depend on copper's corrosion resistance for protection.

In Transportation Industry, copper is used in various components. According to ICSG (International Copper Study Group), World Copper Factbook 2024, most conventional cars contain an average of 23 kg copper and hybrid electric vehicles contain about 40 kg copper. Copper is extensively used in industrial machinery and equipment. It is used in a number of consumer products, such as, coinage, utensils, fixtures, etc. Large quantities of copper are consumed in making copper-based alloys, such as, brass and bronze.

As per ICSG, the share of Electrical and Telecommunication Industry in total consumption is 56%, followed by Transport (8%), Consumer Durables (7%), Building & Construction (7%), General Engineering Goods (6%) and other industries including Process Industries (16%).

According to the ICSG, World Copper Factbook 2024, China was the largest consumer of refined copper in 2023, with an apparent usage of around 15.5 million tonnes, accounting for about 58% of world's total copper consumption i.e. 26.5 million tonnes.

The apparent availability of copper for internal consumption in various industries has been computed on the basis of production of refined copper (cathodes) and from the imports and exports data of copper (refined). Copper is also traded in the form of alloys but has not been considered for arriving at apparent availability of copper. During 2023-24, the imports of refined copper were more than the exports same as in 2022-23. The apparent consumption of refined copper increased by 20% from 7,04,734 tonnes in 2022-23 to 8,44,375 tonnes in 2023-24 (Table-14).

Table – 14: Apparent Consumption of Copper
(Based on Production of Refined Copper, Imports and Exports)

Item	(Quantity in tonnes)	
	2022-23	2023-24
Total Production* (Cathodes)	554241	509429
Total Imports (copper refined)	180622	372953
Total Exports (copper refined)	30129	38007
Apparent Consumption	704734	844375

(*): Primary

SUBSTITUTES

Copper is vulnerable for substitution on grounds of price, technical superiority or weight. Aluminium is used as substitute for copper in various products such as automobile radiators, cooling and refrigeration tube, electrical equipment and power cable. Optical fibre substitutes for copper in telecommunication applications and plastics substitute for copper in water pipe, plumbing fixtures and many structural application.

WORLD REVIEW

The world reserves of copper metal have been assessed at 1,000 million tonnes of copper content. Chile has the largest share, accounting for about 19% of world reserves, followed by Peru (12%), Australia (10%), Russia & Dem. Rep. of Congo (8% each), Mexico & USA (5% each), China (4%) and Poland (3%). The remaining 26% was contributed by other countries collectively (Table-15).

Table – 15: World Reserves of Copper
(By Principal Countries)

Country	(Quantity in '000 tonnes of copper content)	
	Reserves	
World: Total (rounded off)	1000000	
Chile	190000	
Peru	120000	
Australia ^(a)	100000	
Russia	80000	
Congo, Dem. Rep. of	80000	
Mexico	53000	
USA	50000	
China	41000	
Poland	34000	
Other countries	252600	

Source: USGS, Mineral Commodity Summaries, 2024.

(a): For Australia, Joint Ore Reserves Committee-Compliant or equivalent reserves were about 27 million tonnes.

The world mine production of copper was 23 million tonnes of metal content in 2023 which increased by about 4% from 22.2 million tonnes in 2022. Chile continued to be the largest producer of copper in 2023 with 23% share followed by Democratic Republic of Congo (14%), Peru (12%), China (8%), USA (5%), Indonesia & Russia (4% each), Mexico, Zambia, Australia & Kazakhstan (3% each).

The remaining 18% was contributed by other countries (Table-16).

As per BGS, World Mineral Production, 2019-2023, world refined copper production was 27.6 million tonnes in the year 2023 which showed an increase of 7% from 25.9 million tonnes in the 2022. China was the largest producer of world's total refined copper with 47% followed

by Chile & Dem. Rep. of Congo (8% each), Japan (5%), Russia (4%), USA (3%), Rep. of Korea, Germany, Poland & India (2% each). The remaining 17% was contributed by other countries. World copper smelter production was 21.5

million tonnes in the year 2023 which showed an increase of 5% from 20.4 million tonnes in the 2022. China accounted for 55% of world copper smelter production, followed by Japan & Chile (5% each).

Table – 16: World Mine Production of Copper, (2021-23)

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World Total (rounded off)	21400000	22200000	23000000
Chile	5624900	5330500	5372694
Congo, Dem. Rep. of	1924400	2515800	3148567
Peru	2329886	2445271	2755139
China	1818000	1899200	1815010
USA	1230000	1230000	1100000*
Indonesia	758356	952668	922143
Russia	938800	917900	910000
Mexico	782136	828180	778839
Australia	801052	826271	778369
Zambia	803747	763550	698566
Kazakhstan	511900	594100	576900
Other countries	3369654	3936999	4100432

Source: BGS, World Mineral Production, 2019-2023.

Note (s)

(a): Years ended 31st March following that stated.

*: estimated"

(In tonnes of metal content)

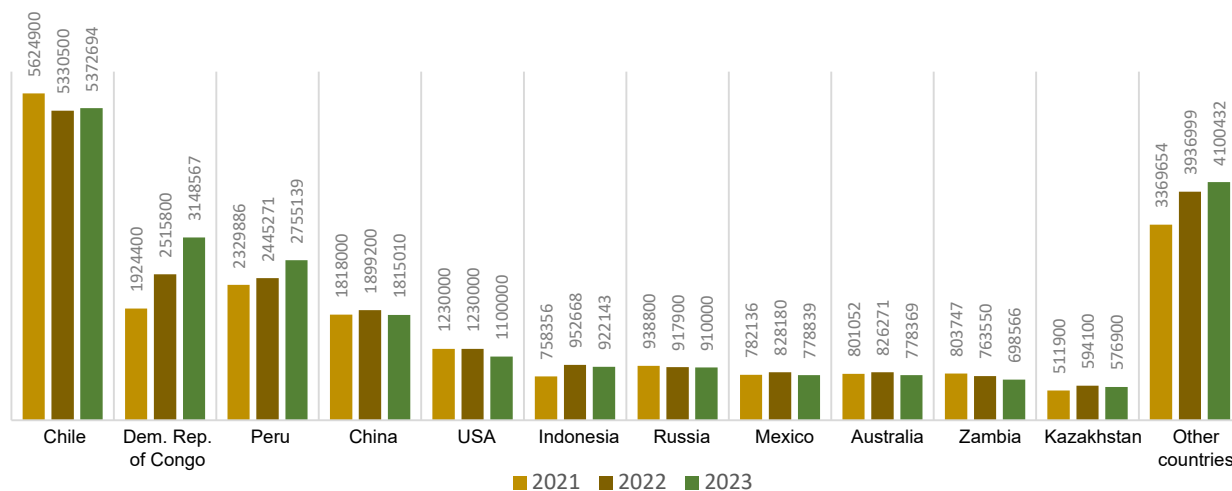


Fig. 1: Countrywise Mine Production of Copper

FOREIGN TRADE

Exports

The exports of copper from India are in various forms such as copper ores & concentrates, cement copper precipitated, copper & alloys, refined copper, brass & bronze, scrap, mattes and powder & flakes etc.

Exports of copper ores & concentrates decreased by 12% to 23,188 tonnes during 2023-24 from 26,336 tonnes in 2022-23. The exports were solely to China. However,

exports of cement copper precipitated were negligible during 2023-24 as compared to 293 tonnes during 2022-23. Exports of copper and alloys increased by 63% to 2,12,103 tonnes during 2023-24 from 1,30,105 tonnes in 2022-23. The exports were mainly to Saudi Arabia (47%) followed by China (14%), USA (8%). Exports of refined copper increased by 26% to 38,007 tonnes during 2023-24 from 30,129 tonnes in 2022-23. Exports of copper (scrap) also increased by 14% to 14,859 tonnes during 2023-24 from 12,995 tonnes in 2022-23 (Tables-17 to 20).

Table – 17: Export of Copper Ores & Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	26336	2435371	23188	2430311
China	26279	2433187	23188	2430309
Bangladesh	--	--	++	2
Cambodia	57	2184	--	--

Figures rounded off

Table – 18: Export of Copper (Cement Copper Precipitated)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	293	19034	++	41
USA	--	--	++	25
Malaysia	--	--	++	16
China	293	18955	--	--
UAE	++	63	--	--
Thailand	++	16	--	--

Figures rounded off

Table – 19: Export of Copper and Alloys

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	130105	101156433	212103	163463265
Saudi Arabia	30006	21922840	99206	70926438
USA	15576	19056236	15953	19596471
China	31117	20261430	29977	18647823
UAE	5371	4492495	11245	8797626
Germany	3800	3931575	4177	3983321
Indonesia	1560	1495192	3730	3618466
Malaysia	799	687092	3521	2653286
Nepal	2239	1544604	3093	2271218
Korea, Rep. of	7920	1307379	9063	2264074
Singapore	165	211611	3147	2246145
Other countries	31552	26245979	28991	28458397

Figures rounded off

Table – 20: Export of Copper & Alloys (Excl.Brass & Bronze and scrap)

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All items	143100	109121445	226962	171244191
Copper (scrap)	12995	7965012	14859	7780926
Copper Alloys:unwrought Excl. Brass and bronze	4690	3336064	4847	3177527
Copper and Alloys:worked (bars, rods, plates, etc)	17224	15938764	17664	16384551
Copper and Alloys: worked, nes	34094	35523195	34682	40204736
Copper Mattes	9303	1329375	10231	1282190
Copper Powder and Flakes	161	126380	458	176033
Copper Refined Copper Worked	33863	24532219	105748	76025432

Table- 20 (Concl'd)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Electroplated Anode of Nickel	++	2860	5	4637
Master Alloys of Copper	641	471314	461	345815
Refined Copper	30129	19896262	38007	25862344

Figures rounded off

Imports

The imports of copper to India are in various forms such as copper ores & concentrates, cement copper precipitated, copper & alloys, refined copper, brass & bronze, scrap, mattes and powder & flakes etc.

Imports of copper ores & concentrates decreased by 14% to 10,16,303 tonnes during 2023-24 from 11,78,921 tonnes in 2022-23. The imports were mainly from Indonesia (32%) followed by Chile (28%) and Peru (10%). Similarly, imports of cement copper precipitated increased

by 14% to 1,188 tonnes during 2023-24 from 1,039 tonnes in 2022-23. The imports were solely from Rep. of Korea. Imports of copper and alloys increased by 29% to 8,47,478 tonnes during 2023-24 from 6,57,430 tonnes in 2022-23. The imports were mainly from Japan (32%) followed by Tanzania (20%) and Veitnam (7%). Similarly, imports of refined copper sharply increased by two-folds to 3,72,953 tonnes during 2023-24 from 1,80,622 tonnes in 2022-23. Imports of copper (scrap) also increased by 45% to 1,97,224 tonnes during 2023-24 from 1,35,906 tonnes in 2022-23 (Tables-21 to 24).

Table – 21: Import of Copper Ores & Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1178921	273744293	1016303	259512343
Indonesia	262835	73145069	324177	92028584
Chile	400156	72498499	281284	55756059
Australia	102379	36276452	74457	25014611
Peru	162446	31185149	99397	17563022
Papua New Guinea	49105	13439081	58936	16783158
Philippines	--	--	30591	11621142
Canada	29878	8136590	38514	11206745
Panama	90887	16406274	60460	11131282
Tanzania	--	--	8701	6062977
Brazil	20418	3786715	18673	4293477
Other countries	60817	18870464	21113	8051286

Figures rounded off

Table – 22: Import of Copper (Cement Copper Precipitated)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1039	7114	1188	9618
Korea, Rep. of	418	2322	1188	9614
UAE	--	--	++	4
Belgium	621	4789	--	--
China	++	3	--	--

Figures rounded off

Table – 23: Import of Copper and Alloys

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	657430	463197337	847478	609554615
Japan	156905	100136286	272531	183087743
Tanzania	144721	99638388	169112	121270524
Vietnam	64071	50908674	56719	44639158
Thailand	38823	29222246	46533	34352641
UAE	38537	27072774	46492	33201951
Malaysia	33004	25329885	41317	31414874
China	36147	23589267	45771	28660085
Namibia	--	--	32853	23917380
Mozambique	27489	18385812	32485	22863556
South Africa	89938	62614860	26861	19401025
Other countries	27795	26299145	76804	66745678

Figures rounded off

Table – 24: Import of Copper & Alloys (excl.Brass & Bronze and Scrap)

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All items	793336	529223688	1044702	698997987
Copper (scrap)	135906	66026351	197224	89443372
Blister and Other Unrefined Copper	78368	55626124	501	344318
Copper Alloys:unwrought Excl. Brass Andbronze	193	313323	129	302533
Copper and Alloys:worked (bars, rods, plates, etc)	122225	91604756	126216	92341591
Copper and Alloys: worked, nes	10394	11855688	9914	11466980
Copper Mattes	1	1017	++	2
Copper Powder and Flakes	1094	1160069	1451	1409117
Copper Refined Copper Worked	103540	77045951	131894	98428250
Electroplated Anode of Nickel	160756	108987699	204222	152226864
Master Alloys of Copper	237	237624	198	225721
Refined Copper	180622	116365086	372953	252809239

Figures rounded off

FUTURE OUTLOOK

Global copper demand is set to rise steadily over the coming decades, driven by electrification and clean-energy systems, with the IEA's Net Zero Emissions (NZE) scenario projecting demand to reach 31–33 million tonnes by 2030 and around 40 million tonnes by 2050 due to large additions in renewable power, EV manufacturing, battery storage and extensive T&D grid expansion. The ICSG (Sept. 2024) outlook supports this trajectory: world mine production is expected to grow 1.7% in 2024 and 3.5% in 2025, while refined output is forecast to increase 4.2% in 2024 and 1.6% in 2025, with global usage rising 2.2% in 2024 and 2.7% in 2025. Although refined surpluses of 0.469 million tonnes in

2024 and 0.194 million tonnes in 2025 are projected, these may mask tightening conditions arising from declining ore grades, long project lead times and concentrate constraints, which ICSG indicates could shift the market toward deficit. Recycling will play a supporting role secondary refined output reached 4.55 million tonnes in 2023 but gains remain modest and insufficient to close the structural gap under NZE pathways. Demand drivers will continue to be EVs, renewable installations, grid reinforcement and digital infrastructure, with EVs alone expected to add ~1.7 million tonnes by 2027.

In India, copper consumption is projected to grow 6–7% annually, supported by major government initiatives

including Make in India, Smart Cities, industrial corridors, defence production, national highway and railway expansion, green-energy corridors and large renewable energy targets. Rising per-capita usage from a relatively low base further strengthens long-term domestic demand. On the supply side, HCL is implementing a major capacity expansion plan to increase ore production from 4 MTPA to 12.2 MTPA by FY 2028–29, across its Malanjkhand, Khetri and Kolihan operations. The company is also reopening the

Kendadih and Rakha mines in the Singhbhum Copper Belt, supported by Jharkhand's recent lease extensions. These initiatives will help of boost domestic copper output and to a certain extent reduce dependence on imports.

Overall, the global copper market is expected to tighten gradually, while India's outlook remains strongly positive due to sustained structural demand and expanding domestic capacity.



16. Gallium

Gallium is recovered as a by-product while producing alumina.

3.2 lakh kg
World production of high-purity refined gallium in 2023.

The demand for gallium is likely to increase with the growth of Electronic Industry in the country.

Gallium is a soft, silvery-white strategic metal, predominantly used in electronics. There is no primary source of gallium in the country. Gallium does not occur as a free element in nature. It usually occurs as a trace component in zinc & bauxite ores. It is generally recovered from sodium aluminate liquors obtained in Bayer's alumina process during aluminium production and from residues obtained during zinc processing in some countries. It can also be extracted from polymetallic ores by leaching and also from coal ash and coal. Gallium is also recycled from scrap generated from industries that manufacture Gallium arsenide (GaAs) and Gallium nitride (GaN) based devices. Though India is endowed with bauxite ores in abundance due to limitation in the viability of producing gallium economically, no production has been reported in the recent past.

Ministry of Mines, Govt. of India, in June 2023, released the list of 30 Critical Minerals for India which also included Gallium-bearing minerals. Consequent to this effect, the Central Government of India has amended the Mines and Minerals Development and Regulation Act, 1957 (MMDR Act, 1957) through the MMDR Amendment Act, 2023, whereby 24 critical and strategic minerals (including Gallium-bearing minerals) have been inserted in Part D to the Schedule-1 of the MMDR Act, 1957 which have been identified as critically and strategic important minerals for the country. Further, the Amended Act has also empowered the Central Government to auction critically and strategically minerals blocks.

USES & CONSUMPTION

Gallium is predominantly used in the Electronic Industry. It has an unusual property that it expands by 3.1 % on solidification. Gallium-based compounds, such as, Gallium arsenide (GaAs) and Gallium nitride (GaN) are used in the production of semiconductors for use in Electronic Industry. GaAs and GaN are increasingly used in the production of light-emitting diodes (LEDs), solar panels and laser diodes. It is also used in the manufacture of memory cells and other optoelectronic devices, such as, photo-detectors and solar cells. Use of GaAs is expected to increase especially

in Electronics & Communication Industry. Increased use of cellular communications and direct broadcast satellite applications are expected to inflate the demand for gallium.

Gallium is increasingly used in the manufacture of new gallium nitride devices used in high density data storage (compact disk players and digital video disk players), high-quality laser printing, communications and lighting purposes. Gallium nitride power transistors operate at high voltages and with higher power density than current GaAs devices. Gallium salts, such as, gallium citrate and gallium nitrate are used in medical imaging as radio contrast agents.

The plutonium used in nuclear weapon pits is machined by alloying with gallium to stabilise its phase. It is used as the alloying element in the Magnetic-shape-memory alloy "Ni-Mn-Ga". Gallium Gadolinium Garnet (GGG) is used as substrate for a bubble memory device. Gallium is used in some high temperature thermometers and an eutectic alloy of gallium, indium and tin is widely utilised in fever thermometers, replacing mercury. It is also used as a component in low melting alloys and in creating brilliant mirrors.

In 2022, the largest end-use consumption of gallium was used in for lighting and accounted for a share of 44%. In that year, integrated circuits and photovoltaics accounted for 36% and 7%, respectively, of the global gallium end-use consumption.

MINING LEASE & PRODUCTION

Gallium is recovered as a by-product while producing alumina. Two plants, namely, Hindalco Industries Ltd, at Renukoot, Uttar Pradesh and National Aluminium Co. Ltd. at Damanjodi Alumina Refinery, Odisha, had recovered gallium in the past. NALCO was reportedly in the process of sourcing environment-friendly technology for establishing a gallium extraction plant. NALCO has plans to set up 10 tpy gallium extraction plant at its Alumina Refinery at Damanjodi in Odisha. It targets to produce gallium metal with a purity of 99.99%. In December 2015, NALCO signed R&D agreement with Chalco, China for separation of iron concentrate from Red Mud and extraction of gallium from Bayer's Liquor.

As there is no primary source of gallium in the country, no Mining Lease has been granted for exploration or mining of Gallium metal in India as on date.

RESEARCH & DEVELOPMENT

An MoU was signed between NALCO and Bhabha Atomic Research Centre (BARC) in May 2016 for various R&D works like extraction of Gallium and other rare earth elements from Bayer Process liquor and Alumina waste and studies on the suitability of red mud for sacrificial core catcher material. In line of the same, the project works are currently in progress.

The Centre for Materials for Electronics Technology (C-MET), under the Ministry of Electronics & IT, has developed indigenous zone-refining and vacuum-heat processes to upgrade 3N/4N grade gallium to 6N purity. The multi-step purification includes hydrochemical treatment, vacuum heating at ~900 °C, zone refining, and fractional crystallization to remove trace impurities, aiming device-grade quality for semiconductor and optoelectronic applications.

Furthermore, the Bhabha Atomic Research Centre (BARC) has developed process protocols to synthesize 99.999% purity Trimethylgallium (TMG). TMG is a critical precursor for MOVPE-based growth of compound

semiconductors like GaAs, GaN, GaP, used in LEDs, high-frequency electronics and solar tech, supporting Make-in-India semiconductor goals.

SUBSTITUTES

Liquid crystals made from organic compounds are used in visual displays as substitutes for LEDs. Silicon-based complementary metal-oxide semiconductor power amplifiers compete with GaAs power amplifiers in midtier 3G cellular handsets. Indium phosphide components can be substituted for GaAs-based infrared laser diodes in some specific wavelength applications. The GaAs competes with helium-neon lasers in visible laser diode applications. Silicon is the principal competitor for GaAs in solar cell applications. GaAs-based integrated circuits are used in many defense applications because of their unique properties and there are no effective substitutes for GaAs in these applications. In some bipolar transistor applications, silicon-germanium is used as substitute for GaAs. Researchers are working to develop organic-based LED that may compete with GaAs in future.

WORLD REVIEW

Gallium occurs in very small concentrations in ores of other metals. Gallium contained in world resources of bauxite is estimated to exceed 1 million tonnes and a considerable quantity could be contained in world zinc resources.

The production of gallium was reported mainly from countries like China, Russia, Japan, Ukraine and the Republic of Korea. Imports of gallium into Japan and the United States, two leading consuming countries, were initially used as the basis for estimating world gallium production.

China reported that its primary low-purity gallium production capacity increased by 2,50,000 kilograms per year in 2022 to 10,00,000 kilograms per year. This latest increase followed a series of expansions from a capacity of 1,40,000 kilograms per year in 2010. China accounted for approximately 89% of worldwide primary low-purity gallium production capacity of an estimated 11,00,000 kilograms per year. China accounted for 98% of worldwide primary low-purity gallium production.

The remaining primary low-purity gallium producers outside of China included Japan, the Republic of Korea, and Russia. Germany, Hungary, and Kazakhstan ceased primary production in 2016, 2015, and 2013, respectively. Ukraine most likely ceased primary production in 2022. Owing to China's 2023 gallium export controls, the United States and other countries began considering the start or restart of domestic primary gallium production.

World's high-purity refined gallium production in 2023 was estimated to be about 3,20,000 kilograms, a 3% increase from the revised estimated figure of 3,10,000 kilograms in 2022. Canada, China, Japan, Slovakia and the United States were the known principal producers of high-purity

refined gallium. The United Kingdom ceased high-purity refined gallium production in 2018. Gallium was recovered from new scrap in Canada, China, Japan, Slovakia, and the United States. World high-purity refined gallium production capacity was an estimated 3,40,000 kilograms per year and secondary high-purity gallium production capacity was an estimated 2,80,000 kilograms per year. The world demand has been strongest in optoelectronic applications, particularly, in light-emitting displays. The enhanced properties of GaAs-based integrated circuits have enabled its use as substitute for silicon in many defence

applications. The cellular telephone market was principally responsible for growth in gallium consumption in the past few years.

During 2023, China, with 600 tonnes production, was the largest producer of Gallium (primary) in the World followed by Russia (15 tonnes), Japan (3 tonnes), Ukraine (2 tonnes) and the Republic of Korea (2 tonnes) (Table-1). In 2022, the largest end-use consumption of gallium was used in for lighting and accounted for a share of 44%. In that year, integrated circuits and photovoltaics accounted for 36% and 7% of the global gallium end-use consumption.

Table – 1: World Production of Gallium (Primary)
(By Principal Countries)

Country	(In tonnes)		
	2021	2022	2023
China	423	600	600*
Russia	13*	14*	15*
Ukraine	4*	2*	2*
Japan	3	3*	3*
Korea, Rep. of	2*	2*	2*

Source: British Geological Survey, World Mineral production, 2019-23

Note(s)

(1) Accurate data for gallium is difficult to obtain because it is produced only as a by-product of bauxite or zinc processing using proprietary techniques. As a consequence, this table may not be complete.

(2) Lower grade production is often subsequently refined to higher grade material which complicates the collection of data as doublecounting is a significant risk.

(3) In addition to the above primary production, there is also believed to be significant production of refined gallium from scrap recycling.

* estimated

FUTURE OUTLOOK

The demand for gallium is likely to increase with the growth of Electronic Industry in the country. Strategic importance of gallium has raised the imperative demand for development of indigenous technology and also the need for collaboration with foreign countries for refining and improving production of gallium. Zinc deposits, as an alternative source, may attract attention in the future, when the present accessible sources would deplete.

India has potential for increasing alumina production with greenfield export-oriented plants which can contribute substantially in meeting the domestic demand of gallium by establishment of gallium recovery units.

Smartphones are a fundamental structural shift in mobile communications, offering services not available on standard cellular telephones, such as, internet access, video streaming, computer programme applications (“apps”), and global positioning systems. Smartphones, which use up to 10 times the amount of GaAs-rich RF content than 2G cellular telephones, are expected to account for 87% of all worldwide handset sales by near future. Overall installation of 5G mobile networks in India is expected to further increase sales of smartphones. Additional increases in GaAs demand will also result from new Wi-Fi applications, such as, point-to-point communications, smart meters and tablet personal computer technologies.

Further, the higher switching capacities combined with low energy losses make gallium nitride an ideal material for power converters. These are an integral part of cell phones and are already often made of gallium nitride. The fact that chargers today are much smaller than they were ten years ago is also due to gallium nitride. 6G is not yet feasible with existing technologies. Gallium nitride is the key to achieving the efficiency and performance needed for 6G mobile communications.

Recent research has also shown that gallium nitride technology aims to replace silicon and boost electric vehicle (EV) performance while cutting costs. GaN power transistors are being brought into greater focus due to their large power-handling capabilities, high-switching frequencies, and higher voltage capabilities.

The gallium market is also expected to see exponential growth in the next few years. It may grow to \$5.38 billion in 2028 at a CAGR of 23.4%. The growth in the forecast period can be attributed to renewable energy, electric vehicles, advanced materials, nanotechnology. Major trends in the forecast period include gallium nitride dominance, recycling and sustainability, hybrid solar cells, healthcare innovations.



17. Gold

518.23 million tonnes
of total reserves/resources of gold ore
as on 1.4.2020 have been established.

760 thousand tonnes
production of gold ore have been
reported in 2023-24.

36,19,767 kg
of gold ores & concentrates were
imported in 2023-24.

Gold, in its purest form, is a bright, slightly reddish yellow, dense, soft malleable and ductile metal. It is one of the least reactive chemical elements and is solid under standard conditions. Gold often occurs in free elemental (native) form, as nuggets or grains, in rocks, in vein and in alluvial deposits. Gold dissolves in alkaline solution of cyanide, which is used in mining and electroplating. It also dissolves in mercury, forming amalgam alloys, but this is not a chemical reaction. Gold is resistant to corrosion and to most acid and has unique properties distinct from other metals. Gold is a relatively scarce metal in the world and a scarce commodity in India. The domestic demand is mainly met through imports.

RESERVES/RESOURCES

As per NMI data, based on UNFC system, as on 01.04.2020, the total reserves/resources of gold ore in the country have been estimated at 518.23 million tonnes. Out of these, 23.72 million tonnes were placed under Reserves category and the remaining 494.50 million tonnes under Remaining Resources category. The total reserves/resources of gold (primary), in terms of metal, stood at 607.26 tonnes. Out of these, 92.76 tonnes were placed under Reserves category and 514.50 tonnes under Remaining Resources category. The resources include placer-type gold ore in Kerala, estimated at 26.12 million tonnes containing 5.86 tonnes

gold metal. By States, largest resources in terms of gold ore (primary) are located in Bihar (43%) followed by Rajasthan (24.92%), Karnataka (20%), West Bengal (2.47%), Andhra Pradesh (3.03%) and Jharkhand (2%). The remaining 5.22% resources of ore are located in Chhattisgarh, Madhya Pradesh, Kerala, Maharashtra and Tamil Nadu. Although, Bihar is the leading State in India as far as resources of gold ore are concerned the resource estimate is at preliminary stage and falls under Inferred (333) and Reconnaissance (334) categories. In terms of metal content, Karnataka remained on top followed by Rajasthan, Andhra Pradesh, Bihar, Jharkhand, etc. (Table-1).

Table – 1: Reserves/resources of Gold as on 01.04.2020

(By Grades/States)

(In tonnes)

States/ Grades	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
Ore (Primary)	23728100	494506270.2	518234370.2
Metal (Primary)	92.76	514.5	607.26
Ore (Placer)	--	26121000	26121000
Metal (Placer)	--	5.86	5.86
By States			
Andhra Pradesh			
Ore (Primary)	3258100	12472898	15730998
Metal (Primary)	5.3	41.87	47.17
Bihar			
Ore (Primary)	--	222884860	222884860
Metal (Primary)	--	37.6	37.6
Chhattisgarh			
Ore (Primary)	--	4841033	4841033
Metal (Primary)	--	5.51	5.51
Jharkhand			
Ore (Primary)	--	10076526.7	10076526.7
Metal (Primary)	--	15.43	15.43
Karnataka			
Ore (Primary)	20470000	82538505.51	103008505.5
Metal (Primary)	87.46	163.71	251.17
Kerala			
Ore (Primary)	--	558460	558460
Metal (Primary)	--	0.2	0.2
Ore (Placer)	--	26121000	26121000
Metal (Placer)	--	5.86	5.86
Madhya Pradesh			
Ore (Primary)	--	7692934	7692934
Metal (Primary)	--	8.25	8.25
Maharashtra			
Ore (Primary)	--	1627000	1627000
Metal (Primary)	--	3.64	3.64
Rajasthan			
Ore (Primary)	--	125913720	125913720
Metal (Primary)	--	234.56	234.56
Tamil Nadu			
Ore (Primary)	--	67000	67000
Metal (Primary)	--	1	1
Uttar Pradesh			
Ore (Primary)	--	13000000	13000000
Metal (Primary)	--	2.08	2.08
West Bengal			
Ore (Primary)	--	12833333	12833333
Metal (Primary)	--	0.65	0.65

Figures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on Exploration & Development under "General Reviews".

MINING LEASES AND PRODUCTION

As on 31.03.2024 (provisional), the total number of mining leases in respect of Gold in force was 10 with a lease area of 6934.26 Ha. The production of gold ore at 760 thousand tonnes during 2023-24 increased by 20% as compared to that in the previous year. The quantity of ore treated also increased to 732 thousand tonnes from 615 thousand tonnes in the previous year. There were six reporting mines of gold ore in 2023-24.

The average grade of gold ore produced in India during 2023-24 was 2.48 g/t as against 2.67 g/t in previous year where as that of gold treated were 2.46 g/t in 2023-24 as compared to 2.63 g/t in the previous year.

Production of primary gold in 2023-24 at 1,603 kg increased by 11.55% as compared to that in the previous year. Karnataka was the leading producer of gold accounting for 97% of the total production. The remaining production was reported from Andhra Pradesh and Jharkhand (Tables-2 to 6).

The average daily employment of labour in 2023-24 was 3,063 as against 3,034 in the previous year.

GOLD BULLION

Production of gold bullion in India was reported both in primary and secondary forms and includes gold recovered from imported copper concentrates. Total production of gold bullion during 2023-24 was 16,639 kg which increased by 20%, as compared to 13,854 kg in the previous year (Table- 7).

Table – 2: Principal Producers of Gold, 2023-24

Name & address of producer	Location of the mine	
	State	District
The Hutti Gold Mines Co. Ltd Hutti, Distt. Raichur-584 115, Karnataka	Karnataka	Raichur
Manmohan Industries (P) Ltd Shanti Niketan, 286, New Patliputra Colony, Patna, Bihar	Jharkhand	Saraikela - Kharaswan

Table – 3: Production of Gold Ore, 2022-23 and 2023-24

(By States)

(In tonnes)

State	2022-23 (R)		2023-24 (P)	
	Ore Produced	Avg. Grade (gram/tonne)	Ore Produced	Avg. Grade (gram/tonne)
India	633514	2.67	760252	2.48
Andhra Pradesh	31028	1.38	33687	1.25
Jharkhand	8809	2.84	10079	3.18
Karnataka	593677	2.68	716486	2.52

(P): Provisional

In tonnes

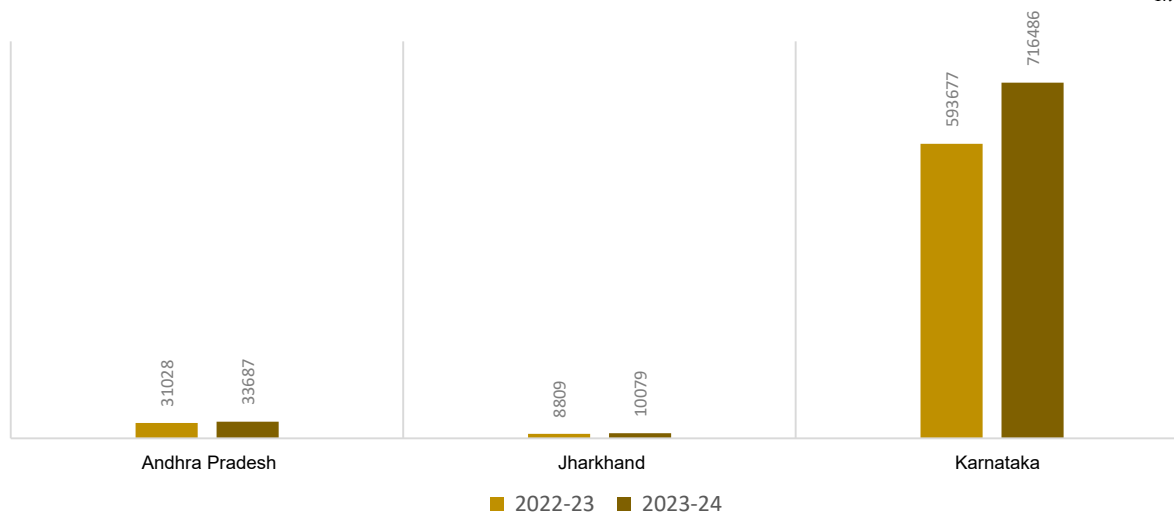


Fig. 1: Production of Gold Ore

Table – 4: Gold Ore Treated, 2022-23 and 2023-24

(By States)

(In tonnes)

State	2022-23 (R)		2023-24 (P)	
	Ore treated	Avg. Grade (gram/tonne)	Ore treated	Avg. Grade (gram/tonne)
Andhra Pradesh	193	1.68	4087	2.09
Jharkhand	8809	2.76	10079	3.18
Karnataka	605976	2.63	718281	2.45

(P): Provisional

Table – 5: Production of Gold, 2021-22 to 2023-24

(By State)

(Quantity in kg; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	1407	6761018	1437	7655220	1603	9670820
Primary Gold	1407	6761018	1437	7655220	1603	9670820
Andhra Pradesh	-	-	4	582	22	25957
Jharkhand	12	56268	22	119607	28	166140
Karnataka	1395	6704750	1411	7535031	1553	9478723

(P): Provisional

Table – 6: Production of Gold, 2021-22 to 2023-24

(By Sectors/States/Districts)

(Quantity in kg; Value in ₹ '000)

State/District	No. of Mines	2022-23 Production		No. of Mines	2023-24 (P) Production	
		Quantity	Value		Quantity	Value
India	5	1437	7655220	6	1603	9670820
Public Sector	4	1415	7535613	4	1575	9504680
Private Sector	1	22	119607	2	28	166140
Primary Gold	5	1437	7655220	6	1603	9670820
Andhra Pradesh	1	4	582	1	22	25957
Kurnool	1	4	582	1	22	25957
Jharkhand	1	22	119607	1	28	166140
Saraikela-Kharaswan	1	22	119607	1	28	166140
Karnataka	3	1411	7535031	4	1553	9478723
Raichur	3	1411	7535031	3	1553	9478723

*: Only labour reported

(P): Provisional

Table – 7: Production of Gold Bullion,

(2021-22 to 2023-24)

(Quantity in kg; Value in ₹ '000)

Year	Quantity	Value
2021-22	10087	48426018
2022-23	13854	73341570
2023-24 ^(P)	16639	100994820

(P): Provisional

Note: Includes gold recovered as by-product from copper concentrates by Hindalco Industries Ltd. in Gujarat.

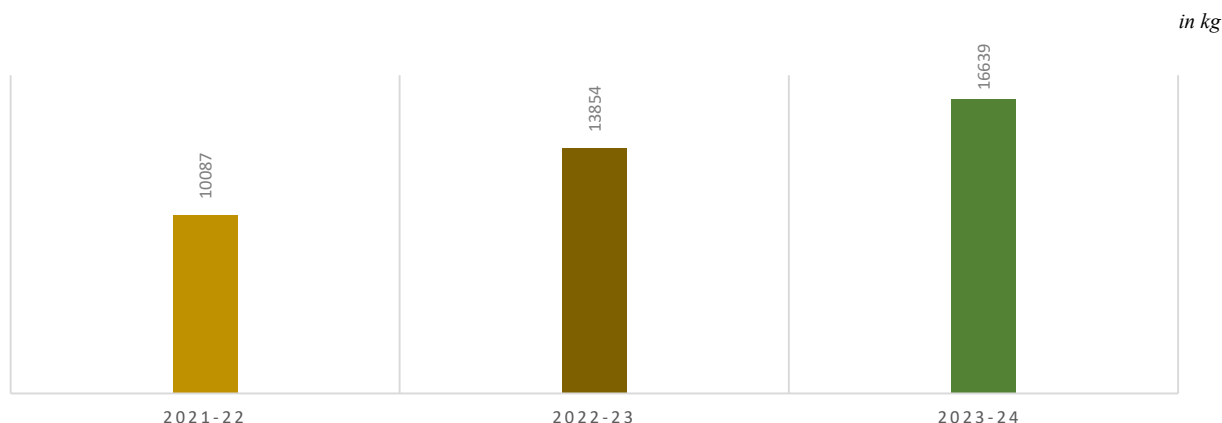


Fig. 2: Production of Gold Bullion

MINING & MILLING

Presently, HGML is the only Public Sector Company producing gold in the country. While in the Private Sector, Manmohan Mineral Industries Pvt. Ltd is engaged in mine production of gold at Kunderkocha in Singhbhum East district, Jharkhand by underground method of mining. Geomysore Services (India) Pvt. Ltd has been granted a mining lease over an area of 597.82 ha for gold mining in Village Jonnagiri in Kurnool district of Andhra Pradesh. HGML operates mines at Hutti and Hira-Buddini in Raichur district, Karnataka. Sub-level and LDBH Stopping methods are adopted in exploitation of gold ore. In the Uti mine, mining was carried out by opencast method till the year 2006 and thereafter by underground method. The ore from this mine is transported to Hutti Mine by road for processing at the mills. Underground exploratory mining too is in progress. Several operations at Hira-Buddini old unit, such as, exploratory mine development and deepening and re-equipping of main shaft are in progress. Exploratory mine development using compressed air jackhammer, drilling and electrical hoist in the shaft is presently underway. Based on the developmental work and feasibility, the locomotive loaders, wagon drills and other required machinery are likely to be used to increase the ROM.

The new ore processing plant based on modern technology (SAG and Ball Mill) with a capacity of 2,000 TPD has been operational at Hutti underground gold mine since 2010. At the Hutti Mineral Treatment Plant, the r.o.m. of -8" size is crushed. The final product from crushing plant, i.e. -10 mm size is stored in a 1,500 tonnes capacity fine ore bin for subsequent treatment, i.e., grinding. The Milling/Grinding process of gold ore employs two distinct grinding techniques. The first technique involves grinding done in two stages, i.e., primary grinding followed by secondary grinding for adequate comminution. The processes involve one primary mill and three tube mills which constitute one stream of grinding in which pebbles and smaller size balls are used as composite grinding media.

There are two such streams and strake tables for collection of coarse gold as concentrate for this circuit. In the second technique, grinding is done by four ball mills of different sizes and each of them is an independent circuit in which large size balls are used as grinding media. In these circuits, Knelson concentrator is used to collect coarse gold as concentrate. In all the milling techniques, cyclones are in closed circuit with the mills so as to get the required sizes (80% passing 75 micron) for the subsequent treatment process.

The concentrate collected from both the techniques is upgraded on James Table. The upgraded concentrates are roasted, magneted and finally smelted into bullion buttons. All the cyclone overflow, i.e., finely-ground ore in the form of slurry from the two streams of first technique and 4 streams of second technique join together in a distributor box from which finely-ground ore slurry is fed to High Rate Thickener for thickening purpose. The thickened pulp (60% solid w/w) thus obtained from thickeners is subjected to cyanidation process in which cyanide accessible gold in slurry makes complexes with cyanide in presence of oxygen and dissolves in solution at high ph. To increase the oxygen potential of slurry, H_2O_2 is added in addition to compressed air. The cyanidation or leaching process is carried out in a series of mechanically agitated agitators of different sizes. The cyanide leached pulp is then fed to two Carbon in Pulp (CIP) plants. The CIP plants are of 1,000 tpd size each and are parallel in circuit. The objective of CIP plant is to absorb the dissolved gold in activated carbon from the solution.

The gold-loaded carbon is removed from the CIP plant periodically, subjected to acid and alkaline wash and then eluted in four elution columns with 1.0% NaOH and 0.1% NaCN solution at 95 °C for a period of 60 hours. The solution is then passed through four electrowinning cells in which gold is deposited on steel wool cathodes. The gold loaded steel wool cathodes are manually removed periodically and subsequently subjected to acid digestion, drying and smelting to obtain bullion buttons. The bullion buttons thus obtained from table concentrate and steel

wool are cast into bullion bars weighing 4 to 11 kg and then dispatched for sales.

In the past, gold was produced by the Central Government undertaking, namely, BGML. BGML earlier mined and processed the ore from Chigargunta reef in Chittoor district, Andhra Pradesh; Mysore Mines of Kolar Gold Fields in Karnataka; and Yeppamana mine in Anantapur district, Andhra Pradesh. All activities of BGML were stopped and BGML was closed w.e.f. 01.03.2001 under Section 25 (O) of the Industrial Disputes Act, 1947 in terms of Ministry of Labour, Government of India's Order dated 29.1.2001.

Gold is sometimes recovered from the pregnant (Simple gold-bearing solution) solutions by adding zinc to form soluble zinc cyanide and precipitate of gold & silver. The pregnant solution can also be passed through activated carbon which absorbs dissolved gold. Gold from either process is cast into bars, bullion and dore (when it contains silver), which must be further refined to remove impurities, such as, mercury, arsenic and copper. Some ores cannot be treated by cyanide processing as gold in them is in small inclusions or even by solid solutions in minerals, such as, pyrite. This gold is generally recovered by roasting which converts pyrite into porous iron oxides containing small grains of gold that can be dissolved by cyanide.

DEVELOPMENT

The Deccan Gold Mines Ltd (DGML) is listed gold exploration company with deep roots in the Exploration and Mining sector. DGML's exploration activities are mainly in Karnataka and Andhra Pradesh States. Within the States of Karnataka, Andhra Pradesh and Kerala, DGML has explored several regions spanning 6,574 sq. km. in Dharwar-Shimoga Greenstone belt, Hutti-Maski Greenstone Belt, Mangalur Schist Belt and Ramagiri Schist Belt. The main prospects for gold at Ganajur and Karajgi have progressed into advanced stages of exploration and existence of high-grade gold-bearing zones in the prospect has been established. DGML has entered into MoU with Government of Karnataka to establish a Gold Mining industry in this project area.

Exploration is being conducted in Hutti belt at various prospects, viz., in Hutti Mine north prospect, Hirenagnur prospect, southern & northern continuity of Uti mine lodes, Uti Temple prospect, Chincherggi prospect, Buttapur prospect and Yatkall prospect. In south Hutti RP block, investigations are going on in Tuppadhur-Buddini prospect, Maski prospect, Ashoka prospect and Sanbal prospect.

Birla Copper Complex of Hindalco Industries Ltd situated at Dahej, district Bharuch, Gujarat has an installed capacity of 15 tpy for gold recovery from imported copper concentrates. HCL recovers by-product secondary gold from indigenous copper ores at its ICC plant in Jharkhand,

which has an installed capacity of 698 kg per annum gold recovery plant. This plant, however, has not reported production since 2007-08.

NMDC has secured a Bulyang /Ombe gold prospect in Tanzania. The gold prospect has a total area of 38.83 sq.km. Initial studies of NMDC revealed that Bulyang/Ombe had a prospect for good concentration where gold values have shown a maximum of 7.2 gram per tonne, which is close to the top quality standard of 8 to 10 gram per tonne set by the World Gold Council. The Company is in process of setting up of a pilot-scale processing plant for gold in Tanzania. NMDC has submitted the proposal for Bhukia Gold block to DMG, Govt. of Rajasthan, for over an area of 24 sq. km in Dist. Banswara. The Company has been allocated 3 Gold blocks, 2 in Karnataka and 1 in Madhya Pradesh. Legacy Iron Ore Ltd (NMDC holding 90.05% equity stake) based in Perth, Australia is concentrating in exploration of gold in Mount Ceila where good occurrence of gold is observed. Mount Celia gold project has identified two gold occurrences, namely, Kangaroo Bore and Blue Peter deposit. Currently exploration drilling and mining study is being carried out to get the confidence & complete the feasibility study. In Jharkhand, NMDC has submitted application to DMG, Jharkhand, for proposal to reserve 24.80 sq. km area in Kuchai Tehsil, District - Saraikela - Kaswan, Jharkhand under Section 17A (2A) of MM(D&R) Amendment Act, 2015 for prospecting and mining operation of gold & associated minerals. The matter is being pursued by NMDC with Govt. of Jharkhand for reservation. NMDC has applied for Peravali-Betapalli Block for gold. NMDC has requested the Government of Andhra Pradesh to reserve the block in favour of NMDC under section 17A (2A) of MM (D&R) Amendment Act, 2015 for prospecting & mining. NMDC has been allotted 3 gold blocks (2 in Karnataka & 1 in Madhya Pradesh) by Ministry of Mines, for G4 level exploration under NMET. NMDC has completed exploration of 5 blocks and submitted Geological Report to NMDC. NMDC has applied for prospecting of various minerals (diamond, gold, PGE, nickel, etc.) in Jabalpur, Katni Block (563 sq. km). Tripartite MoU among GoMP (MRD, through DGM) MPSMCL & NMDC was signed for geological and geophysical exploration for various districts of Madhya Pradesh.

A committee on Transforming India's Gold Market was constituted by NITI Aayog to recommend measures for Transforming the Gold Market Ecosystem in the country. The major recommendations of the committee are structured into five key areas. These are making in India Gold, finalisation of Gold, Tax and duty structure, Regulatory Infrastructure and Skill Development & Technology Upgradation.

The Recommendations of the Committee are summarised as follows:

Gold Mining

1. Make gold mining viable and attractive to investors by promoting ease of doing business with single window clearances.
2. Government may consider making available the risk capital for long-term capital-intensive mining projects to attract investments.
3. The mining policy should consider availability of suitable exit option. Aspects related to brownfield exploration may also be considered.
4. Improve the quality and availability of digital data, covering geological database (which includes quality and scale of maps and ease of access to information).
5. A comprehensive taxation policy should be formulated to align India's taxation framework with the strategic needs of the gold mining sector.

POLICY

Foreign Direct Investment (FDI) up to 100% in Mining Sector has been allowed. In the revised Export-Import Policy, comprised in the Foreign Trade Policy (FTP), 2015-20, gold ores and concentrates are under freely importable category. Under Heading No. 7108, the import of non-monetary gold metal also falls under free category subject to RBI regulations, while import of gold metal in monetary form is restricted.

DEMAND & CONSUMPTION

Jewellery sector accounted for major consumption of gold. The Industrial demand especially in the Electrical

Sector for gold is mainly due to excellent thermal and electrical properties. Besides, a significant amount is consumed in dentistry and medicine. Continuing research has discovered new applications for gold as catalyst and in Nano-technology. There is increase in demand from Ornamental and Electronic Sectors. Gold is considered a valuable asset, for investments and bank reserves.

SUBSTITUTES

Platinum and palladium substitute gold to some extent, but their use is influenced by price relationship and by an established consumer preference for gold. Silver can be a substitute, but it offers less resistance to corrosion. Gold plated palladium and bright tin-nickel can be used in electronics. Titanium and chromium-based alloys can be used in dental work. High prices encourage substitutes, particularly base metal clad with gold in Electronic & Electrical Industry and in Jewellery products. No metal or alloy substitute has all the properties of gold, and therefore, the emphasis is only on reduction of gold content rather than substitution.

WORLD REVIEW

The estimated world reserves of gold were about 59,000 tonnes of metal content. The gold reserves are mainly located in Australia, Russia, South Africa, USA, Indonesia, Brazil and Peru. The world reserves of gold are provided in Table-8. The world mine production of gold was estimated at 3,300 tonnes in 2023. China contributed about 11% to the world's total mine production of gold followed by Russia & Australia (9% each), Canada (6%), USA (5%), Mexico & Kazakhstan (4% each) and Uzbekistan & Peru (3% each) (Table-9).

Table – 8: World Reserves of Gold
(By Principal Countries)

(In tonnes of gold content)	
Country	Reserves
World: Total (rounded off)	59,000
Australia	12000 ^(a)
Russia	11100
South Africa	5000
USA	3000
China	3000
Indonesia	2600
Brazil	2400
Canada	2300
Peru	2300
Uzbekistan	1800
Other countries	9200

Source: USGS, Mineral Commodity Summaries, 2024.

(a): For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 4,600 tonnes.

Table – 9: World Mine Production of Gold
(By Principal Countries)

Country	(In kg)		
	2021	2022	2023
World:Total (rounded off)	3300000	3300000	3300000
China ^(a)	329000	372048	375155
Russia	313830	302000	302000*
Australia	307577	302787	296053
Canada	196730	195226	198335
USA	187000	173000	170000*
Mexico	133340	147700	131842
Kazakhstan	114843	129794	131204
Ghana	87677	113252	125549
Usbekistan	100360	105540	107120
Peru	97190	96966	99916
Others Countries	1383606	1295108	1347042

Source: BGS, World Mineral Production, 2019-2023.

(a):-Metal production

*: Estimated

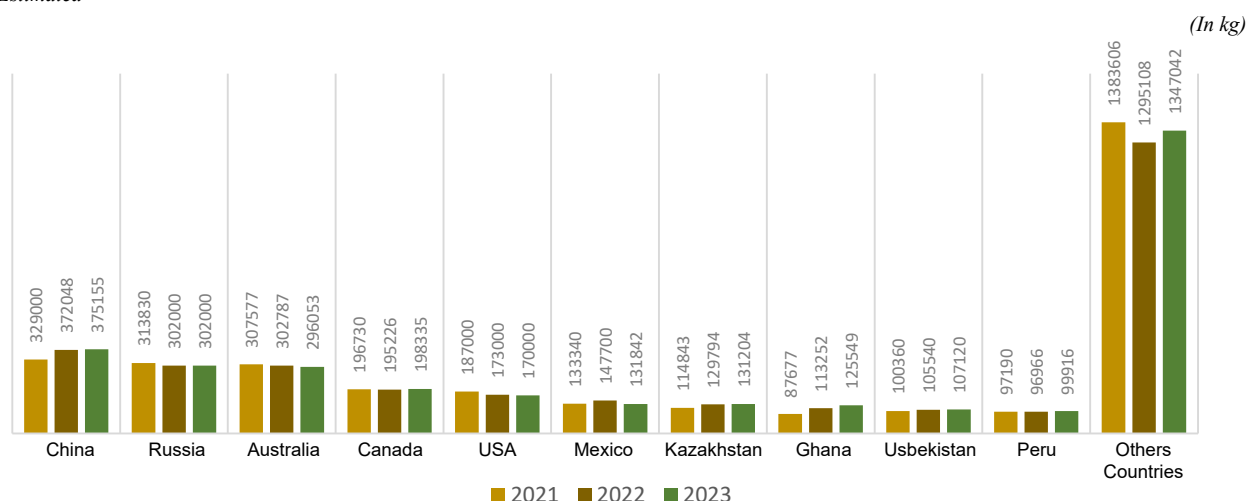


Fig. 3: Countrywise World Production of Gold

FOREIGN TRADE

Exports

During the year 2023-24, the exports of gold ores & conc. were nil and same as that in the preceding year. Out of the total exports of Gold (non-monetary & monetary), the

share of non-monetary was almost 100% while the share of monetary was negligible. The exports of gold (non-monetary and monetary) decreased by 42% to 1,521 kg in 2023-24 from 2,623 kg in 2022-23 (Table- 10).

Table – 10: Export of Gold (Non-monetary and Monetary)
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
Export of Non-Monetary and Monetary	2623	11920184	1521	8162909
Export of Gold – Non-monetary	2623	11920184	1521	8162909
Export of Gold – monetary	++	++	++	++

Imports

The imports of gold ores & concentrates increased by 47% to 36,19,767 kg during 2023-24 from 24,64,467 kg in the preceding year. Imports of gold (monetary and non-monetary) increased substantially by 17% to 7,95,246 kg

in 2023-24 from 6,78,299 kg in 2022-23. Out of the total imports of gold (non-monetary & monetary), the share of non-monetary was almost 100% whereas the share of monetary was negligible (Tables-11 & 12).

Table – 11: Import of Gold (Non-monetary and Monetary)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	678299	2804792414	795246	3772489677
Switzerland	220073	1014901340	278246	1479733272
UAE	52808	248620208	119788	634087071
South Africa	56480	268756428	72900	382021380
Peru	44508	141669149	59875	231535278
Australia	16250	77517319	31901	165230566
USA	39589	149844571	21488	103864835
Bolivia	47959	203211076	18124	89101846
Ghana	26655	112365460	17501	85762631
Tanzania	19862	72266246	19061	78277587
Guinea	31967	135781049	15730	77761757
Other Countries	122148	379859568	140632	445113454

Table – 12: Import of Gold Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	2464467	7336354	3619767	8675743
Colombia	2463049	6964298	3358635	7853511
Bolivia	1417	372055	2724	353450
Australia	--	--	13405	282190
Taiwan	--	--	181779	107290
Malaysia	1	1	63092	69566
Peru	--	--	103	9566
Laos	--	--	11	77
UK	--	--	1	41
Ghana	--	--	2	36
Germany	--	--	9	8
Other Countries	++	++	6	8

FUTURE OUTLOOK

Historically, purchase of gold was considered to be a safe haven, hedge against economic failures, portfolio diversifier and store of wealth.

India is a traditional and stable market for gold consumption. The present production of gold is insufficient and does not meet the ever increasing demand. Therefore, efforts will be required to reduce the gap between production and demand.

The recommendations of the Committee on Transforming India's Gold Market (Constituted by NITI Aayog) contributes for fulfilling the transformational vision

for India's gold market seeking to double its contribution in GDP, enhance employment opportunities, increase FDI inflow and increase the gold market size, without negatively impacting upon India's Current Account Deficit.

Further, to reduce dependence on gold imports, it is necessary to boost domestic supply which has to happen through the 'Make in India' initiative for mining, recycling & refining and increased monetisation. The policies around gold mining may need to be revisited with regard to the auctioning process, providing for single window clearance for the pending proposals and increasing co-operation between the States and the Centre.



18. Iron, Steel & Scrap

144.299 million tonnes
production of crude steel have been
reported in the country in 2023-24.

₹ 1,55,071 crore
value of exports of iron & steel were
reported in 2023-24.

₹ 1,86,460 crore
value of imports of iron & steel were
reported in 2023-24.

Iron & steel are decidedly the vital components of a country's economy and are considered pivotal amongst the driving forces of modernisation. The level of per capita consumption of steel is treated as one of the important indicators of socio-economic development and living standards in any country. Steel continues to be the foremost of engineering materials, which is not only environment-friendly but also is recyclable

The total finished steel (alloy+stainless+non-alloy) production in India has grown from a mere 1.1 million tonnes in 1951 to 139.15 million tonnes (Crude Steel Equivalent) in 2023-24. Out of this, 76.46 million tonnes was Non-Flat steel and the remaining 62.69 million tonnes was Flat steel. The contribution of non-alloy finished steel, alloy finished steel and stainless steel segment is 129.748 million tonnes, 6.019 million tonnes and 3.386 million tonnes, respectively.

Steel exports from India began in 1964. Exports in the first five years were mainly as a result of low demand in the domestic Iron and Steel market. Exports subsequently declined due to revival of domestic demand. India once again started exporting steel in 1975 which subsequently registered a slump due to rising domestic demand. Post liberalisation, rejuvenation in the Steel Sector resulted in large-scale exports of iron and steel. The total finished steel exports stood at 7.487 million tonnes in 2023-24, while imports stood at 8.32 million tonnes.

LIBERALISATION OF THE INDIAN STEEL SECTOR

At the time of independence in 1947, India had only three steel plants—the Tata Iron & Steel Company, the Indian Iron & Steel Company and Visveswaraya Iron & Steel Ltd and a few electric arc furnace-based plants. In the period till 1947, the viable steel producers in the country that operated with a capacity of about 1 million tonnes were wholly under the Private Sector. The provisions of the economic policy

implemented during different phases of time engendered several marked changes in Indian Steel Industry. From the fledgling one million tonnes capacity status at the time of independence, India has now risen to be the 2nd largest crude steel producer in the world and the largest producer of Sponge Iron. From a negligible global presence, the Indian Steel Industry is now globally acknowledged for its product quality.

The rapid pace of growth of the Industry and the

observed market trends called for certain guidelines and framework. Thus, the concept of the National Steel Policy was born with the aim to provide a roadmap of growth and development for the Indian Steel Industry. The National Steel Policy (NSP), 2005 was announced in November 2005 as a basic blueprint for the growth of a self-reliant and globally competitive Steel Sector. The long-term objective of the National Steel Policy, 2005 was to ensure that India has a modern and efficient Steel Industry of world standards, catering to diversified steel demand. The focus of the policy was to attain levels of global competitiveness in terms of global benchmarks of efficiency and productivity. Then, after a detailed review in 2017, the Government released the National Steel Policy, 2017, which laid down the broad roadmap for encouraging long-term growth for the Indian Steel Industry, both on demand and supply sides, by 2030-31, with a vision to create a technologically advanced and globally competitive Steel Industry that which would promote economic growth. At the same time, as a facilitator in the present-day de-regulated, liberalised economic/market scenario, the Government also announced a policy for providing preference to domestically manufactured Iron & Steel products in Government procurement. This policy seeks to accomplish the Hon'ble Prime Minister's vision of 'Make in India' with the objective of nation building and to

encourage domestic manufacturing and is applicable on all Government tenders where price bid is yet to be opened. To ensure quality scrap for the Steel Industry, the Govt. of India came out with a Steel Scrap Recycling Policy that aims to reduce imports, conserve resources and save energy.

The structure of the Indian Steel industry in 2023-24 along with the production for 2022-23 to 2023-24 is furnished in Table-1. Production of iron & steel, crude steel, pig iron and finished steel (alloy + non-alloy) by SAIL, TSL Group, RINL, AM/NS (erstwhile Essar Steel), JSWL, JSPL and other producers along with production of crude steel from oxygen route, electric arc furnace route and induction furnace route during the year 2019-20 to 2023-24 is reflected in Table-2 along with the production of sponge iron through gas-based & coal-based units during the year 2019-20 to 2023-24. The production of iron & steel by Public and Private Sectors during 2019-20 to 2023-24 is furnished in Table-3. The details on plant-wise capacity and production of crude/liquid steel are listed out in Table-4. Table-5 elucidates the production of crude/liquid steel by BOF and Electric route (EAF/IF). Crude Steel Scenario Region/State-wise covering number of Units, Annual Capacity and Production in respect of 2023-24 is shown in Table-6. Prices of steel are provided in Table-7.

Table – 1: Structure of the Indian Steel Industry, 2022-23 & 2023-24

(Capacity/Production: In million tonnes)

Sector	Annual Capacity 2023-24	Production	
		2022-23	2023-24
Crude Steel	179.515	127.2	144.29
(A) Producer-wise			
SAIL, TSL GROUP, RINL, AM/NS, JSW Steel Ltd, JSW Group, JSPL	96.712	80.06	85.37
Other Producers	82.803	47.14	58.92
(B) Sector-wise			
Public Sector	30.932	22.42	24.192
Private Sector	148.583	104.78	120.11
Hot Metal	NA	81.16	87.04
Pig iron	NA	5.86	7.36
Sponge Iron	NA	43.62	51.56
Total Finished Steel (Non alloy + Alloy + stainless)	NA	123.19	139.15
Total (None - Flat)		66.91	76.46
(Total Flat)		56.28	62.69
1) Finished Steel (Non-alloy)	NA	113.55	129.748
A) Non-Flat Products	NA	61.78	70.774
Bars & Rods		51.67	58.924
Structurals		8.60	10.306
Rly Materials		1.49	1.544
B) Flat Products	NA	51.77	58.974
PM Plates		5.33	5.751
HR Coil/Strip		46.43	53.223
2) Finished Steel (Alloy)	NA	6.87	6.019
A) Non-Flat Products	NA	4.28	4.659

Table- 1 (Concl'd.)

(Capacity/Production: In million tonnes)

Sector	Annual Capacity 2023-24	Production	
		2022-23	2023-24
B) Flat Products	NA	2.58	1.359
3) Finished Steel (Stainless)	NA	2.77	3.386
A) Non-Flat Products	NA	0.845	1.027
B) Flat Products	NA	1.92	2.36

Source: Annual Statistics, 2023-24 of JPC

Note: Finished steel data are reported in terms of Crude Steel Equivalent. This change is due to change in reporting system of JPC as approved by Ministry of Steel and Industry Experts.

Figures rounded off.

Table – 2: Production of Iron and Steel, 2019-20 to 2023-24

(In '000 tonnes)

Item/producers	2019-20	2020-21	2021-22	2022-23	2023-24
I. Pig Iron : Total	5421	4877	6262	5861	7364
SAIL,TSL GROUP , RINL, JSWL, JSPL	1193	1413	1462	1184	1909
Other Producers	4227	3464	4801	4677	5455
II. Sponge Iron: Total	37102	34376	39200	43621	51560
Gas based	6564	6175	8866	8007	9785
Coal based	30539	28201	30334	35614	41776
III. Crude Steel: Total	109137	103545	120293	127197	144299
Integrated steel Plants (SAIL, TSL GROUP, RINL, AM/NS, JSWL and JSPL)					
Oxygen Route	46735	43299	52515	56665	58881
EAF Units	21647	21754	22359	23389	26490
Other Producers					
Oxygen Route	1838	1786	2070	2127	2726
EAF Route	6719	7653	8138	4815	5121
IF Route	32198	29052	35211	40201	51081
IV. Total Finished Steel (Non alloy +Alloy + Stainless)	102621	96204	113597	123196	139153
SAIL,TSL GROUP , RINL, AM/NS, JSWL, JSPL	61286	55339	65055	72265	77698
Other Producers	41336	40865	48542	50931	61455

Source: Annual Statistics, 2023-24 of JPC

Note:

1. Finished steel data are reported in terms of Crude Steel Equivalent. This change is due to change in reporting system of JPC as approved by Ministry of Steel and Industry Experts; Figures rounded off.

2. TSL Group includes Bhushan Steel Limited, Tata Steel Long Products Limited & BMW - Gamharia (Jharkhand) along with TSL plants in Jamshedpur & Kalinganagar.

Table – 3: Production of Iron and Steel, 2019-20 to 2023-24

(By Sectors)

(In '000 tonnes)

Item/producers	2019-20	2020-21	2021-22	2022-23	2023-24
I. Pig iron: Total	5421	4877	6262	5861	7364
Public sector (SAIL+RINL)	614	669	634	401	418
Private sector (JSW, JSPL, TSL Group)	579	744	828	783	1490
Other Blast Furnace /Corex Unit)	4227	3464	4801	4677	5455
II. Crude steel /SEMIS: Total	109137	103545	120293	127197	144299
Public sector	20905	19515	22636	22429	24192
Private sector	88232	84030	97658	104768	120107
III. Finished steel (Non-Alloy+Alloy+ Stainless): Total	102621	96204	113597	123196	139153
Public sector (SAIL+RINL)	16029	13783	17579	18925	19986
Private sector (TSL + AM/NS (ESL)	86592	82421	96018	104271	119167

+JSWL+JSPL+Other Producers}

Source: Annual Statistics, 2023-24 of JPC

Note: Finished steel data are reported in terms of Crude Steel Equivalent. This change is due to change in reporting system of JPC as approved by Ministry of Steel and Industry Experts; Figures rounded off

Table – 4: Capacity and Production of Crude/Liquid Steel, 2021-22 and 2023-24

(By Principal Producers)

(In '000 tonnes)

Unit	Annual installed capacity Crude/Liquid steel	Production: Crude/Liquid steel		
		2021-22	2022-23	2023-24
SAIL	20632	17363	18292	19241
Rashtriya Ispat Nigam Ltd (Andhra Pradesh)	7300	5272	4137	4411
Private Sector				
JSW Group	28080	18023	23623	25608
TSL Group	21500	19464	19805	20783
AM/NS (Essar Steel Ltd)	9600	7295	6688	7683
Jindal Steel & Power Ltd	9600	7458	7509	7645
Others	79802	45419	47143	58928
Other BOF	6177	2070	2127	2726
Other EAF	7828	8138	4815	5121
IF Units	68797	35211	40201	51081

Source: Annual Statistics, 2023-24 of JPC

Table – 5: Production of Crude/Liquid Steel, 2019-20 to 2023-24

(By Route)

(In '000 tonnes)

Route/plant	2019-20	2020-21	2021-22	2022-23	2023-24
All Routes: (A+B) Total	109137	103545	120293	127197	144299
A. Oxygen Route: Total	48573	45085	54585	58792	61607
SAIL	15946	15054	17153	18055	18980
RINL	4749	4302	5272	4137	4411
TSL Group	16399	15163	17215	17514	18335
JSW group	9641	8780	10380	14236	14530
Other Oxygen Route	1838	1786	2070	2127	2726
B. Electric Route: Total	60564	58460	65708	68405	82692
Electric Arc Furnace	28367	29407	30498	28204	31611
SAIL	210	158	210	237	261
TSL Group	2126	1392	2249	2290	2448
AM/NS (Essar Steel Ltd)	7121	6696	7295	6688	7683
JSW group	6329	6080	7643	9387	11079
Jindal Steel & Power Ltd	5861	6859	4963	4786	5020
Lloyds Steel Ltd	495	452	681	538	Not available
Jindal Stainless Ltd	1418	1458	1812	1577	Not available
Bhushan Power & Steel Ltd	2901	3754	2720	-	Not available
Other Electric Arc Furnace	1905	2638	2926	2701	Not available
Electric Induction Furnace	32198	29052	35211	40201	Not available

Source: Annual Statistics, 2023-24 of JPC

Note:- TSL Group includes Bhushan Steel Ltd, Tata Steel Long Products, TSL Jamshedpur & TSL Kalinganagar

Figures rounded off

**Table – 6: Region / Statewise Crude Steel Scenario in respect of No. of Units,
Annual Capacity and Production: 2023- 24**

State	No. of Units				Annual Capacity ('000 tonnes)				Annual Production ('000 tonnes)			
	BOF	EAF	IF	TOTAL	BOF	EAF	IF	TOTAL	BOF	EAF	IF	TOTAL
TOTAL	20	39	1032	1091	71195	39522	68797	179515	61607	31611	51081	144299
Eastern Region	10	11	166	187	36477	11267	19349	58906	35263	9104	14467	58834
Arunachal Pradesh	0	0	3	3	0	0	130	130	0	0	46	46
Assam	0	0	12	12	0	0	237	237	0	0	110	110
Bihar	0	0	11	11	0	0	794	794	0	0	553	553
Jharkhand	3	1	26	30	17477	1000	3273	21750	16010	797	2513	19321
Meghalaya	0	0	7	7	0	0	236	236	0	0	62	62
Odisha	5	6	48	59	14300	9686	5102	29088	14418	7826	3496	25740
Tripura	0	0	2	2	0	0	60	60	0	0	17	17
West Bengal	2	4	57	63	4700	581	9516	14796	4834	481	7670	12985
Western Region	3	16	325	344	15000	23832	24437	63269	10917	18249	17899	47066
Chhattisgarh	2	5	95	102	10000	6003	10413	26415	6215	4162	8741	19119
Dadra and Nagar Haveli and Daman Diu	0	0	19	19	0	0	375	375	0	0	321	321
Goa	0	0	10	10	0	0	562	562	0	0	445	445
Gujarat	0	4	117	121	0	10200	5130	15330	0	7834	2761	10595
Madhya Pradesh	0	0	20	20	0	0	1394	1394	0	0	909	909
Maharashtra	1	7	64	72	5000	7629	6565	19194	4702	6254	4722	15678
Northern Region	0	9	318	327	0	1644	15193	16837	0	1293	9717	11010
Delhi	0	0	1	1	0	0	7	7	0	0	8	8
Haryana	0	4	19	23	0	852	260	1112	0	700	199	899
Himachal Pradesh	0	0	28	28	0	0	2495	2495	0	0	1372	1372
Jammu & Kashmir	0	0	8	8	0	0	213	213	0	0	151	151
Punjab	0	5	135	140	0	792	6623	7415	0	593	4326	4919
Rajasthan	0	0	33	33	0	0	1301	1301	0	0	891	891
Uttar Pradesh	0	0	52	52	0	0	2542	2542	0	0	1868	1868
Uttarakhand	0	0	42	42	0	0	1751	1751	0	0	902	902
Southern Region	7	3	223	233	19718	2780	9818	32316	15428	2965	8997	27390
Andhra Pradesh	2	0	25	27	7600	0	2246	9846	4705	0	2071	6776
Karnataka	4	1	22	27	11118	2500	1149	14767	9876	2759	1171	13805
Kerala	0	0	28	28	0	0	490	490	0	0	414	414
Puducherry (UT)	0	0	10	10	0	0	451	451	0	0	486	486
Tamil Nadu	1	1	107	109	1000	180	3104	4284	847	153	2635	3635
Telangana	0	1	31	32	0	0	2379	2479	0	54	2220	2274

Source: Annual Statistics, 2023-24 of JPC

Table – 7: Prices of Steel, 2021-22 to 2023-24

(Domestic Markets)

(In ₹ per tonne)

Grade	Market	2021-22	2022-23	2023-24 [#]
TMT Bars (ISI, 8mm)	Delhi	46180	45044	61650
MS Squares (8mm)	Delhi	49383	63694	61666
MS Angles (25 x 3mm)	Delhi	51011	63289	61751
Channels (75 x 40 mm)	Delhi	51335	64184	61609
Joists (150 x 75 mm)	Delhi	52416	64492	60489
Melting Scrap	Delhi	40515	42856	37785
Induction Ingots	Delhi	47258	50911	46581
TMT Bars (local 8 mm)	Mumbai	50682	63725	61495
MS Rounds (8 mm)	Mumbai	48358	61391	59432
MS Angles(40 x 6 mm)	Mumbai	50893	63665	61429
Joists (150 x 75 mm)	Mumbai	49986	61070	58209
Melting Scrap (Foundry Grade)	Mumbai	45980	52645	47712
Melting Scrap (Steel Grade)	Mumbai	43625	51779	47944
Melting Scrap (CRCA)	Mumbai	46201	50103	44467
Induction ingots	Mumbai	47950	51953	46748
Arc Ingots	Mumbai	47827	51700	46381
Concast Billet ingots	Mumbai	46627	51906	46848
TMT Bars (ISI, 8 mm)	Kolkata	51752	59127	57508
MS Squares (8 mm)	Kolkata	51161	58572	55178
MS Angles (25 x 3 mm)	Kolkata	51530	63410	61310
Channels (75 x 40 mm)	Kolkata	51318	63313	61742
Joists (150 x 75 mm)	Kolkata	50396	61082	57495
Induction Ingots	Kolkata	47740	51462	46606
Arc Ingots	Kolkata	48231	52311	46573
Concast Billet Ingots	Kolkata	47525	51730	46881
Induction ingots (round)	Gobind (Punjab)	46456	50325	45550
Blooms (SAIL, 150 mm)	Gobind (Punjab)	47052	51192	46610
Old Ship Breaking Scrap	Gobind (Punjab)	39898	49351	49099
Melting Scrap (rolling)	Gobind (Punjab)	41169	45768	40246
MS Rounds (10 mm)	Gobind (Punjab)	58165	57108	49385
MS Squares (8 mm)	Gobind (Punjab)	59723	59819	54509
MS Angles (25 x 3 mm)	Gobind (Punjab)	56954	58903	51055
MS Sponge Iron	Gobind (Punjab)	37985	40042	33224
MS Flat (3 x 20 mm)	Gobind (Punjab)	60044	59549	53026
Pig Iron Foundry Grade – A*	Mumbai	49710	57416	50513
Pig Iron Foundry Grade – B**	Punjab	48285	54195	46833
Pig Iron Steel Grade	Punjab	46050	50813	41556

[#]: Calculated by averaging monthly prices from April 2023 to March 2024

A*: Low Sulphur/Phosphorus i.e. 0.09% Max which is used in critical automotive engine components and specialize casting.

B**: High Sulphur / Phosphorus i.e. above 0.09% which is used in Non-Critical castings.

Source: Minerals & Metals Review.

A. Crude Steel

At 144.299 million tonnes (mt) in 2023-24, India's crude steel production increased by 13.44% as compared to 127.197 million tonnes in 2022-23. Given the above production for 2023-24 and with capacity at 179.515 million tonnes, crude steel capacity utilisation stood at 80% during 2023-24. The Crude Steel working Capacity and Capacity Utilisation during the last five years are furnished below in Table-8.

With 83% share, the Private Sector (120.107 million tonnes, led the crude steel production in 2023-24. In fact, India's crude steel production has been consistently led by the Private Sector in the last five years ending 2023-24.

The following are the two primary routes of crude steel production:

- BF/BOF route also called the Oxygen route.
- Electric route comprising of Electric Arc Furnace and Electric Induction Furnace.

Table – 8: Production and Working Capacity of Crude Steel (2019-20 to 2023-24)*(Quantity in million tonnes)*

Year	Working capacity	Production	% Utilisation
2019-20	142.299	109.137	77%
2020-21	143.914	103.545	72%
2021-22	154.062	120.293	78%
2022-23	161.299	127.197	79%
2023-24	179.515	144.299	80%

*Figures rounded off.**Source: Annual Statistics, 2023-24 of JPC***Basic Oxygen Furnace (BOF)**

Presently, there are around 20 Basic Oxygen Furnace units which are available in the Indian Iron & Steel Sector with a total capacity of 71.195 million tonnes. These units reported production of 61.6 million tonnes of crude steel through BOF route in 2023-24 at 86.53% of its capacity utilisation.

Electric Arc Furnace (including corex & MBF/EOF)

Crude steel produced in the Electric Arc Furnace (including corex & MBF/EOF) is mostly by recycling of steel scrap using Electric Arc Furnace (EAF). Electric Arc Furnace units, which are popularly known as mini steel plants, are significantly contributing to the production of steel in the country. Presently, in the Electric Arc Furnaces, there are 39 working units with total capacity of 39.52 million tonnes which produced 31.61 million tonnes crude steel through EAF route in the year 2023-24 at 80% of its capacity utilisation. The recent developments in EAF technology, viz, to increase oxygen consumption, to reduce power consumption and to reduce tap time have led to increase in metal production. The development of thin slab casting has made EAF route more productive. This route enables slab strips rolling at lesser cost, facilitating production of cheaper strips/sheets than those that can be achieved through BF/BOF route.

Induction Furnace (IF)

In case of the Induction Furnace (IF) segment, there are presently 1,032 IF working units with total capacity of 68.797 million tonnes which produced 51.081 million tonnes crude steel through IF route in 2023-24 at 74% of its capacity utilisation. These units are better than their EAF counterparts mainly because of their low cost of production and other factors mainly related to local market supply-demand conditions. Over the time, the IF sector has witnessed considerable technological upgradation with better charge-mix of DRI and refining facilities.

An analysis of the production of crude steel through various process routes indicates that the above performance has been contributed largely by the strong trends in growth of the electric route of steel making, particularly, the induction furnace route (encouraged by strong growth in sponge iron). This is reflected in Table-9.

On further analysis of the relative shares of the various routes in total production of crude steel, electric furnace route is dominated by the Induction Furnace route, which has emerged as a key driver of crude steel production in the country (Table-10).

Table – 9: Crude Steel Production (By Route)*(In million tonnes)*

Process Route	2019-20	2020-21	2021-22	2022-23	2023-24
Oxygen	48.573	45.085	54.585	58.792	61.607
EAF	28.366	29.407	30.498	28.204	31.611
IF	32.198	29.052	35.211	40.201	51.081
Total	109.137	103.545	120.293	127.197	144.299

*Source: Annual Statistics, 2023-24 of JPC***Table– 10: Process Route Share in total Production (In%)**

Process Route	2019-20	2020-21	2021-22	2022-23	2023-24
Oxygen	45	44	45	46	43
EAF	26	28	25	22	22
IF	29	28	30	32	35

Source: Annual Statistics, 2023-24 of JPC

B. Hot Metal

At 87.04 million tonnes in 2023-24, domestic hot metal production increased by 7.24% over 2022-23.

C. Total Finished Steel (Crude Steel Equivalent)

1. Trend in Production of Total Finished Steel

In the year 2023-24, the production of finished steel, stood at 139.153 million tonnes.

An analysis of the broad divisions in terms of the total production of finished steel reflects the following —

- Contribution of the Non-alloy Finished Steel stood at 129.74 million tonnes.
- Contribution of the Alloy finished Steel stood at 6.091 million tonnes.
- Contribution of the Stainless Steel stood at 3.386 million tonnes.

2. Trends in Consumption of Total Finished Steel

Finished Steel consumption stood at 136.291 million

tonnes in 2023-24 as compared to 119.893 million tonnes during 2022-23, notching down a growth of 13.67%. Data on overall consumption of total finished steel (non-alloy + alloy + stainless) during the last five years is provided in Table-11 and these indicate a steady growth in domestic steel consumption during the said period. However, reflecting on the year-on-year (yoy) rate of growth it could be said that such a trend, impacted and shaped largely by macroeconomic factors and prevailing steel market conditions, has followed a zig-zag track (Table-12).

Table-13 highlights the growth pattern yoy along with share of domestic total finished steel consumption, in terms of its two broad components – flat steel and non-flat/long steel – in 2023-24 and 2022-23. Both these components include non-alloy, alloy and stainless steel constituents as well.

Table-14 shows detailed consumption data for major categories of finished steel in 2023-24 over 2022-23 in terms of crude steel equivalent of finished steel as per the present reporting system.

**Table – 11: Total Finished Steel Consumption (Non-alloy + Alloy + Stainless steel)
(Crude Steel Equivalent) (2019-20 to 2023-24)**

(In '000 tonnes)

Year	Total Finished Steel Consumption	% yoy change
2019-20	100171	1.5
2020-21	94891	-5.3
2021-22	105752	11.44
2022-23	119893	13.37
2023-24	136291	13.67

Source: Annual Statistics, 2023-24 of JPC

**Table – 12: Total Finished Steel Consumption - Non-alloy / Alloy / Stainless Steel wise
Crude Steel Equivalent (2022-23 to 2023-24)**

(In '000 tonnes)

ITEM	2022-23		2023-24	
	Qty	% share	Qty	% share
Total Finished Steel (Non alloy+ alloy + stainless)	119893	-	136291	-
Non-alloy Finished Steel	110421	92.1	125617	92.16
Alloy Finished Steel	6038	5.04	6922	5.07
Stainless steel	3435	2.87	3752	2.75

**Table – 13: Total Finished Steel Consumption — Non-Flat/Flat wise
Crude Steel Equivalent (2022-23 to 2023-24)**

(include non-alloy + alloy + stainless)

(In '000 tonnes)

ITEM	2022-23		2023-24	
	Qty	% share	Qty	% share
Total Finished Steel (Non Flat+ Flat)	119893	-	136291	-
Non-Flat Finished Steel	65294	54.46	74415	54.60
Flat Finished Steel	54599	45.53	61876	45.40

Source: Annual Statistics, 2023-24 of JPC

Table – 14: Detailed Consumption Major Categories of Total Finished Steel in 2022-23 over 2023-24

(Crude Steel Equivalent)

(In '000 tonnes)

ITEM	2022-23	2023-24	% yoy Change
Total Finished Steel (Non-alloy + Alloy + Stainless)	119893	136291	13.7
1. Finished Steel (Non-alloy)	110422	125617	13.8
a) Non-Flat Products	60403	68970	14.2
Bars & Rods	50530	57344	13.5
Structurals	8319	10080	21.2
Rly Materials	1554	1545	(minus) 0.5
b) Flat Products	50020	56647	13.3
PM Plates	4800	5387	12.2
HR Coil/Strip & equivalent	45220	51260	13.4
2. Finished Steel (Alloy)	6038	6922	14.7
a Non-flat Products	4215	4597	9.1
b) Flat Products	1823	2326	27.6
3. Finished Steel (Stainless)	3435	3752	9.2
a) Non-flat Products	678	849	25.2
b) Flat Products	2757	2903	5.3

Source: Annual Statistics, 2023-24 of JPC

D. Pig Iron

Pig iron is a product in solid (lumpy) form obtained upon solidification of Hot Metal in Pig Casting Machine. It is called Pig or Pig Iron because of its typical humpy shape. It is a basic input for making iron casting, which finds application in industrial and other sectors of economy. Pig iron is one of the basic raw materials required by the Foundry & Casting Industry for manufacturing various types of castings for the engineering section. In advanced countries pig iron is also used as a partial substitute of melting scrap in the charge mix of Electric Arc Furnaces. Pig Iron is mainly classified into two grades, 'Basic Grade' used for making steel and 'Foundry Grade' used for manufacturing iron castings. Domestic production of pig iron lags behind and is not in tandem with the demand. Efforts were, therefore, made to increase pig iron manufacturing facilities in the Secondary Sector. Production of pig iron in merchant units

in the Secondary Sector got its first major boost in 1992. Thereafter, the growth of this Sector accelerated greatly as Foundry-grade pig iron fast became the preferred raw material for the quality conscious foundries.

The working capacity of hot metal & pig iron during 2023-24 was reported as 95.83 million tonnes. The location and capacity of principal pig iron/ hot metal as well as State-wise capacity and production of hot metal and pig iron units are furnished in Table-15 & Table-16 respectively. The domestic production of pig iron was at 7.36 million tonnes in 2023-24.

As a result of various policy initiatives taken by the Government, the Private Sector showed considerable interest in setting up new pig iron units, especially in the post-liberalised period. This has resulted in drastic change in the contribution of Private Sector producers.

Table – 15: Location and Capacity of Principal Pig Iron Units

(In '000 tonnes)

Sl.No.	Plants/unit	Location	Capacity
1	Adhunik Metaliks Ltd, Odisha	Odisha	70
2	Ankit Metal and Power Ltd	West Bengal	12
3	Aparant Iron and Steel Pvt. Ltd	Goa	125
4	Arcelor Mittal Nippon Steel India Ltd, Surat	Gujarat	3490
5	Arjas Steel Pvt Ltd (Gerdau Steel)	Andhra Pradesh	300
6	Atibir Industries Co. Ltd (Unit II)	Jharkhand	600
7	B R G Iron and Steel Co. Pvt. Ltd	Odisha	120
8	Balmukund Sponge and Iron Pvt. Ltd	Jharkhand	40
9	Bhushan Power and Steel Ltd, Odisha	Odisha	2500
10	Electro Steels Ltd, Jharkhand	Jharkhand	1450
11	Electrosteel Castings Limited, Khardah	West Bengal	250

Table- 15 (Concl'd.)

(In '000 tonnes)

Sl.No.	Plants/unit	Location	Capacity
12	Electrotherm (India) Ltd	Gujarat	277
13	Ispat Damodar Ltd	West Bengal	15
14	J S W Steel Ltd, Salem (Siscol)	Tamil Nadu	1000
15	J S W Steel Ltd, Vijaynagar	Karnataka	12000
16	Jai Balaji Industries Ltd West Bengal Unit-3	West Bengal	429
17	Jai Balaji Industries Ltd West Bengal Unit-4	West Bengal	81
18	Jai Balaji Industries Ltd - I	West Bengal	30
19	Jayaswals Neco Inds Ltd	Chhattisgarh	650
20	Jindal Steel and Power Ltd, Chhattisgarh	Chhattisgarh	2125
21	Jindal Steel and Power Ltd, Odisha	Odisha	3200
22	Jsw Ispat Special Products Ltd, Raigarh	Chhattisgarh	613
23	Jsw Steel Ltd, Dolvi	Maharashtra	3500
24	K I C Metaliks Ltd	West Bengal	165
25	Kalyani Steels Ltd	Karnataka	480
26	Kirloskar Ferrous Inds Ltd	Karnataka	385
27	Kohinoor Steels Ltd	Jharkhand	48
28	Makers Casting India Pvt Ltd	Jharkhand	2
29	Mideast Integrated Steels Ltd	Odisha	460
30	Narsingh Ispat Ltd	Jharkhand	83
31	Neelachal Ispat Nigam Ltd	Odisha	1099
32	Neo Metaliks Ltd	West Bengal	188
33	Niranjan Hi-Tech Ltd	Jharkhand	15
34	Rashmi Metaliks Limited	West Bengal	170
35	SAIL–Bhilai Steel Plant	Chhattisgarh	3925
36	SAIL–Bokaro Steel Plant	Jharkhand	4360
37	SAIL– Durgapur Steel Plant	West Bengal	1802
38	SAIL– IISCO Steel Plant	West Bengal	2500
39	SAIL–Rourkela Steel Plant	Odisha	4400
40	SAIL–Visveswaraya Iron And Steel Ltd, Bhadrabati	Karnataka	118
41	Sathavahana Ispat Ltd	Andhra Pradesh	240
42	Satyam Ferro Tech Ltd	Jharkhand	30
43	Shyam SEL and Power Ltd, Jamuria	West Bengal	60
44	SLR Metaliks Ltd	Karnataka	240
45	Sree Metaliks Ltd	Odisha	36
46	Sri Kalahasthi Pipes Limited (Lanco)	Andhra Pradesh	300
47	Suraj Products Ltd	Odisha	24
48	Swati Concast and Power Pvt Ltd	Jharkhand	43
49	Tata Metaliks Ltd, West Bengal	West Bengal	600
50	Tata Steel BSL Ltd, Odisha	Odisha	3919
51	Tata Steel Long Products Limited	Jharkhand	650
52	Tata Steel Ltd, Jamshedpur Works	Jharkhand	9600
53	Tata Steel Ltd, Kalinganagar Works	Odisha	3000
54	Uttam Galva Metaliks Ltd	Maharashtra	600
55	Vedanta Limited	Goa	625
56	VISA Steel Ltd	Odisha	225
57	VIZAG Steel Plant	Andhra Pradesh	6300

Table – 16: State-wise Capacity and Production of Hot Metal and Pig Iron (2023-24)*(capacity & production in '000 tonnes)*

State	No.of working Units	Working Capacity	Annual Production	
			Hot metal	Pig Iron
Total	55	95832	87045	7364
Andhra Pradesh	3	7900	5467	471
Chhattisgarh	6	15197	10550	677
Goa	1	900	0	831
Gujarat	3	4267	4218	227
Jharkhand	8	16963	17938	775
Karnataka	6	13403	12730	990
Maharashtra	3	9200	8672	102
Odisha	11	19797	20314	1525
Tamil Nadu	1	1000	1079	15
Uttarakhand	1	16	2	4
West Bengal	12	7190	6076	1747

Source: Annual Statistics, 2023-24 of JPC

E. Sponge Iron

India is the largest producer of sponge iron in the world. Sponge iron means porous iron produced by direct reduction (DR) process which may be either gas-based or coal-based. This is a solid-state reaction process (i.e., solid-solid or solid-gas reaction) by which removable oxygen is removed from the iron ore, using coal or reformed natural gas as reductants, below the melting and fusion point of the lump ore or agglomerates of fine ore. The external shape of the ore remains unchanged. Due to removal of oxygen, there is about 27 to 30 per cent reduction in weight, a honey combed microstructure remains which is called Sponge Iron (means solid porous iron, lumps/pellets, with many voids filled with air). It is also known as Direct Reduced Iron (DRI).

Iron & steel scrap is one of the essential requirements for manufacture of steel in Mini-steel Industry. It is also consumed by some major steel plants. Scrap, especially from the Ship Breaking Industry supplies substantial quantity of re-rollable steel as well as steel scrap for the Iron & Steel Industry and Secondary Sector, such as, Micro, Small & Medium Enterprises (MSME). Other form of scrap, i.e., end-of-life cycle scrap or obsolete scrap these get generated in large quantities and form substantial reserves of scrap that could be efficiently put to use. Iron scrap is available in the country in the form of pressed bundles, a mixture of used steel components (called as a commercial scrap), turnings & borings and heavy melting scrap. These are generated by industries of all sectors like automobiles, railways and engineering workshops.

The collection and processing of scrap in an organised manner is undertaken by a few units in the country. In the local market, scrap is supplied by dealers who in turn

arrange to have scrap collected manually or through sub-dealers.

The consumption of scrap is mainly reported by Induction Furnace & Electric Arc Furnace units, Integrated Steel Plants and Alloy Steel & Foundry industries. Scraps are used in the Steel Sector after recycling. Recycling of one tonne of steel scrap saves about 1.1 tonnes of iron ore, 0.6 to 0.7 tonnes of coking coal and around 0.2 to 0.3 tonnes of fluxes. Besides, saving of energy by about 16–17%. It also reduces the water consumption and GHG emission by 40% and 58% respectively. Recycling scrap helps in conservation of energy as remelting of scrap requires much less energy than production of iron or steel from iron ore. Also, the consumption of iron and scrap by remelting reduces the burden on landfill disposal facilities and prevents the accumulation of abandoned steel products in the environment. It increases the availability of semi-finished material, which otherwise would have to be produced using the ore. Thus, it helps in conservation of natural resources.

Ship Breaking

Ship breaking has been a major source of scrap generation. Ship breaking activities are carried out at various places on the Indian coast, the largest concentration being in the West coast. Private entrepreneurs handle the task of ship breaking in India. It is a labour-intensive job, and in India, it is a cost-efficient activity. Locations of present ship breaking activities are:

- (i) Alang and Sosiya yards in Bhavnagar district, Gujarat,
- (ii) Sachana district, Gujarat
- (iii) Mumbai and
- (iv) Kolkata

Alang & Sosiya yards account for 98% concentration of the Ship Breaking Industry in India. The yard has the capacity to recycle about 450 ships per year generating re-rollable steel of > 4.5 million tonnes per annum. There are total of 167 plots available for ship recycling spread over 10 km stretch along the coast of Alang.

The NGO Shipbreaking Platform is a global coalition of environmental, human and labour rights organisations working to promote safe and environmentally sound ship recycling practices. The coalition quickly evolved from being a European Platform to a global one, including NGOs based in the major shipbreaking countries, such as, India, Bangladesh, Pakistan and Turkey. It now has 18 member organisations and six partners in 12 countries. The Platform is recognised by United Nations agencies, the European Union and leading media outlets as the preeminent international civil society advocacy organisation on ship recycling.

Today, Alang possibly represents the single largest concentration of Ship-breaking Industry in the world. The average life of an ocean-going ship is about 25–30 years. About 40% of the ships broken are dry cargo ships, while the remaining 60% of the ships broken are wet cargo, tanker–specialised ships etc. These recyclable steels mainly as steel scrap provide feed to Steel and Foundry Industry in India. The steel generated from ship recycling contributes to around 1% to 2% of the domestic steel demand.

MSTC Ltd

(Formerly Metal Scrap Trade Corp. Ltd)

MSTC Limited was incorporated as “Metal Scrap Trade Corporation Limited”, under the provisions of the then Companies Act, 1956 on September 9, 1964 at Kolkata for regulating export of ferrous scrap from India. The status of the Company underwent a change in February 1974 when it was made a subsidiary of Steel Authority of India (SAIL). In the year 1982-83, the Corporation was converted into an independent PSU under administrative control of Ministry of Steel. It was the canalising agency for import of carbon steel melting scrap, sponge iron, hot briquetted iron and re-rollable scrap till February 1992.

Presently, the Company has diversified mainly into providing e-auction/e-procurement services. Under this segment, the Company undertakes disposal of ferrous and non-ferrous scrap arisings, surplus stores, condemned plants, minerals, Agri & forest produce etc. from Public Sector Undertakings and Government Departments

including private companies. The Trading Division is engaged in import as well as domestic sourcing of bulk industrial raw material for actual users as well as traders. This Division looks after sourcing, purchase and sale of industrial raw materials like low ash metallurgical coke, HR coil, naphtha, crude oil, coking coal, steam coal, line pipes etc. on behalf of customers across steel, oil & gas, power sectors under Private and Public Sector. It also undertakes trading of items within the country in competition with any other private trader.

Ferro Scrap Nigam Ltd (FSNL)

FSNL is a wholly owned subsidiary of MSTC Ltd under the Ministry of Steel. The Company undertakes the recovery and processing of scrap from slag and refuse dumps in the nine steel plants at Bhilai, Bokaro, Burnpur, Durgapur, Rourkela, Visakhapatnam, Dolvi, Duburi and Haridwar and also at Rail Wheel factory Bengaluru. The scrap so recovered is returned to the steel plants for recycling/disposal and the Company pays processing charges on the quantity recovered at varying rates depending on the category of scrap. Scrap is generated during iron & steel making and also in the rolling mills. In addition, the Company provides steel mill services, such as, scarfing of slabs, handling of BOF slag, etc.

TRADE POLICY

As per the ITC (HS), 2022 Schedule 2 Import Policy of DGFT, the imports of primary forms of pig iron, spiegeleisen, sponge iron, ferroalloys, stainless steel, remelting scrap, as also the semi-finished products of iron, non-alloy steel or stainless steel (such as flat-rolled products, bars, rods, coils and wires), primary and semi-finished forms of other alloy-steels, etc. are unrestricted. Similarly, the exports are also allowed freely.

WORLD REVIEW

The world production of pig iron in 2023 was about 1434 million tonnes which increased by 1.12% as against 1418 million tonnes in 2022. China (61%), India (9%) and Japan & Russia (4% each), were the main producers of pig iron including sponge iron and direct reduced iron (DRI) (Table-17).

The world crude steel production in 2023 marginally increased by 0.32% to 1,877 million tonnes from 1871 million tonnes in 2022. China was the top producer accounting for 54% of world's crude steel production, followed by India (7%), Japan (5%) and USA (4%) (Table-18).

Table – 17: World Production of Pig Iron

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World Total	1450000000	1418000000	1434000000
China	868567800	863827800	871013000
India	116600000	122200000	135900000
Japan	70344478	64146549	63041584
Russia	61600000	59300000	62400000
Korea, Rep. of	46440000	43400300	45200000
Iran	34300000	36400000	36900000
Brazil	28530000	26700000	25710000
Germany	25500000	23700000	23600000
USA	18200000	22200000	22500000
Taiwan	15200000	13400000	12200000
Other countries	164793636	143143898	135199054

Figures rounded off ;

Source: BGS, World Mineral Production, 2019-2023

Note: The data in this table include sponge iron and direct reduced iron (DRI), where these have been separately identified.

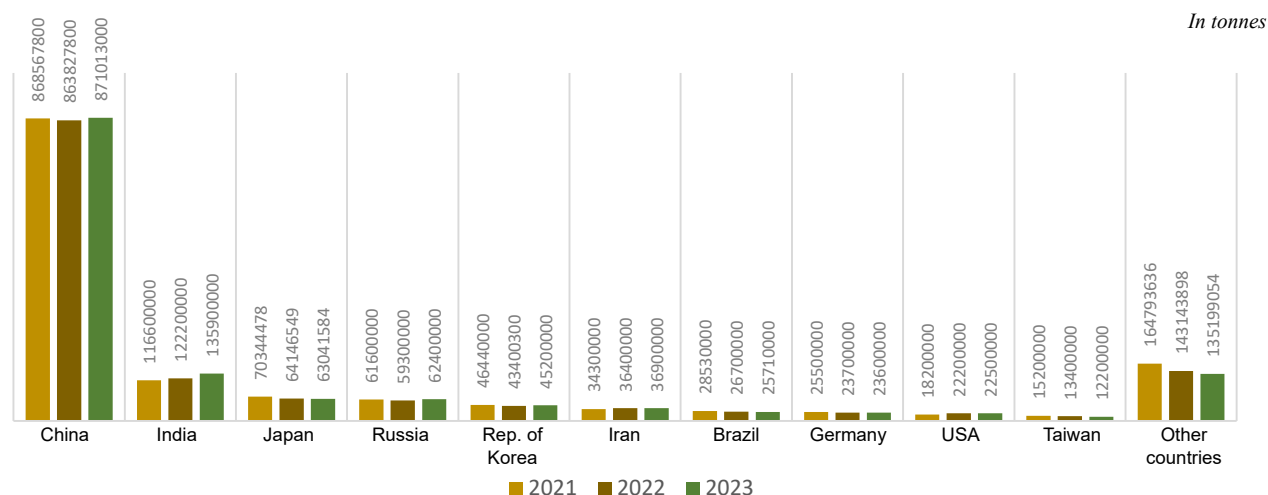


Fig. 1: Countrywise Production of Pig Iron

Table – 18: World Production of Steel Crude

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World: Total	1949000000	1871000000	1877000000
China	1032788000	1013003000	1019081000
India	118201400	125377200	140706000
Japan	96181132	88946808	86748669
USA	85791400	80535000	81392200
Russia	77560000	71930000	77060000
Korea, Rep. of	70418000	65846000	66683000
Germany	40241200	36860400	35438300
Turkey	40360000	35133800	33713700
Brazil	36071000	34089000	32030000
Italy	24412700	21598100	21054100
Other countries	326624189	297487220	283539020

Source: BGS, World Mineral Production, 2019-2023

Figures rounded off;

Note(s)

(1) unless otherwise indicated, these figures include production from scrap.

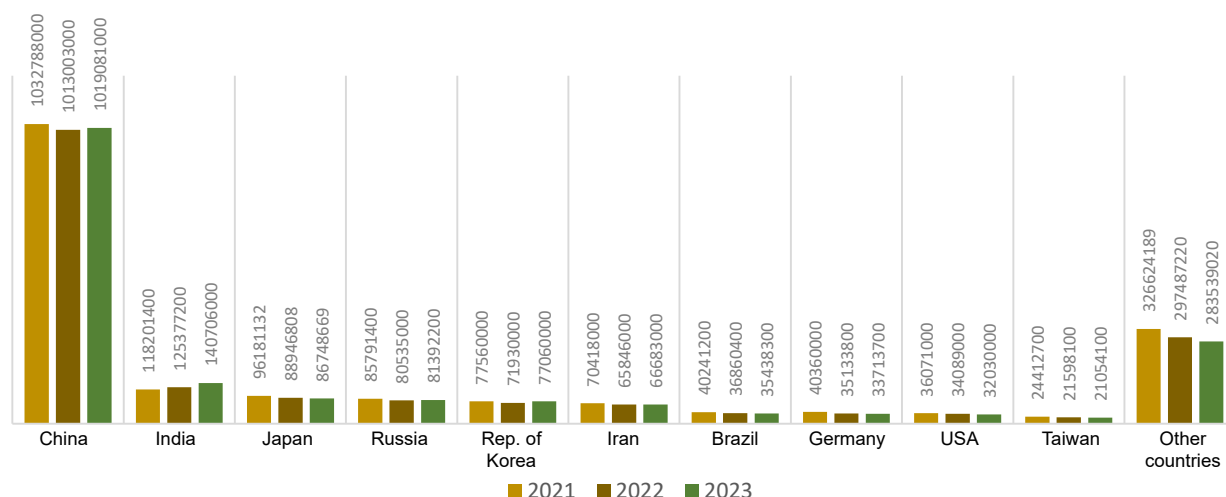


Fig. 2: Countrywise Production of Steel Crude

FOREIGN TRADE

Exports

In terms of value, exports of iron & steel (total) decreased by 0.01% to ₹ 1,55,071 crore in the year 2023-24 from ₹ 1,55,087 crore in the previous year. Iron & Steel exports in 2023-24 comprised mainly of Semi-finished Steel

(including Steel Ingots) with ₹ 40,606 crore and Finished Steel Including Cold Rolled Sheet with ₹ 48,730 crore and Other Finished Steel, NES with ₹ 55,054 crore. In terms of value, exports of iron & steel in the year 2023-24 were mainly to USA (17%), Italy (11%), UAE & Belgium (5% each) and Nepal (4%) (Tables- 19 to 30).

Table 19: Export of Iron and Steel

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	1550875589	**	1550719423
USA	**	283827657	**	256353538
Italy	**	141886768	**	166200340
UAE	**	91275743	**	84054628
Belgium	**	73303313	**	72926293
Nepal	**	79862785	**	68302680
UK	**	44558953	**	60618085
Spain	**	44440622	**	60358996
Germany	**	53853202	**	55015603
Saudi Arabia	**	39439169	**	53119281
Netherlands	**	41065479	**	35400557
Other countries	**	657361898	**	638369422

Figures rounded off

**Table – 20: Export of Iron and Steel
(Finished Steel Inc.Cold-Rolled Sheet)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5226893	524940706	4278693	487304517
USA	461107	91274448	312392	61952621
Italy	375527	36205274	440950	41245632
Belgium	285744	30194861	333501	30686199
Saudi Arabia	111104	11599668	275197	29058573
Spain	192711	19408599	307879	26964198
UAE	420228	35982591	203414	25514860
UK	124528	12489161	237709	18846318

Table- 20 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Russia	35727	8122561	78398	17529464
Poland	49673	7855796	159114	15583170
Korea, Rep. of	11652	3471829	60793	13134901
Other countries	3158892	268335918	1869346	206788581

Figures rounded off

In tonnes

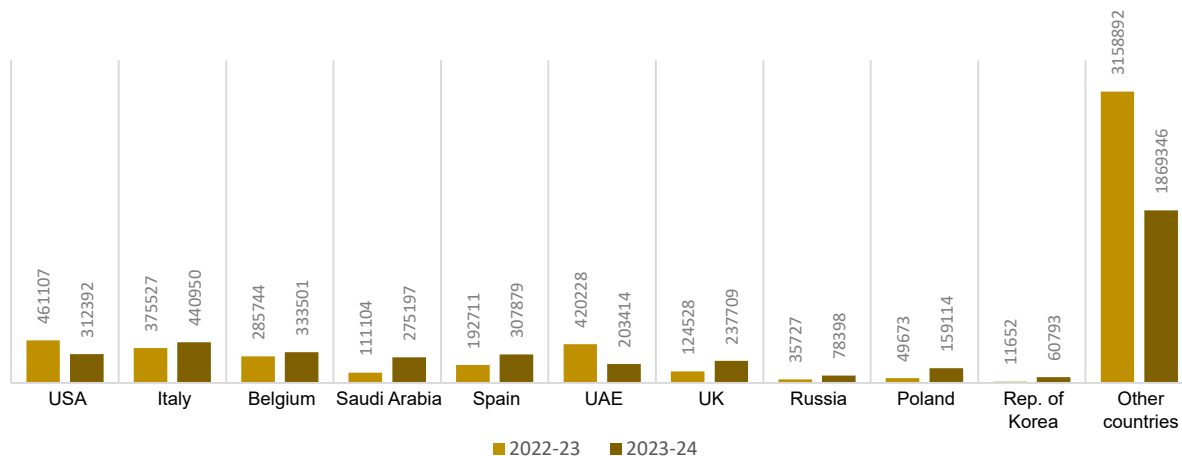


Fig. 3: Countrywise Export of Iron and Steel

Table – 21: Export of Iron and Steel (Steel Wire)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	182786	43621658	176995	37812544
USA	24000	7934488	21038	5885160
Netherlands	19115	4533008	17920	3557125
Russia	9227	2902616	9441	2902338
Turkey	9698	2476907	9330	2193060
Germany	8870	2845487	6184	1992471
Italy	7166	2220777	6495	1829701
UAE	11081	1695242	11201	1721071
France	7327	1866017	7429	1690377
UK	6339	1488491	7098	1452143
Brazil	6707	1400069	8310	1446401
Other countries	73256	14258556	72549	13142697

Figures rounded off

Table – 22: Export of Iron and Steel (Other Finished Steel, Nes)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	526170184	**	550543220
USA	**	158015751	**	170362639
Germany	**	31216174	**	29268368
UAE	**	22481945	**	29079141
UK	**	22038065	**	23794773
Saudi Arabia	**	16238323	**	20359256
Italy	**	14053793	**	16280682
Netherlands	**	16343187	**	15930244

Table- 22 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Australia	**	13221923	**	13960946
Mexico	**	8884466	**	12284213
Canada	**	13015464	**	11250677
Other countries	**	210661093	**	207972281

Figures rounded off

Table – 23: Export of Iron and Steel (Semi-finished Steel Incl. Steel Ingots)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	4545495	383068371	5665550	406065232
Italy	1212844	87561865	1744940	105460964
Belgium	384180	34692546	537969	36116510
Spain	204795	16162267	392799	24511459
UAE	376597	28789893	323720	24220246
Nepal	290068	16544797	466561	23823481
Vietnam	289625	15561918	404480	20127907
UK	87622	8459032	274021	16393213
Turkey	324042	25716509	156521	14335319
Germany	39704	11193959	50888	14035200
USA	130303	15282195	84825	11674622
Other countries	1205715	123103390	1228826	115366311

Figures rounded off

Table – 24: Export of Iron and Steel: Alloy Steel (Granules)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	487	39652	575	42857
Oman	56	3965	142	10115
UAE	196	13406	81	6629
Nepal	38	3358	155	5300
Italy	--	--	22	3944
Qatar	34	3159	42	3202
Saudi Arabia	46	5815	37	2796
Ghana	--	--	27	2309
Bangladesh	30	2619	27	2287
Netherlands	--	--	1	1958
USA	++	363	++	972
Other countries	87	6967	41	3345

Figures rounded off

Table – 25: Export of Iron and Steel: Alloy Steel (Powder)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	78	29057	155	46454
Taiwan	8	3723	88	22476
Turkey	65	23102	36	13566

Table- 25 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
China	--	--	28	8843
Sweden	--	--	2	837
Thailand	--	--	1	327
Indonesia	2	830	++	192
Bangladesh	2	893	++	82
Spain	--	--	++	46
Netherland	--	--	++	22
UAE	--	--	++	22
Other countries	1	509	++	41

Figures rounded off

Table – 26: Export of Iron and Steel (Scrap)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	14610	1547783	9306	1112446
Sweden	2408	799743	2345	737293
Bhutan	9326	270414	4610	175040
China	788	254855	216	31462
USA	9	3269	74	28826
UAE	276	22128	240	23798
Malaysia	56	26127	98	23717
Singapore	652	14429	658	18036
Oman	++	101	327	12991
Taiwan	--	--	174	12833
Korea, Rep. of	48	3269	120	8274
Other countries	1047	153448	444	40176

Figures rounded off

Table – 27: Export of Iron and Steel (Sponge Iron)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1096806	38009852	1323458	39551146
Nepal	614660	21223905	849235	24399612
Bangladesh	380570	13180774	381008	12200355
Bhutan	88154	3009656	85683	2565564
Madagascar	2553	103939	3270	123570
USA	588	79755	651	101334
Sri Lanka	320	11177	1139	37663
Sudan	2764	94333	1099	36979
Djibouti	--	--	760	25571
Netherlands	32	7714	60	15737
Korea, Rep. of	56	10017	76	13211
Other countries	7109	288582	477	31550

Figures rounded off

Table – 28: Export of Iron and Steel (Stainless Steel)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	95970	18120498	63120	11564385
USA	37600	7834120	15832	2920187
Canada	40653	5660267	21669	2357586
Egypt	1438	229086	5876	1264949
UAE	1683	496694	2145	637055
Saudi Arabia	915	628603	1171	483600
Italy	3343	600670	2243	449838
Oman	1154	139535	2113	366812
Brazil	21	8897	3273	322485
Costa Rica	4	94715	10	203374
Nigeria	919	196296	838	196184
Other countries	8240	2231615	7950	2362315

Figures rounded off

Table – 29: Export of Iron and Steel Material

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	162394	15327828	173718	16676622
USA	32719	3402840	35633	3427177
UAE	18943	1792464	36874	2850612
Tanzania	219	28355	8378	2547506
Italy	12549	1244378	11166	929567
Canada	8572	692107	13285	909636
Belgium	16498	1400361	8204	651956
Mexico	1421	109551	6550	574910
Oman	6059	666086	4963	468649
Puerto Rico	4537	364835	5207	409018
Bhutan	4330	330315	4727	339189
Other countries	56547	5296536	38731	3568402

Figures rounded off

Table – 30: Export of Pig and Cast Iron Incl. Spiegeliesen

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	672227	36969622	429042	17001783
USA	423938	25461746	157828	6514402
Turkey	31310	1251499	79042	3052426
Bangladesh	16722	955646	61896	2185282
Nepal	46375	2185608	52375	2121966
UAE	29662	1363527	30350	999714
Japan	10564	590310	10571	543964
China	17921	285148	20500	354514
Thailand	4554	303832	1874	186873
Korea, Rep. of	1342	119223	1471	152508
Germany	337	118673	282	127393
Other countries	89502	4334410	12853	762741

Figures rounded off

Imports

In terms of value, imports of iron & steel (total) in 2023-24 increased by 9.27% to ₹ 1,86,460 crore from ₹ 1,70,638 crore in the previous year. Iron & steel imports in 2023-24 comprised mainly of finished Steel Including Cold Rolled

Sheet with ₹ 55,837 crore, Scrap with ₹ 52,255 crore, Other Finished Steel, NES with ₹ 28,908 crore and Semi-finished Steel including Ingots with ₹ 39,227 crore. In terms of value, the imports of iron & steel in 2023-24 were mainly from China (20%) followed by Republic of Korea (13%), Japan (8%), Indonesia (7%) and USA (6%) (Tables- 31 to 41).

Table – 31: Import of Iron and Steel

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	1706382409	**	1864604839
China	**	297091401	**	374639775
Korea, Rep. of	**	239963048	**	248922376
Japan	**	133961940	**	156112945
Indonesia	**	119794804	**	132424572
USA	**	115363297	**	108652897
Vietnam	**	47560919	**	73816916
UK	**	70053856	**	69669898
UAE	**	86646387	**	61463003
Germany	**	54787808	**	56072106
Singapore	**	44068167	**	49735043
Other countries	**	497090782	**	533095308

Figures rounded off

Table – 32: Import of Iron and Steel (Finished Steel Inc. Cold-Rolled Sheet)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3980055	546607218	4510768	558374413
China	1377934	171819045	1635284	190564549
Korea, Rep. of	892193	94445032	1040951	99860014
Japan	645996	92437503	591703	87601717
Vietnam	115550	14833763	223272	24878505
Hong Kong	146643	12903266	203455	20839534
Germany	34850	11521292	38497	12946884
USA	21081	8263052	28516	10129170
Belgium	72333	11846448	51444	9370251
Russia	41950	8921310	73975	9308082
Indonesia	135105	34318939	48179	8875478
Other countries	496420	85297568	575492	84000229

Figures rounded off

In tonnes

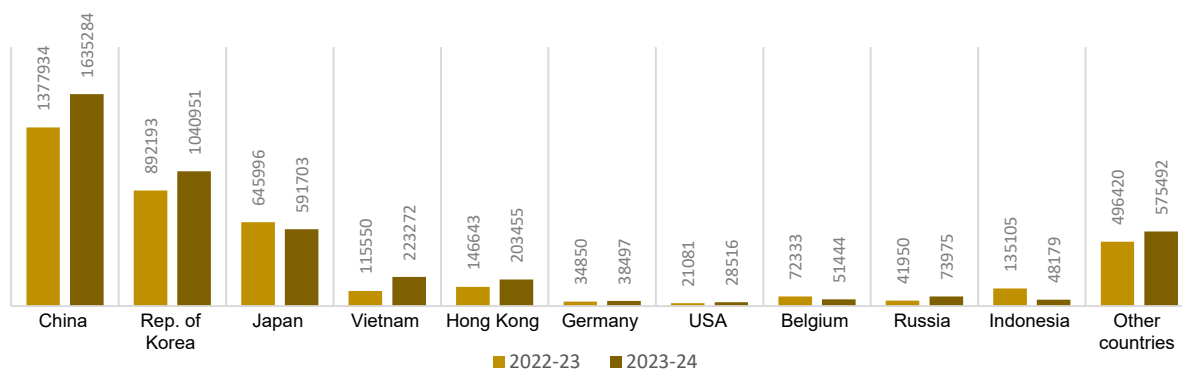


Fig. 4: Countrywise Import of Iron and Steel

Table – 33: Import of Iron and Steel (Steel Wire)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	149052	22150739	170246	22056913
China	60734	7916035	66745	7254121
Korea, Rep. of	21301	3104788	33142	4225231
Japan	7941	2696480	6870	2462376
Malaysia	15926	1906342	16246	1600235
Vietnam	10138	1167374	11569	1102348
Thailand	4065	548920	7419	831454
UAE	12786	810605	10864	709250
Germany	1623	830210	844	655006
USA	354	383393	782	505562
Italy	2197	379738	2486	441513
Other countries	11987	2406854	13279	2269817

Figures rounded off

Table – 34: Import of Iron and Steel (Other Finished Steel, Nes)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	262095115	**	289088811
China	**	91971668	**	105823057
Japan	**	17130007	**	21890836
Germany	**	21263515	**	20983763
Korea, Rep. of	**	20521192	**	20182466
Thailand	**	13839545	**	18304773
USA	**	16002604	**	16321367
Singapore	**	8638584	**	11472615
Italy	**	9975975	**	11035847
Malaysia	**	5212273	**	7982672
Vietnam	**	6813171	**	6004514
Other countries	**	50726581	**	49086901

Figures rounded off

Table – 35: Import of Iron and Steel (Semi-Finished Steel Incl. Steel Ingots)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2932868	265021615	5119221	392271023
Indonesia	408879	78617329	739019	116323780
Korea, Rep. of	1387829	97902759	1602016	98641106
China	176313	14772291	939815	55195101
Japan	253164	17279607	646047	37389889
Vietnam	173283	9263446	587394	31689466
Romania	1842	341634	84905	10832206
Russia	308936	16881744	206580	8667330
Sweden	13307	5501093	12638	5689854
UAE	23094	2023495	54413	3431682
Belgium	12783	1726028	23093	3337623
Other countries	173438	20712189	223301	21072986

Figures rounded off

Table – 36: Import of Iron and Steel: Alloy Steel (Granules)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	18666	1586757	25155	1871320
China	5991	420009	12414	697030
France	5471	460529	4165	335253
Thailand	2899	236105	3426	253400
Germany	1702	181063	1994	238531
Netherlands	729	69669	918	94794
Japan	73	33346	561	74361
Turkey	566	49689	727	54503
Italy	++	147	236	44200
Taiwan	440	53271	329	38416
Spain	649	61067	189	16547
Other countries	146	21862	196	24285

Figures rounded off

Table – 37: Import of Iron and Steel: Alloy Steel (Powder)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3473	1042530	3289	1147378
Sweden	1536	238674	1474	250898
Japan	33	196373	38	242662
UK	159	155420	210	236626
China	499	139941	483	137660
Canada	1082	139026	902	101790
Belgium	61	51220	55	61325
USA	88	73811	58	57509
Singapore	7	30231	8	34068
Germany	3	10542	7	12544
Netherlands	5	6111	3	4774
Other countries	++	1181	51	7522

Figures rounded off

Table – 38: Import of Iron and Steel (Scrap)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	10375501	542058265	11165464	522554515
USA	1843160	88548058	1634796	79428166
UK	1482268	60489964	1626411	61182976
UAE	1491070	73260896	772725	37362587
Singapore	526859	30465612	620558	30704894
Poland	304362	11830062	577480	20610831
Malaysia	274808	20925673	298429	20147853
Brazil	126449	5503740	533271	20030810
Canada	327863	17545974	370601	17968759
Belgium	189646	11079070	288589	14587527
Netherlands	343300	21273519	250326	13237958
Other countries	3465716	201135697	4192278	207292154

Figures rounded off

Table – 39: Import of Iron and Steel (Sponge Iron)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	304299	9750521	581563	19752020
UAE	102850	3285498	228662	8811369
Russia	--	--	257656	8045825
Libya	--	--	49472	1534465
South Africa	30880	1021729	23270	692450
Baharain	1047	38690	15110	470067
UK	910	28890	3377	103496
Egypt	494	7587	2031	38623
Kuwait	1642	46538	1061	32901
Italy	++	5	337	8883
Singapore	--	--	275	6371
Other countries	166476	5321584	312	7570

Figures rounded off

Table – 40: Import of Iron and Steel (Stainless Steel)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	57317	12897261	64315	10296537
China	31261	5293702	46341	5867852
Vietnam	14150	3660517	6078	1436919
Italy	3915	900971	3685	826657
UAE	1676	515226	2914	719754
USA	311	201844	1036	266005
Korea, Rep. of	289	166727	575	249902
Singapore	350	94679	433	132330
Germany	294	239665	156	118703
Japan	2785	1111580	192	102426
Kuwait	496	20709	1438	87407
Other countries	1790	691641	1467	488582

Figures rounded off

Table – 41: Import of Iron and Steel Material

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	441732	43172388	598887	47191909
Korea, Rep. of	165287	14266627	210751	14938353
Taiwan	111955	8194491	146940	9139665
China	47230	4736495	110708	8943689
Germany	65824	8054881	48520	6014387
Vietnam	20332	3861536	44195	3826469
Russia	7098	792248	7432	915135
Japan	4639	637893	7811	794230
Turkey	2714	250100	3533	571033
France	3271	467968	3039	495385
Singapore	4066	415466	4860	359774
Other countries	9316	1494683	11098	1193789

Figures rounded off

FUTURE OUTLOOK

Steel is one of the most important products of the modern world and of strategic importance to any industrial nation. Steel finds its way into a wide variety of applications from construction, industrial machinery to consumer products. The Indian Steel Industry has entered into a new development stage, post deregulation, riding high on the resurgent economy and rising demand for steel. Besides being the 2nd largest global Crude Steel producer, India has also made a mark globally in the production of Sponge Iron/Direct Reduced Iron (DRI). The growth of coal-based sponge iron units in key mineral-rich pockets of the country resulted in rapid increase of domestic Sponge Iron production, enabling the country to achieve and maintain the number one position in the global market. With several expansion projects at different stages of implementation, the future of the Indian Steel Industry is progressive. Steel consumption shows a strong correlation with GDP and is indicative of environmentally sustainable economic

development of any Nation. In India particularly, the Steel Industry can play an important role to make the country one of the largest economies in the world.

Globally, India made noticeable strides — the country remained the second largest producer of crude steel, the 2nd largest consumer of total finished steel and the largest DRI producer during the year 2024, as per ranking released by World Steel Association.

As per World Steel Association, construction is one of the most important steel-using industries, accounting for more than 50% of world steel demand. Buildings, from houses to car-parks to schools and skyscrapers, rely on steel for their strength. Steel is also used on roofs and as cladding for exterior walls. The world's population is expected to increase by 2 billion persons in the next 30 years, from 7.7 billion currently to 9.7 billion in 2050, according to United Nations report launched in 2019. Commensurately, the demand and consumption of steel are expected to soar exponentially across all verticals.



19. Lead & Zinc

16.52 million tonnes
production of lead and zinc ore have
been reported in 2023-24.

58 tonnes
of lead ores and concentrates were
exported in 2023-24.

4,877 tonnes
of lead ores and concentrates were
imported in 2023-24.

Lead is a soft, heavy, toxic and highly malleable metal. It is bluish white when freshly cut, but tarnishes to dull grey when exposed. Both lead & zinc are found to occur together in ore along with other metals like silver and cadmium. Zinc is a silvery blue-grey metal with a relatively low melting and boiling point. The largest single use of lead worldwide today is in the manufacture of lead-acid storage batteries which is about 74%, while the single largest use for zinc is in the galvanizing industry which is about 50%.

Zinc is the fourth most widely used metal across the globe, trailing only steel, aluminium and copper. The country has the self-sufficiency in respect of zinc. In contrast, there is short supply of lead vis-a-vis the demand in the country.

The ever increasing demand for lead especially from Lead Acid Battery Sector is met by the thriving market of lead scrap recycling. The Government of India enacted the Batteries (Management and Handling) Rules, (BHMR), 2002, which were superseded by the Battery Waste Management Rules, 2022, in order to enable further increase in the availability of scrap from the organized sector. It is estimated that 56% of refined lead produced worldwide is from recycled material. Producing lead through this route requires around one-third of the energy needed to extract it from its ores. Recovery of secondary zinc and lead is economically more attractive because of certain advantages. Besides lower energy consumption, it also entails low capital cost, less environmental hazards and high metal contents.

RESERVES/RESOURCES

The total reserves/resources of lead and zinc ore as on 1.4.2020 as per NMI database based on UNFC system have been estimated at 766.49 million tonnes. Of these, 103.27 million tonnes (13.47%) fall under 'Reserves' category while the balance 663.22 million tonnes (86.53%) are classified as 'Remaining Resources'. The total/resources of ore containing +10% Pb & Zn were estimated at 97.53 million tonnes (12.72%), ore containing 5 to 10% Pb & Zn were 280.05 million tonnes (36.53%) and ore containing less than 5% Pb & Zn were 388.91 million tonnes (50.73%). The total metal content in total/ resources of lead is 12.87

million tonnes and that of zinc is 33.17 million tonnes and for lead & zinc metal is 0.14 million tonnes. In terms of reserves, 1.90 million tonnes of lead metal and 7.44 million tonnes of zinc metal have been estimated. Rajasthan is endowed with the largest reserves/resources of lead – zinc ore amounting to 684.66 million tonnes (89.32%), followed by Andhra Pradesh 22.69 million tonnes (2.96%), Madhya Pradesh 19.07 million tonnes (2.48%), Bihar 11.44 million tonnes (1.49%) and Maharashtra 9.27 million tonnes (1.20%). Resources are also established in Gujarat, Meghalaya, Odisha, Sikkim, Tamil Nadu, Uttarakhand and West Bengal (Table – 1).

Table – 1: Reserves/Resources of Lead & Zinc Ores as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India			
Ore	103275	663222	766497
Lead metal	1900.19	10969.8	12869.99
Zinc metal	7438.05	25732.32	33170.37
Lead & Zinc metal	—	143.13	143.13
By Grades			
Ore with (+) 10% Pb & Zn	39100	58429	97529
Ore with 5-10% Pb & Zn	50490	229569	280059
Ore with (-) 5% Pb & Zn	13685	375225	388909
Lead metal	1900.19	10969.8	12869.99
Zinc metal	7438.05	25732.32	33170.37
Lead & Zinc metal	—	143.13	143.13
By States			
Andhra Pradesh			
Ore	—	22689	22689
Lead metal	—	836.88	836.88
Zinc metal	—	63.16	63.16
Bihar			
Ore	—	11435	11435
Lead metal	—	24	24
Zinc metal	—	38.75	38.75
Gujarat			
Ore	—	5689	5682
Lead metal	—	208.45	208.45
Zinc metal	—	261.4	261.4
Lead & Zinc metal	—	0.9	0.9
Madhya Pradesh			
Ore	—	19067	19067
Lead metal	—	36.29	36.29
Zinc metal	—	470.53	470.53
Maharashtra			
Ore	—	9272	9272
Lead metal	—	589.67	589.67
Meghalaya			
Ore	—	880	880
Lead metal	—	16.5	16.5
Zinc metal	—	14	14
Odisha			
Ore	—	1750	1750
Lead metal	—	76.96	76.96
Rajasthan			
Ore	103275	581381	684656
Lead metal	1900.19	9431.73	11331
Zinc metal	7438.05	23827.97	31266.02
Lead & Zinc metal	—	142.23	142.23
Sikkim			
Ore	—	950	950

Table – 1 (Concl'd.)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Lead metal	—	8.58	8.58
Zinc metal	—	20.07	20.07
Tamil Nadu			
Ore	—	790	790
Lead metal	—	7.74	7.74
Zinc metal	—	36.52	36.52
Uttarakhand			
Ore	—	5620	5620
Lead metal	—	182.6	182.6
Zinc metal	—	266.83	266.83
West Bengal			
Ore	—	3706	3706
Lead metal	—	140.07	140.07
Zinc metal	—	143.42	143.42

MINING LEASES & PRODUCTION

Lead & Zinc Ores and Concentrates

As on 31.03.2024 (P), there were a total of 11 mining leases for lead and zinc ore, having an area of 6,890.49 hectares.

The production of lead and zinc ore at 16.52 million tonnes in 2023-24 decreased by 1.3% as compared to that in the previous year.

The metal content of lead and zinc in the ore produced in 2023-24 works out to 2,77,271 tonnes and 9,41,050 tonnes respectively as against the corresponding figures of 2,77,003 tonnes and 9,27,616 tonnes in the previous year. During the year under review, 16.51 million tonnes of lead

& zinc ore was treated as against 16.68 million tonnes in previous year (Tables -2 to 4).

The production of lead concentrates in 2023-24 at 3,80,800 tonnes increased by 1.09% as compared to that in the previous year. Entire production of lead concentrate was reported from Rajasthan (Tables-5 & 6). The production of zinc concentrates increased from 16,70,208 tonnes in previous year to 17,10,144 tonnes in the 2023-24. Entire production of zinc concentrates was also reported from Rajasthan (Tables-7 & 8). The entire output of lead & zinc ore and concentrates in both the years was reported by mines owned by Hindustan Zinc Ltd. (HZL), a private sector unit

Table – 2: Producer of Lead & Zinc Ore, Concentrates & Metals 2023-24

Name & address of producer	Location of the mine	
	State	District
Hindustan Zinc Ltd., Yashad Bhavan, Udaipur – 313004, Rajasthan.	Rajasthan	Ajmer Bhilwara Rajsamand Udaipur

Table – 3: Production of Lead and Zinc Ore, 2022-23 and 2023-24

(By State)

(In tonnes)

State	2022-23			2023-24 (P)		
	Ore	Metal Content		Ore	Metal Content	
	Produced	Lead (Pb)	Zinc (Zn)	Produced	Lead (Pb)	Zinc (Zn)
India	16744082	277003	927616	16521284	277271	941050
Rajasthan	16744082	277003	927616	16521284	277271	941050

(p): Provisional

Table – 4: Lead and Zinc Ore Treated, 2022-23 and 2023-24

(By State)

(In tonnes)

State	2022-23			2023-24 (P)		
	Ore	Metal Content		Ore	Metal Content	
	Treated	Lead (Pb)	Zinc (Zn)	Treated	Lead (Pb)	Zinc (Zn)
India	16687711	274649	921909	16511990	277163	939638
Rajasthan	16684113	274649	921909	16511990	277163	939638

(p): Provisional

Table – 5: Production of Lead Concentrates, 2021-22 to 2023-24

(By State)

(Qty. in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	368040	22025747	376665	24488505	380800	25473522
Rajasthan	368040	22025747	376665	24488505	380800	25473522

(p): Provisional

Table – 6: Production of Lead Concentrates, 2022-23 and 2023-24

(By Sector/State/Districts)

(Qty. in tonnes; Value in ₹ '000)

State/District	2022-23				2023-24 (P)			
	No. of Mines	Quantity	Pb%	Value	No. of Mines	Quantity	Pb%	Value
India	10	376665	58.86	24488505	10	380800	58.58	25473522
Public Sector	2	-	-	-	2	-	-	-
Private Sector	8	376665	58.86	24488505	8	380800	58.58	25473522
Rajasthan	10	376665	58.86	24488505	10	380800	58.58	25473522
Ajmer	1*	-	-	-	1*	-	-	-
Bhilwara	1	84902	62.19	4052380	1	90106	61.7	4218109
Rajsamand	3	194781	54.91	12263592	3	191168	54.35	12852864
Sirohi	1*	-	-	-	1*	-	-	-
Udaipur	4	96982	63.87	8172533	4	99526	63.88	8402549

(p): Provisional

*: Employment reported without any production.

Table – 7: Production of Zinc Concentrates, 2021-22 to 2023-24

(By State)

(Qty. in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	1594085	78727814	1670208	91219638	1710144	85654297
Rajasthan	1594085	78727814	1670208	91219638	1710144	85654297

Table – 8: Production of Zinc Concentrates in India, 2022-23 and 2023-24

(By Sector/State/Districts)

(Quantity in tonnes; Value in ₹ '000)

State/District	2022-23				2023-24 (P)			
	No. of Mines	Production Quantity	Zn%	Value	No. of Mines	Production Quantity	Zn%	Value
India	@	1670208	50.4	91219638	@	1710144	50.13	85654297
Private Sector	@	1670208	50.4	91219638	@	1710144	50.13	85654297
Rajasthan	@	1670208	50.4	91219638	@	1710144	50.13	85654297

Table – 8 (Concl'd.)

(Quantity in tonnes; Value in ₹'000)

State/District	2022-23				2023-24 (P)			
	No. of Mines	Production Quantity	Zn%	Value	No. of Mines	Production Quantity	Zn%	Value
Bhilwara	@	1029008	50.64	47020403	@	1084225	50.62	43504835
Rajsamand	@	442675	49.1	28110009	@	403996	47.79	25255254
Sirohi	@	-	-	-	@	-	-	-
Udaipur	@	198525	52.01	16089226	@	221923	52	16894208

(p): Provisional

(@) Associated mines with lead concentrates.

Grade Analysis

All-India average metal content of ore treated during 2023-24 worked out to 7.36% (1.67% Pb and 5.69% Zn) as against 7.17% (1.64% Pb and 5.53% Zn) in previous year. The metal content of ore treated at Rampura Agucha mine in Bhilwara district of Rajasthan was the highest at 12.41% (1.54% Pb and 10.86% Zn). The lead concentrates produced in Rajasthan during 2023-24 was of grade 58.58% Pb as against 58.86% Pb in previous year. Metal content of zinc

concentrates produced in Rajasthan worked out to 50.13% Zn in 2023-24 as against 50.4% Zn in the previous year.

Stocks

Mine-head closing stock of lead concentrates in 2023-24 was 45,613 tonnes as against 42,572 tonnes in previous year (Table – 9).

Mine-head closing stock of zinc concentrates in 2023-24 was 45,353 tonnes as against 33,208 tonnes in previous year (Table – 10).

Table – 9: Mine-head Closing Stocks of Lead Concentrates, 2022-23 and 2023-24

(By State)

(In tonnes)

State	2022-23	2023-24 (P)
India	42572	45613
Rajasthan	42572	45613

(p):Provisional

Table – 10: Mine-head Closing Stocks of Zinc Concentrates, 2022-23 and 2023-24

(By State)

(In tonnes)

State	2022-23	2023-24 (P)
India	33208	45353
Rajasthan	33208	45353

(p):Provisional

Employment

The average daily labour employed in lead and zinc mines during the year under review was 12,254 as against 10,249 in previous year.

Lead and Zinc Metals

The production of primary lead in 2023-24 increased to

2,15,983 tonnes from 2,10,690 tonnes during the previous year. The entire output of primary lead was contributed by Chanderiya and Dariba smelters of Hindustan Zinc Ltd. The production of zinc ingot metal at 8,17,058 tonnes in 2023-24 decreased by 0.5% as compared to that in the previous year. Hindustan Zinc Ltd. contributed 100% of the total output (Tables-11 to 14).

Table – 11: Production of Lead Metal, 2021-22 to 2023-24

(Quantity in tonnes; Value in ₹'000)

Year	Lead Primary	
	Quantity	Value
2021-22	191185	34944605
2022-23	210690	39567243
2023-24 (P)	215983	40696202

(p):Provisional

Table – 12: Production of Zinc Metal, 2021-22 to 2023-24

(Quantity in tonnes; Value in ₹'000)

Year	Zinc Ingots	
	Quantity	Value
2021-22	775808	202092090
2022-23	820899	241626520
2023-24 ^(P)	817058	180301114

(p): Provisional

Table – 13: Production of Lead (Primary), 2022-23 and 2023-24

(By State/Plants)

(Quantity in tonnes ; Value in ₹'000)

State	Plant	2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value
India		210690	39567243	215983	40696202
Rajasthan	HZL Chanderiya/ Dariba	210690	39567243	215983	40696202

(p):Provisional

Table – 14: Production of Zinc (Ingots), 2022-23 and 2023-24

(By State/Plants)

(Quantity in tonnes ; Value in ₹'000)

State	Plant	2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value
India		820899	241626520	817058	180301114
Rajasthan	HZL Chanderiya/ Dariba/ Debari	820899	241626520	817058	180301114

(p):Provisional

MINING & MILLING

HZL is the only integrated lead and zinc metal producer in the country. Its operations can be classified into mining and smelting. At present, HZL has eight mines and all mining operations are located in Rajasthan. The eight mines are Rampura Agucha mine (Bhilwara district), Kayad mine (Ajmer district), Rajpura Dariba mine, Sindesar Khurd mine (both in Rajsamand district) and Zawar group of mines (4 mines in Udaipur district i.e. Mochia, Balaria, Zawarmala and Baroi), Rajasthan. All the mines of HZL undertake underground mining operations.

Rampura Agucha mine, along with Kayad mines, has an existing annual production capacity of 6.5 million tonnes of lead and zinc ore. Sindesar Khurd mine is highly mechanised and the largest ore producing underground mine with existing annual production capacity of 6.5 million tonnes. The other mines viz, Rajpura Dariba and Zawar group of mines (Mochia, Balaria, Zawarmala and Baroi) are underground mines with an existing annual production capacity of 1.1 million tonnes and 5.2 million tonnes of lead & zinc ore respectively (Table – 15).

Table – 15: Ore Production Capacity of HZL Mines

(By State/Plants)

Mines	Ore	Capacity (million tonnes per year)
Total		19.3
Zawar Mines, Distt Udaipur, Rajasthan	Zinc-lead	5.2
Rajpura-Dariba, Distt Rajsamand, Rajasthan	Zinc-lead	1.1
Sindesar Khurd Mine, Distt Rajsamand, Rajasthan	Zinc-lead	6.5
Rampura Agucha & Kayad, Distt Bhilwara & Ajmer, repectively, Rajasthan	Zinc-lead	6.5

Source: HZL- Investor Presentation

Zawar group of mines is a cluster of four underground mines viz. Mochia, Balaria, Zawarmala and Baroi mines and one beneficiation plant for all mines were commissioned in the year 1966. Zawar group of mines, one of the oldest mines is located about 40 km south of Udaipur. Lead & Zinc ore of the mines is divided into stope blocks which are drilled and blasted using sub-level open stoping mining method. Loading and transportation are done using combination of LHDs, LPDTs, LOCO and shaft hoisting to surface. The ore is further crushed and passed through a flotation process to produce concentrate. Zawarmala Mine has started extraction from pillar posts and filled the voids. In FY 2022-23, new lenses were discovered at shallow depth in Bowa Magra and eastern Zawarmala, using new methods like Mobile Carrier Rig (MCR) for shallow angle surface drilling. Opening of new production front South of Baroi by UG exploration and ore body (OB) delineation is under progress to increase production in Baroi Mine and COP reduction by less hauling distance. In 2023-24, the Zawar group of mines produced 4.03 million tonnes ore as compared to 4.3 million tonnes in the year 2022-23. It aims to expand its capacities further to achieve 6.50 Mtpa in the coming years. As on 31.03.2024, Zawar group of mines have total ore reserve of 42.2 million tonnes with Zn grade of 2.6% and Pb grade of 1.5%.

The Rajpura Dariba mine is an underground lead & zinc mine where mining operations began in 1983. It is located at 75 kilometers north-east of Udaipur, Rajasthan. Mining is carried out by using Blasthole stoping method with hydraulic filling. Mined out stopes are backfilled with cement tailings. During the year 2023-24, Rajpura Dariba Mine continued to improve its performance metrics with ore production of 1.34 million tonnes as compared to 1.39 million tonnes in the year 2022-23. Presently, the mine is accessed via decline and two shafts. As RDM ore body also allows for multiple production centres, the mine is poised to produce at the capacity of 1.75 million tonnes per annum of ore in FY 2024-25 with new level of mechanisation & automation. Mine has achieved highest truck filling hours and exploration drilling, resulting in its readiness for expansion. Rajpura Dariba mine has upgraded the Mill from 0.9 to 1.1 Mtpa capacity, through crusher house upgradation. As on 31.03.2024, Rajpura Dariba mine has total ore reserve of 47.1 million tonnes with Zn grade of 5.4% and Pb grade of 1.7%.

Rampura Agucha mine commissioned in 1991, is located at 230 km north of Udaipur in Bhilwara district, Rajasthan. The mine has the richest zinc-lead deposit in India. It is the largest underground zinc producing mine globally and the lowest cost zinc producer in the world. In 2023-24, the production of ore was carried out by underground mining at 4.93 million tonnes as compared to 4.78 million tonnes in the year 2022-23. Until 2017-18 the production of ore from the mine was by surface mining and the mine has been fully transformed into an underground mine. Two ventilation shafts (North and South) each with

diameter of 7.5 m and 450 m depth are in operation. The main hoisting shaft of 7.5 m diameter has been sunk and furnished to its final depth of 950 m and winders, skip loading & surface conveyors have been installed, while off-shaft development activities to commission underground crushing and conveying system are under progress. The shaft hoisting capacity at RAM showed a marked improvement, from 6,500 tonnes per day (tpd) to 8,000 tpd, at the back of multiple debottlenecking and digitalization initiatives, further actions are in progress to reach 10,000 tpd capacity. Paste-Fill plant has been established in the mine area. Underhand stoping method has been adopted which will use mine tailings to fill stopes after ore extraction. As on 31.03.2024, Rampura Agucha mine have total ore reserve of 44.4 million tonnes with Zn grade of 11% and Pb grade of 1.2%.

Sindesar Khurd mine is located at 6 km NNE of Rajpura Dariba mine and 82 km north-east of Udaipur. It is an underground mine, commissioned in 2006. The mine lies on the same geological belt as the Rajpura Dariba mine. Sindesar Khurd has access to the mine through inclined ramps and underground shaft. Mining method is blast hole stoping with back filling. Introduced shaft partition at Sindesar Khurd mine to increase the shaft hoisting from 2.6 Mtpa in FY 2022-23 to 3.1 Mtpa in FY 2023-24. During the year 2023-24, Sindesar Khurd mine reported ore production of 5.65 million tonnes as compared to 5.6 million tonnes in the year 2022-23. As on 31.03.2024, Sindesar Khurd mine have total ore reserve of 40.1 million tonnes with Zn grade of 3.1% and Pb grade of 2%.

Kayad mine is a newly developed underground mine near Ajmer, Rajasthan. It started operations in 2011 and commenced ore production from FY 2012-13. In Kayad mine, 21 tonnes capacity of LHD and 65 tonnes capacity of LPDT are used to produce the ore. The mine has access through a single decline from surface portal to the top of the ore body and split in two declines 75 meters below surface. Longitudinal Long-Hole Open Stopping method is used for the steeper and thinner portion of ore body and Transverse Long-Hole Open Stopping method for flatty dipping and thick portion of the ore body. In the year 2023-24, the Kayad mine produced 0.56 million tonnes ore as compared to 0.66 million tonnes in the year 2022-23. The ore from Kayad mine is treated at Rampura Agucha's beneficiation plant. As on 31.03.2024, Kayad mine has total ore reserve of 1.3 million tonnes with Zn grade of 5.2% and Pb grade of 0.7%.

SMELTING

Primary lead was produced entirely by HZL which operated smelter at Chanderiya and Dariba having capacity of 90,000 tonnes and 1,20,000 tonnes per annum of lead metal, respectively. Thus, the smelting capacity for lead (primary) in the country presently is 2,10,000 tonnes per annum. Company-wise smelting capacity of lead and zinc smelters is furnished in Table – 16.

Table – 16: Companywise Capacity and Production of Primary Lead and Zinc

(Quantity in tonnes)

Company	Lead Capacity (tpy)	Production		Zinc Capacity (tpy)	Production	
		2022-23	2023-24 (P)		2022-23	2023-24 (P)
Hindustan Zinc Ltd	210000	210690	215983	913000	820899	817058
Edayar Zinc Ltd	-	-	-	38000	-	-
Total	210000	210690	215983	951000	820899	817058

(P): Provisional

The smelting capacity of HZL for zinc is distributed between three smelters at Debari (88,000 tonnes), Chanderiya (5,85,000 tonnes) and Dariba (2,40,000 tonnes). Edayar Zinc Ltd's (EZL) plant at Binanipuram (Aluva), Kerala, has capacity of 38,000 tonnes per annum. Thus, the smelting capacity for zinc in the country is 9,51,000 tonnes per annum. EZL produced zinc from imported concentrates but since the Company has been declared as sick unit, it did not operate its plant. Besides lead & zinc capacities, HZL has capacities to produce 800 tonnes per annum of silver. HZL is India's largest manufacturer of sulphuric acid which is by-product of its smelting operations.

Chanderiya lead-zinc smelting complex is located at 110 km North of Udaipur in Chittorgarh district, Rajasthan. It was commissioned in 1991 with an initial production capacity of 70,000 tonnes per annum. Chanderiya lead-zinc smelting complex comprises one zinc Pyro-metallurgical smelter having production capacity of 1,05,000 tonnes zinc, one lead Pyrometallurgical smelter having production capacity of 90,000 tonnes lead and a Hydro-metallurgical zinc smelter having production capacity of 4,80,000 tonnes zinc. It employs Roast-Leach Electro-winning technology in its Hydro-metallurgical smelters, Imperial Smelting process in lead-zinc smelter and Top Submerged Lance Technology (Designed by M/s Ausmelt Ltd, Australia) coupled with Cansolv Technology for its lead smelter.

Zinc fuming is a process that recovers valuable metals from zinc residue and will help in higher recovery of zinc, lead and silver. At the Hydro plant, Hydro-2 cell house was upgraded from 192 kA to 200 kA for enhancing zinc output.

Zinc Smelter Debari was commissioned in the year 1968 with an initial production capacity of 18,000 tonnes per annum of zinc. The present capacity has now reached 88,000 tonnes per annum of zinc. It is located at about 13 km north of Udaipur, Rajasthan. Zinc smelter Debari employs Roast-leach Electro-winning Technology at its Hydrometallurgical zinc smelter. The plant has three roasting facilities, leaching & purification section, electrolysis, melting and casting sections. It produced surplus calcine, an intermediate product, which is supplied to the rest of the Hydro-metallurgical zinc smelter.

Dariba smelting complex is located at 75 km north-east of Udaipur near Rajpura Dariba mine and 7 km from Sindesar Khurd mine in Rajsamand district, Rajasthan. The zinc smelter at Dariba was commissioned in March 2010

and has a capacity of 2,40,000 tonnes per annum while lead smelter was commissioned in July, 2011 and has a capacity of 1,20,000 tonnes per annum. Dariba smelting complex employs Roast-Leach Electro-winning technology at its Hydro-metallurgical zinc smelter. The plant has two roasting facilities, a leaching & purification section and a cell house. The lead smelter employs SKS bottom blowing technology. The plant consists of SKS furnace-bottom blowing, blast furnace, electric arc furnace & fuming furnace and electro-refining. Fuming furnace is also installed to produce zinc oxide from blast furnace slag. Dariba Smelting Complex lead plant steam was utilised in Dariba Smelting Complex Captive Power Plant (CPP) for reducing the auxiliary steam consumption.

During FY 2023–24, HZL achieved an overall refined metal production of 817.1 thousand tonnes of zinc and 216 thousand tonnes of lead. These figures reflect the combined output from HZL's three major smelting complexes located at Chanderiya (Chittorgarh), Dariba (Rajsamand), and Debari (Udaipur), all in the state of Rajasthan.

The product range of HZL constitutes two grades, namely, Special High Grade (SHG) zinc containing 99.995% Zn (min.) and Prime Western (PW) containing 98.65% Zn (min.). Both these products are available in the form of slabs weighing 25 kg, SHG Jumbo weighing 1,000 kg and PW Jumbo weighing 600 kg. Lead is available as HZL Grade containing 99.99% Pb (min.) in the form of slab weighing 24 kg. In October, 2016, a new zinc alloy value added product, HZDA or Hindustan Zinc Die-cast Alloy was added to the Company's portfolio from Chanderiya Lead-Zinc Smelter. HZL produce refined silver; recovered as a by-product of zinc-lead facility, it has high quality silver bullion having a minimum purity 99.9% of silver is listed on LBMA Good delivered List (LGD) and supply silver in the form of Standard 30 kg bars, 1 kg bars and silver powder. HZL also produce 98% concentrated sulphuric acid at production facilities in Chanderiya, Debari and Dariba in the state of Rajasthan.

RECYCLING OF LEAD & ZINC

Lead

The storage battery scrap is the main source of secondary lead production. Lead is one of the highest recycled metals. It can be re-melted any number of times, and provided enough processes to remove impurities. The final product (termed secondary lead) is indistinguishable from primary

lead produced from ore. The amount of lead recycled is about 75% of the total lead production in India. More than 80% of lead consumed in the country goes for manufacturing of lead batteries.

The Government of India enacted Battery Waste Management Rules, 2022 and issued Gazette Notification No. S.O. 3984(E) dated 22nd August, 2022 to organise the recycling of lead acid batteries and to make available raw material to the lead reproducers. The said Rules are in supersession of the Batteries (Management and Handling) Rules, 2001, published vide number S.O.1035(E) dated 16th May 2001. These Rules shall apply to Producer, dealer, consumer, entities involved in collection, segregation, transportation, re-furbishment, recycling of Waste Battery and all types of batteries regardless of chemistry, shape, volume, weight, material composition and use, except battery used in equipment connected with the protection of the essential security interests including arms, ammunitions, war material, those intended specifically for military purposes and equipment designed to be sent into space. At present, 672 units are registered as lead recycling units with CPCB for processing used lead batteries with the production capacity of 3.53 million tonnes per annum in India. CPCB has also developed online web-based application “Batteries (Importer) Registration Management” for registration/renewal of registration for import of new lead acid batteries.

Lead when used as metal in batteries, cable sheathing and sheathing for containing radiation is fully recyclable and it does not lose its properties. There is indeed a thriving industry that recycles lead in the country. However, due to the health risk involved in lead recycling the Central

Pollution Control Board issues licences to the lead reproducers to ensure adherence to environmental norms.

Zinc

The largest consumer of zinc is the galvanizing industry. Zinc once used for galvanizing, as well as in brass making, is not recoverable. Hence, the quantity of zinc recycling is comparatively smaller than that of lead recycling. Zinc-bearing wastes, such as zinc ash, residues, and dross, are categorized as recyclable hazardous materials under Schedule IV of the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016. Secondary zinc is recovered from pure zinc scrap in the form of sheet cuttings, zinc roofing sheets, old zinc anodes, and alloys containing zinc as a major constituent.

USES & CONSUMPTION

Consumption of lead and zinc in various industries is not available readily. However, it is known that lead and zinc are consumed in the form of metals as well as in the form of compounds and oxides.

Lead

The Battery Industry consumes about 80% of lead and remaining 20% is consumed in pigments & compounds, rolled & extruded products, alloys, cable sheathing and other industries. The apparent consumption of lead during the year 2022-23 and 2023-24 was calculated on the basis of production of lead (primary) and imports & exports of refined lead (unwrought). The apparent consumption thus arrived at was 1,44,361 tonnes in 2022-23 and 30,449 tonnes in 2023-24 (Table – 17). In addition to this, it is understood that large quantities of recycled lead were also consumed in certain other industries.

Table – 17: Apparent Consumption of Lead (Based on the Production of Lead (Primary) and Import & Export of Refined Lead, Unwrought)

Item	(In tonnes)	
	2022-23	2023-24
Total Production Lead (Primary)	210690	215983
Total Import	63422	84411
Total Export	129751	269945
Apparent Consumption Lead (Primary)	144361	30449

Zinc

Owing to its corrosion resistance in varied types of environment, zinc is used for protecting steel from corrosion by way of galvanising. The Galvanising Industry alone consumes about 58% of zinc, followed by approximately 14% is used in the production of zinc die casting alloys. Nearly 9% of the zinc is also utilized for oxides and chemicals and about 10% is used in Brass semis and castings. The apparent consumption of zinc during the year 2022-23 and 2023-24

was calculated on the basis of production of zinc, import & export of Zinc, not alloyed: -- Containing by weight 99.99 percent or more of zinc. The apparent consumption, thus arrived at was 6,05,614 tonnes in 2022-23 and 7,48,161 tonnes in 2023-24 (Table – 18). The data on trade of Zinc, not alloyed: -- Containing by weight 99.99 percent or more of zinc was taken from DGCI&S (HS Code: 79011100). In addition to this, some quantities of recycled zinc are also consumed in certain other industries.

Table – 18: Apparent consumption of Zinc
(Based on the Production of Zinc (Ingots) and Import & Export of Zinc,
not alloyed:– Containing by weight 99.99 percent or more of zinc)

(In tonnes)

Item	2022-23	2023-24
Total Production Zinc	820899	817058
Total Import*	90897	138475
Total Export*	306182	207372
Apparent Consumption	605614	748161

* DGCI & S Kolkata

SUBSTITUTES & TECHNICAL POSSIBILITIES

Lead

Battery replacements include batteries of nickel-zinc, zinc lithium chloride, sulphide or nickel lithium hydride. The large-scale commercial use of any of these four possible substitutes was so far precluded by cost and operating problems. Polyethylene and other materials work as substitute in some cable applications.

In construction applications, in place of galvanised sheets, copper and aluminium are alternatives. In corrosive chemical environment, stainless steel, titanium, plastics and cements are substitutes. Tin, glass, plastics and aluminium are alternatives in tubes and containers; iron & steel or bismuth in shots for ammunition; and tin in solder. In Electronic Industry, there has been a move towards leadfree solders with varying compositions of tin, bismuth, silver and copper.

Environmental concerns for lead are limiting the uses, particularly in gasoline, where its use as an anti-knocking agent was phased out by the introduction of catalytic converters. Storage batteries for industrial load levelling, mains power management and electric vehicles have growing markets. The continued search for weight reduction is reducing the amount of lead per battery, and battery lives are being extended. Possible new developments include the use of lead as an anti-oxidant in asphalt, as a shielding material in nuclear waste, in protection of buildings against radon gases and as a sound buffer. Environmental

legislation will inhibit the growth of new uses and possibly eliminate lead from many existing uses. The Organisation for Economic Cooperation & Development (OECD) is actively examining possible restrictions on uses of lead. New techniques to recover lead from concentrates and from scrap are being developed and are bound to become more important in future. Recycling of lead and zinc through environmentally safe processes needs to be encouraged as the growing use of lead and zinc in railway electrification as well as in road transport vehicles have created shortage of these metals in the country.

Zinc

Aluminium, magnesium and plastic compete in some die-casting applications. Ceramic and plastic coatings, electroplated cadmium & aluminium and special steel compete in some galvanising applications. Aluminium, magnesium and titanium can replace zinc in chemicals and pigments. Zirconium is an alternative in ceramic and enamel applications. New alloys, e.g. superplastic alloys of zinc and aluminium could be developed. Many elements are substitutes for zinc in chemical, electronic and pigment uses.

WORLD REVIEW

Reserve Lead

The world's reserves of lead were estimated at 95 million tonnes in terms of lead content. Australia possesses 37% of the world's reserves followed by China (21%), Russia (9%), Mexico (6%), Peru & USA (5% each) and Iran, India, Sweden, Bolivia & Turkey (2% each) (Table – 19).

Table – 19: World Reserves of Lead
(By Principal Countries)

(In '000 tonnes of lead content)

Country	Reserves
World: Total (rounded off)	95000
Australia ^(a)	35000
China	20000
Russia	8700
Mexico	5600
Peru	5000
USA	4600
Iran	2000
India*	1900

		(In '000 tonnes of lead content)
Country	Reserves	
Sweden	1700	
Bolivia	1600	
Turkey	1600	
Tajikistan	NA	
Other countries	5900	

Source: USGS, Mineral Commodity Summaries, 2024

(a) For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 10 million tonnes.

* India's total reserve/resources of lead & zinc as per National Mineral Inventory based on UNFC as on 01.04.2020 are 766.49 million tonnes.

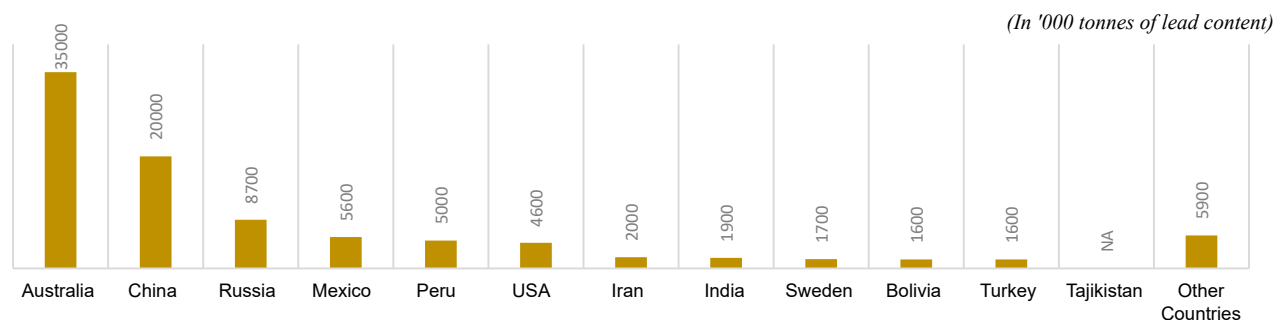


Fig.1: Countrywise Reserve of Lead

Zinc

The world's reserves of zinc were estimated at 220 million tonnes in terms of zinc content. Australia accounts for 29%

of world's zinc reserves, followed by China (20%), Russia (11%), Peru (10%), Mexico (6%), India, Kazakhstan, USA & South Africa (3% each) and Sweden (2%) (Table – 20).

Table – 20: World Reserve of Zinc
(By Principal Countries)

		(In '000 tonnes of zinc content)
Country	Reserves	
World: Total (rounded off)	220000	
Australia ^(a)	64000	
China	44000	
Russia	25000	
Peru	21000	
Mexico	14000	
India*	7400	
Kazakhstan	6700	
USA	6600	
South Africa	6200	
Sweden	4100	
Bolivia	NA	
Other countries	25000	

Source: USGS, Mineral Commodity Summaries, 2024

(a) For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 21 million tonnes.

*India's total reserve/resources of lead & zinc as per National Mineral Inventory based on UNFC as on 01.04.2020 are 766.49 million tonnes.

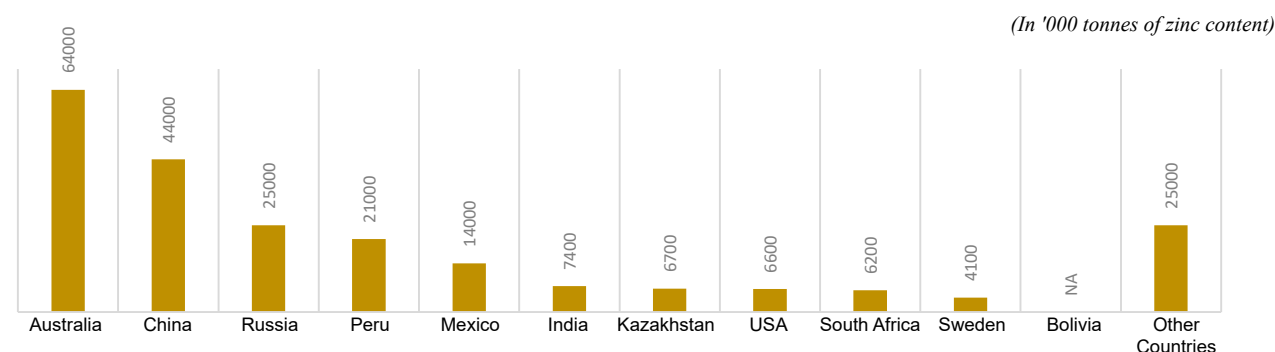


Fig. 2: Countrywise Reserve of Zinc

PRODUCTION

Lead

World mine production of lead in terms of metal content was about 4.5 million tonnes in the year 2023, showing no change as compared to 4.50 million tonnes in the previous

year. China is foremost amongst producing countries with about 1.96 million tonnes (44%) followed by Australia (10%), Peru & USA (6% each), India & Russia (5% each), Mexico (4%) and Bolivia, Iran & Turkey (2% each) (Table – 21).

Table – 21: World Mine Production of Lead

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World: Total (Rounded off)	4600000	4500000	4500000
China	1964400	1945900	1960000
Australia	485487	437825	472242
Peru	264427	255333	273305
USA	294000	273000	262000
India	212654 ^(b)	213320	222334
Russia	212300	229300	220000*
Mexico	272231	272988	182996
Bolivia	93246	90115	102959
Iran	70000*	80000*	80000*
Turkey	93700*	81700*	71100*
Other countries	632097	632966	636211

Source: BGS, World Minerals Production, 2019-23

(*) Estimated

(b) Years ended 31st March following that stated

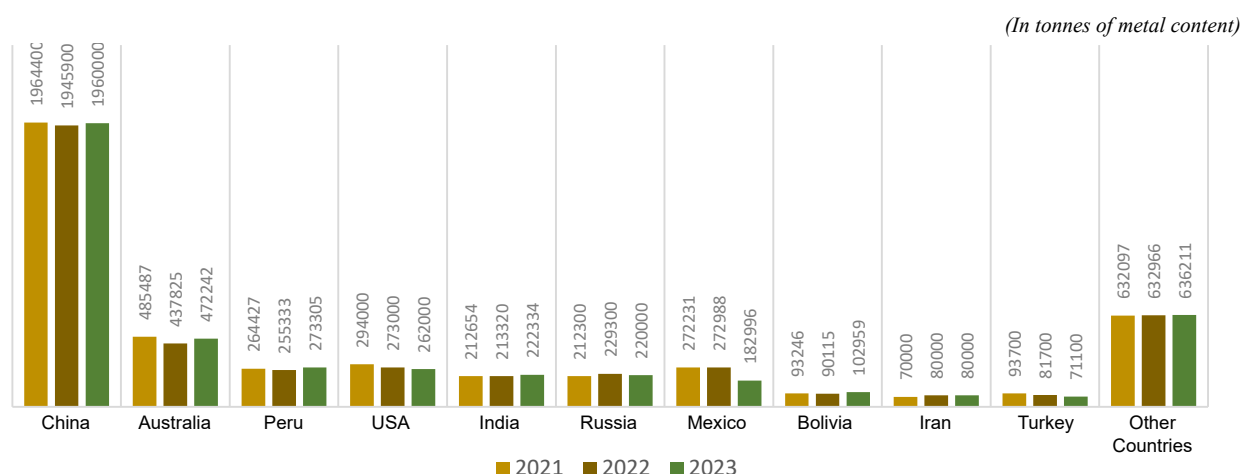


Fig. 3: Countrywise Production of Lead

Zinc

World mine production of zinc ore was at 11.60 million tonnes in terms of zinc content in the year 2023 which was decreased by 10% from 12.90 million tonnes in the year

2022. China is at top position with 3.27 million tonnes (28%) followed by Peru (13%), Australia (9%), India (7%), Mexico (6%), USA (5%), Bolivia (4%), Kazakhstan & Russia (3% each), and Iran (2%) (Table – 22).

Table – 22: World Mine Production of Zinc

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World: Total (rounded off)	12900000	12900000	11600000
China	4136000	4041000	3266000
Peru	1533135	1369649	1468994
Australia	1315160	1237473	1094398
India	797044 ^(a)	832194	853507

Table – 22 (Concl.)

Country	(In tonnes of metal content)		
	2021	2022	2023
Mexico	742926	877000	744285
USA	704000	763000	618400*
Bolivia	499536	518480	491768
Kazakhstan	318400	315100	352900
Russia	287900	304250	305000
Iran	195100	228700	235000*
Other countries	2365719	2371444	2169996

"Source: BGS, World Minerals Production, 2019-23

(*) Estimated

(a) Years ended 31st March following that stated

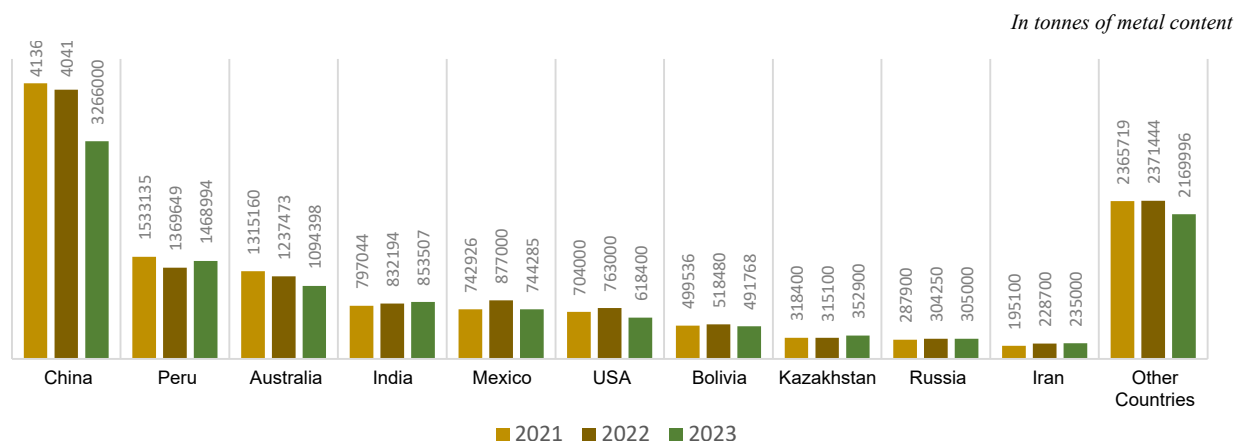


Fig. 4: Countrywise Production of Zinc

FOREIGN TRADE

Lead

Exports

Exports of lead from the country were in the form of ore & concentrates, lead & alloys including scrap, lead waste & scrap, lead unrefined, refined lead unwrought, lead & alloys worked and others. Exports of lead ores and concentrates which were 137 tonnes in 2022-23 decreased to 58 tonnes in 2023-24. U.A.E (95%), Bangladesh (3%) and Iraq (2%) were

the main country which imported lead ores & concentrates from India (Table – 23). Exports of lead & alloys including scrap increased by 69.46% to 3,42,724 tonnes during 2023-24 as compared to 2,02,248 tonnes in the preceding year. Similarly, export of lead and alloys also increased by 69.47% to 3,42,723 tonnes in 2023-24 as compared to 2,02,238 tonnes in the previous year. Exports of refined lead unwrought also increased by 108% to 2,69,945 tonnes in 2023-24 as compared to 1,29,751 tonnes in the previous year (Table – 24).

Table – 23: Export of Lead Ores and Conc.

Country	(By Countries)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	137	7521	58	2867
UAE	54	2255	55	2254
Bangladesh	5	640	2	260
Iraq	--	--	1	237
Sri Lanka	--	--	++	50
South Africa	--	--	++	36
Canada	++	48	++	30
Iran	78	4576	--	--
New Zealand	++	2	--	--

Figure rounded off

Table – 24: Export of Lead & Alloys

Item	(By Items)		2023-24 (P)	
	2022-23 (R)			
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Items	606744	116257659	1028172	194101707
Lead And Alloys: Worked (Bars,Rods,Plates, Etc)	2637	685798	1784	502011
Antimonial Lead	22852	4426688	26882	5236315
Lead Waste And Scrap	10	1558	1	112
Lead And Alloys Unwrought Nes	46997	9747217	44106	9542782
Lead Unrefined ,Nes	1	329	6	1139
Refined Lead, Unwrought	129751	23890963	269945	49418210
Lead And Alloys Incl. Scrap	202248	38752553	342724	64700569
Lead And Alloys	202238	38750995	342723	64700457
Lead (Scrap)	10	1558	1	112

Figure rounded off

Imports

Imports of lead in India are in the form of lead ores & concentrates, lead & alloys including scrap, refined lead/unwrought, pig lead, refined lead & alloys (bars, rods, plates, etc.) and lead unrefined etc. Imports of lead ores & concentrates decreased marginally to 4,877 tonnes in 2023-24 as compared to 5,558 tonnes in 2022-23. Imports were mainly from UAE (57%), Ethiopia (8%), Ghana & Morocco

(6% each), Australia (5%), Singapore (4%), Montenegro, Sudan & Saudi Arabia (2% each) and Zambia (1%) (Table – 25). The total imports of lead (scrap) increased by 17.47% in 2023-24 to 94,801 tonnes as compared to 80,698 tonnes during 2022-23. Imports of lead and alloys during 2023-24 were 4,03,554 tonnes as compared to 2,69,596 tonnes in 2022-23, showing an increase of 49.68% in the import of lead and alloys (Table – 26).

Table – 25: Import of Lead Ores and Concs.

Country	(By Countries)		2023-24 (P)	
	2022-23 (R)			
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5558	294381	4877	363290
UAE	1731	91371	2758	192326
Australia	--	--	239	31414
Morocco	544	37934	275	24298
Singapore	384	23786	182	22411
Ethiopia	405	13050	378	21791
Ghana	413	20466	280	14730
Montenegro	--	--	97	10204
Sudan	740	36195	118	7194
Saudi Arabia	23	1298	108	6786
Zambia	--	--	51	6327
Other countries	1318	70281	391	25809

Figure rounded off

Table – 26: Import of Lead & Alloys

(By Items)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Items	1050882	188128983	1495065	271878954
Lead (Scrap)	80698	12941635	94801	15843519
Lead Waste And Scrap	80698	12941635	94801	15843519
Lead And Alloys:Worked(Bars,Rods,Plates, Etc)	552	337853	866	380735
Lead And Alloys Unwrought Nes	38773	8047405	34099	7341312
Lead Unrefined ,Nes	107437	18888066	181974	32218986
Lead: Pig Lead	2416	428023	1788	334751
Lead And Alloys	269596	49768026	403554	74782799
Refined Lead, Unwrought	63422	11795402	84411	16167963
Lead and Alloys incl. Scrap	350294	62709661	498355	90626318
Antimonial Lead	56996	10271277	100416	18339052

Figure rounded off

Zinc

Exports

Exports of zinc are in the form of ores & concentrates, zinc & alloys including scrap and zinc & alloys in the form of bars, rods & plates etc. Exports of zinc ores & concentrates decreased to 19,902 tonnes in 2023-24 as against 43,877 tonnes in the previous year. Republic of Korea was the main export destination of zinc ores & concentrates and accounted for 98% of all the exports of zinc ores &

concentrates followed by China (2%) and UAE (<1%) (Table – 27). Exports of zinc & alloys including scrap during 2023-24 were 2,63,413 tonnes as against 3,54,217 tonnes in the preceding year. Exports of zinc (scrap) during 2023-24 were 40 tonnes as against 1 tonnes in the preceding year. Exports of zinc and alloys during 2023-24 were 2,63,373 tonnes as compared to 3,54,216 tonnes in 2022-23. Exports of zinc or spelter during 2023-24 were 2,37,466 tonnes as compared to 3,38,705 tonnes in 2022-23 (Table – 28).

Table – 27: Export of Zinc Ores And Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	43877	182437	19902	64126
Korea, Rep. of	42260	157557	19420	60226
China	--	--	456	2668
UAE	--	--	26	1231
Australia	1	2	++	1
Iran	265	19790	--	--
Vietnam	1350	5080	--	--
Canada	++	4	--	--
USA	1	4	--	--

Figure rounded off

Table – 28: Export of Zinc & Alloys

(By Items)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Items	1059751	317386254	784468	179165672
Zinc And Alloys Incl. Scrap	354217	106078261	263413	60177420
Zinc (Scrap)	1	245	40	6056
Zinc And Alloys	354216	106078016	263373	60171364
Zinc And Alloys:Worked (Bars,Rods,Plates, Etc)	10993	3442426	18653	4625763
Zinc And Alloys Nes	1619	1064013	1523	947115
Zinc Or Spelter	338705	100723293	237466	53237954

Figure rounded off

Imports

Imports of zinc in the country are in the form of zinc ores & concentrates, zinc & alloys including scrap, zinc or spelter and zinc & alloys in the forms of bars, rods, plates, etc. Imports of zinc ores & concentrates during 2023-24 were at 455 tonnes as against 1,041 tonnes during the previous year. Imports were mainly from Malaysia (52%) and USA

(48%) (Table – 29). Imports of zinc & alloys during 2023-24 were at 2,28,045 tonnes as compared to 1,88,566 tonnes in 2022-23. Imports of zinc (scrap) were 65,612 tonnes during 2023-24 as compared to 91,406 tonnes in 2022-23. Imports of zinc or spelter were at 1,97,697 tonnes in 2023-24 as compared to 1,67,705 tonnes during the previous year (Table – 30).

Table – 29: Import of Zinc Ores and Concs.

Country	(By Countries)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1041	39415	455	10666
Malaysia	--	--	235	7565
USA	--	--	220	2514
Brazil	--	--	++	485
Kenya	--	--	++	102
UAE	457	21303	--	--
Ethiopia	532	16224	--	--
Tanzania	52	1883	--	--
Nigeria	++	5	--	--

Figure rounded off

Table – 30: Import of Zinc & Alloys

Item	(By Items)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Items	735622	196643842	795033	177083102
Zinc And Alloys Incl. Scrap	279972	73134370	293657	64537932
Zinc (Scrap)	91406	18814024	65612	11726817
Zinc And Alloys	188566	54320346	228045	52811115
Zinc And Alloys:Worked (Bars,Rods,Plates, Etc)	5082	1644842	6574	1573870
Zinc And Alloys Nes	2891	1320521	3448	1283304
Zinc Or Spelter	167705	47409739	197697	45150064

Figure rounded off

FUTURE OUTLOOK

The International Lead and Zinc Study Group (ILZSG) report indicates that both the lead and zinc markets are expected to face notable surpluses through 2026. For lead, global demand is forecast to grow by 1.8% in 2025, reaching 13.25 million tonnes, and by 0.9% in 2026, reaching 13.37 million tonnes. World lead mine supply is projected to increase by 0.7% to 4.57 million tonnes in 2025 and by 2.2% to 4.67 million tonnes in 2026. Refined lead supply is anticipated to rise by 2% to 13.34 million tonnes in 2025, and by 1% to 13.47 million tonnes in 2026. The ILZSG expects that global supply of refined lead metal will exceed demand by 91,000 tonnes in 2025, with a slightly larger surplus of 1,02,000 tonnes projected for 2026. Lead production is expected to increase due to higher mine

output in Europe, China, Australia, and the USA, with supply growth outpacing demand.

Similarly, zinc demand is forecasted to increase by 1.1% to 13.71 million tonnes in 2025 and by 1% to 13.86 million tonnes in 2026. Zinc demand is expected to rise in India, Japan, Saudi Arabia, Thailand, and Vietnam, but decline in Brazil and the Republic of Korea.

India is likely to witness substantial growth in the demand for lead batteries, given that several sectors, including automotive, telecommunication, railways, and defense, are set to expand in the coming years. As a battery ingredient, lead is increasingly used in inverters, UPS, and similar energy storage devices. The zinc mining market in India is expected to reach a projected revenue of US\$ 2,819.8 million by 2030, with a compound annual growth

rate (CAGR) of 1.7% from 2023 to 2030. The lead acid battery market in India is projected to reach a revenue of US\$ 9,594.2 million by 2030, with a CAGR of 8.3% from 2024 to 2030.

To meet domestic demand amid a low production trend and higher imports, recycling can be a viable alternative.

As every major national plan projects continuous growth in the country's power generation capacity, the demand for galvanized transmission towers is also expected to increase. This, along with the rising necessity for erecting mobile towers, will require higher investments in infrastructure. Railways will play a leading role in the use of galvanized steel.



20. Lithium

421.38 tonnes

of lithium carbonates were exported in 2023-24.

1,146.1 tonnes

of lithium carbonates were imported in 2023-24.

Lithium is the lightest of all metals and falls in Group IA of periodic table along with sodium and potassium. Lithium is an alkaline metal and its symbol is Li. It is obtained from lithium-bearing minerals in pegmatites and in brines of certain lakes. Lithium (Li) does not occur freely in nature. Lithium is a white metal, which tarnishes slowly in moist air. The chief commercial lithium minerals are Spodumene (lithium pyroxene) – $\text{LiAl}(\text{SiO}_3)_2$ containing about 8% of Li_2O , Lepidolite (Lithium Mica)– $\text{KLi}_2(\text{Si}_4\text{O}_{10})(\text{OH},\text{F})_2$ containing about 6 to 7% of Li_2O , Amblygonite– $\text{Li}(\text{F},\text{OH})\text{AlPO}_4$ containing about 10% of Li_2O and Zinnwaldite (lithium biotite)– $\text{K}(\text{Mg},\text{Li},\text{Fe},\text{Al},\text{Si})_3\text{O}_{10}(\text{OH}_2\text{F})_2$ in the form of lithium-iron mica. All the lithium minerals occur as constituents of granite pegmatites, and are commonly formed by the replacement of earlier formed minerals, especially feldspars, of the pegmatite by hydrothermal solutions. Spodumene is greyish or greenish in colour with pearly lustre while Lepidolite is rose red, lilac, violet, grey and white in colour with pearly lustre. Lithium-bearing minerals are prescribed as Critical and Strategic Minerals in Schedule I Part D of the Mines & Minerals (Development and Regulation) Amendment Act, 2023.

USES & CONSUMPTION

Lithium is mostly used for making rechargeable batteries (phones, computers, cameras and EVs). Lithium and its compounds have several other important applications, including hydrogen fuel storage; metal alloys (military ballistic armour; aircraft, bicycle and train components); specialised glass and ceramics; drying and air conditioning systems; steam or other vapour generating boilers; pharmaceuticals, medicinal chemical and botanical products; refined petroleum products etc.

MINING LEASE & PRODUCTION

Presently, there is no production of lithium mineral

reported in the country from primary lithium resources. The demand for lithium is usually met through imports. As per available information on the website of Ministry of Mines, 01 CL block of lithium and REE was auctioned in the year 2024-25 in the State of Chhattisgarh. Further, 01 ML block of tungsten, lithium and associated minerals (rubidium, niobium, tin) was declared in the NIT released on 16 September 2025 for RAMB Degana Tungsten, Lithium and Associated Mineral Block, Rajasthan.

RECYCLING

As per Notification No.: F. No. 28/15/2024-CMM-Part (1), consequently upon the approval of the Cabinet, the Ministry of Mines, Government of India, on 24/06/2025

notified that the National Critical Mineral Mission (NCMM) will be set up for seven years from 2024-25 to 2030-31. To foster a Circular Economy through recycling, the Mission will aim to recycle at least 10% of the nation's annual consumption of critical minerals — this initiative will enable reinforcing a circular economy model and reducing the dependency on primary raw materials.

WORLD REVIEW

The world lithium reserves are estimated at 28 million tonnes. Lithium reserves are mainly located in Chile which

contributes 33% to the total reserves followed by Australia (22%), Argentina (13%), China (11%), USA (4%), Canada (3%) and Brazil & Zimbabwe (1% each). The countrywise world reserves of lithium are furnished in Table –1. The world production of lithium minerals in Li content was 209.8 thousand tonnes in 2023 which increased by about 30% as compared to that of the previous year. Australia dominated the world production by accounting for 44% output which was followed by Chile (22%), China (16%), Zimbabwe (9%), Argentina (5%), Brazil (3%) and USA (1%). The details are furnished in Table –2.

Table –1: World Reserves of Lithium

<i>(In tonnes of metal content)</i>	
Country	Reserves
World total (rounded)	28000000
Chile	9300000
Australia *	6200000
Argentina	3600000
China	3000000
USA	1100000
Canada	930000
Brazil	390000
Zimbabwe	310000
Portugal	60000
Other countries	2800000

Source: USGS Mineral Commodity Summaries, 2024

* For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 4.8 million tonnes.

Table – 2: World Production of Lithium

<i>(In tonnes)</i>				
Country	Minerals/Li content	2021	2022	2023
Australia	Spodumene	1966744	2754727	3295041
	(Li content)	54754	76692	91734
Chile	Carbonate	150348	201950	220256
	(Li content)	28265	37967	41408
	Hydroxide	12129	15763	23071
	(Li content)	2001	2601	3807
China	(Li content)	14100*	22600*	33000*
Zimbabwe	Lepidolite	--	--	3446
	(Li content)	--	--	10
	Petalite	40726	86330	310004
	(Li content)	775	1642	5726
Argentina	Spodumene	--	--	465636
	(Li content)	--	--	12963
	Carbonate	47917	50000*	50000*
	(Li content)	9008	9400*	9400*
Brazil	Chloride	6017	6200*	6200*
	(Li content)	981	1011*	1011*
USA ^(a)	Spodumene	102938	143720	263876
	(Li content)	2579	3601	6612
Namibia	(Li content)	5000*	2440*	2300*
	Lepidolite	--	325400	300000
	Li content	--	951	877

Table – 2 (Concl'd.)

(In tonnes)

Country	Minerals/Li content	2021	2022	2023
Nigeria	Petalite	133	130*	130*
	(Li content)	2*	2*	2*
	Spodumene	--	3420	15495
	(Li content)	--	95	431
Canada	Spodumene	--	82426	6000*
	(Li content)	--	2409	179
Bolivia	Carbonate	540	617	949
	(Li content)	102	116	178
Portugal	Lepidolite	18533	19000*	19000*
	(Li content)	129	132	132
World Total	(Li content)	117700	161700	209800

Source: BGS, World Mineral Production, 2019-2023.

(a): carbonate *: estimated

FOREIGN TRADE

Exports

Exports of lithium carbonates decreased by 11.71% and stands at 421.38 tonnes in 2023-24 as compared to 477.28 tonnes in the previous year. The shares of export of lithium carbonates in terms of quantity were mainly to Vietnam (54%), China (15%), Turkey (5%), Russia (4%), UAE (3%), Thailand (3%) and USA (1%). The shares of export of lithium carbonates in terms of value were mainly to China (33%), Turkey (22%), UAE (12%), Russia (8%), USA (6%), Vietnam

(4%) and Thailand (2%). Besides, export of lithium oxide and hydroxide also decreased by 45.20% to 117.65 tonnes in 2023-24 from 214.67 tonnes in the preceding year. The share of export of lithium oxide and hydroxide in terms of quantity were mainly to UAE (72%), Lebanon (5%), Saudi Arabia & France (4% each), Tanzania & Indonesia (3% each), USA (1%). Export of lithium oxide and hydroxide in terms of value were mainly to UAE (68%), Lebanon (6%), Saudi Arabia & Tanzania (5% each) and USA, Indonesia, France & Taiwan (3% each) (Tables-3 & 4).

Table – 3: Countrywise Export of Lithium Carbonates
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	477280	781498	421383	611295
China	67002	273214	65119	202465
Turkey	53	286	20900	134947
UAE	200	728	12301	75316
Russia	85800	382435	18031	47509
USA	1824	25395	2962	33682
Vietnam	280800	28314	226500	24011
Sri Lanka	76	664	2200	21642
Korea, Rep. of	10	101	1054	18448
Thailand	8332	31473	11412	11193
Uruguay	1100	7412	1050	8802
Other countries	32083	31476	59854	33280

Figures rounded off

Table – 4: Countrywise Export of Lithium Oxide and Hydroxide
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	214666	1177440	117645	341894
UAE	102000	554996	84605	231799
Lebanon	--	--	6200	19656
Saudi Arabia	1700	12665	5025	18311
Tanzania	--	--	3600	16193

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
USA	103	971	1024	10868
Indonesia	--	--	3000	10305
France	600	3921	4800	10136
Taiwan	--	--	2700	9298
Nepal	1203	8114	2085	6708
Kenya	1500	10787	250	2345
Other countries	107560	585986	4356	6275

Figures rounded off

(--): Nil

Imports

Imports of lithium carbonates increased by 11.81% to 1,146.1 tonnes in 2023-24 from 1,025.02 tonnes in the previous year. The shares of import of lithium carbonates in terms of quantity were mainly to Ireland (35%), Belgium (15%), China (11%), Argentina (8%), USA (5%) and Germany (3%). The shares of import of lithium carbonates in terms of value were mainly to USA (25%), Belgium (19%), China (16%), Argentina (15%), Brazil & Germany (7% each) and Slovenia (4%). Besides, import of lithium

oxide and hydroxide also increased marginally by 2.47% to 1,147.51 tonnes in 2023-24 from 1,119.78 tonnes in the preceding year. The share of import of lithium oxide and hydroxide in terms of quantity were mainly to Belgium (33%), Russia (26%), China (20%), UAE (7%), Germany & Japan (3% each) and Hong Kong & Latvia (2% each). Import of lithium oxide and hydroxide in terms of value were mainly to Russia (31%), Belgium (30%), China (17%), UAE (10%), Germany (5%) and Japan & Hong Kong (2% each) (Tables-5 & 6).

Table – 5: Countrywise Import of Lithium Carbonates

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1025026	1790126	1146083	2036479
USA	75081	331083	52300	517262
Belgium	227000	953345	166860	386132
China	14	217	126032	327230
Argentina	43000	119053	95000	312655
Brazil	50	37	28600	143754
Germany	23080	59083	38010	134067
Slovenia	13000	47342	5000	90500
UK	40200	228993	20000	60911
Chile	3600	23368	13600	30426
Ireland	200000	9976	400000	19006
Other countries	400001	17629	200681	14536

Figures rounded off

Table – 6: Countrywise Import of Lithium Oxide and Hydroxide

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1119783	5525301	1147506	2480417
Russia	186000	887619	304002	768376
Belgium	537575	2789743	383925	746350
China	51252	322511	228290	432466
UAE	156170	893344	85065	248849
Germany	41187	101054	32149	118565
Japan	--	--	36009	54810
Hong Kong	--	--	20001	50198
Latvia	20000	87951	20000	32218

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
USA	22377	52332	13938	27536
Tanzania	--	--	24000	905
Other countries	105222	390747	127	144

Figures rounded off

(--): Nil

FUTURE OUTLOOK

The future global economy will be underpinned by technologies that depend on minerals, such as, lithium, graphite, cobalt, titanium, rare earth elements etc. These are essential for the advancement of many sectors, including high-tech electronics, telecommunications, transport and defence. The demand for lithium has risen exponentially especially in the manufacturer of rechargeable batteries used in mobile phones, computers, EVs etc. Lithium and its compounds have several other important applications like hydrogen fuel storage; metal alloys for military ballistic armour; aircraft; specialised glass and ceramics and several other uses. Lithium is also vital to power the global transition to a low carbon emissions economy, and the renewable energy technologies that will be required to meet the 'Net Zero' commitments of an increasing number of countries around the world. India's focus on clean energy and lower emissions has further amplified the significance of critical minerals. From electric vehicles to renewable energy systems, these minerals play a pivotal role in accelerating the transition towards a greener and more sustainable future.

To fulfill the domestic requirement of critical and strategic minerals including lithium mineral, under the aegis of Ministry of Mines, Government of India, a Joint Venture Company, namely, Khanij Bidesh India Limited (KABIL) was formed for the three Government Enterprises, namely, NALCO, HCL and MECL. KABIL will strive to identify, explore, acquire, develop, mine, process; procure strategic minerals from foreign sources for supplying primarily to India, to meet its domestic requirements.

On 15th January 2024, KABIL has signed an Agreement with the State-owned company of Catamarca, Argentina for exploration and development of 5 Brine type Lithium Blocks in the Catamarca Province of Argentina, covering an area of 15,703 hectare. With this Agreement, KABIL has obtained Exclusivity Right to explore, develop and exploit for commercial production, after establishment of potential, in the 5 blocks, namely, Cortadera-I, 2. Cortadera-VII, 3. Cortadera-VIII, 4. Cateo-2022-01810132 and 5. Cortadera-VI. KABIL is in the process of obtaining statutory approvals to start exploration activities in the blocks. Besides, due diligence for selection of propositions for long-term investment in Lithium and Cobalt Projects of Australia is in progress.



21. Molybdenum

27.20 million tonnes
Total reserves/resources of
molybdenum ore have been
established as on 1st April, 2020.

76 tonnes
of molybdenum ores & concentrates
were exported in 2023-24.

13,481 tonnes
of molybdenum ores & concentrates
were imported in 2023-24.

G Molybdenum (Mo) is a refractory metal used principally as an alloying agent in steel, cast iron & superalloys to enhance strength and resistivity to wear & corrosion. It does not occur in nature in free state. Usually, it is found in chemically combined form with other elements. Molybdenite (MoS_2) is the principal ore of molybdenum. About two-thirds of global molybdenum production is as by-product of copper mining and only about one-third is obtained from primary molybdenum mines. In India, by-product concentrates of molybdenum are produced intermittently from uranium ore of Jaduguda mine belonging to Uranium Corporation of India Ltd (UCIL) in Jharkhand. The internal demand for molybdenum and its products is met mostly through imports.

RESERVES/RESOURCES

India, though, is endowed with rich mineral wealth, there are several critical minerals that it lacks and one of them is molybdenum. In India, molybdenum is associated generally with copper, lead and zinc ores. Rakha copper deposit in Jharkhand contains 45 to 48 ppm molybdenum. Malanjkhand copper deposit in Madhya Pradesh contains 0.04% recoverable molybdenum. Dariba-Rajpura lead-zinc deposit in Rajasthan contains molybdenum besides bismuth, arsenic and cadmium. The multimetal deposit at Umpyrtha in Khasi and Jaintia Hills, Meghalaya, reportedly

contains molybdenum in association with copper, lead and tungsten. Molybdenum deposit in Karadikuttam in Madurai district, Tamil Nadu, contains 0.02 to 0.14% recoverable molybdenum.

As per NMI database as on 1.4.2020, based on UNFC System, the resources of molybdenum ore in the country have been estimated at 27.20 million tonnes containing about 16,891 tonnes MoS_2 . The above resources of ore are located in Tamil Nadu (17.88 million tonnes), Madhya Pradesh (8 million tonnes) and Karnataka (1.32 million tonnes) (Table-1).

Table – 1: Reserves/Resources of Molybdenum as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
Ore	0	27203398	27203398
Contained MoS ₂	0	16890.56	16890.56
By States			
Karnataka			
Ore	0	1320900	1320900
Contained MoS ₂	0	1718.7	1718.7
Madhya Pradesh			
Ore	0	8000000	8000000
Contained MoS ₂	0	5020	5020
Tamil Nadu			
Ore	0	17882498	17882498
Contained MoS ₂	0	10151.86	10151.86

Figures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on "Exploration & Development" under "General Reviews".

SUBSTITUTES

There is hardly any substitution for molybdenum in its major application, viz, as an alloying element in steel and cast irons. Owing to the non-availability of molybdenum, there was an apparent need to develop new materials that could be a suitable substitute vis-a-vis the alloying properties of the metal. Potential substitutes for molybdenum include chromium, vanadium, niobium (columbium) & boron in alloy steels; tungsten in tool steels; graphite, tungsten & tantalum for refractory materials in high temperature electric furnaces and chrome-orange, cadmium-red & organic-orange pigments for molybdenum orange.

USES AND CONSUMPTION

Usually, molybdenum is used in the form of roasted concentrates, oxide or ferromolybdenum in the Defence

industries. The production of ferromolybdenum increased from 581 tonnes in 2022-23 to 766 tonnes in 2023-24 (Table-2).

Non-ferrous Technology Development Centre at the Defence Metallurgical Research Laboratory, Hyderabad, has a pilot plant for producing molybdenum powder. Institute of Minerals and Materials Technology (formerly RRL), Bhubaneswar, has been undertaking basic research on recovery of molybdenum from spent catalysts and alloy scraps.

Moly Metal LLP, a leading manufacturer of Molybdenum alloys ferromolybdenum (FeMo) and molybdenum disulphide (MoS₂), commenced production in 2007 at a new manufacturing plant in the Union Territory of Daman. RUBAMIN, a Gujarat-based Company, reportedly has a capacity of 1,500 tonnes per annum sodium molybdate and 800 tonnes per annum ammonium molybdate, both of which are derivatives of molybdenum.

**Table – 2: Production of Ferro-molybdenum
2019-20 to 2023-24**

(In tonnes)

Year	Production
2019-20	527
2020-21	428
2021-22	435
2022-23 ^(P)	581
2023-24 ^(P)	766

Source: Monthly Statistics of Mineral Production, March, 2024, IBM

WORLD REVIEW

The world reserves of molybdenum are at 15 million tonnes, located mainly in China (39%), USA (23%), Peru (10%), Chile (9%) and Russia (7%) (Table-3).

The world mine production of molybdenum in terms

of metal content increased by 5% to 2.79 lakh tonnes in 2023 from 2.65 lakh tonnes in 2022. China with 39% production was the main producer of molybdenum in the world followed by Chile (16%), Peru & USA (12% each) and Mexico (10%) (Table-4).

Table – 3: World Reserves of Molybdenum

(By Principal Countries)

(In '000 tonnes of molybdenum content)

Country	Reserves
World: Total (rounded off)	15000
China	5800
USA	3500
Peru	1500
Chile	1400
Russia	1100
Australia	690
Armenia	150
Mexico	130
Argentina	100
Canada	72
Other countries	124

Source: USGS, Mineral Commodity Summaries, 2024.

Table – 4: World Mine Production of Molybdenum

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World: Total (rounded off)	279000	265000	279000
China	106000*	106000*	110000*
Chile	49403	45551	44127
USA	41100	34600	34000
Peru	34148	31588	33476
Mexico	19894	19330	28313
Armenia	11310	11390	11480
Iran ^(a)	8275	6700	6700
Kazakhstan	396	2005	3727
Other countries	8744	7572	7507

Source: BGS, World Mineral Production, 2019-23

Note (s):

(a) years ended 20th March following that stated.

(1) In addition to the countries listed, Australia, Georgia, India, Democrate P.R. of Korea, Romania and Tajikistan are believed to produce molybdenum.

(In tonnes of metal content)

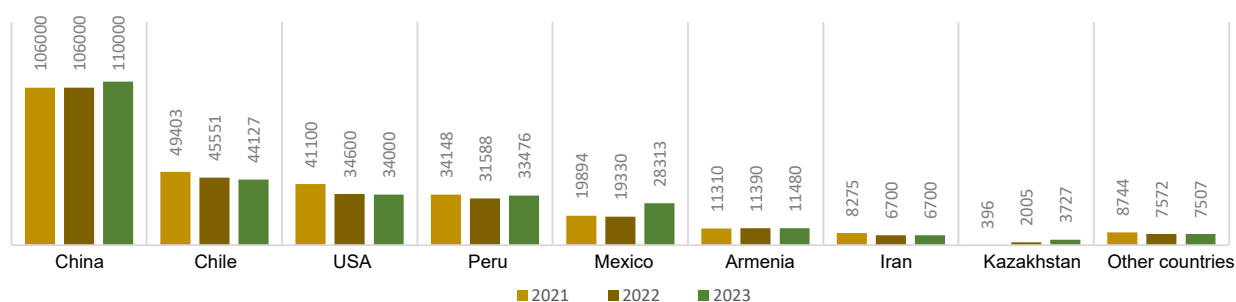


Fig. 1: Countrywise Production of Molybdenum

FOREIGN TRADE

Exports

Exports of molybdenum ores & concentrates increased by 27% to 76 tonnes in 2023-24 from 60 tonnes in 2022-23. Exports were mainly to Republic of Korea (Table-5).

Table – 5: Exports of Molybdenum Ores And Conc.

Country	(By Countries)			
	2022-23(R)		2023-24(P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	60	223955	76	185324
Korea, Rep. of	60	223768	76	185316
UK	--	--	++	8
USA	++	173	--	--
Mexico	++	10	--	--
Indonesia	++	4	--	--

Figures rounded off

Imports

Imports of molybdenum ores & concentrates increased by 29% to 13,481 tonnes in 2023-24 from 10,436 tonnes in 2022-23. Imports were mainly from Chile (39%) , Thailand (24%), UAE (14%) and USA (5%) (Table-6).

Table – 6: Imports of Molybdenum Ores and Conc.

Country	(By Countries)			
	2022-23(R)		2023-24(P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	10436	23316669	13481	34312389
Chile	4259	9596166	5202	14083626
Thailand	2791	6167116	3190	7827782
UAE	625	1821013	1867	5038729
Switzerland	239	931880	582	1820101
USA	741	1475082	627	1791194
Luxembourg	120	298445	320	797404
China	241	549414	198	523771
Korea, Rep. of	515	943227	204	512056
Netherlands	263	579702	161	495965
Taiwan	33	117415	122	345158
Other Countries	609	837209	1008	1076603

Figures rounded off

FUTURE OUTLOOK

The principal uses for molybdenum are expected to continue to be as catalysts in chemicals and as an additive in steel manufacturing, most importantly alloy and stainless steel. Molybdenum plays a vital role in the Energy Industry, and it may become an increasingly important factor in environmental protection technology, where it is used in high-strength steels for automobiles to reduce weight and improve fuel economy and safety. Molybdenum-based catalysts have a number of important applications in the Petroleum and Plastics industries. A major use is in the hydrodesulfurisation of petroleum, petrochemicals, and coal-derived liquids. Catalysts are estimated to account for more than 70% of chemical uses of molybdenum. Molybdenum not only allows for economical fuel refining,

it also contributes to a safer environment through lower sulphur emissions. Analysts expect global demand for molybdenum-based catalysts to continue its increasing trend as there are no practical alternatives to molybdenum in many of the catalytic applications. The need for companies to reduce carbon dioxide emissions from coal-fired power stations will require plants to run at higher temperatures, resulting in greater demand for higher grade molybdenum-bearing steels. Increase in molybdenum use is expected to continue in stainless steels and full alloy steels mainly in the consumer product and transportation industries.

According to the International Molybdenum Association 2021/2022, SMR predicts that over the next 10 years the Chemical Processing Industry and the Oil and Gas Industry will be significant growth areas for molybdenum

demand. Transport vehicle production with stainless steel bipolar plates (BPP) containing 2% molybdenum is forecasted to be the main technology in the future for hydrogen fuel cells. While these will be used in electric vehicles and buses, the main demand is expected to be in the truck sector.

There are many exciting opportunities on the horizon which are likely to lead to an increase in demand

for molybdenum, for example, molybdenum alloying is particularly valuable to special steels used in wind power generation. Due to the growth of the sector and the increasing size and power output of wind turbines, such special steels are reaching high tonnages. The potential molybdenum use, based on likely wind power capacity scenarios, is estimated to amount to 300,000 metric tonnes by 2050.



22. NICKEL

189 million tonnes

Total reserves/resources of nickel ore have been established as on 1st April, 2020.

5,917 tonnes

Nickel and Alloys incl. scrap were exported in 2023-24.

53,574 tonnes

Nickel and Alloys incl. scrap were imported in 2023-24.

Nickel is a lustrous, silvery-white metal. Nickel is the fifth most common element found in earth's crust. Nickel does not occur in native state. Pure nickel is obtained by reduction of its oxides or by the Mond process which consists of the formation of volatile nickel carbonyl produced by passing carbon monoxide over heated nickel oxide, and the dissociation of this compound at higher temperature into nickel and carbon monoxide, which can be used again. It has a melting point of 1,453°C, relatively low thermal & electrical conductivities, high resistance to corrosion & oxidation, excellent strength & toughness at high temperatures and is capable of being magnetised. It is attractive and very durable as a pure metal and alloys readily with other metals. Nickel is not produced from primary sources in the country and the entire demand is met through imports.

RESERVES & RESOURCES

Nickel occurs principally as oxides, sulphides and silicates in India. Important occurrence is nickeliferous limonite in the overburden of chromite in Sukinda Valley, Jajpur district, Odisha. In addition, nickel is found associated with uranium deposits at Jaduguda, Jharkhand and a processing plant is being developed for its recovery. Resources are spread over in Singhbhum East district of Jharkhand and Jajpur, Keonjhar & Mayurbhanj districts of Odisha.

As per the NMI database as on 1.4.2020, based on

UNFC, Resources of nickel are estimated at 189 million tonnes. The entire resources fall under Remaining Resources category. The State of Odisha is endowed with the largest share of resources of nickel ore in the country at 175 million tonnes (93%) followed by Jharkhand and Nagaland. These resources mainly occur in three districts, namely, Jajpur (140 million tonnes), Mayurbhanj (27 million tonnes) and Keonjhar (8 million tonnes). Jharkhand has 9 million tonnes (5%) of resources, most of which are in Singhbhum (East) district. Nagaland has 5 million tonnes (3%) of resources, which predominantly occur in Kiphire district (Table-1).

Table – 1: Reserves/Resources of Nickel Ore as on 01.04.2020

(In million tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India : Total		189	189
By Grades			
+ 0.9% Ni	—	42	42
0.5 to 0.9% Ni	—	94	94
(+) 0.5% Ni, unclassified	—	53	53
Not-known	—	0.23	0.23
By States			
Jharkhand	—	9	9
Karnataka	—	0.23	0.23
Nagaland	—	5	5
Odisha	—	175	175

Figures rounded off.

MINING LEASE & PRODUCTION

Vedanta, one of the world's largest natural resources conglomerates has acquired Nicomet, a leading Nickel and Cobalt producer based in Goa. With this acquisition, Vedanta has become India's sole producer of Nickel. Nicomet has emerged as a certified producer of high-quality battery-grade Nickel Sulphate Crystals used for manufacturing of batteries for electric vehicles globally. India's demand for nickel is currently pegged at 45 KTPA which is entirely met through imports. At present, Nicomet's plant has a capacity to produce 7.5 KTPA Nickel & Cobalt. With an ambitious growth plan in place, Vedanta is well poised to meet 50% of the country's total Nickel demand.

During the year 2022-23, 2 blocks in respect of Ni, Cr and associated PGE minerals have been successfully auctioned in Chhattisgarh.

RESEARCH & DEVELOPMENT

India's first facility to produce nickel, a metal for which the country is completely dependent on imports, has been launched by the Hindustan Copper Limited (HCL) at its Indian Copper Complex (ICC) at Ghatsila in Jharkhand. The new facility "Nickel, Copper and Acid Recovery Plant" is the first facility in India to produce nickel metal of London Metal Exchange (LME) grade from a primary resource.

NMDC has submitted an application to DMG, Govt. of Odisha for proposal to reserve 8 sq. km. area in Jajpur district, Odisha, under Section 17 A (2A) of MM (D&R) Amendment Act, 2015 for prospecting and mining operations of Nickel. CSIR-NML has developed CSIR's first complete and holistic TRL- 4 Process for Extraction and Separation of Nickel and other metals from spent lithium batteries of mixed origin. Considering the need and significance of the problem related to energy materials, CSIR-IMMT has carried out a study on the synthesis of nickel-based alloys by Pyro metallurgical recycling of spent Nickel Metal hydride (Ni-MH) batteries. Key findings of the study were (a) the recovered nickel and cobalt in the

form of alloys (99% pure) could be reused in the Battery Industry, (b) Also, 25% of the nickel requirement in steel making is met from secondary sources, so this alloy can fulfill some of the requirement.

USES & CONSUMPTION

Sectorial uses of nickel metal are in the areas of stainless steel making catalysis in chemical industries, as an electroplating material; heat resistant alloys; as an alloying element for non-ferrous metals; space, defence & rocket industries; and nickel-cadmium batteries. Nickel is used in many specific and recognisable industrial and consumer products including stainless steel, alnico magnets, coinage, for filters & binders, rechargeable batteries, foundry, electric guitar strings, microphone capsules and special alloys. It is also used for plating and as a green tint in glass. Nickel is predominantly an alloy metal & its chief use is in the nickel steel & nickel cast iron, of which there are many varieties. It is also widely used in many other alloys, such as, nickel-bronze & brass and alloys with copper, chromium, aluminum, lead, cobalt, silver & gold. It is used as a catalyst, which is key to several important reactions including the hydrogenation of vegetable oils, reforming of hydrocarbons and in the production of fertilizers, pesticides and fungicides.

Nickel sulphate is an important compound used commercially in the country in nickel plating, in dip baths for enameling, in preparation of nickel compounds and as a catalytic nickel. Nickel-based alloys, like stainless steel with higher nickel content are used for more demanding applications, such as, in gas turbines and some chemical plants.

World over, about 65% of nickel is used in the manufacturing of stainless steel and 20% in other steel and non-ferrous (including super alloys) components often used for highly specialised industrial, aerospace and military applications. About 9% is used in plating and 6% in other uses, including coins and a variety of nickel chemicals.

SUBSTITUTES

Aluminum, coated steels, plain chromium steels and plastics are the common substitutes that could replace stainless steel to a limited extent in many construction and transportation applications. Low-nickel, duplex, or ultra-chromium stainless steels are being substituted for austenitic grades in construction. Nickel-free specialty steels are sometimes used in place of stainless steel within the power-generating, petrochemical and petroleum industries. Titanium alloys or specialty plastics are in use as materials that could substitute nickel metal or nickel-based alloys in applications to resist corrosion in highly corrosive

chemical environments. Lithium-ion batteries are replacing nickel-metal hydride batteries in many applications.

WORLD REVIEW

The world reserves of nickel are estimated at 130 million tonnes of metal content. Indonesia (42%), Australia (18%), Brazil (12%), Russia (6%) and New Caledonia (5%) are the major countries having reserves of Nickel. Globally, nickel resources have been estimated to contain more than 350 million tonnes of nickel, with 54% in laterites and 35% in magmatic sulphide deposits. Extensive nickel resources are also found in manganese crusts and as nodules in the ocean floor (Table-2).

Table – 2: World Reserves of Nickel
(By Principal Countries)

(In metric tonnes of nickel content)

Country	Reserves
World: Total (rounded off)	130000000
Indonesia	55000000
Australia ^(a)	24000000
Brazil	16000000
Russia	8300000
New Caledonia ^(b)	7100000
Philippines	4800000
China	4200000
Canada	2200000
USA ^(c)	340000
Other countries	9100000

Source: USGS, Mineral Commodity Summaries, 2024.

(a): For Australia, Joint Ore Reserve Committee - compliant or equivalent reserves were 8.6 million tonnes.

(b): Overseas territory of France.

(c): Includes reserve data for three projects. An additional three domestic projects have defined resources but have not yet defined reserves.

In 2023, world mine production of nickel increased considerably to 3.75 million tonnes as compared to 3.26 million tonnes of metal content in the previous year. The

principal producers of nickel in the world in 2023 were Indonesia (54%), Philippines (10%), Russia (7%), New Caledonia (6%), Canada (4%), Australia (4%) and China (3%) (Table-3).

Table – 3: World Mine Production of Nickel
(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World: Total (rounded off)	2800000	3261000	3756000
Indonesia*	1069000	1579000	2030000
Philippines	386359	360000	387000
Russia	251040	267980	256320
New Caledonia	186284	199951	231217
Canada	159145	122115	158668
Australia	150876	155007	149090
China	103900	109400	112600

Table- 3 (Concl'd.)

(In tonnes of metal content)			
Country	2021	2022	2023
Brazil	75988	64084	72398
Colombia	43800	46400	42400
Finland ^(a)	42163	44921	41941
Other countries	330949	312589	274449

Source: BGS, World Mineral Production, 2019-2023

(*): Estimated

(a): In addition some nickel is produced as a by-product of talc operation. The amount is negligible.

(In tonnes of metal content)

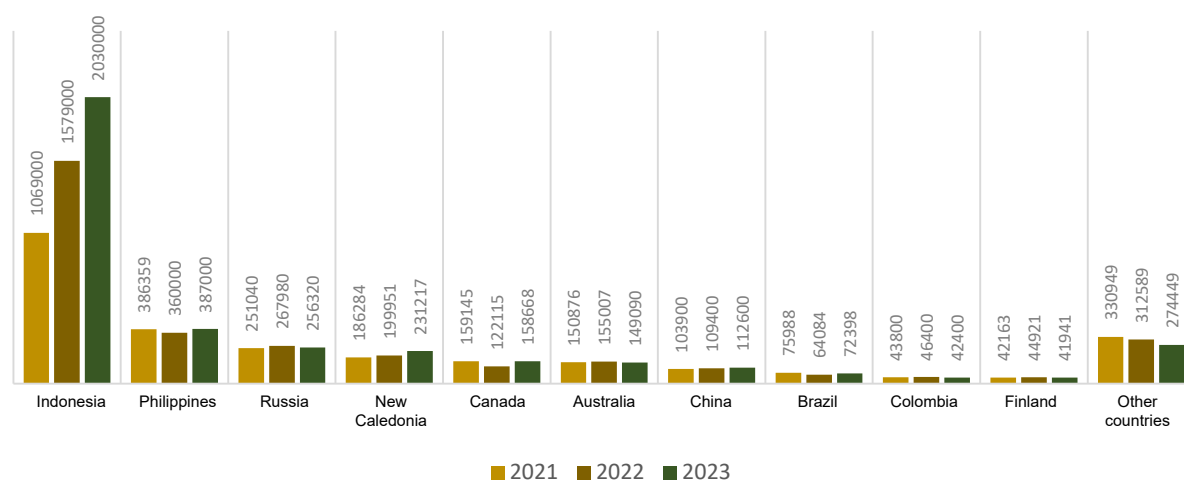


Fig. 1: World Mine Production of Nickel

FOREIGN TRADE

Exports

Exports of nickel ores and concentrates were 19 tonnes in 2023-24, as compared to the negligible quantity in the preceding year. On the other hand, exports of Nickel and Alloys including Scrap decreased by 14% to 5,917 tonnes in 2023-24 from 6,885 tonnes in the previous year. Exports of

Nickel and Alloys were 3,918 tonnes in the year 2023-24, as compared to 4,996 tonnes in the preceding year. Out of the total Nickel and Alloys including Scrap exported in 2023-24, nickel & alloys were 3,918 tonnes, while Nickel Waste & Scrap were 1,999 tonnes. Exports of Nickel and Alloys including Scrap were mainly to UK (21%), USA (9%), Mexico (8%), Republic of Korea (6%) and Saudi Arabia (3%) (Tables-4 to 6)

Table – 4: Export of Nickel Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	19	2971
Belgium	--	--	19	2969
Nepal	--	--	++	2

Figures rounded off.

Table – 5: Export of Nickel and Alloys Incl. Scrap

(By Items)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total (All items)	6885	16098361	5917	13667911
Nickel Waste and Scrap	1889	2255404	1999	2289256
Nickel and Alloys	4996	13842957	3918	11378655

Figures rounded off.

Table – 6: Export of Nickel and Alloys Incl. Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	6885	16098361	5917	13667911
UK	974	1372645	1216	1860596
USA	415	1318016	530	1781983
Mexico	447	1217116	494	1269940
Saudi Arabia	118	630734	200	862098
Korea, Rep. of	205	443813	367	746763
Brazil	147	406415	256	692032
UAE	272	662316	207	613993
Turkey	142	324440	281	605942
China	282	991722	322	521522
Germany	139	364345	196	488706
Other countries	3744	8366799	1848	4224336

Figures rounded off.

Imports

Imports of Nickel Ores & Concentrates were negligible in the year 2023-24. Imports of Nickel & Alloys including Scrap were at 53,574 tonnes in 2023-24 which increased by 10% from that of 48,896 tonnes in the previous year. Out of the total alloys and scrap imported in 2023-24, Nickel & Alloys were at 49,224 tonnes as compared to 45,782 tonnes in the previous year, while Nickel Waste & Scrap were 4,350 tonnes as compared to 3,114 tonnes in the previous year.

Imports of Nickel and Alloys including Scrap in 2023-24 were mainly from China (18%), Japan (10%), Norway (11%), Netherlands (9%) and USA (6%) (Tables-7 to 9).

As per Schedule-1 of Foreign Trade Policy, ITC (HS), 2022, Import Policy of nickel ores & concentrates (ITC-HS Code no. 2604) and Nickel Waste & Scrap (ITC-HS Code no. 75030010) are allowed free. However, some forms of metal waste & scrap (ITCHS Code No. 75030090) are restricted.

Table – 7: Import of Nickel Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	20	369	++	11
USA	--	--	++	11
Ethiopia	20	369	--	--

Figures rounded off.

Table – 8: Import of Nickel and Alloys Incl. Scrap

By Items

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total (All items)	48896	104929668	53574	111013637
Nickel And Alloys Total	45782	102238388	49224	107346581
Nickel Waste and Scrap	3114	2691280	4350	3667056

Figures rounded off.

Table – 9: Import of Nickel and Alloys Incl. Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	48896	104929668	53574	111013637
China	6895	13718087	9562	19885156
Japan	4191	9181433	5443	11060271
USA	2041	7208080	3400	10702223
Norway	4725	10713861	5878	10070781

Table- 9 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Netherlands	5614	11172185	4919	8737907
Singapore	1898	5525799	3445	8293070
UK	1987	5343861	1968	5904883
Canada	1828	3876777	3225	5680066
Germany	1395	4194923	1209	3675563
UAE	4637	6677028	1965	2675351
Other Countries	13685	27317634	12560	24328366

Figures rounded off.

FUTURE OUTLOOK

Primarily, world nickel demand arises from the production of stainless steel which consumes about 65% of global nickel production. Nickel contributes 10 to 20% of the input cost in stainless steel production, depending on the nickel content. The future outlook for nickel depends mainly on the production of stainless steel which is one of the main drivers for nickel consumption. Batteries and the ongoing Electric Vehicle revolution could prove to be a transformational event, as NCA and NCM are still predominantly used. However, Li-ion technology is gaining popularity and is increasingly getting established as the battery of choice.

Nickel has been under constant demand from the Ferroalloys and Alloy/Stainless Steel Industry. Nickel is practically unavailable in the country and the entire quantity of unwrought and other forms of the nickel needs to be imported. Hence, acquisition of assets globally must

be encouraged to maintain a steady supply to the industry. Simultaneously, R&D to extract Nickel from the lateritic ore overburden available in Sukinda Valley, Odisha could serve as an added advantage.

India will have no option but to depend on imports for this metal until a technology to recover nickel from the overburden of chromite ore in Odisha is established on a commercial scale. The process developed by HCL for the production of primary nickel from waste generated during copper refining will be a breakthrough in the area of nickel production in the country.

India imports as well as exports nickel scrap covered by ISRI code, Aroma, Barly, Dandy, Daunt, Delta, Decov, Depth, Hitch, House, Ideal, Indian, Junto, Lemon, Lemur which are covered under HS code 75030010. But there is hardly any data available or reported for recycling and recovery of nickel from scrap. The recycling of nickel-bearing scrap in Organised Sector will be another source for meeting the demand.



23. Platinum and Palladium

20.92 tonnes
of metal content of Total Resources
have been established in the country.

1,044 kg
of platinum alloys and related metals
were exported in 2023-24.

10,410 kg
of platinum alloys and related metals
were imported in 2023-24.

Platinum Group of Metals (PGMs) is a family of 6 metals—platinum, palladium, rhodium, iridium, osmium and ruthenium. They have similar physical and chemical properties and tend to occur together in the same mineral deposits. These six elements are classified into two groups with reference to the specific gravity of gold (19.2 g/cm^3). The elements, Ru, Rh, Pd (sp. gr. $12\text{--}12.4 \text{ g/cm}^3$) are lighter, while the other three elements, Os, Ir and Pt are heavier than gold but within the range of $21.0\text{--}21.5 \text{ g/cm}^3$. Platinum is an extremely rare metal occurring at a concentration of only 0.005 ppm in earth's crust. Major applications of platinum and palladium are in Automotive Sector for emission control and in chemical and petroleum refining.

RESERVES AND RESOURCES

Reserves/Resources of PGMs in the country as on 01.04.2020 as per NMI Database, based on UNFC System, are placed at 20.92 tonnes of metal content. By State, Odisha alone accounts for 67% of country's resources of PGE followed by Uttar Pradesh (13%) and Tamil Nadu (8%) (Table-1).

Boula–Nausahi, a 3 km-long belt, 170 km NE of Bhubaneswar, Odisha, is the only proven Platinum Group of Metals (PGMs) deposit in the country. Preliminary assessment of PGMs in Sukinda ultramafic field indicated isolated anomalous values in chromite. Platinum values of 2 to 400 ppb and palladium values of 1 to 500 ppb were established on analysis. The limonite cappings over ultramafic rocks showed combined platinum and palladium values between 40 and 290 ppb. In Boula–Nuasahi ultramafic complex, the easternmost chromite

band known as Shankar–Ganga load, investigations revealed potential PGMs mineralisation. In Sittampudi Complex, Salem district, Tamil Nadu, analysis of chromite bands showed 0.03 to 0.75 ppm Pt and 0.1 to 1.0 ppm Pd, whereas amphibolite samples showed 0.03 to 0.05 ppm Pt and 0.03 to 0.5 ppm Pd. A platinum-rich chromite-ferrochromite breccia zone stretching to about hundred metres in gabbroic matrix was identified in the southern extension of the already known Boula–Nuasahi area in Kendujhar district, Odisha. In Usgaon area, Southern Goa, PGMs samples analysed up to 0.03 ppm Pt and 0.03 to 0.15 ppm Pd. In recent past, occurrences of PGE mineralisation were reported in mafic-ultramafic complex of Shivamogga schist belt in Davangere district of Karnataka. Three zones having 10 to 830 ppb of platinum and 50 to 1,500 ppb of palladium were established.

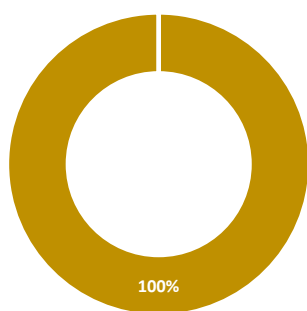
Table – 1: Reserves/Resources of PGM as on 01.04.2020

(By Grades/States)

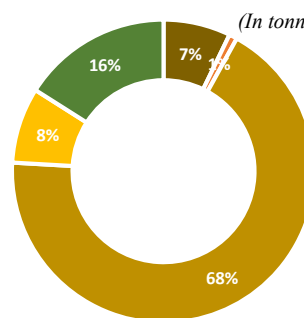
(In tonnes of metal content)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
India	-	20.92	20.92
Karnataka	-	1.5	1.5
Kerala	-	0.18	0.18
Odisha	-	14.2	14.2
Tamil Nadu	-	1.69	1.69
Uttar Pradesh	-	3.35	3.35

Figures rounded off



■ Total Resources



■ Total Resources ■ Kerala ■ Odisha ■ Tamil Nadu ■ Uttar Pradesh

Fig. 1: Resources of PGM

USES & CONSUMPTION

China and India are moving forward with large-scale plans to reduce the amount of carbon emission in their respective countries. Currently, more than half of platinum and palladium mineral goes into making catalytic converters in automobiles. Automobiles that run on diesel predominantly use platinum for catalytic conversion. Platinum-cured silicones are used to coat and protect automotive air bags from their explosive system. The air bags contain an initiator sensor, which uses a fine platinum wire coated with explosive material to facilitate release of the air bag. The chemical inertness and refractory properties of these metals are conducive for their applications in electrical, electronics, dental, medical fields and in the Glass Industry. These metals are also used as catalyst in various chemical processes, viz, in organic synthesis in hydrogenation, de-hydrogenation & isomerisation, production of nitric acid, the raw material for the manufacture of fertilizers, explosives & polymers and fabrication of laboratory equipment.

In addition, platinum, palladium and a variety of complex gold-silver-copper alloys are used as dental restorative materials. The non-corrosive and non-allergic properties of platinum find varied applications in the medical field. Platinum's excellent compatibility with living tissue unaffected by the oxidising reaction of blood enables its utility in pacemakers.

The primary usage of PGMs is in chemotherapy for treatment of cancer. It has the ability to prevent division of certain living cells, a remarkable characteristic which finds profound application in treatment of cancer. Besides,

platinum-iridium alloys are extensively used in prosthetics and biomedical devices.

Platinum's excellent conductivity lends itself for use in the electrodes of phosphoric acid fuel cells for generating electricity. Another significant use of platinum and its alloys, in cast or wrought form is in jewellery. Platinum-iridium alloys find major application in making crucibles for growing crystals. Glass made with platinum and rhodium is used in housing construction, flat screen televisions, computer monitors, display panels, automobile displays, factory monitoring equipment, etc. Recently, a new metallic glass featuring micro-alloys of palladium with silicon, germanium, silver, etc. was reportedly developed at University of California. The glass is characterised by strength and toughness. Platinum is used to enhance storage capacity of devices, such as, computer hard discs, cellphones, digital cameras and personal music players. Recently, palladium-silver resistors have been used in secondary lightning surge protection devices. In Electronic Industry, palladium's use is for Multi-Layer Ceramic Capacitors (MLCC). The effect of miniaturisation of MLCC has not reduced the quantum of palladium used as more number of MLCC are required for the same electronic device. Platinum-based fuel cells are proving to be more cost effective, cleaner and more reliable than alternatives, such as, diesel generators.

Rhodium usage is also on the rise in the Automotive Industry apart from fibre glass. Platinum is the catalyst used by fuel cells to convert hydrogen and oxygen to electricity.

Palladium is also likely to play a role in fuel cells.

Platinum acts as an effective and durable catalyst in hydrogen-powered Fuel Cell Electric Vehicles (FCEVs).

PGMs are also used in electronics, chemical manufacturing, jewelry and renewable energy sectors. India relies heavily on imports to meet its PGM needs, as domestic production is negligible. The country's growing focus on reducing emissions and increasing industrial applications is driving up the demand for these metals.

SUBSTITUTES

Platinum and palladium are two of the most expensive metals on the planet. It is usually easier to substitute metals of the platinum group for one another, especially in alloys, than to use alternative materials, which is evident from the total dominance of ruthenium-based resistors over the palladium–silver resistors for high-powered applications. Substitutes in electrical use include tungsten, nickel, silver, gold and silicon carbide. Alternative catalysts include nickel, molybdenum, tungsten, chromium, cobalt, vanadium, silver and rare earths. Rhenium, however, has been used most satisfactorily as substitute for platinum as a catalyst in petroleum refining. Stainless steel and ceramics can be substituted where resistance to corrosion is the primary concern. Some motor vehicle manufacturers have substituted platinum by palladium in catalytic converters, especially for petrol engines. Particulate matter and residual sulphur contaminate palladium and hence, it was excluded from catalysts used in diesel vehicles. A new technology

now allows up to 25% substitution of platinum in diesel catalytic converters with palladium.

Similarly, manufacturers of electronic parts are also reducing the average palladium content of the conductive pastes used to form the electrodes of multi-layer ceramic capacitors, substituting base metals or silver–palladium pastes which contain significantly less palladium.

Rhenium, tungsten and molybdenum as substitute for platinum in aromatics hydrogenation catalysts have been investigated. Recently, a new type of iron and carbon-based catalysts has been discovered which is stable and active in both acidic and alkaline media and may even eliminate the need for platinum in catalysts and this therefore is likely to revolutionise the Proton Exchange Membrane Fuel Cell (PEFC) Industry.

WORLD REVIEW

The world reserves of Platinum-Group Metals (PGMs) are estimated at 71,000 tonnes concentrated mostly in South Africa (89%) followed by Russia (8%), Zimbabwe (2%) and USA (1%) (Table-2).

World mine production of PGMs increased slightly by 1% to 440 tonnes of metal content in 2023 from 436 tonnes of metal content in 2022 (Table-3).

South Africa, which accounted for 58% of the total PGMs mine production in 2023, was followed by Russia (24%), Zimbabwe (9 %), Canada (5%) and USA (3%) while other countries contributed the remaining 1 per cent.

Table – 2: World Reserves of Platinum Group of Metals
(By Principal Countries)

(In kilograms of PGM content)	
Country	Reserves
World: Total (rounded off)	71000000
South Africa	63000000
Russia	5500000
Zimbabwe	1200000
Canada	310000
USA	820000
Other countries	NA

Source: USGS, Mineral Commodity Summaries, 2024.

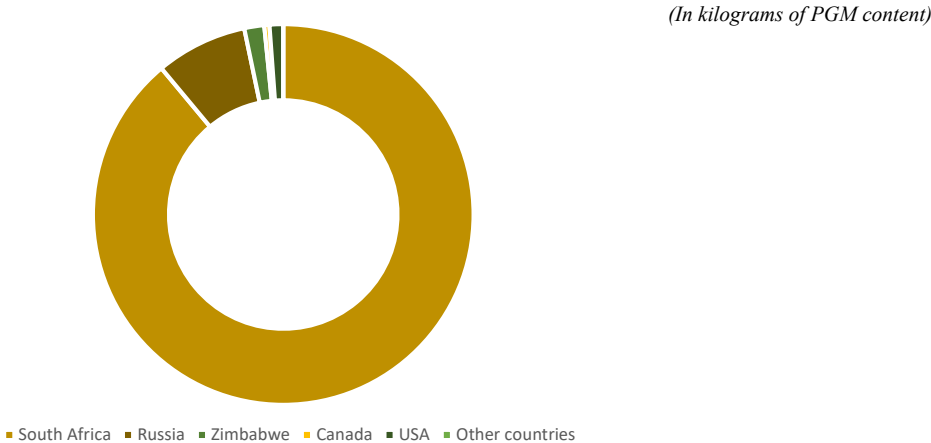


Fig. 2: Countrywise Reserves of PGM

Table – 3: World Mine Production of PGMs

(By Principal Countries)

(In kilograms of metal content)

Country	Metal	2021	2022	2023
World Total (Rounded Off)	All PGMs	467000	436000	440000
Finland	Platinum	1447	1243	933
	Palladium	1036	960	762
Poland	Platinum	8*	5*	5*
	Palladium	4*	4*	2*
Russia	Platinum	19935	20246	20650
	Palladium	81358	86769	83721
	Iridium	200*	200*	200*
	Rhodium	2300	2300	2300
	Ruthenium	300*	300*	300*
Serbia	Platinum	30*	10*	10*
	Palladium	100*	100*	100*
Ethiopia	Platinum ^(a)	4*	30*	30*
South Africa	Platinum	141626	124401	124870
	Palladium	84336	73104	74917
	Iridium	7006	6160	6222
	Rhodium	22256	18800	18549
	Ruthenium	31463	27900	28478
Zimbabwe	Platinum	14732	16460	19180
	Palladium	12619	13935	15820
	Iridium	612	602	958
	Rhodium	1333	1461	1672
	Ruthenium	1239	1341	1331
Canada	Platinum	4888*	5301*	5174*
	Palladium	15275	16565	16114*
	Rhodium	204*	221*	—
USA	Platinum	4017	2737	2769
	Palladium	13725	9202	9448
	Rhodium	100*	—	—
Colombia	Platinum	618	501	893
China	Platinum	2800*	2800*	2800*
	Palladium	1300*	1300*	1300*
Uzbekistan	Palladium	160	93	90*
Australia	Platinum	90	100	102*
	Palladium	380	392	410

Source: BGS, World Mineral Production, 2019-2023

Note (s):

(1) Wherever possible, figures relate to quantities of platinum group metals thought to be recovered from ones originating in the country stated.

(2) South Africa thought to be the only country to produce osmium, however; data is unavailable.

(3) Figures for metal production are only given for countries where recovery is thought to be based predominantly on domestic materials or on imported materials which have not been recorded as mine production elsewhere in the table.

(4) In addition to the production listed, concentrates produced in Spain, and possibly other countries, are believed to be processed in China to recover platinum group metals.

(a) Years ending 7th July of that stated

*: established

FOREIGN TRADE

Exports

Exports of platinum alloys and related metals decreased by about 21% to 1,044 kg valued at ₹ 280.491 crore in 2023-24 from 1,328 kg valued at ₹ 76.976 crore in the previous year.

Export of Platinum (Powder, Unwrought and Others) were mainly confined to USA (39 %) followed by UK (31%), Italy (21%) and Malaysia (8%). Exports in 2023-24 also included of platinum (unwrought) at 657 kg which decreased by 45% as compared to 1,191 kg in the previous year. During

2023-24, exports of other metals of platinum group rose significantly by 179.56% to 383 kg as compared to 137 kg during the preceding year while export of platinum-powder

was 4 kg in 2023-24 as compared to negligible quantity in the previous year (Tables- 4 to 9)

Table – 4: Export of Platinum Alloys & Related Metals: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1328	769761	1044	2804912
Italy	95	348990	289	793256
UK	551	297016	243	501525
USA	640	86101	391	914063
France	3	26720	2	29143
Japan	++	5580	1	9977
UAE	1	2378	++	308
Switzerland	++	920	25	297978
Malaysia	1	107	55	57
Germany	--	--	23	258513
Saudi Arabia	--	--	++	38
Other Countries	37	1949	15	54

Figures rounded off

(In kg)

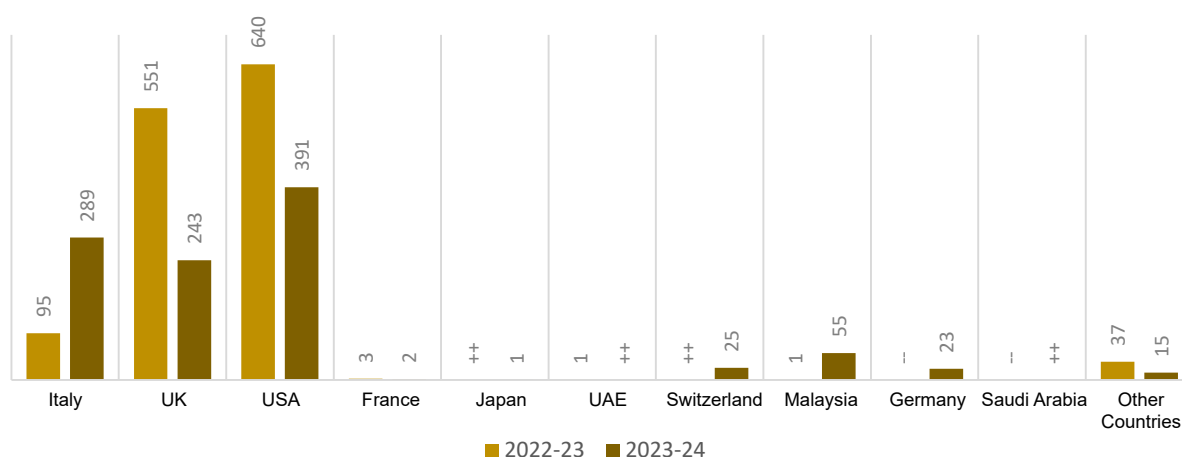


Fig 3: Countrywise Exports of Platinum Alloys & Related Metals

Table – 5: Export of Platinum (Powder, Unwrought and Others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1191	457116	661	1336834
UK	498	294465	202	479152
USA	635	83989	258	400070
Italy	31	68778	138	337860
Germany	--	--	8	119370
UAE	1	2378	++	303
Malaysia	++	2	55	57
Korea, Rep. of	1	47	++	22
Japan	++	5580	--	--
Tanzania	20	1385	--	--
South Africa	5	156	--	--
Other Countries	++	336	++	++

Figures rounded off

Table – 6: Export of Platinum Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1191	451536	657	1322042
UK	498	294465	198	464360
USA	635	83989	258	400070
Italy	31	68778	138	337860
Germany	--	--	8	119370
UAE	1	2378	++	303
Malaysia	++	2	55	57
Korea, Rep. of	1	47	++	22
Tanzania	20	1385	--	--
South Africa	5	156	--	--
Kenya	++	104	--	--
Other Countries	++	232	++	++

Figures rounded off

Table – 7: Export of Platinum Powder

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	++	5580	4	14792
UK	--	--	4	14792
Japan	++	5580	--	--

Figures rounded off

Table – 8: Export of Other Metals of Platinum Group

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	137	312645	383	1468078
USA	5	2112	133	513993
Italy	64	280212	151	455396
Switzerland	++	920	25	297978
Germany	--	--	15	139143
France	3	26720	2	29143
UK	53	2551	41	22373
Japan	--	--	1	9977
Saudi Arabia	--	--	++	38
Israel	--	--	1	23
Maldives	--	--	++	5
Other Countries	12	130	14	9

Figures rounded off

Table – 9: Export of Platinum-Clad Base /Precious Metal

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	3	9	++	42
USA	--	--	++	42
Singapore	2	8	--	--
Canada	1	1	--	--

Figures rounded off

Imports

Imports of platinum alloys and related metal decreased significantly by 67 % to 10,410 kg valued at ₹ 2,433 crore during 2023-24 as compared to 31,498 kg valued at ₹ 12,289 crore in the previous year. Imports were mainly confined from UK (33 %), UAE (23%), South Africa (16 %) and USA (12 %). Imports in 2023-24 also included of platinum (powder, unwrought & others) at 5,318 kg and other metals of platinum group at 5,092 kg. Imports of

other metals of platinum group were mainly from UK (53%), South Africa (23%), USA (9%), Italy (8%) and Hong Kong (5%). During 2023-24, imports of platinum-clad base/ precious metals increased by 22% to 9,315 kg as compared to 7,665 kg in the previous year. Imports were mainly from China (98%) and Italy (1%). During 2023-24, imports of platinum powder were at 250 kg as compared to 266 kg in the preceding year. Imports were mainly from USA (100 %) (Tables- 10 to 15).

Table – 10: Import of Platinum Alloys and Related Metals

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	31498	122892033	10410	24329799
UAE	21844	93444553	2404	1029407
UK	3731	16444447	3467	11227244
USA	1420	3930061	1283	3455369
South Africa	1142	3432451	1629	4661058
Germany	507	3234980	207	875644
Italy	615	1367126	630	1144251
Hong Kong	420	799514	615	1226492
Korea, Rep. of	1751	51071	83	429533
Russia	1	9425	8	90069
Japan	3	3909	25	88858
Other Countries	64	174496	59	101874

Figures rounded off

(In kg)

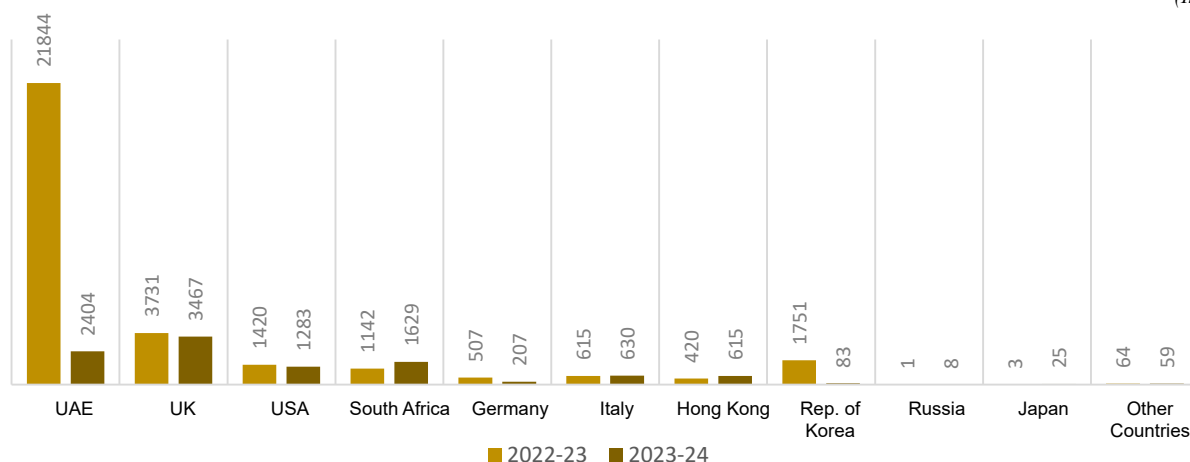


Fig. 4: Countrywise Imports of Platinum Alloys & Related Metals

Table – 11: Import of Platinum (Powder, Unwrought and Others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	25040	101387439	5318	8951957
UAE	21829	93400281	2377	1005003
UK	818	2001048	751	2122082
USA	784	1911347	848	2094588
South Africa	761	1881642	447	1142038
Germany	333	814786	133	324565
Hong Kong	255	631517	383	975582
Italy	225	581833	244	627322

Table- 1 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
Korea, Rep. of	10	47667	83	429533
Russia	1	9425	8	90069
Japan	2	2006	24	64981
Other Countries	22	105887	20	76194

Figures rounded off

Table – 12: Imports of Platinum—Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	24774	100762587	5068	8314886
UAE	21829	93400281	2377	1005003
UK	807	1976662	751	2121278
South Africa	756	1870556	447	1142038
USA	534	1321967	598	1458321
Germany	333	814786	133	324565
Hong Kong	255	631517	383	975582
Italy	225	581833	244	627322
Korea, Rep. of	10	47667	83	429533
Russia	1	9425	8	90069
Japan	2	2006	24	64981
Other Countries	22	105887	20	76194

Figures rounded off

Table – 13: Import of Platinum Powder

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	266	624852	250	637071
USA	250	589380	250	636267
UK	11	24386	++	804
South Africa	5	11086	--	--

Figures rounded off

Table – 14: Import of Other Metals of Platinum Group

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	6458	21504594	5092	15377842
UK	2913	14443399	2716	9105162
South Africa	381	1550809	1182	3519020
USA	636	2018714	435	1360781
Germany	174	2420194	74	551079
Italy	390	785293	386	516929
Hong Kong	165	167997	232	250910
UAE	15	44272	27	24404
Japan	1	1903	1	23877
Singapore	33	56206	5	19822
China	4	5281	34	4777
Other Countries	1746	10526	++	1081

Figures rounded off

Table – 15: Import of Platinum-Clad Base /Precious Metal

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	7665	17093	9315	26718
Italy	96	7980	95	7238
Spain	40	2442	29	1680
Belgium	30	1658	--	--
China	7455	1442	9100	821
USA	12	1214	21	7371
Germany	17	1185	--	--
Israel	8	751	--	--
Portugal	7	421	16	917
Switzerland	--	--	41	7887
Netherlands	--	--	13	804

Figures rounded off

FUTURE OUTLOOK

India meets its entire demand for PGMs through imports. According to the Report of the Sub-Group for the 12th Plan period, domestic demand for PGEs is projected to reach 120 tonnes by 2025. As per Johnson Matthey's 2024 PGM Market Report, the demand for platinum, palladium, and rhodium is expected to continue outpacing supply, with

platinum facing its largest supply shortfall in a decade. This growing demand, coupled with increased investment in platinum, is likely to lead to a global supply deficit. In India, platinum jewellery demand is expected to expand further, driven by industry-led marketing campaigns promoting platinum jewellery sets as wedding gifts and as a symbol of enduring value.



24. Selenium & Tellurium

36,000 tonnes

world reserves of tellurium contained in cooper resources as on 2023.

165 tonnes

of selenium were exported in 2023-24.

506 tonnes

of selenium were imported in 2023-24.

Selenium and tellurium are rare elements widely distributed within the Earth's crust. They do not occur in concentrations high enough to justify mining solely for their content. They are recovered as by-products, mostly from anode mud or slime obtained during electrolytic refining of copper. Tellurium is found mostly in tellurides associated with metals, such as, bismuth, lead, gold and silver. It is found with selenium in the anode slime from electrolytic copper refineries.

EXTRACTION

Selenium and tellurium metals were being recovered as allied products at Ghatsila Copper Smelter of HCL in Jharkhand, where the annual licensed capacity was 10,000 kg while annual installed capacity to produce selenium was 14,600 kg. HCL has not reported production of selenium since 2006-07 and that of tellurium since 2004-05. HCL has developed its own Precious Metal Recovery Plant at ICC successfully. As per the Annual Report 2022-23 of Hindalco Industries Ltd, in the electrolytic refining step in the copper

manufacturing process, anode slime is generated as a by-product including tellurium, selenium. During the copper removal stage in slime leachate, about 50-60% of tellurium in anode slime gets co-dissolved. The remaining tellurium is present in a solid residue, resulting in the loss of this valuable element. At HIC Copper, Hindalco developed a process to recover this tellurium from slime leachate in the form of Copper Telluride (Cu_2Te) powder at its Dahej Smelter in Gujarat.

USES & CONSUMPTION

Selenium

Selenium is used as a decolourising agent in the Glass Industry. Selenium decolorises the green tint caused by iron impurities in glass bottles. Approximately, 1 kg selenium is used for about 150 tonnes of glass production. It is also used in architectural plate glass to reduce solar heat transmission. High purity selenium compounds were used principally as photoreceptors on the drums of older plain paper copiers which are gradually being replaced by newer models that do not use selenium in the reproduction process. Dietary supplement for livestock is the largest agricultural usage of selenium. Also, selenium is known to be added to fertilizer to enrich selenium-poor soils.

Selenium is added to steel, copper and lead alloys to improve machinability which enables faster production with better surface finish and casting properties. Selenium is added to low antimony-lead alloys used in the support grids of lead acid storage batteries. The addition of 0.02% selenium by weight as a grain refiner improves the casting and mechanical properties of alloy. Metallurgical applications of selenium also include its use in the production of Electrolytic Manganese Metal (EMM) as a current efficiency enhancer wherein about 2 kg of SeO_2 is required per tonne of electrolytic manganese metal produced. Due to this high absorption coefficient values, selenium has been found to be useful solar PV material.

Chemical uses of selenium are in industrial and pharmaceutical applications. The principal pharmaceutical use of selenium is in anti-dandruff hair shampoos. Selenium is also used as a human dietary supplement. Other industrial chemical uses are as lubricant, rubber compounding catalysts and as a promoter in the reformation of naphtha.

In pigment applications, selenium is used to produce colour changes in cadmium sulphide-based pigments. Sulphoselenide pigments have good heat stability, resistant to light and chemical attack and hence are used in ceramics, plastics, paints, inks and enamels. Selenium is used in catalysts to enhance selective oxidation and in plating solutions to improve appearance and durability. It is also used in blasting caps and gun bluing.

The use of selenium in glass has increased due to higher colourless glass production. The use of selenium in fertilizer and supplements in the plant-animal human chain and as human vitamin supplements increased as its health benefits were documented. The use of selenium in copper-indium-gallium-diselenide (CIGD) solar cell has increased.

Selenium is recovered from used electronic and photocopier components and recycled. The estimated global use of selenium was in metallurgy (40%); glass (25%); agriculture, chemicals/ pigments, electronic (10% each); and other industries (5%).

Since, selenium is primarily used in the glass, electronics,

and health supplements industries. Its consumption has been relatively stable, with growing interest due to its role in solar energy technologies. India imports most of its selenium, primarily for use in industries like metallurgy and electronics.

Tellurium

Tellurium (Te) element lies on the borderland between metals and non-metals. It is added to non-ferrous metals like aluminium, tin, copper and lead to modify certain physical properties, like ductility, hardness, machinability, toughness, strength and resistance. Tellurium (Te) demonstrates properties similar to those elements known to be toxic to humans and has application in industrial processes, which is rapidly growing in importance and scale. Tellurium is used principally as an alloying element in the production of free-machining low carbon steel, where an addition up to 0.1% tellurium greatly improves machinability. It is also used as a minor additive in copper alloys to improve machinability without reducing conductivity. Tellurium catalysts are used chiefly for the oxidation of organic compounds and also in hydrogenation and halogenation reactions. Tellurium chemicals are used as vulcanising and accelerating agents in processing of rubber compounds. It finds use as a component of catalysts for synthetic fibre production that is increasingly used in cadmium-tellurium-based solar cells. In plain paper copiers and in thermoelectric and photoelectric devices, tellurium is used along with selenium. Mercury-cadmium telluride is used as a sensing material for thermal imaging devices. Tellurium is also used as an ingredient in blasting caps and as a pigment to produce colours in glass and ceramics. High purity tellurium is used in alloys for electronic applications. India's tellurium consumption is modest, with demand linked to the renewable energy sector, which is gradually growing as the country expands its solar energy capacity.

Both minerals, i.e., selenium and tellurium are largely imported since domestic production is minimal and their consumption trends are closely tied to advancements in technology and renewable energy initiatives in India.

SUBSTITUTES

The use of selenium as an alloy to substitute for lead in plumbing continued to increase in response to requirements of Public Law for safe drinking Water Act Amendment 1996. High-purity silicon has replaced selenium in high-voltage rectifiers and is the major substitute for selenium in low and medium voltage rectifiers and solar photovoltaic cells. Other inorganic semi-conductor materials, such as, silicon, cadmium, tellurium, gallium and arsenic as well as organic photoconductors are the substitutes for selenium in photoelectric applications. Cerium oxide is one substitute of selenium used as a colorant or decolorant in glass. Amorphous silicon and organic photoreceptors are substitutes of selenium in plain paper photocopiers. Sulphur dioxide can be used as a replacement for selenium

dioxide in the production of electrolytic manganese metal.

Several materials can replace tellurium in most of its uses, but usually with loss in production efficiency or product characteristics. Bismuth, calcium, lead, phosphorus, selenium and sulphur can be used in place of tellurium in many free-machining steels. Several of the chemical process reactions catalysed by tellurium can be carried out with other catalysts or by means of non-catalytic processes. The chief substitutes for tellurium were selenium and sulphur in rubber compound applications and selenium, germanium and organic compounds in electronic applications.

WORLD REVIEW

Selenium

The world reserves of selenium at 95,000 tonnes only cover the estimated selenium contents of copper reserve, with

the exception of China where selenium was obtained as a by-product with copper. Substantial resources also exist in association with other metals and in uneconomic copper deposits. Selenium reserves are mainly found in Russia (27%), Peru (20%), USA (12%), Canada (6%), China (5%) and Poland (3%) (Table-1).

The world production of refined selenium is furnished in Table-2. The chief producers of selenium in the world in 2023 were China, Japan, Germany, Belgium, and Russia. In addition to the countries listed, Mexico, Finland, Canada and the USA are believed to produce refined selenium. Global selenium and tellurium output cannot be determined easily because not all companies or countries report production and because trade in scrap and semi-refined products may be included with refined metal trade data.

Table – 1: World Reserves of Selenium
(By Principal Countries)

(In tonnes of Selenium content)	
Country	Reserves
World:(Total) (Rounded Off)	95,000
Russia	26,000
Peru	19,000
USA	11,000
Canada	6,000
China	5,000
Poland	3,000
India	500
Sweden	500
Other countries	24,000

Source: USGS, Mineral Commodity Summaries, 2024.

Table – 2: World Production of Selenium, Refined
(By Principal Countries)

(In tonnes)			
Country	2021	2022	2023
China	1260	1290	1,500
Japan	750	750	750
Germany ^(a)	200	200	200
Belgium	200	200	200
Russia	194	190	190
Mexico	135	141	131
Finland	100	130	122
Canada	100	110	120
Chile	100	100	100
Other countries	345	381	320

Source: BGS, World Mineral Production, 2019-2023

Note (s):

(1) In addition to the countries listed, Australia, Iran, the Republic of Korea, and Zimbabwe are believed to produce refined selenium.

(a) Includes selenium produced from imported material.

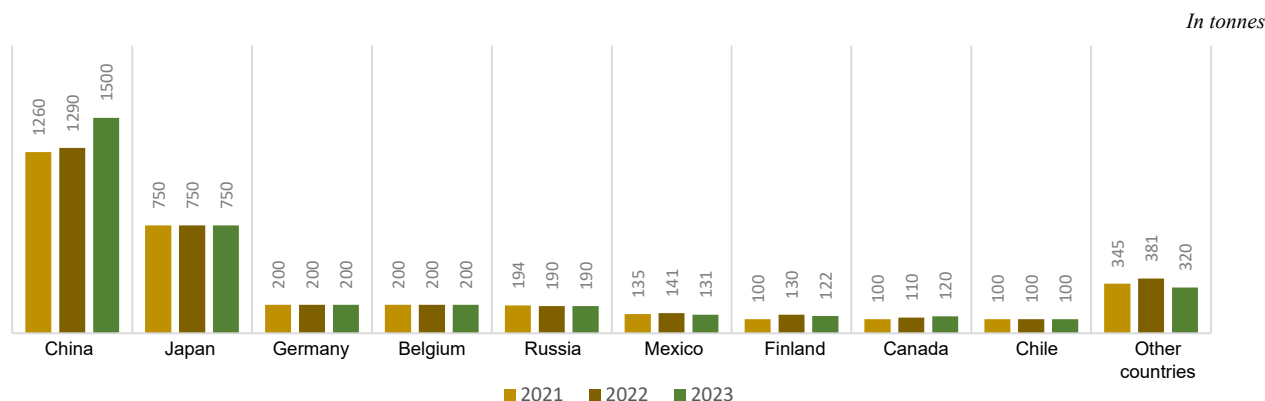


Fig. 1: Countrywise Production of Selenium

Tellurium

The world reserves of tellurium were at 36,000 tonnes contained in copper resources. Tellurium reserves are mainly located in Russia (16%), USA (11%) and China (9%). In addition to the countries listed Bulgaria, Canada, Japan, South Africa, Sweden and Uzbekistan. Concentration of tellurium could also be found in lead and gold deposits. The quantity of tellurium in deposits of coal, copper and other metals that are of sub-economic grade are several times the amount of tellurium contained in identified economic copper deposits (Table-3).

More than 90% of tellurium is produced from anode slimes as a by-product of electrolytic copper refining, and the remainder was derived from skimmings at lead

refineries and from flue dusts and gases generated during the smelting of bismuth, copper and lead-zinc ores. Other potential sources of tellurium include bismuth telluride and gold telluride ores. These anode slimes of copper and tellurium could also be found in lead and gold deposits. The quantity of tellurium in deposits of coal, copper, and other metals that are of sub-economic grade are several times the amount of tellurium contained in identified economic copper deposits (Table-3). Lead refineries normally contain about 3% tellurium. The chief producers of refined tellurium in the world in 2023 were China, Russia, Japan, Sweden, Canada, Uzbekistan and Bulgaria. These countries together contributed an estimated 647 tonnes to the world production in 2023 as compared to 589 tonnes produced in 2022 (Table-4).

Table – 3: World Reserves of Tellurium
(By Principal Countries)

		(In tonnes of tellurium content)
Country	Reserves (Rounded)	
World: Total (Rounded off)	36,000	
Russia	5,800	
USA	4,000	
China	3,100	
Canada	800	
South Africa	800	
Sweden	700	
Bulgaria	NA	
Japan	-	
Uzbekistan	NA	
Other countries	21,000	

Source: USGS, Mineral Commodity Summaries, 2024.

Table – 4: World Production of Tellurium, Refined
(By Principal Countries)

		(In tonnes)		
Country	2021	2022	2023	
China	330*	380*	430*	
Japan	75*	75*	75*	
Russia	56*	70*	75*	
Sweden	41	33	36	
Canada	25*	25*	27*	

Table- 4 (Concl'd.)

Country	(In tonnes)		
	2021	2022	2023
Uzbekistan	48	3*	3*
Bulgaria	3	3	1

Source: BGS, World Mineral Production, 2019-2023

Note (s)

(1) In addition to the countries listed, Germany and Belgium are believed to produce refined tellurium.

*: Estimated

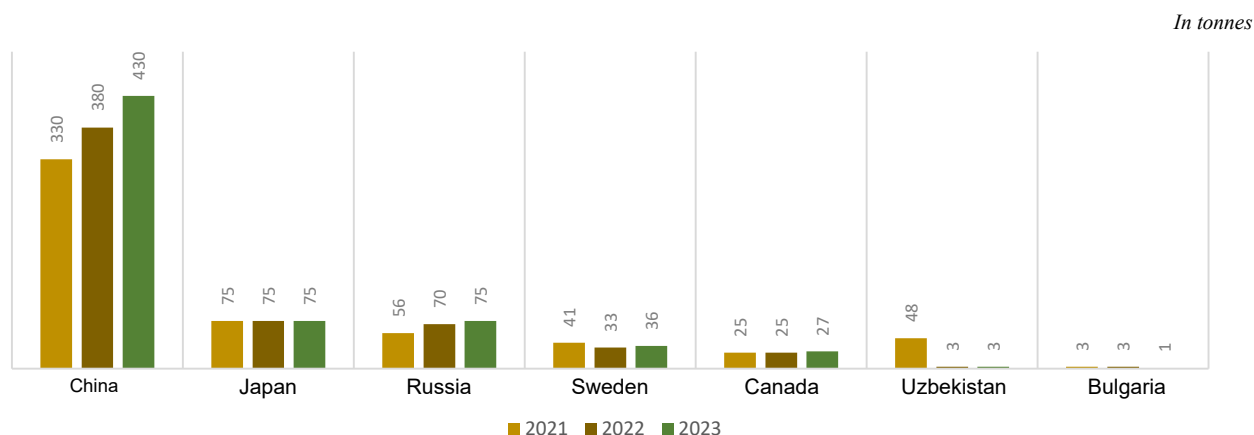


Fig. 2: Countrywise Production of Tellurium

FOREIGN TRADE

Exports

Exports of selenium during 2023-24 decreased marginally by 10% to 165 tonnes from 184 tonnes in 2022-23. Exports

were mainly to Hong Kong (79%), Italy (11%), Belgium (3%) and UAE (2%). There were negligible amount of exports of selenium to USA, South Africa, Nepal, Nigeria. Canada and Armenia. Exports of tellurium were negligible during 2023-24 (Tables-5 & 6).

Table – 5: Exports of Selenium

Country	(By Countries)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	184	2,95,421	165	2,99,169
Hong Kong	151	2,51,531	130	2,37,507
Italy	--	--	18	28,089
Belgium	++	107	5	8,210
USA	--	--	2	7,539
UAE	2	3,095	4	6,996
Canada	++	1,169	++	1,984
South Africa	++	16	1	1,872
Armenia	9	70	++	1,573
Nepal	++	35	1	1,160
Nigeria	++	868	1	1,150
Other Countries	22	38,530	3	3,089

Figures rounded off

Table – 6: Exports of Tellurium

Country	(By Countries)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2	197	++	320
USA	++	155	++	167
Oman	--	--	++	56
Singapore	--	--	++	45
Tunisia	--	--	++	31

Table- 6 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
UAE	--	--	++	16
Kenya	--	--	++	3
Russia	--	--	++	2
Czech Republic	++	28	--	--
South Africa	++	6	--	--
Korea, Rep. of	++	5	--	--
Other Countries	2	3	++	++

Figures rounded off

Imports

Imports of selenium during 2023-24 increased substantially by 31% to 506 tonnes as compared to 387 tonnes in the preceding year. Imports were mainly from Japan (42%), Republic of Korea (22%), Belgium (17%), Germany (15%) and Netherlands (4%). Imports of tellurium increased

substantially by 250% to 7 tonnes as compared to 2 tonnes in the preceding year. Imports were mainly from China & Japan (43% each) and Luxembourg (14%). Negligible quantities were also contributed from Canada, UK, France, Netherlands countries (Tables-7 & 8).

Table – 7: Imports of Selenium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	387	6,89,860	506	8,87,589
Japan	158	3,50,935	214	4,39,894
Korea, Rep. of	128	1,63,241	111	1,55,760
Belgium	41	55,793	84	1,24,988
Germany	29	47,717	75	1,20,019
Netherlands	--	--	21	35,372
China	3	21,738	1	8,582
Hong Kong	++	268	++	1,565
USA	1	531	++	1,409
Philippines	26	37,849	--	--
Canada	1	11,163	--	--
Other Countries	++	625	++	++

Figures rounded off

Table – 8: Imports of Tellurium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2	20,915	7	45,287
China	2	14,822	3	21,599
Japan	--	--	3	17,990
Luxembourg	--	--	1	4,398
Netherlands	--	--	++	969
France	--	--	++	245
Canada	++	4,136	++	81
UK	++	38	++	5
Germany	++	1,747	--	--
USA	++	172	--	--

Figures rounded off

FUTURE OUTLOOK

The supply of selenium is dependent on the supply of main product from which it is derived, copper and also to a lesser extent by the supply of nickel where the nickel production is from sulphide ore. The selenium prices are often inversely related to the supply of copper and nickel.

China has been purchasing large quantities of crude selenium. As this material becomes scarce, the prices for standard grade selenium may rise. The combination of these two factors, the decline of selenium containing concentrates from North America and the growth of Chinese demand, should firm up the prices for selenium in the short term.

Demand for selenium in photoreceptors is likely to see further decline as the cost of substituting organic compounds decreases. The Photoreceptor Industry which was once a major consumer of selenium and tellurium has reached the replacement stage. Selenium has been substituted

by alternative material in newer models. Further, use of selenium in cancer prevention and other health applications may eventually lead to increased consumption of the metal. Dosages taken directly for human consumption will not affect the demand for the metal because small quantities are necessary for effective therapy.

The demand and supply of tellurium has remained fairly balanced for a decade. In short term, significant increases are not anticipated in either consumption or production, although reduction in copper production may have a bearing on tellurium supply. An increase in demand for high purity tellurium for cadmium telluride solar cells might have a major impact on tellurium consumption. The use of tellurium alloys in DVD's consumes only small amounts of tellurium and will, therefore, have minimal impact on tellurium demand.



25. Silver

7,45,980 kg
production of silver have been
reported in 2023-24.

132 tonnes
of silver were exported in 2023-24.

7,337 tonnes
of silver were imported in 2023-24.

Silver is soft and lustrous metal that is grouped in the category of noble metals. Its brilliant white colour, malleability and resistance to atmospheric oxidation have enhanced its value as a highly desired precious metal which is used in many industrial applications. Apart from its monetary and decorative uses, silver is known to have the highest electrical conductivity amongst all metals that enhances its potential in modern age applications, viz, for printed electric circuits, coating for electronic conductors and in alloys of gold & copper for electrical contacts. Its chloride and iodide are light-sensitive and hence used in photographic material. Silver is typically used (in paste form) on solar cells. It means the photovoltaics (PV) market has become one of the most important areas of silver demand. These two major uses have contributed to the increase in supply of scrap of silver contained products. Silver, which is the least expensive of the precious metals, is the whitest element and has the highest electrical and thermal conductivity among all the metals.

In India, there are no native silver deposits except the small and unique Bharak deposit in Rajasthan. It occurs generally with lead, zinc, copper (especially in their sulphide ore) and gold ores and is extracted as a by-product from electrolysis or chemical methods. It is usually extracted by melting silver-bearing lead ore (ore containing argentiferous galena).

Silver is recovered as a co-product as well as a by-product in the country. Silver was recovered in the past as a co-product in gold refining at KGF Complex and Hutti Gold Mines in Karnataka and as a by-product in smelting and refining of lead, zinc and copper concentrates at Chanderiya and Debari smelters in Rajasthan, Tundoo and Moubandar (Ghatsila) smelters in Jharkhand and at Visakhapatnam smelter in Andhra Pradesh. The present production of silver comes from Chanderiya lead-zinc smelter of HZL and from gold refinery of HGML.

In addition, Hindalco extracts silver as a by-product during smelting of imported copper concentrates at Dahej in Gujarat.

RESERVES/RESOURCES

As per the NMI database based on UNFC system as on 1.4.2020, all India total reserves/resources of silver ore in the country have been estimated at about 568.64 million tonnes. Of the total resources, 170.44 million tonnes

are Reserves and 398.20 million tonnes fall under the Remaining Resources category. The total reserves/resources of silver in the country as on 1.4.2020, in terms of metal content, were estimated at about 30,267 tonnes, of which 7,707 tonnes were classified as 'Reserves' and 22,560 tonnes as 'Remaining Resources'.

By States, Rajasthan accounted for about 86.29% of reserves/resources in terms of ore, followed by Karnataka (4.56%), Jharkhand (4.19%), Andhra Pradesh (2.98%), Uttarakhand (0.60%), Madhya Pradesh (0.57%), Odisha (0.31%), Sikkim (0.17%), Meghalaya (0.15%), Tamil Nadu (0.14%) and Maharashtra (0.04%).

By States, Rajasthan accounted for about 97.93% of reserves/resources in terms of metal, followed by Madhya Pradesh (0.53%), Uttarakhand (0.46%), Andhra Pradesh (0.42%), Odisha (0.21%), Sikkim (0.19%), Tamil Nadu (0.14%), Meghalaya (0.07%), Karnataka (0.03%), Jharkhand (0.02%) and a negligible amount from Maharashtra (Table-1).

Table – 1: Reserves/Resources of Silver as on 01.04.2020

(By Grades/States)

(In tonnes)

State/Grade	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
Ore	170446020	398197732	568643752
Metal	7707.07	22560.84	30267.91
By State			
Andhra Pradesh			
Ore	0	16950000	16950000
Metal	0	128.13	128.13
Jharkhand			
Ore	0	23840000	23840000
Metal	0	5.22	5.22
Karnataka			
Ore	22120000	3813612	25933612
Metal	5.43	4.29	9.72
Madhya Pradesh			
Ore	0	3216000	3216000
Metal	0	159.86	159.86
Maharashtra			
Ore	0	235000	235000
Metal	0	0.23	0.23
Meghalaya			
Ore	0	880000	880000
Metal	0	19.8	19.8
Odisha			
Ore	0	1749500	1749500
Metal	0	64.91	64.91
Rajasthan			
Ore	148326020	342383997	490710017
Metal	7701.64	21940.57	29642.21
Sikkim			
Ore	0	949623	949623
Metal	0	56.69	56.69
Tamil Nadu			
Ore	0	790000	790000
Metal	0	42.55	42.55
Uttarakhand			
Ore	0	3390000	3390000
Metal	0	138.59	138.59

Figures rounded off

MINING LEASES & PRODUCTION

Currently, there are no working mining leases for silver in the country for production purpose. Production of silver is reported as a by-product from lead and zinc concentrates, copper smelter and as a co-product of gold refining. As per Annual Report of HZL 2023-24, silver refining capacity of the company is 800 tonnes per annum. HZL is also operating a plant for processing and refining of zinc, lead and silver at SIDCUL, Pantnagar, Uttarakhand since 2011. However, this facility does not add to the overall smelting capacity.

During the year 2023-24, the production of silver at 7,45,980 kg, increased by 4.51% as compared to the previous year. The production of silver from gold refining was 150 kg in 2023-24 as against 148 kg in previous year. One private sector and one public sector undertakings reported production of silver during the current year (Tables-2 to 4). In addition, Hindalco Industries Limited reported production of 73,745 kg and 80,663 kg silver from imported copper concentrates in 2023-24 and 2022-23 respectively.

Table – 2: Principal Producers of Silver, 2023-24

Name & address of producer	Name of plant	Location of the plant	
		State	District
Hindustan Zinc Ltd., Yashad Bhavan, Udaipur-313004, Rajasthan.	Chandaria	Rajasthan	Chittorgarh
The Hutti Gold Mines Co. Ltd., Hutti, Distt. Raichur-584115, Karnataka.	Hutti	Karnataka	Raichur

**Table – 3: Production of Silver*,
2022-23 and 2023-24
(By Sectors/States/Districts)**

(Qty in kg; Value in ₹ '000)

State/District	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
India	713768	43814376	745980	53559386
Public Sector	148	9299	150	10841
Private Sector	713620	43805077	745830	53548545
Karnataka	148	9299	150	10841
Raichur	148	9299	150	10841
Rajasthan	713620	43805077	745830	53548545
Chittorgarh	713620	43805077	745830	53548545

(P): Provisional

*: Silver as a by-product–

(i) In Karnataka, it is recovered at Raichur while refining of gold at Hutti and Uti gold mines.

(ii) In Rajasthan, it is recovered at Chandaria lead- zinc smelter of HZL.

(iii) Excludes by-product recovery of 80,663 kg and 73,745 kg silver from imported copper concentrates in 2022-23 and 2023-24 respectively.

**Table – 4: Production of Silver*,
2021-22 to 2023-24
(Bv States)**

(Quantity in kg, Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	647156	42126921	713768	43814376	745980	53559386
Karnataka	143	11503	148	9299	150	10841
Rajasthan	647013	42115418	713620	43805077	745830	53548545

(P): Provisional

*: Excludes by-product recovery of silver by Hindalco Industries Ltd at Dahej, Gujarat from imported copper concentrates.

USES AND CONSUMPTION

Silver is used for various purposes such as physical investment (bars), electrical and electronics application, coins and medals, photovoltaics (PV), jewellery and silverware, brazing and solder and other industrial uses

and photography. Other applications of silver include use in antimicrobial bandages, clothing, pharmaceuticals, and plastics; batteries; bearings; catalytic converters in automobiles; electroplating; inks; mirrors; photography; photovoltaic solar cells; water purification; wood treatment;

and processing of spent ethylene oxide catalysts. Mercury and silver, the main components of dental amalgam, are biocides, and their use in amalgam inhibits recurrent decay.

In 2023, global consumption of silver was estimated at 36,000 tonnes, a 6% decrease from that in 2022 but still higher than consumption before the COVID-19 pandemic. Coin and bar consumption decreased by 7% in 2023 owing to most key markets declining after a strong 2022. In 2023, consumption of silver for industrial uses was estimated to have increased by 4% compared with that in 2022 owing to the installation of fifth-generation (5G) telecommunications infrastructure and power grids, gross domestic product (GDP) growth, increased production of PV, higher consumer electronics demand and continued rising vehicle output. Consumption of silver in jewellery and silverware was estimated to have decreased by 15% and 24%, respectively. The decline was due to demand reverting to pre-pandemic levels for both. In India, the apparent consumption of silver was 7,951 tonnes in 2023-24.

RECYCLING

Recycling is a significant factor in the supply of many of the metals used in modern society. It provides environmental benefits, such as, energy saving, reduction in emission etc. Photographic wastes, spent catalysts and electronic scrap are the major sources of materials for silver recycling. Other recyclable silver-bearing materials include dental alloys, jewellery and silverware. Cell phones have become one of the major sources for recycled silver recovery.

As per USGS Report titled "Recycled Cell Phones — A Treasure Trove of Valuable Metals", references on data

offered by the Falconbridge Ltd, indicate that one tonne of obsolete cellphones (exclusive of batteries) contains an average 3.14 kilograms of silver metal.

As per World Silver Survey 2024 report, Global silver recycling rose by just 1% to a 10-year high of 5,556 tonnes (178.6 Moz) in 2023. Every key segment of scrap supply rose except photography, which suffered further structural losses. Industrial scrap benefited from growth in ethylene oxide (EO) change-outs and electrical supplies. It was also observed that higher silver prices contributed to a rise in jewellery and silverware scrap supply especially from India.

SUBSTITUTES

Digital imaging, film with reduced silver content, silverless black-and-white film, and xerography substitute for traditional photographic applications for silver. Surgical pins and plates may be made with stainless steel, tantalum, and titanium in place of silver. Stainless steel may be substituted for silver flatware. Nonsilver batteries may replace silver batteries in some applications. Aluminum and rhodium may be used to replace silver that was traditionally used in mirrors and other reflecting surfaces. Silver may be used to replace more costly metals in catalytic converters for off-road vehicles.

WORLD REVIEW

The total reserves of silver in terms of metal content is estimated at 6,10,000 tonnes. Peru held 18% of world's total reserves followed by Australia & Russia (15% each), China (12%), Poland (10%), Mexico (6%), Chile, USA & Bolivia (4% each), Argentina & India (1% each). The remaining 9% was contributed by other countries (Table-5).

Table – 5: World Reserves of Silver
(By Principal Countries)

(In tonnes of silver content) ¹	
Country	Reserves
World: Total (rounded off)	610000*
Peru	110000
Australia	94000 ^a
Russia	92000
China	72000
Poland	63000*
Mexico	37000
Chile	26000
USA	23000
Bolivia	22000
India	8000
Argentina	6500
Kazakhstan	NA
Other countries	57000

Source: USGS, Mineral Commodity Summaries, 2024.

NA- Not Available

1: one tonne (1,000 kilograms)=32,150.7 troy ounces,

a: For Australia, Joint Ore Reserves Committee-compliant reserve were 27,000 tonnes.

*: Correction posted on March 5, 2024.

The total world mine production of silver in metal content was reported at 26,691 tonnes during the year 2023 which increased by 0.71% as compared to 26,502 tonne in the preceding year. Mexico was the leading producer with 27% share in the total production followed by China (13%), Peru

(11%), Russia & Poland (6% each), Bolivia & Chile (5% each), Australia & USA (4% each) and Kazakhstan (3%). The remaining 16% was contributed by other countries. World mine production of silver is furnished in Table- 6.

Table – 6: World Mine Production of Silver
(By Principal Countries)

(In kilograms of metal content)

Country	2021	2022	2023
World:Total	26680000	26502000	26691000
Mexico	6492180	7067580	7277840
China	3511554	3477400	3399600
Peru	3375645	3084039	3040137
Russia ^b	1517000	1567000	1536000
Poland	1522000	1533000	1532000
Bolivia	1292190	1219040	1344630
Chile	1383041	1274363	1262287
Australia	1329718	1169252	1032993
USA	1020000	1010000	1000000*
Kazakhstan	1004789	1013069	894807
Other countries	4231883	4087257	4370706

Source: BGS, World Mineral Production, 2019-23.

b: Smelter and/or refinery production

*: estimated

(In kilograms of metal content)

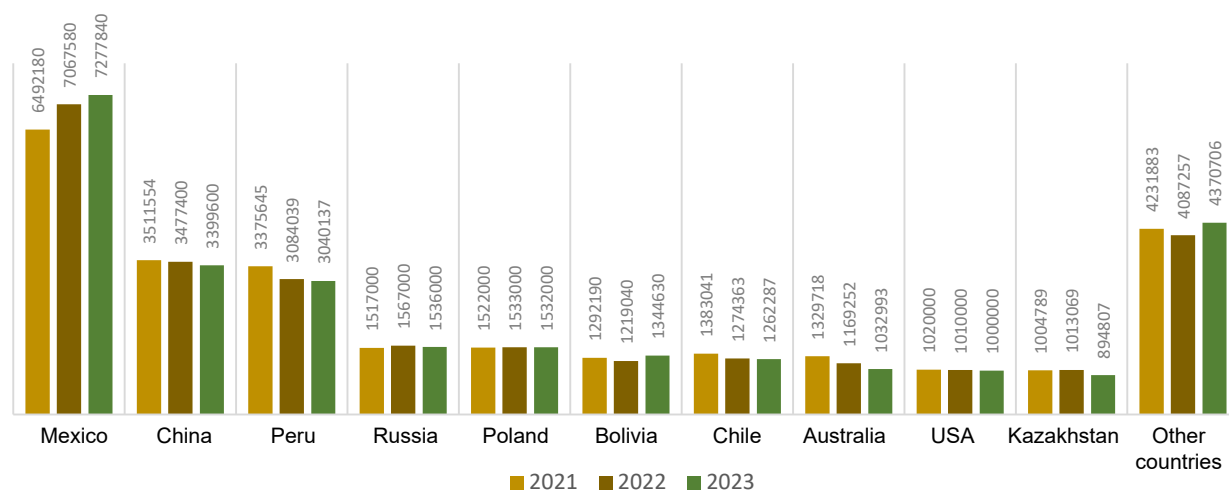


Fig. 1: Countrywise Production of Silver

FOREIGN TRADE

Exports

Exports of silver during 2023-24 stood at 132 tonnes representing an increase of 1.54% from 130 tonnes in the previous year. Exports were mainly to Switzerland (45%), UK (19%), USA (16%), Germany (7%), Canada, Turkey, Russia & Denmark (2% each), Puerto Rico & Italy (1% each). Exports of silver ores and conc. were negligible during 2023-24 as compared with 26,000 kg in 2022-23.

Exports of silver-clad base metals decreased by 3% to 2,777 kg during 2023-24 from 2,861 kg in 2022-23. Exports of Semi-manufactured silver increased by 1% to 128 tonnes in 2023-24 as compared with 127 tonnes in the preceding year. Exports of silver-unwrought were 3 tonnes during the year 2023-24 as compared with negligible in previous year. Exports of silver powder decreased by 67% to 1 tonne during 2023-24 as compared with 3 tonnes in the preceding year (Tables-7 to 12).

Table – 7: Export of Silver

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	130	5737293	132	5252653
Switzerland	--	--	59	3976565
USA	21	464869	21	476792
Germany	9	159084	9	196998
UK	81	4560685	25	79854
Canada	3	70201	3	72858
Puerto Rico	5	184977	1	66974
Turkey	1	51853	2	57630
Russia	--	--	2	50697
Denmark	3	31554	3	34272
Italy	2	56582	1	26621
Other Countries	5	157488	6	213392

Figures rounded off

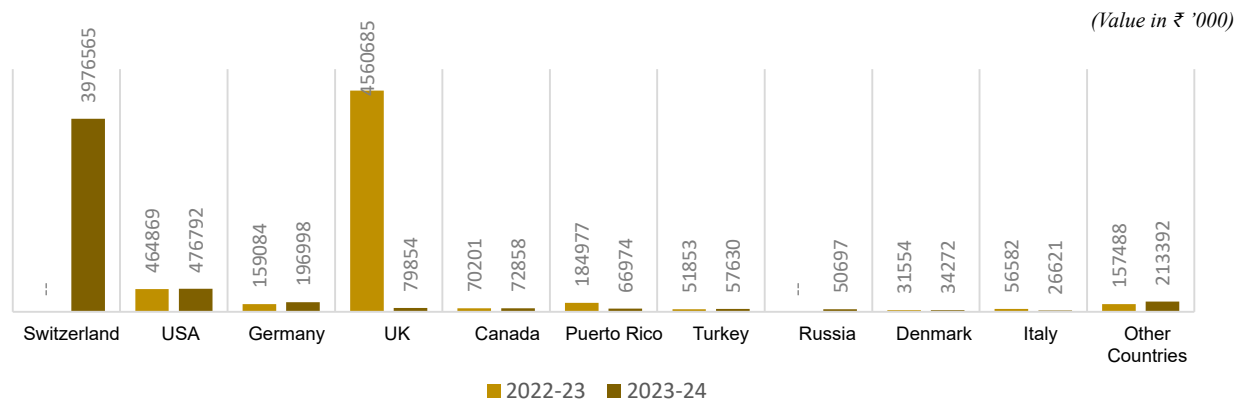


Fig. 2: Countrywise Export of Silver

Table – 8: Export of Silver Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	26000	269	++	++
Thailand	26000	269	--	--

Figures rounded off

Table – 9: Export of Silver Clad Base Metals

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	2861	13326	2777	14655
Sri Lanka	2672	11017	2654	12361
USA	102	2271	86	2201
Australia	--	--	12	53
Botswana	--	--	++	20
Nepal	--	--	25	20
Kenya	3	21	--	--
Bangladesh	74	11	--	--
Canada	7	5	--	--
Netherlands	3	1	--	--
Seychelles	++	++	--	--

Figures rounded off

Table – 10: Export of Silver: Semi-Manufactured

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	127	5567938	128	5168361
Switzerland	--	--	59	3976565
USA	21	460961	21	476792
Germany	9	159084	9	196998
Canada	3	69463	3	72858
UK	81	4560664	22	70697
Turkey	1	51853	2	57630
Russia	--	--	2	50697
Denmark	3	31554	3	34272
Italy	2	56036	1	26058
Poland	++	23675	++	24111
Other Countries	7	154648	6	181683

Figures rounded off

Table – 11: Export of Silver: Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	7931	3	20481
UK	++	21	3	9157
Korea, Rep. of	--	--	++	8041
Bangladesh	++	923	++	2157
Hong Kong	++	49	++	1101
Ethiopia	--	--	++	13
Ireland	--	--	++	11
France	--	--	++	1
USA	++	3824	--	--
Sri Lanka	++	1932	--	--
Canada	++	738	--	--
Other Countries	++	444	++	++

Figures rounded off

Table – 12: Export of Silver: Powder

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3	161424	1	63811
Puerto Rico	3	159609	1	61783
Malaysia	--	--	++	1319
Italy	++	546	++	563
Finland	--	--	++	41
Dominica	--	--	++	36
Romania	--	--	++	25
France	--	--	++	24
Hong Kong	++	6	++	16
Thailand	--	--	++	4
Germany	--	--	++	++
Other Countries	++	1263	++	++

Figures rounded off

Imports

Imports of silver during 2023-24 stood at 7,337 tonnes representing decrease of 10% from 8,156 tonnes in the previous year. Imports were mainly from UK (37%), UAE (31%), Hong Kong (21%), Switzerland (2%), Uzbekistan, Singapore, Kazakhstan, China, Peru & Republic of Korea (1% each). Imports of silver ores and conc. declined sharply to 1 kg during 2023-24 from 77,361 kg in 2022-23. Imports of silver-clad base metals increased by 54% to 2,060 kg

during 2023-24 from 1,342 kg in 2022-23. Imports of Semi-manufactured silver decreased by 40% to 4,786 tonnes in 2023-24 as compared with 7,927 tonnes in the preceding year. Imports of silver-unwrought rose sharply, increasing thirteen-fold to 2513 tonnes during 2023-24 as compared with 191 tonnes in previous year. Imports of silver powder remains unchanged at 38 tonnes during 2023-24, the same as in the previous year (Tables-13 to 18).

Table – 13: Import of Silver
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	8156	422370924	7337	451647495
UK	4113	213432868	2729	165597515
UAE	44	2314003	2248	144391793
Hong Kong	1749	91713085	1531	93468950
Switzerland	275	14248619	142	8619351
Uzbekistan	20	1055877	100	6401134
Singapore	50	2592239	81	6082570
Kazakhstan	73	3644069	76	4771780
China	973	51185299	99	4612624
Peru	21	1181624	47	3361384
Korea, Rep. of	39	2041109	43	2680256
Other Countries	799	38962132	241	11660138

Figures rounded off

In tonnes

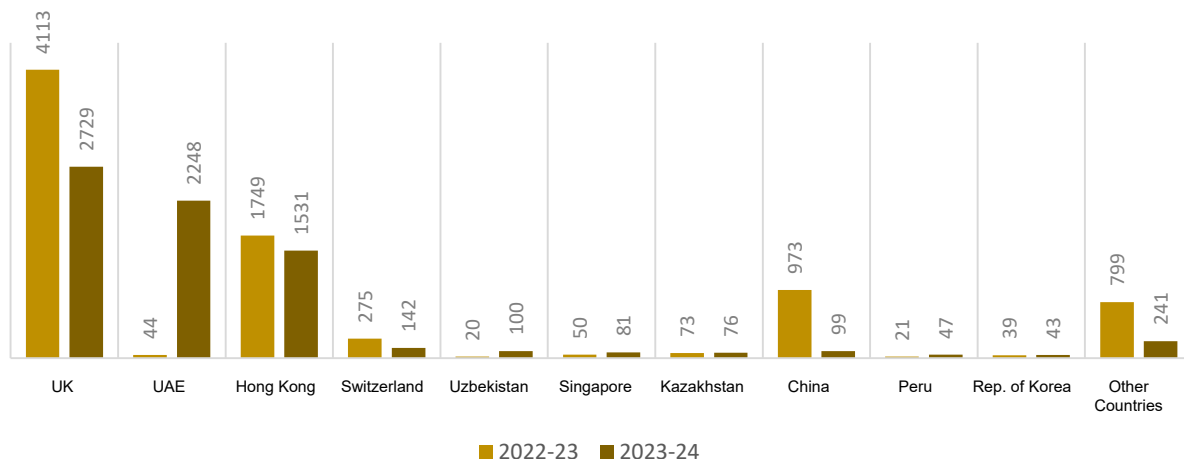


Fig. 3: Countrywise Import of Silver

Table – 14: Import of Silver Ores and Conc.
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	77361	10001	1	++
China	--	--	1	++
Peru	77361	10001	--	--

Figures rounded off

Table – 15: Import of Silver Clad Base Metal

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1342	26471	2060	24771
UK	61	3952	205	9116
Thailand	525	6853	1340	6247
USA	188	5956	106	3419
France	400	2953	209	2548
Malaysia	32	2101	24	1609
Italy	--	--	150	1090
Netherlands	36	3964	6	321
Germany	--	--	5	303
China	--	--	15	118
Belgium	100	691	--	--
Other Countries	++	1	++	++

Figures rounded off

Table – 16: Import of Silver: Semi-Manufactured

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	7927	411350242	4786	289204423
UK	4113	213432868	2728	165551140
Hong Kong	1724	90227888	1453	88736224
Uzbekistan	20	1055877	100	6401134
Switzerland	267	13800074	97	5820806
Kazakhstan	73	3644069	76	4771780
China	961	50961230	63	3895791
Peru	21	1181624	46	3302340
Korea, Rep. of	30	1592001	43	2680256
Russia	243	12148339	40	2378997
Turkey	4	219428	26	1503785
Other Countries	471	23086844	114	4162170

Figures rounded off

Table – 17: Import of Silver: Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	191	10617695	2513	162095110
UAE	16	881495	2241	143952321
Singapore	41	2249064	78	6036418
Hong Kong	25	1485197	78	4732726
Switzerland	8	448545	45	2798545
Italy	28	1648392	27	1770750
Germany	15	917606	16	1103948
China	4	183728	10	591706
Thailand	++	7650	9	550935
USA	8	478106	7	412875
Peru	--	--	1	59044
Other Countries	46	2317912	1	85842

Figures rounded off

Table – 18: Import of Silver: Powder

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	38	402987	38	347962
Brazil	3	169194	2	144482
China	8	40341	26	125127
USA	23	153124	7	44751
Singapore	4	28432	3	17540
Germany	++	11421	++	16062
Italy	++	467	--	--
Korea, Rep. of	++	8	--	--

Figures rounded off

FUTURE OUTLOOK

The overall demand for silver fell by 7% to 1,195 Moz in 2023, after a record-breaking year in 2022. The jewellery, silverware, and price-sensitive physical investment industries were mostly responsible for this decline. On the other hand, industrial demand surged to a new high in 2023, driven mostly by the electrical and electronics industry, which expanded by 20% to 445.1 Moz. This increase was the result of silver's expanding and vital usage in photovoltaics, which reached a new high of 193.5 Moz, increasing by a staggering 64% over 2022 total of 118.1 Moz.

In India, silver is mainly used for jewellery purposes. India's silver consumption in the industrial sector is relatively low as compared to global levels. However, the demand from the industrial sector is expected to increase significantly due to new-age technological developments, like EVs and 5G, in the coming days. As per Metal Focus in

2023, Indian silver demand was around 6,800 tonnes, and is expected to grow to around 7,400 tonnes in 2024.

HZL is planning to scale up the silver production to a target capacity of 1,000 MT by 2029-30. As per Market Research Future Analysis, the global silver market was valued at 20.45 USD Billion in 2023 and is projected to grow to 30 USD Billion by 2035, at a CAGR of 3.24% from 2025 to 2035. Key drivers include rising demand in electronics, renewable energy applications and increased investment in precious metals.

In view of the sustained requirement of silver in solar energy, electronics, automotive electrification and emerging high-technology applications, combined with limited growth in primary supply and continued reliance on secondary sources, the medium-term outlook for silver remains favourable.



26. Strategic and Critical Minerals

8.23 thousand tonnes
of beryl production was reported
globally in 2023.

958 tonnes
were the estimated world refinery
production of Indium in 2023.

54 tonnes
were the estimated world production
of rhenium in 2023.

Critical minerals from the foundation of modern technology. These minerals have become the building blocks of the new age economy. Most countries in the world have identified critical minerals based on their priorities and future requirements. From solar panels to semiconductors, wind turbines to advanced batteries for storage and transportation, the world needs critical minerals to build these products. As the demand of these minerals continues to rise, it is vital to have a clear understanding of their availability, extraction methods and application prospects to effectively secure India's economic, technological and environmental future. The future of India's economy will be underpinned by technologies that depend on minerals such as lithium, graphite, cobalt, titanium and rare- earth elements. India's future economic prosperity will depend on how well we can use our vast energy and mineral resources and how well we can adapt to the global shift towards zero emissions.

A mineral is labelled as critical when the risk of supply shortage and associated impact on the economy is higher than for other raw materials. The risk of supply shortage would ideally capture import dependence, recycling potential, and substitutability of the mineral in question. Strategic Minerals are those minerals in which a country has negligible domestic resource and depends upon external resources for supply.

The Ministry of Mines vide order No. 11/1/2022-IC dated 01.11.2022 constituted a Committee for the identification of critical and strategic minerals. Based on a three-stage assessment process that considered important parameters such as resource/ reserve position in the country, production, import dependency, use for future technology/ clean energy, requirement of fertilizer minerals in an agrarian economy, the Committee identified a set of 30 critical minerals. These are **Antimony, Beryllium, Bismuth, Cobalt, Copper, Gallium, Germanium, Graphite, Hafnium, Indium, Lithium, Molybdenum, Niobium, Nickel, PGE, Phosphorous, Potash, REE,**

Rhenium, Silicon, Strontium, Tantalum, Tellurium, Tin, Titanium, Tungsten, Vanadium, Zirconium, Selenium and Cadmium. Further, MMDR (Amendment) Act 2023, Schedule I Part D prescribed 24 Critical and Strategic Minerals. These minerals are:

1. Beryl and other beryllium-bearing minerals
2. Cadmium-bearing minerals
3. Cobalt-bearing minerals
4. Gallium-bearing minerals
5. Glauconite
6. Graphite

7. Indium-bearing minerals
8. Lithium-bearing minerals
9. Molybdenum-bearing minerals
10. Nickel-bearing minerals
11. Niobium-bearing minerals
12. Phosphate (without uranium)
13. Platinum group of elements-bearing minerals
14. Potash
15. Minerals of the “rare earths” group not containing Uranium and Thorium
16. Rhenium-bearing minerals
17. Selenium-bearing minerals
18. Tantalum-bearing minerals
19. Tellurium-bearing minerals
20. Tin-bearing minerals
21. Titanium-bearing minerals and ores (ilmenite, rutile and leucoxene)
22. Tungsten-bearing minerals
23. Vanadium-bearing minerals
24. Zirconium-bearing minerals and ores including zircon

The Review covers 5 minerals, namely, beryl and other beryllium-bearing minerals, Indium-bearing minerals, Niobium-bearing minerals, Rhenium-bearing minerals and Tantalum-bearing minerals. The remaining 19 minerals have been already covered in other reviews of the Indian Minerals Yearbook 2024.

BERYL AND OTHER BERYLLIUM-BEARING MINERALS RESOURCES & PRODUCTION

As per National Mineral Inventory (NMI) as on 01.04.2020, no resources were identified for Beryl and other beryllium-bearing minerals. Further, there is no production of Beryl and other beryllium-bearing mineral in India and entire requirement is met through imports.

USES

It has various uses in electronics for its conductivity and in aerospace for its light weight strength. These minerals are also used in nuclear reactors, aerospace applications and as moderators in some reactors due to their ability to absorb neutrons.

These minerals are commonly found in pegmatites, granites as well as in hydrothermal veins associated with granitic rock. Additionally, beryllium can be found in certain types of sedimentary rocks and as a trace element in some soil and clays.

WORLD REVIEW

As per USGS, Mineral Commodity Summaries 2024, the world's identified beryllium resources have been estimated to be more than 1,00,000 tonnes. About 60% of these resources are in the United States; by tonnage, the Spor Mountain area in Utah, the McCullough Butte area in Nevada, the Black Hills area in South Dakota, the Sierra Blanca area in Texas, the Seward Peninsula in Alaska, and the Gold Hill area of Utah account for most of the resources.

The global beryl production was estimated at 8.23 thousand tonnes in 2023. The USA dominated the world production by accounting for 53% of total output, followed by China (22%) and Mozambique (12%) (Table-1).

Table – 1: World Production of Beryl
(By Principal Countries)

Country	2021	2022	2023
USA ^(b)	4370	4330	4400*
China	1750*	1750*	1850*
Mozambique	330	1285	1000*
Nigeria	200*	349	978
Zambia	5	5	4
Namibia	15	1	1*
Madagascar ^(a)	1*	1*	1*
Uganda	473	--	--*

"Source: BGS, World Mineral Production, 2019-2023.

** Estimate; (a) Include ornamental and industrial products*

(b) Includes bertrandite ore, calculated as equivalent to beryl containing 11% beryllium oxide."

Foreign Trade

Exports & Imports

Trade data for beryllium mineral for 2023-24 is not available.

INDIUM-BEARING MINERALS

Resources & Production

As per the National Mineral Inventory (NMI) as on 01.04.2020, no resources were identified for Indium bearing mineral. Further, there is no production of Indium bearing mineral in India and current requirement is made through imports.

Uses

It is primarily used in electronics especially in the production of flat panel display, touch screens and solar panel. It is also used in semiconductors, solders and as thin fill for lubrication. It has application in medical imaging and as a coating for bearings in high performance engine.

It is primary found in zinc ores, tin, iron and copper

ores. However, commercial extraction typically occurs as a byproduct of zinc smelting in India.

World Review

As per U.S. Geological Survey, Mineral Commodity Summaries, 2024, quantitative estimates of reserves were not available. Further, Indium is most commonly recovered from the zinc-sulfide ore mineral sphalerite. The indium content of zinc deposits from which it is recovered ranges from less than 1 part per million to 100 parts per million. Although the geochemical properties of indium are such that it occurs in trace amounts in other base-metal sulfides—particularly chalcopyrite and stannite—indium recovery from most deposits of these minerals is not economic.

The world production of Indium refinery was estimated at 958 tonnes in 2023. China dominated the world production by accounting for 72% of total output, followed by Republic of Korea (10%), Japan (7%), Canada (4%) and Belgium & France (2% each) (Table-2).

Table – 2: World Refinery Production of Indium

Country	(In tonnes)		
	2021	2022	2023
China	600	667	690*
Korea, Rep. of	100*	100*	100*
Japan	70	70*	70*
Canada	44	39*	40*
Belgium	20*	20*	20*
France	38*	20*	20*
Russia	7*	7*	7*
Italy	5*	5*	5*
Brazil	5*	5*	5*
Uzbekistan	1	1	1*
Peru	12*	-*	-*

"Source: BGS, World Mineral Production, 2019-2023

*: Estimated"

Foreign Trade

Exports & Imports

Trade data for indium mineral for 2023-24 is not available.

RHENIUM-BEARING MINERALS

Resources & Production

As per the National Mineral Inventory (NMI) as on 01.04.2020, no resources were identified for Rhenium bearing minerals. Further, there is no production of Rhenium bearing mineral in India and current requirement is met through imports.

Uses

These minerals are valuable for their application in high temperature environments. They are primarily used in super alloys for jet engine components, as catalyst for

petroleum refining and in production of lead free gasoline. It has application in thermo couples for measuring high temperature.

They are rare in nature and often found as trace elements in various minerals associated with molybdenum and copper deposits. These can also be found in some PGM including sulphites. However, commercial production involves by-product recovery from the refining of molybdenum and copper ores.

World Review

The world reserves of rhenium are estimated at about 2,314.2 tonnes. Rhenium reserves are mainly in Chile which contributes (56 %) to the total reserves followed by USA (17%), Russia (13%), Kazakhstan (8%) and Aemenia (4%) (Table- 3).

Table – 3: World Reserves of Rhenium

(In tonnes)

Country	Reserves
World Total	Large
Chile*	1300
USA	400
Russia	310
Kazakhstan	190
Armenia	95
China	19.2
Korea Rep of	NA
Poland	NA
Uzbekistan	NA

": Estimated; rhenium recovered from roaster residues from Belfium, Chile, Mexico and Peru.

Source: USGS, Mineral Commodity Summaries, 2024"

The world production of rhenium was estimated at 54 accounting for 56% output which was followed by USA & tonnes in 2023. Chile dominated the world production by Poland (17% each) and China (6%) (Table- 4).

Table – 4: World Production of Rhenium

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
Chile	30*	29*	30*
USA	9	9	9*
Poland	9	6	9
China	3*	3*	3*
Russia	2*	2*	2*
Kazakhstan	1*	1*	1*

"Source: BGS, World Mineral Production, 2019-2023.

* Estimated"

Foreign Trade

Exports & Imports

Trade data for Rhenium mineral for 2023-24 is not available.

NIOBIMUM-BEARING MINERALS

Resources & Production

As per National Mineral Inventory (NMI) as on 01.04.2020, no resources were identified for Niobium bearing mineral. Further, there is no production of Niobium bearing minerals in India and current requirement is met through imports.

Uses

Niobium is primarily used in the production of high strength, low alloy steels for infrastructure pipelines and automotive applications. It is also used in jet engines, rockets, construction beams, building girders, oil rigs and pipelines, superconducting magnets, MRI scanners, NMR equipment, eyeglasses, titanium niobium oxide anode materials, nuclear power plants, production of electronic components. It has application in medical implants due to its biocompatibility and corrosion resistance properties.

Niobium-bearing minerals are often associated with

complex pegmatites, carbonatites, and rare-earth elements. These are also found in some alluvial deposits and as trace elements in various igneous and metamorphic rocks.

World Review

The world reserves of Niobium are estimated to be about 17 million tonnes. Niobium reserves are mainly in Brazil which contributes (94 %) to the total reserves (Table-5). The world production of tantalum and niobium minerals is given in Table-6.

Table – 5: World Reserves of Niobium (Columbium)

(In tonnes)

Country	Reserves
World Total	>17000000
Brazil	16000000
Canada	1600000
USA	210000
Congo Dem, Rep. of	NA
Russia	NA
Rwanda	NA

Source: USGS, Mineral Commodity Summaries, 2024

Table – 6: World Production of Tantalum and Niobium Minerals

(In tonnes)

Country	Sub-commodity	2021	2022	2023
Australia	Tantalite	213	184	159
	Ta content	105	90	78
Bolivia	Tantalite	2*	9*	4*
	Ta content	1*	4*	2*
Brazil	Columbite	11201	10978	10977
	Nb content	3915	3837	3837
	Pyrochlore	198777	200940	207208
	Nb content	75036	76896	76896
	Tantalite	419	--	--
	Ta content	65	--	--
Burundi	Columbite–tantalite	42	10	6
	Nb content	16	4	2
	Ta content	7	2	1
Canada	Pyrochlore	13900*	12100*	8500*
	Nb content	5200*	4500*	3240*
China	Columbite–tantalite	304*	300*	316*
	Nb content	118*	117*	123*
	Ta content	47*	46*	49*
Congo, Dem. Rep.	Columbite–tantalite	2422	2500*	2500*
	Nb content	711	734*	734*
	Ta content	694	717*	717*
Ethiopia ^(a)	Nb content	12*	15*	15*
	Ta content	19*	19*	19*
Malaysia	Struverite	-	35	49
	Ta content	-	3	5
Mozambique	Tantalite	178	211	205
	Ta content	88	104	101
Nigeria	Columbite	1378	1400*	1860
	Nb content	537	545*	724
	Ta content	213	217*	288
	Tantalite	44	4	5
	Ta content	22	2	2
	Nb content	454	373	362
Russia	Ta content	34	17	29
	Nb content	435	435	655
Rwanda ^{(b) (c)}	Columbite–tantalite	1118	1116	1682
	Nb content	435	435	655
	Ta content	173	173	260
Spain	Ta content	12	16	20
Uganda	Columbite–tantalite	404	400*	400*
	Nb content	161	159*	159*
	Ta content	50	49*	49*
Zimbabwe	Columbite–tantalite	-	-	-
	Nb content	-	-	-
	Ta content	-	-	-

Source: BGS, World Mineral Production, 2019-2023.

* Estimated; (a): years ended 7th July of that stated;

(b): years ended 30th June of that stated

(c) Exports

Foreign Trade

Exports

A meagre quantity of one tonne of Ferro-columbium exported to Peru in 2023-24. Export of Ferro-niobium

during 2023-24 decreased by 55% to 5 tonnes from 11 tonnes in the previous year. The export of Ferro-niobium were mainly to Malaysia (80%) followed by Turkey (20%) (Tables-7 & 8).

Table – 7: Countrywise Export of Ferrocolumbium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	1	4273
Peru	--	--	1	4273

++: Negligible '--: Nil;
(R): Revised; (P): Provisional

Table – 8: Countrywise Export of Ferroniobium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	11	32186	5	17150
Malaysia	6	15611	4	11585
Turkey	++	697	1	3840
Thailand	--	--	++	1576
UK	++	317	++	149
UAE	2	6872	--	--
Russia	2	5906	--	--
South Africa	1	1807	--	--
Egypt	++	935	--	--
Chile	++	41	--	--

++: Negligible '--: Nil;
(R): Revised; (P): Provisional

Imports

Import of Ferro-niobium during 2023-24, by quantity, decreased marginally by 3% to 5.08 thousand tonnes from 5.25 thousand tonnes in the previous year. The value of ferro-niobium during 2023-24 increased by 5%

to ₹ 1289.54 crore from ₹ 1233.64 crore in the previous year. The share of import of Ferro-niobium were mainly to Singapore (57%) followed by Brazil (32%), Canada (6%), USA (2%) and Switzerland (1%). (Table-9).

Table – 9 : Countrywise Import of Ferroniobium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5245	12336415	5079	12895411
Singapore	3249	8127677	2911	8037363
Brazil	1649	3332583	1639	3420950
Canada	259	681310	316	829652
USA	20	41326	112	299820
Switzerland	16	55351	62	183323
China	--	--	17	73012
Netherlands	21	41871	20	47681
Korea, Rep. of	--	--	2	3609
UK	--	--	++	1
UAE	19	42276	--	--
Other Countries	12	14021	++	++

++: Negligible '--: Nil;
(R): Revised; (P): Provisional

TANTALUM-BEARING MINERALS

Resources & Production

As per the National Mineral Inventory (NMI) as on 01.04.2020, no resources were identified for Tantalum bearing minerals. Further, there is no production of Tantalum bearing mineral in India and current requirement is met through imports.

Uses

Tantalum is primarily used in electronic devices like smart phones and laptops due to high capacitance and reliability. It is also used in jet engine components, turbine blades and in cutting tools for its high melting points and resistance to corrosion and wear. Its applications in Medical industry is also significant.

Tantalum-bearing minerals are associated with pegmatites, granites and certain rare-metal minerals like lithium-bearing spodumene and beryl. They are also found to occur in alluvial deposits.

World Review

Tantalum reserves are mainly in China which contributes 62% to the total reserve followed by Australia (28%) and Brazil (10%) (Table-10).

The world production of tantalum and niobium minerals in Nb content was 86.40 thousand tonnes & Ta content was 1.60 thousand tonnes in 2023. Brazil dominated the world production of Nb content by accounting for 93% and Congo dominated the world production of Ta content accounting 45% output. (Table-6).

Table – 10: World Reserves of Tantalum

(In tonnes)

Country	Reserves
World Total	NA
China	240000
Australia ^(a)	110000
Brazil	40000
USA	NA
Burundi	NA
Congo Dem, Rep. of	NA
Nigeria	NA
Russia	NA
Rwanda	NA

Source: USGS, Mineral Commodity Summaries, 2024

(a): For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 28,000 tonnes.

Foreign Trade

Exports

Export of tantalum alloys (other) during 2023-24 decreased by 33% to 6 tonnes from 9 tonnes in the previous year. The exports were mainly to Germany & Bangladesh (50% each). The value of export of tantalum waste and scrap (powder) during 2023-24 increased manifold from 2 thousand rupees

in previous year to 156 thousand rupees in 2023-24. The entire export of tantalum waste and scrap (powder) was to Denmark. The value of export of tantalum unwrought during 2023-24 also increased by 102% as compared to previous year. The shares of value of export of tantalum unwrought were mainly to Saudi Arabia (88%) and UK (12%). (Tables-11 to 13).

Table – 11: Countrywise Export of Tantalum Alloys (Other)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	9	170552	6	52449
Germany	9	168711	3	48298
Liechtenstein	++	1518	++	1681
USA	--	--	++	1276
UK	++	2	++	409
Japan	++	14	++	282
Saudi Arabia	--	--	++	281
Bangladesh	--	--	3	154
Denmark	++	288	++	52
Bhutan	--	--	++	12

Table- 11 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Bahrain	--	--	++	4
Other Countries	++	19	++	++

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Table – 12 : Countrywise Export of Tantalum Waste and Scrap (powder)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	2	++	156
Denmark	++	2	++	156

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Table – 13: Countrywise Export of Tantalum Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	460	++	928
Saudi Arabia	++	9	++	820
UK	--	--	++	108
USA	++	315	--	--
UAE	++	136	--	--

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Imports

Import of tantalum alloys (other) during 2023-24, by quantity, decreased by 75% from 4 tonnes in the previous year to 01 tonne in 2023-24. The value of tantalum alloys (other) also decreased from ₹ 16.57 crore in previous year to ₹ 12.72 crore during 2023-24. By value, the imports were mainly from China (34%), UK (30%), Austria (16%) and Germany (13%). The value of tantalum and scrap also decreased by 33% to ₹ 13.39 crore from ₹ 20.11 crore in the

previous year. The share of value of import of tantalum and scrap were mainly to China (37%), UK (29%), Austria (15%) and Germany (12%). The value of tantalum unwrought decreased drastically by 80% to ₹ 0.68 crore from ₹ 3.42 crore in the previous year. The share of value of import of tantalum unwrought were mainly from China (92%) and USA (8%). In 2023-24, very small quantity of tantalum bars were imported. (Table-14 to 17).

Table – 14: Countrywise Import of Tantalum Alloys (Other)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	4	165736	1	127185
China	2	45700	1	42979
UK	++	10158	++	38265
Austria	1	53469	++	20121
Germany	++	16441	++	16287
USA	++	1393	++	8099
Denmark	++	735	++	557
Switzerland	--	--	++	303
Taiwan	--	--	++	233
Netherlands	--	--	++	99
Canada	--	--	++	80
Other Countries	1	37840	++	162

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Table – 15: Countrywise Import of Tantalum and Scrap

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	6	201090	1	133946
China	2	54921	1	49177
UK	++	10164	++	38275
Austria	1	53469	++	20121
Germany	++	16441	++	16287
USA	1	4424	++	8652
Denmark	++	735	++	557
Switzerland	--	--	++	303
Taiwan	--	--	++	233
Netherlands	--	--	++	99
Canada	--	--	++	80
Other Countries	2	60936	++	162

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Table – 16 : Countrywise Import of Tantalum Unwrought

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2	34236	++	6761
China	++	9221	++	6198
USA	1	1913	++	553
UK	++	6	++	10
Estonia	1	23096	--	--

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

Table – 17: Countrywise Import of Tantalum Bars

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	1118	++	++
USA	++	1118	--	--

++: Negligible '--: Nil;

(R): Revised; (P): Provisional

FUTURE OUTLOOK

India has set a goal to achieve 'Net Zero' emissions by 2070. Thus, future consumption of these critical minerals will increase as they are vital to power the transition to a low-emission economy and for renewable technologies.

The future global economy will be underpinned by technologies that depend on minerals such as lithium, graphite, cobalt, titanium, rare earth elements etc. These are essential for the advancement of many sectors, including high-

tech electronics, telecommunications, transport, and defence.

To full-fill the domestic requirement of critical and strategic minerals, under the aegis of Ministry of Mines, Government of India, formed the Khanij Bidesh India Limited (KABIL), a Joint Venture Company among the three Government Enterprises namely NALCO, HCL and MECL. KABIL is to identify, explore, acquire, develop, mine, process, procure strategic minerals outside India for supplying primarily to India, to meet domestic requirements due to its non or meager availability in the country.



27. Tin

83.72 million tonnes

Total reserves/resources of tin ore have been established as on 1st April, 2020.

22,345 kg

production of tin concentrates have been reported in 2023-24.

Tin is a silvery-white, soft and highly malleable non-ferrous metal, known for its chemical inertness, non-toxicity and exceptional alloying ability, which endows it with vital applications in coatings, soldering and in alloying for bronze (used since the dawn of civilization). Although tin does not occur naturally in elemental form, its chief ore, cassiterite (SnO_2), offers a theoretical tin content of 78.77% and is commonly mined from placer and vein deposits. Lesser-known ores, such as, stannite ($\text{Cu}_2\text{SnFeS}_4$) also contribute to tin supply, albeit in smaller quantities. Typical impurities niobium, tantalum, zirconium, scandium, tungsten and iron can complicate processing. Despite its low abundance (~2 ppm) in Earth's crust, tin remains indispensable in applications ranging from food-safe coatings and glass floatation to high-performance superalloys and soldering for electronics and renewable-energy infrastructure. The Ministry of Mines, Government of India, in June, 2023 listed 'Tin' among 30 critical minerals, highlighting its high strategic importance and supply-chain risk.

RESERVES/RESOURCES

In India, tin ore occurs in association with granite, pegmatites, quartz veins, and also in placer deposits. The key regions where tin resources are found include the Bastar and Dantewada districts in Chhattisgarh, the Tosham deposit in Bhiwani district of Haryana and the Malkangiri district in Odisha. According to the National Mineral Inventory (NMI) data, based on the UNFC classification

system as of 1st April 2020, the total tin ore reserves/resources in the country stand at 83.72 million tonnes, containing approximately 1,03,757 tonnes of metal. Of this, around 2,101 tonnes of ore, containing 974 tonnes of metal, fall under the 'Reserves' category, while the remaining 83.72 million tonnes, with about 1,02,783 tonnes of metal, are classified as 'Remaining Resources' (Table-1).

Table – 1: Reserves/Resources of Tin as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India			
Ore	2101	83720794	83722895
Metal	973.99	102782.91	103756.9
By States			
Chhattisgarh			
Ore	2101	29795176	29797277
Metal	973.99	15909.58	16883.57
Haryana			
Ore	-	53910000	53910000
Metal	-	86220.6	86220.6
Odisha			
Ore	-	15618	15618
Metal	-	652.73	652.73

Figures rounded off

Exploration & Development

The exploration and development details, if any, are covered in the Review on 'Exploration and Development' placed under 'General Reviews' of the title.

Mining Leases & Production

As on 31st March 2024, a total of 17 mining leases with a cumulative area of 338.183 ha. were granted.

Tin Concentrates

The production of tin concentrates in 2023-24 was at 22,345 kg as against 45,432 kg in the preceding year. One Public

Sector and three Private Sector mines reported production in 2023-24. All these mines are located in Chhattisgarh. The Chhattisgarh Mineral Development Corporation Ltd (CMDC) purchases tin concentrates from local tribals, allowing them to collect it from the lease area. Hence, no labour was reported by the mine owned by the CMDC Ltd, whereas Precious Minerals and Smelting Ltd employed 8 workers in the current year and 7 workers in the previous year (Tables- 2 to 5).

The mine-head closing stock of tin concentrates was 4,903 kg in 2023-24 as against 313 kg in 2022-23.

Table – 2: Producers of Tin Concentrates (2023-24)

Name & address of producer	Location of the mine	
	State	District
Chhattisgarh Mineral Dev. Corpn. Ltd, 27/520, New Shanti Nagar Shankar Nagar Road, Raipur-462011, Chhattisgarh.	Chhattisgarh	Dantewada
Precious Minerals and Smelting Ltd, Semi Urban Industrial Estate, Frezerpur, Jagdalpur- 494001, Chhattisgarh	Chhattisgarh	Dantewada

* Producing as an associated mineral with kyanite

Table – 3: Production of Tin Concentrates, 2021-22 to 2023-24

(By State)

(Qty in kg; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	26301	32619	45432	52285	22345	17341
Chhattisgarh	26301	32619	45432	52285	22345	17341

(P): Provisional

Table – 4: Production of Tin Concentrates, 2022-23 and 2023-24**(By Sectors/State/District)***(Qty in kg; Value in ₹ '000)*

State	No . of mines	2022-23		No . of mines	2023-24 (P)	
		Quantity	Value		Quantity	Value
India	4	45432	52285	4	22345	17341
Public sector	1	44899	51057	1	21946	16635
Private sector	3	533	1228	3	399	706
Chhattisgarh	4	45432	52285	4	22345	17341
Dantewada	4	45432	52285	4	22345	17341
Sukma	-	-	-	-	-	-

*(P): Provisional***Table – 5: Mine-head closing stocks of Tin Concentrates, 2022-23 & 2023-24****(By State)***(In kg)*

State	2022-23	2023-24 (P)
India	313	4903
Chhattisgarh	313	4903

*(P): Provisional***Tin Metal**

The plant owned by Precious Minerals and Smelting Ltd reported production of tin metal at 9,249 kg in 2023-24

as against 17,160 kg in the preceding year. The plant is located at Jagdalpur in Dantewada district of Chhattisgarh (Table-6).

Table – 6: Production of Tin Metal, 2021-22 to 2023-24**(By State)***(Quantity in kg; Value in ₹ '000)*

Year	Quantity	Value
2021-22	4868	7307
2022-23	17160	35435
2023-24 (P)	9249	19885

*(P): Provisional***USES & CONSUMPTION**

Tin is a versatile, sustainable metal widely used in eco-friendly packaging, especially as tinplate for edible oils, paints, pesticides, foods, and beverages. In pure form, tin finds use in chemical storage tanks, capacitors, fuse-wires, ammunition and tinned iron sheets for protecting perishables. It forms key alloys like solders (with lead) for electronics and plumbing, bronzes for gears and bearings, and fusible alloys for safety devices, such as, sprinklers and fuses. Tin enhances cast iron microstructure and plays a role in advanced materials like zirconium–tin (nuclear cladding) and niobium–tin (superconductors). Tin oxide catalysts support air purification and CO₂ lasers; organotin compounds are vital in agrochemicals and antifouling paints. In the Float Glass Industry, molten tin ensures flatness and fire-polish, while tin oxide coatings enable conductive glass for de-icing and alarms. Tin drives innovation in lead–free solar cells, fire retardants, longer-lasting batteries and animal health products. As a fuel catalyst, it cuts emissions and saves energy. Electrolytic

tinplate continues to be prominently used in food-grade packaging.

RESEARCH & DEVELOPMENT

Tin Diselenide: Scientists have identified tin diselenide (SnSe₂) as a highly promising electrode material for next-gen lithium-ion, sodium-ion batteries, and supercapacitors due to its high theoretical capacity, fast ion diffusion and structural stability. Its layered structure and selenium content help overcome tin's volume expansion issues, making it ideal for efficient and durable energy storage.

Indium Tin Oxide (ITO): Researchers have demonstrated how tin, in the form of indium tin oxide (ITO), plays a crucial role in improving solid-state sodium battery performance. The tin–based ITO interlayer enhances interface stability, enables uniform sodium deposition and prevents dendrite formation positioning tin as a vital material in advancing clean energy technologies.

Tin–Zinc Electrodes: Researchers have developed a tin–zinc electrode that efficiently converts CO₂ into formate

using seawater as the electrolyte. The tin-based design, enhanced with a tannic acid coating, resists corrosion and unwanted reactions, achieving high efficiency and stability showcasing tin's role in sustainable carbon capture and utilisation technologies.

Tin Oxide Nanoparticles: Scientists have developed a green synthesis method to produce tin (IV) oxide nano particles (SnO₂NPs) using plant extracts, offering a sustainable alternative to chemical methods. The eco-friendly SnO₂NPs showed potential in high-tech applications and even terminated over 50% of breast cancer cells, highlighting tin's emerging biomedical role.

POLICY

As per the Import Policy, under Schedule I of ITC (HS) 2022 and Export Policy under Schedule II of ITC (HS) 2022, there are no restrictions on the export and import of tin ores and concentrates under the HS code 26090000.

Tin, in June, 2023, has been identified as critical minerals for the country. The National Critical Minerals Mission, notified vide Notification F.No.28/15/2024-CMM-Part (1) dated 24/06/2025, is a dedicated mission which aims to secure a long-term sustainable supply of critical minerals including Tin and strengthen India's critical mineral value chains encompassing all stages from mineral exploration and mining to beneficiation, processing, and recovery from end-of-life products.

SUBSTITUTES

Tin is most commonly used for packing goods since it is environmentally beneficial. Alternatives to tin include aluminium, glass, paper, plastic and tin-free steel. A variety of materials can be utilised to replace tin in its many applications, including tetrapack for liquid food items, plastic/polycontainers for solid and semi-solid food, besides, aluminium, glass and tin-free steel have become essential substitutes for production of cans and containers. Tin-Free Steel (TFS) is electrolytic chrome-plated steel with a thin layer of chromium and a layer of chromium oxide formed on the steel substrate, resulting in a stunning, lustrous metallic finish on both sides. TFS provides excellent corrosion resistance, lacquer-adhesion and printability. Other substitutes for tin include new epoxy resins for tin solders, bronze-aluminum alloys, copper-based alloys, and plastic; plastic for bearing metals that contain tin; and lead-sodium compounds for various tin chemicals.

WORLD REVIEW

The world reserves of tin metal estimated in 2024 were 4.3 million tonnes, located mainly in China (26%), Myanmar (16%), Australia(14%), Russia (11%), Brazil (10%) and Bolivia (9%). The world reserves of tin by principal countries are furnished in Table-7.

Table – 7: World Reserves of Tin
(By Principal Countries)

In tonnes of tin content	
Country	Reserves
World total (rounded off)	43,00,000
China	11,00,000
Myanmar	7,00,000
Australia	6,20,000
Russia	4,60,000
Brazil	4,20,000
Bolivia	4,00,000
Peru	1,30,000
Congo (Kinshasa)	1,20,000
Vietnam	11,000
Other countries	3,39,000

Figures rounded off
Source: USGS , Mineral Commodity Summaries, 2024.

The world mine production increased marginally by 1.4% during 2023 to 2,91,000 tonnes as compared to that 2,87,000 tonnes in the preceding year (Table- 8). China continued to be the largest producer of tin in 2023 with contribution of about 23% share in the total world production followed by Indonesia (17%) and Myanmar (16%).

The tin supply chain begins with the mining of tin ore, primarily from countries, such as, China, Indonesia and

Myanmar etc. After extraction, the ore is processed into refined tin, which is then used in a wide range of industries, including electronics, packaging, energy storage and construction. Alongside primary production, recycling of tin from used products contributes significantly to global supply, reducing dependence on new mining and supporting sustainability. Maintaining a stable tin supply involves monitoring production costs, tracking development of

new mining projects, and ensuring responsible sourcing practices throughout the chain-from ore extraction to end-product manufacturing. Challenges, such as, resource

depletion, geopolitical risks and environmental concerns make long-term investment in exploration, innovation and recycling infrastructure essential for securing future supply.

Table – 8: World Mine Production of Tin
(By Principal Countries)

Country	In tonnes (metal content)		
	2021	2022	2023
World total	272000	287000	291000
China	83144	67365	66500
Indonesia	52467	57735	50669
Myanmar	32000*	48838	47081
Peru ^(c)	26995	28231	26230
Brazil	8203	9799	25121
Bolivia	19669	17557	18515
Congo, Democratic Republic	15963	20420	18460
Australia	8772	9001	9852
Vietnam	7666	8139	9079
Nigeria	6900	8121	6863
Other Countries	42221	11797	79130

Figures rounded off

Source: BGS, World Mineral Production, 2019-2023

(c) Recoverable

* Estimated

FOREIGN TRADE

Exports

There were negligible exports of Tin ores & concentrates

during the year 2023-24 (Table-9). Productwise export details are provided in Table-10.

Table – 9: Export of Tin Ores and Conc.
By Countries

Product	2022-23(R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	17	++	++
New Zealand	++	17	--	--

Figures rounded off

Table – 10: Export of Tin (Productwise)

Product	2022-23(R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Tin And Alloys Incl. Scrap	797	975946	1025	1374631
Tin Alloys Nes	88	214103	514	716854
Tin: Anode, Cathode, Etc of Tin Unwrought	121	329981	83	199809
Tin Blocks	++	140	2	2103
Tin And Alloys Worked	588	431569	418	449420
Tin Waste And Scrap	++	153	8	6445

Figures rounded off

Imports

The imports of Tin ores & concentrates in 2023-24 were negligible and remained the same as that of the previous

year (Table-11). Productwise import details are provided in Table-12.

Table – 11: Import of Tin Ores and Conc.

(By Countries)

Product	2022-23(R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	366	++	242
USA	++	75	++	242
South Korea	++	285	--	--
Uganda	++	4	--	--
Singapore	++	2	--	--

Figures rounded off

Table – 12: Import of Tin (Productwise)

Product	2022-23(R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Tin and Alloys Incl. Scrap	13792	32096144	12672	28051073
Tin and Alloys: Worked	591	1136247	705	1407384
Tin Waste and Scrap	20	19136	++	++
Tin Alloys Nes	105	261511	94	249038
Tin: Anode, Cathode, Etc. of Tin Unwrought	13064	30647451	11865	26376255
Tin Blocks	12	31799	8	18396

Figures rounded off

FUTURE OUTLOOK

Global demand for tin is rapidly expanding, driven by its essential role as solder in electronics accounting for nearly half of consumption—the emerging green technologies like solar panels and electric vehicles have spurred demand significantly. According to the International Tin Association (ITA), demand is projected to reach approximately 4,20,000 tonnes annually by 2030, creating a potential 40,000–tonne deficit by the end of the decade due to declining output from aging mines, slow progress of new projects and limited gains in recycling. While recycling has grown, secondary supply rose 6.8% in 2024. However, this alone won't close the gap as new primary production continues to lag. Supply disruptions in major producers, such as China, Indonesia and Myanmar exacerbate the issue; despite a narrow deficit in 2024, ITA forecasts this could widen post-2027, challenging secure and affordable tin supply.

India is positioning tin as a strategically critical mineral and has recently included it in its official list of 30 critical minerals with an aim to secure a long-term sustainable supply. This is expected to strengthen India's critical mineral value chains encompassing all stages from mineral exploration and mining to beneficiation, processing, and recovery from end-of-life products. In conclusion, the global tin market is set for steady demand growth, but faces potential supply deficits without increased investment in mining and recycling. India, with its growing demand, supportive policies and resource potential, is well-positioned to strengthen its role in the global tin supply chain. International bodies like UNCTAD and the IEA highlight the need for sustainable, diversified and ethical sourcing of critical minerals like tin to ensure resilient supply chains for clean energy and technological development.



28. Tungsten

89.43 million tonnes
total reserve/resources of tungsten as
on 1st April, 2020

624 tonnes
tungsten and alloys were exported in
2023-24.

74 tonnes
tungsten ores and concentrates were
imported in 2023-24 .

Tungsten, also referred to as 'wolfram', is a dense, lustrous metal that ranges in colour from greyish white to steel grey. In June, 2023, an updated list of 30 critical minerals was issued by Ministry of Mines which listed tungsten as a critical mineral for the security and overall economic development of the nation. The inclusion comes after the committee for identifying critical minerals identified tungsten as a mineral of both high economic value and high supply risk. It is intrinsically brittle which makes it more challenging to work with. Tungsten is a crucial metal that is essential to the industrial development of the country. Tungsten does not occur in nature as a free metal. Scheelite (Calcium tungstate, CaWO_4) and wolframite [combination of ferrous tungstate and manganese tungstate, $[(\text{Fe},\text{Mn})\text{WO}_4]$] are the primary sources of tungsten, both of which are hydrothermal in origin. Tungsten has the highest melting point of any metal at 3,422 °C. It is also resistant to all acids at normal temperatures. It possesses excellent corrosion resistance, thermal and electrical conductivity, and a low coefficient of expansion. It is elastic, ductile, has high tensile strength, and can be formed into tiny wires. Tungsten is considered the most important metal for thermo-emission applications not only because of its high electron emissivity but also because of its high thermal and chemical stability. As tungsten has extremely high melting point and is ductile, it is widely used in filaments of light bulbs and vacuum tubes, and for heating elements in electrical furnaces. When exposed to air, tungsten forms a protective oxide on its surface. However, it can be totally oxidised at higher temperatures. When tungsten is added to steel in modest amounts, it significantly boosts its hardness. Imports are the primary source of tungsten and related derivatives for domestic consumption. Recycling tungsten and its alloys, including waste and scrap, helps recover a considerable amount of the metal.

RESERVES/RESOURCES

As per NMI as on 01.04.2020, India's total tungsten ore resources are assessed to be 89.43 million tonnes, with a WO_3 content of 1,44,650 tonnes. These resources are included under the 'Remaining Resources' category. Resources of tungsten-bearing minerals are mainly distributed in Karnataka (41%), Rajasthan (27%), Andhra Pradesh (17%) and Maharashtra (11%). The remaining 4%

resources are in Haryana, Tamil Nadu, Uttarakhand and West Bengal (Table-1). Incidences of WO_3 in tungsten ore have been reported from different areas of the country. At Degana, Rajasthan, in a total of 7 blocks, the minimum and maximum values of WO_3 noticed were 0.09% and 1.62%, respectively. At Balda of Sirohi district, Rajasthan, the average WO_3 content was found to range from 0.24 to 0.48 percent. In Dewa-Ka-Bera of Sirohi district, the average

WO₃ is 0.03% and in Udwarya of Sirohi, it is 0.27%. In West Bengal, Bankura deposit contains an average of 0.1% WO₃. In Kuhi-Khobana, Agargaon belt of Maharashtra, GSI has estimated resources in Sakoli basin in the district of Bhandara and Nagpur. The analysis showed 0.01% to

0.19% WO₃ in Kuhi block, 0.13% to 0.38% WO₃ in Khobana block and 0.48% WO₃ in Pardi Dahegaon-Pipalgaon block. Gold ore at Mysore mine of BGML in Karnataka has been reckoned as a potential source of scheelite. The tailing dumps at Kolar Gold Fields contain about 0.01% to 0.05% WO₃.

Table – 1: Reserves/Resources of Tungsten as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India : Total			
Ore	-	89432464	89432464
Contained WO₃	-	144650	144650
By States			
Andhra Pradesh			
Ore	-	14802300	14802300
Contained WO ₃	-	20263	20263
Haryana			
Ore	-	2230000	2230000
Contained WO ₃	-	3568	3568
Karnataka			
Ore	-	36677818	36677818
Contained WO ₃	-	6235	6235
Maharashtra			
Ore	-	10122250	10122250
Contained WO ₃	-	18591	18591
Rajasthan			
Ore	-	23928294	23928294
Contained WO ₃	-	93708	93708
Tamil Nadu			
Ore	-	250000	250000
Contained WO ₃	-	50	50
Uttarakhand			
Ore	-	658000	658000
Contained WO ₃	-	705	705
West Bengal			
Ore	-	763802	763802
Contained WO ₃	-	1531	1531

Figures rounded off.

EXPLORATION & DEVELOPMENT

The details on exploration and development, if any, are covered in the Review on Exploration & Development under "General Reviews"

MINING LEASES & PRODUCTION

As on 31.03.2024, no mining lease for mining of tungsten was granted and, as such, there was no reported production of tungsten ore/concentrate during 2023-24. However, in the past, production of tungsten was reported from Degana in Rajasthan and Chendapathar in West Bengal.

Currently, there is no mining activity happening for tungsten in the country. Degana in Rajasthan and

Chendapathar in West Bengal were the only mines of tungsten in India that had produced meager quantities of concentrate. These mines, owing to economic non-viability, had to be closed down. However, NMDC is involved in overseas tungsten exploration project through Legacy Iron Ore Ltd (Legacy) in Western Australia, which focuses mainly on gold, iron ore and base metals. Legacy holds 24 prospective tenements across its Mount Bevan, South Laverton and East Kimberley project areas in Western Australia. The tenements are in various stages of exploration for a host of commodities, including iron (1), gold (20), base metals and tungsten (3).

USES & CONSUMPTION

Tungsten is mainly used in the form of ferrotungsten in making of special and alloy steels and military applications. Ferrotungsten typically contains between 25% and 75% tungsten. The other principal use of tungsten is in the manufacture of tungsten carbide, one of the hardest synthetic materials used in various industries like construction, metalworking, mining and oil drilling. It is also consumed for strategic and security purposes, such as, production of hard materials like high penetration alloy for weaponry, rockets and missiles. Tungsten alloys and tungsten composites are used as a substitute for lead in bullet and shot. It is used widely in the manufacture of cutting tools & devices and in wear-resistant materials, particularly, those that need to be operated at high temperatures. In making this, cobalt or nickel metal powder is used as a binder to hold together the tungsten carbide grains. Tungsten compounds are used in dyes and pigments; manufacture of paints & printing ink; and also in Ceramic Industry for producing yellow tint. Other alloys bearing tungsten have wide range of applications, i.e., ornaments, heat sinks, radiation shielding, weights & counter-weights, super alloys for turbine parts, tool steels wear-resistant alloy parts & coatings, etc. Tungsten is used as filament in incandescent light bulbs and cathodes for electronic tubes, cell phones, television set, HID lamps and other electrical consumer products. The metal is used in superalloys with copper or silver and in Chemical Industry. Tungsten carbide is often used in armour-piercing ammunition.

The entire domestic requirement of tungsten ore/concentrates is met by imports. Sandvik Asia Pvt. Ltd, Pune, Maharashtra; Widia (India) Ltd, Bengaluru, Karnataka; Rapicut Carbides Ltd, Ankleshwar, Gujarat; Mishra Dhatu Nigam Ltd, Hyderabad, Telangana; and Sunflag Iron & Steel Co. Ltd, Bhandara, Maharashtra were the important

consumers of ferrotungsten for production of alloy steel. However, the current information regarding consumption of ferrotungsten by these companies are not available. Mining Machinery Industry is the main consumer of the imported ore/concentrates.

Institute of Minerals & Materials Technology (IMMT), Bhuvneshwar has completed projects on recovery of tungsten and molybdenum from secondary resources, such as, metal scrap and spent catalyst during the year 2021-23.

SUBSTITUTES

Tungsten is still primarily used for the manufacturing of filaments, electrodes and contacts in lamp & lighting applications. However, an electrodeless non-tungsten lamp is offered as an option for commercial and industrial applications. Titanium, tantalum and niobium carbides can all be used in wear-resistant applications. Molybdenum and tungsten tool steels can be used interchangeably. Bulk ceramic is an option for certain cutting tool applications.

In some instances, substitution would almost always result in an increase in cost or a decline in product performance.

TRADE POLICY

As per the Import Policy – ITC (HS) 2022 and Export Policy – ITC (HS) 2022, the imports and exports of tungsten ores and concentrates (HS Code 26110000) are allowed free.

WORLD REVIEW

Global tungsten resources are widely distributed. China leads the world in tungsten resources and reserves, with some of the largest deposits. The total worldwide tungsten reserves in terms of metal content is approximately 4.4 million tonnes, with China accounting for 52%, Australia (13%), Russia (9%), Vietnam (1.7%) and Spain (1.5%) (Table-2).

Table – 2: World Reserves of Tungsten
(By Principal Countries)

(In tonnes of Tungsten content)	
Country	Reserves
World: Total (rounded)	4400000
China	2300000
Australia ^(a)	570000
Russia	400000
Vietnam	74000
Spain	66000
Korea, Rep. of	29000
Austria	10000
Portugal	4000
Other countries	950000

Figures rounded off.

Source: USGS, Mineral Commodity Summaries, 2024.

a: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 2,20,000 tonnes.

The world mine production of tungsten in terms of metal content decreased by 14% to 85,700 tonnes in 2023 from 1,00,200 tonnes in the previous year. China was the leading

producer (72%) followed by Vietnam (14%) and Russia & Rwanda (2% each) (Table-3).

Table – 3: World Mine Production of Tungsten

(By Principal Countries)

(In tonnes of metal content)

Country	2021	2022	2023
World: Total (rounded off)	99300	100200	85700
China	75000*	75000*	62000*
Vietnam	15858	15477	12227
Russia	2143	1846	1900
Rwanda ^{(b)(c)}	1281	1808	1786
Korea, Rep. of	500*	1500*	1600*
Bolivia	1194	1038	1150
Spain	643	741	897
Austria	919	910	835
Portugal ^(a)	502	448	500*
Other countries	1210	1403	1778

Figures rounded off

Source: BGS, World Mineral Production, 2019-23.

Symbols:

(a): Wolframite and scheelite

(b): Years ended 30th June of that stated

(c): Exports

*: Estimated

FOREIGN TRADE

Exports

drastically decreased by 44% to 624 tonnes in 2023-24 from

The exports of Tungsten and Alloys Including Scrap 1,117 tonnes in the previous year. (Table-4).

Table – 4: Export of Tungsten and Alloys Incl. Scrap

(By Product)

Product	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
Tungsten & Alloys Incl. Scrap (Total)	1117211	2746384	623881	2298988
Tungsten and Alloys Unwrought	440134	1009240	248839	698803
Tungsten Filament	3941	67887	3487	73208
Tungsten Unwrought	101094	204750	473	3772
Tungsten Waste and Scrap	551199	988690	348356	695776
Tungsten Wire	20843	475817	22726	827429

Figures rounded off.

Imports

Imports of Tungsten Ores and Concentrates decreased significantly by 83% to 74 tonnes in 2023-24 from 423 tonnes in the previous year. Imports were mainly from

Netherlands (78%) followed by Lithuania (15%) and Canada (7%) (Table-5). Imports of Tungsten and Alloys Including Scrap drastically increased by 42% to 601 tonnes in 2023-24 from 422 tonnes in the previous year (Table-6).

Table – 5: Import of Tungsten Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	423	142414	74	25679
Canada	1	3467	5	13720
Netherlands	343	113766	58	9590
Lithuania	--	--	11	2368
Bolivia	--	--	++	1
Japan	31	11985	--	--
Belgium	40	10377	--	--
Latvia	8	2819	--	--

Figures rounded off.

Table – 6: Import of Tungsten and Alloys Incl. Scrap

(By Product)

Product	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
Tungsten and Alloys Incl. Scrap (Total)	422270	2478727	601162	3362809
Tungsten and Alloys Unwrought	223702	1315264	270809	1470227
Tungsten Filament	1365	33353	3367	44188
Tungsten Unwrought Total	83145	285025	208373	778661
Tungsten Waste and Scrap	14429	23342	11457	22529
Tungsten Wire	99629	821743	107156	1047204

Figures rounded off.

FUTURE OUTLOOK

The Tungsten Industry is projected to grow from USD 5.13 billion in 2024 to USD 9.49 billion by 2032, exhibiting a compounded annual growth rate (CAGR) of 8.00% during the forecast period (2024–2032). The tungsten market is experiencing strong expansion due to increased demand for downstream products in industries, such as, automotive, industrial engineering, energy and aviation. The tungsten market is seeing expansion due to the discovery of new uses in industries, such as, medicine, defence and electronics. In India, the entire demand of tungsten can only be met by imports and recycling, as there is no indigenous production of tungsten ore & concentrates. High content of WO_3 in the tailing dumps of Kolar can be worked on priority basis to meet the demand. World tungsten supply was dominated

by production in China and exports from China. Chinese Government regulated its Tungsten Industry by limiting the number of mining and export licences, imposing quotas on concentrate production, and placing constraints on mining and processing. Scrap continued to be an important source of raw material for the Tungsten Industry worldwide. China was the world's leading tungsten consumer.

In view of growing demand of tungsten in key sectors including defence & security and high supply chain risks, the Government of India notified tungsten as one of the 30 critical minerals. Khanij Bidesh India Limited (KABIL) under the aegis of Ministry of Mines, Government of India, has been established to ensure supply side assurance of critical and strategic minerals and mineral security of the nation.



29. Vanadium

24.63 million tonnes
Total reserves/resources of vanadium ore have been established as on 1st April, 2020.

26 tonnes
of ferrovanadium were exported in 2023-24.

1,206 tonnes
of ferrovanadium were imported in 2023-24.

Vanadium was discovered in 1801 by a Mexican Mineralogist Andres Del Rio. It is a scarce element, hard, silvery grey, ductile and malleable transition metal with good structural strength. It is a versatile metal with a melting point of 1,910 °C. Vanadium metal and its compounds are gaining tremendous importance in the rapidly advancing field of science & technology. It occurs naturally in about 65 different minerals, including patronite, vanadinite, roscelite and carnotite. It is also present in bauxite and in fossil fuel deposits. It occurs in association with titaniferous magnetite and is recovered as a by-product during iron & steel manufacture. Vanadium is also concentrated in many end-products of organic material including coal, crude oil, shale and tar sands. It is found in small percentages in meteoritic stones. In addition, vanadium present in bauxite can be recovered as vanadium sludge from red mud generated during the production of alumina. Vanadium is widely used in green technology applications, especially in battery technology.

Vanadium has the property of increasing the tensile strength of steel. Its high strength to weight ratio meets fuel efficiency requirements in the Automotive and Aerospace Industries.

The Ministry of Mines, Government of India, in June 2023, released the list of 30 Critical Minerals for India, which also included vanadium-bearing minerals in it. In line with this, the Central Government amended the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act, 1957) through the MMDR Amendment Act, 2023, whereby 24 critical and strategic minerals (including vanadium-bearing minerals) have been inserted in Part D to the Schedule-1 of the MMDR Act, 1957 identifying them as critical and strategic minerals for the country. Further, the amended Act has also empowered Central Government to auction critical and strategic minerals blocks.

In addition, the MMDR Amendment Act, 2023, also introduced exploration licence for deep-seated and critical minerals, such as, gold, silver, copper, zinc, lead, nickel, cobalt, platinum group of minerals, diamonds, vanadium-bearing minerals etc. which have been placed under the 7th Schedule of the MMDR Act.

RESERVES/RESOURCES

In India, vanadium is associated with titaniferous magnetite which contains 0.8 to 3% V₂O₅. It also occurs in significant amounts in association with chromite, laterite, bauxite and ferromagnesian-rich rocks, such as, pyroxenite, base anorthosite and gabbro.

As per NMI database, based on UNFC system, the total estimated reserves/resources of vanadium ore as on 1.4.2020 are placed at 24.63 million tonnes with an estimated V₂O₅ content of 64,594 tonnes. The entire resources of vanadium are placed under Remaining Resources category (Table-1).

Table – 1: Reserves /Resources of Vanadium as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total			
By Grades			
Ore	0	24633855	24633855
Contained V ₂ O ₅	0	64594.01	64594.01
By States			
Karnataka			
Ore	0	19384430	19384430
Contained V ₂ O ₅	0	49497.55	49497.55
Maharashtra			
Ore	0	384630	384630
Contained V ₂ O ₅	0	1538.52	1538.52
Odisha			
Ore	0	4864795	4864795
Contained V ₂ O ₅	0	13557.94	13557.94

Figures rounded off

MINING LEASE & PRODUCTION

Vanadium is recovered from slag generated during the processing of vanadiferous magnetite ore, where iron and steel are the principal products. The processing of gas and petroleum products is also an important source of vanadium, which is recovered both from the raw material and from the recycling of vanadium-bearing catalysts. RIL has developed a low-cost & low-temperature hybrid green process to extract vanadium from gasifier slag. The green process is being scaled up from lab to pilot level. Vanadium sludge is separated as a by-product during the Bayer process for production of alumina hydrate. NALCO has completed lab-scale studies to recover vanadium sludge from various Bayer Liquors. Vedanta's Lanjigarh Alumina Refinery has developed an in-house process for extraction of V₂O₅ (Vanadium Pentoxide) present in bauxite, involving simple operations with very low energy consumption. Thus, it is considered innovative, cost-effective and environment-friendly besides ensuring optimum utilisation of natural resources. This project was undertaken as part of Vedanta's ambitious project-Zero Discharge and Zero Waste Alumina Refinery. Vanadium recovery will make the country self-sufficient in meeting its demand as most of these hitherto were imported to make ferrovanadium. Such vanadium recovery projects also contribute to additional revenue generation. The vanadium sludge obtained at BALCO's Korba plant contains 6 to 10% V₂O₅, while that at Hindalco's Renukoot plant contains 8.2% V₂O₅ and Muri & Belagavi plants 6 to 20% V₂O₅.

As per the auction data extracted from the website of Ministry of Mines, five mineral blocks of graphite and vanadium and one block of vanadium and graphite

(amorphous) have successfully been auditioned so far as on 27.05.2025. NITs for two blocks of titanium, bauxite, and aluminous laterite associated minerals (vanadium, gallium etc.); two blocks of vanadium & graphite; and two blocks of vanadium, titanium & aluminous laterite are under process.

USES & CONSUMPTION

Vanadium is used primarily as an alloying element in Iron & Steel Industry and to some extent as a stabiliser in titanium and aluminum alloys which are used in aerospace applications. It imparts toughness and strength to steel, alloys and also acts as scavenger for oxygen. Vanadium is consumed in the Steel Industry in the manufacture of a wide-range of products, from low carbon flat rolled steels, high strength plates & structural steels to pipes, reinforcing bars, forging steels, rail steels and tool steels. About 80% vanadium is used in the form of ferrovanadium as a means of introducing vanadium into steel. There are two groups of vanadium steel alloys—vanadium high-carbon steel alloys which contain 0.15% to 0.25% vanadium and high speed steels (HSS), where the vanadium content ranges from 1% to 5%. HSS steel is used in surgical instruments and other tools. The content of vanadium in ferrovanadium varies from 45 to 50% and in some cases up to 80%, depending upon the demand. The 45% to 50% grade is produced from slag and other vanadium containing materials by silicothermic reduction of vanadium pentoxide (V₂O₅) in presence of steel scrap or by direct reduction in an electric arc furnace. The resultant vanadium steels can be divided into micro-alloy or low-alloy steels (with less than 0.15% vanadium) and high-alloy steels (with up to 5% vanadium). Non-metallurgical applications include its use as catalyst and in ceramic, chemical, pigments, health preparations

and electronic industries. It is also used to produce super conductive magnets with a field of 1,75,000 gauss. The most common oxide of vanadium, i.e., vanadium pentoxide (V_2O_5), is used as a catalyst in the manufacture of sulphuric acid. In biological context, vanadium is a micro-nutrient found naturally in mushroom, shellfish, black pepper, parsley, dill, grains and grain products. It exists as both vanadyl sulphate, the form most commonly used in food supplements and vanadate.

Modern applications of vanadium include its use as vanadium secondary batteries for power plants and rechargeable vanadium redox batteries (VRB) for commercial applications. The main advantages of VRB are that it can offer almost unlimited capacity simply by using sequentially larger storage tanks; can be left completely discharged for long periods with no ill-effects; can be recharged by replacing the electrolyte if no power source is available and suffer no permanent damage if the electrolytes are accidentally mixed. The VRB has also been shown to have the least ecological impact among all energy storage technologies.

Indian power utility National Thermal Power Corporation (NTPC) is progressing towards the commissioning and integrating a 600 kW/ 3,000 kWh Vanadium Redox Flow Battery (VRFB) system for long-duration energy storage (LDES) at NTPC Energy Technology Research Alliance (NETRA) center in Greater Noida. Hence, the increased demand of vanadium for vanadium Redox Flow Batteries will lead to increase in the domestic exploration of vanadium.

SUBSTITUTES

Substitution of vanadium in steel with niobium, chromium, titanium, manganese, molybdenum and tungsten is possible, although at higher cost or with lower performance. Heat-treated carbon steels can replace vanadium steels in certain applications. Platinum and nickel can be used in some catalytic processes but at higher cost. Presently, there is no acceptable substitute for vanadium in titanium alloys used in aerospace application.

WORLDREVIEW

The world reserves of vanadium were estimated at about 19 million tonnes of metal located mainly in Australia (45%), Russia (26%), China (23%), South Africa (4%) and the US & Brazil (with negligible share) (Table-2). Titaniferous magnetite is the most important source accounting about 85% of current world V_2O_5 production from which vanadium could be extracted as a by-product of iron. The resources are also available in crude oil (in the Caribbean basin, parts of the Middle East and Russia), tar sands (in Western Canada), phosphate rock uraniferous sandstone and siltstone. In all these cases, extraction depends on the economic recoverability of the product. The world production of vanadium in 2023 was 1,03,000 tonnes of metal content. This includes vanadium in slag product but excludes vanadium recovered as a by-product during the refining and burning of heavy oils. Major producing countries were China (66%), Russia (19%) and South Africa (9%) (Table-3). Remaining countries together contributed 6% of the total world mine production. Most of the world's vanadium supply originates from primary sources or co-production.

Table – 2: World Reserves of Vanadium
(By Principal Countries)

(in `000 tonnes of vanadium content)	
Country	Reserves
World: Total (rounded off)	19,000
Australia [#]	8,500
Russia	5,000
China	4,400
South Africa	750
Brazil	120
USA	45

Source: USGS, Mineral Commodity Summaries, 2024
[#] For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 3 million tonnes.

Table – 3: World Mine Production of Vanadium
(By Principal Countries)

(In tonnes of metal content)			
Country	2021	2022	2023
World: Total (rounded off)	112000	107000	103000
China	70300	66900	68000*
Russia	20058	20000*	20000*
South Africa	15424	8870	9100*
Brazil	5779	10436	5421
India ^(a)	400	400*	400*

Source: BGS, World Mineral production, 2019-23

Note:

(1) This table includes vanadium in slag products but excludes vanadium recovered as a by-product of the refining and burning of heavy oils.

(a) Years ending 31st March following that stated.

* estimated

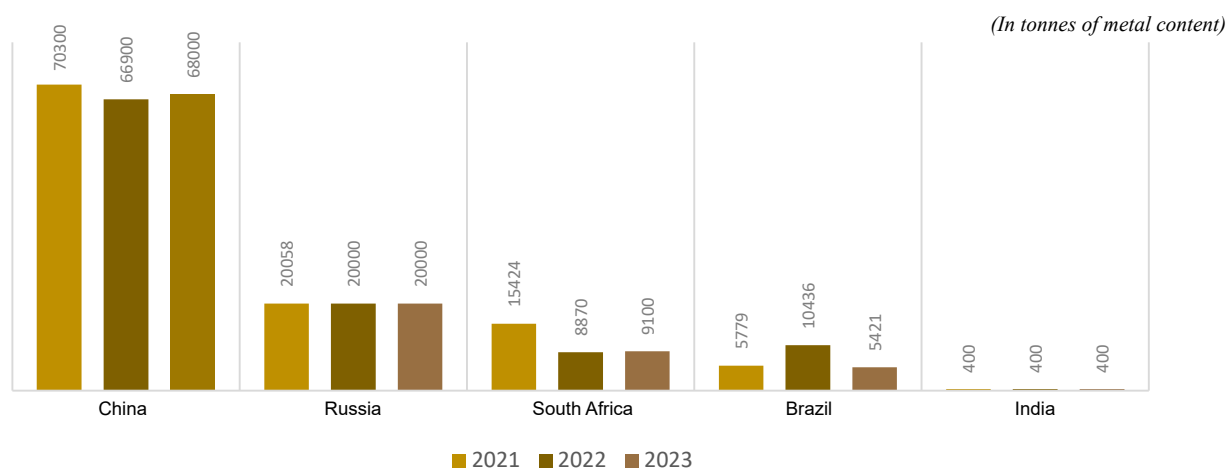


Fig.1: Countrywise Production of Vanadium

FOREIGN TRADE

Exports

In 2023-24, the exports of ferrovanadium decreased drastically by 69% to 26 tonnes from 85 tonnes in the previous year. Exports were mainly confined to Thailand (54%), UAE (38%) and Turkey (8%)(Table-4). The exports of vanadium ore and concentrates were reported as Nil during both the years, i.e., 2022-23 and 2023- 24 (Table-5).

Table – 4: Export of Ferrovanadium
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	85	233314	26	69330
Thailand	31	85566	14	31794
UAE	11	34895	10	30226
Turkey	1	3600	2	4354
Indonesia	++	271	++	1110
Bangladesh	--	--	++	782
Malaysia	++	564	++	532
Saudi Arabia	--	--	++	251
Kenya	++	8	++	121
Qatar	--	--	++	76
UK	--	--	++	64
Other Countries	42	108410	++	20

Figures rounded off

-- Nil

++ Negligible

(In tonnes)

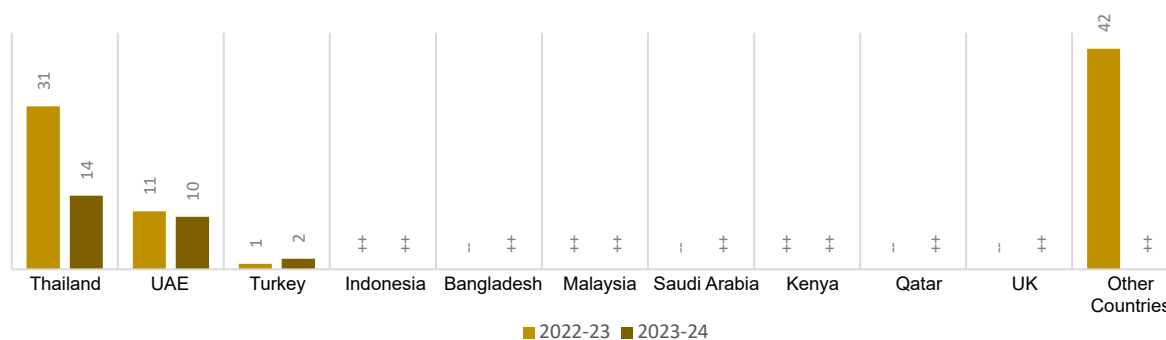


Fig. 2: Countrywise Export of Ferrovandium

Table – 5: Export of Vanadium Ores & Conc.**By Countries**

Product	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	--	--	--	--

Nil

Imports

In 2023-24, the imports of ferrovanadium decreased by 0.4% to 1,206 tonnes from 1,211 tonnes in the previous year. The imports were mainly from Republic of Korea (28%), Russia (21%), Japan (21%), Germany (10%) and Turkey (5%). The

imports of vanadium ores and concentrates during 2023-24 decrease drastically by 91% to 325 tonnes as compared to 3,661 tonnes in the previous year. The imports were mainly from Germany (37%), Russia (33%), Mexico (13%), Taiwan (6%) and USA (4%) (Tables 6 & 7).

Table – 6: Import of Ferrovandium**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1211	2795214	1206	2519991
Korea, Rep. of	546	1340803	335	696566
Russia	40	80359	248	613324
Japan	299	570201	254	433814
Germany	92	213941	118	258364
Turkey	--	--	60	123731
Austria	--	--	32	96956
Czechia	10	28099	30	60030
UAE	100	239797	16	51941
Slovenia	12	38481	20	51929
Netherlands	10	25503	18	45153
Other Countries	102	258030	75	88183

Figures rounded off

-- Nil

(In tonnes)

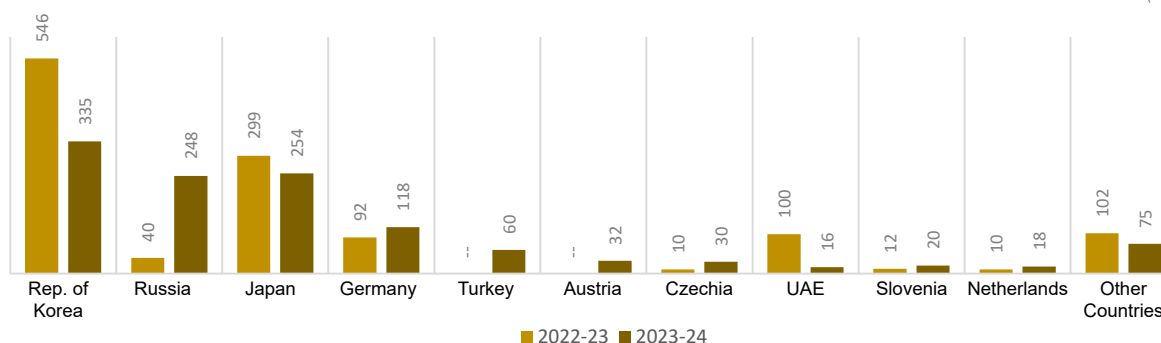


Fig. 3: Countrywise Imports of Ferrovandium

Table – 7: Import of Vanadium Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3661	209238	325	77069
Germany	--	--	119	23282
Russia	--	--	107	20198
UAE	--	--	9	8173
Mexico	114	7564	43	7496
USA	37	3595	14	7095
Taiwan	33	19165	20	6285
Spain	--	--	13	4540
Canada	3417	176087	--	--
Korea, Rep. of	60	2738	--	--
France	++	89	--	--

Figures rounded off

-- Nil

++ Negligible

FUTURE OUTLOOK

The worldwide demand for vanadium is directly linked to the demand for steels specially high strength steel. In vanadium batteries, the consumption is expected to increase in future. On the other hand, with growth of Automobile and Casting Sectors, demand for ferrovanadium is expected to increase and this will have to be met through imports. The accelerated growth in the Forging Industry and increased demand for die steels and tool steels have paved way for increased vanadium consumption. Utilisation of the huge vanadium-bearing titaniferous ores available in the States viz, Karnataka, Maharashtra and Odisha, will require R&D efforts targeted to meet the domestic demand of vanadium

pentoxide and ferrovanadium. Imperatives for utilisation of the huge vanadium-bearing titaniferous ores available in the States viz, Karnataka, Maharashtra and Odisha, through R&D efforts will have to be initiated to meet the domestic demand of vanadium pentoxide and ferrovanadium.

As per Vanadium Market Size, Share, Analysis & Forecast Report [2032] of Fortune Business Insights, the global vanadium market is expected to grow from \$4.28 billion in 2023 to \$4.89 billion by 2032, at a compound annual growth rate (CAGR) of 4.4%. Increasing demand for vanadium in Automobile, Medical and Steel Industry, as well as in Vanadium Redox Flow Batteries, are expected to stimulate the sales of vanadium globally.



Mineral Reviews

30. Apatite and Rock Phosphate

1,558 thousand tonnes
production of phosphorite have been
reported in 2023-24.

487 tonnes
of rock phosphate (ground) were
exported in 2023-24.

3.84 million tonnes
of rock phosphate (ground) were
imported in 2023-24 .

Apatite is a crucial group of phosphate minerals, first named by German Geologist Abraham Werner in 1786. It is the most abundant crystalline phosphate mineral and commonly occurs as an accessory mineral in nearly all types of igneous rocks. It may be concentrated in pegmatites, metallic veins, and magmatic deposits, and it also occurs in metamorphic rocks and as a secondary mineral in sedimentary phosphatic rocks.

The term "apatite" encompasses three main varieties: hydroxyapatite, fluorapatite and chlorapatite. Among these, fluorapatite ($\text{Ca}_5(\text{PO}_4)_3\text{F}$) is the most prevalent and also serves as a secondary source of fluorine. Another form, Collophane, is an amorphous or cryptocrystalline calcium phosphate complex.

Rock Phosphates, or phosphorites, are sedimentary deposits rich in various calcium phosphates, primarily hydroxyapatite, carbonateapatite, and fluorapatite. These rocks are essential for global phosphate production, as they contain sufficient purity and quantity of phosphatic minerals to be used directly or after processing for commercial applications.

Moreover, rock phosphate is a valuable source of by-product fluorine with materials containing 3-4% CaF_2 , particularly useful for recovering fluorite. Hydrofluorosilicic acid is generated as a by-product in phosphoric acid plants during rock phosphate processing. Additionally, phosphate rocks are also considered a significant secondary resource for uranium.

India is deficient in Apatite & Rock Phosphate availability. In case of apatite, the country is entirely dependent on imports, while the domestic Rock Phosphate production has been reported only from two States, namely, Rajasthan and Madhya Pradesh.

RESERVES/RESOURCES

Apatite

The total Reserves/Resources of apatite as per NMI data, based on UNFC system as on 1.4.2020 are estimated at 21.11 million tonnes. Out of these resources, the Reserves are placed at 0.029 million tonnes, while 21.08 million tonnes

are placed under Remaining Resources category. West Bengal accounts for the bulk of 50% of the total reserves/resources, followed by Jharkhand (34%) and Meghalaya (6%). The remaining 10% resources are located in Rajasthan, Andhra Pradesh, Gujarat and Tamil Nadu. Grade wise, soil reclamation grade accounts for 45% followed by

beneficiale grade (32%), Low/Non-beneficiale grade grades (7%). The resources of Chemical Fertiliser grade are (15%) and remaining Blendable, Unclassified & Not-known about one per cent (Table-1).

Table – 1: Reserves/Resources of Apatite as on 01.04.2020
(By Grades/States)

(In tonnes)

State/Grade	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	29395	21080904	21110299
By Grades			
Chemical Fertilizer	29395	230163	259558
Soil Reclamation	–	9513002	9513002
Low/Non-beneficiale	–	3083006	3083006
Beneficiale	–	6799902	6799902
Blendable	–	264831	264831
Unclassified	–	1000000	1000000
Not-known	–	190000	190000
By States			
Andhra Pradesh	29395	200163	229558
Gujarat	–	351000	351000
Jharkhand	–	7270000	7270000
Meghalaya	–	1300000	1300000
Rajasthan	–	1067521	1067521
Tamil Nadu	–	240000	240000
West Bengal	–	10652220	10652220

Figures rounded off

Rock Phosphate

The total reserves/resources of rock phosphate as per NMI data, based on UNFC system as on 1.4.2020 has been placed at 311.25 million tonnes. Out of these, the reserves constitute only 30.87 million tonnes while 280.37 million tonnes are under Remaining Resources category. Of the total reserves/resources, 34% are in Jharkhand, 30% in Rajasthan, 19% in

Madhya Pradesh, 8% each in Uttar Pradesh & Uttarakhand, respectively. Meagre quantities of resources are also located in Gujarat and Meghalaya. Gradewise, Low-grade account for 37%, followed by Beneficiale (22%), Soil Reclamation (15%) Blendable (10%), Chemical Fertiliser (9%) and remaining Unclassified and Not-known grades (about 7%) (Table-2).

Table – 2: Reserves/Resources of Rock Phosphate as on 1.4.2020
(By Grades/States)

(In tonnes)

State/Grade	Total Reserves	Total Remaining Resources	Total Resources
All India: Total	30876093	280377392	311253485
By Grades			
Chemical Fertilizer	22442415	6005357	28447772
Blendable	-	32399429	32399429
Soil Reclamation	-	46851282	46851282
Beneficiale	5978874	61674385	67653259
Low Grade	-	115547549	115547549
Unclassified	2454804	14324390	16779194
Not known	-	3575000	3575000
By States			
Gujarat	-	314820	314820
Jharkhand	-	107370000	107370000
Madhya Pradesh	9031093	49425938	58457031
Meghalaya	-	1311035	1311035
Rajasthan	21845000	72003769	93848769

Table- 2 (Concl'd.)

(In tonnes)

State/Grade	Total Reserves	Total Remaining Resources	Total Resources
Uttar Pradesh	-	25773444	25773444
Uttarakhand	-	24178386	24178386

Figures rounded off

MINING LEASES & PRODUCTION

Apatite

As on 31.03.2024(P), one mining lease has been granted for mining of apatite covering a total mining lease area of 13.47 Ha. No production of apatite was reported since 2017-18.

The mine-head closing stocks at the end of 2023-24 were 6,306 tonnes, same as in 2022-23 (Table-3). The average daily labour employed in apatite mines during 2023-24 was nil, same as in the previous year.

Table – 3: Mine-head Closing Stocks of Apatite, 2022-23 & 2023-24
(By States)

(In tonnes)

State	2022-23	2023-24(P)
India	6306	6306
Andhra Pradesh	-	-
West Bengal	6306	6306

Phosphorite/Rock Phosphate

As on 31.03.2024 (P), seven mining leases have been granted for mining of Rock phosphate covering a total mining lease area of 1,534.237 Ha. The total production of rock phosphate/phosphorite was at 1,558 thousand tonnes in 2023-24, a decrease of 21.26% as compared to the previous year.

state contributing 84.79% of the total production and the rest was contributed by Madhya Pradesh (Tables - 4 to 6).

The mine-head closing stock of phosphorite in the year 2023-24 was 2,950 thousand tonnes as compared to 2,426 thousand tonnes in 2022-23.

The average daily labour employed in phosphorite mines in 2023-24 was 895 as against 698 in the previous year (Table-7).

There were 8 reporting mines in 2023-24 same as in 2022-23. Rajasthan continued to be the principal producing

Table – 4: Principal Producers of Phosphorite/Rock Phosphate, 2023-24

Name & address of producer	Location of the mine	
	State	District
Rajasthan State Mines & Minerals Limited, C-89-90, Janpath, Lal, Kothi Scheme, Jaipur-302 015, Rajasthan	Rajasthan	Udaipur
M.P. State Mining Corporation Ltd Prayas Bhavan Block No 1(A) Second Floor, Jail Road, Arera Hills, Bhopal, Madhya Pradesh	Madhya Pradesh	Jhabua
Khajuraho Stones (India) Pvt. Ltd. Sagar Road Dhadari, Chhatarpur - 471001	Madhya Pradesh	Chhatarpur

Table – 5: Production of Phosphorite, 2021-22 to 2023-24

(By State)

(Qty. in tonnes; Value in ₹'000)

Name & address of producer	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	1394959	6648070	1978450	12354618	1557783	11309762
Madhya Pradesh	113730	111393	444985	483255	236964	331722
Rajasthan	1281229	6536677	1533465	11871363	1320819	10978040

Table – 6: Production of Phosphorite, 2022-23 and 2023-24
(By Sector/State/Districts)

(Qty. in tonnes; Value in ₹ '000)

State/District	2022-23						2023-24(p)							
	No. of Mines	Grade: P ₂ O ₅ content			Total		No. of Mines	Grade: P ₂ O ₅ content			Total			
		Above 30%	25% to 30%	20% to 25%	Up to 20%	Qty.		Value	Above 30%	25% to 30%		20% to 25%	Up to 20%	Qty.
India	8	234280	601296	-	1142874	1978450	12354618	8	258886	531269	-	767628	1557783	11309762
Public Sector	7	234280	601296	-	1043004	1878580	12260740	7	258886	531269	-	669764	1459919	11142823
Private Sector	1	-	-	-	99870	99870	93878	1	-	-	-	97864	97864	166939
Madhya Pradesh	6	-	-	-	444985	444985	483255	6	-	-	-	236964	236964	331723
Chhatarpur	2	-	-	-	253845	253845	263250	2	-	-	-	97864	97864	166939
Jhabua	3	-	-	-	151565	151565	176472	3	-	-	-	139100	139100	164783
Sagar	1	-	-	-	39575	39575	43533	1	-	-	-	-	-	-
Rajasthan	2	234280	601296	-	697889	1533465	11871363	2	258886	531269	-	530664	1320819	10978040
Udaipur	2	234280	601296	-	697889	1533465	11871363	2	258886	531269	-	530664	1320819	10978040

Table – 7: Mine-head Closing Stocks of Phosphorite, 2022-23 and 2023-24
(By States/Grades)

(In tonnes)

State	2022-23					2023-24(p)				
	Grade: P ₂ O ₅ content					Grade: P ₂ O ₅ content				
	Above 30%	25% to 30%	20% to 25%	Upto 20%	Total	Above 30%	25% to 30%	20% to 25%	Upto 20%	Total
India	264882	424572	900	1735294	2425648	2827803	5981438	764315	19926106	29499662
Madhya Pradesh	-	1340	900	248576	250816	-	16081	10399	3048865	3075345
Rajasthan	264882	423232	-	1486718	2174832	2827803	5965357	753916	16877241	26424317

(p) provisional

MINING

Production of phosphorite/rock phosphate in India was reported from eight mines. In Rajasthan, the ore body at Jhamarkotra mine of M/s Rajasthan State Mines and Minerals Limited (RSMM Ltd) extends over a strike length of more than 10 km and the average width of phosphate bed is about 15 m with an average inclination of about 550 from the vertical. The height of the bench is maintained up to 10 m. Shovels (6.1 cu. m) and dumpers (85 tonnes) are used for removal of ore and overburden. The mine has an annual rock handling capacity of about 20 million tonnes.

The beneficiation plant of RSMM Ltd at Jhamarkotra has a capacity of 9 lakh MT per annum to treat run-of-mine low-grade ore, with an average 16% P_2O_5 .

RSMM Ltd produces the following products:

- 1) (+) 30% P_2O_5 crushed -1/2" size high-grade rock phosphate (for SSP manufacturing units).
- 2) 31.5% P_2O_5 high-grade rock phosphate Chips (for DAP/Phos Acid manufacturing units).
- 3) 31.54% P_2O_5 BRP Grade (for SSP & DCP Manufacturing units, PROM, etc.)
- 4) 18% P_2O_5 ground low-grade beneficiated rock phosphate (RAJPHOS) (direct application to acidic soils).

The Meghnagar mine in Jhabua district and Hirapur mines in Chhatarpur and Sagar districts of Madhya Pradesh are worked by opencast method and both the mines are operated by Madhya Pradesh State Mining Corporation Ltd. Compressed-air jack hammers are deployed for drilling. The present run-of-mine capacity of Jhabua mine is 1,50,000 tonnes per year. The production of Meghnagar Mine is used in Fertiliser and Phosphorus Industries. The BRP plant at Hirapur mine is operated by Madhya Bharat Agro Industries Ltd. The processed ore from the plant is predominantly sold to manufacturers of phosphatic fertilisers and chemicals. Some parts of the ore are also internally consumed for fertiliser production.

INDUSTRY

As per the Ministry of Chemicals and Fertilizers, Department of Fertilizer's Annual Report 2023-24, among the major fertilizer products, the estimated production of urea during the year 2023-24 was 31.41 million tonnes, Diammonium Phosphate (DAP) 4.29 million tonnes, Single Super Phosphate (SSP) 4.43 million tonnes.

The major phosphatic fertilizer plants in Public Sector are Fertilizers and Chemicals (Travancore) Ltd (FACT) at Udyogamandal, Kochi (Kerala); Rashtriya Chemicals and Fertilizer Ltd (RCF) at Trombay, Mumbai (Maharashtra); Madras Fertilizer Limited at Chennai (Tamil Nadu); Brahmaputra Valley Fertilizer Corporation Ltd (BVFCL) at Namrup (Assam); National Fertilizers Ltd (NFL) at Noida (U.P.); FCI Aravalli Gypsum and Minerals

India Ltd (FAGMIL) at Jodhpur (Rajasthan); Projects and Development India Limited (PDIL) at Noida (U.P.); Fertiliser Corporation of India Limited (FCIL) in New Delhi; and Hindustan Fertilizer Corporation Ltd (HFCL) in New Delhi.

The plants in Private Sector are Gujarat State Fertilizer Company Ltd (GSFC) at Vadodara (Gujarat); Zuari Agro Chemicals Ltd in Goa; Mangalore Chemicals and Fertilizers Ltd at Mangaluru (Karnataka); Gujarat Narmada Valley Fertilizers & Chemicals Ltd (GNFC) at Bharuch (Gujarat); Nagarjuna Fertilizers and Chemicals Ltd (NFCL) at Kakinada (Hyderabad); Chambal Fertilisers and Chemicals (CFCL) at Gadepan (Rajasthan); Tata Chemicals Ltd (TCL) at Bardala (Uttar Pradesh); Kanpur Fertiliser and Cements Ltd (KFCL) at Kanpur (Uttar Pradesh); Indo-Gulf Fertilizers Limited at Jagdishpur, Uttar Pradesh, etc.

The plants in the Cooperative Sector that manufacture phosphatic fertilizers are Indian Farmers Fertiliser Cooperative Ltd (IFFCO) at Kandla (Gujarat) and Krishak Bharti Cooperative Ltd (KRIBHCO) at Surat (Gujarat). The two (2) plants of IFFCO are in Gujarat (Kalol and Kandla), two (2) in Uttar Pradesh (Phulpur and Aonla) and one (1) is located in Odisha (Paradeep).

Besides, RSMMML has a beneficiation plant in Jhamarkotra in Rajasthan, while Krishna Phoschem Ltd has set up a 600 tpd rock phosphate beneficiation plant at Meghnagar in Jhabua district of Madhya Pradesh. The Company has long-term tie-up with Madhya Pradesh State Mining Corporation Ltd.

The other associated industries utilising rock phosphate include Coimbatore Pioneer Fertilizers Ltd and Rashtriya Chemicals & Fertilizers Ltd (RCF), Mumbai which have domestic plants that recover by-product fluorine from rock phosphate in the form of hydrofluorosilicic acid, sodium silico-fluoride; and aluminium fluoride. Department of Atomic Energy has issued sanctions for establishment of 2 units for recovery of uranium from rock phosphatic sources and these are Rashtriya Chemicals & Fertilizers, Mumbai in association with Heavy Water Board (HWB); and Southern Petrochemical Industries Corporation Ltd (SPIC), Thoothukudi in association with IREL (India) Limited.

RCF is also setting up a rapidwall plant for manufacture of a unique building material using phospho-gypsum as a raw material which is the by-product of phosphoric acid plant. The project is estimated to cost ₹ 75 crore.

Red phosphorus is manufactured mainly by United Phosphorus Ltd. Red phosphorus is consumed in Match Industry. It also has applications as fumigant in Agriculture Industry and as a flame retardant.

JOINT VENTURES ABROAD

India's dependency on import at present is to the extent of 25% for urea, 90% in case of phosphates (either as raw

material or finished fertilisers) (DAP/MAP/TSP) and 100% in case of potash.

The Government has been encouraging Indian companies to establish joint ventures in countries which are rich in fertilizer resources with arrangements for buy back and to enter into long-term agreement for supplying fertiliser to India. The Department of Fertilizers has undertaken joint ventures abroad with 5 countries in the previous years. Although, during the year 2023-24, no joint venture with any country was signed.

RESEARCH AND DEVELOPMENT

RSMML has developed an organic fertilizer called Phosphate Rich Organic Manure (PROM) by using high-grade rock phosphate with farmyard waste and other organic matter. The field trials conducted through different agricultural universities in the country have shown that the agronomic efficacy of this new P-fertiliser is higher than that of the complex phosphatic fertilisers available in the market today. 'PROM' is suitable for neutral and alkaline soils, which will prove to be a boon to the Indian farmers. RSMML with the help of GSFC is making efforts for utilization of secondary Ore accumulated at Jhamarkotra mine either through beneficiation route or its direct conversion into phosphoric acid.

1. R&D efforts in the following areas strengthened the Company's operation through technology absorption, adaptations & innovation:
 - a) Productivity studies of HEMM at Jhamarkotra Mines.
 - b) Beneficiation of secondary rock phosphate.
2. Benefits derived as a result of R&D:
 - a) Strengthening of market share.
 - b) Converting waste into useful product.
 - c) Conservation of Mineral.
 - d) Future plan of action-Energy efficient processes.
3. The Company has developed a low cost organic fertilizer "PROM".
4. Two patents have been filed by the Company jointly with Mohan Lal Sukhadiya University, Udaipur, which got approved under the title -
 - i) "process for making slow release phosphate fertiliser".
 - ii) "An eco-friendly process for making EPSOM and Gypsum".
5. The Company has introduced 30% crushed Rock phosphate replacing 31.5% CRP, which has improved mineral conservation.
6. Benefits derived as a result of the efforts are product improvement, cost reduction, product development, import substitution, etc.

These efforts helped in satisfying the consumer needs as well as business requirements by introducing innovative products.

ENVIRONMENTAL CONCERNS

There are apparent concerns regarding phospho-gypsum which is formed as a by-product during manufacturing of phosphoric acid. It contains about 1% P_2O_5 , 1% F and 10-30 times higher concentrations of radon, none of which is desirable. Environmental Protection Agency (EPA) of USA stipulated in 1989 that phospho-gypsum is unsuitable for sale as common gypsum. Production of each tonne of P_2O_5 yields about five tonnes of phospho-gypsum. EPA has prescribed stringent measures for storage, transport and disposal of phospho-gypsum. In India, however, by-product phospho-gypsum is used widely in cement manufacturing.

The use of phosphate also attracts scrutiny. Much attention has been paid to its role in stimulating the growth of algae and other organisms on surface water, the process known as eutrophication. This process is deleterious because it causes blooms of algae which consume dissolved oxygen in lakes and even in shallow, isolated arms of the ocean. Phosphate fertilisers are probably not the only cause of phosphate-induced eutrophication. Another concern is that fertilizer phosphate does not leach readily from soil. One of the best ways to remove this phosphate is through the addition of lime which causes precipitation of apatite. However, this procedure, being relatively costly, has not been applied widely. Other applications where the use of phosphate has been discouraged is in manufacturing of detergents.

USES & CONSUMPTION

Rock Phosphate is primarily mined for fertilizer production yet, its applications extend far beyond agriculture. It is used in animal feed supplements and as a source for elemental phosphorus and phosphoric chemicals, which have diverse uses in detergents, insecticides, pharmaceuticals, and even soft drinks and toothpaste. Additionally, phosphate compounds play a role in glassmaking, photographic films, and pyrotechnics, including fireworks and military applications.

Apatite, a mineral form of phosphate, is noteworthy for its transparent specimens, which can exhibit vibrant colours like green, blue, yellow, and pink, making them suitable for faceted gemstones. Furthermore, due to their stability and structure, apatite and other phosphates are being researched as potential materials for nuclear waste storage, highlighting their importance in both industrial and environmental contexts.

As per annual report 2023-24 of The Fertilizer Association of India (FAI), apparent consumption of rock phosphate (P_2O_5) in 2023-24 was 8.31 million.

SPECIFICATIONS

Elemental Phosphorus and Phosphoric Acid

BIS has prescribed the IS:11224-1985, reaffirmed 2010 specifications for rock phosphate used in the manufacture of elemental phosphorus (Type-I) and phosphoric acid (Type-II).

Single Superphosphate

The P_2O_5 content in rock phosphate for manufacturing single superphosphate should be a minimum of 31%. Silica up to 8% can be tolerated. Iron and alumina, i.e., R_2O_3 should not exceed 3.5%. Higher R_2O_3 may cause reversion of available P_2O_5 (water soluble P_2O_5). Carbonate up to 5% improves the reactivity of rock phosphate by increasing the reaction temperature and making the mass porous.

Direct Application of Rock Phosphate as Fertiliser

In India, the finely-ground rock phosphate containing 16% P_2O_5 is used for direct application to the soil for soil amendment. This application is dependent upon the structure and chemical composition of the rock. Direct application is suited mostly for pastures and forage crops and for acidic soils. According to PPCL the following specifications are considered for utilising any rock phosphate as phosphatic fertiliser for direct application in acidic soils.

1. Absolute citrate solubility index 7% (max.)
2. Apatite to carbonate ratio 0.035 CO_2
3. Origin of rock phosphate Sedimentary
4. Mesh size 100
5. Hydroxyl ion in crystal lattice 2 is higher indicating substitution of OH for $PO_4:H_2O$
6. Grade of rock phosphate powder: 16% P_2O_5 citrate soluble fraction
7. Iron (as Fe_2O_3) 5%
8. CaO to P_2O_5 ratio: 1:8

The use of rock phosphate for direct application as fertilizer depends on its level of solubility in acidic soil.

POLICY

Imports of natural calcium phosphates (including apatite), natural aluminium-calcium phosphates and phosphatic chalk are allowed 'free' under Heading No. 2510 as per the Import Policy ITC (HS) 2022. All chemical fertilisers except urea continue to be decontrolled. The Government of India has been implementing a scheme of concession fixing indicative maximum retail price (MRP) for enabling sales of decontrolled phosphatic and potassic fertilisers at reasonable prices.

In case of Phosphate Fertilizer Industry, the scarcity of domestic raw material constrains the attainment of self-sufficiency in the country. A policy has, therefore, been adopted which involves the following three options:

1. Domestic production based on indigenous imported rock phosphate and imported sulphur.
2. Domestic production based on imported intermediates, viz, phosphoric acid.
3. Imports of finished fertilizers.

The Government of India notified New Urea Policy extended for the period 01.06.2015 to 30.09.2020 for existing gas-based urea manufacturing units. Thereafter the target energy norms of NUP-2015 have been enforced on these urea units from 1st October, 2020 onwards.

WORLD REVIEW

The world reserves of phosphate rock are about 74 billion tonnes, located mainly in Morocco (68%), China (5%), Egypt (4%), Tunisia, Russia & Algeria (3% each), Brazil, South Africa & Saudi Arabia (2% each) and Australia (1%). The remaining 7% is located in other countries (Table-8).

Table – 8: World Reserves of Phosphate Rock
(By Principal Countries)

Country	Reserves
World total (rounded off)	74000000
Morocco	50000000
China [#]	3800000
Egypt	2800000
Tunisia	2500000
Russia	2400000
Algeria	2200000
Brazil	1600000
South Africa	1500000
Saudi Arabia	1400000
Australia ^(a)	1100000
Other countries	4922000

Source: USGS, Mineral Commodity Summaries, 2024

Note: India's total reserves/ resources of rock phosphate as per National Mineral Inventory as on 1.4.2020 are 311.25 million tonnes.

a: For Australia, Joint Ore Reserves Committee compliant or equivalent reserves were 120 million tonnes.

#: Production data for large mines only, as reported by the National Bureau of Statistics of China.

The world production of phosphate rock increased by 1% to 237 million tonnes in 2023 from 234 million tonnes in 2022. China (44%), Morocco (12%), USA (8%), Russia (6%), and Jordan & Peru (5% each) were the major producers (Table-9).

Table – 9: World Production of Phosphate Rock
(By Principal Countries)

Country	(In '000 tonnes)		
	2021	2022	2023
World: Total (rounded off)	241000	234000	237000
China	104107	104745	105309
Morocco	38122	30456	28675
USA	21600	19800*	19600*
Russia	14100	14400	14400*
Peru	10776	10955	11920
Jordan	10015	11258	11454
Saudi Arabia	8400	8700*	9900*
Brazil [#]	5029	5307	4200
Egypt ^(a)	3098	3000*	4000*
Tunisia	3726	3563	3093
Other countries	22026	21816	24449

Source: BGS, World Mineral Production, 2019-23.

#: Including beneficiated and directly shipped material.

*: estimated

(a): Years ended 30th June of that stated.

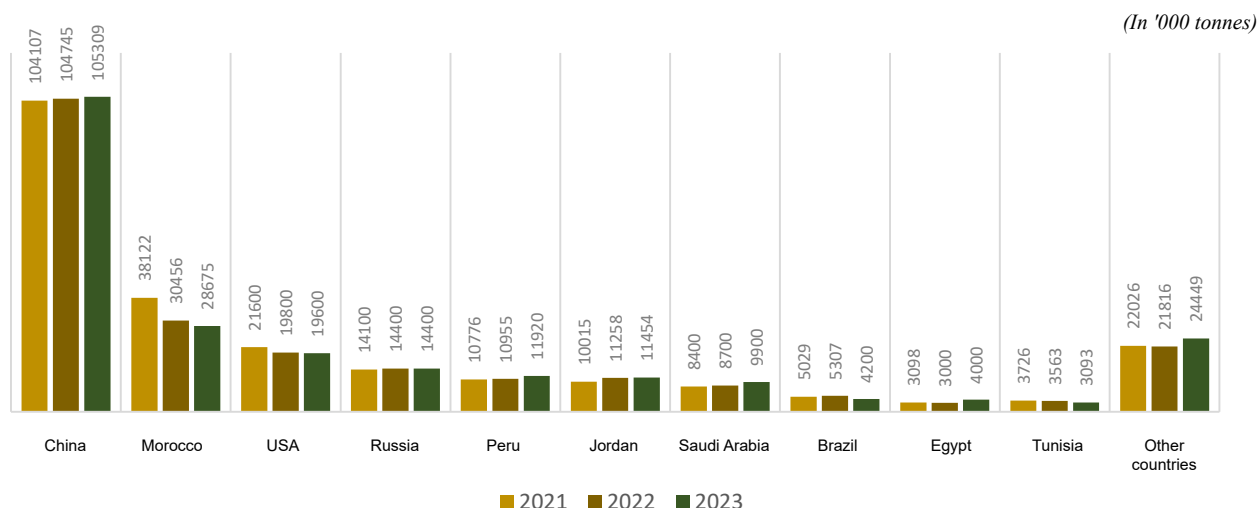


Fig. 1: Countrywise Production of Phosphate Rock

FOREIGN TRADE

Exports

In 2023-24, exports of Rock Phosphate (ground) increased by 31% to 487 tonnes from 371 tonnes in the previous year. Rock Phosphate (ground) was exported mainly to Nepal

(100%). Exports of Rock Phosphate (unground) decreased by 71% to 47 tonnes in 2023-24 from 161 tonnes in the previous year. Rock Phosphate (unground) was exported mainly to Nepal (89%), Australia (6%) and Spain & Belgium, (2% each) (Table – 10 & 11).

Table – 10: Export of Rock Phosphate (Ground)

Country	(By Countries)			
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	371	3602	487	2536
Nepal	45	228	487	2523
Bhutan	--	--	--	13
UAE	325	3362	--	--
Belgium	1	12	--	--

Figures rounded off

Table – 11: Export of Rock Phosphate (Unground)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	161	15757	47	635
Nepal	60	1964	42	432
Australia	--	--	3	138
Spain	--	--	1	54
Belgium	--	--	1	11

Figures rounded off

Imports

Imports of rock phosphate (ground) decreased by 2% to 3.84 million tonnes in 2023-24 from 3.91 million tonnes in the previous year. Imports of rock phosphates (ground) were mainly from Morocco (28%), Togo (20%), Algeria (18%), UAE (11%), Egypt (9%), Senegal (5%), Jordan (4%), Cyprus & Lebanon (1% each) and other countries (1%).

Imports of rock phosphate (unground) decreased by 4% to 4.97 million tonnes in 2023-24 from 5.18 million tonnes in the previous year. Imports of rock phosphates (unground) were mainly from Jordan (59%), Egypt (17%), Lebanon (10%), Togo (7%), Morocco (5%), Peru, Syria & UAE (1% each) (Tables – 12 & 13).

Table – 12: Import of Rock Phosphate (Ground)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3910889	71626208	3838444	60256247
Morocco	1198687	27618130	1091648	19555572
Togo	631513	12590466	783271	14375258
Algeria	641278	9545510	701521	9482215
UAE	428207	6197885	438576	5757895
Egypt	390804	5524255	363967	4451749
Senegal	66190	1078283	175265	2495127
Jordan	338945	5230654	172546	2472091
Cyprus	28875	448799	52350	649646
Lebanon	52931	699194	33110	416686
Syria	--	--	13600	176289
Other countries	133459	2693032	12590	423719

Figures rounded off

Table – 13: Import of Rock Phosphate (Unground)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5180702	79743794	4972510	66573353
Jordan	3587912	54744691	2939360	38078329
Egypt	733319	9287089	833366	9552736
Togo	245375	6005529	358155	8703438
Lebanon	240172	3644013	485740	6174557
Morocco	92940	1887969	224786	2779009
Peru	119149	1716650	55000	583542
Syria	--	--	35196	389435
UAE	--	--	36056	248673
Sri Lanka	--	--	2808	53772
Singapore	--	--	2000	8376
Other countries	161835	2457853	43	1486

Figures rounded off

FUTURE OUTLOOK

There is no substitute for phosphorus in agriculture and it is essential element for ensuring food security in our country. The country is deficient in all fertiliser minerals. The reserves/resources of chemical and fertiliser grades apatite and rock phosphate in India are very limited. Therefore, detailed exploration is necessary for conversion of remaining resources into reserves. Secondly, the search for apatite and rock phosphate may have to be intensified in Andhra Pradesh, Rajasthan, Madhya Pradesh, Jharkhand, Tamil Nadu, Meghalaya, Gujarat, Uttar Pradesh, Uttarakhand, West Bengal, etc. Till the domestic resources of these two minerals are improved, the country has no alternative but to depend on imports. Concerted efforts should be made by way of constituting consortia of public-private companies to acquire assets abroad specifically in countries like Uzbekistan, Jordan, etc. Only about 10-15% requirement of raw material for phosphate fertiliser production is met through indigenous sources. The remaining requirement is met through imports in the form of rock phosphate, phosphoric acid and direct fertilizers.

Private Sector participation in rock phosphate mining needs to be promoted in order to make available the two minerals to reduce import dependence for promotion of fertilisers for Agricultural sector.

Demand for phosphatic fertilizer is expected to increase gradually in tandem with the growth in population and corresponding increase in food requirements. The Government has been encouraging Indian Companies to establish joint ventures abroad in countries which are rich in fertiliser resources.

In India, most of the existing phosphatic fertilizer and phosphoric acid plants have been designed for high-grade imported rock phosphate, mainly from Morocco and Jordan. Unfortunately Indian deposits are particularly of low-grade variety. Therefore, the fertilizer and phosphoric acid plants those are likely to be set up as replacement of the existing plants may have to be designed to accept indigenous ores as feed. In addition, beneficiation of domestic low-grade ores would be a step in the right direction and should be promoted persuasively.



31. Asbestos

22.90 million tonnes
Total reserves/resources of asbestos have been established as on 1st April, 2020

1,206 tonnes
of asbestos were exported in 2023-24

4,65,449 tonnes
of asbestos were imported in 2023-24

Asbestos is a group of six naturally occurring fibrous silicate minerals. The physical properties, besides its fibrous character, such as, fineness, flexibility, tensile strength & length of fibers, infusibility, low heat conductivity and high resistance to electricity, sound & corrosion by acids, make asbestos commercially important. Commercial asbestos is classified into two main mineralogical groups: serpentine asbestos or chrysotile asbestos and amphibole asbestos. The latter includes asbestos minerals, such as, tremolite, actinolite, anthophyllite, amosite and crocidolite. Commercially, chrysotile asbestos is far superior in physical properties and hence more valuable than amphibole asbestos.

Inhalation of asbestos fibers increases the risk of various dangerous lung conditions, including Mesothelioma, Asbestosis and lung cancer. India's asbestos requirement is met through imports from Russia, Kazakhstan, Brazil, China and other countries.

RESERVES/RESOURCES

As per NMI database, based on UNFC system, the total reserves/resources of asbestos in the country as on 1.4.2020 has been placed at 22.90 million tonnes. A total of 22.90 million tonnes of asbestos are placed under Remaining Resources. Out of the total resources, Rajasthan

accounts for 13.61 million tonnes (59%) and Karnataka for 8.28 million tonnes (36%). The remaining five percent of resources are estimated in the States of Jharkhand, Andhra Pradesh, Odisha and Uttarakhand (Table-1).

Table-2 summarises the mineralogical varieties of asbestos occurring in various parts of the country.

Table – 1: Reserves/Resources of Asbestos as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	-	22908067	22908067
By Grades			
Chrysotile	-	831905	831905
Amosite	-	4463667	4463667
Tremolite	-	4200109	4200109
Chrysotile mixed with others	-	22516	22516
Mixed Amphibole	-	12383632	12383632
Actinolite	-	34311	34311
Anthophyllite	-	20000	20000
Others	-	432134	432134
Not-known	-	463091	463091
Unclassified	-	56701	56701
By States			
Andhra Pradesh	-	79799	79799
Jharkhand	-	154893	154893
Karnataka	-	8282457	8282457
Odisha	-	56700	56700
Rajasthan	-	13615710	13615710
Uttarakhand	-	311	311

Figures rounded off

Table – 2: Occurrences of Asbestos in India

State	District	Mineralogical variety
Andhra Pradesh	Anantapur	Chrysotile
	Cuddapah	
Jharkhand	Singhbhum (East)	Chrysotile, tremolite, chrysotile mixed with other minerals
	Singhbhum (West)	
Karnataka	Chikkamagaluru	Amosite
	Hassan	Anthophyllite
	Mandya	Mixed amphibole minerals
	Mysuru	Chrysotile
	Shivamogga	Amosite
Odisha	Kendujhar	-
	Ajmer	Mixed amphibole minerals
Rajasthan	Bhilwara	-do-
	Dungarpur	-do-
	Paliw	Tremolite, chrysotile mixed with other amphibole minerals
	Rajsamand	Tremolite, actinolite and mixed amphibole minerals
	Udaipur	Chrysotile, tremolite and mixed amphibole minerals
Uttarakhand	Chamoli	Others

PRODUCTION

No production of asbestos was reported in 2023-24 as well as in the previous year and there were no reporting mines in 2023-24 as well as in the preceding year.

Similarly, the mine-head closing stocks of asbestos also remained 'Nil' for the year 2023-24 as well as in the preceding year 2022-23. The average daily employment of labour for both the years was also 'Nil.'

MINING & MILLING

Presently there is no working mine of asbestos in India. The usual method of mining chrysotile in Pulivendla Tehsil, Cuddapah district, Andhra Pradesh, was by opening an incline along the dip, varying from 200° to 250°, keeping the trap as floor and limestone as roof. Two or three such inclines were converted into a regular underground mine by developing levels and winzes, connecting them and adopting board-and-pillar system of development. In almost all the mines, operations like blasting, hole drilling, hoisting, pumping and ventilation were mechanised.

The run-of-mine was subjected to manual sorting of asbestos-bearing rock (ABR). ABR was then hand-combed for chipping off the asbestos-bearing portion in small pieces of about 2.5 cm to produce asbestos concentrates. From ABR, serpentine was removed as a waste. The asbestos concentrate was loaded manually into hopper of a hammer mill. In hammer mill, asbestos and other minerals were separated and then processed into double-deck screen having 10 to 40 mesh sieves. The screening gives three fractions: (a) oversize, (b) middling and (c) tailing. Tailing was considered as a waste which generally did not contain appreciable quantity of asbestos. The oversize was recycled in the hammer mill, and the middling fiber was drawn out by a cyclone and collected.

Grading & Marketing

Small fibers recovered through milling process account for nearly two-third of the total asbestos production. The general grading system adopted is as follows

Grade	Fibre Size	Method
Grade - As	45 mm and above	Hand-sorted
Grade - A	Between 25 and 45 mm	
Grade - B	Between 12 and 25 mm	
Grade - C	Above 16 mesh	
Grade - D3	24 mesh	Mill-processed
Grade - D4	40 mesh	
Grade - D6	60 mesh	

Note: Producers of amphibole asbestos sell their output as crude or fluff and powder

CLASSIFICATIONS

Various classifications of chrysotile asbestos followed in India are largely based, on fibre length:

(1) Grade A or

A Special	- 25.4 mm fibres or larger
As 1	- 25.4 mm and larger fibres but brittle compared to As or A Special
A	- 19.05 to 25.4 mm fibres
A1	- 19.05 to 25.4 mm fibres but brittle compared to A
A2	- 19.05 to 25.4 mm fibres but brittle compared to A1

Grade B - 6.35 to 19.05 mm fibers

B1	- 6.35 to 19.05 mm fibres but brittle compared to B
B2	- 6.35 to 19.05 mm fibres but brittle compared to B1
C	- Below 6.35 mm fibres

(2) Grade A Special - Above 31.5 mm

A	- Between 19 and 31.5 mm
B	- Between 6.3 and 19 mm
C	- Below 6.3 mm including powder
D	- Dust

3) Quebec standard asbestos testing machine classification of chrysotile asbestos according to groups is given below

Crude Asbestos

Group No. 1	Crude No. 1	Consists basically of crude, 3/4 inch and longer staple
Group No. 2	Crude No. 2	Consists basically of crude, 3/8 to 3/4 inch staple.

Milled Asbestos

Standard designation of grade	Guaranteed minimum spinning test
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Group No. 3 (spinning fibres)

3 D	10.5-3.9-1.3-0.3
3 Z	0-8-6-2

Group No. 4 (shingle fibres)

4 D	0-7-6-3
4 Z	0-1.5-9.5-5

Group No. 5 (paper fibres)

5 D	0-0.5-10.5-5
5 R	0-0-10-6

Group No. 6 (waste)

6 D	0-0-7-9
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Group No. 7 (shorts or refuse)

7 D	0-0-8-11
7 W	0-0-0-16

Group No. 7 (floats)*

7 RF	No test
7 TF	No test

Group No. 8 (sand & gravel)

8 S	Less than 50 lb per cu. ft loose measure
8 T	Less than 75 lb per cu. ft loose measure

Group No. 9 (gravel & stone)

9 T	More than 75 lb cu. ft loose measure
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* The suffix 'F' designates 'floats' in the case of 7R and 7T grades.

USES

Industrial use of asbestos is linked to the type of asbestos. Chrysotile asbestos, being more fibrous & possessing better tensile strength than amphibole variety is used in the manufacture of asbestos fabrics, cement sheets, pipes, building materials (roofing shingles, ceiling & floor tiles,

paper products) and allied products. It is also used in brake linings, insulation and fire proof clothing. Short fibres are used with cement as binders for manufacturing asbestos cement products. Amphibole asbestos generally finds use in heat insulation and treatment of acids. Anthophyllite and tremolite fibres, although of good length, are too weak and brittle to be spun. They are, therefore, used for boiler lagging, hard-setting magnesia composition and as filler in asbestos paints & various asbestos-moulded articles.

SUBSTITUTION

Materials substituted for asbestos include calcium silicate, carbon fibres, fibres of cellulose, ceramic, glass & steel, wollastonite and several organic fibres like aramid, polyethylene, polypropylene and polytetrafluoroethylene. where reinforcement properties of fibres are not required, several non-fibrous minerals are also considered for possible substitution. Some of the most common alternatives to asbestos are polyurethane foam, thermoset plastic, fiberglass, mineral wool, cellulose and natural minerals. However, no single substitution has been found to be as versatile or as cost-effective as asbestos.

ENVIRONMENTAL IMPACT OF ASBESTOS

Asbestos, used as a part of construction materials due to its flame retardant quality, poses a major risk to human health and environment. Asbestos has been linked to a number of serious medical conditions. These include lungs and respiratory problems because asbestos is made of tiny fibres that, when released into the air, can settle in the lungs upon prolonged inhalation and irritate the tissues in the chest cavity. Mesothelioma is a rare form of cancer of the lungs and digestive tract which is most commonly caused by exposure to asbestos polluted air. Besides health hazards, asbestos also has negative impacts on the environment. A study presented in 2006 at the International Conference on Health, Environment and Justice found that asbestos dust can easily travel through the air and into the water supply. It can also settle on the surface of the soil instead of getting absorbed into the ground, which means that it can still be picked up by the wind and inhaled into the human respiratory system.

However, as per the report of the 'Asbestos Cement Products Manufacturers' Association in India, only chrysotile (white) asbestos fibre is used for the manufacture of asbestos-cement sheets which contain a very small quantity of chrysotile fibre (only 8–10%) and asbestos-cement pipes The other raw materials used are cement 45%, fly ash 30-35% and wood pulp. The asbestos fibres are firmly locked-in or encapsulated within the cement matrix during manufacture so that fibres cannot be emitted into the atmosphere under normal use and thus, they pose no health risk to the general public or the environment. Several studies abroad have concluded that the use of chrysotile in the manufacture of Asbestos Cement Products under controlled conditions is safe for workers, environment and

the general public.

India has again opposed the listing of chrysotile asbestos as a hazardous substance under the Rotterdam Convention at the eighth meeting of the Conference of Parties (COP) held in Geneva from 24th April to 5th May 2017.

TRADE POLICY & LEGISLATION

No restrictions have been imposed on the exports of asbestos in the Foreign Trade Policy, 2015-20. As per the prevailing Foreign Trade Policy, asbestos under Heading 2524 can be imported freely with the exception of amosite which is restricted. However, the imports of crocidolite, actinolite, anthophyllite, amosite and tremolite are restricted in terms of Interim Prior Informed Consent (Interim PIC) Procedure of Rotterdam Convention for Hazardous Chemicals and Pesticides.

The Ministry of Environment and Forest, vide Notification dated 13.10.1998, under Sections 3 (1) and 6 (2) (d) of Environment (Protection) Act, 1986 and Rule 13 of Environment (Protection) Rules, 1986, has prohibited the imports of waste asbestos (dust and fibre), on account of it being a hazardous waste detrimental to human health and environment.

WORLD REVIEW

Large reserves of Asbestos are located mainly in Kazakhstan, Russia, China and Brazil. Russia was the leading producer with 600 thousand tonnes, followed by Kazakhstan (255 thousand tonnes), China (200 thousand tonnes) and Brazil (189 thousand tonnes) (Tables-3 and 4).

Table – 3: World Reserves of Asbestos
(By Principal Countries)

(In million tonnes)	
Country	Reserves
World: Total	Large
Kazakhstan	Large
Russia	110
China	18
Brazil	11
USA	Small

Source: USGS, Mineral Commodity Summaries, 2024.
Note - India's total reserves/resources of asbestos as per NMI database, based on UNFC system, as on 1.04.2020 have been estimated at 22.90 million tonnes.

Table – 4: World Production of Asbestos
(By Principal Countries)

(In '000 tonnes)			
Country	2021	2022	2023
World: Total	1300	1300	1200
Russia	750	630	600
Kazakhstan	250	250	255.3
China	160	196	200
Brazil	172	183	189.34

Source: BGS, World Mineral Production, 2019-2023.
Note: In addition to the countries listed, asbestos may also be produced in Afghanistan, North Korea, Romania & Slovakia.

FOREIGN TRADE

Exports

Exports of asbestos decreased substantially by 63% to 1,206 tonnes in 2023-24 as compared to 3,286 tonnes in the previous year. Exports were mainly to Bangladesh (79.4%) and Sri Lanka (20.6%). Exports of asbestos (fibre

products) were at 47,128 tonnes in 2023-24 as compared to 42,269 tonnes in the previous year. Exports of asbestos fibre (raw, beaten or washed or graded) were at 1,206 tonnes during the year 2023-24 as compared to 3,220 tonnes in the preceding year. Exports of asbestos-cement products were 1,19,975 tonnes in 2023-24 as compared to 1,36,028 tonnes in the preceding year (Table-5 to 8).

Table – 5: Exports of Asbestos

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3286	217449	1206	81470
Bangladesh	3220	217275	958	64856
Sri Lanka	--	--	248	16607
Nepal	66	174	++	7
Brazil	++	++	--	--

Figures rounded off

Table – 6: Exports of Asbestos

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Asbestos	3286	217449	1206	81470
Asbestos (others Nes)	1	7	--	--
Asbestos Fibre (raw, beaten or washed or graded)	3220	217275	1206	81465
Asbestos Flakes or Powder	65	164	--	--
Asbestos Other in Rock Form	++	3	++	5

Source: DGCIS, Kolkata

P: Provisional

--: Nil

++: Negligible

Table – 7: Exports of Asbestos Fibre Products

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	42269	6866139	47128	7689004
USA	13681	2029580	14311	2113225
UAE	3572	450985	3906	512769
Turkey	574	276867	1306	410883
Egypt	1623	150622	3256	334129
Brazil	1033	317549	599	319367
Russia	772	191000	1236	301145
Canada	1362	184054	1951	273033
Kenya	1429	206440	1642	247245
Saudi Arabia	1699	267394	1543	229205
Nepal	602	169164	762	207328
Other Countries	15922	2622484	16616	2740675

Figures rounded off

Table – 8: Exports of Asbestos Cement Products

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	136028	2530491	119975	2345999
UAE	48341	812305	41127	720046
Nepal	23268	434018	18972	364370
Qatar	23141	381973	17155	288525
UK	12485	251086	10668	231501
Oman	4834	82290	5357	95317
South Africa	1357	25635	4475	94023
Maldives	2914	73763	2948	66678
Saudi Arabia	3083	60387	3539	65637
USA	401	35947	462	56580
Turkey	1404	28657	2772	54048
Other Countries	14800	344430	12500	309274

Figures rounded off

Imports

Imports of asbestos at 4,65,449 tonnes in 2023-24, increased by 15% as against 4,04,154 tonnes in the previous year. Almost entire import was that of asbestos fibre (raw, beaten or washed or graded). Imports of asbestos were mainly from Brazil (33%), Russia (28%), Lithuania (21%), Turkey & Kazakhstan (5% each) and Georgia (4%) during 2023-24. A total of 23,687 tonnes asbestos-cement products

were also imported in 2023-24 as against 24,361 tonnes in the previous year. Imports of asbestos fibre products were 2,952 tonnes during the year 2023-24 as compared to 2,838 tonnes in previous year. Imports of Asbestos fibre (raw, beaten or washed or graded) decreased to 4,64,688 tonnes during the year 2023-24 as compared to 4,03,293 tonnes in the preceding year (Tables-9 to 12).

Table – 9: Imports of Asbestos

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	404154	22940349	465449	23260379
Brazil	175994	9374000	155438	8233082
Russia	96718	5459089	129926	7723275
Lithuania	2720	151892	97879	2135646
Turkey	49992	3420470	23492	1669490
Georgia	42526	2563287	20808	1259002
Kazakhstan	20317	1165382	21133	1232649
Poland	3894	199364	9108	531855
UAE	4905	309199	4280	287405
China	889	45818	1967	100471
South Africa	400	26307	540	39772
Other Countries	5799	225541	878	47732

Figures rounded off

Table – 10: Imports of Asbestos

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Asbestos	404154	22940349	465449	23260379
Asbestos Fibre (raw, beaten or washed or graded)	403293	22894231	464688	23213394
Asbestos Flakes or Powder	479	26362	582	37040
Asbestos Other in Rock Form	--	--	179	9945
Asbestos (others Nes)	382	19756	--	--

Figures rounded off

Table – 11: Imports of Asbestos Fibre Products
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2838	3237762	2952	3671629
Japan	490	1309613	575	1665222
Germany	199	310312	545	895849
China	625	170918	1004	232853
USA	229	366961	150	189653
Korea, Rep. of	96	150919	68	138151
Czechia	65	69716	61	118586
UK	277	85851	202	97905
Denmark	528	534195	30	54386
Spain	10	14853	31	40909
France	82	46386	71	40660
Other Countries	237	178038	215	197455

Figures rounded off

Table – 12: Imports of Asbestos Cement Products
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	24361	646965	23687	549428
Thailand	22127	499435	20033	362672
China	343	28411	1302	95524
Indonesia	1400	41813	1630	47646
Japan	43	4754	514	11399
Italy	6	5014	13	10779
Germany	34	31837	6	5563
Czech Republic	--	--	6	4895
Vietnam Soc. Rep.	2	196	156	3441
USA	2	1820	3	3264
UAE	2	114	4	1810
Other Countries	402	33571	20	2435

Figures rounded off

FUTURE OUTLOOK

Consumption of asbestos minerals in India & other countries of the world will decline steadily in near future. This decline will be due to health and liability issues associated with asbestos use, leading to the displacement of asbestos from traditional domestic markets by substitutes, alternative materials and technological advancements. While the economic impact of asbestos mining in India

is minimal, mining operations do adversely affect human and environmental health. Globally, asbestos-cement products are expected to continue to be the leading market for asbestos. India's imports of asbestos fibre (raw, beaten or washed or graded) have also been showing significant growth. Owing to continued demand for asbestos products in many regions of the world, global production is likely to remain steady at approximately 2.0 million tonnes per year for the near future, as per USGS report on asbestos.



32. Bauxite

23,968 thousand tonnes
production of bauxite have been
reported in 2023-24

164 thousand tonnes
of bauxite were exported in 2023-24

4.49 million tonnes
of bauxite were imported in 2023-24

Bauxite is basically an aluminous rock that contains hydrated aluminium oxide as the main constituent and iron oxide, silica & titania as minor constituents present in varying proportions. Hydrated aluminium oxides present in the bauxite ore are diasporite and boehmite, $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$ (Al_2O_3 :85%; Al: 45%); gibbsite or hydrargillite, $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ (Al_2O_3 : 65.4%; Al: 34.6%), and bauxite (containing colloidal alumina hydrogel), $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ (Al_2O_3 :73.9%; Al: 39.1%). The iron oxide in bauxite ore is present as haematite or goethite; silica as clay; and free quartz & titania as leucosene or rutile. Bauxite is the principal ore of aluminium which is one of the most important non-ferrous metals used in the modern industry. It is also an essential ore for Refractory and Chemical industries.

RESERVES/RESOURCES

As per the NMI data, based on UNFC system as on 1.4.2020, all India reserves/resources of Bauxite have been placed at 4,958 million tonnes. These resources include 646 million tonnes Reserves and 4,311 million tonnes Remaining Resources. The country is endowed with huge quantities of metallurgical grade bauxite. By grades, about 79% resources are of Metallurgical grade (I, II & Mixed). The

resources of Refractory and Chemical grades are limited and together account for about 4%. By States, Odisha alone accounts for 41% of country's resources of bauxite followed by Chhattisgarh (20%), Andhra Pradesh (12%), Gujarat (8%), Jharkhand (6%), Maharashtra (5%) and Madhya Pradesh (4%). Major bauxite resources are concentrated in the East Coast bauxite deposits in Odisha and Andhra Pradesh (Table-1).

Table – 1: Reserves/Resources of Bauxite as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	6,46,493	43,11,754	49,58,248
(By Grades)			
Chemical	6,639	15,405	22,044
Refractory	42,363	1,00,709	1,43,072
Chemical/Refractory Mixed with Others	1,575	24,786	26,362
Metallurgical-1	5,07,527	26,45,493	31,53,020
Metallurgical-2	37,100	5,85,786	6,22,886
Metallurgical Mixed	8,783	1,46,366	1,55,149
Low grade	26,649		5,48,122
Beneficiable	832	79,050	79,883
Mixed grade Excluding Chem./Refrac.	8,403	66,576	74,979
Abrasive	-	3,076	3,076
Others	4,047	43,257	47,304
Unclassified	2,576	57,500	60,076
Not known	-	22,276	22,276
By States			
Odisha	4,09,740	16,47,284	20,57,024
Chhattisgarh	23,695	9,68,860	9,92,555
Andhra Pradesh	0	6,15,267	6,15,267
Gujarat	1,01,230	2,95,797	3,97,027
Jharkhand	39,972	2,49,272	2,89,244
Maharashtra	38,472	1,93,958	2,32,430
Madhya Pradesh	18,564	1,67,695	1,86,259
Other states	14,819	1,73,622	1,88,441

Figures rounded off

MINING LEASES AND PRODUCTION

The production of bauxite at 23,968 thousand tonnes in 2023-24 increased by 1% as compared to the previous year.

There were 160 reporting mines in 2023-24 as against 150 in the previous year. The production of bauxite was reported as an associated mineral by 8 mines during the year. In all, 59 producers reported production of bauxite in 2023-24. Ten principal producers having a total of 37 mines contributed 92% of the total production.

NALCO is the leading producer of bauxite contributed 31% of the total production. The share of Public Sector mines was about 46% of the total production in 2023-24 as same as in the previous year.

About 81% of the total production of bauxite was of

40%-45% Al_2O_3 grade followed by 10% of Cement grade, 7% of 45%-50% Al_2O_3 grade and remaining production was reported in other grades except one grade (60% Al_2O_3 grade) during the year 2023-24. Odisha emerged as the leading producing state accounting for about 73% of the total production during 2023-24.

Mine head closing stocks of bauxite in 2023-24 were 17,524 thousand tonnes as compared to 17,320 thousand tonnes in the previous year. About 85% of total stock was held in Gujarat at the end of the year (Tables-2 to 6).

As on 31.03.2024 (P), the total number of mining leases granted for mining of bauxite is 310 covering mining lease area of 23,524.36 hectare. The average daily employment of labour in bauxite mines was 6,243 in 2023-24 as against 7,857 in the previous year.

Table – 2: Principal Producers of Bauxite,

2023-24

Name & address of producer	Location of the mine	
	State	District
National Aluminium Co. Ltd., NALCO Bhawan, P-1, Nayapalli, Bhubaneswar –751061, Odisha	Odisha	Koraput
Utkal Alumina International Ltd., J-6, Jayadev-Vihar, Bhubaneswar-751013, Odisha	Odisha	Rayagada
M/s The Orissa Mining Corporation Limited, OMC Office, Unit-V, Bhubaneswar- 751001	Odisha	Koraput
M/s Hindalco Industries Limited, Ahura Centre, First Floor, B Wing, Mahakali Caves Road, Andheri(East), Mumbai-400030, Maharashtra	Jharkhand, Chhattisgarh	Gumla, Lohardaga, Surguja
M/s Minerals & Minerals Limited, Court Road, Lohardaga, Lohardaga – 835302, Jharkhand	Jharkhand	Gumla, Lohardaga
Mr Alimiya Imamali Saiyed, FF/16, Samruddhi Complex, Near L.I.C Office, Himmatnagar, District- Sabarkantha- 383001, Gujarat	Gujarat	Sabarkantha
M/s Ashapura Minechem Limited, Ashapura Minechem Limited, Jeevan Udyog Building, 3rd Floor, 278, D N Road, Fort, Mumbai-400001	Maharashtra	Ratnagiri
M/s Gujarat Mineral Development Corporation Limited, GMDC Limited, "Khanij Bhavan", 132 Feet Ring Road, Vastrapur - Ahmedabad, PIN:380 052	Gujarat	Jamnagar
Mr Sanjaykumar Trikamlal Shah, 2, Arbudanagar-1, Ratnakar Mata Road, Kapadwanj, Kheda-387620, Gujarat	Gujarat	Kheda
Mr Bharatesh Construction Company, 34 Corporation Complex, Belgaum, Kolhapur-416012, Maharashtra	Maharashtra	Kolhapur

Table – 3: Production of Bauxite, 2021-22 to 2023-24 (P)

(By States)

(Qty in tonnes; Value in ₹ '000)

States	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	22494049	25284169	23844337	27371722	23967866	26903449
Chhattisgarh	968248	1079736	1056576	1215499	1033176	1230964
Gujarat	2018462	1692854	1667043	1470137	1643130	1677753
Jharkhand	1808750	2352953	2214879	3108731	2043734	2946196
Madhya Pradesh	608848	486524	601911	552530	597278	640503
Maharashtra	640345	387799	900608	590438	1058581	619010
Odisha	16449396	19284303	17403320	20434387	17591967	19789023

(P): Provisional

Table – 4A: Gradewise Production of Bauxite, 2022-23
(By Sectors, States and Districts)

(Qty in tonnes; Value in ₹ '000)

State / District	For use in Alumina & Aluminium extraction: Al ₂ O ₃ content							For use other than Alumina & Aluminium extraction				(Total)	
	No. of Mines	60% & above	55% - 60%	50% -55%	45% -50%	40%-45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Production	Value
India	150(8)	-	-	-	1479332	19544476	12042	2346156	112386	227883	122062	23844337	27371722
Public Sector	19	-	-	-	0	10658613	12042	106445	0	101380	115059	10993539	15381035
Private Sector	131(8)	-	-	-	1479332	8885863	-	2239711	112386	126503	7003	12850798	11990687
Chhattisgarh	11	-	-	-	885050	159484	12042	-	-	-	-	1056576	1215499
Kabirdham	2	-	-	-	0	3008	-	-	-	-	-	3008	2963
Kondagaon	-	-	-	-	0	0	-	-	-	-	-	0	0
Surguja	9	-	-	-	885050	156476	12042	-	-	-	-	1053568	1212536
Gujarat	75	-	-	-	-	21353	-	1296983	103388	130260	115059	1667043	1470137
Arvalli	2	-	-	-	-	0	-	30	0	0	0	30	5
Devbhoomi Dwarka	1	-	-	-	-	0	-	21	0	24	0	45	29
Jamnagar	46	-	-	-	-	5169	-	141128	103346	100668	0	350311	453011
Junagadh	1	-	-	-	-	0	-	0	0	500	0	500	330
Kachchh	11	-	-	-	-	16184	-	0	0	28492	115059	159735	199470
Kheda	9	-	-	-	-	0	-	521418	0	0	-	521418	396253
Porbandar	4	-	-	-	-	0	-	186096	0	410	-	186506	156790
Sabar Kantha	1	-	-	-	-	0	-	448290	42	166	-	448498	264249
Jharkhand	21	-	-	-	529882	1618894	-	-	-	66103	-	2214879	3108731
Gumla	14	-	-	-	529882	649131	-	-	-	66103	-	1245116	1828849
Latehar	-	-	-	-	-	-	-	-	-	0	-	0	0
Lohardaga	7	-	-	-	-	969763	-	-	-	0	-	969763	1279882
Madhya Pradesh	23(8)	-	-	-	-	-	-	563388	-	31520	7003	601911	552530
Anuppur	-	-	-	-	-	-	-	0	-	0	0	0	0
Jabalpur	2	-	-	-	-	-	-	75021	-	0	0	75021	63336
Katni	8(4)	-	-	-	-	-	-	284966	-	1106	648	286720	221366
Rewa	6	-	-	-	-	-	-	32912	-	1670	3273	37855	22897
Satna	2(4)	-	-	-	-	-	-	26935	-	23061	3082	53078	119582
Shahdol	2	-	-	-	-	-	-	125774	-	0	-	125774	91647
Sidhi	3	-	-	-	-	-	-	17780	-	5683	-	23463	33702

Table- 4A (Concl.)

(Qty in tonnes; Value in ₹ '000)

State / District	For use in Alumina & Aluminium extraction: Al ₂ O ₃ content							For use other than Alumina & Aluminium extraction				(Total)	
	No. of Mines	60% & above	55% - 60%	50% -55%	45% -50%	40%-45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Production	Value
Maharashtra	14	-	-	-	64400	341425	-	485785	8998	0	-	900608	590438
Kolhapur	6	-	-	-	64400	248050	-	0	0	0	-	312450	316813
Raigad	3	-	-	-	-	93375	-	74719	4549	0	-	172643	49637
Ratnagiri	5	-	-	-	-	0	-	411066	4449	0	-	415515	223988
Odisha	6	-	-	-	-	17403320	-	-	-	0	-	17403320	20434387
Koraput	4	-	-	-	-	10502138	-	-	-	-	-	10502138	14856230
Raygada	1	-	-	-	-	6901182	-	-	-	-	-	6901182	5578157
Sundargarh	1*	-	-	-	-	-	-	-	-	-	-	-	-

(*): Only labour reported
Figures in parentheses indicate number of associated mines.

Table – 4B: Gradewise Production of Bauxite, 2023-2024
(By Sectors/ States/Districts)

(Qty in tonnes; Value in ₹ '000)

State / District	For use in Alumina & Aluminium extraction: Al ₂ O ₃ content							For use other than Alumina & Aluminium extraction				(Total)	
	No. of Mines	60% & above	55% - 60%	50% -55%	45% -50%	40%-45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Production	Value
India	160(8)	-	7300	4000	1599022	19350266	15718	2494259	33500	349480	114321	23967866	26903449
Public Sector	22	-	-	-	-	10590812	4750	116534	12519	129606	112361	10966582	14942606
Private Sector	138(8)	-	7300	4000	1599022	8759454	10968	2377725	20981	219874	1960	13001284	11960843
Chhattisgarh	15	-	-	-	944100	77326	11750	-	-	-	-	1033176	1230964
Kabirdham	2	-	-	-	-	35902	7000	-	-	-	-	42902	43518
Kondagaon	2	-	-	-	-	-	-	-	-	-	-	-	-
Surguja	11	-	-	-	944100	41424	4750	-	-	-	-	990274	1187446
Gujarat	75	-	-	-	-	56119	-	1197030	33500	244120	112361	1643130	1677753
Amreli	1	-	-	-	-	-	-	0	-	0	-	0	0
Arvalli	1	-	-	-	-	-	-	5	-	-	-	5	4
Devbhoomi Dwarka	1	-	-	-	-	-	-	48658	-	84552	-	133210	218771
Jamnagar	47	-	-	-	-	9157	-	211092	33217	114611	-	368077	473019
Junagadh	1	-	-	-	-	-	-	-	-	0	-	0	0
Kuchchh	11	-	-	-	-	46962	-	30854	283	44857	112361	235317	296443

Table- 4B (Concd.)

(Qty in tonnes; Value in ₹ '000)

State / District	For use in Alumina & Aluminium extraction: Al ₂ O ₃ content						For use other than Alumina & Aluminium extraction					(Total)	
	No. of Mines	60% & above	55% - 60%	50% -55%	45% -50%	40%-45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Production	Value
Kheda	7	-	-	-	-	-	-	243415	-	-	-	243415	182318
Porbandar	5	-	-	-	-	-	-	215306	-	100	-	215406	188292
Sabarkantha	1	-	-	-	-	-	-	447700	-	-	-	447700	318906
Jharkhand	21	-	7300	4000	650422	1318645	3968	-	-	59399	-	2043734	2946196
Gumla	14	-	7300	4000	650422	374705	-	-	-	59399	-	1095826	1679286
Lohardaga	7	-	-	-	-	943940	3968	-	-	-	-	947908	1266910
Madhya Pradesh	24(8)	-	-	-	-	-	-	549357	-	45961	1960	597278	640503
Jabalpur	2	-	-	-	-	-	-	40915	-	2978	-	43893	44602
Katni	8(4)	-	-	-	-	-	-	335903	-	-	-	335903	268001
Rewa	6	-	-	-	-	-	-	50088	-	610	830	51528	36205
Satna	3(4)	-	-	-	-	-	-	20785	-	21898	1130	43813	127212
Shahdol	2	-	-	-	-	-	-	55531	-	-	-	55531	45661
Sidhi	3	-	-	-	-	-	-	46135	-	20475	-	66610	118822
Maharashtra	19	-	-	-	4500	306208	-	747873	-	-	-	1058581	619010
Kolhapur	7	-	-	-	4500	283600	-	-	-	-	-	288100	219889
Raigad	7	-	-	-	-	22608	-	212784	-	-	-	235392	129792
Ratnagiri	5	-	-	-	-	-	-	535089	-	-	-	535089	269329
Odisha	6	-	-	-	-	17591967	-	-	-	-	-	17591967	19789023
Koraput	4	-	-	-	-	10547721	-	-	-	-	-	10547721	14375448
Raygada	1	-	-	-	-	7044246	-	-	-	-	-	7044246	5413575
Sundargarh	1	-	-	-	-	-	-	-	-	-	-	-	-

Figures in parantheses indicate number of associated mines

Table – 5: Production of Bauxite, 2022-23 and 2023-24 (P)

(By Frequency Groups)

(Qty in tonnes; Value in ₹ '000)

Production Group	No. of Mines		Production		Percentage to total production		Cumulative Percentage	
	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)
Total	150(8)	160(8)	23844337	23967866	100	100	-	
Upto 1000	62(1)	73(1)	5695	1260	0.02	0.01	0.02	0.01
1001 – 3000	3	10(1)	4409	19956	0.02	0.08	0.04	0.09
3001 – 5000	5(1)	4(1)	19686	19724	0.08	0.08	0.12	0.17
5001 – 10000	7(1)	12	55933	85912	0.23	0.36	0.35	0.53
10001– 25000	23(4)	11(3)	425201	234954	1.78	0.98	2.13	1.51
25001 – 50000	15(1)	14(2)	555542	587334	2.33	2.45	4.46	3.96
50001 and above	35	36	22777871	23018726	95.54	96.04	100	100

Figures in parantheses indicate number of associated mines

P: Provisional

Table – 6A: Mine-head Closing Stocks of Bauxite, 2022-23

(By States / Grades)

(Quantity in tonnes)

State	For use in Alumina & Aluminium metal Extraction: Al ₂ O ₃ Content							For use in other than Alumina & Aluminium metal extraction			
	60% & above	55% - 60%	50% -55%	45% -50%	40% -45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Total
India	-	-	-	136658	1242692	229872	14738080	544291	326393	102426	17320412
Chhattisgarh	-	-	-	36750	159118	47700		95			243663
Gujarat	-	-	-	9171	445150	46500	13670256	544196	300261	69578	15085112
Jharkhand	-	-	-	55233	163336	12858	0				231427
Madhya Pradesh	-	-	-	0	421	1028	786958		26133	32849	847388
Maharashtra	-	-	-	22315	175779	111649	280866				590609
Odisha	-	-	-	13189	298887	10137					322213

Table – 6B: Mine-head Closing Stocks of Bauxite, 2023-24 (P)

(By States / Grades)

(Quantity in tonnes)

State	For use in Alumina & Aluminium metal Extraction: Al ₂ O ₃ Content							For use in other than Alumina & Aluminium metal extraction			
	60% & above	55% - 60%	50% -55%	45% -50%	40% -45%	Below 40%	Cement	Abrasive	Refractory	Chemical	Total
India	-	-	3300	129849	1278397	211049	14958176	540113	338779	64298	17523962
Chhattisgarh	-	-	-	36767	135912	35097	-	95	1255	1261	210387
Goa	-	-	-	-	-	-	-	-	-	-	-
Gujarat	-	-	-	9171	434006	46501	13559330.47	540019	295010	31485	14915522
Jharkhand	-	-	3300	51145	241779	12772	-	-	-	-	308996
Karnataka	-	-	-	-	-	-	-	-	-	-	-
Madhya Pradesh	-	-	-	-	421	1036	954566	-	42513	31552	1030088
Maharashtra	-	-	-	19577	127190	111649	444279.997	-	0	-	702696
Odisha	-	-	-	13189	339089	3995	-	-	0	-	356273

P: Provisional

USES & CONSUMPTION

Bauxite is primarily used to produce alumina through the Bayer process. Aluminium Industry normally uses bauxite containing minimum 40% Al_2O_3 . However, slightly inferior grades with a suitable blend are also used depending upon other characteristics, such as, solubility in caustic soda and absence of silica. The BIS has specified IS: 5953-1985 (Reaffirmed 2008 & 2014) specifications for metallurgical grade bauxite. Details of the industries are provided in a separate Review 'Aluminium and Alumina'.

In Steel Industry, bauxite is used as a slag corrector in place of fluorite and generally bauxite; containing 45 to 54% Al_2O_3 and 5% SiO_2 (max.) is consumed. Size preference is 25 to 125 mm with a tolerance of 5% (max.) for -25 mm & +100 mm fractions.

BIS has prescribed the specifications of bauxite as 'IS: 10817-1984 (Reaffirmed in 2020) for Refractory Industry. The IS specifications of bauxite for consumption in Chemical and Petroleum industries are given in 'IS: 3605-1984 (Reaffirmed 2020).

Apart from the chemical specifications, the physical requirements are that the material passing through 90-micron IS sieve but retained on 212- micron IS sieve should be 90% maximum; that passing through 300-micron IS sieve shall be 1% by mass maximum; and that passing through 212- micron IS sieve but retained on 300-micron IS sieve should be 10% maximum.

The other specifications laid down by BIS are 'IS: 8228-1976 (Reaffirmed 2020)' for bauxite sand and 'IS: 8988-1978 (Reaffirmed 2019)' for bauxite powder for foundry washes. As per Ministry of Mines Notification dated 25th April 2018, the threshold value of bauxite mineral has been classified into the following two categories:

- (i) For Aluminous laterite: Al_2O_3 –20 % (min.)
- (ii) For Bauxite: Al_2O_3 –30 % (min.) and SiO_2 (Total)–7 % (max.)

In 2023-24, the apparent consumption of bauxite was about 28.30 million tonnes as against 27.26 million tonnes during the preceding year, showing an increase of about 4%.

SUBSTITUTES

There is no substitute for bauxite as source for aluminium metal extraction carried out on a large scale. However, calcined clay can be substituted for refractory bauxite but only with reduction in time and stock resistance. Sillimanite, alumina, silicon carbide, magnesite–chromite and carbon-magnesite refractories are the other alternatives for high-alumina material but these would entail higher cost. Silicon carbide and diamonds can substitute for fused aluminium oxide in abrasive use but these would entail again at higher cost. Synthetic mullite is a probable substitute for

bauxite-based refractories which is produced from Kyanite, Sillimanite and Kaolin.

Silicon carbide and alumina-zirconia are costlier substitutes for bauxite-based abrasives. The raw material like alunite, anorthosite, coal wastes and oil shales are other potential sources of alumina. The extraction, however, would require new plants with different technology. These non-bauxitic materials could satisfy the demand for primary metal, refractories, aluminium chemicals and abrasives.

TRADE POLICY

As per the Foreign Trade Policy 2015-2020 and policy on export and import, imports of aluminium ores and concentrates including natural bauxite, calcined and activated bauxite and others are permitted free. There are no policy restrictions on the export of bauxite.

WORLD REVIEW

The world bauxite reserves are estimated at 30 billion tonnes and are located mainly in Guinea (25%), Vietnam (19%), Australia (12%), Brazil (9%), Jamaica (7%) Indonesia (3%), China, India & Russia (2% each). Countrywise reserves of bauxite are furnished in Table-7.

Table –7: World Reserves of Bauxite
(By Principal Countries)

(In '000 tonnes)	
Country	Reserves
World total (rounded)	30000000
Guinea	7400000
Vietnam	5800000
Australia*	3500000
Brazil	2700000
Jamaica	2000000
Indonesia	1000000
China	710000
India	650000
Russia	480000
Saudi Arabia	180000
Kazakhstan	160000
Turkey	63000
Other countries	5100000

Figures rounded off

Source: USGS, Mineral Commodity Summaries, 2024

**: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 1.6 billion tonnes.*

The world production of bauxite increased by 4% to 416 million tonnes in 2023 as compared to 400 million tonnes in the preceding year.

Guinea and Australia together are the top producers which accounted for about 55% (Guinea 30% and Australia 25%) share in the total production, followed by China (16%), Brazil (8%), Indonesia (7%) and India (6%) as mentioned in Table-8.

Table – 8: World Production of Bauxite, 2021 - 2023

(By Principal Countries)

Country	(In tonnes)		
	2021	2022	2023
World total	350100000	400100000	416300000
Guinea ^(b)	87438557	103525513	123067415
Australia	103266404	100478354	103741376
China	40000000*	74300000	65516300
Brazil	35949600	34215200	34472600
Indonesia	25781187	28808674	29000000
India ^(e)	22493947	23843329	23360552
Russia	6529000	6630000	7031000
Jamaica	5937320	4364729	5988079
Saudi Arabia	5488000	5930000	5900000
Kazakhstan	4370100	4177800	4559900
Other Countries	12798939	13859896	13664257

Note(s)

(1) This table includes production of refractory bauxite.

(2) In addition to the countries listed, some bauxite continues to be produced in Mexico but it is all used as construction aggregates and figures are not reported.

(b): No adjustment has been made for moisture content

(e): Years ended 31 March following that stated

* :Estimated

(In tonnes)

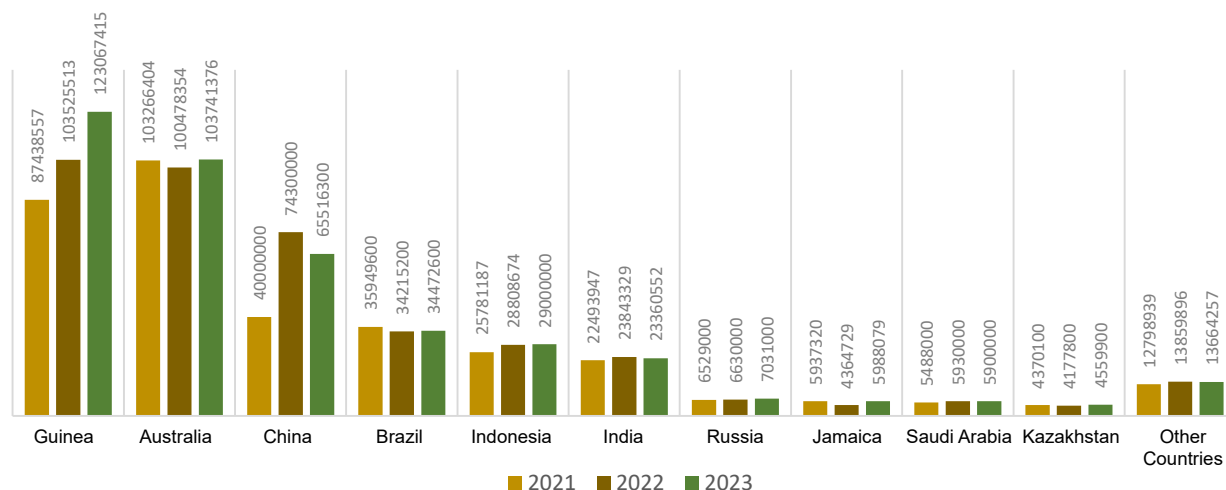


Fig. 1: Countrywise Production of Bauxite

FOREIGN TRADE

Exports

During 2023-24, exports of bauxite from India decreased by about 8% in quantity to 164 thousand tonnes, valued

at ₹ 24.35 crore as compared to 177 thousand tonnes valued at ₹ 47.90 crore in 2022-23. Exports were mainly to Nepal (67%), Kuwait (31%) and Slovenia (2%). Exports to Cameroon, Thailand and Sudan, though small in absolute terms, showed a rising trend (Tables-9 & 10).

Table – 9: Export of Bauxite

(By Country)

Country/Region	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	177535	479019	164177	243541
Kuwait	67300	121398	50755	89985
Nepal	85474	75855	109321	81450
Slovenia	11220	199431	3085	55906
Cameroon	54	1451	218	5977
Thailand	40	1222	252	3207

Table- 9 (Concl'd.)

Country/Region	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Sudan	—	—	83	2179
Bangladesh	131	3049	100	2052
UAE	91	1980	105	1122
Malaysia	5	284	25	703
Korea, Rep. of	330	1168	220	657
Other Countries	12890	73181	13	303

Figures rounded off

(In tonnes)

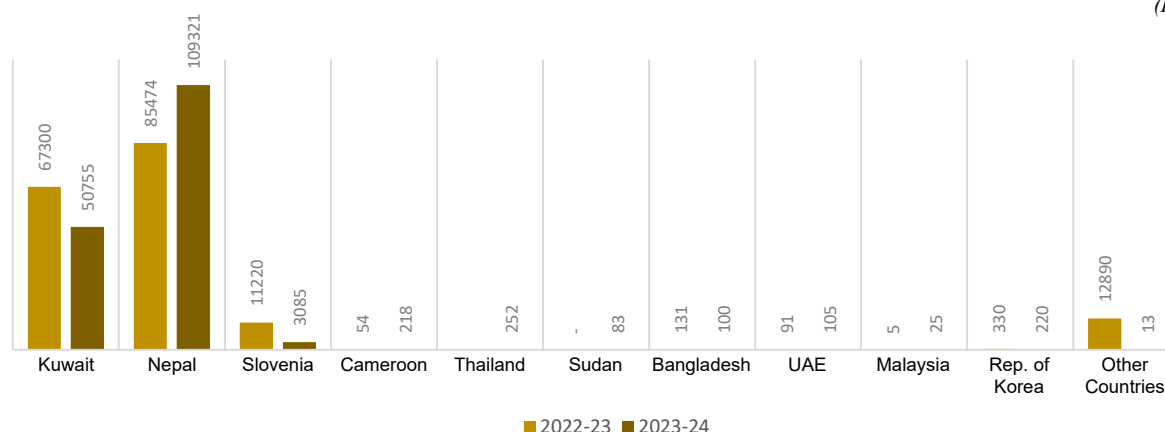


Fig. 2: Countrywise Export of Bauxite

Table – 10: Export of Bauxite: (Aluminium and Concentrate)
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	177535	479019	164177	243541
Kuwait	67300	121398	50755	89985
Nepal	85474	75855	109321	81450
Slovenia	11220	199431	3085	55906
Cameroon	54	1451	218	5977
Thailand	40	1222	252	3207
Sudan	--	--	83	2179
Bangladesh	131	3049	100	2052
UAE	91	1980	105	1122
Malaysia	5	284	25	703
Korea, Rep. of	330	1168	220	657
Other Countries	12890	73181	13	303

Figures rounded off

Imports

During 2023-24, imports of bauxite increased by about 25% in quantity to 4.49 million tonnes, valued at ₹ 2,820 crore, as compared to 3.60 million tonnes valued at ₹ 2,342 crore in 2022-23. Most imports were of aluminium ores

and concentrates. Guinea remained the principal supplier, contributing about 96% of total imports (4.32 million tonnes), followed by China with around 3% share. Minor quantities were also imported from Trinidad, USA, and several European countries (Tables-11 & 12).

Table –11: Import of Bauxite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3596098	23423723	4492899	28197412
Guinea	3150600	17945643	4322948	23016058
China	76199	2809072	117667	4123591
Trinidad	27555	524940	17741	326689
USA	15	6667	9828	169299
Hong Kong	7953	279906	5208	157474
Switzerland	--	--	7799	142669
Guyana	3598	64523	7741	139389
Spain	--	--	653	30523
France	--	--	1710	27740
Netherlands	409	24609	405	27171
Other Countries	329769	1768363	1199	36809

Figures rounded off

In tonnes

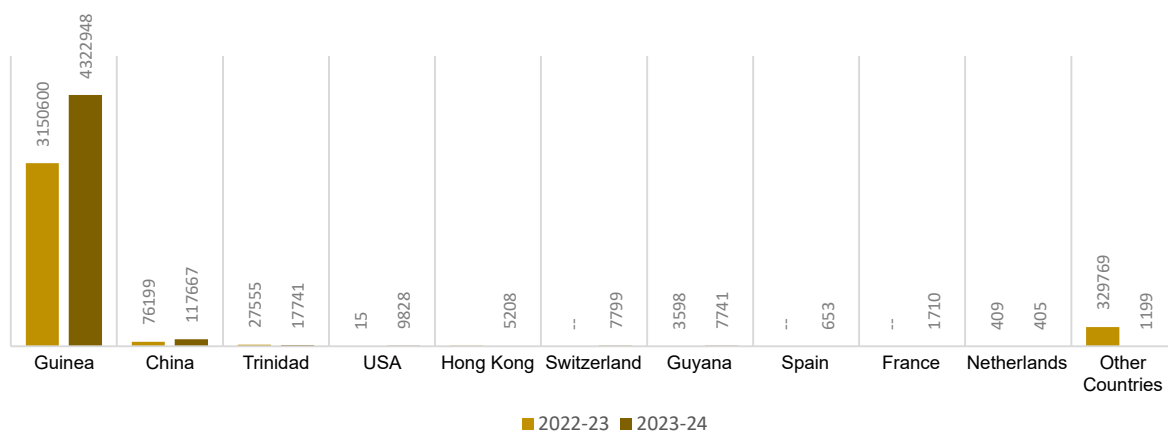


Fig. 3: Countrywise Import of Bauxite

Table – 12: Import of Bauxite: (Aluminium and Concentrate)

(By Countries)

Country/Region	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3596098	23423723	4492899	28197412
Guinea	3150600	17945643	4322948	23016058
China	76199	2809072	117667	4123591
Trinidad	27555	524940	17741	326689
USA	15	6667	9828	169299
Hong Kong	7953	279906	5208	157474
Switzerland	--	--	7799	142669
Guyana	3598	64523	7741	139389
Spain	--	--	653	30523
France	--	--	1710	27740
Netherlands	409	24609	405	27171
Other Countries	329769	1768363	1199	36809

Figures rounded off

FUTURE OUTLOOK

The total resources of bauxite that comprise various grades, as found to occur in the country as on 1.4.2020, are estimated at 4,958 million tonnes. The resources of Metallurgical grade bauxite are adequate while those of the Chemical and Refractory grade bauxite are relatively limited considering the future requirements. India's strength in aluminium production is due to its rich reserve of bauxite, a core resources used in production of aluminium. As per provision made in Mineral (Auction) Rules 2015, a total of 8 bauxite blocks (including blocks of Bauxite and other associated minerals) were auctioned in FY 2023-24 in the State of Gujarat (4 blocks), Karnataka (02 blocks), Maharashtra (01 block) and Jharkhand (01 block).

As per the FITCH solution Report 2024, the production of bauxite is estimated to grow to 50.7 million tonnes by 2027. Aluminium industry is one of the leading segments of the Indian economy and is expected to play a significant role in the country's future growth. Apart from its potentially large growing market, India is endowed with large deposits

of high-quality bauxite ore, resources for power generation (coal) and formidable pool of manpower, both skilled and unskilled. Indian aluminium industry is forging ahead with rapid expansion in both primary metal and downstream sectors.

The demand for aluminium is expected to move upward with shifting demand of consumers towards electric vehicles. Also, the recyclable nature of the metal puts up with goal of reducing carbon emission of many countries. The demand for aluminium will be supported from sectors like aviation, construction, renewable energy production, consumer goods, defence etc.

In the recent trend towards clean technology for need of climate action and sustainable lifestyles, aluminium is one of the critical metals which will help to cater the emerging demand of clean energy solutions, green technologies and sustainable systems. Aluminium is a lightweight material, which is ideal for use in EVs, 'Green Buildings' and power cabling.



33. Boron Minerals

74,204 tonnes

Total reserves/resources of borax have been established as on 1st April, 2020.

3,190 tonnes

of borax (total) were exported in 2023-24.

2,02,250 tonnes

of borax (total) were imported in 2023-24.

Boron minerals occur mostly as borates which are deposited from volcanic gases or hot springs near volcanic activities. The deposits, predominantly of borax and sassolite are formed as a result of drying up of shallow saline and alkaline tertiary lakes called 'Playa'. The principal boron minerals are borax—hydrated sodium borate ($\text{Na}_2\text{O} \cdot 2\text{B}_2\text{O}_3 \cdot 10\text{H}_2\text{O}$), kernite (rasorite)—hydrated sodium borate ($\text{Na}_2\text{O} \cdot 2\text{B}_2\text{O}_3 \cdot 4\text{H}_2\text{O}$), colemanite—hydrated calcium borate ($\text{Ca}_2\text{B}_6\text{O}_{11} \cdot 5\text{H}_2\text{O}$) and ulexite—hydrated sodium calcium borate ($\text{NaCaB}_5\text{O}_9 \cdot 8\text{H}_2\text{O}$). Besides the above four boron minerals of commercial importance, two other boron-bearing minerals, viz., sassolite (H_3BO_3) – the natural boric acid and boracite ($\text{Mg}_3\text{B}_7\text{O}_{13}\text{Cl}$) are also well known but have relatively lesser commercial importance.

Borax is, presently, not produced in India. However, it was obtained since ancient times from the lakes in Jammu & Kashmir in India. The domestic requirements of boron minerals are met solely through imports of crude borate which is refined in the country for producing borax and boric acid.

RESERVES/RESOURCES

Economically viable deposits of borax have not been established in the country so far. The only deposit of little economic significance is reported from Puga Valley in Leh district, Jammu & Kashmir. As per NMI data, based on

UNFC system, the total reserves/resources of borax as on 01.04.2020, has been estimated at 74,204 tonnes in Jammu & Kashmir. All resources are of Reconnaissance category viz., UNFC Code 334 (Table-1).

Table – 1: Reserves/Resources of Borax as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	-	74204	74204
By Grades			
Unclassified	-	74204	74204
By UT			
Jammu & Kashmir	-	74204	74204

Figures rounded off.

USES & CONSUMPTION

Glass and Porcelain industries are the major consumers of borax and boric acid. It is an essential component of heat-resisting boro-silicate glass, glass fibres and industrial & optical glass. In glass, enamels and ceramics, it controls thermal expansion, improves durability, assists melting processes and adds to inorganic colours & decorations.

Borax is used in medicine (boric powder), leather processing, adhesive, corrosion inhibition, ferrous wire manufacture, flame-proofing and timber preservation.

Borax is used as a flux in brazing, welding, soldering and in the manufacture of artificial gems like, cubic boron nitride, (commercially called 'Borazon') which is equal to diamond in hardness. Boron carbide, titanium boride and tungsten boride are other boron compounds that are superhard materials which are next to diamond in hardness and find applications on account of their good electrical conductivity and thermal stability.

Its easy solubility and property to soften hard water find applications in soaps, cleaners & detergents and for water treatment. Its mild alkalinity and germicidal nature, enable its use in manufacturing toothpastes and mouth washes. Borax is used as an antiseptic and emulsifying agent in Cosmetics Industry. As a decolourising agent, it is used in Vanaspati Industry. In Textile Industry, borax is used as a decolourising agent as well as for maintaining the alkalinity of solutions used for producing rayons. It prevents mould formation in citrus fruits. In agriculture, borax is used as an essential plant nutrient.

Boron compounds are used in the production of fertilizers, algicides, herbicides and insecticides. Borax and boric acid are used in fire-retardant treatment and as food grain preservative, respectively.

Borate ester is used as dehydrating agent, special solvent and catalyst in Chemical Industry. In nuclear reactor, boron acts as neutron absorber. "Boron neutron capture therapy", a form of radiochemotherapy, is becoming increasingly important for treatment of certain forms of cancers and boron neutron capture synovectomy for treatment of arthritis.

Borates are consumed mainly in glass fibre for insulations and as textile-grade fibre. They are also used as anti-knock agents in gasoline. Diborane (gas), pentaborane (liquid) and decaborane (solid) are potential jet and rocket engine fuels. Boron hydride also has potential value as rocket fuel. The high energy fuel value imparted by the addition of boron compounds has given considerable strategic significance to borates. Another use of borates is the invention of oxgano-sodium borate (liquibor) for use in hydraulic brake fluids.

Ferroboron is a boron ferroalloy containing 0.2% to 24% boron used primarily to introduce small quantities of boron into speciality steels.

SUBSTITUTES

Substitutes in applications, such as, soaps, detergents, enamels and insulations are available. In detergents, boron compounds can be replaced with chlorine and enzymes. Lithium compounds can be used to make enamels and glass products. Insulation substitutes include cellulose, foams and mineral wools. Substitution of borosilicate glass by plastic materials may reduce the use of boron.

Sodium percarbonate can replace borates in detergents and requires lower temperatures to undergo hydrolysis, which is an environmental consideration. Some enamels can use other glass-producing substances, such as, phosphates. In soaps, sodium and potassium salts of fatty acids can act as cleaning and emulsifying agents.

TECHNICAL POSSIBILITIES

A proprietary process called 'Hydrogen on Demand' has been developed using water and sodium borohydride. Hydrogen from the system can be used in fuel cells or internal combustion engines. A longer-life battery based on boron has also been designed. Synthetic diamond containing about 3% boron which is normally a semiconductor becomes superconductor at 4 K. Boron-doped diamond, thus, has numerous possible applications as it can carry electricity without resistance.

Improvements made in evaporating brine solutions are widening the choice of source. Production of boric acid through solution mining of colemanite is a possibility.

INDUSTRY

In borax manufacturing process, crude sodium borate is dissolved in water, charged, oxidised, crystallised and centrifuged. Centrifuged material is then dried to get borax decahydrate.

Crude calcium borate lumps are crushed and wet-ground with mother liquor to make slurry. This slurry is decomposed with sulphuric acid to give calcium sulphate and boric acid. Boric acid is separated by filtration, purified, cooled and centrifuged to produce boric acid granules which are powdered as per demand.

As per the Annual Report, 2024-25 of M/s DMCC

Speciality Chemicals Limited, formerly known as the Dharamsi Moraji Chemical Company Limited, at the Dahej Plant, Gujarat, the capacity of boric acid production has increased from 10 MTD (Metric Tonnes per day) to 21 MTD. As per the Annual Report, 2024-25 of M/s National Peroxide Limited, formerly known as NPL Chemicals Limited, NPL is the country's largest manufacturer of Hydrogen peroxide, boasting an impressive installed capacity of 15,000 MTPA (on 50% w/w basis).

WORLD REVIEW

The world reserves of boron are furnished in Table-2. Turkey was the leading producer of borates followed by USA, Kazakhstan, Chile, Bolivia (Table-3).

Table – 2: World Reserves of Boron

(By Principal Countries)

(In '000 tonnes)

Country	Reserves
World:Total⁽¹⁾	xx
Turkey, refined borates	950000
USA	48000
Russia, datolite ore	40000
Chile, ulexite	35000
China, boric oxide equivalent	20000
Peru, crude borates	4000
Argentina, crude ore	NA
Bolivia, ulexite	NA
Germany, compounds	NA

Source: USGS, Mineral Commodity Summaries, 2024.

1: World totals can not be calculated because production and reserves are not reported in a consistent manner by all countries.

Table – 3: World Production of Borates

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
Turkey	4057299	2452485	1793746
USA ^(a)	1300000*	1300000*	1300000*
Kazakhstan	500000*	500000*	500000*
Chile	363032	421935	450239
Bolivia	312906	295515	327253
China ^(b)	205000*	200000*	200000*
Peru	246362	292986	179938
Argentina	130000*	130000*	130000*
Russia	80000*	80000*	80000*
Iran ^(c)	2000	2735	2700

Source: BGS, World Mineral Production, 2019-23

*: Estimated

a: Sold or used by producers.

b: B₂O₃ equivalent.

c: Years ended 20th March following that stated."

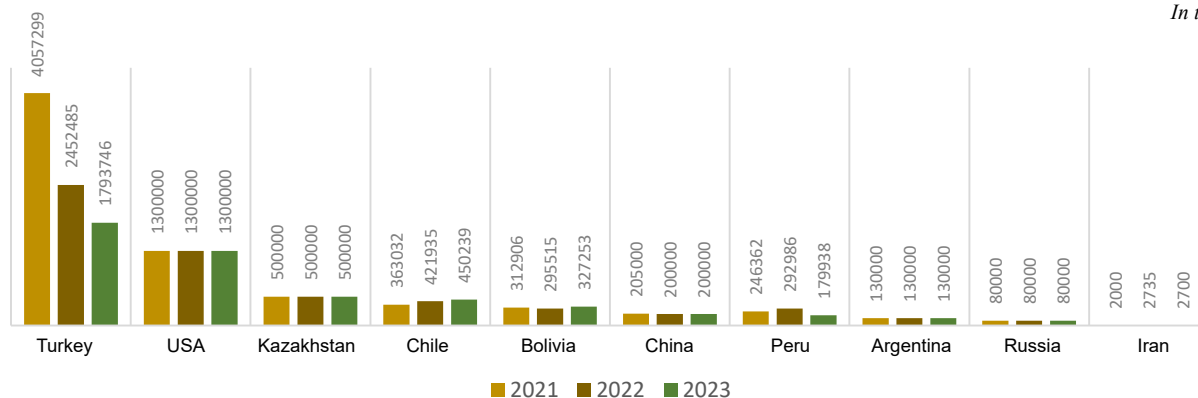


Fig .1: Countrywise Production of Borates

FOREIGN TRADE

Exports

Exports of borax (total) decreased by 23% to 3,190 tonnes in 2023-24 from 4,151 tonnes in the previous year. Exports of natural borate in 2023-24 decreased by 19% to 76 tonnes from 94 tonnes in the previous year. In 2023-24, exports of

sodium borate were at 769 tonnes and other borates at 2,345 tonnes. Exports of Borax (total) were mainly to USA (43%), Italy (10%), Nepal (7%) and Bangladesh (5%). Exports of boric acid decreased by 60% to 1,383 tonnes in 2023-24 from 3,456 tonnes in the previous year. Exports of boric acid were mainly to Nigeria (18%), Kenya (9%) and Sri Lanka & Nepal (8% each) (Tables- 4 to 10).

Table – 4: Exports of Boron

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	6402	1	1497
France	--	--	++	365
Jordan	--	--	1	361
USA	++	6368	++	328
Bangladesh	--	--	++	243
Korea, Rep. of	++	34	++	123
Nepal	--	--	++	32
Egypt	--	--	++	23
UAE	--	--	++	15
Colombia	--	--	++	4
Australia	--	--	++	3

Figures rounded off

Table – 5: Exports of Borax

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	4151	818305	3190	627776
USA	1614	454037	1369	364799
Italy	599	115008	320	59388
Bangladesh	181	28248	145	26853
Poland	100	19821	120	22038
Oman	38	10516	78	20494
Nepal	154	12653	228	16894
Spain	100	20935	80	14688
South Africa	101	21708	90	13886
Australia	30	7864	105	13646
Sri Lanka	70	6494	98	9235
Other countries	1164	121021	557	65855

Figures rounded off

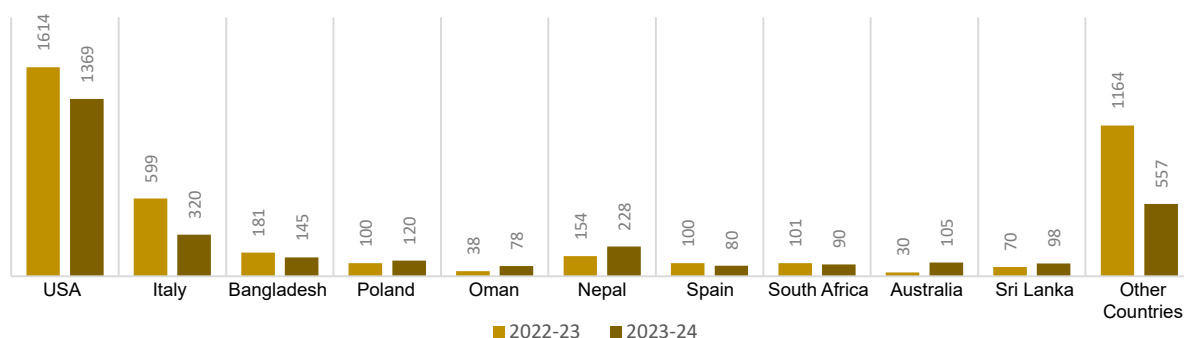


Fig 2: Countrywise Exports of Borax

Table – 6: Exports of Natural Borate**By Countries**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	94	6417	76	5099
Algeria	--	--	60	3673
Kuwait	20	857	16	1424
UAE	2	199	++	2
Maldives	--	--	++	++
Indonesia	72	3495	--	--
Bangladesh	++	1478	--	--
USA	++	364	--	--
Tanzania	++	24	--	--

*Figures rounded off***Table – 7: Export of Boric Acid****By Countries**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3456	414727	1383	194800
Nigeria	225	31084	249	31263
USA	131	35692	65	22669
Kenya	112	15879	129	15054
Nepal	177	22812	112	12527
Sri Lanka	14	2045	117	10554
Iraq	++	74	70	10286
Angola	71	9427	49	6705
UAE	641	67780	34	5844
Ethiopia	35	4955	44	5828
Djibouti	70	9744	37	5510
Other Countries	1980	215235	477	68560

*Figures rounded off***Table – 8: Export of Boric Acid****By Countries**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3456	414727	1383	194800
Nigeria	225	31084	249	31263
USA	131	35692	65	22669

Table- 8 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Kenya	112	15879	129	15054
Nepal	177	22812	112	12527
Sri Lanka	14	2045	117	10554
Iraq	++	74	70	10286
Angola	71	9427	49	6705
UAE	641	67780	34	5844
Ethiopia	35	4955	44	5828
Djibouti	70	9744	37	5510
Other countries	1980	215235	477	68560

Figures rounded off

Table – 9: Export of Sodium Borate

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	894	87107	769	83078
USA	20	26410	31	23218
Nepal	91	6895	187	13125
Jordan	5	343	121	8875
Australia	++	125	72	4941
Sri Lanka	37	2044	77	4911
Nigeria	3	254	50	4838
Myanmar	256	16296	48	2911
Algeria	--	--	37	2898
Angola	4	372	33	2704
South Africa	++	4	27	1746
Other countries	478	34364	86	12911

Figures rounded off

Table – 10: Export of Borax: Other Borates

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3163	724781	2345	539599
USA	1594	427263	1338	341581
Italy	599	115008	320	59388
Bangladesh	140	25788	144	26552
Poland	100	19821	120	22038
Oman	37	10458	78	20489
Spain	100	20935	80	14688
South Africa	101	21704	63	12140
Australia	30	7739	33	8705
Sri Lanka	33	4450	21	4324
Nepal	63	5758	41	3769
Other countries	366	65857	107	25925

Figures rounded off

Imports

Imports of borax (total) increased slightly by 2% to 2,02,250 tonnes in 2023-24 from 1,98,719 tonnes in the previous year. Imports of natural borate also increased by 30% to 91,918 tonnes as compared to 70,532 tonnes in the previous year. In 2023-24, imports of sodium borate were at 1,00,925 tonnes and other borates 9,407 tonnes. Borax (total) was

mainly imported from Turkey (64%), USA (20%), Bolivia (10%) and Spain (3%). Imports of boric acid decreased to 2,562 tonnes in 2023-24 from 5,478 tonnes in the previous year. Boric acid was imported mainly from Singapore (68%) and Turkey (32%). Import of boron was negligible in both current and the previous year (Tables-11 to 16).

Table – 11: Imports of Borax

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	198719	10161598	202250	10661154
Turkey	152023	7304367	128725	6515133
USA	36581	2061857	41101	2753611
Bolivia	--	--	19893	418740
Spain	5122	251770	6782	349969
China	1100	156426	1317	205017
Djibouti	--	--	1575	73286
Peru	534	68720	437	52792
Singapore	1522	103386	584	40199
Netherlands	134	23210	175	39233
Italy	61	11043	362	30112
Other countries	1642	180819	1299	183062

Figures rounded off

In tonnes

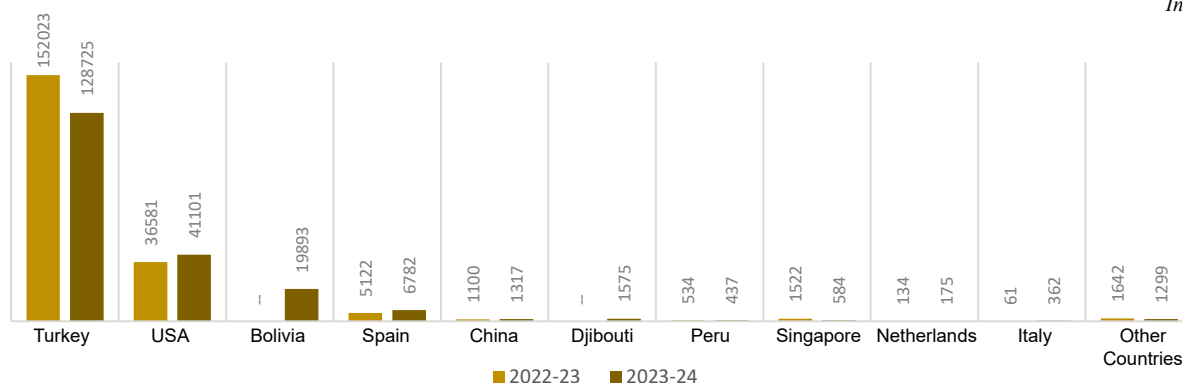


Fig 3: Countrywise Imports of Borax

Table – 12: Imports of Natural Borate

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	70532	2894632	91918	3514354
Turkey	65351	2647895	65208	2744018
Bolivia	--	--	19893	418740
Spain	4760	222398	6671	334375
USA	53	6113	138	16019
Oman	--	--	8	1183
Japan	--	--	++	19
Argentina	368	18226	--	--

Figures rounded off

Table – 13: Imports of Boron

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	6094	++	8793
China	++	3026	++	4874
USA	++	2675	++	3780
UK	--	--	++	125
Turkey	--	--	++	14
Germany	++	393	--	--

Figures rounded off

Table – 14: Import of Borax: Other Borates

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	11489	1248036	9407	1174537
Turkey	6496	602512	5230	552793
USA	2461	272430	1668	226505
China	835	127918	1257	198084
Peru	448	57804	280	33448
Austria	119	23802	92	19374
Slovenia	128	11768	190	18137
Australia	1	5264	2	16145
Spain	362	29372	111	15594
Israel	--	--	100	14779
Italy	58	10539	111	13856
Other countries	581	106627	366	65822

Figures rounded off

Table – 15: Import of Sodium Borate

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	116698	6018930	100925	5972263
Turkey	80176	4053960	58287	3218322
USA	34067	1783314	39295	2511087
Djibouti	--	--	1575	73286
Netherlands	134	22780	175	38964
Singapore	1139	59628	501	33414
UK	540	31477	340	22743
Peru	86	10916	157	19344
Italy	3	504	251	16256
Chile	--	--	119	7278
China	265	28508	60	6933

Figures rounded off

Table – 16: Import of Boric Acid

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5478	409638	2562	230157
Singapore	1743	139900	1743	158369
Turkey	3675	261961	818	67165
Germany	++	237	1	2229
USA	59	5480	++	1935
Italy	--	--	++	171
UAE	++	93	++	114
Spain	--	--	++	91
Unspecified	--	--	++	63
Japan	++	50	++	20
China	1	1693	--	--
Other countries	++	224	++	++

Figures rounded off

FUTURE OUTLOOK

Consumption of borates is expected to increase, spurred by strong demand in agriculture, ceramic and glass markets in Asia and South America. Continued investment in new refines and technologies and the continued increase in demand were expected to fuel growth in world production for the foreseeable future. Consumption of boron-based fertilizers is expected to increase as the demand for food and bio-fuel crops is on the rise. Higher crop prices have enabled farmers to invest in advanced farming techniques and higher grade fertilizers.

Consumption of boron nitride is expected to increase owing to the development of high-volume production techniques coupled with the creation of new technologies requiring boron nitride. The properties intrinsic to cubic boron nitride, such as, hardness (second only to diamond), high thermal conductivity and oxidation resistance, make it an ideal material for a variety of emerging applications. Hexagonal boron nitride is used in producing ceramics, creating intermetallic composites, imparting thermal shock resistance, improving machine ability and reducing friction.



34. Chromite

3,148 thousand tonnes
production of chromite have been
reported in 2023-24

2.12 thousand tonnes
of chromite (total) were exported
in 2023-24

172.41 thousand tonnes
of chromite (total) were imported
in 2023-24.

Chromite is the single commercially viable ore of chromium (Cr) which is chemically known as iron chromium oxide (FeCr_2O_4). The name of the element is derived from the Greek word 'chroma', meaning colour, because many of its compounds are intensely coloured. It is a steely-grey, lustrous, hard and brittle metal which takes high polish, resists tarnishing and has a high melting point. The properties of chromium that make it most versatile and indispensable are its resistance to corrosion, oxidation, wear & galling and enhancement of hardenability. The main use of chromite is in the metallurgical, refractory and chemical industries. Chromium is an important alloying metal in ferrous metallurgy, perhaps next only to manganese. It is used in the manufacture of alloys along with other metals, such as, nickel, cobalt, molybdenum, copper, titanium, zirconium, vanadium, columbium and selenium. Chromium is traded primarily as chromium ore or as an alloy of chromium and iron, namely, ferrochrome or charge chrome.

RESERVES/RESOURCES

As per NMI database based on UNFC system, the total reserves/resources of chromite in the country as on 01.04.2020 has been estimated at 332 million tonnes with 79 million tonnes as "Reserves" (24%) and 253 million tonnes as "Remaining Resources" (76%). More than 96% resources of chromite are located in Odisha, mostly in Jajpur, Kendujhar and Dhenkanal districts. Minor deposits are scattered over

Manipur, Nagaland, Karnataka, Jharkhand, Maharashtra, Tamil Nadu, Telangana and Andhra Pradesh. Gradewise, Charge chrome grade accounts for 28% resources followed by Beneficiable grade (24%), Ferrochrome grade (17%), Refractory grade (16%) and Unclassified grade (11%). Low, Others, and Not-known grades together account for remaining 4% (Table-1).

Table – 1: Reserves/Resources of Chromite as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	78535	253150	331685
By Grades			
Refractory	26663	25675	52338
Charge chrome	31896	61815	93711
Low	4480	6284	10764
Beneficiable	15113	65413	80526
Ferrochrome	-	57429	57429
Others	-	15	15
Unclassified	384	35793	36177
Not-known	-	725	725
By States			
Andhra Pradesh	-	-	-
Jharkhand	-	736	736
Karnataka	499	1317	1817
Maharashtra	5	533	538
Manipur	-	6657	6657
Nagaland	-	3200	3200
Odisha	78031	240237	318269
Tamil Nadu	-	282	282
Telangana	-	186	186

Figures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on Exploration & Development.

MINING LEASES PRODUCTION AND STOCKS

As on 31.03.2024, the total number of mining leases in respect of chromite in force was 22, covering a total area of 5620.859 ha. The production of chromite at 3,148 thousand tonnes during 2023-24 decreased by 12% as compared to that in the previous year. The number of reporting mines in 2023-24 as well as in the preceding year was 18. The share of Public Sector in total production was 42% in 2023-24 as compared to 29% in the previous year.

The share of captive mines in the total production stood at 23% in 2023-24 as compared to 17% in the

previous year. Odisha continued to be the major producing state of Chromite, accounting for the entire production during 2023-24 and nil production was reported from the Karnataka state. Grade wise analysis of production during 2023-24 reveals that about 49% of total production of Chromite accounted for 40% - below 52% Cr₂O₃ (Lumps & fines), followed by 27% accounted for Below 40% Cr₂O₃ (Lumps & fines), followed by 21% accounted for 52% & above Cr₂O₃ (fines), and the remaining 3% accounted for concentrates grade.

Mine-head closing stocks of Chromite in 2023-24 were 2,624 thousand tonnes as compared to 2,940 thousand tonnes in 2022-23. The average daily employment of labour in Chromite mines during 2023-24 was 6,253 as against 5,911 in the previous year (Tables-2 to 8).

Table – 2: Principal Producers of Chromite, 2023-24

Name & address of producer	Location of the mine	
	State	District
Tata Steel Mining Ltd, Bombay House, 24, Homi Mody Street, Fort, Mumbai – 400 001, Maharashtra	Odisha	Jajpur
The Odisha Mining Corporation Ltd, 'OMC House', Unit 5, Post Box No. 34, Bhubaneswar – 751 001, Odisha	Odisha	Jajpur Kendujhar

Indian Metals & Ferro Alloys Ltd,
IMFA Building, Bomikhal,
P.O.-Rasulgarh,
Rasulgarh, Bhubaneswar – 751 010,
Odisha

Odisha

Jajpur

Ferro Alloys Corporation Ltd,
Charge Chrome Plant,
D.P.Nagar, Randia, Bhadrak-756135
Odisha

Odisha

Jajpur

M/s. Jindal Stainless Ltd,
O.P. Jindal Marg,
Hisar- 125005
Haryana

Odisha

Jajpur

Table – 3: Production of Chromite, 2021-22 to 2023-24

(By Countries)

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Qty	Value	Qty	Value	Qty	Value
India	3785625	47969167	3557289	46633415	3147954	48863434
Karnataka	-	-	-	-	-	-
Odisha	3785625	47969167	3557289	46633415	3147954	48863434

(P): Provisional

Table – 4: Gradewise Production of Chromite, 2022-23

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	No. of Mines	Production by Grade: Cr ₂ O ₃ content						Concen trates	Total	
		Below 40%		40–52%		52% & Above			Quantity	Value
		Lumps	Fines	Lumps	Fines	Lumps	Fines			
India	18	22537	858445	204748	1328121	-	1093937	49501	3557289	46633415
Public sector	7	1593	327453	4809	476480	-	232838	-	1043173	14106824
Private sector	11	20944	530992	199939	851641	-	861099	49501	2514116	32526591
Odisha	18	22537	858445	204748	1328121	-	1093937	49501	3557289	46633415
Dhenkanal	3*	-	-	-	-	-	-	-	-	-
Jajpur	14	20944	858445	199939	1327810	-	1093937	49501	3550576	46513675
Kendujhar	1	1593	-	4809	311	-	-	-	6713	119740

* Only labour reported

Table – 5: Gradewise Production of Chromite, 2023-24 (P)

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	No. of Mines	Production by Grade: Cr ₂ O ₃ content						Concen trates	Total	
		Below 40%		40–52%		52% & Above			Quantity	Value
		Lumps	Fines	Lumps	Fines	Lumps	Fines			
India	18	34419	800454	283268	1271697	-	674064	84052	3147954	48863434
Public sector	7	7824	367950	9906	783486	-	142230	101	1311497	24356720
Private sector	11	26595	432504	273362	488211	-	531834	83951	1836457	24506714
Odisha	18	34419	800454	283268	1271697	-	674064	84052	3147954	48863434
Dhenkanal	3*	-	-	-	-	-	-	-	-	-
Jajpur	14	26595	798267	273362	1270283	-	674064	84052	3126623	48475058
Kendujhar	1	7824	2187	9906	1414	-	-	-	21331	388376

* Only labour reported.

Table – 6: Production of Chromite, 2022-23 and 2023-24

(By Frequency Groups)

(Quantity in tonnes)

Production Group	No. of mines		Production for the group		Percentage to total production		Cumulative percentage	
	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)
Total	18	18	3557289	3147954	100	100	-	-
Up to 10000	9	7	6713	0	0.19	0	0.19	0.00
10001–100000	3	4	174132	124422	4.90	3.95	5.09	3.95
100001 – 200000	1	1	156240	165827	4.39	5.27	9.48	9.22
200001 – 300000	2	1	560666	299997	15.76	9.53	25.24	18.75
300001 and above	3	5	2659538	2557708	74.76	81.25	100.00	100.00

Table – 7: Mine-head closing stocks of Chromite, 2022-23

(By States/Grades)

(In tonnes)

State	Stocks by Grades: Cr ₂ O ₃ Content							
	Below 40%		40–52%		52% & Above		Quantity	
	Lumps	Fines	Lumps	Fines	Lumps	Fines	Concen trates	Total
India	19883	2080919	13778	403973	285	399515	21615	2939968
Odisha	19883	2080919	13778	403973	285	393515	21615	2939968

Table – 8: Mine-head Closing Stocks of Chromite, 2023-24

(By States/Grades)

(In tonnes)

State	Stocks by Grades: Cr ₂ O ₃ Content							
	Below 40%		40–52%		52% & Above		Concen trates	Total Quantity
	Lumps	Fines	Lumps	Fines	Lumps	Fines		
India	40084	1903244	28951	289672	285	353406	8146	2623788
Odisha	40084	1903244	28951	289672	285	353406	8146	2623788

USES & CONSUMPTION

The apparent consumption of chromite decreased by about 9% to 3.32 million tonnes in 2023-24 from 3.63 million tonnes in 2022-23.

In metallurgy, chromite is mainly used in the manufacture of ferrochrome, silicochrome, charge chrome and chromium metal. Chromium imparts additional strength, hardness and toughness to its alloys. It also shows resistance to corrosion & prevents steel abrasion, reduces oxidation and flow of electricity. Stainless steel, high-speed tool steel and corrosion & heat-resistant steel are some of the important varieties of chromium steel.

Chromite is used in Refractory Industry because of its high chemical stability, its high temperature resistance and corrosion resistant properties. Further, its high melting point, ability to withstand sudden temperature changes, its chemically neutral character, moderate thermal expansion and mechanical strength besides, abundant availability and reasonable price are added advantages for use in Refractory Industry.

Chromite is used for manufacturing important

chromium compounds like chromates and bichromates of sodium and potassium, chromium pigments like chromic oxide green and chromic acid, which in turn, are used in chromium-plating solution.

Chromium is an essential trace element for human health. However, some of its compounds are highly toxic and carcinogenic. Environment concerns have reduced the use of chromite refractories and chromium chemicals.

SUBSTITUTES

Development of substitutes for chromium tends to be deterred by cost performance or the customer appeal for chromium. There are no substitutes for chromium in stainless steel or superalloys. Boron, manganese, nickel and molybdenum can be substituted in alloy steels and cast irons. Chromium containing scrap can substitute for ferrochromium in some metallurgical uses. Dolomite is an alternative for some refractory bricks. Cadmium yellow is one of the several alternative pigments. However, it is not environmentally acceptable and nickel and zinc are possible substitutes for the protection of decorative coatings.

INDUSTRY

Chromite is mainly used in Metallurgical Industry for manufacture of ferroalloys, e.g., ferrochrome, charge chrome and silicochrome which are used as additives in making stainless steel and special alloy steel. Ferroalloys are the essential ingredients for the production of high quality special alloy steel as well as mild steel. The demand for ferroalloys is associated with the production of alloy steel. Production of ferrochrome/charge chrome was mainly reported by Ferro Alloys Corp. Ltd, Shri Vasavi Industries Ltd, Balasore Industries Ltd, Tata Steel Ltd, Indian Metals & ferro-Alloys Ltd and Indian Charge-chrome Ltd (merged with Indian Metals & Ferro alloys Ltd in 2006) were amongst the major producers of charge chrome in India. Charge chrome contains 50 to 60% chromium and 6 to 8% carbon. Hard lumpy chromite is used for high-carbon ferrochrome while friable ores and fine briquettes are used for low-carbon ferrochrome. Briquette fines along with lumpy ores were also consumed in charge chrome plants.

The production of ferro-chrome was 1,126 thousand tonnes and 1,322 thousand tonnes during 2022-23 and 2023-24 respectively. The country's top three ferrochrome players are IMFA, Tata Steel and Balasore Alloys. Ferrochrome consumption has shown only limited growth. A notable development in the sector was the acquisition of Ferro Alloys Corporation (FACOR) by Vedanta in September 2021, which is expected to contribute to a more competitive Private Sector landscape and foster greater efficiency within the ferro-alloys Industry.

The important plants which produce chromite based refractories are Tata Steel Ltd (formerly OMC Alloys), Orissa Industries Ltd, Bhilai Refractories Ltd, Burn Standard Co. Ltd, Joglekar Refractories and Ceramics (P) Ltd and Associated Ceramics Ltd.

Ferrochrome when added to steel imparts hardness, strength and augments its stainless characteristics. Carbon content classifies the ferrochrome alloy into high-carbon (6–8%), medium-carbon (3–4%) and low-carbon (1.5 – 3%), although chromium content in all the three grades is around 60–70 per cent. Around 2.5 tonnes chrome ore with an estimated power consumption of about 4,500 kWh is required to produce one tonne of ferrochrome.

Commercially, chrome ore can be divided into three categories: (i) high-grade, containing >48% chromite, (ii) medium-grade with > 40% chromite and (iii) low-grade containing less than 40% chromite.

Chromium metal and the alloy ferrochromium are commercially produced from chromite by silicothermic or aluminothermic reactions, or by roasting and leaching processes. Chromium metal assumes high value due to its properties, such as, high corrosion resistance and imparting of hardness.

The discovery that steel could be made highly resistant to corrosion and discolouration by adding metallic chromium to form stainless steel led to major developments in the Steel Sector. This application, along with chrome plating (electroplating with chromium) are currently the major commercial use for the element. The element also finds application in the production of chromium compounds, albeit to a minor extent.

The strengthening effect of forming stable metal carbides at the grain boundaries and the strong increase in corrosion resistance has made chromium an important alloying material for steel. The high-speed tool steels contain between 3 and 5% chromium. Stainless steel, the main corrosion proof metal alloy is formed when chromium is added to iron in sufficient concentrations usually above 11%.

WORLD REVIEW

World reserves of Shipping-grade chromite are about 560 million tonnes in terms of chromium content. Countries that possess sizeable quantities of reserves are Kazakhstan (41%) and South Africa (36%). These two countries together hold about 77% of world's chromite reserves. India possesses 14% while Turkey accounts for 5% of the world reserves of chromite. The available data on world reserves of chromite (Shipping-grade) is furnished in Table-9.

The world mine production of chromite ores & concentrates increased by 2.18% to 37.50 million tonnes in 2023 from 36.70 million tonnes recorded in the previous year. South Africa was the leading producer contributing about 52% to the total world production followed by Kazakhstan (16%), India (9%), Turkey (7%), Zimbabwe & Finland (3% each), Albania & Russia (2% each) and Brazil (1%)(Table-10).

Upgradation of technology and advancement in beneficiation processes, such as, agglomeration of ore, preheating and pre-reduction of furnace feed, closed-furnace technology and recovery of chromium from slags have brought about significant changes and are now followed worldwide.

Table – 9: World Reserves of Chromite
(Shipping Grade)**
(By Principal Countries)

(In '000 tonnes of chromium content)

Country	Reserves
World: Total (rounded off)	560000
Kazakhstan	230000
South Africa	200000

(In '000 tonnes of chromium content)

Country	Reserves
India ^e	79000
Turkey	27000
Finland	8300
USA	630
Other countries	NA

Source: USGS, Mineral Commodity Summaries, 2024

Note: Reserves/resources of chromite in the country as on 1.4.2020 as per NMI database based on UNFC system have been placed at 332 million tonnes.

**: Units are thousand metric tons of shipping-grade chromite ore, which is deposit quantity and grade normalized to 45% Cr₂O₃, except for the USA, where grade is normalized to 7% Cr₂O₃, and Finland, where grade is normalized to 26% Cr₂O₃.

e: Estimated; NA-Not available.

Table – 10: World Production of Chromium Ores and Concentrates

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World:Total (rounded off)	36900000	36700000	37500000
South Africa	18435250	19061754	19683351
Kazakhstan	6192000	5716900	6108000
India*	3785625	3470110	3243545
Turkey	2779467	2562533	2581916
Zimbabwe	1244300	1154158	1204755
Finland ^(a)	1464738	1208006	1148905
Albania	650200	932686	909669
Russia	554000	809000	809000
Brazil	627735	565575	522869
Other countries	1189566	1236520	1264674

Source: BGS, World Mineral Production, 2019-23

(a): Concentrate;

*:Production of chromite in India during 2021-22, 2022-23 and 2023-24 (P) was 3.79 million tonnes, 3.56 million tonnes and 3.15 million tonnes respectively.

FOREIGN TRADE

Exports

Exports of chromite (total) decreased drastically by 94% to 2.12 thousand tonnes in 2023-24 from 33.88 thousand tonnes in the previous year. Out of total chromite exported in 2023-24, the share of chrome ore (others) was almost 100% while very meagre quantity of Chromium and Alloys (Including Scrap) and Chromium Unwrought: Powders were exported. Exports of Chrome ore (others) were almost

fully to China in 2023-24. Export of Chrome ore (other) decreased to 2,117 tonnes in 2023-24 from 4,303 tonnes in the preceding year. In 2023-24, 215 tonnes of Chromium & alloys (Including Scrap) were exported to various countries. The share of export of Chromium & alloys (Including Scrap) were mainly to USA (88%). The export of Chromium Unwrought: Powders was also recorded as 215 tonnes which increased in 2023-24 from 156 tonnes in the preceding year (Tables-11 to 14).

Table – 11: Exports of Chromite: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	33881	904154	2117	99434
China	33875	903780	2115	99359
Kenya	--	--	2	75
Oman	5	187	--	--
Canada	1	169	--	--
Finland	++	18	--	--

Figures rounded off

++: Negligible; --: Nil

Table – 12: Exports of Chrome Ore Concentrates
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	29578	780686	++	++
China	29578	780686	--	--

Figures rounded off

++: Negligible; '--: Nil

Table – 13: Exports of Chrome Ore (Others)
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	4303	123468	2117	99434
China	4297	123094	2115	99359
Kenya	--	--	2	75
Oman	5	187	--	--
Canada	1	169	--	--
Finland	++	18	--	--

Figures rounded off

++: Negligible; '--: Nil

Table – 14: Itemwise Exports of Chromium
(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Chromium and Alloys (Including Scrap)	157	164945	215	210654
Chromium Articles , Nes	1	723	++	5006
Chromium Unwrought: Powders	156	164070	215	205648
Chromium Waste and Scrap	++	152	++	++
Chrome Ore Lumps	--	--	--	--

Figures rounded off

++: Negligible; '--: Nil

Imports

Imports of chromite (total) increased by 55% to 172.41 thousand tonnes in 2023-24 from 111.29 thousand tonnes in the previous year. Imports were mainly from South Africa (93%), Mozambique (3%) and Turkey (2%). Out of total quantity of chromite imported in 2023-24, Chrome ore (others) accounted for 67%, while Chrome ore lumps accounted for 28% and Chrome ore concentrate accounted about 5%. More than 98% imports of chrome ore (others) were from South Africa while Mozambique contributed

1%. Almost 83% of the imports of Chrome ore lumps were from South Africa only and 6% from Turkey. Imports of Chrome ore concentrate in 2023-24 were at 7,776 tonnes as compared to 7,911 tonnes in the previous year. Imports of Chromium & alloys (Including Scrap) in 2023-24 were at 1,976 tonnes as compared to 1,344 tonnes in the previous year. The import of Chromium & alloys (Including Scrap) were mainly from China (35%), Russia (23%) and Netherlands (16%). Imports of Chromium & scrap were not reported (Tables-15 to 19).

Table – 15: Imports of Chromite: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	111291	2969916	172411	4563915
South Africa	80492	2193902	159881	4143050
Mozambique	25904	621937	5616	182337
Turkey	4050	103587	2711	72658
China	49	17167	1349	72369
Albania	--	--	1835	52108
Netherlands	242	13262	216	13176
UAE	116	3589	337	11291
Oman	383	12844	312	10314
France	--	--	60	2809
Saudi Arabia	--	--	81	2699
Other countries	55	3628	13	1104

Figures rounded off

++: Negligible; '--: Nil

In tonnes

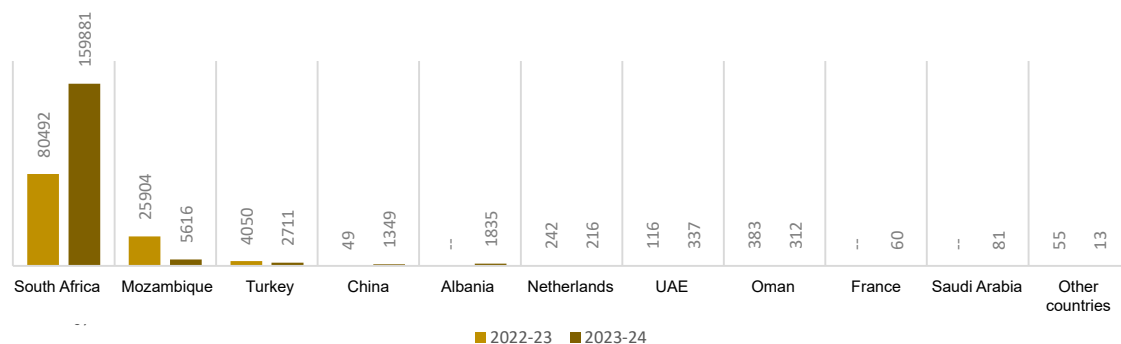


Fig. 1: Countrywise Imports of Chromite

Table – 16: Imports of Chrome Ore Concentrate

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	7911	215038	7776	240354
South Africa	5049	138077	5211	160912
Mozambique	2862	76961	2565	79442

Figures rounded off

Table – 17: Imports of Chrome Ore (Others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	84361	2179213	115752	2904035
South Africa	63010	1664798	114127	2840606
Mozambique	20936	477603	864	29193
Netherlands	242	13262	216	13176
UAE	94	3139	337	11291
China	49	17167	54	3157
France	--	--	60	2809
Saudi Arabia	--	--	81	2699
Germany	8	809	8	805

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Brazil	2	171	5	299
USA	20	2167	--	--
Other countries	++	97	++	++

Figures rounded off

++: Negligible; '--: Nil

Table – 18: Imports of Chrome Ore Lump

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	19019	575665	48883	1419526
South Africa	12433	391027	40543	1141532
Mozambique	2106	67373	2187	73702
Turkey	4050	103587	2711	72658
China	--	--	1295	69212
Albania	--	--	1835	52108
Oman	383	12844	312	10314
UAE	22	450	--	--
Belize	24	301	--	--
Taiwan	1	83	--	--

Figures rounded off

--: Nil

Table – 19: Itemwise Imports of Chromium

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Chromium and Alloys (Including Scrap)	1344	1382782	1976	1670445
Chromium Articles , Nes	77	209670	167	204861
Chromium Unwrought: Powders	1267	1173112	1809	1465584
Chromium Waste and Scrap	--	--	--	--

Figures rounded off

++: Negligible

FUTURE OUTLOOK

The current status of chromite production and consumption is on anticipated lines, but the consumption could increase enormously in the coming years and the country may have to depend on imports even for the domestic needs of chromite. National Steel Policy 2017 has laid down the broad roadmap for encouraging long-term growth for the Indian Steel Industry, both on demand and supply sides. The requirement of Chromite ore is forecasted as 5 million tonnes by 2030-31. Indian Steel Sector is facing challenges due to limited availability of some of the essential raw material including chromite. Besides, India is expected to show strong growth in usage of steel in the coming years because of its robust economy, massive infrastructure needs and expansion of industrial production.

Depletion of reserves is bound to create a serious problem for the future of the Chromite Industry in the country. An Expert Committee constituted by the Ministry

of Steel, Government of India, has recommended for detailed exploration of chromite in all the potential areas in Odisha, Karnataka and in the ophiolite belt of North-Eastern region with a view to prognosticate resources to a depth of 500 m in Sukinda belt and estimate resources in all other potential areas. Further addressing concerns in ferrochrome production which is energy intensive segment is also essential. Setting up of such plant must strike a cost balance between raw materials and electrical energy supply. There are other imminent issues like consistent supply of chrome ore at the right cost, steady power supply and other input materials like low phosphorous metallurgical coke and good market conditions that need redressal in respect of the continuous and unscrupulous exploitation of chromite. Adherence to stringent pollution control norms, innovations in the process technology and plant equipment design would become inevitable for the future of the industry.



35. Cryolite

343 tonnes
of cryolite and chiolite (artificial) were exported in 2023-24

1,944 tonnes
of cryolite and chiolite (artificial) were imported in 2023-24

Cryolite is a double fluoride of sodium and aluminium and has a stoichiometry very near the formula Na_3AlF_6 and a melting point of about $1,010^\circ\text{C}$. It was found to occur in substantial quantities in Greenland, and was mined extensively there in the early twentieth century, but the mine is now essentially exhausted. Synthetic cryolite can be produced by reacting hydrofluoric acid with an alkaline sodium aluminate solution. It is colourless to white but occurs in other shades too, for instance brown, red and sometimes black. It has a specific gravity of about 2.5 to 3. It has a low index of refraction close to that of water. Synthetic cryolite is used as an electrolyte in the reduction of alumina to aluminium due to non-availability of natural cryolite. Composition and properties of synthetic cryolite are the same as that of natural cryolite, but synthetic cryolite is often deficient in sodium fluoride.

INDUSTRY

Synthetic cryolites are obtained by adopting several processes. The selection of the process depends upon the availability and cost of raw materials. The simplest and most common method of obtaining synthetic cryolite is by reacting hydrofluoric acid with soda ash and alumina hydrate. Hydrofluoric acid is produced by reacting acid-grade fluorspar with sulphuric acid and this process also yields gypsum as by-product. In the secondary reaction between hydrofluoric acid and sodium chloride brine, sodium fluoride and hydrochloric acid are produced. In the primary reaction, dry aluminium hydroxide reacts with hydrofluoric acid to produce aluminium fluoride which reacts with sodium fluoride produced earlier and forms synthetic cryolite.

Besides fluorspar, fluorine gas produced as by-product

at plants that produces phosphatic fertilizer and phosphoric acid, has emerged as an important alternative source for hydrofluoric acid and other fluorine chemicals including cryolite and aluminium fluoride. Rock phosphate usually contains 7–8% CaF_2 . In terms of fluorine, it works out to 3–4% which is liberated at the time of acidulation of rock phosphate with sulphuric acid. Fluorine combines with silica to form silicon tetrafluoride which when scrubbed with water forms fluorosilicic acid. By recycling, 18–24% fluorosilicic acid is obtained, which serves as a raw material for manufacturing various fluoro-chemicals, including synthetic cryolite. From fluorosilicic acid, fluorine values are precipitated as sodium fluorosilicate by treating it with sodium salts. Sodium fluorosilicate becomes starting point for the production of synthetic cryolite.

For manufacturing synthetic cryolite from sodium

fluorosilicate, two routes are generally adopted in the country. In the first route, sodium fluorosilicate is reacted with ammonia and in other route; sodium fluorosilicate is reacted with soda ash.

Important known units producing synthetic cryolite are highlighted below:

1. Navin Fluorine International Ltd, Udhana-Navasari Road, Surat, Gujarat-395 023.
2. Navin Fluorine International Ltd, Agra-Mumbai Road, Dewas, Madhya Pradesh-455 002.
3. Tanfac Industries Ltd, Kudikadu, Cuddalore, Tamil Nadu-607 005.
4. Harshil Industries Sarigam, Vapi, Gujarat-396 195.
5. Triveni Interchem Pvt. Ltd, GIDC, Vapi, Gujarat.
6. S.B. Chemicals, GIDC, Char Rasta, Vapi, Gujarat-396 195.
7. J.N. Chemicals, GIDC, Vapi, Gujarat.
8. Balaji Amines Ltd, Solapur, Maharashtra.
9. Arti Chemicals, Ankleshwar, Gujarat.

Navin Fluorine International Ltd is one of the largest manufacturers of speciality fluoro chemicals comprising synthetic cryolite Aluminium Fluoride & Fluorocarb gases.

Some important units engaged in manufacturing of synthetic cryolite are -Triveni Chemicals, S.B. Chemicals, Jay Intermediates & Chemicals (Vapi, Gujarat), Madras Fluorine Pvt. Ltd (Manali, Chennai and Tamil Nadu) and Tarun Fluo-Chem Pvt. Ltd (Delhi), they also produce other fluorine chemicals. These units also manufacture potassium cryolite (K_3AlF_6) which is a foundry flux and used in welding, chemicals and explosives.

The total installed capacity of aluminum fluoride in Organised Sector was about 25,600 tonnes per annum

USES AND CONSUMPTION

The Indian Standard specifications of cryolite for use in Aluminium Industry defined vide IS - 5893: 1989 (Second Revision; reaffirmed 2024) are as follows:

Constituents (on dry basis)	Specification
F	53% min.
Na	31 to 34%
Al	13 to 15%
SiO ₂	0.20% max.
Fe ₂ O ₃	0.10% max.
CaF ₂	0.06% max.
Al ₂ O ₃	1.00% max.
SO ₃	0.50% max.
P ₂ O ₅	0.01% max.
Loss on Ignition (LOI)	0.50% max.
NaF/AlF ₃ (by mass)	1.45 max. (Ratio required for maintaining in acidic region)

Note: i) LOI is to be determined at 550°C for 60 minutes.

ii) Moisture should not be more than 0.20% when determined at 110±5 °C

The consumption of cryolite is nowadays not estimated because many industries prefer the use of synthetic cryolite instead of natural cryolite. However, consumption was reported earlier in bonded abrasives as a filler, insecticides, porcelainous glass and salts of sodium & aluminium.

The commercial application of cryolite is confined mainly to aluminium metallurgy where it is used as an electrolyte in the reduction of alumina to aluminium metal by the Hall-Heroult process. Alumina is a bad conductor of electricity and its melting point is 2,348 °C. It is very expensive to carry out electrolysis at this temperature. To facilitate electrolysis, alumina is dissolved in molten cryolite as it lowers the melting point. Further, addition of certain additives, such as, aluminium fluoride improves the physical and electrical properties of the electrolyte, besides lowering the melting point. The amount that is added is, however, limited as it also causes reduction in electrical conductivity. Addition of calcium fluoride (CaF₂) further depresses the melting point with less adverse effect on conductivity. In contrast to this advantage, too much CaF₂ raises the density of the melt closer to that of liquid aluminium metal, thus inhibiting the separation of metal from electrolyte. The substituent, sodium fluoride, though is known to improve the density and conductivity, it also affects current efficiency.

A compromise made on all these factors has led to the following general composition of the bath to be in use —80–85% cryolite, 5–7% AlF₃, 5–7% CaF₂, 0–7% LiF and 2–8% Al₂O₃. The electrolyte bath tends to deplete AlF₃ content of cryolite during the process. Hence, the composition of the electrolyte has to be adjusted regularly by addition of AlF₃.

In aluminium refining, high density electrolyte capable of floating aluminium is required. For this purpose, barium fluoride is used to raise density. Aluminium fluoride can be used to improve current efficiency of cryolite bath.

Cryolite is obtained as a by-product during the production of phosphatic fertilizer/phosphoric acid. When utilised in the Aluminium Industry, necessary precautions are observed as even 0.01% P in the electrolyte could cause 1–1.5% reduction in current efficiency in the production process of aluminium.

Other metallurgical uses of cryolite are in aluminising steel, in compounding of welding rod coatings and as fluxes. In glass, cryolite functions as a powerful flux because of its excellent solvent power for oxides of silicon, aluminium & calcium and for its ability to reduce melt viscosity at lower melting temperatures. Cryolite is used as filler for resin-bonded grinding wheels in Abrasive Industry to impart longer life. Sodium fluoride (NaF) or fluorosilicic acid is also used for this purpose. Cryolite is used in certain nitrocellulose-based gun propellants required in small-calibre weapons, cannons and small & large rockets.

FOREIGN TRADE

Exports

In 2023-24, exports of cryolite and chiolite (artificial) increased substantially by 1,083% to 343 tonnes from 29 tonnes in the previous year. Taiwan (79%), USA (12%), Georgia (9%) were the main buyers from India in 2023-24 (Table-1).

Table – 1: Export of Cryolite and Chiolite (artificial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	29	4274	343	12043
Taiwan	--	--	272	5890
USA	25	3414	40	4861
Georgia	--	--	30	789
Indonesia	--	--	1	301
Saudi Arabia	--	--	++	90
France	--	--	++	72
Kenya	++	1	++	18
Germany	--	--	++	18
China	--	--	++	3
Denmark	--	--	++	1
Other Countries	4	859	++	++

Figures rounded off

Source: DGCI&S, Kolkata

"++ : Negligible

--: Nil

Imports

In 2023-24, imports of cryolite and chiolite (artificial) increased by 11% to 1,944 tonnes from 1,747 tonnes in the previous year. Imports were mainly from China (56%), Bahrain (37%) and Belgium (3%) (Table-2).

Table – 2: Import of Cryolite and Chiolite (artificial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1747	121744	1944	122318
China	836	79306	1094	99068
Belgium	21	4359	63	10008
Bahrain	264	4303	724	6334
Germany	23	4944	17	3621
Hungary	7	2294	9	2738
Taiwan	260	12570	14	330
Qatar	--	--	23	195
USA	--	--	++	24
Netherlands	63	9960	--	--
Mozambique	196	2878	--	--
Other Countries	77	1130	++	++

Figures rounded off

Source: DGCI&S, Kolkata

"++ : Negligible

--: Nil

FUTURE OUTLOOK

The future of cryolite is dependent upon its use in the Aluminium Industry. Increased usage of aluminium and

high performance fluoro polymers in automobiles will drive growth in inorganic and specially fluoro chemicals segments



36. Diamond

349 carats
production of diamond have been
reported in 2023-24

₹ 1,39,513 crore
value of exports of diamond were
reported in 2023-24

₹ 1,78,731 crore
value of imports of diamond were
reported in 2023-24

Diamond has been the most valuable among gems for more than 2,000 years. Diamond occurs in two types of deposits, primarily in igneous rocks of basic or ultrabasic composition and in alluvial deposits derived from the primary sources. Its composition is pure carbon and has cubic crystal system and common form octahedron. India is known for its diamond cutting & polishing business especially for small-sized diamonds. Most of the world's diamond cutting and polishing business comes to India, particularly to Surat in Gujarat. India depends largely on imports of rough gem diamonds for its Cutting and Polishing Industry as there is no notable production except for two producers in Madhya Pradesh whose limited production is too sparse to meet the Cutting and Polishing Industry's requirements. The cut and polished diamonds are predominantly re-exported.

Broadly, industrial diamonds have three varieties, viz, 'ballas' which is mass of minute diamond crystals difficult to cleave; 'bort' is yellowish grey to black colour and massive, flawed or irregular in shape and 'carbonado' is black, very hard, opaque and without cleavage.

RESERVES/RESOURCES

Diamond occurrences are reported since prehistoric times in the country. Presently, diamond fields of India are grouped into four regions:

1. South Indian tract of Andhra Pradesh, comprising parts of Anantapur, Kadapa, Guntur, Krishna, Mahabubnagar and Kurnool districts;
2. Central Indian tract of Madhya Pradesh, comprising Panna belt;

3. Behradin-Kodawali area in Raipur district and Tokapal, Dugapal, etc. areas in Bastar district of Chhattisgarh; and
4. Eastern Indian tract mostly of Odisha, lying between Mahanadi and Godavari valleys.

As per the NMI data, based on UNFC system as on 1.4.2020, all India reserves/resources of diamond have been placed at 31.72 million carats with an R/R ratio of 2.75%. By States, Madhya Pradesh accounts for about 90.14% resources followed by Andhra Pradesh 5.74% and Chhattisgarh 4.11% (Table-1).

Table –1: Reserves/Resources of Diamond as on 01.04.2020

(By Grades/States)

(Quantity in carats)

Grade/State	Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	847559	30876432	31723991
By Grades			
Gem	297692	756765	1054457
Industrial	254559	840823	1095382
Unclassified	295308	29278844	29574152
By States			
Andhra Pradesh	0	1822955	1822955
Chhattisgarh	0	1304000	1304000
Madhya Pradesh	847559	27749477	28597036

Figures rounded off

MINING LEASES & PRODUCTION

As on 31.03.2024 (provisional), the total number of mining leases granted for Diamond was 2 covering an area of 275.963 ha. There were three reporting mines, all under Public Sector located in Panna district of Madhya Pradesh. Production of diamond in 2023-24 was 349 carats which decreased by 10 % as against 388 carats in the previous year.

The principal producers of diamond was Directorate of Geology & Mining, (Diamond Project), Govt. of Madhya Pradesh (Tables-2 to 4).

Out of the total output, Gem variety covering rough & uncut constituted 38%, 35% was of Industrial grade and 27% was of other varieties (Table 4). The average daily employment of labour during 2023-24 was 645 as against 871 in 2022-23.

Table – 2: Principal Producers of Diamond, 2023-24

Name & Address of Producer	Location of the mine	
	State	District
Directorate of Geology & Mining (Diamond Project), Government of Madhya Pradesh, Khanij Bhavan, 29-A, Arera Hills, Bhopal-462016, Madhya Pradesh	Madhya Pradesh	Panna

Table – 3: Production of Diamond, 2021-22 to 2023-24

(By States)

(Quantity in carats; Value in ₹ '000)

States	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	266	18051	388	61473	349	23858
Madhya Pradesh	266	18051	388	61473	349	23858

(p):Provisional

Table – 4: Production of Diamond, 2022-23 & 2023-24

(By Sector/State/District/Grades)

(Quantity in carats; Value in ₹ '000)

State/District	2022-23						2023-24 (P)					
	Quantity						Quantity					
	No. of Mines	Gem (Rough & uncut)	Industrial*	Others	Total	Value	No. of Mines	Gem (Rough & uncut)	Industrial	Others	Total	Value
India	3	289	28	71	388	61473	3	131	123	95	349	23858
Public Sector	3	289	28	71	388	61473	3	131	123	95	349	23858
Madhya Pradesh	3	289	28	71	388	61473	3	131	123	95	349	23858
Panna	3	289	28	71	388	61473	3	131	123	95	349	23858

(p): provisional

*: Includes off color and dark brown varieties of diamond.

Table – 5: Mine-head Closing Stocks of Diamond, 2022-23 & 2023-24**(By State)***(In tonnes)*

State	2022-23	2023-24 (P)
India	260	412
Madhya Pradesh	260	412

(P): Provisional

MINING & PROCESSING

Majhgawan in Madhya Pradesh is a fully mechanised mine operated by National Mineral Development Corporation Ltd. It is worked by opencast method in tuff rock by deploying 4.1 cu. m hydraulic shovel and 40 tonnes dumpers in combination. The mine benches have been designed with a height of about 10 m. A few benches were of 4-5 m in height. Drilling is done by 4-inch diameter drills and charged with slurry explosives, and about 40–50 holes are blasted at a time with delay pattern. At present, the capacity of the mine is about 1,00,000 carats per year. Diamonds are also recovered from conglomerate and gravel beds at shallow depths by small operations on the basis of annual permits granted by Diamond Officer, Government of Madhya Pradesh. At Majhgawan, kimberlite rock, after mining the ore is stockpiled for weathering action and then is fed to crushing plant. It is processed through Heavy Media Separation System in processing plant for recovery of diamond. Recently, X-ray diamond sorter has been installed for sorting of diamonds from ore and this has increased the recovery of raw diamonds to 98%.

Diamond Mining Factors

Grade

Grade is the weight of diamond expressed as carats per tonne (ct/t) of ore. It varies widely from one mine to another but generally falls somewhere between 0.3 and 1.3 ct/t. One carat is equivalent to 0.2 grams.

Size (weight) of Rough Diamonds in Deposit

Individually, rough diamonds can range from micro weight to stones weighing more than 1,000 carats. Depending on the mine, the average size of rough diamond recovered can weigh from 0.01 ct (about 1 mm) to more than 0.7 ct. Many mines in the world show an average of about 0.4 to 0.5 ct per tonne.

INDUSTRY

Indian Diamond Industry enjoys credible standing and reputation in the world market, particularly for small diamonds used in jewellery. Indian diamond manufacturing standards are reckoned as the best in the world. Workmanship & skill of Indian artisans at polishing small diamonds economically and efficiently has been

widely acknowledged. Surat in Gujarat is the main centre of the Cutting and Polishing Industry.

The Indian Diamond Industry thrives in the atmosphere of secrecy and informality that envelops the diamond trade and has for long been labeled as an unorganised sector of the economy. However, it resembles a close-knit community composed of thousands of small, medium and large sized CPD (Cut and Polished Diamonds) units and has grown to become one of the highest foreign exchange earners for the country. An in-depth study of the Industry reveals that the so called unorganised sector is in fact highly organised and has great potential to offer useful insights to the field of management in terms of new forms of organising, networking, business processing and conducting international business.

India's predominance as leader in the world market is due to a combination of pragmatic policies of the Government and sustained efforts of exporters. Policy changes, such as, creation of Special Economic Zones (SEZ) is expected to boost the export prospects further. Several diamond polishing companies have already established offices in India for trading in rough & polished diamonds. India obtains rough diamonds from Belgium, UK, Hong Kong, UAE, Israel etc. Indian diamond traders seek opportunities to establish direct trade ties with mining companies. The expectations of the Indian Diamond Industry are to access rough diamonds at competitive rates directly from the producers to maintain its lead in the world market.

Tripartite MoU among Government of Madhya Pradesh (MRD, through DGM), MPSMCL & NMDC were signed for geological and geophysical exploration in various districts of Madhya Pradesh. In this regard, Government of Madhya Pradesh issued Gazette Notifications of 20 prospective diamond blocks which could be applied under MoU for exploration and subsequent reservation.

Remote Sensing Studies have been completed in collaboration with NRSC in Chhattarpur Panna Block 1, Chhattarpur Panna Block 2 and Damoh Block, Panna (5 Prospective Blocks) and Additional 12 Diamond Blocks.

Several target areas were established after conducting ground magnetic, VLF-EM & Gravity surveys and stream

sediment sampling and several additional anomalous areas were demarcated in the diamond blocks by processing the Raw Aeromagnetic Data acquired/provided by GSI in Diamond Blocks.

Forest applications for drilling has been prepared and submitted. The matter is being pursued with Forest Department for obtaining permission for drilling.

Proposals were submitted by NMDC to the Government of Chhattisgarh for Baloda–Belmundi Diamond Block over an area of 156.80 km² in Saraipali tehsil, Mahasamund district for reservation under Section 17(A) (2A) of MM (D&R) Amendment Act, 2015 for undertaking prospecting and exploitation operations. Ministry of Mines, Government of India vide G.S.R.744(E) dt 14.10.2021 granted reservation of Baloda–Belmundi Diamond Block for PL or ML in favour of NMDC-CMDC limited under Section 17A(1A) of MMDR Act, 1957. NCL vide letter dated 13.01.2022, submitted application to Secretary, MRD, Government of Chhattisgarh for grant of Prospecting Licence for Baloda–Belmundi Diamond Block.

Main Mining Lease of Diamond Mining Project, Panna for total excavation of 13.90 LTPA (ROM kimberlite ore @ 8 LTPA and Overburden removal @ 5.90) Environment Clearance obtained from Ministry of Environment, Forest & Climate Change (MOEF&CC) on 22.02.2024. Supplementary Mining Lease of Diamond Mining Project, Panna for 0.72 LTPA Over burden removal and processing of Kimberlite ore @ 8 LTPA for production of 1.00 lakh carats of Diamonds per annum Environment Clearance obtained from MOEF&CC on 22.02.2024.

USES & CONSUMPTION

Industrial diamonds are mostly consumed by manufacturers of drill bits, grinding tools and

stone cutting & polishing machines and demand of industrial diamonds is mostly met by imports. There are many small-scale sector units that operate in cutting & polishing trade.

SUBSTITUTES

Synthetic Diamond

Today, market for industrial diamond is dominated by synthetic stones, first developed in 1950s. Synthetic diamonds manufactured using high pressure and high temperature methods compete as an abrasive mineral with natural industrial diamonds and also with manufactured materials like Silicon Carbide (SiC), alumina (Al₂O₃), Tungsten Carbide (WC) and Carbide Boron Nitrate (CBN). Synthetic diamonds being marketed are mostly 0.6 to 0.8 mm and smaller in size. Synthetic Diamond Abrasives (SDA) are used for sawing, drilling or milling hard stones, concrete

aggregate, refractory materials, masonry and asphalt. In general, large crystals are used for cutting softer materials and smaller crystals for tougher jobs. Synthetic diamonds now account for bulk supply of industrial diamonds and are preferred over natural diamonds because their quality can be controlled to suit customer's requirements.

Synthetic diamonds were produced earlier by using graphite with a metal catalyst under very high pressure & temperature.

Of late a new process, such as, Chemical Vapour Deposition (CVD) has been evolved which requires relatively low pressure for production of synthetic diamonds. This process involves depositing tiny crystals of diamond on a film which can be built in complicated shapes and used at desired places or instruments, such as, machine part, heat conductors in micro circuit, shortwave UV, microwave sources and radiation detectors. In future, CVD can be a substitute for silicon in Computer Industry. In USA, developments have taken place in CVD method of growing 100% pure diamond using microwave plasma technology. This method is more economical and also enables production of larger crystals.

TRADE POLICY

Import of diamond under HS Code 7102, Diamonds, whether or not worked, but not mounted or set, fall under 'Free' category as per the Import Policy ITC(HS), 2022, Schedule 1. Foreign Direct Investment (FDI) in diamond mining up to 100% is admissible for automatic approval of Reserve Bank of India.

WORLD REVIEW

The world reserves of industrial diamond are about 1,700 million carats located mainly in Russia (51%), Botswana (16%), Dem. Rep. of Congo and Angola (9% each), and South Africa (6%). The world reserves of industrial diamond are furnished in Table-6.

The total world production of diamond decreased by about 9 % to 111.60 million carats in 2023 from 122.2 million carats in 2022.

The principal producers were Russia (33 %), Botswana (22 %), Canada (14%), Angola (9 %), Dem. Rep. of Congo (7 %), South Africa (5 %) and Zimbabwe (4%). During the year 2023, increase in diamond production was observed in Botswana, Angola, Zimbabwe, Namibia etc. while decrease in production was observed in Russia, Canada, Dem. Rep. of Congo and South Africa (Table-7).

Natural diamonds are cut in about 52 countries. The major diamond cutting centres in the world are Antwerp in Belgium, Ramat Gan in Israel, New York in USA, Surat in India and Guangzhou & Shenzhen in China.

Table – 6: World Reserves of Diamond (Industrial)

(By Principal Countries)

(In million carats)

Country	Reserves
World: Total (rounded off)	1700
Russia	860
Botswana	280
Angola	150
Dem. Rep. of Congo	150
South Africa	95
Zimbabwe	NA
USA	NA
Other countries	120

Source: Mineral Commodity Summaries, USGS, 2024

Table – 7: World Production of Diamond

(By Principal Countries)

(In '000 carats)

Country	2021	2022	2023
World total	117000	122200	111600
Russia	39116.97	41923.91	37316.754
Botswana	22696	24479	25095
Canada	16865.216	16217.233	16160.688
Angola	8723.069	8763.309	9754.31
Democratic Republic of Congo	12179	11683	8306
South Africa	9723.811	9646.3	5891.865
Zimbabwe	4224.3	4844.019	4913.593
Namibia	1523.12	2188.075	2384.975
Sierra Leone	829.33	680.432	540.72
Other Countries	1119.184	1774.722	1236.095

Note (s):

(1) So far as possible the amounts shown include estimates for artisanal and/ or illegal production.

(2) The above data does not include production of synthetic diamond.

Source: BGS, World Mineral Production, 2019-2023

(In '000 carats)

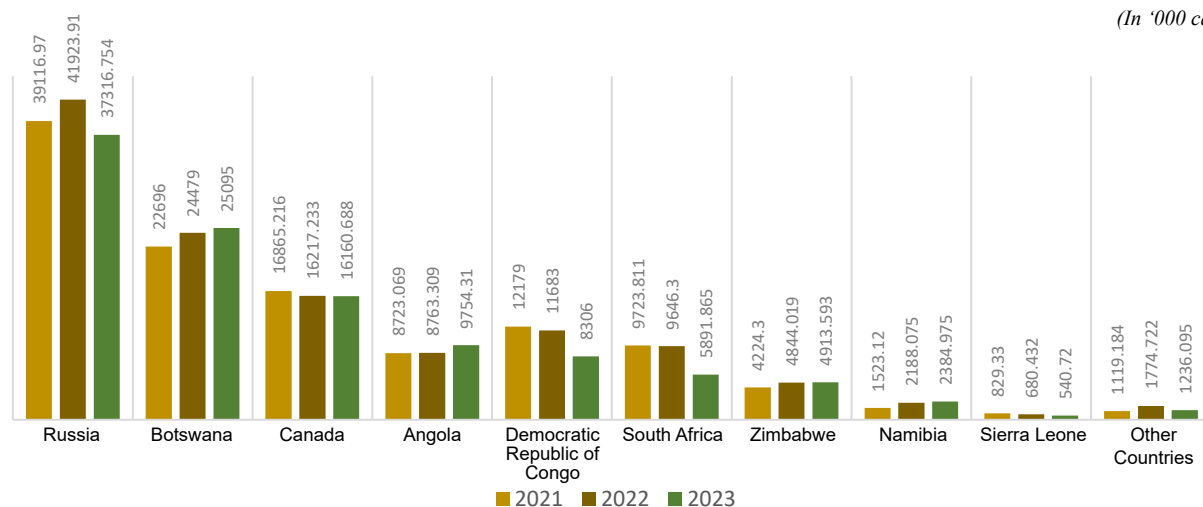


Fig. 1: Countrywise Production of Diamond

FOREIGN TRADE

Exports

The value of exports of diamond decreased by 24% to ₹ 1,39,513 crore in 2023-24 against ₹ 1,83,728 crore in the previous year. Diamond (mostly cut) alone accounted for almost 100% percent of total value of exports during

2023-24. The share of industrial diamonds and diamond powder in terms of value was about ₹ 60 crore and ₹ 29 crore, respectively in 2023-24. In terms of value, exports were mainly to USA (33%), Hong Kong (26 %), UAE (12%) and Belgium (12%) (Tables- 8 to 9).

Table – 8: Exports of Diamond

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	1837280521	**	1395132973
USA	**	637384858	**	463720271
Hong Kong	**	445586325	**	361282201
UAE	**	179476958	**	171172580
Belgium	**	205899789	**	164973192
Israel	**	98138821	**	62627261
Thailand	**	73636392	**	41657819
Switzerland	**	21483071	**	22365852
Japan	**	20960492	**	16389240
UK	**	15658219	**	13923652
France	**	10213315	**	9648636
Other Countries	**	128842281	**	67372269

Note: '**'Not additive
Figures rounded off.

Table – 9: Exports of Diamond

By Items

Items	2022-23 (R)		2023-24 (P)	
	Qty (Carats)	Value (₹ '000)	Qty (Carats)	Value (₹ '000)
Diamond (industrial)	12527927	901704	10217771	596444
Diamond (mostly Cut)	32096573	1836070859	26579149	1394245154
Diamond Powder	**	307958	**	291375

(P): Provisional

Imports

In 2023-24, imports of diamond, in terms of value, decreased by 14 % to Rs 1,78,731 crore from Rs 2,07,769 crore in the previous year. Diamond (mostly cut) accounted for the bulk, i.e., almost 99% percent of the imports in terms

of value. Imports of industrial diamond was about 17.98 million carats and valued at Rs 1538 crore and diamond powder was valued at Rs 25 crore. In terms of value, imports were mainly from UAE (41 %), USA (15%), Hong-Kong (14 %) Belgium (13 %), and Russia (5%) (Tables- 10 to 11).

Table – 10: Imports of Diamond

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty	Value (₹ '000)	Qty	Value (₹ '000)
All Countries	**	2077695939	**	1787312382
UAE	**	782076891	**	741015604
USA	**	409814405	**	276076620
Hong Kong	**	145599794	**	256255461
Belgium	**	373374794	**	227996603
Russia	**	96485452	**	95392018
South Africa	**	67268829	**	41533688
Israel	**	59362451	**	38773032
Canada	**	277239	**	31608746
Botswana	**	36637526	**	26608908

Table- 10 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty	Value (₹ '000)	Qty	Value (₹ '000)
Switzerland	**	11097174	**	16144977
Other Countries	**	95701384	**	35906725

Note: '**'Not additive
Figures rounded off.

Table – 11: Imports of Diamond

By Items

Items	2022-23 (R)		2023-24 (P)	
	Qty (Carats)	Value (₹ '000)	Qty (Carats)	Value (₹ '000)
Diamond (industrial)	20011091	16783936	17981091	15377560
Diamond (mostly Cut)	120861334	2057858821	137631888	1771687398
Diamond Powder	**	3053182	**	247424

FUTURE OUTLOOK

The Diamond Industry in the country currently employs over 8 lakh artisans who are experts in cutting and polishing of small diamonds and are now in a position to process full range of sizes and qualities of gemstones using latest technology. With production of about 90% of the world's polished diamonds, the Indian Diamond Industry is at the doorstep of major expansion today.

The Chinese Government has begun to initiate multi-billion dollar deals for rough diamonds in exchange for things that China produces like medicines, oils and industrial goods & services. Also, China's investment in Africa is a large threat to the Indian Diamond Cutting and Polishing Industry. There is a growing preference for polishing diamonds in countries where the diamonds are mined, like in Africa. It means that the Indian sector may

face problems as India is not a large producer, and depends on import of rough diamonds from Africa. Low profit margin in the cutting and polishing segment has boosted midstream players' interest in synthetic diamonds, but synthetics have to be contented with only limited acceptance among jewellery retailers and end consumers. Albeit, this technological advancement resulting in enhanced appeal for synthetic diamonds may propel demand in future.

With the support in the form of increasing urbanisation, middle-class expansion and appeal as engagement rings, India will emerge as a third largest market for diamond jewellery leaving behind Europe and Japan. The ever growing jewellery segment in India has prospects to become the market for the world and it can also play a crucial role in Indian economy. Meanwhile, China and the US are expected to remain as the leading diamond jewellery markets.



37. Diatomite

2.89 million tonnes
Total reserves/resources of diatomite have been established as on 1st April, 2020

2,658 tonnes
of diatomite were exported in 2023-24

2,396 tonnes
of diatomite were imported in 2023-24

Diatomite is a chalk-like, soft, friable, fine-grained siliceous sedimentary rock that is light in colour (white and yellowish)—white if pure, commonly buff to grey in situ. It is composed of fossilized remains of unicellular aquatic algae-like plants called diatoms which are both marine & lacustrine in origin. Diatoms are microscopic, single-celled organisms, often appearing as colonial aquatic plants (algae). Diatom cells contain an elaborate internal siliceous skeleton. More than 10,000 living diatom species have been identified, in addition to another 10,000 known diatom fossil forms. Diatomite deposits are the remains from the cell wall of diatoms, which are composed of amorphous hydrous silica. Diatomite has the ability to absorb soluble silica to form a highly porous skeletal framework. The dead remains of these diatoms over the ages have fossilized and formed the deep-bedded deposits in ocean & lake floors. The special properties of diatomite, i.e., light weight (low in density), high porosity and high absorptivity have facilitated its application as filter medium and absorbents. It is chemically inert & highly stable. Diatomite consists of approximately 90 per cent silica and the remainder consists of compounds, such as, aluminum and iron oxides. It is also called 'Kieselguhr'. Diatomaceous earth is a common alternate name, but the term is more appropriate for unconsolidated or less lithified rock of the same origin.

Another lesser known mineral, namely, siliceous earth has many similarities with diatomite in character. Siliceous earth is, however, an inorganic material which has chemical composition of more than 80% of amorphous silica. This amorphous phase is very rare and used widely in industrial-scale because of its high porosity, fine particle size, very low density and its high surface area. Its chemical and physical features are same as diatomite which is also amorphous silica consisting of fossilized remains of diatoms, a type of hard-shelled algae. Siliceous earth differs from diatomite in its origin and seems to be formed from volcanic ash. However, due to the similarities in both materials, siliceous earth finds similar uses as diatomite, and therefore, the same has been included in the Review.

RESERVES/RESOURCES

The occurrences of diatomite are reported from Gujarat, Rajasthan, Tamil Nadu, Andhra Pradesh and Camorta & Trincat Islands in Andaman and Nicobar archipelago. As per NMI database, based on UNFC system, the total

resources of diatomite as on 01.04.2020 has been estimated at 2.89 million tonnes, all of which fall under Remaining Resources. The resources are distributed in Rajasthan (72%) and Gujarat (28%) (Table-1).

Table – 1: Reserves/Resources of Diatomite as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	–	2885	2885
By Grade			
Unclassified	–	2885	2885
By States			
Gujarat	–	811	811
Rajasthan	–	2074	2074

Figures rounded off.

MINING LEASES & PRODUCTION

Diatomite

Production of diatomite has not been reported since 1991-92. Pandava and Khadriliya areas in Bhavnagar district, Gujarat, were the producing areas prior to 1991-92.

Siliceous Earth

As on 31.03.2024(P), the total number of mining leases granted for mining for mining of Siliceous Earth was 17

covering a total mining lease area of 168.4086 ha. The production of siliceous earth was 58,047 tonnes during 2023-24 (P) as against 32,070 tonnes in 2022-23 (Tables- 2 to 4). There were 10 reporting mines in the year 2023-24 (P) as against 12 reporting mines in 2022-23. Mine-head closing stocks of siliceous earth in the year 2023-24 (P) were 502,633 tonnes as against 38,543 tonnes in 2022-23 (Table-5). The average daily employment of labour in 2023-24 was 66 as against 99 in the previous year.

Table – 2: Principal Producers of Siliceous Earth, 2023-24

Name & address of producer	Location of the mine	
	State	District
Mr. Babu Lal Mali Akali Shiv, Barmer - 344701 Rajasthan	Rajasthan	Barmer
Mr. Narayan Singh Rathore Chandan Singh ki Dhani, Shiv, Barmer -344001 Rajasthan	Rajasthan	Barmer
P.P. Enterprises C/o Saturday Shoppe, E Sector, Shastri Nagar, Kalpataru Shoeing Center, Jodhpur -342003 Rajasthan	Rajasthan	Barmer
M/s. Ratan Lal Khatri Office No. 12, 2nd Floor Sethiya Complex, Near Hotel Kailash International, Choathan Road, Barmer -344001 Rajasthan	Rajasthan	Jaisalmer
M/s. Seema Minerals & Metals, Prop. Ramesh Chandra Ojha 203, A, Mewar Industrial Area, Madri, Udaipur -313003 Rajasthan	Rajasthan	Jaisalmer
Mr. Ishwar Singh Rathore Village- Jayani, Post- Jayal, Nagaur-341001 Rajasthan	Rajasthan	Jaisalmer

Table – 3: Production of Siliceous Earth, 2021-22 to 2023-24 (P)

(By States)

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	33898	22837	32070	17360	58047	33735
Rajasthan	33898	22837	32070	17360	58047	33735

(P): Provisional

(In tonnes)

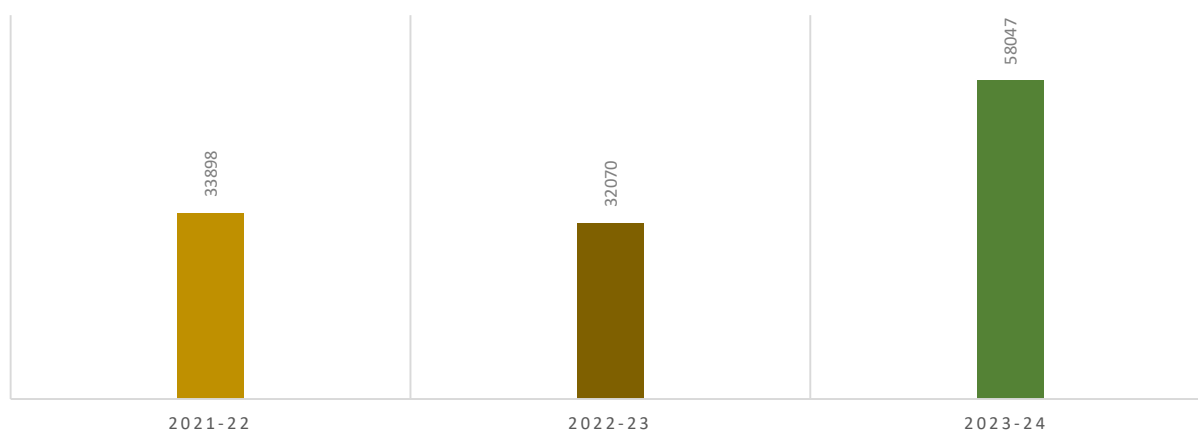


Fig. 1: Production of Siliceous Earth

Table – 4: Production of Siliceous Earth, 2022-23 and 2023-24 (P)
(By Sector/States/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	12	32070	17360	10	58047	33735
Private sector	12	32070	17360	10	58047	33735
Rajasthan	12	32070	17360	10	58047	33735
Barmer	6	22650	13390	4	36240	21729
Jaisalmer	6	9420	3970	6	21807	12006

(P): Provisional

Table – 5: Mine-head Closing Stocks of Moulding Sand, 2022-23 and 2023-24 (P)
(By States)

State	2022-23	2023-24 (P)
India	38543	502633
Rajasthan	38543	502633

(P): Provisional

USES & CONSUMPTION

Diatomite is commonly used after calcination in plate and frame filter units. Processed diatomite finds a wide range of applications due to its properties like porosity, low bulk density, soluble impurities, and high absorptive capacity for liquids, large surface area, low thermal conductivity, mild abrasive nature and chemical inertness.

Diatomite is an excellent filtering material for many liquids especially beverages, fruit juices, soft drinks, beer and wine. It is used in chemicals like sodium hydroxide, sulphuric acid and gold salts. Filtration of cooking oils (vegetable and animal) and sugar (cane, beet and corn) is an application where diatomite is predominantly used. It is also used as an absorbent of vegetable oil, polyethylene, rayon liquors and as a flattening agent in paint, plastic, rubber, drugs, toothpaste, polishes and chemicals. Diatomite is utilized for safe handling and storage of hazardous chemicals like sulphuric acid. Besides, diatomite is also used as an abrasive in metal polishing in automobiles and toothpastes;

as pozzolanic admixtures in Cement Industry; as animal feed stuff conditioners; and in explosives. It is also used as a coating material in the manufacture of ammonium nitrate fertilizer which is hygroscopic. Diatomite clay is the new revolution in hydroponics. In pharmaceuticals, it is used to filter syrups and other bulk drugs in liquid form. It is used as a facial exfoliator to promote skin health. Research has also shown that food-grade diatomaceous earth may offer positive benefits for controlling cholesterol levels which in turn would improve heart health. The cleaning of grease and oils continued to be the largest end-use for diatomite. In Oil Industry, before packing, it is used for filtering oil which not only gives it a shine but also helps in removing any suspended impurity. Wine and beer are filtered through diatomite filters before packing to remove molasses. Filter candles are made from diatomite filter aids for drinking water purification. Processed diatomite granules 15 to 50 mm, are used in denim wash (commonly known as stonewash) to give it shine and design.

Diatomite is also used as caking agent in fertilizers, plastics and as a natural insecticide for organic pest control. Potable water treatment and biological filtration are areas of expansion in diatomite consumption. Siliceous earth is mainly used as filler & filter, heat & sound-resistant material and in Ceramic Industry. Filtration and cleaning of vegetable oils and animal fats and manufacturing of medicines are other uses. In advance applications, it is used as carrier for catalyst in chemical processes and for mineral fertilizers & herbicides, pesticides and fungicides. It is also used as raw material for refinement and filtration as well as constituent of synthetic molding mass. It is increasingly used for the removal of microbial contaminants, such as, bacteria, protozoa and viruses, from public water systems. Recently, diatomite derived products included filter aids (50 per cent), filters (15 per cent), absorbents (5 per cent), and other minor uses, including specialized pharmaceutical and biomedical applications (less than 1 per cent).

Siliceous earth, on the other hand, is used in powder form which may vary from 80 mesh to 500 mesh powder. It is also used in granule form in some specific operations and in paint, filler, rubber, catalyst, fertilizer, pesticides, agriculture and many other industries.

Diatomite is increasingly used in agriculture segment in fungicides, insecticides and rodenticide owing to its dehydration properties. Other diatomite applications include its use as a growing medium in hydroponic gardens wherein its characteristics, such as, inertness, water holding capability and porosity allow the soil to breathe.

PROCESSING

Diatomite deposits are usually mined as open pit operations. If necessary, the mined crude is dried and crushed. Dried diatomite is collected in cyclones and fed through air separators to remove coarse material and impurities. Calcination and flux calcination are used to thermally volatilise organic material and oxidise iron. Calcination is also used to increase diatomic hardness, specific gravity and refractive index. The fusing of small diatomite particles into clusters can also be accomplished through calcination, which results in increased pore size and volume. Diatomite products are sold as various grades of calcined powders.

INDUSTRY

M/s Seema Minerals & Metals, Udaipur, Rajasthan, produces various grades of diatomaceous earth of which some grades are as follows:

- Diatomaceous Earth for insulation in Fire Proof Cabinets and safes.
- Diatomaceous Earth for calcium silicate boards.
- Diatomaceous Earth for water purification.

- Diatomaceous Earth for animal feed.
- Diatomaceous Earth for agriculture crops.
- Diatomaceous Earth for filtration.
- Calcined Diatomaceous Earth.
- Diatomaceous Earth Oil absorbents.
- Diatomaceous Earth for insecticides.

SUBSTITUTION

Many materials can be substituted for diatomite. However, the unique properties of diatomite assure its continued use in many applications. Expanded perlite and silica sand compete for filtration. Filters made from manufactured materials, notably ceramic, polymeric, or carbon membrane filters and filters made with cellulose fibers are becoming competitive as filter media. Alternate filler materials include clay, ground limestone, ground mica, ground silica sand, perlite, talc, and vermiculite. For thermal insulation, materials, such as, various clays, exfoliated vermiculite, expanded perlite, mineral wool and special brick can be used. Transportation costs will continue to determine the maximum economic distance that most forms of diatomite may be shipped. Diatomite still remains competitive despite availability of alternative materials. Many alternatives exist for diatomite as a pozzolan; however, its use as an ingredient of Portland cement has not diminished but on the contrary has increased in recent years. The encroachment of natural and synthetic substitute materials into diatomite markets has not been significant.

TRADE POLICY

As per the Foreign Trade Policy, 2015-2020, the imports and the exports of siliceous fossil meals (kieselguhr, tripolite, and diatomite) and similar siliceous earth, whether or not calcined under ITC (HS) Code 251200 (25121010, 25120020, 25120030 and 25120090) are free.

WORLD REVIEW

The World reserves of diatomite are adequate for the foreseeable future. The USA has the largest reserves at 250 million tonnes, followed by China with 170 million tonnes, Spain with 57 million tonnes and Turkey with 44 million tonnes. World's largest producing district in terms of volume is near Lompoc, California in USA (Table- 6).

The world's diatomite production was 2.40 million tonnes in 2023. The USA, with 849 thousand tonnes, dominated the world production by accounting for 35% outputs which was followed by China (15%), Turkey (6%), Republic of Korea (5%), Mexico, Denmark (moler), Argentina & Peru (4% each) and France & Spain (3% each). Production in Denmark was mostly of moler, an impure diatomite containing a large proportion of clay (Table- 7).

Table – 6: World Reserves of Diatomite

(By Principal Countries)

(In '000 tonnes)

Country	Reserves
World: Total (rounded)	Large
United States ^(a)	2,50,000
Argentina	NA
China	1,70,000
Czech Republic	NA
Denmark (Processed) ^(b)	NA
France	NA
Germany	NA
Korea, Rep. of	NA
Mexico	NA
Mozambique	NA
Peru	NA
Russia	NA
Spain	57,000
Turkey	44,000
Other countries	NA

Source: USGS, Mineral Commodity Summaries, 2024

NA - Not available

(a): Processed ore sold or used by producers.

(b): Include sales of moler production.

Table – 7: World Production of Diatomite

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World: Total	2500000	2400000	2400000
USA ^(d)	998000	827000	849000
China	317000	368000	370000*
Turkey	207732	237267	154444
Korea, Rep. of	49616	11880	120941
Mexico	140928	107060	102906
Denmark (moler) ^(a)	120625	143125	101875
Argentina	100000*	100000*	100000*
Peru	94860	104660	98872
France	90000*	80000*	80000*
Spain ^(b)	55982	87095	73997
Other countries	358175	372906	332639

Source: BGS, World Mineral Production, 2019-2023.

Note (S):

(1) In addition to the countries listed, Georgia and vietnam may produce daitomite.

(a): Moler is an impure diatomite containing a large proportion of clay

(b): Including the mineral Tripoli

(d): Sold or used by producers

*: Estimated

FOREIGN TRADE

Exports

Exports of diatomite increased by 40% to 2,658 tonnes in 2023-24 from 1,901 tonnes in the previous year. Exports were mainly to Vietnam (36%), Taiwan (27%), Philippines

(5%), Sri Lanka & Algeria (4% each), Tanzania, Singapore & Poland (3% each) and France (2%).

On the other hand, exports of Kieselguhr decreased substantially by 37% to 26 tonnes in 2023-24 from 41 tonnes in the previous year. Exports were mainly to Switzerland (100%) (Table 8-9).

Table – 8: Exports of Diatomite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1901	62622	2658	69569
Vietnam	238	4518	944	15171
Taiwan	717	11493	717	11353
Congo D. Rep.	--	--	40	6157
Tanzania	49	3550	75	5627
France	84	10464	61	5172
Algeria	--	--	104	4003
Singapore	68	3485	72	3943
Poland	15	534	70	2562
Philippines	--	--	136	2342
Sri Lanka	325	7934	108	2197
Other Countries	405	20644	331	11042

Figures rounded off

In tonnes

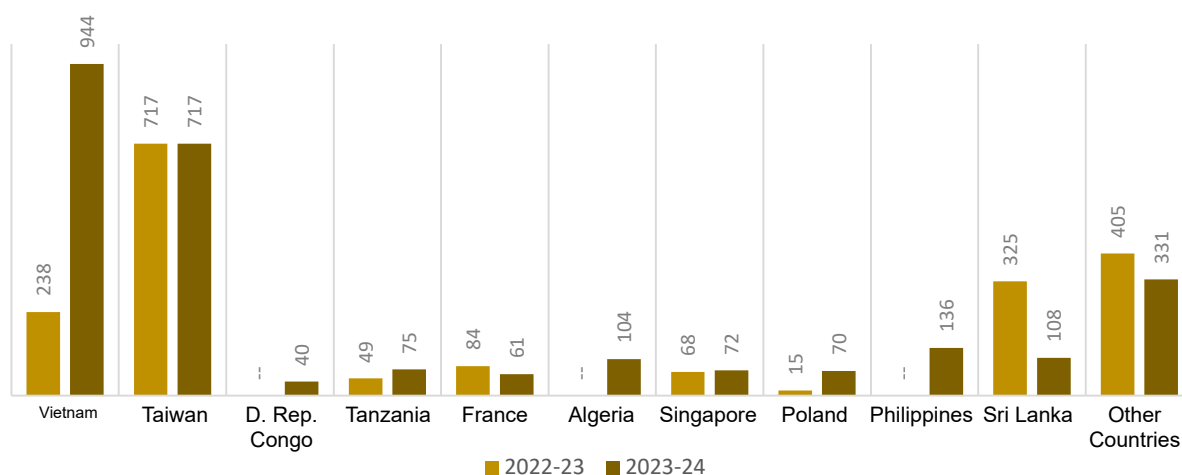


Fig 2: Countrywise Exports of Diatomite

Table – 9: Exports of Kieselguhr

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	41	3329	26	584
Switzerland	26	483	26	544
Philippines	++	25	++	34
Colombia	++	4	++	3
Malaysia	--	--	++	2
Russia	--	--	++	1
Bangladesh	15	2812	--	--
Uzbekistan	++	5	--	--

Figures rounded off

Imports

Imports of diatomite decreased by 36% to 2,396 tonnes in 2023-24 from 3,722 tonnes in the previous year. Imports were mainly from USA (43%), China (35%), Mexico (15%), Netherlands, Australia & Spain (2% each), France, Belgium & Turkey (1% each).

Imports of Kieselguhr were at 26 tonnes in 2023-24 which increased by 333% as compared to 6 tonnes in the previous year. Imports of Kieselguhr were mainly from China (77%) and USA (23%).

Imports of Tripoli Earth were 23 tonnes in 2023-24 as compared to negligible in the previous year (Table10-12).

Table – 10: Imports of Diatomite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3722	243082	2396	166678
USA	1452	102623	1022	82049
China	1625	88052	841	39921
Mexico	616	47695	369	30403
Spain	10	953	35	3450
Netherlands	3	594	40	3378
France	8	1889	19	2813
Turkey	--	--	11	2180
Belgium	--	--	19	1475
Australia	--	--	40	997
Korea, Rep. of	++	14	++	12
Other Countries	8	1262	++	++

Figures rounded off

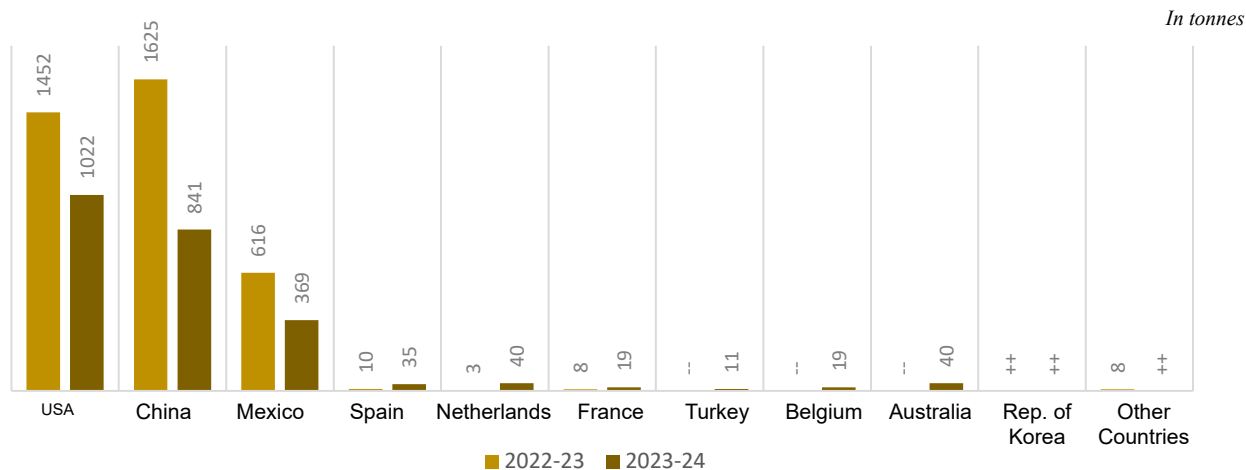


Fig. 3: Countrywise Imports of Diatomite

Table – 11: Imports of Kieselguhr

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	6	520	26	2413
China	--	101	20	1772
USA	5	388	6	343
Germany	--	--	++	248
Italy	--	--	++	27
UK	++	25	++	12
Belgium	++	4	++	11
Spain	--	2	--	--

Figures rounded off

Table – 12: Imports of Tripoli Earth

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	2	23	987
Spain	++	2	23	987

Figures rounded off

FUTURE OUTLOOK

Diatomite market is expected to witness substantial growth owing to increasing use in industrial applications including filter aid and filler material. Natural filtering characteristics due to its unique honeycomb structure and its rising use in filtration application, in food and beverage industry as well as human blood plasma and other biotechnical applications are key drivers for the growth of diatomite market.

Growing use of diatomite in industries, such as filtration, paints, plastics, insecticides, pharmaceuticals, chemicals, adhesives, sealants, paper etc. is expected to propel market growth over the forecast period. In addition, its use in Plastic Industry as an anti-blocking agent assists in separation of plastic parts during manufacturing and separation of plastic bags. Growing health and safety concerns associated with inhalation of crystalline silica during diatomite processing is expected to affect market growth.

North America accounted for largest market share in terms of demand owing to large amount of diatomite

reserves available in the region. In addition, increasing demand from various industrial applications including water treatment, absorbents and crop protection chemicals is expected to augment market growth. Asia Pacific accounted for second largest market share owing to increasing demand from crop protection chemicals and industrial applications, particularly in China. Furthermore, demand from countries, including India, Republic of Korea, Japan and Australia are expected to provide better opportunity for diatomite market. Europe contributed for a significant share of diatomite market owing to increasing demand from crop protection and water treatment particularly in Spain, Germany and United Kingdom. Middle East is expected to witness significant growth owing to increasing use of diatomite in various stages of oil & gas exploration.

The economic stability of diatomite was largely on account of its use as a viable filtration medium. The demand for diatomite as filtration medium still remains strong, particularly in the filtration of spirit as well as human blood plasma and in other biotechnical applications.



38. Emerald

55.87 tonnes total reserves/resources of emerald have been established as on 1 st April, 2020	₹ 1,935.58 crore value of exports of emerald (cut & uncut) were reported in 2023-24	₹ 3,473.90 crore value of imports of emerald (cut & uncut) were reported in 2023-24
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Emerald, commonly called as 'Panna' in Hindi, is a variety of mineral beryl and is chemically a complex silicate of aluminium and beryllium $[\text{Be}_3\text{Al}_2(\text{SiO}_3)_6]$. The hardness of emerald is about 7.5 to 8 on Moh's scale. Emeralds are rare & precious gemstones that command immense value on account of their cut, colour, clarity and carat. Emeralds occur in hues varying from yellow-green to blue-green. However, emeralds with unique velvety green colour and high degree of transparency are considered as most precious among gemstones. The rose variety of beryl is called morganite, and the golden yellow variety is termed 'golden beryl'. If the colour is bluish-green or blue then it is called 'aquamarine'. When emerald contains chromium as an impurity, it imparts green colour. Emerald is generally found at the contact of pegmatites intruding schists, mainly amphibole schists.

RESERVES/RESOURCES

As per NMI data, as on 01.04.2020 based on UNFC system, the total resources of emerald in the country is estimated at 55.87 tonne. All resources are placed under Reconnaissance category and unclassified grade resources have been reported in the State of Jharkhand (Table-1). Occurrences of emerald are also reported from Rajasthan, Odisha and Chhattisgarh. However, resources have not yet been estimated. In Rajasthan, emeralds occur at a number of places in districts of Rajsamand and Ajmer. In Ajmer–Rajsamand belt of Rajasthan, emerald occurrences

are confined to the upper fringes of pegmatites. Emerald deposit is found in the 221 km long belt stretching from Gam Gurha in Rajsamand district to Bubani and Muhami in Ajmer district. Important localities are Rajgarh, Tikhi and Kalaguman (Rajsamand). The occurrences are highly sporadic and variable. In Odisha, occurrences of emerald are reported from Bira–Mohorajpur belt in district Bolangir. In Chhattisgarh, occurrences of emerald are reported from Deobhog area in Raipur district. Sporadic and irregular crystals of emerald, aquamarine and amethyst are also reported in a few localities in district Coimbatore, Tamil Nadu.

Table – 1: Reserves/Resources of Emerald as on 01.04.2020

(By Grades/States)

(In kg.)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	-	55869	55869
By Grade			
Unclassified	-	55869	55869
By State			
Jharkhand	-	55869	55869

Figures rounded off

PRODUCTION

No production of emerald was reported during the years 2023-24 and 2022-23. The mine-head closing stocks of emerald at the end of the years 2023-24 and 2022-23 were 20 kg each (Table-2). The average daily employment was nil in both the years.

Table – 2: Mine-head Stocks of Emerald, 2022-23 & 2023-24

(By State)

(In kg)

State	2021-22	2022-23 (P)
India	20	20
Rajasthan	20	20

(p):provisional

MINING AND MARKETING

Mining of emerald was mostly undertaken manually by opencast method. The pits were worked at shallow depths. The stones collected during mining were sorted according to size, shape and quality. The stones so collected were deposited with the State Government under the supervision of DMG officials and mine owners. Worldwide, crude emerald is sold through public auctions from time to time.

The beauty of rough stone is enhanced by skillful cutting and polishing into faceted or rounded forms for use in jewellery. The rough stones are sorted out to determine the angles at which facets can be cut. They may then be sawed or polished in any direction according to shape, size and colour to be retained. After the shape and size are determined, the next process of 'pre-shaping' is done. The third and important process called 'calibration' gives the pre-shaped stone a definite proportion and size. This is followed by 'faceting' and 'polishing'. The aim of this process is to achieve maximum internal reflection to enhance the beauty of the stones. Emeralds are generally given a step-cut or cabochon-cut.

India is a leading exporter of cut and polished gems, but continues to be import-dependent for raw materials. The Diamond and Gem Development Corporation of

India has set up Diamond and Imported Gem Parks for cutting and polishing of gems in different States. Jaipur in Rajasthan is the major centre for processing emeralds. Like diamonds, uncut emerald is imported and part of it is exported after processing. Emerald ranks next to diamond (uncut) amongst precious and semi-precious stones, being imported and re-exported after cutting and polishing. Prices are governed by many factors including beauty, clarity, defect, demand, durability and rarity. Prices of precious stones also vary over time.

WORLD REVIEW

There is no information available regarding world production in World Mineral Production 2019–23, published in 2025 by the British Geological Survey (BGS), and also no information regarding world reserves in Mineral Commodity Summaries 2024, published by the United States Geological Survey (USGS).

FOREIGN TRADE

MMTC is an authorised agency of the Government of India for import of precious & semi-precious stones including emerald and for supplying these items to jewellers for domestic distribution and exports.

Exports

In 2023-24, exports of emerald (cut & uncut) by value increased by 13.21% to ₹ 1,935.58 crore from ₹ 1,709.69 crore in the previous year. Exports were mainly to Hong Kong (42.21%), USA (21.76%), Thailand (15.02%), Italy (4.54%), Switzerland (4.37%), Belgium (2.48%), France (2.32%), UAE (1.80%), Armenia and UK (less than 1% each). Out of the total export value of emerald (cut & uncut), the share of the export value of emerald (cut) was about 99.48% and emerald (uncut) about 0.52%. The export value of emerald (cut) increased by 13.25% to ₹ 1,925.47 crore in 2023-24 from ₹ 1,700.22 crore in the preceding year. In terms of quantity, the export of emerald (cut) decreased by nearly 2.03% to 7,255.72 thousand carat in 2023-24 from 7,406.27 thousand carat in the preceding year. On the other hand, the share of the export value of emerald (uncut) was negligible, i.e., less than 1%. The export value of emerald (uncut) increased by 6.83% to ₹ 10.11 crore in 2023-24 from ₹ 9.47 crore in the preceding year (Tables-3 & 4).

Table – 3: Exports of Emerald (Cut & Uncut): Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	17096852	**	19355783
Hong Kong	**	4621626	**	8170241
USA	**	4358699	**	4212713
Thailand	**	3200452	**	2907951
Italy	**	1115218	**	880597
Switzerland	**	943427	**	846482
Belgium	**	310673	**	480717
France	**	460211	**	450202
UAE	**	511070	**	350037
Armenia	**	48422	**	184734
UK	**	242666	**	181841
Other Countries	**	1284388	**	690268

Figures rounded off

** : Not additive

(P):provisional

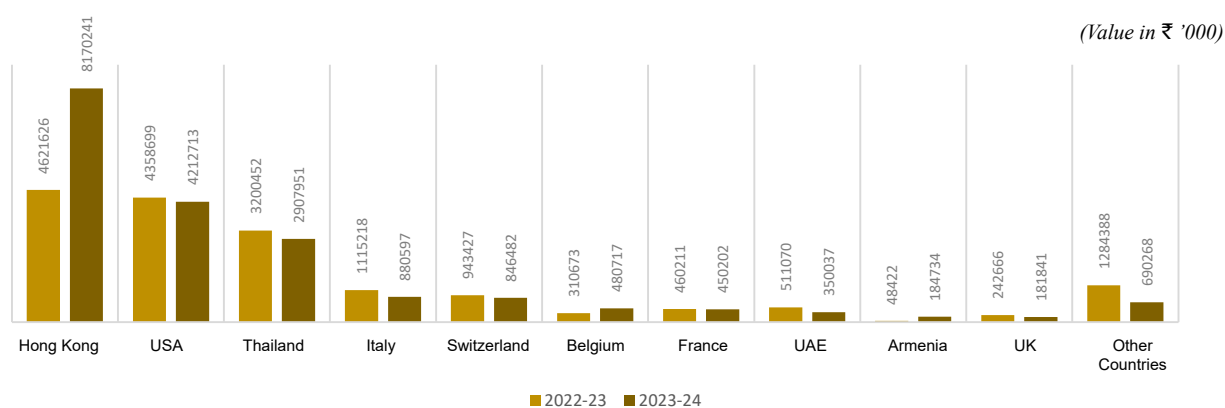


Fig. 1: Countywise Exports of Emerald (Cut & Uncut)

Table – 4: Exports of Emerald

(By Items)

Item	2022-23 (R)		2023-24 (P)	
	Qty ('000'carat)	Value (₹ '000)	Qty ('000'carat)	Value (₹ '000)
Emerald (Cut)	7406273	17002198	7255717	19254658
Emerald (Uncut)	2	94654	++	101125

Figures rounded off

++ : Negligible

(p):provisional

Imports

In 2023-24, the total imports of emerald (cut and uncut) by value increased by 7.44% to ₹ 3,473.90 crore from ₹ 3,233.40 crore in the preceding year. Imports were mainly from Zambia (35.60%), Hong Kong (31.92%), USA (9.87%), UAE (6.70%), Thailand (5.36%), Brazil (2.24%), Switzerland (2.19%), Belgium (1.41%), Columbia (1.37%) and France (0.81%). In the total import value of emerald (cut & uncut), the share of import value of emerald (cut) was 34.51%, while the share of import value of emerald

(uncut) was 65.5%. The import value of emerald (cut) increased by 32.43% to ₹ 1,199.10 crore in 2023-24 as compared to ₹ 905.46 crore in the preceding year. In terms of quantity, the imports of emerald (cut) increased by 3466% to 94,568.93 thousand carat in 2023-24 from 2,651.67 thousand carat in the preceding year. The import value of emerald (uncut) decreased by 2.3% to ₹ 2,274.80 crore in 2023-24 as compared to ₹ 2,327.94 crore in the preceding year. In terms of quantity, the imports of emerald (uncut) also decreased slightly by 1.57% to 63 tonnes in 2023-24 from 64 tonnes in the preceding year (Tables-5 & 6).

Table – 5: Import of Emerald (Cut and Uncut): Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	32333990	**	34739081
Zambia	**	12918166	**	12360201
Hong Kong	**	5406474	**	11087783
USA	**	3435993	**	3429637
UAE	**	2543346	**	2330841
Thailand	**	2173187	**	1860990
Brazil	**	707479	**	779136
Switzerland	**	983918	**	759395
Belgium	**	1102653	**	490161
Colombia	**	186489	**	476445
France	**	266385	**	283046
Other Countries	**	2609900	**	881446

Figures rounded off

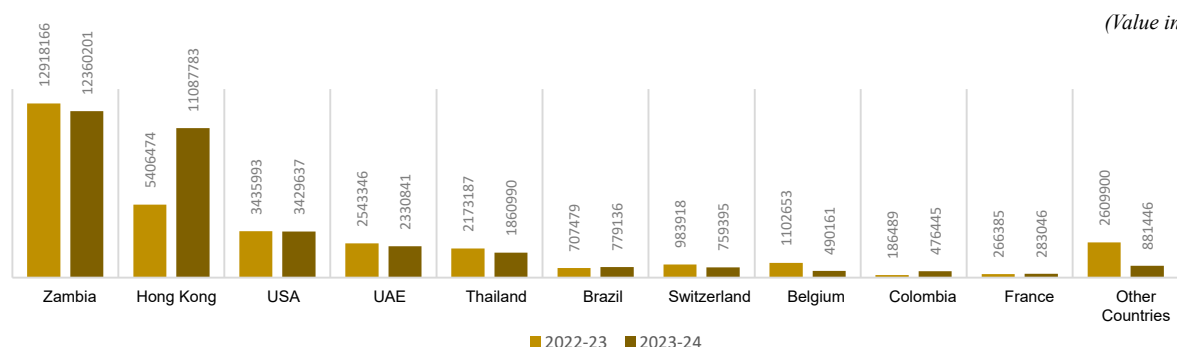


Fig. 2: Countrywise Value of Import of Emerald (Cut & Uncut)

Table – 6: Imports of Emerald

(By Item)

Item	2022-23 (R)		2023-24 (P)	
	Qty ('000'carat)	Value (₹ '000)	Qty ('000'carat)	Value (₹ '000)
Emerald (Cut)	2651674	9054590	94568931	11991030
Emerald (Uncut)	64	23279400	63	22748051

Figures rounded off

(P):provisional

FUTURE OUTLOOK

The Gems and Jewellery industry of India contributes 7% to India's Gross Domestic Product (GDP). This Sector is export oriented and labour intensive, contributing significantly to foreign exchange earnings and employment of over more than five million skilled and semi-skilled workforces in the country. The Sector contributes about 15.71% of India's total merchandise exports, accounting for the third largest commodity share. With a view to strengthen the Gems & Jewellery Industry in the country, the Government has taken a number of steps, such as, establishment of Special Notified Zone (SNZ); announcement of separate ITC HS Code for lab-grown diamonds; launch of Gold Monetisation Scheme; amendment of wastage and value addition norms for gems and jewellery items; approval of a scheme for setting up of Common Facility Centres for Gem and Jewellery Sector; and provision of financial assistance for

participation in international fairs, organisation of buyer-seller meets etc. under Market Development Assistance (MDA) and Market Access Initiative (MAI) Schemes of the Department of Commerce. The demand for emerald gemstones in India is projected to see considerable growth due to the rising purchasing power of the growing affluent class in the country. The Coloured Gemstones market in India is estimated to be valued at USD 0.9 billion in 2025 and is projected to reach USD 2.3 billion by 2035, growing at a CAGR of 10.5% from 2025 to 2035. Among various coloured gemstones, emeralds are the leading segment in demand, projected to hold approximately 41.7% of total market demand by 2025 within the stone type category. Growth is driven by emerald's prominence in fine jewelry, cultural symbolism, perception as a luxury stone, and its adoption as an investment asset. India, with its domestic cutting and polishing capabilities, have established itself as a significant player in the emerald craftsmanship domain.



39. Fluorite

6,042 tonnes
production of fluorite (graded) have
been reported in 2023-24

1,358 tonnes
of fluorite were exported in 2023-24

2,56,110 tonnes
of fluorite were imported in 2023-24

Fluorite (CaF_2), commercially known as fluorspar, occurs naturally in a wide range of colours across the spectrum. It is one of the most colour-varied minerals in the mineral kingdom. Pure fluorite is colourless, and its wide range of colours results from various impurities. The colours may vary from purple, blue, green, yellow, colourless, and brown to pink, black, and reddish-orange. The rich purple variety of fluorite is the most well-known and popular colour of the mineral. Fluorite serves as an important commercial source of fluorine. It plays a vital role in various manufacturing industries, with major consumers being the chemical, cement, iron and steel, and electrode industries. It is also used in the production of synthetic cryolite, which is essential for aluminium extraction.

Fluorite is commonly formed through hydrothermal processes associated with igneous intrusions. Mineralisation typically occurs as vein or replacement deposits, either through the filling of cavities and fissures or by replacement of the host rock, generally carbonates.

Mainly two grades of fluorite are recognised in consumption and trade, namely, the acid grade (acid spar), containing more than 97% CaF_2 , and the sub-acid grade, containing 97% or less CaF_2 . The sub-acid grade includes metallurgical (60–85% CaF_2) and ceramic (85–95% CaF_2) grades and is generally referred to as metallurgical grade (met spar). Fluorite production in the country is meagre compared with global production.

In addition to natural fluorite, synthetic fluorite is recovered as a by-product during uranium processing, petroleum alkylation, and stainless steel pickling. Furthermore, fluorosilicic acid, a by-product obtained from phosphoric acid plants during the processing of phosphate rock, also serves as an alternative source of fluorine, supplementing natural fluorite.

RESERVES/RESOURCES

As per NMI database, based on the UNFC system, the total reserves/resources of fluorite in the country as on 1.4.2020 has been estimated at 20.99 million tonnes. Out of these, 0.40 million tonnes are placed under Reserves category (of which 0.23 million tonnes are under Proved category and 0.17 million tonnes under Probable category). The Remaining Resources comprise 20.59 million tonnes.

By states, Gujarat accounts for 68% of the total reserves/resources having 14.36 million tonnes, followed by Rajasthan with 5.60 million tonnes (27%), Chhattisgarh 0.54 million tonnes (3%) and Maharashtra 0.49 million tonnes (2%). Gradewise, the resources are classified into Marketable grade which accounted for 82% of the total resources followed by Low grade (15%), Unclassified grade (2%) and Not-known grade around 2% (Table-1).

Table – 1: Reserves/Resources of Fluorite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	404241	20588239	20992480
By Grades			
Marketable	404241	16731425	17135666
Not-Known	-	332057	332057
Low	-	3169481	3169481
Unclassified	-	355276	355276
By States			
Chhattisgarh	-	545455	545455
Gujarat	-	14355280	14355280
Maharashtra	386142	100000	486142
Rajasthan	18099	5587504	5605603

Figures rounded off

MINING LEASES & PRODUCTION

As per information available, as on 31.3.2024 (P), there were a total of 7 fluorite mining leases covering an area of 416.158 hectares.

The production of fluorite (graded) at 6,042 tonnes increased by 423% in 2023-24 as compared to that in the previous year.

There were three reporting mines in 2023-24 and four

in 2022-23. The 87% (5,260 tonnes) of production was reported by one private sector mine located in Rajasthan and 13% (782 tonnes) of production was reported by two public sector mines located in Maharashtra.

The mine-head closing stocks of fluorite (graded) was 11,95,593 tonnes in 2023-24 as against 98177 tonnes in 2022-23 (Table-2 to 5).

The average daily labour employed in fluorite mines in 2023-24 and 2022-23 was 55 and 52 respectively.

Table – 2: Producers of Fluorite, 2023-24

Name & address of producer	Location of the mine	
	State	District
Maharashtra State Mining Corporation Ltd, Plot No. 7, Ajni Square, Wardha Road, Nagpur-440 015, Maharashtra.	Maharashtra	Chandrapur
Mr. Bhanvar Lal Bisnoi Lumbanio ki Dhani Digaon, Raniwara, Jalore-343040, Rajasthan"	Rajasthan	Jalore

Table – 3: Production of Fluorite (Graded), 2021-22 to 2023-24

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	1237	9430	1155	6224	6042	5716
Gujarat	-	-	100	330	-	-
Maharashtra	1237	9430	910	5850	782	4658
Rajasthan	-	-	145	44	5260	1058

(p): Provisional

Table – 4: Production of Fluorite, 2022-23 & 2023-24**(By Sector/States/Districts)***(Qty in tonnes, Value in ₹'000)*

State/District	2022-23			2023-24 (p)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	4	1155	6224	3	6042	5716
Public Sector	3	1010	6180	2	782	4658
Private Sector	1	145	44	1	5260	1058
Gujarat	2	100	330	-	-	-
Vadodara	2	100	330	-	-	-
Maharashtra	1	910	5850	2	782	4658
Chandrapur	1	910	5850	2	782	4658
Rajasthan	1	145	44	1	5260	1058
Jalore	1	145	44	1	5260	1058

*(p): Provisional;***Table – 5: Mine-head closing stocks of Fluorite, 2022-23 and 2023-24****(By States)***(In tonnes)*

State	2022-23	2023-24 (P)
India	98177	1195593
Gujarat	84396	1012684
Maharashtra	13636	159944
Rajasthan	145	22965

(p): Provisional

BENEFICIATION

Fluorite (CaF_2) is primarily beneficiated through hand sorting, followed by gravity concentration methods such as heavy media separation, jigs, and tables. These processes effectively separate calcite and silicate mineral impurities, enhancing the quality of the fluorite concentrate. The low-grade fluorite produced is subsequently utilised in various industries after beneficiation. GMDC operates a beneficiation plant with a capacity of 500 tonnes per day (TPD) located at Kadipani, Gujarat. This plant processes low-grade fluorspar ore, typically containing 23–25% CaF_2 , upgrading it to produce; Acid Grade Fluorite with CaF_2 content of 96%, and Metallurgical Grade Fluorite with a CaF_2 content of 90%. Additionally, it has the facility to produce MFC & MET grade powder analysing 75 to 85% CaF_2 & 85 to 92.5% CaF_2 , respectively and other products, such as, starch briquettes (81% CaF_2 min.) and silicate briquettes (79% CaF_2 min.).

USES & CONSUMPTION

Acid grade fluorite is used as a principal feedstock in the manufacture of hydrofluoric acid (HF), which in turn is essential for producing aluminium fluoride (AlF_3). The major use of HF is for the production of a wide range of fluorocarbon chemicals, including hydrofluorocarbons (HFCs), hydrochlorofluorocarbons (HCFCs), and fluoropolymers. However, owing to environmental concerns, a part of chlorofluorocarbons (CFCs) has been replaced by hydrochlorofluorocarbons (HCFCs). HF is used

in the manufacture of uranium tetrafluoride, an important ingredient used for producing nuclear fuel and fission explosives. It is also used in stainless pickling, petroleum alkylation, glass etching, and oil & gas well treatment and as etcher/cleaner in Electronic Industry.

HF is used in the manufacture of a host of fluorine chemicals used in dielectrics, metallurgy, wood preservatives, herbicides, mouthwashes, decay-preventing dentifrices, plastics and water fluoridation. Aluminium fluoride serves as a flux in the electrolytic recovery of aluminium metal. On an average, worldwide consumption of fluorides is about 21 kg for every tonne of aluminium produced. This ranged from 10 to 12 kg per tonne in a modern pre-baked aluminium smelter and about 40 kg in an old Soderberg cell without scrubber.

Ceramic grade fluorite containing 85 to 95% CaF_2 is used in Ceramic Industry as a flux and as an opacifier in the production of flat glass, white or opal glass and enamels. The addition of 10–30% Ceramic grade fluorspar to glass makes it opaque, white and opalescent. It is also used in the manufacture of magnesium, some manganese chemicals and welding rod coating.

Metallurgical grade fluorite is used primarily as fluxing agent by Steel Industry. It is added to slag to make it more reactive through increased fluidity. Fluorite of varying grades (including ceramic and metallurgical) finds application in the manufacture of aluminium, cement, and

glass fibres. It is also used in the melt shop by Foundry Industry.

The apparent consumption of fluorite was about 2,60,794 tonnes in 2023-24, as against the 3,10,304 tonnes in 2022-23.

INDUSTRY

Many fluorine-based chemicals like hydrofluoric acid, aluminium fluoride, cryolite, sodium silicofluoride and hydrofluorosilicic acid are produced by Chemical and Fertilizer industries in the country. In addition to material produced indigenously, substantial quantity of high grade fluorite was also imported to meet the demand of the fluorine-based Chemical Industries.

Tanfac Industries Limited is established as a joint sector collaboration between Anupam Rasayan India Limited and Tamil Nadu Industrial Development Corporation (TIDCO) at Cuddalore, Tamil Nadu. It is engaged in the manufacture of fluorine chemicals, such as, aluminium fluoride, anhydrous hydrofluoric acid, sodium silicofluoride, ammonium bifluoride, potassium fluoride, and various other fluorine-based chemicals. It is the largest producers of Anhydrous Hydrofluoric Acid and Aluminium Fluoride in India. The Company has an annual installed capacity of 15,600 tonnes each of aluminium fluoride and anhydrous hydrogen fluoride, 67,200 tonnes of sulphuric acid, and 3,400 tonnes of speciality fluorides. The successful commissioning of the state-of-the-art hydrofluoric acid unit in October 2024 has doubled the capacity from 14,850 TPA to 29,700 TPA, making the Cuddalore site one of the largest in the country. A landmark achievement was the commissioning of a 10,000 TPA solar-grade dilute hydrofluoric acid facility in June 2025. This plant is the first of its kind in India, supporting the domestic solar manufacturing value chain and establishing the company as a key supplier within the clean energy ecosystem. The company plans to expand its production capacity and diversify its product offerings to meet the increasing demand for speciality chemicals. Strategic partnerships, such as the recent agreement with a leading Japanese firm for refrigerant gas supply, will further strengthen Tanfac's market position.

Navin Fluorine International Ltd (NFIL) is an Indian

manufacturers of speciality fluorochemicals, established in 1967. Its headquarter is in Mumbai, and manufacturing units in Surat (Gujarat) and Dewas (Madhya Pradesh), along with one unit of Navin Fluorine Advanced Sciences Limited, wholly-owned subsidiary, in Dahej (Gujarat). It belongs to the Padmanabh Mafatlal Group – one of India's oldest industrial houses. NFIL has an installed capacity of about 22,000 tpy of hydrofluoric acid. The Company produces a number of fluorine chemicals, namely, hydrofluoric acid, cryolite, aluminium fluoride and various other organic and inorganic fluorine-based chemicals.

Apatite and rock phosphate containing 3 to 4% CaF_2 was another useful source for recovery of fluorine.

SUBSTITUTES

There is a strong global trend to replace fluorite and fluorite-based chemicals with more environment friendly alternatives in many industrial applications. There is the possibility to replace fluorite-based chemicals with hydrocarbon-based refrigerants such as butane and propane. In the Steel and Iron industry, Olivine or dolomitic limestone can be used as a substitute of fluorite. Other possible alternatives are calcium aluminate/chloride, aluminium smelting dross, borax, iron oxides, manganese ore, silica sand and titanium oxide. In electrolysis-based aluminium production, fluorspar cannot be substituted; however, fluorspar-free carbothermic reduction processes are considered conceivable. The by-product of fluorosilicic acid from phosphoric acid production could also be used as a substitute in aluminium fluoride production.

WORLD REVIEW

The world total reserves of fluorite/fluorspar were at 280 million tonnes. World reserves are concentrated mainly in Mexico & China (24% each), South Africa (15%), Mongolia (12%), Spain (5%), Iran (2%) and Vietnam (1%) (Table- 6).

World production of fluorite/fluorspar in 2023 increased marginally by 1.22% to 8.30 million tonnes as compared to 8.2 million tonnes in the previous year. China (69%), Mexico (12%), South Africa (5%), Mongolia (3%), Thailand, Vietnam & Spain (2% each) and Iran, France & Morocco (1% each) were the principal producing countries of fluorite/ fluorspar in 2023 (Table-7).

Table – 6: World Reserves of Fluorite
(By Principal Countries)

<i>(In '000 tonnes)</i>	
Country	Reserves
World total (rounded)	2,80,000
Mexico	68,000
China	67,000
South Africa	41,000
Mongolia	34,000
Spain	15,000
Iran	4,500

Table- 6 (Concl'd.)

(In '000 tonnes)

Country	Reserves
Vietnam	3,400
United States	NA
Germany	NA
Pakistan	NA
Other countries	50,000

Source: U.S.G.S., Mineral Commodity Summaries, 2024

NA: Not Available

Table – 7: World Production of Fluorite
(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World total	8300000	8200000	8300000
China	5700000	5700000	5700000*
Mexico	100718	990079	988972
South Africa	403000	406000	410000*
Mongolia	118300	122900	222400
Thailand	9766	59243	187566
Spain	187225	203768	146975
Vietnam	215027	217975	146043
Iran ^(c)	110000*	110000*	110000*
France	80000*	84112	81301
Morocco	189012	77035	77000*
Other countries	1186952	228888	239743

Source: BGS, World Mineral Production, 2019-23

*: Estimated

c: Year ended 20th March following that stated

In tonnes

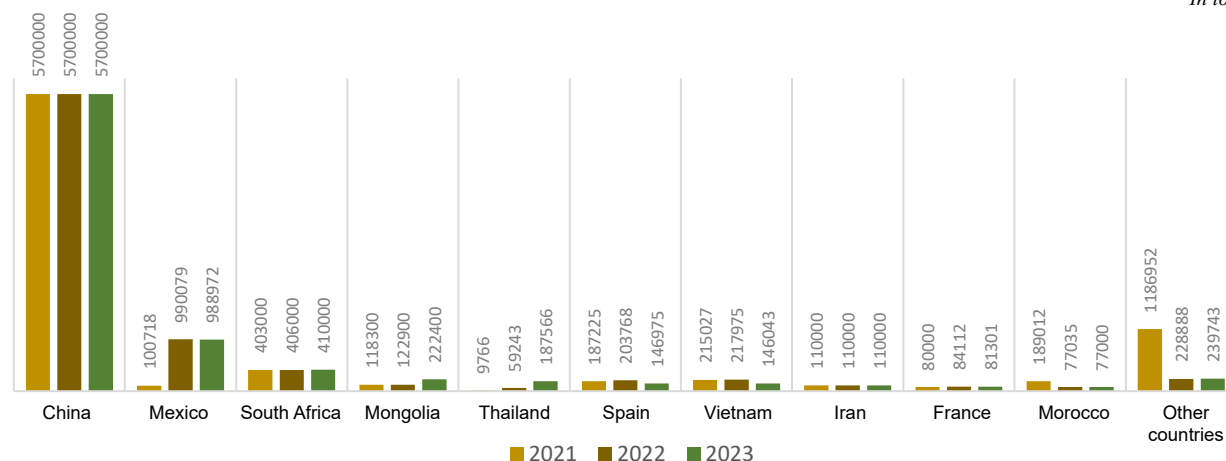


Fig. 1: Countrywise Production of Fluorite

FOREIGN TRADE

Exports

Exports of fluorite increased by 32.62% to ,1358 tonnes in 2023-24 from 1,024 tonnes in the previous year. Exports were mainly to Indonesia (43%), Bangladesh (17%), U.A.E & Egypt (12% each), Kuwait (3%), Turkey, Kenya & Nigeria (2% each) and Malaysia & Brazil (1% each). While exports of aluminium fluoride increased drastically by 440.92%

to 2,115 tonnes in 2023-24 as compared to 391 tonnes in the previous year. Exports were mainly to Turkey (82%), U.A.E (14%) and Japan (4%). Exports of hydrofluoric acid increased by 56.67% to 1,291 tonnes in 2023-24 as compared to 824 tonnes in the preceding year. Exports were mainly to U.S.A. (50%), Thailand (28%), Saudi Arabia (7%), Republic of Korea, Japan & Australia (3% each), Turkey (2%) and Russia, Taiwan & Malaysia (1% each) (Tables-8 to 10).

Table – 8: Export of Fluorite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1024	65899	1358	81974
Indonesia	427	29879	581	38370
UAE	19	2490	161	11192
Egypt	163	10920	160	10576
Bangladesh	227	9426	233	8426
Turkey	1	63	31	2001
Kenya	33	1907	28	1896
Nigeria	--	--	28	1592
Kuwait	--	--	40	1514
Malaysia	--	--	20	1178
Brazil	29	2554	13	1121
Other Countries	125	8660	63	4108

Figures rounded off

--; Nil

In tonnes

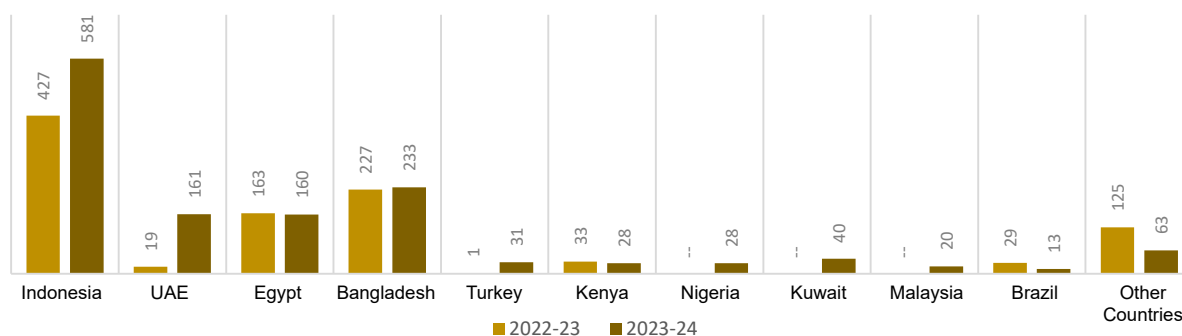


Fig .2: Countrywise Export of Fluorite

Table – 9: Export of Aluminium Fluoride

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	391	50222	2115	50925
Turkey	10	1055	1725	29903
Japan	180	24217	90	11953
UAE	200	24513	300	8746
USA	--	--	++	297
Germany	1	373	++	21
Bhutan	--	--	++	5
Thailand	--	--	++	++
Belgium	++	62	--	--
Uganda	++	2	--	--
Other Countries	++	++	++	++

Figures rounded off

--; Nil

++; Negligible

Table–10: Export of Hydrofluoric Acid**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	824	113899	1291	167582
USA	98	11482	651	74767
Thailand	441	63476	366	50499
Saudi Arabia	--	--	90	9124
Russia	--	--	12	8326
Korea Rep.of	--	--	36	6184
Taiwan	--	--	11	4985
Japan	--	--	34	3610
Australia	--	--	37	2620
Turkey	206	31719	21	1953
Malaysia	--	--	7	1695
Other Countries	79	7222	26	3819

*Figures rounded off**--; Nil***Imports**

Imports of fluorite decreased by 17.43% to 2,56,110 tonnes in 2023-24 as compared to 3,10,173 tonnes in the previous year. Imports were mainly from South Africa (58%), China (16%), Thailand (11%), Hong Kong (8%), Luxembourg & Morocco (2% each) and Spain, Belgium, Vietnam & Turkey (each <1%). However, imports of aluminium fluoride, decreased slightly by 0.5% to 54,864 tonnes in 2023-24 from

55,141 tonnes in the previous year. Imports were mainly from Italy (41%), China (19%), Bahrain (12%), Indonesia (7%), Mexico (4%), Sweden, UAE & Jordan (3% each) and Croatia & U.K (1% each). While imports of hydrofluoric acid increased significantly by 78.68% to 6,663 tonnes in 2023-24 from 3,729 tonnes in the preceding year. Imports were mainly from China (86%), Taiwan (7%), UAE & Malaysia (3% each) and Indonesia (1%) (Tables- 11 to 13).

Table –11: Import of Fluorite**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	310173	11509548	256110	10416654
South Africa	223139	8265794	149395	6304582
China	35394	1565500	40190	1743548
Hong Kong	9403	393720	19456	838755
Thailand	30296	890976	28274	803982
Luxembourg	2458	107292	5928	264671
Morocco	126	4722	5845	263647
Spain	1015	27736	1335	44617
Belgium	988	34161	607	29698
Vietnam	2879	68286	1006	29481
Turkey	1843	37551	1294	28734
Other Countries	2632	113810	2780	64939

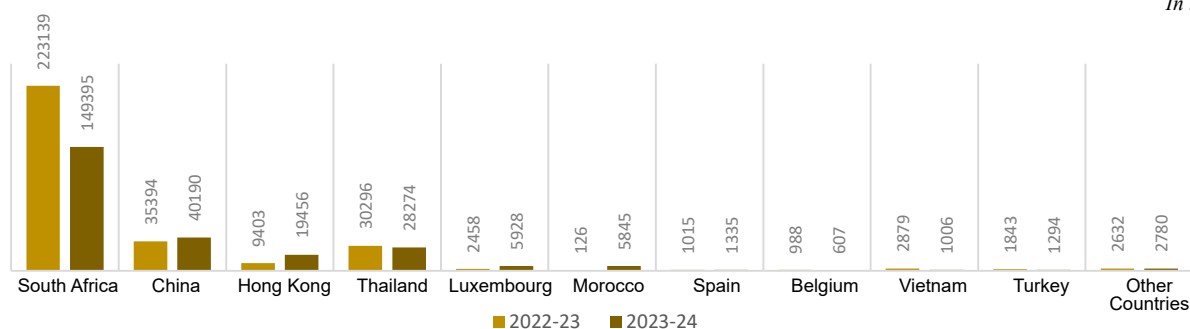
*Figures rounded off**In tonnes***Fig. 3: Countrywise Import of Fluorite**

Table – 12: Import of Hydrofluoric acid

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3729	440788	6663	734419
China	2340	288521	5706	622829
Taiwan	1035	101288	490	43484
UAE	107	22631	217	41985
Malaysia	--	--	182	16324
Indonesia	--	--	49	5996
Sri Lanka	247	24602	19	2067
Germany	++	2292	++	759
France	++	438	++	637
Sweden	++	132	++	267
USA	++	449	++	71
Other Countries	++	435	++	++

Figures rounded off

--; Nil

++; Negligible

Table – 13: Import of Aluminium Fluoride

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	55141	5621747	54864	5120665
Italy	2862	403569	22275	2440990
China	30255	3613047	10609	1220185
Indonesia	5220	506585	3780	371831
Mexico	4625	485342	2350	232875
Sweden	235	31821	1614	175467
Bahrain	2331	21569	6680	168450
UAE	3126	438934	1438	160343
Jordan	--	--	1528	128187
Croatia	22	1692	700	65971
UK	--	--	540	51591
Other Countries	6465	119188	3350	104775

Figures rounded off

--; Nil

FUTURE OUTLOOK

India's chemical industry is a key pillar of the country's manufacturing and economic architecture, contributing approximately 7% to the national gross domestic product. It supplies essential raw materials to several high-impact industries, including agriculture, pharmaceuticals, textiles, automotive, and construction. With its strategic significance well established, India has emerged as the sixth-largest producer of chemicals globally and the third-largest in Asia. In 2023, the domestic chemicals market was valued at approximately USD 220 billion. The sector is expected to grow significantly, reaching between USD 400 and 450

billion by 2030, and is further projected to reach between USD 850 and 1,000 billion by 2040, subject to continued policy support and infrastructure development. While this future growth outlook is promising, India's global footprint in the chemicals space remains relatively limited. As of 2023, the country's share in global chemicals consumption stood at 3–3.5%.

With steady growth and demand across semiconductors, electronics, and other sectors like fluorocarbons and inorganic fluorine, the global fluorochemicals market is anticipated to grow at a 4% CAGR over the next 10 years, reaching USD 36 billion from the current level of USD 25

billion, with the Asia Pacific region accounting for over 40% of global demand, buoyed by increased demand from India due to large investments in infrastructural developments and growth of the automotive and aerospace industries.

During the next 10 years, the Indian fluorochemicals market is expected to grow at over 10% CAGR due to the factors mentioned above and the 'Make in India' initiative. Rapid urbanisation, a rising middle class, and the recent announcement of GST rationalisation to boost consumption across sectors are expected to further fuel growth in the demand for refrigerant gas. Moreover, the rising application of aluminium in the automotive industry is spurring the demand for fluorochemicals. In India, the

market growth of fluorochemicals is driven by downstream sectors like automobile, air conditioning, refrigeration, construction, cold storage and pharma / life science segments and this demand is currently fulfilled by mostly imports which contribute close to 95% of demand.

As on 01.04.2020, the resources of fluorite in India are 20.99 million tonnes which is considered to be limited. Hence, to meet the requirements, the domestic Chemical Industry will have to depend (both qualitatively and quantitatively) on imported fluorite in the coming years, both for direct use and for blending with the domestic Acid grade fluorite.



40. Garnet

23,838 tonnes
production of garnet (abrasive) have
been reported in 2023-24.

1,46,915 tonnes
of garnet (abrasive) were exported in
2023-24.

3,086 tonnes
of garnet (abrasive) were imported in
2023-24.

Garnet is a collective name for a group of minerals which crystallise in the cubic system with different chemical composition. The principal members of the Garnet group are Almandine (Fe-Al), Pyrope (Mg-Al), Spessartine (Mn-Al), Grossularite (Ca-Al), Andradite (Ca-Fe) and Uvarovite (Ca-Cr). Among these, Almandine is the hardest variety and is often used for abrasive purposes. Garnet is dense & hard with sharp angular chisel-edged fracture, containing small amounts of free silica and exhibits high resistance to physical and chemical attacks. It is used both as semi-precious stone and as an abrasive. The hardness of garnet varies from 6.5 to 7.5 on Mohs scale which makes it an effective abrasive.

RESERVES / RESOURCES

In India, garnet deposits suitable for use in Abrasive Industry occur in Andhra Pradesh, Chhattisgarh, Jharkhand, Kerala, Odisha, Rajasthan, Tamil Nadu and Telangana. Gem variety of garnet occurs in Ajmer, Bhilwara, Jhunjhunu, Sikar and Tonk districts, Rajasthan; Nellore and Srikakulam districts, Andhra Pradesh; Khammam district, Telangana and Coimbatore, Ramanathapuram, Tirunelveli, Kanyakumari, Tiruchirappalli and Tiruvarur districts, Tamil Nadu. Garnet also occurs in beach sands along with ilmenite, rutile, sillimanite, etc. in the States of Kerala, Odisha and Tamil Nadu.

The total reserves/resources of garnet in India as on 01.04.2020, as per UNFC system have been placed at 56.01 million tonnes of which Reserves under Proved and Probable categories together constitute 8.60 million tonnes. Of the total resources, about 20.87 million tonnes are of Abrasive grade, whereas resources of Semi-precious grade are mere 8,468 tonnes. Tamil Nadu alone accounted for about 46% of the total resources followed by Andhra Pradesh (31%), Odisha (17%) and Telangana (3%). The remaining States together shared less than 3% (Table- 1).

Table – 1: Reserves/Resources of Garnet as on 01.04.2020

(By Grades/States)

(In tonnes)

Grades/States	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	8590472	47416654	56007126
By Grades			
Gem	1	46158	46160
Abrasive	8537296	12336249	20873545
Semi-precious	637	7831	8468
Others	-	260982	260982
Unclassified	52538	34325967	34378505
Not-known	-	439466	439466
By States			
Andhra Pradesh	-	17267129	17267129
Chhattisgarh	-	28800	28800
Jharkhand	-	110071	110071
Kerala	-	198861	198861
Odisha	8330046	1177318	9507364
Rajasthan	207888	842923	1050811
Tamil Nadu	52538	25831356	25883894
Telangana	-	1960196	1960196

Figures rounded off

EXPLORATION & DEVELOPMENT

The details about the exploration and development, if any, are covered in the Review on “Exploration and Development” under “General Reviews”.

MINING LEASES & PRODUCTION

Garnet (Abrasive)

As on 31.03.2024 (provisional), the total number of mining lease granted for the mineral garnet was 20 covering a total area of 118.7838 ha. Production of garnet (abrasive) was at 23,838 tonnes during 2023-24 and 9,859 tonnes in the preceding year. There were 8 reporting mines during

2023-24 and 5 reporting mines in 2022-23. Three principal producers accounted for about 100% of the total output during the year (Table-2 to 4).

Mine-head closing stock of Garnet (abrasive) for the year 2023-24 was 4,54,737 tonnes as against 13,895 tonnes for the previous year (Table-5).

The average daily employment of labour during 2023-24 was 124 as against 64 in the previous year.

Garnet (Gem)

No production of garnet (gem) was reported since 2018-19.

Table – 2: Principal Producer of Garnet (Abrasive), 2023-24

(By Grades/States)

Name & address of producer	Location of the mine	
	State	District
"Arun Bagdiya, C/o, Shri Ramdev Bagdiya, Resi No-110, Kendriya Vihar, Sector-8, Bidyadhar Nagar, Jaipur-302039, Rajasthan."	Rajasthan	Ajmer
"Ummed Singh Ranawat, Vill-Basda, P.O. Gundali Bhilwara-311001, Rajasthan."	Rajasthan	Bhilwara
"AKD Gem Garnet Mines, F-203, Near Mahapragya Circle, Azad Nagar, Bhilwara - 311 001, Rajasthan."	Rajasthan	Bhilwara
Shree Mahadev, 1, Agrawal Beej Udyog Building, Near Roadways Bus Stand, Bhilwara-311001, Rajasthan	Rajasthan	Rajsamand

*Producing as an associated mineral with sillimanite.

Table – 3: Production of Garnet (Abrasive), 2021-22 to 2023-24

(By State)

(Qty. in tonnes; Value in ₹'000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	8182	24660	9859	36217	23838	95837
Odisha	-	-	-	-	-	-
Rajasthan	8182	24660	9859	36217	23838	95837
Tamil Nadu	-	-	-	-	-	-
Telangana	-	-	-	-	-	-

(P): Provisional

Table – 4: Production of Garnet (Abrasive), 2022-23 & 2023-24

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹'000)

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	5	9859	36217	8	23838	95837
Public Sector	-	-	-	-	-	-
Private Sector	5	9859	36217	8	23838	95837
Rajasthan	5	9859	36217	8	23838	95837
Ajmer	2	2328	7026	3	6469	21917
Bhilwara	3	7531	29191	3	16316	70148
Rajsamand	-	-	-	1	1050	3763
Tonk	-	-	-	1	3	9
Tamil Nadu	-	-	-	-	-	-
Tiruchirapalli	-	-	-	-	-	-
Telangana	-	-	-	-	-	-
Khammam	-	-	-	-	-	-

(P): Provisional

Table – 5: Mine-head closing stocks of Garnet (Abrasive), 2022-23 & 2023-24

(By State)

(In tonnes)

State	2022-23	2023-24 (P)
India	13895	454737
Andhra Pradesh	-	-
Odisha	-	-
Rajasthan	3142	47661
Tamil Nadu	10753	402989
Telangana	-	4087

(P): Provisional

USES & CONSUMPTION

The most important industrial use of garnet in the form of garnet sand is as an abrasive. About 90% of the total production of abrasive garnet is used for manufacturing of garnet-coated papers, clothes and discs. Garnet-coated abrasives are used in the form of belts, covers for drums, discs or as small sheets. It is used for cleaning spark plugs, paints, polishing and grinding of plate-glass. The remaining

10% output is used in the form of loose grains for surfacing and polishing soft stones (marble, slate, soapstone, etc.). Clear, flawless and rich-coloured crystals of garnet are used as semi-precious stones. The principal varieties among them are pyrope, deep-crimson almandine, orange-yellow grossularite, etc. Other applications include use in Electronic and Television Industry for polishing glass and TV tubes. Garnet granules are used in 'abrasive blasting

commonly called ‘sand blasting’ in order to smoothen, clean and remove oxidation products from metals, stone and other material. MMTC’s specifications of garnet sand used for sand blasting/jet cutting/other uses for exports to USA, Europe, Middle East and Taiwan are as follows: Al_2O_3 : 20.8 to 21.2%, Bulk density; 2.17 kg/m³, Hardness in Mohs scale should be 7.5 to 8.

Water jet cutting machines generally use finely-ground 80-120 mesh size garnet as cutting medium propelled by high pressure water. Owing to its inertness to a wide range of chemicals and relatively high specific gravity, it is used as a filter medium for water and other liquids.

SUBSTITUTES

Other natural and manufactured abrasives can substitute to some extent for all major end uses of garnet. In many cases, however, using the substitutes would entail compromise in quality or cost. Fused aluminium oxide and staurolite compete with garnet as a sandblasting material. Ilmenite, magnetite and plastics compete as filtration media. Corundum, diamond and fused aluminium oxide compete for lens grinding and for many lapping operations. Emery is a substitute for non-skid surfaces. Fused aluminium oxide, quartz sand and silicon carbide compete for the finishing of plastics, wooden furniture and other products.

WORLD REVIEW

Garnet group of minerals are found throughout the world in metamorphic, igneous and sedimentary rocks.

World resources of garnet are large and occur in a wide variety of rocks, particularly, gneisses and schists. Garnet also occurs in contact metamorphic deposits in crystalline limestones, pegmatites, serpentinites and in vein

deposits. In addition, alluvial garnet is present in many heavy mineral sand and gravel deposits throughout the world. Large domestic resources of garnet are concentrated in coarsely-crystalline gneiss near North Creek, NY; other significant domestic resources of garnet occur in Idaho, Maine, Montana, New Hampshire, North Carolina, and Oregon. In addition to those in the United States, major garnet deposits exist in Australia, Canada, China, India and South Africa, where they are mined for foreign and domestic markets; deposits in Russia and Turkey have also been mined in recent years, primarily for internal markets. Additional garnet resources are located in Chile, Czechia, Pakistan, Spain, Thailand and Ukraine; small mining operations have been reported in most of these countries.

In 2023, Australia accounted for about 40% of the total global production of industrial garnet, followed by China (32%), South Africa (19%), USA (7%), India (1.5%), Pakistan (0.2%), & Czechia (0.1%) and the remaining 0.5% was contributed by other countries. Russia and Turkey are also mining garnet for domestic markets. Garnet is also mined in Canada, Chile, Spain, Thailand and Ukraine.

Worldwide, the end uses of garnet and market shares are: abrasive blasting media 30%, abrasive grains for water jet cutting 35%, water filtration 20%, abrasive powder 10% and other end uses 5 per cent.

In the current scenario, China has around 37 Million tonnes of reserves followed by, India (8.6 Million tonnes), USA (5 Million tonnes), South Africa (0.3 Million tonnes), Australia (moderate to large) and also occurrences were reported in Pakistan & Czechia. The world reserves/resources and production of industrial garnet are furnished in (Tables-6 & 7).

Table – 6: World Reserves of Garnet (Industrial)
(By Principal Countries)

(In tonnes)	
Country	Reserves
World: Total (rounded)	Moderate to Large
China	37000000
India [#]	8600000
United States	5000000
South Africa	320000
Australia	Moderate to Large
Pakistan	NA
Czechia	NA

Source: USGS Mineral Commodity Summaries, 2024

[#]: In India as per NMI data based on UNFC system the total reserves/resources of garnet as on 1.4.2020 are estimated at 56.01 million tonnes.

Table – 7: World Production of Garnet (Industrial)

(By Principal Countries)

(In tonnes)

Country	2022	2023
World: Total (rounded)	971000	970000
Australia	388000	390000
China	310000	310000
South Africa	179000	180000
USA	76400	68000
India*	15000	15000
Pakistan	1870	1900
Czechia	500	500
Others	230	4600

Source: USGS, Mineral Commodity Summaries 2024

Note: Figures are rounded off

*: India's production of garnet (abrasive) during 2020-21, 2021-22, 2022-23 and 2023-24 were 7,114; 8,182; 9,859 and 23,838 tonnes, respectively.

FOREIGN TRADE

Exports

In 2023-24, exports of garnet (abrasive) increased by 42% to 1,46,915 tonnes from 1,03,398 tonnes in the previous year. Exports were mainly to UAE (29%), Qatar (12%), USA (10%), Saudi Arabia (9%), Malaysia (7%), Kuwait & Italy (6% each), Singapore (4%), Thailand (3%), Canada (2%) and other countries (12%). Exports in terms of value in respect of cut & uncut garnet variety increased by 15% to ₹ 55.53 crore in 2023-24 from ₹ 48.40 crore in the previous year. In terms of value, exports of garnet (cut & uncut) were mainly to Thailand & Hong Kong (30% each), USA (15%),

Armenia (6%), Japan (4%), UAE (3%), Germany & Italy (2% each), Australia and UK (1% each) & other countries (4%).

Out of the total exports, in terms of value of cut & uncut garnet, in 2023-24, cut variety of garnet accounted for 93% share and the remaining 7% was contributed by the uncut garnet. In terms of value, exports of cut variety were mainly to Thailand (30%), Hong Kong (27%), USA (16%), Armenia (7%), Japan & UAE (4% each), Germany (3%), Italy (2%), UK & Australia (1% each) & other countries (5%). Similarly, exports of uncut garnet were mainly to Hong Kong. (Tables- 8 to 11).

Table – 8: Exports of Garnet (Abrasive)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	103398	1903765	146915	2771501
UAE	29736	541717	42116	800925
Qatar	5992	110643	17802	309402
USA	12320	215213	14110	246744
Saudi Arabia	13575	265459	13634	273872
Malaysia	7700	140756	9714	184962
Kuwait	3975	73499	9238	183050
Italy	4620	80356	8540	155680
Singapore	5571	109762	6219	131395
Thailand	6268	120195	4126	82887
Canada	1454	25547	2738	51935
Other Countries	12187	220618	18678	350649

Figures rounded off

In tonnes

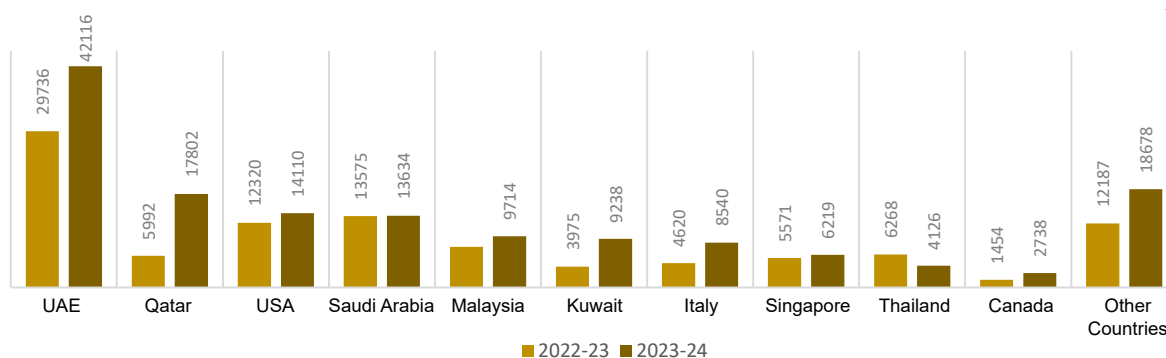


Fig. 1: Countrywise Exports of Garnet (Abrasive)

Table – 9: Exports of Garnet (Cut & Uncut)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	484011	**	555370
Hong Kong	**	111227	**	167827
Thailand	**	135679	**	166839
USA	**	77865	**	82019
Armenia	**	14316	**	34233
Japan	**	37731	**	23284
UAE	**	25452	**	19254
Germany	**	16284	**	13539
Italy	**	11967	**	11819
UK	**	8671	**	6282
Australia	**	5171	**	6058
Other Countries	**	39648	**	24216

Note: ** - Not additive. The total may not tally.

Figures rounded off

Table – 10: Exports of Garnet (Cut)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (Ctm)	Value (₹ '000)	Qty (Ctm)	Value (₹ '000)
All Countries	60229611	452710	58423044	518357
Thailand	2554519	129615	2760898	157867
Hong Kong	1522398	89080	2737894	140360
USA	5629617	77281	2961052	81969
Armenia	199697	14284	469013	34233
Japan	428126	37703	539000	23257
UAE	136312	25447	295758	19254
Germany	274464	16229	219712	13350
Italy	399114	11967	117646	11819
UK	229821	8275	115568	6282
Australia	46831021	5171	47760111	6058
Other Countries	2024522	37658	446392	23908

Figures rounded off

Table – 11: Exports of Garnet (Uncut)

Country	(By Countries)		2023-24 (P)	
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	3	31301	4	37013
Hong Kong	2	22147	4	27467
Thailand	++	6064	++	8972
Germany	++	55	++	189
Canada	++	4	++	172
Greece	--	--	++	63
USA	++	584	++	50
Switzerland	--	--	++	43
Japan	++	28	++	27
Korea, Rep. of	--	--	++	16
Indonesia	--	--	++	9
Other Countries	1	2419	++	5

Figures rounded off

Imports

In 2023-24, imports of abrasive garnet increased by 10187% to 3086 tonnes from 30 tonnes in the previous year. Imports were from UAE (99.999%) and Australia (0.001%). Imports in terms of value in respect of cut & uncut garnet variety increased by 61% to ₹ 48.87 crore in 2023-24 from ₹ 30.31 crore in the previous year. In terms of value, imports were mainly from Hong Kong (50%), Thailand (15%), USA (9%), China & Tanzania (7% each), Kenya (6%), South Africa, Mozambique, Sri Lanka (1% each) & other countries (2%) and unspecified.

Out of the total imports in terms of value of cut & uncut garnet in 2023-24, uncut variety of garnet accounted for 71% share and the remaining 29% was contributed by the cut garnet. In terms of quantity, imports of uncut variety were mainly from Hong Kong (48%) and Thailand (14%), China (10%), Tanzania (9%) & Kenya (8%). Similarly, imports of cut garnet were mainly from Hong Kong (54%), USA (19%), Thailand (15%) and Sri Lanka (4%) (Tables-12 to 15).

Table – 12: Imports of Garnet (Abrasive)

Country	(By Countries)		2023-24 (P)	
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	30	929	3086	67440
UAE	30	929	3084	67190
Australia	--	--	2	250

Figures rounded off

Table – 13: Imports of Garnet (Cut & Uncut)

Country	(By Countries)		2023-24 (P)	
	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	**	303072	**	488685
Hong Kong	**	57679	**	242938
Thailand	**	88744	**	71139
USA	**	20665	**	45445
China	**	34	**	33579
Tanzania	**	49873	**	31958
Kenya	**	40247	**	27992
South Africa	**	19275	**	7311
Mozambique	**	4558	**	5996
Sri Lanka	**	5169	**	5412
Unspecified	**	83	**	5371
Other Countries	**	16745	**	11544

Note: ** - Not additive. The total may not tally.

Figures rounded off

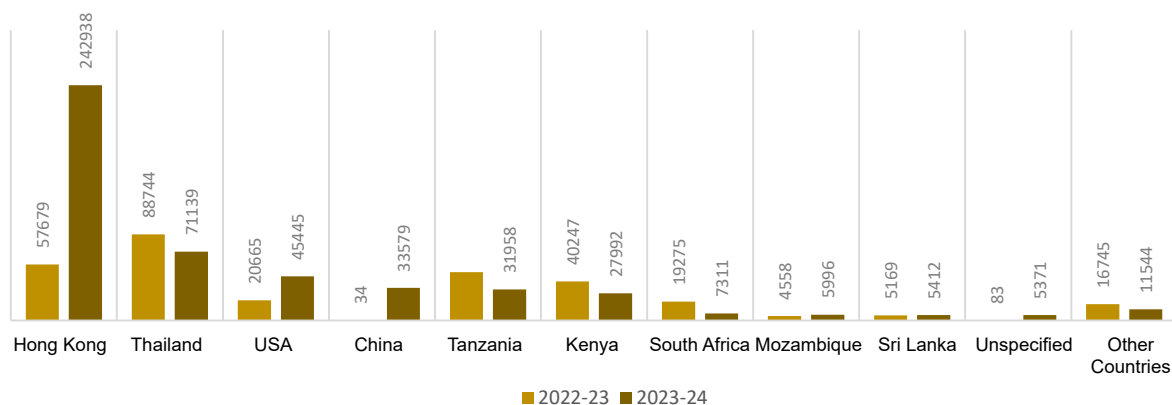


Fig. 2: Countrywise Value of Garnet (Cut & Uncut)

Table – 14: Imports of Garnet (Cut)

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (Ctm)	Value (₹ '000)	Qty (Ctm)	Value (₹ '000)
All Countries	1102403	90487	1990073	142256
Hong Kong	251306	24192	799390	77217
USA	147062	19538	323611	27557
Thailand	454690	29957	445268	20946
Unspecified	37	83	31	5371
Sri Lanka	109698	5169	368059	5296
Japan	13649	356	3577	2533
Germany	15429	559	1561	711
UAE	1571	130	1487	689
Switzerland	461	959	315	522
Canada	2751	449	4977	515
Other Countries	105749	9095	41797	899

Figures rounded off

Table – 15: Imports of Garnet (Uncut)

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	52	212585	13	346429
Hong Kong	14	33487	1	165721
Thailand	2	58787	2	50193
China	++	16	++	33498
Tanzania	2	49291	2	31958
Kenya	1	40247	2	27992
USA	++	1127	++	17888
South Africa	1	19275	++	7311
Mozambique	14	4558	1	5996
Madagascar	16	2881	4	2702
Nigeria	++	1094	++	838
Other Countries	2	1822	1	2332

Figures rounded off

FUTURE OUTLOOK

Garnet has a wide range of applications, such as, in production of abrasives, sand blasting, water filtration materials, abrasive blasting media and water-jet cutting. Garnet is expected to continue replacing silica sand blasting media, owing to latter's associated occupational health risks. Moreover, garnet is safer for the environment and cheaper to dispose of after recycling. Hence, the worldwide demand for garnet is expected to increase, especially for water jet

cutting and for abrasive blasting media. China and India are expected to steadily increase garnet production and will become significant garnet sources for other countries. The garnet market is very competitive. To increase profitability and remain competitive with imported material, production may be restricted to only high-grade garnet ores as a by-product of other saleable mineral products that occur with garnet, such as, kyanite, marble, metallic ores, mica minerals, sillimanite, staurolite or wollastonite.



41. Graphite

1,68,341 tonnes
production of graphite have been
reported in 2023-24.

708 tonnes
of graphite (natural) were exported in
2023-24.

54,784 tonnes
of graphite (natural) were imported
in 2023-24.

Graphite, also known as plumbago or black lead or mineral carbon, is a stable form of naturally occurring carbon. Structurally, graphite is known to crystallise in hexagonal system and occurs in layered & lamellar form with grey-to-black metallic lustre and a greasy feel. Natural graphite is categorised into two commercial varieties (i) crystalline (flaky) graphite and (ii) amorphous graphite. Both flaky and amorphous varieties of graphite are produced in India. The quality of graphite depends upon its physical qualities and carbon content. Besides natural graphite, there is synthetic or artificial graphite which is manufactured on a large-scale in electric furnaces, using anthracite or petroleum coke as raw feed.

RESERVES/RESOURCES

Graphite occurrences are reported from various States but the deposits of economic importance are located in Chhattisgarh, Jharkhand, Odisha, Kerala and Tamil Nadu.

As per NMI database, based on the UNFC system, the total reserves/resources of graphite as on 1.4.2020 has been placed at about 211.62 million tonnes, out of which 8.56 million tonnes are in the Reserves category and 203.6 million tonnes are placed under Remaining Resources

category. Total resources containing +40% fixed carbon constitute about 2.91 million tonnes and total resources of 10–40% fixed carbon grade constitute 43.98 million tonnes. The balance 164.73 million tonnes fall under Beneficiable 'Others', 'Unclassified' and 'Not-known' grades. Arunachal Pradesh accounts for 36% of the total resources which is followed by Jammu & Kashmir (30%), Jharkhand & Orissa (9% each), Madhya Pradesh (6%) and Tamil Nadu (5%). However, in terms of reserves, Tamil Nadu has the leading share of about 36% followed by Jharkhand (30%) and Odisha (33%) of the total reserves (Table-1).

Table – 1: Reserves/Resources of Graphite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	8563411	203060176	211623587
By Grades			
+ 40% F.C.	1387851	1527904	2915755
10-40% F.C.	7175560	36808911	43984471
Beneficiable	-	793330	793330
Others	-	19153359	19153359
Unclassified		69486325	69486325
Not-known	-	75290347	75290347
By States			
Andhra Pradesh	-	1138275	1138275
Arunachal Pradesh	-	76318257	76318257
Chhattisgarh	5282	1330	6612
Gujarat	-	3355805	3355805
Jammu & Kashmir	-	62740555	62740555
Jharkhand	2604079	17402288	20006367
Karnataka	-	992632	992632
Kerala	15443	1419532	1434975
Madhya Pradesh	-	12640000	12640000
Maharashtra	-	1160000	1160000
Odisha	2838414	17142707	19981121
Rajasthan	-	1913554	1913554
Tamil Nadu	3100193	6605086	9705279
Telangana	-	219455	219455
Uttarakhand	-	10700	10700

Figures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on Exploration & Development under "General Review".

MINING LEASES & PRODUCTION

As on 31.03.2024 (P), the total number of mining leases in respect of graphite in force were 27 with area covered 1199.04 ha.

Production of graphite at about 1,68,341 tonnes in 2023-24 increased by 77.60% as compared to the previous year. There were 10 reporting mines in 2023-24 as compared

to 11 in the previous year. Five principal producers accounted for 90.29% during the year. About 79.24% of the total production in 2023-24 was accrued from three mines, each producing more than 5,000 tonnes annually, while 20.76% was contributed by seven mines in the production range up to 5,000 tonnes per annum. Tamil Nadu was the leading producing state contributing 47.08% to the total output during 2023-24, followed by Odisha. Mine-head closing stock in the year 2023-24 was 2,71,933 tonnes as against 1,91,057 tonnes in the previous year. The average daily employment of labour during 2023-24 was 126 against 133 in the previous year (Tables - 2 to 6).

Table – 2: Principal Producers of Graphite, 2023-24

Name & Address of Producer	Location of the mine	
	State	District
TAMIL NADU MINERALS LIMITED 31, Kamarajar Salaitwad House, Chepauk, Chennai-600 005, Tamil Nadu	Tamil Nadu	Sivaganga
MR. SIBANANDA PRADHAN A 19 Trishna Residency Phase 3, Kiit College Road, Patia, Bhubaneswar	Odisha	Rayagada

Table- 2 (Concl'd.)

Name & Address of Producer	Location of the mine	
	State	District
PRADHAN INDUSTRIES House No 1, Telenga Bazar, Cuttack Sadar – 753009, Odisha	Odisha	Rayagada
M/s. PARIJAT MINING INDUSTRIES (INDIA) PVT LTD Town hall road, opp. shivajee maidan, daltonganj (medininagar), Pamalu, Jharkhand, 822101	Jharkhand,	Latehar
Mr. PRAMOD KUMAR AGRAWAL Shantikunj farm road, modipara sambalpur, Odisha-768002	Odisha	Nuapada

Table – 3: Production of Graphite, 2021-22 to 2023-24 (P)

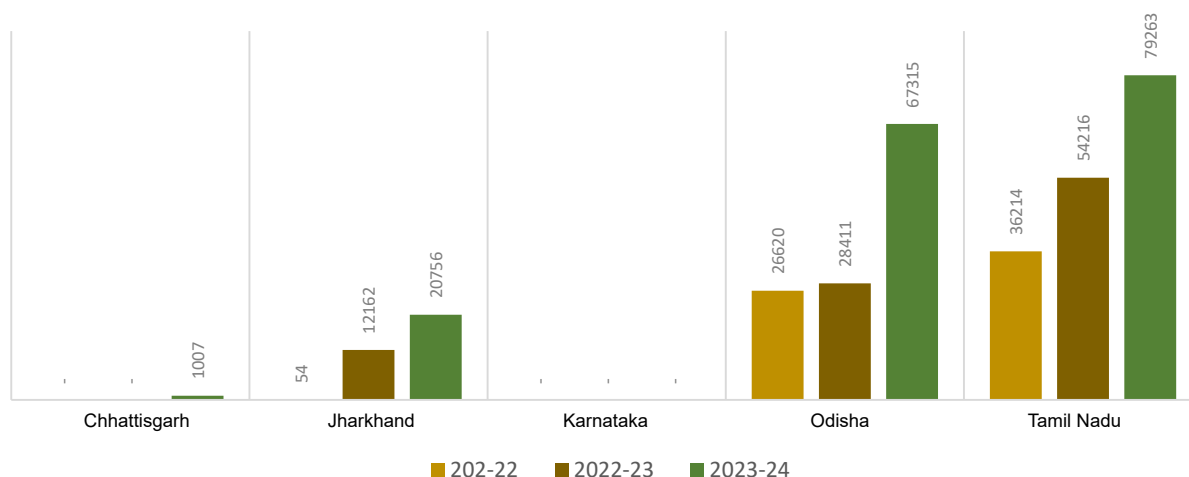
(By States)

(Quantity in tonnes; Value in ₹ '000)

States	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	62888	122147	94789	180152	168341	224542
Chhattisgarh	-	-	-	-	1007	1316
Jharkhand	54	59	12162	10615	20756	19533
Karnataka	-	-	-	-	-	-
Odisha	26620	69216	28411	103302	67315	149102
Tamil Nadu	36214	52872	54216	66235	79263	54591

(P)-provisional

(In tonnes)



■ 2021-22 ■ 2022-23 ■ 2023-24

Fig. 1: State-wise Production of Graphite

Table – 4: Production of Graphite, 2022-23 and 2023-24 (P)

(By Sectors/States/Districts/Grades)

(Qty in tonnes; Value in ₹ '000)

State/District/ Sector	2022-23						2023-24 (P)						
	No. of Mines	Grade: Fixed Carbon content			Total		No. of Mines	Grade: Fixed Carbon content				Total	
		80% or more	40% or more but less than 80%	Less than 40%	Qty	Value		80% or more	40% or more but less than 80%	20% or more but less than 40%	Less than 20%	Qty	Value
India	11	1407	1848	91534	94789	180152	10	1401	1308	1007	164625	168341	224542
Public Sector	1			54216	54216	66235	1				79263	79263	54591
Private Sector	10	1407	1848	37318	40573	113917	9	1401	1308	1007	85362	89078	169951
Chhattisgarh	1						1			1007		1007	1316
Surguja	1						1			1007		1007	1316

Table- 4 (Concl'd.)

(Qty in tonnes; Value in ₹ '000)

State/District/ Sector	2022-23						2023-24 (P)						
	No. of Mines	Grade: Fixed Carbon content			Total		No. of Mines	Grade: Fixed Carbon content				Total	
		80% or more	40% or more but less than 80%	Less than 40%	Qty	Value		80% or more	40% or more but less than 80%	20 % or more but less than 40%	Less than 20%	Qty	Value
Jharkhand	2	-	-	12162	12162	10615	2	-	-	-	20756	20756	19533
Latehar	1	-	-	9356	9356	9122	1	-	-	-	12734	12734	12416
Palamau	1	-	-	2806	2806	1493	1	-	-	-	8022	8022	7117
Karnataka	1*	-	-	-	-	-	-	-	-	-	-	-	-
Mysore	1*	-	-	-	-	-	-	-	-	-	-	-	-
Odisha	6	1407	25156	25157	28411	103302	6	1401	1308	-	64606	67315	149102
Nawapara	2	-	-	8865	8865	10470	2	-	-	-	10017	10017	12551
Raygada	3	1407	1848	12287	15541	87814	3	1401	1308	-	52000	54709	133307
Bolangir	1	-	-	4005	4005	5018	1	-	-	-	2589	2589	3244
Tamil Nadu	1	-	-	54216	54216	66235	1	-	-	-	79263	79263	54591
Sivagangai	1	-	-	54216	54216	66235	1	-	-	-	79263	79263	54591

(P): Provisional

Table – 5: Production of Graphite, 2022-23 and 2023-24 (P)

(By Frequency Groups)

(Qty in tonnes)

Frequency Group	No. of Mines		Production		Percentage to total production		Cumulative Percentage	
	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)
India	11	10	94789	168341	100	100	-	-
Up to 1000	2	-	-	-	-	-	-	-
1001 to 2000	1	1	1150	1007	1.21	0.6	1.21	0.6
2001 to 5000	4	3	13411	9370	14.15	5.57	15.36	6.17
5001 to 10000	2	2	14875	13967	15.7	8.3	31.06	14.47
Above 10000	2	4	65353	143997	68.94	85.53	100	100

(P): provisional

Table – 6: Mine-head closing stocks of Graphite, 2022-23 and 2023-24 (P)

(By States/Grades)

(In tonnes)

State/District	2022-23					2023-24 (P)				
	Grade: Fixed Carbon content					Grade: Fixed Carbon content				
	80% or more	40% or more but less than 80%	Less than 40%	Less than 20%	Total	80% or more	40% or more but less than 80%	20% or more but less than 40%	Less than 20%	Total
India	205	258	8000	182594	191057	212	232	9007	262482	271933
Chattisgarh	-	-	-	-	6258	-	-	-	7265	7265
Jharkhand	-	-	-	11267	11267	-	-	-	11440	11440
Karnataka	-	-	1742	-	1742	-	-	-	1742	1742
Odisha	205	258	-	26119	26582	212	232	-	79812	80256
Tamil Nadu	-	-	-	145208	145208	-	-	-	171230	171230

(P): provisional

MINING & MARKETING

Graphite mines, barring a few underground mines, are mostly small and opencast. Active mining centres of graphite are in Palamu district in Jharkhand; Nawapara & Balangir districts in Odisha; and Madurai & Sivagangai

districts in Tamil Nadu. Disseminated deposits of flaky graphite containing 5 to 20% Fixed Carbon (F.C.) are found in Palamu district of Jharkhand. In Odisha, areas in and around Balangir are the chief mining centres where

several graphite grades are produced. At Balangir, a few opencast workings are deeper than 45 m from surface and the ROM from such mines generally contains 10 to 20% F.C. Sargipalli underground mine in Sambalpur district, operated by T.P. Mineral Industries (TPMI), produced graphite that analysed up to 40% F.C. in the past. Water seepage beyond 6 m depth is the main problem faced by almost all mine owners in Odisha. Graphite of Balangir district is utilised mostly by the Graphite Crucible Industry. The technological changes in recent years have considerably reduced the use of graphite as a lubricant. However, recycled graphite is still used in production of clay bonded graphite crucibles. The Sivagangai graphite is of flaky variety with 14% average Fixed Carbon (F.C.) used in the manufacture of refractory bricks, expanded graphite, crucibles and carbon brushes. It is being mined by opencast mining method. The mined graphite is subjected to size reduction by crushing, grinding, flotation and dewatering to upgrade the graphite concentrate from other gangue minerals. Mining is considered to be easy and safe as regards graphite deposits in view of their comparatively soft nature and presence of hard rocks on either side. In order to expose graphite deposit, thickness of 1 to 2 meters of top lateritic soil is dozed out using dozer or removed by excavator and loaded through dumper and transported to separate dump yard located in non-mineralised zone in the lease area. The graphite ore obtained usually is transported to stock yard for blending. In stock yard, both high-grade and low-grade ores are stacked separately. Depending on plant requirements, blending work is carried out and blended ore is dispatched for consumption. Tamil Nadu Minerals Ltd (TAMIN) has over 600 acres of graphite-bearing areas in Pudupatti, Kumaripatti and Senthidayanathapuram of Sivagangai district, Tamil Nadu.

BENEFICIATION

During graphite beneficiation one of the challenges is to maximise the recovery of flaky graphite from low grade graphite ore without breaking the flakes of graphite. This is because flaky graphite has a huge industrial demand due to its distinct properties such as excellent lubricity and higher thermal conductivity. Graphite occurs generally admixed with country rocks, and hence, it requires beneficiation for obtaining desired grade for various end-uses. Processes for graphite beneficiation depend upon nature and association of gangue minerals present. The common processes adopted are washing, sorting, tabling, and acid leaching and froth flotation. Amongst these, froth flotation process is used widely as it helps in producing a fairly high-grade graphite concentrate. Sometimes, beneficiated concentrate is further enriched by chemical treatment (acid leaching, chlorination, etc.) to obtain a very high-grade concentrate containing 98% to 99% F.C. Prominent beneficiation plants for graphite in India are Chota Nagpur Graphite

Industries and Carbon & Graphite Products, Daltonganj; Agrawal Graphite Industries, Gandhamardhan Graphite Udyog and T.P. Minerals Private Limited, Sambalpur; Tamil Nadu Minerals Ltd (TAMIN), Sivagangai, etc. The ROM, containing an average of about 10% F.C. has to be invariably beneficiated before marketing. Indigenously fabricated equipment is used generally to upgrade the ROM to produce marketable grade graphite which contains normally 70% to 80% F.C. About 92% F.C. product has been obtained by many producers after repeated cycles of beneficiation. A few plant owners have claimed to have obtained product containing as high as 95% F.C. Beneficiation plants in Odisha seem to have been designed for treating +10% F.C. graphite (ROM). In practice, it is seen that lower grade graphite having +5% F.C. is blended with higher grades to meet the requirements of beneficiation plant, i.e., +10% F.C. Thus, low-grade ore analysing +5% F.C. also gets used. Tamil Nadu Minerals Ltd (TAMIN) produces flaky graphite from a mine in Sivagangai district in Tamil Nadu. The beneficiation plant located adjacent to the mine site is designed to produce 8,400 tpy of natural graphite concentrate containing 96% F.C. with 92% recovery from ROM.

USES & CONSUMPTION

Traditional uses of graphite are in crucibles, foundries, pencils, etc. More sophisticated applications of graphite are in refractories that are used in the manufacture of steel, cement and glass, expanded graphite-based sealing gaskets, graphitised grease, braid, brushes, brake lining, etc. It is also used for specialty applications, such as, in the Nuclear Industry, soil conditioners and graphite foils, which is used for sealing in the Chemical and Petrochemical Industries as well as in the Energy, Engineering and Automotive Industries. It is also used in minor amounts as a vital additive for producing foundry coatings to prevent fusion of liquid metal with sand at the mould or core face. Such coatings are either applied by spraying or painting in the form of suspension or by dusting or by rubbing as dry powders. Graphite used for coating is of high quality which does not peel off as flakes on drying and imparts a smooth surface to the casting. Graphite, a major additive to many coating systems, is known for its multifarious functions, such as, refractory, lubricant, thermal conductor, electrical conductor, UV shield, electromagnetic pulse shield, corrosion shield and pigment. It is also used as moderator in nuclear reactors and in Lithium-ion (Li-ion) batteries which is used in the electric vehicles that require high purity flake graphite in their anode material. Li-ion battery anode are typically made of graphite, which can absorb and hold the lithium ions between the layers in its atomic structure, while at the same time conducting electric charge.

The apparent consumption of various grades of graphite during 2023-24 was 2,22,417 tonnes which was increased by 61% as compared to 1,38,544 tonnes in the previous year.

SUBSTITUTION

In principle, it is possible to substitute graphite by either synthetic graphite, produced primarily from high carbon precursors, such as, petroleum coke and coal tar pitch (e.g. in batteries or for increasing the carbon in steel) or by replacing the product as in the case of pencils or by other compounds as in high temperature applications (e.g. refractories). In the later case, it is difficult to fully substitute graphite as it is tough to replicate the same level of performance that graphite provides. There is a limit to how much charge graphite can store and lithium moves relatively slowly through graphite. Therefore, while considering the improvement required for large scale EV, tin and silicon may be the future competitor because of higher charge capacity and also tin conducts lithium-ion faster.

WORLD REVIEW

The world resources of graphite are believed to exceed 800 million tonnes of recoverable graphite. However, world reserves of graphite have been placed at 280 million tonnes of which China accounts for 28% followed by Brazil (26%), Madagascar & Mozambique (9% each), Tanzania (6%), Russia (5%), India (3%), Turkey & Canada (2% each) and other countries (9%) (Table-7).

World production of graphite was 1.70 million tonnes in 2023 as compared to 1.8 million tonnes in 2022. China was the leading producer, with a share of about 72% which is followed by India (8%), Mozambique (6%), Madagascar & Brazil (4% each), Turkey, France and other countries (2% each) (Table-8).

Table – 7: World Reserves of Graphite (Natural)

(By Principal Countries)

(In tonnes)

Country	Reserves
World: Total (rounded off)	28,00,00,000
China	7,80,00,000
Brazil	7,40,00,000
Mozambique	2,50,00,000
Madagascar	2,40,00,000
Tanzania	1,80,00,000
Russia	1,40,00,000
India	86,00,000
Turkey	69,00,000
Canada	57,00,000
Other Countries	2,58,00,000

Source: USGS Mineral Commodity Summaries, 2024

Table – 8: World Production of Graphite

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World Total (Rounded off)	1600000	1800000	1700000
China ^(e)	1189000	1212000	1230000*
India	57264 ^(f)	85873	128949
Mozambique	77116	165932	100000*
Brazil ^(d)	78555	79371	66346
Madagascar ^(c)	91100	116700	60000*
France	40000*	48385	42319
Turkey	28336	27715	27974
Russia	16200	18800	18800*
Austria ^{(a)(b)}	17000*	17000*	17000*
Other Countries	41395	31643	32388

Source: BGS World Mineral Production, 2019-23.

Note: This table include all forms of amorphous and crystalline graphite but excludes synthetic material.

a: Crude

b: Sales

c: Exports

d: Including beneficiated and directly shipped materials.

e: Including Flaky graphite.

f: Year ended 31st March, following that stated.

*: estimated

(In tonnes)

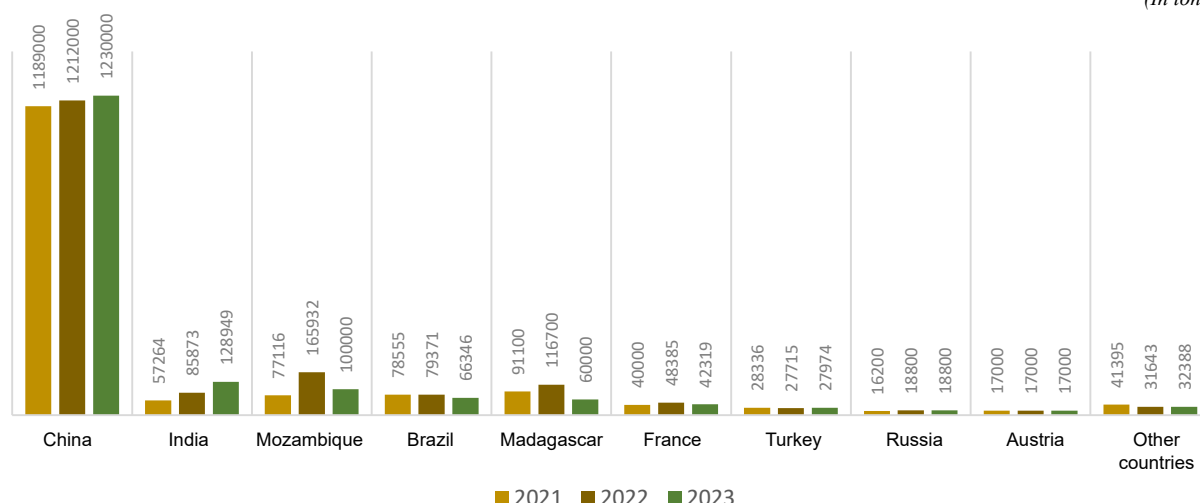


Fig. 2: World Production of Graphite

FOREIGN TRADE

Exports

In 2023-24, exports of graphite (natural) decreased by 68% to 708 tonnes as compared to 2,239 tonnes in the previous year. Graphite (natural) was exported mainly to UAE (43%), Turkey (11%), Tanzania (8%), Saudi Arabia (6%), South Africa & Nigeria (4% each) and Germany, Mexico & Bangladesh (2% each).

The exports of graphite (artificial) increased by 13% to 33,484 tonnes in 2023-24 from 29,522 tonnes in the previous year. Graphite (artificial) was exported mainly to Bhutan (17%), USA and Saudi Arabia (14% each), UAE (12%), Bangladesh & Germany (7% each), Oman (5%) and Turkey (4%) (Tables - 9 & 10).

Table – 9: Exports of Graphite (Natural)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2239	120029	708	63308
UAE	289	20386	303	21930
Turkey	191	13639	78	6812
Germany	++	5	12	5038
Tanzania	64	4160	59	3322
South Africa	2	165	30	2950
Nigeria	1027	33533	31	2882
Saudi Arabia	++	1041	45	2559
Bangladesh	3	515	17	1974
Malaysia	431	24958	10	1886
Mexico	14	1251	17	1554
Other Countries	218	20376	106	12401

Figures rounded off

(In tonnes)

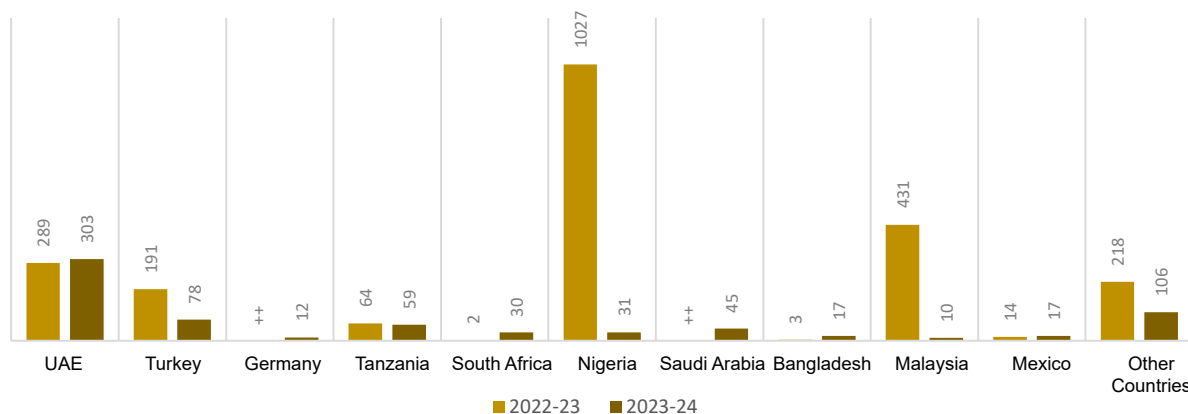


Fig. 3: Countrywise Exports of Graphite (Natural)

Table – 10: Exports of Graphite (Artificial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	29522	4136910	33484	4901183
USA	4166	714833	4588	1114938
Germany	1967	740350	2278	876335
Saudi Arabia	4702	592070	4564	676901
Bhutan	6051	519734	5850	412151
UAE	2325	237991	4069	343148
Bangladesh	1699	152288	2334	145765
Kuwait	55	58094	172	128343
Oman	1565	136210	1741	115330
Zambia	205	55219	421	114156
Turkey	1622	136212	1477	99342
Other Countries	5165	793909	5990	874774

Figures rounded off

Imports

Imports of graphite (natural) increased by 19% to 54,784 tonnes in 2023-24 from 45,994 tonnes in the preceding year. Graphite (natural) was imported mainly from Madagascar (45%), China (39%), and Mozambique (10%). Imports of

graphite (artificial) increased by 23% to 1,01,889 tonnes in 2023-24 from 82,721 tonnes in the previous year. Imports of graphite (artificial) were mainly from China (66%), Germany & UAE (5% each), UK (4%) and remaining 20% share was contributed by other countries (Tables- 11 & 12).

Table – 11: Imports of Graphite (Natural)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	45994	2939533	54784	3203146
Madagascar	13778	940394	24429	1427892
China	20576	1131934	21326	1055808
Mozambique	8525	503134	5548	360488
USA	186	42368	356	53443
Japan	57	20136	56	43632
Austria	288	31198	253	36301
Vietnam	--	--	878	35990
Tanzania	1500	80522	767	31059
UAE	20	1259	454	28906
Sweden	86	27792	77	28132
Other Countries	978	160796	640	101495

Figures rounded off

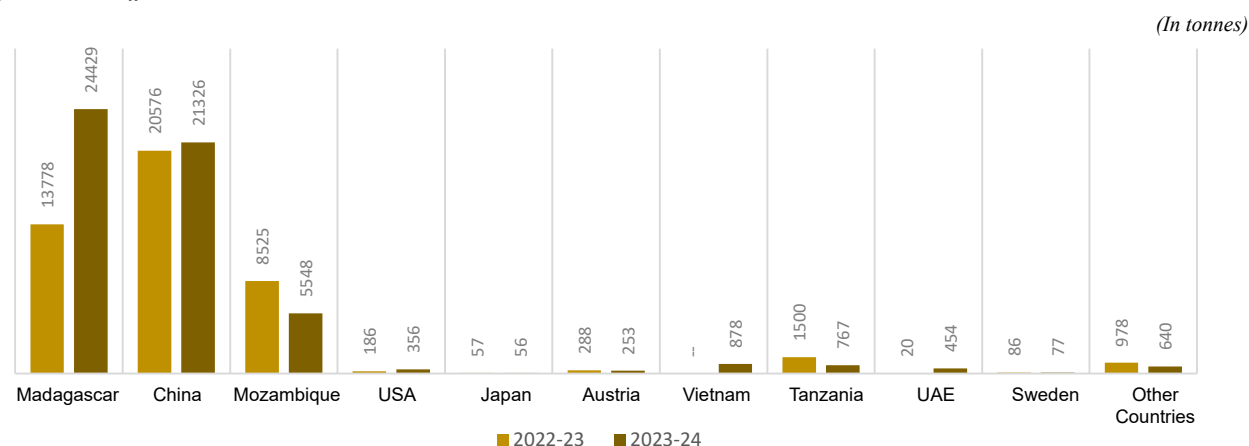


Fig. 4: Countrywise Imports of Graphite (Natural)

Table – 12: Imports of Graphite (Artificial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	82721	12008937	101889	12796588
China	49331	6034478	67088	6364656
Germany	9699	2521680	5435	1905876
UK	3094	594958	3832	1130681
Poland	2998	397432	3233	825769
France	1398	328454	1257	383399
USA	379	291362	269	363068
Netherlands	982	244086	1507	307712
Norway	2059	217809	2425	256758
Japan	2908	783832	436	234676
UAE	5550	131136	5353	172674
Other Countries	4323	463710	11054	851319

Figures rounded off

FUTURE OUTLOOK

Worldwide demand for combined natural and synthetic graphite is expected to rise along with improvements in the global economic conditions. Demand is also expected to augment further with the development of non-carbon energy applications, such as, batteries used in electric vehicles, electric devices and energy storage devices that use graphite. The graphite reserves having +40% Fixed Carbon are rather limited in the country. Detailed exploration of graphite deposits in Odisha, Jharkhand, Jammu & Kashmir and Kerala should be carried out. Cost-effective beneficiation technologies for low-grade graphite ore need to be developed. Silicon-carbide-graphite crucibles are being diversified and manufactured to improve upon the use of inferior grade material with less quantity and at the same time ensuring longer life of crucible.

Of late, a few emerging & important specialised applications of exfoliated graphite have been reported especially in the manufacture of sealings, gaskets, braids and brushes. New products of synthetic graphite, such as, graphite fibres/ropes and graphite insulation blankets have been introduced. In the world scenario, there seems to be

a rapid diversification in respect of potential large-volume end-use for natural graphite, such as, in heat sinks, also called spreader shield, which is a graphite foil material that conducts heat only in two directions. It has thermal conductivity above aluminium and almost equal to copper. These are used for dissipating heat in laptop computers, flat-panel displays, wireless phones, digital video cameras, etc. Such emerging & high growth applications of graphite are certainly causing noticeable impacts on the demand & consumption patterns within the country & globally. The demand for high purity graphite is increasing exponentially due to increase in the demand of lithium-ion batteries for electric vehicles, laptops, smartphones, home/ business applications and traditional uses for expanded graphite foils are also the potential areas that are expected to be major drivers for graphite consumption. It represents 23% of global flake graphite demand. The demand for graphite in the Battery segment is forecasted to double in the next six years. Graphite is mentioned in Part D of the First Schedule to the MMDR Act for critical & strategic minerals which is essential for our country's economic development and national security.



42. Ilmenite and Rutile

403 thousand tonnes
production of Ilmenite have been
reported in 2023-24.

0.22 million tonnes
of titanium ores & conc. were
exported in 2023-24.

89,920 tonnes
of titanium ores & conc. were
imported in 2023-24.

India is endowed with large resources of heavy minerals which occur mainly along coastal stretches of the country and also in inland placers. Heavy mineral sands comprise a group of seven minerals, viz. Ilmenite leucoxene (brown Ilmenite), Rutile, Zircon, Sillimanite, Garnet and Monazite. Ilmenite ($\text{FeO} \cdot \text{TiO}_2$) and rutile (TiO_2) are the two chief minerals of titanium. Titanium dioxide occurs in polymorphic forms as rutile, anatase (octahedrite) and brookite. Though brookite is not found on a large-scale in nature, it is an alteration product of other titanium minerals. Leucoxene is an alteration product of ilmenite and is usually found associated with ilmenite.

RESERVES/RESOURCES

Ilmenite and Rutile along with other heavy minerals are important constituents of beach sand deposits found right from Saurashtra coast (Gujarat) in the west to Digha coast, West Bengal in the east. These minerals are concentrated in five well-defined zones:

1. Over a stretch of 22 km between Neendakara and Kayamkulam, Kollam district, Kerala (known as 'Chavara' deposit after the main mining centre).
2. Over a stretch of 6 km from the mouth of River Valliyar to Colachal, Manavalakurichi and little beyond in Kanyakumari district, Tamil Nadu (known as MK deposit).
3. On Chatrapur coast stretching to about 18 km between Rushikulya river mouth and Gopalpur lighthouse with an average width of 1.4 km in Ganjam district, Odisha (known as 'OSCOM' deposit after IREL's Orissa Sands Complex).

4. Brahmagiri deposit stretches for 30 km from Giralanala to Village Bhabunia with an average width of 1.91 km in Puri district, Odisha.
5. Bhavanapadu coast between Nilarevu and Sandipeta with 25 km length and 700 m average width in Srikakulam district, Andhra Pradesh.

The AMD of the Department of Atomic Energy has been carrying out exploration of these mineral deposits. Of the total, coastal length of 5,921 km spread in Odisha, Andhra Pradesh, Tamil Nadu, Kerala, Karnataka, Maharashtra, Goa, Gujarat and West Bengal about 451 km, 1,873 km, and 112 km have been covered by detailed exploration, general exploration and preliminary exploration respectively. Coastal lengths of 2,272 km have not been covered due to various reasons viz., mangrove, port activity etc., leaving an unexplored coastal length of 1,214 km. The distributions of area coverage (sq km) in different geological domains are Beach & Dune (1845), Inland Sand Body (180), Terrace

sediments (368), River Channel (32), Inland alluvium (646) and Lake & Sea Beds (38). The Ilmenite resource estimation for the areas explored up to March, 2020 has been completed and the resources are up from 629.57 million tonnes (including leucoxene) in the year 2016 to 687.57 million

tonnes in the year 2020. The resources include Measured, Indicated and Inferred categories. Though, the latest state-wise /deposit-wise details are not available, the state-wise/deposit-wise details as received from DAE in the year 2018 are furnished in Tables-1 & 1A.

Table – 1: Ilmenite Reserves/Resources in India

(In million tonnes)

State/Deposit	Ilmenite reserves
Andhra Pradesh	
1. Bhavanapadu	10.18
2. Kakinada (Phase I-VIII)	13.81
3. Kalingapatnam	7.03
4. Narasapur	2.92
5. Nizampatnam	19.26
6. Srikurman (South)	8.6
7. Visakhapatnam (Bhimunipatnam)	2.88
8. Amalapuram (Phase I-IV)	4.72
9. Pandurangapuram-Voderevu (Bapatla-Chirala coast)	10.38
10. Vetapalem Coast (Chirala coast)	5.31
Kerala	
1. Chavara Barrier beach	13.17
2. Chavara Eastern Extension (Phase-I)	17.02
3. Chavara Eastern Extension (Phase-II)	49.26
4. Trikkunnapuzha-Thotapally Beach & Eastern Extension	9.5
5. Alapuzha-Kochi	5.88
Maharashtra	
1. Ratnagiri	3.68
2. Munge-Achra-Malvan	1.12
3. Vijayadura-Mithbav	0.7
Gujarat	
1. Moti Daman-Umbrat coast	2.77
Odisha	
1. Brahmagiri (Phase I-V and NW extension)	86.04
2. Chatrapur	26.72
3. Gopalpur	6.42
Tamil Nadu	
1. Kudiraimozhi	22.86
2. Ovari-Periyatalai-Manapadu (Teri)	24.01
3. Sattankulam Teris	41.26
4. Cuddalore-Pudupattuchavadi (beach sand)	4.67
5. Vayakallur (beach sand)	4.52
6. Manavalakurichi (beach sand)	3.07
7. Midalam	1.64

Source: As per letter received from Department of Atomic Energy, Mumbai dated 26/07/2018

Table – 1A: Reserves/Resources of Ilmenite and Rutile

(In million tonnes)

State	Total in situ*
Ilmenite*: Total	629.57
Andhra Pradesh	156.17
Jharkhand	0.73
Gujarat	2.77
Kerala	144.02
Maharashtra	5.5
Odisha	150.62
Tamil Nadu	167.7
West Bengal	2.06
Rutile: Total	33.95
Andhra Pradesh	10.55
Jharkhand	0.01
Gujarat	0.02
Kerala	8.74
Maharashtra	0.01
Odisha	6.58
Tamil Nadu	7.85
West Bengal	0.19

"Source: As per letter received from Department of Atomic Energy, Mumbai dated 26/07/2018. The resources of beach sand minerals (BSM) viz. Ilmenite, Rutile, Zircon, Garnet, leucoxene, monazite and Sillimanite were last updated in the year 2016 by AMD. # Inclusive of Indicated, Inferred and Speculative categories.

* Including leucoxene."

PRODUCTION AND PRICES

Ilmenite

The production of Ilmenite at 403 thousand tonnes in 2022-23 increased by 3% as compared to that in the preceding year. Odisha was the leading producer of Ilmenite during the year under review, contributing 61% of the total production followed by Kerala (30%) and Tamil Nadu (9%).

Rutile

The production of rutile at 14 thousand tonnes in 2022-23 increased by 3% as compared to that in the previous year. Odisha was the leading producer of rutile accounting for 66% of the total production followed by Kerala (20%) and Tamil Nadu (14%) (Table- 2).

The prices of Rutile and Ilmenite, as per Department of Atomic Energy (DAE), are furnished in Tables-3 & 4 respectively.

Table – 2: Production of Ilmenite and Rutile

(By States)

(In tonnes)

State	2020-21	2021-22	2022-23
ILMENITE			
India: Total	351387	390638	402686
Kerala	88110	110654	118906
Odisha	230040	234132	245961
Tamil Nadu	33237	45852	37819
RUTILE			
India: Total	12845	13283	13659
Kerala	2197	2790	2790
Odisha	9409	8856	8974
Tamil Nadu	1239	1637	1895

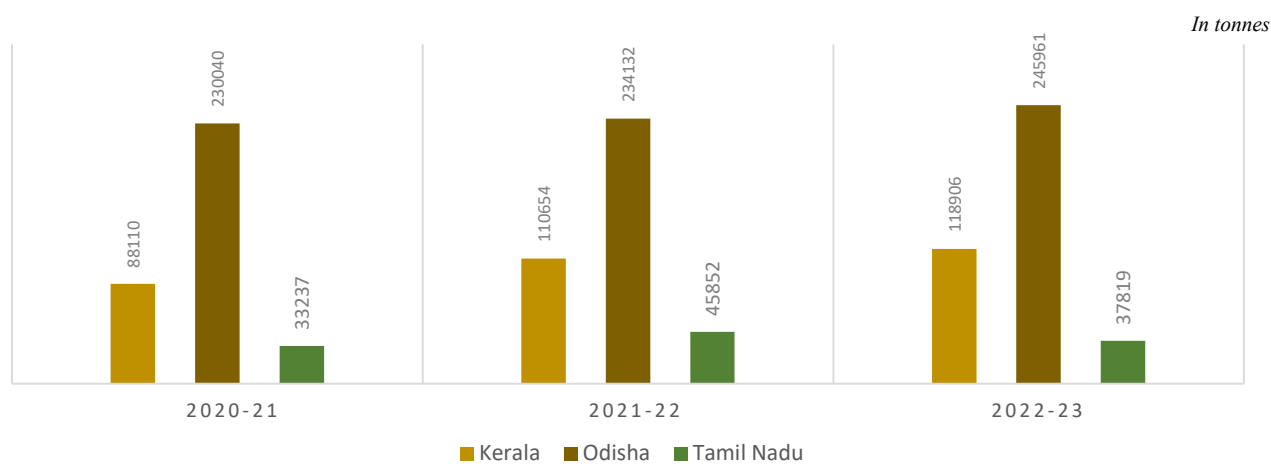


Fig. 1: Statewise Production of Ilmenite

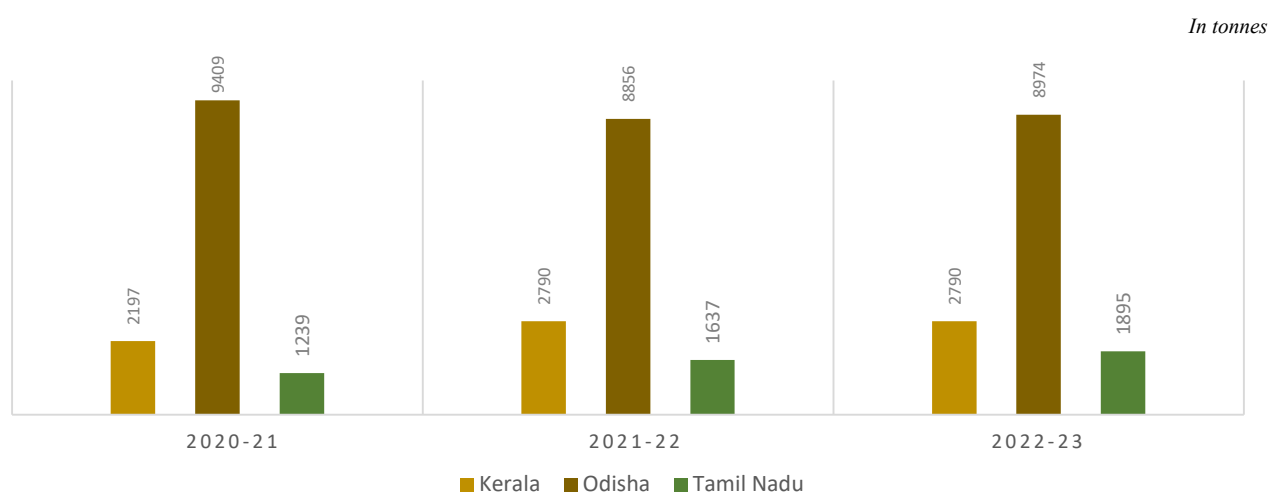


Fig. 2: Statewise Production of Rutile

Table – 3: Prices of Rutile
(2017-18 to 2019-20)

		₹ per tonne)	
Year	Grade	Price	Remarks
IREL			
2017-18**	Q	60748	Ex-works, Bagged
	MK	-	Ex-works, Bagged
2018-19		61070	Ex-works, Bagged
	Q	76995	Ex-works, Bagged
	MK	79627	Ex-works, Bagged
2019-20	OR	76798	Ex-works, Bagged
	Q	92138	Ex-works, Bagged
	MK	93052	Ex-works, Bagged
	OR	92788	Ex-works, Bagged
KMML			
2017-18	-	52083	Average
2018-19	-	66916	Average
2019-20	-	NA	-
V.V. Mineral			
2017-18	-	NA	Average
2018-19	-	NA	-
2019-20	-	-	-

Source: Department of Atomic Energy, Mumbai

Note: Q: Quilon; MK: Manavalakurichi; OR: Odisha

**During the financial year 2017-18 MK unit not despatches any material due to non- availability of transport permit.

Table – 4: Prices of Ilmenite
(2017-18 to 2019-20)

(₹ per tonne)

Period	Grade	Price	Remarks
IREL			
2017-18**	Q*	11922	Ex-works, loose
	MK*	-	Ex-works, loose
	OR*	11618	Ex-works, loose
2018-19	Q*	12978	Ex-works, loose
	MK*	12667	Ex-works, loose
	OR*	12512	Ex-works, loose
2019-20	Q*	14618	Ex-works, loose
	MK*	14235	Ex-works, loose
	OR*	13167	Ex-works, loose
KMML			
2017-18		NA	
2018-19		NA	
2019-20		NA	
V.V. Mineral			
2017-18	-	-	-
2018-19	-	-	-
2019-20	-	-	-
BMC			
2017-18		NA	
2018-19		NA	
2019-20		NA	
DCW Ltd			
2017-18	-	8423	
2018-19	-	14489	
2019-20	-	15265	

Source: Department of Atomic Energy, Mumbai.

Note: Q: Quilon; MK: Manavalakurichi; OR: Odisha

Ilmenite is usually sold on NAW (naked at works) basis from all production center

*: Regular Price

MINING & PROCESSING

Mining and processing of beach sand is carried out by IREL, a Government of India Undertaking and KMML, a Kerala State Government Undertaking. Exploitation work of beach sand deposits located at Chavara in Kerala, Gopalpur in Odisha and Manavalakurichi in Tamil Nadu by IREL is under progress.

At IREL, Chavara, Beach Sand was collected over a stretch of 23 km between Neendakara and Kayamkulam in Kerala and was transported to plant site. The unit has adopted wet mining operations involving use of two Dredge and Wet Concentrator (DWC) of 100 tph capacity each to exploit the inland deposits away from the beaches. Chavara Ilmenite is rich in TiO₂ content (60% TiO₂ Ilmenite) and has great demand in India and abroad for manufacture of pigments.

At Manavalakurichi, the deposit is spread over 300 hectares at Thuthoor–Ezudesam villages, Vilavancode

tehsil, district Kanyakumari, Tamil Nadu. All the raw sand required for the mineral separation plant to operate to its full capacity is collected from nearby beaches. Deposits are also exploited by DWC of 100 tph capacity. Manavalakurichi is next to Chavara in terms of TiO₂ content which is more than 55%.

The sand deposits of OSCOM at Chatrapur in district Ganjam extend along the coast of Bay of Bengal with an average width of 1.4 km and average depth of 7.5 m. Mining operations involve suction dredging to 6 m depth below water level on a much larger scale (500 tph) augmented by a smaller sized (100 tph) supplementary. The ilmenite from OSCOM is inferior in grade in terms of TiO₂ content (50%) in comparison to Chavara and Manavalakurichi. The Synthetic Rutile Plant of OSCOM is presently not operational. As a result, the majority of OSCOM Ilmenite produced of late is routed to the international market as feedstock for production of both slag grade and anatase grade pigment.

In dry mining, beach washings laden with 40–70% Heavy Minerals (HM) are collected through front-end loaders and bulldozers for further concentration to 90% HM at land-based concentrators. Though dry mining is very simple and economic, there is considerable opposition by local people for this form of mining for reasons that removal of sand would expose the land area to sea erosion. Therefore, collection of beach washings has reduced significantly in recent past.

As an alternate approach, IREL has adopted wet mining involving dredging and wet concentration (DWC) from inland areas away from the beach lines. In this mode, an artificial pond is created, the sand bed is cut and the slurry is pumped to spiral concentrator for removal of quartz. Manavalakurichi was the first plant to install a DWC (100 tph) followed by one (500 tph) at OSCOM and two (each 100 tph) at Chavara. The concentrate (90% HM) of beach washing plant from DWC is further upgraded to 97% HM grade at a Concentrate Upgradation Plant (CUP) before sending it to Mineral Separation Plant (MSP).

Execution of Supplementary Mining Lease deed for OSCOM Mines till a period up to the year 2047 has been completed under the provision of AMCR 2016. IREL received LoI on 11.01.2023 for grant of mining lease for 1,487.746 ha from the State Govt of Odisha for Brahmagiri deposit in Puri district of Odisha.

MK unit has received LOI from Govt. of Tamil Nadu on 28.06.2022 towards grant of mining lease for 1,144.0618 ha. The Government is showing keen interest in exploiting the resources through a joint venture between IREL and TAMIN, a State PSU. This initiative would be a breather for MK operations as the mineable land within the mining

leasehold areas are on the verge of exhaustion. Further, it will also pave way for formation of a new subsidiary of IREL in the same line as that of IREL and IDCOL.

KMML collects seasonal accretions of heavy mineral sand from the beach front. The pit so formed gets filled by fresh accretions of heavy mineral sand. The mineral sand is collected using bulldozers and wheel loaders and transported in tippers to Mineral Separation Plant.

The mineral separation plants use variety of equipment, such as, gravity concentrators, high tension electrostatic separators and magnetic separators. Making use of difference in physical properties like electrical conductivity, magnetic susceptibility and difference in specific gravity, etc., individual minerals like ilmenite, rutile, zircon, sillimanite and garnet are separated. The mined beach sands are pre-concentrated and dried after sieving (30-mesh) to separate the heavies from rejects. The heavy minerals are passed through electrostatic separators where conducting minerals – ilmenite and rutile – are separated from other non-conducting minerals. Ilmenite and Rutile are further subjected to low-intensity magnetic separators where magnetic fraction of ilmenite is separated from rutile. Similarly, non-conducting fractions are subjected to high-intensity magnetic separators where weak magnetic fraction (monazite and garnet) is separated from non-magnetic fraction (zircon and sillimanite). The fractions are further processed on wind tables to separate garnet from monazite and sillimanite from zircon.

Installed capacity and production of ilmenite, rutile and other associated heavy minerals by various separation plants are furnished in Table-5.

Table – 5: Installed Capacity & Production of Ilmenite, Rutile and Other Heavy Minerals
(2017-18 to 2019-20)

(In tonnes)

Company/Location	Mineral/Product	Installed capacity	Production		
		(tpy)	2017-18	2018-19	2019-20
Indian Rare Earths Ltd					
Manavalakurichi, # Distt Kanyakumari, Tamil Nadu.	Ilmenite	90000	-	25745	320985
	Rutile	3500	-	938	11674
	Zircon	10000	-	2190	11490
	Sillimanite	8778	-	-	26052
	Monazite	6000	-	-	69748
	Garnet	10000	-	7425	5034
Chavara, Distt Kollam, Kerala.	Ilmenite	200000	43253	48694	-
	Rutile	11400	1515	1723	-
	Zircon	17500	2649	3072	-
	Rare Earths	4500*	-	-	-
	Sillimanite	10000	6826	7953	-
Orissa Sands Complex, Distt Ganjam, Odisha	Ilmenite	220000	184657	191492	-
	Rutile	7400	7860	8384	-
	Sillimanite	13000	16698	17930	-
	Garnet	20000	34170	31332	-

Table- 5 (Concl'd)

(In tonnes)

Company/Location	Mineral/Product	Installed capacity	Production		
		(tpy)	2017-18	2018-19	2019-20
Kerala Minerals & Metals Ltd					
Chavara, Distt Kollam, Kerala.	Ilmenite	61600	56757	26140	30009
	Rutile	4400	2454	1548	1428
	Zircon	6500	4844	4762	4110
	Sillimanite	3600	701	271	1329
V.V. Mineral					
Distt Thoothukudi, Tamil Nadu.	Ilmenite	450000	-	-	-
	Rutile	12000	-	-	NA
	Zircon	18000	-	-	-
	Zircon-sillimanite	24000	-	-	-
Beach Minerals Co. Pvt. Ltd					
Kuttam, Distt Tirunelveli, Tamil Nadu.	Ilmenite	150000	-	-	-
V.V. Titanium Pigments Pvt. Ltd					
Distt Thoothukudi, Tamil Nadu.	Titanium Dioxide	18000	13801	11902	9419

Source: Department of Atomic Energy, Mumbai and IREL.

* In terms of rare earths chloride. '-' Not Available # During the year 2017-18, Manavalkurichi Plant was non-operating from Jan 2017 because of non-availability of environmental clearance (EC). V.V.Mineral mine is not in operation since 2017-18

INDUSTRY

For manufacturing titanium dioxide pigment, ilmenite is first treated chemically to obtain upgraded ilmenite, commonly called as Synthetic Rutile. There are two major pigment production processes, namely, chloride process and sulphate process depending on different operating characteristics and feedstock requirements. Plants employing chloride process consume high TiO_2 content feedstocks like synthetic Rutile and chloride slag. On the other hand, plants employing the sulphate process use lower-grade Ilmenite and sulphate slag.

Ilmenite obtained from Mineral Separation Plant (MSP) is chemically treated to remove impurities, such as, iron to obtain synthetic Rutile (90% TiO_2) in Synthetic Rutile Plant (SRP). Indian SRP is based on reduction roasting followed by acid leaching with or without generation of hydrochloric acid. Plants of IREL (OSCOM) and KMML depend on acid regeneration from the leach liquor while those of Cochin Minerals & Rutile Ltd (CMRL) and DCW use fresh acid and recover ferric chloride from the leach liquor for its use in water purification.

At OSCOM plant of IREL, reduction-roasting of ilmenite with coal is followed by leaching with HCL to separate iron as soluble ferrous chloride. The leached ilmenite is calcined to yield Synthetic Rutile and the acidic leach liquor is treated in an acid regeneration plant to recover HCL for recycling with iron oxide as waste. The unit stopped production in 1997, as it was not viable economically. Against the Request for Proposal floated to set up titanium slag plant under Build-Own-Operate (BOO) model, a CPSE has shown keen interest and based on their request, the due date of submission has been extended.

Considering the formidable investment and difficulties in sourcing technology in the field, the progress, though slow is in the right direction.

Environment Clearance for setting up nanotitania/zirconia facilities have been received. However, in consideration of the stringent norms of Zero Effluent Discharge, work has been taken up to use alternate feed material to meet the stipulations.

The DCW Ltd procures ilmenite from Manavalakurichi which is then roasted with coke fines to convert Fe_2O_3 into FeO . The reduced ore is leached with concentrated hydrochloric acid to remove oxides of iron and other metals. The leached ore is washed and calcined to get upgraded Ilmenite which contains more than 95% TiO_2 . The upgraded ilmenite is micronised to 2 microns by using high-pressure steam. This is marketed as Titox. The liquor from ilmenite leaching process contains fine TiO_2 particles and chlorides. The TiO_2 recovered by filtration & washing in filter process is marketed as Utox. The Company has plans to increase the plant capacity to 48,000 tpy and also to install facilities for the manufacture of ferrite grade iron oxide from the effluent of the ilmenite plant.

Cochin Minerals and Rutile Ltd (CMRL), which began production at its 10,000 tpy synthetic Rutile plant in Kerala in 1990 as a 100% EOU, has gradually raised the production capacity to around 45,000 tpy since 2008-09 for exports. It also has ferric chloride & ferrous chloride plants having capacities of 24,000 tpy & 72,000 tpy, respectively.

The Travancore Titanium Products Ltd (TTPL), a Kerala State Government Undertaking, manufactures titanium dioxide pigment by sulphate process at its plant at

Kochuveli, Thiruvananthapuram. Ilmenite is reacted with sulphuric acid in digesters and a porous cake is formed. The mass in the solid form is dissolved in dilute sulphuric acid to get titanium in solution as titanium oxysulphate along with other metallic ingredients in Ilmenite as their sulphate. The liquor is reduced using scrap iron, when ferric iron gets completely reduced to the ferrous state. The liquor is clarified, concentrated and boiled to precipitate the titanium content as hydrated titania which is then filtered by vacuum filters and calcined. Sulphuric acid required for captive consumption is produced at site using elemental sulphur. Till recently, TTPL was the only unit producing anatase grade titanium dioxide pigment in India. TTPL has capacity to produce 17,000 tpy of titanium dioxide, and with plans to modernise and diversify in stages, the Company has chalked out targets to produce both anatase and Rutile grades titanium dioxide pigment.

Present domestic titanium metal production is negligible. KMML has setup a 500 tpy titanium sponge plant with Defence Metallurgical Research Laboratory (DMRL) technology and first batch of titanium was delivered in September 2011. The plant will be further expanded to 1,000 tpy. IREL is to setup a 10,000 tpy titanium sponge plant at OSCOM for which proposals have been invited on "build, operate and own" basis. Two forward looking MoUs have been entered by IREL, one with UKTMP, Kazakhstan for setting up of facility for production of titanium slag using ilmenite produced from OSCOM mines, while the other one is with Sultanate of Oman for Co-operation in the field of rare earths. Titanium sponge is imported by Mishra Dhatu Nigam Ltd (MIDHANI) for further processing in the country.

The available data on plant-wise installed capacities of synthetic Rutile and TiO_2 pigment are furnished in Table-6.

Table – 6: Installed Capacity of Synthetic Rutile/Titanium dioxide Pigment

(In tonnes)

Plant	Location	Specification	Installed capacity (tpy)
IREL	Orissa Sands Complex, Distt Ganjam, Odisha.	90.5% TiO_2 (min.)	100000 (Synthetic rutile)
KMML	Chavara, Distt Kollam, Kerala	92-93% TiO_2	55000 (Synthetic rutile) 60000 (TiO_2 – Chloride Process)
DCW Ltd	Sahupuram, Distt Thoothukudi, Tamil Nadu.	95% TiO_2	42000 (Synthetic rutile)
CMRL	Edayar, Distt Ernakulam, Kerala.	96.5% TiO_2	50000 (Synthetic rutile)
TTPL	Kochuveli, Distt Thiruvananthapuram, Kerala.	97.5% TiO_2	17000 (TiO_2 –Sulphate Process)
VVTi Pigments Pvt. Ltd* (formerly Kilburn Chemicals)	Thoothukudi, Tamil Nadu.	98% TiO_2 (min.)	18000 (TiO_2 –Sulphate Process)
Kolmark Chemicals Ltd	Kalyani, Distt Nadia, West Bengal.	NA	4800 (TiO_2 – Sulphate Process)

Source: Department of Atomic Energy, Mumbai and individual companies

Note: KMML captively consumes synthetic rutile while CMRL and DCW export synthetic rutile

*: Including Kilburn Chemicals

USES & CONSUMPTION

About 90% of the world's titanium mineral production is used in the manufacturing of white titanium dioxide pigment. The unique combination of superior properties of high refractive index, low specific gravity, high hiding power and opacity and non-toxicity enable titanium dioxide in its application in the manufacture of all types of white and pastel shades of paints, white-walled tyres, glazed papers, plastics, printed fabrics, flooring materials like linoleum, pharmaceuticals, soaps, face powders and other cosmetic products. Besides, its non-toxic nature facilitates its use in cosmetics, pharmaceuticals, and even in foodstuffs as well as in toothpastes. Titanium dioxide is used in the manufacture of many sunscreen lotions and creams because of its non-toxicity and ultraviolet absorption properties. Synthetic Rutile is used for coating welding electrodes as flux component and for manufacture of titanium tetrachloride which in turn is used in making

titanium sponge. Synthetic Rutile is also used as ingredient of special abrasives. Titanium metal is a versatile material with exceptional characteristics. The lightness, strength and durability of the metal make it an essential metal for the Aerospace Industry. It is also used in desalination and power generation plants and corrosive chemical industries because of its inertness and resistance to corrosion and high thermal conductivity. Its non-reactive property makes titanium metal one of the few materials that can be used in the human body for orthopaedic use and in pacemakers.

The consumption of ilmenite was placed at 1,64,800 tonnes in 2019-20 which is lower as compared to the previous year. The bulk of Ilmenite was consumed by the Chemical Industry (99 %). Moderate proportion was consumed by Electrode Industry. The consumption of Rutile in 2018-19 was 10,500 tonnes as compared to 9,700 tonnes in 2017-18. The entire consumption was reported from Electrode Industry since last two years (Table-7).

Table – 7: Consumption* of Ilmenite and Rutile 2017-18 to 2019-20

(By Industries)

(In tonnes)

Industry	2017-18	2018-19 (R)	2019-20 (P)
Ilmenite			
All Industries	295000 (36)	222900 (36)	164800 (26)
Chemicals	294100	222000	163900
Electrode	800	800	900
Others (Ceramic, Ferroalloys)	100	100	-
Rutile			
All Industries	9700	10500	NA
Electrode	9700	10500	NA
Paint	-	-	NA

Figures rounded off

*: Includes actual reported consumption and/or estimates made wherever required. Paucity of data, hence, the coverage may not be complete.

() : Number of plants reported/estimated.

POLICY

The specific measures of systemic improvements adopted by the Department of Atomic Energy (DAE) to curb the illegal mining activities in India:

1. In the year 2015, the Govt. of India has brought amendment in MMDR Act, 1957 and a number of new rules were introduced. Under Section 11B of the MMDR Act, the Atomic Minerals Concession Rules, 2016 [AMCR, 2016] was notified. The concept of “Threshold Value” for Atomic Minerals was introduced in the AMCR. In the said notification the
2. All beach sand minerals including Garnet and Sillimanite were notified as ‘atomic minerals’ under Part B of First Schedule of MMDR Act, 1957 in 2016.
3. The revised Mineral Conservation Development Rules, 2017 designates Director, Atomic Minerals Directorate for Exploration and Research (AMD) as competent authority for Atomic Minerals above the threshold values specified in AMCR, 2016 (Table-8).

Table – 8: Particulars of Threshold Value for Atomic Minerals

[See Rule 2 (1)(m) and Rule 36]

Uranium-bearing tailings left over from ores after extraction of copper and gold, ilmenite and other titanium ores. 60 ppm U_3O_8 and/or 250 ppm ThO_2 .

Zirconium-bearing minerals and ores including zircon.

All cases of zirconium-bearing minerals occurring in Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e., the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade. In other cases, zircon containing less than 2,000 ppm of Hafnium.

Beach Sand Minerals, i.e., economic heavy minerals found in the teri or beach sand, which include ilmenite, rutile, leucoxene, garnet, monazite, zircon and sillimanite.

All cases of Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e., the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade.

4. Monazite Test Certification (MTC) by AMD has been reintroduced for every export consignments of beach sand minerals as per the Standard Operating Procedures (SOP) for Canalisation.
5. As the mining of beach sand minerals by private parties has been terminated, AERB has also not been renewing the license for operation of mineral separation plants by these private parties under Atomic Energy (Radiation Protection) Rules, 2004 from radiological safety considerations.
6. In view of stockpiled monazite enriched tailings with private BSM operators, AERB has been renewing the approval of Radiological Safety Officer (RSO)
- in these facilities so as to ensure the radiation surveillance of the plant premises and submission of reports of radiation monitoring to Atomic Energy Regulatory Board (AERB).
7. For licensing of Beach Sand Minerals processing facilities which are not involved in direct mining but procure the raw material from IREL or other countries, AERB issues license from radiation safety considerations only after the facility been licensed by DAE under Atomic Energy (Working of Mines, Minerals and handling of Prescribed Substance) Rules, 1984 for handling of monazite tailings.
8. Rule 45 of the Mineral Conservation and Development

- Rules, (MCDR) 2017 makes it mandatory for all miners, traders, stockiest, exporters and end-users of minerals to register and report on the production, trade and utilisation of minerals to the State Government(s) and Indian Bureau of Mines. It facilitates end-to end national-scale accounting of all minerals produced in the country from pithead to its end-use, reducing the scope for illegal mining, royalty evasion, etc.
9. Penalty for illegal mining was made more stringent by amendment of the MMDR Act. The penalties have been increased from T 25 thousand per hectare to T 5 Lakh per hectare and 1 year term of imprisonment has been increased from 2 years to 5 years for contravention of Section 4 (1) and 4 (1A) of the Act. Further, any rule made under the Act may provide that any contravention thereof shall be punishable with imprisonment for a term which may extend to two years or with fine which may extend to five lakh rupees, or with both, and in the case of a continuing contravention, with additional fine which may extend to fifty thousand rupees per day during which such contravention continues after conviction for the first such contravention. Section 30B of the Act provides for constitution of Special courts and 30C of the Act provides that such Special Courts shall be deemed to be a Court of Session.
 10. State Governments were requested to set up Task Forces at State and District Level to control illegal mining. Subsequently, 22 States including Tamil Nadu have set up Task Forces to control illegal mining and review the action taken by member departments for checking the illegal mining activities at state and district levels so far.
 11. The Ministry of Mines (MoM) through Indian Bureau of Mines has developed the mining Surveillance System (MSS), a satellite based monitoring system which aims to establish a regime of responsive mineral administration, through public participation, by curbing instances of illegal mining activity through automatic remote sensing detection technology.
 12. 21 State Governments including Tamil Nadu have framed rules under Section 23C of MMDR Act, 1957 to curb illegal mining.
 13. Issue of Notification by Ministry of Mines of Atomic Mineral Concession Rules, 2016 (AMCR-2016) towards protecting and conserving the "Atomic Minerals". "Beach Sand Minerals" (which include Ilmenite, Rutile, Zircon, Monazite, Sillimanite, Leucoxene and Garnet) are included to the list of "Atomic Minerals" declared under the MMDR Act, 1957.
 14. Directorate General of Foreign Trade, Department of Commerce, Ministry of Commerce and Industry, Government of India issued Notification No. 26/2015-2020 dated 21.08.2018 on export policy on BSM under which export of BSM have been brought under State Trading Enterprise and shall be canalised through IREL, a CPSE under DAE.
 15. MoM has issued Notification No. G.S.R. 134(E) dated 20.02.2019 towards ensuring complete government control over "Monazite" and "Zircon" occurring within the "Beach sand Minerals", by amending the Threshold value of "Monazite" for BSM occurring in beach sand or placer deposits as "0.00%" in Total Heavy Minerals (THM), which resulted in bringing the overall BSM deposits of the Country under the complete control of the AMCR, 2016 irrespective of monazite content.
 16. In the interest of regulation of mines & minerals development and conservation of BSM resources, MoM vide Order No. 1/1/2019-M.VI dated 01.03.2019 requested all the State Governments to take necessary action for pre-maturely terminating the existing private BSM mining leases under Section 4A(1) & 4A(3) of the MMDR Act 1957 towards conservation of Zirconium, Hafnium, Rare Earths and Thorium bearing strategic beach sand minerals.
 17. A Notification dated July 27, 2019 was issued for reserving the prospecting and mining rights of offshore minerals under Offshore Areas Mineral (Development and Regulation) Act, 2002 exclusively to Government and Government owned companies for curbing illegal mining of atomic minerals by private parties.
 18. MoM issued Notifications vide Nos. S.O. 2805 (E) dated 12.07.2021 and S.O. 2807 (E) dated 12.07.2021 under Section 22 (Cognizance of Offences) and Section 24 (Power of entry and inspection) of MMDR Act, 1957 specified under Part B of First Schedule to the MMDR Act, 1957. The above Notifications authorised Director, AMD and other officers of AMD for the purpose.

SUBSTITUTES

There are no cost-effective substitutes for titanium dioxide pigments. Synthetic Rutile made from ilmenite can be substituted for natural Rutile. Nickel steels, stainless steels and some non-ferrous metal alloys can sometimes replace titanium alloys in industrial uses although at the expense of performance or economics. Tungsten carbide competes with titanium carbide for surface cutting machine tools. Titanium slag competes with ilmenite and rutile.

Environmental awareness indicates that titanium dioxide plants are likely to use chloride technology in future as it produces much less quantity of waste products. Synthetic

Rutile or slag (made from ilmenite) is likely to be used as feed in increasing amount. There is also a strong pressure to reduce the radioactive content of feed stocks because it affects the marketability of beach sand ilmenite. Titanium alloys could be replaced in aerospace applications by lithium-aluminium alloys or carbon-epoxy composites.

WORLD REVIEW

World reserves of ilmenite are estimated at 690 million tonnes in terms of TiO_2 content. Major reserves are in China with 210 million tonnes (30%) followed by Australia

with 180 million tonnes (26%), India with 85 million tonnes (12%), Canada with 52 million tonnes (8%), Brazil with 43 million tonnes (6%), Norway with 37 million tonnes (5%), South Africa with 28 million tonnes & Madagascar with 27 million tonnes (4% each) and Mozambique with 22 million tonnes (3%). The world reserves of rutile are 55 million tonnes in terms of TiO_2 content. Major rutile reserves are located in Australia with 35 million tonnes (64%), followed by India with 7.4 million tonnes (13%) and South Africa with 6.1 million tonnes (11%) (Tables -9 & 9A).

Table – 9: World Reserves of Ilmenite
(By Principal Countries)

(In '000 tonnes of contained TiO_2)	
Country	Reserves
World: Total (Rounded off)	690000
China	210000
Australia ^a	180000
India	85000
Canada ^b	52000
Brazil	43000
Norway	37000
South Africa ^b	28000
Madagascar ^b	27000
Mozambique	22000
Other countries	6000

Source: USGS, Mineral Commodity Summaries, 2024

a: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were estimated to be 43 million tonnes for ilmenite TiO_2 content.

b: Mine production of titaniferous magnetite is primarily used to produce titaniferous slag.

Table – 9A: World Reserves of Rutile
(By Principal Countries)

(In '000 tonnes of contained TiO_2)	
Country	Reserves
World: Total (Rounded off)	55000
Australia ^a	35000
India	7400
South Africa ^b	6100
Sierra Leone	2900
Ukraine	2500
Madagascar ^b	520
Mozambique	720
Other countries	5260

Source: USGS, Mineral Commodity Summaries, 2024

a: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were estimated to be 11 million tonnes for rutile TiO_2 content.

b: Mine production of titaniferous magnetite is primarily used to produce titaniferous slag.

World production of ilmenite and rutile concentrates was 10.9 million tonnes and 0.50 million tonnes, respectively, in 2023. China contributed 3.1 million tonnes (28%) and Canada 1.9 million tonnes (17%) of ilmenite production, followed by South Africa with 1 million tonnes and Mozambique 0.98 million tonnes.

Australia produced 0.177 million tonnes of rutile, contributing 36% of world rutile output, followed by Sierra Leone & South Africa with 0.1 million tonnes each (20% each), Kenya with 0.067 million tonnes (14%) and Ukraine with 0.018 million tonnes (4%) (Table-10 & 10A).

Table – 10: World Production of Ilmenite
(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World total Ilmenite (wt of concs)	14100000	14200000	10900000
China	5714000	6300000*	3100000*
Canada [#]	1700000*	2100000*	1900000*
South Africa	1000000	900000	1000000*
Mozambique	1119400	1088334	986300
Korea, Rep. of	322581	137976	645949
Madagascar	582000	545000	613200*
Australia ^b	519875	496805	502360
Senegal	543061	498000	421000
Norway	545000	492000	401000
USA	200000*	300000*	300000*
India ^a	265000	265000	265000
Other countries	1589083*	1076885*	765191*

Source: BGS, World Mineral Production, 2019-23.

Notes-

(1) The figures in this table refer to gross tonnage of titanium concentrates.

(2) Some ilmenite is converted to synthetic rutile in Australia, India, Japan and USA.

[#]: Canada produces some ilmenite which is sold as such and not processed into slag, but tonnages are small

^a: Years ended 31st March

^b: Years ended 30th June

*: Estimated"

Table – 10A: World Production of Rutile
(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World total Rutile (wt of concs)	600000	600000	500000
Australia ^b	138906	99995	177961
Sierra Leone	131311	135818	100970
South Africa	95000	100000	100000*
Kenya	69844	163242	67604
Ukraine	100300*	58600*	18600*
India ^a	11000*	11000*	11000*
Mozambique	8914	8870	8400
Madagascar	6000*	6000*	6000*
Malaysia	11513	8359	3708
Sri Lanka	1881	740	2150
Other countries	25331	7376	3607

Source: BGS, World Mineral Production, 2019-23.

*: Estimated

The figures in this table refer to gross tonnage of titanium concentrates.

^a: Years ended 31st March

^b: Years ended 30th June

Metal

Commercial production of titanium metal involves the chlorination of titanium-containing mineral concentrates to produce titanium tetrachloride (TiCl₄), which is reduced with magnesium (Kroll process) or sodium (Hunter process) to produce a commercially pure form of titanium metal. The metal formed has a porous appearance and is referred to as sponge. Titanium ingot and slab are produced

by melting titanium sponge or scrap or a combination of both, usually with various other alloying elements.

Pigment

Global TiO₂ pigment production capacity was estimated to be 9.8 million tonnes per year. TiO₂ pigment produced is categorised by crystal form as either anatase or rutile. Rutile pigment is less reactive with the binders in paint when exposed to sunlight than the anatase pigment and

is preferred substance in outdoor paints. Anatase pigment has a bluer tone than rutile and is somewhat softer and is used mainly in indoor paints and in paper manufacturing. Depending on the manner in which it is produced and subsequently finished, TiO₂ pigment can exhibit a wide range of functional properties, including dispersion, durability, opacity and tinting.

FOREIGN TRADE

Exports

Exports of titanium ores & conc. increased to 0.22 million tonnes during 2023-24 from 0.14 million tonnes in the preceding year. Exports were mainly to China (41%) and Japan (33%).

Exports of titanium and alloys (including waste & scrap) were at 392 tonnes in 2023-24, registering an increase of 56% from 252 tonnes in the previous year. Exports were mainly to USA (27%) and Republic of Korea (20%). Exports of titanium oxide and dioxide (total) decreased by 10% to 31,871 tonnes in 2023-24 from 35,393 tonnes in the preceding year. Exports of Titanium Articles were at 49 tonnes in 2023-24, registering a decrease of 60% from 123 tonnes in the previous year. Exports were mainly to Nigeria (12%) and Singapore (10%). Exports of Ferrotitanium were at 2,044 tonnes, registering an increase of 16% from 1,765 tonnes in the previous year. Exports were mainly to UAE (25%), Japan & Netherlands (13% each) and Republic of Korea (12%) (Tables-11 to 18).

Table – 11: Export of Titanium Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	143607	7288953	224756	7915287
Japan	56422	3318834	74868	3320553
China	37778	1543591	93064	2844920
Korea, Rep. of	11000	290098	34000	880131
Netherlands	20000	574130	20000	522594
Malaysia	2548	364424	2496	293547
Saudi Arabia	--	--	216	32748
Italy	--	--	112	20786
UAE	--	--	++	8
Taiwan	5859	877685	--	--
Turkey	10000	320190	--	--
Other Countries	++	1	++	++

In tonnes

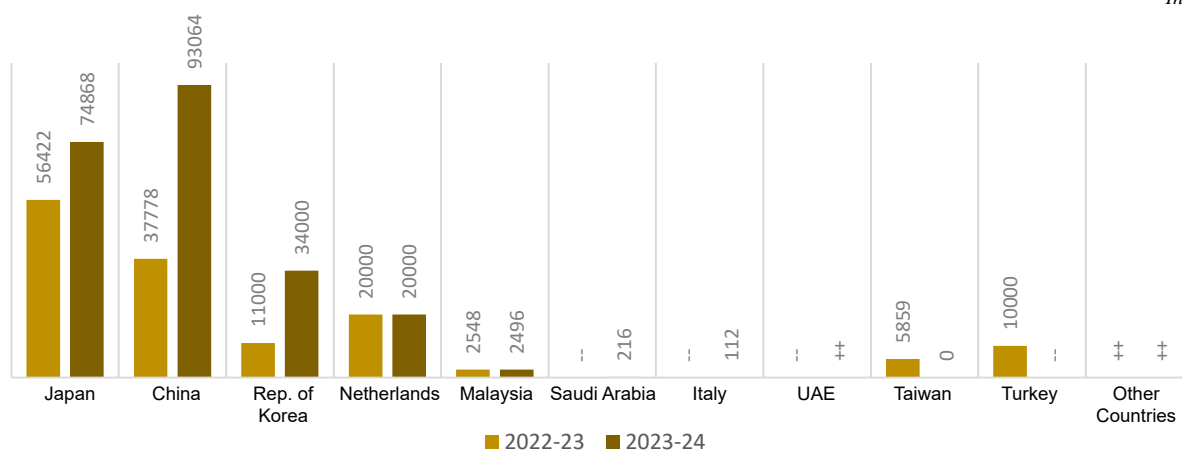


Fig. 3: Countrywise Export of Titaninum Ores & Conc.

Table– 12: Export of Titanium Ores and Conc. (Ilmenite)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	143607	7288953	220344	7437190
Japan	56422	3318834	72012	3002995
China	37778	1543591	92252	2783841
Korea, Rep. of	11000	290098	34000	880131
Netherlands	20000	574130	20000	522594
Malaysia	2548	364424	2080	247621
UAE	--	--	++	8
Taiwan	5859	877685	--	--
Turkey	10000	320190	--	--
Russia	++	1	--	--

Table– 13: Export of Titanium Ores and Conc. (Rutile)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	4300	457311
Japan	--	--	2856	317558
China	--	--	812	61079
Malaysia	--	--	416	45926
Saudi Arabia	--	--	216	32748

Table – 14: Export of Titanium Ores and Conc. (others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	112	20786
Italy	--	--	112	20786

Table – 15: Export of Titanium and Alloys (incl. Waste and Scrap)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	252	576880	392	625949
UK	5	61606	43	109000
USA	92	111981	107	100298
Korea, Rep. of	23	70160	78	59164
Japan	7	4341	56	48603
Saudi Arabia	1	10733	4	36183
Finland	2	31239	2	33624
Singapore	++	1696	20	30380
France	3	30178	3	30000
Canada	++	151	23	25371
Israel	1	5902	17	22135
Other Countries	118	248893	39	131191

Table – 16: Export of Titanium Oxide and Dioxide –Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	35393	5961366	31871	3802435
Japan	3582	420677	12231	1346428
China	8195	1065077	8661	764146
Russia	4	913	5277	560442
USA	9199	1848649	1899	400331
Malaysia	300	21133	1414	142240
Italy	664	162858	461	101402
UAE	79	19096	441	83414
Nepal	150	42468	170	41660
Brazil	76	21968	116	35315
Bangladesh	85	24898	123	32967
Other Countries	13059	2333629	1078	294090

Table – 17: Export of Titanium Articles

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	123	368318	49	242339
Finland	2	31239	2	33624
Saudi Arabia	1	10628	3	33602
USA	2	52406	3	28598
Singapore	++	1404	5	23532
Poland	--	--	3	10904
UAE	4	11875	3	10372
Turkey	17	68681	2	10240
Nigeria	++	528	6	8905
Germany	4	17513	1	8895
Lithuania	--	--	++	8144
Other Countries	93	174044	21	65523

Table – 18: Export of Ferrotitanium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1765	909712	2044	719361
UAE	122	55671	517	200032
Japan	260	177873	260	94969
Korea, Rep. of	292	153431	253	83928
Netherlands	394	154314	267	82801
UK	62	29780	151	58508
Spain	192	107938	123	42698
Saudi Arabia	249	132208	120	38346
Oman	132	60150	110	35080
USA	++	1	39	17863
Turkey	12	6754	57	14154
Other Countries	50	31592	147	50982

Imports

Imports of titanium ores & conc. increased by 32% to 89,920 tonnes in 2023-24 from 68,000 tonnes in the preceding year. Imports were mainly from Mozambique (44%), Malaysia & Netherlands (15% each), Australia (6%) and China (5%).

Imports of titanium and alloys (including waste & scrap) were 11,022 tonnes in 2023-24 as compared to 9,970 tonnes in the previous year. Imports were mainly from China (27%), USA (17%), Japan (14%), Singapore (9%), Taiwan (6%) and Republic of Korea (5%). Imports of titanium oxide and dioxide (total) were 17,786 tonnes

in 2023-24 as compared to 18,654 tonnes in the preceding year. Imports were mainly from Republic of Korea (35%), China (27%), Japan (11%), Belgium (10%) and Germany (7%). Imports of Titanium Articles were at 2,545 tonnes in 2023-24, registering an increase of 5% from 2,427 tonnes in the previous year. Imports were mainly from China (71%) and Germany (6%). Imports of Ferrotitanium were at 1,528 tonnes, registering an exponential increase of 171% from 564 tonnes in the previous year. Imports were mainly from Netherlands (41%), Estonia (25%), UK (14%) and Germany & Canada (7% each) (Tables - 19 to 26).

Table – 19: Import of Titanium Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	68000	4969119	89920	6409133
Malaysia	8607	795381	13924	1878489
Mozambique	20003	870731	39946	1341708
Australia	4888	677059	5522	743562
China	4932	769209	4811	635886
Netherlands	13143	336467	13232	364481
Senegal	1540	221855	1900	258345
Vietnam	3741	331216	1836	246095
Thailand	3503	236277	1876	199642
South Africa	1400	224759	1152	179210
UAE	2147	111503	1097	143465
Other Countries	4096	394662	4624	418250

In tonnes

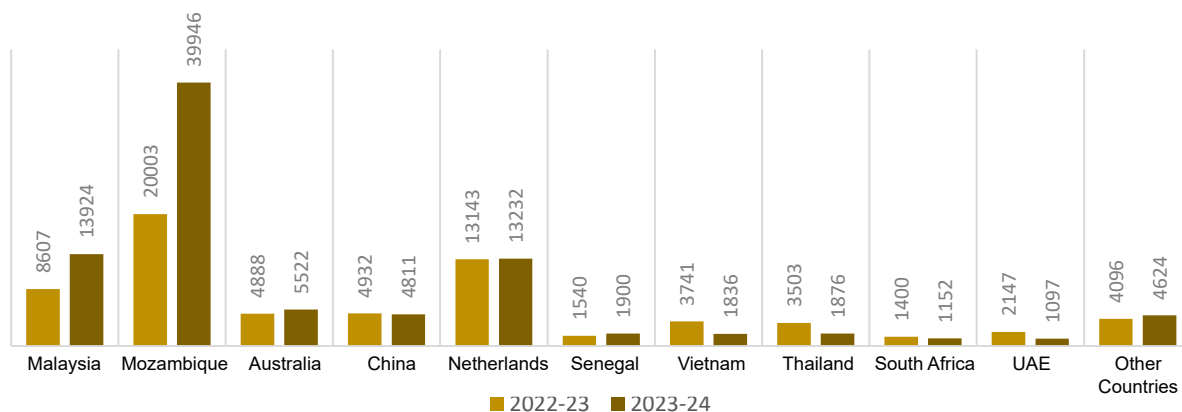


Fig. 4: Countrywise Imports of Titanium Ores & Conc.

Table – 20: Import of Titanium Ores and Conc. (Ilmenite)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	32126	1398364	44530	1533498
Mozambique	20003	870731	39920	1337344
Malaysia	4450	143491	2288	86104
Australia	--	--	416	50575
China	101	8466	707	27979
Egypt	--	--	459	12962
Thailand	2496	107340	364	11653

Table- 20 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
UAE	1997	71399	115	3135
New Zealand	--	--	257	2719
USA	3	695	4	1027
Sri Lanka	2227	152602	--	--
Other Countries	849	43640	++	++

Table – 21: Import of Titanium Ores and Conc. (Rutile)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	19468	2774144	25525	3721434
Malaysia	2745	422971	9062	1389897
Australia	4134	572022	4716	635278
China	4544	722369	2644	388892
Vietnam	3122	304633	1809	242754
Thailand	1007	128937	1512	187989
South Africa	1400	224759	1126	175158
Senegal	940	135038	1200	162263
UAE	--	--	820	126486
Kenya	281	51648	674	103007
Hong Kong	--	--	520	76492
Other Countries	1295	211767	1442	233218

Table – 22: Import of Titanium Ores and Conc. (Others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	16406	796611	19865	1154201
Malaysia	1412	228919	2574	402488
Netherlands	12675	268168	12954	306467
China	287	38374	1460	219015
Senegal	600	86817	700	96082
Australia	754	105037	390	57709
Germany	--	--	1492	35455
UAE	150	40104	162	13844
Ukraine	--	--	52	9557
Mozambique	--	--	26	4364
South Africa	--	--	26	4052
Other Countries	528	29192	29	5168

Table – 23: Import of Titanium and Alloys (incl. Waste and Scrap)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	9970	10442098	11022	11138422
China	2825	4183002	3009	4425979
USA	1194	1466434	1831	1777830
Germany	211	743307	340	1105893
Japan	1325	590550	1573	546302
France	56	409088	67	544784

Table- 23 (Concl'd.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
UK	265	555951	159	471807
Singapore	1247	538896	988	368617
Taiwan	986	458815	657	317870
Italy	59	111220	343	310296
Korea, Rep. of	638	322153	518	217263
Other Countries	1164	1062682	1537	1051781

Table – 24: Import of Titanium Oxide and Dioxide – Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	18654	5508817	17786	4862334
Korea, Rep. of	5540	1419606	6150	1231413
China	5771	1440940	4861	1094516
Belgium	808	310687	1775	749140
Germany	1941	752775	1159	524386
Japan	1446	475343	1941	509950
Netherlands	2102	735064	1139	445855
Taiwan	249	90639	283	114083
Canada	18	8337	198	90412
France	25	28676	44	35092
Singapore	66	21989	40	11991
Other Countries	688	224761	196	55496

Table – 25: Import of Titanium Articles

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2427	5874108	2545	6880354
China	1732	2869268	1803	3107954
Germany	141	671804	162	1021132
USA	111	864087	74	889927
France	53	394059	67	544072
UK	39	390372	77	417713
Italy	30	103143	85	239959
UAE	95	91816	25	112245
Taiwan	22	57793	33	90127
Russia	8	27943	34	85954
Japan	36	81511	36	72533
Other Countries	160	322312	149	298738

Table – 26: Import of Ferrotitanium

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	564	252271	1528	530315
Netherlands	17	9414	625	211230
Estonia	100	40790	379	126678
UK	142	55831	213	86744
Germany	16	17604	100	33158

Table- 26 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Canada	--	--	100	31083
Russia	20	7964	50	15674
Switzerland	22	5743	40	13596
Belgium	--	--	17	9415
China	169	69173	2	1897
Sweden	--	--	2	840
Other Countries	78	45752	++	++

FUTURE OUTLOOK

As per Technology Vision Document 2035, the demand for titanium in India would be approximately 1,000 tonnes by 2035. The contribution to the demand would be from Space Sector (100 tonnes), General Engineering (50 tonnes), Atomic Energy (125 tonnes), Aeronautical (50 tonnes), Power Generation (150 tonnes), Petroleum Refinery (50 tonnes) and Chemical Industry (475 tonnes).

Construction activity has gathered momentum and the Real Estate Industry too is witnessing accelerated growth with the Government from its end planning and stepping up infrastructure spends. These are likely to have a positive impetus on paint, ceramics, welding electrodes, steel, cement and allied industries, which will have positive spin off effect on demand for ilmenite in domestic market.

As per data available, the defence, atomic energy and space research – which are critical sectors – have been assigned targets to increase communication set-up, safeguard India's security with modern arms, ammunitions & control and to increase power generation by three-fold.

For meeting these targets, Indian Engineering Industry will depend on input materials like titanium sponge. Titanium sponge was not available in India till 2012, the first ever commercially indigenously made Ti-sponge was released in the year 2013 at KMML, Kerala, with the support of ISRO.

Titanium Sponge Industry, which suffered setbacks during the pandemic, has been steadily regaining its lost ground as the demand from the Aviation Sector has shown a strong recovery and resurgence in terms of long-term growth prospects. Global demand growth for TiO₂ is expected to trend with the prospects of economic growth and production of paint, paper and plastics.

Aerospace, defence and industrial uses are expected to strongly influence the consumption of titanium metal in the near future.

The impetus by the Government for renewal energy and infrastructure, however, augurs well for demand for steel and that for IREL's Rutile which finds application in production of welding electrodes.



43. Iron Ore

277.332 million tonnes
production of iron ore have been
reported in 2023-24.

46.37 million tonnes
of iron ore were exported in 2023-24.

4.96 million tonnes
of iron ore were imported in 2023-24.

Iron & steel is fundamental and a key factor for modernisation and industrial development. The vitality of the Iron & Steel Industry largely influences a country's economic status. The mining of iron ore, an essential raw material for Iron & Steel Industry, is arguably of prime importance among all mining activities undertaken by any country. With the total resources of over 35.29 billion tonnes of haematite (Fe_2O_3) and magnetite (Fe_3O_4), India is one of the leading producers of iron ore in the world.

RESERVES/RESOURCES

Haematite and magnetite are the most important iron ores in India. About 79% haematite ore deposits are found in the Eastern Sector (Assam, Bihar, Chhattisgarh, Jharkhand, Odisha & Uttar Pradesh) while about 93% magnetite ore deposits occur in Southern Sector (Andhra Pradesh, Goa, Karnataka, Kerala & Tamil Nadu). Karnataka alone contributes 72% of magnetite deposit in India. Of these, haematite is considered to be superior because of its higher grade. Indian deposits of haematite belong to the Precambrian Iron Ore Series and the ore is within banded iron ore formations occurring as massive, laminated, friable and also in powdery form.

As per NMI database based on UNFC system, the total reserves/resources of haematite as on 1.4.2020 have

been estimated at 24,057 million tonnes of which 6,209 million tonnes (25.80%) are under 'Reserves' category and the balance 17,848 million tonnes (74.20%) are under 'Remaining Resources' category. By grades, Lumps constitute about 45% followed by Lumps with Fines 27%, Fines 13%, and the remaining 15% are Black Iron ore, Beneficiable grade; Others; unclassified, Not-known; Lumps & fines & blue dust and low grade; Lumps & fines & blue dust unclassified. Major reserves/resources of haematite are located in Odisha (9,409 million tonnes or 39%), Jharkhand (4,710 million tonnes or 20%), Chhattisgarh (4,592 million tonnes or 19%), Karnataka (2,835 million tonnes or 12%) and Goa (1,197 million tonnes or 5%). The balance 5% resources of haematite are spread in Andhra Pradesh, Assam, Bihar, Madhya Pradesh, Maharashtra, Meghalaya, Rajasthan, Telangana and Uttar Pradesh (Table-1).

Table – 1: Reserves/Resources of Iron Ore (Haematite) as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	6209034	17848870	24057905
By Grades			
Lump, high grade	832185	940095	1772279
Lump, medium grade	1432585	4188788	5621372
Lump, low grade	686331	2388547	3074877
Lump, unclassified grade	210	246248	246459
Fines, high grade	146830	67192	214022
Fines, medium grade	147829	1204201	1352029
Fines, low grade	148301	1208318	1356619
Fines, unclassified grade	490	115885	116375
Lumps & fines high grade	471678	478709	950387
Lumps & fines medium grade	1016424	1141559	2157983
Lumps & fines low grade	698544	2435597	3134141
Lumps & fines unclassified	188144	153661	341805
Black iron ore	-	19106	19106
Beneficiable grade	270521	1118438	1388959
Others	23905	53585	77490
Unclassified	86138	329906	416044
Not-known	1569	1728782	1730351
Lumps & fines & blue dust low grade	-	1847	1847
Lumps & fines & blue dust unclassified grade	57351	28408	85759
By States			
Andhra Pradesh	44744	350872	395616
Assam	-	30890	30890
Bihar	-	55	55
Chhattisgarh	1593732	2998379	4592111
Goa	117235	1080322	1197557
Jharkhand	534677	4175469	4710146
Karnataka	1043212	1792781	2835992
Madhya Pradesh	54129	302870	356999
Maharashtra	15241	286304	301544
Meghalaya	-	225	225
Odisha	2798749	6610582	9409331
Rajasthan	7314	28166	35480
Telangana	-	105627	105627
Uttar Pradesh	-	86330	86330

figures rounded off

Magnetite is another principal iron ore that also occurs in the form of oxide, either in igneous or metamorphosed banded magnetite-silica formation. As per NMI database based on UNFC system, the total reserves/resources of magnetite as on 01.04.2020 have been estimated at 11,227 million tonnes of which 'Reserves' constitute 202 million tonnes while 11,024 million tonnes are placed under 'Remaining Resources'. Classification on the basis of grades shows that 20% resources are of Metallurgical grade while 80% resources belong to grades that are categorised as Coal washery, Foundry, Beneficiable, Others, Unclassified and

Not-known. The resources of Others and Foundry grades constitute meagre proportions. India's 96.70% magnetite reserves/resources are located in five States, namely Karnataka (7,802 million tonnes or 69.50%) followed by Andhra Pradesh (1,472 million tonnes or 13.10%), Rajasthan (794 million tonnes or 7.10%), Tamil Nadu (528 million tonnes or 4.70%) and Goa (266 million tonnes or 2.30%). Assam, Bihar, Chhattisgarh, Jharkhand, Kerala, Maharashtra, Meghalaya, Nagaland, Odisha and Telangana together account for the remaining 3.30% resources (Table-2).

Table – 2: Reserves/Resources of Iron Ore (Magnetite) as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	202823	11024791	11227614
By Grades			
Metallurgical	315	2238244	2238559
Coal washery	118678	124368	243045
Foundry	-	836	836
Beneficiable	-	36798	36798
Others	1351	6923	8274
Unclassified	82479	8264559	8347038
Not-known	-	353064	353064
By States			
Andhra Pradesh	-	1472383	1472383
Assam	-	15380	15380
Bihar	-	49439	49439
Chhattisgarh	75876	30045	105921
Goa	4990	261345	266336
Jharkhand	-	10667	10667
Karnataka	318	7801853	7802171
Kerala	-	83435	83435
Maharashtra	578	1210	1788
Meghalaya	-	3380	3380
Nagaland	-	5280	5280
Odisha	-	242	242
Rajasthan	121060	673866	794926
Tamil Nadu	-	528901	528901
Telangana	-	87366	87366

Figures rounded off

MINING LEASES & PRODUCTION

As on 31.03.2024(P), the total number of mining leases granted for mining of iron ore is 337 covering a total mining lease area of 56,920.61665 ha. The production of iron ore constituting lumps, fines, calibrated lump ore (CLO) and concentrates was at 277.332 million tonnes in the year 2023-24, showing an increase of about 7.36% as compared to that in the preceding year.

There were 202 mines that reported production in 2023-24 as against 207 in the previous year. Among them, 43 mines were in the Public Sector and 159 in Private Sector. Besides, production of iron ore was reported as associated mineral by 15 mines in 2023-24 which is one greater than that of the year 2022-23. The contribution of Public Sector to the total production was about 41.03% as against about 41.47% in the preceding year. The remaining 58.97% of the production in 2023-24 was from Private Sector. Among 43 iron ore mines in Public Sector, 24 iron ore mines each producing more than one million tonnes annually accounted for about 96.98 % of the total output in Public Sector during 2023-24. Out of 159 iron ore mines under Private Sector, 35 iron ore mines each producing more than one million tonnes annually accounted for about 83.01% of the total output of Private Sector during the year. Thus, 59 iron ore mines, each producing more than one million tonnes of iron ore annually, contributed about 88.75% of

the total output in 2023-24. The captive mines reported production of 102.737 million tonnes comprising about 37.04% of total production and non-captive mines reported production of 174.595 million tonnes, i.e., about 62.96% during 2023-24.

Gradewise analysis of the current year's output reveals that, out of total output of 277.332 million tonnes, iron ore lumps constituted 26.842 million tonnes (i.e., about 9.68%), fines constituted 202.309 million tonnes (i.e., about 72.95%), CLO constituted 46.335 million tonnes (i.e., about 16.71%) and concentrates constituted 1.846 million tonnes (i.e. about 0.67%). Among the States, Odisha recorded the highest production of 148.959 million tonnes, i.e., about 53.71% of the country's production in 2023-24. Chhattisgarh was at the second place with a production of 46.043 million tonnes, i.e., about 16.60% of the total production followed by Karnataka with 40.996 million tonnes, i.e., about 14.78 % of the country's production. The remaining 41.33 million tonnes, i.e., (14.91%) production was reported from Andhra Pradesh, Madhya Pradesh, Maharashtra, Rajasthan, Goa and Jharkhand (Table-3 to 6).

STOCKS AT MINE-HEAD

The mine-head closing stock of iron ore for year 2023-24 was 83.05 million tonnes as compared to 97.36 million tonnes in 2022-23 [Table-7(A) & 7 (B)].

Table – 3: Principal Producers of Iron Ore, 2023-24

(By States)

Name & address of producer	Location of the mine	
	State	District
National Mineral Corporation Ltd., 10-3-311/A, Khanji Bhavan, Castle Hills, Masab Tank, Hyderabad-500028"	Chhattisgarh Karnataka	Dantewada Bellary
Steel Authority of India, Ispat Bhavan, Lodhi Road, (West), New Delhi- 11000	Chhattisgarh Jharkhand Odisha	Durg Singhbhum Keonjhar Sundargarh
Odisha Mining Corporation Ltd, OMC House, Unit-5, P.B.No.34, Distt.Khurda, Bhubaneswar-751001, Odisha	Odisha	Keonjhar Sundargarh
Tata Steel Ltd, (West) Bombay House, 24, Homi Mody Street, Fort, Mumbai-440001, Maharashtra	Jharkhand Odisha"	Singhbhum Keonjhar
JSW Steel Limited, JSW Centre, Bandra Kurla Complex, Bandra (East), Mumbai-400051, Maharashtra	Karnataka Odisha"	Bellary Keonjhar
Rungta Sons (P) Ltd, 8A Express Tower, 42A Shakespeare Sarani, Kolkata-700017 West Bengal	Odisha	Keonjhar
M/s Lloyds Metals and energy limited A-2, 2 nd Floor, Madhu Estate, Pandurang Budhkar Marg, Lower Parel, Mumbai-400013	Maharashtra	Gadchiroli
Jindal Steel & Power Ltd., O.P. Jindal Marg, Delhi Road, Hissar-125005, Haryana	Odisha	Sundargarh
Vedanta Ltd, Sesa Ghor, EDC Complex, Patto, Panaji, Tisavadi-403001, Goa	Goa Karnataka	North Goa South Goa Chitradurga
M/s Arcelor Mittal Nippon steel India limited Raheja Towers, 6th & 7th Floor, BKC, Bandra East, Mumbai-400051, Maharashtra, India	Odisha	Sundargarh
M/s Arcelor Mittal India Private Limited Regus Business Centre, Office No. 317, 2&3 Floor, Privillion Block B 202 203 204 East Wing , Sarkhej Gandhinagar Highway Ahmedabad, Gujarat, India – 380054	Odisha	Keonjhar
M/s ESL Steel Limited, Lohanchal Colony, Plot No.10, Beside Sector 12, Bokaro Steel City - 827013	Odisha	Sundargarh
Geetarani Mohanty C 59 Palashpalli, Bhubaneswar 751020	Odisha	Keonjhar
Mr Prabodh Mohanty Weigh Bridge Road, Barbil, Dist - Keonjhar, Odisha, Pin - 758035	Odisha	Sundargarh
M/s Jayaswal Neco Industries Limited f-8, M.I.D.C. Hingna Industrial Area, Hingna, Nagpur-440016	Chhattisgarh Maharashtra	Kanker Narayanpur Gondia
M/s Godawari Power & Ispat LTD Plot No. 428/2, Phase I, Industrial Area, Raipur District Chhattisgarh 493111	Chhattisgarh Odisha	Kanker Sundargarh

Table – 4: Production of Iron Ore, 2021-22 to 2023-24 (P)

(By States/Grade)

(Quantity in '000 tonnes; Value in ₹ '000)

States	Grade	2021-22		2022-23		2023-24 (P)	
		Quantity	Value	Quantity	Value	Quantity	Value
India	Total	254099	1008264608	258312	834001723	277332	998247994
	Lumps	72872	367124174	31702	119239059	26842	95501984
	Fines	179833	633761585	182223	517694387	202309	683860125
	CLO	-	-	42678	189627528	46335	210469947
	Concentrate	1394	7378849	1709	7440749	1846	8415938
Andhra Pradesh	Total	319	237855	185	126441	125	107592
	Lumps	197	166143	95	77605	84	85340
	Fines	122	71712	90	48835	41	22252
	CLO	-	-	-	-	-	-
	Concentrate	-	-	-	-	-	-
Chhattisgarh	Total	41333	222498723	42487	170877445	46043	186363948
	Lumps	14564	89901192	12271	51711017	9673	33877931
	Fines	26769	132597531	27250	99839036	30479	113731096
	CLO	-	-	2966	19327392	5891	38754921
	Concentrate	-	-	-	-	-	-
Goa	Total	-	-	-	-	12	17421
	Lumps	-	-	-	-	2	8085
	Fines	-	-	-	-	10	9326
	CLO	-	-	-	-	-	-
	Concentrate	-	-	-	-	-	-
Jharkhand	Total	24728	55168902	23091	48359178	22896	54829400
	Lumps	5680	13628848	3456	8497035	3059	8970279
	Fines	19048	41540054	17342	35830534	17904	42065098
	CLO	-	-	2293	4031609	1933	3794023
	Concentrate	-	-	-	-	-	-
Karnataka	Total	40333	160324539	39822	115421232	40996	140433482
	Lumps	11707	57045697	11312	40054829	11754	42424709
	Fines	28626	103278842	28510	75366403	29242	98008773
	CLO	-	-	-	-	-	-
	Concentrate	-	-	-	-	-	-
Madhya Pradesh	Total	7375	4727826	4365	2475698	6047	3812315
	Lumps	1173	756699	810	479676	544	397286
	Fines	6202	3971127	3555	1996022	5499	3395134
	CLO	-	-	-	-	-	-
	Concentrate	-	-	-	-	4	19895
Maharashtra	Total	1984	6653399	6109	20708375	10727	40921814
	Lumps	742	4264350	2082	11345048	1693	9678310
	Fines	1242	2389049	4027	9363327	9034	31243504
	CLO	-	-	-	-	-	-
	Concentrate	-	-	-	-	-	-
Odisha	Total	136792	553095800	140932	469762857	148959	564457411
	Lumps	38675	201277779	1669	7053936	23	33992
	Fines	97821	349910387	101447	295248211	110094	395379858
	CLO	-	-	37419	166268527	38511	167921003
	Concentrate	296	1907634	397	1192183	331	1122558
Rajasthan	Total	1235	5557564	1321	6270497	1527	7304611
	Lumps	134	83466	7	19912	10	26052
	Fines	3	2883	2	2019	6	5074
	CLO	-	-	-	-	-	-
	Concentrate	1098	5471215	1312	6248566	1511	7273485

(P): provisional

Table – 5 (A): Production of Iron Ore, 2022-23
(By Sectors / States / Districts / Grades)

(Quantity in '000 tonnes; Value in ₹ '000)

State/ District	No. of mines	Lumps						Fines						Concentrates			Calibrated Lump Ore (CLO)								Total	Value												
		Below 45% Fe (For Mag- netite Ore)	45% to below 55% Fe	55% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% to below and above 65% Fe	Total	Value	Below 45% Fe (For Magne- tite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% to below and above 65% Fe	Total	Value	Below 62% Fe (CLO any size)	62% to below 65% Fe (5-18 40 mm size CLO)	62% to below 65% Fe (10- 40 mm size CLO)	65% to below abo- ve Fe (5-18 mm size CLO)	65% and above 65% Fe (10- 40 mm size CLO)	65% and above 65% Fe (10- 40 mm size CLO)			65% and above 65% Fe (10- 40 mm size CLO)											
India	207(14)	93	1794	2860	1157	3410	4542	10533	7313	31702	11923	9059	50	3876	15047	26847	13452	47840	52129	22982	182223	517694	387	1709	7440749	13575	3949	14968	27	133	10026	-	42678	189627528	258312	83400	1723	
Public Sector	42	-	20	355	312	1346	3545	6769	7122	19469	77223	785	-	12	615	2605	4764	29280	25915	5995	69186	227964	670	-	-	960	1864	12222	-	11	3432	-	18489	98735009	107144	403923	464	
Private Sector	165(14)	93	1774	2505	845	2064	997	3764	191	12233	4201	5274	50	3864	14432	24242	8688	18560	26214	16987	113037	289729	717	1709	7440749	12615	2085	2746	27	122	6594	-	24189	90892519	151168	430078	259	
Andhra Pradesh	8 (1)	-	21	60	14	-	-	-	-	95	77606	-	22	34	34	-	-	-	-	-	-	90	48835	-	-	-	-	-	-	-	-	-	-	-	-	185	126441	-
Kurnool	7 (1)	-	21	23	14	-	-	-	-	58	47253	-	22	10	34	-	-	-	-	-	-	66	34756	-	-	-	-	-	-	-	-	-	-	-	-	124	82009	-
Y.S.R.	1	-	-	37	-	-	-	-	-	37	30353	-	-	24	-	-	-	-	-	-	24	14079	-	-	-	-	-	-	-	-	-	-	-	-	61	44432	-	
Chhattis-garh	20	-	28	657	257	506	1327	2622	6874	12271	5171	1017	-	250	1904	1126	1243	3435	12674	6618	27250	998390	36	-	-	-	-	52	-	2914	-	2966	19327392	42487	17087	7445	-	
Dantewada	6	-	-	-	-	44	-	132	6874	7050	43050	831	-	-	22	24	492	12033	5739	18310	83503	450	-	-	-	-	-	52	-	2914	-	2966	19327392	28326	14588	1673	-	
Durg	4	-	20	69	32	90	1236	2406	-	3853	6528	572	-	12	128	277	157	2581	604	0	3759	69400	36	-	-	-	-	-	-	-	-	-	-	-	7612	13468	608	-
Kanker	6	-	8	410	118	53	79	46	-	714	7373	90	-	238	1241	695	569	308	37	879	3967	7269	189	-	-	-	-	-	-	-	-	-	-	4681	8006	579	-	
Narayanpur	2	-	-	-	35	290	6	38	++	369	653	112	-	-	229	84	477	51	-	-	841	1126	795	-	-	-	-	-	-	-	-	-	-	1210	1779	907	-	
Rajnand-gaon	2	-	-	178	72	29	6	++	-	285	741	112	-	-	306	48	16	3	++	-	373	999	566	-	-	-	-	-	-	-	-	-	-	658	1740	678	-	
Goa	8	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
North Goa	3*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
South Goa	5*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Jharkhand	12	-	-	188	110	808	1527	795	28	3456	8497	035	-	-	8	2011	2770	3200	2146	7207	17342	35830	534	-	-	813	-	-	-	-	1480	-	2293	4031609	23091	48359	178	-
West Singhbhum	12	-	-	188	110	808	1527	795	28	3456	8497	035	-	-	8	2011	2770	3200	2146	7207	17342	35830	534	-	-	813	-	-	-	-	1480	-	2293	4031609	23091	48359	178	-
Karnataka	51(2)	93	1038	1596	676	1725	1479	4330	375	11312	4005	4829	43	1539	5005	6819	3975	3442	7046	641	28510	75366	403	-	-	-	-	-	-	-	-	-	-	-	39822	11542	1232	-
Bagalkote	2	-	9	95	-	-	-	-	-	104	268	457	-	10	11	84	-	-	-	-	105	151	239	-	-	-	-	-	-	-	-	-	-	209	419	966	-	
Ballari	43	93	997	1254	441	1061	1344	4131	375	9696	3296	4721	43	1480	3047	2293	3445	3398	7046	641	21393	58713	247	-	-	-	-	-	-	-	-	-	-	31089	91677	968	-	
Chitradurga	5(1)	-	30	240	232	663	135	199	-	1499	679	4637	-	11	1947	4432	530	44	-	-	6964	16444	149	-	-	-	-	-	-	-	-	-	-	8463	2323	876	-	
Tumakuru	1(1)	-	2	7	3	1	-	-	-	13	270	14	-	38	-	10	-	-	-	-	48	57	768	-	-	-	-	-	-	-	-	-	-	61	84	782	-	
Madhya Pradesh	24(10)	-	697	113	++	-	-	-	-	810	479	676	-	1069	2431	40	10	5	-	-	3555	1996	022	-	-	-	-	-	-	-	-	-	-	4365	247	5698	-	-
Chhatarpur	1	-	-	90	-	-	-	-	-	90	40	590	-	-	8	-	-	-	-	-	8	34	12	-	-	-	-	-	-	-	-	-	-	98	44	002	-	
Gwalior	1*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	

Table- 5A (Concl'd.)

(Quantity in '000 tonnes; Value in ₹ '000)

State/ District	No. of mines	Lumps										Fines										Concentrates			Calibrated Lump Ore (CLO)										Total	Value																																																																																																																																																																																																																																																																																																																																																																																						
		Below 45% Fe (For Mag-netite Ore)					45% to 55% Fe					55% to 60% Fe					60% to 65% Fe					Total	Value	Below 62% Fe (CLO any size)	62% to 65% Fe (10-40 mm size CLO)	62% to 65% Fe (10-40 mm size CLO)	65% to 65% and above Fe (10-40 mm size CLO)	65% and above Fe (10-40 mm size CLO)	Total	Value																																																																																																																																																																																																																																																																																																																																																																																												
		Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe										Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe			65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe	60% to 65% Fe	65% to 65% and above Fe	Below 45% Fe (For Mag-netite Ore)	45% to 55% Fe	55% to 60% Fe

*: Only labour reported ; ++: Negligible

(): Figure in parenthesis indicates mines of Dolomite, Manganese Ore, Ochre and Laterite as an associate mineral.

Table – 5 (B): Production of Iron Ore, 2023-24 (P)
(By Sectors / States / Districts / Grades)

State/District	No. of mines	Lumps										Fines										Concentrates		Calibrated Lump Ore (CLO)										Total	Value	
		Below 45% Fe (For Magnetite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Value	Below 45% Fe (For Magnetite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Value	Below 62% Fe (CLO any size)	62% to below 65% Fe (5-18 mm size CLO)	62% to below 65% Fe (10-40 mm size CLO)	62% to below 65% Fe (CLO others)	65% and above Fe (5-18 mm size CLO)	65% and above Fe (10-40 mm size CLO)	65% and above Fe (CLO others)	Total	Value						
India	202 (15)	112 2272 2887	1446 3011 3851	8631 4632 26842	95501984	133 8519	21008 26144	14144 48263	68415 15683	68415 15683	202309 683860125	1846	8415938	15701 3113	16363	1 237	10920	-	46335 210469947	277332	998247994															
	Public Sector	43	- 97 96	443 1938	2474 6391	4469 15908	60709167	- 184 892	2402 4760	33056 30144	4819 76257	282927205	-	- 1116 1702	12348	- 1	6462	-	21629 110998987	113794	454635359															
	Private Sector	159 (15)	112 2175 2791	1003 1073 1377	2240 163	10834 34792817	133 8335	20116 23742	9384 15207	38271 10864	126052 400932920	1846	8415938	14585 1411	4015	1 236	4458	-	24706 99470960	163538	543612635															
Andhra Pradesh	9(1)	- 2 82	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	
	Kurnool	7(1)	- 2 32	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	
	Y.S.R.	2	- - 50	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	
Chhattisgarh	21	- 95 609	226 644 764	3293 4042 9673	33877931	- 694 1836	1329 918	5212 15915	4575 30479	113731096	-	-	2	91	-	5798	-	5891 38754921	46043	186363948																
	Dantewada	6	- - -	- 113 1 401	4026 4541	23024148	- - -	1 119	1891 14576	3930 20517	93353958	-	-	91	-	5798	-	5891 38754921	30949	155133027																
	Durg	4	- 85 10	35 151 651	2566 - 3498	8134644	- 66 186	267 6	2566 1338	- 4429	10706248	-	-	-	-	-	-	-	7927	18840892																
Kanker	7	- 10 504	91 8 76	40 16 745	737027	- 628 1159	725 375	534 - 645	4066 7204637	-	-	-	-	-	-	-	-	4811	7941664																	
	Narayanpur	2	- - -	- 27 353	36 286 - 702	1556369	- 0 369	292 404	214 - 1279	2039679	-	-	-	-	-	-	-	-	1981	3596048																
	Rajnandgaon	2	- - 95	73 19 ++ -	- - -	- 187 425743	- 0 122	44 14 7	1 - 188	426574	-	-	-	-	-	-	-	-	375	852317																
Goa	1	- - 2	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	12	17421																
	North Goa	1	- - 2	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	12	17421																
	Jharkhand	9	- - 10	17 1061 1030	613 328 3059	8970279	- 24 581	2416 3813	2296 8774	17904 42065098	-	-	456	-	-	1477	-	1933 3794023	22896	54829400																
West Singhbhum	9	- - 10	17 1061 1030	613 328 3059	8970279	- 24 581	2416 3813	2296 8774	17904 42065098	-	-	456	-	-	1477	-	1933 3794023	22896	54829400																	
	Karnataka	51(2)	112 1687 2053	1142 1281 1597	3620 262 11754	42424709	46 2437 6944	7530 3168	2350 6156	611 29242	98008773	-	-	-	-	-	-	-	40996	140433482																
	Bagalkote	2	- 69 28	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	247	270928															
Ballari	42	112 1527 1190	863 1217 1515	3447 262 10133	35890282	46 2221 2743	4173 3168	2310 6156	611 21428	69742217	-	-	-	-	-	-	-	-	31561	105632499																
	Chitradurga	6(1)	- 70 834	278 64 82	173 - 1501	6322375	- 82 4164	3356 - 40 -	7642 28112235	-	-	-	-	-	-	-	-	9143	34434610																	
	Tumakuru	1(1)	- 21 1	1 - -	- - -	- 23 57766	- 21 -	1 - -	- - -	22 37679	-	-	-	-	-	-	-	-	45	95445																
Madhya Pradesh	25(9)	- 459 85	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	6047	3812315																
	Chhatarpur	1	- - 19	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	19	9085																
	Gwalior	1*	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	-	-																
Jabalpur	22(9)	- 454 66	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	5994	3783495																
	Sagar	1	- 5	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	34	19735																

Table- 5B (Concl.d.)

(Quantity in '000 tonnes; Values in ₹ '000)

State/District	No. of mines	Lumps										Fines										Concentrates		Calibrated Lump Ore (CLO)										Total	Value		
																						Total	Value													Total	Value
		Below 45% Fe (For Magnetite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Value	Below 45% Fe (For Magnetite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Value			Below 62% Fe (CLO any size)	62% to below 65% Fe (5-18 mm size CLO)	62% to below 65% Fe (10-40 mm size CLO)	62% to below 65% Fe (CLO others)	65% and above Fe (5-18 mm size CLO)	65% and above Fe (10-40 mm size CLO)	65% and above Fe (CLO others)							
Maharashtra	10	-	24	20	59	25	460	1105	-	1693	9678310	16	53	153	1299	-	-	7455	58	9034	31243504	-	-	-	-	-	-	-	-	-	-	10727	40921814				
	1	-	14	-	5	25	8	-	-	52	214586	16	20	-	-	-	-	-	-	36	26264	-	-	-	-	-	-	-	-	-	-	88	240850				
	1	-	-	-	-	-	452	1105	-	1557	9270937	-	-	-	929	-	-	7455	58	8442	30102072	-	-	-	-	-	-	-	-	-	-	9999	39373009				
	3	-	-	12	-	-	-	-	-	12	20542	-	-	4	-	-	-	-	-	4	3144	-	-	-	-	-	-	-	-	-	16	23686					
Sindhudurg	5	-	10	8	54	-	-	-	-	72	172245	-	33	149	370	-	-	-	-	552	1112024	-	-	-	-	-	-	-	-	-	-	624	1284269				
Odisha	66(3)	-	-	23	-	-	-	-	-	23	33992	71	1091	10759	15385	7642	36888	36593	1665	110094	395379858	331	1122568	15243	3113	16272	1	237	3645	-	38511	167921003	148959	564457411			
Kendujhar	38(1)	-	-	19	-	-	-	-	-	19	28592	0	225	6106	2919	4303	20802	24012	924	59291	209224913	-	-	5274	2060	12993	-	18	2941	-	23286	99375071	82596	308628576			
Mayurbhanj	3	-	-	-	-	-	-	-	-	-	-	71	683	358	105	-	13	++	-	1230	1659762	-	-	226	-	-	-	-	-	226	993603	1456	2653365				
Sundargarh	25(2)	-	-	4	-	-	-	-	-	4	5400	-	183	4295	12361	3339	16073	12581	741	49573	184485183	331	1122568	9743	1053	3279	1	219	704	-	14999	67552329	64907	253175470			
Rajasthan	10	-	5	3	2	-	-	-	-	10	26052	-	6	++	-	-	-	-	-	6	5074	1511	7273485	-	-	-	-	-	-	-	-	-	1527	7304611			
Bhilwara	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1511	7273485	-	-	-	-	-	-	-	-	1511	7273485				
Jaipur	3	-	4	-	-	-	-	-	-	4	8171	-	3	-	-	-	-	-	-	3	2545	-	-	-	-	-	-	-	-	-	8	10716					
Jhunjhunu	4	-	1	++	-	-	-	-	-	1	2053	-	-	++	-	-	-	-	-	++	849	-	-	-	-	-	-	-	-	-	1	2902					
Sikar	1	-	-	3	2	-	-	-	-	5	15828	-	3	-	-	-	-	-	-	3	1680	-	-	-	-	-	-	-	-	-	7	17508					

(P): provisional

*: Only labour reported ; ++: Negligible

(): Figure in parenthesis indicates mines of Dolomite, Manganese Ore, Ochre and Laterite as an associate mineral.

Table – (6): Production of Iron Ore, 2022-23 and 2023-24 (P)

(By Frequency Groups)

Production Group (In tonnes)	No. of Mines		Production (In '000 tonnes)		Percentage in total production		Cumulative Percentage	
	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)
Total	207(14)	202(15)	258312	277332	100	100	-	-
Up to 50000	74(11)	65(13)	720	764	0.3	0.3	0.3	0.3
50001 to 100000	14(2)	14	1294	1215	0.5	0.4	0.8	0.7
100001 to 500000	41 (1)	45(2)	11141	13736	4.3	5.0	5.1	5.7
500001 to 1000000	18	19	14173	15489	5.5	5.6	10.6	11.3
1000001 to 1500000	14	9	17429	11839	6.7	4.3	17.3	15.5
1500001 to 2000000	7	7	12510	11640	4.8	4.2	22.2	19.7
Above 2000000	39	43	201045	222650	77.8	80.3	100.0	100.0

(P): provisional

(): Figures in parenthesis indicate mines of Dolomite, Manganese Ore, Ochre and Laterite as an associate mineral.

Table – 7 (A): Mine-head Closing Stocks of Iron Ore, 2022-23
(By States/Grade)

(In '000 tonnes)

State	Lumps										Fines										Concen- trates	Calibrated Lump Ore (CLO)										Total																																																																																																																																																																																																																																				
	Below 45% Fe (For Mag-netite Ore)	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% to below 65% Fe	65% to below 65% Fe	Total	Below 45% Fe (For Mag-netite Ore), Fines	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% to below 65% Fe	65% to below 65% Fe	Total		Below 62% Fe (CLO any size)	62% to below 65% Fe (5-18 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)		65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)	65% to below 65% Fe (10-40 mm size CLO)

++: Negligible
(P): provisional

Table – 7 (B): Mine-head Closing Stocks of Iron Ore, 2023-24 (P)
(By States/Grade)

(In '000 tonnes)

State	Lumps										Fines										Concen- trates	Calibrated Lump Ore (CLO)										Total
	Below 45% Fe (For Mag-netite Ore)	45% to below 51% Fe	51% to 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Below 45% Fe (For Magnetite Ore),Fines	45% to below 51% Fe	51% to below 55% Fe	55% to below 58% Fe	58% to below 60% Fe	60% to below 62% Fe	62% to below 65% Fe	65% and above Fe	Total	Below 62% Fe any size	62% to below 65% Fe (5-18 mm size CLO)		62% to below 65% Fe (10-40 mm size CLO)	65% and above Fe (5-18 mm size CLO)	65% and above Fe (10-40 mm size CLO)	65% and above Fe (10-40 mm size CLO)	Total						
India	26	2290	2587	707	684	440	1020	1317	9071	78	3371	9695	10686	4575	11025	10909	2331	52670	22	9764	2094	7912	52	41	1304	1232	21290	83053				
Andhra Pradesh	-	85	253	20	-	-	-	-	358	-	98	963	35	-	-	-	-	1096	-	-	-	-	-	-	-	-	-	1454				
Chhattisgarh	-	++	297	23	312	2	335	1157	2126	-	410	180	194	110	612	1388	1746	4640	-	-	-	20	-	-	388	-	408	7174				
Goa	-	-	19	-	-	-	-	-	19	-	10	23	-	-	-	-	-	33	-	-	-	-	-	-	-	-	-	52				
Jharkhand	-	-	14	253	61	128	94	74	624	-	-	815	5404	1930	1174	1486	235	11044	-	430	-	-	-	5	-	38	-	473	12141			
Karnataka	26	1961	1348	348	267	235	509	34	4728	42	1589	975	592	172	110	468	67	4015	-	-	-	-	-	-	-	-	-	-	8743			
Madhya Pradesh	-	192	513	30	9	17	-	-	761	-	458	1403	9	15	3	-	-	1888	-	-	-	-	-	-	-	-	-	2649				
Maharashtra	-	25	20	26	28	11	1	-	111	13	33	94	17	15	++	++	++	172	-	-	-	-	-	-	-	-	-	283				
Odisha	-	6	121	7	7	47	81	52	321	23	755	5240	4435	2333	9126	7567	28329762	21	9334	2094	7892	47	41	878	1232	20409	50513					
Rajasthan	-	21	2	-	-	-	-	-	23	-	18	2	-	-	-	-	-	20	1	-	-	-	-	-	-	-	-	44				

(P): Provisional
++: negligible

EMPLOYMENT

The average daily employment of labour was 59,620 during 2023-24 as against 56,102 in the preceding year.

MINING, MARKETING & TRANSPORT

Iron ore mining is carried out by opencast method through manual, semi-mechanised and mechanised operations.

The method of mining and deployment of machinery varies from place to place depending upon the characteristics of iron ore as per geological setup. Large mechanised mines are mostly in the Public Sector. Manual and semi-mechanised mines are mainly in Private Sector. Some mechanised mines in Jharkhand and Odisha are also operated by the Private Sector.

Manual Mines

Generally, these mines are confined to float ores where mining is done by digging the ore with pick axes, crow bars, chisels and spades. The mined material is screened manually to separate +10 mm float ore which is then stacked separately. The waste is backfilled into the pits. In some reef workings, 35–40 mm diameter holes are drilled to 0.6 m depth by hand-held jackhammers at a spacing of about 0.6 m and each hole is charged with 150–200 g of gunpowder or special gelatine cartridges. Blasted tonnage per kg of gunpowder is usually 2.5–3 tonnes. Blasted ore is manually loaded into trucks for transport to either railway sidings or to buyer's destination directly.

Mechanised Mines

Most of the mechanised mines are captive belongings of different steel plants and have been developed to cater to specific requirements. Mining is done by formation of systematic benches in overburden and ore. The height of the benches normally varies from 10 to 12 m and width up to 20 m in the ore. Drilling holes of 300 mm diameter and till 12 m depth by crawler drills and use of explosives, such as, ANFO, SMS and emulsion explosives for blasting are in practice. Loading is done by earth-moving machinery powered by diesel or electric engines, such as, hydraulic excavators in the range from 1.9 cu. m to 10 cu. m. Ripper dozers and motor graders are also deployed for excavation and levelling purposes.

In mines, where ore is predominantly in powdery form, hydraulic shovels with boom height of around 9 m are used for excavation and loading. Heavy-duty Ripper-Dozers are preferred for such mining as the ores are soft. Height of the benches is restricted to 7 m for safe and efficient operations. Width of working benches is maintained at more than 15 m and the bench slope is maintained at about 80°. The ore produced is transported to short distances by dumpers up to 40 tonnes capacity. For longer distances and barge loading, dumpers/trucks up to 10 tonnes capacity are used. The barges carry the ore to harbours. The ore from the barges is loaded onto ships either through berth or through transshippers.

Almost all the Public Sector mines including Kiriburu, Barsua, Gua, Bailadila, Donimalai, Daitari and Dalli-Rajhara operated by SAIL, NMDC and OMC are fully mechanised. In Private Sector, most of the captive mines are mechanised. Approximately, 90% iron ore production comes from mechanised mines. NMDC operates a couple of large mechanised iron ore mines in the country at Bailadila (Chhattisgarh) and Donimalai (Karnataka). The Company has three highly mechanised iron ore mine complexes. Two are located in Chhattisgarh and one in Karnataka.

The processing of iron ore in the country involves crushing, screening, washing and in some cases beneficiation and agglomeration. Crushing and screening are adopted mainly for sizing the ore and also for removing the adherent gangue minerals. Dry and wet grinding is also resorted to in some cases.

The lumps and fines of iron ore are marketed after screening and beneficiation. Fines are converted into sinters for use in steel plants while pellets made from concentrates/fines are exported and also are utilised for internal consumption in domestic iron & steel industries.

ENVIRONMENTAL FACTORS

Afforestation, waste dump management, topsoil management, management of sub-grade minerals, mechanical beneficiation, dust suppression, monitoring of water & air quality, vibration survey, publicity and propaganda are some common environmental restoration efforts pursued by all mechanised and semi-mechanised iron ore mines. Mining and beneficiation of ores carried out on a large-scale cause environmental problems. A specific problem in iron ore mining is the disposal of tailings and other deleterious silica minerals and phosphorous. To safeguard the environment and prevent ecological degradation, thrust has been laid on green belt development, solid waste management, monitoring of liquid & air effluents, and other crucial environmental parameters.

Goa region is prone to siltation of agricultural fields, nallahs, riverbeds and creeks due to wash off from iron ore dumps in rainy season. Loss in crop yield and reduction in fish population in streams and navigation difficulties are the problems caused by silting. To overcome these problems, check dams and water filter beds at higher contours have been constructed. Tailing ponds are also being maintained at some mines. Afforestation is the mainstay in reclaiming the mined out areas in Goa. In a few cases, pits are used as water reservoir for pisciculture. But, in early 2018, the apex court had quashed 88 mining leases for violation of mining procedures and asked the State Government of Goa to issue fresh leases instead of renewing existing ones. [February 2021 marked three years since India's apex court stopped iron ore mining in Goa].

In Ballari–Hosapete area, Karnataka, dust concentration (suspended particulate matter) is the main environmental

problem. In Bailadila Sector, Chhattisgarh, forest is fairly widespread and dense, supported by good rainfall and rich flora and fauna. The deforestation taking place due to mining and waste dumping needs to be compensated continuously by afforestation at suitable slopes and in township areas. In Jharkhand, afforestation of land is the main recourse adopted for reclamation of degraded lands or improvement in land uses.

INDUSTRY

Iron ore is the basic raw material used for making pig iron, sponge iron and finished steel. The iron ore is used mainly in blast furnaces, mini-blast furnaces (MBF), DRI & sintering and pelletisation plants.

Pelletisation

In general, the pelletisation process involves mixing of iron ore and required limestone with water which later is ground in ball mills to the desired size. The discharged slurry from ball mills is filtered in pressure filters. The filter cake from filters is then mixed with dry-ground coke fines to which bentonite is mixed in suitable proportion to form green pellets in pelletising discs. The coke fines and bentonite are ground separately. The green pellets are then dried, heated and fired in indurating machine to produce iron ore pellets. There is an increasing trend for utilisation of pellets or sinters in the recent years. The use of pellets as feed in the blast furnace has several advantages because of their uniform size, known composition and strength. Iron ore pellet is a kind of agglomerated fines which has better tumbling index as compared to that of parent ore and can be used as a substitute used in blast furnaces in countries where lump ore is not available.

As per the Pellet Manufacturers Association of India, the fifty-five pelletisation plants in the country about which information is available have a total capacity of 146.1 million tonnes per annum. The JSW Steel Ltd has a manufacturing capacity of 17.20 million tonnes of pellets annually at Vijayanagar, Bellary, Karnataka. Amba River Coke Limited, a wholly owned subsidiary company of JSW Steel, has set up a 13.3 million tpy pellet plant (Dolvi units 1 and 2) at Dolvi. The pellets produced are primarily supplied to the Dolvi units of the company. During the year 2022-23, all India production of pellets as per the Pellet Manufacturers Association of India, was 83.96 million tonnes.

With a strong belief in prudent forward and backward integrations, JSPL established India's largest 10 MTPA Pelletisation Complex at Barbil, Odisha. The plant includes dry grinding facility that harnesses recuperation type of straight grate technology. The Company's Barbil Plant is India's largest single-location pellet manufacturing facility with 4.5 MTPA Dry Grinding Unit and a 4.5 MTPA Wet Grinding Unit. The JSPL pellet plant helps to process low-cost iron ore fines as against expensive lumps, thereby handing a cost advantage. Laced with state-of-the-art technology backed with proximity to iron ore access, JSPL's

Barbil Plant has emerged as India's largest pellet exporter in recent years.

Arcelor Mittal Nippon Steel India, i.e., 'AM/NS India' (formerly known as Essar Steel as was acquired jointly by Arcelor Mittal and Nippon Steel in December 2019) has 8 MTPA iron ore pellet plant in Visakhapatnam, Andhra Pradesh, to cater to the pellet requirements of the HBI plant in Hazira, Gujarat. The plant has an assured supply of high-quality iron ore from the beneficiation plant at Bailadilla, Chhattisgarh. The plant is capable of producing both DR and BF grade pellets and is linked to the Visakhapatnam port through conveyors to enable easy material movement in and out of the plant. The plant is located strategically near a deep draft, all-weather port that ensures the movement of large vessels to supply pellets throughout the year to the Hazira steel-making facility. A 6 million tpy pellet plant is located at Paradip in the iron ore rich State of Odisha. The plant has an assured supply of high-quality iron ore from the beneficiation plant at Dabuna, Odisha. The Paradip Pellet plant is likely to add another 6 million tpy to its capacity which is under completion. After completion of this plant AM/NS India's total pelletisation capacity at Paradip would get augmented to 12 million tpy and supported by a 20-million-tonnes pellet-making capability, the company is on its way to become the largest pellet producer in India.

NMDC has forayed into pellet-making through setting up of a 1.2 MTPA pellet plant at Donimalai. Another 2 MTPA pellet plant is in the process of being set up at Nagarnar, Chhattisgarh.

KIOCL is currently engaged in the business of manufacturing and selling of iron ore pellets. The state-of-the-art pelletisation plant with 3.5 million tpy rated capacity and 0.216 million tpy Blast Furnace Unit is located at Mangaluru.

Steel plants are likely to increase usage of pellets in their production process to reduce pollution and increase productivity. Moreover, the forecast of spike in growth in Infrastructure, Real Estate and Automobile Sectors in the ensuing years are expected to augment demand for steel, which in turn would raise the demand and prices of pellets in the near future.

Sintering

In sintering process, iron ore fines, other iron-bearing wastes and coke dust are blended and combusted. The heat fuses the fines into course lumps that can be charged to a blast furnace. There are about thirty-nine sintering plants in the country about which information is available and have a total capacity of about 96.268 million tonnes per annum. Most of the Integrated Steel Plants (ISP) in the country have their own sintering plants. Sinter plants receive raw material mostly from their captive mines. Information on capacity and production of pellets and sintering plants is provided in Table-8.

Table – 8: Name, Location and Installed Capacity of Pellets and Sinters Plants

(By Plants)

(In million tonnes)

Name & Location of Plant	Annual Installed Capacity
A) Pellet Plants	
1. Amba River Coke Ltd, (A wholly owned subsidiary co. of JSW Steel Ltd), Dolvi, Maharashtra	12
2. Atibir Industries Co. Ltd. Unit-II, Bhorandiha, Jharkhand	0.3
3. Ardent Steel Ltd, Phulj Keonjhar, Odisha	0.8
4. Arya Iron and Steel Company (AISCO) Barbil, Odisha	1.2
5. Pellet Sponge Iron Plant BMM Ispat, Karnataka	2.4
6. Arcelor Mittal Nippon Steel India, Visakhapatnam, Andhra Pradesh	8
7. Arcelor Mittal Nippon, Steel India, Paradip Port, Odisha.	12
8. Godawari Power & Ispat Ltd Siltara, Chhattisgarh	2.7
9. Jindal Steel & Power Ltd, Barbil, Odisha	10
10. Jindal Saw Ltd, Bhilwara, Rajasthan	1.5
11. JSW Steel Ltd, Tornagallu, Toranagally, Ballari, Karnataka	17.2
12. Jayaswal Neco Industries Ltd, Siltara, Raipur, Chhattisgarh	1.5
13. KIOCL Ltd, Panambur, Mangaluru, Karnataka	3.5
14. Mandovi Pellets Ltd, Near Borim Bridge, Shiroda, Goa – 403 103	1.8
15. Minera Steel & power Private Ltd, Ballari, Karnataka	0.6
16. MSP Steel & Power Ltd, Raigarh, Chhattisgarh	0.9

Table- 8(Cont.)

(In million tonnes)

Name & Location of Plant	Annual Installed Capacity
17. NMDC Ltd, Donimalai, Karnataka.	1.2
18. Orissa Metalics Private Ltd, Paschim Mednapore, West Bengal	3.6
19. Rashmi Metaliks Ltd, Shyamraipur, Gokulpur, West Midnapore, West Bengal	1.2
20. Sarda Energy and Minerals Ltd, Siltara, Mandhar, Raipur, Chhattisgarh	0.8
21. Shri Bajarang Power & Ispat Ltd, Borjhara, Tilda & Gondwara, Raipur, Chhattisgarh	1.4
22. Tata Steel Limited, Jamshedpur, East singbhum, Jharkhand	8
23. Xindia Steels Ltd, Kunikere & Hirebaganal Giniger, Koppal, Karnataka	0.8
24. Bhushan Power & Steel Ltd Sambalpur, Odisha	3.85
25. Shyam Metalics And Energy Ltd Sambalpur, Sambalpur, Odisha	3
26. MSP Metalics Ltd, Jharsuguda, Odisha	0.6
27. River Pellets Limited, jaipur, Odisha	4
28. Sree Metaliks Ltd Keonjhar, Odisha	0.6
29. MSP Sponge Iron Ltd, Keonjhar, Odisha	0.75
30. Rungta Mines, kamanda, Sundergarh, Odisha	3
31. Essel Mining & Industries Ltd. Keonjhar, Odisha	1
32. Shri Jagannath Steels & Power Ltd (KJS Group), Keonjhar, Odisha	1.2
33. Shri Mahavir Ferro Alloys Ltd Sundergarh, Odisha	1.6
34. MSPL Limited, Koppal, Karnataka	1.2

Table- 8(Cont.)

(In million tonnes)	
Name & Location of Plant	Annual Installed Capacity
35. Janki Corporation Ltd Bellari, Karnataka	0.6
36. Rashmi Udyog Private Ltd (Rashmi Group), Paschim Midnapur, West Bengal	1.2
37. Orissa Alloy Steel Pvt. Ltd. (Rashmi Group), Kharagpur, West Bengal	3
38. Shakambhari steel, Purulia, West Bengal	2
39. Super Smelters Sponge (P) Ltd, Jamuria, Paschim Bardhman West Bengal	1.2
40. Bravo Sponge Iron (P) Ltd Purulia, West Bengal	0.85
41. Shyam Sel And Power Ltd (Jamuria), Paschim Bardhman, West Bengal	3
42. Ankit Metal And Power Ltd, Bankura, West Bengal	0.6
43. Tata Sponge Iron Ltd Main Plant-1& 2, Gamharia, Jharkhand	1.2
44. Rungta Mines, Chaliyama, Saraikela Jharkhand	3
45. Amalgam Steel Private Limited (Formerly Adhunik Alloys), Saraikela\ Jharkhand	1.2
46. JSW Ispat Special Products Ltd (Raigarh), Raigarh, Chhattisgarh	2.2
47. Raipur Power & Steel Durg, Durg, Chhattisgarh	0.6
48. Rashi Steel And Power Ltd, Bilaspur, Chhattisgarh\	0.4
49. Sal Steel Ltd (Shah Alloys Ltd), Gandhinagar, Gujarat	0.6
50. Vinayak Steels Ltd, Mehbubnagar, Telangana	0.07
51. Gullantt Ispat Ltd, Gorakhpur, Uttar Pradesh	0.79

Table- 8(Cont.)

(In million tonnes)	
Name & Location of Plant	Annual Installed Capacity
B) Sintering Plant	
1. Atibir Industries Co. Ltd, Unit-II, Bhorandiha, Jharkhand	680
2. Bokaro Steel Plant, Jharkhand	6900
3. Bhilai Steel Plant, Bhilai, Durg, Chhattisgarh.	6334
4. Bhushan Steel Ltd, Dhenkanal, Odisha	6680
5. Durgapur Steel Plant, West Bengal	3009
6. Electrosteel Casting Ltd, Khardah, Barrackpore, West Bengal	365
7. Electrosteels Ltd, Siyaljori, Jharkhand	2980
8. Gerdau Steel India Ltd, Tadipatri, Anantpur, A.P.	470
9. IISCO Steel Plant, SAIL Burnpur, West Bengal	3880
10. Jayaswal Necco Industries Ltd, Siltara Growth Centre, Raipur-493 221, Chhattisgarh	729
11. Jindal Steel & Power Ltd, Raigarh, Chhattisgarh	2300
12. Jindal Saw Ltd, Mundra, Gujarat	900
13. JSW Steel Ltd, Tornagallu, Toranagallu, Ballari, Karnataka	12950
14. JSW Steel Ltd, Dolvi Works, Raigad, Maharashtra	5400
15. JSW Steel Ltd Salem works, Mkalipatti, Metturdam, Tamil Nadu	1106
16. Jai Balaji Industries Banskopa, West Bengal	608
17. Kalyanigerdua Steels Ltd, formerly SJK Steel Plant, Jambulapadu, Tadipatri, Andhra Pradesh	500

Table- 8(Cont.)

(In million tonnes)	
Name & Location of Plant	Annual Installed Capacity
18. Kirloskar Ferrous Industries Ltd, Bevinahalli, Koppal, Karnataka.	500
19. KIC Metaliks Ltd, Raturia, Angadpur, Durgapur. West Bengal	336
20. Monnet Ispat and Energy Raigarh, Chhattisgarh	962.3
21. Mukund Ltd, M/s Hospet Steel Ltd, Ginigera, Koppal, Karnataka	500
22. Neometaliks Ltd, Gopalpur, Durgapur, West Bengal	316
23. Neelachal Ispat Nigam Ltd, Kalinga Nagar, Industrial Complex, Duburi-755 026, Distt Jajpur, Odisha.	1710
24. Rashmi Metaliks Ltd, Shyamraipur, Gokulpur, West Midnapore, West Bengal.	1440
25. RINL, Visakhapatnam Steel Plant No. -1& 2 , Visakhapatnam, Andhra Pradesh	5256
26. RINL, Visakhapatnam Steel Plant No. -3, Andhra Pradesh	3600
27. Rourkela Steel Plant, Odisha	5300
28. SBQ Steel Ltd, Gudur, Nellore, Andhra Pradesh	240

Pig Iron

Pig iron is one of the basic raw materials required by Foundry and Casting Industry for manufacturing various types of castings for the engineering section. The post-liberalisation regime has witnessed Expression of Interest from a large number of entrepreneurs for setting up mini-blast furnaces for production of hot metal/pig iron. Commissioned pig iron units are mostly of stand-alone type.

The production of pig iron has increased from 1.6 million tonnes in 1991-92 to 7.36 million tonnes in 2023-24. As per Ministry of Steel Annual Report 2023-24, the production of pig iron in 2022-23 was 5.8 MT, an increase of 25.59% over that of last year. As per the Annual Report-2023-24 of Ministry of Steel, the Private Sector accounted

Table- 8 (Concl.)

(In million tonnes)	
Name & Location of Plant	Annual Installed Capacity
29. Sri Kalahasthi Pipes Ltd, Chittoor, Andhra Pradesh	500
30. SLR Metaliks Ltd, Ballari, Karnataka	350
31. Sesa Goa Ltd, Vedanta Ltd, North Goa	1000
32. Sunflag Iron & Steel Co. Ltd, Warrthy, Bhandara, Maharashtra	450
33. Tata Steel Ltd, Jamshedpur, Jharkhand	8000
34. Tata Metaliks Ltd, Kharagpur, West Bengal	528
35. Tata Steel Ltd, Kalingnagar, Odisha	5750
36. Usha Martin Ltd (Usha Alloys and Steel Division), Jamshedpur.	715
37. Uttam Galva, Metallics Ltd, Wardha, Maharashtra	887
38. Vedanata Ltd. Amona, Goa	1000
39. Value Added business, Amona, Goa	1000

for 90% (6.646 MT) while the Public Sector accounted for the remaining 10 % (0.71.718 MT) of the total production of pig iron (7.36 MT) in the country in 2023-24. As per National Steel Policy 2017, the demand for pig iron for merchant use, such as, for castings and supplementary metallic in the electric arc or induction furnaces is projected to increase to 17 million tonnes by 2030-31.

Sponge iron

India is the world's largest producer of sponge iron or Direct Reduced Iron (DRI) with a host of coal-based units located in the mineral-rich States of the country. Over the years, the coal based route has emerged as a key contributor and accounted for 81% of total sponge iron production in the country in 2023. India has been the world's largest sponge

iron producer every year since 2003. The growth of Sponge Iron Industry during the last few years in terms of capacity has been substantial. The installed capacity of sponge iron increased from 1.52 million tonnes per annum in 1990-91 to around 47.85 million tonnes in 2019-2020. Production has increased from 0.9 million tonnes in 1990-91 to 51.56 million tonnes in 2023-24. As per National Steel Policy 2017, the demand for sponge iron is projected to increase to 80 million tonnes by 2030-31. It is projected that the sponge iron capacity may increase to 114 million tonnes by 2030-31 with around 30% share of gas-based capacities on account of increased environmental considerations and long-term availability of gas.

Sponge iron is a good substitute for scrap which is required by the electric arc furnaces and induction furnaces or mini-steel plants in the country. The availability of indigenous metal scrap is scarce, and therefore, to meet the domestic demand, scrap is usually imported. Sponge iron is a viable alternative for scrap and is produced by direct reduction of high-grade iron ore or pellets to metallic iron ore in solid state by using coal or natural gas as reductant. It is also known as Direct Reduced Iron (DRI) or Hot Briquetted Iron (HBI).

Iron & Steel

The details of the Iron & Steel Industry are provided in the Review on "Iron, Steel & Scrap and Slag".

Ferroalloys

Iron is an important constituent of ferroalloys, like ferromanganese (high carbon, medium carbon and low carbon), ferrosilicon, ferrochrome (high carbon and low carbon)/charge chrome, ferromolybdenum, ferrovanadium, ferrotungsten, ferro-silicon-magnesium, ferroaluminium, ferro-silicon-zirconium, ferrotitanium, etc. Ferroalloys are used in Steel Industries to impart some special qualities in steel making process also. They are consumed in domestic industries and are also exported. The details about the Ferroalloys Industry are provided in the Review on 'Ferroalloys'.

Cement

Iron ore lumps and powder containing +58% Fe are normally used in the Cement Industry as they improve burning properties, impart colour and balance the composition of the mix. Further details about the Cement Industry are provided in the Review on 'Cement'.

USES & CONSUMPTION

Iron ore is mainly used for manufacturing pig iron, sponge iron and steel. It is also used in Cement, Coal Washeries, Ferroalloys, Foundry and Glass Industries. In 2023-24, the apparent consumption of iron ore was about 235.916 million tonnes, as against 238.934 million tonnes in the previous year. The specifications of iron ore consumed by major sponge iron plants & steel iron plants are furnished in (Tables-9 and 10).

Table – 9: Specifications of Iron Ore Consumed by Major Sponge Iron Plants

Sl. No.	Name of the Plant	Specifications				
		Size	Fe	Al ₂ O ₃ + SiO ₂	P	S
1	Orissa Sponge Iron Plant	5–18 mm	65% min.	4.5% max.	0.03% max.	N. A.
2	Welspun Max Steel Ltd	9–16 mm	66%	2.6% max.	0.05%	0.01%
3	Sunflag Iron & Steel Ltd	5–20 mm	67.50%	–	–	–
4	NMDC Ltd (Sponge iron unit)	6–20 mm	55–58% &	–	–	–
					64–66%	
5	Essar Steel Ltd	10–40 mm	67%	2.60% max.	0.05%	0.01%
6	Jindal Steel & Power Ltd	10–30 mm	65% min.	3% max. (SiO ₂)	0.05%	–
7	Tata Sponge Iron Ltd	5–18 mm	65% min.	5% max.	–	–
8	Steel Exchange India Ltd	10–40 mm	62%	–	–	–
9	Sarda Energy & Minerals Ltd	5–18 mm	65–66%	–	–	–
10	OCL Iron & Steel Ltd	Sized	62% min.	–	–	–
11	Nalwa Steel & Power Ltd	5–20 mm	63% min.	–	–	–
12	Shri Bajrang Power & Ispat Ltd	5–18 mm	64% min.	–	–	–
13	Jai Balaji Industries Ltd	5–18 mm	65%	5%	0.05%	0.03%
		10–30 mm	–	–	–	–
		10–150 mm	–	–	–	–

Table – 10: Specifications of Iron ore

(By Steel Plants)

(In '000 tonnes)	
Steel plant	Specifications
Bokaro Steel Plant, Bokaro, Jharkhand	Lumps: Fe-63.40%, SiO ₂ : 2.25%, Al ₂ O ₃ 2.39%, Size: 10-40 mm Fines: Fe - 62.24%, SiO ₂ - 3.36%, Al ₂ O ₃ - 3.45%
Durgapur Steel Plant, Durgapur, West Bengal	Lumps: Fe - 62.48%, Al ₂ O ₃ - 2.42%, Size: 10-50 mm Fines: Fe - 62.8%, SiO ₂ - 2.28%, Size: -10 mm
IISCO Steel Plant, Burnpur, West Bengal	Lumps: Fe - 62.86%, SiO ₂ - 2.56%, Al ₂ O ₃ - 2.56% (max.), Size: 10-40 mm
Bhilai Steel Plant, Chhattisgarh	-

Table- 10 (Concl.d.)

(In '000 tonnes)

Steel plant	Specifications
Rourkela Steel Plant SAIL, Rourkela, Odisha	-
JSW Steel Ltd—Dolvi Works Raigad, Maharashtra.	-
JSW Steel Ltd Tornagallu, Sandur, Ballari Karnataka	-
JSW Steel Ltd—Salem Works, Mkalipatti, Metturdam, Tamil Nadu	-
Tata Steel Limited, Jamshedpur	-
RINL Vishakhapatnam Steel Plant, Andhra Pradesh	Lumps: Fe 65.5 % min. SiO ₂ 2.25 % max., Al ₂ O ₃ 2.25 % max.. Fines: Fe 64.5 % min. SiO ₂ 3.00 % max. Al ₂ O ₃ 3.00 % max.

TRADE POLICY

To ensure easy availability of raw material in domestic market at reasonable prices, export duty on iron ore is at 30% for both lumps and fines varieties of 58% Fe content and above. The export duty is at 0% for both lumps and fines varieties of iron ore less than 58% Fe content. Export of (i) Iron ore of Goa origin when exported to China, Europe, Japan, South Korea and Taiwan, irrespective of

the Fe content; (ii) Iron ore of Redi origin to all markets, irrespective of the Fe content; and (iii) all iron ore of Fe content up to 64%, is free. Exports of Iron ore other than those specified under Free Category can only be exported by STE.

As per the ITC(HS), 2022 Schedule 1 Import Policy, the present import policy scenario is furnished below in brief. As per the policy, imports of iron ore lumps, fines, concentrates and agglomerated pellets are freely allowed.

HS Code	Item	Import Policy
2601	Iron ore and concentrates, including roasted iron pyrites	Free
260111	Iron ore and concentrates, other than roasted iron pyrites: Non-agglomerated	Free
26011111	60% Fe or more but below 62% Fe	Free
26011112	62% Fe or more but below 65% Fe	Free
26011119	65% Fe and above	Free
26011121	Iron ore lumps (below 60% Fe, including black iron ore containing up to 10 % Mn)—Iron Ore lumps below 55% Fe	Free
26011122	Iron ore lumps (below 60% Fe, including black iron ore containing up to 10 % Mn) – Iron Ore lumps 55% Fe or more but below 58% Fe	Free
26011129	Iron ore lumps (below 60% Fe, including black iron ore containing up to 10 % Mn) – Iron Ore lumps 58% Fe or more but below 60% Fe	Free
26011131	Iron ore fines (62% Fe or more)— 62% Fe or more but below 65% Fe	Free
26011139	Iron ore fines (below 62% Fe or more)— 65% Fe and above	Free
26011141	Iron ore fines (below 62% Fe) – below 55% Fe	Free
26011142	Iron ore fines (below 62% Fe) –55% Fe or more but below 58% Fe	Free
26011143	Iron ore fines (below 62% Fe) – 58% Fe or more but below 60% Fe	Free
26011149	Iron ore fines (below 62% Fe) – 60% Fe or more but below 62% Fe	Free
26011150	Iron ore concentrates	Free

HS Code	Item	Import Policy
26011190	Others	
260112	Iron ore and concentrates other than roasted iron pyrites: Agglomerated	Free
26011210	Iron ore pellets	Free
26011290	Other	Free
26012000	Roasted iron pyrites	Free

WORLD REVIEW

The world reserves of crude iron ore are estimated to be around 190 billion tonnes. In terms of iron content, the iron ore reserves are estimated to be around 87 billion tonnes. The world reserves of crude iron ore and iron content by principal countries are furnished in Table - 11.

In 2023, the world production of iron ore was 2,507 million tonnes as against 2,436 million tonnes in the

previous year. Australia 952 million tonnes (38%), Brazil 436 million tonnes (17%), India 277 million tonnes (11%), China 250 million tonnes (10%), Russia 90 million tonnes (4%) Iran 79 million tonnes (3%), South Africa 63 million tonnes (3%), Canada 59 million tonnes (2%) Kazakhstan 47 million tonnes (2%) and USA 44 million tonnes (2%) were the principal producers. These ten countries accounted for about 92% of the world production of iron ore and remaining 10% was contributed by the other countries. The world production of iron ore is provided in Table-12.

Table – 11: World Reserves of Iron Ore
(By Principal Countries)

Country	Reserves (In million tonnes)	
	Crude ore	Iron content
World: Total (rounded off)	1,90,000	87,000
Australia ^(a)	58000	27000
Brazil	34000	15000
Russia	29000	14000
China	20000	6900
India*	5500	3400
Canada	6000	2300
Ukraine ^(b)	6500	2300
Iran	3300	1500
USA	3100	1300
Peru	2600	1200
Kazakhstan	2500	900
South Africa	990	620
Sweden	1300	600
Turkey	152	99
Chile	NA	NA
Mauritania	NA	NA
Mexico	NA	NA
Other countries	18000	9500

Source: USGS, Mineral Commodity Summaries, 2024.

(a): For Australia Joint Ore Reserves Committee compliant reserves were about 23 billion tonnes for crude ore and 11 billion tonnes for iron content.

(b): For Ukraine, reserves consist of the A and B categories of the Soviet reserves classification system.

* As per UNFC system as on 1.4.2020, India's reserves/resources of Iron ore (Haematite) and Iron ore (Magnetite) were estimated at 24,057 million tonnes and 11,227 million tonnes respectively.

NA - Not available.

Table – 12: World Production of Iron Ore

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World: Total (rounded off)	2552000000	2436000000	2507000000
Australia	922158512	944966398	952510097
Brazil	423185073	403024748	436842189
India [#]	253973000 ^(h)	252317000	277955000
China	2660000000	2526000000	2500000000*
Russia	1006000000	951000000	908000000
Iran ⁽ⁱ⁾	75039588	70974267	79705135
South Africa ^(e)	73090918	63711254	63187576
Canada ^(f)	58267104	65032476	59421934
Kazakhstan	64089700	53623800	47582800
USA ^{(f)(g)}	47500000	37300000	44700000
Other countries	268176181	197570974	205954664

Source: BGS World Mineral Production, 2019-2023

Note: World Total may not tally as data has been rounded off

(h): Years ended 31st March following that stated

(i): Years ended 20th March following that stated

(f): Including by-product iron ore

(e): Including by-product magnetite;

(g): Including beneficiated and direct shipping ore

[#]: India's production of iron ore in 2021-22, 2022-23 and 2022-24 was 253.97 million tonnes, 252.31 million tonnes and 277.95 million tonnes, respectively.

*: estimated.

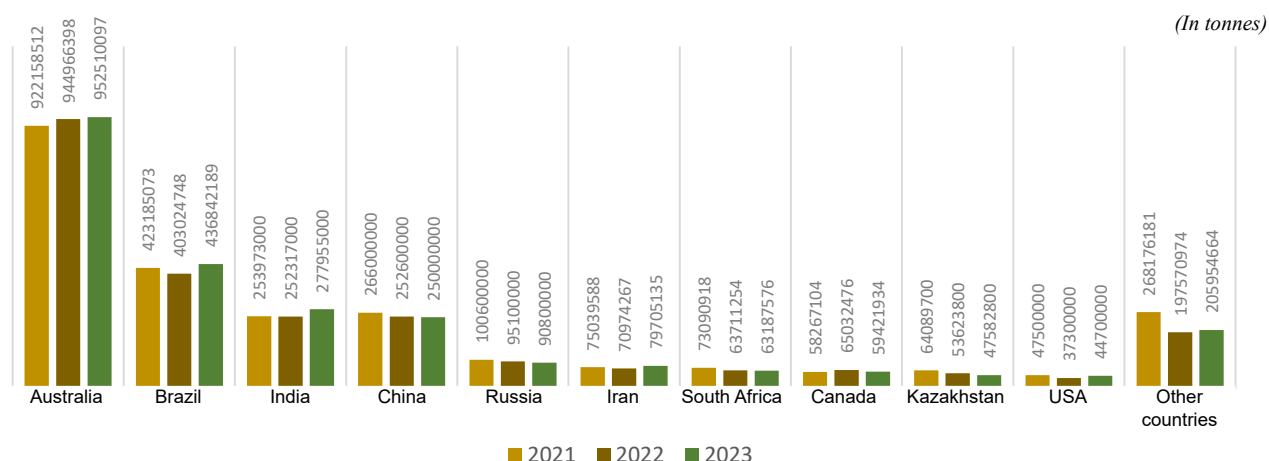


Fig. 1: Countrywise Production of Iron Ore

FOREIGN TRADE

Exports

Exports of iron ore increased by 119 % to 46.37 million tonnes in 2023-24 from 21.16 million tonnes in the previous year. Exports were mainly to China (94%) and Indonesia

(3 %). The total exports of iron ore in 2023-24, in terms of quantity comprised iron ore fines 31.38 million tonnes (68%), iron ore pellets 12.24 million tonnes (26%), iron ore lumps 2.52 million tonnes (5%) and negligible quantity of iron ore non-agglomerated concentrate and iron ore pyrites (Tables- 13 to 18).

Table – 13: Exports of Iron Ore

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	21168	144299670	46376	324332446
China	18286	113191493	43718	301320441
Indonesia	662	5927616	1232	11022892
Malaysia	337	2537472	466	4156918
UK	125	1525705	136	1336880
Korea, Rep. of	69	1004616	113	1288313

Table- 13 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
Netherlands	112	1547345	78	1159159
France	--	--	76	1042971
Kenya	48	55457	53	625349
Italy	131	1775512	53	600055
Qatar	161	740549	139	589436
Other Countries	1237	15993905	312	1190032

Figures rounded off

Table – 14: Exports of Iron Ore: Lumps

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	183	696000	2524	16063986
China	130	637341	2410	15630831
Indonesia	--	--	58	335520
Qatar	52	55025	54	82158
Malaysia	--	--	2	11972
Germany	++	19	++	2781
USA	--	--	++	707
Brazil	++	2	++	9
Netherlands	--	--	++	8
Tanzania	++	2047	--	--
UAE	1	1476	--	--
Other Countries	++	90	++	++

Figures rounded off

Table – 15: Exports of Iron Ore: Fines

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	14457	71125032	31380	183311744
China	13767	68584846	30716	180368011
Indonesia	177	876643	240	1308692
Qatar	109	685489	85	507146
Saudi Arabia	++	106	82	489551
Malaysia	194	787012	81	467343
Nepal	149	110737	176	170984
Germany	++	18	++	9
Australia	--	--	++	5
South Africa	--	--	++	3
Kenya	48	55068	--	--
Other Countries	13	25113	++	++

Figures rounded off

Table – 16: Exports of Iron Ore: Pyrites

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	++	32492	++	20133
Thailand	++	3551	++	7345

Table- 16 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
Netherlands	++	5982	++	4425
UAE	++	4365	++	1792
Bangladesh	++	1952	++	1779
Turkey	++	847	++	1099
Lebanon	--	--	++	971
Denmark	--	--	++	947
Afghanistan	--	--	++	389
Egypt	--	--	++	298
Senegal	++	57	++	246
Other Countries	++	15738	++	842

Figures rounded off

Table – 17: Exports of Iron Ore: Concentrates Non-agglomerated

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	102	913160	230	1510622
China	++	32	229	1509536
Nepal	--	--	1	705
UAE	--	--	++	180
Malawi	--	--	++	168
Australia	--	--	++	16
Congo, P Rep	--	--	++	11
Belgium	--	--	++	5
Canada	--	--	++	1
Germany	++	5	++	++
USA	++	2	++	++
Other Countries	102	913121	++	++

Figures rounded off

Table – 18: Exports of Iron Ore: Pellets

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000t)	Value (₹ '000)	Qty ('000t)	Value (₹ '000)
All Countries	6426	71532986	12242	123425961
China	4389	43969274	10363	103812063
Indonesia	485	5050973	934	9378680
Malaysia	143	1749438	383	3677519
UK	125	1525644	136	1336732
Korea, Rep. of	69	1003214	113	1288313
Netherlands	112	1541363	78	1154726
France	--	--	76	1042971
Kenya	--	--	53	625349
Italy	131	1775512	53	600055
Brazil	119	1627939	53	509551
Other Countries	853	13289629	++	2

Figures rounded off

Imports

Imports of iron ore increased to 4.96 million tonnes in 2023-24 from 1.79 million tonnes in the previous year. Imports of iron ore were from Australia (70%), Brazil (12%), Malaysia(7%), Norway & South Africa (4% each),

Oman (3%), Canada (2%) and negligible amount from other countries. The total imports in 2023-24 comprised iron ore fines (79%) non-agglomerated concentrates (17%) and iron ore lumps (4%) (Tables-19 to 23).

Table – 19: Imports of Iron Ore: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	1790	15947473	4960	43968382
Australia	367	1262142	3467	27494383
Brazil	696	6366826	577	7000054
Malaysia	--	--	323	3289374
Norway	--	--	191	1792564
South Africa	63	606072	176	1774603
Oman	--	--	144	1616338
Canada	483	5108079	77	874747
Turkey	4	79605	4	70866
Croatia	2	39105	1	15451
Italy	++	7397	++	10526
Other Countries	175	2478247	++	29476

Figures rounded off

Table – 20: Imports of Iron Ore: Concentrates Non-agglomerated

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	1316	13603285	845	9672717
Brazil	696	6366720	577	7000054
Norway	--	--	191	1792564
Canada	483	5108079	77	874747
Sweden	++	26034	++	2339
France	--	--	++	2302
USA	137	2102145	++	633
Spain	--	--	++	42
Russia	--	--	++	18
Australia	--	--	++	14
Namibia	--	--	++	3
Other Countries	++	307	++	1

Figures rounded off

Table – 21: Imports of Iron Ore: Pyrites

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	6	138772	5	113202
Turkey	4	79605	4	70865
Croatia	2	39105	1	15451
Italy	++	7397	++	10526
China	++	3970	++	9102
Peru	--	--	++	5751
UAE	--	--	++	779
USA	++	634	++	669
Zambia	--	--	++	41
Tanzania	--	--	++	18
Russia	++	7954	--	--
Other Countries	++	107	++	++

Figures rounded off

Table – 22: Imports of Iron Ore Lumps**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	131	1262348	176	1782376
South Africa	63	606072	176	1774603
Sweden	++	1113	++	3451
Japan	--	--	++	2963
France	++	3463	++	1163
Mali	--	--	++	154
Mozambique	--	--	++	37
Zimbabwe	--	--	++	3
Qatar	--	--	++	2
Uruguay	38	332386	--	--
Australia	30	319074	--	--
Other Countries	++	240	++	++

*Figures rounded off***Table – 23: Imports of Iron Ore: Fines****(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty ('000 t)	Value (₹ '000)	Qty ('000 t)	Value (₹ '000)
All Countries	337	943068	3934	32400087
Australia	337	943068	3467	27494369
Malaysia	--	--	323	3289374
Oman	--	--	144	1616338
Germany	--	--	++	6

Figures rounded off

FUTURE OUTLOOK

India is one of the leading producers of iron ore in the world. Among the consuming industries, Cement Industry is the second major consumer of iron ore after Iron & Steel Industry (including Sponge Iron Industry).

The Ministry of Steel under Government of India has introduced the new National Steel Policy, 2017 and with the roll-out of the National Steel Policy, 2017 and the DMI & SP policy, it is envisaged that the industry can be steered with appropriate policy support in creating an environment for promoting domestic steel and thereby ensuring a scenario where production meets the anticipated pace of growth in consumption. Thus, the Indian Steel Sector is all set to achieve its vision thereby setting a global benchmark in terms of quality, standards and technology. It is anticipated that crude steel capacity of 300 million tonnes will be required by 2030-31 and to fulfill this capacity, about 437 million tonnes of iron ore is required. However, achieving crude steel capacity up to 300 million tonnes will require

extensive mobilisation of natural resources, finances, manpower and infrastructure including land. To address the concerns regarding availability of raw material (iron ore), intensive & deeper exploration would have to be promoted for augmentation of resource base. Eco-friendly viable underground mining techniques for optimal utilisation of magnetite ore deposits locked in Western Ghats would also have to be explored in collaboration with mining research institutes. The Government has already promulgated the Mines and Minerals (Development and Regulation) Amendment Act, 2015 and therein has laid great emphasis on time bound mine development with increased stress on mineral exploration and sustainable mining operations which may support ore output growth.

The Act has brought clarity on mine allocation process (through auction) and procedures for mining lease renewal. The Act, further, provides for reservation of any particular mine for a particular end use and put conditions permitting auction among such eligible end-users.



44. Kyanite, Sillimanite and Andalusite

3,324 tonnes

production of kyanite have been reported in 2023-24.

771 tonnes)

Production of sillimanite have been reported in 2023-24.

199 tonnes

of kyanite were exported in 2023-24.

Kyanite, sillimanite and andalusite are naturally occurring anhydrous aluminosilicate minerals with the same chemical formula (Al_2SiO_5) but different crystal structures and physical properties making them mineral polymorphs. When calcined at high temperatures around 1,350 °C to 1,380 °C for kyanite and slightly higher for andalusite and sillimanite, these minerals are converted to mullite, ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$) and silica (SiO_2), both of which are refractory minerals.

Synthetic mullite is made by heating mixtures of alumina and silica or bauxite and kaolin at around 1,550°C to 2,000°C. Refractories are heat resistant materials used in high temperature applications, such as, furnaces, ladles, kilns, in the metallurgical, glass, chemical, cement and other industries.

Kyanite, sillimanite and andalusite are metamorphic minerals, which are formed in aluminium-rich rocks by orogenic or contact metamorphism. Andalusite is generally preferred over kyanite, because it can be used without calcination, thereby saving energy. Most of the world production of andalusite comes from South Africa and France.

RESERVES/RESOURCES

Kyanite

As per NMI database based on UNFC system as on 1.4.2020, total reserves/resources of kyanite in India have been placed at about 105.68 million tonnes. Of the total resources, only 0.84 million tonnes are classified as Reserves and 104.84 million tonnes fall under the Remaining Resources. Gradewise, high and medium-grade resources together account for 1.44%, low-grade for 8.18%,

mixed-grade for 0.53%, quartz kyanite rock, kyanite gneiss rock & kyanite schist for 88.21% and granular, unclassified, others & not-known grades for 1.63%. Statewise, share of Telangana is 45.75% of the total resources followed by Andhra Pradesh (30.28%), Karnataka (12.47%) and Jharkhand (7.83%). The remaining 3.67% resources are in Kerala, Maharashtra, Rajasthan, Tamil Nadu and West Bengal collectively (Table –1).

Table – 1: Reserves/Resources of Kyanite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	846865	104835455	105682321
By Grades			
High Grade	-	438700	438700
Medium Grade	368562	716745	1085307
Low Grade	67554	8577045	8644599
High & Medium Mixed	-	141390	141390
Medium & Low Mixed	-	48000	48000
High, Medium & Low Mixed	-	368378	368378
Granular	1620	247829	249449
Quartz Kyanite Rock	330202	82550736	82880938
Kyanite Gneiss Rock	-	5370800	5370800
Kyanite Schist	-	4974625	4974625
Unclassified	63	-	63
Others	78865	990854	1069719
Not-known	-	410353	410353
By States			
Andhra Pradesh	-	32004228	32004228
Jharkhand	331193	7943367	8274560
Karnataka	181600	12991460	13173060
Kerala	-	184733	184733
Maharashtra	332389	3063615	3396004
Rajasthan	-	23703	23703
Tamil Nadu	1683	247829	249512
Telangana	-	48350000	48350000
West Bengal	-	26520	26520

Figures rounded off

Sillimanite

As per NMI database based on UNFC system as on 1.4.2020, total reserves/resources of sillimanite in India have been placed at about 72.27 million tonnes. Of the total resources, 8.26 million tonnes fall under Reserves Category, while 64.01 tonnes fall under the Remaining Resources. Gradewise, 73.89% are granular high-grade, while quartz sillimanite rocks and sillimanite-bearing rocks are about

21.03%. Resources of massive sillimanite of all grades constitute about 4.80% and other, unclassified & not-known account for about 0.28%. Statewise, the resources are located mainly in Odisha (24.50%), Tamil Nadu (24.01%), Uttar Pradesh (15.84%), Andhra Pradesh (15.31%), Kerala (9.57%) and Assam (6.37%). The remaining 4.39% resources are in Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Meghalaya, Rajasthan and West Bengal (Table –2).

Table – 2: Reserves/Resources of Sillimanite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	8262300	64005091	72267391
By Grades			
Massive High Grade	163557	11903	175460
Massive Medium Grade	62703	33705	96408
Massive Low Grade	38000	3139605	3177605

Table – 2 (Cont...)

Table – 2 (Concl'd.)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
Massive High & Medium	-	19800	19800
Massive High, Medium & Low	-	38	38
Granular High	7994582	45404791	53399373
Quartz Sillimanite Rock	-	3748000	3748000
Sillimanite-bearing Rock	-	11450000	11450000
Others	-	11070	11070
Unclassified	3458	84000	87458
Not-known	-	102180	102180
By States			
Andhra Pradesh	1670025	9395739	11065764
Assam	-	4604700	4604700
Jharkhand	-	83000	83000
Karnataka	-	982725	982725
Kerala	553000	6366167	6919167
Madhya Pradesh	-	101600	101600
Maharashtra	181748	30580	212328
Meghalaya	82512	55807	138319
Odisha	5640985	12062208	17703193
Rajasthan	-	819	819
Tamil Nadu	134030	17218747	17352777
Uttar Pradesh	-	11450000	11450000
West Bengal	-	1653000	1653000

Figures rounded off

Andalusite

As per NMI database based on UNFC system as on 1.4.2020, total reserves/resources of andalusite in the country have been placed at 126.05 million tonnes. There

is no reserve for andalusite and 126.05 million tonnes fall under the Remaining Resources. Gradewise, all resources of andalusite are of low grade only. Statewise, resources of andalusite are located mainly in Uttar Pradesh (90.64%) followed by Jharkhand (9.36%) (Table –3).

Table – 3: Reserves/Resources of Andalusite as on 01.04.2020
(By Grades/States)

(In '000 tonnes)

State	Total Reserves	Remaining Resources	Total Resources
All India: Total	-	126050	126050
By Grades			
Low	-	126050	126050
By States			
Jharkhand	-	11800	11800
Uttar Pradesh	-	114250	114250

Figures rounded off

MINING LEASES AND PRODUCTION

Kyanite

As on 31.03.2024 (Provisional), the total number of mining leases granted for mining of Kyanite was 15 covering mining lease area of 288.982 Hectares. Since 2015, only one

ML block of kyanite, i.e., Girola (Jamdi) in Maharashtra, with an area of 6.55 hectares, reserves of 0.393 million tonnes, final bid of 32.05%, preferred bidder, namely, Shanti GD Ispat & Powers Pvt Ltd has been successfully auctioned on 11.01.2022. The production of kyanite was 3,324 tonnes in 2023-24, an increase of 20% compared to the previous

year. There were 5 reporting mines in 2023-24, same as the previous year. Three producers contributed the entire production of kyanite during the year. Maharashtra is the principal producing state contributing 93.68 % of the total production as rest was contributed by Jharkhand in the year 2023-24. About 57.88% of the total production was reported

by the public sector. Mine-head closing stock of kyanite for 2023-24 was 12,643 tonnes as against 12,194 tonnes in 2022-23 (Tables- 4 to 7). The average daily employment of labour was 62 in 2023-24 as against 86 in the previous year

Table – 4: Principal Producers of Kyanite, 2023-24

Name & address of producer	Location of the mine	
	State	District
Maharashtra State Mining Corporation Ltd., Plot No. 7, Ajani Chowk, Wardha Road, Nagpur- 440 015, Maharashtra.	Maharashtra	Bhandara
Pavri Kyanite Mines, A-1 Indra Sagar, Ravindranath Tagore Marg, Civil Lines, Nagpur- 440001, Maharashtra	Jharkhand	East Singhbhum
J S M D C LTD Khanij Bikash Nigam, Nepal House Area Doranda,Ranchi-834002, Jharkhand	Jharkhand	East Singhbhum

Table – 5: Production of Kyanite, 2021-22 to 2023-24

(By State)

(Qty in tonnes; Value in ₹ '000)

Name & address of producer	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	9320	17267	2765	5319	3324	10515
Jharkhand	2785	5152	2071	3831	210	389
Karnataka	5075	9084	360	644	-	-
Maharashtra	1460	3031	334	844	3114	10126

(P): Provisional

In tonnes

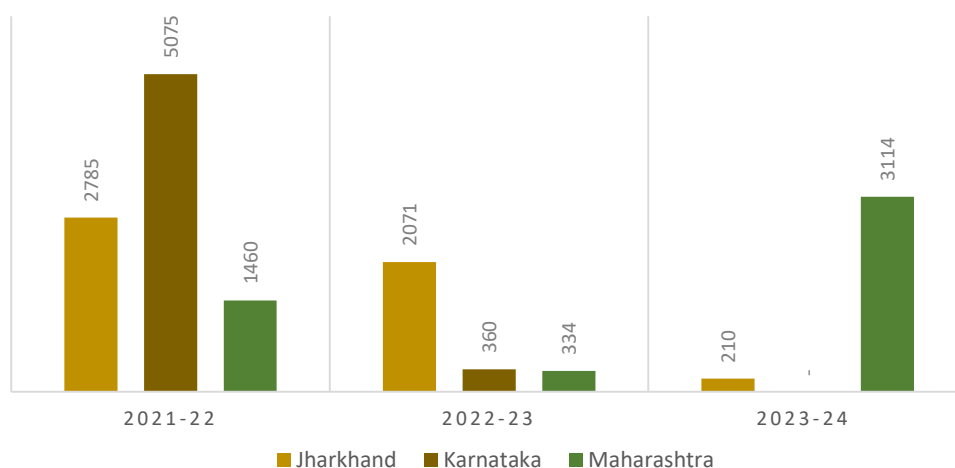


Fig. 1: Production of Kyanite in India

Table – 6: Production of Kyanite, 2022-23 and 2023-24

(By Sectors/States/Districts/Grades)

(Quantity in tonnes; Value in ₹ '000)

State/District	2022-23					2023-24 (P)				
	Quantity					Quantity				
	No. of mines	40% Al ₂ O ₃ & above	Below 40% Al ₂ O ₃	Total	Value	No. of mines	40% Al ₂ O ₃ & above	Below 40% Al ₂ O ₃	Total	Value
India	5	-	2765	2765	5319	5	1400	1924	3324	10515
Public Sector	2	-	2405	2405	4675	2	-	1924	1924	4717
Private Sector	3	-	360	360	644	3	1400	-	1400	5798
Jharkhand	1	-	2071	2071	3831	1	-	210	210	389
Singhbhum East	1	-	2071	2071	3831	1	-	210	210	389
Karnataka	1	-	360	360	644	1	-	-	-	-
Mysore	1	-	360	360	644	1	-	-	-	-
Maharashtra	3	-	334	334	844	3	1400	1714	3114	10126
Bhandara	3	-	334	334	844	3	1400	1714	3114	10126

(P): Provisional

Table – 7: Mine-head closing stocks of Kyanite, 2022-23 and 2023-24

(By States/Grades)

(In tonnes)

State	2022-23			2023-24 (P)		
	40% Al ₂ O ₃ & above	Below 40% Al ₂ O ₃	Total	40% Al ₂ O ₃ & above	Below 40% Al ₂ O ₃	Total
India	321	11873	12194	1721	10922	12643
Jharkhand	-	1541	1541	-	1366	1366
Karnataka	-	9629	9629	-	9252	9252
Maharashtra	321	703	1024	1721	304	2025

(P): Provisional

Sillimanite

As on 31.03.2024 (Provisional), one mining lease has been granted for sillimanite, covering mining lease area of 4.64 Hectares. The production of sillimanite decreased by 46.34% to 771 tonnes in 2023-24 as compared to 1437 tonnes in the previous year. Two mines reported production

of sillimanite as an associated mineral with kyanite during the year 2023-24. In 2023-24, the whole production of sillimanite was reported by private sector. Maharashtra was the only state contributed to the total production of Sillimanite during the year 2023-24. Mine-head closing stock for the year 2023-24 was 4,193 tonnes as against 3,898 tonnes in the previous year (Table – 8 to 11).

Table – 8: Principal Producers of Sillimanite, 2023-24

Name & address of producer	Location of the mine	
	State	District
*Pavri Kyanite Mine, A-1, Indra Sagar, Ravindranath Tagore Marg, Civil Lines, Nagpur-440001, Maharashtra.	Maharashtra	Bhandara
*Dighori Kyanite Mine, Apna Nagar, Tkiya Ward, Nagpur Road Bhandara, Bhandara-441904, Maharashtra.	Maharashtra	Bhandara

* Producing as an associated mineral with kyanite

Table – 9: Production of Sillimanite, 2021-22 to 2023-24

(By States)

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	3432	8283	1437	3757	771	2648
Andhra Pradesh	-	-	-	-	-	-
Kerala	-	-	-	-	-	-
Maharashtra	3432	8283	1437	3757	771	2648
Meghalaya	-	-	-	-	-	-
Odisha	-	-	-	-	-	-

(P): Provisional

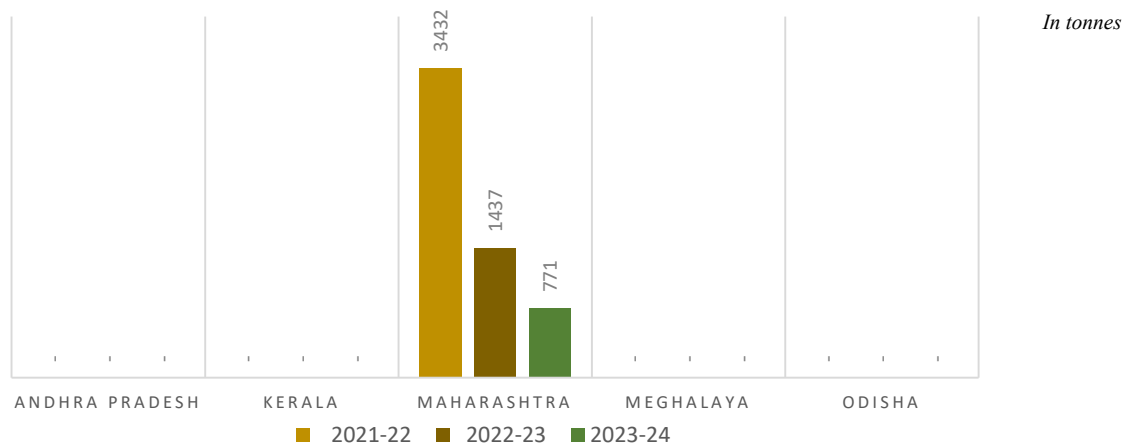


Fig. 2: Production of Sillimanite

Table –10: Production of Sillimanite, 2022-23 and 2023-24

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	No. of mines	2022-23		No. of mines	2023-24 (P)	
		Quantity	Value		Quantity	Value
India	(2)	1437	3757	(2)	771	2648
Public Sector	-	-	-	-	-	-
Private Sector	(2)	1437	3757	(2)	771	2648
Maharashtra	(2)	1437	1437	(2)	771	2648
Bhandara	(2)	1437	1437	(2)	771	2648
Meghalaya	-	-	-	-	-	-
Khasi Hills West	-	-	-	-	-	-

(P): Provisional

(): Figures in parenthesis indicate the number of associate mines with kyanite.

Table – 11: Mine-head Closing Stocks of Sillimanite, 2022-23 and 2023-24

(By States)

(In tonnes)

State	2022-23	2023-24 (P)
India	3898	4193
Andhra Pradesh	-	-
Kerala	-	-
Meghalaya	188	188
Maharashtra	3710	4005

(P): Provisional

Andalusite

No production of andalusite has been reported since 1988.

USES & CONSUMPTION

Kyanite, sillimanite and andalusite are primarily used in refractories and ceramic products due to their ability to form mullite phase at high temperatures. Kyanite is used to produce refractories for the production of steel and nonferrous (non-iron-bearing) metals. Some amount is used in the manufacture of refractories for glass and heat-resistant ceramics. Additionally, kyanite is used in the production of spark plugs and in various non-refractory applications. Mullite is an essential component of high-alumina refractories forming the inner lining of furnaces and high temperature vessels widely used in the production of metals, ceramics, glass and cement. It is used in manufacturing refractory products such as dense bricks, insulating bricks, monolithics, and castables. Sillimanite refractory bricks are extensively used in the steel and glass industries, as well as in ceramics, cement kilns, heat treatment furnaces, and the petrochemical industry.

Based on the production, imports and exports, the apparent consumption of kyanite during 2023-24 was 4,816 tonnes.

WORLD REVIEW

World reserves of kyanite and related minerals are largely located in the USA. Significant resources of andalusite are known to exist in China, France, Peru and South Africa. Kyanite resources have been identified in Brazil, India and Russia. While Sillimanite has been identified in India (Table –12). The main producers and exporters of andalusite are South Africa and France while USA and India are the main producers of kyanite. Also, India is the leading producer of sillimanite. World production of kyanite and related minerals is indicated in Table –13.

The availability of inexpensive refractory-grade bauxite from China has increased the demand for alternative raw materials, such as andalusite. Demand for refractories in iron and steel production is expected to rise more significantly in countries with higher growth rates in steel production. Additionally, increased demand is also anticipated for refractories used in the production of other metals and in the industrial mineral market, due to the growing production of cement, ceramics, glass, and other mineral products.

Table – 12: World Reserves of Kyanite, Sillimanite & Andalusite
(By Principal Countries)

		(In tonnes)
Country	Reserves	
Worls: Total (rounded off)		NA
United States (kyanite)		Large
India (kyanite and sillimanite)		72,00,000
France (andalusite)		NA
China (andalusite, crude ore)		50,00,000
Peru (andalusite)		NA
South Africa (andalusite)		NA

Source: USGS, Mineral Commodity Summaries, 2024

Table – 13: World Production of Kyanite, Sillimanite & Andalusite
(By Principal Countries)

		(In tonnes)		
Country	2021	2022	2023	
France				
Andalusite	139789 ^(b)	162434 ^(b)	141816 ^(b)	
South Africa				
Andalusite	170000*	170000*	170000*	
USA				
Kyanite ^(c)	105000	85900	82400	
India				
Kyanite	9432 ^(d)	5507	2652	
Sillimanite	3432 ^(d)	1828	910	

Table – 13 (Concl'd.)

(In tonnes)

Country	2021	2022	2023
Nepal			
Kyanite ^(e)	–	–	1*

Source: BGS, World Mineral Production, 2019-2023.

Notes:

(1) A number of other countries produce sillimanite minerals but details of output are not reported.

(2) In addition to the countries listed above as producing sillimanite minerals, synthetic mullite is believed to be produced in Brazil, Germany and Hungary.

(3) Production capacity for synthetic mullite also exists in china, India, Japan, the USA and the UK but output is believed to be zero.

(b) Andalusite, kyanite and sillimanite

(c) Including related minerals.

(d) Years ended 31st March following that stated.

(e) Years ending 15th July of that stated.

* estimated

FOREIGN TRADE

Exports

Exports of kyanite during 2023-24 at 199 tonnes decreased by 33% from 297 tonnes in the previous year. Exports were mainly to Turkey (53%), Greece (36%), UAE (6%), Bhutan (3%), Saudi Arabia (2%) and Sri Lanka (1%). On

the other hand, exports of sillimanite increased by 52% to 3,293 tonnes in 2023-24 from 2,161 tonnes in the previous year. Sillimanite was exported mainly to China (90%), Japan (8%) and Thailand, Indonesia and Nepal (1% each). Exports of andalusite during 2023-24 decreased by 53% to 9 tonnes from 19 tonnes in the previous year. Andalusite was exported solely to UAE (Tables - 14 to 16).

Table – 14: Exports of Kyanite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	297	8631	199	8627
Greece	71	1859	72	3076
Bhutan	--	--	5	2625
Turkey	112	3123	105	2323
UAE	12	450	12	572
Saudi Arabia	--	--	4	22
Sri Lanka	--	--	1	6
Madagascar	--	--	++	2
Nepal	--	--	++	1
France	52	2103	--	--
UK	20	671	--	--
Other Countries	30	425	++	++

Figures rounded off

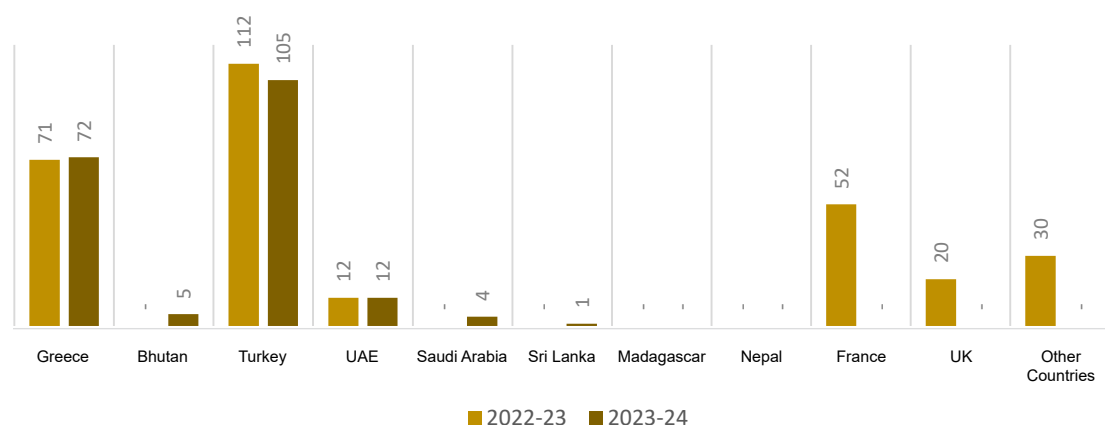


Fig 3: Countrywise Exports of Kyanite

Table – 15: Exports of Sillimanite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	2161	73954	3293	115853
China	1568	56393	2968	104869
Japan	394	13509	248	8971
Thailand	25	765	25	783
Indonesia	--	--	24	718
Nepal	109	1422	18	251
Iraq	--	--	10	88
France	--	--	++	74
UK	--	--	++	39
Netherlands	--	--	++	27
Bangladesh	--	--	++	12
Other Countries	65	1865	++	21

Figures rounded off

Table – 16: Exports of Andalusite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	19	1515	9	894
UAE	19	1515	9	894

Figures rounded off

Imports

In 2023-24, imports of kyanite were at 1,691 tonnes as against 1,354 tonnes in the previous year registering an increase of 25%. The imports of kyanite were from USA (92%) and China (8%). Imports of sillimanite were at 13 tonnes, which decreased by 78% during 2023-24 as compared to 58 tonnes in the previous year. Poland (31%),

USA & China (23% each), Estonia, Japan & Spain (8% each) were the main suppliers of sillimanite. Imports of andalusite at 14,454 tonnes increased by 2% during 2023-24 from 14,168 tonnes in previous year. South Africa (86%), Peru (7%), France (5%) and USA (1%) were the main suppliers of andalusite (Tables - 17 to 19).

Table – 17: Imports of Kyanite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	1354	63110	1691	81361
USA	1276	61455	1559	78309
China	78	1655	132	3052

Figures rounded off

Table – 18: Imports of Sillimanite
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	58	3424	13	3128
Estonia	--	--	1	1011
Japan	++	255	1	586
Spain	++	2	1	473
Poland	4	343	4	352
USA	4	1849	3	327
Germany	++	229	++	303
China	++	25	3	66
Saudi Arabia	--	--	++	9
UAE	--	--	++	1
Somalia	50	451	--	--
Other Countries	++	270	++	++

Figures rounded off

Table – 19: Imports of Andalusite
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Quantity (t)	Value (₹ '000)	Quantity (t)	Value (₹ '000)
All Countries	14168	599795	14454	847210
South Africa	12192	497613	12448	724155
Peru	82	3895	1038	58371
France	1749	91716	763	55512
USA	140	6150	200	8264
Austria	--	--	5	908
UAE	5	421	--	--

Figures rounded off

FUTURE OUTLOOK

The global outlook for kyanite, sillimanite, and andalusite is promising, driven by growing demand from the steel, cement, glass, and refractory industries. Andalusite is expected to continue steady growth, reflecting its critical role in high-temperature industrial applications. Kyanite's market is poised for expansion, supported by its widespread use in advanced refractories, while sillimanite continues

to see strong demand as a precursor to mullite. The Asia-Pacific region, with India, China, and Southeast Asian countries at the forefront, is emerging as a prominent growth corridor, supported by rapid industrialization and the ongoing expansion of steel and cement production. Overall, the outlook remains positive, with robust growth expected in key markets worldwide, particularly across Asia-Pacific.



45. Limestone

452 million tonnes

Production of limestone have been reported in 2023-24.

1.6 million tonnes

of limestone were exported in 2023-24.

33.8 million tonnes

of limestone were imported in 2023-24.

INTRODUCTION

Limestone is a calcareous sedimentary rock primarily composed of the mineral calcite (CaCO_3). The majority of cave systems are through limestone bedrock, which makes up around 10% of sedimentary rocks. Dolomite and calcite are the two main components of limestone. Magnesite (MgCO_3) combined with calcite or dolomite ($\text{CaMg}(\text{CO}_3)_2$) are two common forms of magnesium carbonate found in limestone.

'Dolomitic' or 'magnesian' limestone is the name given to such rocks. Stones transformed by dynamic or contact metamorphism become crystalline and are called "crystalline limestone" and "marble." Other common forms of limestone include travertine, onyx, hydraulic limestone, lithographic limestone, "marl", "oolite" (oolitic limestone), shelly limestone, algal limestone, coral limestone, pisolitic limestone, and so forth. Nonetheless, the limestone that is utilized in large quantities by enterprises is sedimentary limestone of the bedded variety.

The thick calcareous shells of molluscs that are deposited as beds and found in shallow oceans and ancient lakes are another type of calcareous material utilized in the industry. This material is known as "limeshell." The quantity of silt and clay in "marl", a mud rich in lime, varies.

"Flagstone" is the name given to a limestone rock that splits well along the strata into a slab that is only a few centimeters thick. Dimensional limestone is utilized in both architectural and decorative stone applications.

RESERVES/RESOURCES

As on 01.04.2020, the total Reserves/Resources of limestone of all grades and categories according to the NMI database, based on the UNFC system have been estimated at 2,27,589 million tonnes, of which 2,08,561 million tonnes (92%) are classified as Remaining Resources and 19,028 million tonnes (8%) as Reserves.

With 24% of the total resources, Karnataka leads all other states, including Andhra Pradesh (13%), Rajasthan (12%), Gujarat & Meghalaya (10% each), Telangana (7%), Chhattisgarh (5%) and Madhya Pradesh (4%). Other states shared the remaining 15 percent. Regarding grades, Portland's Cement grade has the largest proportion at roughly 68%, followed by Unclassified grades (11%), BF

grade (6%) and other grades. The remaining 15% is shared by various other grades [Table-1 (A)].

According to the NMI database based on UNFC system, the total reserves/resources of marl of all categories and

grades as on 1.4.2020 have been estimated at 99.20 million tonnes in Gujarat, of which 68.15 million tonnes (69%) are under Reserves category and 31.05 million tonnes (31%) are under Remaining Resources category [Table-1 (B)].

Table – 1(A): Reserves/Resources of Limestone as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Reserves Total (A)	Remaining Resources Total (B)	Total Resources (A+B)
All India: Total	19028470	208560789	227589259
By Grades			
Chemical	254835	5197825	5452660
S.M.S.(O.H.)	103570	4845398	4948968
S.M.S.(L.D.)	27379	591243	618622
S.M.S.(O.H. & L.D. mixed)	143912	236642	380554
B.F.	746646	13426453	14173099
S.M.S. & B.F. mixed	21580	1240926	1262506
Cement (portland)	16712957	139641159	156354115
Cement (white)	28006	46798	74804
Cement (portland& white)	55411	1044164	1099575
Cement (blendable)	588507	7051343	7639850
B.F. & cement mixed	19864	617408	637273
S.M.S., chemical & paper	182	1234096	1234278
Paper	56274	841137	897411
Blendable (CaO 34-38%)	-	883884	883884
Others	48714	3440357	3489071
Unclassified	165092	26382988	26548080
Not-known	55542	1838969	1894511
By States / Union Territory			
Andhra Pradesh	3256690	26582132	29838822
Arunachal Pradesh	-	482796	482796
Assam	188130	1683540	1871670
Bihar	11807	994188	1005995
Chhattisgarh	1486351	11724867	13211218
Daman & Diu	-	128670	128670
Gujarat	903115	21929169	22832284
Haryana	-	74677	74677
Himachal Pradesh	1022012	5597134	6619146
*Jammu & Kashmir	185490	2242071	2427561
Jharkhand	10687	610078	620765
Karnataka	2271221	53899236	56170457
Kerala	10540	184059	194599
Madhya Pradesh	1692431	7960747	9653178
Maharashtra	701349	3107044	3808392
Manipur	-	46053	46053
Meghalaya	251043	23583945	23834988
Nagaland	-	1752200	1752200
Odisha	468580	1727424	2196004
Puducherry	-	15732	15732
Rajasthan	4804154	24157095	28961249
Sikkim	-	2380	2380
Tamil Nadu	547024	1577025	2124049
Telangana	1214127	16438327	17652454
Uttar Pradesh	3720	439723	443443
Uttarakhand	-	1575771	1575771
West Bengal	-	44706	44706

Figures rounded off

* Notified as Union Territory and is to be known as Union Territory of Jammu & Kashmir comprising the territory of the existing State of Jammu & Kashmir (Gazette Notification No. 53, New Delhi, Friday, August 9, 2019)

Table – 1 (B): Reserves/Resources of Marl as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Reserves Total (A)	Remaining Resources Total (B)	Total Resources (A+B)
All India: Total	68145000	31053477	99198477
By Grade			
Unclassified	68145000	31053477	99198477
By State			
Gujarat	68145000	31053477	99198477

Figures rounded off

MINING LEASE AND PRODUCTION

Limestone

As on 31.03.2024 (Provisional), the total no. of leases granted for mining of limestone is 1,812 covering an overall mining lease area of 1,68,264.605 Ha. The production of limestone at 452 million tonnes in 2023-24 increased by about 11.42% as compared to that of the previous year. There were 735 reporting mines in 2023-24 as against 703 during the previous year. Forty four mines, each producing more than 3 million tonnes per annum contributed 51.85 percent of the total production of limestone in 2023-24. The share of 22 mines, each in the production range of 2 to 3 million tonnes was 12.45% of the total production. About 16.22% of the total production was contributed by 51 mines, each producing 1 to 2 million tonnes annually. The remaining 19% of the total production was reported by 242 mines during the year. Ten principal producers contributed about 61% of the total production. About 4.63% of the

production was reported by public sector mines as against 4.96% in the previous year. About 97.47% of the total production of limestone during 2023-24 was of cement grade and remaining 2.52% by other grades.

Rajasthan was the leading producing state accounting for 22.9% of the total production of limestone, followed by Madhya Pradesh (13.8%), Andhra Pradesh (13.5%), Chhattisgarh (10.9%), Karnataka (9%), Tamil Nadu (5.9%), Telangana (5.8%), Gujarat (5.2%), and remaining 13% was collectively contributed by Maharashtra, Himachal Pradesh, Meghalaya, Odisha, Uttar Pradesh, Assam, Bihar, Kerala, Jharkhand and Union Territory of Jammu and Kashmir.

Mine-head closing stocks of limestone for the year 2023-24 were 30.74 million tonnes and for the year 2022-23, it was 21.52 million tonnes. Average daily labour employment in limestone mines in 2023-24 was 21,765 as against 21,940 in the previous year (Tables- 2 to 6).

Table – 2: Principal Producers of Limestone, 2023-24

Name & address of producer	Location of the Mine	
	State	District
M/s UltraTech Cement Limited 'B' Wing, Ahura Centre, 2nd Floor, Mahakali Caves Road, Andheri (E), Mumbai - 400 093, Maharashtra.	Andhra Pradesh	Krishna
	Andhra Pradesh	Kurnool
	Chhattisgarh	Baloda Bazar
	Chhattisgarh	Raipur
	Gujarat	Amreli
	Gujarat	Bhavnagar
	Gujarat	Kachchh
	Himachal Pradesh	Solan
	Karnataka	Kalaburagi
	Madhya Pradesh	Dhar
	Madhya Pradesh	Katni
	Madhya Pradesh	Neemuch
	Madhya Pradesh	Rewa
	Madhya Pradesh	Satna
	Madhya Pradesh	Sidhi
	Maharashtra	Chandrapur
	Rajasthan	Chittorgarh
	Rajasthan	Jaipur
	Rajasthan	Nagaur
	Rajasthan	Sirohi

Table 2 (Cont.)

Name & address of producer	Location of the Mine	
	State	District
M/s Shree Cement Limited Post Box No.33, Bangur Nagar, Beawar - 305 901, Rajasthan.	Tamil Nadu	Pali
	Tamil Nadu	Ariyalur
	Uttar Pradesh	Perambalur
	Uttar Pradesh	Sonbhadra
	Andhra Pradesh	Guntur
	Chhattisgarh	Baloda Bazar
	Chhattisgarh	Raipur
	Gujarat	Kachchh
	Karnataka	Kalaburagi
	Rajasthan	Ajmer
M/s Ambuja Cements Limited Elegant Business Park, MIDC Cross Road B, Off Andheri- Kurla Road, Andheri East, Mumbai - 400 059, Maharashtra.	Rajasthan	Jhunjhunu
	Rajasthan	Pali
	Chhattisgarh	Baloda Bazar
	Chhattisgarh	Raipur
	Gujarat	Junagadh
	Himachal Pradesh	Solan
	Maharashtra	Chandrapur
	Rajasthan	Nagaur
	Rajasthan	Pali
	Chhattisgarh	Bilaspur
M/s ACC Limited Cement House, 121, Maharshi Karve Road, Mumbai - 400 020, Maharashtra.	Chhattisgarh	Durg
	Himachal Pradesh	Bilaspur
	Jharkhand	West Singhbhum
	Karnataka	Kalaburagi
	Madhya Pradesh	Katni
	Maharashtra	Yavatmal
	Odisha	Bargarh
	Rajasthan	Bundi
	Tamil Nadu	Coimbatore
	Andhra Pradesh	Y.S.R.
M/s Dalmia Cement (Bharat) Limited Dalmiapuram, Main Road, Kallakudi, Lalgudi, Thiruchirapalli - 621 651, Tamil Nadu.	Assam	Dima Hasao
	Bihar	Rohtas
	Karnataka	Belagavi
	Madhya Pradesh	Satna
	Meghalaya	East Jaintia Hills
	Odisha	Sundargarh
	Tamil Nadu	Arilayur
	Tamil Nadu	Tiruchirappalli
	Karnataka	Bagalkote
	Madhya Pradesh	Neemuch
M/s J.K. Cement Limited Kamla Tower, Kanpur - 208 001, Uttar Pradesh	Madhya Pradesh	Panna
	Rajasthan	Chittorgarh
	Rajasthan	Jaisalmer
	Rajasthan	Nagaur
	Andhra Pradesh	Krishna
	Andhra Pradesh	Kurnool
	Karnataka	Chitradurga
	Karnataka	Tumakuru
	Tamil Nadu	Ariyalur
M/s The Ramco Cement Limited 5th Floor, Auras Corporate Centre, 98-A, Dr.Radhakrishanan Salai, Mylapore, Chennai - 600 004,		

Table- 2 (Concld.)

Name & address of producer	Location of the Mine	
	State	District
Tamil Nadu.	Tamil Nadu	Perambalur
	Tamil Nadu	Tuticorin
	Tamil Nadu	Virudhunagar
M/s Wonder Cement Limited	Rajasthan	Chittorgarh
17, Old Fatehpura ,Near Seva Mandir,Udaipur-313 004	Rajasthan	Jaisalmer
Rajasthan		
M/s Hettinad Cement Corporation Private Limited	Andhra Pradesh	Guntur
Rani Seethai Hall ,No. 603,	Karnataka	Kalaburagi
Annasalai, Chennai, Tamil Nadu, India - 600 006.	Tamil Nadu	Ariyalur
	Tamil Nadu	Dindigul
	Tamil Nadu	Karur
	Tamil Nadu	Perambalur

Table – 3: Production of Limestone, 2021-22 to 2023-24

(By States)

(Qty in '000 tonnes; Value in ₹ '000)

State/UT	2021-22		2022-23		2023-24 (P)	
	Production	Value	Production	Value	Production	Value
India	392034	102022622	405664	111239856	452009	118909140
Andhra Pradesh	50278	10865618	53526	12016234	60877	13124556
Assam	1681	540157	1812	609080	1996	634849
Bihar	987	348399	929	327984	999	396419
Chhattisgarh	41889	11523823	44777	13064371	49279	14021421
Gujarat	23250	5212323	22852	5493989	23405	5767770
Himachal Pradesh	13810	2983883	11879	2751753	13706	3118011
Jammu and Kashmir	1156	286892	1237	297134	1185	310646
Jharkhand	72	103729	3	952	3	998
Karnataka	39395	7893558	38942	8129116	40832	8129362
Kerala	379	370969	347	317936	381	304893
Madhya Pradesh	49807	15833338	50597	17526113	62556	18974820
Maharashtra	15758	3884495	16656	3998958	19320	4562484
Meghalaya	6399	2974757	8111	4127730	10070	4765751
Odisha	7059	2574534	6551	2533383	8019	2939254
Rajasthan	87483	23336293	93964	26050938	103644	27433019
Tamil Nadu	21323	6505610	24434	7642296	26777	8203516
Telangana	28499	5932307	26464	5662826	26082	5524833
Uttar Pradesh	2809	851937	2583	689063	2878	696538
Uttarakhand	-	-	-	-	-	-

(p): Provisional

Table – 4: Production of Limestone, 2022-23 & 2023-24 (P)

(By Frequency Groups)

(Qty in '000 tonnes)

Production Group	No. of Mines		Production for the Group		Percentage in Total Production		Cumulative Percentage	
	2022-23	2023-24	2022-23	2023-24	2022-23	2023-24	2022-23	2023-24
All Groups	703 (6)	735 (3)	405664	452009	100	100	-	-
Up to 10000	224 (6)	242(3)	337	342	0.08	0.08	0.08	0.08
10001 – 50000	117	105	3319	2848	0.82	0.63	0.90	0.71
50001 – 100000	64	81	4722	6081	1.16	1.35	2.07	2.05
100001 – 200000	56	48	8176	7141	2.02	1.58	4.08	3.63

Table- 4 (Concl'd.)

(Qty in '000 tonnes)

Production Group	No. of Mines		Production for the Group		Percentage in Total Production		Cumulative Percentage	
	2022-23	2023-24	2022-23	2023-24	2022-23	2023-24	2022-23	2023-24
200001 – 300000	35	39	8953	9684	2.21	2.14	6.29	5.77
300001 – 400000	24	26	8367	9125	2.06	2.02	8.35	7.79
400001 – 500000	17	17	7656	7548	1.89	1.67	10.24	9.46
500001 – 600000	9	10	5079	5628	1.25	1.25	11.49	10.71
600001 – 700000	14	15	9132	9901	2.25	2.19	13.74	12.90
700001 – 800000	17	13	12948	9871	3.19	2.18	16.93	15.08
800001 – 900000	8	12	6773	10258	1.67	2.27	18.60	17.35
900001 – 1000000	7	10	6784	9658	1.67	2.14	20.27	19.49
1000001– 2000000	52	51	76161	73302	18.77	16.22	39.05	35.70
2000001 – 3000000	24	22	60348	56274	14.88	12.45	53.92	48.15
3000001 & above	35	44	186909	234348	46.08	51.85	100.00	100.00

(p): provisional

(): Figure in parenthesis indicates mines of chalk, dolomite & shale with limestone as an associate mineral.

No. of Mines does not include those mines that have reported only labour, without reporting any production (only mines reporting production are included)

Table – 5 (A): Production of Limestone, 2022-23

(By Sectors/States/Districts/Grades)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
India	703(6)	394706	8662	2296	405664	111239856
Public Sector	31(1)	15966	3907	269	20142	8214861
Private Sector	672(5)	378740	4755	2027	385522	103024995
Andhra Pradesh	76(2)	53167	359	-	53526	12016234
Anantapur	6(1)	3798	17	-	3815	720599
Guntur	12	5704	-	-	5704	1203803
Krishna	10	13240	125	-	13365	3548173
Kurnool	41(1)	16249	217	-	16466	3304102
Palnadu	1	657	-	-	657	133671
Y.S.R.	6	13519	-	-	13519	3105886
Assam	3	1812	-	-	1812	609080
Dima Hasao	2	1621	-	-	1621	544189
Karbi Anglong	1	191	-	-	191	64891
Bihar	1	929	-	-	929	327984
Rohtas	1	929	-	-	929	327984
Chhattisgarh	72	43803	974	-	44777	13064371
Baloda Bazar	6	8652	-	-	8652	2320674
Bastar	11	2	1	-	3	2459
Bilaspur	3	249	671	-	920	398863
Durg	28	7260	302	-	7562	2732300
Janjgir-Champa	3	1405	-	-	1405	495226
Kabirdham	1	32	-	-	32	20329
Raipur	20	26203	-	-	26203	7094520
Gujarat	84	20827	-	2025	22852	5493989
Amreli	4	5080	-	-	5080	1154560
Bhavnagar	2	++	-	-	++	12
Jamnagar	19	1714	-	295	2009	465172

Table 5 (A) (Cont.)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
Junagadh	30	7828	-	457	8285	1998616
Kachchh	3	4074	-	0	4074	875877
Porbandar	24	1878	-	1273	3151	891844
Rajkot	1	2	-	-	2	586
Surat	1	251	-	-	251	107322
Himachal Pradesh	22	11762	115	2	11879	2751753
Bilaspur	1	2852	-	-	2852	622097
Sirmaur	19	956	115	2	1073	465014
Solan	2	7954	-	-	7954	1664642
Jammu and Kashmir	16	1237	-	-	1237	297134
Anantnag	7	++	-	-	++	10
Pulwama	6	641	-	-	641	119527
Srinagar	3	596	-	-	596	177597
Jharkhand	2	3	-	-	3	952
Ranchi	2	3	-	-	3	952
Karnataka	53	38566	376	-	38942	8129116
Bagalkote	33	3341	361	-	3702	1038009
Belagavi	5	2346	15	-	2361	577486
Chitradurga	3	31	-	-	31	10605
Kalaburagi	10	32848	-	-	32848	6503016
Shivamogga	1*	-	-	-	-	-
Tumakuru	1*	-	-	-	-	-
Kerala	1	347	-	-	347	317936
Palakkad	1	347	-	-	347	317936
Madhya Pradesh	168(3)	46191	4406	-	50597	17526113
Damoh	1	3492	-	-	3492	982365
Dhar	9	4988	-	-	4988	916923
Indore	2	391	-	-	391	90845
Katni	60(2)	4820	3431	-	8251	3441666
Narsinghpur	1	-	64	-	64	31210
Neemuch	6	4562	-	-	4562	1003179
Panna	3	853	-	-	853	434133
Rewa	11	3318	++	-	3318	1770591
Satna	72(1)	21269	911	-	22180	7669420
Sidhi	3	2498	-	-	2498	1185781
Maharashtra	26	16656	-	-	16656	3998958
Chandrapur	13	13376	-	-	13376	3095718
Yavatmal	13	3280	-	-	3280	903240
Meghalaya	17	8111	-	-	8111	4127730
East Jaintia Hills	15	4658	-	-	4658	1447519
East Khasi Hills	2	3453	-	-	3453	2680211
Odisha	7	6498	53	-	6551	2533383
Bargarh	1	699	-	-	699	426786
Koraput	1	98	-	-	98	40851
Sundargarh	5	5701	53	-	5754	2065746

Table- 5(A) (Concld.)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
Rajasthan	41	91317	2378	269	93964	26050938
Ajmer	2	2378	-	-	2378	664079
Banswara	1	1389	-	-	1389	344542
Bundi	1	996	-	-	996	274722
Chittorgarh	10	36813	-	-	36813	9458564
Jaipur	2	4329	-	-	4329	1489634
Jaisalmer	3	1312	2378	-	3690	2001221
Jhunjhunu	1	1	-	-	1	208
Kota	1	3001	-	-	3001	870198
Nagaur	9	4922	-	269	5191	1577389
Pali	6	21182	-	-	21182	5135643
Sirohi	3	13051	-	-	13051	3700089
Udaipur	2	1943	-	-	1943	534649
Tamil Nadu	81	24432	2	-	24434	7642296
Ariyalur	35	13872	2	-	13874	3847183
Dindigul	4	2434	-	-	2434	849954
Karur	1	593	-	-	593	203683
Perambalur	15	2341	-	-	2341	722184
Salem	2	527	-	-	527	198679
Tiruchirappalli	10	2834	-	-	2834	823086
Tirunelveli	5	379	-	-	379	174240
Tuticorin	5	1243	-	-	1243	704386
Virudhunagar	4	209	-	-	209	118901
Telangana	30	26464	-	-	26464	5662826
Adilabad	2	3331	-	-	3331	822048
Hyderabad	2	1298	-	-	1298	276174
Karimnagar	2	1150	-	-	1150	522882
Nalgonda	19	16961	-	-	16961	3232688
Ranga Reddy	4	3721	-	-	3721	808172
Suryapet	1	3	-	-	3	862
Uttar Pradesh	3	2583	-	-	2583	689063
Sonbhadra	3	2583	-	-	2583	689063
Uttarakhand	(1)	-	-	-	-	-
Bageshwar	(1)	-	-	-	-	-

(++) : Negligible

() : Figure in parenthesis indicates mines of chalk, dolomite and shale with limestone as an associate mineral.

(*) Only labour reported.

() : Figure in parenthesis indicates mines of chalk, dolomite and shale with limestone as an associate mineral.

Table – 5 (B): Production of Limestone, 2023-24

(Bv Sectors/States/Districts/Grades)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
India	735(3)	440614	8883	2512	452009	118909140
Public Sector	30	16335	4279	349	20963	8267652
Private Sector	705(3)	424279	4604	2163	431046	110641488
Andhra Pradesh	78(1)	60521	356	-	60877	13124556
Anantapur	6(1)	2740	10	-	2750	548472
Guntur	14	8363	-	-	8363	1638309

Table 5 (B) (Cont.)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
Krishna	11	14505	223	-	14728	3699787
Kurnool	40	19828	123	-	19951	3908480
Palnadu	1	709	-	-	709	156907
Y.S.R.	6	14376	-	-	14376	3172601
Assam	4	1996	-	-	1996	634849
Dima Hasao	3	1832	-	-	1832	579024
Karbi Anglong	1	164	-	-	164	55825
Bihar	1	999	-	-	999	396419
Rohtas	1	999	-	-	999	396419
Chhattisgarh	63	47798	1481	-	49279	14021421
Baloda Bazar	6	9736	-	-	9736	2572270
Bastar	9	2	++	-	2	1260
Bilaspur	3	368	1081	-	1449	650724
Durg	25	7628	400	-	8028	3016543
Janjgir-Champa	3	1401	-	-	1401	483689
Kabirdham	1	12	-	-	12	8307
Raipur	16	28651	-	-	28651	7288628
Gujarat	89	21244	-	2161	23405	5767770
Amreli	4	5711	-	-	5711	1312593
Bhavnagar	2	1	-	-	1	284
Jamnagar	21	1906	-	304	2210	515692
Junagadh	29	6669	-	564	7233	1840685
Kachchh	3	4635	-	-	4635	973497
Porbandar	27	2041	-	1293	3334	1010454
Rajkot	2	116	-	-	116	43653
Surat	1	165	-	-	165	70912
Himachal Pradesh	26	13623	81	2	13706	3118011
Bilaspur	1	3733	-	-	3733	726026
Mandi	1*	-	-	-	-	-
Sirmaur	22	807	81	2	890	476906
Solan	2	9083	-	-	9083	1915079
Jammu and Kashmir	19	1185	-	-	1185	310646
Anantnag	10*	-	-	-	-	-
Pulwama	6	555	-	-	555	97670
Srinagar	3	630	-	-	630	212976
Jharkhand	2	3	-	-	3	998
Ranchi	2	3	-	-	3	998
Karnataka	53	40267	565	-	40832	8129362
Bagalkote	34	3413	499	-	3912	1055153
Belagavi	5	2375	66	-	2441	624507
Chitradurga	2*	-	-	-	-	-
Kalaburagi	10	34479	-	-	34479	6449702
Shivamogga	1*	-	-	-	-	-
Tumakuru	1*	-	-	-	-	-
Kerala	1	381	-	-	381	304893
Palakkad	1	381	-	-	381	304893

Table 5 (B) (Cont.)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
Madhya Pradesh	191(2)	58863	3693	-	62556	18974820
Damoh	1	3577	-	-	3577	958471
Dhar	18	7892	-	-	7892	1264683
Indore	2	180	-	-	180	30300
Katni	64(2)	7675	3441	-	11116	3867223
Narsinghpur	1	-	76	-	76	39568
Neemuch	8	4370	-	-	4370	1051012
Panna	4	4482	-	-	4482	1088052
Rewa	10	3910	-	-	3910	1603429
Satna	79	24075	176	-	24251	7869674
Sidhi	4	2702	-	-	2702	1202408
Maharashtra	31	19320	-	-	19320	4562484
Chandrapur	13	13928	-	-	13928	3051514
Yavatmal	18	5392	-	-	5392	1510970
Meghalaya	17	10070	-	-	10070	4765751
East Jaintia Hills	15	6445	-	-	6445	1889850
East Khasi Hills	2	3625	-	-	3625	2875901
Odisha	7	7828	191	-	8019	2939254
Bargarh	1	406	-	-	406	223837
Koraput	1	99	-	-	99	47624
Sundargarh	5	7323	191	-	7514	2667793
Rajasthan	44	100806	2489	349	103644	27433019
Ajmer	2	2885	-	-	2885	779272
Banswara	1	1348	-	-	1348	330535
Bundi	1	883	-	-	883	358342
Chittorgarh	11	38479	-	-	38479	9608073
Jaipur	2	4685	-	-	4685	1447775
Jaisalmer	4	1198	2489	-	3687	2193889
Jhunjhunu	1	1064	-	-	1064	301126
Kota	1	3278	-	-	3278	950721
Nagaur	10	5846	-	349	6195	1818109
Pali	6	25531	-	-	25531	5692315
Sirohi	3	12696	-	-	12696	3104804
Udaipur	2	2913	-	-	2913	848058
Tamil Nadu	76	26751	26	-	26777	8203516
Ariyalur	35	14792	26	-	14818	3936097
Dindigul	4	2473	-	-	2473	889414
Karur	1	779	-	-	779	291514
Perambalur	16	2294	-	-	2294	669988
Salem	2	472	-	-	472	158713
Tiruchirappalli	10	3582	-	-	3582	1029065
Tirunelveli	2	423	-	-	423	194573
Tuticorin	5	1631	-	-	1631	877166
Virudhunagar	1	305	-	-	305	156986
Telangana	30	26082	-	-	26082	5524833
Adilabad	2	3625	-	-	3625	841193
Hyderabad	2	778	-	-	778	195092

Table- 5(B) (Concl'd.)

(Qty in '000 tonnes; Value in ₹ '000)

State/District	No. of Mines	Cement	LD, SMS & BF	Chemical	Qty	Value
Karimnagar	2	1240	-	-	1240	521065
Nalgonda	19	17013	-	-	17013	3215326
Ranga Reddy	4	3423	-	-	3423	750184
Suryapet	1	3	-	-	3	1973
Uttar Pradesh	3	2878	-	-	2878	696538
Sonbhadra	3	2878	-	-	2878	696538

(p): provisional

(++) : Negligible

(): Figure in parenthesis indicates mines of chalk, dolomite and shale with limestone as an associate mineral.

(*) Only labour reported.

Table – (6): Mine-head Closing Stocks of Limestone, 2022-23 and 2023-24 (P)

(in '000 tonnes)

State/UT	2022-23				2023-24 (P)			
	Grades				Grades			
	Cement	LD, SMS & BF	Chemical	Total	Cement	LD, SMS & BF	Chemical	Total
India	16806	3015	1699	21519	24882	3768	2089	30739
Andhra Pradesh	340	110	-	450	528	79	-	607
Assam	29	-	-	29	27	-	-	27
Bihar	-	-	-	-	-	-	-	-
Chhattisgarh	2161	96	-	2257	1752	273	-	2025
Gujarat	1265	-	1648	2913	1533	-	2016	3548
Himachal Pradesh	317	14	-	331	194	10	-	204
Jammu and Kashmir	792	-	-	792	770	-	-	770
Jharkhand	3	-	-	3	11	++	-	12
Karnataka	3026	492	-	3519	2730	545	-	3275
Kerala	5	-	-	5	3	-	-	3
Madhya Pradesh	5819	1782	-	7600	6454	1893	-	8346
Maharashtra	495	-	-	495	433	-	-	433
Meghalaya	42	-	-	42	87	-	-	87
Odisha	566	434	-	999	585	408	-	993
Rajasthan	966	80	51	1097	8675	462	73	9210
Tamil Nadu	654	7	++	660	669	98	++	767
Telangana	326	-	-	326	432	-	-	432
Uttar Pradesh	-	-	-	-	-	-	-	-
Uttarakhand	-	-	-	-	-	-	-	-

(p): provisional

(++) : Negligible

(): Figure in parenthesis indicates mines of chalk, dolomite and shale with limestone as an associate mineral.

Limeshell

As on 31.03.2024 (Provisional), the total no. of mining leases granted for mining of Limeshell is 18 covering an overall mining lease area of 873.958 Ha. The production of Limeshell was 891 tonnes during 2023-24 and 250 tonnes in the preceding year. There were 2 reporting mines in 2023-24

as well as in 2022-23. Mine-head closing stocks of limeshell in the year 2023-24 were 9,162 tonnes as against 696 tonnes in the previous year. The average daily employment of labour during the year 2023-24 was 17 as against 1 in the previous year (Tables-7 to 9).

Table – (7): Production of Limeshell, 2021-22 to 2023-24

(By State)

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	100	231	250	656	891	2633
Karnataka	100	231	250	656	891	2633
Kerala	-	-	-	-	-	-
Tamil Nadu	-	-	-	-	-	-

(p): Provisional

Table – (8) Production of Limeshell, 2022-23 and 2023-24

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	2	250	656	2	891	2633
Public sector	-	-	-	-	-	-
Private sector	2	250	656	2	891	2633
Karnataka	2	250	656	2	891	2633
Uttara Kannada	2	250	656	2	891	2633
Kerala	-	-	-	-	-	-
Kottayam	-	-	-	-	-	-
Tamil Nadu	-	-	-	-	-	-
Cuddalore	-	-	-	-	-	-

(p): Provisional

Table – (9): Mine head Closing Stocks of Limeshell, 2022-23 and 2023-24

(By States)

(In tonnes)

State/District	2022-23	2023-24 (P)
India	696	9162
Karnataka	696	9162
Kerala	-	-
Tamil Nadu	-	-

(p):Provisional

Marl

The production of marl during 2023-24 was 1,606 thousand tonnes as compared to 1,462 thousand tonnes in the preceding year. The entire production of marl was reported as associated mineral with limestone in both the years. There were 20 associate mines reporting production of marl

during 2023-24 as well as in 2022-23. The entire production was reported by private sector mines. Entire production of marl during 2023-24 was reported from Gujarat and Tamil Nadu states. Mine-head stocks at the end of 2023-24 were 672 thousand tonnes as against 478 thousand tonnes in the previous year (Table- 10 to 13).

Table – 10: Principal Producers of Marl, 2023-24

Name & Address of Producer	Location of the Mine	
	State	District
*UltraTech Cement Limited, 'B' Wing, 2nd Floor, Ahura Center, Mahakali Caves Road, Andheri (E), Mumbai- 400 093"	Gujarat	Amreli
Saurashtra Cements Ltd, Near railway station, Ranavav, Gujarat- 360560	Gujarat	Porbandar

Table- 10 (Concl'd.)

Name & Address of Producer	Location of the Mine	
	State	District
The Ramco Cements Ltd, 3rd floor, Auras Corporate centre 98-A, Dr. Radhakrishnan Salai, Mylapore, Chennai- 600004	Tamil Nadu	Ariyalur
M/s Chettinad Cement Corporation Pvt. Ltd, Rani Seethai Hall, #603, Anna Salai, Chennai - 600 006	Tamil Nadu	Ariyalur
Ultra Tech Cement Ltd, 'B' Wing, 2nd floor, Ahura Center, Mahakali Caves Road, Andheri (E), Mumbai- 400 093	Gujarat	Amreli

* Producing as an associated mineral with Limestone

Table – 11: Production of Marl, 2021-22 to 2023-24 (P)

(By State)

(Qty in tonnes, Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	1853481	326498	1461668	316762	1606380	301431
Gujarat	900560	133211	875308	184289	1081896	192182
Tamil Nadu	952921	193287	586360	132473	524484	109249

(p): Provisional

Table – 12: Production of Marl, 2022-23 & 2023-24 (P)

(By Sector/States/Districts)

(Qty in tonnes, Value in ₹ '000)

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	(20)	1461668	316762	(20)	1606380	301431
Private Sector	(20)	1461668	316762	(20)	1606380	301431
Gujarat	(7)	875308	184289	(7)	1081896	192182
Amreli	(2)	725895	155438	(2)	598395	119926
Jamnagar	(1)	7458	281	(1)	9829	2303
Junagadh	(3)	63901	13165	(3)	21856	4069
Porbandar	(1)	78054	15405	(1)	451816	65884
Tamil Nadu	(13)	586360	132473	(13)	524484	109249
Ariyalur	(9)	586360	132473	(9)	524484	109249
Perambalur	(1)	0	0	(1)	0	0
Tiruchirappalli	(3)	0	0	(3)	0	0

(): Figures in Parentheses indicate associated mines with limestone.

(p): Provisional

Table – 13: Mine-head Stocks of Marl, 2022-23 & 2023-24 (P)

(By States)

(In tonnes)

State	2022-23	2023-24 (P)
India	477696	672423
Gujarat	477696	672424
Tamil Nadu	258751	453478

"(): Figures in Parentheses indicate associated mines with limestone.

(p): Provisional"

USES & CONSUMPTION

Limestone, plays a pivotal role in India's mineral economy due to its vast reserves and diverse industrial applications. In India, limestone is mostly used in the cement industry, iron and steel flux, and chemicals. It is also used in fertilizers, paper, sugar and agriculture.

As per Mines and Minerals (Development and Regulation) Act, 1957, limestone used for industrial purpose falls under 'major mineral', while the use of limestone in lime kilns and for building purposes comes under 'minor mineral' category.

The threshold value of limestone as per the revised Notification issued by IBM vide No.C- 284/3/CMG/2017 dated 25th April 2018 is CaO 34% (min.) and MgO 5% (max.).

The principal use of limestone is in the Cement Industry. The broad chemical specifications of cement grade limestone tabulated in Table-14. Other important uses are as raw material in the manufacture of quicklime (calcium oxide), slaked lime (calcium hydroxide) and mortar. Pulverised limestone is used as a soil conditioner to neutralize acidic soils (agricultural lime). It is used in sculptures because of its suitability for carving. It is often found in medicines and cosmetics. In some circumstances, limestone is used for glass making. As a reagent in fuel-gas desulphurization, it reacts with sulphur dioxide which enables air pollution control. It can suppress methane explosions in underground coal mines. It is added to

toothpaste, paper, plastic, paint, tiles and other materials as both white pigment and cheap filler. In blast furnaces, limestone binds with silica and other impurities and facilitates their removal from iron.

Lime is prepared by heating limestone in kilns up to 1,000 OC. The CO released is effluxed and 'quicklime' (CaO) formed remains as hard white lumps. This, when slaked with water and mixed with sand, forms mortar or plaster.

Commonly, the commercial lime is prepared as dry hydrated lime Ca(OH)₂ by adding to quicklime the right amount of water (18 parts to 56 parts of CaO). The value of lime for most purposes depends upon its CaO (or CaO + MgO) content.

The manufacture of metallic calcium is one of the latest uses of lime. Calcium is used in reducing organic compounds, desulphurising petroleum, debismuthising lead production of hard lead alloys and calcium-silicon alloys, and in the manufacture of calcium hydride which is further used as an efficient hydrogen carrier.

Limeshell is used mainly in Chemical and White Cement Industries. It is also used in the manufacture of polyfibre and in Tanning Industry. Marl is used as lithographic stone.

In the year 2023-24, the apparent consumption of limestone was 484.15 million tonnes which increased by 12% from 432.88 million tonnes in the preceding year.

Table – 14: Broad Chemical Specifications of Cement Grade (Run-of-Mine) Limestone

(Clause 6.1.1)

Oxide components/ other constituents	Acceptable range for manufacture of portland cement (33, 43 and 53 grade) (per cement)	Limiting values taking into consideration other types of cements scope of beneficiation and blending (per cement)
CaO	44-52	40 (min.)
MgO	3.5 (max.)	5.0 (max.)
SiO ₂	To satisfy LSF silica	-
Al ₂ O ₃	Modules and alumina	-
Fe ₂ O ₃	Modules	-
TiO ₂	<0.5	<1.0
Mn ₂ O ₃	<0.5	<1.0
R ₂ O (Na ₂ O + K ₂ O)	<0.6	<1.0
Total S as SO ₃	<0.6	<0.8
P ₂ O ₅	<0.6	<1.0
Cl	0.015	<0.05
Free Silica	<8.0	<10.0

Source: Report on Norm for limestone deposits for cement manufacture by National Council for Cement and Building Materials, New Delhi, May 2001

FOREIGN TRADE

Exports

Exports of limestone decreased by 34% to 1.6 million tonnes in 2023-24 from 2.5 million tonnes in the previous year. Limestone in bulk was exported mainly to Bangladesh

(98%). On the other hand, exports of chalk decreased by 19% to 857 tonnes in 2023-24 from 1,057 tonnes in the previous year. Chalk was exported mainly to Nepal (88%), Egypt & Bangladesh (5% each) and UAE (1%).

Exports of bleaching powder increased by 15% to

33,028 tonnes in 2023-24 as compared to 28,840 tonnes in the previous year. Bleaching powder was exported mainly to Bangladesh (79%), Sri Lanka (6%), Nepal (5%) and USA (3%).

In the year 2023-24, exports of calcium carbide decreased by 39% to 450 tonnes as compared to 735 tonnes in the previous year. Exports were mainly to Bangladesh (89%), and Nepal & Bhutan (6% each) (Tables-15 to 18).

Table – 15: Export of Limestone
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2515714	3124865	1662127	1899997
Bangladesh	2476791	2618881	1626045	1373410
Nepal	15082	105736	14144	99987
South Africa	334	7840	4996	95183
UK	4186	67550	3865	60813
UAE	478	26521	689	33903
China	1798	37989	1688	31696
USA	1254	26495	721	24629
Bhutan	1237	20762	1319	20886
Ireland	2162	29272	998	15853
Congo, D. Rep. of	1134	12955	1017	14941
Other Countries	11258	170864	6645	128696

Figures rounded off

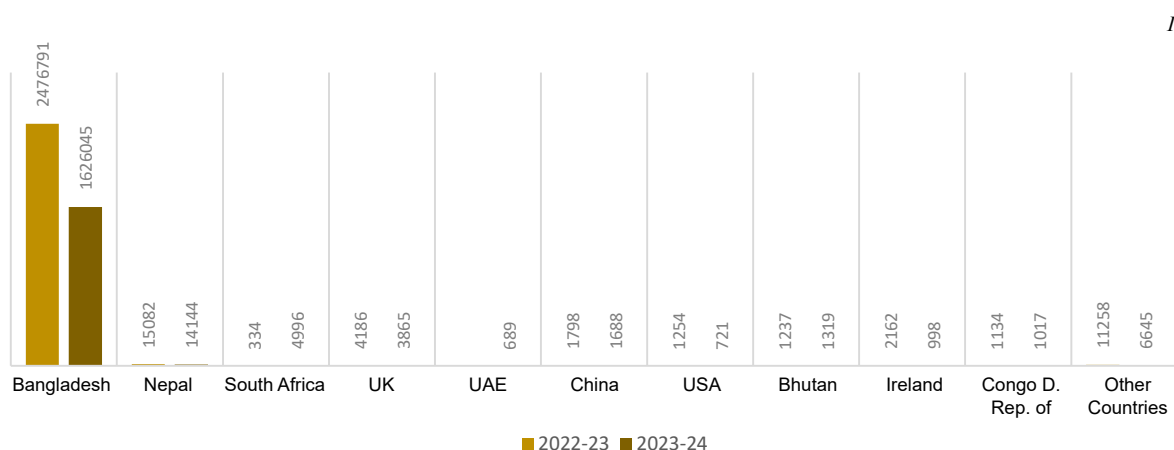


Fig. 1: Countrywise Export of Limestone

Table – 16: Export of Chalk
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1057	7330	857	6374
Nepal	979	5595	758	4546
Egypt	16	388	45	719
Bangladesh	38	286	39	378
UAE	7	207	9	217
Mozambique	++	2	1	108
Nigeria	++	9	++	58
USA	++	53	++	48
Bhutan	++	36	1	46
Maldives	1	31	1	45
Australia	++	13	++	44
Other Countries	16	710	3	165

Figures rounded off

Table – 17: Export of Bleaching Powder

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	28840	1014781	33028	1038960
Bangladesh	22642	620889	25967	602665
USA	1752	206219	1041	137202
Sri Lanka	1087	43556	1934	61084
UAE	343	40778	587	57404
Vietnam	12	1204	503	40691
Nepal	1855	26152	1634	21185
Tanzania	--	--	189	18121
Canada	43	6759	139	17572
Egypt	81	8361	110	14626
Malaysia	359	16770	254	12471
Other Countries	666	44093	670	55939

Figures rounded off

Table – 18: Export of Calcium Carbide

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	735	84151	450	40597
Bangladesh	519	62245	399	36976
Bhutan	47	5354	25	2474
Nepal	46	2545	26	1084
Qatar	++	1	++	58
Maldives	++	13	++	2
Turkey	--	--	++	2
Guyana	--	--	++	1
Angola	87	10069	--	--
Congo, P. Rep,	36	3804	--	--
Germany	++	80	--	--
Other Countries	++	40	++	++

Figures rounded off

Imports

Imports of limestone increased by 14% to 33.8 million tonnes in 2023-24 from 29.73 million tonnes in the previous year. On the other hand, imports of chalk in the year 2023-24 decreased by 73.1% to 11 tonnes as against 41 tonnes in the previous year.

Limestone was imported mainly from UAE (81%), Oman (15%), and Malaysia (2%), whereas chalk was

imported mainly from Germany(73%), Spain (18%) and Denmark (9%).

Imports of calcium carbide increased by 14% to 25,223 tonnes in 2023-24 from 22,133 tonnes in the previous year. Calcium carbide was imported mainly from China (97%) and Indonesia (3%). The imports of bleaching powder during 2023- 24 decreased by 82% to 3 tonnes as against 17 tonnes in the previous year. Imports were made completely from Argentina (100%) (Tables-19 to 22).

Table –19: Import of Limestone

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	29736036	62593636	33809129	66157693
UAE	24851743	48778721	27423729	50611602
Oman	3836422	8243536	5145231	9953195
Malaysia	674307	3428226	654130	3013531
Vietnam	102092	626056	268936	956198
Egypt	131341	560984	179155	625787
China	12151	296565	13163	217300
Jordan	12300	54470	25408	191916
Bhutan	39731	101344	50042	143130
Thailand	8742	156213	9498	121202
UK	26363	133859	3262	113103
Other Countries	40844	213662	36575	210729

Figures rounded off

In tonnes

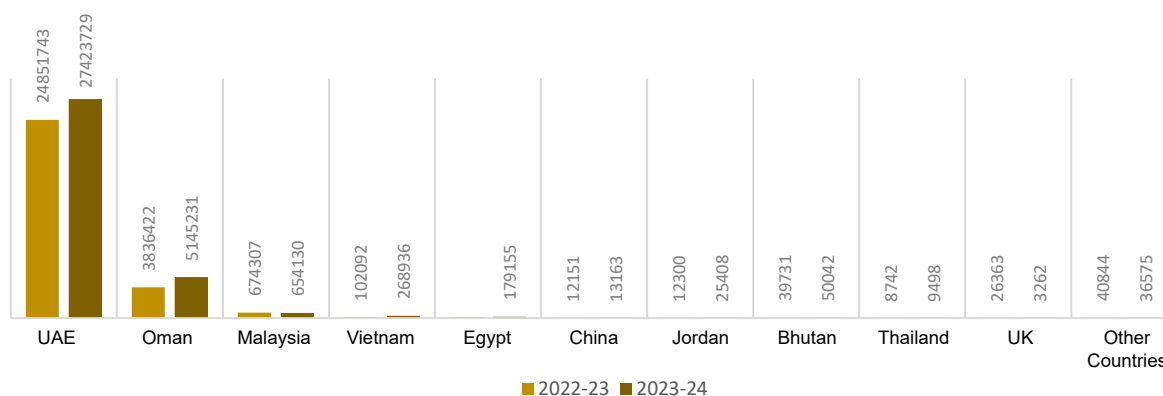


Fig. 2: Countrywise Import of Limestone

Table – 20: Import of Chalk

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	41	2191	11	975
Germany	1	100	8	762
Denmark	--	--	1	112
Taiwan	++	248	++	98
Spain	20	1118	2	3
France	20	647	--	--
Italy	++	62	--	--
UK	++	16	--	--

Figures rounded off

Table – 21: Import of Calcium Carbide

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	22133	1715011	25223	1334734
China	19550	1529340	24362	1303465
Indonesia	2583	183783	861	31269
Switzerland	++	1850	--	--
Germany	++	38	--	--

Figures rounded off

Table – 22: Import of Bleaching Powder**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	17	5096	3	2318
Argentina	5	3187	3	2318
USA	12	1909	--	--

Figures rounded off

FUTURE OUTLOOK

India has huge resources of limestone distributed over different parts of the country. It is comfortably placed in terms of annual capacity and production of cement. Cement-grade limestone occurs in all the limestone-bearing areas, while SMS, BF and Chemical-grade limestones occur in selective areas. Concerted efforts to locate SMS and BF grade limestone along with Cement-grade limestone are imperative to meet the growing demand.

The demand of raw materials for cement, such as, limestone and gypsum is expected to cause disruptive growth in the next few decades. According to India Brand Equity Foundation (IBEF), a Trust established by the Department of Commerce, Ministry of Commerce and

Industry, Govt. of India, the Indian Cement Industry is expected to continue to grow rapidly and reach an installed capacity of 850 million tonnes per annum by 2030 and 1,350 million tonnes per annum by FY50.

The demand for paper in India is expected to rise at a healthy rate mainly due to the Packaging Industry and the increasing number of schools. The increasing number of construction projects is expected to lead to a thriving Building and Construction Industry in India. This is expected to contribute 10% to the GDP of India. Additionally, with the rising growth in Indian Pharmaceutical and Food & Beverage industries, the consumption of calcium carbonate (limestone) in India is expected to increase.



46. Magnesite

1,32,005 tonnes
Production of magnesite have been reported in 2023-24.

3,842 tonnes
of magnesite were exported in 2023-24.

4,79,521 tonnes
of magnesite were imported in 2023-24.

Magnesite (MgCO_3) is a magnesium carbonate mineral that occurs mainly in serpentinised ultramafic rocks, dolomitic limestones, and other magnesium-rich host rocks, typically forming as bedded deposits or irregular veins. It is found in crystalline, amorphous, or massive forms, with common impurities, such as, calcium, silica, iron oxide and alumina influencing its quality. As the principal raw material for producing magnesia (MgO), magnesite is indispensable to refractory applications due to its high melting point, thermal stability and chemical inertness, making it a cornerstone for the steel, cement, glass and non-ferrous metallurgy industries. In industrial practice, the term 'magnesite' also encompasses its calcined products, including caustic magnesia (lightly calcined at 700–1,000 °C and used in cements and animal feed), dead-burnt magnesite (calcined at 1,500–1,800 °C for refractory bricks) and fused magnesia (ultra-high temperature calcination yielding dense periclase). These processed forms, especially dead-burnt and fused magnesia, dominate refractory consumption, while caustic magnesia finds wider use in specialty cements, agriculture and environmental applications. The combination of its geological occurrence and industrial versatility underscores magnesite's importance as a critical mineral resource.

RESERVES AND RESOURCES

According to the NMI database based on the UNFC system, the total reserves/resources of magnesite as of 1st April, 2020 are approximately 459 million tonnes, with 66 million tonnes in Reserves and 393 million tonnes in Remaining Resources. Uttarakhand holds the most resources (52%),

followed by Tamil Nadu (34%) and Rajasthan (12%). Resources are also found in Andhra Pradesh, Himachal Pradesh, Jammu & Kashmir, Karnataka and Kerala. Occurrences of magnesite in Tamil Nadu are low in lime and high in silica, whereas those of Uttarakhand are high in lime and low in silica. The gradewise and statewide reserves and resources of magnesite are furnished in Table -1.

Table – 1: Reserves/Resources of Magnesite as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	66070	393047	459118
By Grades			
High Grade	-	3336	3336
Medium Grade	63849	106968	170818
Beneficiable/Low	2154	153537	155691
High & Medium Mixed	-	2339	2339
Medium & Low Mixed	-	115910	115910
Others	6	7258	7264
Unclassified	-	83	83
Not-known	60	3617	3677
By States			
Andhra Pradesh	-	80	80
Himachal Pradesh	-	298	298
Jammu & Kashmir	-	4145	4145
Karnataka	1027	4516	5543
Kerala	-	40	40
Rajasthan	-	54091	54091
Tamil Nadu	55084	100402	155486
Uttarakhand	9959	229476	239434

Fig.ures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on Exploration & Development under "General Review".

MINING LEASES & PRODUCTION

As on 31.03.2024, the total number of mining leases granted for Magnesite was 33 with a cumulative lease area of 2,301.721 ha. "Production of magnesite in 2023-24 was 1,32,005 tonnes, registering an increase of 23% over the previous year. There were 10 reporting mines in 2023-24, the same as in 2022-23. The five principal producers accounted for about 94.83% of the total output during 2023-24. Out of total production, about 53% of magnesite was contributed by the Public Sector and the remaining 47% by

Private Sector during 2023-24. Major magnesite producing mines in Salem area belong to Tamil Nadu Magnesite Ltd (TANMAG, a State Government Undertaking), Ponkumar Magnesite Mines, Mysore Minerals, Dalmia Magnesite Corporation (a Private Sector Enterprise) and SAIL Refractory Co. Ltd (a Central Government Undertaking). Tamil Nadu continued to be the principal producing State contributing 60% of the total production and the rest was contributed by Uttarakhand and Karnataka. Mine-head closing stock of magnesite for the year 2023-24 was 44,057 tonnes as against 46,033 tonnes in the previous year. The average daily employment of labour in magnesite mines during the year was 664 as against 675 in the previous year (Tables- 2 to 5).

Table – 2: Principal Producers of Magnesite, 2023-24

Name & address of producer	Location of the mine	
	State	District
Tamil Nadu Magnesite Limited 5/53, Omalur Main Road Jagirampalayam, Salem-636 302, Tamil Nadu.	Tamil Nadu	Salem
Almora Magnesite Ltd, Village Matela, P.O. Billori, Distt. Bageshwar-263 630, Uttarakhand .	Uttarakhand	Bageshwar
S. Sundararajan, 5/22-A, Periyakollapatti Kannankuruchi, Post – Gorimedu, Distt. Salem -636 008, Tamil Nadu.	Tamil Nadu	Salem

Table- 2 (Concl'd.)

Name & address of producer	Location of the mine	
	State	District
India Magnesia Product Ltd. 11/239 Ramakrishna Road, Balaji Towers 3 rd floor, Salem - 636 007 Tamil Nadu	Tamil Nadu	Salem
Pithoragarh Magnesite And Minerals WB-4Transport Nagar, Haldwani, Nainital, Uttarakhand 263 139	Uttarakhand	Pithoragarh

Table – 3: Production of Magnesite, 2021-22 to 2023-24

(By States)

(Qty. in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (p)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	113497	554772	107524.99	502586.9478	132005	611500
Karnataka	7057	51412	8195	65923	6816	45429
Rajasthan	-	-	1	1	-	-
Tamil Nadu	81013	454667	62123	363065	78085	452154
Uttarakhand	25427	48693	37206	73598	47104	113917

(p): provisional

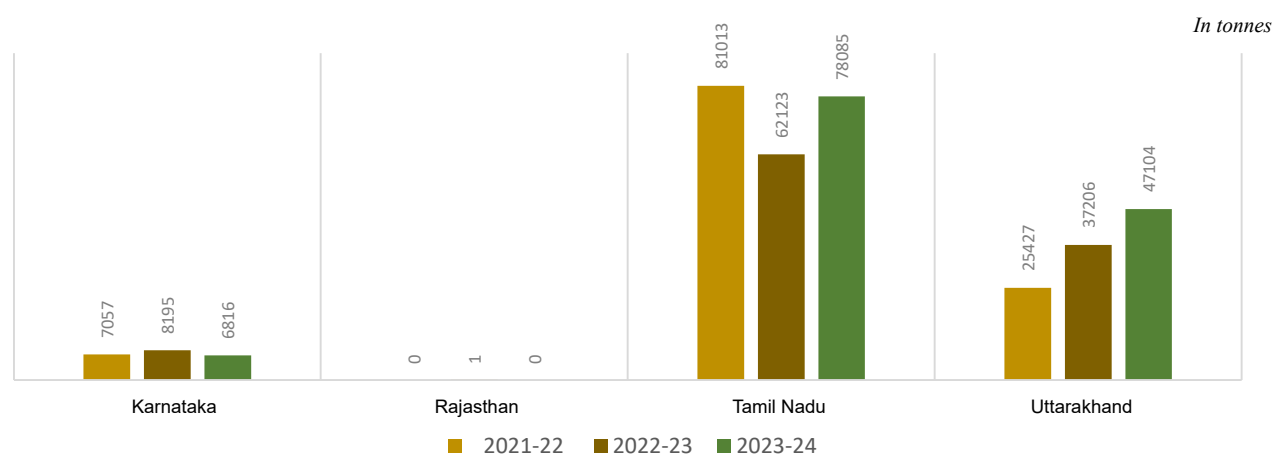


Fig. 1: Statewise Production of Magnesite

Table – 4: Production of Magnesite, 2022-23 and 2023-24

(By Sector/State/Districts)

(Qty. in tonnes; Value in ₹ '000)

State/District	2022-23			2023-24 (p)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	10	107525	502587	10	132005	611500
Public Sector	5	60462	252013	5	69288	353231
Private Sector	5	47063	250574	5	62717	258269
Karnataka	3	8195	65923	3	6816	45429
Mysore	3	8195	65923	3	6816	45429
Tamil Nadu	5	62123	363065	5	78085	452154
Salem	5	62123	363065	5	78085	452154
Uttarakhand	1	37206	73598	2	47104	113917
Bageshwar	1	37206	73598	1	32357	96854
Pithoragarh	-	-	-	1	14747	17063
Rajasthan	1	1	1	-	-	-
Pali	1	1	1	-	-	-

(p): Provisional

**Table – 5: Mine-head Closing Stocks of Magnesite,
2022-23 and 2023-24**

(By States)

(₹ per tonne)

State	2022-23	2023-24 (P)
India	46033	44057
Karnataka	3613	4154
Rajasthan	31	31
Tamil Nadu	40514	35472
Uttarakhand	1875	4400

(p): provisional

USES AND CONSUMPTION

Magnesite (MgCO_3) is a strategically important industrial mineral with extensive high-temperature and chemical applications. The bulk mined magnesite undergoes calcination to produce dead-burnt magnesite (DBM) or fused magnesia, which serve as critical feedstocks for high-grade refractory materials in steelmaking, cement production, non-ferrous metallurgy and other high-temperature industrial processes. Raw magnesite is also utilised in magnesium chemical production, magnesium metal manufacturing, fertilizers and food-processing applications. In India, magnesite is predominantly consumed by the Refractory Industry, where raw material is dead-burned to yield DBM, which is subsequently processed into basic refractory bricks, refractory mortars, ramming mass, tar/pitch-impregnated magnesia, magnesia-carbon bricks, slide-gate plates and other specialty refractory components. As per the Industries Department of the Tamil Nadu Government's Policy Note 2016-17, Industrial production of one tonne of DBM typically requires 2.7 tonnes of raw magnesite and 220 litres of furnace oil. Strict control of major impurities, such as, SiO_2 , CaO , Fe_2O_3 and Al_2O_3 is essential to meet specifications for different refractory grades. Regionally, Asia-Pacific dominates magnesite consumption due to extensive industrialisation and infrastructure development, while demand in the Middle East and Africa is rising in line with construction and industrial expansion. Overall, the Refractory Sector remains the principal consumer of magnesite, with consumption patterns driven by industrial growth, regional infrastructure investment and technological advances in processing & beneficiation. The apparent consumption for the year 2023-24 is 6,07,684 tonnes as against 6,05,099 tonnes in the preceding year, a marginal increase of 0.4%.

INDUSTRY

Dead Burnt Magnesite (DBM)

Raw magnesite when calcined at temperatures in the range of 1,660–1,800 °C in the rotary kiln, carbon dioxide gets expelled completely and a dense product 'Dead Burnt Magnesite' is obtained. Dead Burnt Magnesite refers to the

magnesite that is chemically unreactive or 'dead', therefore, enabling it to be used in brick making or monolithic hearths without undue difficulty arising out of hydration or shrinkage.

Caustic Calcined Magnesite (CCM)

Low calcined magnesite also known as Caustic Calcined Magnesite is obtained by calcining magnesite in a shaft or rotary kiln at temperature ranging between 800 °C and 1,000 °C. The incomplete dissociation causes retention of 8 to 10% carbon dioxide as carbonate. Low calcined magnesia when mixed with water forms a feebly plastic paste. Industries like paper, rubber, ceramic, asbestos products, glass, etc. use caustic magnesia.

Fused Magnesia

Fused magnesia is produced by the fusion of the high-grade magnesite in Higgin's or electric arc tilt furnaces between 2,500 °C and 3,000 °C. It is resistant to the action of molten metals, and fluxes and high temperatures. It is used in the form of moulded vessels and as compressed material for covering resistant elements of the furnaces used in the melting of lead, tin, etc. As per the available information, presently there are seven major plants that manufacture Dead Burnt Magnesite, while there are four plants that produce calcined magnesite and one that produces fused magnesia (Table-7). By-product magnesium carbonate and other magnesium salts were also produced during salt manufacturing from sea water. Dalmia Magnesite Corporation and Tamil Nadu Magnesite Ltd are the major producers of DBM and caustic calcined grades. As per Annual Report of National Mineral Development Corporation (NMDC) 2018-19, J&K Mineral Development Corporation Ltd, a subsidiary of NMDC, will be setting up a 30,000 TPA DBM plant at Panthal, Jammu. The Ministry of Environment, Forest and Climate Change granted Environment Clearance vide letter dated 03.05.2011. However, MoEFCC, subsequently, vide their letter dated 28.10.2016 had withdrawn the EC thereby reversing the position to the earlier status.

Sea Water Magnesia (SWM)

Sea water or lake bitters are an alternative source to obtain magnesia by chemical reaction. The main raw materials

required other than sea water are dolomite or limestone, fresh water and sulphuric acid. The magnesia content of sea water is about 0.2%, and even by enrichment with dolomite, around 300 kilograms sea water need to be processed to obtain one kilogram of magnesia. The sea water magnesia can be used to manufacture Dead Burnt Magnesite, caustic magnesia and other magnesium compounds.

Marine By-products

Carbonates, chlorides and sulphates of magnesium are obtained as by-products in the production of common salt by solar evaporation. Salt Commissioner, Jaipur, reported 3,552.50 tonnes production of magnesium chloride and nil production of by-product magnesium sulphate in 2022-23. The production is normally reported from the salt pans in Jamnagar–Gandhidham, Gujarat.

Magnesium Metal

Magnesium metal is a fairly strong, silvery-white, lightweight metal (about one-third lighter than aluminium). It is traditionally produced in ingot form of approximately 7 kg each with purity close to 99.9%. Its chief applications are, in die casting (alloyed with zinc), to remove sulphur in the production of iron and steel, for production of titanium in the Kroll process. The other application field of magnesium is in electronic devices. Defence equipment and nuclear reactor materials also consume magnesium. Magnesium technology and its commercial production in India are still at its infancy. India has developed silicothermic reduction process as well as fused salt electrolytic process, with capacity of 600 t/year for each process. However, the cost of production is very high as compared to the landed cost of imported magnesium metal. Hence, its production has been stopped by one of the companies. The production is only about 15–20% of the rated capacity.

SUBSTITUTES

In some refractory applications, alumina, chromite and silica are used as substitutes for magnesium.

TRADE POLICY

In accordance with the ITC (HS) 2022 import policy in Schedule-I and export policy in Schedule-II, natural magnesium carbonate (magnesite), fused magnesia, dead burned (sintered) magnesia, whether or not containing small quantities of other oxides added before sintering, and other magnesium oxide, whether or not pure, are allowed freely with no restrictions.

WORLD REVIEW

Global magnesite reserves have been updated to approximately 7,700 million tonnes in terms of magnesium oxide content, excluding significant quantities of magnesium-bearing replacements, such as, dolomite, brucite and olivine. In addition, magnesium compounds can be economically extracted from well and lake brines, as well as from seawater, which contains about 0.13% magnesium by weight—making it an important source of magnesium metal and derivatives. Russia holds the largest share of global magnesite reserves, representing nearly 30%, followed by Slovakia with around 16%, China with about 8%, Australia & Greece both at roughly 4%, Brazil with nearly 3% and Turkey at about 1.4%. Other countries collectively hold about 34% of global reserves. In 2023, China continued to dominate global magnesite production, accounting for nearly 54% of the world's total output. Australia followed as the second largest producer, contributing about 15 %. Russia produced approximately 11% of global magnesite, making it the third largest contributor. Turkey nearly 6%, while Austria and Spain accounted for approximately 3% each. Other countries collectively made up approximately 6% of total production.

Table – 6: World Reserves of Magnesite
(By Principal Countries)

(In '000 tonnes of magnesium oxide (MgO) content)

Country	Reserves
World: Total (rounded off)	7700000
Russia	2300000
Slovakia	1200000
China	580000
Australia*	280000
Greece	280000
Brazil	200000
Turkey	110000
India	66000
Austria	49000
Spain	35000
Other countries	2600000

Source: USGS, Mineral Commodity Summaries, 2024

*For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 37 million tonnes.

Table – 7: World Production of Magnesite

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World Total (rounded off)	2720000*	2580000	2420000
China	1400000*	1300000*	1300000*
Australia ^(b)	2739388	2571152	3525221
Russia	260000*	260000*	260000*
Turkey	1927280	1820068	1333728
Austria	842679	707267	669400
Spain	703834	676699	660108
Slovakia	576700	512000	392000
Finland	179781	324226	298025
Brazil ^(a)	200000*	200000*	200000*
Other countries	1630338	1588588	1521518

Source: BGS, World Mineral Production, 2019-23.

* Estimated

(a) Including beneficiated and directly shipped material

(b) Years ended 30th june of that stated

FOREIGN TRADE

Exports

The export of magnesite decreased by 27% to 3,842 tonnes in 2023-24 from 5,263 tonnes in the preceding year. Exports were mainly to Bangladesh (22%), UAE (14%) and Cameroon (7%). Out of the total exports in 2023-24, those of fused magnesia was at 3 tonnes as compared to 1 tonne in

the preceding year; calcined magnesite were at 375 tonnes as compared to 401 tonnes; non-calcined magnesite were at 1,096 tonnes as compared to 460 tonnes; other magnesite 663 tonnes as compared to 2,279 tonnes; and magnesium oxide 1,293 tonnes as compared to 1,260 tonnes in the preceding year. The export data of magnesite is furnished in Tables-8 & 9.

Table – 8: Exports of Magnesite

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5263	239139	3842	240820
UK	72	24144	118	40508
South Africa	17	3128	156	29359
UAE	936	39773	526	24860
Bangladesh	747	27251	837	24361
Thailand	244	17140	224	16745
Cameroon	162	6613	270	11306
Singapore	140	7250	101	7748
Turkey	67	6852	39	7308
Iran	45	11903	25	6916
France	++	337	15	6303
Other countries	2833	94748	1531	65406

In tonnes

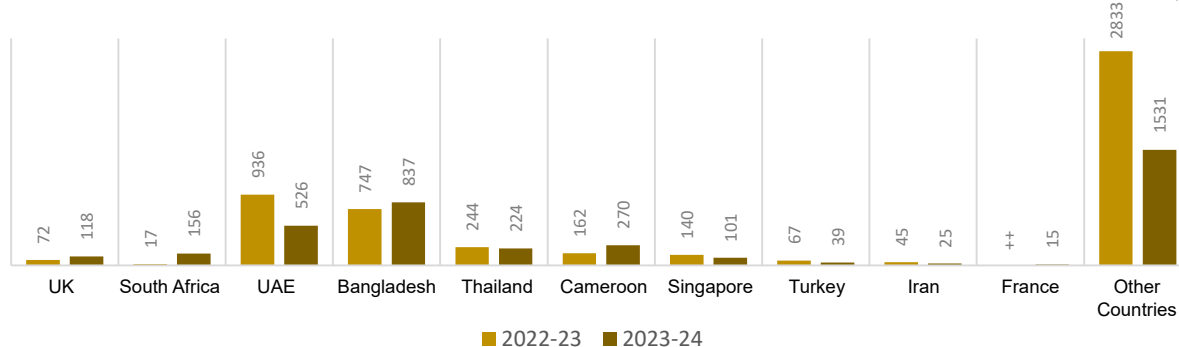


Fig. 2: Contrywise Exports of Magnesite

Table – 9: Exports of Magnesite

(Item Wise)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total	5263	239139	3842	240820
Magnesium Oxide	1260	139762	1293	164718
Magnesite (not Calcined)	460	10164	1096	22982
Magnesite:dead-burntMagnesia	862	37034	412	15717
Magnesite (calcined)	401	14462	375	14634
Magnesia (fused)	1	781	3	2102
Other	2279	36936	663	20667

Imports

The imports of magnesite decreased by 4.6% to 4,79,521 tonnes in 2023-24 from 5,02,837 tonnes in the year 2022-23. Imports were mainly from China (82%), UAE (4%) and Turkey & Australia (3% each). Out of the total imports in 2023-24, those of fused magnesia were at 67,441 tonnes as compared to 47,972 tonnes in the preceding year; magnesite (calcined) were at 76,606 tonnes as compared to

48,967 tonnes in the previous year; non-calcined magnesite were at 7,215 tonnes as compared to 27,279 tonnes in the previous year; other magnesite 56,980 tonnes as compared to 1,35,823 tonnes; magnesium oxide 67,751 tonnes as compared to 63,286 tonnes; and dead burnt magnesite were at 2,03,528 tonnes as compared to 1,79,510 tonnes in the year 2022-23. The import data of magnesite is furnished in Tables-10 & 11.

Table – 10: Imports of Magnesite

By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	502837	15431256	479521	14872734
China	298601	10055774	392091	10219717
Australia	26339	1524297	13777	1169294
Netherlands	2179	170450	9982	1068746
Turkey	28795	1047026	15407	591696
Saudi Arabia	12108	290079	10405	281883
Japan	3359	388153	2014	268592
UAE	34385	220036	18290	209559
Hong Kong	5827	138745	5066	138479
USA	2224	216903	1291	138109
Belgium	324	103439	348	126776
Other Countries	88696	1276354	10850	659883

In tonnes

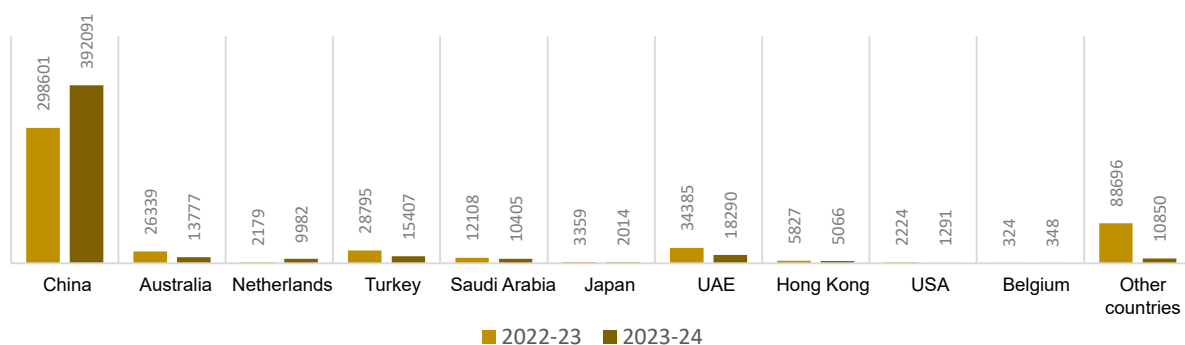


Fig. 3: Contrywise Imports of Magnesite

Table – 11: Imports of Magnesite

(Item Wise)

Item	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total	502837	15431256	479521	14872734
Magnesite:dead-burnt Magnesia	179510	6135407	203528	5522817
Magnesite (calcined)	48967	1109909	76606	2095096
Magnesium Oxide	63286	2443589	67751	2035870
Magnesia (fused)	47972	3095460	67441	3240102
Magnesite (not Calcined)	27279	140613	7215	73505
Other	135823	2506278	56980	1905344

FUTURE OUTLOOK

In recent years, the global magnesite market has been steadily growing due to rising demand from various industries. The market size was estimated at USD 8.86 billion in 2021 and is expected to grow to USD 13.98 billion in 2030, at a Compound Annual Growth Rate (CAGR) of 5.2% during the forecast period (2023–2030). Magnesite demand is expected to remain broadly positive over the coming decade, driven by its continued indispensability in high-temperature refractories, cement production and other industrial applications that underpin steelmaking and heavy industry. Global assessments indicate that magnesite and magnesia supply is geographically concentrated and regionally uneven, which makes supply security, diversification and sustainable resource management high on policy agendas. Frameworks on critical raw materials emphasise the importance of responsible resource development, recycling and adoption of lower-carbon

processing routes for industrial minerals. Regionally, sustained infrastructure and industrial investment in Asia, the Middle East and Africa is likely to underpin steady demand for refractories and construction-related uses of magnesite, although local growth rates and financing conditions may vary. In India, these trends are particularly significant— the steel and refractory industries are expanding in line with the country's infrastructure and development agenda, making magnesite a key raw material to support future industrial growth. Strengthening domestic exploration, improving beneficiation technologies, and aligning with international best practices for sustainable production will be crucial to meet rising demand and reduce dependence on external supply chains. Taken together, these factors point to a medium-term outlook of steady, structurally driven demand for magnesite-derived magnesia, with policy focused on supply-chain resilience, responsible production and decarbonisation of magnesia manufacturing to meet both industrial and climate goals.



47. Manganese Ore

3,381 thousand tonnes
production of manganese ore have
been reported in 2023-24.

271 tonnes
of manganese ore (total) were
exported in 2023-24.

5.59 million tonnes
of manganese ore (total) were
imported in 2023-24.

Manganese occurs as silvery grey in colour and is very hard and brittle in nature. It is always available in combination with iron, laterite and other minerals. Manganese in alloy form is an essential input in steel making and steel is one of the most important indicators of growth in the industrial economy of any country. In recent years, the trade volume of manganese ore has grown world over. Presently, India is one of the major importers of manganese ore in the world. Manganese ores of major commercial importance are: (i) Pyrolusite (MnO_2 , Mn about 63.2%); (ii) Psilomelane (manganese oxide, containing water and varying amounts of oxides of Ba, K and Na as impurities; Mn commonly 45–60%); (iii) Manganite ($\text{Mn}_2\text{O}_3 \cdot \text{H}_2\text{O}$, Mn about 62.4%); and (iv) Braunite ($3\text{Mn}_2\text{O}_3$, MnSiO_8 , Mn about 62% and SiO_2 about 10%).

Indian manganese ore deposits occur mainly as metamorphosed bedded sedimentary deposits associated with Gondite Series (Archaean) of Madhya Pradesh (Balaghat, Chhindwara & Jhabua districts), Maharashtra (Bhandara & Nagpur districts), Gujarat (Panchmahal district), Odisha (Sundargarh district) and with Kodurite Series (Archaean) of Odisha (Ganjam & Koraput districts) and Andhra Pradesh (Srikakulam & Visakhapatnam districts).

RESERVES/RESOURCES

The total reserves/resources of manganese ore in the country as on 1.04.2020 has been placed at 503.62 million tonnes as per NMI database, based on UNFC system. Out of these, 75.04 million tonnes are categorised as Reserves and the balance 428.58 million tonnes are in the Remaining Resources category. Grade-wise, Ferromanganese grade accounts for 8%, Medium grade 7%, BF grade 30% and the remaining 55% are of Mixed, Low, Others, Unclassified, and

Not-known grades including 0.03% of Battery/Chemical grade.

State-wise, Odisha tops the total reserves/resources with 34% share followed by Karnataka (25%), Madhya Pradesh & Maharashtra (12% each), Goa (7%), Andhra Pradesh (6%) and Jharkhand (3%). Rajasthan, Gujarat, Telangana and West Bengal together shared the remaining 2% resources (Table-1).

Table – 1: Reserves/Resources of Manganese Ore as on 01.04.2020

(By Grades/States)

(In '000 tonnes)

State/Grade	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	75041	428583	503624
By Grades			
Battery/Chemical	-	167	167
Ferromanganese	16816	25251	42067
Medium	3850	30581	34430
BF	9892	138545	148437
Mixed	1517	23848	25365
Medium & BF mixed	6207	44877	51084
Ferromanganese medium & BF mixed	24512	59696	84208
Ferromanganese & BF	2039	19197	21236
Low (-) 25% Mn	2271	28485	30756
Beneficiable	2506	26826	29332
Others	4167	14772	18939
Unclassified	1265	14160	15425
Not-known	-	2178	2178
By States			
Andhra Pradesh	8088	21756	29844
Goa	65	34436	34501
Gujarat	695	2180	2875
Jharkhand	1059	13691	14749
Karnataka	15464	108508	123972
Madhya Pradesh	19558	40499	60057
Maharashtra	17733	41303	59036
Odisha	11469	160058	171528
Rajasthan	568	1790	2359
Telangana	342	4162	4503
West Bengal	-	200	200

Figures rounded off

EXPLORATION & DEVELOPMENT

The exploration and development details, if any, are covered in the Review on "Exploration & Development" under "General Reviews".

MINING LEASES & PRODUCTION

As on 31.03.2024 (p), the total number of mining leases granted for mining of manganese ore is 215 covering a total mining lease area of 11,482.96 Ha.

The production of manganese ore at 3,381 thousand tonnes during 2023-24 increased by 20% as compared to 2,827 thousand tonnes in the previous year. There were 138 reporting mines during the year 2023-24 as against 132 in the previous year. Besides, manganese ore production was reported as associated mineral by 13 mines in 2023-24.

In 2023-24, thirty-three (33) Public Sector mines jointly accounted for around 52% of the total production. The contribution of captive mines was 12.46% of the total production.

As regards grade-wise composition of production in 2023-24, around 69% of the total production was of lower grade (Below 35% Mn), around 23% of medium grade (35% - below 46% Mn) and about 7% was of higher grade (Above 46% Mn). Production of manganese dioxide was 25,903 tonnes (0.77%) during the year.

Madhya Pradesh and Maharashtra were the leading producing States of manganese ore which accounted for a total production of 61.10% (30.55% each) in 2023-24, followed by Odisha (19.25%), Karnataka (10.50%) and Andhra Pradesh (8.81%) (Tables – 2 to 6).

The mine-head closing stock of manganese ore was 3,468 thousand tonnes for the year 2023-24 as against 3,102 thousand tonnes for the previous year [Tables-7(A) & 7 (B)]. The average daily employment of labour in manganese ore mines was 13,422 in 2023-24 as against 12,960 in the previous year.

Table – 2: Principal Producers of Manganese Ore, 2023-24

Name & Address of Producer	Location of mine	
	State	District
MOIL Ltd, MOIL Bhavan, 1A – Katol Road, Nagpur - 440 013, Maharashtra.	Madhya Pradesh	Balaghat
	Maharashtra	Bhandara Nagpur
Tata Steel Ltd, Bombay House, 24, Homi Mody Street, Fort, Mumbai – 400 001 Maharashtra.	Odisha	Kendujhar
The Sandur Manganese & Iron Ores Ltd, Satyalaya, Door No. 266 (Old no.80), Ward No. 1, Behind Taluk office, Sandur- 583 119, Karnataka.	Karnataka	Ballari
R.B.S.S. Durga Prasad & F.N.Das, Mor Bhavan, Ramdaspath, Nagpur-440010, Maharashtra.	Andhra Pradesh	Vizianagaram
Mr. Debabrata Behera, 1234/P Govind Prasad, Bomikhal, Rasulgarh, Bhubaneswar-751 010, Odisha.	Odisha	Kendujhar
Yazdani Steel & Power Ltd, 5th Floor, Fortune Tower, A- Wing, Bhubaneswar-751023, Odisha.	Odisha	Sundargarh
Patnaik Minerals Pvt. Ltd, Boneikalajoda, Barbil - 758 038 Distt. Kendujhar, Odisha.	Odisha	Sundargarh
S.K. Sarawagi & Co. Private Limited, 10/1/31, Signature Towers, Level-4, Waltair Uplands, Visakhapatnam - 530 003, Andhra Pradesh.	Andhra Pradesh	Vizianagaram
Krishnaping Alloys Ltd, Pinnacle Corporate Park, A-402, Next to Trade Centre, BKC, Bandra East, Mumbai – 400 051, Maharashtra.	Madhya Pradesh	Chhindwara
A.P. Trivedi Sons, Main Road, Balaghat – 481 001, Madhya Pradesh.	Madhya Pradesh	Balaghat

Table – 3: Principal Producers of Manganese Dioxide Ore, 2023-24

Name & Address of Producer	Location of mine	
	State	District
MOIL Ltd, MOIL Bhavan, 1 A – Katol Road, Nagpur - 440 013, Maharashtra.	Maharashtra	Bhandara

Table – 4: Production of Manganese Ore, 2021-22 to 2023-24

(By States)

(Quantity in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	2692408	22067826	2826964	22842633	3380889	24787707
Andhra Pradesh	204002	1115138	213790	1387367	297537	1748784
Karnataka	380004	3167390	344731	2592147	355057	2934535
Madhya Pradesh	845351	6842478	855874	7437348	1033424	7703313
Maharashtra	732018	8425784	751104	8288184	1032992	9117126
Odisha	512591	2429734	644218	3033883	650856	3222884
Rajasthan	8008	25626	6437	22528	4368	20661
Telangana	10434	61676	10810	81176	6055	40404

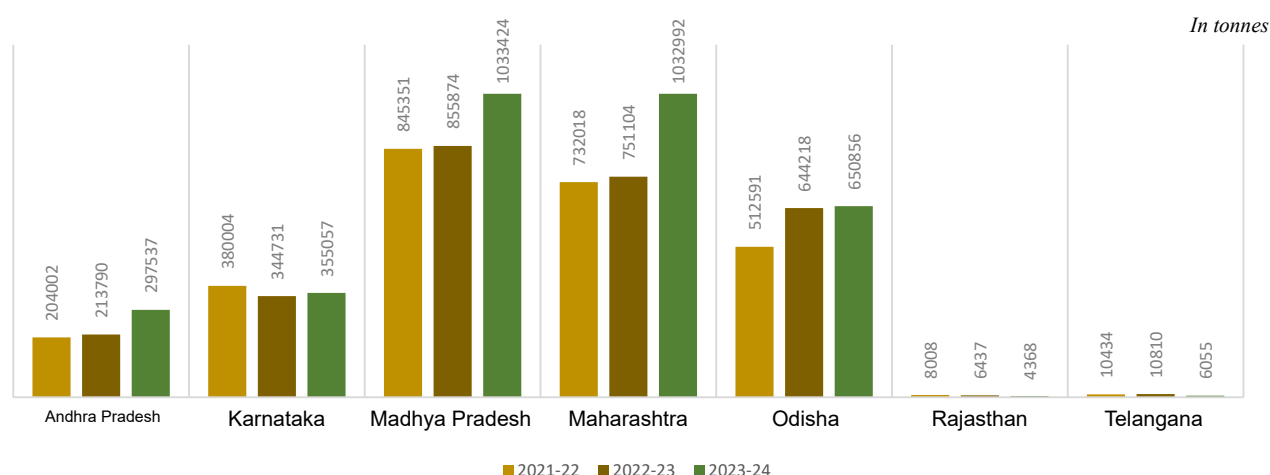


Fig. 1: Statewise Production of Manganese Ore

Table – 5 (A): Grade-wise Production of Manganese Ore, 2022-23

(By Sectors/States/Districts)

(Quantity in tonnes; Value in ₹'000)

State/District	No. of mines	Production by Grades: Mn Content					Total	
		MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	Quantity	Value
India	132 (11)	24706	221112	614180	1185768	781198	2826964	22842633
Public Sector	30	24581	145318	383652	645213	110521	1309285	13988821
Private Sector	102 (11)	125	75794	230528	540555	670677	1517679	8853812
Andhra Pradesh	18	-	-	46970	34182	132638	213790	1387367
Vizianagaram	18	-	-	46970	34182	132638	213790	1387367
Goa	2*	-	-	-	-	-	-	-
South Goa	2*	-	-	-	-	-	-	-

Table- 5A (Concl'd.)

(Quantity in tonnes; Value in ₹'000)

State/District	No. of mines	Production by Grades: Mn Content					Total	
		MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	Quantity	Value
Karnataka	10 (2)	125	105	55584	184668	104249	344731	2592147
Ballari	2 (2)	-	105	55584	182039	56344	294072	2483079
Chitradurga	4	125	-	-	874	2000	2999	13713
Davangere	2	-	-	-	1755	36005	37760	75149
Tumakuru	2	-	-	-	-	9900	9900	20206
Madhya Pradesh	50 (5)	-	118483	95437	410702	231252	855874	7437348
Alirajpur	1	-	-	-	-	766	766	3105
Balaghat	38	-	118085	91132	363296	132543	705056	6785674
Chhindwara	4	-	398	2870	17608	24840	45716	257043
Jabalpur	6 (5)	-	-	-	-	73103	73103	197872
Jhabua	1	-	-	1435	29798	-	31233	193654
Maharashtra	29	24581	27756	316948	342964	38855	751104	8288184
Bhandara	5	24581	2557	241955	179543	14185	462821	5250839
Nagpur	24	-	25199	74993	163421	24670	288283	3037345
Odisha	15 (4)	-	74768	99241	198128	272081	644218	3033883
Kendujhar	9 (3)	-	74768	98167	179366	182172	534473	2563278
Raygada	1	-	-	1	75	-	76	542
Sundargarh	5 (1)	-	-	1073	18687	89909	109669	470063
Rajasthan	1	-	-	-	6437	-	6437	22528
Banswara	1	-	-	-	6437	-	6437	22528
Telangana	7	-	-	-	8687	2123	10810	81176
Adilabad	7	-	-	-	8687	2123	10810	81176

Figures in parentheses indicate associated mines of iron ore, laterite, limestone.

* Only labour reported.

Table – 5 (B): Grade-wise Production of Manganese Ore, 2023-24 (P)

(By Sectors/States/Districts)

(Quantity in tonnes; Value in ₹'000)

State/District	No. of mines	Production by Grades: Mn Content					Total	
		MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	Quantity	Value
India	138 (13)	25903	245986	792169	1386168	930663	3380889	24787707
Public Sector	33	25903	163653	542288	765062	259313	1756219	14982641
Private Sector	105 (13)	-	82333	249881	621106	671350	1624670	9805066
Andhra Pradesh	21	-	-	80490	46725	170322	297537	1748784
Vizianagaram	21	-	-	80490	46725	170322	297537	1748784
Goa	1*	-	-	-	-	-	-	-
South Goa	1*	-	-	-	-	-	-	-
Karnataka	11 (2)	-	105	51401	191783	111768	355057	2934535
Ballari	2 (2)	-	105	51401	191253	58248	301007	2844539
Chitradurga	5	-	-	-	25	50	75	358
Davangere	2	-	-	-	505	45470	45975	73222

Table- 5B (Concld.)

(Quantity in tonnes; Value in ₹'000)

State/District	No. of mines	Production by Grades: Mn Content					Total	
		MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	Quantity	Value
Tumakuru	2	-	-	-	-	8000	8000	16416
Madhya Pradesh	54 (6)	-	141422	110136	482298	299568	1033424	7703313
Alirajpur	1	-	-	-	441	119	560	2637
Balaghat	41	-	137209	102896	427989	242209	910303	7044944
Chhindwara	5	-	4213	7240	20019	10233	41705	316521
Jabalpur	6 (5)	-	-	-	-	47005	47005	145202
Jhabua	1	-	-	-	33849	-	33849	194002
Katni	0 (1)	-	-	-	-	2	2	7
Maharashtra	31	25903	26649	463291	424532	92617	1032992	9117126
Bhandara	5	25903	1514	380077	274710	62798	745002	6539392
Nagpur	26	-	25135	83214	149822	29819	287990	2577734
Odisha	16 (5)	-	77810	86851	230116	256079	650856	3222884
Kendujhar	10 (4)	-	77810	86564	205928	194537	564839	2808316
Rayagada	1	-	-	-	122	-	122	938
Sundargarh	5 (1)	-	-	287	24066	61542	85895	413630
Rajasthan	1	-	-	-	4968	-	4968	20661
Banswara	1	-	-	-	4968	-	4968	20661
Telangana	3	-	-	-	5746	309	6055	40404
Adilabad	3	-	-	-	5746	309	6055	40404

Figures in parenthesis indicate production/labour reporting associated mines with Iron ore and Laterite.

* Only labour reported.

(p) Provisional.

Table – 6: Production of Manganese Ore, 2022-23 and 2023-24 (P)

(Quantity in tonnes)

Production Group	No. of Mines		Production		Total Production (%)		Cumulative percentage	
	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)	2022-23	2023-24 (P)
Total	132 (11)	138 (13)	2826964	3380889	100	100	-	-
Up to 1000	57 (1)	65 (7)	8666	12046	0.31	0.36	0.31	0.36
1001 – 5000	33 (4)	24 (3)	95537	67736	3.38	2	3.69	2.36
5001 – 10000	11 (1)	10	92583	63710	3.27	1.88	6.96	4.24
10001 – 20000	8 (1)	11	133941	153529	4.74	4.54	11.7	8.78
20001 – 30000	1 (1)	5	41524	131429	1.47	3.89	13.17	12.67
30001 – 40000	5 (1)	7 (1)	205535	278178	7.27	8.23	20.44	20.9
40001 – 50000	4	2	189918	91502	6.72	2.71	27.16	23.61
50001 and above	13(2)	14 (2)	2059260	2582759	72.84	76.39	100	100

Figures in parentheses indicate associated mines of iron ore, laterite, limestone and clay (others)

Table – 7 (A): Mine-head Closing Stocks of Manganese Ore, 2022-23

(By States/Grades)

(In tonnes)

State	Grades: Mn content					Total Quantity
	MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	
India	20511	24869	68043	338709	2649966	3102098
Andhra Pradesh	-	-	762	29096	73844	103702
Goa	-	-	-	-	-	-
Jharkhand	-	-	-	28	-	28
Karnataka	125	-	2112	35455	296158	333850

Table- 7A (Concld.)

(In tonnes)

State	Grades: Mn content					Total Quantity
	MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	
Madhya Pradesh	-	15653	21891	48573	929929	1016046
Maharashtra	20178	5737	14593	41855	50692	133055
Odisha	208	3479	28685	182022	1298522	1512916
Rajasthan	-	-	-	1078	-	1078
Telangana	-	-	-	602	821	1423

Table – 7 (B): Mine-head Closing Stocks of Manganese Ore, 2023-24 (P)

(By States/Grades)

(In tonnes)

State	Grades: Mn content					Total Quantity
	MnO ₂	46% and above	35% to below 46%	25% to below 35%	below 25%	
India	4465	15900	71743	425545	2950767	3468420
Andhra Pradesh	-	-	2208	45007	175119	222334
Goa	-	-	-	-	-	-
Jharkhand	-	-	-	28	-	28
Karnataka	125	-	12577	50907	297178	360787
Madhya Pradesh	-	6515	22619	94378	982754	1106266
Maharashtra	4132	4077	20705	44891	110826	184631
Odisha	208	5308	13634	183442	1383957	1586549
Rajasthan	-	-	-	5048	-	5048
Telangana	-	-	-	1844	933	2777

MINING, PROCESSING, MARKETING & TRANSPORT

Manganese ore mining in the country is carried out by opencast as well as by underground methods of the 138 reporting mines, 8 are underground (3 in Madhya Pradesh and 5 in Maharashtra). Seven underground mines were operated by Manganese Ore (India) Limited a Public Sector company, and one by M/s J. K. Minerals, Balaghat (Madhya Pradesh), a private company. All the underground mines are mechanised or semi-mechanised and adopt cut-and-fill method of stoping. In Kandri mine, hydraulic sand stowing has been introduced in place of manual filling system. This system is faster, cheaper and requires less manpower. Conventional timber supports are replaced by cable bolting pre-mining support to increase safety and productivity. In Balaghat underground mechanised mine, overhand flat back cut-and-fill method with rock bolting support and sand stowing is being practised to fill up the voids with a level interval of 30 m and size of stope block as 30 m x 30 m to 60 m x 30 m. Side Dump Loaders (SDL) of 0.66 cu. m bucket capacity were also deployed in underground levels for mechanised loading of run-of-mine (r.o.m.) in stopes. Tyre mounted Rocker shovel was also introduced in Balaghat mine for mechanised loading of ore from ore drive at stripping area. Deepening of high speed vertical shaft up to 750 m in Balaghat, up to 169 m in Chikla mine, up to 245 m in Kandri mine and up to 160 m in Mansar mine of MOIL is in progress. Sinking of high speed vertical shafts

up to 160 m, 330 m and 324 m is in progress at Mansar, Gumgaon and Ukwa mines, respectively. MOIL has also plans to increase its production from present level of 1.2 million tonnes to 3 million tonnes by 2030.

The open-pits are usually worked manually by benching method, using portable compressors, jackhammers and dumper trucks. Tirodi mine of MOIL is worked by opencast mechanised method. Height of the benches in overburden is kept at 7.5 m and that in the ore at 6 m. Drills of 100 mm diameter with 0.9 to 1.7 m³ capacity of shovels and 20–25 tonnes dumpers are used for production, loading and transport.

The workings vary from shallow depth in lateritoid-type deposits in Odisha, Karnataka and Goa to deep operations in deposits of a more regular nature found in Madhya Pradesh, Maharashtra and Andhra Pradesh. Bulldozers are used where the overburden is soft. In a few cases, tramways are laid up to the working face and loaded tubs are pushed manually to the dumping ground. In Odisha, Goa and Karnataka, ore is worked by loosening the ground either with crowbars or by blasting. After picking up manganese ore, the waste is removed to the dumping ground. Mining of bedded ore in Madhya Pradesh and Maharashtra is generally carried out by drilling and blasting.

Hand sorting and visual grading are adopted widely to upgrade the ore. Scrubber is also used for washing the ore at some mines. Manual as well as mechanised jigging is done in a few mines.

MOIL has set up an integrated manganese ore beneficiation plant at Dongri Buzurg Mine in Bhandara district, Maharashtra, with 4 lakh tonnes annual capacity to process r.o.m. The plant is equipped with handling, crushing, wet screening, drying and magnetic separation facilities in one complex. It has also installed a manganese ore beneficiation plant of 5,00,000 tonnes per annum capacity at Balaghat mine in order to conserve mineral and profitably utilise the low/medium-grade ore. The plant facilities include crushing, wet screening, classification and jigging operations.

The plant upgrades the low/medium-grade fines into high-grade and the value addition is around 3-4 times, in case of low-grade fines. The Company has plans to set up a sintering plant for agglomeration of these fines. After agglomeration, these fines will be utilised in ferroalloys production.

Most of the producers market manganese ore directly to the industrial units. In a few cases, especially in case of supplies of special type of ore or a semi-processed product, middlemen are found to be involved in marketing. Ore from mines is usually sold to the domestic consumers, either at the rail-head or ex-plant. In cases of Integrated Iron & Steel and Ferromanganese Industry, the units draw their supplies largely from captive mines. However, special ore types for specific purposes are obtained from other producers. In case of ore meant for exports, producers other than MOIL Ltd supply it to Metals and Minerals Trading Corporation of India Limited the canalising agency, either at rail-head or at the port.

Transport of manganese ore from mines to rail-head is generally done by trucks from where it is transported to ports by rail wagons. From the mine of MOIL in Balaghat district, Madhya Pradesh, the ore is transported by aerial ropeways to the loading bins at the rail-heads. Battery loco was introduced for underground transport of r.o.m. tub from ore pass chute to skip bunker. In Goa, ore in bulk is carried by road-cum-river routes up to Mormugao harbour and in a few cases by rail where the mines are close to the railways. The ore loading at river-head into barges is carried out mechanically.

USES & CONSUMPTION

Manganese ore is an important material in iron and steel metallurgy, where it is used both in the ore form and as ferromanganese. Manganese improves strength, toughness, hardness and workability of steel, acts as a deoxidiser and desulphuriser, and also helps in getting ingots free from blowholes. About 90 to 95% world production of manganese ore is used in metallurgy of iron and steel. High amount of phosphorous makes the manganese ore unsuitable for its metallurgical use, whereas, high phosphorous and high iron contents make it unsuitable for Battery Industry. Manganese has no satisfactory substitute in its major applications.

For manganese ore used in Ferromanganese Industry, besides manganese content, other important considerations are high manganese to iron ratio and a very low content of deleterious phosphorous. Manganese dioxide is used for manufacturing dry cell batteries in which it functions as a depolariser of hydrogen. Suitability of ore depends not only on manganese dioxide content but also on its crystallographic structure. Ore having predominant gamma structure is required. The ore must have high manganese dioxide and low iron content, a certain degree of porosity and moderate hardness. It should be free from metallic compounds, such as, copper, nickel, cobalt, arsenic, lead and antimony, which are electronegative to zinc (container).

Pyrolusite is used generally to impart glaze to the pottery and to make coloured bricks. It also finds use as driers for oils, varnishes and paints. Manganese sulphide is used in the manufacture of salts and in calico printing. Manganese chloride is used in cotton textile as a bronze dye. Manganese salts are used in photography and in leather and matchbox industries.

The apparent consumption of manganese ore was about 8.97 million tonnes in 2023-24 as against 7.46 million tonnes in 2022-23.

INDUSTRY

Manganese alloys are the largest produced ferroalloys in the world with a share of about 41% of the global production of ferroalloys. For production of one tonne of ferromanganese, about 2.6 tonnes of manganese ore, 0.5 tonne of reductant and 3 MWh of electricity inputs are required. As per Indian Ferro Alloys Producers' Association (IFAPA), the total installed capacity of manganese alloys including ferromanganese/silico-manganese in the country was estimated to be around 3.16 million tonnes per annum. During the year 2023-24, Balaghat Mine of MOIL has produced 10,163 tonnes of High Carbon Ferro Manganese (77%±1% Mn).

MOIL has a High Intensity Magnetic Separation Plant and 1,500 tpy Electrolytic Manganese Dioxide (EMD) Plant at Dongri Buzurg Mine, Bhandara, Maharashtra which has produced 1,413 tonnes of EMD during 2023-24 against 1,100 tonnes in 2022-23. Dry battery Industry also consumes EMD along with natural manganese dioxide ore.

SUBSTITUTES

Cost and technology militate against substitution in major applications. However, for economic reasons, there is only limited substitution in minor applications in Chemical and Battery industries. The Steel Industry has, however, made great strides in economising the use of manganese, largely through changes in steel-making techniques.

TECHNICAL POSSIBILITIES

The deep-sea nodules can be a potential resource of manganese in the next decades to come. There is a trend

towards using lower grades of ores in ferromanganese production. New steel-making practices and techniques are reducing the amount of manganese consumed in the process. However, counter balancing this to some extent is a trend towards higher manganese specifications for modern steels.

TRADE POLICY

Export Policy

As per the ITC (HS), 2022 Schedule 2 Export Policy, the present export policy scenario of Manganese ore is furnished below in brief.

HS Code	Item Description	Export Policy	Policy Condition
26020010	Manganese ore (46 percent or more)	Restricted	
26020020	Manganese ore (44 percent or more but below 46 percent)	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020030	Manganese ore (40 percent or more but below 44 percent)	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020040	Manganese ore (35 percent or more but below 40 percent)	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020050	Manganese ore (30 percent or more but below 35 percent)	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020060	Ferruginous (10 percent or more but below 30 percent)	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020070	Manganese ore sinters, agglomerated	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.
26020090	Other	STE (State Trading Enterprise)	Export is allowed through Manganese Ore India Limited (MOIL) only.

Import Policy

Imports of manganese ore and concentrates including ferruginous manganese ores and concentrates containing 20% or more manganese (calculated on dry weight basis), agglomerated manganese ore sinters, etc. are freely allowed.

HS Code	Item description	Import Policy
260200	Manganese ores and concentrates including ferruginous Manganese ores and concentrates with Manganese content of 20% or more calculated on the dry wet	Free

WORLD REVIEW

The total world reserves of manganese ore is approximately 1,900 million tonnes of metal content as per Mineral Commodity Summary, 2024 of USGS. Reserves are located in South Africa (32%), Australia (26%), China (15%), Brazil (14%) and Ukraine (7%). Only a small fraction of global manganese reserves is economical. This fact continues to prompt and support interests in deep-sea manganese nodules, which constitute an enormous untapped resource (Table-8).

World mineral production of manganese ore as per British Geological Survey has been reported to be around 53 million tonnes in 2023 as compared to 54.6 million tonnes in 2022. South Africa was the leading producer contributing about 36% followed by Gabon (19%), Australia (12%), China (8%) and India & Ghana (6% each) (Table-9). The production of manganese ore is linked with the production of steel. The Steel Industry consumes it in the form of ore and manganese alloys.

Table – 8: World Reserves of Manganese Ore
(By Principal Countries)

(In '000 tonnes of metal content)	
Country	Reserves
World: Total (rounded off)	1900000
South Africa	600000
Australia ^(a)	500000
China	280000
Brazil	270000
Ukraine (concentrate)	140000
Gabon	61000
India*	34000
Ghana	13000
Kazakhstan (concentrate)	5000
Mexico	5000
Other countries	small

Source: USGS, Mineral Commodity Summaries, 2024

(a) Joint Ore Reserve Committee compliant or equivalent reserves were 110 million tonnes

* India's total reserves/resources of manganese ore as per NMI database based on UNFC system have been estimated at 503.62 million tonnes as on 1.4.2020.

Table – 9: World Production of Manganese Ore

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
World: Total	5,71,00,000	5,46,00,000	5,30,00,000
South Africa	19156479	18966594	19076583
Gabon	9241000	10116000	10289000
Australia	6188587	6322670	6503469
China	8000000	6500000	4500000
Ghana	3336273	3171722	3001736
India	2695991 ^(a)	2798694	3293409
Brazil	1435561	1344155	1697881
Kazakhstan	1248700	349000	930700
Cote D' Ivoire	961267	929705	850000
Ukraine	1850000*	900000*	200000
Other countries	3014447	3087705	2634957

Source: BGS, World Mineral Production, 2019-23.

(a) Years ended 31st March following that stated.

* Estimated

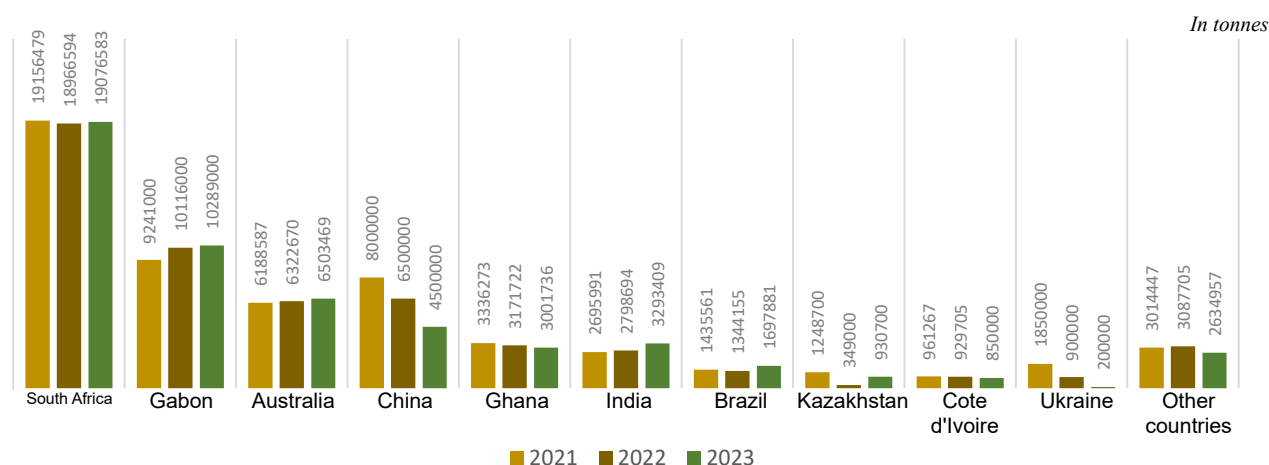


Fig. 2: Countrywise Production of Manganese Ore

FOREIGN TRADE

Exports

Exports of manganese ore (total) decreased by around 6 times from 1,557 tonnes in 2022-23 to 271 tonnes in 2023-24. Out of the total exports in 2023-24, 46 tonnes of

manganese ore having 40% or more but below 44% Mn was exported. Exports of manganese ore (others) were at 225 tonnes as compared to 190 tonnes in the preceding year. Exports of 271 tonnes of manganese ore (total) were to Nepal alone (Tables- 10 to 14).

Table – 10: Exports of Manganese Ore: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1557	21566	271	2204
UAE	1367	20371	--	--
Nepal	190	1164	271	2204
Tanzania	++	24	--	--
Spain	++	7	--	--

Figures rounded off

Table – 11: Exports of Manganese Ore (46% or more Mn)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	24	++	++
Tanzania	++	24	--	--

Figures rounded off

Table – 12: Exports of Manganese Ore (Others)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	190	1164	225	1808
Nepal	190	1164	225	1808
China			--	--
Indonesia			--	--
Germany			--	--
Tanzania			--	--

Figures rounded off

**Table – 13: Exports of Manganese Ore
(35% or More but Below 40% Mn)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1367	20378	++	++
UAE	1367	20371	--	--
Spain	++	7	--	--

Figures rounded off

**Table – 14: Exports of Manganese Ore
(40% or More But Below 44% Mn)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	++	46	396
Nepal	--	--	46	396

Figures rounded off

Imports

Imports of manganese ore (total) increased by 20% to 5.59 million tonnes in 2023-24 from 4.64 million tonnes in the previous year. South Africa (43%), Gabon (20%), UAE (12%), Singapore (9%) and France (5%) were the main suppliers of manganese ore in 2023-24. Out of the total

manganese ore imported, the contribution of manganese ore having +46% Mn was 27,418 tonnes (0.49%), manganese ore having 35-40% Mn was 17,41,076 tonnes (31%), manganese ore having 30 to 35% Mn was 7,30,981 tonnes (13%) and manganese ore (40% or more but below 44% Mn) was 11,03,971 tonnes (20%) (Tables-15 to 20).

Table – 15: Imports of Manganese Ore: Total

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	4641809	82621009	5592880	77503284
South Africa	1843078	30508646	2388153	31493083
Gabon	831403	15971671	1144635	17905632
Australia	443527	10544492	236685	3724316
Singapore	427995	8674202	475523	7603311
UAE	582442	8642386	663916	7487031
France	141396	3037668	300796	5009732
Hong Kong	169403	2167328	161009	1778956
Brazil	68911	1296699	54993	769306
Cote D' Ivoire	79232	863790	108751	932918
Namibia	--	--	28313	321668
Other Countries	54422	914127	30106	477331

Figures rounded off

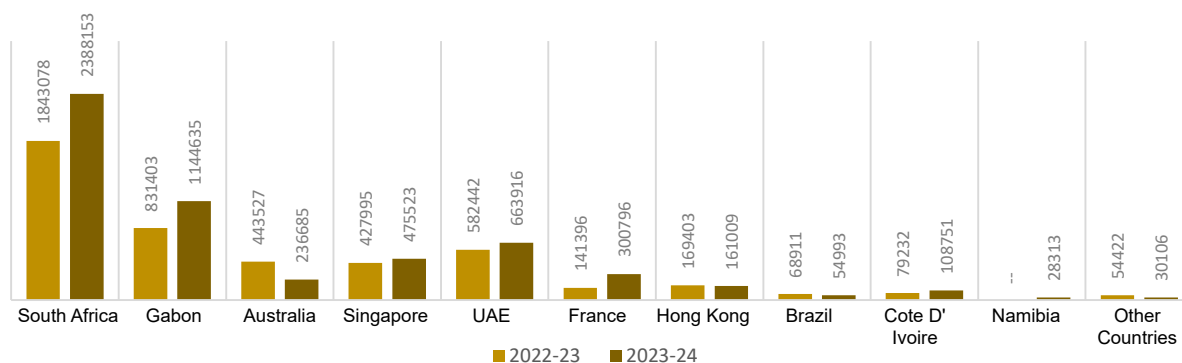


Fig. 3: Countrywise Imports of Manganese Ore

Table – 16: Imports of Manganese Ore (46% or more Mn)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	47028	1042963	27418	520328
Gabon	--	--	11944	201535
Singapore	--	--	10537	177227
Tanzania	7834	197825	2228	67351
Zambia	1421	38655	1088	26080
Turkey	--	--	528	19452
UAE	83	2249	437	11870
Brazil	8402	184115	472	6919
Morocco	50	1736	90	5492
Peru	322	13939	67	2887
China	162	10758	27	1515
Other Countries	28754	593686	++	++

Figures rounded off

Table – 17: Imports of Manganese Ore (35% or more but below 40% Mn)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1654178	23587098	1741076	20184175
South Africa	897065	12803158	1090870	12741479
UAE	317121	4458280	307810	3532381
Hong Kong	114281	1479337	120936	1374830
Gabon	155350	2253897	110327	1225226
Singapore	40544	623747	62941	715066
Namibia	--	--	28313	321668
UK	--	--	18109	249335
Brazil	46337	863048	1744	24025
Kenya	537	6824	26	165
Cote D' Ivoire	38503	464026	--	--
Other Countries	44440	634781	++	++

Figures rounded off

**Table – 18: Imports of Manganese Ore
(30% or more but below 35% Mn)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	189188	2136109	730981	7393098
Gabon	89854	1052936	115567	1240020
Hong Kong	38597	447670	40073	404126
Cote D' Ivoire	38500	366705	108751	932918
South Africa	20252	250675	181318	1824109
Kenya	1911	17380	--	--
Brazil	74	743	--	--
UAE	--	--	259392	2717636
Switzerland	--	--	4020	37828
Singapore	--	--	21860	236461

Figures rounded off

**Table – 19: Imports of Manganese Ore
(40% or more but below 44% Mn)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	796039	16758263	1103971	16255347
South Africa	268498	5582994	580530	8438262
Australia	254548	5792453	214092	3353387
Gabon	44694	821362	175120	2575063
Brazil	3403	57135	50815	707959
UAE	108042	2173168	49078	655660
Singapore	107291	2147569	21852	351668
France	--	--	10120	134664
Tanzania	1891	34662	2117	35316
Zambia	216	4465	247	3368
Hong Kong	5909	121308	--	--
Other Countries	1547	23147	++	++

Figures rounded off

**Table – 20: Imports of Manganese Ore
(Ferruginous, 10% or more but below 30% Mn)**

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	180387	1667126	66037	552002
UAE	110702	991446	21749	177919
South Africa	59448	569295	38695	321163
Seychelles	6957	71553	--	--
Hong Kong	3280	34832	--	--
Singapore	--	--	5443	52161
Oman	--	--	150	759

Figures rounded off

FUTURE OUTLOOK

As per World Steel Association, India's per capita steel use is about 102.6 kg as against the world's per capita consumption of 214.7 kg in 2024. This difference in the per capita consumption of steel in itself reflects opportunities that are bound to occur for Steel Industry which in turn would positively impact the demand for manganese ore. The apparent Steel use of India is 8.5% in comparison to 49.2% of China. Even though India stands in top of the world's population list, it is pertinent to mention that the steel usage is far lesser which reduces its carbon footprint compared to China.

Production of crude steel is the single most important

factor that influences the demand for manganese ore. India has set a production capacity target of 300 million tonnes of steel by 2030-31. The demand for manganese ore is expected to raise commensurately to about 10 million tonnes per year in the coming years. India's largest manganese ore producing company "MOIL Ltd." has targets to increase its production to 3 million tonnes by 2030, the gap in the demand will continue to be filled by imports in years to come.

India, through its recent announcements is moving towards Green steel and given taxonomy for Green Steel aiming to decarbonise the steel sector in alignment with net-zero emission intensity target by 2070.



48. Perlite

2.41 million tonnes
total resources of perlite have been
established as on 1st April, 2020.

1,067 tonnes
of perlite were exported in 2023-24.

48,240 tonnes
of perlite were imported in 2023-24.

Perlite is a hydrated, amorphous, siliceous volcanic glass of rhyolitic composition, exhibiting a perlitic texture with concentric fracturing and vitreous lustre. It forms in subaerial, high-silica volcanic environments through secondary hydration of obsidian or pitchstone, typically within felsic lava domes, pyroclastic flows, and volcanic domes associated with continental or island-arc volcanism. The hydration occurs at low temperatures over geologic time due to groundwater interaction. Crude perlite is generally grey to black, while expanded perlite, produced by heating at 850–1,150 °C, appears white to off-white and expands up to 20 times its original volume due to vaporisation of the 2–5% bound water. Bulk density is reduced from ~ 1,100 kg/m³ (crude) to 30–150 kg/m³ (expanded). Industrially, perlite is used in two primary forms: (i) expanded perlite, applied in insulation (thermal and cryogenic), lightweight concrete, horticultural substrates and filtration systems; and (ii) crude perlite, crushed and screened for further processing. Based on colour, lustre, and expansion ratio, it is classified as high-grade (glassy black, >15–20 times expansion), medium-grade (dull with rhyolitic admixture), and low-grade (grey, vitrified, poor expansion).

RESERVES & RESOURCES

As per the National Mineral Inventory (NMI) database, the total Remaining Resources of perlite in India, as on 01.04.2020, are estimated at 2.41 million tonnes. Of this, approximately 12% is classified as High-Grade, 12% as Medium-Grade, and 6% as Low-Grade, while the remaining 70% falls under the Unclassified category. It is important

to note that no reserves of perlite have been reported; the entire quantity is accounted for under the Remaining Resources category (Table-1). The only known deposit of perlite in the country is located at Village Patanvav in the Rajkot district of Gujarat, where it occurs in the form of a discontinuous sill on Osam Hill.

Table – 1: Reserves/Resources of Perlite in India as on 01.04.2020

(By Grades/State)

(In '000 tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	0	2406	2406
By Grades			
High	0	283	283
Medium	0	300	300
Low	0	152	152
Unclassified	0	1671	1671
By State			
Gujarat	0	2406	2406

Source: National Mineral Inventory, 2020

MINING LEASE & PRODUCTION

As on 31/03/2023, a single mining lease covering an area of 144.88 hectares was granted in the Rajkot district of Gujarat. No production of perlite has been reported in the current financial year and no stocks were reported during the year 2023-24.

USES AND CONSUMPTION

Perlite's versatility stems from its physical traits—lightweight, porous, inert, non-combustible and expansive structure. Its applications are traditionally segmented into three categories—Construction, Horticultural and Industrial.

Construction Applications

Expanded perlite is widely deployed in lightweight insulation systems, including loose-fill cavity-fill & masonry insulation, plaster aggregates, ceiling & roof insulation boards and fire-rated building components. Its thermal and acoustic insulating capabilities, along with fire resistance and pest deterrence, render it suitable for roof decks, portland cement based plasters, beams, chimney linings and partition walls.

Horticultural Applications

In horticulture and soilless cultivation, expanded perlite is used in potting media mixtures to enhance root aeration, drainage and moisture retention. It maintains neutral pH, is sterile and weed-free, and remains lightweight making it suitable for seed starting, hydroponics, container growth and green roofing systems. It also serves as a carrier for fertilizers pesticides, and pelletised seeds.

Industrial Applications

Perlite's industrial roles are extensive, including:

- Filter media for food, beverages, pharmaceuticals, chemicals, swimming pool and municipal water treatment systems
- Fillers in plastics, paints, sealants, adhesives & packaging materials for shock absorption, lightweight bulk and fire retardancy

- Abrasive media for soaps, polishes and cleaning products
- Thermal insulation in high-temperature and refractory systems (e.g. molten metal topping, refractories, pipe insulation)
- Cryogenic insulation for LNG oxygen and nitrogen storage
- Foundry aid, used in slag coagulation, molding sands, refractory bricks and mortar formulations
- Oil & gas well cement slurries, where it reduces density and improves insulation characteristics

Policy

As per the Import Policy under Schedule 1 of ITC (HS), 2022 and the Export Policy under Schedule 2 of ITC (HS), 2022, there are no restrictions on the export and import of perlite.

SUBSTITUTES

Several materials can serve as substitutes for perlite across its primary end-use sectors without significantly compromising performance. In construction applications, materials, such as, diatomite, expanded clay, expanded shale, pumice and blast furnace slag are commonly used alternatives. These materials offer comparable properties in terms of thermal insulation, lightweight aggregate function and fire resistance, making them suitable replacements in plasters, concrete and insulation boards. For horticultural use, perlite is often substituted with vermiculite, coco coir, wood pulp and pumice, which have similar properties, such as, enhancing aeration, water retention and root development in soilless media and as potting mixes. These alternatives are widely used either independently or in combination with perlite. In the Animal Feed sector, bentonite and zeolite are effective substitutes due to their excellent binding, adsorption and flow control characteristics, especially in mineral-based feed formulations. According to various mineral assessments, these substitute materials are readily available and, in many

cases, selected based on cost-effectiveness, local availability, or specific functional requirements.

WORLD REVIEW

Reliable global resource estimates for perlite are limited, however, according to the USGS Mineral Commodity Summaries 2024, reported reserves include approximately 180 million tonnes in Greece, 50 million tonnes in the USA,

32 million tonnes in China, 30 million tonnes in Slovakia and 15 million tonnes in Iran (Table-2). As per BGS World Mineral Production, 2019-2023, China led the global perlite production with 1,500 thousand tonnes, followed by Turkey at 1,174 thousand tonnes and Greece with 850 thousand tonnes. The USA produced 410 thousand tonnes, while smaller contributions came from other countries (Table-3).

Table – 2: World Reserves of Perlite
(By Principal Countries)

		(In '000 tonnes)
Country	Reserves	
World: Total	NA	
Greece ^(e)	180000	
USA	50000	
China ^(e)	32000	
Slovakia	30000	
Iran ^(e)	15000	
Other countries ^(e)	NA	

Source: USGS, Mineral Commodity Summaries, 2024.

e- Estimated

NA-Not available

Table – 3: World Production of Perlite
(By Principal Countries)

		(In tonnes of metal content)		
Country	2021	2022	2023	
China	1500	1500	1500	
Turkey	1429	1319	1 174	
Greece	975	850	850	
USA	491	442	410	
Hungary	84	86	78	
Georgia	31	52	64	
Italy	60	60	60	
Russia	45	45	45	
Slovakia	35	38	39	
Argentina	18	32	32	
others	115	66	76	

Source: BGS, World Mineral Production, 2019-2023.

FOREIGN TRADE

As per information available on DGFT website, under the HS Code 25301020, India's exports of perlite rose significantly from 627 tonnes in 2022-23 to 1,067 tonnes in 2023-24, marking a robust growth of 70%. In contrast, perlite imports declined from 55,692 tonnes in 2022-23 to 48,240 tonnes in 2023-24, reflecting a negative growth of 13%.

FUTURE OUTLOOK

As per international commodity assessments including those by agencies, such as, the USGS and BGS, the global perlite industry is poised for sustained growth in 2024.

Building upon an estimated market size of USD 1.83 billion in 2023, the sector is projected to reach USD 1.95 billion in 2024, advancing at a compound annual growth rate (CAGR) of 6.90%, with long-term growth projected to reach USD 3.34 billion by 2032.

The Asia-Pacific region, particularly driven by India and other emerging economies, is anticipated to be the fastest-growing market for expanded perlite. The region's rapid industrialisation, urbanisation, and focus on infrastructure development are key growth drivers. In India, government-led initiatives in housing, smart cities and infrastructure are expected to boost domestic demand for perlite-based construction materials.



49. Potash

23,091 million tonnes
total reserves/resources of potash
have been established as on 1st April,
2020.

29,114 tonnes
of potash fertilizer were exported in
2023-24.

3.43 million tonnes
of potash fertilizer were imported in
2023-24 .

The term 'Potash' originally referred to only potassium carbonate and potassium salts. However, the scope of the term has now been extended to cover a variety of potassium-bearing minerals, including its refined products. The potassium content of an ore/mineral is expressed in terms of $K_2O\%$, though it (K_2O) neither occurs in nature nor is manufactured, but it serves as a denominator for commercial composition of all potassium compounds.

It plays a crucial role as one of the three primary agricultural nutrients (nitrogen, phosphorus and potassium, or N-P-K), with over 90% of its usage as fertilizer. Commercial potash deposits primarily originate from evaporite formations, often located deep underground.

Commercial potash deposits of evaporite origin are not available while glauconitic sandstones (a potassium phyllosilicate mineral) are in plenty, therefore, could be of great economic significance as these could be used as an alternative, indigenous resource for potash. Glauconite is essentially a complex hydrous silicate of iron and potassium chiefly with ferric oxide and partly with ferrous oxide. It contains about 4–7% K_2O . Its direct use as potash fertilizer has been found to be eco-friendly, because the glauconitic sand mixes homogeneously with the soil and contributes towards its fertility by providing potash as a major plant nutrient together with many other micro-nutrients. Glauconitic sand also helps in improving the soil texture, porosity and permeability due to its more or less uniform grain size. The residual mineralogical constituents (quartz, feldspar, calcite and clay minerals) of the glauconitic sand after releasing the useful nutrients get homogeneously mixed with the soils of the agricultural fields.

RESERVES & RESOURCES

As per National Mineral Inventory (NMI) database, based on UNFC system, the total resources of potash as on 01.04.2020 have been estimated at 23,091 million tonnes,

all of which are placed under Remaining Resource category. Rajasthan alone contributes 89% to the total resources, followed by Madhya Pradesh (5%) and Uttar Pradesh (4%). The remaining states together contribute only 2% (Table-1).

Table – 1: Reserves/Resources of Potash as on 01.04.2020

(By Grades/State)

(In million tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	0	23091	23091
By Grades			
Glaucanite	0	3149	3149
Polyhalite	0	16164	16164
Sylvite	0	2572	2572
Unclassified	0	1206	1206
By States			
Bihar	0	230	230
Jharkhand	0	152	152
Madhya Pradesh	0	1244	1244
Rajasthan	0	20572	20572
Uttar Pradesh	0	893	893

Figures rounded off

OCCURRENCES

Potash is found naturally in evaporite (marine & non-marine), sea, lake and sub-surface Brine, from Insoluble Potash Minerals and in the form of by-product potash, from fly ash, from molasses residue, etc. The occurrence of marine potash-bearing evaporite sequence is recorded from the northern and northwestern parts of Rajasthan and the adjoining States of Haryana and Punjab. This potash-bearing evaporite sequence comprises halite, polyhalite, sylvinitic, sylvite, langbeinite, carnallite, etc.

In India, notable potash occurrences have been identified in several regions, including:

- Nagaur–Ganganagar Evaporite Basin, Rajasthan
- Tso–Kar lake Leh, Ladakh
- Kuda, Kharagoda – Jhinjvada, Malia, Santalpur and Adosar of Little Rann of Kachchh, Gujarat
- Rohtak & Sirsa districts, Haryana

Notable glauconite occurrences have been identified as follows; in Madhya Pradesh, glauconitic sandstones/shales occur in Sidhi & Satna districts; in Uttar Pradesh, glauconitic sandstones/shales occur in Banda & Mirzapur districts; in Rajasthan, glauconitic sandstones/shales occur in Chittorgarh, Kota, Karauli, Jaisalmer and Barmer districts; in Gujarat, glauconite is found in Ukra Formation at Guneri in Kachchh district. In Himachal Pradesh, glauconite of hydrothermal origin is found in Kumla–Kathwar area of Sirmaur district; in Karnataka, glauconite is found in Gulberga and Bijapur districts; and in Kerala, glauconite occurs in Quilon Limestone and seabed sediments of Thiruvananthapuram coast.

USES & CONSUMPTION

Potash refers to various water-soluble potassium compounds, historically derived from leaching wood ashes,

hence the name "pot-ash." It includes key fertilizers like potassium chloride (KCl), which is rich in potassium (60–62% K₂O equivalent) and essential for enhancing nutrient value and disease resistance in crops. Other notable potassium salts used in agriculture are potassium sulfate, potassium magnesium sulfate, and potassium nitrate.

Potassium chloride and potassium nitrate also find applications beyond agriculture, including in glass and ceramics manufacturing, soap production, dye, synthetic rubber, and explosives.

In terms of fertilizers, two main forms are common—Muriate of Potash (MOP) and Sulphate of Potash (SOP). SOP is a chloride-free option preferred for high-value crops like fruits and leafy vegetables, while MOP is typically used for carbohydrate-rich crops like wheat. Potash fertilizers enhance plant health and can significantly increase crop yields.

According to the Fertiliser Association of India (FAI), the all India consumption of Potassic fertilizer (in K₂O content) was at 1.88 million tonnes during 2023-24, whereas it was 1.71 million tonnes in the previous year.

WORLD REVIEW

The world reserves are estimated at approximately 3,600 million tonnes of K₂O content. Reserves are located mainly in Canada (31%), Belarus (21%), Russia (18%), USA (6%), China (5%), Germany (4%) and Chile (3%) (Table-2).

The world production of potash in 2023 was 44.4 million tonnes in terms of K₂O content as against 39.6 million tonnes in 2022. Canada is the leading producer of potash with 32% share in total production in 2023, followed by Russia (16%), China (14%), Belarus (10%), Laos (7%), Germany (6%), Israel (5%) and Jordan (4%)(Table-3).

Table – 2: World Reserves of Potash
(By Principal Countries)

(In '000 tonnes of K₂O equivalent)

Country	Reserves
World: total (rounded off)	3600000
Canada	1100000
Belarus	750000
Russia	650000
USA ¹	220000
China	180000
Germany	150000
Chile	100000
Laos	75000
Spain	68000
Brazil	2300
Other countries	300000

Figures rounded off

Source: USGS, Mineral Commodity Summaries, 2024

¹Data are rounded to not more than two significant digits to avoid disclosing company proprietary data

Table – 3: World Production of Potash
(By Principal Countries)

(In '000 tonnes of K₂O content)

Country	2021	2022	2023
World: total (rounded off)	47500	39600	44400
Canada (Chloride)	14276	14593	14043
Russia (Chloride)	9746	4828	7102
China	5200	5900	6000
Belarus	8000	3200	4500
Laos	819	1484	3308
Germany (Potassic salts)	2793	2709	2536
Israel (Chloride)	2406	2475	2356
Jordan	1563	1637	1696
UK (Polyhalite)	789	953	1009
Chile (Chloride)	879	672	742
USA (Potassic salts)	480	430	390
Brazil (Chloride)	213	300	363
Spain (Chloride)	376	430	360

Source: BGS, World Mineral Production, 2019-23.

FOREIGN TRADE

Exports

Exports of potash fertilizer increased exponentially to 29,114 tonnes in 2023-24, as compared to 5,347 tonnes during the previous year. Exports were mainly to Nepal (79%), Croatia (9%), UAE (5%), Mozambique (2%), South Africa and Serbia

(1% each). Exports of potassium nitrate increased by 75% to 1,639 tonnes in 2023-24 from 936 tonnes in the previous year (2022-23). Exports were mainly to Thailand (40%), Germany (18%), USA (11%), Slovakia (10%), UAE (5%), Bangladesh (4%), Indonesia (4%), South Africa (2%), Egypt (2%) and Republic of Korea (1%)(Tables- 4 & 5).

Table – 4: Exports of Potash Fertilizers

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	5347	237161	29114	1224475
Nepal	25	3029	22960	834960
Croatia	--	--	2574	106470
UAE	2060	81032	1417	64947
USA	2	1133	73	37686
South Africa	++	1	330	28157
Mozambique	69	4675	535	27374
Serbia	--	--	351	24919
Peru	8	2936	61	21932
Vietnam	90	7162	125	14404
Mexico	25	3122	100	9498
Other Countries	3068	134071	588	54128

Figures rounded off

(In tonnes)

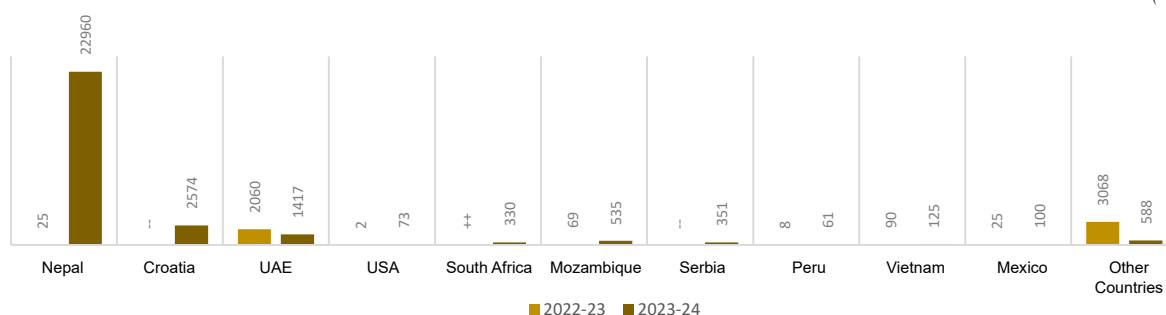


Fig .1: Countrywise Exports of Fertilizers

Table – 5: Exports of Potassium Nitrate

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	936	251715	1639	340277
Thailand	359	59902	657	93534
USA	312	132307	177	67375
Germany	++	4	295	67112
Slovakia	2	328	158	40846
Bangladesh	24	5003	60	14327
Indonesia	58	9931	65	12587
UAE	21	3826	85	10201
Egypt	52	13531	36	9555
South Africa	54	12017	28	7379
Korea, Rep. of	13	3540	22	5877
Other Countries	41	11326	56	11484

Figures rounded off

Imports

Imports of potash fertilizer increased by 34% to 3.43 million tonnes in 2023-24 as compared to 2.56 million tonnes during the previous year. Imports were mainly from Canada (34%), Russia (32%), Jordan (13%), Israel (8%), Turkmenistan (5%), Germany (2%), United Kingdom (2%)

and United States of America (1%). Imports of potassium nitrate increased to 1,473 tonnes in 2023-24 from 524 tonnes in the previous year. China (100%) was the main supplier of potassium nitrate in 2023-24 along with Italy that supplied minor quantities (Tables- 6 & 7).

Table – 6: Imports of Potash Fertilizers
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2563414	120342931	3430279	104684672
Canada	1142719	54381046	1172783	36298497
Russia	42900	1580967	1091731	31831993
Jordan	416847	19878688	444759	13321240
Israel	562389	28755426	280481	9082627
Turkmenistan	69726	2898231	180798	5641482
Germany	33035	2151056	82506	2957226
UK	76207	1681402	61501	1294367
USA	29420	1465082	33945	1236840
Taiwan	7768	614324	16225	791071
Hong Kong	--	--	15000	433929
Other Countries	182403	6936709	50550	1795400

Figures rounded off

(In tonnes)

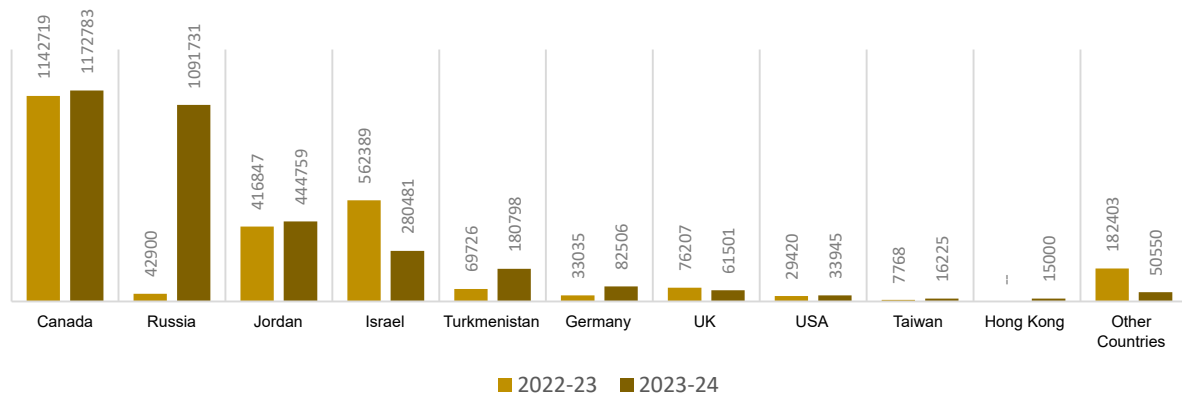


Fig. 2: Countrywise Imports of Fertilizers

Table – 7: Imports of Potassium Nitrate
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	524	59500	1473	139736
China	515	56228	1471	137818
Italy	1	452	2	1633
USA	++	13	++	170
France	++	39	++	94
Germany	++	246	++	17
UK	++	11	++	4
Netherlands	8	1773	--	--
Singapore	++	738	--	--

Figures rounded off

FUTURE OUTLOOK

Agriculture is the backbone of India's economy and declining soil fertility hugely impacts crop productivity. The appropriate application of fertilizer is a key factor in enhancing soil fertility and productivity for overcoming potassium depletion.

India is almost entirely dependent on imports for potash— meeting over 95% of its annual requirement of muriate of potash (MOP). This makes the country vulnerable to global price volatility and geopolitical disruptions, given that supply is concentrated in Canada, Russia and Belarus. Demand is expected to rise steadily in line with agricultural intensification, balanced fertilizer policies and higher cropping intensity.

A strategic shift began in recent years with the successful auction of potash blocks in Rajasthan's Nagaur – Ganganagar basin & glauconite blocks in Chhattisgarh, Uttar Pradesh and Bihar. These are India's first steps toward domestic potash production, leveraging evaporite deposits identified by the Geological Survey of India. Domestic production could reduce the fertilizer import bill, but it will require major investments in solution mining, brine processing and downstream fertilizer blending capacity. Policies, such as, accelerated auctions, investment incentives and technology partnerships will be critical to shorten lead times and attract Private and Public sector players.

Looking ahead, India's potash security will hinge on a dual strategy — scaling up Rajasthan's deposits while maintaining diversified imports and long-term contracts to hedge against global risks.



50. Rare Earths

12.73 million tonnes
Total resource deposits of monazite have been established as on March, 2021.

17.25 thousand tonnes
of ores & concentrates of Rare Earth metals were exported in 2023-24.

18.61 thousand tonnes
of Rare Earth other compounds oxides were imported in 2023-24.

INTRODUCTION

The term "rare earths" denote those metals which require elaborate processing of mineral compounds to obtain to a form of reasonable purity for use in different applications. Occurring mostly as common oxide-type minerals (earths), the rare-earth minerals are not found in concentrated forms but in highly dispersed state making their economic exploitation rather cumbersome and expensive. Gadolinite extracted from a mine in the village of Ytterby, Sweden is reportedly the first rare-earth mineral to be discovered.

The Rare-Earth Elements (REEs) are a group of 17 elements, namely, yttrium (Y), scandium (Sc) and lanthanides (15 elements in the periodic table with atomic numbers 57 to 71, namely, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb) and lutetium (Lu). They are divided into two groups: La to Eu as Light REE (LREE) and Gd to Lu as Heavy REE (HREE). Both Y and Sc are clubbed with HREE.

They are moderately abundant in earth's crust but not concentrated enough to make them economically exploitable. However, with the exception of the highly unstable promethium, rare-earth elements are found in relatively high concentrations in the earth's crust with cerium being 25th most abundant element in the earth's crust at 68 parts per million. REEs are characterised by high density, high melting point, high conductivity and high thermal conductance. A number of rare-earth minerals contain thorium and uranium in variable amounts, but they do not constitute essential components in the composition of the minerals.

In recent time there is a very fast growing demand for REE due to their applications in various advanced technological fields. Rare earths of Primary products are mainly used as raw materials for high-purity individual rare-earth chemicals, and in the making of petroleum and environment protection catalysts, misch metal and polishing powders. Specific fields of their applications include metallurgy, permanent magnets, terfenol, magnetic refrigeration, magneto-optic materials, ceramics, electronics, chemical, optical, phosphors, nuclear, hydrogen storage, superconductor, etc.

The main REE resources identified in India are contained in inland riverine placer deposits, beach placer deposits and carbonatite & alkaline rocks. The first one is mixed resource of both Y-HREE and LREE, whereas the last two form resources mainly for LREE. The principal sources of REEs are bastnaesite (a fluorocarbonate which occurs in carbonatites and related igneous rocks), xenotime (yttrium phosphate) commonly found in mineral sand deposits, loparite which occurs in alkaline igneous rocks and monazite (a phosphate). The rare earths occur in many other minerals and are recoverable as by-products from phosphate rock and from spent uranium leaching. In

India, monazite is the principal source of rare earths and thorium. Monazite is a complex phosphate of thorium and Rare-earth minerals [(Ce, La, Nd, Th, Y) PO₄] and this is radioactive in nature.

RESERVES/RESOURCES

As per NMI database, based on UNFC system, the total resources of REEs as on 01.04.2020 has been estimated at around 460 thousand tonnes, all of which are placed under Remaining Resource category. Gujarat alone contributes 92% to the total resources, followed by Odisha (6%) and remaining 2% resources are contributed by Karnataka, Maharashtra, Bihar, Uttar Pradesh & Jharkhand (Table- 1).

Table – 1: Reserves/resources of Rare Earth Elements as on 01.04.2020

(In tonnes)

State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	-	459727.49	459727.49
Bihar	-	1458.7	1458.7
Gujarat	-	424000	424000
Jharkhand	-	3.59	3.59
Karnataka	-	3734	3734
Maharashtra	-	2090.2	2090.2
Odisha	-	25493	25493
Uttar Pradesh	-	2948	2948

Figures rounded off

OCCURRENCES

Rare Earth ore deposits have been classified into nine categories in context with Indian scenario – these are hosted with Carbonatites, Agpaitic Nepheline Syenite & Peralkaline-Alkaline Felsic Rocks, Hydrothermal veins & Pegmatites, Quartz-Pebble Conglomerates (QPC), Stream Placers & Beach Sands, Residual/Supergene Weathering, Ion Adsorption Clays, Iron-Oxide Copper-Gold (IOCG) Type and Other Types.

Beach sand hosted deposits are observed in State of Odisha, Andhra Pradesh, Tamil Nadu, Maharashtra and Gujarat. Inland Stream placer deposits are located in Ranchi, Simdega districts of Jharkhand; Purulia district of West Bengal; Jashpur, Sarguja, Balrampur districts of Chhattisgarh; Sonbhadra district of Uttar Pradesh; Bastor Cratonic Terrain; Vadodara, Panchmahal district of Gujarat; southern States of Telangana, Karnataka & Tamil Nadu; East Kameng district of Arunachal Pradesh; and Garo Hills of Meghalaya.

Carbonatite hosted REE occurrences are found in Chhota Udaipur district of Gujarat; Barmer, Sirohi & Udaipur districts of Rajasthan; Karbi Anglong district of Assam; Jaintia Hills district of Meghalaya; Dharampuri, Salem, North Arcot districts of Tamil Nadu; Prakasham, Anantpur district of Andhra Pradesh and Purulia district of West Bengal.

Peralkaline-Alkaline Felsic Rocks hosted occurrences

are seen in Barmer district of Rajasthan; Jamnagar, Junagarh, Mehsana district of Gujarat; Nanded, Yeotmal district of Maharashtra; Prakasham district of Andhra Pradesh and in Dongargarh Granite, Chhattisgarh; Pala Lahar Granite, Kumarkunti–Jharnomal in Odisha; Kuilapal Granite in West Bengal; Nongpoh Granite in Meghalaya; Karbi Hills Granite in Assam and in Himalayan Granites. Hydrothermal Veins hosted occurrences are observed in Singhbhum Shear zone in Jharkhand; South Purulia Shear Zone in West Bengal; Kashipatnam, Vishakhapatnam districts of Andhra Pradesh; Sendra Complex in Rajasthan; Albitite Belts of Rajasthan & Haryana and Gogi, Bhima Basin in Karnataka.

MINING LEASES & PRODUCTION

Indian Rare Earth Limited (IREL), a Mini Ratna Company, is a Government of India Undertaking under the Department of Atomic Energy (DAE) and KMML, a Kerala State Government Undertaking, are actively engaged in mining and processing of beach sand minerals from placer deposits. IREL is the only entity processing monazite to produce Rare-earth (RE) compounds. Some REE which are available in India, such as, Lanthanum, Cerium, Neodymium, Praseodymium, Samarium, etc. are in supply surplus while Dysprosium, Terbium, Europium which are classified as HREE are having supply constraint. These HREEs are not available in Indian deposits in extractable quantity. Government is actively engaged in capacity building for consumption of LREE.

The annual installed mining, production and processing capacities are as under:

- (i) Mining: 10 million tonnes
- (ii) Processing capacity in terms of rare-earth concentrate: 11,200 tonnes
- (iii) Refining capacity in terms of Total Rare-earth Oxide (TREO): 5,000 tonnes
- (iv) Rare-earth Concentrate Production: 5,040 tonnes
- (v) Refining in terms of TREO: 2,000 tonnes (both Government and Private sector)

INDUSTRY

Several decades of exploration has proved substantial reserves of LREE contained in monazite and other associated minerals in the beach sands of Kerala, Tamil Nadu, Odisha and Andhra Pradesh. Some of these deposits, namely, Chavara in Kerala, Manavalakurichi in Tamil Nadu and Chhatrapur in Odisha, are being mined by IREL, a Public Sector undertaking of DAE, for REE since last few decades. About 12 million tonnes of monazite resource are estimated along the coasts of India.

IREL has a plant at Udyogamandal, Aluva, located in Ernakulam district, Kerala, wherein the monazite obtained from Manavalakurichi, is chemically treated to separate rare earths in its composite chloride form and thorium as hydroxide upgrade.

IREL has set up a Rare-Earth Extraction plant (REEP) producing mixed Rare-earth chloride (MRCL) Tri-sodium phosphate etc., at its unit in OSCOM Odisha. MRCL produced from the above plant is processed at IREL's plant at Rare Earth Division (RED) in Aluva, Kerala, for producing separated High Pure Rare Earth (HPRE) Oxides/Compounds.

IREL implemented flow sheet developed by BARC and has produced 50 kg Nuclear Grade Gadolinium Oxide (99.99%). The same can be converted into Gadolinium Nitrate which is used by NPCIL. The Company invited research projects pertaining to products in the value chain of ilmenite, zircon and rare-earth compounds for improvement in recovery energy efficiency, etc.

IREL is actively pursuing setting up of Greenfield operations in Kanyakumari district of Tamil Nadu and Bramhagiri district of Odisha. The proposal for harnessing the beach sand mineral deposits in Tamil Nadu by constituting a Joint Venture Company with the nominated State Government Agency, TAMIN, is under active consideration by the State Government.

Ambadungar RE Project has been initiated to harness the carbonatite deposit explored by AMD in the State of

Gujarat. Initially, about 1.55 Ha of the deposit is proposed to be harnessed, which will be further extended based on the exploratory results of AMD. Action has been initiated to establish the technical feasibility and financial viability of the project.

POLICY

In order to safeguard the strategic interest of the nation it is expedient in larger national interest to prohibit the grant of operating rights in terms of any reconnaissance permit exploration license or production lease of atomic minerals as defined in Part-B of the first schedule of the MM(DR) Act, 1957.

As per MMDR Act, 1957, Minerals of the "rare earths" group not containing uranium and thorium are mentioned in Part D of First Schedule to the Act for Critical & Strategic minerals. As per PIB Press Release dated 28.11.2023 by Ministry of Mines, Critical minerals are essential for our country's economic development and national security. The lack of availability of these minerals or concentration of their extraction or processing in a few countries may lead to supply chain vulnerabilities.

Other minerals under Code 2617 are freely exportable, except those which have been notified as prescribed substances and controlled under Atomic Energy Act 1962.

As per the Foreign Trade Policy, import policy under ITC(HS), 2022 Schedule 1, the import policy on the imports of ores & concentrates of rare-earth metals (under HS Code 25309040) are permitted 'freely' whereas export policy under ITC(HS) 2018 Schedule 2, the export policy on the exports of ores and concentrates of rare-earth metals (under HS Code 25309040) are permitted 'freely'.

Export of Beach Sand Minerals have been brought under STE and shall be canalised through Indian Rare Earths Limited (IREL). Beach sand minerals, permitted anywhere in the export policy, will now be regulated in terms of policy stated under at Sl. No. 98A of Chapter 26 of Schedule 2 of the Export Policy.

As per Gazette Notification No : GSR.134 (E) dated 20.2.2019, the particulars of threshold values for atomic minerals in respect of Beach Sand Minerals (BSM) shall be regulated as Schedule A [Rule 2(1)(m) and Rule 36].

Many projects of IREL have been initiated, such as, capacity expansion of Mineral Separation Plant of OSCOM, Rare Earth Permanent Magnet Plant and Rare Earth Theme Park. Agencies to implement these projects are in place and the projects will be commissioned within the next 2-3 years. New areas of operations, such as, Rare Earths in Ambadungar, Gujarat or Atomic Minerals in Odisha and Tamil Nadu are also in advanced stage of development.

During the year 2020-21, Licence to Operate (LTO) office has been established for centralised monitoring & managing the regulatory compliances, required to achieve and sustain the targeted production plans. Centralised monitoring and control of statutory compliances is enabled through database dashboards. Units are sensitised for complying with the statutory requirements at regular intervals.

Manavalakurichi and OSCOM units are continuing their operations in a sustainable manner. OSCOM has deposited the statutory fees as desired under Stage-I Forest Clearance towards obtaining the Stage-II Forest Clearance. Chavara Unit is in the final stage of receiving the environment clearances for its mining lease areas. LTO office establishes the processes & tools for proactive information gathering and structuring of regulatory requirements.

Rare Earth Permanent Magnet plant will be set up in BARC campus, Achitapuram, Vizag, for production of Samarium–Cobalt permanent magnet for use in Atomic Energy, Defence and Space sectors.

Environment clearance and clearance from Design Safety Review Committee and Safety Committee, BARC for carrying out site activities have been received. Detailed engineering has been completed and appointment of contractor to commence site activities on Engineering Procurement Construction (EPC) model too has been done.

Subsidiary IREL–IDCOL Limited (IIL), the Joint Venture company has been established to harness the beach sand mineral deposit in the State which in turn will widen the footprint of the Company. Ground work towards declaration of the precise area is in the final stages at the State after completion of DGPS survey and preparation of cadastral map. DPR for the project has been prepared.

USES & CONSUMPTION

The Rare Earth Permanent Magnet (REPM) in Vizag and Rare Earth and Titanium Theme Park (RETTP) in Bhopal have kick started with the funding assistance of Government of India, which will enhance the visibility of IREL in the strategic and niche sector.

Environmental clearance for REPM project has been received from MoEF&CC and M/s MECON Limited, Bengaluru, has been appointed as consulting firm for detailed engineering. As regards RETTP project, lease deed execution towards land has been completed. A Letter of Understanding has been inked with BARC towards developing and transferring laboratory-scale technologies in the value chain of Rare Earths which will be suitably upscaled by IREL to pilot-scale and installed in the theme park.

Execution of Supplementary Mining Lease deed for OSCOM Mines till a period up to the year 2047 has been completed under the provisions of AMCR 2016. Communication on precise area of the Bramhagiri Mineral Sands Deposit in Puri district under AMCR, 2016 is in the final stages of issuance by the Government of Odisha.

Rare-earth materials are utilised in a wide range of critical products enabling many emerging green energy technologies, high-tech applications and defence systems, such as, hybrid cars, plug-in hybrid electric vehicles (PHEVs), the latest generation of efficient wind power turbines, computer disc drives, missile guidance systems, etc.

The lanthanide elements as a group have magnetic, chemical and spectroscopic properties that have led to their application in a wide range of end-uses. Cerium finds application in polishing of glass items like lenses & display screens of cathode-ray tubes, liquid-crystal displays & plasma-display panels, in petrol & diesel fuels as fuel additive and along with lanthanum for replacement of cadmium in red pigments. Mixed salts of the cerium group of elements, other than fluorides are used in medicine, non-irritating antiseptic dressings, waterproofing agents and fungicides in textile manufacture. The principal uses of commercially pure cerium compounds that are in the form of nitrate are in the manufacture of incandescent gas mantles and cerium compounds as oxide. It also finds usage as a polishing agent of glass. Cerium compounds are also used in ceramic and glass as colouring pigments and also as catalysts in Chemical Industry.

To produce samarium–cobalt (Sm–Co) magnet for meeting national objectives, a Special Purpose Vehicle (SPV) has been formed. Production of Sm–Co metal and magnet is based on technologies developed by BARC, Mumbai & DMRL, Hyderabad. Activities for firming up the investment, plant location etc., are under progress. Supply of Nuclear Grade Ammonium diuranate (NGADU) from new source, i.e, the newly commissioned monazite processing plant at OSCOM, Odisha, has commenced. Subsequent to identification and development of conditions for dissolution of Rare Earths from fly ash generated at lignite coal fired thermal power plant at Neyveli, Tamil Nadu, studies were taken up to understand the overall process efficiency that would enable precipitate dissolved rare earths in purified form. Cerium, lanthanum and neodymium are used as glass additives in optical lenses and display screens, as catalysts in automobiles to reduce sulphur dioxide emission, in multilayer capacitors and along with yttrium in magnesium, aluminium and hydrogen storage alloys. Mischmetal which is an alloy of cerium with small amounts of other rare-earth metals is used in lighter flints, for desulphurisation in steel

and foundry, and with lanthanum alloys, in batteries and hydrogen storage systems meant for electronics and hybrid cars. Cerium oxide is used in glass polishing industries.

Lanthanum oxide and neodymium compounds are used in special glass manufacture. Lanthanum finds application in X-ray films as phosphors; yttrium in advanced ceramics like nitrides, Y-stabilised ceramics, etc., and gadolinium in magnet alloys.

Yttrium, europium and terbium are used as phosphors in displays of computers, TV, etc. and with lanthanum, cerium & gadolinium as phosphors in fluorescent and halogen lamps. Neodymium, samarium, dysprosium, praseodymium and terbium have application as high intensity magnets in electronics, electric motors and audio equipment. Lanthanum, erbium and ytterbium have application in fibre optics and lasers.

Lanthanum and yttrium find application in solid oxide fuel cells. Scandium is used mainly in aluminium alloys for sporting goods. Scandium in minor amounts is used in semiconductors and special lighting, including halogen bulbs. Mixed rare-earth products are used as catalysts in petroleum refining and fluid cracking. Neodymium is used in welding in heavy industries and also in MRI scanners. Praseodymium is not a primary element for any specific use, but finds use as a substitute for neodymium in magnets.

Samarium is used essentially for the Sm-Co magnets. Europium is a primary component of phosphorus and is responsible for white light in compact fluorescent lamps when used with terbium compounds.

Erbium used as fibre optic has emerged in the nineties as a remarkable tool for communication technology through which high quality rapid data in tight pulses can be transferred in speed unthinkable in the past. The main application for neodymium-iron-boron (Nd-Fe-B) magnets are in automobiles for anti-lock brakes, and in computer hard disk drives, videos, CD-ROMs used in many small-size electronic consumer products, such as, digital cameras, where major advantage is their small sizes. Nickel metal hydride (Ni-MH) batteries, containing mischmetal, a mixture of rare-earth compounds, are used mainly in portable electronic equipment, such as, laptops, camcorders and mobile phones. Though, the market for batteries for portable electronic equipment is growing strongly, the Ni-MH batteries are increasingly replaced by lithium-ion batteries. Ground monazite is digested with caustic soda lye to produce trisodium phosphate (TSP) and mixed hydroxide slurry. This slurry is used for production of diverse rare-earth compounds. Elaborate solvent extraction

and ion exchange facilities were built to produce individual RE oxides, like oxides of Y, Ce, Nd, Pr and La of specific purities. India is the second largest supplier of yttrium in the world and the maximum production is reported from the plant in Kerala. Uranium values presenting monazite which are recovered in the form of nuclear grade ammonium diuranate (ADU) are vital supplement to the indigenous supply of uranium. Thorium is separated in its pure oxalate form. A part of it is taken to OSCOM for further processing by solvent extraction to produce thorium nitrate. A small part of the purified thorium nitrate is converted to nuclear grade thorium oxide powder for supply to Bhabha Atomic Research Centre (BARC) and Nuclear Fuel Complex (NFC) for developing thorium-based fuel for nuclear reactors. IREL has built a large stockpile of impure thorium hydroxide upgrade associated with rare earths and unreacted materials.

Monazite contains about 25.28% P_2O_5 which can be recovered as a by-product for manufacture of fertilizers and production of elemental phosphorus or its salts. Besides, rare earths, thorium is also recovered from monazite. It is a source of atomic energy. An important use of thorium is for addition to tungsten in minute quantity (about 0.75%) to increase the ductility of tungsten wire and thus to facilitate its drawing into filaments used in electric lamps. Metallic thorium is also used in photoelectric cells and X-ray tubes and in certain alloys. Thorium is used as catalytic agent for various processes. Amongst thorium salts, thorium nitrate is used largely in the manufacture of incandescent gas mantles. Mesothorium, the chief radioactive element recovered as a by-product in the chemical treatment of monazite, is marketed usually in the form of its bromide and used in self-luminous paints or enamels. Mesothorium is also used in the treatment of certain types of cancer and skin diseases.

WORLD REVIEW

The total world reserves of rare earths are estimated at 110 million tonnes of rare-earth oxide (REO) equivalent content, of which China alone accounts for 44 million tonnes (40%) followed by Vietnam (20%), Brazil (19%), Russia (9%), India (6%), Australia (5%) and USA (2%) (Table- 2).

China holds the leading position among producers of rare-earth oxides with 240 thousand tonnes. The other major producers are Myanmar, USA, Australia, India, Madagascar, Russia, Vietnam and Malaysia (Table-3).

Table – 2: World Reserves of Rare Earths

(By Principal Countries)

(In tonnes of REO equivalent content)

Country	Reserves
World: Total (rounded off)	110000000
China	44000000
Vietnam	22000000
Brazil	21000000
Russia	10000000
India	6900000
Australia*	5700000
USA	1800000
Greenland	1500000
Tanzania	890000
Canada	830000
South Africa	790000
Thailand	4500

Source: USGS, Mineral Commodity Summaries, 2024

* For Australia, Joint Ore Reserves Committee-complaint or equivalent reserves were 3.3 million tonnes.

Table – 3: World Production of Rare-Earths Oxides

(By Principal Countries)

(In tonnes)

Country	2021	2022	2023
China ^{(a)*}	180000	210000	240000
Myanmar*	35000	24000	72000
USA*	25200	25200	25800
Australia ^(c)	16000	16000	17000
India ^(b)	4200	4200	4200
Russia	2276	2038	1964
Madagascar*	2000	960	960
Vietnam*	440	1200	600
Malaysia	29	52	134

Source - BGS, World Mineral Production, 2019-23

*Estimated

a :Includes production from iron ore extraction, bastnaesite concentrates and ion absorption clays.

b : Year ending 31st March following that stated.

c : Year ending 30th June following that stated.

In tonnes

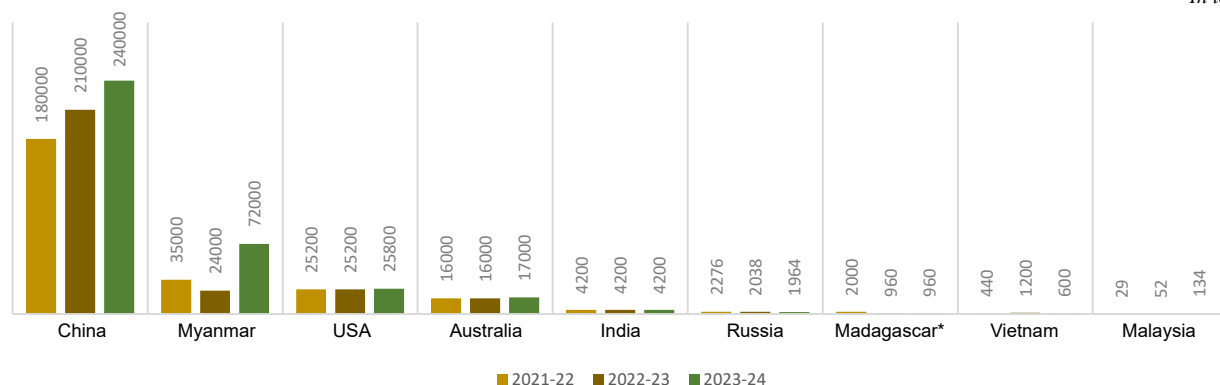


Fig. 1: Countrywise Production of Rare-Earths Oxides

FOREIGN TRADE

Exports

The Export of Other Compounds Inorganic/Organic of Rare Earth Materials increased by 13% to 1,397.06 tonnes in 2023-24 as compared to 1,233.384 tonnes in 2022-23. Japan (44%), Vietnam (27%), China (22%), Italy (5%) and France (1%) were the main buyers from India. Export of Rare Earth Chlorides NES increased to 18 tonnes in 2023-24 from 3 kg in the previous year. UAE (100%) was the main buyer from India. The export details of rare earth fluorides are tabulated in Table-6. Export of Rare Earth Oxides

decreased by 51% to 19 kg in 2023-24 as compared to 39 kg in 2022-23. Tunisia (53%), South Africa (21%), Zimbabwe (16%) and Bangladesh & Turkey (5% each) were the main buyers from India. Export of Rare Earth Metals Scandium & Yttrium W/N Intermixed/Interallloyed was substantially increased by 119% to 11 tonnes in 2023-24 as compared to 5.02 tonnes in 2022-23. UAE (95%) and Bhutan (5%) were the main buyers from India. Export of Ores & Concentrates of Rare Earth Metals was 17.25 tonnes in 2023-24. Qatar (75%) and Sri Lanka (25%) were the main buyers from India (Tables- 4 to 9).

Table – 4: Export of Other Compounds Inorganic/Organic of Rare Earth Materials
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	1233384	5426942	1397062	3840855
Vietnam	402000	3507417	374501	1895267
Japan	431982	1064846	612946	1396971
China	229401	565441	301402	313145
Italy	50541	84837	76013	83349
UAE	18750	2994	6627	80024
France	1001	8392	7999	38910
Turkey	2443	20200	1194	9988
Spain	14011	114041	1000	5779
Slovenia	--	--	4000	5065
Taiwan	++	199	170	3011
Other countries	83255	58575	11210	9346

In kg

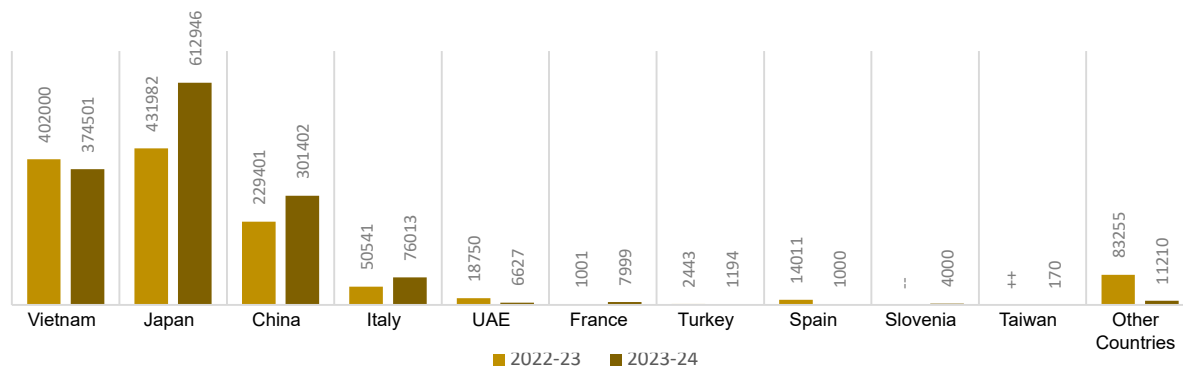


Fig. 2: Countrywise Export of Other Compounds Inorganic/Organic of Rare Earth Materials

Table – 5: Export of Rare Earth Chlorides NES
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	3	36	18002	121789
UAE	--	--	18000	121725
Belgium	--	--	2	47
Korea, Rep. of	1	7	++	16
Uganda	--	--	++	1
Tanzania	2	19	--	--
Ethiopia	++	6	--	--
Zambia	++	4	--	--

Table – 6: Export of Rare Earth Fluorides NES

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	++	31	++	++
Germany	++	31	--	--

Table – 7: Export of Rare Earth Oxides NES

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	39	130	19	83
Russia	15	42	++	31
Tunisia	--	--	10	28
Zimbabwe	--	--	3	9
South Africa	++	4	4	5
Turkey	4	12	1	3
Bangladesh	--	--	1	3
Philippines	--	--	++	3
Uganda	--	--	++	1
Saudi Arabia	10	21	--	--
Ecuador	1	19	--	--
Other Countries	9	32	++	++

Table – 8: Export of Rare Earth Metals Scandium & Yttrium W/N Intermixed/Interallloyed

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	5020	1827	11001	3662
UAE	--	--	10500	3126
Bhutan	5000	1700	500	504
USA	--	--	1	31
Peru	--	--	++	1
Qatar	++	81	--	--
Korea, Rep. of	++	25	--	--
Denmark	20	21	--	--

Table – 9: Export of Ores & Concentrates of Rare Earth Metals

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	--	--	17250	300
Qatar	--	--	13000	100
Sri Lanka	--	--	4250	200

Imports

Import of Other Compounds Inorganic/Organic of Rare Earth Materials decreased by 10% to 354.85 tonnes in 2023-24 as compared to 392.250 tonnes in 2022-23. China (74%), Japan (13%), USA (7%), Austria (3%) and France, Slovenia & Switzerland (1% each) were the main suppliers to India. Import of Rare Earth Chlorides NES at 20 tonnes in 2023-24 remained almost the same as that of 2022-23. China (100%) was the main supplier to India. Import of Rare Earth Fluorides NES substantially fell by 98% to 5 kg in 2023-

24 from 200 kg in 2022-23. China (100%) was the main supplier to India. Import of Rare Earth Oxides increased by 33% to 18.612 tonnes in 2023-24 as compared to 14.021 tonnes in 2022-23. China (100%) was the main suppliers to India. Import of Rare Earth Metals Scandium & Yttrium W/N Intermixed/Interalloyed drastically increased by 50% to 1,184.781 tonnes in 2023-24 as compared to 792.082 tonnes in 2022-23. China (59%), Hong Kong (20%), Japan (16%) and Mongolia (5%) were the main suppliers to India (Tables-10 to 14).

Table – 10: Import of Other Compounds Inorganic/Organic of Rare Earth Materials

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	392250	436273	354846	416735
China	279431	107168	261445	210013
Japan	61034	153299	45493	101220
USA	17815	43204	24052	40271
Austria	10029	39898	11469	32687
France	12305	39327	4237	9982
Slovenia	4000	6943	4000	7581
Korea, Rep. of	750	2774	60	6847
Hong Kong	193	1752	1073	3022
Singapore	410	1979	543	2224
Switzerland	256	2269	2220	1242
Other Countries	6027	37660	254	1646

In kg

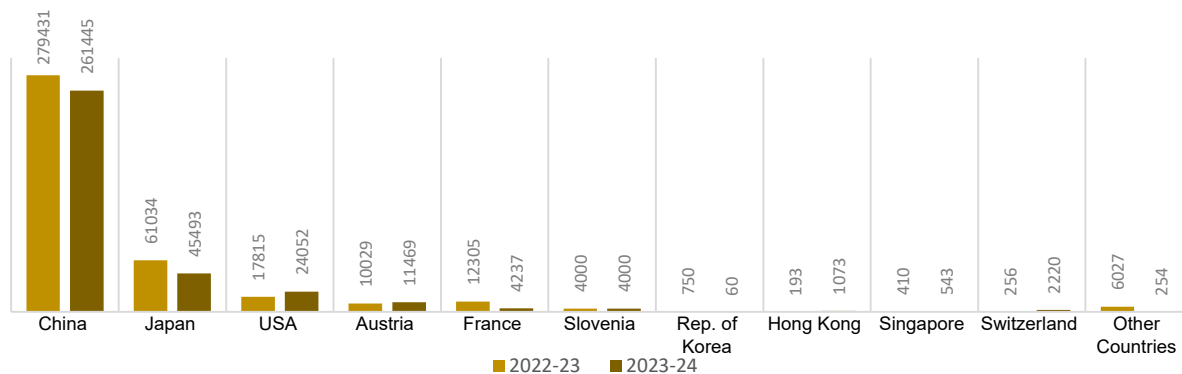


Fig. 3 : Countrywise Import of Other Compounds Inorganic/Organic of Rare Earth Materials

Table – 11: Import of Rare Earth Chlorides NES

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	20001	3140	20002	2473
China	20000	2770	20000	2132
USA	1	370	2	341

Table – 12: Import of Rare Earth Fluorides NES

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	200	1671	5	167
China	200	1657	5	167
Germany	++	14	--	--

Table – 13: Import of Rare Earth Oxides NES

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	14021	24019	18612	46599
China	11800	18549	18600	46074
UK	--	--	++	276
USA	++	6	12	249
Japan	2175	4719	--	--
Germany	11	677	--	--
Spain	35	68	--	--

Table – 14: Import of Rare Earth Metals Scandium & Yttrium W/N Intermixed/Interalloyed

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (kg)	Value (₹ '000)	Qty (kg)	Value (₹ '000)
All Countries	792082	288800	1184781	436836
China	708809	236090	698545	203656
Japan	42000	33935	192100	159845
Hong Kong	20000	5715	234000	55663
Mongolia	--	--	60000	13740
USA	1089	3726	26	2811
UK	184	3225	108	877
Canada	--	--	2	244
Singapore	20000	5953	--	--
Germany	++	156	--	--

FUTURE OUTLOOK

Worldwide explorations for economic deposits of REEs and efforts to bring them into production have increased substantially since the year 2000. Recovery of REEs can be complex because they occur in minerals as a group of similar elements, and at many deposits the REEs are hosted within more than one mineral. The success and timeliness of rare earth mining projects, and the rare-earth elements industry in general, is difficult to predict and will be continuously monitored and studied by various countries in the world.

Rare earths comprise of 17 elements and are classified as light RE elements (LREE) and heavy RE elements (HREE). Some REEs which are available in India, such as, Lanthanum, Cerium, Neodymium, Praseodymium, Samarium, etc. are in supply surplus while Dysprosium, Terbium, Europium

which are classified as HREE are having supply constraint. These HREE are not available in Indian deposits in extractable quantity. Entire Chhottanagpur Granite Gneiss Complex terrain offers tremendous scope for exploration inputs which may result in identifying additional HREE resources. Godhra–Idar Granite offers enormous scope for exploration inputs for Xenotime bearing placers in Gujarat & adjoining areas. There is a scope for exploration inputs in younger Potassic granites and associated soils in their vicinity to augment HREE resources in parts of Telangana.

The Government is actively engaged in capacity building for consumption of LREEs. The RE resources in India are fifth largest in the world. Indian resource is significantly lean with reference to grade and it is tied with radioactivity making the extraction long drawn, complex and expensive.

Production of REEs depends on deposits and end-Industry consuming the products. India is one of the pioneers in processing of RE and these capabilities are available in terms of capacity, technology and skill. The Government has targeted increasing REO producing capacity by 3 times by the year 2032. Also, in order to enhance consumption of RE in Indian industries, especially Electric Vehicles, the Government had announced a PLI scheme vide Item No. 6 page 44 of Notification No. S.O. 4632(E) dated 9th November 2021 through Ministry of Heavy Industries.

IREL has provision for expanding the capacity of processing rare-earth-mineral to 20,000 tonnes per annum in near future. The EV car projects which were expected to

boost demand for Rare-earth Magnets are likely to be put on the back burner as the industry will take some time to come back on track. IREL is actively pursuing setting up of Greenfield operations in Kanyakumari district of Tamil Nadu and Bramhagiri district of Odisha. The proposal for harnessing the beach sand mineral deposits in Tamil Nadu by constituting a Joint Venture Company with the nominated State Government Agency, TAMIN, is under active consideration by the State Government. As the country is gearing up towards e-mobility, green energy, e-office and other niche sectors, there is need to secure rare-earth mineral resources in areas beyond the boundaries of the country. IREL has the mandate to explore and acquire rare-earth resources abroad and efforts in this direction are actively underway.



51. Salt

12.78 million tonnes
Total reserves/resources of rock salt have been established as on 1st April, 2020.

13.47 million tonnes
of salt (other than common salt & rock salt) were exported in 2023-24.

94,868 tonnes
of salt (other than common salt & rock salt) were imported in 2023-24.

Salt is the most significant of all the industrial minerals. Occurrence of rock salt in India is scanty. At the time of Independence, there used to be shortfall in production of salt and the same was met through imports. Since then, India has made tremendous progress in production of salt, achieving self-sufficiency in 1953 and even started exporting salt to other countries.

Common salt, when pure, is mineral halite which is an essential item for human consumption. The per-capita consumption of salt in the country is estimated to be about 14 kg, which includes edible and industrial salt. The current annual requirement of salt in the country is estimated to be 63 lakh tonnes for edible use (including requirement for cattle) and 118 lakh tonnes for industrial use.

"Salt is a Central Subject in the Constitution of India and appears as Item No. 58 in the Union List of the 7th Schedule, which reads:

- (a) "Manufacture, Supply and Distribution of salt by Union Agencies; and
- (b) "Regulation and control of manufacture, supply and distribution of salt by other agencies".

The Central Government is responsible for controlling and regulating all aspects of the Salt Industry. The Salt Commissioner's Organisation (SCO), Jaipur, Rajasthan, an attached office under the Ministry of Commerce & Industry (Department of Industrial Policy & Promotion), Government of India, is the authority entrusted with the above task. The SCO is responsible for monitoring the production, distribution, quality, prices, supply and administration of Salt Cess Act, 1953 and the Rules, made thereunder.

RESERVES/RESOURCES

As per NMI database, based on UNFC system, the total reserves/resources of rock salt as on 01.04.2020 has been estimated at 12.78 million tonnes in Mandi, Himachal Pradesh. The entire resources fall under Remaining Resources category.

MINING LEASES & PRODUCTION

As on 31.03.2024 the total number of mining leases granted for mining of Rock Salt is 01, covering an overall mining lease area of 8.122 Ha. The production of Rock-Salt was nil in 2023-24, as compared to 1002 tonnes in the previous year. However, in 2023-24, only dispatches were reported by

the single public sector mine based in Himachal Pradesh. 2023-24 were 295 tonnes as against 561 tonnes in 2022-23 (Tables-1 to 3). The mine-head closing stocks at the end of (Table-4).

Table – 1: Producers of Rock- Salt, 2023-24

(By States)

Name & address of producer	Location of the mine	
	State	District
Hindustan Salts Ltd., B- 427, Pradhan Marg, Malviya Nagar, Jaipur -302 017, Rajasthan.	Himachal Pradesh	Mandi

Table – 2: Production of Rock- Salt, 2021-22 to 2023-24

(By State)

(Qty in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	286	712	1002	4849	0	0
Himachal Pradesh	286	712	1002	4849	0	0

Table – 3: Production of Rock-Salt, 2022-23 and 2023-24

(By Sector/State/Districts)

(Qty in tonnes; Value in ₹ '000)

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	1	1002	4849	1	0	0
Public Sector	1	1002	4849	1	0	0
Private Sector	-					
Himachal Pradesh	1	1002	4849	1	0	0
Mandi	1	1002	4849	1	0	

Table – 4: Mine-head Closing Stocks of Rock-Salt, 2022-23 and 2023-24

(By States/Grades)

(In tonnes)

State	2022-23	2023-24 (P)
India	561	295
Himachal Pradesh	561	295

The average daily labour employed in Rock-Salt mine during 2023-24 was 21 as against 22 in the previous year.

India is the 3rd largest salt (common) producing country in the world after China & USA with a production of about 280 lakh tonnes during the year 2022-23. Production of salt involves extensive use of renewable sources of energy. Weather conditions play an important role in production of salt.

Salt (common) is manufactured mainly by solar evaporation of sea water. Sea salt constitutes about 82% of the total salt produced in the country. Salt manufacturing activities are carried out in the coastal states of Gujarat, Tamil Nadu, Andhra Pradesh, Maharashtra, Karnataka, Odisha, West Bengal, Goa and hinterland State of Rajasthan. Gujarat (87%) was the leading State followed by Tamil Nadu (4.7%) and Rajasthan (6%). Private Sector contributed significantly in the production of salt (common) and accounted for 97.12% of the production.

Co-operative Sector contributed 1.77% and the rest was from the Public/Joint Sector which contributed 1.11% during 2022-23. The major by-products recovered in the Salt Industry are gypsum, bromine, magnesium chloride, magnesium sulphate, ammonium bicarbonate, soda ash, caustic soda and soda bicarbonate.

MINING

In a recent development, Drang Mohal (Bhatog) Rock Salt Mine, Village Mohal Bhatog, District Mandi, Himachal Pradesh, of M/s Hindustan Salts has started production and it produced 1,002 tonnes of rock salt during 2022-23, but in 2023-24, nil production was reported. Mining is carried out by underground semi-mechanised method. The entire mining area is hill (Government waste land) region with no forest land cover.

USES & CONSUMPTION

Salt, in addition to its culinary and domestic uses, is also used in meat packing, fish curing & its preservation,

dairying, preservation & processing of hides, manufacturing of soaps, detergent & dyestuff, dyeing & finishing of fabrics, refrigeration, glazing earthenware, explosives, leather industries, bakery products, for soil stabilisation, in manufacturing artificial rubber and as wood preservative. In agriculture, salt is used for treating hay and as a fertilizer for certain crops. In countries with cold weather conditions, salt is used in deicing, i.e., removal of snow and ice, from streets and highways.

Industrial Applications

Salt is an important raw material used in Chemical Industry. It is used in the production of basic chemicals like sodium carbonate (soda ash), caustic soda, hydrochloric acid, chlorine, bleaching powders, chlorates, sodium sulphate (salt cake) and sodium metal. These basic chemicals, in turn, are used in the preparation of various end-products, such as, soaps, detergents, chlorinated hydrocarbons and carbon tetrachloride. Other important applications where salt is widely used are in food processing; as freezing-point depressant in refineries & milk supply schemes; treatment of industrial wastes; purification of drinking water; and manufacture of synthetic indigo, explosives, papers, etc. The rock salt produced from Mandi mines contains 67.81% NaCl which is not suitable for human consumption as the content of sodium chloride is low from the required 96%. However, this salt finds application as essential supplement in cattle feeds.

Iodised Salt

The SCO is the nodal agency for monitoring the production and quality of iodised salt at the production level and ensuring its equitable distribution in the country. Production of iodised salt continued its progressive trend and the Universal Salt Iodisation (USI), a programme under National Iodine Deficiency Disorder Control Programme (NIDDCP) reached new heights.

Iodised salt is produced by mixing potassium iodate with salt using spray, drip feed, dry mixing and submersion processes. Iodisation of salt is carried out in plants operated by Sambhar Salts Ltd (a subsidiary of Hindustan Salts Ltd), Kharagoda, Gujarat. SCO has facilitated establishment of 523 salt iodization units including 121 refineries & washeries (capacity 149.5 lakh tonnes) with an annual installed capacity of 218.5 lakh tonnes up to March 2022. The Bureau of Indian Standards (BIS) has revised the specifications of iodised salt and formulated new specifications for refined iodised salt and Vacuum Evaporated iodised salt under IS 7224:2006. BIS has prescribed IS 797:1982 (Reaffirmed 2012) as specification of common salt for chemical industries.

Potassium Iodate Manufacture

Potassium iodate, used for fortifying salt with iodine, is produced indigenously. However, iodine, the basic raw material for its production, is imported. There are about 13 potassium iodate manufacturing units registered with the Salt Commissioner. State-wise production of salt (common) during the year 2021-22 is furnished in Table-5.

Table – 5: Statewise Production of Salt (Common), 2021-22

(By States)

(In lakh tonnes)

States	2021-22
India	270.24
Gujarat	227.68
Tamil Nadu	21.15
Rajasthan	16.9
Andhra Pradesh	3.3
Maharashtra	0.975
Karnataka	0.13
West Bengal	0.072
Odisha	0.029
Goa	-

Source: Salt Commissioner, Govt. of India, Jaipur, Annual Report, 2021-22

CONSUMPTION

Salt is extensively used in various industries like Caustic Soda, Soda Ash, Soaps & Detergents, Chemicals, Water softening plants, dyes etc. The Chlor-Alkali Industries, however, are the major consumer of salt.

TRADE POLICY

Exports of salt have been brought under Open General Licence (OGL) with effect from 25th August, 1987 vide

Ministry of Commerce, New Delhi, Export Trade Control Public Notice No. 26-ETC(PN)/87 dated 25th August, 1987. The Ministry of Commerce, Government of India, vide Notification No. 482(E) dated 25th July, 1991 has exempted certain categories of export from compulsory pre-shipment inspection. As per Foreign Trade Policy (FTP) 2015-2020, exports and imports of rock salt and common salt (including iodised salt) under Heading no. 2501 are allowed as 'Free'.

FOREIGN TRADE

Exports

During 2023-24, the exports of salt (other than common salt & rock salt) marginally increased by about 0.5% to 13.47 million tonnes from 13.40 million tonnes in the previous

year. Exports were mainly to China (47%), Rep. of Korea (20%), Japan (9%) and Qatar, Indonesia & Taiwan (5% each). On the other hand, the exports of rock salt increased by about 37% to 310.1 thousand tonnes in 2023-24 from 225.6 thousand tonnes in 2022-23. The exports were mainly to Bangladesh (92%) and Vietnam (6%) (Tables-6 to 7).

Table – 6: Export of Salt (Other than Common Salt & Rock Salt)

(By Countries)

Country	2022-23(R)		2023-24 (P)	
	Qty(t)	Value (₹ '000)	Qty(t)	Value (₹ '000)
All Countries	13406235	23673306	13473826	24463925
China	6499944	11746555	6382960	10650706
Korea, Rep. of	1906271	2547547	2656809	4518082
Japan	1270137	1881207	1175908	2219246
Qatar	653969	1322897	693315	1310288
Indonesia	844604	1322100	607980	1265188
Taiwan	529928	788176	631630	1256139
Vietnam	696513	1474152	504773	1066228
Bangladesh	387849	893657	263227	517287
Oman	140690	264546	172671	322899
Thailand	123789	250760	170526	272212
Other Countries	352541	1181709	214027	1065650

Figures rounded off

In tonnes

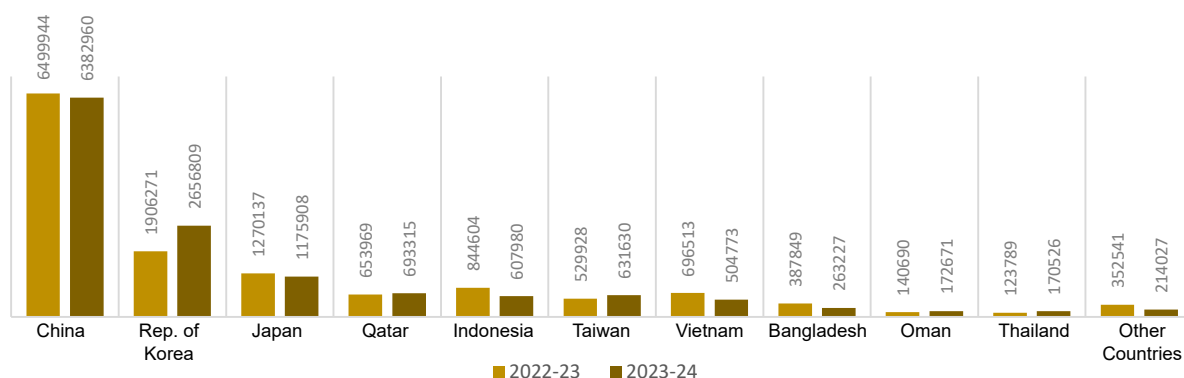


Fig. 1: Countrywise Export of Salt

Table – 7: Export of Rock salt

(By Countries)

Country	2022-23(R)		2023-24 (P)	
	Qty(t)	Value (₹ '000)	Qty(t)	Value (₹ '000)
All Countries	225595	621445	310132	811417
Bangladesh	206034	500205	286212	618800
Vietnam	17121	33666	19802	35689
USA	236	17039	355	30973
Canada	140	10061	335	22168
Nepal	644	8610	1270	17878
Singapore	149	5255	227	14103
UAE	315	8088	420	12213
Australia	53	3069	105	8907
UK	51	3339	69	5597
Saudi Arabia	128	2841	245	4064
Other Countries	724	29272	1092	41025

Figures rounded off

Imports

The imports of salt (other than common salt & rock salt) increased by 76% to 94,868 tonnes in 2023-24 from 53,880 tonnes in the previous year. Imports were mainly from Iran (48%), UAE (44%), China (3%) & Turkey (2%). Similarly,

import of rock salt decreased by 16 % to 37,760 tonnes in 2023-24 from 45,106 tonnes in 2022-23. The imports were mainly from UAE (71%), Iran (27%) and Malaysia (1%). (Tables-8 to 9).

Table – 8: Import of Salt (Other than Common Salt & Rock Salt)

(By Countries)

Country	2022-23(R)		2023-24 (P)	
	Qty(t)	Value (₹ '000)	Qty(t)	Value (₹ '000)
All Countries	53880	545064	94868	926054
UAE	10124	65346	41321	370644
Iran	38877	209612	45972	250285
USA	375	118931	231	140493
China	1764	30678	2960	34815
Thailand	1069	33155	887	26157
Turkey	75	3554	2224	22250
France	30	18295	52	18971
Germany	423	22197	304	17976
Netherlands	304	6459	428	13425
UK	691	17971	175	7757
Other Countries	148	18866	314	23281

Figures rounded off

In tonnes

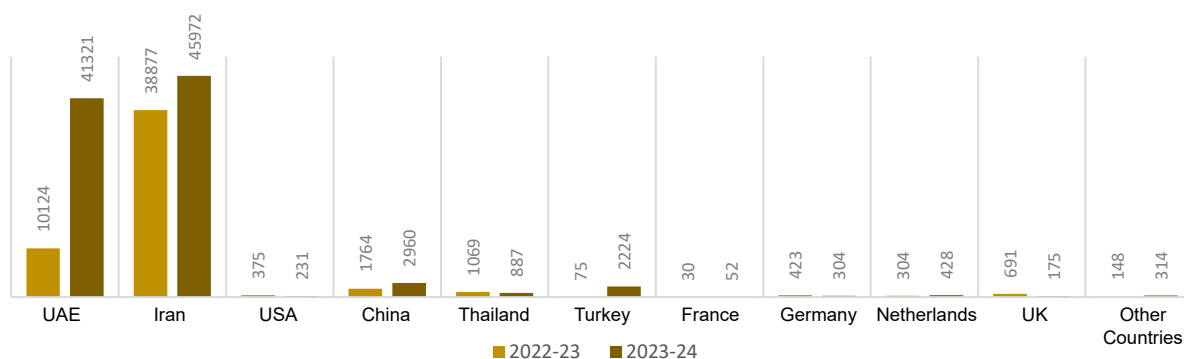


Fig. 2: Countrywise Import of Salt

Table – 9: Import of Rock salt

(By Countries)

Country	2022-23(R)		2023-24 (P)	
	Qty(t)	Value (₹ '000)	Qty(t)	Value (₹ '000)
All Countries	45106	363389	37760	339688
UAE	31213	271046	26831	263523
Iran	11209	73312	10360	65071
Malaysia	286	6550	389	8233
Oman	61	607	100	779
Sri Lanka	53	710	70	689
Germany	--	--	8	576
USA	222	1847	++	367
Taiwan	++	23	2	220
UK	++	20	++	117
Belgium	1	42	++	107
Other Countries	2061	9232	++	6

Figures rounded off

FUTURE OUTLOOK

Sambhar Salt Ltd (SSL) approximately produces 2 lakh tonnes of raw salt every year. In addition, approximately 40,000 tonnes of processed salt are also produced. Schemes are also under preparation to enhance further capacity of raw salt production to 10 lakh tonnes per annum.

Hindustan Salts Limited has been awarded the long term contract through open tender in May-2021 for large scale commercial mining of rock salt at Mandi, wherein

modern trackless underground mining is proposed to be used to enhance production up to 50,000 MT/annum. This proposal forms a part of Hon'ble PM's vision of Atmanirbhar Bharat wherein it is planned to produce adequate quantity of Himalayan Rock Salt so that complete import of Rock Salt both from Iran and Pakistan gets stopped within 2 years. The company also intends to install Salt Refinery for processing of Rock Salt with latest technology as part of contract.



52. Sulphur and Pyrites

1,674 million tonnes
Total reserves/resources of pyrites have been established as on 1st April, 2020.

977 thousand tonnes
production of Sulphur have been reported in 2023-24.

11,85,636 tonnes
sulphur (excl. sublimed precipitated and colloidal) were exported in 2023-24 .

Sulphur is an essential raw material for many chemical industries and is essentially used for the production of sulphuric acid which in turn is used for the production of chemical fertilizers, textiles, dyestuffs, pickling and galvanising of steel, storage batteries, refining of petroleum, explosives and other acids.

In India, presently there are no mineable elemental sulphur reserves. Sulphur combines directly with almost all the elements with the exception of gold, platinum and the noble gases. In its native form, sulphur is a yellow crystalline solid. It can be found as a pure element or as sulphate or sulphide minerals. The crystallography of sulphur is complex. Depending on the specific conditions, the sulphur allotropes form several distinct crystal structures, with rhombic and monoclinic (S_8) best known.

Pyrites is naturally occurring mineral comprised of the elements iron and sulphur (FeS_2). It is used for manufacture of sulphuric acid, and as direct feed for soil conditioning. Pyrite is a fairly ubiquitous mineral and it occurs most commonly in sedimentary rocks. Pyrite has a brass yellow colour, brownish black streak, metallic lusture and occurs as cubic crystals. Pyrites includes a range of sulphide materials, such as, marcasite, pyrite and pyrrhotite. Marcasite usually occurs in low temperature metasediments and sedimentary rocks. Pyrrhotite occurs usually in magmatic or contact metasomatic deposits associated with basic igneous rocks and high temperature sulphide veins and is often nickeliferous. Pyrites was used as a substitute for sulphur in the manufacture of sulphuric acid. However, there was no production of pyrites since 2003.

Native sulphur deposit has been reported in Puga

Valley of Leh district in the Union Territory of Jammu & Kashmir. The grade of the deposit ranges from 9% to 24% of sulphur. Small occurrences of native sulphur are also reported from Barren Island of Bay of Bengal. Sulphur along with hot springs were reported from various parts of Chamoli, Rudraprayag, Uttarkashi districts in Garhwal & Kumaun divisions of Uttarakhand. In Andhra Pradesh, native sulphur occurs in granular form with clay and silt in coastal areas of Krishna and East Godavari districts. Occurrences have also been reported from Alappuzha district of Kerala and Kangra district of Himachal Pradesh.

Sulphide occurs naturally in mineral ores, oil and coal deposits. Natural waters containing elevated concentrations of hydrogen sulphide are used for therapeutic baths and have been consumed for medical purposes. Hydrogen sulphide (H_2S), which exists as a colourless gas under

normal conditions, has a characteristic odour of rotten eggs and occurs naturally in coal, natural gas, oil, volcanic gases and sulphur springs and lakes; H₂S is a central participant in the sulphur cycle, the biogeochemical cycle of sulphur on earth. Sulphides form an indispensable link in the sulphur cycle (the reversible interconversion of sulphide and sulphate) in nature.

Petroleum refineries and gas processing plants extract H₂S when making “clean fuels” and use it as a feed stock to produce sulphur and water. The domestic production of elemental sulphur is limited to by-product recoveries from petroleum refineries and fuel oil used as feedstock for manufacturing fertilizer. Tar sands-natural sand (Oil sands) formations contain about 10% bitumen with high hydrogen sulphide content.

The sulphide ores contain sulphur and during the production of metal from sulphide ores, sulphur is released as SO₂ which is used to produce sulphuric acid. The sulphuric acid thus produced contains about 32.7% of sulphur and contributes in the industries which otherwise would have used elemental sulphur.

RESERVES/ RESOURCES

The total reserves/resources of pyrites in the country as per NMI data, based on UNFC system as on 1.4.2020 has been placed at 1,674 million tonnes. There are no reserves and all resources are grouped under 'Remaining Resources' category. Out of these, about 27 million tonnes are under Feasibility (STD211) category. Out of the total resources, Beneficiable grade resources are 62 million tonnes, Low grade 1,555 million tonnes and Soil Reclamation grade resources are about 6 million tonnes. The balance of about 51 million tonnes resources fall under Unclassified/Not-known grades. Major reserves/resources are located in Bihar (94%) and Rajasthan (5%) (Table - 1).

Reserves/resources of sulphur (native) have been estimated in the Inferred (STD333) category only. Entire resources are located in Jammu & Kashmir and are placed at 0.21 million tonnes as on 1.4.2020 as per NMI data, based on UNFC System (Table-2).

Table – 1: Reserves/Resources of Pyrites as on 01.04.2020
(By Grades/States)

(In '000 tonnes)

Grades/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	–	1674401	1674401
By Grades			
Soil Reclamation	–	6024	6024
Beneficiable	–	61628	61628
Low	–	1555330	1555330
Unclassified	–	51419	51419
By States			
Andhra Pradesh	–	880	880
Bihar	–	1574561	1574561
Himachal Pradesh	–	2560	2560
Karnataka	–	3000	3000
Rajasthan	–	90876	90876
Tamil Nadu	–	24	24
West Bengal	–	2500	2500

Figures rounded off

Table – 2: Reserves/Resources of Sulphur (Native) as on 01.04.2020
(By Grades/States)

(In '000 tonnes)

Grades/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	–	210	210
By Grades			
Sulphur (Native)	–	210	210
By States/ UTs			
Jammu & Kashmir	–	210	210

Figures rounded off

MINING LEASES AND PRODUCTION

Sulphur (By-product)

The production of Sulphur recovered as by product from fertilizer plant and oil refineries were 977 thousand tonnes in 2023-24 as against 926 thousand tonnes in the 2022-23.

The oil refineries in public sector reported production of sulphur. During the year 2023-24, the contribution of IOCL, BPCL and NRL was 72.26%, 27.31% and 0.43% respectively. Among the states, Odisha accounted for 26% of the total sulphur production followed by Kerala 22%, Haryana 17%, Gujarat 14%, West Bengal 8%, Uttar Pradesh

5%, Maharashtra 5%, and remaining production was contributed by Bihar and Assam.

In addition, refineries of Hindustan Petroleum Corp. Ltd, RIL and Essar Oil also reported recovery of by-product sulphur which in turn is used as feedstock in manufacturing fertilizers and pharmaceuticals. The Vadinar refinery of Essar Oil Ltd is also reported to produce by-product sulphur. In Fertilizer Industry, the sulphuric acid is further used for manufacturing phosphoric acid and single super phosphate (SSP) from rock phosphate (Tables-3 to 5).

Table – 3: Principal Producers of, Sulphur (By-product)

Name & Address of Producer	Location of the Mine	
	State	District (S)
Indian Oil Corporation Ltd, (Refineries Division) Scope Complex, Core-II, 7, Institutional Area, Lodhi Road, New Delhi – 110 003	Assam	Kamrup Metro, Tinsukia
	Bihar	Chirang
	Gujarat	Begusarai
	Haryana	Vadodra
	Odisha	Panipat
	Uttar Pradesh	Jagatsinghpur
	West Bengal	Mathura
Numaligarh Refinery Limited, 122S, G.S. Road Christanbasti, Distt- Guwahati, Assam – 781 005	Assam	Purba Medinipur
		Golaghat
Bharat Petroleum Corporation Ltd, Bharat Bhavan, 4 & 6, Currimbhoy Road, Ballard Estate, Mumbai- 400 001, Maharashtra	Maharashtra	Mumbai
	Kerala	Ernakulam

Note: Sulphur is recovered as by-product from fertilizer plants and oil refineries (excluding units working under Private Sector).

Table – 4: Production of Sulphur (By-product) 2021-22 to 2023-24

(By State)

(In tonnes)

State	2021-22	2022-23	2023-24 (P)
	Quantity	Quantity	Quantity
India	880858	925663	976615
Assam	6545	7705	9148
Bihar	8160	14652	16240
Gujarat	117588	146401	135697
Haryana	178740	157228	163930
Kerala	182352	182398	216097
Maharashtra	53165	46283	50577
Odisha	207831	228518	257338
Uttar Pradesh	60307	55411	53193
West Bengal	66170	87067	74395

Figures rounded off

(P): Provisional

(In tonnes)

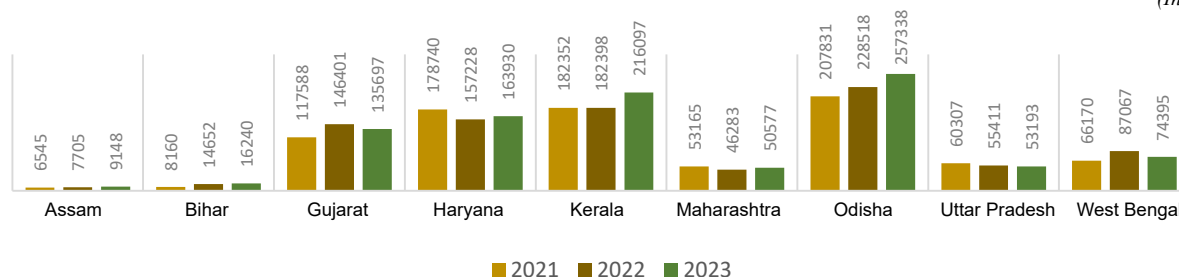


Fig. 1: Statewise Production of Sulphur (By-product)

Table – 5: Production of Sulphur (By-product) 2022-23 and 2023-24

(By Sectors/States/Districts)

(In tonnes)

Sector/State/District	2022-23 (R)		2023-24 (P)	
	No. of units	Quantity	No. of units	Quantity
India	12	925663	12	976615
Public Sector	12	925663	12	976615
Assam	4	7705	4	9148
Chirang	1	2114	1	3916
Tinsukia	1	335	1	334
Kamrup Metro	1	653	1	673
Golaghat	1	4603	1	4225
Bihar	1	14652	1	16240
Begusarai	1	14652	1	16240
Gujarat	1	146401	1	135697
Vadodara	1	146401	1	135697
Haryana	1	157228	1	163930
Panipat	1	157228	1	163930
Kerala	1	182398	1	216097
Ernakulam	1	182398	1	216097
Maharashtra	1	46283	1	50577
Mumbai	1	46283	1	50577
Odisha	1	228518	1	257338
Jagatsinghpur	1	228518	1	257338
Uttar Pradesh	1	55411	1	53193
Mathura	1	55411	1	53193
West Bengal	1	87067	1	74395
Purba Medinipur	1	87067	1	74395

Figures rounded off

(P): Provisional

Pyrites

Pyrites Phosphates and Chemicals Ltd (PPCL) had two pyrites production units located at Amjhore (Bihar) and Saladipura (Rajasthan) besides, phosphorite division in Dehradun. The Government approved closure and hiving off of these two units in July 2002 and Amjhore unit in June 2003 and since then no activity has been reported.

Petroleum Refining

In fossil fuels, sulphur is naturally present as an impurity when fuel is burnt, the sulphur is released as sulphur dioxide, an air pollutant. Hydro desulphurisation (HDS) is a catalytic chemical process widely used to remove sulphur from natural gas and from refined petroleum products, such as, gasoline or petrol, jet, fuel, kerosene, diesel and fuel oils. Sulphur is a by-product produced in various refineries processing high sulphur crude oil. Sulphur is produced from the sulphur-rich fuel gas as a process to reduce the emission level of sulphur in the atmosphere along with flue gases from the furnaces. Mathura refinery started

production of sulphur from beginning itself and sulphur recovery units have been commissioned in Haldia, Koyali, Panipat, Mathura and Guwahati refineries.

Refinery-wise Sulphur (by-product) production capacity of Indian Oil Corporation Ltd is as under:

Unit	Production Capacity ('000 MTPA)
Mathura	48
Haldia	24
Koyali	18
Panipat	144
Barauni	12
Guwahati	0.6

Specification of sulphur at Mathura, Panipat, Haldia, Barauni & Guwahati refineries is as under:

Property	Specification
Purity	99.9
Colour	Yellow
Shape	Lump

USES & CONSUMPTION

Flowers of Sulphur (sublimed sulphur)

Powdered form of sulphur produced by sublimation process that may contain up to 30% of the amorphous allotrope are generally used in rubber vulcanisation, agricultural dusts, pharmaceutical products and stock feeds.

Sulphur dioxide (SO₂)

Sulphur dioxide is a by-product gas generated during processing of sulphide ores as well from other industries. It is used in many industrial processes such as, chemical preparation, refining, pulp-making and solvent extraction and also is the feed stock to manufacture sulphuric acid. Sulphur dioxide is also used in the preparation and preservation of food because it prevents bacterial growth and browning of fruit.

Sulphuric Acid

Sulphuric acid is a strong mineral acid with the formula H₂SO₄. It is soluble in water at all concentrations. Sulphuric acid has many applications and is produced in greater amounts than any other chemical besides water. Principal uses include ore processing, fertilizer manufacturing, oil refining, waste water processing and chemical synthesis.

Miscellaneous

One of the direct uses of sulphur is in vulcanisation of rubber. Sulphur is a component of gunpowder. It reacts directly with methane to give carbon disulphide, which is used in the manufacturing of cellophane and rayon.

Elemental sulphur is mainly used as a precursor to other chemicals. Most of the sulphur is converted to sulphuric acid (H₂SO₄), which is of prime importance to the world economy.

The production and consumption of sulphuric acid are an indicator of a nation's industrial development. The principal use of sulphuric acid is in the manufacture of phosphatic fertilizer.

Other applications of sulphuric acid include oil refining, waste water processing and mineral extraction. Sulphur compounds are also used in detergents, fungicides, dyestuffs and agrichemicals. In silver based photography, sodium and ammonium thio-sulphate are used as "fixing agents". Sulphites, derived from burning sulphur, are used to bleach paper. They are also used as preservatives in dried fruit and processed fruit products.

Sulphur is used as a light-generating medium in the rare lighting fixtures known as "sulphur lamps". The sulphur lamp is a highly efficient full-spectrum electrodeless lighting system whose light is generated by sulphur plasma that has been excited by microwave radiation.

Nitrogen (N), phosphorus (P) and potassium (K) are critical components of a well-fertilized crop. But to achieve yields and more nutritious foods, crops need sulphur (S). It improves protein and oil percentage in seeds, cereal quality

for milling and baking, marketability of dry coconut kernel (copra), quality of tobacco, nutritive value of forages, etc. It is associated with special metabolisms in plant and the structural characteristics of protoplasm. Judicious application in sulphur-deficient soils is a cost-effective way to produce more food and feed.

Concrete binder made with sulphur is an eco-efficient alternative to conventional Portland cement for paving stones, sidewalks and building foundations. In road construction, sulphur technology can replace up to 30 per cent of asphalt binder, a high energy-intensive input in blacktop roads. Sulphur-enhanced roads and parking lots offer a longer life cycle.

INDUSTRY

Coromandel International Ltd, formerly Coromandel Fertilizers Limited (CFL), is a leading manufacturer of a wide range of fertilizers & pesticides. The fertilizer manufacturing units in Visakhapatnam, Kakinada, and Ennore operate with multiple rock and acid combinations, allowing them to produce over 10 products. Additionally, there is a water-soluble fertilizer plant at Kakinada. Through its 8 SSP plants across the country, the company offers a low-cost source of phosphate-based nutrition products.

Coromandel holds the top position as the largest producer of Single Super Phosphate (SSP) in the country, with an annual production capacity of approximately 4.5 million tonnes of complex fertilizer and SSP. According to Coromandel International Ltd's Annual Report for 2022-23, the commissioning of its new 1,650 MT per day sulphuric acid plant was on track and was likely to be completed during the year 2023. With this addition, the company plans to increase its sulphuric acid production by 5,00,000 MT per annum from the current level of 6,00,000 MT, resulting in a combined capacity of 1.1 million MT.

The Fertilisers and Chemicals Travancore Limited (FACT) was incorporated in 1943 as India's first largescale fertilizer plant in Udyogamandal, Kochi, Kerala. The Udyogamandal plant has an installed capacity of 76,050 tonnes of N and 29,700 tonnes of P₂O₅. It produces fertilizers like Ammonium Sulphate and Ammonium Phosphate Sulphate (FACTAMFOS 20:20:0:13) as well as intermediate chemicals such as Sulphuric Acid, Anhydrous Ammonia, Phosphoric Acid, Sulphur Dioxide, and Oleum. The FACT Kochi facility produces 4,85,000 TPA of Complex Fertilizer (FACTAMFOS 20:20:0:13) 3,30,000 TPA of Sulphuric Acid, and 1,15,200 TPA of Phosphoric Acid.

According to the 2022-23 Annual Report of FACT, the Udyogamandal Complex produced 2,08,146 MT of Factamfos (NP 20:20:0:13), which is 140% of its installed capacity, and 2,44,732 MT of Ammonium Sulphate, which is 109% of its installed capacity. The Cochin Division produced 6,20,050 MT of Factamfos (128% of installed capacity), 1,24,010 MT each of nutrient nitrogen and nutrient P₂O₅ (127.85% of installed capacity), 2,34,330 MT

of Sulphuric Acid, and 63,050 MT of Phosphoric Acid.

At present Company has a tie-up with BPCL-Kochi Refinery for sourcing about 60% of its annual requirement minimising import. Company also imports Sulphuric Acid to meet its requirement for fertilizer production, mainly from metallurgical industry, where it is a waste/by-product during processing.

According to the Annual Report of Gujarat State Fertilizers & Chemical Limited, (GSFC), the company is planning to set up a 600 tonnes per day Sulphuric Acid Plant at its Vadodara unit. Additionally, it has plans to install a 600 tonnes per day Phosphoric Acid Plant and an 1,800 tonnes per day Sulphuric Acid Plant at its Sikka unit. To expand GSFC's Agro-product portfolio, company is considering to enter into organic fertilizers by setting up 2x200 tonnes per day Phosphate Rich Organic Manure (PROM) plant at Polymer unit by using available existing infrastructure.

HZL produce 98% concentrated Sulphuric Acid at Chanderia, Debari and Dariba plants in the state of Rajasthan. Hindustan Zinc Ltd, the second largest zinc miner, is also the largest producer of sulphuric acid, which is sold in the domestic market, mainly to fertiliser and sulphonation industries. Rock phosphate, sulphuric acid, ammonia and potash are the major ingredients for production of DAP/ NPK. Currently, HZL is operating total 8 sulphuric acid plants in Rajasthan and is planning to commission the fertiliser plant in Chanderiya, enabling better utilisation of sulphuric acid generated in operations through production of fertilisers, catering to the demand in the Indian market and delivering international quality products, while commanding a premium in the domestic market. Sulphuric Acid is used in production of Single Super Phosphate Fertilizers/ Zinc Sulphate/ Phosphoric Acid/LABSA for detergent/Chemical Gypsum for Cement Industries/ Metal Industry/Speciality Chemicals/ Dyes etc. for all spectrum of Industries.

TRADE POLICY

Imports of sulphur of all kinds other than sublimed sulphur, precipitated sulphur and colloidal sulphur under Heading No. 2503 are allowed free under the Foreign Trade Policy (FTP), 2015-20. Similarly, the imports of unroasted iron pyrites under Heading No. 2502 are allowed free.

WORLD REVIEW

The world sulphur industry was composed of two sectors—discretionary and nondiscretionary. In the discretionary sector, the mining of sulphur or pyrites is the sole objective; this voluntary production of either sulphur or pyrites (mostly naturally occurring iron sulphide) is based on the mining of discrete deposits, with the objective of obtaining as nearly a complete recovery of the resource as economic conditions permit. In the non-discretionary sector, sulphur or sulphuric acid is recovered as an involuntary byproduct; the quantity of output is subject to demand for the primary product and environmental regulations that limit atmospheric emissions of sulphur compounds irrespective of sulphur demand. Discretionary sources (Frasch, native, and pyrites), once the primary sources of sulphur in all forms, represented only 8% of the sulphur produced worldwide in 2018.

The Frasch process is the term for hot-water mining of native sulphur associated with the caprock of salt domes and in sedimentary deposits; in this mining method, the native sulphur is melted underground with superheated water and brought to the surface by compressed air. The United States, where the Frasch process was developed early in the 20th century, was the leading producer of Frasch sulphur until 2000. Poland, with 6,60,000 tonnes, was the only country that produced more than 3,00,000 tonnes of native sulphur by using either the Frasch process or conventional mining methods. Small quantities of native sulphur were produced in Asia, Europe, and South America. The importance of pyrites to the world's sulphur supply has significantly decreased; China and Finland were the top producers of sulphur from pyrites with China accounting for 95% of the world pyrite production.

Native sulphur production, including production of Frasch sulphur at Poland's last operating mine, was estimated to be about the same as that in 2017. Recovered elemental sulphur production and byproduct from metallurgy was slightly higher than that in 2017. Globally, production of sulphur from pyrites was estimated to have been the same as that in 2017. Pyrites is a less attractive alternative to elemental sulphur for sulphuric acid production, primarily because the environmental remediation cost of mining pyrites is high (Table-6).

Table – 6: World Production of Sulphur & Pyrite
(By Principal Countries)

Country	(Intonnesofsulphurcontent)		
	2021	2022	2023
World(total): Pyrites	5500000	5500000	5500000
World(total): Frasch	400000	500000	500000
World(total): Recovered	66100000	74200000	67300000
World(total): Sulphur	50000	40000	40000
Austria			
(Recovered) ^(a)	28000*	26000*	24000*

Table 6 (Contd.)

Country	(Intonnesofsulphurcontent)		
	2021	2022	2023
Belarus			
(Recovered) ^(a)	66800*	62000*	58000*
Belgium			
(Recovered) ^{(b) (a)}	400000*	400000*	400000*
Bosnia&Herzegovina			
(Recovered)	3700*	3700*	3700*
Bulgaria			
(Recovered) ^(b)	425188	504467	422494
(Recovered) ^(a)	35000*	65000*	55000*
Croatia			
(Recovered) ^(a)	8659	7274	7729
CzechRepublic			
(Recovered) ^(a)	21000*	19000*	17000*
Denmark			
(Recovered) ^(a)	4090	3727	2700
Finland			
(Pyrites)	182000	143000	140000*
(Recovered) ^(b)	338400	341100	314600
(Recovered) ^(a)	120000*	120000*	120000*
France			
(Recovered) ^(a)	300000*	300000*	300000*
(Recovered) ^(c)	55000*	55000*	5500*
Germany			
(Recovered) ^(a)	382049	370664	276876
(Recovered) ^(c)	235769	227281	167783
Greece			
(Recovered) ^(a)	500000	390000	390000
Hungary			
(Recovered) ^(a)	74000*	75000*	77000*
Italy			
(Recovered) ^(d)	550000*	500000*	480000*
Lithuania			
(Recovered) ^(a)	72988	64498	54538
Netherlands			
(Recovered) ^(b)	100000*	100000*	100000*
(Recovered) ^(a)	390000*	400000*	400000*
Norway			
(Recovered) ^(b)	70948	69013	69982
(Recovered) ^(a)	21000*	20000*	19000*
Poland			
(Frasch)	449100	476900	472020
(Recovered) ^(b)	280000*	280000*	280000*
(Recovered) ^(a)	24240	22670	22730
Portugal			
(Recovered)	21000*	*21000	21000*
Romania			
(Recovered)	45000*	*42000*	40000*
Russia			
(Pyrites)	71000*	*71000*	71000*
(Recovered) ^(a)	6270000*	*6420000*	6500000*

Table 6 (Contd.)

Country	(Int tonnes of sulphur content)		
	2021	2022	2023
(Recovered) ^(c)	954000*	954000*	954000*
Sulphure ore	8000*	28000*	28000*
Serbia			
(Recovered) ^(b)	29400*	29400*	29400*
Slovakia			
(Recovered) ^(b)	4900*	4900*	4900*
(Recovered) ^(a)	25000*	15000*	5000*
Sweden			
(Recovered) ^(b)	172700	179900	185100
(Recovered) ^(a)	38248	28170	28000*
Turkey			
(Pyrites)	39681	41788	39929
(Recovered) ^{(a) (c)}	98000*	102000*	117000*
UK			
(Recovered) ^(a)	109000	123000	114000
Algeria			
(Recovered) ^(a)	12000*	13000*	13000*
Egypt			
(Recovered) ^(a)	64000*	65000*	65000*
Libya			
(Recovered) ^(a)	139000	139500	138200
Morocco			
(Recovered)	60000*	60000*	60000*
Namibia			
(Recovered)	65500	64700*	63900
South Africa			
(Recovered) ^{(b) (a)}	32044	70640	63695
Zambia			
(Recovered) ^(b)	691900	640800	600000*
Canada			
(Recovered) ^(b)	555000*	550000*	550000*
(Recovered) ^(a)	4193003	4348607	4431431
Cuba			
(Recovered)	2846	2951	2653
(Recovered) ^(a)	20000*	20000*	20000*
Mexico			
(Recovered) ^(b)	556000*	556000*	556000*
(Recovered) ^(a)	182871	290121	255452
Trinidad & Tobago			
(Recovered) ^(a)	11000*	11000*	10000*
USA			
(Recovered) ^(b)	600000	636000	640000
(Recovered) ^(a)	692000	7490000	760000
Argentina			
(Recovered) ^(b)	20000	*20000	20000
Brazil			
(Pyrites)	20000*	20000	20000*
(Recovered) ^(b)	292000*	292000*	292000*
(Recovered) ^(a)	239000*	239000	239000*
Chile			
(Recovered) ^(b)	1400388	1334713	1258888

Table 6 (Contd.)

Country	(In tonnes of sulphur content)		
	2021	2022	2023
Colombia			
(Recovered) ^(a)	5600*	5700*	5800*
Ecuador			
(Recovered) ^{(a)(e)}	5400*	5500*	5400*
Peru			
(Recovered) ^(d)	556000*	556000*	556000*
Venezuela			
(Recovered) ^(a)	110000*	115000*	135000*
Bahrain			
(Recovered) ^(a)	110000*	110000*	110000*
China			
(Pyrites)	5200000*	5200000*	5200000*
(Recovered)	18800000*	18800000*	19000000*
India			
(Recovered) ^{(b)(f)}	1200000*	1200000*	1200000*
(Recovered) ^(a)	(f)880858	926270	981260
Indonesia			
(Recovered) ^(b)	160000*	160000*	160000*
(Recovered) ^(a)	85000*	80000*	80000*
Iran			
(Recovered) ^(d)	1900000*	2000000*	2400000*
Iraq			
(Recovered) ^(a)	200800*	221300*	213000*
Israel			
(Recovered) ^(a)	43664	40000*	40000*
Japan			
(Recovered) ^(b)	1575032	1655704	1535117
(Recovered) ^(a)	1420094	1542244	1367438
Jordan			
(Recovered) ^(a)	490000*	490000*	490000*
Kazakhstan			
(Recovered) ^(b)	604000*	604000*	604000*
(Recovered) ^(a)	2492000*	2445000*	2612000*
Korea, Rep. of			
(Recovered) ^(b)	1078000*	1078000*	1078000*
(Recovered) ^(a)	2000000	2000000	2000000*
Kuwait			
(Recovered) ^(a)	780000*	875000*	839000*
Oman			
(Recovered) ^(a)	48400*	53000*	52200*
Pakistan			
(Sulphure) ^(g)	19398	12442	11690
Philippines			
(Recovered) ^(b)	169000*	169000*	169000*
(Recovered) ^(a)	22600*	20700*	15250*
Qatar			
(Recovered) ^(a)	1345200*	1381300*	1407900*
Saudi Arabia			
(Recovered) ^(a)	3605000*	4019000*	3722000*

Table 6 (Contd.)

Country	(Intonnesofsulphurcontent)		
	2021	2022	2023
Singapore			
(Recovered) ^(a)	300000*	300000*	300000*
Syria			
(Recovered) ^(a)	5000*	5000*	5000*
Taiwan			
(Recovered)	192227	162571	122238
Thailand			
(Recovered) ^(a)	49200*	41000*	39700*
Turkmenistan			
(Recovered) ^(a)	252000*	244000*	193000*
UAE			
(Recovered) ^(a)	2300000*	2500000*	2500000*
Uzbekistan			
(Recovered) ^(b)	131000*	131000*	131000*
(Recovered) ^(a)	227000*	227000*	216000*
Australia			
(Recovered) ^(b)	810000*	810000*	810000*
(Recovered) ^(a)	60000*	55000*	50000*
New Zealand			
(Recovered) ^(a)	14000*	12000*	14000*

Source: BGS, World Mineral Production, 2019-23

Note: In addition to the countries listed, Dem. P.R. of Korea may also produce pyrites and may recover sulfur from metal processing; Ukraine may also produce sulfur.

(a): From petroleum refining and/or natural gas

(b): From metal sulphide processing

(c): Other

(d): Sulphur; all forms

(e): Including Frasch

(f) : Years ended 31st March following that stated

(g) : Years ended 30th June of that stated

*: Estimated

FOREIGN TRADE

Exports

The exports of sulphur (excluding sublimed, precipitated, and colloidal) decreased by 24% to 11,85,636 tonnes in 2023-24 as compared to 15,54,999 tonnes in the preceding year. Exports were mainly to China (98%), Tanzania (1%), and Morocco (0.42%). On the other hand, exports of sulphur (sublimed, precipitated, and colloidal) marginally increased by 1% to 14,599 tonnes in 2023-24 as compared

to 14,515 tonnes in the preceding year. Exports were mainly to the USA (18%), Netherlands & Brazil (15% each), Thailand (9%), Indonesia & Russia (7% each), Portugal & South Africa (5% each), Slovenia (4%) and Italy (3%). Exports of sulphur (sublimed) were at 14,598 tonnes for the year 2023-24. Exports of sulphur (colloidal) and sulphur (precipitated) were negligible for both the years 2022-23 and 2023-24 (Tables-7 to 11).

Table – 7: Exports of Sulphur (exc. Sublimed Precipitated and Colloidal)
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1554999	33267617	1185636	10346076
China	1224865	26488190	1165600	9597902
Tanzania	5673	285410	6116	227792
Turkey	1741	206739	910	127077
Brazil	145	15059	1143	126968
Morocco	126000	1904754	5000	71972
South Africa	171543	3714608	1987	52427

Table 7 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Sri Lanka	1278	54667	1300	31504
Nepal	1042	25631	859	16152
Djibouti	67	2586	668	15727
France	++	1	105	9248
Other Countries	22645	569972	1948	69307

Figures rounded off

In tonnes

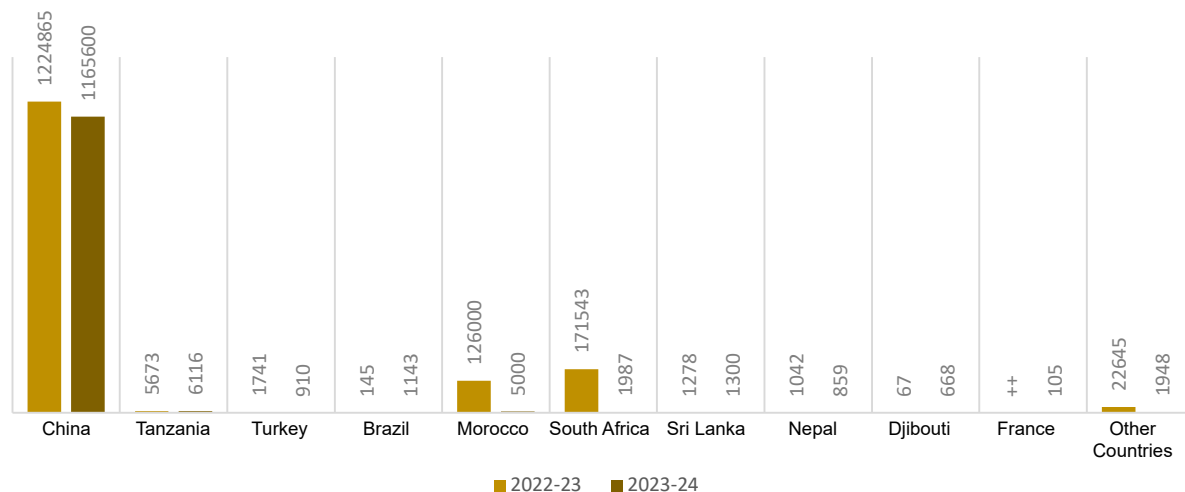


Fig. 2: Countrywise Exports of Sulphur (exc. Sublimed Precipitated and Colloidal)

Table – 8: Exports of Sulphur (Sublimed Precipitated and Colladial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	14515	2059103	14599	1922318
USA	2491	349843	2630	356020
Netherlands	1613	262811	2205	336395
Brazil	1657	243930	2215	276260
Thailand	1130	147880	1313	150263
Russia	1217	144710	1016	114526
Indonesia	1180	154906	1001	112330
Portugal	648	102326	748	109614
South Africa	852	128511	670	93803
Slovenia	455	67472	587	79727
Italy	675	95092	476	70599
Other Countries	2597	361622	1738	222781

Figures rounded off

Table – 9: Exports of Sulphur (Colloidal)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	312	++	307
Sri Lanka	++	312	++	307

Figures rounded off

Table – 10: Exports of Sulphur (Sublimed)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	14515	2052830	14598	1921435
USA	2491	349841	2630	355944
Netherlands	1613	262811	2205	336395
Brazil	1657	243930	2215	276260
Thailand	1130	147880	1313	150263
Russia	1217	144710	1016	114526
Indonesia	1180	154906	1001	112330
Portugal	648	102326	748	109614
South Africa	852	128511	670	93803
Slovenia	455	67472	587	79727
Italy	675	95092	476	70599
Other Countries	2597	355351	1737	221974

Figures rounded off

Table – 11: Exports of Sulphur (Precipitated)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	++	5961	1	576
Gambia	--	--	1	476
USA	++	2	++	76
Mauritius	--	--	++	9
Bahrain	--	--	++	7
UAE	++	++	++	3
Lebanon	--	--	++	2
Turkey	--	--	++	2
Saudi Arabia	--	--	++	1
Sri Lanka	++	102	++	++
Russia	++	++	++	++
Other Countries	++	5857	++	++

Figures rounded off

Imports

Imports of sulphur (excluding sublimed, precipitated, and colloidal) increased by 28% to 17,08,664 tonnes in 2023-24 from 13,37,789 tonnes in the previous year. Imports were mainly from UAE (46%), Oman (15%), Qatar (11%), Kuwait (9%), Saudi Arabia (7%), Japan (4%), Russia and Singapore (3% each). Imports of sulphur (sublimed, precipitated, and

colloidal) decreased by 15% to 881 tonnes in 2023-24 from 1,036 tonnes in the previous year. Imports were mainly from Taiwan (70%), China (23%), Malaysia and Germany (2% each). Imports of sulphur (sublimed) and sulphur (precipitated) were at 843 tonnes and 19 tonnes respectively during the year 2023-24 (Table-12 to 16).

Table – 12: Imports of Sulphur Sulphur (exc. Sublimed Precipitated and Colloidal)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1337789	26766634	1708664	17365951
UAE	443186	7686919	785554	7888124
Oman	186816	4781107	248377	2446228
Qatar	431741	8485668	191013	2216791
Kuwait	91398	1828376	152711	1534800

Table 12 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Saudi Arabia	47976	1108595	126175	1222901
Japan	37999	1107462	74809	698732
Russia	51819	870827	55000	617267
Singapore	8144	107387	54169	513928
Bahrain	--	--	6459	74473
Iraq	12943	306763	4342	49027
Other Countries	25767	483530	10055	103680

Figures rounded off

In tonnes

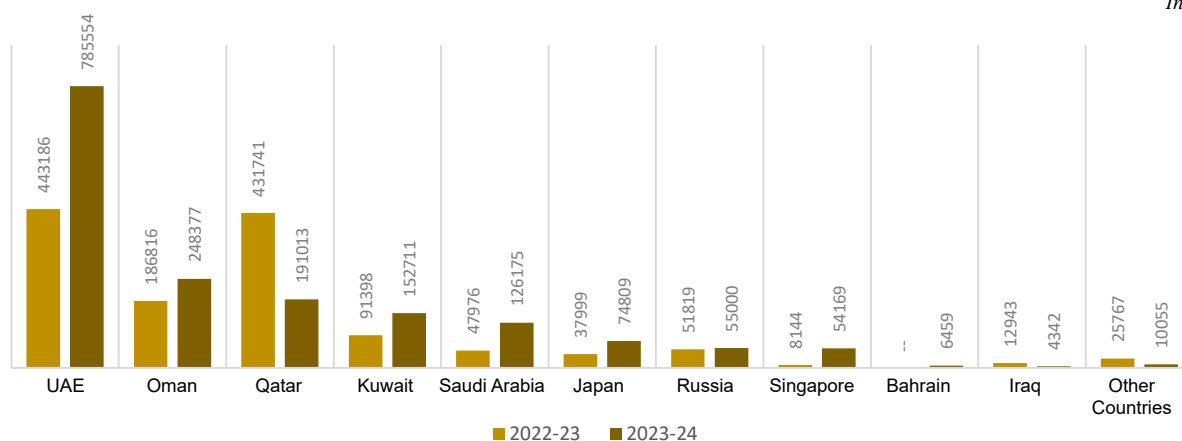


Fig. 3: Countrywise Imports of Sulphur (exc. Sublimed Precipitated and Colloidal)

Table – 13: Imports of Sulphur (Sublimed Precipitated And Collodial)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	1036	290426	881	270187
Taiwan	546	203812	615	229104
China	159	21130	207	22589
Netherlands	4	1812	11	5819
Belgium	7	3899	8	3247
Germany	37	8957	16	3134
Malaysia	256	40209	16	2639
Korea, Rep.of	8	3147	3	1352
Thailand	--	--	2	979
Japan	4	1071	3	699
USA	++	183	++	603
Other Countries	15	6206	++	22

Figures rounded off

Table – 14: Imports of Sulphur (Precipitated)

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	252	40331	19	3356
Germany	28	4141	16	2700
Japan	4	1018	3	656
Malaysia	220	35109	--	--
USA	++	63	--	--

Figures rounded off

Table – 15: Imports of Sulphur (Colloidal)**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	27	10921	19	9661
Netherlands	4	1812	11	5819
Belgium	7	3899	8	3247
USA	--	--	++	488
Japan	++	53	++	43
Germany	9	4732	++	42
UK	++	15	++	22
Malaysia	7	395	--	--
Bahrain	++	15	--	--

*Figures rounded off***Table – 16: Imports of Sulphur (Sublimed)****(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	757	239174	843	257170
Taiwan	546	203812	615	229104
China	159	21130	207	22589
Malaysia	29	4705	16	2639
Korea, Rep. of	8	3147	3	1352
Thailand	--	--	2	979
Germany	++	84	++	392
USA	++	120	++	115
Vietnam	15	6170	--	--
France	++	6	--	--

Figures rounded off

FUTURE OUTLOOK

Country is deficient in sulphur and pyrites which are essential for Fertilizer Industry. Recovered sulphur output was expected to increase significantly worldwide. Refineries in developing countries are anticipated to improve & adapt more efficient environmental protection measures and eventually match with the environmental standards & guidelines of plants in Japan, North America and Western Europe in future. Higher sulphur recovery is likely to result from several factors; higher refining rates, higher sulphur content in crude oil, lower allowable sulphur content in finished fuels and reduced sulphur emissions mandated by regulations.

World consumption of natural gas is expected to maintain strong growth, and sulphur recovery from that sector is likely to maintain an increasing trend. Some of the future gas production is expected to come from unconventional natural gas resources, such as, shale gas and coal-bed methane.

In the near term, increased global production and continued demand will keep the sulphur market balanced, which is expected to be followed in the long term by a surplus worldwide. International sulphur trade is expected to increase significantly, driven by demand for sulphuric acid in industrial sectors (particularly new ore-leaching operations) and a modest increase in demand for fertilizers.



53. Vermiculite

1,129 tonnes
production of vermiculite have been reported in 2023-24.

843 tonnes
of vermiculite were exported in 2023-24.

627 tonnes
of vermiculite were imported in 2023-24.

Vermiculite is a term applied commercially to a group of micaceous minerals, essentially hydrated silicates of Al, Mg and Fe that are typically alteration products of biotite or phlogopite micas. They are formed through the removal of alkalis and addition of water. Vermiculite differs from mica in its characteristic property of exfoliation. Crude vermiculite is therefore always exfoliated before use. Flakes of raw vermiculite concentrate are micaceous in appearance and contain interlayer water in their structure. When the flakes are heated rapidly at a temperature above 870° C, the water flashes into steam, and the flakes expand into accordion-like particles. This process is called exfoliation, or expansion, and the resulting lightweight material is chemically inert, fire resistant, and odourless. In lightweight plaster and concrete, vermiculite provides good thermal insulation. Vermiculite can absorb such liquids as fertilizers, herbicides, and insecticides, which can then be transported as free-flowing solids.

RESERVES/RESOURCES

The total reserves/resources of vermiculite as on 1.4.2020 as per NMI data, based on UNFC system has been estimated at 2.36 million tonnes, of which about 1.60 million tonnes (68%) are placed under the Reserves category and the remaining 0.77 million tonnes (32%) are placed under the

Remaining Resources category. Total resources are mainly located in Tamil Nadu which reported 1.86 million tonnes (79%) followed by Andhra Pradesh with 0.19 million tonnes (8%), Karnataka with 0.16 million tonnes (7%), Rajasthan with 0.10 million tonnes (4%) and Jharkhand with 0.03 million tonnes (1%). Nominal resources are located in Gujarat, Madhya Pradesh and West Bengal (Table-1).

Table – 1: Reserves/Resources of Vermiculite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India: Total	1590996	765227	2356223
By Grades			
Refractory	38752	807	39559
Unclassified	1552244	764420	2316664
By States			
Tamil Nadu	1516803	343051	1859854
Andhra Pradesh	74193	117908	192101
Karnataka	-	162240	162240
Rajasthan	-	104125	104125
Jharkhand	-	30048	30048
Gujarat	-	1960	1960
Madhya Pradesh	-	329	329
West Bengal	-	5566	5566

Figures rounded off

MINING LEASES & PRODUCTION

As on 31.03.2024 (Provisional), a total of 50 numbers of mining leases have been granted for mining of vermiculite, covering a total mining lease area of 890.9683 hectares. Production of vermiculite was at 1,129 tonnes in 2023-24 which decreased by 51% as compared to 2,303 tonnes in the year 2022-23. There were 10 reporting mines including two associated mines in 2023-24 as compared

to 9 reporting mines including three associate mines in the previous year. Andhra Pradesh is the leading producer of vermiculite in 2023-24, which accounted for 99.82% of the total production (Tables-2 to 4). Mine-head closing stocks of vermiculite at the end of the year 2023-24 were 11,139 tonnes as against 7,871 tonnes in the previous year (Table-5). The average daily employment of labour during this year was 74 as against 52 in the previous year.

Table – 2: Principal Producers of Vermiculite, 2023-24

Name & address of producer	Location of mine	
	State	District
T. Meenatchi Sundaram, Plot No. 2, Industrial Estate, Gudur P.O., (Mandal), SPSR Nellore - 524 101, Andhra Pradesh.	Andhra Pradesh	Nellore
Sadhana Minerals, 1-116, Masthanvali Complex, Anil Nagar, Bye Pass Road , Gudur, Chillakur, Post- Gudur Dist. Nellore-524412, Andhra Pradesh	Andhra Pradesh	Nellore
Modi Minerals 13/128,Gangammagudi Street, Gudur Mandal SPSR NELLORE 524101 Andhra Pradesh	Andhra Pradesh	Nellore
Eswara Enterprises 13/128,Gangammagudi Street,, GUDUR-(P.O), SPSR Nellore Dt,(A.P) SPSR Nellore 524101 Andhra Pradesh	Andhra Pradesh	Nellore

Table – 3: Production of Vermiculite, 2021-22 to 2023-24

(By States)

(Qty in tonnes; Value in ₹'000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	3060	3765	2303	1610.2	1129	781
Andhra Pradesh	2370	1481	2303	1610.2	1127	780
Karnataka	-	-	-	-	-	-
Tamil Nadu	690	2284	-	-	2	1

(P): Provisional

Table – 4: Production of Vermiculite, 2021-22 and 2022-23

(By Sectors/States/Districts)

(Qty in tonnes; Value in ₹'000)

State/District	No. of Mines	2022-23 Production		No. of Mines	2023-24 (P) Production	
		Quantity	Value		Quantity	Value
India	9	2303	1610	10	1129	781
Public Sector	1	-	-	1	2	1
Private Sector	8	2303	1610	9	1127	780
Andhra Pradesh	7	2303	1610	8	1127	780
Nellore	7	2303	1610	8	1127	780
Karnataka	1	-	-	1	-	-
Mysuru	1	-	-	1	-	-
Tamil Nadu	1	-	-	1	2	1
Vellore	1	-	-	1	2	1

Table – 5: Mine-head Closing Stocks of Vermiculite 2022-23 and 2023-24

(By States)

(Qty in tonnes)

State	2022-23	2023-24 (P)
India	7871	11139
Andhra Pradesh	2699	5965
Karnataka	-	-
Tamil Nadu	5172	5174

MINING AND INDUSTRY

In Andhra Pradesh, vermiculite is available in the district of Nellore, where three working mines have been reported during 2023-24. Vermiculite is mined and refined using a variety of techniques and supplied commercially in a range of particle-size grades of vermiculite concentrate. In Tamil Nadu, good-quality deposits of vermiculite are mostly found in the North Arcot regions, where mines remain non-working during the year. In Karnataka, vermiculite is found in the districts of Hassan, Mandya and Mysuru. Vermiculite, when heated to a high temperature, exfoliates and expands 8–14 times in volume, yielding exfoliated vermiculite by the loss of water molecules. The chemical composition shows average moisture content of 7.89%, loss on ignition of 11.05%, SiO₂ of 30.52%, Fe₂O₃ of 16.32%, and TiO₂ of 2.63%. The strong basal cleavage exhibits exfoliation at right angles which forms the basis for its commercial use. A change in colour is observed during the heating process, depending on the composition of the vermiculite and the furnace temperature.

USES & CONSUMPTION

Vermiculite is known for its horticultural applications. It is a common component in potting soils. Unfoliated (unexpanded) vermiculite has only minor uses, such as, for circulation in drilling muds and in the annealing of steel. In order to convert raw vermiculite into a product suitable for industrial use, it must be exfoliated or expanded by heating, a process termed 'exfoliation'. Vermiculite is chemically

inert, fireproof, non-conductor of electricity and a good insulator against heat (both radiant and conducted), cold and sound. Unlike cork and other organic lightweight insulating materials, it neither rots nor is attacked by vermin and has a fair mechanical strength.

Vermiculite is added to soil for conditioning and lightening either alone or in conjunction with peat or compost which facilitates the plant growth by enabling the absorption of the NPK nutrients and promoting anchorage for tender young root systems. It is also used as a carrier in fertilizers, herbicides and insecticides. Cementing mixtures of exfoliated vermiculite and binding agents, such as, gypsum and plaster, have been important industrial products and are applied to structural steel members in commercial buildings.

The mineral is used in various types of building boards and pollution control applications. Fine-sized, untreated vermiculite concentrates are included in the preparation of fireproof plaster boards. The exfoliated product forms the basis of some lightweight plasterboard, while ground, exfoliated vermiculite is used in various refractory board products. The principal uses of expanded vermiculite are based on its thermal insulating quality (due to presence of innumerable air cells), low-density, fire-proof nature and granular form. Larger vermiculite granules are used as a loose fill for thermal insulation for homes, industrial structures, cold storage, and refrigeration and high & low temperature industrial equipment.

Vermiculite is also used for refractory and high temperature insulation as it can withstand hot face temperature of 1,000°C, i.e., it can act as a thermal insulator.

The high absorbency and chemical inertness of exfoliated vermiculite have made it suitable for a wide range of absorbent packing materials as well as for packaged units used for the containment of oil and similar liquids. Like perlite, vermiculite is a mined mineral that is heated to yield a soilless ingredient of potting mixes. Unlike perlite, vermiculite absorbs and retains water and nutrients. This quality is desirable especially in container gardens because of their quicker evaporation rates compared to in-ground plantings. On heating, vermiculite swells up and exfoliates into thin sheets, expanding up to twenty times or more. The swelled-up vermiculite is very light and thus may be used as an ingredient in light cement or plaster. Vermiculite is also used in the Packing Industry and for the making of insulators against heat and sound. Its colour may be yellow, brown or blue. It is very soft and slippery like soap. Pure vermiculite is also used for plaster aggregates as fireproof insulating material in steel/concrete structures. It is widely used in lightweight insulating material for roof/ floor deck system, insulation fill for homes and commercial building, packaging material, bitumen-coated vermiculite screens, acoustic insulation, etc.

The apparent consumption of vermiculite has significantly decreased for the year 2023-24, to 913 tonnes, as against 3,606 tonnes in 2022-23.

SUBSTITUTES

Expanded perlite is a substitute for exfoliated vermiculite in lightweight concrete and plaster. Other denser but less

costly substitutes in these applications are expanded clay, shale, slag and slate. Alternate materials for loose-fill fireproofing insulation include fibreglass, perlite and slag wool. In agriculture, substitutes include bark and other plant materials, peat, perlite, sawdust and synthetic soil conditioners. Finely-ground pine bark also called "pine fines" is a principal ingredient of most potting mixes. As an alternative to vermiculite, pine fines also have water retentive qualities, especially in smaller particle sizes. Cotton gin waste comprises gin leavings, such as, stems, leaves and hulls. Cotton gin compost renders these waste products into a viable alternative to vermiculite, also because of its ability to increase water-holding qualities in mixes. Resembling sphagnum peat moss, coir is the finished product of ground coconut husks. The University of Arkansas Co-operative Extension Service reports that coir can retain up to nine times its weight in water.

POLICY

As per foreign trade policy 2015-20, the imports and exports of vermiculite (unexpanded) [ITC (HS) Code 25301010] and vermiculite insulation bricks [ITC (HS) Code 69029030] are allowed 'free'.

WORLD REVIEW

Significant deposits have been reported in Australia, China, Russia, Uganda and some other countries, but information on reserves and resources is derived from multiple sources and in most cases, it is not clear whether the numbers refer to vermiculite alone or to vermiculite along with other minerals, host rock and overburden or production data. The details of world reserves and production of vermiculite are reflected in Tables- 6 & 7.

Table – 6: World Reserves of Vermiculite
(By Principal Countries)

		(In '000 tonnes)
Country	Reserves	
World: Total (Rounded off)		NA
USA		25000
South Africa		14000
Turkey		11000
Brazil		6600
China		2900
India		1600

Source: USGS, Mineral Commodity Summaries, 2024

Table – 7: World Production of Vermiculite
(By Principal Countries)

		(In tonnes)		
Country	2021	2022	2023	
Country	2021	2022	2023	
South Africa	217109	171898	169797	
USA ^(a)	100000	100000	100000*	

Table- 7 (Concl'd.)

Country	(In tonnes)		
	2021	2022	2023
Brazil	50000	50000	50000*
China	39000	39000	39000*
Russia	30000 *	29000	29000*
Zimbabwe	27713	26380	28276
Uganda	21754	22947	23000*
Turkey	3300	21230	10500
Bulgaria	10000 *	10000	10000*
Uzbekistan	2000 *	3000*	3000*
India	3061 ^(b)	2091	1537
Kenya	400 *	400	400*
Mexico	-	-	76

"Source: BGS, World Mineral Production, 2019-23

* Estimated

(a): Sold or used by producers

(b): Years ended 31st March following that stated"

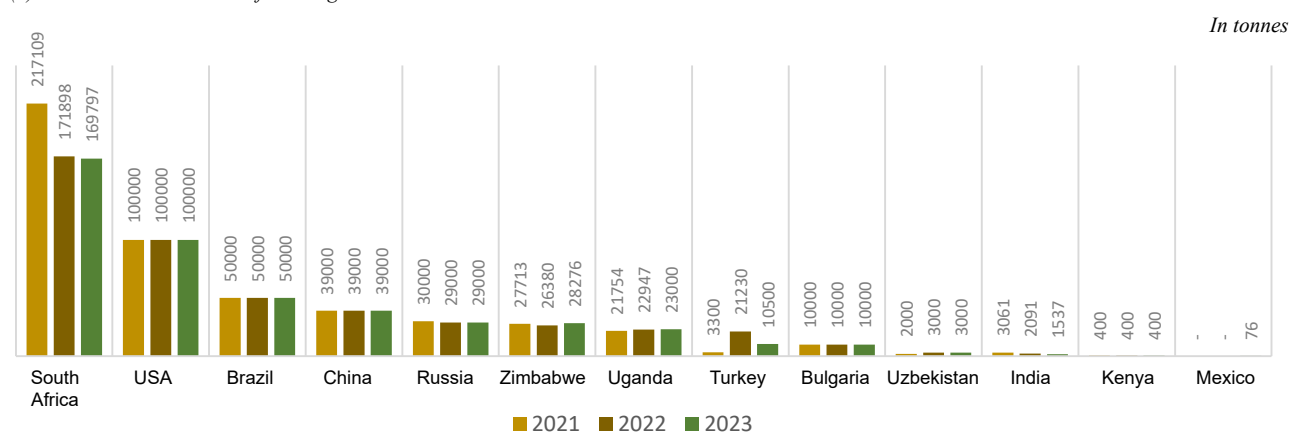


Fig. 1: Countrywise Production of Vermiculite

FOREIGN TRADE

Exports

Exports of vermiculite decreased by 14.76% to 843 tonnes in 2023-24 as compared to 989 tonnes in 2022-23. Exports were mainly to UAE (53%) and Japan (37%) during 2023-24. During 2022-23, Nepal was in first place

with 42% share while this year it has only 0.1% share in the list. While Japan has maintained the same ranking when compared to 2022-23 (Table-8). In 2022-23, Nepal ranked first with a 42% share, whereas in 2023-24 its share declined sharply to 0.1%. Japan maintained the same ranking as in the previous year.

Table – 8: Exports of Vermiculite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	989	12888	843	17551
UAE	175	3601	451	10239
Japan	338	5634	315	5930
Taiwan	--	--	24	525
Slovenia	--	--	25	461
Jordan	--	--	25	246
Sri Lanka	--	--	2	68
Nepal	418	2250	1	40
Maldives	--	--	++	33
Bhutan	++	22	++	9
Norway	48	1083	--	--
Other Countries	10	298	++	++

Figures rounded off

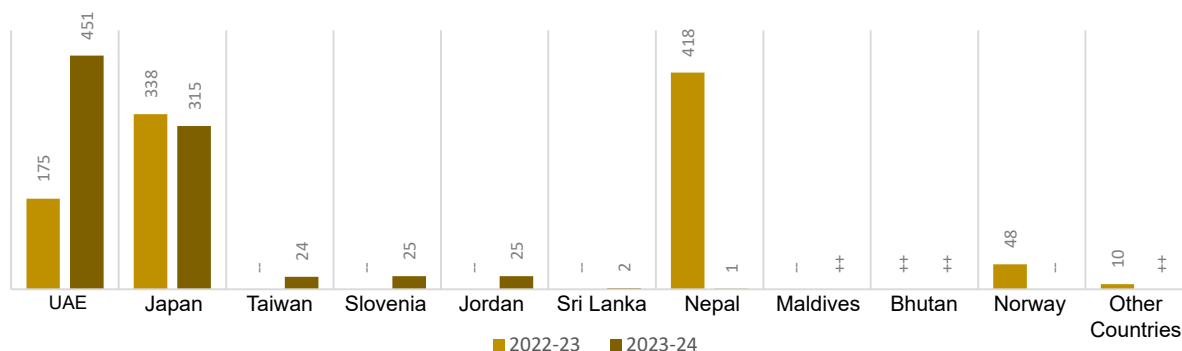


Fig. 2: Countrywise Exports of Vermiculite

Imports

Imports of vermiculite decreased significantly by 72.64% to 627 tonnes in 2023-24 from 2,292 tonnes in 2022-

23. Imports were mainly from Kenya (80.70%), South Africa (19.14%) and Uganda (0.16%) for the year 2023-24 (Table-9).

Table – 9: Imports of Vermiculite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	2292	73873	627	19588
Kenya	1012	26931	506	14339
South Africa	224	7767	120	5233
Uganda	--	--	1	16
Mozambique	740	26237	--	--
Brazil	315	12856	--	--
Iran	1	79	--	--
Turkey	++	3	--	--

Figures rounded off

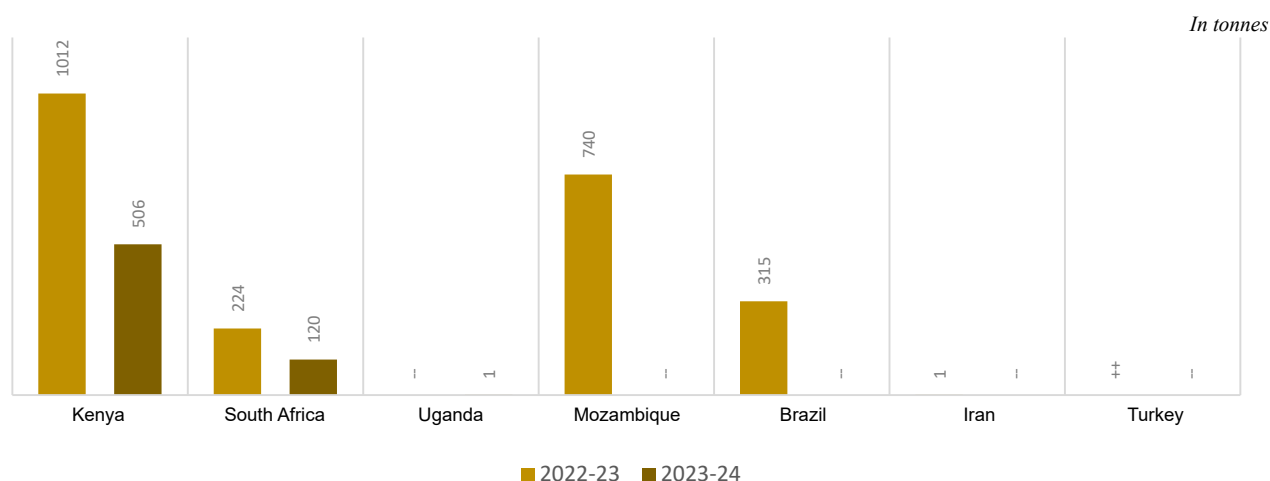


Fig. 3: Countrywise Imports of Vermiculite

FUTURE OUTLOOK

Global vermiculite (fine, medium and coarse grade) demand has increased over time due to a number of factors, such as industrial applications, building activity, economic conditions, and developments in agriculture. The increased demand for these larger grades is projected to drive exploration and development of vermiculite resources particularly with medium, large and premium

(coarser) grades, primarily in China and South Africa. It is anticipated that operations in Brazil and the United States are expected to help maintain regional and global supplies of fine, superfine and micron grades. Innovative applications have continued to emerge including the use of fine to micron-sized grades of vermiculite to combat air pollution and absorb water in mines, replacing zeolites in ion-exchange columns, purifying waste water and for containing or removing nuclear waste.



54. Wollastonite

1,10,596 tonnes

Production of wollastonite have been reported in 2023-24.

12,285 tonnes

of wollastonite were exported in 2023-24.

62,410 tonnes

of wollastonite were imported in 2023-24 .

Wollastonite is a chemically simple mineral named in honour of English Mineralogist and Chemist Sir W.H.Wollaston. Wollastonite is composed of calcium and silica with a chemical formula CaSiO_3 . Wollastonite may contain impurities like iron, potassium, manganese, etc. Though normally wollastonite is bright white in colour, the impurities can produce grey, cream, brown or red colour in it. Wollastonite is formed when limestone/dolomite is subjected to high temperature and pressure in the presence of silica-bearing fluid as in skarn deposits or metamorphic rocks. It occurs as aggregates of bladed or needle-like crystals with hardness of 4.5 to 5 on Mohs scale. The uses of wollastonite in applications other than as filler include marine wallboard, paint, plastic, in refractory liners in steel mills and as a partial replacement for short-fibre asbestos in certain applications.

RESERVES/RESOURCES

Major deposits of wollastonite have been found in Ajmer, Dungarpur, Pali, Sirohi and Udaipur districts of Rajasthan. Besides, in Ghoda area, Banaskantha district in Gujarat and in Dharmapuri and Tirunelveli districts in Tamil Nadu, occurrences of a few deposits have been reported. As on 1.4.2020, the reserves/resources of wollastonite, as per NMI database, based on UNFC system are placed at 25.11 million tonnes of which Reserves under Proved and

Probable categories together constitute 2.68 million tonnes (11%) and Remaining Resources constitute for the balance 22.43 million tonnes (89%). Out of the total resources, about 92% (23.11 million tonnes) including 2.68 million tonnes reserves are located in Rajasthan and the remaining about 8% resources (1.99 million tonnes) in Gujarat. Meagre resources are also located in Tamil Nadu (3,533 tonnes) (Table-1).

Table – 1: Reserves/Resources of Wollastonite as on 01.04.2020

(By Grades/States)

(In tonnes)

Grade/State	Total Reserves (A)	Remaining Resources (B)	Total Resources (A+B)
All India : Total	2680978	22427488	25108466
By Grades			
Marketable	1790818	9161800	10952618
Unclassified		10662900	11302845
Not-known	250215	2602788	2853003
By States			
Gujarat	0	1990000	1990000
Rajasthan	2680978	20433955	23114933
Tamil Nadu	0	3533	3533

Fig.ures rounded off

MINING LEASES & PRODUCTION

As per information available in the Mining Lease Directory (as on 31.03.2023), the total number of Mining Leases granted for wollastonite is 13 covering an overall mining lease area of 258.69 hectares.

Production of wollastonite at 1,10,596 tonnes during 2023-24 decreased by less than 1% as compared to 1,10,793 tonnes in the preceding year. The major producers of

wollastonite are located in Ajmer and Udaipur districts of Rajasthan. Provisionally, there were three reporting mines in 2023-24, as compared to five in the previous year. The entire production was reported only from Private Sector mines located in the State of Rajasthan (Table -2 to 4). The mine-head closing stocks at the end of 2023-24 were 1,65,968 tonnes as against 1,57,198 in 2022-23 (Table-5). The average daily labour employed in wollastonite mines during 2023-24 was 239 against 222 in the previous year.

Table – 2: Producers of Wollastonite, 2023-24

Name & address of producer	Location of the mine	
	State	District
Wolkem Industries Ltd., P.B. 21, E-101, Mewar Industrial Area, Madri, Udaipur – 313 003, Rajasthan.	Rajasthan	Udaipur
Renu Atre, C- 378, Pradhan Marg, Malviya Nagar, Jaipur – 302 017, Rajasthan.	Rajasthan	Ajmer

Table – 3: Production of Wollastonite, 2021-22 to 2023-24

(By State)

(Qty. in tonnes; Value in ₹ '000)

State	2021-22		2022-23		2023-24 (P)	
	Quantity	Value	Quantity	Value	Quantity	Value
India	108335	149156	110793	151761.448	110596	156107.612
Rajasthan	108335	149156	110793	151761.448	110596	156107.612

(P) : Provisional

(In tonnes)

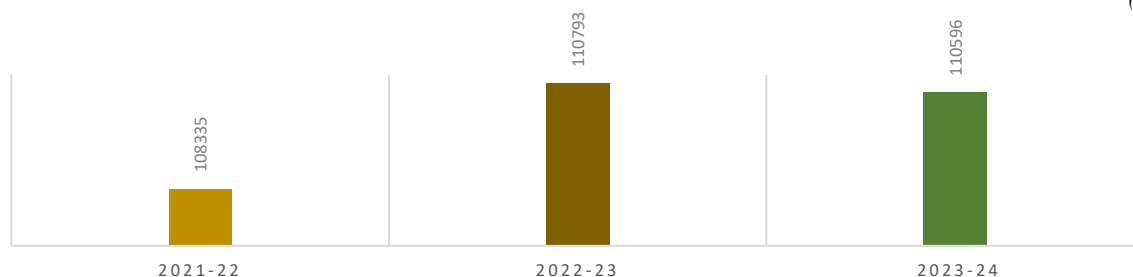


Fig. 1: Production of Wollastonite in India

Table – 4: Production of Wollastonite, 2022-23 and 2023-24**(By Sector/State/Districts)***(Qty. in tonnes; Value in ₹ '000)*

State/District	2022-23			2023-24 (P)		
	No. of mines	Quantity	Value	No. of mines	Quantity	Value
India	5	110793	151761	3	110596	156108
Private Sector	5	110793	151761	3	110596	156108
Rajasthan	5	110793	151761	3	110596	156108
Ajmer	3	8326	5234	1	7610	4870
Pali	1	0	0	1	0	0
Udaipur	1	102467	146527	1	102986	151238

*(P) : Provisional***Table –5: Mine-head closing stocks of Wollastonite, 2022-23 and 2023-24****(By State)***(In tonnes)*

State	2022-23	2023-24 (P)
India	157198	165968
Rajasthan	157198	165968

(P) : Provisional

USES & CONSUMPTION

The use of wollastonite depends on the acicularity or the aspect ratio, i.e., ratio between length and width of a crystal, chemical composition, brightness and fibre length. Wollastonite having aspect ratio in the range from 3:1 to 5:1 has little potential for reinforcing applications. Hence, market is primarily confined to ceramic, metallurgical fluxes and simple filler and coating applications. Wollastonite reduces the volume of the expensive plastic or resin medium and contributes to physical and chemical properties of the finished products. It improves tear strength, dielectric properties and retains mechanical properties at elevated temperatures. Wollastonite is used primarily in automobile brakes, ceramics, metallurgical processing, paper, paint, plastic, cosmetics, adhesives and as a replacement of asbestos in asbestos-cement boards and sheets. Some of the properties that make it so useful are high brightness & whiteness, low moisture & oil absorption, low volatile content and the acicular nature of some wollastonite. A better compatibility between the polymer and the filler is achieved by chemical surface treatment of the mineral filler. Wollastonite results improved flexural modules in polypropylene and improved reinforcement in nylon. It

is also used as performance additive in a wide range of construction material (concrete, stucco and adhesives). Bulk of the demand for wollastonite in the country mainly lies in the Ceramic Industry for the manufacture of floor and wall tiles. In ceramics, wollastonite decreases shrinkage and gas evolution during firing. Small quantities are used in asbestos-cement products as a partial replacement for short fibre asbestos, paint, insecticide, marine wallboard and welding rod industries. In metallurgical applications, wollastonite serves as a flux for welding, a source for calcium oxide, as slag conditioners and to protect the source of molten metal during the continuous casting of steel. The addition of wollastonite to metallurgical fluxes provides ready fusibility, good insulating qualities and low viscosity. A new and promising development is the ability of wollastonite as a sequestration mineral for carbon dioxide, a major contributor in global warming. In contrast to alternative approaches, sequestration by wollastonite is permanent and it produces a mixture of precipitated calcium carbonate and silica that may have filler applications in paper, plastics and rubber. The estimated consumption of wollastonite in 2023–24 was 1,60,721 tonnes, an increase of 15.3% as compared to 1,39,382 tonnes in 2022–23 (Table-6).

Table – 6: Estimated Consumption of Wollastonite, 2021-22 to 2023-24**(All Industries)***(Qty. in tonnes)*

Industry	2021-22	2022-23	2023-24 (P)
All Industries	127255	139382	160721

*Fig.ures rounded off**(P): Provisional*

SUBSTITUTE

The acicular nature of many wollastonite products allow it to compete with other acicular materials, such as, ceramic fibre, glass fibre, steel fibre and several organic fibres, such as, aramid, polyethylene, polypropylene, and polytetrafluoroethylene in products where improvements in dimensional stability, flexural modulus and heat deflection are sought. Wollastonite also competes with several non-fibrous minerals or rocks, such as, kaolin, mica & talc, which are added to plastics to increase flexural strength and such minerals as baryte, calcium carbonate, gypsum and talc, which impart dimensional stability to plastics. In ceramics, wollastonite competes with carbonates, feldspar, lime and silica as a source of calcium and silica. Its use in ceramics depends on the formulation of the ceramic body and the fixing method.

WORLD REVIEW

World resources of wollastonite are estimated to exceed 100 million tonnes. Many deposits have been identified but have not been surveyed sufficiently to estimate their reserves. The large deposits of wollastonite have been identified in China, Finland, Mexico and the United States. Smaller but significant deposits were in Canada, Chile, Kenya, Namibia, South Africa, Spain, Sudan, Tajikistan, Turkey and Uzbekistan. In 2023, the global sales of wollastonite were estimated to be in the range of 8,00,000 to 1,000,000 tonnes, the same as those in 2022. In 2023, China was the largest producer of wollastonite with a production of 900 thousand tonnes. Mexico with 155 thousand tonnes, India with 115 thousand tonnes and USA with 50 thousand tonnes were the other major producers. In addition, small quantities of wollastonite were also produced in Spain, Australia and Finland. India stands at 3rd position globally in production of wollastonite in 2023-24. The country-wise production of wollastonite by principal countries from 2021-2023 is furnished in Table-7.

Table – 7: World Production of Wollastonite
(By Principal Countries)

Country	In tonnes		
	2021	2022	2023
China	890000*	900000*	900000*
Mexico	200588	162410	154934
India	108383 ^(a)	104294	115458
USA	50000*	50000*	50000*
Spain	17246	18741	23456
Finland	11000*	11000*	11000*
Australia ^(b)	4495	2000	2000

Source: BGS, World Mineral Production, 2019-2023

Note(s)

(1) In addition to the countries listed, Canada also produces wollastonite (since 2014).

(a) Years ended 31 March following that stated

(b) Years ended 30 June of that stated

* estimated

FOREIGN TRADE

Exports

In 2023-24, exports of wollastonite decreased marginally by 4.37% to 12,285 tonnes from 12,847 tonnes in the previous year. Exports were mainly to Belgium (52%), Hungary (22%), Germany (10%), Japan (8%), UK & Bangladesh (2% each) and Saudi Arabia, Turkey & France (1% each) (Table-8).

Table – 8: Exports of Wollastonite
By Countries

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	12847	365656	12285	348673
Belgium	8093	246137	6376	195031
Hungary	1452	32927	2662	65351
Germany	1250	29674	1231	29685
Japan	812	19376	965	25038
UK	240	7661	272	9928

Table- 8 (Concl.d.)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Bangladesh	163	5218	191	5673
France	180	5866	99	3757
Mexico	82	4468	67	3720
Turkey	127	3544	100	2996
Saudi Arabia	101	871	172	2080
Other Countries	347	9914	150	5414

Fig.ures rounded off

R: Revised

P: Provisional

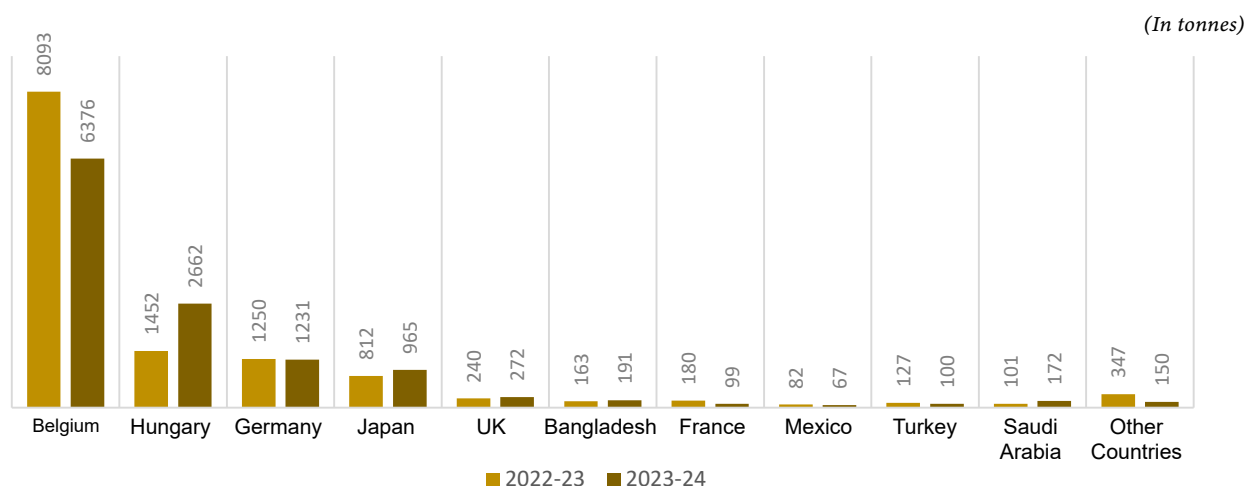


Fig. 2: Countrywise Exports of Wollastonite

Imports

Imports of wollastonite in 2023-24 increased substantially by 50.62%, reaching 62,410 tonnes as compared to 41,436

tonnes in the previous year. Imports were almost entirely from China (99%) and the remaining 1% were from Mexico, Canada, Hong Kong and other countries (Table-9).

Table – 9: Imports of Wollastonite

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	41436	877192	62410	1037583
China	41091	837980	61840	986306
Mexico	112	10396	220	17557
Canada	60	4851	160	12258
Germany	20	3423	23	6146
Belgium	37	6516	22	5121
USA	94	9130	45	4978
Netherlands	19	3023	20	3883
Hong Kong	1	456	79	1129
Korea, Rep. of	--	--	1	91
Japan	1	406	++	45
Other Countries	1	1011	++	69

Fig.ures rounded off

R: Revised

P: Provisional

++ Negligible

-- Nil

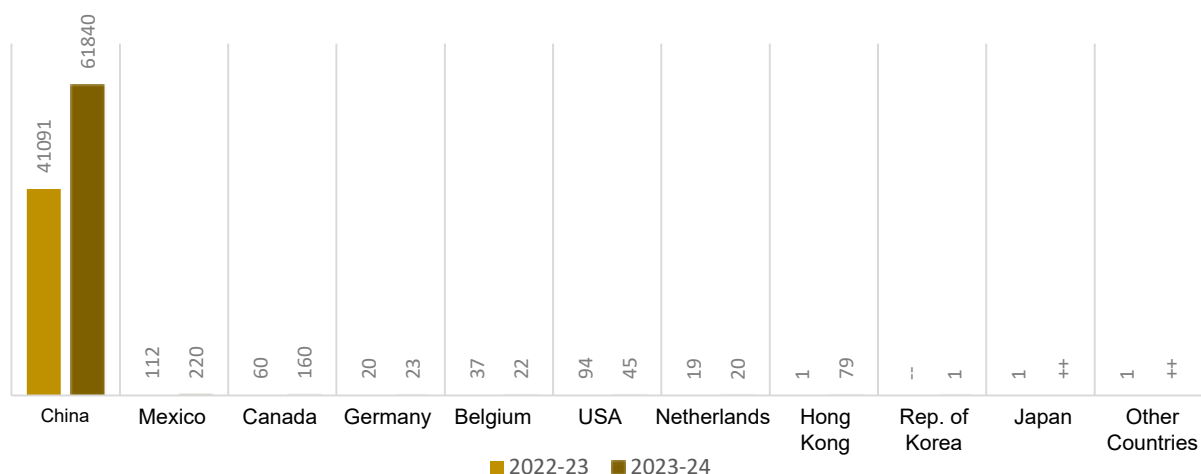


Fig. 3: Countrywise Imports of Wollastonite

FUTURE OUTLOOK

Presently, India is world's third largest producer of wollastonite after China and Mexico. The existing mines in the country are, to some extent, able to meet the domestic requirements of the ceramic industry, supplemented by some imports. There is an increasing demand for wollastonite in the international markets, in industries, such as, ceramic, metallurgy, paint, construction and as a

asbestos substitute. The exports of processed wollastonite with high aspect ratio and powdered wollastonite may have to be encouraged for the betterment of export of value-added products. As a result of augmentation of resources of wollastonite in the States of Tamil Nadu and Gujarat, India would end up being in a formidable position and would be in a position to cope with any futuristic demand.



55. Zircon

198 tonnes

of zirconium ores and concentrates were exported in 2023-24.

82,257 tonnes

of zirconium ores and concentrates were imported in 2023-24 .

Zirconium is the twentieth most abundant element in the Earth's crust and occurs in a variety of rock types and geological environments, most often, in the form of zircon (ZrSiO_4). It is present as a constituent in heavy mineral sand assemblages, which include ilmenite, rutile, leucoxene, monazite and garnet in varying proportions. Zircon sand and baddeleyite (an oxide- ZrO_2) are used through their salts to extract zirconium and hafnium. Normally, all zirconium compounds contain between 1.4% and 3% hafnium. Zircon is very stable at high temperature and has excellent thermal shock resistance, low thermal conductivity and chemical inertness. It is mainly used in industries like ceramic, refractory, abrasive, foundry, chemical and specialty alloys. Gem variety of zircon is used in jewellery. Ministry of Mines vide its Notification No. S.O. 2356 (E) dated 11.07.2016 inserted Zircon (covered under beach sand minerals) as entry 12 of Part B of the First Schedule to the MMDR Act, 1957. However, later on, as per (1) Mines and Minerals (Development and Regulation) Amendment Act, 2023 (vide notification no. 16 of 2023 dated 09.08.2023) and (2) Atomic Minerals Concession (Amendment) Rules, 2024 (vide notification no. 97 dated 14.02.2024), six minerals, namely, (i) Beryl and other beryllium-bearing minerals (ii) Lithium-bearing minerals, (iii) Niobium-bearing minerals, (iv) Titanium bearing minerals and ores, (v) Tantalum-bearing minerals and (vi) Zirconium-bearing minerals and ores have been delisted from the list of atomic minerals specified in Part-B of the First Schedule to the Act.

RESERVES/RESOURCES

Zircon occurs in close association with other heavy minerals, such as, ilmenite, rutile and monazite in beach sands, along the coastal tracts of the country. Its concentration in the deposits is about 0.6–18.7% of the total heavy minerals. Indian zircons analyse 63–66% ZrO_2 . Atomic Minerals Directorate for Exploration and Research (AMD) has carried out reconnaissance investigation for zircon in various parts of Gujarat, Maharashtra, Karnataka, Andhra Pradesh, Tamil Nadu, Kerala, Odisha and West Bengal. As per the latest information available from the Department of

Atomic Energy (as of December 2024), one of its constituent unit, AMD, has established a total of 1,309.42 million tonnes (Mt) of beach sand mineral resources (including inland placers). These resources comprise 13.15 Mt of monazite, 706.24 Mt of ilmenite, 278.48 Mt of sillimanite, 217.83 Mt of garnet, 38 Mt of zircon, 35.98 Mt of rutile and 19.75 Mt of leucoxene. However, the latest state-wise break-up is not available so far. The state-wise breakup of the 33.71 million tonnes of resources of zircon up to 2016, is furnished in Table-1

Table – 1: Resources of Zircon

(In million tonnes)

State	Resources [#]
Total	33.71
Andhra Pradesh	11.94
Bihar/Jharkhand	0.08
Gujarat	0.01
Kerala	7.83
Maharashtra	0.01
Odisha	3.25
Tamil Nadu	10.2
West Bengal	0.39

Source: Department of Atomic Energy, Mumbai.

: Inclusive of indicated, inferred and speculative categories.

As per letter received from Department of Atomic Energy dated 26.07.2018.

MINING LEASES AND PRODUCTION

Production of zircon increased to 15,600 tonnes in 2019-20 from 11,906 tonnes in the preceding year. The production of zircon is provided in Table-2. Prices of zircon, as furnished by Indian Rare Earths Limited (IREL) and Kerala Minerals and Metals Limited (KMML), are detailed in Table- 3.

Due to strategic reasons, the mining leases for zircon in the country were hitherto granted only to the government company, namely, IREL and the state government undertaking, namely, KMML. IREL has mining lease areas in the Neendakara–Kayamkulam (NK) belt of Kerala whereas KMML has mining lease areas in the Neendakara–Kayamkulam (NK) belt of Kerala. As per the Annual Report 2023-24 of IREL, zircon produced by IREL fulfills the entire need of the Nuclear Power Sector in India along with meeting the domestic requirement of the Ceramics, Refractories and Foundry sector.

**Table – 2: Production of Zircon
2017-18 to 2019-20**

(In tonnes)

Year	Production of Zircon*
2017-18	9107
2018-19	11906
2019-20	15600

*As reported by Indian Rare Earths Ltd.

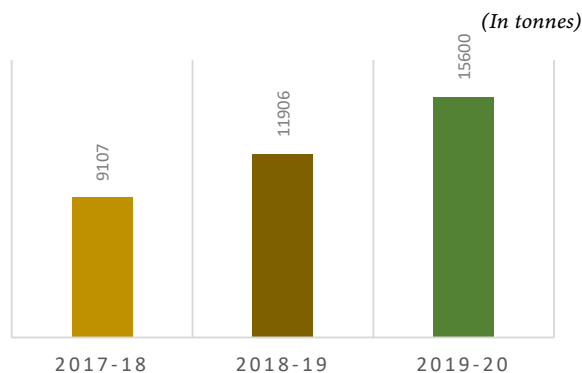


Fig. 1: Production of Zircon in India

**Table – 3: Prices of Zircon,
2017-18 to 2019-20**

(₹ per tonne)

Period	Grade	Price	Remarks
IREL			
2017-18	Q	70,215	-
	MK	-	-
	OR	61,638	-
2018-19	Q	1,05,245	-
	MK	1,03,099	-
	OR	90,149	-
2019-20	Q	1,06,790	-
	MK	1,07,032	-
	OR	95,297	-
KMML			
2017-18	Zircon Gr.I	85,167	Average
	Zircon Gr.II	80,167	Average
	Zircon Gr.III	76,167	Average
	Zircon Gr.IV	Nil	
2018-19	Zircon Gr.I	1,09,750	Average
	Zircon Gr.II	1,04,750	Average
	Zircon Gr.III	1,00,750	Average
	Zircon Gr.IV	98,500	Average
2019-20	Zircon Gr.I	1,15,000	Average
	Zircon Gr.II	1,10,000	Average
	Zircon Gr.III	1,06,000	Average
	Zircon Gr.IV	1,02,000	Average

Source: Department of Atomic Energy, Mumbai.

INDUSTRY

IREL has set up a dry grinding mill at Chavara, Kerala to produce Zirflour for its application in the Ceramic Industry. A wet grinding mill was also set up at Chavara to produce micro-zir for its specialised application as opacifier. Besides, IREL established a small chemical plant at Manavalakurichi, Tamil Nadu to produce zircon frit, zirconium chloride, etc.,

primarily for supply of zircon frit to Department of Atomic Energy's Nuclear Fuel Complex (NFC), Hyderabad. A pilot plant (3.5 tpy capacity) was set up at Orissa Sand Complex (OSCOM) to produce a whole range of zirconia stabilised with CaO, MgO and rare earths.

The NFC manufactures and supplies fuel bundles for Pressurised Heavy Water Reactors (PHWRs) and Boiling Water Reactors (BWRs) of NPCIL. The NFC, Hyderabad, has different types of production facilities which include the zirconium oxide plant for processing of zircon to pure zirconium oxide and zirconium sponge plant for conversion of zirconium oxide to pure sponge metal and Zirconium Complex (ZC) at Pazhayakayal, near Thoothukudi, Tamil Nadu. NFC Hyderabad produced 585 tonnes, 470 tonnes and 605 tonnes of zirconium oxide during 2019-20, 2020-21 and 2021-22, respectively. ZC, Pazhayakayal, produced 454 tonnes, 408 tonnes and 535 tonnes of ZrO₂ in 2019-20, 2020-21 and 2021-22, respectively. Besides, the Zircaloy Fabrication Plant produces various zirconium alloy tubings and sheet, rod and wire products. The plant also

has facilities for reclamation of zircaloy mill-scrap. Zircon sand is processed through caustic fusion, dissolution, solvent extraction (to remove hafnium), precipitation and calcination to obtain zirconium oxide. The pure oxide is then subjected to high temperature chlorination, reactive metal reduction and vacuum distillation to obtain homogeneous zirconium sponge. The sponge is briquetted with alloying ingredients and melted in vacuum to produce zircaloy ingots. The alloy ingots are extruded to convert into seamless tubes, sheets and bars. The total installed capacity and production of zirconium oxide and zirconium sponge plants at NFC and ZC are furnished in Tables 4 & 5.

Besides, Bhalla Chemical Works Pvt. Ltd operates three plants—two of which are located in Gurgaon, Haryana, to manufacture zirconium derivatives (ZrO₂) based on imported zircon ore (capacity 10,000 tpy) and zirconium silicate opacifiers (capacity 5,000 tpy). The third plant of the Company in Rajasthan manufactures zirconium oxychloride crystals and special zirconias (capacity 10,000 tpy).

Table – 4: Plant-wise Capacity and Production of Zircon Ore, 2017-18 to 2019-20
(By States)

Company	Location	Specification	Installed capacity** (tpy)	Production (In tonnes)		
				2017-18	2018-19	2019-20
Total			39000	13951	-	
Indian Rare Earths Ltd	Manavalakurichi, Kanniyakumari distt, Tamil Nadu	65% ZrO ₂ +HfO ₂ (min)	10000	*	2190	
	Chavara, Kollam distt, Kerala	65% ZrO ₂ +HfO ₂ (min.)	17500	2649	3072	11490
	Orissa Sand Complex, Ganjam distt, Odisha	64.25% ZrO ₂ (min)	5000	6458	6694	
Kerala Minerals & Metals Ltd	Chavara, Kollam distt, Kerala	Zircon Gr.I 64.0% (min.) Zircon Gr.II 62% (min.)	6500	4844	4762	4110

Source: Respective Producers and Department of Atomic Energy, Mumbai.

* During 2017-18 plant was not operating from Jan. 17 because of non-availability of environment clearance.

** Excluding the V.V.Mineral's installed capacity of 18,000 tpy vide their letter and e-mail dated 09.09.2019 regarding non-operation of mining.

Table – 5: Production of Zirconium Oxide and Sponge Plants of DAE at NFC and ZC, 2019-20 to 2021-22

Company	Installed capacity** (tpy)	Production (In tonnes)		
		2019-20	2020-21	2021-22
Zirconium Oxide Plant, NFC, Hyderabad	600	585	470	605
Zirconium Sponge Plant, NFC, Hyderabad	400	-	-	-
Zirconium Oxide Plant, ZC, Pazhayakayal	500	454	408	535
Zirconium Sponge Plant, ZC, Pazhayakayal	250	-	-	-

Source: Department of Atomic Energy, Mumbai.

USES & CONSUMPTION

Zircon's exceptional qualities of hardness and durability makes it a must-use for the manufacture of ceramics and refractory tiles and also for a range of other high-tech applications, such as, armour plating on military aircraft, heat shield in space shuttles, potentially as solid oxide fuel

cells in hydrogen- powered vehicles and in many industrial & chemical applications. Owing to its chemical inertness, very low heat conductivity, high specific gravity, low expansion, good resistance to abrasion, high melting point and no shrinkage on being heated up to 17,500 °C, zircon is found to be an outstanding refractory material. Zircon finds its application in ceramics, zirconia, chemicals, refractory and

foundry & castings which accounts for zircon's total world estimated consumption. Zirconia and Zirconium chemicals can be used for a variety of uses. Yttria-stabilised zirconia (YSZ) is used in the manufacture of oxygen sensors that control combustion in automobile engines and furnaces. In Foundry Industry, zircon is used as facing for foundry moulds as it increases the resistance to metal penetration and accords a uniform finish to castings. Zircon sand is preferred to silica sand because of its uniform size, higher melting point, low thermal expansion and resistance to molten metal, acidic chemicals, slag, etc. Zircon containing 64% ZrO_2 is used generally for foundry applications. In Ceramic Industry, finely ground high-grade zircon and zirconium dioxide are used as opacifier in melts for vitreous enamelling and as pigment in ceramic glazes. Zirconium oxide is considered as a potential ceramic material for high temperature applications like engine components. Usually, zircon containing 65% ZrO_2 is preferred in ceramics. The toughened zirconia finds its use in ceramic coatings in jet aircraft engines and in other applications where strength and high temperature oxidation resistance are important. Zirconia ceramics are also used in automobile sensors for the microprocessor control of engines. In Chemical Industry, its property of high resistance to corrosion is used where dry chlorine, hydrochloric acid and caustic alkalis are involved. Abrasive and grinding wheels made from zircon sands are used for polishing optical glasses. Zircon powder is used as a medium in waterjet cutting machines.

Zirconium and zirconium powders are used in ammunition, primers, detonation caps, flashlight mixtures, radio tubes and in various heating elements. Hafnium-free zirconium metal is used as cladding material in atomic reactors due to its low absorbing cross section for thermal neutron. Green, blue, indigo, red, orange coloured zircon is used as a natural gemstone and also processed to produce cubic zirconia - a synthetic gemstone resembling diamond. Zircon compounds have a very low toxicity and are not perceived as a potential environmental hazard. They are even said to have some medicinal properties and are now increasingly preferred in the manufacture of food products and pharmaceuticals too. It is also widely used in television and computer screens, resistance to corrosion and erosion makes zircon products ideal for use in the Chemical Industry and in desalination plants. Zircon flour is manufactured by milling zircon sand. It is used in ceramic frits, foundry mould coatings, ceramic shells for investment casting, refractories, friction products, insulating fibres and glass. Zircon opacifier is used in refractories and friction products. Zirconium metal or sponge is used mainly in the Nuclear Industry with a requirement for minimum content of hafnium.

Consumption of zircon/zirflour decreased considerably by 17% to 13,829 tonnes in 2019-20 from 16,600 tonnes in 2018-19. The consumption of zircon during the year 2019-20 was reported only by IREL. Consumption of zircon/zirflour during the year 2017-18 to 2019-20 is furnished in Table- 6.

**Table – 6: Estimated Consumption* of Zircon
2017-18 to 2019-20
(By Industries)**

Industry	(In tonnes)		
	2017-18	2018-19 (R)	2019-20 (P)
All Industries	17500	16600	13829[#]
Ceramic	7200	6300	3963
Refractory	9700	9400	1153
Others (alloy steel, iron & steel, chemical, foundry, paint and cement)	600	900	8751

Figures rounded off.

* : Includes actual reported consumption and/or estimates made wherever required. Due to paucity of data, coverage may not be complete.

[#]: only reported by IREL.

RESEARCH & DEVELOPMENT

Various R&D studies have been conducted by R&D Laboratory, OSCOM and Department of Atomic Energy to improve the metallurgical performance of mining and mineral separation plants. As a result, throughput of the plants has increased and the grade and recovery of heavy minerals in concentrate has been enhanced. The two most outstanding contributions of this laboratory in the areas of value addition are development of new process flow sheets for the preparation of synthetic rutile from OSCOM ilmenite and zirconia from zircon.

IREL Technology Development Council (IRELTDC) has been formed with an objective of promoting industrial scale R&D that would be beneficial to the overall programme of DAE in both strategic and non-strategic fields utilising mineral & value added products of IREL.

During 2023-24, IREL carried out an in-house R&D project on “Upgradation of escaped Monazite in Hydrosizer overflow fraction at Monazite Upgradation Section (MUS) by using additional hydrosizer”. As defined, the hydrosizer overflow consists of around 30-31% Monazite & 19-20%

Zircon and hence, it is reprocessed in Heavy Upgradation Section (HUS). In order to recover the heavy mineral values in the MUS circuit, the hydrosizer overflow was collected and passed through lab model hydrosizer for finalising the parameters. Around 94% Monazite of 60% grade was recovered in the underflow. Additionally, 85-86% of zircon was recovered in overflow along with other minerals.

POLICY

Zircon was earlier classified as a 'prescribed substance', as per notifications issued under Atomic Energy Act, 1962. From the revised list notified vide S. O. No. 61(E), dated 20.1.2006, zircon was deleted, subject to the condition that the mineral shall remain a prescribed substance till the policy on exploration of beach sand minerals notified on 6.10.1998, is adopted/revised/modified by Ministry of Mines or till 1.1.2007, whichever occurs earlier and shall cease to be so thereafter. Ministry of Mines, vide Notification No. S.O. 2356 (E) dated 11.07.2016, inserted zircon, covered under beach sand minerals, as entry 12 after entry 11 of Part B of the First Schedule to the MMDR Act, 1957. As per the Foreign Trade Policy, 2015-20, the export and import of zirconium ores and concentrates under ITC (HS) Code 26151000 are freely allowed. The notification reads:

- S.O. 1592(E)—In pursuance of Clauses (f) and (g) of Sub-section (1) of Section 2 and Section 3 of the Atomic Energy Act, 1962 (No. 33 of 1962) and in supersession of the notifications of the Government of India in the Department of Atomic Energy vide Number S.O. 61(E)

dated 20th January 2006, the Central Government hereby notifies the substances, equipment and technology specified in the Schedule appended hereto as 'Prescribed Substances, Prescribed Equipment and Technology'.

- Under Category 0A303 and under Heading Nuclear materials, nuclear-related other materials, equipment and technology — "Zirconium with hafnium content of less than 1 part to 500 parts of zirconium by weight (i.e., less than 2,000 ppm) in the form of metal, alloys containing more than 50% zirconium by weight, compounds, manufactures thereof, waste or scrap of any of the foregoing".
- G.S.R. 134(E), dated 20.02.2019 — In exercise of the powers conferred under Section 11B of the Mines and Minerals (Development and Regulation) Act, 1957 (67 of 1957) and Rule 36 of the Atomic Minerals Concession Rules, 2016, the Central Government hereby makes the following amendments further to amend the Atomic Minerals Concession Rules, 2016, namely—
 - (1) These rules may be called the Atomic Minerals Concession (Second Amendment) Rules, 2019.
 - (2) They shall come into force on the date of their publication in the Official Gazette.

In the Atomic Minerals Concession Rules, 2016, for Schedule A, the following Schedule has been substituted, namely –

PARTICULARS OF THRESHOLD VALUE FOR ATOMIC MINERALS

[See Rule 2 (1) (m) and Rule 36]

1.	Beryl and other beryllium-bearing minerals	0.1% BeO (1000 ppm BeO) of the rock or 10 kg/tonne Beryl in excavated material.
2.	Lithium-bearing minerals	0.5% (5000 ppm) Li ₂ O in ore, except brine (200 ppm Li, i.e. 200 g/tonne Li).
3.	Minerals of the 'rare earths' group containing uranium and thorium	60 ppm U ₃ O ₈ and/or 250 ppm ThO ₂ in ore
4.	Niobium-bearing minerals	100 ppm (Nb+Ta) ₂ O ₅ (100 g/tonne) in ore.
5.	Phosphorites and other phosphatic ores containing uranium	60 ppm U ₃ O ₈ in ore.
6.	Pitchblende and other uranium ores	60 ppm U ₃ O ₈ in ore, except in Singhbhum Shear zone in Jharkhand where the threshold value will be 150 ppm U ₃ O ₈ in ore.
7.	Titanium bearing minerals and ores (ilmenite, rutile and leucosene)	In case of titanium-bearing minerals occurring in hard rock, 60 ppm U ₃ O ₈ and/or 250 ppm ThO ₂ in the rock. All cases of titanium-bearing minerals occurring in Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e. the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade.
8.	Tantalum-bearing minerals	100 ppm (Ta+Nb) ₂ O ₅ (100 g/tonne) in ore.
9.	Uraniferous allanite, monazite and other thorium minerals	60 ppm U ₃ O ₈ and/or 250 ppm ThO ₂ All cases of Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e. the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade.

10	Uranium-bearing tailings left over from ores after extraction of copper and gold, ilmenite and other titanium ores.	60 ppm U ₃ O ₈ and/or 250 ppm ThO ₂ .
11.	Zirconium-bearing minerals and ores including zircon.	All cases of zirconium-bearing minerals occurring in Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e. the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade. In other cases, zircon containing less than 2000 ppm of Hafnium.
12.	Beach Sand Minerals i.e. economic heavy minerals found in the teri or beach sand, which include ilmenite, rutile, leucoxene, garnet, monazite, zircon and sillimanite	All cases of Beach Sand Minerals and other placer deposits in association with monazite are notified as above threshold (i.e. the threshold is 0.00% monazite in Total Heavy Minerals), irrespective of monazite grade.

As per (1) Mines and Minerals (Development and Regulation) Amendment Act, 2023 (vide Notification no. 16 of 2023 dated 09.08.2023) and (2) Atomic Minerals Concession (Amendment) Rules, 2024 (vide Notification no. 97 dated 14.02.2024), six minerals, namely, (i) Beryl and other beryllium-bearing minerals (ii) Lithium-bearing minerals, (iii) Niobium-bearing minerals, (iv) Titanium-bearing minerals and ores, (v) Tantalum-bearing minerals and (vi) Zirconium-bearing minerals and ores have been delisted from the list of atomic minerals specified in Part-B of the First Schedule to the Act. These minerals have various applications in space industry, electronics, communications, energy sector, electric batteries and are critical in net-zero emission commitment of India. Due to their inclusion in the list of atomic minerals, their mining

and exploration is reserved for government entities. Due to removal of these minerals from the said list, exploration and mining of these minerals has been opened up for the Private Sector as well. As a result, exploration and mining of these minerals is expected to increase significantly in the country.

WORLD REVIEW

World reserves of zirconium are placed at 74 million tonnes in terms of ZrO₂. The world's largest reserves are with Australia (74%), South Africa (8%) and Senegal (4%). The world production of zirconium minerals was estimated at 1.3 million tonnes in 2023 which is same as that of in the previous year. Australia (36%), South Africa (31%) and Mozambique & Indonesia (7% each) are the principal producers of zirconium minerals (Tables-7 & 8).

Table – 7: World Reserves of Zirconium
(By Principal Countries)

<i>(In thousand metric tonnes, ZrO₂ content)¹</i>	
Country	Reserves
World: Total (rounded)	74000
Australia [#]	55000
South Africa	5600
Senegal	2600
Madagascar	2300
Mozambique	1500
USA	500
Sierra Leone	290
China	72
Kenya	18
Indonesia	NA
Other countries	5700

Source: USGS, Mineral Commodity Summaries, 2024.

¹: Calculated ZrO₂ content as 65% of gross weight.

[#]: For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 20 million tonnes, ZrO₂ content.

Table – 8: World Production of Zirconium Minerals

(By Principal Countries)

(In metric tonnes)

Country	2021	2022	2023
World: Total	1200000	1300000	1300000
Australia ^(f)	427565	493992	462126
South Africa	320000	320000	*400000
Mozambique	100158	103553	96800
Indonesia ^(e)	*54000	*97000	*95000
China	*35000	*50000	*50000
Senegal	64000	57000	48000
USA	*35000	*40000	*40000
Madagascar ^(b)	23100	26800	*34400
Kenya	25928	90698	21094
Other countries	115249	20957	52580

Source: BGS, World Mineral Production, 2019-23.

Note: In this table the term 'zirconium minerals' is understood to mean zircon, unless otherwise stated.

(b): Exports,

(e): Conservative BGS estimates, based on exports,

(f): Years ended 30th June of that stated.

(*): Estimate.

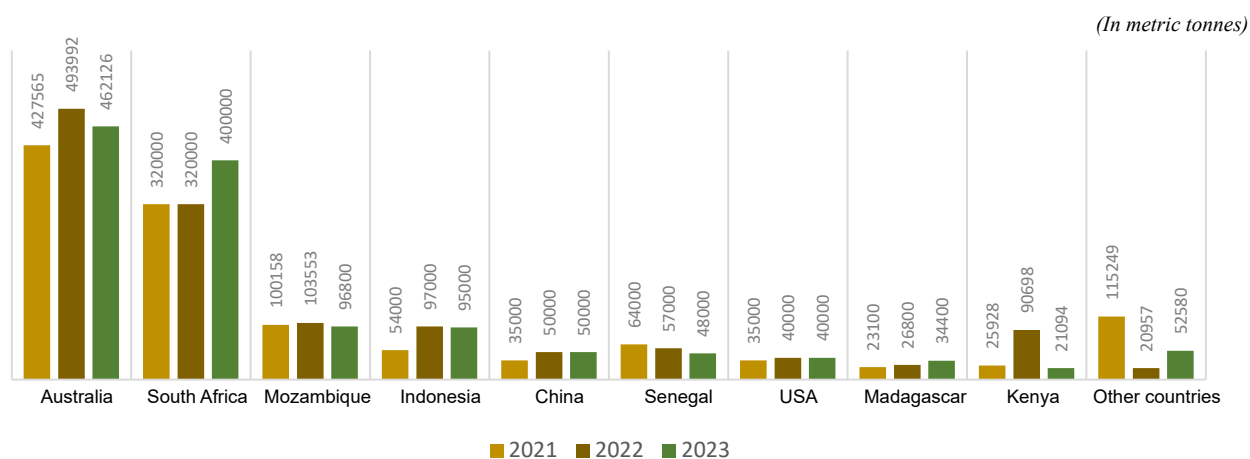


Fig. 2: Countrywise Production of Zirconium Minerals

FOREIGN TRADE

Exports

In 2023-24, exports of zirconium ores and concentrates increased tremendously by 360% to 198 tonnes as compared to 43 tonnes in the previous year. The export was mainly

confined to UAE (50%), Italy (24%), USA (19%) and Kuwait (7%). The product-wise export of zirconium in 2023-24 was negligible as compared to 18 tonnes in the previous year. The export of Ferro-Zirconium increased by 19% to 38 tonnes in 2023-24 as compared to 32 tonnes in the previous year (Table-9 to 11).

Table – 9: Export of Zirconium Ores and Conc.

(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	43	15779	198	50059
UAE	43	15649	99	26140
USA	--	--	37	14618
Kuwait	--	--	14	6297
Italy	--	--	48	3002
Seychelles	--	--	++	2
Iran	++	++	++	++
Japan	--	--	++	++
Germany	++	130	--	--

Figures rounded off

Table – 10: Exports of Zirconium**(By Product)**

Product	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total	18	41576	++	54088
Zirconium and Scrap	9	20788	++	27044
Zirconium Other	9	18937	++	26746
Zirconium Unwrought: Other	++	1851	++	298

*Figures rounded off***Table – 11: Export of Ferro-zirconium****(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	32	10215	38	12535
South Africa	--	--	13	4067
Netherlands	--	--	4	2969
France	--	--	10	2545
Germany	26	6128	8	2179
Saudi Arabia	--	--	2	508
Bhutan	--	--	1	260
China	--	--	++	7
Malaysia	3	2763	--	--
UAE	3	857	--	--
Brazil	++	304	--	--
Other Countries	++	163	++	++

Figures rounded off

Imports

In 2023-24, imports of zirconium ores and concentrates decreased by nearly 1% to 82,257 tonnes from 82,831 tonnes in the previous year. The main suppliers were Indonesia (36%), Australia (19%) and Malaysia & South Africa (13% each). The product-wise import of zirconium in 2023-24

was 64 tonnes as compared to 80 tonnes in the previous year. Further, the Import of Ferro-zirconium decreased by 21% to 522 tonnes in 2023-24 as compared to 658 tonnes in the previous year. The import was mainly from China (99%) (Tables-12 to 14).

Table – 12: Imports of Zirconium Ores and Conc.**(By Countries)**

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	82831	15163201	82257	13937396
Indonesia	22741	4367177	29239	4741094
Australia	22330	3951295	15397	2854364
Malaysia	16338	2970528	10639	1863055
South Africa	11089	1891198	10444	1842485
USA	3591	704551	5771	1021369
Vietnam	404	87778	3482	535586
Senegal	1971	385678	2320	389435
Thailand	1764	320304	1638	231445
UAE	--	--	1145	171424
China	2372	422312	878	122400
Other Countries	231	62380	1304	164739

Figures rounded off

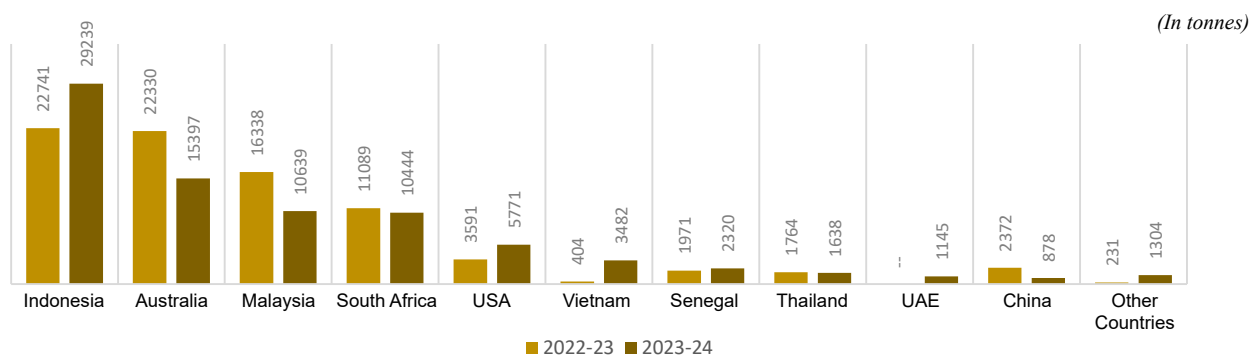


Fig. 3: Countrywise Imports of Zircon Ores & Conc.

Table – 13: Imports of Zirconium
(By Product)

Product	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
Total	80	282026	64	189088
Zirconium and Scrap	40	141013	32	94544
Zirconium Other	36	135267	32	91126
Zirconium Unwrought: Other	4	5746	++	3418

Figures rounded off

Table – 14: Imports of Ferro-zirconium
(By Countries)

Country	2022-23 (R)		2023-24 (P)	
	Qty (t)	Value (₹ '000)	Qty (t)	Value (₹ '000)
All Countries	658	162218	522	116181
China	637	157656	517	114372
Poland	--	--	5	1809
Hong Kong	10	4106	--	--
UK	11	456	--	--

Figures rounded off

FUTURE OUTLOOK

As per Annual Report 2023-24 of IREL, the collapse of Chinese construction industry has led to demand contraction for Zircon which finds use in tiles and sanitaryware. Though the Chinese Government has taken measures to boost investment in the construction sector, lack of financial support from local governments means the weakness in the market will be prolonged. In Europe, a major consumption hub of Zircon outside of China, producers of tiles are lowering the blend of zircon to reduce the input costs. In contrast, the initiative of the Government of India towards Pradhan Mantri Awas Yojna (PMAY), i.e., Housing for All, Gati Shakti Scheme and Smart Cities have resulted in robust demand for zircon and will continue to be drivers for the sale of zircon. However, the impact of negative global factors are likely to heavily weigh down on the demand for zircon as the year progresses.

The growing use in construction and industrial applications is a significant driver for the zirconium market in the future. Further, zirconium's role in the Nuclear Industry is pivotal, as it is used for cladding fuel rods in

reactors. Its excellent resistance to corrosion by water and its transparency to neutrons makes it an ideal material for nuclear power stations. The demand for zirconium in this sector supports substantial market growth. Zirconium alloys, formed by alloying with metals, such as, nickel, titanium, or iron, find applications in aerospace, automotive and chemical processing industries. The versatility and demand for these alloys across multiple sectors further drive market expansion. Rapid urbanisation and infrastructure development in emerging economies are boosting the demand for zirconium-based materials, especially in tiles, sanitaryware and architectural applications.

During 2024, the demand for zircon sand has been soft, mainly affected by the decline in the Chinese construction industry that accounts for 50% of the global zircon demand. As per report namely "Zirconium Market Size, Share, Industry, Forecast and Outlook (2024-2031)", the global zirconium market reached USD 1.4 billion in 2022 and is expected to reach 2.5 billion by 2030, growing with a CAGR of 6.9% during the forecast period 2024-2031.



INDIAN MINERALS YEARBOOK

2024